The Claimed Complex Structure on S^6
Validate, Repair, or Disprove

Working mathematical notebook following the three-stream proof audit

4 September 2026

Abstract

This paper is the consolidated successor to the former 56-page workbench: it incorporates the nonduplicate proofs from the subsequent 18-page repair branch and the later Reviewer B/Codex validation work. It validates, repairs, or attempts to disprove a reconstructed manuscript claiming a compact complex threefold diffeomorphic to S^6 and of algebraic dimension one. In its present form the workbench gives complete proofs of the simultaneous elliptic-orientation normalization. Let $\Delta \simeq (\mathbb{Z}/3) * (\mathbb{Z}/4)$ act on $\mathfrak{h}_z$, let g_1, g_2 have fixed points z_1, z_2, put $g_0 = (g_1 g_2)^{-1}$, and let $\tau : \mathfrak{h}_z \to \mathfrak{h}$ be the normalized period lift. Define

$$\varphi : \Delta \longrightarrow \mathrm{PSL}_2(\mathbb{Z}), \qquad \tau \circ g = \varphi(g) \circ \tau,$$

and

$$\mathfrak{S}_1(w) = \frac{w-1}{w}, \qquad \mathfrak{S}(w) = -\frac{1}{w}, \qquad \mathfrak{P}(w) = w - 1.$$

Then

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = -i, \qquad g_0 \text{ is parabolic},$$

and

$$\varphi(g_1) = \mathfrak{S}_1, \qquad \varphi(g_2) = \mathfrak{S}, \qquad \varphi(g_0) = \mathfrak{P}, \qquad \mathfrak{S}_1\mathfrak{S}\mathfrak{P} = 1.$$

The sign at z_2 is fixed by the simultaneous parabolic normalization; the quadratic ramification calculation alone gives only $(g_2'(z_2))^2 = -1$. The resulting nondegenerate period family descends, and the middle quotient is a proper holomorphic submersion with compact torus fibres. The toric cusp filling, both finite logarithmic fillings, all three punctured-overlap biholomorphisms, and the global analytic gluing yield a connected compact Hausdorff smooth complex threefold and a proper surjective holomorphic map $f : X \to \mathbb{P}^1$ with connected fibres. Its full cusp fibre W is reduced, irreducible, nonnormal, and normal crossings, with normalization dP_6; the other two full scheme-theoretic fibres have multiplicities three and four. The workbench also gives complete proofs of integral nearby-cycle specialization, the global Leray page and three primitive transgressions, the equivariant toric collapse, its integral cellular map and its equality with specialization, the based van Kampen presentation, sphere recognition, and the very-general-fibre Néron–Severi and meromorphic-descent calculation. For the constructed threefold it therefore verifies

$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = 2, \qquad \mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1.$$

The same verified object satisfies the stated hypotheses of Campana–Demailly–Peternell Theorem 2.2 and contradicts its conclusions, so the workbench records a counterexample to that theorem as stated as well as two exact defects in its printed dependency chain. It also proves the strongest corrected conclusion that survives those defects: existence of one L_0 with $H^2(X, T_X \otimes L_0) = 0$ implies generic off-degree-one vanishing and $c_3(X) \leq 0$; for reduced full fibres the invalid singular-fibre shortcut is replaced by injectivity of the exact conductor connecting map. Because $c_3(X) = 2$, contraposition gives $H^2(X, T_X \otimes L_0) \neq 0$ for every line bundle L_0; independently, the conductor calculation gives $R^2 f_*(T_X \otimes L) \neq 0$ for every L, so the stronger relative sufficient condition also fails. The independent alternate certificate is now complete as well: the actual finite-band maps are $\alpha_q = (\pi_{1*}, -\pi_{2*})$ in primitive integral bases, the cusp attachment reduces to blocks with determinant $\pm p$, and the resulting

obstruction squares commute with all three Leray transgressions. The candidate's Proposition 10.8 is also verified: the structure-sheaf direct images, degree-one base change, Picard line, exact finite normal characters, and uniform period/countability argument are reconstructed in full without using the alternate collapse/Mayer–Vietoris route. Independently of the global topology calculation, the conductor construction also proves the required direct-image nonvanishing for every $L \in \mathrm{Pic}^0(X)$. On the cusp model the invariant form

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2}$$

first proves intrinsically that $\omega_{N_0} \simeq \mathcal{O}_{N_0}$. After global gluing, the open embedding $N_0 \hookrightarrow X$ gives

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}.$$

Write $\nu : \widetilde{W} \simeq \mathrm{dP}_6 \to W$ for the normalization. For arbitrary line bundles the exact obstruction is the restriction homomorphism $\nu^* i^* : \mathrm{Pic}(X) \to \mathrm{Pic}(\mathrm{dP}_6)$. A direct Leray-edge calculation gives $e(c_1(L)) \in E_\infty^{0,2} \subseteq \ker d_2^{0,2} = 0$; proper-base-change naturality then makes $\nu^* i^* L$ trivial for every L. This avoids invoking the packaged theorem $H^2(X;\mathbb{Z}) = 0$ but reuses its decisive integral Leray transgression. A second all-Pic proof now avoids every global Leray differential. Put $q := uw + 6\gamma\delta \in \bigwedge^2 V = H^2(F;\mathbb{Z})$. If $E = c_1(L|_F)$, the order-four filling forces $E = 2nq$. The pair-of-pants defect classes satisfy

$$\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0, \qquad \bar{d}_1 = 4n[\delta], \quad \bar{d}_2 = -3n[\delta], \quad \bar{d}_0 = 0,$$

where the first two values come from the exact logarithmic-cover norm maps and the last from a literal cusp solid torus. Hence $n = 0$; local integral specialization then gives $\nu^* i^* L \simeq \mathcal{O}$. The cusp-local map is nevertheless nonzero: a descended line bundle on the cusp neighbourhood has generic Chern class $-2q$ and normalized central degree vector $(-12, 2, 0, -12, 2, 0)$. The third global certificate therefore incorporates both finite fillings analytically. For $c_1(L|_F) = kq$, its $\mathcal{O}^*$-factor defects satisfy

$$(\bar{d}_1, \bar{d}_2, \bar{d}_0) = (2k, -\tfrac{3}{2}k, 0)[\delta], \qquad \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0,$$

The finite norm identity is computed in the typed affine deck group

$$\Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda \rangle$$

Its extended defect satisfies

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I)a_j, \qquad \widetilde{d}_j(\tau_v) = \iota_v E,$$

and therefore

$$N_{T_j} d_j = \widetilde{d}_j((Q_{j,\log})^{m_j}) = \widetilde{d}_j(\tau_{v_j}) = \iota_{v_j} E.$$

Thus the displayed balance gives $k = 0$; the cusp equality comes from the argument principle in a Stein chart. The comparison uses the actual toric-open direction $U_{\rho_v} = D(z_w z_u) \subset U_\sigma$ and a contractible Stein chart S_σ, so adjacent central frames have zero horizontal winding. If μ_λ^+ is multiplication by χ^{m_λ} on $\mathcal{O}(D_a)$, the required map on the negative divisor is the typed inverse-dual

$$\mu_\lambda^- = ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

Put $M = L|_{N_0}$. If (d_1, d_2, d_3) are the independent normalized central degrees, then

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix},$$

$$E_M((\lambda, 0), (0, m)) = -B(B_0\lambda, m).$$

The preceding balance gives $k = 0$, hence $E_M = c_1(L|_F) = 0$. Unimodularity of B_0 now forces $B = 0$ and all six degrees to vanish. This third proof avoids both global integral Leray and local integral specialization/bounding-chain input. The workbench also completes the Hodge calculation left open in the manuscript. The torsion-free absolute cotangent direct images are

$$R^0 f_* \Omega_X^1 \simeq \mathcal{O}(-2), \quad R^1 f_* \Omega_X^1 \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}, \quad R^2 f_* \Omega_X^1 \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

Hence $h^{1,2} = h^{2,1} = 1$, $h^{1,1} = h^{2,2} = 2$, and every Hodge number is known. The Frölicher spectral sequence does not degenerate at E_1: its two middle $d_1 = \partial$ rows are exact

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathbb{C}^2 \longrightarrow \mathbb{C} \longrightarrow 0.$$

Thus $\dim_{\mathbb{C}} E_1 = 10$, while $E_2 = E_\infty$ and $\dim_{\mathbb{C}} E_2 = 2$. Put

$$\Sigma = p_0 + p_1 + p_2, \quad D = f^{-1}(\Sigma)_{\mathrm{red}}, \quad \mathcal{L} = \Omega^1_{\mathbb{P}^1}(\log \Sigma), \quad \mathcal{E} = \Omega^1_X(\log D)/f^*\mathcal{L}, \quad \mathcal{G} = R^1 f_* \mathcal{O}_X.$$

Then

$$f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1), \quad R^1 f_* \mathcal{E} \simeq \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2}, \quad R^2 f_* \mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1),$$

and all three sheaves are locally free and commute with base change. The global calculation uses the actual cyclic invariant lattices, the simple divisor $p_1 + p_2$ of the logarithmic Kodaira–Spencer determinant, and the normalized global scalar at the order-four point. At a finite fibre write $t = s^m$, with $(m, a) = (3, 2)$ or $(4, 1)$, and, after removing the common ds from $D_{\mathrm{KS}} = (d_{rs})$, put

$$A = \frac{s}{m} d_{00}, \quad B = \frac{s^{1-a}}{m} d_{01}, \quad C = \frac{s^{1-a}}{m} d_{10}, \quad D_0 = \frac{s^{m+1-2a}}{m} d_{11}.$$

In the invariant source basis $(e_0, s^{m-a}e_1)$, target basis $(b_0, s^a b_1)\ell$, and ordered degree-one source basis $e_0 b_0, s^a e_0 b_1, s^{m-a} e_1 b_0, e_1 b_1$,

$$[\kappa_0] = \begin{pmatrix} A & tB \\ C & D_0 \end{pmatrix}, \qquad [\kappa_1] = (-C, A, -D_0, B).$$

The exact finite residue cancellation is

$$\mathrm{res}_{1,j}(x b_0 + y s^a b_1) = m_j x [b_0], \qquad \mathrm{res}_{1,j}\, \kappa_0 = 0,$$

$$\ker \mathrm{res}_{1,j} = \mathbb{C}\, s^a b_1 = \mathrm{im}(\kappa_0 \bmod t).$$

At the cusp put

$$Q_{\mathrm{cusp}} = \nu_* \mathcal{O}_{\mathrm{dP}_6}/\mathcal{O}_W, \qquad \eta_i = d \log x_i, \qquad q_1 = (-1, 0, 1), \qquad q_2 = (0, -1, -1),$$

and, for the normalization connecting maps, set

$$\alpha_i = \delta_0(q_i), \qquad \theta = -\delta_1^{-1}(\alpha_1 \smile \alpha_2).$$

The standard Čech sign gives

$$\delta_1\big(\rho_*(\alpha \otimes \eta_i)\big) = -\alpha \smile \alpha_i,$$

hence the exact row $(0, 1, -1, 0)$ with target $\mathbb{C}\theta$. Let

$$\xi = \frac{\partial}{\partial \zeta_2}, \qquad h = (a(z), 1) \begin{pmatrix} d\zeta_1 \\ d\zeta_2 \end{pmatrix}, \qquad h(\xi) = 1$$

be the unique global lift. At the order-four point, in the genuine invariant frame $\omega_2 = s\, u_2(s)(\gamma \wedge u)\ell$, $u_2(0) = 1$, one has $a(z_2) = \frac{1}{2}$, $\dot\tau(z_2) = 0$, $K(z_2) = 6\dot\mu(z_2)^2 \neq 0$, and

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\big(a(z_2)\dot\tau(z_2) + \dot\mu(z_2)\big) = -\tfrac{1}{4}\dot\mu(z_2) \neq 0.$$

The stalks at $t = xy$ and $t = xyz$ show that $\Omega^1_{X/B}$ is torsion-free but non-locally-free at the cusp. On a finite cover $t = s^{m_j}$, its exact local module is

$$R\, dy_1 \oplus R\, dy_2 \oplus \big(R/(s^{m_j - 1})\big) ds,$$

so its torsion is supported exactly at the two multiple fibres. No step assumes that direct image commutes with duality across a multiple fibre. McHugh's compact-threefold identities then determine the remaining Bott–Chern and Aeppli invariants. In total degrees 0 through 6,

$$(h^k_{BC}) = (1, 0, 2, 2, 2, 2, 1), \qquad (h^k_A) = (1, 2, 2, 2, 2, 0, 1),$$

and the non-$\partial\bar\partial$ degrees are $(0,2,4,4,4,2,0)$. Thus no Bott–Chern or Aeppli dimension remains open. On a marked smooth fibre,

$$\eta = u\wedge w + 6\gamma\wedge\delta, \qquad \eta^2 = 12\,\gamma\wedge u\wedge w\wedge\delta.$$

If Q_F is its displayed Dolbeault representative, then

$$\int_{(F_b,\mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F^2 = -12;$$

the marked and complex orientations are opposite because $\det_{\mathbb{R}}\Pi < 0$.

Let $\mathfrak{g}\in H^{0,1}_{\bar\partial}(X)$ be the distinguished degree-one class and let $\mathfrak{q}\in H^{1,1}_{\bar\partial}(X)$ be a chosen coherent-Leray lift of η. The latter is determined only modulo $\mathbb{C}\,d_1\mathfrak{g}$. Choose global $\bar\partial$-closed smooth representatives

$$G\in\mathcal{A}^{0,1}(X), \qquad Q\in\mathcal{A}^{1,1}(X)$$

of $\mathfrak{g},\mathfrak{q}$. Their marked smooth-fibre cohomology classes have the exact period-coordinate formulas proved below; no literal equality of an arbitrary global Q with the displayed coefficient formula is used. With $\omega = c_1(\mathcal{O}_{\mathbb{P}^1}(1))$,

$$\exp(12\mathfrak{g}) = f^*\mathcal{O}_{\mathbb{P}^1}(1), \qquad c_{1,BC}(\exp(z\mathfrak{g})) = z\mathsf{A} + \bar z\,\mathsf{B},$$
$$c_{1,BC}(f^*\mathcal{O}_{\mathbb{P}^1}(1)) = 12(\mathsf{A}+\mathsf{B}).$$

Put $\mathbf{1} = [1]_{BC}$, and choose a smooth top form ϑ with $\int_X\vartheta = 1$, so $\mathsf{T} = [\vartheta]_{BC}$. Then

$$\mathsf{A} = [\partial G],\ \mathsf{B} = [\bar\partial\bar G],\ \mathsf{C} = [\partial Q],\ \overline{\mathsf{C}},\ \mathsf{D} = [\partial(G\wedge Q)],\ \overline{\mathsf{D}},\ \mathsf{E} = [\partial(Q^2)],\ \overline{\mathsf{E}},\ \mathsf{T}$$

are bases of all nonzero positive-degree Bott–Chern groups; together with $\mathbf{1}$, they give every nonzero group. The two load-bearing fibre pairings in the complex orientation are

$$\int_{F_b} Q_F^2 = -12, \qquad \int_{F_b} Q_F\wedge\Gamma\wedge d\zeta_2 = -1.$$

They prove the required Dolbeault nonvanishing, while the constant fibre coefficients give the typed identity

$$i_b^*(d_1[G\wedge Q]_{\bar\partial}) = [\partial_F(\Gamma\wedge Q_F)]_{\bar\partial} = 0.$$

If N_{BC} is the span of the nine displayed positive-degree classes, then

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1}\oplus N_{\mathrm{BC}}, \qquad N_{\mathrm{BC}}^2 = 0.$$

Thus every product of two positive-degree Bott–Chern classes vanishes; see theorems 43.4 and 43.5. The result does not select a preferred literal global coefficient atlas for G,Q through the finite fillings and cusp; that presentation choice is not an open mathematical premise. The marked positive radial cusp also carries an exact auxiliary flat-metric calculation. With

$$L_T = T + \operatorname{Im} h_T, \qquad q_T = L_T - \operatorname{Im} b_T, \qquad m_T = \operatorname{Im}\mu_T, \qquad D_T = L_T q_T + 6m_T^2,$$

where $H_{\mathrm{cusp}} = \operatorname{Im} h(0)$ and $B_{\mathrm{cusp}} = \operatorname{Im} b(0)$, the complete dual lattice is $P_T^{-\top}\mathbb{Z}^4$, and its shortest nonzero vector gives

$$\lambda_1(F_T,g_T) = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H_{\mathrm{cusp}} + 2\min\{B_{\mathrm{cusp}},0\}}{T} + O(T^{-2})\right),$$

while the unit-volume metric $\widehat g_T = D_T^{-1/2}g_T$ gives

$$\lambda_1(F_T,\widehat g_T) = \frac{4\pi^2}{T}\big(1+O(T^{-1})\big) \longrightarrow 0.$$

For fixed $E\ge 0$, the scalar eigenvalue count with Fourier multiplicity is

$$N_T(E) := \#\{k\in\mathbb{Z}^4 : 4\pi^2\|P_T^{-\top}k\|^2\le E\} = \pi D_T\sum_{n\in\mathbb{Z}^2}\left(\frac{E}{4\pi^2} - \|n\|^2\right)_+ + O_E(T).$$

In particular, for $0 < E < 4\pi^2$, $N_T(E) = ED_T/(4\pi) + O_E(T)$: the number of soft modes grows quadratically, rather than only one eigenvalue tending to zero. The mean-zero vertical direct integral is a single injective self-adjoint operator with 0 in its essential spectrum. The unmodified regular four-torus fibres F_T have native pointed limit exactly $\mathbb{T}^2 \times \mathbb{R}^2$. A separate ancillary construction changes both the fibre lattice and the cusp base: for an integer $N \geq 1$, after the added operations

$$t_c = v^N, \qquad \Lambda^{(N)} = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta} \rangle,$$

the punctured degree-N^2 fibre covers have pointed local limit $\mathbb{R}^4$ and momentum lattices becoming dense in all four directions. The extension across the central fibre is explicit. With source residual term $C_N(v) = C(v^N)/N$, put

$$f_N : N_0^{\langle N \rangle} \longrightarrow \Delta_v, \qquad W_N = (f_N^{-1}(0))_{\mathrm{red}}.$$

Toric multiplication descends to

$$\kappa_N : N_0^{\langle N \rangle} \longrightarrow N_0, \qquad \deg \kappa_N = N^3, \qquad \kappa_N^* W = NW_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N.$$

Factoring through the normal base change $N_0[N] = N_0 \times_{\Delta_t} \Delta_v$ gives a generic degree-N^2 map Q_N; for $N > 1$ it is quasi-étale but nonflat exactly over the double and triple strata. On the normalized central fibre it is the toric map $[N] : \mathrm{dP}_6 \to \mathrm{dP}_6$. This is a completed *local* cusp morphism, not a cover of the entire base or of X.

Independently, the compact cusp lattice $K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle$ is not preserved by the finite monodromies, but its invariant closure is $L_\Lambda = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$. The invariant character lattice is $\mathbb{Z}\varepsilon_\gamma$; fix the displayed primitive generator. Its real group quotient has exact circle circumference $\ell_T = D_T/\sqrt{L_T^2 + m_T^2} = T + H_{\mathrm{cusp}} - B_{\mathrm{cusp}} + O(T^{-1})$. Four synchronized copies therefore give a genuine $\mathbb{T}^4$-quotient over the full regular base whose pointed cusp limit is $\mathbb{R}^4$, and four copies are minimal for this invariant-kernel mechanism. The two finite fillings extend to explicit Seifert targets after simultaneous-root normalization, with central degrees 27 and 64. At the cusp, two paths with the same limit and different character values still prove that the *literal fixed-target* quotient cannot extend; no finite base cover repairs the native two-copy kernel.

A different, exact compactification changes the codomain rather than the character values. After choosing a base point, collapsing the inner and outer cusp cross-sections gives a based quotient (denoted q_2 in the detailed construction, and written $\rho_{2,\mathrm{cone}}$ here to avoid collision with an earlier Čech vector)

$$\rho_{2,\mathrm{cone}} : (X, x_0) \longrightarrow (\mathcal{A}_{\mathrm{cone}}, *) = \big(\mathrm{Susp}(S^1 \times \overline{\mathbb{T}}^2), *\big).$$

Its Cartesian square restricts on synchronized radial fibres to $\pi_T^{\times 2} : F_T^{\times 2} \to Q_T^{\times 2}$, and basedness induces $\rho_{2,\mathrm{cone}}^{\wedge 2} : X^{\wedge 2} \to \mathcal{A}_{\mathrm{cone}}^{\wedge 2}$. Likewise the regular and finite-filled circle targets glue to a compact three-manifold $\mathcal{M}_{\mathrm{Seif}}$ with boundary $\mathbb{T}^2$; coning that boundary gives a based quotient (denoted q_1 in the detailed construction) $\rho_{1,\mathrm{cone}} : (X, x_0) \to (\mathcal{C}_{\mathrm{cone}}, *)$ whose fourth Cartesian power restricts to $\chi_{\gamma,T}^{\times 4} : F_T^{\times 4} \to C_T^{\times 4}$, and hence induces a fourth smash power.

Separately, composing $\rho_{2,\mathrm{cone}}$ and the cusp-local circle quotient $q_{1,\mathrm{cusp}}$ with their respective base-angle-forgetting maps gives continuous surjective quotient maps

$$\widehat{\rho}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{\rho}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8,$$

after a based choice $X \cong_{\mathrm{diff}} S^6$. Both full sphere carriers are based null-homotopic, by two distinct calculations. In the detailed notation $\widehat{\rho}_\gamma = \widehat{q}_\gamma$ and $\widehat{\rho}_3 = \widehat{q}_3$. For any based map $f : S^6 \to S^2$, the smash square has the exact factorization

$$S^{12} = S^6 \wedge S^6 \xrightarrow{f \wedge \mathrm{id}_{S^6} = \Sigma^6 f} S^2 \wedge S^6 = S^8 \xrightarrow{\mathrm{id}_{S^2} \wedge f} S^2 \wedge S^2 = S^4.$$

Freudenthal stability and the vanishing fourth stable stem give $\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0$, so $f \wedge f$ is based null-homotopic. Associativity then writes $f^{\wedge r} = (f \wedge f) \wedge f^{\wedge (r-2)}$, and hence $[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r})$ for every $r \geq 2$. In particular, the entire quotient map $\widehat{\rho}_\gamma^{\wedge 4} : S^{24} \to S^8$, not merely its moving four-sphere restriction, is based null-homotopic.

For the other carrier, let $g : S^6 \to S^3$ be any based map. The exact smash factorization is

$$g \wedge g = (\Sigma^3 g) \circ (\Sigma^6 g) : S^{12} \longrightarrow S^9 \longrightarrow S^6.$$

Write st for stabilization. The exact low-stem data are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \qquad \pi_3^{\mathrm{S}} = \mathbb{Z}/24\langle\nu\rangle, \qquad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\langle\nu^2\rangle,$$

$$\pi_{12}(S^6) = \mathbb{Z}/2\langle\nu_6 \circ \nu_9\rangle, \qquad \mathrm{st} : \pi_{12}(S^6) \xrightarrow{\cong} \pi_6^{\mathrm{S}}.$$

The full groups are tabulated in [9, p. 339]; the signed smash formula and the unstable stabilization calculation are [12, Propositions 3.1 and 5.11]. Since $12[g] = 0$, its stable image has the form $\mathrm{st}([g]) = 2a\nu$ for some $a \in \mathbb{Z}/12$. Stable smash multiplication therefore gives

$$\mathrm{st}([g \wedge g]) = \mathrm{st}([g])^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

The displayed stabilization isomorphism now forces $[g \wedge g] = 0 \in \pi_{12}(S^6)$. Hence every $g^{\wedge r}$ with $r \geq 2$ is based null-homotopic as well. In particular,

$$[\widehat{\rho}_3^{\wedge 2}] = [\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{\rho}_\gamma^{\wedge 4}] = [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

The one-copy classes $[\widehat{q}_3] \in \pi_6(S^3)$ and $[\widehat{q}_\gamma] \in \pi_6(S^2)$ remain uncomputed. The unstable classes of the conical-target maps $\rho_{2,\mathrm{cone}}^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $\rho_{1,\mathrm{cone}}^{\wedge 4}$ also remain **[OPEN]**; nullity after postcomposition to a sphere does not prove nullity before that postcomposition. See theorems 9.21, 9.22, 9.25 and 9.27.

Their targets contain moving topologically embedded four-spheres, equipped separately with their intrinsic quotient length metrics, $S_T^2 \wedge S_T^2$ and $C_T^{\wedge 4}$. With their quotient length metrics and midpoint base points, every radius-R ball is exactly Euclidean once $T - C_{\mathrm{sys}} > 4R$, respectively once $\ell_T > 4R$; thus both moving four-spheres converge pointed local-isometrically and Gromov–Hausdorff to $\mathbb{R}^4$, with smooth convergence on those Euclidean neighborhoods. Deleting the common smash basepoint gives literal nonclosed topological four-cells

$$(Q_T \setminus K_T^{(1)})^2 \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad (C_T \setminus \{0\})^4 \cong \mathbb{R}^4 \hookrightarrow S^8.$$

The based inclusion maps

$$S_T^2 \wedge S_T^2 \hookrightarrow S^6, \qquad C_T^{\wedge 4} \hookrightarrow S^8$$

are null-homotopic. These slice-inclusion null-homotopies are separate from the global equalities $[\widehat{\rho}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6)$ and $[\widehat{\rho}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8)$. Two copies are rank-minimal for the native cusp-local quotient, while four are rank-minimal for a primitive full-monodromy-invariant quotient with four expanding directions. Here $C_{\mathrm{sys}} > 0$ is a fixed bound for the holomorphic error in the radial period lattice.

Exact five-manifolds with boundary interpolate the positive four-torus slices with a boundary fibre $\mathbb{R}^4$. There are continuous surjective comparison maps to the compact cone carriers which are homeomorphisms away from that boundary and collapse it to the cone vertex; however, these maps are nonproper and nonquotient. Thus they are not topological blow-downs or resolutions, and the compact carriers are not the quotients of the five-manifolds by their $\mathbb{R}^4$ boundary fibres via these maps. For every unrescaled fixed compatible target metric the radial slices collapse to the cone point. A fixed cusp-adapted metric can recover the slice limits only after the blow-up factors T/r_T and ℓ_T/r_T; the corresponding full ambient blow-ups are $\mathbb{R}^8$ and $\mathbb{R}^{12}$, not $\mathbb{R}^4$. Thus the sphere constructions yield exact carrier slices with intrinsic $\mathbb{R}^4$ limits and a decompactification boundary, but not $\mathbb{R}^4$ as the whole compact sphere or compact target. Moreover $X^{\times 4} \cong (S^6)^{\times 4} \not\simeq S^{24}$, and Borel–Serre still excludes an almost-complex structure on either S^{12} or S^{24} [62]. The distinct first-coordinate holomorphic quotient is now classified exactly: $\mathcal{A}_z \to \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$ exists precisely when $6\mu = a\tau + b$ for $(a, b) \in \mathbb{Z}^2$, has connected kernel $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}(a\mu - \beta))$, and its marked labels obey $g_0(a, b) = (a, a + b)$, $g_2(a, b) = (-b, 6 + a)$. This locus has finite image in the completed orbifold base and misses a punctured cusp neighborhood, so it cannot be the cusp gap-closing mechanism; the theorem does not exclude other elliptic quotients or smooth integral magnetic fluxes. The reduced classical Yang–Mills Hessian has the same scalar factor g_{YM}^{-2}, and the finite cubical model is diagonalized in every mesh regime. On the degree-M_{cov}^2 cover, an independent N_{mesh}^4-vertex cubical subdivision gives a correctly typed holonomy map to $SU(2)^{E_{N_{\mathrm{mesh}}}}$. For $H_{\mathrm{link}} = \mathrm{diag}(i, -i)$, the explicit unit-flux link field has

$$U_{12}(n) = \exp(2\pi H_{\mathrm{link}}/N_{\mathrm{mesh}}^2), \qquad S_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}^{\log} = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}.$$

On the reduced finite Coulomb space, the Gaussian oscillator has gap

$$\sqrt{\min_{k\neq 0}\frac{1}{g_{\mathrm{YM}}^2}\left\|P_T^{-\top}S_{M_{\mathrm{cov}}}^{-1}\mathfrak{d}_{N_{\mathrm{mesh}}}(k)\right\|^2}.$$

The joint sequence $T_j^{\mathrm{rad}}=j^2,\ M_{\mathrm{cov},j}=j,\ N_{\mathrm{mesh},j}=j^3$ simultaneously gives the pointed cover limit $\mathbb{R}^4$, mesh diameter $O(j^{-1})$, Gaussian gap $O(j^{-2})$, and logarithmic action $O(j^{-2})$. This oscillator lives on a linear Coulomb space with Lebesgue measure; it is not the compact-link Haar/Gauss-law Hilbert space. For a nondegenerate integral magnetic form $\mathsf{E}_{\mathrm{mag}}$, the exact oscillator frequencies satisfy

$$b_1b_2=\frac{4\pi^2|\operatorname{Pf}(\mathsf{E}_{\mathrm{mag}})|}{D_T},\qquad \operatorname{gap}(\Delta_{\mathsf{E}_{\mathrm{mag}},T})=2\min\{b_1,b_2\}\le 4\pi\sqrt{\frac{|\operatorname{Pf}(\mathsf{E}_{\mathrm{mag}})|}{D_T}}=O(T^{-1}),$$

for the Friedrichs magnetic Laplacian on the corresponding Hermitian line bundle over the marked four-torus. Independently, the explicit reducible $SU(2)$ critical point has

$$\mathrm{YM}_T=\frac{2\pi^2}{g_{\mathrm{YM}}^2D_T},\qquad \mathcal{J}_Ta=-\frac{4\pi}{D_T}a.$$

These are geometric and classical operator theorems. No positive-metric physical Yang–Mills Hilbert space, reconstructed Hamiltonian, or scale-controlled state-and-energy maps have been constructed, so the interacting four-dimensional mass-gap implication is **[OPEN]**.

The same native full-flux construction extends exactly to every nonzero trace-zero integral weight vector $\mathbf{m}=(m_1,\dots,m_r)$ in a split Cartan of $SU(r)$. Writing $s_2(\mathbf{m})=\sum_i m_i^2$, the resulting reducible bundle and connection satisfy

$$c_2(\mathscr{E}_{\mathbf{m},T})=-\frac{s_2(\mathbf{m})}{2}[\eta_T]^2,\qquad \langle c_2,[F_T]_{x\text{-complex}}\rangle=6s_2(\mathbf{m}),$$

and

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T})=\frac{12\pi^2s_2(\mathbf{m})}{g_{\mathrm{YM}}^2}+\frac{\pi^2s_2(\mathbf{m})S_T^2}{g_{\mathrm{YM}}^2D_T}.$$

Its charged Coulomb–Jacobi bands, coincident-band multiplicities, wall nullities, and Morse index are computed root by root; on the deep cusp tail the real index is $36rs_2(\mathbf{m})$. These are fibrewise classical Yang–Mills theorems. The nonzero c_2 also proves that the bundles do not extend over any proposed carrier with $H^4=0$, including spheres of dimensions (6), (12), and (24); changing the gauge group therefore does not create a four-dimensional interacting quantum theory.

Finally, for $r=(r_1,r_2,r_3)\in\mathbb{R}^3$, put

$$Q_K(r)=\frac{2}{3}(r_1+r_2+r_3)^2-(r_1^2+r_2^2+r_3^2)=h^2-x^2-y^2,$$

where

$$h=\frac{r_1+r_2+r_3}{\sqrt{3}},\qquad x=\frac{r_1-r_3}{\sqrt{2}},\qquad y=\frac{r_1-2r_2+r_3}{\sqrt{6}}.$$

The real-linear coefficient isomorphism

$$(h,x,y)\longmapsto(t,m,\varrho)=\left(h+y,\frac{x}{\sqrt{6}},h-y\right)$$

identifies the Koide cone $Q_K=0$ with $t\varrho-6m^2=0$ inside the three-dimensional imaginary-coefficient space of the canonical period ansatz. It does not identify that cone with the full real period discriminant, whose three real coefficients remain free. For every choice of those real parts, $\det P=t\varrho-6m^2=Q_K(r)$. Hence each Koide-cone point gives $\det P=0$, whereas the actual positive radial cusp has $\det P_T=-D_T<0$ and $\det P_T\sim -T^2$; the Koide cone does not parametrize regular fibres.

Independently, write

$$\Delta(x,y,z)=4(xy+xz+yz)-(x^2+y^2+z^2)$$

and, for $Y \geq 0$, define

$$R_{\mathrm{K}}(Y) = \#\{(x,y,z) \in \mathbb{Z}^3_{>0} : x \leq y \leq z \leq Y,\ \Delta(x,y,z) = -1,\ z \geq 4(x+y)\}.$$

The integral coordinates

$$n = z - 4(x+y), \qquad q = y - x, \qquad a = x + y - 4n$$

give a bijection onto

$$\{(a,q,n) \in \mathbb{Z}^3 : a > 0,\ q,n \geq 0,\ a + 4n > q,\ a^2 - 3q^2 - 18n^2 = -2\},$$

with height $z = 4a + 17n$. Calibrated homogeneous counting gives unconditionally

$$R_{\mathrm{K}}(Y) \sim C_0 Y, \qquad C_0 = \frac{9}{2\pi^2} \log(2+\sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right).$$

The stronger estimate $R_{\mathrm{K}}(Y) - C_0 Y = O_\delta(Y^{1/2+\delta})$ for every $\delta > 0$ remains **[OPEN]**.

There is also an exact arithmetic bridge to the earlier Erdős–Straus divisor-shell construction. Fix a prime $p \equiv 1 \pmod{12}$ and $\varepsilon \in \{0,1\}$, and put

$$L = \frac{p-1}{4}, \qquad 0 \leq j < L, \qquad R_j = 4j + 3, \qquad a_j^{\mathrm{ES}} = \frac{p + R_j}{4}.$$

For $u \in \mathbb{Z}_{>0}$ satisfying $u \mid (a_j^{\mathrm{ES}})^2$ and $R_j \mid (4u + p^\varepsilon)$, set

$$T_{\mathrm{sel}} = \frac{4u + p^\varepsilon}{R_j}, \qquad \ell_0 = \frac{T_{\mathrm{sel}} + 1}{4}, \qquad \ell_1 = u - jT_{\mathrm{sel}}, \qquad \ell_2 = 1.$$

Then $T_{\mathrm{sel}} \in \mathbb{Z}_{>0}$, $12\ell_0 - 4\ell_1 - 3 = p^\varepsilon$, and

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T_{\mathrm{sel}} \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1 - 4\ell_0 \end{pmatrix}.$$

For fixed (p, ε), the marked assignment $(j,u) \mapsto (j, \ell_0, \ell_1)$ is a bijection onto the triples satisfying

$$0 \leq j < L, \quad 12\ell_0 - 4\ell_1 - 3 = p^\varepsilon, \quad T_{\mathrm{sel}} = 4\ell_0 - 1 > 0, \quad u = \ell_1 + jT_{\mathrm{sel}} > 0, \quad u \mid (a_j^{\mathrm{ES}})^2.$$

This is an integral-relator equivalence, not a geometric map between the reciprocal cubic and X.

The higher-rung calculation now has several exact, sharply separated outputs. For two complex columns, left Gram reduction gives

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})/SU(3) \cong \mathrm{Herm}_2(\mathbb{C})_{\geq 0}, \qquad S^{12}_{\mathrm{col}}/SU(3) \cong B^4,$$

whose orbit-type boundary is S^3. Here S^{12}_{col} is the one-point compactification of the column representation; it is not the cusp-smash S^{12} constructed elsewhere in the workbench. For $\mathscr{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C})$, left Gram reduction and the alternating determinant give

$$\mathscr{V}_3/U(3) \cong \mathrm{Herm}_3(\mathbb{C})_{\geq 0}, \qquad \mathscr{V}_3/SU(3) \cong \{(H,z) : H \geq 0,\ |z|^2 = \det H\} \cong \mathbb{R}^{10}.$$

Consequently

$$S^{17}/U(3) \cong B^8, \quad \mathscr{V}_3^+/U(3) \cong B^9, \quad S^{17}/SU(3) \cong S^9, \quad \mathscr{V}_3^+/SU(3) \cong S^{10}.$$

The compactified rank-deficient locus is the standard locally flat $S^8 \subset S^{10}$, and $S^{10} \setminus S^8 \cong \mathbb{R}^8 \times \mathbb{C}^* \simeq S^1$, with generator $\det A/|\det A|$. The full rank-three cubic Jordan cone is not itself the rank-two quadratic Lorentz cone; the marked angular quotient nevertheless supplies a split positive observation onto that cone while retaining all five complementary directions.

For

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad u = (1,1,1)^{\mathsf{T}}, \quad R = P^*(PP^*)^{-1}, \quad \Pi = uu^*/3,$$

define

$$\mathcal{O}_P(H) = PHP^*, \qquad \jmath_P(G) = RGR^*.$$

Then

$$\mathcal{O}_P\jmath_P = \mathrm{id}, \qquad \jmath_P\mathcal{O}_P(H) = (I_3 - \Pi)H(I_3 - \Pi),$$

and the positive zero fibre is exactly $\{\lambda uu^* : \lambda \geq 0\}$. With $X = (R\ u)$, every Hermitian H has unique coordinates

$$H = X \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} X^*, \qquad \det H = \lambda \det G - \psi^* \operatorname{adj}(G)\psi,$$

where

$$H \succeq 0 \iff G \succeq 0, \quad \psi \in \operatorname{im} G, \quad \lambda \geq \psi^* G^+ \psi.$$

Thus

$$\mathrm{Herm}_3(\mathbb{C}) \cong \mathrm{Herm}_2(\mathbb{C}) \oplus \mathbb{C}^2 \oplus \mathbb{R}, \qquad 9 = 4 + 4 + 1.$$

The faithful embedding $T_P(g) = RgP + \Pi = X(g \oplus 1)X^{-1}$ acts by $(G, \psi, \lambda) \mapsto (gGg^*, g\psi, \lambda)$. The three summands are the Lorentz four-vector, the defining complex spin representation, and a real scalar; the vector action kills $\{\pm I_2\}$, whereas the full action retains the spin double cover because $-I_2$ negates ψ. On positive cones, $\jmath_P$ properly compactifies to $B^4 \hookrightarrow S^8$, while the reverse boundary observation is nonproper. The determinant-phase equation is

$$|\omega|^2 = \lambda \det G - \psi^* \operatorname{adj}(G)\psi,$$

and $\omega = 0$ is the rank-loss locus compactifying to $S^8 \subset S^{10}$. The four-branch chart supplies the exact seed

$$K(z) = \operatorname{diag}(z_0 z_3, z_1 z_2), \qquad \det K = z_0 z_1 z_2 z_3,$$

whose positive $SL_2(\mathbb{C})$-orbit fills the future causal cone. Finally, with retained $\alpha, c > 0$,

$$p_\alpha(Z) = \alpha Z^* Z, \qquad m^2_{\alpha,c}(Z) = \frac{\alpha^2}{c^2} \sum_{a<b} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2$$

is the complete $SU(3)$-orbit momentum, and $m^2_{\alpha,c}(rZ) = r^4 m^2_{\alpha,c}(Z)$. Hence the unrestricted pointwise carrier is gapless. The complete presymplectic reduction of the displayed worldline action is now proved. For $m_0 > 0$, the rank-two colour action is free, $DP_{\alpha,Z}$ is surjective with kernel $\mathfrak{su}(3) \cdot Z$, and the characteristic quotient is

$$\big(\mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}\big)/\big((x,p) \sim (x + sp, p)\big) \cong_{\mathrm{symp}} T^*\mathcal{H}^+_{m_0,c};$$

thus this stratum is exactly the future-timelike spinless relativistic particle. For $m_0 = 0$, the intrinsic nonzero rank-one quotient similarly gives $T^*\mathcal{N}^\times_+$, but the written multiplier action is irregular because

$$\det(\alpha Z^* Z) = \alpha^2 \sum_{a<b} |\Delta_{ab}(Z)|^2, \qquad d(\det \circ P_\alpha)_Z = 0 \quad (\mathrm{rank}_{\mathbb{C}}\, Z \leq 1).$$

Its rank-one equation is $\dot{x} = \lambda p$, with no equation $\lambda = e$; four real transverse rank-opening directions are first-order null but are not colour gauge, and the apex $Z = 0$ admits arbitrary $x(\tau)$. On the later hard-constraint carrier with $\omega_0 \neq 0$, the momentum projection has rank four for $w \neq 0$ and rank three at $w = 0$; direct variation forces $\dot{x} = 0$ strictly above the sharp threshold and gives $\dot{x} \in \mathbb{R}p$ only at the threshold. Its determinant bound and multiplication/Fock spectra therefore remain exact algebraic and free-operator results, but the multiplication operator is not the Hamiltonian generating the displayed worldline action and the threshold is not a dynamically generated interacting gap. See theorems 21.2 to 21.7, 22.6, 22.9, 22.12 and 22.16.

The same marked data now carry an exact field-theoretic continuation on the affine normal slice

$$\mathcal{N}_P = \{Y \in \mathrm{Herm}_3(\mathbb{C}) : Yu = 0\} = \jmath_P\big(\mathrm{Herm}_2(\mathbb{C})\big).$$

Polarization of $\det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G$ gives a constant metric of signature $(1,3)$, and the congruence action induces the spin cover $SL_2(\mathbb{C}) \to SO^+(1,3)$. The standard Moore–Penrose metrics select

$$W_P = R^*R = C^{-1}, \qquad K_P = \{a \in SL_2(\mathbb{C}) : a^* W_P a = W_P\} \cong SU(2),$$

acting on the real normal field as $\mathbf{1} \oplus \mathbf{3}$. Gauging this pointwise action yields an explicit nonnegative classical gauge–scalar action; the centre acts trivially on the normal field, so it cannot screen a fundamental centre charge without the retained centre-nontrivial spinor or another added field. The candidate inverse-dual matrix remains outside the positive Gram momentum cone because

$$\det G_{\eta,T} = -\frac{2}{3} D_T < 0, \qquad \operatorname{im} p_\alpha = \operatorname{Herm}_2(\mathbb{C})_{\geq 0}.$$

On this same Lorentzian host, the Klein–Gordon/Fock construction has exact gap m_{KG} for an independently supplied $m_{\mathrm{KG}} > 0$, is gapless at $m_{\mathrm{KG}} = 0$, and the massless compact gauge Hessian has lowest nonzero Gaussian frequency $2\pi/L \to 0$. Thus neither the determinant identity, the Gram cone, nor entanglement alone chooses an interacting mass gap. A full-domain comparison satisfying a uniform norm lower bound and energy-form upper bound would instead imply the quantitative estimate $\operatorname{gap}(A_{\mathrm{phys}}) \geq \gamma c_R^2 / C_R$; an intertwiner on only one gapped subspace does not. See theorems 23.2, 23.5, 23.6, 23.8, 23.10, 23.12 and 23.16.

The selectively corrected Jordan determinant/Euler audit identifies an exact family of Lorentz sections inside the marked Hermitian cone. With $[a]_\times v = a \times v$, one has for every real symmetric S

$$\det\bigl(S + i[a]_\times\bigr) = \det S - a^{\mathsf{T}} S a.$$

Hence every marked positive monodromy form B defines

$$\ell_B(\lambda, a) = \lambda B + i[a]_\times, \qquad \ell_B^{-1}(\operatorname{Herm}_3(\mathbb{C})_{\geq 0}) = \{\lambda \geq 0 : a^{\mathsf{T}} B a \leq \lambda^2 \det B\}.$$

There is also a real-linear order-cone bijection from $\operatorname{Herm}_2(\mathbb{C})_{\geq 0}$ onto the positive part of $\mathbb{R} I_3 \oplus i\mathfrak{so}(3)$ for which

$$\det_3 \Theta(H) = \frac{\operatorname{tr} H}{2} \det_2 H;$$

it sends positive-semidefinite ranks $2, 1, 0$ to $3, 2, 0$. If P is a rank-one projection, then $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not idempotent, which separately proves that Θ is not a Jordan homomorphism. These are exact algebraic and order-theoretic maps, not holomorphic toric, period-lattice, or filling maps. In particular, the toric limit $z_0 z_1 z_2 z_3 = t \to 0$ keeps $B = I_3$ fixed and is not the operation $\det B \to 0$.

The corresponding rank-three toroidal degeneration is now constructed as a separate analytic object. Its smooth fibre is E_τ^3, its deepest local equation is $t = z_0 z_1 z_2 z_3$, and its irreducible central fibre has stratum counts

$$(f_0, f_1, f_2, f_3) = (1, 7, 12, 6), \qquad \chi(W_3) = 6.$$

Its normalization has orbit vector $(1, 14, 36, 24)$; the strata of positive codimension map with degrees $2, 3, 4$ onto the $7, 12, 6$ quotient strata. Thus any real S^8-total-space fibration retaining W_3 must have its remaining singular fibres contribute Euler sum -4. More generally, an injective integral shear $\mathcal{B} : \bar{L}_3 \hookrightarrow L_3$ gives

$$\chi(W_{3,\mathcal{B}}) = 3!\,[L_3 : \mathcal{B}\bar{L}_3] = 3!\,|\det \mathcal{B}|.$$

The toroidal object is not merely analogous to the reconstructed candidate cusp: the explicit epimorphisms

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

give $\Phi = P \oplus Q : H_1(T^6) \twoheadrightarrow H_1(T^4)$ with

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3, \qquad P Z_3(t) = Z_2(t) Q.$$

This extends to a holomorphic map of local cusp families with connected elliptic kernel. An explicit unimodular change of basis identifies the rank-three cusp matrix with $M_F \oplus \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, and compatible order-three and order-four affine peripheral actions exist on the added elliptic factor. Under the marked A_2-Cartan identification, the six Kuhn tetrahedron orbits map equivariantly to the six diagonal Weyl chambers. This last correspondence is fan-theoretic: it neither identifies the four-branch toroidal germ with the quadratic Gram-rank germ nor supplies the still-open analytic gluing over both finite special fibres.

The free rank-six object native to this construction is now identified without conflating it with a cyclic quotient or with the different defect group on the original regular locus:

$$\Lambda_{\mathrm{tor}} = H_1(T^6;\mathbb{Z}) = \mathbb{Z}^3_{\mathrm{angular}} \oplus \mathbb{Z}^3_{\mathrm{deck}}.$$

With diagonal coordinate permutations it fits into the exact sequence of integral S_3-modules

$$0 \longrightarrow \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,k) \longrightarrow \Lambda_{\mathrm{tor}} \xrightarrow{P\oplus Q} A_F \oplus B_F \longrightarrow 0, \qquad k = (1,1,1).$$

The sequence splits integrally with monodromy but not over $\mathbb{Z}[S_3]$; its quotient copies are A_2 weight lattices, with the root lattices sitting at index three. The rank identity $\operatorname{rk}\Lambda_{\mathrm{tor}} = 6$, the chamber count $|S_3| = 3! = 6$, and $\chi(W_{3,\mathcal{B}}) = 3!|\det\mathcal{B}|$ are thereby separated: the six homology basis vectors form two three-element S_3-orbits, whereas the six Kuhn/Weyl chambers form one regular orbit. That regular orbit does occur exactly in the associated rank-nine carrier:

$$\operatorname{Hom}(\overline{L}_3, L_3) = \bigoplus_i \mathbb{Z}E_{ii} \oplus \bigoplus_{i\neq j} \mathbb{Z}E_{ij},$$

where the off-diagonal matrix units form $\mathbb{Z}[S_3]$, and the chamber $\pi = (\pi_1\pi_2\pi_3)$ corresponds to E_{π_1,π_3}. Equivalently, the chambers are the six mixed angular–deck basis pairs with zero alternating pairing; the three excluded diagonal pairs have pairing one. The real-linear S_3-isomorphism

$$X \longmapsto \frac{X + X^{\mathsf{T}}}{2} + i\frac{X - X^{\mathsf{T}}}{2}$$

identifies the ambient matrix representation with $\mathrm{Herm}_3(\mathbb{C})$, but has integral index 2^3 in the symmetric/skew coordinates and sends every off-diagonal E_{ij} to an indefinite Hermitian matrix with eigenvalues $2^{-1/2}, -2^{-1/2}, 0$. It is therefore an exact representation and chamber-incidence bridge, not a positive, Jordan, determinant, or toric morphism. The separate map into the Gram boundary is only the nonadditive piecewise-linear chamber map $\Phi_{\mathrm{G}} \circ \jmath \circ P$ on the angular cocharacters; it ignores the deck half and does not descend to the torus. See theorems 19.1 to 19.3, 20.1 and 20.4. This successful lift is distinct from the diagonal local-system candidate

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T, \qquad M_1M_2M_0 = I.$$

Although its finite factors have orders 4 and 6, each $I - M_j : T^6 \to T^6$ is an isogeny for $j = 1, 2$; consequently every affine lift of either finite action has a fixed point.

A literal complex-fourfold analogue on S^8 is impossible: an almost complex tangent bundle would have $c_1 = c_2 = c_3 = 0$, $c_4 = 2u$, hence $p_2 = 4u$, while the signature formula forces $p_2 = 0$. A nonsingular T^6-bundle over S^2 is likewise impossible; any real singular replacement must derive its actual filling maps and satisfy the displayed Euler, van Kampen, Wang, and Mayer–Vietoris tests. For its regular-piece monodromy ρ, the corrected homological Serre filtration is

$$0 \longrightarrow (\textstyle\bigwedge^q \Lambda)_\rho \longrightarrow H_q(E;\mathbb{Z}) \longrightarrow H_1(B^\circ; \bigwedge^{q-1}\Lambda) \longrightarrow 0,$$

while the dual edge map is $H^q(E;\mathbb{Z}) \twoheadrightarrow (\bigwedge^q \Lambda^\vee)^\rho$. Thus the filling maps must preserve the coinvariant subobject and induce the required map on the base-one-cycle quotient; using the reverse filtration invalidates the map-level test. Every effective torus action on S^8 has rank at most four, and the bound is sharp for the coordinate action

$$S^8/T^4 \cong \mathrm{Susp}(\Delta^3) \cong B^4.$$

This is a real two-pole singular compactification with generic T^4 orbit, not a three-special-fibre T^6-fibration over S^2.

The exact bundle-theoretic alternative occurs on the regular locus $U = X \setminus (W \sqcup S_1 \sqcup S_2)$:

$$H^4(U;\mathbb{Z}) \cong H_1(W;\mathbb{Z}) \oplus H_1(S_1;\mathbb{Z}) \oplus H_1(S_2;\mathbb{Z}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2.$$

Principal $SU(3)$-bundles on U are classified by this complete c_2-class and extend across the three deleted fibres exactly when it vanishes. This is a free rank-six defect lattice, not a cyclic group of order six. The formulas from this point through the normal-holonomy and common-centre comparison are

exact results of the dated fibre-restriction, defect-lattice, and common-centre auxiliary audits. Their full theorem–proof source has not yet been integrated into a substantive research fragment of this workbench, so they are recorded here without a **[VERIFIED]** status claim. In the marked basis

$$(a_0, b_0, a_1, b_1, a_2, b_2) = (\bar{\gamma}, \bar{u}, g_1, h_1, g_2, h_2),$$

the actual collar matrix gives the primitive fibre-degree functional

$$[\deg_F] = (12, 0, 4, 0, -3, 0), \qquad \ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle.$$

Define

$$q_{\mathrm{col}} : L \longrightarrow H_1(X^\circ; \mathbb{Z}), \qquad a_0 \mapsto c, \quad a_1 \mapsto \bar{g}_1, \quad a_2 \mapsto \bar{g}_2, \quad b_0, b_1, b_2 \mapsto 0,$$

and let $\Phi_1 : H_1(X^\circ; \mathbb{Z}) \to \mathbb{Z}$ have row $(12, 4\ell_1, 3\ell_2)$ in the basis $(c, \bar{g}_1, \bar{g}_2)$. Then the typed identity is $\deg_F = \Phi_1 \circ q_{\mathrm{col}}$, and $\Phi_1(j_*[\widetilde{a}_0]) = p$. The order-three and order-four normal holonomies combine, with the candidate's relative sign, to characters

$$\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 = \chi_{12}^2 = \rho_6 \circ \deg_F, \qquad \rho_m(k) = e^{-2\pi i k/m}.$$

Thus the natural finite quotients are $H^4(U)/\ker \chi_{12} \cong \mathbb{Z}/12$ and $H^4(U)/\ker \chi_6 \cong \mathbb{Z}/6$; the latter is not inferred from the rank. Ordinary coefficient reduction instead gives $H^4(U; \mathbb{Z}/6) \cong (\mathbb{Z}/6)^6$; this is not the single distinguished cyclic quotient selected by χ_6. Equivalently, for a principal $SU(3)$-bundle $P \to U$,

$$\chi_6(c_2(P)) = \exp\left(-\frac{2\pi i}{6}\langle c_2(P)|_F, [F]\rangle\right).$$

Tensor product supplies a separate exact centre statement:

$$SU(2) \times SU(3) \hookrightarrow SU(6), \qquad (A, B) \longmapsto A \otimes B, \qquad \mu_2 \times \mu_3 \xrightarrow{\sim} Z(SU(6)) = \mu_6.$$

On the five-dimensional CW model of U, c_2 classifies $SU(n)$-bundles for $n \geq 3$, and $c_2(E_2 \otimes E_3) = 3c_2(E_2) + 2c_2(E_3)$. Hence every chosen $h \in H^4(U; \mathbb{Z})$ is represented by an $SU(6)$-bundle admitting a tensor-subgroup reduction; its c_2-factorizations are

$$(e, f) = (h + 2t, -h - 3t), \qquad t \in H^4(U; \mathbb{Z}),$$

so the geometry selects neither a nonzero h nor a preferred reduction. Separately,

$$H^5(U; \mathbb{Z}) \cong \ker\{\mathbb{Z}^3 \xrightarrow{(1,1,1)} \mathbb{Z}\} \cong Q(A_2)$$

for the combinatorial form making the three support components orthonormal; this is not a cup-product form and does not identify $\ker \deg_F$ with A_5. The finite-group identification $H^4(U)/\ker \chi_6 \cong Z(SU(6))$ is not an $SU(6)$-connection or a gauge-transition cocycle. The tensor map is injective and does not factor through $(SU(2) \times SU(3))/\mu_6$, because its central μ_6 maps to nontrivial scalars. The distinct common-centre map is

$$\Gamma_6 = \{((\lambda\mu)^{-1}, \lambda I_2, \mu I_3) : \lambda \in \mu_2,\ \mu \in \mu_3\}, \qquad \frac{U(1) \times SU(2) \times SU(3)}{\Gamma_6} \hookrightarrow U(6), \quad [z, A, B] \longmapsto z(A \otimes B),$$

where $\Gamma_6 \cong \mu_6$. Its determinant is z^6, and its image meets $SU(6)$ in exactly the tensor subgroup $\iota(SU(2) \times SU(3))$; it is not generally contained in $SU(6)$.

Fibre evaluation therefore gives a canonical surjection

$$\bar{\kappa}_F : H^4(U; \mathbb{Z}) \twoheadrightarrow H^4(T^4; \mathbb{Z}) \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}.$$

Identifying this degree-four quotient with a degree-two common-centre bundle obstruction remains unproved. Global $SU(3)$-bundles on S^6 instead carry the separate clutching class $c_3 = \pm 2ku$. The Lorentz-observation module and its exact presymplectic reduction identify the massive particle, the punctured massless quotient, the irregular massless multiplier/apex branches, and the strict-interior versus threshold dynamics of the hard-constraint packet. These carrier, reduction, and bundle theorems

do not supply a Yang–Mills mass-gap conclusion: the required field connection, interacting operator and domain, gauge-invariant spectral sector, and continuum comparison morphism remain to be constructed.

Reviewer B independently reviewed the reconstruction existing up to its evidentiary cutoff and provisionally agreed on the non-collapse route; the later Codex proofs here are separate. In a 26 August follow-up, Reviewer B also independently supplied the finite-complex/UCT proof of top-degree base change and confirmed the conductor–duality nonvanishing from the three named candidate inputs; see [35]. Throughout, **[VERIFIED]** means that a complete auditable proof is written in this document, not that specialist replication or community acceptance has already occurred. The candidate construction and its original claims remain attributed to their reported source. This paper is an AI-produced verification aid: its own contribution is to expose the relevant morphisms, signs, lattices, comparison maps, and singular-fibre arguments as complete checkable proofs, thereby reducing the reconstruction work required of human experts. It makes no claim of priority for the candidate and does not replace independent specialist adjudication.

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1 Results and evidence boundary

The target is a reconstructed manuscript claiming a compact complex threefold X and holomorphic map $f : X \to \mathbb{P}^1$ with

$$X \cong_{\text{diff}} S^6, \qquad a(X) = 1, \qquad \langle c_3(T_X), [X] \rangle = 2.$$

As of 4 September 2026, the workbench records the following mathematical status. Here **[VERIFIED]** means that a complete proof is written in this workbench; it is not a proxy for a model verdict or for community acceptance.

1. **[VERIFIED]** Let $\Delta \simeq (\mathbb{Z}/3) * (\mathbb{Z}/4)$ act on $\mathfrak{h}_z$, let g_1, g_2 be its elliptic generators with chosen fixed points z_1, z_2, and put $g_0 = (g_1 g_2)^{-1}$. For the normalized period lift $\tau : \mathfrak{h}_z \to \mathfrak{h}$, define

$$\varphi : \Delta \longrightarrow \mathrm{PSL}_2(\mathbb{Z}), \qquad \tau \circ g = \varphi(g) \circ \tau.$$

To avoid confusing the modular transformations with the finite reduced fibres S_j below, write

$$\mathfrak{s}_1(w) = \frac{w-1}{w}, \qquad \mathfrak{s}_2(w) = -\frac{1}{w}, \qquad \mathfrak{p}(w) = w - 1.$$

The finite generators are normalized simultaneously by

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = -i, \qquad g_0 \text{ parabolic},$$

and the exact modular deck certificate is

$$\varphi(g_1) = \mathfrak{s}_1, \qquad \varphi(g_2) = \mathfrak{s}_2, \qquad \varphi(g_0) = \mathfrak{p}, \qquad \mathfrak{s}_1\mathfrak{s}_2\mathfrak{p} = 1.$$

The sign at z_2 is forced by the simultaneous parabolic normalization; the quadratic ramification calculation alone gives only $(g_2'(z_2))^2 = -1$. The cusp normalization proof now records its covering levels explicitly: it chooses a g_0-stable component $U_0 \subset \pi^{-1}(D_0^*)$ and a component $K \subset j^{-1}(\{|w| > R\})$ containing $\tau(U_0)$. For $P_{\text{cusp}} := \varphi(g_0)$, the auxiliary conjugation

$$\tau' = \gamma_0^{-1}\tau|_{U_0}, \qquad P' = \gamma_0^{-1} P_{\text{cusp}} \gamma_0, \qquad \gamma_0(\infty) = c,$$

where c is the rational boundary cusp of K, proves that $P' \neq 1$ and preserves the selected cusp at infinity, hence is parabolic. The proof then discards γ_0, returns to the original $\mathfrak{s}_1$-normalized frame, performs only the allowed final $\mathfrak{s}_1$-power conjugation, and obtains the displayed modular certificate. Only after that return does it define anew the cusp exponential

$$q_{\text{cusp}} := e^{2\pi i \tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0;$$

this q_{cusp} is distinct from the integral class q below. The resulting nondegenerate period family descends, and the middle quotient

$$J = (\mathcal{T}|_{\mathfrak{h}_z^\circ})/\Delta \longrightarrow B_{\text{mid}} := \mathfrak{h}_z^\circ/\Delta$$

is a proper holomorphic submersion with compact torus fibres. The toric cusp filling, the two finite logarithmic fillings, all three punctured-overlap biholomorphisms, and the global gluing produce a connected compact Hausdorff smooth complex threefold and a proper surjective holomorphic map with connected fibres

$$f : X \longrightarrow B \simeq \mathbb{P}^1.$$

Its full scheme-theoretic cusp fibre $X_{p_0} = W$ is reduced, irreducible, nonnormal, and normal crossings, with local equations

$$z_0, \qquad z_0 z_1, \qquad z_0 z_1 z_2,$$

and normalization $\nu : E_0 \simeq \mathrm{dP}_6 \to W$; it is not a simple-normal-crossings divisor. For the finite fibres, let

$$A_0^{(j)} = (\Lambda \otimes_{\mathbb{Z}} \mathbb{R})/\Lambda \simeq F$$

be the marked central torus on the covering family, and put

$$v_1 = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad v_2 = -(\widehat{\gamma} + 3\widehat{u} - 3\widehat{w}),$$

so that

$$S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle, \qquad f^*(p_j) = m_j S_j, \qquad (m_1, m_2) = (3, 4).$$

The quotient covering and its codomain are, explicitly,

$$\pi_j : A_0^{(j)} \longrightarrow S_j, \qquad x \longmapsto [x].$$

The radial retraction $N_j \to S_j$ identifies finite-fibre specialization exactly with

$$\mathrm{sp}^r_{p_j} = \pi_j^* : H^r(S_j; \mathbb{Z}) \longrightarrow H^r(A_0^{(j)}; \mathbb{Z}) \xrightarrow[\text{marking}]{\sim} H^r(F; \mathbb{Z}),$$

with no unrecorded conjugacy or index. This stalk identification is theorem 8.8; see also theorems 8.1, 8.2, 8.4 to 8.7 and 8.9.

2. **[VERIFIED]** The monodromy, cusp invariant lattices, global coinvariants, finite-filling stalk maps, terminal group presentation, and terminal degree-two lattice are proved over $\mathbb{Z}$ in theorems 5.1 to 5.7.

3. **[VERIFIED]** The intrinsic integral nearby-cycle specialization at the cusp is injective with saturated image

$$\mathrm{im}(\mathrm{sp}^q) = H^q(F; \mathbb{Z})^{T_0} \qquad (0 \le q \le 4),$$

where the degree-two invariant class is $q := uw + 6\gamma\delta \in \bigwedge^2 V = H^2(F; \mathbb{Z})$, and the finite stalks and parabolic sheaf calculation give the complete global Leray page. The canonical truncation triangle identifies the explicit Cech discrepancy cocycle with Leray d_2, and multiplicativity then gives

$$d_2(12\gamma) = \omega, \qquad d_2(2q) = \omega[\delta], \qquad d_2(2\gamma uw) = \omega[\gamma\delta].$$

Consequently

$$H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = H^3(X; \mathbb{Z}) = 0, \qquad H_2(X; \mathbb{Z}) = 0;$$

see theorems 39.1 to 39.5.

4. **[VERIFIED]** The candidate's independent toric-collapse construction is continuous across every stratum and descends equivariantly to a strong deformation retraction $r : N_0' \to W$. Its fixed-fibre restriction $c = r|_F$ has the exact primitive integral sequence

$$0 \longrightarrow \mathrm{im}(\textstyle\bigwedge^q M_0 - I) \longrightarrow \bigwedge^q \Lambda \xrightarrow{c_*} H_q(W; \mathbb{Z}) \longrightarrow 0,$$

and proper-base-change functoriality gives, in every degree,

$$c^* = i_F^* r^* = \mathrm{sp}^q .$$

Thus the former C1–C3 obligations are closed; see theorems 44.5 to 44.7.

5. **[VERIFIED]** With zero-section whiskers and the fixed clockwise meridians, the actual overlap inclusions give

$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1.$$

Together with the preceding integral cohomology this proves

$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = 2;$$

see theorems 39.6 and 39.7.

6. **[VERIFIED]** The very-general fibre has

$$\eta := q = u \wedge w + 6\gamma \wedge \delta, \qquad \mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad a(F_b) = 0.$$

The square is the oriented wedge identity

$$\eta \wedge \eta = 12\,\gamma \wedge u \wedge w \wedge \delta,$$

so, for the orientation determined by the ordered marking (γ, u, w, δ) and for the complex orientation respectively,

$$\int_{(F_b, \mathrm{or}_{\mathrm{mark}})} \eta \wedge \eta = 12, \qquad \int_{(F_b, \mathrm{or}_{\mathbb{C}})} \eta \wedge \eta = -12.$$

The repaired meromorphic-descent argument gives

$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

with f the algebraic reduction; see theorems 40.2 and 40.4.

7. **[REFUTED]** The p. 692 sheaf-theoretic inference of Campana–Demailly–Peternell is false in the stated generality. If $\mathscr{K}_S = \ker(\Omega^1_X|_S \to \widetilde{\Omega}^1_S)$, then

$$0 \longrightarrow N^*_{S/X} \longrightarrow \mathscr{K}_S \longrightarrow \mathrm{tors}\,\Omega^1_S \longrightarrow 0,$$

and the two printed vanishings do not control sections mapping to the conductor-supported quotient. The proper nodal-cubic-times- $\mathbb{P}^1$ family in theorem 46.6 is an explicit counterexample to that inference.

8. **[VERIFIED]** The conductor section, nonnormal Cartier-divisor duality, flatness, and top-degree analytic base change are complete in theorems 46.5 and 46.7 to 46.11. On the cusp model the invariant nowhere-vanishing form

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2}$$

first proves intrinsically that $\omega_{N_0} \simeq \mathcal{O}_{N_0}$. After the global gluing, the open embedding $N_0 \hookrightarrow X$ gives the exact restriction chain

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}.$$

Let $\nu : E_0 \simeq \mathrm{dP}_6 \to W$ be the normalization. Consequently, for every $L \in \mathrm{Pic}^0(X)$,

$$\nu^*((L^\vee \otimes \omega_X)|_W) \simeq \mathcal{O}_{\mathrm{dP}_6},$$

and the conductor–duality–base-change chain gives

$$R^2 f_*(T_X \otimes L) \neq 0$$

without using global topology; see theorems 8.4, 8.9, 46.12 and 46.22.

9. **[VERIFIED]** For arbitrary line bundles the exact trivial-pullback obstruction is

$$r = \nu^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6).$$

The conductor argument applies uniformly to all line bundles exactly when $r = 0$; see theorem 46.13. For the constructed candidate, the Leray edge image of every $c_1(L)$ lies in $E_\infty^{0,2} \subseteq \ker d_2^{0,2}$, while $d_2^{0,2}(2q) = \omega[\delta]$ is an isomorphism. Proper-base-change naturality therefore gives $c_1(L|_W) = 0$, and injectivity of c_1 on $\mathrm{Pic}(\mathrm{dP}_6)$ forces $r = 0$; see theorem 46.14. Combining this with the analytic inputs gives

$$R^2 f_*(T_X \otimes L) \neq 0 \qquad \text{for every } L \in \mathrm{Pic}(X).$$

Thus the topology-free Pic^0 route and the all-Pic route are compatible rather than competing statuses. There are now three all-Pic certificates. The first is the preceding Leray-edge proof. The second uses no global Leray differential: the order-four specialization lattice writes $c_1(L|_F) = 2nq$, and the pair-of-pants crossed defect has boundary values

$$\bar{d}_1 = 4n[\delta], \qquad \bar{d}_2 = -3n[\delta], \qquad \bar{d}_0 = 0.$$

The exact boundary relation forces $n = 0$, and local integral cusp specialization then forces $r(L) = 0$; see theorems 46.15 to 46.18. This second proof is independent of the global transgressions, but it still uses one local cusp bounding chain and the local specialization theorem.

10. **[VERIFIED]** The local cusp Picard map itself is nonzero. The explicit equivariant Cartier divisor

$$\mathcal{D}_Q = \sum_{v \in \mathbb{Z}^2} (v_1^2 - 6v_2^2) E_v$$

descends to $\mathcal{L}_Q \in \mathrm{Pic}(N_0)$ with

$$c_1(\mathcal{L}_Q|_{F_t}) = -2q, \qquad \left(\deg_{C_i} \nu^*(\mathcal{L}_Q|_W)\right)_{i=1}^{6} = (-12, 2, 0, -12, 2, 0).$$

Thus neither the cusp geometry nor analytic relative-Picard representability alone can force normalized central restriction to vanish; see theorem 41.8.

11. **[VERIFIED]** After the global coinvariant calculation writes

$$E := c_1(L|_F) = kq \in \mathrm{Alt}^2(\Lambda, \mathbb{Z}) = \bigwedge^2 V,$$

the third all-Pic certificate completes the argument by holomorphic factors and toric support functions. Put $V_{\mathbb{R}} = \mathrm{Hom}(\Lambda, \mathbb{R})$, let $P \subset B^\circ$ be the compact pair-of-pants core with clockwise boundary meridians ρ_1, ρ_2, ρ_0, and let $d : \pi_1(P) \to V_{\mathbb{R}}$ be the factor-of-automorphy crossed homomorphism, put $d_i = d(\rho_i)$, and let a bar denote passage to $V_G \otimes \mathbb{R} = \mathbb{R}[\delta]$. Logarithms of the actual $\mathcal{O}^*$-factors give

$$\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0, \qquad \bar{d}_1 = 2k[\delta], \quad \bar{d}_2 = -\tfrac{3}{2}k[\delta], \quad \bar{d}_0 = 0.$$

The finite norm identities now include the fully typed factor-extension repair (5.17-A14c)–(5.17-A15), rather than identifying maps living on different covering levels. Put

$$\widetilde{J}_j = \Delta_j \times \mathbb{C}^2, \qquad r_j : \widetilde{J}_j \longrightarrow J'_j,$$

where r_j is the lattice quotient with deck translations τ_λ. On this vector cover, and only there, define

$$Q_{j,\log} = \widetilde{g}_j^{\,v_j/m_j}, \qquad Q_{j,\mathrm{lin}} = \widetilde{g}_j^{\,0}.$$

Their exact relations are

$$Q_{j,\log}\tau_\lambda Q_{j,\log}^{-1} = \tau_{A_j\lambda}, \quad (Q_{j,\log})^{m_j} = \tau_{v_j}, \qquad Q_{j,\mathrm{lin}}\tau_\lambda Q_{j,\mathrm{lin}}^{-1} = \tau_{A_j\lambda}, \quad (Q_{j,\mathrm{lin}})^{m_j} = 1.$$

Let $q_{j,\log}, q_{j,\mathrm{lin}}$ be their descended order-m_j automorphisms of J'_j, put $J'^*_j = J'_j|_{\Delta_j^*}$, and define

$$p_{j,\log} : J'^*_j \to N_j^* := J'^*_j/\langle q_{j,\log}\rangle, \qquad p_{j,\mathrm{lin}} : J'^*_j \to J^* := J'^*_j/\langle q_{j,\mathrm{lin}}\rangle.$$

The branch-independent fibre translation $h_j : J'^*_j \to J'^*_j$ and its quotient biholomorphism $\bar{h}_j : N_j^* \to J^*$ satisfy

$$h_j q_{j,\log} h_j^{-1} = q_{j,\mathrm{lin}}, \qquad p_{j,\mathrm{lin}} \circ h_j = \bar{h}_j \circ p_{j,\log},$$

and the gluing gives the typed line-bundle comparison

$$p_{j,\log}^{*}(L|_{N_j^{*}}) \simeq h_j^{*} p_{j,\mathrm{lin}}^{*}(L|_{J^{*}}).$$

For the filled factor extension return to the vector cover. With

$$P_{j,\log} : J_j' \to N_j, \qquad \pi_j^{\mathrm{vec}} := P_{j,\log} \circ r_j : \widetilde{J}_j \to N_j, \qquad \Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda \rangle,$$

where $P_{j,\log}$ is the full quotient and $P_{j,\log}|_{J_j'^{*}} = p_{j,\log}$, pull back $L|_{N_j}$ by π_j^{vec}, push its integral factor extension out along $\mathbb{Z} \hookrightarrow \mathbb{R}$, and choose normalized vertical lifts $U_\lambda^{(j)}$. For $\alpha \in \Gamma_j^{\log}$, define $A_\alpha \in \mathrm{Aut}(\Lambda)$ and $T_\alpha \in \mathrm{Aut}(V_{\mathbb{R}})$ by

$$\alpha \tau_\lambda \alpha^{-1} = \tau_{A_\alpha \lambda}, \qquad T_\alpha = (A_\alpha^{-1})^{\mathsf{t}}.$$

For a factor-extension lift $\widetilde{G}_\alpha$, define $\widetilde{d}_j(\alpha) \in V_{\mathbb{R}}$ by

$$\widetilde{G}_\alpha U^{(j)}_{A_\alpha^{-1}\lambda} \widetilde{G}_\alpha^{-1} = \mathbf{c}\Big(\langle \widetilde{d}_j(\alpha), \lambda \rangle\Big)\, U_\lambda^{(j)}.$$

The normalized product gives

$$\widetilde{d}_j(\alpha\beta) = \widetilde{d}_j(\alpha) + T_\alpha \widetilde{d}_j(\beta).$$

Compatible basepoints in the quotient square identify the clockwise boundary lift successively with $q_{j,\mathrm{lin}}$, $q_{j,\log}$, and finally the vector-cover deck transformation $Q_{j,\log}$. The induced extension isomorphism then proves, for some $a_j \in V_{\mathbb{R}}$,

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I)a_j, \qquad \widetilde{d}_j(\tau_v) = \iota_v E, \qquad (\iota_v E)(\lambda) = E(v, \lambda).$$

Consequently, with $N_{T_j} = I + T_j + \cdots + T_j^{m_j - 1}$,

$$N_{T_j} d_j = N_{T_j} \widetilde{d}_j(Q_{j,\log}) = \widetilde{d}_j((Q_{j,\log})^{m_j}) = \widetilde{d}_j(\tau_{v_j}) = \iota_{v_j} E,$$

because $N_{T_j}(T_j - I) = 0$. This proves the finite boundary values, including their signs, inside the actual filled logarithmic deck group and closes the factor-extension gap; the certificate therefore remains **[VERIFIED]**. Changes of line-bundle frame, central lift, normalized vertical system, or logarithm branch alter only the displayed principal character term and leave $h_j, \bar{h}_j$, the quotient square, and (5.17-A15) unchanged. The cusp equality follows from the argument principle in a Stein chart, not from a Chern evaluation on a bounding chain; hence $k = 0$. For adjacent rays in a maximal cone $\sigma = \{v, w, u\}$, the actual toric-open inclusions are

$$U_{\rho_v} = D(z_w z_u) \subset U_\sigma, \qquad U_{\rho_w} = D(z_v z_u) \subset U_\sigma, \qquad U_{\rho_v} \cap U_{\rho_w} = U_0.$$

The contractible Stein comparison chart

$$S_\sigma = \{(z_v, z_w, z_u) \in \mathbb{C}^3 : |z_v z_w z_u| < \epsilon\}$$

has $\mathrm{Pic}(S_\sigma) = 0$, so its frames give zero horizontal winding along the connected one-skeleton. For the support-function Cartier divisor D_a, the exact deck-translate identity

$$\Psi_\lambda^{*} D_a - D_a = \mathrm{div}(\chi^{m_\lambda})$$

shows that multiplication by χ^{m_λ} extends from the dense torus to an isomorphism

$$\mu_\lambda^{+} : \Psi_\lambda^{*} \mathcal{O}(D_a) \xrightarrow{\sim} \mathcal{O}(D_a).$$

The map on the negative divisor required by the descent tensor product is the inverse-dual

$$\mu_\lambda^- = ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

Here the cusp shear is the typed lattice isomorphism

$$\Lambda_{\mathrm{tor}} := \ker(M_0 - I) = \mathrm{im}(M_0 - I) = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad \overline{\Lambda} := \Lambda / \Lambda_{\mathrm{tor}},$$

$$B_0 : \overline{\Lambda} \xrightarrow{\sim} \Lambda_{\mathrm{tor}}, \qquad B_0 \overline{\widehat{\gamma}} = -\widehat{\delta}, \qquad B_0 \overline{\widehat{u}} = \widehat{w}.$$

Thus its matrix is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ from the ordered basis $(\overline{\widehat{\gamma}}, \overline{\widehat{u}})$ to $(\widehat{w}, \widehat{\delta})$; in particular B_0 is a map between two named lattices, not an untyped endomorphism, and $\det B_0 = 1$. For $M = L|_{N_0}$, if (d_1, d_2, d_3) are the independent degrees of $\nu^*(M|_W)$, then

$$\left(\deg_{C_i} \nu^*(M|_W)\right)_{i=1}^6 = (d_1, d_2, d_3, d_1, d_2, d_3),$$

and their support function has polarization

$$B \in \mathrm{Sym}^2(\Lambda_{\mathrm{tor}}^*), \qquad [B]_{(\widehat{w}, \widehat{\delta})} = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}.$$

For the Appell–Humbert form $E_M := c_1(M|_F) \in \mathrm{Alt}^2(\Lambda, \mathbb{Z})$, one has

$$E_M((\lambda, 0), (0, m)) = -B(B_0 \lambda, m) \quad (\lambda \in \overline{\Lambda},\ m \in \Lambda_{\mathrm{tor}}).$$

The already proved antecedent $k = 0$ gives $E_M = c_1(L|_F) = 0$; only then does the unimodularity of B_0 force $B = 0$, then $d_1 = d_2 = d_3 = 0$, and hence $\nu^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}$. Beyond the preceding coinvariant input, this proof uses neither a global integral Leray transgression, local integral nearby-cycle specialization, nor Chern-class evaluation on a cusp bounding chain; see theorems 46.19 to 46.21.

12. **[REFUTED]** The separate split-extension calculation imported from CDP98 replaces monodromy coinvariants by the entire fibre homology. The correct formula is

$$(K \rtimes Q)^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q.$$

13. **[VERIFIED]** A corrected replacement theorem now survives both printed defects. If one $L_0 \in \mathrm{Pic}(X)$ satisfies

$$H^2(X, T_X \otimes L_0) = 0,$$

then generic endpoint vanishing, semicontinuity, and Hirzebruch–Riemann–Roch give

$$H^i(X, T_X \otimes M) = 0 \ (i \neq 1), \qquad \chi(X, T_X \otimes M) = \tfrac{1}{2} c_3(X) \leq 0$$

for general M. It is enough that $R^2 f_*(T_X \otimes L) = 0$ for one L. For reduced full fibres, the invalid p. 692 shortcut is replaced exactly by injectivity of

$$\delta_c : H^0(W_c, \mathrm{tors}\, \Omega^1_{W_c} \otimes A_c) \longrightarrow H^1(W_c, N^*_{W_c/X} \otimes A_c)$$

together with the two endpoint vanishings. See theorems 46.28 to 46.30. The constructed candidate fails the global and relative conditions for separate reasons. Since $c_3(X) = 2$, the contrapositive of theorem 46.28 gives $H^2(X, T_X \otimes L_0) \neq 0$ for every L_0. Independently, its conductor makes $R^2 f_*(T_X \otimes L) \neq 0$ for every L, so the stronger relative sufficient condition also fails. The original CDP20 theorem therefore remains refuted.

14. **[VERIFIED]** The constructed X satisfies the stated hypotheses of CDP20 Theorem 2.2 but has both $c_3(X) = 2 > 0$ and $a(X) = 1$. Thus it is a counterexample to that theorem and its ensuing S^6 corollary as stated. This theorem-level conclusion uses the complete construction, topology, and algebraic-reduction proofs above; it is not inferred merely from the two printed proof-route defects.

15. **[VERIFIED]** The candidate's Proposition 10.8 conclusion holds. The workbench proves

$$R^1 f_* \mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad \mathrm{Pic}(X) \simeq \mathbb{C},$$

and the finite-filling Hodge-line calculation now distinguishes the descended canonical section from the direct-image frame. In the notation of theorem 41.1,

$$\mathrm{div}(\overline{\sigma}_j) = k_j S_j, \qquad f_{j*}\omega_{N_j/D_j} = \mathcal{O}_{D_j} \cdot s_j^{k_j} \vartheta_j', \qquad k_j = m_j - 1 - a_j.$$

Thus $\overline{\sigma}_2$ has divisor $2S_2$ and is not itself a nowhere-vanishing frame, while the displayed invariant module still has the claimed generator, a unit times $s_j^{-a_j} e$, over the punctured cover. This removes a local wording defect without changing the Hodge line or any subsequent direct-image calculation. The workbench also proves degree-one base change at every scheme-theoretic fibre, and $\mathrm{ord}(\mathcal{O}_X(S_j)|_{S_j}) = m_j$. The smooth-fibre period condition is exactly $c\tau + d\mu - b = 0$; hence every parameter outside $\mathbb{R}$ and a countable set gives one line bundle non-torsion on every full fibre and on both S_j. The cusp extension is proved intrinsically through nearby cycles and does not depend on the separately reconstructed cellular collapse; see theorems 41.4 and 41.6.

16. **[VERIFIED]** The alternate Mayer–Vietoris route is now closed. The finite-band inclusions are exactly $\alpha_q = (\pi_{1*}, -\pi_{2*})$ in named primitive bases; the second cusp attachment has degree-one determinant $-p$ and reduced degree-two block $\mathrm{diag}(-p, 1)$; and the dual obstruction squares commute with the three verified Leray transgressions. For $p = -1$ both global blocks are unimodular, giving an independent integral sphere certificate; see theorems 44.10, 44.11, 44.13 and 44.18.

17. **[VERIFIED]** The Hodge diamond and Frölicher sequence are now exact. Here are the typed logarithmic objects used in the calculation:

$$B = \mathbb{P}^1, \qquad \Sigma = p_0 + p_1 + p_2, \qquad D = f^{-1}(\Sigma)_{\mathrm{red}} = W + S_1 + S_2,$$
$$\mathcal{L} = \Omega^1_B(\log \Sigma) \simeq \mathcal{O}_B(1), \qquad \mathcal{A} = \Omega^1_X(\log D), \qquad \mathcal{E} = \mathcal{A}/f^*\mathcal{L}, \qquad \mathcal{G} = R^1 f_* \mathcal{O}_X.$$

They occur in the locally free logarithmic conormal sequence

$$0 \longrightarrow f^*\mathcal{L} \longrightarrow \mathcal{A} \longrightarrow \mathcal{E} \longrightarrow 0,$$

whose connecting maps have the exact types

$$\kappa_0 : f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes \mathcal{G}, \qquad \kappa_1 : R^1 f_* \mathcal{E} \longrightarrow \mathcal{L} \otimes R^2 f_* \mathcal{O}_X \simeq \mathcal{O}_B.$$

The logarithmic relative direct images are

$$f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^1 f_* \mathcal{E} \simeq \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2}, \qquad R^2 f_* \mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1),$$

and are locally free and compatible with base change.

On a finite cover $t = s^m$, with $(m, a) = (3, 2)$ or $(4, 1)$, choose equivariant relative-cotangent frames (e_0, e_1) and structure-sheaf frames (b_0, b_1) with semilinear characters $(1, \zeta^a)$ and $(1, \zeta^{-a})$, respectively. The actual $\mathbb{C}\{t\}$-bases are

$$\begin{aligned} f_*\mathcal{E} :&\quad e_0,\ s^{m-a} e_1, \\ R^1 f_* \mathcal{E} :&\quad e_0 b_0,\ s^a e_0 b_1,\ s^{m-a} e_1 b_0,\ e_1 b_1, \\ R^2 f_* \mathcal{E} :&\quad s^a e_0 (b_0 \wedge b_1),\ e_1 (b_0 \wedge b_1). \end{aligned}$$

Let $\ell = dt/t$, and let $D_{\mathrm{KS}} = (d_{rs})_{0\le r,s\le 1}$ be the marked Kodaira–Spencer matrix after its common factor ds is removed. Put

$$A = \frac{s}{m}d_{00}, \qquad B_1 = \frac{s^{1-a}}{m}d_{01}, \qquad C = \frac{s^{1-a}}{m}d_{10}, \qquad D_0 = \frac{s^{m+1-2a}}{m}d_{11}.$$

In source basis $(e_0, s^{m-a}e_1)$, target basis $(b_0, s^a b_1)\ell$, ordered degree-one source basis $(e_0b_0, s^a e_0 b_1, s^{m-a}e_1b_0, e_1b_1)$, and target generator $s^a(b_0 \wedge b_1)\ell$, the coarse maps are exactly

$$[\kappa_0] = \begin{pmatrix} A & tB_1 \\ C & D_0 \end{pmatrix}, \qquad [\kappa_1] = (-C, A, -D_0, B_1).$$

The finite residue is the typed map

$$\mathrm{res}_{1,j} : (\mathcal{L}\otimes\mathcal{G})|_{p_j} \longrightarrow H^1(S_j, \mathcal{O}_{S_j}), \qquad \mathrm{res}_{1,j}((xb_0 + ys^a b_1)\ell) = m_j x[b_0] \quad (x, y \in \mathbb{C}),$$

and the derived residue square gives

$$\mathrm{res}_{1,j}\,\kappa_0 = 0, \qquad \mathrm{im}(\kappa_0 \bmod t) = \ker \mathrm{res}_{1,j} = \mathbb{C}\, s^a b_1.$$

At the cusp put

$$\mathcal{Q}_{\mathrm{cusp}} := \nu_*\mathcal{O}_{\mathrm{dP}_6}/\mathcal{O}_W,$$

and let $\rho : \mathcal{E}|_W \to \mathcal{Q}_{\mathrm{cusp}}$ be the relative normalization–residue quotient map. Retain the connecting isomorphisms

$$\delta_0 : H^0(\mathcal{Q}_{\mathrm{cusp}}) \xrightarrow{\sim} H^1(W, \mathcal{O}_W), \qquad \delta_1 : H^1(\mathcal{Q}_{\mathrm{cusp}}) \xrightarrow{\sim} H^2(W, \mathcal{O}_W).$$

For

$$q_1 = (-1, 0, 1), \qquad q_2 = (0, -1, -1), \qquad \eta_i = d\log x_i, \qquad \alpha_i = \delta_0(q_i),$$

let $\theta \in H^1(\mathcal{Q}_{\mathrm{cusp}})$ be the unique class satisfying

$$\delta_1(\theta) = -\alpha_1 \smile \alpha_2.$$

The standard Čech differential $(\delta c)_{i_0\ldots i_{r+1}} = \sum_k (-1)^k c_{i_0\ldots\widehat{i_k}\ldots i_{r+1}}$ fixes the sign

$$\delta_1\big(\rho_*(\alpha\otimes\eta_i)\big) = -\alpha \smile \alpha_i.$$

Consequently the two cusp maps, with their domains and codomains, are

$$H^0(\mathcal{E}|_W) \xrightarrow{\sim} H^0(\mathcal{Q}_{\mathrm{cusp}}),$$

$$H^1(\mathcal{E}|_W) = \mathbb{C}\langle \alpha_1\eta_1, \alpha_1\eta_2, \alpha_2\eta_1, \alpha_2\eta_2\rangle \xrightarrow{(0,1,-1,0)} H^1(\mathcal{Q}_{\mathrm{cusp}}) = \mathbb{C}\theta;$$

the second is surjective.

Finally, the invariant section $\xi = \partial/\partial\zeta_2$, nowhere zero in the locally free relative sheaf $T_{X/B} = \mathcal{E}^\vee$, gives the split evaluation sequence

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow f_*\mathcal{E} \xrightarrow{\mathrm{ev}_\xi} \mathcal{O} \longrightarrow 0.$$

There is a unique lift $h \in H^0(B, f_*\mathcal{E})$ of $1 \in H^0(B, \mathcal{O})$, because $H^0(B, \mathcal{O}(-1)) = 0$. On the marked cover it has the uniquely determined form

$$h = (a(z), 1)\begin{pmatrix} d\zeta_1 \\ d\zeta_2 \end{pmatrix}, \qquad a(z_2) = \frac{1}{2}.$$

Use the genuine invariant target frame

$$\omega_2 = s\,u_2(s)(\gamma \wedge u)\ell, \qquad u_2(0) = 1.$$

The determinant and final global scalar are

$$\operatorname{div}(\det \kappa_0) = p_1 + p_2, \qquad \kappa_1(h \otimes \gamma)|_{p_2} = -\frac{1}{4}\dot{\mu}(z_2)\,\omega_2|_{p_2} \neq 0 \quad \text{in } (\mathcal{L} \otimes R^2 f_* \mathcal{O}_X)|_{p_2}.$$

This uses $\dot{\tau}(z_2) = 0$ and $K(z_2) = 6\dot{\mu}(z_2)^2 \neq 0$.

Thus, with $\mathcal{K} = \ker \kappa_1$,

$$\mathcal{K} \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2},$$

and the torsion-free absolute cotangent direct images are

$$R^0 f_* \Omega^1_X \simeq \mathcal{O}(-2), \quad R^1 f_* \Omega^1_X \simeq \mathcal{K}, \quad R^2 f_* \Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

Therefore

$$h^{1,2} = h^{2,1} = 1, \qquad h^{1,1} = h^{2,2} = 2.$$

In full, with row index q and column index p, the Hodge table is

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	1
1	1	2	1	0
0	1	0	0	0.

The two middle d_1-rows are exact $0 \to \mathbb{C} \to \mathbb{C}^2 \to \mathbb{C} \to 0$, and

$$\dim_{\mathbb{C}} E_1 = 10, \qquad \dim_{\mathbb{C}} E_2 = 2, \qquad E_2 = E_\infty.$$

See theorems 42.5 to 42.7, 42.9 and 42.10.

18. **[REFUTED]** Remark 1.3's assertion that the relative differential sheaves have torsion along all three singular fibres is false already for $\Omega^1_{X/B}$. The $t = xy$ and $t = xyz$ modules inject into the free logarithmic relative module, so $\Omega^1_{X/B}$ is torsion-free but non-locally-free near the cusp. At a point of the smooth finite cover $q_j : J'_j \to N_j$, with $R_j = \mathcal{O}_{J'_j,x} \simeq \mathbb{C}\{s, y_1, y_2\}$, and $t = s^{m_j}$, the exact module is

$$(q_j^* \Omega^1_{N_j/B})_x \simeq \Omega^1_{J'_j/B,x} \simeq R_j\,dy_1 \oplus R_j\,dy_2 \oplus \left(R_j/(s^{m_j-1})\right)ds,$$

and hence

$$(q_j^* \operatorname{tors} \Omega^1_{N_j/B})_x = \left(R_j/(s^{m_j-1})\right)ds.$$

Faithful flatness descends this module and its support. Thus the torsion support is exactly the two multiple fibres and is absent at the cusp; see the local calculation (11.17f).

19. **[VERIFIED]** The Bott–Chern and Aeppli dimensions are now exact as well. McHugh's threefold formula gives

$$h^{1,2}_{BC} = h^{1,2} + \dim E_2^{3,1} - k^{2,0} = 1 + 0 - 0 = 1,$$

because the de Rham image of the pure doubly closed forms surjects onto $K^{2,0}$, so $k^{2,0} \leq b_2 = 0$. The complete Bott–Chern rows $q = 3, 2, 1, 0$ are

$$(0,0,1,1), \quad (0,1,2,1), \quad (0,2,1,0), \quad (1,0,0,0),$$

and Aeppli duality gives

$$(0,0,0,1), \quad (0,1,2,0), \quad (1,2,1,0), \quad (1,1,0,0).$$

See theorem 42.11; no dimension remains open.

20. **[VERIFIED]** Global representatives and the complete Bott–Chern product structure are now known. The two Dolbeault inputs must not be conflated. The distinguished class

$$\mathfrak{g} \in H^{0,1}_{\bar\partial}(X) = H^1(X, \mathcal{O}_X)$$

is the unique coherent-Leray class mapping to the invariant section $\gamma^{0,1} \in H^0(B, R^1 f_* \mathcal{O}_X)$. In contrast, the marked integral fibre class is

$$q = \eta = u \wedge w + 6\gamma \wedge \delta \in H^2(F_b; \mathbb{Z}),$$

whose holomorphic extension is a section $\boldsymbol{\eta} \in H^0(B, R^1 f_* \Omega^1_X)$. Choose a coherent-Leray lift

$$\mathfrak{q} \in H^{1,1}_{\bar\partial}(X) \quad \text{with edge value} \quad \boldsymbol{\eta}.$$

This lift is not distinguished: its exact ambiguity is

$$\mathfrak{q} \longmapsto \mathfrak{q} + c\, d_1\mathfrak{g}, \qquad c \in \mathbb{C},$$

because the kernel of the edge map is $H^1(B, f_*\Omega^1_X) = \mathbb{C}\, d_1\mathfrak{g}$.

Choose global $\bar\partial$-closed smooth representatives

$$G \in \mathcal{A}^{0,1}(X), \quad [G]_{\bar\partial} = \mathfrak{g}, \qquad Q \in \mathcal{A}^{1,1}(X), \quad [Q]_{\bar\partial} = \mathfrak{q}.$$

For a marked smooth fibre put

$$y = \operatorname{Im}\tau, \qquad m_\mu = \operatorname{Im}\mu, \qquad r = \operatorname{Im}\beta, \qquad \Delta_F = r - \frac{6m_\mu^2}{y} < 0,$$

and define the translation-invariant forms

$$\Gamma = \frac{i}{2\Delta_F}\left(d\bar\zeta_2 - \frac{m_\mu}{y} d\bar\zeta_1\right), \qquad U = \frac{i}{2y\Delta_F}\left(r\, d\bar\zeta_1 - 6m_\mu\, d\bar\zeta_2\right),$$

$$Q_F = -d\zeta_1 \wedge U - 6 d\zeta_2 \wedge \Gamma.$$

The restriction statements are equalities of typed Dolbeault classes, not claims about a preferred literal coefficient gauge:

$$i_b^*\mathfrak{g} = [\Gamma]_{\bar\partial} \in H^{0,1}_{\bar\partial}(F_b), \qquad i_b^*\mathfrak{q} = [Q_F]_{\bar\partial} \in H^{1,1}_{\bar\partial}(F_b).$$

After fixing $\mathfrak{g}$ and one $\mathfrak{q}$, changing smooth representatives by $G \mapsto G + \bar\partial\phi$ and $Q \mapsto Q + \bar\partial\Psi$ changes the derived representatives only by $\partial\bar\partial$-exact terms. Thus no preferred literal global coefficient atlas through the singular fillings is asserted or needed. Put

$$\mathsf{A} := [\partial G]_{BC}, \qquad \mathsf{B} := [\bar\partial \bar G]_{BC} \quad \text{in } H^{1,1}_{BC}(X).$$

Let

$$\exp : H^1(X, \mathcal{O}_X) \longrightarrow \operatorname{Pic}(X)$$

be induced by $z \mapsto e^{2\pi i z}$, and orient $\omega \in H^2(B; \mathbb{Z})$ by $\omega = c_1(\mathcal{O}_B(1))$. The integral Picard normalization and Bott–Chern first Chern class are

$$\exp(12\mathfrak{g}) = f^*\mathcal{O}_B(1), \qquad c_{1,BC}(\exp(z\mathfrak{g})) = z\mathsf{A} + \bar z\, \mathsf{B},$$

$$c_{1,BC}(f^*\mathcal{O}_B(1)) = 12(\mathsf{A} + \mathsf{B}).$$

The absolute factor is 12; choosing the opposite primitive orientation replaces $\mathfrak{g}$ by $-\mathfrak{g}$ in the pullback identity. The exact fibre pairings, with the orientation stated, are

$$\int_{(F_b,\mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F^2 = -12,$$

$$\int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F \wedge \Gamma \wedge d\zeta_2 = -1.$$

They prove the needed Dolbeault nonvanishing, while the constant fibre coefficients give the typed vanishing

$$i_b^* d_1 [G \wedge Q]_{\bar\partial} = i_b^* [\partial (G \wedge Q)]_{\bar\partial} = 0 \quad \text{in } H^{2,2}_{\bar\partial}(F_b).$$

Choose $\upsilon \in \mathcal{A}^{3,3}(X)$ with $\int_X \upsilon = 1$. Together with the unit $\mathbf{1} = [1]_{BC} \in H^{0,0}_{BC}(X)$, define

class		bidegree
$\mathsf{A} = [\partial G]_{BC},$	$\mathsf{B} = [\bar\partial \bar G]_{BC}$	$(1,1)$
$\mathsf{C} = [\partial Q]_{BC},$	$\overline{\mathsf{C}} = [\bar\partial \bar Q]_{BC}$	$(2,1),(1,2)$
$\mathsf{D} = [\partial (G \wedge Q)]_{BC},$	$\overline{\mathsf{D}} = [\bar\partial (\bar G \wedge \bar Q)]_{BC}$	$(2,2),(2,2)$
$\mathsf{E} = [\partial (Q^2)]_{BC},$	$\overline{\mathsf{E}} = [\bar\partial (\bar Q^2)]_{BC}$	$(3,2),(2,3)$
$\mathsf{T} = [\upsilon]_{BC},$	$\int_X \mathsf{T} = 1$	$(3,3).$

These are bases of every nonzero positive-degree Bott–Chern group. For their span N_{BC},

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1} \oplus N_{\mathrm{BC}}, \qquad N_{\mathrm{BC}}^2 = 0.$$

See theorems 43.4 and 43.5. The lift and smooth-gauge ambiguities above change neither the verified dimensions nor the square-zero product algebra.

21. **[VERIFIED]** The positive radial cusp now has an explicitly declared flat quotient metric and an exact scalar eigenvalue calculation. Write

$$\mu_T = \alpha_T + i m_T, \qquad h_T = \eta_T + i H_T, \qquad b_T = \xi_T + i B_T, \qquad c_T = \xi_T - \eta_T,$$

$$L_T = T + H_T, \qquad q_T = L_T - B_T, \qquad D_T = L_T q_T + 6 m_T^2.$$

Set also

$$m = \operatorname{Im} \mu(0), \qquad H = \operatorname{Im} h(0), \qquad B = \operatorname{Im} b(0).$$

In the ordered lattice basis $(\widehat\gamma, \widehat u, \widehat w, \widehat\delta)$ and real coordinate order (x_1, y_1, x_2, y_2), the real period matrix, its oriented determinant, and its positive covolume are

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}, \qquad \det P_T = -D_T, \qquad \operatorname{covol}(P_T \mathbb{Z}^4) = D_T > 0, \qquad D_{\mathrm{can},T} = -\frac{D_T}{L_T} < 0.$$

The full inverse-dual certificate is

$$P_T^{-\mathsf{T}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \dfrac{m_T}{D_T} & \dfrac{q_T}{D_T} & \dfrac{-6m_T\alpha_T - q_T\eta_T}{D_T} & \dfrac{-m_T c_T - q_T \alpha_T}{D_T} \\ 0 & 0 & 0 & 1 \\ -\dfrac{L_T}{D_T} & \dfrac{6m_T}{D_T} & \dfrac{6(L_T\alpha_T - m_T\eta_T)}{D_T} & \dfrac{L_T c_T - 6 m_T \alpha_T}{D_T} \end{pmatrix}.$$

Thus $(P_T\mathbb{Z}^4)^* = P_T^{-\mathsf{T}}\mathbb{Z}^4$, and for $k = (r, s, n_1, n_2)^{\mathsf{t}}$, with

$$R_T(k) = r - 6\alpha_T n_1 - c_T n_2, \qquad S_T(k) = s - \eta_T n_1 - \alpha_T n_2,$$

its member indexed by k is exactly

$$\left(n_1, \frac{m_T R_T(k) + q_T S_T(k)}{D_T}, n_2, \frac{-L_T R_T(k) + 6m_T S_T(k)}{D_T}\right).$$

For the declared Euclidean quotient metric g_T, the first positive scalar eigenvalue is, for all large T,

$$\lambda_1(F_T, g_T) = 4\pi^2 \min\left\{\frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2}\right\} = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right).$$

The metric normalization is recorded separately and gives

$$\begin{aligned} \widehat{g}_T &= D_T^{-1/2} g_T, & \mathrm{Vol}(F_T, \widehat{g}_T) = 1, \\ \lambda_1(F_T, \widehat{g}_T) &= \frac{4\pi^2}{T}\left(1 + \frac{-H - B/2 + 2\min\{B, 0\}}{T} + O(T^{-2})\right), \\ \lambda_1(F_T, \widehat{g}_T) &\longrightarrow 0. \end{aligned}$$

Normalized Haar measure on the fixed angular torus is only the unitary measure convention used below; it is not the metric rescaling $g_T \mapsto \widehat{g}_T$. See theorems 9.1 to 9.3.

22. **[VERIFIED]** The degree- N^2 regular-fibre cover extends across the completed local toric cusp, with the base-change and total degrees kept distinct. Retain

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s},$$

let $Y = Y_\Sigma$ be the toric threefold of the height-one translation-invariant A_2 fan, and let the target action be

$$\Psi_{\bar\lambda}(x, t) = (e^{2\pi i C(t)\bar\lambda} t^{B_0\bar\lambda} x, t), \qquad f_0 : N_0 := Y/\bar\Lambda \longrightarrow \Delta_t.$$

For every integer $N \geq 1$, after shrinking the disc, put

$$\rho_N(v) = v^N, \qquad C_N(v) = \frac{1}{N} C(v^N), \qquad \Psi_{\bar\lambda}^{(N)}(\xi, v) = (e^{2\pi i C_N(v)\bar\lambda} v^{B_0\bar\lambda}\xi, v).$$

This is a free properly discontinuous action on a copy $Y^{(N)}$ of Y, and

$$f_N : N_0^{\langle N\rangle} := Y^{(N)}/\bar\Lambda \longrightarrow \Delta_v$$

is a proper holomorphic map from a smooth threefold, with reduced irreducible nonnormal central fibre W_N normalized by dP_6. Multiplication by N on the three-dimensional cocharacter lattice gives

$$P_N : Y^{(N)} \longrightarrow Y, \qquad (\xi_1, \xi_2, v) \longmapsto (\xi_1^N, \xi_2^N, v^N),$$

whose maximal-chart ring map is

$$\mathbb{C}\{z_0, z_1, z_2\} \longrightarrow \mathbb{C}\{w_0, w_1, w_2\}, \qquad z_i \longmapsto w_i^N.$$

It is finite flat of degree N^3, is equivariant for the two displayed actions, and descends to a finite flat degree- N^3 map

$$\kappa_N : N_0^{\langle N\rangle} \longrightarrow N_0, \qquad f_0 \circ \kappa_N = \rho_N \circ f_N, \qquad \kappa_N^* W = N W_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N.$$

For the normal base change $N_0[N] = N_0 \times_{\Delta_t} \Delta_v$, the unique factorization

$$N_0^{\langle N \rangle} \xrightarrow{Q_N} N_0[N] \longrightarrow N_0$$

has generic degree N^2. If $N > 1$, Q_N is quasi-etale and nonflat: its non-etale and nonflat loci are precisely the inverse images of the double and triple strata, where the closed-fibre lengths are respectively $2N - 1 > N$ and $3N^2 - 3N + 1 > N^2$. On reduced central fibres the generic degree is N^2, and the normalization square is

$$\begin{array}{ccc} \mathrm{dP}_6 & \xrightarrow{[N]} & \mathrm{dP}_6 \\ {\scriptstyle \nu_N}\downarrow & & \downarrow{\scriptstyle \nu} \\ W_N & \longrightarrow & W, \end{array}$$

where $[N]$ has degree N^2 and ramification index N on each of the six toric boundary curves. Over $v \neq 0$, Q_N is the cover for

$$\mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta} \rangle \subset \Lambda, \qquad \mathrm{Deck}(Q_N) \cong (\mathbb{Z}/N\mathbb{Z})^2,$$

and, with $S_N = \mathrm{diag}(1, 1, N, N)$,

$$S_N^{-1} M_0^N S_N = M_0.$$

This is a local completed-cusp morphism, not a degree- N^2 self-map over the unchanged disc, not a normalization, and not a cover over the entire base; see theorems 9.8 to 9.10. Its use in four-dimensional finite-cover unspooling is an ancillary construction: it changes both the fibre lattice and the cusp base. It does not answer the native S^{12}/S^{24} sphere-route problem, whose solution begins from the unaltered quotient fibres Q_T and C_T.

23. **[VERIFIED]** The native cusp quotient and the invariant-circle quotient give distinct exact conical smash-sphere carriers. With

$$K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle = \mathrm{im}(M_0 - I), \qquad \overline{\Lambda} = \Lambda / K_\Lambda, \qquad \overline{T}^2 = (\overline{\Lambda} \otimes \mathbb{R})/\overline{\Lambda},$$

there is a fibrewise real-analytic group epimorphism

$$\overline{\pi} : N_0 \setminus W \longrightarrow D_0^* \times \overline{T}^2$$

with connected kernel and trivial target local system. On $F_T = f^{-1}(e^{-2\pi T})$, it is π_T after the isometry

$$\iota_T : \overline{T}^2 \xrightarrow{\sim} Q_T, \qquad [(a, b)] \longmapsto [Y_T(a, b)^{\mathsf{T}}], \qquad Y_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix},$$

and, for a fixed constant $C_{\mathrm{sys}} > 0$, $\mathrm{sys}(Q_T) \geq T - C_{\mathrm{sys}}$. For

$$\mathcal{A} = \frac{[0, \epsilon] \times S^1 \times \overline{T}^2}{\{0\} \times S^1 \times \overline{T}^2 \sim v_0,\ \{\epsilon\} \times S^1 \times \overline{T}^2 \sim v_1} \cong \mathrm{Susp}(T^3),$$

the map

$$q_2 : (X, x_0) \longrightarrow (\mathcal{A}, v_0), \qquad q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)]$$

for $t(f(x)) = re^{i\phi}$, with $\overline{\pi}_f = \mathrm{pr}_{\overline{T}^2} \circ \overline{\pi}$, $q_2(W) = v_0$ and $q_2(X \setminus \mathrm{int}\, N_0) = v_1$, is a continuous surjective based quotient. Hence

$$q_2^{\wedge 2} : X \wedge X \cong S^{12} \longrightarrow \mathcal{A} \wedge \mathcal{A}$$

is global, and its synchronized radial restriction is exactly

$$\pi_T \times \pi_T : F_T \times F_T \longrightarrow Q_T \times Q_T, \qquad (Q_T \times Q_T, 0) \xrightarrow[T\to\infty]{\text{pointed}} (\mathbb{R}^4, 0)$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

The cusp-supported one-circle route uses

$$\mathcal{B} = \frac{[0,\epsilon] \times S^1 \times S^1}{\{0\} \times S^1 \times S^1 \sim b_0,\ \{\epsilon\} \times S^1 \times S^1 \sim b_1} \cong \operatorname{Susp}(T^2)$$

and the continuous surjective based quotient

$$q_{1,\text{cusp}} : (X, x_0) \longrightarrow (\mathcal{B}, b_0), \qquad q_{1,\text{cusp}}(x) = [r, e^{i\phi}, \chi_\gamma(x)]$$

on $N_0 \setminus W$, with $q_{1,\text{cusp}}(W) = b_0$ and the exterior collapsed to b_1. Its fourth smash power is

$$q_{1,\text{cusp}}^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{B}^{\wedge 4},$$

and its synchronized radial restriction is $\chi_{\gamma,T}^{\times 4} : F_T^4 \to C_T^4$.

For the full-base circle route, let

$$P = B \setminus \bigcup_{j=0}^{2} \operatorname{int} D_j, \qquad \mathcal{M} = (P \times S^1) \underset{h_1^{S^1}}{\cup} C_1 \underset{h_2^{S^1}}{\cup} C_2, \qquad \partial\mathcal{M} = \partial D_0 \times S^1,$$

and $\mathcal{C} = \mathcal{M} \cup_{\partial\mathcal{M}} \operatorname{Cone}(\partial\mathcal{M})$, with cone vertex c_0. The continuous surjective based quotient

$$q_1 : (X, x_0) \longrightarrow (\mathcal{C}, c_0)$$

equals (f, θ_γ) over P, equivalently $\chi_\gamma|_{f^{-1}(P)}$, and equals the finite-filling maps $N_j \to C_j$ for $j = 1, 2$, sends $x \in N_0 \setminus W$ to $[r, (e^{i\phi}, \theta_\gamma(x))]$, and sends W to c_0. Its fourth smash power is

$$q_1^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{C}^{\wedge 4},$$

whose synchronized radial restriction is exactly

$$\begin{aligned} \chi_{\gamma,T}^{\times 4} &: F_T^4 \longrightarrow C_T^4, \\ C_T &= \mathbb{R}/\ell_T\mathbb{Z}, \\ \ell_T &= \frac{D_T}{\sqrt{L_T^2 + m_T^2}} = T + H - B + O(T^{-1}), \end{aligned}$$

so

$$(C_T^4, 0) \xrightarrow[T\to\infty]{\text{pointed}} (\mathbb{R}^4, 0)$$

in the same three pointed topologies.

The exponents 2 and 4, and hence the source-sphere dimensions 12 and 24, are optimal for these two native quotient problems with the unscaled quotient of the marked Euclidean fibre metric. Indeed, if a primitive kernel $N_{\ker} \subset \Lambda^{\oplus r}$ defines a connected torus quotient of r cusp fibres whose quotient systole tends to infinity, then every uniformly bounded period must lie in $N_{\ker}$. Thus

$$K_\Lambda^{\oplus r} \subset N_{\ker}, \qquad \operatorname{rank}(\Lambda^{\oplus r}/N_{\ker}) \le 2r.$$

For a quotient over the full regular base, or over any finite cover of that base, monodromy forces the additional sublattice $\mathbb{Z}\langle\widehat{u}\rangle^{\oplus r}$ into $N_{\ker}$: one uses $A_1\widehat{w} = \widehat{u} - \widehat{w}$, and on a finite cover the exact formula

$$(A_2A_1)^m\widehat{w} = \widehat{w} + m(\widehat{u} - \widehat{\delta})$$

together with primitivity of $N_{\ker}$. Consequently

$$\operatorname{rank}(\Lambda^{\oplus r}/N_{\ker}) \leq r.$$

A rank-four expanding quotient therefore needs at least two copies cusp-locally and at least four copies globally; the displayed Q_T^2 and C_T^4 constructions attain the two bounds.

After collapsing the coordinate one-skeleton

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})) \subset \overline{T}^2,$$

the two native routes also give continuous surjective based quotient maps

$$\widehat{q}_3 : X \twoheadrightarrow S^3, \qquad \widehat{q}_\gamma : X \twoheadrightarrow S^2,$$

and hence, using the fixed based choice $X \cong_{\mathrm{diff}} S^6$,

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8.$$

Both full sphere carriers are based null-homotopic. First let $f = \widehat{q}_\gamma : S^6 \to S^2$ after the chosen based identification. There is an exact factorization

$$S^{12} = S^6 \wedge S^6 \xrightarrow{f\wedge \mathrm{id}_{S^6} = \Sigma^6 f} S^2 \wedge S^6 = S^8 \xrightarrow{\mathrm{id}_{S^2}\wedge f} S^2 \wedge S^2 = S^4.$$

Freudenthal stability gives $\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0$, so $f \wedge f$ is based null-homotopic. Since $f^{\wedge r} = (f \wedge f) \wedge f^{\wedge(r-2)}$, it follows that

$$[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r}) \qquad \text{for every } r \geq 2.$$

In particular, the displayed surjective quotient $\widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$ is itself based null-homotopic.

For the other carrier, let $g : S^6 \to S^3$ be any based map. Its exact smash factorization is

$$g \wedge g = (\Sigma^3 g) \circ (\Sigma^6 g) : S^{12} \longrightarrow S^9 \longrightarrow S^6.$$

Write st for stabilization. The exact low-stem data are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \quad \pi_3^{\mathrm{S}} = \mathbb{Z}/24\langle\nu\rangle, \quad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\langle\nu^2\rangle,$$

$$\pi_{12}(S^6) = \mathbb{Z}/2\langle\nu_6 \circ \nu_9\rangle, \qquad \mathrm{st} : \pi_{12}(S^6) \xrightarrow{\cong} \pi_6^{\mathrm{S}}.$$

The full groups are tabulated in [9, p. 339]; the signed smash formula and the unstable stabilization calculation are [12, Propositions 3.1 and 5.11]. Since $12[g] = 0$, one has $\mathrm{st}([g]) = 2a\nu$ for some $a \in \mathbb{Z}/12$. By multiplicativity,

$$\mathrm{st}([g \wedge g]) = \mathrm{st}([g])^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

The stabilization isomorphism forces $[g \wedge g] = 0 \in \pi_{12}(S^6)$. Thus $g^{\wedge r}$ is based null-homotopic for every $r \geq 2$. Taking $g = \widehat{q}_3$ proves

$$[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Both displayed maps remain surjective quotient maps. The one-copy classes $[\widehat{q}_3] \in \pi_6(S^3)$ and $[\widehat{q}_\gamma] \in \pi_6(S^2)$ have not been computed. Their synchronized radial restrictions contain based topologically embedded four-spheres, equipped separately with the intrinsic quotient length metrics

$$\mathbb{S}^4_{12,T} = (Q_T/K_T^{(1)})^{\wedge 2} \hookrightarrow S^6, \qquad \mathbb{S}^4_{24,T} = C_T^{\wedge 4} \hookrightarrow S^8,$$

obtained from, respectively,

$$F_T^2 \xrightarrow{\pi_T^{\times 2}} Q_T^2 \xrightarrow{c_T^{\times 2}} (S_T^2)^2 \longrightarrow \mathbb{S}^4_{12,T}, \qquad F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \longrightarrow \mathbb{S}^4_{24,T},$$

where the last arrows collapse the fat wedges. The full maps from F_T^2 and F_T^4 retain the nontrivial kernels of their first arrows; only the quotient-stage maps are injective off their explicitly collapsed skeleton/fat-wedge loci. Equip the displayed products with their Euclidean ℓ^2-product metrics and give each four-sphere the induced quotient length metric. If z_T is the image of $[(1/2, 1/2)]$ and

$$\zeta_{12,T} = [z_T, z_T], \qquad \zeta_{24,T} = [\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2],$$

then every radius-R ball about these points is exactly a Euclidean four-ball once, respectively,

$$T - C_{\mathrm{sys}} > 4R, \qquad \ell_T > 4R.$$

Therefore

$$(\mathbb{S}^4_{12,T}, \zeta_{12,T}) \longrightarrow (\mathbb{R}^4, 0), \qquad (\mathbb{S}^4_{24,T}, \zeta_{24,T}) \longrightarrow (\mathbb{R}^4, 0)$$

in pointed local-isometric and pointed Gromov–Hausdorff topology, with smooth convergence on the eventual exact Euclidean neighborhoods of the displayed base points. Both based inclusion maps

$$i_{12,T} : \mathbb{S}^4_{12,T} \hookrightarrow S^6, \qquad i_{24,T} : \mathbb{S}^4_{24,T} \hookrightarrow S^8$$

are null-homotopic: sliding their fixed interior suspension coordinate to a suspension endpoint contracts them through based maps. Thus $(i_{12,T})_*[S^4] = 0 \in H_4(S^6; \mathbb{Z})$ and $(i_{24,T})_*[S^4] = 0 \in H_4(S^8; \mathbb{Z})$. These slice-inclusion null-homotopies are distinct from the global equalities $[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6)$ and $[\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8)$.

Deleting the smash basepoint nevertheless gives literal nonclosed topologically embedded copies of $\mathbb{R}^4$:

$$(Q_T \setminus K_T^{(1)})^2 \cong \mathbb{S}^4_{12,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad (C_T \setminus \{0\})^4 \cong \mathbb{S}^4_{24,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^8.$$

They are open only in the embedded four-spheres and are not globally Euclidean-isometric under a fixed compatible compact-target metric.

There are also Hausdorff, second-countable topological five-manifolds with boundary $\mathfrak{E}_{12}$ and $\mathfrak{E}_{24}$ whose positive fibres are $Q_{1/s}^{\times 2}$ and $C_{1/s}^{\times 4}$, respectively, and whose zero fibres are literally $\mathbb{R}^4$. Continuous surjections $\beta_\nu : \mathfrak{E}_\nu \to \mathcal{D}_\nu$ identify every positive fibre with its radial carrier slice and collapse the zero fibre to the cone vertex. They are homeomorphisms off the zero fibre but are nonproper and nonquotient; thus they are comparison/collapse maps, not topological blow-downs or resolutions, and $\mathcal{D}_\nu$ is not the quotient $\mathfrak{E}_\nu/\mathbb{R}^4$ via β_ν.

The metric quantifiers are essential. For every fixed metric compatible with the compact conical target topology, the target slices $j_T(Q_T^2)$, $k_T^{\mathcal{B}}(C_T^4)$, and $k_T^{\mathcal{C}}(C_T^4)$ converge in Hausdorff, hence Gromov–Hausdorff, distance to the smash basepoint. A fixed cusp-adapted metric can realize the intrinsic limits only after a moving-point blow-up: with $r_T = e^{-2\pi T}$, the factors T/r_T and ℓ_T/r_T give the exact induced slice metrics

$$\left(\frac{T}{r_T}\right)^2 j_T^* g_{Z_{12}} = g_{Q_T} \oplus g_{Q_T}, \qquad \left(\frac{\ell_T}{r_T}\right)^2 (k_T^{\mathcal{D}})^* g_{Z_{24}^{\mathcal{D}}} = g_{C_T}^{\oplus 4}, \quad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}.$$

The slice bubbles are $\mathbb{R}^4$, but the corresponding full ambient bubbles are $\mathbb{R}^8$ for the S^{12} route and $\mathbb{R}^{12}$ for either S^{24} route. More generally, if $\lambda_T > 0$ and $\beta_T = \lambda_T r_T$, the slice limit is a point when $\beta_T \to 0$, a square flat four-torus of side L when $\beta_T \to L \in (0, \infty)$, and $\mathbb{R}^4$ when $\beta_T \to \infty$; in the last case the cone vertex is at scaled distance $\beta_T \to \infty$. Hence the $\mathbb{R}^4$ limits are neither vertex tangent cones nor ambient limits of fixed smooth metrics on the source spheres, whose ordinary tangent dimensions are 12 and 24.

The original maps $q_2^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $q_1^{\wedge 4}$ have the displayed conically stratified codomains. The repeated-smash null theorems for the sphere-valued composites $\widehat{q}_3$ and $\widehat{q}_\gamma$ do not supply null-homotopies before the final one-skeleton/base-angle quotients. The unstable classes of

$$q_2^{\wedge 2} : S^{12} \to \mathcal{A}^{\wedge 2}, \qquad q_1^{\wedge 4} : S^{24} \to \mathcal{C}^{\wedge 4}$$

remain **[OPEN]**: their zero ordinary Hurewicz images and absence of homotopy sections do not decide nullity. The class of $q_{1,\mathrm{cusp}}^{\wedge 4} : S^{24} \to \mathcal{B}^{\wedge 4}$ has likewise not been computed. The sphere codomains occur only after the additional one-skeleton and base-angle collapses defining $\widehat{q}_3$ and $\widehat{q}_\gamma$. All identifications of the sources with S^{12} and S^{24} use the chosen based diffeomorphism $X \cong_{\mathrm{diff}} S^6$. See theorems 9.11 to 9.14, 9.16, 9.18 to 9.25, 9.27, 9.28, 9.34 to 9.36 and 9.38.

24. **[REFUTED]** The assertion that the literal fixed-target map

$$\chi_\gamma : J \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z})$$

extends continuously across W, or that

$$\chi_\gamma^{[4]} : J^{[4]} \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z})^4$$

extends with the stated regular-fibre restriction to the compact fourfold fibre product or to X^4, is false: two explicit paths have the same toric limit in W and constant character values $3/4$ and $1/2$. The cone map above changes the codomain by collapsing $\partial D_0 \times S^1$ to c_0, so it does not contradict this nonextension. Moreover

$$X^2 \simeq S^6 \times S^6 \not\simeq S^{12}, \qquad X^4 \simeq (S^6)^4 \not\simeq S^{24};$$

only the smash products satisfy $X^{\wedge 2} \cong S^{12}$ and $X^{\wedge 4} \cong S^{24}$. Neither smash sphere admits an almost-complex structure, and neither route constructs a compact complex sphere; see theorems 9.7 and 9.18.

25. **[VERIFIED]** The exceptional first-coordinate elliptic quotient is now classified on the actual period image, rather than on an arbitrary ambient path. For

$$\mathcal{A}_z = \mathbb{C}^2/\mathbb{Z}\langle (6\mu, \beta), (\tau, \mu), (1, 0), (0, 1)\rangle,$$

the first coordinate induces a surjective holomorphic homomorphism

$$\mathcal{A}_z \longrightarrow \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$$

if and only if $6\mu = a\tau + b$ for a unique $(a, b) \in \mathbb{Z}^2$. Its connected kernel has period $\tau_f = a\mu - \beta$ and $\operatorname{Im} \tau_f = -D_{\mathrm{can}} > 0$. The exact marked-label action is

$$g_0 : (a, b) \mapsto (a, a + b), \qquad g_2 : (a, b) \mapsto (-b, 6 + a).$$

The union of these loci has finite image in the completed orbifold base and is absent from a punctured cusp neighborhood. Consequently this particular elliptic/Landau sector does not produce the cusp soft modes. This statement does not classify all elliptic quotients, all Néron–Severi classes, or all smooth integral magnetic fluxes. See theorems 9.39 to 9.41.

26. **[VERIFIED]** The fibrewise small eigenvalues are assembled into one specified gapless operator. On

$$\mathscr{H}_{\mathrm{aux}} = L^2([T_0,\infty), dT; L^2_0(\mathbb{R}^4/\mathbb{Z}^4, dy)) \cong \bigoplus_{k\in\mathbb{Z}^4\setminus\{0\}} L^2([T_0,\infty), dT),$$

put

$$\lambda_k(T) = 4\pi^2 |P_T^{-\mathsf{T}} k|^2, \qquad (\mathscr{L}_{\mathrm{aux}} u)_k(T) = \lambda_k(T) u_k(T),$$

$$D(\mathscr{L}_{\mathrm{aux}}) = \left\{ u : \sum_{k\neq 0} \int_{T_0}^{\infty} \lambda_k(T)^2 |u_k(T)|^2 \, dT < \infty \right\}.$$

This multiplication operator is nonnegative and self-adjoint, $\ker \mathscr{L}_{\mathrm{aux}} = 0$, and the escaping-support Weyl sequence in the $k = (1,0,0,0)$ channel proves

$$0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}}).$$

This is a spectral-gap theorem for the displayed auxiliary direct integral, not for the complex threefold without a metric and not for a quantum gauge Hamiltonian; see theorem 38.1.

27. **[VERIFIED]** The cover index and the cubical mesh are independent. For integers $M_{\mathrm{cov}} \geq 1$ and $N_{\mathrm{mesh}} \geq 2$, put

$$S_{M_{\mathrm{cov}}} = \mathrm{diag}(1,1,M_{\mathrm{cov}},M_{\mathrm{cov}}), \qquad P_{T,M_{\mathrm{cov}}} = P_T S_{M_{\mathrm{cov}}},$$

$$\widetilde{F}_{T,M_{\mathrm{cov}}} = \mathbb{R}^4/P_{T,M_{\mathrm{cov}}}\mathbb{Z}^4 \longrightarrow F_T, \qquad \mathrm{Deck} \cong (\mathbb{Z}/M_{\mathrm{cov}}\mathbb{Z})^2.$$

This cover has degree M_{cov}^2, Gram matrix and covolume

$$G_{T,M_{\mathrm{cov}}} = P_{T,M_{\mathrm{cov}}}^{\mathsf{T}} P_{T,M_{\mathrm{cov}}}, \qquad \mathcal{D}_{T,M_{\mathrm{cov}}} = M_{\mathrm{cov}}^2 D_T.$$

For the cubical complex write, once and for all,

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1,2,3,4\}.$$

The diffeomorphism

$$\Phi_{T,M_{\mathrm{cov}}} : \mathbb{R}^4/\mathbb{Z}^4 \xrightarrow{\sim} \widetilde{F}_{T,M_{\mathrm{cov}}}, \qquad [y] \longmapsto [P_{T,M_{\mathrm{cov}}} y]$$

transports the whole standard N_{mesh}-cubical complex; its vertex map is precisely

$$(\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4 \xrightarrow{\sim} \widetilde{F}_{T,M_{\mathrm{cov}}}[N_{\mathrm{mesh}}], \qquad [n] \longmapsto \left[\frac{P_{T,M_{\mathrm{cov}}} n}{N_{\mathrm{mesh}}} \right],$$

so the finite set is the torsion subgroup, not the entire torus. Its oriented edge is

$$\gamma_{n,j}(t) = \left[\frac{P_{T,M_{\mathrm{cov}}}(n + t\mathbf{e}_j)}{N_{\mathrm{mesh}}} \right], \qquad 0 \leq t \leq 1.$$

For a compact matrix Lie group G, the parallel-transport convention

$$U'(t) = -A_{\gamma(t)}(\dot{\gamma}(t)) U(t), \qquad U(0) = 1$$

defines the typed map

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g}) \longrightarrow G^{E_{N_{\mathrm{mesh}}}}, \qquad A \longmapsto (U_A(\gamma_{n,j}))_{n,j},$$

and for $A^g = g^{-1}Ag + g^{-1}dg$ its exact endpoint law is

$$\mathrm{Hol}(A^g)_{n,j} = g(\gamma_{n,j}(1))^{-1} \, \mathrm{Hol}(A)_{n,j} g(\gamma_{n,j}(0)).$$

See equation (318) and theorems 11.1 and 11.2.

28. **[VERIFIED]** The compact-link SU(2) flux uses the inverse-transport convention and has an exact metric-weighted action. With

$$H_{\mathrm{link}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = 1, \qquad N = N_{\mathrm{mesh}},$$

set

$$\begin{aligned} U_1(n) &= \exp(-2\pi n_2 H_{\mathrm{link}}/N^2), \\ U_2(n) &= \begin{cases} 1, & 0 \le n_2 \le N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}}/N), & n_2 = N-1, \end{cases} \\ U_3(n) &= U_4(n) = 1. \end{aligned}$$

For

$$U_{ij}(n) = U_i(n) U_j(n+\mathbf{e}_i) U_i(n+\mathbf{e}_j)^{-1} U_j(n)^{-1},$$

including all wrap-around edges,

$$\begin{aligned} U_{12}(n) &= \exp(2\pi H_{\mathrm{link}}/N^2), \\ U_{ij}(n) &= 1 \quad (i<j,\ (i,j) \neq (1,2)). \end{aligned}$$

The principal logarithm is unambiguous because $0 < 2\pi/N^2 \le \pi/2$, and

$$\mathcal{F}_N^{\log}(n) = N^2 \operatorname{Log}(U_{12}(n))\, dy^1 \wedge dy^2 = 2\pi H_{\mathrm{link}}\, dy^1 \wedge dy^2.$$

With the full inverse Gram metric,

$$S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2 N^4} \sum_{n \in V_{N_{\mathrm{mesh}}}} \|\mathcal{F}_N^{\log}(n)\|^2_{G^{-1}_{T,M_{\mathrm{cov}}} \otimes \mathfrak{su}(2)} = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}.$$

There is no canonical scalar Wilson action for a general nonorthogonal metric obtained by independent scalar plaquette weights. For this configuration the explicitly defined $(1,2)$-sector functional has

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = (G^{-1}_{T,M_{\mathrm{cov}}})_{11}(G^{-1}_{T,M_{\mathrm{cov}}})_{22} - (G^{-1}_{T,M_{\mathrm{cov}}})^2_{12} = D_T^{-2},$$

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}} \kappa^{12}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2} \sum_{n \in V_{N_{\mathrm{mesh}}}} (2 - \operatorname{Re}\operatorname{tr} U_{12}(n)) = \frac{M_{\mathrm{cov}}^2 N^4}{2g_{\mathrm{YM}}^2 D_T} \left(2 - 2\cos\frac{2\pi}{N^2}\right)$$

Its exact mesh expansion is

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T} \left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right),$$

so its $N \to \infty$ limit is $2\pi^2 M_{\mathrm{cov}}^2/(g_{\mathrm{YM}}^2 D_T)$. See theorem 11.3.

29. **[VERIFIED]** The finite Gaussian oscillator is defined on the real reduced Coulomb space with the physical inverse-Gram exterior metric. For

$$\vartheta^j_{T,M_{\mathrm{cov}}} = P^{-\mathsf{T}}_{T,M_{\mathrm{cov}}} \mathbf{e}_j, \qquad \mathfrak{d}_{N_{\mathrm{mesh}}}(k)_j = N_{\mathrm{mesh}}(e^{2\pi i k_j/N_{\mathrm{mesh}}} - 1),$$

let

$$C^p_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}}) = \operatorname{Map}\left((\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \bigwedge^p (\mathbb{R}^4)^* \otimes \mathfrak{su}(2)\right).$$

The exterior metric is

$$\left\langle \vartheta^{i_1}\wedge\cdots\wedge\vartheta^{i_p},\vartheta^{j_1}\wedge\cdots\wedge\vartheta^{j_p}\right\rangle=\det\left((G^{-1}_{T,M_{\mathrm{cov}}})_{i_aj_b}\right)_{a,b=1}^{p},$$

and the cochain inner product is

$$\langle a,b\rangle_p=\frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{N_{\mathrm{mesh}}^4}\sum_{n\in V_{N_{\mathrm{mesh}}}}\langle a(n),b(n)\rangle.$$

Writing $\mathsf{T}_j a(n)=a(n+\mathbf{e}_j)$, put

$$D_j^+=N_{\mathrm{mesh}}(\mathsf{T}_j-1),\qquad D_j^-a(n)=N_{\mathrm{mesh}}(a(n)-a(n-\mathbf{e}_j)),$$

and then

$$d_N=\sum_{j=1}^4\varepsilon(\vartheta^j)D_j^+,\qquad d_N^*=-\sum_{j=1}^4\iota_{(\vartheta^j)^\sharp}D_j^-,$$

$$\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}=\ker(d_N^*:C^1\to C^0)\cap(\ker\Delta_N^{(1)})^\perp,\qquad\mathcal{L}=g_{\mathrm{YM}}^{-2}d_N^*d_N.$$

The positive eigenvalue on the Fourier fibre $k\in(\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4\setminus\{0\}$ is

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k)=\frac{1}{g_{\mathrm{YM}}^2}\left\|P_T^{-\mathsf{T}}S_{M_{\mathrm{cov}}}^{-1}\mathfrak{d}_{N_{\mathrm{mesh}}}(k)\right\|^2,$$

with complex transverse multiplicity 9, and

$$\lambda(\mathbf{e}_1)=\frac{4N_{\mathrm{mesh}}^2\sin^2(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}^2}\frac{L_T^2+m_T^2}{D_T^2},$$

which is independent of M_{cov}. On $L^2(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}},da)$, the Friedrichs realization of

$$\widehat{H}^{\mathrm{Gauss}}=-\frac{1}{2}\Delta_{\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}}+\frac{1}{2}\langle a,\mathcal{L}a\rangle$$

has a unique Gaussian ground state and

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}})=\sqrt{\min_{k\neq0}\lambda(k)}\le\frac{2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}}\frac{\sqrt{L_T^2+m_T^2}}{D_T}.$$

For the explicit sequence

$$T_j=j^2,\qquad M_{\mathrm{cov},j}=j,\qquad N_{\mathrm{mesh},j}=j^3,$$

one has simultaneously

$$\mathrm{sys}(\widetilde{F}_{T_j,M_{\mathrm{cov},j}})\to\infty,\quad\max_r\frac{\|P_{T_j,M_{\mathrm{cov},j}}\mathbf{e}_r\|}{N_{\mathrm{mesh},j}}\to0,\quad\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}})\to0,\quad S^{\log}_{T_j,M_{\mathrm{cov},j},N_{\mathrm{mesh},j}}(U)\to0.$$

See theorems 11.4 and 11.5.

30. **[VERIFIED]** The componentwise exponential

$$\mathrm{Exp}_{N_{\mathrm{mesh}}}:C^1\longrightarrow\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}},\qquad a\longmapsto(\exp(a_e/N_{\mathrm{mesh}}))_e$$

is only a local diffeomorphism at zero. Its derivative is

$$D\,\mathrm{Exp}_{N_{\mathrm{mesh}}}(0)a=N_{\mathrm{mesh}}^{-1}a,$$

and, if a smooth compact-link action S_W has the identity as a critical point, then exactly

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0[a,b] = N_{\mathrm{mesh}}^{-2} D^2(S_W)_{\mathbf{1}}[a,b].$$

It is not globally injective: a cochain supported on one edge with value $2\pi N_{\mathrm{mesh}} H_{\mathrm{link}}$ has the same image as zero. Thus

$$\mathcal{H}^{\mathrm{Gauss}} = L^2(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}, da) \quad \text{and} \quad \mathcal{H}^{\mathrm{inv}}_{N_{\mathrm{mesh}}} = L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU)^{\mathrm{SU}(2)^{V_{N_{\mathrm{mesh}}}}}$$

are not identified by a global coordinate chart or a constructed unitary map; moreover $\delta_U \notin L^2$ for product Haar measure. No nonlinear compact-link Hamiltonian, Gauss-law spectral problem, time direction, reflection-positive transfer matrix, renormalized-scale limit, or Osterwalder–Schrader reconstruction is supplied by these results. See theorem 11.6 and equations (370) and (371).

31. **[VERIFIED]** The reducible direct-sum connection on $\mathscr{E}_T = \mathscr{L}_T \oplus \mathscr{L}_T^{-1}$ is the non-flat Yang–Mills critical point which, in the lifted splitting over $\mathbb{R}^4$, has the connection form

$$\nabla^T = \mathrm{d} + 2\pi u_1 \, \mathrm{d}u_2 \, H_{\mathrm{mag}}, \qquad H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},$$

$$F_{\nabla^T} = \frac{2\pi}{D_T} H_{\mathrm{mag}} \, \mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad \mathrm{d}^*_{\nabla^T} F_{\nabla^T} = 0.$$

The underlying principal $SU(2)$-bundle is smoothly trivial; after choosing any of the proved global $SU(2)$-trivializations, the displayed lifted-splitting form is carried to its gauge transform on that trivial bundle. It is not asserted to remain literally $\mathrm{d} + 2\pi u_1 \, \mathrm{d}u_2 \, H_{\mathrm{mag}}$ in an arbitrary global trivialization. For $\mathrm{YM}_T(\nabla) = (2g_{\mathrm{YM}}^2)^{-1} \int_{F_T} |F_\nabla|^2 \, \mathrm{d}\,\mathrm{vol}_{F_T}$, its action is

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T} = \frac{2\pi^2}{g_{\mathrm{YM}}^2 T^2} \left(1 + \frac{B - 2H}{T} + O(T^{-2})\right).$$

On the background-gauge slice its classical Jacobi operator is

$$(\mathcal{J}_T a)_\nu = -\sum_{\mu=1}^{4} \nabla_{e_\mu} \nabla_{e_\mu} a_\nu - 2 \sum_{\mu=1}^{4} [F_{\nu\mu}, a_\mu],$$

and an explicit smooth real unit vector satisfies

$$\mathrm{d}^*_{\nabla^T} a = 0, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a, \qquad \mathrm{YM}''_T \,|_{\nabla^T}(a,a) = -\frac{4\pi}{g_{\mathrm{YM}}^2 D_T}.$$

Thus the displayed reducible classical critical point is a saddle for every finite T; this is not a statement about a quantum Yang–Mills Hamiltonian. See theorems 12.3, 12.4 and 12.8.

32. **[VERIFIED]** The monodromy-invariant full-flux class gives an exact split-Cartan $SU(r)$ theorem for every nonzero

$$\mathbf{m} = (m_1, \ldots, m_r) \in \mathbb{Z}^r, \qquad \sum_i m_i = 0, \qquad s_2(\mathbf{m}) = \sum_i m_i^2.$$

On the regular native fibre define

$$\mathscr{E}_{\mathbf{m},T} = \bigoplus_{i=1}^{r} \mathscr{L}_{\eta,T}^{\otimes m_i}, \qquad H_{\mathbf{m}} = i \, \mathrm{diag}(m_1, \ldots, m_r).$$

The ordered splitting canonically and parallelly trivializes the determinant, and, retaining the two opposite orientations used by the native construction,

$$c_1(\mathscr{E}_{\mathbf{m},T}) = 0, \qquad c_2(\mathscr{E}_{\mathbf{m},T}) = -\frac{s_2(\mathbf{m})}{2}[\eta_T]^2,$$

$$\langle c_2, [F_T]_{\mathrm{mark}}\rangle = -6s_2(\mathbf{m}), \qquad k_{\mathbf{m}} := \langle c_2, [F_T]_{x\text{-complex}}\rangle = 6s_2(\mathbf{m}).$$

Here $s_2(\mathbf{m})$ is even, so this split family realizes precisely a subfamily with $k_{\mathbf{m}} \in 12\mathbb{Z}_{>0}$. The diagonal connection

$$\nabla^{\mathbf{m},T} = \mathrm{d} - 2\pi(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3)H_{\mathbf{m}}, \qquad F_{\mathbf{m},T} = -2\pi H_{\mathbf{m}}\eta_T$$

is a reducible non-flat Yang–Mills critical point and has exact action

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T}) = \frac{2\pi^2 k_{\mathbf{m}}}{g_{\mathrm{YM}}^2} + \frac{\pi^2 s_2(\mathbf{m}) S_T^2}{g_{\mathrm{YM}}^2 D_T}.$$

For the positive root $i < j$, put $q_{ij} = |m_i - m_j|$ and

$$\rho_T^* = \frac{\max\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}}{\min\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}}, \qquad N_-(T) = \#\{n \geq 0 : 2n+1 < \rho_T^*\}.$$

Its complete negative normalized Coulomb spectrum is

$$\nu_{ij,n}(T) = q_{ij}\big((2n+1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big), \qquad 0 \leq n < N_-(T),$$

with real multiplicity $12q_{ij}^2$ before summing coincident numerical bands. Consequently

$$\mathrm{ind}_{\mathbb{R}}\,\mathcal{J}_{\mathbf{m},T} = 12rN_-(T)s_2(\mathbf{m}),$$

and on the deep cusp tail, where $N_-(T) = 3$, this is $36rs_2(\mathbf{m}) = 6rk_{\mathbf{m}}$. The module also computes the centralizer harmonic kernel and every odd-frequency wall nullity; negative eigenspaces are descent directions, whereas only the zero-mode walls are possible Lyapunov–Schmidt bifurcation parameters. Finally, naturality of c_2 proves that this nonzero-flux bundle cannot extend over a carrier with $H^4(-;\mathbb{Z}) = 0$, in particular not over a space diffeomorphic to S^6, S^{12}, or S^{24}. See theorems 14.1, 14.2 and 14.5 to 14.7.

33. **[VERIFIED]** The Koide relation has an exact ambient quadratic-form interpretation, and it is rigorously separated from the regular canonical cusp family. For

$$Q_K(r) = \frac{2}{3}(r_1+r_2+r_3)^2 - (r_1^2+r_2^2+r_3^2), \qquad \begin{pmatrix} h \\ x \\ y \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix},$$

one has $Q_K(r) = h^2 - x^2 - y^2$. The real-linear coefficient-space isomorphism

$$(h,x,y) \longmapsto (t,m,\varrho) = \left(h+y, \frac{x}{\sqrt{6}}, h-y\right)$$

satisfies, with

$$(t,m,\varrho) = (\mathrm{Im}\,\tau, \mathrm{Im}\,\mu, \mathrm{Im}\,\beta),$$

for the canonical realification of $Z = \left(\begin{smallmatrix} 6\mu & \tau \\ \beta & \mu \end{smallmatrix}\right)$,

$$\det P(\tau,\mu,\beta) = t\varrho - 6m^2 = \det \begin{pmatrix} t & \sqrt{6}\,m \\ \sqrt{6}\,m & \varrho \end{pmatrix} = Q_K(r).$$

For $r \in \mathbb{R}^3_{\geq 0} \setminus \{0\}$, the Koide equation $Q_K(r) = 0$ makes this symmetric matrix positive semidefinite of rank one and makes P singular. Its columns therefore do not form a full lattice and do not define a regular compact complex two-torus. In contrast, the actual positive radial cusp has

$$\det P_T = -D_T < 0, \qquad D_T = T^2 + (2H - B)T + O(1), \qquad \det P_T \sim -T^2.$$

Thus the result is an isomorphism of ambient coefficient spaces with their quadratic discriminants, not a map from Koide triples to fibres of the constructed S^6 family; see theorems 24.1 to 24.3.

34. **[VERIFIED]** Independently of that discriminant comparison, the positive defect-one shell has an exact integral parametrization and a power-saving count with its exact constant. Put

$$\Delta(x, y, z) = 4(xy + xz + yz) - (x^2 + y^2 + z^2)$$

and let $R(X)$ count triples $0 < x \leq y \leq z \leq X$ with $\Delta = -1$ on the terminal Vieta wall $z \geq 4(x + y)$. The integral change of variables

$$n = z - 4(x + y), \qquad q = y - x, \qquad a = x + y - 4n$$

is a bijection onto

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0,\ q, n \geq 0,\ a + 4n > q, \qquad z = 4a + 17n.$$

Write

$$\mathscr{X} = \{(a, q, n) \in \mathbb{A}^3 : a^2 - 3q^2 - 18n^2 = -2\}.$$

The exact local densities are

$$\sigma_2 = \frac{3}{2}, \qquad \sigma_3 = 2, \qquad \sigma_p = \frac{1 - p^{-2}}{1 - (\frac{3}{p})p^{-1}} \qquad (p \geq 5),$$

and the Poincaré-residue measure of the real chamber has leading factor

$$I_\infty = \frac{1}{2\sqrt{3}} \log\left(\frac{3}{2\sqrt{2}}\right).$$

After verifying the Gorodnik–Oh–Shah sector hypotheses and Borovoi's adelic sign coefficient $c_{\mathscr{X}} = 1$, one obtains unconditionally

$$R(X) \sim C_0 X, \qquad C_0 = \frac{9}{2\pi^2} \log(2 + \sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right).$$

The exact ideal parametrization now proves more. For $n \geq 1$, put

$$\begin{aligned} m_n &= 18n^2 - 2 = 2(3n - 1)(3n + 1), \\ \mathcal{P}(m_n) &= \{(C, D, E) \in \mathbb{Z}^2_{>0} \times \mathbb{Z} : \\ &\qquad C^2 D = m_n, \quad 0 \leq E < D, \quad E^2 \equiv 3 \pmod{D}\}. \end{aligned}$$

The map

$$(C, D, E) \longmapsto C(D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z})$$

is a bijection to the ideals of $\mathbb{Z}[\sqrt{3}]$ of norm m_n. Each ideal has a unique seed $a_0 + q_0\sqrt{3}$ with $a_0 > 2q_0 \geq 0$, and every solution in its orbit is

$$a_k + q_k\sqrt{3} = (a_0 + q_0\sqrt{3})(2+\sqrt{3})^k, \qquad k \geq 0.$$

The inverse is finite and integral: apply $(a, q) \mapsto (2a - 3q, 2q - a)$ while $a \leq 2q$, then recover the unique HNF triple. Consequently every point has multiplicity one and the number of unit orbits at level n is

$$\#\mathcal{P}(m_n) = \rho(m_n) = \sum_{d|m_n} \chi_{12}(d) = \rho(3n-1)\rho(3n+1).$$

The omitted $n = 0$ face is separate and exact:

$$a_k + q_k\sqrt{3} = (1+\sqrt{3})(2+\sqrt{3})^k, \qquad k \geq 1,$$

subject to the unchanged strict wall. Its contribution is the floor count $R_0(X)$ of equation (1034), and $R_0(X) = O(\log(2+X))$.

The exact sharp-height floor formula and its Hecke-phase Fourier expansion are written in theorems 29.3 and 30.2. A direct verification of Benoist–Oh effective well-roundedness for the literal five-wall family, followed by their Theorem 13.2 [53, pp. 1931–1932] on each of the finitely many integral spin-group orbits, strengthens the asymptotic to

$$\boxed{R(X) = C_0 X + O(X^{1-\eta})} \qquad \text{for some fixed } \eta > 0.$$

Neither Benoist–Oh's theorem nor its proof supplies $\eta \geq \frac{1}{2}$. The primary-source comparison is exact: Liu–Sarnak give the smooth standard-radial exponent $39/64$ for this congruence arithmetic lattice, while Good and Chatzakos–Petridis give the sharp standard-radial $2/3$ frontier after the relevant small-spectrum terms are displayed. Neither theorem counts the unchanged five-wall region with height $4a + 17n$, and neither proves its pointwise square-root discrepancy; see section 37.1.

The exact census gives $R(10^8) = 3{,}536{,}002$, after an independent six-coordinate replay of all 3,536,002 rows. Its largest $|R(X) - C_0X|/\sqrt{X}$ on $10^4 \leq X \leq 10^8$ is $0.17140019280880145597\ldots$, at $X = 867{,}240$; this is finite evidence only. The stronger proposed estimate $R(X) - C_0X = O_\varepsilon(X^{1/2+\varepsilon})$ is neither asserted nor used in this status item; the proved conclusion is the displayed linear power-saving asymptotic. See theorems 25.4, 25.6, 25.7, 25.11, 25.12 and 32.2 and sections 35 and 37.1.

35. **[VERIFIED]** The same parity-restricted shell has an exact quaternionic realization. In

$$B = (18, -6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18, J^2 + 6, IJ + JI), \qquad K = IJ,$$

put

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}\frac{I+J}{2} \oplus \mathbb{Z}\frac{I-J}{2} \oplus \mathbb{Z}\frac{K}{2}.$$

The complete multiplication table proves that this is an order. Its reduced discriminant is 108, so it has index 18 in a maximal overorder; the integer 18 is not an Eichler level, since the order is nonmaximal at the ramified primes 2 and 3. The algebra is a division algebra over $\mathbb{Q}$, split over $\mathbb{R}$, and ramified exactly at $\{2, 3\}$. The bijection

$$(a, q, n) \longmapsto \gamma(a, q, n) = 2 + \frac{qI + aJ + nK}{2}$$

identifies $a^2 - 3q^2 - 18n^2 = -2$, with its forced parity $a \equiv q \pmod 2$, with the reduced-trace-4, reduced-norm-1 locus of $\mathcal{O}$. Every such γ is primitive hyperbolic, with

$$\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}], \qquad C_{\mathcal{O}^1}(\gamma) = \{\pm\gamma^k : k \in \mathbb{Z}\},$$

eigenvalues $2 \pm \sqrt{3}$, and translation length $2\log(2+\sqrt{3})$. The full shell is a finite union of $\mathcal{O}^1$-conjugacy classes. For the displayed real splitting, the standard radial Huber parameter is exactly

$$X_i(\gamma)^2 = 1 + \frac{(7a + 15\sqrt{2}\, n)^2}{48},$$

whereas the Koide height is $4a + 17n$. Thus the hyperbolic reduction is exact, but the radial theorem does not transfer a discrepancy estimate to the sharp Koide polyhedron; see sections 36, 36.7 and 36.8.

36. **[VERIFIED]** The fixed-prime Erdős–Straus divisor selector is an exact marked slice of the candidate's terminal integral presentations. For $p \equiv 1 \pmod{12}$ prime, put

$$L = \frac{p-1}{4}, \qquad R_j = 4j + 3, \qquad a_j = \frac{p + R_j}{4}, \qquad 0 \le j < L.$$

If $u \in \mathbb{Z}_{>0}$, $u \mid a_j^2$, $\varepsilon \in \{0, 1\}$, and $R_j \mid 4u + p^\varepsilon$, then

$$T = \frac{4u + p^\varepsilon}{R_j}, \qquad (\ell_0, \ell_1, \ell_2) = \left(\frac{T+1}{4}, u - jT, 1\right)$$

is integral and satisfies

$$12\ell_0 - 4\ell_1 - 3\ell_2 = p^\varepsilon,$$

with the exact relation-matrix identity

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1 - 4\ell_0 \end{pmatrix}.$$

The inverse is $T = 4\ell_0 - 1$, $u = \ell_1 + jT$. With p, j, the channel ε, positive orientation, and the divisor gate retained, this is a bijection between the complete two-channel selector and the normalized terminal slice. It proves an exact integral-relator crosswalk, not a geometric morphism from the reciprocal cubic to X; see section 6 and theorem 6.3.

37. **[VERIFIED]** The two-column $SU(3)$ Gram quotient is the four-dimensional positive Hermitian cone:

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})/SU(3) \cong \mathrm{Herm}_2(\mathbb{C})_{\ge 0}.$$

Its one-point compactified representation sphere satisfies

$$S^{12}_{\mathrm{col}}/SU(3) \cong B^4,$$

with orbit-type boundary S^3. This S^{12}_{col} is the column representation sphere, not the cusp-smash S^{12}; see theorems 15.1 and 15.5.

38. **[VERIFIED]** The three-column orbit calculation produces an exact rank-three carrier, not a conjectural dimensional analogy. For $\mathscr{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C})$,

$$\mathscr{V}_3/U(3) \cong \mathrm{Herm}_3(\mathbb{C})_{\ge 0}$$

by the complete Gram invariant $A^\dagger A$, whereas strict $SU(3)$ retains the alternating invariant:

$$\mathscr{V}_3/SU(3) \cong \{(H, z) : H \ge 0,\ |z|^2 = \det H\} \cong \mathbb{R}^{10}.$$

Thus

$$S^{17}/U(3) \cong B^8, \quad \mathscr{V}_3^+/U(3) \cong B^9, \quad S^{17}/SU(3) \cong S^9, \quad \mathscr{V}_3^+/SU(3) \cong S^{10}.$$

The compactified rank-loss locus is a standard topologically locally flat $S^8 \subset S^{10}$, and its complement is

$$S^{10} \setminus S^8 \cong \mathbb{R}^8 \times \mathbb{C}^* \simeq S^1,$$

whose positive winding generator is $\det A/|\det A|$. The rank-one trace-one stratum inside the rank-loss sphere is $\mathbb{C}P^2$. The rank-three determinant is cubic; this cone is not the quadratic Lorentz cone arising from two columns. See theorems 16.4, 16.5, 16.7, 16.9 and 16.11.

39. **[VERIFIED]** The selectively corrected Jordan determinant/Euler module supplies an exact algebraic refinement, but no new geometric globalization. In the fixed orientation, every Hermitian matrix has a unique form $S + i[a]_\times$, and

$$\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.$$

For every marked positive-definite symmetric form B, the linear section

$$\ell_B(\lambda, a) = \lambda B + i[a]_\times$$

has the exact positive inverse image

$$\ell_B^{-1}(\mathrm{Herm}_3(\mathbb{C})_{\geq 0}) = \{\lambda \geq 0 : a^{\mathsf{T}} B a \leq \lambda^2 \det B\},$$

a future $(3+1)$-Lorentz cone. The explicit order-cone lift $\Theta : \mathrm{Herm}_2(\mathbb{C}) \to \mathbb{R}I_3 \oplus i\mathfrak{so}(3)$ satisfies

$$\det_3 \Theta(H) = \frac{\operatorname{tr} H}{2} \det_2 H$$

and sends positive-semidefinite ranks $2, 1, 0$ to $3, 2, 0$. For a rank-one projection P, the matrix $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not idempotent, which separately proves that Θ is not a Jordan homomorphism. Neither Θ nor ℓ_B is a toric, period-lattice, or filling map. The claimed index-three translation compression is also rejected: the proved six-to-two operation is the rank-lowering map $P : \mathbb{Z}^3 \twoheadrightarrow \mathbb{Z}^2$, with the three even and three odd Kuhn labels mapping to the two target triangle orbits. See theorems 18.1, 18.7 and 18.8.

40. **[VERIFIED]** The rank-three toroidal successor and its exact quotient to the reconstructed cusp are now constructed. The local analytic family

$$\pi_3 : N_3 \longrightarrow \Delta$$

has smooth fibre E_τ^3, deepest local equation $t = z_0 z_1 z_2 z_3$, cusp monodromy

$$M_3 = \begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix},$$

and filling homomorphism $a_i \mapsto 0,\ b_i \mapsto e_i$. Its irreducible central fibre has

$$(f_0, f_1, f_2, f_3) = (1, 7, 12, 6), \qquad \chi(W_3) = 6.$$

Its normalization is the permutohedral threefold with orbit vector $(1, 14, 36, 24)$; the positive-codimension strata map with degrees $2, 3, 4$ onto the $7, 12, 6$ quotient strata. Hence a real S^8-total-space fibration retaining W_3 requires its remaining singular fibres to contribute Euler sum -4. For a full-rank integral shear $\mathcal{B} : \bar{L}_3 \hookrightarrow L_3$, the exact generalization is

$$\chi(W_{3,\mathcal{B}}) = 3!\,[L_3 : \mathcal{B}\bar{L}_3] = 3!\,|\det \mathcal{B}|.$$

In the candidate decomposition $\Lambda_F = A_F \oplus B_F$, define

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}.$$

Then

$$\Phi = P \oplus Q : H_1(T^6;\mathbb{Z}) \twoheadrightarrow H_1(T^4;\mathbb{Z})$$

obeys the literal monodromy, filling, and period identities

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3, \qquad P Z_3(t) = Z_2(t) Q.$$

It extends over the toric strata to a holomorphic map of cusp families with connected elliptic kernel. Moreover an explicit $J \in SL_6(\mathbb{Z})$ gives

$$J M_3 J^{-1} \cong M_F \oplus \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and the candidate's free order-three and order-four peripheral actions lift on the added elliptic factor.

The latter successful lifts are distinct from the diagonal candidate

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T, \qquad M_1 M_2 M_0 = I.$$

Although its finite factors have orders 4 and 6, the maps $I - M_j : T^6 \to T^6$ are isogenies for $j = 1, 2$, so every affine lift of either finite action has a fixed point.

The six source tetrahedron orbits also map S_3-equivariantly to the six marked diagonal A_2 Weyl chambers: with $\nu_i = E_{ii} - I_3/3$, the π-th cell maps to the cone

$$\mathbb{R}_{\geq 0}\nu_{\pi_1} + \mathbb{R}_{\geq 0}(-\nu_{\pi_3}) = \{\lambda_{\pi_1} \geq \lambda_{\pi_2} \geq \lambda_{\pi_3}\}.$$

This is an exact fan/Cartan correspondence, not a geometric identification with the Gram determinant singularity. The global period-and-overlap gluing over both finite-order discs remains **[OPEN]**. See theorems 17.3, 17.5, 17.8 and 18.3 to 18.6.

41. **[VERIFIED]** The $\mathbb{Z}^6$ native to the higher-rung cusp is

$$\Lambda_{\mathrm{tor}} = H_1(T^6;\mathbb{Z}) = L_3 \oplus \overline{L}_3,$$

not the cyclic group $\mathbb{Z}/6$ and not the separate group $H^4(U;\mathbb{Z})$. For $k = (1,1,1)$, the exact sequence

$$0 \longrightarrow \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,k) \longrightarrow \Lambda_{\mathrm{tor}} \xrightarrow{P \oplus Q} A_F \oplus B_F \longrightarrow 0$$

is S_3-equivariant and compatible with monodromy. It splits as an integral monodromy lattice but not as a $\mathbb{Z}[S_3]$-module. Each quotient copy is the A_2 weight lattice; its root lattice has index three. The six standard generators form two S_3-orbits of size three, whereas the six Kuhn/Weyl chambers form the regular S_3-orbit. Thus

$$\operatorname{rk} \Lambda_{\mathrm{tor}} = 6, \qquad |S_3| = 3! = 6, \qquad \chi(W_{3,\mathcal{B}}) = 3!|\det \mathcal{B}|$$

are three precisely related but nonidentical statements. The actual map to the Gram boundary is $\Phi_{\mathrm{G}} \circ \jmath \circ P \circ \mathrm{pr}_{\mathrm{ang}}$: it is an S_3-equivariant, piecewise-linear chamber map on the real cocharacter space, not an additive map, a toric fan morphism, or a map of torus fibres.

The regular chamber orbit itself has an exact rank-nine realization. With $\mathcal{M}_{\mathbb{Z}} = \mathrm{Hom}(\overline{L}_3, L_3)$,

$$\mathcal{M}_{\mathbb{Z}} = \bigoplus_i \mathbb{Z}E_{ii} \oplus \bigoplus_{i\neq j} \mathbb{Z}E_{ij},$$

the second summand is $\mathbb{Z}[S_3]$, and $c_{(\pi_1\pi_2\pi_3)} \mapsto E_{\pi_1,\pi_3}$ identifies its basis with the six chambers. These are precisely the six mixed basis pairs with zero alternating pairing; the three diagonal matrix units encode the excluded pairs with pairing one. The real matrix carrier is S_3-equivariantly isomorphic to $\mathrm{Herm}_3(\mathbb{C})$, but the integral symmetric/skew splitting has index 8, and every off-diagonal matrix unit maps to an indefinite Hermitian matrix. Hence this is an exact representation and incidence theorem inside the nine-dimensional carrier, not a positive-semidefinite or toroidal map. See theorems 19.1 to 19.3, 20.1 and 20.4.

42. **[VERIFIED]** The marked angular epimorphism now gives an exact positive observation morphism from the full rank-nine carrier to Minkowski space, while retaining and classifying every fibre direction. Put

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad u = (1,1,1)^{\mathsf{T}}, \quad C = PP^*, \quad R = P^*C^{-1}, \quad \Pi = uu^*/3, \quad X = (R\ u).$$

Then $PR = I_2$, $RP = I_3 - \Pi$, and $\det X = 1$. The maps

$$\mathcal{O}_P(H) = PHP^*, \qquad \jmath_P(G) = RGR^*$$

satisfy

$$\mathcal{O}_P\jmath_P = \mathrm{id}, \qquad \jmath_P\mathcal{O}_P(H) = (I_3 - \Pi)H(I_3 - \Pi),$$

and every $H \in \mathrm{Herm}_3(\mathbb{C})$ has unique coordinates

$$H = X \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} X^*, \qquad \det H = \lambda \det G - \psi^* \operatorname{adj}(G)\psi.$$

The complete positive fibre over $G \succeq 0$ is

$$\psi \in \operatorname{im} G, \qquad \lambda \geq \psi^* G^+ \psi,$$

and the zero fibre is exactly $\{\lambda uu^* : \lambda \geq 0\}$. For $g \in SL_2(\mathbb{C})$,

$$T_P(g) = RgP + \Pi = X(g \oplus 1)X^{-1}$$

acts by $(G, \psi, \lambda) \mapsto (gGg^*, g\psi, \lambda)$. Thus

$$\mathrm{Herm}_3(\mathbb{C}) \cong \text{Lorentz four-vector} \oplus \text{defining complex spin representation} \oplus \text{real scalar}.$$

The vector action has kernel $\{\pm I_2\}$, but the full action is faithful because $-I_2$ sends ψ to $-\psi$. Moreover

$$e_2(\jmath_P(G)) = \tfrac{1}{3}\det G, \qquad \det(uu^* + \varepsilon\jmath_P(G)) = \varepsilon^2 \det G.$$

Consequently $\jmath_P$ extends properly to an embedding $B^4 \hookrightarrow S^8$; the reverse observation on the rank-loss boundary is nonproper and has no infinity-preserving extension. In the strict $SU(3)$-quotient its determinant phase obeys

$$|\omega|^2 = \lambda \det G - \psi^* \operatorname{adj}(G)\psi.$$

Finally, for $Z \in \mathrm{Mat}_{3\times 2}(\mathbb{C})$ and retained $\alpha, c > 0$, the exact colour-invariant Lorentz momentum and mass are

$$p_\alpha(Z) = \alpha Z^* Z, \qquad m^2_{\alpha,c}(Z) = \frac{\alpha^2}{c^2} \sum_{a<b} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2.$$

Rank two is equivalent to positive mass, rank at most one to zero mass, and $m^2_{\alpha,c}(rZ) = r^4 m^2_{\alpha,c}(Z)$; hence this unrestricted kinematic carrier is gapless. Its exact presymplectic reduction is now complete. For $m_0 > 0$,

$$(\mathcal{J}_2 \times \mathcal{H}^+_{m_0,c})/((x,p) \sim (x+sp,p)) \cong_{\mathrm{symp}} T^*\mathcal{H}^+_{m_0,c},$$

so the rank-two system is exactly the future-timelike spinless particle. For $m_0 = 0$, the intrinsic punctured rank-one reduction is $T^*\mathcal{N}^\times_+$, but the composite multiplier action is irregular:

$$d(\det \circ p_\alpha)_Z = 0 \quad (\mathrm{rank}_{\mathbb{C}}\, Z \le 1),$$

so $\dot{x} = \lambda p$ does not imply $\lambda = e$; four transverse real rank-opening directions are first-order null without being colour gauge, and at $Z = 0$ every $x(\tau)$ is stationary. On the hard (λ_0, ω_0)-constraint with $\omega_0 \neq 0$, the momentum map has rank four at $w \neq 0$ and rank three at $w = 0$. Thus its own action forces $\dot{x} = 0$ above the sharp threshold and permits $\dot{x} \in \mathbb{R}p$ only at the threshold. Its determinant inequality and multiplication/Fock spectrum are exact, but the latter is not the Hamiltonian of that action and gives a constraint-selected free spectral gap, not a dynamically generated interacting gap. See theorems 21.2 to 21.7, 22.6, 22.9, 22.12 and 22.16.

43. **[VERIFIED]** The determinant normal slice has now been promoted from an exact Lorentzian analogy to a completely typed host for explicit field models, while the additional choices and the spectral limitations remain visible. With

$$\mathcal{N}_P = \{Y \in \mathrm{Herm}_3(\mathbb{C}) : Yu = 0\}, \qquad \jmath_P(G) = RGR^*,$$

the map $\jmath_P : \mathrm{Herm}_2(\mathbb{C}) \to \mathcal{N}_P$ is a real-linear isomorphism with inverse $Y \mapsto PYP^*$, and

$$\det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G.$$

The polarized determinant is exactly the Minkowski form $tt' - xx' - yy' - zz'$, so the affine slice carries a constant metric of signature $(1,3)$, its positive-semidefinite part is the future cone, and the lifted congruence action realizes

$$PSL_2(\mathbb{C}) \cong SO^+(1,3).$$

The standard Hermitian metrics used to define the Moore–Penrose section select

$$W_P = R^*R = C^{-1}, \qquad K_P = \{a \in SL_2(\mathbb{C}) : a^*W_P a = W_P\} \cong SU(2),$$

and the positive invariant form $E_P(G,K) = \mathrm{tr}(GW_PKW_P)$. The real normal representation is $\mathbf{1} \oplus \mathbf{3}$; its centre acts trivially. Promoting K_P to a local gauge symmetry gives the explicit nonnegative action

$$\int_M \left(\frac{|F_{\mathcal{A}}|^2}{2g^2_{\mathrm{YM}}} + \frac{Z_G}{2}|D_{\mathcal{A}}G|^2_{E_P} + \frac{m^2_G}{2}|G|^2_{E_P} + \frac{\lambda_4}{4}(\det G)^2 \right) d\,\mathrm{vol}_g,$$

and the lift $\widehat{\mathcal{A}} = R\mathcal{A}P$ satisfies $D_{\widehat{\mathcal{A}}}\jmath_P(G) = \jmath_P(D_{\mathcal{A}}G)$. This gauging is an additional physical construction; the normal field cannot screen a fundamental centre charge without the centre-nontrivial retained spinor ψ or another added matter field.

The candidate-specific separation is now exact rather than verbal. The retained inverse-dual cusp matrix obeys

$$G_{\eta,T} = \begin{pmatrix} 2L_T & 2m_T \\ 2m_T & -q_T/3 \end{pmatrix}, \qquad \det G_{\eta,T} = -\frac{2}{3}D_T < 0,$$

whereas $p_\alpha(Z) = \alpha Z^*Z$ ranges over the entire positive-semidefinite cone. Positive rescaling and complex congruence cannot change this determinant sign/inertia, so a further morphism is required to place these two matrices in one physical model. On the Lorentzian host, the free scalar Fock

Hamiltonian has exact gap m_{KG} when the independently supplied coefficient is positive and is gapless when it is zero. For the massless compact gauge Gaussian, the lowest nonzero transverse frequency is $2\pi/L$, which closes as $L \to \infty$. Entangled vacua and sectors occur in both explicitly gapped and explicitly gapless Hamiltonians. Hence none of the determinant identity, the Gram carrier, or entanglement alone proves a nonlinear four-dimensional mass gap. The full-domain comparison theorem identifies a sufficient exact bridge: if $\|Ru\| \geq c_R\|u\|$ and $\mathfrak{b}[Ru] \leq C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}}\|u\|^2)$ throughout the physical excitation form domain and $\mathfrak{b} \geq \gamma$, then

$$\operatorname{gap}(A_{\mathrm{phys}}) \geq \frac{\gamma c_R^2}{C_R}.$$

An exact counterexample proves that an intertwiner defined on only one gapped subspace is insufficient. See theorems 23.1 to 23.3, 23.5 to 23.8, 23.10, 23.12 and 23.16.

44. **[REFUTED]** A literal Fabel–Alpöge-style complex-fourfold construction on a homotopy S^8 cannot exist. If $E = (TS^8, J)$ were complex of rank four, then

$$c_1(E) = c_2(E) = c_3(E) = 0, \qquad c_4(E) = 2u,$$

and therefore $p_2(TS^8) = 2c_4(E) = 4u$. The eight-dimensional signature formula, with $p_1 = 0$ and signature zero, forces $p_2 = 0$, a contradiction. A nonsingular bundle $T^6 \to S^8 \to S^2$ is also impossible: its homotopy sequence would require a surjection $\mathbb{Z} \to \mathbb{Z}^6$, and its Euler characteristic would give $2 = 0$. A merely real singular fibration is not refuted, but it must satisfy $\sum_i \chi(F_i) = 2$ and the exact van Kampen, Wang, and integral Mayer–Vietoris maps proved in the workbench. For regular-piece monodromy ρ, the corrected homological Serre edge sequence is

$$0 \longrightarrow (\textstyle\bigwedge^q \Lambda)_\rho \longrightarrow H_q(E;\mathbb{Z}) \longrightarrow H_1(B^\circ; \bigwedge^{q-1} \Lambda) \longrightarrow 0,$$

whereas dually $H^q(E;\mathbb{Z})$ surjects onto the invariant group $(\bigwedge^q \Lambda^\vee)^\rho$. Thus every filling map must respect the coinvariant subobject and the base-one-cycle quotient; the reversed edge maps cannot be used in the Mayer–Vietoris test. Every effective torus action on S^8 has rank at most four, and the sharp coordinate model is

$$S^8/T^4 \cong \operatorname{Susp}(\Delta^3) \cong B^4.$$

It is a real two-pole singular compactification with generic T^4 orbit, not a three-special-fibre T^6-fibration over S^2. See section 24.1 and equations (892), (895), (897), (899) and (902) to (904).

45. **[VERIFIED]** The original reconstructed S^6 carries a separate, exact six-coordinate bundle problem on its regular locus. With

$$D = W \sqcup S_1 \sqcup S_2, \qquad U = X \setminus D,$$

Alexander duality and the computed homology of the three special fibres give

$$H^4(U;\mathbb{Z}) \cong H_1(W;\mathbb{Z}) \oplus H_1(S_1;\mathbb{Z}) \oplus H_1(S_2;\mathbb{Z}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2.$$

Because U has CW dimension at most five, principal $SU(3)$-bundles on U are classified exactly by their complete $c_2 \in H^4(U;\mathbb{Z})$. The connecting map

$$\delta : H^4(U;\mathbb{Z}) \xrightarrow{\sim} H^5(X, U;\mathbb{Z})$$

shows that such a bundle extends across all three deleted fibres if and only if $c_2 = 0$. This is the free lattice $\mathbb{Z}^6$, not the cyclic group $\mathbb{Z}/6\mathbb{Z}$.

46. **[VERIFIED]** The primitive fibre-degree map, its saturated kernel, the finite normal-holonomy quotients, and the common-centre matrix maps now have a complete theorem–proof module in section 45 and theorems 45.1 to 45.4. The result is exact but stops before any gauge-bundle, connection, confinement, or spectral assertion.

In the marked basis

$$(a_0, b_0, a_1, b_1, a_2, b_2) = (\bar{\gamma}, \bar{u}, g_1, h_1, g_2, h_2),$$

solving the transpose of the verified one-shot collar matrix gives

$$\operatorname{res}_F\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = (12n_0 + 4n_1 - 3n_2)\omega_F, \qquad [\deg_F] = (12, 0, 4, 0, -3, 0),$$

and

$$\ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle.$$

The map is primitive, so restriction to the oriented fibre is surjective. Define

$$q_{\mathrm{col}} : L \longrightarrow H_1(X^\circ; \mathbb{Z}), \qquad a_0 \mapsto c, \quad a_1 \mapsto \bar{g}_1, \quad a_2 \mapsto \bar{g}_2, \quad b_0, b_1, b_2 \mapsto 0.$$

If $\Phi_1 : H_1(X^\circ; \mathbb{Z}) \to \mathbb{Z}$ has row $(12, 4\ell_1, 3\ell_2)$ in the basis $(c, \bar{g}_1, \bar{g}_2)$, then the factorization through the actual collar quotient is

$$\deg_F = \Phi_1 \circ q_{\mathrm{col}}, \qquad \Phi_1(j_*[\widetilde{a}_0]) = p.$$

Let $\vartheta_1(a_1) = e^{-2\pi i/3}$ and $\vartheta_2(a_2) = e^{-2\pi i/4}$ be the normal-line holonomy characters, trivial on the other marked generators. The candidate twists select

$$\chi_{12}\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = \vartheta_1(n_1 a_1 + r_1 b_1)\vartheta_2(n_2 a_2 + r_2 b_2)^{-1} = e^{-2\pi i(4n_1 - 3n_2)/12}.$$

For $\rho_m(k) = e^{-2\pi i k/m}$, the exact typed identities are

$$\boxed{\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 := \chi_{12}^2 = \rho_6 \circ \deg_F\,.}$$

Consequently

$$H^4(U; \mathbb{Z})/\ker \chi_{12} \cong \mathbb{Z}/12\mathbb{Z}, \qquad H^4(U; \mathbb{Z})/\ker \chi_6 \cong \mathbb{Z}/6\mathbb{Z},$$

whereas ordinary coefficient reduction gives

$$H^4(U; \mathbb{Z}/6) \cong (\mathbb{Z}/6)^6.$$

Thus this six-coordinate coefficient group is not the single distinguished cyclic quotient selected by χ_6. A principal $SU(3)$-bundle $P \to U$ satisfies

$$\chi_6(c_2(P)) = \exp\left(-\frac{2\pi i}{6}\langle c_2(P)|_F, [F]\rangle\right).$$

The order-six quotient is therefore the reduction of a geometrically computed degree and of the combined local normal holonomy, not a torsion subgroup inferred from the rank-six notation.

Tensor product gives an independent group-theoretic match:

$$\iota : SU(2) \times SU(3) \hookrightarrow SU(6), \qquad (A, B) \longmapsto A \otimes B, \qquad \mu_2 \times \mu_3 \xrightarrow{\sim} Z(SU(6)) = \mu_6.$$

Since U has homotopy dimension at most five, c_2 classifies $SU(n)$-bundles for $n \geq 3$. For determinant-trivial bundles of ranks two and three,

$$c_2(E_2 \otimes E_3) = 3c_2(E_2) + 2c_2(E_3).$$

Thus every chosen $SU(6)$-bundle class $h \in H^4(U;\mathbb{Z})$ admits a tensor-subgroup reduction, with all c_2-factorizations

$$(e, f) = (h + 2t, -h - 3t), \qquad t \in H^4(U;\mathbb{Z}).$$

Neither h nor a reduction is preferred. The separate support group is

$$H^5(U;\mathbb{Z}) \cong \ker\{\mathbb{Z}^3 \xrightarrow{(1,1,1)} \mathbb{Z}\} \cong Q(A_2),$$

for the combinatorial form making the three distinguished support basis vectors orthonormal; this is not a cup-product or intersection form. It is not an intrinsic $A_2 \otimes \mathbb{Z}^2$ inside $H^4(U)$, and the rank-five kernel of $\deg_F$ is not thereby an A_5 root lattice.

The finite-group isomorphism $H^4(U)/\ker\chi_6 \cong Z(SU(6))$ supplies neither transition functions nor a connection. The injective tensor map does not factor through $(SU(2) \times SU(3))/\mu_6$, because its central μ_6 maps to nontrivial scalar matrices. The separate common-centre construction is

$$\begin{aligned} \Gamma_6 &= \{((\lambda\mu)^{-1}, \lambda I_2, \mu I_3) : \lambda \in \mu_2,\ \mu \in \mu_3\}, \\ \frac{U(1) \times SU(2) \times SU(3)}{\Gamma_6} &\hookrightarrow U(6), \\ [z, A, B] &\longmapsto z(A \otimes B), \end{aligned}$$

where $\Gamma_6 \cong \mu_6$. Its determinant is z^6, and its image meets $SU(6)$ in exactly $\iota(SU(2) \times SU(3))$; the image is not generally contained in $SU(6)$. No present morphism identifies this degree-four quotient with the degree-two lifting obstruction of a common-centre gauge quotient. Global $SU(3)$-bundles on S^6 are instead classified by $k \in \pi_5(SU(3)) \cong \mathbb{Z}$, with $c_3 = \pm 2ku$, and all restrict trivially to U. See equations (910), (912) and (914) to (916).

47. **[VERIFIED]** The exact operator-theoretic bridge required for any physical no-gap inference is proved as a typed implication. Let $H \geq 0$ be self-adjoint on $\mathcal{H}_{\text{phys}}$, put

$$P_0 = \mathbf{1}_{\{0\}}(H), \qquad \mathcal{H}_{\text{exc}} = (I - P_0)\mathcal{H}_{\text{phys}}, \qquad \mathfrak{h}[\psi] = \|H^{1/2}\psi\|^2, \qquad \mathcal{Q}(H) = D(H^{1/2}).$$

For a nonnegative quadratic form $\mathfrak{q}_n$ and nonzero $u_n \in D(\mathfrak{q}_n) \subset \mathcal{K}_n$, and for $J_n : \operatorname{span}\{u_n\} \to \mathcal{Q}(H)$, define $\psi_n = (I - P_0)J_n u_n$. If, for some $c > 0$, $C_n \geq 0$, and $\varepsilon_n \geq 0$,

$$\|\psi_n\| \geq c\|u_n\|, \qquad \mathfrak{h}[\psi_n] \leq C_n \mathfrak{q}_n[u_n] + \varepsilon_n \|u_n\|^2,$$

$$C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n \longrightarrow 0,$$

then the self-adjoint restriction

$$H_{\text{exc}} = H\big|_{D(H) \cap \mathcal{H}_{\text{exc}}}$$

satisfies

$$\inf \sigma(H_{\text{exc}}) = 0, \qquad 0 \in \sigma_{\text{ess}}(H_{\text{exc}}).$$

The full spectral projection P_0, rather than orthogonality to one chosen vacuum, is essential unless vacuum uniqueness is independently proved; see theorem 38.4.

48. **[OPEN]** The hypotheses needed to apply that bridge to four-dimensional quantum Yang–Mills theory have not been constructed here. In particular, the workbench has no common gauge-invariant positive-metric physical Hilbert space $\mathcal{H}_{\text{phys}}$, no reconstructed quantum Hamiltonian H, and no maps J_n satisfying the displayed norm and energy estimates with control through gauge reduction, reflection-positive reconstruction, renormalization, and the continuum and infinite-volume limits. The exact radial scalar, continuum Hessian, compact-link flux, finite cochain, Gaussian-oscillator, magnetic, and split-Cartan $SU(r)$ Jacobi calculations are **[VERIFIED]** auxiliary geometric, finite-dimensional, or classical statements. They neither prove nor disprove a mass gap for a physical four-dimensional quantum Yang–Mills Hamiltonian; see theorems 10.12 and 38.6.

External validation note. A further context-isolated AI derivation independently recovered the cyclic invariant lattices (11.4), determinant divisor (11.6), finite residue cancellation (11.17), scalar test (11.19)–(11.20), all three absolute cotangent direct images, and $h^{1,2} = 1$, $h^{1,1} = 2$. Independent human specialist replication has not been recorded. This is an external-validation boundary, not a mathematical premise or an unresolved step of the Hodge theorem; see theorem 43.7.

Independent review checkpoint. Reviewer B independently examined the reconstruction existing up to that report's evidentiary cutoff and agreed that the intrinsic nearby-cycle, Leray, and based van Kampen route does not depend on the then-open collapse route. Its disposition was provisionally favourable with focused verification requests. The later Codex proofs recorded here are separate work: they are not retroactively attributed to Reviewer B, and its report remains useful as an independent comparison against the argument it actually saw. Reviewer B's 26 August analytic follow-up also independently proves the uniform top-degree base-change corollary and the conductor–duality nonvanishing from CAND-X, CAND-W, and TOP-H2 [35]; those inputs are not certified by that follow-up itself.

Status language. **[VERIFIED]** means proved here or reproduced by an explicit certificate. **[CONDITIONAL]** means a conclusion depending on an explicitly named established or assumed hypothesis not proved in the present argument; it never means doubtful or politely false. A completely proved implication whose antecedents are part of its exact statement is **[VERIFIED]**. The current candidate-specific ledger has no **[CONDITIONAL]** row. **[OPEN]** means that no complete proof or counterexample is recorded. **[REFUTED]** means an exact assertion has a complete contradiction or counterexample. The status applies to the immediately following assertion, not to an entire paper by contagion.

2 Mathematical results so far

This is the short route through the current mathematics. It states what is proved in this workbench and exhibits the load-bearing formulas. Both the intrinsic nearby-cycle/Leray certificate and the independent collapse/Mayer–Vietoris certificate are now complete for the candidate.

1. **[VERIFIED] The analytic threefold exists in the reconstructed coordinate model.** The descended modular periods give

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix},$$

with

$$\det_{\mathbb{R}} \Pi = -\det \operatorname{Im} Z = \operatorname{Im}\tau \left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} \right) < 0.$$

The Cayley coordinates $s_j(z) = (z - z_j)/(z - \bar{z}_j)$, the two elliptic orientations, and the modular deck normalization are fixed simultaneously:

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = e^{-2\pi i/4} = -i, \qquad s_1 \circ g_1 = e^{-2\pi i/3} s_1, \quad s_2 \circ g_2 = -i s_2.$$

The period-deck homomorphism φ is defined by $\tau \circ g = \varphi(g) \circ \tau$. Its cusp image is proved in a fixed covering component, not by an untyped single-valuedness assertion: choose

$$U_0 \subset \pi^{-1}(D_0^*), \qquad g_0 U_0 = U_0, \qquad \tau(U_0) \subset K \subset j^{-1}(\{|w| > R\}).$$

For $P = \varphi(g_0)$ one has $P(K) = K$. If c is the boundary cusp of K, choose $\gamma_0(\infty) = c$ and put

$$\tau' = \gamma_0^{-1} \tau|_{U_0}, \qquad P' = \gamma_0^{-1} P \gamma_0, \qquad \tau' \circ g_0 = P' \circ \tau'.$$

If $P' = 1$, then τ' descends to $\bar{\tau}' : D_0^* \to \mathfrak{h}$ and $q' = \exp(2\pi i \bar{\tau}')$ simultaneously has a positive-order zero at the cusp and a single-valued holomorphic logarithm, an impossible winding combination. Thus P' and hence P are nontrivial parabolic.

The auxiliary γ_0-frame is then discarded and the proof returns to $\tau(z_1) = \rho$, $\varphi(g_1) = S_1$. For the trace-+2 lift, with primitive fixed vector $(p, r)^{\mathsf{t}}$, $\gcd(p, r) = 1$, and $k \neq 0$,

$$\widetilde{P} = I + k \begin{pmatrix} -pr & p^2 \\ -r^2 & pr \end{pmatrix},$$

and

$$\widetilde{S}_1 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \operatorname{tr}(\widetilde{S}_1^{-1} \widetilde{P}^{-1}) = 1 + k(p^2 - pr + r^2) = 0.$$

Thus projective order two of $S_1^{-1} P^{-1}$ gives $-k(p^2 - pr + r^2) = 1$, so $k = -1$ and $p^2 - pr + r^2 = 1$; conjugating the whole deck representation by a power of S_1 leaves S_1 fixed and gives the final normalization below. Rename the resulting period map τ and define anew, in this cusp paragraph,

$$q = e^{2\pi i \tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0.$$

For $S_1(w) = (w - 1)/w$, $S(w) = -1/w$, and $P(w) = w - 1$,

$$\varphi(g_1) = S_1, \qquad \varphi(g_2) = S, \qquad \varphi(g_0) = P, \qquad S_1 S P = 1 \quad \text{in } \mathrm{PSL}_2(\mathbb{Z}).$$

At z_2, comparison of quadratic leading terms gives

$$\big(g_2'(z_2)\big)^2 = S'(i) = -1;$$

the simultaneous parabolic normalization selects $g_2'(z_2) = -i$. The periodic toric cusp action is free and properly discontinuous; its bounded-chart argument uses $|B_t \bar{\lambda}| \geq \frac{1}{2} |\bar{\lambda}|$. Exponentiation identifies the punctured toric quotient with the family of compact two-dimensional complex tori, not with an abelian-surface family. For the finite logarithmic actions the opposite translation signs are retained:

$$v_1 = \varepsilon, \qquad v_2 = -\varepsilon', \qquad \gamma(v_1) = 1, \qquad \gamma(v_2) = -1,$$

and

$$(z, x) \longmapsto (g_j z, A_j x + v_j / m_j).$$

These actions are free by the invariant covector criterion. On an overlap, the compatible logarithm is

$$\ell'(w) = \ell(g_j^{-1} w) - \frac{2\pi i}{m_j}, \qquad h_j \widetilde{g}_j^{\log} h_{\bar{j}}^{-1} = \widetilde{g}_j^{0},$$

so no branch sign is suppressed. Their radial stalk maps are the printed quotient maps, and the three fillings glue to a compact connected smooth complex threefold X with a proper surjective holomorphic map

$$f : X \longrightarrow \mathbb{P}^1.$$

On the cusp cover the toric form

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2} = \pm dz_0 \wedge dz_1 \wedge dz_2$$

extends nowhere vanishingly and satisfies $\Psi_{\bar{\lambda}}^* \Omega = \Omega$. This first proves $\omega_{N_0} \simeq \mathcal{O}_{N_0}$; only then does the open embedding $N_0 \hookrightarrow X$ give the typed two-stage chain

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}.$$

The complete chain is theorems 8.2, 8.4, 8.5 and 8.7 to 8.9.

2. **[VERIFIED] The cusp fibre has the exact geometry needed in the topology and conductor arguments.** Its scheme-theoretic local equations are

$$z_0, \qquad z_0z_1, \qquad z_0z_1z_2.$$

It is reduced, irreducible, nonnormal, and normal crossings (not simple normal crossings). Its normalization is dP_6, with opposite sides of the anticanonical hexagon identified, and its double locus consists of three rational curves through two triple points.

3. **[VERIFIED] The algebraic reduction is computed directly and does not use the topology or CDP20.** For a very general smooth fibre,

$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad \eta = u \wedge w + 6\gamma \wedge \delta, \qquad \eta \wedge \eta = 12\gamma \wedge u \wedge w \wedge \delta.$$

The inverse-dual matrix G_η associated with the Appell–Humbert form has determinant

$$\det G_\eta = \frac{2}{3} \operatorname{Im}\tau \left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} \right) < 0,$$

and the same inertia as that form, hence signature $(1,1)$ and $a(F_b) = 0$. The repaired meromorphic-descent argument then gives

$$\mathcal{M}(X) = f^*\mathbb{C}(t), \qquad a(X) = 1,$$

and f is the algebraic reduction; see theorems 40.2 and 40.4.

4. **[VERIFIED] The intrinsic nearby-cycle/Leray route proves the integral topology without using the cellular collapse.** For the cusp filling, the integral nearby-cycle map satisfies

$$\mathrm{sp}^r : H^r(W;\mathbb{Z}) \xrightarrow{\sim} H^r(F;\mathbb{Z})^{T_0} \qquad (0 \le r \le 4)$$

as saturated lattices. With $q := uw + 6\gamma\delta \in \bigwedge^2 V = H^2(F;\mathbb{Z})$, the global sheaf calculation gives

$$H^0(R^1) = \mathbb{Z}(12\gamma), \quad H^0(R^2) = \mathbb{Z}(2q), \quad H^0(R^3) = \mathbb{Z}(2\gamma uw).$$

The canonical truncation triangle

$$R^0f_*\mathbb{Z}_X \longrightarrow \tau_{\le 1}Rf_*\mathbb{Z}_X \longrightarrow R^1f_*\mathbb{Z}_X[-1] \xrightarrow{\kappa} R^0f_*\mathbb{Z}_X[1]$$

identifies the explicitly computed Cech discrepancy with Leray d_2. For $(\ell_0,\ell_1,\ell_2) = (0,1,-1)$, hence $p = -1$, orientation of ω may be chosen so that

$$d_2(12\gamma) = \omega, \qquad d_2(2q) = \omega[\delta], \qquad d_2(2\gamma uw) = \omega[\gamma\delta].$$

Therefore

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0.$$

5. **[VERIFIED] The based van Kampen maps and sphere recognition close the topology theorem.** Zero-section whiskers fix the common basepoint, the finite logarithmic overlaps impose the primitive relations with one common sign, the cusp inclusion is $\Lambda \to \bar{\Lambda}$, and the cusp meridian bounds the explicit toric zero-section disc with no translation drift. Hence

$$\pi_1(X) = \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1.$$

Poincaré duality, universal coefficients, Hurewicz, and the smooth homotopy-sphere recognition theorem now give

$$X \cong_{\mathrm{diff}} S^6, \qquad \langle c_3(T_X), [X] \rangle = \chi_{\mathrm{top}}(X) = 2.$$

See theorems 39.1 to 39.7.

6. **[REFUTED] The printed CDP dependency has two exact failures.** For a reduced nonnormal fibre,

$$0 \longrightarrow N^*_{S/X} \longrightarrow \ker(\Omega^1_X|_S \to \widetilde{\Omega}^1_S) \longrightarrow \operatorname{tors}\Omega^1_S \longrightarrow 0,$$

so the vanishings used on p. 692 of [2] do not control the conductor-supported middle term. A proper nodal-cubic-times-$\mathbb{P}^1$ family is a counterexample to that inference. Separately, a split extension satisfies

$$(K \rtimes Q)^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q,$$

not the direct sum with all of K^{ab} unless monodromy acts trivially.

7. **[VERIFIED] The singular-fibre nonvanishing has a topology-free Pic^0 proof and three exact all-Pic certificates.** The invariant toric three-form first trivializes ω_{N_0}, and the open embedding then gives $\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}$, as recorded above. Let $\nu : \widetilde{W} \simeq \mathrm{dP}_6 \to W$ be the normalization. For $L \in \mathrm{Pic}^0(X)$, naturality of c_1 and injectivity of

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \hookrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

give $\nu^*((L^\vee \otimes \omega_X)|_W) \simeq \mathcal{O}_{\mathrm{dP}_6}$. The conductor section, Ramis–Ruget duality on the compact Cartier fibre, and top-degree analytic base change then prove

$$R^2 f_*(T_X \otimes L) \neq 0 \qquad (L \in \mathrm{Pic}^0(X))$$

without any global homology calculation. For arbitrary L, the exact trivial-pullback obstruction is the homomorphism

$$r : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6), \qquad r([L]) = [\nu^* i^* L].$$

For the candidate, the Leray edge image of $c_1(L)$ lies in $E_\infty^{0,2} \subseteq \ker d_2^{0,2}$, but $d_2^{0,2}(2q) = \omega[\delta]$ is an isomorphism. Hence the edge section is zero; proper base change and naturality give $c_1(L|_W) = 0$, and injectivity of c_1 on dP_6 forces

$$\nu^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}, \qquad r([L]) = 0.$$

This proves the conclusion for every $L \in \mathrm{Pic}(X)$ without citing the packaged $H^2(X;\mathbb{Z}) = 0$ theorem, although it reuses the same integral Leray transgression.

The second certificate uses no global Leray differential. The exact order-four filling lattice forces $c_1(L|_F) = 2nq$. Subtracting the invariant vertical form on the pair-of-pants bundle gives a crossed homomorphism with global-coinvariant boundary values. The logarithmic translations and their cyclic norm maps give

$$\bar{d}_1 = 4n[\delta], \qquad \bar{d}_2 = -3n[\delta],$$

while the flat zero-section cusp sweep bounds a literal toric solid torus, so $\bar{d}_0 = 0$. The based relation gives $\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0$, hence $n = 0$. Naturality and injectivity of the local integral cusp specialization then give $c_1(L|_W) = 0$ and

$$\nu^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}, \qquad r([L]) = 0.$$

This is independent of global d_2, though not of local cusp topology. See theorems 46.12 to 46.14, 46.18, 46.22 and 46.25.

The precise limit of a cusp-only analytic replacement is also computed. For $Q(v) = v_1^2 - 6v_2^2$, the locally finite toric divisor $\mathcal{D}_Q = \sum_v Q(v)E_v$ has an explicit equivariant descent $\mathcal{L}_Q \in \mathrm{Pic}(N_0)$ satisfying

$$c_1(\mathcal{L}_Q|_{F_t}) = -2q, \qquad (\deg_{C_i} \nu^*(\mathcal{L}_Q|_W)) = (-12, 2, 0, -12, 2, 0).$$

Thus $\mathrm{Pic}(N_0) \to \mathrm{Pic}(\mathrm{dP}_6)$ is nonzero, so no cusp-local representability argument can prove the global restriction theorem by itself. The completed third certificate adds the two finite fillings analytically and retains the prior coinvariant input

$$(\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \qquad q = u \wedge w + 6\gamma \wedge \delta, \qquad V_G \otimes \mathbb{R} = \mathbb{R}[\delta].$$

Thus, for a global line bundle L, $E = c_1(L|_F) = kq$. The finite norm proof keeps every covering level distinct. On the vector cover

$$\widetilde{J}_j = \Delta_j \times \mathbb{C}^2 \xrightarrow{r_j} J'_j$$

define the vector-cover automorphisms

$$Q_{j,\log}, Q_{j,\mathrm{lin}} \in \mathrm{Aut}(\widetilde{J}_j), \qquad Q_{j,\log} = \widetilde{g}_j^{\,v_j/m_j}, \quad Q_{j,\mathrm{lin}} = \widetilde{g}_j^{\,0}.$$

These are not the quotient automorphisms: on the vector cover,

$$\begin{aligned} Q_{j,\log}\tau_\lambda Q_{j,\log}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\log})^{m_j} = \tau_{v_j}, \\ Q_{j,\mathrm{lin}}\tau_\lambda Q_{j,\mathrm{lin}}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\mathrm{lin}})^{m_j} = 1. \end{aligned}$$

The sign in $(Q_{j,\log})^{m_j} = \tau_{v_j}$ is positive because m_j clockwise iterations of $s_j \circ g_j = e^{-2\pi i/m_j} s_j$ produce translation by $+v_j$. Let

$$q_{j,\log}, q_{j,\mathrm{lin}} \in \mathrm{Aut}(J'_j)$$

be their descents. They have order dividing m_j by these relations and exactly m_j because they cover g_j. On $J'^*_j = J'_j|_{\Delta_j^*}$ define

$$\begin{aligned} p_{j,\log} &: J'^*_j \longrightarrow N^*_j = J'^*_j/\langle q_{j,\log}\rangle, \\ p_{j,\mathrm{lin}} &: J'^*_j \longrightarrow J^*_j = J'^*_j/\langle q_{j,\mathrm{lin}}\rangle. \end{aligned}$$

The branch-independent translation and its descended biholomorphism are typed by

$$h_j : J'^*_j \longrightarrow J'^*_j, \qquad \bar{h}_j : N^*_j \longrightarrow J^*_j,$$

and satisfy

$$h_j \circ q_{j,\log} \circ h_j^{-1} = q_{j,\mathrm{lin}}, \qquad p_{j,\mathrm{lin}} \circ h_j = \bar{h}_j \circ p_{j,\log},$$

and the filled line bundle gives the typed pullback comparison

$$p^*_{j,\log}(L|_{N^*_j}) \simeq h^*_j p^*_{j,\mathrm{lin}}(L|_{J^*_j}).$$

For the factor extension return to the filled vector cover:

$$P_{j,\log} : J'_j \longrightarrow N_j, \qquad \pi_j^{\mathrm{vec}} := P_{j,\log} \circ r_j : \widetilde{J}_j \longrightarrow N_j, \qquad P_{j,\log}|_{J'^*_j} = p_{j,\log},$$

$$\Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda \rangle.$$

The pullback $(\pi_j^{\mathrm{vec}})^*(L|_{N_j})$ defines the filled factor extension. Over Δ_j^* the quotient square identifies its commutator form with E; that integer-valued form is locally constant on the connected disc, so it remains E on the filled cover. After trivializing this pullback, choose normalized vertical lifts

$$U_\lambda^{(j)} U_\mu^{(j)} = \mathbf{c}\Big(\frac{1}{2}E(\lambda,\mu)\Big)\, U_{\lambda+\mu}^{(j)},$$

write $\alpha\tau_\lambda\alpha^{-1} = \tau_{A_\alpha\lambda}$ and $T_\alpha = (A_\alpha^{-1})^{\mathsf{t}}$, and define $\widetilde{d}_j(\alpha) \in V_{\mathbb{R}}$ by

$$\widetilde{G}_\alpha U^{(j)}_{A_\alpha^{-1}\lambda} \widetilde{G}_\alpha^{-1} = \mathbf{c}\Big(\langle \widetilde{d}_j(\alpha), \lambda\rangle\Big)\, U_\lambda^{(j)}.$$

It is a crossed homomorphism:

$$\widetilde{d}_j(\alpha\beta) = \widetilde{d}_j(\alpha) + T_\alpha \widetilde{d}_j(\beta).$$

To compare the boundary factors, use the actual common cover and its restricted lattice quotient

$$\widetilde{J}_j^* := \Delta_j^* \times \mathbb{C}^2, \qquad r_j^* := r_j|_{\widetilde{J}_j^*} : \widetilde{J}_j^* \longrightarrow J_j'^*.$$

The two quotient maps from this common domain are

$$\begin{aligned} \Phi_{j,\log} : \widetilde{J}_j^* &\longrightarrow N_j^*, \quad \Phi_{j,\log} := p_{j,\log} \circ r_j^*, \\ \Phi_{j,\mathrm{glob}} : \widetilde{J}_j^* &\longrightarrow J_j^*, \quad \Phi_{j,\mathrm{glob}} := p_{j,\mathrm{lin}} \circ h_j \circ r_j^* = \bar{h}_j \circ \Phi_{j,\log}. \end{aligned}$$

Both maps are $\Gamma_j^{\log}$-invariant. Translation invariance follows from $r_j^* \circ \tau_\lambda = r_j^*$. For every $\widetilde{x} \in \widetilde{J}_j^*$, invariance under the nontrivial generator is the point-valued calculation

$$\begin{aligned} \Phi_{j,\log}(Q_{j,\log}(\widetilde{x})) &= p_{j,\log}(q_{j,\log}(r_j^*(\widetilde{x}))) = \Phi_{j,\log}(\widetilde{x}), \\ \Phi_{j,\mathrm{glob}}(Q_{j,\log}(\widetilde{x})) &= p_{j,\mathrm{lin}}(h_j(q_{j,\log}(r_j^*(\widetilde{x})))) \\ &= p_{j,\mathrm{lin}}(q_{j,\mathrm{lin}}(h_j(r_j^*(\widetilde{x})))) = \Phi_{j,\mathrm{glob}}(\widetilde{x}). \end{aligned}$$

Thus $\widetilde{J}_j^*$ is the same $\Gamma_j^{\log}$-cover of the two biholomorphic quotients N_j^* and J_j^*. Pulling back the gluing isomorphism gives

$$\Theta_j : \Phi_{j,\log}^*(L|_{N_j^*}) \xrightarrow{\sim} \Phi_{j,\mathrm{glob}}^*(L|_{J_j^*}).$$

It intertwines descent and hence identifies the pushed-out filled and global $\Gamma_j^{\log}$-factor extensions while fixing every central $\mathbf{c}(r)$, $r \in \mathbb{R}$.

The loop and its three successive lifts are fixed pointwise as follows. Let

$$\gamma_j^J : [0,1] \longrightarrow J_j^*$$

be the zero-section image of the unwhiskered clockwise boundary meridian, so $\gamma_j^J(1) = \gamma_j^J(0)$. Choose the global zero-section basepoint $\widehat{b} \in J|_P$ and the fixed zero-section whisker

$$\omega_j : [0,1] \longrightarrow J|_P, \qquad \omega_j(0) = \widehat{b}, \qquad \omega_j(1) = \gamma_j^J(0),$$

where the boundary copy of J_j^* is identified with its image in $J|_P$. The retained based total-space loop is the class

$$\widehat{\rho}_j = [\omega_j * \gamma_j^J * \omega_j^{-1}] \in \pi_1(J|_P, \widehat{b}).$$

Choose $x_j \in J_j'^*$ such that $p_{j,\mathrm{lin}}(h_j(x_j)) = \gamma_j^J(0)$, put $x_{j,\mathrm{lin}} = h_j(x_j)$, and let $\widetilde{\gamma}_{j,\mathrm{lin}} : [0,1] \to J_j'^*$ be the $p_{j,\mathrm{lin}}$-lift of γ_j^J beginning at $x_{j,\mathrm{lin}}$. For the chosen clockwise generator,

$$\begin{aligned} p_{j,\mathrm{lin}} \circ \widetilde{\gamma}_{j,\mathrm{lin}} &= \gamma_j^J, \\ \widetilde{\gamma}_{j,\mathrm{lin}}(0) &= x_{j,\mathrm{lin}}, \\ \widetilde{\gamma}_{j,\mathrm{lin}}(1) &= q_{j,\mathrm{lin}}(x_{j,\mathrm{lin}}). \end{aligned}$$

Set

$$\gamma_j^N := \bar{h}_j^{-1} \circ \gamma_j^J : [0,1] \longrightarrow N_j^*, \qquad \widetilde{\gamma}_{j,\log} := h_j^{-1} \circ \widetilde{\gamma}_{j,\mathrm{lin}} : [0,1] \longrightarrow J_j'^*.$$

Then the quotient square gives the point-valued identities

$$\begin{aligned} p_{j,\log} \circ \widetilde{\gamma}_{j,\log} &= \gamma_j^N, \\ \widetilde{\gamma}_{j,\log}(0) &= x_j, \\ \widetilde{\gamma}_{j,\log}(1) &= q_{j,\log}(x_j). \end{aligned}$$

Choose $\widetilde{x}_j \in \widetilde{J}_j^*$ with $r_j^*(\widetilde{x}_j) = x_j$, and let $\widetilde{\gamma}_{j,\mathrm{vec}} : [0,1] \to \widetilde{J}_j^*$ be the r_j^*-lift of $\widetilde{\gamma}_{j,\log}$ beginning at $\widetilde{x}_j$. On adjacent sheets choose compatible logarithm branches

$$\ell'(w) = \ell(g_j^{-1} w) - \frac{2\pi i}{m_j}.$$

The point-valued vector-cover endpoint is

$$\begin{aligned} r_j^* \circ \widetilde{\gamma}_{j,\mathrm{vec}} &= \widetilde{\gamma}_{j,\log}, \\ \widetilde{\gamma}_{j,\mathrm{vec}}(0) &= \widetilde{x}_j, \\ \widetilde{\gamma}_{j,\mathrm{vec}}(1) &= Q_{j,\log}(\widetilde{x}_j). \end{aligned}$$

excluding both $Q_{j,\log}^{-1}(\widetilde{x}_j)$ and every $\tau_\lambda Q_{j,\log}(\widetilde{x}_j)$ with $\lambda \neq 0$. Fibre loops remain the pointwise deck paths from $\widetilde{x}_j$ to $\tau_\lambda(\widetilde{x}_j)$, since h_j is a fibre translation.

The factor of L along the fixed lift of ω_j identifies the target factor over the local boundary segment with the globally based factor. Let $G_{\rho_j}^{(j)}$ be the target lift of the local boundary deck element whose whiskered image is the chosen global lift G_{ρ_j}. It satisfies

$$G_{\rho_j}^{(j)} U_{A_j^{-1}\lambda} (G_{\rho_j}^{(j)})^{-1} = \mathbf{c}(\langle d_j, \lambda \rangle) U_\lambda, \qquad d_j = d(\rho_j).$$

Consequently, for some $a_j \in V_{\mathbb{R}}$ and $b_j \in \mathbb{R}$,

$$\Theta_{j,*}(U_\lambda^{(j)}) = \mathbf{c}(\langle a_j, \lambda \rangle) U_\lambda, \qquad \Theta_{j,*}(\widetilde{G}_{Q_{j,\log}}) = \mathbf{c}(b_j) G_{\rho_j}^{(j)}.$$

The central factor $\mathbf{c}(b_j)$ disappears under conjugation. Therefore

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I) a_j, \qquad \widetilde{d}_j(\tau_v) = \iota_v E.$$

Here $A_{Q_{j,\log}} = A_j$ and $T_{Q_{j,\log}} = T_j$. Consequently the exact, sign-sensitive norm chain is

$$\begin{aligned} N_{T_j} d_j &= N_{T_j} \widetilde{d}_j(Q_{j,\log}) = \widetilde{d}_j((Q_{j,\log})^{m_j}) \\ &= \widetilde{d}_j(\tau_{v_j}) = \iota_{v_j} E. \end{aligned}$$

Finally, on a simply connected $U \subset \Delta_j^*$ the logarithm branch ℓ gives the point-typed vector-cover lift

$$H_j^\ell : U \times \mathbb{C}^2 \longrightarrow U \times \mathbb{C}^2, \qquad H_j^\ell(z, \zeta) = \left(z, \zeta + \frac{\ell(z)}{2\pi i} \Pi(z) v_j \right)$$

of h_j. A branch change $\ell \mapsto \ell + 2\pi i n$ gives

$$H_j^{\ell + 2\pi i n} = \tau_{n v_j} \circ H_j^\ell.$$

At factor-extension level this gives

$$\Theta'_{j,*} = \mathrm{Ad}(U_{n v_j}) \circ \Theta_{j,*}.$$

Thus the comparison character changes by $a_j \mapsto a_j + n \iota_{v_j} E$. Since

$$T_j(\iota_{v_j} E) = \iota_{v_j} E,$$

both displayed $\widetilde{d}_j$ and norm identities are unchanged; the descended $h_j, \bar{h}_j$ and quotient square are branch independent. The exact contractions are

$$\iota_{v_1} q = 2\eta_1, \quad \iota_{v_2} q = -3\eta_2, \quad [\eta_1] = 3[\delta], \quad [\eta_2] = 2[\delta],$$

and logarithms of the actual $\mathcal{O}^*$ factors give

$$\bar{d}_1 = 2k[\delta], \qquad \bar{d}_2 = -\tfrac{3}{2}k[\delta], \qquad \bar{d}_0 = 0, \qquad \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0,$$

where the cusp equality is an argument-principle calculation in a Stein chart. Hence $k = 0$ and $E = 0$. The toric support-function calculation uses the explicitly typed unimodular map

$$B_0 : \bar{\Lambda} \simeq \mathbb{Z}^2 \longrightarrow \Lambda_{\mathrm{tor}} \simeq \mathbb{Z}^2, \qquad [B_0] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \det B_0 = 1.$$

For adjacent rays $\rho_v, \rho_w < \sigma = \{v, w, u\}$ the affine-open directions are, exactly,

$$U_{\rho_v} = D(z_w z_u) \subset U_\sigma, \qquad U_{\rho_w} = D(z_v z_u) \subset U_\sigma, \qquad U_{\rho_v} \cap U_{\rho_w} = D(z_v z_w z_u) = U_0.$$

If $\mu_\lambda^+ : \Psi_\lambda^* \mathcal{O}(D_a) \to \mathcal{O}(D_a)$ is multiplication by χ^{m_λ}, the map needed on the negative divisor is the inverse-dual, not the ordinary dual:

$$\mu_\lambda^- := ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

Now set, with its type retained,

$$M = L|_{N_0} \in \mathrm{Pic}(N_0), \qquad E_M = c_1(M|_{F_t}) = E = 0.$$

For $A = \nu^*(M|_W)$ and $(\deg_{C_1} A, \dots, \deg_{C_6} A) = (d_1, d_2, d_3, d_1, d_2, d_3)$, the normalized degrees determine

$$[B] = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}, \qquad E_M((\lambda, 0), (0, m)) = -B(B_0\lambda, m).$$

The antecedent $E_M = 0$ and unimodularity of B_0 force $B = 0$, hence all six degrees vanish and

$$\nu^*(L|_W) = \nu^*(M|_W) \simeq \mathcal{O}_{E_0}; \qquad r([L]) = [\mathcal{O}_{E_0}] = 0 \quad \text{in } \mathrm{Pic}(E_0) \simeq \mathrm{Pic}(\mathrm{dP}_6).$$

This proof uses neither global Leray nor local integral specialization. See theorems 41.8 and 46.19 to 46.21.

8. **[VERIFIED] The constructed threefold is a theorem-level counterexample to CDP20 Theorem 2.2 as stated.** It has

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0, \qquad a(X) = 1, \qquad f : X \to \mathbb{P}^1 \text{ a holomorphic algebraic reduction,}$$

while

$$c_3(X) = 2 > 0, \qquad X \cong_{\mathrm{diff}} S^6.$$

These are precisely the relevant hypotheses and the opposite of the published conclusions $c_3(X) \le 0$ and $a(X) = 0$. This conclusion uses the verified candidate proof, not merely the fact that defects were found in the published proof.

9. **[VERIFIED] The strongest corrected CDP conclusion is now proved, with its exact extra hypothesis.** For any compact complex threefold in the CDP holomorphic-reduction setting, existence of a single L_0 with

$$H^2(X, T_X \otimes L_0) = 0$$

implies, for general M,

$$H^i(X, T_X \otimes M) = 0 \quad (i \neq 1), \qquad \chi(X, T_X \otimes M) = \frac{1}{2} c_3(X) \le 0.$$

The relative condition $R^2 f_*(T_X \otimes L) = 0$ for one L implies this global hypothesis by projection formula, Serre vanishing on the base, and Leray. On a reduced full fibre W_c, after the two endpoint vanishings, the missing ambient-cotangent vanishing is equivalent to injectivity of

$$H^0(W_c, \operatorname{tors} \Omega^1_{W_c} \otimes A_c) \xrightarrow{\delta_c} H^1(W_c, N^*_{W_c/X} \otimes A_c).$$

This repairs the proof route without asserting the false original theorem. Because $c_3(X) = 2$, the contrapositive of the corrected global theorem gives $H^2(X, T_X \otimes L_0) \neq 0$ for every L_0. Independently, the candidate has nonzero $\ker \delta_c$ at its cusp and $R^2 f_*(T_X \otimes L) \neq 0$ for every L. See theorems 46.28 to 46.30.

10. **[VERIFIED] The Proposition 10.8 uniform non-torsion branch is complete.** The reconstructed direct-image calculation gives

$$R^1 f_* \mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad \operatorname{Pic}(X) \simeq \mathbb{C},$$

with degree-one base change on all scheme-theoretic fibres. At a finite filling, the invariant canonical section and the direct-image generator are not conflated: if $k_j = m_j - 1 - a_j$, then

$$\operatorname{div}(\overline{\sigma}_j) = k_j S_j, \qquad f_{j*}\omega_{N_j/D_j} = \mathcal{O}_{D_j} \cdot s_j^{k_j} \vartheta'_j, \qquad s_j^{k_j} \vartheta'_j = \frac{e^{\varphi(s_j)}}{m_j} s_j^{-a_j} e \quad \text{on } \Delta_j^*.$$

In particular $k_2 = 2$, so the canonical section is not a frame there; the invariant-module identity is the exact statement that proves the Hodge line used below. The finite character calculation gives $\operatorname{ord}(\mathcal{O}_X(S_j)|_{S_j}) = m_j$. For $\lambda = a\gamma + bu + cw + d\delta$, membership of its Dolbeault image in the global Picard line is exactly

$$c\tau + d\mu - b = 0.$$

Linear independence of $1, \tau, \mu$ makes the rank-two exceptional fibres countable, while rank-one fibres contribute bad parameters only in $\mathbb{R}$. The three full special fibres and the reductions S_j each add only countably many parameters. Thus every $z \in \mathbb{C} \setminus (\mathbb{R} \cup \text{a countable set})$ supplies one L_z non-torsion on every full fibre and on both S_j. The cusp step was first repaired using intrinsic nearby cycles and therefore does not depend on the later cellular-collapse verification.

11. **[VERIFIED] The equivariant cellular collapse and its exact comparison with nearby cycles are now complete.** The positive toric quotient is a compact 3-manifold cobordism between two tori, its central boundary is a strong deformation retract, and the lifted retraction spreads by compact phases and conjugates back to the original twist. On the fixed fibre, the resulting map $c = r|_F$ has

$$0 \longrightarrow \operatorname{im}(\textstyle\bigwedge^q M_0 - I) \longrightarrow \bigwedge^q \Lambda \xrightarrow{c_*} H_q(W; \mathbb{Z}) \longrightarrow 0.$$

In degree two, with the primitive orientations fixed in theorem 44.6,

$$[c_*]_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix},$$

and its kernel is $\mathbb{Z}\langle \widehat{w}\widehat{\delta}, \widehat{\gamma}\widehat{w} + \widehat{u}\widehat{\delta} \rangle$. The neighborhood realization of the nearby-cycle unit then gives, for all q,

$$\mathrm{sp}^q = i_F^* (i_W^*)^{-1} = i_F^* r^* = c^*.$$

Thus C1–C3 are verified integrally, including saturation and the degree-three orientation.

12. **[VERIFIED] The global Mayer–Vietoris realization and its Leray comparison are complete.** The earlier audit conflated the two stages. On $X^\circ = X_1 \cup X_2$, the verified radial markings give the actual maps

$$\alpha_q = (\pi_{1*}, -\pi_{2*}).$$

Their degree-one and degree-two matrices have saturated rank three. On the second attachment $X = N_0' \cup X^\circ$, primitive splittings reduce the global maps to

$$D_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -12 & 0 & -p \end{pmatrix}, \qquad D_2^{\mathrm{red}} = \begin{pmatrix} -p & 0 \\ 0 & 1 \end{pmatrix}.$$

For $p = -1$ they are unimodular. Dualizing the same oriented collar maps gives commuting obstruction squares and

$$d_2(12\gamma) = -p\omega, \quad d_2(2q) = -p\omega[\delta], \quad d_2(2\gamma uw) = -p\omega[\gamma\delta].$$

Thus the alternate route independently gives the integral homology of S^6; see theorems 44.10, 44.11, 44.13 and 44.18.

13. **[VERIFIED] The Hodge diamond and Frölicher sequence are computed completely.** Put

$$\Sigma = p_0 + p_1 + p_2, \qquad D = f^{-1}(\Sigma)_{\mathrm{red}} = W + S_1 + S_2, \qquad \mathcal{L} = \Omega^1_B(\log \Sigma) \simeq \mathcal{O}_B(1).$$

The logarithmic relative bundle $\mathcal{E} = \Omega^1_X(\log D)/f^*\Omega^1_B(\log \Sigma)$ has

$$f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^1 f_*\mathcal{E} \simeq \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2}, \qquad R^2 f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

All three are locally free and commute with base change. At a finite point write $t = s^m$ and $\ell = dt/t = m\, ds/s$, with $(m, a) = (3, 2)$ at p_1 and $(m, a) = (4, 1)$ at p_2, and let $D_{\mathrm{KS}} = (d_{rs})$ be the marked Kodaira–Spencer matrix after removing ds. Dots in the raw period determinant mean d/ds, and

$$K = \det(\dot{Z}) = 6\dot{\mu}^{\,2} - \dot{\tau}\dot{\beta}.$$

With $\zeta_m = e^{-2\pi i/m}$, choose relative-cotangent eigenframes (e_0, e_1) with characters $(1, \zeta_m^a)$ and $R^1\mathcal{O}$ eigenframes (b_0, b_1) with characters $(1, \zeta_m^{-a})$. The connecting maps are typed as

$$\kappa_0 : f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^1 f_*\mathcal{O}_X, \qquad \kappa_1 : R^1 f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^2 f_*\mathcal{O}_X \simeq \mathcal{O}.$$

In the actual invariant source basis $(e_0, s^{m-a}e_1)$ and target basis $(b_0, s^a b_1)\ell$, put

$$A = \frac{s}{m} d_{00}, \quad B = \frac{s^{1-a}}{m} d_{01}, \quad C = \frac{s^{1-a}}{m} d_{10}, \quad D_0 = \frac{s^{m+1-2a}}{m} d_{11}.$$

The two connecting maps then have the exact matrices

$$[\kappa_0] = \begin{pmatrix} A & tB \\ C & D_0 \end{pmatrix}, \qquad [\kappa_1] = (-C, A, -D_0, B),$$

where the ordered source of the row is $(e_0b_0, s^a e_0 b_1, s^{m-a} e_1 b_0, e_1 b_1)$ and its target generator is $s^a(b_0 \wedge b_1)\ell$. Thus

$$\mathrm{div}(\det \kappa_0) = p_1 + p_2, \qquad \mathrm{ord}_{z_1} K = 2, \quad \mathrm{ord}_{z_2} K = 0, \qquad \kappa_1 \text{ is sheaf-surjective.}$$

The residue normalizations are also exact:

$$\mathcal{O}(1) \longrightarrow \mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \qquad \sigma \longmapsto (\sigma(p_0), 3\sigma(p_1), 4\sigma(p_2)),$$

$$\operatorname{res}_{1,j}(xb_0 + ys^a b_1) = m_j x[b_0],$$

and

$$\operatorname{im}(\kappa_0 \bmod t) = \ker \operatorname{res}_{1,j} = \mathbb{C}\, s^a b_1.$$

Hence the two finite cokernel skyscrapers map isomorphically to $H^1(S_1, \mathcal{O}) \oplus H^1(S_2, \mathcal{O})$ and cancel in the absolute residue sequence.

At the cusp, the quotient from ordinary to logarithmic relative differentials is checked on the xy and xyz modules. In the fixed global branch ordering its two triple-point maps are

$$d_P(a) = (a_0 - a_1, a_0 - a_2, a_1 - a_2), \qquad d_Q(a) = (a_2 - a_1, a_0 - a_1, a_0 - a_2),$$

Put $\eta_1 = d\log x_1$ and $\eta_2 = d\log x_2$. These cusp logarithmic forms lie in the logarithmic-form module, whereas the finite-filling covectors denoted η_j above lie in the filling-dual lattice. The exact morphism used below is the branch-difference map ρ from the former module to the branch-difference lattice; no identification of the two source modules is made. Both branch maps give

$$\rho(\eta_1) = q_1 = (-1, 0, 1), \qquad \rho(\eta_2) = q_2 = (0, -1, -1),$$

which is the degree-zero branch-difference isomorphism. With $\alpha_i = \delta_0(q_i)$ and the standard Čech differential, the total-complex sign is

$$\delta_1\big(\rho_*(\alpha \otimes \eta_i)\big) = -\alpha \smile \delta_0(q_i) = -\alpha \smile \alpha_i.$$

Thus, for $\theta = -\delta_1^{-1}(\alpha_1 \smile \alpha_2)$, the induced row is exactly

$$\rho_* = (0, 1, -1, 0) : \mathbb{C}\langle \alpha_1\eta_1, \alpha_1\eta_2, \alpha_2\eta_1, \alpha_2\eta_2\rangle \longrightarrow \mathbb{C}\theta.$$

The same local modules show that $\Omega^1_{X/B}$ is torsion-free but non-locally-free at the cusp; its torsion is supported precisely on the two multiple fibres.

For the remaining scalar, rescale the Cayley coordinate to $s = (z_2 - \bar{z}_2)s_2$, so that $s \circ g_2 = -is$ and $ds/dz(z_2) = 1$; write a dot for d/ds, and put $\ell = dt/t = 4\, ds/s$. In the marked fibre coordinate $\zeta = (\zeta_1, \zeta_2)$, the invariant section $\xi = \partial/\partial\zeta_2$ is nowhere zero in the locally free relative sheaf $T_{X/B} = \mathcal{E}^\vee$ (although its ambient tangent-space value can vanish at a toric fixed point), and the unique lift of its dual unit is $h = (a(z), 1)(d\zeta_1, d\zeta_2)^{\mathsf{t}}$ with $a(z_2) = 1/2$. At the order-four lift the genuine invariant target frame is

$$\omega_2 = s\, u_2(s)(\gamma \wedge u)\ell, \qquad u_2(0) = 1;$$

here b_1 is normalized by $\gamma \wedge b_1 = u_2(s)(\gamma \wedge u)$. $s(\gamma \wedge u)\ell$ alone is not invariant away from $s = 0$. In this frame the signed coarse value is

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\big(a(z_2)\dot{\tau}(z_2) + \dot{\mu}(z_2)\big).$$

Since $\operatorname{ord}_{z_2}(\tau - i) = 2$, one has $\dot{\tau}(z_2) = 0$ and $K(z_2) = 6\dot{\mu}(z_2)^2 \neq 0$; hence the scalar is exactly $-\frac{1}{4}\dot{\mu}(z_2) \neq 0$. Therefore

$$f_*\Omega^1_X \simeq \mathcal{O}(-2), \quad R^1 f_*\Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1)^2, \quad R^2 f_*\Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1),$$

and the exact Hodge table is

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	1
1	1	2	1	0
0	1	0	0	0.

The two middle Frölicher rows are exact copies of $0 \to \mathbb{C} \to \mathbb{C}^2 \to \mathbb{C} \to 0$. Thus the sequence does not degenerate at E_1, while $\dim E_1 = 10$, $E_2 = E_\infty$, and $\dim E_2 = 2$. See theorems 42.5 to 42.7, 42.9 and 42.10. The internal calculation is **[VERIFIED]**, and a further context-isolated AI derivation has independently recovered the invariant lattices, determinant divisor, residue cancellation, scalar test, direct-image splittings, and the same values $h^{1,2} = 1$, $h^{1,1} = 2$. The separate H2 audit record remains **[OPEN]** only because independent human specialist replication is still outstanding; that external replication boundary is not a premise of the Hodge theorem.

14. **[VERIFIED] Every Bott–Chern and Aeppli dimension is fixed.** The auxiliary spaces in McHugh's compact-threefold formulas satisfy $k^{p,q} \le b_{p+q}$: the de Rham image of the pure, doubly closed forms surjects onto $K^{p,q}$. Hence they vanish in all middle degrees for this X; also $E_2^{3,1} = 0$. Thus

$$h^{1,1}_{BC} = 2, \quad h^{1,2}_{BC} = h^{2,1}_{BC} = 1, \quad h^{2,2}_{BC} = 2, \quad h^{2,3}_{BC} = h^{3,2}_{BC} = 1,$$

besides the two endpoint copies of $\mathbb{C}$, and all other Bott–Chern dimensions vanish. Schweitzer duality gives the complete Aeppli diamond. Equivalently,

$$(h^k_{BC}) = (1,0,2,2,2,2,1), \quad (h^k_A) = (1,2,2,2,2,0,1), \quad (\Delta^k) = (0,2,4,4,4,2,0).$$

See theorem 42.11. Coordinate-level representatives and the product structure are certified separately in the next item.

15. **[VERIFIED] The Bott–Chern algebra has explicit generators and square-zero positive ideal.** Choose global $\bar\partial$-closed smooth representatives $G \in \mathcal{A}^{0,1}(X)$ and $Q \in \mathcal{A}^{1,1}(X)$ of the two candidate-specific Dolbeault classes $\mathfrak{g}$ and $\mathfrak{q}$. Here $\mathfrak{g}$ is distinguished, while $\mathfrak{q}$ is a chosen coherent-Leray lift, unique modulo $\operatorname{im} d_1$. Set

$$\mathsf{A} = [\partial G]_{BC}, \qquad \mathsf{B} = [\bar\partial \bar G]_{BC}.$$

On a marked smooth fibre,

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta,$$

put

$$y = \operatorname{Im}\tau, \qquad m = \operatorname{Im}\mu, \qquad r = \operatorname{Im}\beta, \qquad \Delta = r - \frac{6m^2}{y} < 0,$$

and define

$$\Gamma = \frac{i}{2\Delta}\left(d\bar\zeta_2 - \frac{m}{y} d\bar\zeta_1\right), \qquad U = \frac{i}{2y\Delta}\left(r\, d\bar\zeta_1 - 6m\, d\bar\zeta_2\right).$$

These represent the fibre images of γ and u, and the global classes satisfy

$$i_b^*\mathfrak{g} = [\Gamma]_{\bar\partial}, \qquad i_b^*\mathfrak{q} = [Q_F]_{\bar\partial}.$$

The Appell–Humbert matrix itself is

$$H_\eta = \frac{1}{y\Delta}\begin{pmatrix} r & -6m \\ -6m & 6y \end{pmatrix}, \qquad H_\eta = 2G_\eta^{-1}, \qquad \det H_\eta = \frac{6}{y\Delta} < 0.$$

The load-bearing wedge identity is

$$\eta = u \wedge w + 6\gamma \wedge \delta = u \wedge \sigma_1 + 6\gamma \wedge \sigma_2 = -\sigma_1 \wedge u - 6\sigma_2 \wedge \gamma,$$

which gives the exact fibre representative

$$Q_F = -d\zeta_1 \wedge U - 6 d\zeta_2 \wedge \Gamma, \qquad \eta \wedge \eta = 12\gamma \wedge u \wedge w \wedge \delta.$$

The marked and complex orientations are opposite because the real determinant of the marked lattice columns is $y\Delta < 0$. Accordingly,

$$\int_{(F_b,\mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b,\mathrm{or}_{\mathbb{C}})} Q_F^2 = -12.$$

The fibre Dolbeault–Serre pairing is normalized by

$$\int_{F_b} Q_F \wedge \Gamma \wedge d\zeta_2 = -1 \quad \text{for the complex orientation,}$$

so $[\Gamma \wedge Q_F]_{\bar\partial} \neq 0$.

The classes $d_1\mathfrak{g}$ and $\mathfrak{q}$ are independent in $H^{1,1}_{\bar\partial}(X)$: the first restricts to zero on every smooth fibre, whereas the second restricts to the nonzero class of Q_F. Exactness of the first middle Frölicher row therefore gives

$$d_1\mathfrak{q} = [\partial Q]_{\bar\partial} \neq 0,$$

so $\mathsf{C} = [\partial Q]_{BC}$ and its conjugate are nonzero. Moreover, putting

$$v = [G \wedge Q]_{\bar\partial} \in H^{1,2}_{\bar\partial}(X)$$

gives $i_b^* v = [\Gamma \wedge Q_F]_{\bar\partial} \neq 0$. Thus v spans $H^{1,2}_{\bar\partial}(X)$, and injectivity in the second middle row gives

$$d_1 v = [\partial(G \wedge Q)]_{\bar\partial} \neq 0.$$

Consequently $\mathsf{D} = [\partial(G \wedge Q)]_{BC}$ and its conjugate are nonzero.

The global integral normalization is equally explicit. If the base orientation is chosen so that $d_2(12\gamma) = c_1(\mathcal{O}_B(1))$ and $\exp(z) = e^{2\pi i z}$, the holomorphic structure $\bar\partial_z s = -2\pi i z G \otimes s$ on the smooth trivial line has unit-metric Chern connection and curvature

$$\nabla_z = d - 2\pi i(zG + \bar z\,\bar G), \qquad F_{\nabla_z} = -2\pi i(z\,\partial G + \bar z\,\bar\partial \bar G).$$

With the convention $c_{1,BC}(L,h) = [\,iF_{\nabla^h}/(2\pi)\,]_{BC}$, one obtains

$$\exp(12\mathfrak{g}) = f^*\mathcal{O}_B(1), \qquad c_{1,BC}(\exp(z\mathfrak{g})) = z\mathsf{A} + \bar z\,\mathsf{B},$$

and hence

$$c_{1,BC}(f^*\mathcal{O}_B(1)) = 12(\mathsf{A} + \mathsf{B}).$$

With the opposite primitive orientation both occurrences of 12 acquire the common sign -1:

$$\exp(-12\mathfrak{g}) = f^*\mathcal{O}_B(1), \qquad c_{1,BC}(f^*\mathcal{O}_B(1)) = -12(\mathsf{A} + \mathsf{B}).$$

Thus the canonical integral Picard–Leray datum fixes the absolute factor 12; it does not assert that A, B are integral Bott–Chern classes.

Put

$$\mathbf{1} = [1]_{BC}, \qquad \mathsf{T} = [\nu_X]_{BC}, \quad \nu_X \text{ a smooth positive } (3,3)\text{-form with } \int_X \nu_X = 1.$$

Together with the unit, the following nine positive-degree classes form bases of all positive-degree Bott–Chern groups, with T the normalized top class:

$$\mathsf{A} = [\partial G]_{BC},\ \mathsf{B} = [\bar\partial \bar G]_{BC},\ \mathsf{C} = [\partial Q]_{BC},\ \overline{\mathsf{C}},\ \mathsf{D} = [\partial(G \wedge Q)]_{BC},\ \overline{\mathsf{D}},\ \mathsf{E} = [\partial(Q^2)]_{BC},\ \overline{\mathsf{E}},\ \mathsf{T}.$$

If N denotes their span, then

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1} \oplus N, \qquad N^2 = 0.$$

For the last one-dimensional generator, $[Q^2]_{\bar\partial}$ has nonzero fibre restriction, whereas every class in $\mathrm{im}(d_1 : H^{1,2}_{\bar\partial} \to H^{2,2}_{\bar\partial})$ restricts to zero. Exactness therefore gives

$$d_1[Q^2]_{\bar\partial} = [\partial(Q^2)]_{\bar\partial} \neq 0,$$

which certifies E (and conjugation certifies $\overline{\mathsf{E}}$). Equivalently every product of two positive-degree classes is zero. Put $a = \partial G$, $a^\dagger = \bar\partial \bar G$, and $c = \partial Q$. The exact primitive

$$\partial\bar\partial(G \wedge \bar G) = -a \wedge a^\dagger$$

gives $\mathsf{AB} = 0$. Represent $c_{1,BC}(f^*\mathcal{O}_B(1))$ by the pullback Fubini–Study Chern form; its square vanishes pointwise because B is a curve. Squaring the factor-12 identity and using $\mathsf{AB} = 0$ gives $\mathsf{A}^2 + \mathsf{B}^2 = 0$. Moreover

$$[(a^\dagger)^2]_{\bar\partial} = 0, \qquad [a^2]_{\bar\partial} = d_1[G \wedge a]_{\bar\partial} = 0.$$

Injectivity of $d_1 : H^{1,2}_{\bar\partial} \to H^{2,2}_{\bar\partial}$ then gives a form Φ with

$$G \wedge a = \bar\partial\Phi, \qquad a^2 = \partial\bar\partial\Phi,$$

so $\mathsf{A}^2 = \mathsf{B}^2 = 0$. The last below-top product requiring a pairing is disposed of by

$$\int_X a \wedge c \wedge G = -\int_X G \wedge a \wedge c = -\int_X \bar\partial\Phi \wedge c = 0,$$

together with the perfect Dolbeault–Serre pairing and the isomorphism $H^{3,2}_{BC} \to H^{3,2}_{\bar\partial}$; the Bott–Chern–Aeppli perfect pairing alone would not prove this vanishing. Every top-degree product of two positive-degree generators contains a de Rham exact factor, while

$$H^{3,3}_{BC}(X) \xrightarrow{\int_X} \mathbb{C}$$

is an isomorphism, so those products vanish as well. See theorems 43.4 and 43.5. This asserts global smooth representatives, but not a preferred literal coefficient atlas through the two finite fillings and the cusp: $[i_b^* Q]_{\bar\partial} = [Q_F]_{\bar\partial}$ does not require $i_b^* Q = Q_F$ as forms. The chosen Q cannot be real, although its fibre class is real. After fixing the distinguished $\mathfrak{g}$ and one coherent-Leray lift $\mathfrak{q}$, changing the smooth representatives and horizontal corrections changes G, Q by Dolbeault coboundaries and changes none of the displayed classes, pairings, or products. This is a candidate-specific rank-one line-bundle normalization, not a statement about arbitrary gauge theories.

16. **[VERIFIED] The positive radial cusp has an exact flat period, inverse-dual, and scalar-gap certificate.** Put

$$\mu_T = \alpha_T + i m_T, \qquad h_T = \eta_T + i H_T, \qquad b_T = \xi_T + i B_T, \qquad c_T = \xi_T - \eta_T,$$

$$L_T = T + H_T, \qquad q_T = L_T - B_T, \qquad D_T = L_T q_T + 6m_T^2.$$

Set also

$$m = \mathrm{Im}\,\mu(0), \qquad H = \mathrm{Im}\,h(0), \qquad B = \mathrm{Im}\,b(0).$$

In the ordered marked lattice and interleaved real coordinate bases,

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}, \qquad \det P_T = -D_T, \qquad \mathrm{covol}(P_T\mathbb{Z}^4) = D_T > 0, \qquad D_{\mathrm{can},T} = -\frac{D_T}{L_T} < 0,$$

and the complete inverse transpose is

$$P_T^{-\mathsf{T}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \dfrac{m_T}{D_T} & \dfrac{q_T}{D_T} & \dfrac{-6m_T\alpha_T - q_T\eta_T}{D_T} & \dfrac{-m_T c_T - q_T\alpha_T}{D_T} \\ 0 & 0 & 0 & 1 \\ -\dfrac{L_T}{D_T} & \dfrac{6m_T}{D_T} & \dfrac{6(L_T\alpha_T - m_T\eta_T)}{D_T} & \dfrac{L_T c_T - 6m_T\alpha_T}{D_T} \end{pmatrix}.$$

Consequently $(P_T\mathbb{Z}^4)^* = P_T^{-\mathsf{T}}\mathbb{Z}^4$. More explicitly, for $k = (r, s, n_1, n_2)^{\mathsf{t}}$, define

$$R_T(k) = r - 6\alpha_T n_1 - c_T n_2, \qquad S_T(k) = s - \eta_T n_1 - \alpha_T n_2.$$

The corresponding dual vector is

$$\left(n_1, \frac{m_T R_T(k) + q_T S_T(k)}{D_T}, n_2, \frac{-L_T R_T(k) + 6m_T S_T(k)}{D_T}\right),$$

and the scalar eigenvalue is $4\pi^2$ times its squared norm. For every sufficiently large T,

$$\lambda_1(F_T, g_T) = 4\pi^2 \min\left\{\frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2}\right\} = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right).$$

This is the gap of the explicitly declared Euclidean quotient metric. Its unit-volume rescaling is a different operator family and obeys

$$\widehat{g}_T = D_T^{-1/2} g_T, \qquad \lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}\left(1 + \frac{-H - B/2 + 2\min\{B, 0\}}{T} + O(T^{-2})\right) \longrightarrow 0.$$

The degeneration has an exact fixed-energy density, not merely a lowest-mode estimate. Counting Fourier multiplicities by

$$N_T(E) = \#\{k \in \mathbb{Z}^4 : 4\pi^2 \|P_T^{-\mathsf{T}} k\|^2 \le E\}, \qquad R_E^2 = \frac{E}{4\pi^2},$$

one has, for every fixed $E \ge 0$,

$$N_T(E) = \pi D_T \sum_{n \in \mathbb{Z}^2} (R_E^2 - \|n\|^2)_+ + O_E(T + 1), \qquad (x)_+ = \max\{x, 0\}.$$

The sum is finite. In particular, below the first transverse threshold,

$$0 < E < 4\pi^2 \quad \Longrightarrow \quad N_T(E) = \frac{E}{4\pi} D_T + O_E(T + 1) = \frac{E}{4\pi} T^2 + O_E(T).$$

The proof writes the admissible (r, s)-indices for each fixed transverse $n \in \mathbb{Z}^2$ as the lattice points in a translated ellipse of area $\pi D_T(R_E^2 - \|n\|^2)$ and perimeter $O_E(T + 1)$, then applies the elementary planar area–perimeter lattice bound. Thus the remainder is uniform in the otherwise unrestricted real parts of the periods. See theorem 9.4.

Normalized Haar measure in the angular-coordinate direct integral is a unitary measure convention, not this metric rescaling. See theorems 9.1 to 9.4.

17. **[VERIFIED] The native $\mathbb{T}^2 \times \mathbb{R}^2$ limit and the ancillary covered $\mathbb{R}^4$ limit have different exact constructions.** The real projection $\rho(x_1, y_1, x_2, y_2) = (y_1, y_2)$ gives

$$0 \longrightarrow \mathbb{T}^2_{\mathrm{re}} \longrightarrow F_T \longrightarrow Q_T := \mathbb{R}^2/Y_T\mathbb{Z}^2 \longrightarrow 0, \qquad Y_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix}, \qquad \det Y_T = D_T.$$

The quotient systole is at least $T - C$, whereas the kernel retains its two unit periods; this is the native pointed $\mathbb{T}^2 \times \mathbb{R}^2$ degeneration. Adding

$$\mathsf{S}_M = \operatorname{diag}(1, 1, M, M), \qquad t_c = v^M$$

produces a degree-M^2 relative fibre cover over $v \neq 0$, because

$$\mathsf{S}_M^{-1} M_0^M \mathsf{S}_M = M_0.$$

The cover character indexed by $k \in \mathbb{Z}^4$ has momentum

$$P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} k,$$

and it is pulled back from the original fibre exactly when k_3, k_4 are divisible by M. The covering radius of this momentum lattice is $O(T^{-1} + M^{-1})$; hence $T, M \to \infty$ fills all four momentum directions and gives eventual exact Euclidean balls. This argument by itself proves only a pointed local $\mathbb{R}^4$ limit over the punctured base; the modified-action completion across the singular fibre is the distinct local Kummer construction recorded below, and neither argument alone gives a global cover of the completed threefold or an S12/S24-type globalization.

There is, however, an exact four-copy regular-locus construction. The monodromy-invariant closure of the compact cusp lattice is

$$L_\Lambda = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \qquad \Lambda / L_\Lambda \cong \mathbb{Z}\langle \widehat{\gamma} \rangle.$$

The induced real circle quotient has exact circumference

$$\ell_T = \frac{D_T}{\sqrt{L_T^2 + m_T^2}} = T + H - B + O(T^{-1}) \longrightarrow \infty.$$

The fourfold fibre product of $X^\circ = f^{-1}(B^\circ)$ therefore has a real fibrewise-group $\mathbb{T}^4$ quotient over B° whose synchronized radial fibres converge pointed-smoothly to $\mathbb{R}^4$. Two copies are insufficient for this monodromy-invariant-kernel mechanism; four are minimal, and no finite cover of the base repairs the native two-copy quotient. The two finite fillings admit explicit simultaneous-root normalizations over Seifert targets, with degrees 27 and 64 on the reduced central fibres. The toric cusp does not: two paths converging to the same triple point retain different character values $3/4$ and $1/2$, so the literal quotient and its fourfold product have no continuous cusp extension. The Cartesian source is not S^{24}: $X^4 \simeq (S^6)^4$, while that fixed-target fourfold-product map does not factor through the topological smash sphere $X^{\wedge 4} \cong S^{24}$; Borel–Serre also rules out almost-complex structures on S^{12} and S^{24}. Thus this exact regular-locus construction answers the globalization question as far as this mechanism can go and proves its compact-extension obstruction. See theorems 9.6, 9.7 and 9.42.

18. **[VERIFIED] The degree-N^2 regular-fibre cover has an exact completed local toric Kummer model, with direct and relative degrees kept distinct.** Write

$$Z(s) = sB_0 + C(t), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad t = e^{2\pi i s},$$

and let the target lattice action on the toric variety $Y = Y_\Sigma$ be

$$\Psi_{\bar{\lambda}}(x, t) = (e^{2\pi i C(t)\bar{\lambda}} t^{B_0 \bar{\lambda}} x, t), \qquad N_0 = Y / \bar{\Lambda}.$$

After the base change $t = v^N$, the correct source factor and action are

$$C_N(v) = \frac{1}{N} C(v^N), \qquad \Psi_{\bar{\lambda}}^{(N)}(\xi, v) = (e^{2\pi i C_N(v)\bar{\lambda}} v^{B_0 \bar{\lambda}} \xi, v).$$

They extend freely and properly discontinuously to a copy $Y^{(N)}$ of the same toric variety. Thus

$$N_0^{\langle N\rangle} = Y^{(N)}/\bar{\Lambda} \longrightarrow \Delta_v$$

is a proper holomorphic map from a smooth threefold, with reduced irreducible nonnormal central fibre W_N, whose normalization is dP_6. The toric power map

$$P_N(\xi_1, \xi_2, v) = (\xi_1^N, \xi_2^N, v^N)$$

is finite flat of degree N^3, is equivariant for the two displayed actions, and descends to

$$\kappa_N : N_0^{\langle N\rangle} \longrightarrow N_0, \qquad \deg \kappa_N = N^3, \qquad \kappa_N^* W = N W_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N.$$

If $N_0[N] = N_0 \times_{\Delta_t} \Delta_v$, then $N_0[N]$ is normal and

$$N_0^{\langle N\rangle} \xrightarrow{Q_N} N_0[N] \longrightarrow N_0$$

is the unique relative factorization. The generic degree of Q_N is N^2; for $N > 1$ it is quasi-etale but not flat, with non-etale and nonflat loci exactly over the double and triple strata of the base-changed central fibre. It is not a normalization. On reduced central fibres the map has generic degree N^2, and on normalizations it is the toric map

$$[N] : \mathrm{dP}_6 \longrightarrow \mathrm{dP}_6,$$

of degree N^2, ramified with index N along all six toric boundary curves and respecting the paired conductor curves and the two triple-point classes. Over $v \neq 0$, it is exactly the lattice cover

$$\mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, N\widehat{w}, N\widehat{\delta}\rangle \subset \Lambda, \qquad \mathrm{Deck}(Q_N) \cong (\mathbb{Z}/N\mathbb{Z})^2, \qquad S_N^{-1} M_0^N S_N = M_0,$$

where $S_N = \mathrm{diag}(1, 1, N, N)$. The factor $C(v^N)/N$ is essential: reusing the target action would require the generally false equality $e^{2\pi i C(v)\bar{\lambda} N} = e^{2\pi i C(v^N)\bar{\lambda}}$. This is a morphism of completed *local* cusp families. For $N > 1$ the cover lattice is not preserved by the full monodromy, so it gives neither a cover over all of B nor a product- or smash-sphere map. See theorems 9.8 to 9.10.

19. **[VERIFIED] The maps q_2, $q_{1,\mathrm{cusp}}$, and q_1 give exact cone-smash carriers with sources S^{12} and S^{24}, and their intrinsic metrics, ambient metrics, homology, and limitations are all explicit.** Let

$$K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle, \qquad \overline{\Lambda} = \Lambda/K_\Lambda, \qquad \overline{T}^2 = (\overline{\Lambda} \otimes \mathbb{R})/\overline{\Lambda}.$$

Fix $C_{\mathrm{sys}} > 0$ so that, for all sufficiently large T,

$$Y_T = T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + E_T, \qquad \|E_T\|_{\mathrm{op}} \leq C_{\mathrm{sys}}.$$

Since

$$\mathrm{im}(M_0 - I) = K_\Lambda = \ker(M_0 - I),$$

the punctured cusp has a fibrewise group epimorphism

$$\overline{\pi} : N_0 \setminus W \longrightarrow D_0^* \times \overline{T}^2$$

with connected two-torus kernel and trivial target local system. On the radial fibre, $\overline{T}^2$ is isometric to $Q_T = \mathbb{R}^2/Y_T\mathbb{Z}^2$ and has systole at least $T - C_{\mathrm{sys}}$.

Put

$$\mathcal{A} = \operatorname{Susp}(S^1 \times \overline{T}^2) \cong \operatorname{Susp}(T^3).$$

For $t(f(x)) = re^{i\phi}$, the continuous surjective quotient

$$q_2 : (X, x_0) \longrightarrow (\mathcal{A}, v_0), \qquad q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)]$$

on $N_0 \setminus W$, where $\overline{\pi}_f = \operatorname{pr}_{\overline{T}^2} \circ \overline{\pi}$, sends W to v_0 and $X \setminus \operatorname{int} N_0$ to the other suspension vertex v_1. Its smash square

$$q_2^{\wedge 2} : X \wedge X \cong S^{12} \longrightarrow \mathcal{A} \wedge \mathcal{A}$$

restricts on synchronized radial fibres to

$$\pi_T \times \pi_T : F_T^2 \longrightarrow Q_T^2, \qquad (Q_T^2, 0) \longrightarrow (\mathbb{R}^4, 0),$$

where the convergence is pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff, and every fixed radius ball is eventually exactly Euclidean. The nonzero reduced integral homology groups are

$$\widetilde{H}_k(\mathcal{A}) = \begin{cases} \mathbb{Z}^3, & k = 2, 3, \\ \mathbb{Z}, & k = 4, \end{cases} \qquad \widetilde{H}_k(\mathcal{A}^{\wedge 2}) = \begin{cases} \mathbb{Z}^9, & k = 4, \\ \mathbb{Z}^{18}, & k = 5, \\ \mathbb{Z}^{15}, & k = 6, \\ \mathbb{Z}^6, & k = 7, \\ \mathbb{Z}, & k = 8. \end{cases}$$

More precisely,

$$\mathcal{A}^{\wedge 2} \simeq \bigvee^{9} S^4 \vee \bigvee^{18} S^5 \vee \bigvee^{15} S^6 \vee \bigvee^{6} S^7 \vee S^8.$$

The cusp-supported circle version uses

$$\mathcal{B} = \operatorname{Susp}(S^1 \times S^1) = \operatorname{Susp}(T^2)$$

and the continuous surjective based quotient

$$q_{1,\mathrm{cusp}} : (X, x_0) \longrightarrow (\mathcal{B}, b_0), \qquad q_{1,\mathrm{cusp}}(x) = [r, e^{i\phi}, \chi_\gamma(x)], \qquad q_{1,\mathrm{cusp}}(W) = b_0,$$

with the exterior collapsed to b_1. Its fourth smash power has source

$$q_{1,\mathrm{cusp}}^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{B}^{\wedge 4}$$

and restricts to $\chi_{\gamma,T}^{\times 4} : F_T^4 \to C_T^4$. Here

$$\widetilde{H}_k(\mathcal{B}) = \begin{cases} \mathbb{Z}^2, & k = 2, \\ \mathbb{Z}, & k = 3, \end{cases} \qquad \widetilde{H}_k(\mathcal{B}^{\wedge 4}) = \begin{cases} \mathbb{Z}^{16}, & k = 8, \\ \mathbb{Z}^{32}, & k = 9, \\ \mathbb{Z}^{24}, & k = 10, \\ \mathbb{Z}^8, & k = 11, \\ \mathbb{Z}, & k = 12. \end{cases}$$

For the full-base version, glue the regular circle target and the two finite Seifert targets to obtain the compact three-manifold with boundary

$$\mathcal{M} = (P \times S^1) \cup C_1 \cup C_2, \qquad \partial\mathcal{M} = \partial D_0 \times S^1,$$

and cone its boundary:

$$\mathcal{C} = \mathcal{M} \cup_{\partial\mathcal{M}} \mathrm{Cone}(\partial\mathcal{M}).$$

The map $q_1 : X \to \mathcal{C}$ is (f, θ_γ) over P, equivalently $\chi_\gamma|_{f^{-1}(P)}$, is the finite-filling map $N_j \to C_j$ for $j = 1, 2$, and on the cusp is

$$q_1(x) = [r, (e^{i\phi}, \theta_\gamma(x))], \qquad q_1(W) = c_0.$$

It is continuous, surjective, and based. Its fourth smash power is

$$q_1^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow \mathcal{C}^{\wedge 4},$$

and its synchronized radial restriction is again $\chi_{\gamma,T}^{\times 4} : F_T^4 \to C_T^4$, with

$$\mathrm{sys}(C_T^4) = \ell_T = T + H - B + O(T^{-1}), \qquad (C_T^4, 0) \longrightarrow (\mathbb{R}^4, 0).$$

The target homology is

$$\widetilde{H}_k(\mathcal{C}) = \begin{cases} \mathbb{Z}, & k = 2, 3, \end{cases} \quad \widetilde{H}_k(\mathcal{C}^{\wedge 4}) = \begin{cases} \mathbb{Z}, & k = 8, 12, \\ \mathbb{Z}^4, & k = 9, 11, \\ \mathbb{Z}^6, & k = 10. \end{cases}$$

In fact,

$$\mathcal{C}^{\wedge 4} \simeq S^8 \vee \bigvee^4 S^9 \vee \bigvee^6 S^{10} \vee \bigvee^4 S^{11} \vee S^{12}.$$

All omitted degrees in these three homology displays are zero. The conical-target maps $q_2^{\wedge 2}$ and $q_1^{\wedge 4}$ induce zero on all positive-degree integral homology and cohomology and have zero ordinary Hurewicz image; neither has a continuous homotopy section. Their unstable homotopy classes remain **[OPEN]**, because those facts do not decide nullity. The class of the cusp-supported conical-target map $q_{1,\mathrm{cusp}}^{\wedge 4} : S^{24} \to \mathcal{B}^{\wedge 4}$ has likewise not been computed.

The two-copy and four-copy exponents are sharp for the unscaled quotient norm induced by the marked Euclidean fibre metric. If a primitive lattice $N_{\mathrm{quot}} \subset \Lambda^{\oplus r}$ defines a connected quotient over the punctured cusp whose radial systole tends to infinity, then every bounded-length class must vanish, so

$$K_\Lambda^{\oplus r} \subset N_{\mathrm{quot}}, \qquad \mathrm{rank}(\Lambda^{\oplus r}/N_{\mathrm{quot}}) \le 2r.$$

Thus a rank-four quotient needs at least two copies, and equality is attained by $(\Lambda/K_\Lambda)^{\oplus 2}$. Over the full regular base, including after a finite base cover, monodromy invariance and primitivity additionally force $\widehat{u} \in N_{\mathrm{quot}}$: indeed $A_1\widehat{w} = \widehat{u} - \widehat{w}$, while a finite-index subgroup contains some power G_*^m satisfying $G_*^m\widehat{w} = \widehat{w} + m(\widehat{u} - \widehat{\delta})$. Consequently

$$\mathrm{rank}(\Lambda^{\oplus r}/N_{\mathrm{quot}}) \le r,$$

so four copies are necessary and sufficient there. This proves the exact minimality of the S^{12} cusp-local and S^{24} full-base smash sources within this native connected torus-quotient class; it asserts no minimality among arbitrary continuous quotient maps or after changing to a T-dependent rescaling. See theorem 9.19.

There are also sphere targets, not merely conical targets. Let

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})), \qquad c : \overline{T}^2 \twoheadrightarrow \overline{T}^2/K^{(1)} \cong S^2,$$

where c collapses the coordinate one-skeleton. Forgetting the cusp base angle in the reduced suspensions gives continuous surjections

$$\widehat{p}_2 : \mathcal{A} \twoheadrightarrow S^3, \quad [r, e^{i\phi}, u] \longmapsto [r, c(u)], \qquad \widehat{p}_1 : \mathcal{B} \twoheadrightarrow S^2, \quad [r, e^{i\phi}, \theta] \longmapsto [r, \theta].$$

Thus $\widehat{q}_3 = \widehat{p}_2 \circ q_2$ and $\widehat{q}_\gamma = \widehat{p}_1 \circ q_{1,\mathrm{cusp}}$ induce, after the chosen based diffeomorphism $X \cong_{\mathrm{diff}} S^6$, quotient maps

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8.$$

Both full sphere carriers are based null-homotopic, by two distinct exact calculations. First let $f : S^6 \to S^2$ be any based map. The factorization

$$S^{12} = S^6 \wedge S^6 \xrightarrow{f \wedge \mathrm{id}_{S^6} = \Sigma^6 f} S^2 \wedge S^6 = S^8 \xrightarrow{\mathrm{id}_{S^2} \wedge f} S^2 \wedge S^2 = S^4$$

and $\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0$ show that $f \wedge f$ is based null-homotopic. Associativity gives $f^{\wedge r} = (f \wedge f) \wedge f^{\wedge(r-2)}$, so

$$[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r}) \qquad (r \geq 2).$$

Taking $f = \widehat{q}_\gamma$ proves, in particular,

$$[\widehat{q}_\gamma^{\wedge 2}] = 0 \in \pi_{12}(S^4), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Thus the full surjective quotient $S^{24} \twoheadrightarrow S^8$ is based null-homotopic.

For the other carrier, let $g : S^6 \to S^3$ be any based map. Its exact smash factorization is

$$g \wedge g = (\Sigma^3 g) \circ (\Sigma^6 g) : S^{12} \longrightarrow S^9 \longrightarrow S^6.$$

Writing st for stabilization, the low-stem data are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \quad \pi_3^{\mathrm{S}} = \mathbb{Z}/24\langle\nu\rangle, \quad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\langle\nu^2\rangle,$$

$$\pi_{12}(S^6) = \mathbb{Z}/2\langle\nu_6 \circ \nu_9\rangle, \qquad \mathrm{st} : \pi_{12}(S^6) \xrightarrow{\cong} \pi_6^{\mathrm{S}}.$$

The full groups are tabulated in [9, p. 339]; the signed smash formula and the unstable stabilization calculation are [12, Propositions 3.1 and 5.11]. Because $12[g] = 0$, its stable image is $\mathrm{st}([g]) = 2a\nu$ for some $a \in \mathbb{Z}/12$. Stable smash multiplication gives

$$\mathrm{st}([g \wedge g]) = \mathrm{st}([g])^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

The displayed stabilization isomorphism forces $[g \wedge g] = 0 \in \pi_{12}(S^6)$. Associativity again shows that $g^{\wedge r}$ is based null-homotopic for every $r \geq 2$. Taking $g = \widehat{q}_3$ therefore proves

$$[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Both maps retain their point-set surjectivity and quotient-map status. The one-copy classes $[\widehat{q}_3] \in \pi_6(S^3)$ and $[\widehat{q}_\gamma] \in \pi_6(S^2)$ remain uncomputed. Neither full-map null theorem settles any of the conical-target classes named above; see theorems 9.21, 9.22, 9.25 and 9.27. Give $S_T^2 = \overline{T}^2/K^{(1)}$ the quotient length metric transported from Q_T, and give both smash products the quotient length metrics from the Riemannian ℓ^2-product metrics on Q_T^2 and C_T^4. These moving-slice metrics are assigned intrinsically and are not restrictions of one fixed ambient sphere metric. The targets contain moving topologically embedded four-spheres

$$\mathbb{S}^4_{12,T} = S_T^2 \wedge S_T^2 \subset S^6, \qquad \mathbb{S}^4_{24,T} = C_T^{\wedge 4} \subset S^8,$$

and the synchronized radial restrictions are exactly

$$F_T^2 \xrightarrow{\pi_T^{\times 2}} Q_T^2 \xrightarrow{c_T^{\times 2}} (S_T^2)^2 \longrightarrow \mathbb{S}^4_{12,T}, \qquad F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \longrightarrow \mathbb{S}^4_{24,T},$$

where the last arrows collapse the corresponding fat wedges. The full maps from F_T^2 and F_T^4 retain the nontrivial kernels of their first arrows; at the quotient stage, precisely the displayed skeleton/fat-wedge loci are collapsed. At the midpoint basepoints, every radius-R ball is exactly a Euclidean four-ball once $T - C_{\mathrm{sys}} > 4R$, respectively $\ell_T > 4R$. Hence both moving four-spheres converge to $\mathbb{R}^4$ in pointed local-isometric and pointed Gromov–Hausdorff topology; on those eventual Euclidean basepoint neighborhoods the convergence is also smooth. Each moving four-sphere is null-homotopic in its ambient sphere by sliding its suspension coordinate to an endpoint, so its fundamental class dies in the target. See theorems 9.20 and 9.24. Removing the smash basepoint gives exact homeomorphisms

$$(Q_T \setminus K_T^{(1)})^2 \cong \mathbb{S}^4_{12,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad (C_T \setminus \{0\})^4 \cong \mathbb{S}^4_{24,T} \setminus \{*\} \cong \mathbb{R}^4 \hookrightarrow S^8.$$

These are nonclosed topological open cells, not globally Euclidean-isometric subspaces for a fixed compatible compact-target metric; see theorem 9.23. Both sphere-to-sphere quotient maps induce zero on every positive-degree integral homology group, and both full maps are based null-homotopic:

$$[\widehat{q}_3^{\wedge 2}] = 0 \in \pi_{12}(S^6), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8).$$

Their one-copy classes remain uncomputed. These full-map statements are distinct from the two explicit null-homotopies of the moving four-sphere inclusions.

For completeness, the radial target-slice embeddings used below are typed as follows. Put $r_T = e^{-2\pi T}$ and

$$\vartheta_T : \mathbb{R}/\mathbb{Z} \xrightarrow{\cong} C_T, \qquad [a] \longmapsto [\ell_T a].$$

Then

$$j_T : Q_T^2 \hookrightarrow \mathcal{A}^{\wedge 2}, \qquad j_T(y_1, y_2) = \left[[r_T, 1, \iota_T^{-1}(y_1)], [r_T, 1, \iota_T^{-1}(y_2)]\right]_\wedge,$$
$$k_T^{\mathcal{B}} : C_T^4 \hookrightarrow \mathcal{B}^{\wedge 4}, \qquad k_T^{\mathcal{B}}(c_1, \dots, c_4) = \left[[r_T, 1, \vartheta_T^{-1}(c_1)], \dots, [r_T, 1, \vartheta_T^{-1}(c_4)]\right]_\wedge,$$

and, in the cusp-cone piece of $\mathcal{C}$,

$$k_T^{\mathcal{C}} : C_T^4 \hookrightarrow \mathcal{C}^{\wedge 4}, \qquad k_T^{\mathcal{C}}(c_1, \dots, c_4) = \bigwedge_{j=1}^{4} [r_T, (1, \vartheta_T^{-1}(c_j))].$$

These are the embeddings of theorem 9.15.

The intrinsic decompactification can be made into an actual boundary family. Fix T_0 sufficiently large, put $r_0 = e^{-2\pi T_0}$, and define the compact radial carriers

$$\mathcal{D}_{12} = \frac{[0, r_0] \times (\overline{T}^2)^2}{\{0\} \times (\overline{T}^2)^2 \sim d_{12}}, \qquad \mathcal{D}_{24} = \frac{[0, r_0] \times (S^1)^4}{\{0\} \times (S^1)^4 \sim d_{24}}.$$

Both are homeomorphic to $\mathrm{Cone}(\mathbb{T}^4)$. At either cone vertex d_ν,

$$H_k(\mathcal{D}_\nu, \mathcal{D}_\nu \setminus \{d_\nu\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^4, & k = 2, \\ \mathbb{Z}^6, & k = 3, \\ \mathbb{Z}^4, & k = 4, \\ \mathbb{Z}, & k = 5, \\ 0, & \text{otherwise}, \end{cases}$$

so d_ν is not a topological-manifold point, with or without boundary, of any dimension. Set

$$A_T^{(12)} = \operatorname{diag}(Y_T, Y_T), \qquad A_T^{(24)} = \ell_T I_4, \qquad s_0 = T_0^{-1},$$

and, for $\nu \in \{12, 24\}$, quotient $[0, s_0] \times \mathbb{R}^4$ fibrewise by

$$(0, x) \sim_\nu (0, y) \iff x = y, \qquad (s, x) \sim_\nu (s, y) \iff x - y \in A_{1/s}^{(\nu)} \mathbb{Z}^4 \quad (s > 0).$$

The resulting $\mathfrak{E}_\nu$ is a Hausdorff, second-countable topological five-manifold with boundary, with

$$p_{12}^{-1}(s) \cong Q_{1/s}^2, \qquad p_{24}^{-1}(s) \cong C_{1/s}^4 \quad (s > 0), \qquad p_\nu^{-1}(0) \cong \mathbb{R}^4.$$

Its other boundary component, over $s = s_0$, is a compact four-torus. Every fixed Euclidean ball embeds isometrically in all sufficiently small positive fibres. There is a continuous surjective comparison map $\beta_\nu : \mathfrak{E}_\nu \to \mathcal{D}_\nu$ which is a homeomorphism away from the zero fibre, identifies each positive fibre with its radial torus slice, and collapses the entire zero-boundary component $p_\nu^{-1}(0) \cong \mathbb{R}^4$ to the vertex of the compact $\mathcal{D}_\nu \cong \operatorname{Cone}(\mathbb{T}^4)$. Thus the Euclidean space is an exact decompactification boundary, not the whole compact source or target and not a continuous quotient target of a compact source. The map β_ν is neither proper nor a quotient map, so this is not a topological blow-down or a quotient identification $\mathfrak{E}_\nu / \mathbb{R}^4 \cong \mathcal{D}_\nu$. Explicitly, for $a = (1/2, 1/2, 1/2, 1/2)$ and $s_j \downarrow 0$, the points

$$z_j = [s_j, A_{1/s_j}^{(\nu)} a]$$

form a closed subset of $\mathfrak{E}_\nu$. Indeed, in the S^{12} case

$$\inf_{n \in \mathbb{Z}^4} \|A_{1/s_j}^{(12)}(a + n)\| \geq (1/s_j - C_{\mathrm{sys}}) \operatorname{dist}(a, \mathbb{Z}^4) \to \infty,$$

and in the S^{24} case the same infimum equals $\ell_{1/s_j} \operatorname{dist}(a, \mathbb{Z}^4) \to \infty$; hence the sequence has no accumulation in a positive fibre and eventually misses every half-space chart about every zero-fibre point. On the other hand, $\beta_\nu(z_j) = [e^{-2\pi/s_j}, a] \to d_\nu$. Hence the image set is not closed although its inverse image is closed, which proves that β_ν is not quotient; see theorems 9.28 and 9.29.

Unrescaled ambient metrics impose a different conclusion. Put

$$Z_{12} := \mathcal{A}^{\wedge 2}, \qquad Z_{24}^{\mathcal{B}} := \mathcal{B}^{\wedge 4}, \qquad Z_{24}^{\mathcal{C}} := \mathcal{C}^{\wedge 4}.$$

For every fixed compatible metric on these compact targets,

$$\operatorname{diam}_{\mathrm{amb}} j_T(Q_T^2) \to 0, \qquad \operatorname{diam}_{\mathrm{amb}} k_T^{\mathcal{B}}(C_T^4) \to 0, \qquad \operatorname{diam}_{\mathrm{amb}} k_T^{\mathcal{C}}(C_T^4) \to 0,$$

whereas intrinsically

$$\operatorname{diam}(Q_T^2) \geq \tfrac{1}{2}(T - C_{\mathrm{sys}}) \to \infty, \qquad \operatorname{diam}(C_T^4) \geq \tfrac{1}{2}\ell_T \to \infty.$$

There are fixed cusp-adapted compatible length metrics with regular-stratum metric tensors $g_{Z_{12}}$ and $g_{Z_{24}^{\mathcal{Q}}}$, $\mathcal{Q} \in \{\mathcal{B}, \mathcal{C}\}$, for which the exact rescaled first fundamental forms are

$$\left(\frac{T}{r_T}\right)^2 j_T^* g_{Z_{12}} = g_{Q_T} \oplus g_{Q_T}, \qquad \left(\frac{\ell_T}{r_T}\right)^2 (k_T^{\mathcal{Q}})^* g_{Z_{24}^{\mathcal{Q}}} = g_{C_T}^{\oplus 4}, \quad \mathcal{Q} \in \{\mathcal{B}, \mathcal{C}\}.$$

With their induced Riemannian path metrics (not the restricted ambient distance functions), the rescaled slices converge to $\mathbb{R}^4$, but the corresponding full ambient pointed blow-ups converge on their eventual regular-stratum balls to $\mathbb{R}^8$ and $\mathbb{R}^{12}$, respectively. In every rescaling that unrolls a slice to $\mathbb{R}^4$, the cone/fat-wedge basepoint recedes to infinite scaled distance; vertex-based conic tangents

have dimensions 8 and 12 for exact conic target metrics, and unit-volume normalization gives a flat four-torus, not $\mathbb{R}^4$. See theorems 9.34 to 9.38.

Thus coning changes the codomain and does not contradict the proved nonextension of the literal character map to the fixed target $B \times S^1$. Nor do Cartesian powers give spheres:

$$X^2 \simeq S^6 \times S^6 \not\simeq S^{12}, \qquad X^4 \simeq (S^6)^4 \not\simeq S^{24};$$

only the based smash powers are spheres. There is no full-base two-copy four-torus quotient, no fixed T^4 target retaining the four character classes, and no fixed S^4 target extending a degree-one collapse of the radial quotient. No nontrivial finite base cover preserves K_Λ, and the local unspooling cover is not a nontrivial connected unbranched cover of the simply connected compact sources. Finally, S^{12} and S^{24} admit no almost-complex structure, so the smash sources are not compact complex spheres. See theorems 9.12 to 9.14, 9.16, 9.18, 9.26 and 9.27. Every construction in this sphere-route item uses the native quotients of the original, unscaled cusp fibres; none uses the ancillary finite-cover unspooling from the preceding item.

20. **[VERIFIED] The repaired compact-link construction and its finite Gaussian quantization have exact actions, spectra, and a resolved zero-gap diagonal sequence.** Keep the cover index and the mesh index independent:

$$M_{\mathrm{cov}} \in \mathbb{Z}_{\geq 1}, \qquad S_{M_{\mathrm{cov}}} = \mathrm{diag}(1, 1, M_{\mathrm{cov}}, M_{\mathrm{cov}}), \qquad P_{T,M_{\mathrm{cov}}} = P_T S_{M_{\mathrm{cov}}}, \qquad N_{\mathrm{mesh}} \in \mathbb{Z}_{\geq 2}.$$

Then

$$\widetilde{F}_{T,M_{\mathrm{cov}}} = \mathbb{R}^4 / P_{T,M_{\mathrm{cov}}} \mathbb{Z}^4 \longrightarrow F_T$$

has degree M_{cov}^2, Gram matrix and covolume

$$G_{T,M_{\mathrm{cov}}} = P_{T,M_{\mathrm{cov}}}^{\mathsf{T}} P_{T,M_{\mathrm{cov}}}, \qquad \mathcal{D}_{T,M_{\mathrm{cov}}} = \sqrt{\det G_{T,M_{\mathrm{cov}}}} = M_{\mathrm{cov}}^2 D_T.$$

Put

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1, 2, 3, 4\}.$$

Transport of the standard N_{mesh}-cubulation by $[y] \mapsto [P_{T,M_{\mathrm{cov}}} y]$ is an honest cubulation of the whole cover torus; its vertex set is exactly the N_{mesh}-torsion subgroup, via

$$[n] \longmapsto \left[P_{T,M_{\mathrm{cov}}} n / N_{\mathrm{mesh}}\right].$$

Parallel transport along its oriented edges gives the typed gauge-equivariant holonomy map to $\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}$.

Write $M = M_{\mathrm{cov}}$ and $N = N_{\mathrm{mesh}}$ only within the following compact-link formulas. Normalize

$$\langle X, Y \rangle_{\mathfrak{su}(2)} = -\tfrac{1}{2} \operatorname{tr}(XY), \qquad H_{\mathrm{link}} = \mathrm{diag}(i, -i), \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = I_2.$$

The periodic inverse-transport links

$$\begin{aligned} U_1(n) &= \exp(-2\pi n_2 H_{\mathrm{link}}/N^2), \\ U_2(n) &= \begin{cases} I_2, & 0 \leq n_2 \leq N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}}/N), & n_2 = N-1, \end{cases} \\ U_3(n) &= U_4(n) = I_2. \end{aligned}$$

have, including every wrap plaquette,

$$U_{12}(n) = \exp(2\pi H_{\mathrm{link}}/N^2), \qquad U_{ij}(n) = I_2 \quad ((i,j) \neq (1,2),\ i < j),$$

The principal-logarithm action and the explicit continuum-matched $(1,2)$-Wilson sector are exactly

$$S^{\log}_{T,M,N}(U) = \frac{2\pi^2 M^2}{g^2_{\mathrm{YM}} D_T},$$

$$S^{W,12}_{T,M,N}(U) = \frac{M^2 N^4}{2 g^2_{\mathrm{YM}} D_T}\left(2 - 2\cos\frac{2\pi}{N^2}\right) = \frac{M^2}{g^2_{\mathrm{YM}} D_T}\left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right).$$

The inverse-Gram two-form minor producing these formulas is exactly

$$(G^{-1}_{T,M_{\mathrm{cov}}})_{11}(G^{-1}_{T,M_{\mathrm{cov}}})_{22} - (G^{-1}_{T,M_{\mathrm{cov}}})^2_{12} = D_T^{-2}.$$

For $0 \le p \le 4$, define the real cochain spaces

$$C^p_N = \mathrm{Map}\Big(V_{N_{\mathrm{mesh}}}, \bigwedge\nolimits^p (\mathbb{R}^4)^* \otimes \mathfrak{su}(2)\Big).$$

For $\vartheta^j_{T,M} = P^{-\mathsf{T}}_{T,M}\mathbf{e}_j$, let $\mathsf{T}_j a(n) = a(n+\mathbf{e}_j)$ and set

$$D^+_{N,j} = N(\mathsf{T}_j - I), \qquad D^-_{N,j} a(n) = N(a(n) - a(n-\mathbf{e}_j)),$$

$$d_N a = \sum_{j=1}^4 \varepsilon(\vartheta^j_{T,M}) D^+_{N,j} a, \qquad d^*_N = -\sum_{j=1}^4 \iota_{(\vartheta^j_{T,M})^\sharp} D^-_{N,j},$$

and

$$\Delta^{(1)}_N = d_N d^*_N + d^*_N d_N, \qquad \mathscr{C}_{T,M,N} = \ker(d^*_N : C^1_N \to C^0_N) \cap (\ker \Delta^{(1)}_N)^\perp.$$

This last orthogonal complement removes the zero Fourier harmonic cochains. On this real reduced Coulomb space put

$$\mathcal{L}_{T,M,N} = g^{-2}_{\mathrm{YM}} d^*_N d_N, \qquad \mathfrak{d}_N(k)_j = N(e^{2\pi i k_j/N} - 1).$$

For every $k \neq 0$, its eigenvalue and complex transverse multiplicity are

$$\lambda_{T,M,N}(k) = \frac{1}{g^2_{\mathrm{YM}}}\left\| P_T^{-\mathsf{T}} S^{-1}_{M_{\mathrm{cov}}} \mathfrak{d}_N(k)\right\|^2, \qquad 3\dim \mathfrak{su}(2) = 9.$$

In particular,

$$\lambda_{T,M,N}(\mathbf{e}_1) = \frac{4N^2 \sin^2(\pi/N)}{g^2_{\mathrm{YM}}} \frac{L_T^2 + m_T^2}{D_T^2},$$

independently of M. The Friedrichs realization on $L^2(\mathscr{C}_{T,M,N}, da)$ of

$$\widehat{H}^{\mathrm{Gauss}}_{T,M,N} = -\tfrac{1}{2}\Delta_{\mathscr{C}_{T,M,N}} + \tfrac{1}{2}\langle a, \mathcal{L}_{T,M,N} a\rangle$$

has a unique Gaussian ground state and exact gap

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}_{T,M,N}) = \sqrt{\lambda_{\min}(\mathcal{L}_{T,M,N})} \le \frac{2N\sin(\pi/N)}{g_{\mathrm{YM}}} \frac{\sqrt{L_T^2 + m_T^2}}{D_T}.$$

For

$$T_j = j^2, \qquad M_j = j, \qquad N_j = j^3,$$

one has simultaneously

$$\mathrm{sys}(\widetilde{F}_{T_j,M_j}) \to \infty, \qquad \max_r \frac{\|P_{T_j,M_j}\mathbf{e}_r\|}{N_j} \to 0, \qquad \mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}_{T_j,M_j,N_j}) \to 0, \qquad S^{\log}_{T_j,M_j,N_j}(U) \to 0.$$

The exponential map $a \mapsto (\exp(a_e/N))_e$ is only a local diffeomorphism at zero, has derivative $N^{-1}\,\mathrm{id}$, is not globally injective, and, if the compact-link action satisfies $DS_W(\mathbf{1}) = 0$, transports its Hessian with the exact factor

$$D^2(S_W \circ \mathrm{Exp}_N)_0 = N^{-2} D^2(S_W)_{\mathbf{1}}.$$

Consequently the displayed Gaussian Hilbert space is not the Haar L^2-space on compact links or its nonlinear gauge-invariant subspace; no unitary identification, nonlinear compact-link Hamiltonian, Gauss-law spectral problem, transfer matrix, renormalized continuum limit, or Osterwalder–Schrader reconstruction has been constructed. This theorem therefore remains strictly on the auxiliary finite Gaussian side of the physical Yang–Mills boundary. See theorems 11.1 to 11.6 and equations (370) and (371).

21. **[VERIFIED] The exceptional first-coordinate elliptic quotient is finite and does not approach the cusp.** For the marked complex torus

$$\mathcal{A}_z = \mathbb{C}^2/\mathbb{Z}\langle(6\mu, \beta), (\tau, \mu), (1, 0), (0, 1)\rangle,$$

the first coordinate descends onto $E_\tau = \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$ exactly on

$$6\mu = a\tau + b, \qquad (a, b) \in \mathbb{Z}^2.$$

The kernel is connected and equals $\mathbb{C}/(\mathbb{Z} + \mathbb{Z}(a\mu - \beta))$, with $\mathrm{Im}(a\mu - \beta) = -D_{\mathrm{can}} > 0$. Under the two generators the labels transform by

$$(a, b) \overset{g_0}{\longmapsto} (a, a + b), \qquad (a, b) \overset{g_2}{\longmapsto} (-b, 6 + a).$$

The marked union is therefore monodromy invariant; its image in the completed orbifold base is finite and avoids a punctured cusp neighborhood. Thus this specific elliptic quotient cannot realize the cusp gap collapse. It remains logically possible for a fibre to have a different elliptic quotient or an integral magnetic class not induced by this projection. See theorems 9.39 to 9.41.

22. **[VERIFIED] A single specified auxiliary operator has zero essential spectral threshold.** Fourier expansion identifies

$$\mathscr{H}_{\mathrm{aux}} = L^2([T_0, \infty), dT; L^2_0(\mathbb{R}^4/\mathbb{Z}^4, dy)) \cong \bigoplus_{k \in \mathbb{Z}^4 \setminus \{0\}} L^2([T_0, \infty), dT).$$

With

$$\lambda_k(T) = 4\pi^2 |P_T^{-\mathsf{T}} k|^2, \qquad (\mathscr{L}_{\mathrm{aux}} u)_k(T) = \lambda_k(T) u_k(T),$$

the exact domain is

$$D(\mathscr{L}_{\mathrm{aux}}) = \left\{ u : \sum_{k \neq 0} \int_{T_0}^{\infty} \lambda_k(T)^2 |u_k(T)|^2 \, dT < \infty \right\}.$$

The operator is nonnegative and self-adjoint, has zero kernel, and the $k = (1, 0, 0, 0)$ singular sequence supported on $[n, n + 1]$ proves

$$0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}}).$$

The word "gapless" here belongs to this auxiliary Hilbert space and operator; it is not an invariant of the complex threefold alone. See theorems 38.1 and 38.3.

23. **[VERIFIED] The continuum and finite classical gauge fluctuation operators have exact spectra and cusp gaps.** For a compact Lie group with $d_G = \dim_{\mathbb{R}} \mathfrak{g}$, the product-connection Hessian on the reduced Coulomb space is

$$\mathcal{L}_T = g_{\mathrm{YM}}^{-2} \mathrm{d}^* \mathrm{d}, \qquad \lambda_T^{\mathrm{YM}}(k) = \frac{4\pi^2}{g_{\mathrm{YM}}^2} |P_T^{-\mathsf{T}} k|^2$$

on every $k \neq 0$ transverse Fourier fibre, with complex multiplicity $3d_G$. Hence

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2} \min\left\{ \frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2} \right\} = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right).$$

For the finite cubical complex,

$$\mathfrak{d}_N(k)_j = N(e^{2\pi i k_j/N} - 1), \qquad \zeta_{N,T}(k) = P_T^{-\mathsf{T}} \mathfrak{d}_N(k), \qquad \mathcal{L}_{N,T} = g_{\mathrm{YM}}^{-2} \mathrm{d}_N^* \mathrm{d}_N,$$

and the reduced $k \neq 0$ eigenvalue is exactly

$$\lambda_{N,T}^{\mathrm{YM}}(k) = g_{\mathrm{YM}}^{-2} |P_T^{-\mathsf{T}} \mathfrak{d}_N(k)|^2$$

with the same complex multiplicity. Writing $\varrho_N^{\mathrm{spec}} = 4N^2 \sin^2(\pi/N)$, there is a T_1, independent of $N \geq 2$, such that for every $T \geq T_1$ its gap is

$$\gamma_{N,T} = \frac{\varrho_N^{\mathrm{spec}}}{g_{\mathrm{YM}}^2} \min\left\{ \frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36m_T^2}{D_T^2} \right\} = \frac{\varrho_N^{\mathrm{spec}}}{g_{\mathrm{YM}}^2 T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right)$$

uniformly for $N \geq 2$. Thus each fixed finite model is positively gapped, but $\gamma_{N(T),T} \to 0$ for every $N(T) \geq 2$; resolving the varying continuum geometry is the stronger condition $N(T)/T \to \infty$. The finite operator is a linearized Lie-algebra-valued cochain model, not a nonlinear compact-link Wilson action. See theorems 10.5, 10.6, 10.8 and 10.10.

24. **[VERIFIED] Every fixed nondegenerate integral flux on the marked four-torus has an exact magnetic-oscillator spectrum whose first spacing tends to zero along the cusp.** In angular coordinates $u \in \mathbb{R}^4/\mathbb{Z}^4$, let

$$G_T = P_T^{\mathsf{T}} P_T, \qquad \sqrt{\det G_T} = D_T, \qquad \mathsf{E} = \sum_{j<k} E_{jk}\, \mathrm{d}u_j \wedge \mathrm{d}u_k \in \bigwedge^2 (\mathbb{Z}^4)^*.$$

The factor of automorphy and connection

$$J_n^{\mathsf{E}}(u) = \exp\left(-2\pi i \sum_{j<k} E_{jk} n_j u_k\right), \qquad \nabla^{\mathsf{E}} = \mathrm{d} + 2\pi i \sum_{j<k} E_{jk} u_j\, \mathrm{d}u_k$$

define a Hermitian line bundle $\mathscr{L}_{\mathsf{E},T}$ with curvature $F_{\nabla^{\mathsf{E}}} = 2\pi i \mathsf{E}$. The nonnegative closed form

$$q_{\mathsf{E},T}[f] = \int_{[0,1]^4} (G_T^{-1})^{jk} \overline{\nabla_j^{\mathsf{E}} f} \nabla_k^{\mathsf{E}} f\, D_T\, \mathrm{d}^4 u$$

has form domain $H^1(F_T; \mathscr{L}_{\mathsf{E},T})$; its Friedrichs operator $\Delta_{\mathsf{E},T} = (\nabla^{\mathsf{E}})^* \nabla^{\mathsf{E}}$ has domain $H^2(F_T; \mathscr{L}_{\mathsf{E},T})$ and compact resolvent. For nondegenerate E, define $b_1, b_2 > 0$ by

$$\mathrm{Spec}\Big(G_T^{-1/2} (2\pi \mathsf{E}) G_T^{-1/2}\Big) = \{ib_1, -ib_1, ib_2, -ib_2\}.$$

The ladder commutators give the complete set of spectral values

$$\mathrm{Spec}(\Delta_{\mathsf{E},T}) = \{(2n_1 + 1)b_1 + (2n_2 + 1)b_2 : n_1, n_2 \in \mathbb{Z}_{\geq 0}\}, \qquad \mathrm{gap}(\Delta_{\mathsf{E},T}) = 2\min\{b_1, b_2\}.$$

Writing

$$d = |\operatorname{Pf}(\mathsf{E})| = |E_{12}E_{34} - E_{13}E_{24} + E_{14}E_{23}| > 0,$$

the determinant identity is

$$b_1 b_2 = \frac{4\pi^2 d}{D_T},$$

and therefore

$$\operatorname{gap}(\Delta_{\mathsf{E},T}) \le 4\pi\sqrt{\frac{d}{D_T}} = \frac{4\pi\sqrt{d}}{T}\left(1 + \frac{B-2H}{2T} + O(T^{-2})\right) \longrightarrow 0.$$

This is a fibrewise scalar magnetic Laplacian on the marked cusp tori. It does not assert that $[\mathsf{E}]$ is fixed by full monodromy or identify this level spacing with a physical Yang–Mills particle mass. See theorems 12.5 and 12.6.

25. **[VERIFIED] A reducible non-flat $SU(2)$ critical connection has exact action and an exact negative Jacobi eigenvalue.** The rank-two sum $\mathscr{E}_T = \mathscr{L}_T \oplus \mathscr{L}_T^{-1}$ has parallel trivial determinant, $c_1 = c_2 = 0$, and an explicitly nullhomotopic $SU(2)$ clutching map. Thus its principal $SU(2)$-bundle is smoothly trivial. In the lifted line-bundle splitting $\mathscr{L}_T \oplus \mathscr{L}_T^{-1}$, the direct-sum connection and its curvature are

$$H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \nabla^T = \mathrm{d} + 2\pi u_1\, \mathrm{d}u_2\, H_{\mathrm{mag}}, \qquad F_{\nabla^T} = \frac{2\pi}{D_T} H_{\mathrm{mag}}\, \mathrm{d}y_1 \wedge \mathrm{d}y_2,$$

where the Lie-algebra norm is the same one fixed above, $\langle X, Y\rangle = -\frac{1}{2}\operatorname{tr}(XY)$, so that $\|H_{\mathrm{mag}}\|^2 = 1$. In any chosen global $SU(2)$-trivialization of the smoothly trivial principal bundle this matrix is replaced by its corresponding gauge transform. One has $\mathrm{d}^*_{\nabla^T} F_{\nabla^T} = 0$ and

$$\mathrm{YM}_T(\nabla) = \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_T} |F_\nabla|^2\, \mathrm{d}\,\mathrm{vol}_{F_T},$$

so

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T} = \frac{2\pi^2}{g_{\mathrm{YM}}^2 T^2}\left(1 + \frac{B-2H}{T} + O(T^{-2})\right).$$

Let $(e_\mu)_{\mu=1}^4$ be a constant orthonormal frame for the descended flat metric. The background-gauge Jacobi expression

$$(\mathcal{J}_T a)_\nu = -\sum_{\mu=1}^4 \nabla_{e_\mu}\nabla_{e_\mu} a_\nu - 2\sum_{\mu=1}^4 [F_{\nu\mu}, a_\mu]$$

has an explicit smooth real unit eigenvector satisfying

$$\mathrm{d}^*_{\nabla^T} a = 0, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a, \qquad \mathrm{YM}''_T\,|_{\nabla^T}(a,a) = -\frac{4\pi}{g_{\mathrm{YM}}^2 D_T}.$$

This proves a classical saddle direction at every finite T, not a spectral assertion for a quantum Hamiltonian. See theorems 12.3, 12.4 and 12.8.

26. **[VERIFIED] The native full-flux background extends root by root from split $SU(2)$ to split-Cartan $SU(r)$, with exact topology, action, spectrum, index, and wall nullities.** For every nonzero trace-zero $\mathbf{m} \in \mathbb{Z}^r$, put

$$\mathscr{E}_{\mathbf{m},T} = \bigoplus_i \mathscr{L}_{\eta,T}^{\otimes m_i}, \qquad s_2(\mathbf{m}) = \sum_i m_i^2, \qquad k_{\mathbf{m}} = 6 s_2(\mathbf{m}).$$

Then

$$c_2(\mathscr{E}_{\mathbf{m},T}) = -\frac{s_2(\mathbf{m})}{2}[\eta_T]^2, \quad \langle c_2, [F_T]_{\mathrm{mark}}\rangle = -6 s_2(\mathbf{m}), \quad \langle c_2, [F_T]_{x\text{-complex}}\rangle = k_{\mathbf{m}},$$

and the corresponding reducible Yang–Mills connection satisfies

$$\mathrm{YM}_T = \frac{2\pi^2 k_{\mathbf{m}}}{g_{\mathrm{YM}}^2} + \frac{\pi^2 s_2(\mathbf{m}) S_T^2}{g_{\mathrm{YM}}^2 D_T}.$$

For each positive root $i < j$, the charge is $q_{ij} = |m_i - m_j|$; the complete negative normalized Coulomb bands are

$$q_{ij}\big((2n+1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big), \qquad 0 \le n < N_-(T),$$

each contributing real dimension $12q_{ij}^2$, with coincident bands summed. Hence

$$\operatorname{ind}_{\mathbb{R}} \mathcal{J}_{\mathbf{m},T} = 12 r N_-(T) s_2(\mathbf{m}) = 36 r s_2(\mathbf{m})$$

on the deep cusp tail. At $S_T = 0$ the negative spectrum is empty but the root zero modes double, so no strict-stability conclusion is made. These are classical fibrewise theorems, not an interacting quantum spectrum; moreover their nonzero c_2 obstructs extension over every proposed sphere carrier with vanishing fourth integral cohomology. See theorems 14.1, 14.2 and 14.5 to 14.7.

27. **[VERIFIED] The Koide cone is isomorphic only to an ambient period-coefficient discriminant, not to the regular cusp family.** For

$$Q_K(r) = \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2), \qquad \begin{pmatrix} h \\ x \\ y \end{pmatrix} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix},$$

the coordinate change is orthogonal and

$$Q_K(r) = h^2 - x^2 - y^2.$$

The real-linear map

$$(h, x, y) \longmapsto (t, m, \varrho) = \left(h + y, \frac{x}{\sqrt{6}}, h - y\right)$$

is an ambient coefficient-space isomorphism. For the canonical convention

$$Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}, \qquad P = \begin{pmatrix} 6\operatorname{Re}\mu & \operatorname{Re}\tau & 1 & 0 \\ 6m & t & 0 & 0 \\ \operatorname{Re}\beta & \operatorname{Re}\mu & 0 & 1 \\ \varrho & m & 0 & 0 \end{pmatrix},$$

one has, independently of the real parts,

$$\det P = t\varrho - 6m^2 = \det \begin{pmatrix} t & \sqrt{6}\, m \\ \sqrt{6}\, m & \varrho \end{pmatrix} = Q_K(r).$$

For $r \in \mathbb{R}^3_{\ge 0} \setminus \{0\}$, the equation $Q_K(r) = 0$ is equivalent to the displayed symmetric matrix being positive semidefinite of rank one. It also forces $\det P = 0$, so the columns of P do not form a full lattice and define no regular compact complex two-torus. Actual radial cusp fibres instead satisfy

$$\det P_T = -D_T < 0, \qquad D_T = T^2 + (2H - B)T + O(1), \qquad \det P_T \sim -T^2.$$

See theorems 24.1 to 24.3.

28. **[VERIFIED] The positive defect-one Koide shell has an exact ideal-and-unit parametrization and an unconditional power-saving counting law.** Define

$$\Delta(x,y,z) = 4(xy+xz+yz) - (x^2+y^2+z^2),$$

and put

$$\mathcal{R} = \{(x,y,z) \in \mathbb{Z}^3_{>0} : x \le y \le z,\ \Delta(x,y,z) = -1,\ z \ge 4(x+y)\},$$
$$R_{\mathrm{K}}(Y) = \#\{(x,y,z) \in \mathcal{R} : z \le Y\}.$$

The terminal Vieta wall is exact: below it, the other root $4(x+y)-z$ is positive and smaller than y. On the wall, the coordinates

$$n = z - 4(x+y), \qquad q = y - x, \qquad a = x + y - 4n$$

give a bijection with the integral chamber

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0, q, n \ge 0, a + 4n > q, \qquad z = 4a + 17n,$$

whose inverse is

$$(x,y,z) = \left(\frac{a+4n-q}{2}, \frac{a+4n+q}{2}, 4a+17n\right).$$

For

$$F_{\mathrm{K}}(a,q,n) = a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} = \{F_{\mathrm{K}} = -2\},$$

direct Hensel counting and the good-prime formula give

$$\sigma_2 = \frac{3}{2}, \qquad \sigma_3 = 2, \qquad \sigma_p = \frac{1-p^{-2}}{1-\left(\frac{3}{p}\right)p^{-1}} \qquad (p \ge 5),$$

and hence

$$\mathfrak{S}(\mathscr{X}) = \frac{9}{2}\frac{\log(2+\sqrt{3})}{\sqrt{3}\,\zeta(2)}.$$

The exact Poincaré-residue integral over the reduced chamber is

$$I_\infty = \frac{1}{2\sqrt{3}} \log\left(\frac{3}{2\sqrt{2}}\right).$$

The Gorodnik–Oh–Shah sector theorem applies to the genuine norm $\max\{|4a+17n|, |q|, |a|\}$. An explicit element of $\mathrm{SO}(F_{\mathrm{K}}, \mathbb{Z}_2)$ with spinor norm -17 exchanges the two Borovoi sign orbits and proves $c_{\mathscr{X}} = 1$. Calibrating both counting theorems against the same Euclidean balls therefore gives

$$C_0 := \frac{9}{2\pi^2}\log(2+\sqrt{3})\log\left(\frac{3}{2\sqrt{2}}\right)$$

and

$$\boxed{R_{\mathrm{K}}(Y) \sim C_0 Y.}$$

For $n \ge 1$, the exact ideal-and-unit parametrization puts

$$m_n = 18n^2 - 2 = 2(3n-1)(3n+1),$$
$$\mathcal{P}(m_n) = \{(C,D,E) \in \mathbb{Z}^2_{>0} \times \mathbb{Z} : C^2 D = m_n, \quad 0 \le E < D, \quad E^2 \equiv 3 \pmod{D}\}.$$

The map

$$(C,D,E) \longmapsto C(D\mathbb{Z} + (E+\sqrt{3})\mathbb{Z})$$

is a bijection to norm-m_n ideals of $\mathbb{Z}[\sqrt{3}]$. Each ideal has exactly one generator $a_0 + q_0\sqrt{3}$ with $a_0 > 2q_0 \geq 0$, and its positive solutions are

$$a_k + q_k\sqrt{3} = (a_0 + q_0\sqrt{3})(2 + \sqrt{3})^k, \qquad k \geq 0.$$

The inverse descent is $(a, q) \mapsto (2a - 3q, 2q - a)$, iterated while $a \leq 2q$, followed by the unique HNF recovery. Hence multiplicity is one and

$$\#\mathcal{P}(m_n) = \rho(m_n) = \sum_{d|m_n} \chi_{12}(d) = \rho(3n-1)\rho(3n+1).$$

The $n = 0$ face is not absorbed into this positive-norm ideal family: it is exactly

$$a_k + q_k\sqrt{3} = (1 + \sqrt{3})(2 + \sqrt{3})^k, \qquad k \geq 1,$$

with the original strict wall, and contributes the explicit floor count $R_0(Y) = O(\log(2 + Y))$ in equation (1034). For every real $Y \geq 0$, the unchanged weak inequality $4a + 17n \leq Y$ is retained in the exact floor formula of theorem 30.2; its discrepancy splits into the radial shifted divisor term and the exact phase discrepancies. If $S_\ell(m)$ denotes the norm-m coefficient of the unitary Hecke character defined in equation (1043), then the angular data obey the exact shifted identity

$$S_\ell(18n^2 - 2) = (-1)^\ell S_\ell(3n-1)S_\ell(3n+1),$$

with the shifted factorization proved in equation (1049); the endpoint half-weights and all five sharp walls are retained in equation (1051). A direct proof of effective well-roundedness for the unchanged sharp family allows Benoist–Oh, Theorem 13.2 [53, pp. 1931–1932], to be applied orbit by orbit and strengthens the count to

$$\boxed{R_{\mathrm{K}}(Y) = C_0 Y + O(Y^{1-\eta})} \qquad \text{for some fixed, presently unspecified } \eta > 0.$$

The primary-source comparison gives no exact-domain square-root theorem: Liu–Sarnak's 39/64 exponent is for a smooth standard-radial count on the exact congruence lattice, and the sharp standard-radial results of Good and Chatzakos–Petridis have the 2/3 frontier after small-spectrum terms. Neither transfers to the five-wall height $4a + 17n$; see section 37.1. The exact finite census gives $R_{\mathrm{K}}(10^8) = 3{,}536{,}002$, with all rows independently replayed; the largest $|R_{\mathrm{K}}(Y) - C_0 Y|/\sqrt{Y}$ for $10^4 \leq Y \leq 10^8$ is $0.17140019280881455 97\ldots$ at $Y = 867{,}240$.

There is also an exact hyperbolic bridge. For

$$B = (18, -6)_{\mathbb{Q}}, \qquad \mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}\frac{I+J}{2} \oplus \mathbb{Z}\frac{I-J}{2} \oplus \mathbb{Z}\frac{IJ}{2},$$

the map

$$(a, q, n) \longmapsto 2 + \frac{qI + aJ + nIJ}{2}$$

is a bijection from the parity-restricted ternary shell to the reduced-trace-4, reduced-norm-1 locus. The order has reduced discriminant 108, index 18 in a maximal overorder, and is not an Eichler order; B is ramified exactly at $2, 3$. Each image is primitive hyperbolic with centralizer $\{\pm\gamma^k : k \in \mathbb{Z}\}$, eigenvalues $2 \pm \sqrt{3}$, and length $2\log(2 + \sqrt{3})$. The shell is a finite union of conjugacy classes, but its canonical height $4a + 17n$ differs from the exact radial Huber height $7a + 15\sqrt{2}\,n$, since

$$X_i(\gamma)^2 = 1 + \frac{(7a + 15\sqrt{2}\,n)^2}{48}.$$

This theorem is independent of the existence of the proposed complex structure on S^6. The stronger square-root discrepancy $R_{\mathrm{K}}(Y) - C_0 Y = O_\varepsilon(Y^{1/2+\varepsilon})$ is an explicitly **[OPEN]**problem, not a hypothesis used in the asymptotic; see theorems 25.4, 25.6, 25.7, 25.11, 25.12, 29.3 and 32.2 and sections 35, 36.7 and 37.1.

29. **[VERIFIED] The earlier Erdős–Straus divisor selector and the S^6 terminal presentation are the same marked integral matrix slice.** For $p \equiv 1 \pmod{12}$ prime, define

$$L = \frac{p-1}{4}, \qquad 0 \le j < L, \qquad R_j = 4j+3, \qquad a_j = \frac{p+R_j}{4}.$$

For a selector state

$$u \in \mathbb{Z}_{>0}, \qquad u \mid a_j^2, \qquad R_j \mid 4u + p^{\varepsilon}, \qquad \varepsilon \in \{0,1\},$$

set

$$T = \frac{4u+p^{\varepsilon}}{R_j}, \qquad \ell_0 = \frac{T+1}{4}, \qquad \ell_1 = u - jT, \qquad \ell_2 = 1.$$

Then $T \equiv 3 \pmod 4$,

$$12\ell_0 - 4\ell_1 - 3\ell_2 = p^{\varepsilon},$$

and

$$\boxed{\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1-4\ell_0 \end{pmatrix}.}$$

The inverse $T = 4\ell_0 - 1$, $u = \ell_1 + jT$ proves bijectivity once the prime, shell label, channel, orientation, and divisor support are retained. The $\varepsilon = 0$ channel lies over terminal determinant parameter 1, and the $\varepsilon = 1$ channel lies over p. The proof also derives the complete selector from the factorization $(R_j y - pa_j)(R_j z - pa_j) = p^2 a_j^2$. This is a structural equality of integer presentations, not a map of the underlying geometric spaces; see section 6.

30. **[VERIFIED] Two complex columns give the exact four-dimensional $SU(3)$ Gram cone.**

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})/SU(3) \cong \mathrm{Herm}_2(\mathbb{C})_{\ge 0}.$$

For the one-point compactified column representation,

$$S^{12}_{\mathrm{col}}/SU(3) \cong B^4,$$

and the rank-one, zero, and infinity orbit types form its boundary S^3. This representation sphere is not the cusp-smash S^{12}; see theorems 15.1 and 15.5.

31. **[VERIFIED] Three complex columns give an exact cubic rank-three cone with a distinguished rank-loss S^8.** For $\mathscr{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C})$, the complete orbit invariants are

$$\mathscr{V}_3/U(3) \cong \mathrm{Herm}_3(\mathbb{C})_{\ge 0}, \qquad \mathscr{V}_3/SU(3) \cong \{(H,z) : H \ge 0,\ |z|^2 = \det H\}.$$

The second space is $\mathbb{R}^{10}$, so

$$S^{17}/SU(3) \cong S^9, \qquad \mathscr{V}_3^+/SU(3) \cong S^{10}.$$

Inside that S^{10}, the compactified rank-deficient locus is the standard topologically locally flat S^8, with

$$S^{10} \setminus S^8 \cong \mathbb{R}^8 \times \mathbb{C}^* \simeq S^1.$$

The generator is the determinant phase $\det A/|\det A|$, and the trace-one rank-one stratum is $\mathbb{C}P^2$. Quotienting the determinant phase gives $\mathscr{V}_3^+/U(3) \cong B^9$ with boundary S^8. As a whole this is a cubic Jordan/Gram carrier rather than a rank-two quadratic Lorentz cone. The marked angular quotient nevertheless gives an exact split Lorentz observation with every complementary direction retained, as recorded in the next item; see section 16.

32. **[VERIFIED] The marked angular epimorphism gives a split positive Lorentz observation, and its full fibre is retained.** For

$$P=\begin{pmatrix}1&-1&0\\0&1&-1\end{pmatrix},\qquad u=(1,1,1)^{\mathsf T},\qquad R=P^*(PP^*)^{-1}=\frac{1}{3}\begin{pmatrix}2&1\\-1&1\\-1&-2\end{pmatrix},$$

put $\Pi=uu^*/3$, $Q_u=I_3-\Pi$, and

$$\mathcal{O}_P(H)=PHP^*,\qquad \jmath_P(G)=RGR^*.$$

These positive maps satisfy

$$\mathcal{O}_P\jmath_P=\mathrm{id},\qquad \jmath_P\mathcal{O}_P(H)=Q_uHQ_u,$$

with

$$\ker\mathcal{O}_P=\{a\,uu^*+uv^*+vu^*:a\in\mathbb{R},\ v\in u^\perp\},\qquad \mathcal{O}_P^{-1}(0)\cap\mathcal{P}_3=\{\lambda uu^*:\lambda\geq 0\}.$$

Writing $X=(R\ u)$, every Hermitian matrix is uniquely

$$H=X\begin{pmatrix}G&\psi\\\psi^*&\lambda\end{pmatrix}X^*,\qquad (G,\psi,\lambda)=\left(PHP^*,\frac{1}{3}PHu,\frac{1}{9}u^*Hu\right),$$

so

$$\mathrm{Herm}_3(\mathbb{C})\cong\mathrm{Herm}_2(\mathbb{C})\oplus\mathbb{C}^2\oplus\mathbb{R},\qquad 9=4+4+1,$$

$$\det H=\lambda\det G-\psi^*\,\mathrm{adj}(G)\psi,\qquad H\succeq 0\iff G\succeq 0,\psi\in\mathrm{im}\,G,\ \lambda\geq\psi^*G^+\psi.$$

The faithful embedding

$$T_P(g)=RgP+\Pi=X(g\oplus 1)X^{-1}$$

acts by $(G,\psi,\lambda)\mapsto(gGg^*,g\psi,\lambda)$. For $g\in SL_2(\mathbb{C})$, the summands are a Lorentz four-vector, the defining complex spin representation, and a real scalar. The vector action factors through $SO^+(1,3)$, while $-I_2$ negates ψ, so the full carrier retains the spin double cover.

The section is proper and compactifies to

$$\jmath_P^+:B^4\cong\mathcal{P}_2^+\hookrightarrow(\partial\mathcal{P}_3)^+\cong S^8,\qquad e_2(\jmath_P(G))=\frac{1}{3}\det G,$$

and

$$\det(uu^*+\varepsilon\jmath_P(G))=\varepsilon^2\det G.$$

The reverse boundary observation is nonproper. In the strict $SU(3)$ quotient,

$$|\omega|^2=\lambda\det G-\psi^*\,\mathrm{adj}(G)\psi,$$

and $\omega=0$ is the rank-loss cone compactifying to $S^8\subset S^{10}$. The four-branch chart supplies the exact toric seed

$$K(z)=\mathrm{diag}(z_0z_3,z_1z_2),\qquad \det K=z_0z_1z_2z_3,$$

whose positive $SL_2(\mathbb{C})$-orbit fills the future causal cone. For $Z\in\mathrm{Mat}_{3\times 2}(\mathbb{C})$ and retained $\alpha,c>0$,

$$p_\alpha(Z)=\alpha Z^*Z,\qquad m^2_{\alpha,c}(Z)=\frac{\alpha^2}{c^2}\sum_{a<b}|Z_{a1}Z_{b2}-Z_{a2}Z_{b1}|^2.$$

This is the complete $SU(3)$-orbit momentum and is Lorentz covariant. Its positive-mass locus is exactly rank two, and $m^2_{\alpha,c}(rZ)=r^4m^2_{\alpha,c}(Z)$, proving that the unrestricted kinematic carrier has no

positive threshold. The exact reduction separates four statements which the weak Z-equation alone does not. First, for $m_0 > 0$ the colour quotient and reparametrization characteristic give

$$T^*\mathcal{H}^+_{m_0,c}$$

with its canonical symplectic form, exactly the future spinless massive particle. Second, the intrinsic nonzero rank-one quotient gives $T^*\mathcal{N}^\times_+$. Third, the full massless multiplier action is not the standard massless variational system: the determinant constraint has zero differential on rank one, its equation is $\dot{x} = \lambda p$ without $\lambda = e$, four transverse rank-opening directions are non-gauge first-order null directions, and $Z = 0$ admits arbitrary $x(\tau)$. Fourth, on the hard $\omega_0 \neq 0$ packet the momentum differential has rank four above threshold and rank three at threshold, so its displayed action has $\dot{x} = 0$ in the strict interior and $\dot{x} \in \mathbb{R}p$ only at the endpoint. The packet's multiplication operator is not the Hamiltonian generating this action; its positive endpoint is a constraint-selected algebraic/free spectral threshold rather than a dynamically generated interacting gap. See theorems 21.2 to 21.7, 22.6, 22.9, 22.12 and 22.16.

33. **[VERIFIED] The determinant normal slice is an exact Lorentzian host for explicit field models, but it does not itself select a mass gap.** The real-linear isomorphism

$$\jmath_P : \mathrm{Herm}_2(\mathbb{C}) \xrightarrow{\sim} \mathcal{N}_P := \{Y \in \mathrm{Herm}_3(\mathbb{C}) : Yu = 0\}, \qquad G \longmapsto RGR^*,$$

has inverse $Y \mapsto PYP^*$, and

$$\det(uu^* + \varepsilon \jmath_P(G)) = \varepsilon^2 \det G.$$

Writing $G(t,x,y,z) = \left(\begin{smallmatrix} t+y & x+iz \\ x-iz & t-y \end{smallmatrix}\right)$, polarization gives

$$\eta_P(\jmath_P(G), \jmath_P(K)) = tt' - xx' - yy' - zz'.$$

The positive-semidefinite matrices are the future cone, the nonzero rank-one matrices its null boundary, and the lifted congruence action induces $PSL_2(\mathbb{C}) \cong SO^+(1,3)$. The future timelike determinant shell, future-null locus, and candidate spacelike determinant shell are respectively

$$SL_2(\mathbb{C})/SU(2), \qquad SL_2(\mathbb{C}) \cdot \mathrm{diag}(1,0), \qquad SL_2(\mathbb{C})/SU(1,1).$$

The standard metrics already retained by the Moore–Penrose construction select

$$W_P = R^*R = C^{-1}, \qquad K_P = \{a : a^*W_P a = W_P,\ \det a = 1\} \cong SU(2),$$

and the positive form $E_P(G,K) = \mathrm{tr}(GW_PKW_P)$. Under K_P,

$$\mathrm{Herm}_2(\mathbb{C}) \cong \mathbf{1} \oplus \mathbf{3},$$

with trivial central action. If this pointwise symmetry is promoted to a gauge symmetry, the action

$$S[\mathcal{A}, G] = \int_M \left(\frac{|F_\mathcal{A}|^2}{2g^2_{\mathrm{YM}}} + \frac{Z_G}{2}|D_\mathcal{A}G|^2_{E_P} + \frac{m_G^2}{2}|G|^2_{E_P} + \frac{\lambda_4}{4}(\det G)^2 \right) d\,\mathrm{vol}_g$$

is nonnegative and gauge invariant, and $\widehat{\mathcal{A}} = R\mathcal{A}P$ intertwines the covariant derivatives. The normal field alone cannot screen a fundamental centre charge; the full (G, ψ, λ) carrier contains the centre-nontrivial spinor ψ, but its kinetic and interaction theory is not inserted by this construction.

The candidate inverse-dual matrix and the positive Gram momentum are also separated by an exact invariant:

$$\det G_{\eta,T} = -\frac{2}{3}D_T < 0, \qquad \mathrm{im}\, p_\alpha = \mathrm{Herm}_2(\mathbb{C})_{\geq 0}.$$

Neither positive rescaling nor complex congruence can identify their inertia types; a further morphism is required. After a separate length/time dictionary is supplied, the same affine host supports the ordinary Klein–Gordon theory. Its Fock Hamiltonian has excitation gap exactly m_{KG} for $m_{\mathrm{KG}} > 0$ and gap zero for $m_{\mathrm{KG}} = 0$. On a periodic spatial gauge regulator, the massless transverse Gaussian frequency is $2\pi/L \to 0$. Explicit countermodels also show that entanglement occurs in both gapped and gapless systems. A positive auxiliary gap transfers to the physical Hamiltonian only through a map on the entire excitation form domain satisfying

$$\|Ru\| \geq c_R\|u\|, \qquad \mathfrak{b}[Ru] \leq C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}}\|u\|^2),$$

in which case $\mathrm{gap}(A_{\mathrm{phys}}) \geq \gamma c_R^2/C_R$. An intertwiner on a single gapped subspace is not enough. See theorems 23.2, 23.3, 23.5 to 23.10, 23.12 and 23.16.

34. **[VERIFIED] The marked cubic cone contains exact Lorentz sections, but these do not identify it with the toroidal degeneration.** With the oriented cross-product matrix $[a]_\times$,

$$\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.$$

Consequently each marked positive monodromy form B determines the exact linear section

$$\ell_B(\lambda, a) = \lambda B + i[a]_\times, \qquad \ell_B^{-1}(\mathrm{Herm}_3(\mathbb{C})_{\geq 0}) = \{\lambda \geq 0 : a^{\mathsf{T}} B a \leq \lambda^2 \det B\}.$$

This is a future $(3+1)$-Lorentz cone. An explicit real-linear order-cone map Θ from 2×2 Hermitian matrices satisfies

$$\det_3 \Theta(H) = \frac{\operatorname{tr} H}{2} \det_2 H$$

and sends positive-semidefinite ranks $2, 1, 0$ to $3, 2, 0$. It is not a Jordan homomorphism: for a rank-one projection P, the matrix $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not idempotent. Neither map carries a toric period lattice, monodromy, or singular-fibre filling. In particular, the toric limit at fixed $B = I_3$ is not the rank-drop operation $\det B \to 0$. The rejected packet's claimed index-three translation compression is likewise not a constructed map: the actual six-to-two bridge is the saturated rank-lowering quotient $P : \mathbb{Z}^3 \twoheadrightarrow \mathbb{Z}^2$, which sends the even and odd Kuhn labels to the two unimodular target-triangle orbits. See theorems 18.1, 18.7 and 18.8.

35. **[VERIFIED] The rank-three toroidal successor maps onto the candidate cusp with a connected elliptic kernel.** The Kuhn fan gives a proper analytic family

$$\pi_3 : N_3 \longrightarrow \Delta, \qquad N_{3,t} \cong E_\tau^3 \quad (t \neq 0),$$

whose deepest local equation is $t = z_0 z_1 z_2 z_3$, whose cusp matrix is $M_3 = \left(\begin{smallmatrix} I_3 & I_3 \\ 0 & I_3 \end{smallmatrix}\right)$, and whose filling map kills the three angular cycles and retains the three deck cycles. The irreducible central fibre has

$$(f_0, f_1, f_2, f_3) = (1, 7, 12, 6), \qquad \chi(W_3) = 6.$$

Its normalization is the permutohedral threefold with orbit vector $(1, 14, 36, 24)$; the positive-codimension strata map with degrees $2, 3, 4$ onto the $7, 12, 6$ quotient strata. Hence any real S^8-total-space fibration retaining W_3 requires the remaining singular fibres to contribute Euler sum -4. For an injective integral shear $\mathcal{B} : \bar{L}_3 \hookrightarrow L_3$, this becomes the exact determinant/index formula

$$\chi(W_{3,\mathcal{B}}) = 3!\,[L_3 : \mathcal{B}\bar{L}_3] = 3!\,|\det \mathcal{B}|.$$

The explicit matrices

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

define a surjection $\Phi = P \oplus Q : H_1(T^6) \twoheadrightarrow H_1(T^4)$ satisfying

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3, \qquad P Z_3(t) = Z_2(t) Q.$$

It extends over all cusp toric strata to a holomorphic map with connected elliptic kernel. Integrally,

$$M_3 \cong M_F \oplus \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and the order-three and order-four free affine peripheral actions have compatible rank-six lifts. The six Kuhn tetrahedron orbits map S_3-equivariantly to the six marked diagonal Weyl chambers, but only as fan/Cartan data; the four-branch toroidal germ and the quadratic Gram-rank germ remain inequivalent. What remains open is the single holomorphic period-and-overlap family across both finite-order discs. See theorems 17.8 and 18.3 to 18.6.

36. **[VERIFIED] The free rank-six cusp lattice, the six chambers, and the Euler factor six are now separated exactly.** The lattice native to the higher-rung construction is

$$\Lambda_{\mathrm{tor}} = H_1(T^6; \mathbb{Z}) = \mathbb{Z}^3_{\mathrm{angular}} \oplus \mathbb{Z}^3_{\mathrm{deck}}.$$

For $k = (1, 1, 1)$, the map $\Phi = P \oplus Q$ gives an exact S_3- and monodromy-equivariant sequence

$$0 \longrightarrow \mathbb{Z}(k, 0) \oplus \mathbb{Z}(0, k) \longrightarrow \Lambda_{\mathrm{tor}} \longrightarrow A_F \oplus B_F \longrightarrow 0.$$

It splits integrally as a monodromy module but not over $\mathbb{Z}[S_3]$. The quotient copies are A_2 weight lattices; their root sublattices have index three. The six standard homology generators are two permutation orbits of size three, whereas the six Kuhn cells and six marked Weyl chambers form one regular S_3-orbit. Consequently

$$\operatorname{rk} \Lambda_{\mathrm{tor}} = 6, \qquad \#\{\text{chambers}\} = 3! = 6, \qquad \chi(W_{3,\mathcal{B}}) = 3! |\det \mathcal{B}|$$

do not identify six lattice directions with six strata. The regular six-element chamber set instead occurs in the rank-nine bilinear carrier

$$\operatorname{Hom}(\overline{L}_3, L_3) = \bigoplus_i \mathbb{Z} E_{ii} \oplus \bigoplus_{i \neq j} \mathbb{Z} E_{ij} :$$

the off-diagonal units are a regular S_3-set, and $c_{(\pi_1 \pi_2 \pi_3)} \mapsto E_{\pi_1, \pi_3}$ is an equivariant bijection. These are exactly the mixed angular–deck basis pairs with zero alternating pairing; the three excluded diagonal pairs have pairing one. The ambient real matrix representation is equivariantly isomorphic to $\mathrm{Herm}_3(\mathbb{C})$, but the symmetric/skew integral splitting has index 2^3, and the images of the off-diagonal units are indefinite. This is a representation/incidence bridge, not a positive, Jordan, determinant, or toric morphism. The actual nine-dimensional-boundary map is only the piecewise-linear angular map $\Phi_{\mathrm{G}} \circ \jmath \circ P$; the deck lattice is instead used by the period and filling squares. This map does not descend to a torus fibre. The separate bundle-defect lattice $H^4(U; \mathbb{Z}) \cong (\mathbb{Z}^2)^3$ has a different domain and degree. See section 19 and theorems 19.2, 19.3, 20.1 and 20.4.

The successful lifted candidate actions are distinct from the diagonal local-system candidate

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T, \qquad M_1 M_2 M_0 = I.$$

Its finite factors have orders 4 and 6, but $I - M_j : T^6 \to T^6$ is an isogeny for $j = 1, 2$; every affine lift of either finite action therefore has a fixed point.

37. **[REFUTED] There is no literal smooth complex-fourfold analogue on a homotopy S^8.** An almost-complex tangent bundle would have

$$c_1 = c_2 = c_3 = 0, \qquad c_4 = 2u, \qquad p_2 = 2c_4 = 4u,$$

whereas the signature formula forces $p_2 = 0$. A nonsingular T^6-bundle over S^2 with total space S^8 is independently excluded by both the homotopy sequence and Euler characteristic. A real singular T^6-over-S^2 construction remains a legitimate search problem, but its special fibres must obey $\sum_i \chi(F_i) = 2$. For monodromy ρ on the regular piece, its actual filling inclusions must use the corrected Serre filtration

$$0 \longrightarrow (\textstyle\bigwedge^q \Lambda)_\rho \longrightarrow H_q(E;\mathbb{Z}) \longrightarrow H_1(B^\circ; \textstyle\bigwedge^{q-1} \Lambda) \longrightarrow 0,$$

and the dual quotient $H^q(E;\mathbb{Z}) \twoheadrightarrow (\bigwedge^q \Lambda^\vee)^\rho$. Thus the integral Mayer–Vietoris test uses coinvariants as the homological subobject and the base-one-cycle term as quotient, not conversely; see section 24 and equations (899) and (902). Every effective torus action on S^8 has rank at most four, and the sharp coordinate model is

$$S^8/T^4 \cong \mathrm{Susp}(\Delta^3) \cong B^4.$$

This is a real two-pole singular compactification with generic T^4 orbit, not a three-special-fibre T^6-fibration over S^2; see equations (903) and (904).

38. **[VERIFIED] The regular locus of the reconstructed S^6 has a six-coordinate, nonextendable defect lattice.** For $D = W \sqcup S_1 \sqcup S_2$ and $U = X \setminus D$,

$$\boxed{H^4(U;\mathbb{Z}) \cong H_1(W) \oplus H_1(S_1) \oplus H_1(S_2) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2.}$$

Principal $SU(3)$-bundles on U are classified exactly by this c_2-class, and the pair connecting morphism $H^4(U) \xrightarrow{\sim} H^5(X,U)$ proves that extension across all three special fibres occurs exactly for $c_2 = 0$. The displayed group is the free rank-six lattice $\mathbb{Z}^6$, not a cyclic $\mathbb{Z}/6$.

39. **[VERIFIED] The primitive fibre degree has exact order-twelve and order-six normal-holonomy quotients.** Complete proofs, including the one-shot collar matrix, transpose system, saturated kernel, coefficient reduction, tensor-centre map, and common-centre quotient, are given in section 45 and theorems 45.1 to 45.4. Those group and lattice theorems do not themselves define a gauge field or an energy operator.

In the marked basis

$$(a_0, b_0, a_1, b_1, a_2, b_2) = (\bar\gamma, \bar u, g_1, h_1, g_2, h_2),$$

the transpose of the actual one-shot collar matrix gives

$$\boxed{[\deg_F] = (12, 0, 4, 0, -3, 0)},$$

meaning

$$\mathrm{res}_F\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = (12n_0 + 4n_1 - 3n_2)\omega_F.$$

It is primitive, with saturated kernel

$$\boxed{\ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle},$$

Define

$$q_{\mathrm{col}} : L \longrightarrow H_1(X^\circ;\mathbb{Z}), \qquad a_0 \mapsto c, \quad a_1 \mapsto \bar g_1, \quad a_2 \mapsto \bar g_2, \quad b_0, b_1, b_2 \mapsto 0.$$

If $\Phi_1 : H_1(X^\circ;\mathbb{Z}) \to \mathbb{Z}$ has row $(12, 4\ell_1, 3\ell_2)$ in the basis $(c, \bar{g}_1, \bar{g}_2)$, then the factorization through the same collar functional that kills the final cusp is

$$\deg_F = \Phi_1 \circ q_{\mathrm{col}}, \qquad \Phi_1(j_*[\widetilde{a}_0]) = p.$$

Let $\vartheta_1(a_1) = e^{-2\pi i/3}$ and $\vartheta_2(a_2) = e^{-2\pi i/4}$ be the local normal-line holonomies, trivial on the other marked generators. The actual relative twist gives

$$\chi_{12}\left(\sum_{i=0}^{2}(n_i a_i + r_i b_i)\right) = \vartheta_1(n_1 a_1 + r_1 b_1)\vartheta_2(n_2 a_2 + r_2 b_2)^{-1} = e^{-2\pi i(4n_1 - 3n_2)/12}.$$

If $\rho_m(k) = e^{-2\pi i k/m}$, then

$$\boxed{\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 := \chi_{12}^2 = \rho_6 \circ \deg_F\,.}$$

Therefore

$$H^4(U;\mathbb{Z})/\ker\chi_{12} \cong \mathbb{Z}/12\mathbb{Z}, \qquad H^4(U;\mathbb{Z})/\ker\chi_6 \cong \mathbb{Z}/6\mathbb{Z}.$$

Ordinary coefficient reduction instead gives

$$H^4(U;\mathbb{Z}/6) \cong (\mathbb{Z}/6)^6;$$

it is not the single distinguished cyclic quotient selected by χ_6. For a principal $SU(3)$-bundle $P \to U$, the order-six character is exactly

$$\chi_6(c_2(P)) = \exp\!\left(-\frac{2\pi i}{6}\langle c_2(P)|_F, [F]\rangle\right).$$

This is a topological congruence for the fibre second-Chern number, not a line-operator or spectral assertion.

The tensor embedding

$$SU(2) \times SU(3) \hookrightarrow SU(6), \qquad (A, B) \longmapsto A \otimes B$$

restricts to an isomorphism $\mu_2 \times \mu_3 \xrightarrow{\sim} Z(SU(6)) = \mu_6$. Since U has homotopy dimension at most five, c_2 classifies $SU(n)$-bundles for $n \geq 3$, while

$$c_2(E_2 \otimes E_3) = 3c_2(E_2) + 2c_2(E_3).$$

Every chosen $SU(6)$-bundle class $h \in H^4(U;\mathbb{Z})$ therefore admits a tensor-subgroup reduction, and all its c_2-factorizations are

$$(e, f) = (h + 2t, -h - 3t), \qquad t \in H^4(U;\mathbb{Z}).$$

No nonzero h or reduction is canonically selected. Separately,

$$H^5(U;\mathbb{Z}) \cong \ker\{\mathbb{Z}^3 \xrightarrow{(1,1,1)} \mathbb{Z}\} \cong Q(A_2),$$

for the combinatorial form making the three distinguished support basis vectors orthonormal. This is not a cup-product or intersection form, and the support-redistribution lattice does not identify the rank-five kernel of $\deg_F$ with A_5.

The finite-group isomorphism $H^4(U)/\ker\chi_6 \cong Z(SU(6))$ supplies no $SU(6)$ transition functions or connection. The injective tensor map does not factor through $(SU(2)\times SU(3))/\mu_6$, because the central μ_6 maps to nontrivial scalar matrices. The distinct common-centre map is

$$\Gamma_6 = \{((\lambda\mu)^{-1}, \lambda I_2, \mu I_3) : \lambda \in \mu_2,\ \mu \in \mu_3\},$$

$$\frac{U(1) \times SU(2) \times SU(3)}{\Gamma_6} \hookrightarrow U(6),$$

$$[z, A, B] \longmapsto z(A \otimes B),$$

where $\Gamma_6 \cong \mu_6$. Its determinant is z^6, and its image meets $SU(6)$ in exactly $\iota(SU(2) \times SU(3))$; the image is not generally contained in $SU(6)$. The missing step is a degree-changing identification with a degree-two common-centre lifting obstruction and its line-operator pairing. The separate global clutching classes on S^6 have $c_3 = \pm 2ku$, $k \in \pi_5(SU(3)) \cong \mathbb{Z}$, and restrict trivially to U; see section 24.8.

40. **[VERIFIED] The exact typed transfer from auxiliary Rayleigh quotients to a physical excitation spectrum is proved.** Let $H \geq 0$ be self-adjoint on $\mathcal{H}_{\mathrm{phys}}$, and set

$$P_0 = \mathbf{1}_{\{0\}}(H), \qquad \mathcal{H}_{\mathrm{exc}} = (I - P_0)\mathcal{H}_{\mathrm{phys}}, \qquad \mathfrak{h}[\psi] = \|H^{1/2}\psi\|^2, \qquad \mathcal{Q}(H) = D(H^{1/2}).$$

For a nonnegative quadratic form $\mathfrak{q}_n$ and nonzero $u_n \in D(\mathfrak{q}_n) \subset \mathcal{K}_n$, suppose

$$J_n : \mathrm{span}\{u_n\} \longrightarrow \mathcal{Q}(H), \qquad \psi_n = (I - P_0)J_n u_n,$$

and, for $c > 0$, $C_n, \varepsilon_n \geq 0$,

$$\|\psi_n\| \geq c\|u_n\|, \qquad \mathfrak{h}[\psi_n] \leq C_n \mathfrak{q}_n[u_n] + \varepsilon_n \|u_n\|^2, \qquad C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n \longrightarrow 0.$$

Then the self-adjoint excitation restriction

$$H_{\mathrm{exc}} = H\big|_{D(H)\cap\mathcal{H}_{\mathrm{exc}}}$$

satisfies

$$\inf \sigma(H_{\mathrm{exc}}) = 0, \qquad 0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{exc}}).$$

The projection removes the full kernel of H; orthogonality to a single chosen vacuum is equivalent only after vacuum uniqueness is separately proved. This is a proved implication with fully stated antecedents, not a claim that the present geometry supplies them. See theorem 38.4.

41. **[OPEN] The physical Yang–Mills bridge data are absent.** The workbench has not constructed a common gauge-invariant positive-metric $\mathcal{H}_{\mathrm{phys}}$, a four-dimensional quantum Yang–Mills Hamiltonian H, or maps J_n obeying the norm and energy inequalities through gauge reduction, reflection-positive reconstruction, renormalization, and the continuum and infinite-volume limits. Therefore the verified radial scalar spectra, auxiliary direct integral, classical gauge Hessians, finite cochain operators, compact-link flux configurations, finite Gaussian oscillators, magnetic Laplacians, and reducible split-Cartan $SU(r)$ Jacobi operators do not prove or disprove a mass gap for a physical quantum Yang–Mills Hamiltonian. See theorems 10.12 and 38.6.

Exact-computation certificates. The retained exact-arithmetic harnesses passed all 28 full-manuscript checks, all 38 independently implemented core checks, and all 13 cusp-spectral checks, with no failures; the 13-check harness also passed unchanged under `python -O`. The first two harnesses cover finite and cusp monodromy, invariant and coinvariant lattices in exterior degrees 0 through 4, Smith forms and saturation, finite-quotient norm operators, the relation-group determinant, specialization indices, three twisting parameters, period-form identities, and the corrected existential form of Remark 7.28. The 13-check cusp-spectral harness is scoped to the radial period determinant and full inverse transpose, the restricted dual maps and pairings, the integral coframe wedge identity, the ambient Koide discriminant identity, the finite coordinate eigenvalues, and the radial volume and first-correction expansions. The geometric collapse, finite-band and cusp inclusion maps, Leray comparison, Kummer morphisms, cone carriers, compact-link construction, and Koide counting theorem are proved separately rather than inferred from those executions.

Independent review checkpoint. Reviewer B independently reviewed the reconstruction existing up to its evidentiary cutoff. It withdrew the earlier topology-based rejection, distinguished the intrinsic specialization sp^q from the then-open collapse pullback c^*, and found no load-bearing defect in the non-collapse nearby-cycle/Leray and based van Kampen route. Its provisionally favourable disposition and focused verification requests are useful independent evidence about the argument it saw. The later Codex proofs in this workbench are separate and should not be attributed to that review. In its 26 August analytic follow-up, Reviewer B independently proved the finite-complex/UCT top-degree base-change statement on every scheme-theoretic surface fibre and, from CAND-X, CAND-W, and TOP-H2, confirmed the conductor–duality implication $R^2 f_*(T_X \otimes L) \neq 0$ for every L [35]. Those three inputs remain hypotheses in that review record; its cutoff labels do not replace their later workbench classifications.

Current evidence boundary. No mathematical gap is presently recorded in the construction, either topology certificate, sphere recognition, algebraic dimension, or Hodge and Bott–Chern/Aeppli calculations. Exact marked-fibre period-coordinate representatives, global Bott–Chern generators, and their complete product structure are now included. The canonical data do not single out a literal global coefficient atlas for the chosen smooth representatives through the singular fillings; no such gauge is needed for the class or algebra theorem. The new cusp spectral statements apply to the explicitly declared quotient metrics and auxiliary or classical operators. They now include the complete fixed-energy lattice-point law $N_T(E) = \pi D_T \sum_n (E/(4\pi^2) - \|n\|^2)_+ + O_E(T+1)$ and the full-torus integral magnetic bound $\mathrm{gap}(\Delta_{\mathsf{E},T}) \leq 4\pi\sqrt{|\operatorname{Pf}(\mathsf{E})|/D_T}$. The modified-action toric Kummer map now completes the local degree-N^2 regular-fibre cover, while retaining its direct degree N^3, ramification, nonflat relative map, and strictly local scope. The exact maps $q_2^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $q_1^{\wedge 4}$ now provide topological S^{12}- and S^{24}-source carriers only by changing the targets through suspension, collapse, and coning. Two copies are rank-minimal cusp-locally and four are rank-minimal over the full monodromy base. Further quotients $\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6$ and $\widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$ contain moving, null-homotopic metric four-spheres with eventual exact Euclidean basepoint balls. Both full maps are themselves based null-homotopic. More generally, $\widehat{q}_\gamma^{\wedge r}$ and $\widehat{q}_3^{\wedge r}$ are based null-homotopic for every $r \geq 2$. The one-copy homotopy classes of $\widehat{q}_\gamma$ and $\widehat{q}_3$ remain uncomputed. The conical-target classes of $q_2^{\wedge 2}$, $q_{1,\mathrm{cusp}}^{\wedge 4}$, and $q_1^{\wedge 4}$ remain **[OPEN]**. The topological five-manifolds $\mathfrak{E}_{12}, \mathfrak{E}_{24}$ realize $\mathbb{R}^4$ as a decompactification boundary fibre. Unrescaled slices collapse in every fixed compatible ambient metric; for the fixed cusp-adapted metrics, the rescaled slice bubbles are $\mathbb{R}^4$ while the full ambient bubbles are $\mathbb{R}^8$ and $\mathbb{R}^{12}$. This leaves intact the nonextension to the fixed product target and supplies no compact complex structure on either smash sphere. The repaired compact-link calculation keeps cover degree and mesh refinement independent and proves the exact finite Gaussian gap and its zero-gap diagonal sequence, but not a nonperturbative compact-link spectrum. The physical Hilbert space, quantum Hamiltonian, and comparison morphisms required by the typed transfer theorem remain **[OPEN]**; accordingly, these calculations neither prove nor disprove a four-dimensional quantum Yang–Mills mass gap. Separately, the exact three-column quotient is now a rank-three cubic carrier with compactified rank-loss pair (S^{10}, S^8), while a literal almost-complex S^8 is refuted. The toroidal angular map gives that cubic carrier a split positive observation onto the ordinary future Lorentz cone, with exact $9 = 4 + 4 + 1$ four-vector–spin-representation–scalar coordinates and a proper $B^4 \hookrightarrow S^8$ section. Its determinant-phase equation, toric seed, and $SU(3)$-invariant momentum are explicit, and $m_{\alpha,c}^2(rZ) = r^4 m_{\alpha,c}^2(Z)$ proves that the unrestricted kinematic carrier has no positive threshold. On the candidate's own regular locus, the complete $SU(3)$-bundle classification is the free defect lattice $H^4(U;\mathbb{Z}) \cong \mathbb{Z}^2 \oplus \mathbb{Z}^2 \oplus \mathbb{Z}^2$, and every nonzero class fails to extend across the three special fibres. The one-shot collar matrix computes the primitive fibre-degree row $(12, 0, 4, 0, -3, 0)$; the candidate's local normal holonomies are exactly its reductions $\chi_{12} = \rho_{12} \circ \deg_F$ and $\chi_6 = \rho_6 \circ \deg_F$. The latter quotient is abstractly the same μ_6 as the centre of the tensor subgroup $SU(2) \times SU(3) \hookrightarrow SU(6)$, but no degree-changing map to a common-centre lifting obstruction, gauge cocycle, connection, or physical line pairing has been constructed. The carrier theorem advances the physical bridge through an exact colour-orbit momentum and a now fully reduced worldline action: its massive stratum is the standard spinless particle, its punctured massless characteristic quotient is standard

while its multiplier action and apex are irregular, and the hard-constraint packet propagates only on its threshold stratum. Its normal-slice continuation now supplies a constant affine Minkowski metric, the spin and Poincaré actions, an exact Klein–Gordon/Fock theory and finite regulator, an independently chosen classical compact-group Yang–Mills theory, and the metric-selected $K_P \cong SU(2)$ gauge–scalar model. These constructions show both possible gap values: the free scalar gap is the externally supplied m_{KG}, while the massless scalar and thermodynamic Gaussian gauge limits are gapless. They also isolate the candidate cusp matrix as spacelike and outside the positive Gram-momentum image. Dynamics on the retained ψ, λ, ω directions, a dynamically selected gauge-invariant interacting spectral sector, a continuum quantum reconstruction, and a full-domain comparison map to its physical excitation space remain unconstructed. The labels above mean that complete arguments are present here for audit. Specialist replication and external adjudication are separate validation boundaries, neither hypotheses nor gates on these proofs; they remain welcome. The corrected CDP sufficient theorem is now proved in theorems 46.28 and 46.30. Recovery of the original theorem under its original hypotheses would instead require overturning a verified premise of the constructed counterexample.

3 Methodology

The unit of audit is an exact mathematical assertion, not a paragraph's plausibility. Each assertion is recorded with its domain, codomain, coefficient ring, variance convention, and dependencies. The four outcomes used in this paper are literal:

VERIFIED A complete proof from the stated hypotheses is given or a precisely cited theorem is applied after every hypothesis is checked.

CONDITIONAL The implication is proved, but its conclusion depends on an explicitly named established or assumed hypothesis that is not proved in the present argument. This label is never a politeness substitute for OPEN or REFUTED.

OPEN No complete proof or counterexample is presently recorded. This is not a euphemism for false.

REFUTED A complete contradiction or counterexample is given to the exact assertion. A defect in a proof is not, by itself, a disproof of its theorem.

Repair is an operation, not a fifth status: when a printed assertion is false but a corrected statement is proved, the former is recorded as REFUTED and the latter separately as VERIFIED, with every downstream use recomputed.

Computations are converted into checkable certificates: displayed matrices, integer row or column relations, determinant ideals, primitive kernels, or explicit substitutions. A software pass is corroboration, not a proof. Likewise, an algebraically correct matrix calculation does not identify that matrix with a geometric specialization, attaching, or boundary map; that identification receives its own theorem and complete proof.

A theorem whose antecedents are explicitly part of its statement and whose implication is completely proved is VERIFIED as that exact implication. CONDITIONAL is reserved for a conclusion that imports an explicitly named established or assumed hypothesis not proved in the present argument. The current candidate-specific ledger has no such row.

Relevant human literature is searched proactively, not only when a remembered fact seems doubtful. Results are cited at the point of use with theorem, section, or page locators and with their hypotheses intact. Public working LaTeX editions such as the maintained SGA English source [34] may be used as searchable reading layers, but the published SGA volumes [30, 31] remain canonical. Every reference in

the compiled paper is resolvable without access to the editor's computer. The citation and tool-disclosure policy follows the Leiden Declaration [1]; neither a model-generated review nor an internal audit note substitutes for a human mathematical source.

4 Relation to existing literature and exact scope

The theorem-level conflict is with Campana–Demailly–Peternell [2, Theorem 2.2 and Corollary 2.3, p. 680]. Their hypotheses match the now workbench-verified identities $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, algebraic dimension one, and holomorphic algebraic reduction. Their conclusion gives $c_3(X) \leq 0$ and, for a threefold homeomorphic to S^6, algebraic dimension zero. The same constructed threefold instead has $c_3(X) = 2$ and $a(X) = 1$. It is therefore a counterexample to Theorem 2.2 and Corollary 2.3 as stated; this conclusion uses the complete candidate reconstruction, not merely the two printed proof-route defects below, and is not resolved by observing that the candidate proof does not invoke [2].

The first local issue isolated below occurs in the opening reduction of [2, §7, p. 692]. The explicit nodal-cubic family in theorem 46.6 refutes that reduction as a formal sheaf-theoretic inference. A second issue is the replacement of monodromy coinvariants by the full fibre homology in [3, Lemma 3.2, pp. 84–85]; CDP20 later imports the resulting rank argument. Either defect might in principle admit a different repair under stronger hypotheses. The original theorem cannot be repaired under its printed hypotheses because the constructed threefold is a verified counterexample. The strongest intact replacement is now proved in theorem 46.28: one global H^2-vanishing implies the stated generic vanishing and $c_3 \leq 0$. The stronger relative condition in theorem 46.29 is sufficient for such a global vanishing after a sufficiently positive twist from the base. For reduced full fibres, theorem 46.30 replaces the p. 692 shortcut by the exact connecting-map injectivity condition.

For the singular fibre, analytic duality is checked in two independent ways. The proof in theorem 46.7 derives the result from smooth Serre duality [8, Theorem 9.37]; Ramis–Ruget's dualizing-complex theorem provides the singular analytic comparison [4, Theorems 1–2, pp. 81–84, and Proposition 2, p. 91]. Relative duality for a proper analytic morphism is also cited in its precise form as Ramis–Ruget–Verdier, Invent. Math. 13 (1971), 261–283 [5]. Analytic cohomology and base change is applied in the form of Bănică–Stănăşilă, Chapter III, §3, Theorem 3.4 and Corollary 3.10, together with Grauert's local-freeness/continuity theorem in Chapter III, §4, Theorem 4.12 [6]. These locators cover all degrees used by the structure-sheaf direct-image proof, not only its top-degree specialization. The algebraic top-degree analogue in the Stacks Project, Tag 0EX2 [36], is a useful cross-check but is not substituted for the analytic theorem.

The algebraic nearby-cycle formalism in SGA 7 II, Exposé XIII [31], and the semistable/Raynaud framework in SGA 7 I, Exposé IX [30], do not by themselves cover every step for the present proper holomorphic map, which is not assumed algebraic. The needed analytic formalism is nevertheless available. Dimca's Definition 4.2.1 constructs nearby cycles for complex analytic varieties; Proposition 4.2.2 identifies their stalks with Milnor-fibre cohomology; Proposition 4.2.11 commutes nearby cycles with proper analytic direct image; and Theorem 2.3.26 gives proper base change, all for coefficient rings including $\mathbb{Z}$ [32]. Kashiwara–Schapira [33, §8.6] is an independent formal comparison. Thus the general integral analytic framework is verified. None of these references by itself identifies the candidate's specialization image with the full monodromy-invariant lattice. That candidate-specific conclusion, including saturation, is proved in theorem 39.1 for the reconstructed proper cusp model, using the explicit local equations and a supported Gysin calculation. The direct comparison with the cellular collapse is now proved in theorem 44.7: the neighborhood realization of the nearby-cycle unit is

$$\mathrm{sp}^q = i_F^*(i_W^*)^{-1},$$

and the candidate retraction gives $(i_W^*)^{-1} = r^*$, hence $\mathrm{sp}^q = c^*$ over $\mathbb{Z}$ in every degree.

The topological input to the retraction is Brown's collaring theorem for the boundary of a topological manifold [28], followed by Whitehead and the cofibration argument in Hatcher [9]. Clemens's classical

collapse for semistable Kähler degenerations [29] is useful context, but it is not used as a substitute for the candidate-specific proof: the central fibre here is irreducible and nonnormal, so theorems 44.2 to 44.5 constructs and checks the map directly.

For compact-support Poincaré duality with local coefficients and its projection formula we use Bredon [13, Chapter VI], with Hatcher [9, §3.H] as a concise topological overview. This is the precise input behind the multiplicative identification $H^2(B, j_*\mathcal{L}) \simeq M_G$ used in the Leray calculation.

The completed global comparison uses the chain-level Mayer–Vietoris convention of Hatcher [9, §2.2, pp. 149–150]: the inclusion of the intersection complex is $z \mapsto (i_{A*}z, -i_{B*}z)$. This fixes the only global minus sign. The candidate-specific content is not supplied by that citation: the radial markings in the reconstructed finite fillings prove that the two band inclusions retract to π_1 and π_2, while the explicit toric disc and solid tori give the primitive cusp kernels. Those geometric facts produce the matrices and the commuting Leray squares in theorems 44.10, 44.11 and 44.13.

For the Hodge-number question, Ugarte's general restrictions on a hypothetical complex structure on S^6 [50] and Angella's later synthesis [47] provide the correct general comparison class. The candidate-specific calculation is sharper: its reconstructed structure-sheaf direct images and explicit vanishing of holomorphic forms first determine the Hodge diamond up to $m = h^{1,2}$ and force $E_2 = E_\infty$. Neither reference determines m for this family. The workbench now does: the cyclic invariant lattices, logarithmic Kodaira–Spencer divisor, normalized residue sequence, and order-four scalar give $m = 1$, hence $h^{1,1} = 2$. This is a candidate-specific calculation, not a consequence imported from the general literature.

The spectral interpretation has also been audited against the primary physical framework. Jaffe–Witten formulate the four-dimensional Yang–Mills mass gap on $\mathbb{R}^4$ only after construction of the quantum theory: on its physical Hilbert space, the Hamiltonian $H \geq 0$ has the vacuum at energy zero and a mass gap precisely when there is a $\Delta > 0$ such that

$$\sigma(H) \cap (0, \Delta) = \varnothing \qquad (\text{equivalently, } \sigma(H) \subseteq \{0\} \cup [\Delta, \infty)).$$

This is the physical mass-gap condition in [63]. Euclidean correlation data yield such an $\mathcal{H}_{\text{phys}}$, vacuum, and positive time-translation generator only through Osterwalder–Schrader reconstruction; the required input includes Euclidean covariance and symmetry, distributional regularity/growth, reflection positivity, and clustering, with the precise reconstruction qualifications supplied by the two primary papers [67, 68]. Hodge decompositions, periods, Leray terms, and coherent direct images are not Schwinger functions and supply neither reflection positivity nor a physical Hamiltonian. They also do not determine even a geometric Laplace spectrum: the explicit fixed-complex-torus calculation in theorem 38.3 keeps the complex structure, integral cohomology, Hodge decomposition, coherent direct-image data, and total volume fixed while the Kähler metrics

$$g_R = R^2(dx_1^2 + dy_1^2) + R^{-2}(dx_2^2 + dy_2^2)$$

have $\lambda_1(\Delta_{g_R}) = 4\pi^2/R^2 \to 0$. Thus spectra depend on the chosen metric and operator, and no spectral or physical conclusion follows from the cohomological/direct-image data alone. The auxiliary spectral statements in the workbench are direct operator calculations; none is claimed to be established by these references or to constitute an Osterwalder–Schrader reconstruction of quantum Yang–Mills theory.

For the compact-link appendix, the historical scope has now been made explicit. Wilson supplies the plaquette/character lattice action [64]; Lüscher supplies a selfadjoint strictly positive transfer-matrix construction in the standard Wilson lattice theory [65]; and Menotti–Pelissetto establish Osterwalder–Schrader positivity for the Wilson action in the corresponding reflection setup [66]. These citations identify the standard constructions and the physical reconstruction boundary; they do not prove the appendix's metric-dependent sector normalization, exact flux action, finite Hessian spectrum, or Gaussian gap, all of which remain the displayed candidate-specific calculations. Nor do they turn that finite Gaussian model into an Osterwalder–Schrader reconstruction, whose axiomatic requirements remain those of [67, 68].

The general mechanisms used inside that candidate-specific calculation were also checked against the acquired primary archival PDFs (including Deligne's published erratum); no author- or publisher-supplied

primary TeX was found for these historical works. Cartan's analytic quotient theorem identifies the local quotient rings with invariants [38, §1, Theorem 1, pp. 1–2]; Reynolds averaging then gives exactness of finite-group invariants over $\mathbb{C}$. Deligne's logarithmic conormal and normalized-stratum residue formalism covers the $d\log$ bases, relative sequence, and self-crossing residue target [40, Chapter II, §3.2–3.7, pp. 72–78]. Katz–Oda supplies the connecting-map and exterior-product formalism for the Gauss–Manin and Kodaira–Spencer maps [46, §2, equations (5)–(10), pp. 202–204]; Grothendieck and the cited Stacks tags supply the splitting and elementary-modification background on $\mathbb{P}^1$ [45, Theorem 2.1, p. 126]. Finally, Frölicher's original lemmas identify the first and limiting pages with Dolbeault and associated-graded de Rham cohomology [41, p. 643]. None of these sources supplies the printed period derivatives or cusp branch maps; those remain the explicit candidate-specific calculations in the workbench. The subsequent line-by-line audit therefore kept their normalization data explicit: the finite determinant formula is defined up to a nowhere-zero frame unit, and the constant cusp Kodaira–Spencer matrices are special-fibre reductions modulo t. These distinctions leave the divisor, residue, and rank conclusions unchanged.

Once those candidate-specific Hodge and Frölicher data are known, the Bott–Chern and Aeppli dimensions are no longer open. McHugh's general compact-threefold formulas [48, Theorem 6.3, Proposition 6.4, equation (7), and §7.1] express them in terms of Dolbeault dimensions, $E_2^{3,1}$, the de Rham subgroup represented by real $(1,1)$-forms, and the auxiliary spaces $K^{p,q}$. For this candidate $E_2^{3,1}=0$, $H^2_{\mathrm{dR}}(X)=0$, and $\dim K^{p,q}\le b_{p+q}=0$ in every middle degree. The last inequality is not an injection $K^{p,q}\hookrightarrow H^{p+q}_{\mathrm{dR}}$: the image in de Rham cohomology of the pure, ∂- and $\bar\partial$-closed forms surjects onto $K^{p,q}$ because every pure d-exact form lies in McHugh's denominator. Direct substitution therefore yields the Bott–Chern diamond, and Schweitzer's Hodge-star isomorphism [49, §2.c, pp. 9–10] yields the Aeppli diamond, in theorem 42.11. Angella's published synthesis [47, §4] independently confirms the endpoint dimensions, $h^{1,1}_{BC}=2h^{0,1}$, the relation $h^{1,1}_{BC}+h^{2,2}_{BC}=2h^{2,1}_{BC}+2$, and Schweitzer duality. The literature supplies the general identities; the numerical specialization depends on the workbench's candidate-specific proof that $h^{1,2}=1$ and $E_2=E_\infty$.

The multiplicative refinement is also now complete. Schweitzer's definitions give the Bott–Chern algebra, its natural map to Dolbeault cohomology, and the Chern-class convention [49, §2 and §8.a], but they do not determine this candidate's generators or products. The workbench constructs global $\bar\partial$-closed representatives G, Q, proves their exact marked-fibre period-coordinate classes, and uses the two exact Frölicher rows, the factor-12 exponential–Leray comparison, Dolbeault–Serre duality, and top integration to obtain all nine positive-degree Bott–Chern generators and

$$H^{*,*}_{BC}(X)=\mathbb{C}\mathbf{1}\oplus N,\qquad N^2=0.$$

See theorems 43.4 and 43.5. This is a candidate-specific calculation, not a consequence of the general dimension identities. It determines the cohomology classes and every product, while making no claim of a preferred literal global coefficient atlas for the noncanonical smooth representatives through the singular fillings.

The algebraic reduction of a compact complex torus is used in the exact form given by Birkenhake–Lange [18, Chapter 2, §6], with the independent statement in Pareschi–Popa–Schnell [21, §8, p. 2437]. The toric chart facts are compared with Fulton [22]. The exact classical-topology formula $\pi_1(X_\Sigma)\cong N/N_\Sigma$ is cited from Cox–Little–Schenck [23, Theorem 12.1.10], with Danilov's original toric treatment [24, §9] as an independent comparison. All assertions specific to the locally finite infinite fan and the candidate's proper-discontinuity argument are proved directly.

Finally, the sphere-recognition endpoint uses Hurewicz and homological Whitehead exactly as stated by Hatcher [9, Theorem 4.32 and Corollary 4.33], Smale's generalized Poincaré theorem [14, Theorem A], and the computation of the smooth endpoint by Smale [15, Theorem 1.1 and Corollary 1.2]. Kervaire–Milnor provide an independent equivalent check: their opening table gives $|\Theta_6|=1$, while the following Remark identifies Θ_6 with diffeomorphism classes in dimension six [16, pp. 504–505]. Those results close the recognition step after the candidate's fundamental group and integral homology are established in theorems 39.5 and 39.6; they are not used to establish those inputs.

For the normalized cusp-restriction calculation, the public Stacks Project references were checked at the formal spectral-sequence level rather than relying on an unstated edge-map convention. Leray is Tag 01F2; the passage $E_r \mapsto E_{r+1}$ and the stable term are Tags 011N and 011O; functoriality, identification with the associated graded, and convergence are Tags 012O, 012U, and 012W [36]. Together with analytic proper base change [32, Theorem 2.3.26], these results make the evaluation square for $H^2(X;\mathbb{Z}) \to H^0(B, R^2 f_*\mathbb{Z})$ commute with restriction to every fibre. Naturality of the exponential-sequence boundary then sends $c_1(L)$ to $c_1(L|_{X_b})$. The literature supplies this functorial formalism; the injective differential $d_2^{0,2}(2q) = \omega[\delta]$ remains the candidate-specific calculation proved in theorem 39.4.

The independent direct boundary-Picard certificate uses Brown's standard crossed-homomorphism model for cohomology of a group with coefficients in a module [27, Chapter I, §5]. That reference supplies only the identification of H^1 with crossed homomorphisms modulo principal ones. The candidate-specific content is proved directly: the finite-cover norm, the direction of each logarithmic conjugacy, the contraction formula for a translated invariant two-form, the order-four parity lattice, the explicit cusp solid torus, and Chern-class naturality under the local nearby-cycle specialization. These give

$$E = 2nq, \qquad (\bar{d}_1, \bar{d}_2, \bar{d}_0) = (4n, -3n, 0)[\delta], \qquad n = 0.$$

The exact Appell–Humbert source is Birkenhake–Lange, Chapter 1, §2, pp. 5–8, especially Theorem 2.1 on p. 7 [18]; Chapter 2, §6 is instead the separate source for algebraic dimension used above. Schnell's public notes record the precise factor/coboundary and antisymmetrized connecting-cocycle formulas [19, equations (3.3)–(3.10), pp. 3–7], together with the translation contraction $w \mapsto \iota_w E$ [20, equation (6.2), p. 1]. These sources certify the standard fibrewise formalism, not the candidate's chosen lifts, finite norms, signs, or cusp winding; those are proved directly.

The completed third certificate uses only holomorphic $\mathcal{O}^*$ factors, both finite fillings, the cusp argument principle, and the toric support function. Its exact boundary equations are

$$E = kq, \qquad (\bar{d}_1, \bar{d}_2, \bar{d}_0) = (2k, -\tfrac{3}{2}k, 0)[\delta], \qquad k = 0.$$

For normalized central degrees (d_1, d_2, d_3), it then proves

$$B = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}, \qquad E_L((\lambda, 0), (0, m)) = -B(B_0\lambda, m).$$

Fulton's chart, character-divisor, and support-function results [22, §2.1, pp. 28–29; §3.3, pp. 60–63; §3.4, pp. 63–72] and the parallel theorem-numbered statements in Cox–Little–Schenck [23, Example 1.2.21, p. 31; Propositions 4.1.1–4.1.2, p. 171; Theorems 4.2.8, p. 179 and 4.2.12, p. 182] fix the convention: for $D = \sum a_\rho D_\rho$ the standard support function has value $-a_\rho$, while $\operatorname{div}(\chi^m) = \sum \langle m, u_\rho \rangle D_\rho$ has the positive sign used in the proof. Cartan theorem B [39, §2, fundamental theorem, p. 81] and the exponential sequence [8, Proposition 6.28 and equations (6.35)–(6.36), pp. 180–182] trivialize only the residual line bundles whose c_1 has already been proved zero. All infinite-fan local finiteness, descent factors, frame comparisons, and the final cross-block identity remain candidate-specific proofs in section 46.7.

Bingener's analytic representability theorem was checked against the public primary text, not inferred from the algebraic Picard literature. For a proper flat holomorphic map cohomologically flat in dimension zero, the relative Picard functor is represented by a complex space which need not be separated [37, Introduction, p. 318; Satz 6.2, pp. 336–337]. The same introduction states that the direct exponential-sequence quotient works under topological local triviality and cohomological flatness in dimension one, and explains why that quotient argument fails in general. These statements provide representability for the cusp family but do not provide the component-specialization or separatedness assertion required to annihilate the normalized central restriction. The candidate-specific calculation in theorem 41.8 makes this limitation concrete: it constructs a descended local line bundle with generic Chern class $-2q$ and nonzero normalized central degree vector $(-12, 2, 0, -12, 2, 0)$. Thus all three proved global restriction theorems genuinely use data absent from a cusp-local representability argument; no general relative-Picard theorem has silently replaced the candidate-specific finite-filling and common-boundary calculation.

4.1 Targeted corpus cross-check (29 August 2026)

Before extending the validate/repair/disprove chain, the relevant primary texts were reread in the local LaTeX corpus and compared with their public editions. This check adds no unproved hypothesis; it records exactly which formal tools the sources do, and do not, supply.

First, the primary transcription of Campana–Demailly–Peternell (2020), Theorem 2.2 and Corollary 2.3, confirms the printed implication

$$H^1(X,\mathbb{Z}) = H^2(X,\mathbb{Z}) = 0, \quad a(X) = 1 \Longrightarrow c_3(X) \leq 0, \qquad X \simeq S^6 \Longrightarrow a(X) = 0,$$

under the stated compact-threefold and algebraic-reduction hypotheses. Its Proposition 2.4 identifies the obstruction with the vanishing of $R^2 f_*(T_X \otimes L)$, equivalently the fibrewise dual spaces involving $H^0(\Omega^1_X \otimes L^*)$. The workbench's conductor section, absolute-fibre duality, and analytic base-change propositions prove a nonzero singular-fibre contribution for the reconstructed candidate; this is the exact point at which the earlier conditional objection is replaced by a candidate-specific calculation.

The same primary source confirms that Proposition 2.4 is logically separate from the defective attempts to establish its premise. The workbench now proves that premise-to-conclusion implication without importing either defect. If one L has $R^2 f_*(T_X \otimes L) = 0$, projection formula and Serre vanishing on the base produce L_0 with $H^2(X, T_X \otimes L_0) = 0$; generic endpoint vanishing and semicontinuity then give

$$H^i(X, T_X \otimes M) = 0 \quad (i \neq 1), \qquad \chi(X, T_X \otimes M) = \frac{1}{2} c_3(X) \leq 0.$$

This is a corrected sufficient theorem, not a reinstatement of Theorem 2.2: the candidate violates this relative sufficient condition for every L. Separately, $c_3(X) = 2$ and the corrected global theorem imply by contraposition that $H^2(X, T_X \otimes L_0) \neq 0$ for every L_0.

The reread of CDP (1998), Lemma 3.2, confirms a separate printed rank defect: for a split extension with fibre module $H_1(F)$ and monodromy group Q, the abelianization contributes the coinvariants $(H_1(F))_Q$, not the whole $H_1(F)$. The elementary example $\mathbb{Z} \rtimes_{-1} \mathbb{Z}$ has abelianization $\mathbb{Z} \oplus \mathbb{Z}/2$, which is the exact correction used in theorem 5.1.

The SGA 7 II nearby-cycle transcription supplies the specialization–variation sequence and the identity $\mathrm{Var}(\sigma)\alpha = \sigma - 1$; the Stacks/EGA transcriptions supply Leray functoriality and the precise proper/flat base-change hypotheses. Both are algebraic formal references, so neither is silently applied as an analytic existence theorem. The candidate's integral specialization map, saturation, and stalk values remain the explicit local calculations in theorem 39.1.

Finally, the independent Angella and McHugh transcriptions reproduce the known six-sphere Hodge inequalities and the Bott–Chern dimension identities. Substitution of the workbench values

$$(h^{0,1}, h^{0,2}, h^{1,0}, h^{2,0}, h^{1,1}, h^{1,2}) = (1,0,0,0,2,1),$$
$$(h^{1,1}_{BC}, h^{1,2}_{BC}, h^{2,2}_{BC}) = (2,1,2)$$

agrees with the explicit nine-generator square-zero algebra. This is an external consistency check, not a substitute for the candidate-specific quotient, gluing, and period proofs.

4.2 Stable-stem audit of the sphere-valued carriers (3 September 2026)

Let

$$g = \widehat{q}_3 : (S^6, *) \longrightarrow (S^3, *), \qquad f = \widehat{q}_\gamma : (S^6, *) \longrightarrow (S^2, *)$$

be the based sphere-valued quotients obtained after the chosen based identification of X with S^6.

Hatcher's full low-dimensional homotopy-group table [9, p. 339] gives

$$\pi_6(S^3) = \mathbb{Z}/12, \qquad \pi_{12}(S^6) = \mathbb{Z}/2.$$

In particular $12g = 0$. Toda's Proposition 5.11 [12, Proposition 5.11] identifies the latter group as

$$\pi_{12}(S^6) = \mathbb{Z}/2\{\nu_6 \circ \nu_9\}.$$

The same proposition gives $\pi_{n+6}(S^n) = \mathbb{Z}/2\{\nu_n \circ \nu_{n+3}\}$ for $n \geq 5$. Suspension carries each displayed generator to the next displayed generator. Hence the stabilization homomorphism

$$\mathrm{st} : \pi_{12}(S^6) \longrightarrow \pi_6^{\mathrm{S}}$$

is an isomorphism. Isaksen–Wang–Xu's stable-stem table [11, Table 1, p. 113] identifies

$$\pi_3^{\mathrm{S}} = \mathbb{Z}/24\{\nu\}, \qquad \pi_6^{\mathrm{S}} = \mathbb{Z}/2\{\nu^2\}.$$

Since $12\,\mathrm{st}(g) = 0$ in $\mathbb{Z}/24\{\nu\}$, there is an $a \in \mathbb{Z}/12$ such that

$$\mathrm{st}(g) = 2a\nu.$$

Stabilization is multiplicative for smash products, so

$$\mathrm{st}(g^{\wedge 2}) = \mathrm{st}(g)^2 = 4a^2\nu^2 = 0 \in \pi_6^{\mathrm{S}}.$$

Injectivity of the preceding stabilization isomorphism now gives $[g^{\wedge 2}] = 0 \in \pi_{12}(S^6)$. Thus every based map $S^6 \to S^3$, and in particular $\widehat{q}_3$, has based null-homotopic smash square. Smashing this based null-homotopy with further copies of g also gives

$$[g^{\wedge r}] = 0 \in \pi_{6r}(S^{3r}) \qquad (r \geq 2).$$

For the second quotient, Hatcher's stable-range corollary [9, Corollary 4.24, p. 360] says that $\pi_{n+k}(S^n)$ has reached the stable group π_k^{S} when $n > k+1$. Applying this with $n = 8$ and $k = 4$, and then using the empty $k = 4$ row of Isaksen–Wang–Xu's complete stable-stem table [11, Table 1, p. 113], gives the exact equality

$$\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0.$$

Consequently the sixfold reduced suspension

$$\Sigma^6 f = f \wedge \mathrm{id}_{S^6} : S^6 \wedge S^6 \longrightarrow S^2 \wedge S^6$$

is based null-homotopic. Under the canonical associativity and symmetry identifications of based sphere smashes, the smash square factors as

$$f^{\wedge 2} = (\mathrm{id}_{S^2} \wedge f) \circ (f \wedge \mathrm{id}_{S^6}) : S^6 \wedge S^6 \longrightarrow S^2 \wedge S^2.$$

Its right-hand factor is $\Sigma^6 f$, so this factorization supplies a based null-homotopy of $f^{\wedge 2}$. For every $r > 2$, associativity gives

$$f^{\wedge r} = f^{\wedge 2} \wedge f^{\wedge (r-2)},$$

and smashing the preceding based null-homotopy with the based map $f^{\wedge(r-2)}$ proves

$$[f^{\wedge r}] = 0 \in \pi_{6r}(S^{2r}) \qquad (r \geq 2).$$

Hence both already constructed continuous surjective quotients

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$$

are based null-homotopic. Neither calculation determines the one-copy class $[g] \in \pi_6(S^3)$ or $[f] \in \pi_6(S^2)$. Both arguments use the sphere-valued postcompositions essentially, so they give no null-homotopy of any map into a conically stratified target; those based homotopy classes remain open. Point-set surjectivity, quotient topology, the embedded moving four-spheres, and their intrinsic metric limits are unaffected by nullity of the global sphere-valued maps.

4.3 Effective counting for the exact Koide sector

The effective assertion used in the Koide chapter is matched to its literal domain rather than transferred from a smoother or radial problem. After the five defining walls and the height $H(a,q,n) = 4a + 17n$ are checked directly against Definition 13.1, Benoist–Oh, Theorem 13.2 [53], applies to each of the finitely many integral orbits and proves

$$R(X) = C_0 X + O(X^{1-\eta}) \qquad (\eta > 0 \text{ fixed but unspecified}).$$

Neither the statement nor the proof of that theorem supplies $\eta \geq \frac{1}{2}$.

The hyperbolic literature gives two stronger exponents for different domains. Liu–Sarnak, Proposition 3.2 [56], gives the smooth standard-radial exponent $1/2 + 7/64 = 39/64$. The exact order group is a congruence arithmetic Fuchsian group because, for a maximal overorder $\mathcal{M}$,

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1;$$

Jacquet–Langlands and Kim–Sarnak give $\lambda_1 \geq 975/4096$ for that input [57, 58]. Good's sharp radial theorem and its Chatzakos–Petridis formulation [54, 55] instead have a $2/3$ error after the small-spectrum terms are displayed. The exact radial Huber variable is $7a + 15\sqrt{2}\,n$, whereas the Koide count uses $4a + 17n$ together with five sharp walls. Consequently neither radial theorem transfers its exponent to the target count. A pointwise square-root remainder for the unchanged domain remains open; the complete hypothesis-by-hypothesis audit is section 37.1.

4.4 Constrained dynamics and symplectic reduction

The worldline calculation in section 22 uses three primary sources for its general reduction framework. Dirac's original constrained-Hamiltonian paper [70] supplies the distinction between regular and singular constraint dynamics and motivates retaining the multiplier and all primary constraint equations. Gotay–Nester–Hinds [71] supplies the presymplectic constraint algorithm: one first restricts the closed two-form to the constraint locus, determines its characteristic distribution, and checks whether successive consistency conditions or rank jumps occur. Marsden–Weinstein [72] supplies the regular symplectic quotient by a Hamiltonian symmetry action.

Their role here is structural, not candidate-specific evidence. None of these sources computes the Gram morphism $P_\alpha(Z) = \alpha Z^* Z$, its $SU(3)$-orbits and stabilizers, the rank-one zero of $\mathrm{d}(\det \circ P_\alpha)$, the apex branch, or the hard-constrained packet's threshold rank jump. Those data and the exact cotangent sign are established directly in the cited workbench module. In particular, regular Marsden–Weinstein reduction applies to the massive rank-two stratum and to the intrinsic punctured rank-one stratum after their characteristic quotients have been proved regular; it does not authorize an extension of that quotient through the irregular zero-momentum apex.

5 Exact integral algebra of the printed monodromy data

This section replaces opaque computational status reports by certificates that can be checked by hand. Let

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle.$$

Vectors are columns in the displayed ordered bases. The printed monodromy on V is

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{3.1}$$

Put $T_0 = (T_1 T_2)^{-1}$ and let $A_j = (T_j^{-1})^t$, $M_0 = (T_0^{-1})^t$ act on Λ.

Proposition 5.1 (monodromy certificate). ***[VERIFIED]*** *Direct integral multiplication gives*

$$T_1^3 = I, \quad T_2^4 = I, \quad T_0 = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (T_0 - I)^2 = 0,$$

and

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \quad M_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix},$$

with $A_1 A_2 M_0 = I$.

Proof. Multiplying the first two matrices in (3.1) gives

$$T_1 T_2 = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its inverse is the displayed T_0. Thus $T_0 - I$ sends $(a, b, c, d)^t$ to $(d, -c, 0, 0)^t$, and applying it twice gives zero. The complete order certificate is

$$T_1^2 = \begin{pmatrix} 1 & 6 & -6 & -2 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_1^2 T_1 = I,$$

$$T_2^2 = \begin{pmatrix} 1 & 6 & -6 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2^3 = \begin{pmatrix} 1 & 0 & -6 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2^3 T_2 = I.$$

In particular $T_1 \neq I$ and $T_2^2 \neq I$. Transposing the integral inverses gives the three displayed contragredient matrices; finally $A_1 A_2 = (M_0)^{-1}$ by the definitions, hence $A_1 A_2 M_0 = I$. □

5.1 The cusp invariant, image, and coinvariant lattices

Theorem 5.2 (all exterior-power cusp lattices). ***[VERIFIED]*** *The equality*

$$\ker(M_0 - I) = \operatorname{im}(M_0 - I) = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle \tag{3.2}$$

holds as an equality of saturated sublattices of Λ. *On* V,

$$V^{T_0} = \mathbb{Z}\langle \gamma, u \rangle,$$

$$(\textstyle\bigwedge^2 V)^{T_0} = \mathbb{Z}\langle \gamma u, \gamma\delta, uw, \gamma w - u\delta \rangle, \qquad (\textstyle\bigwedge^3 V)^{T_0} = \mathbb{Z}\langle \gamma uw, \gamma u\delta \rangle. \tag{3.3}$$

Every displayed invariant lattice is saturated. Consequently the invariant ranks in degrees $0, 1, 2, 3, 4$ *are* $(1, 2, 4, 2, 1)$.

Proof. From the displayed matrix,

$$(M_0 - I)(a, b, c, d)^t = (0, 0, b, -a)^t.$$

Its kernel is given by $a = b = 0$, and its image consists of every vector $(0, 0, b, -a)^t$; this proves (3.2). The two generators are members of the chosen basis, so the lattice is saturated.

The action of T_0 on basis vectors is

$$T_0\gamma = \gamma, \quad T_0 u = u, \quad T_0 w = w - u, \quad T_0\delta = \delta + \gamma. \tag{3.4}$$

Thus a vector $a\gamma + bu + cw + d\delta$ is fixed exactly when $c = d = 0$. For degree two, write

$$x = a\gamma u + b\gamma w + c\gamma\delta + d\, uw + e\, u\delta + f\, w\delta.$$

Expanding (3.4) and collecting coefficients gives

$$(\textstyle\bigwedge^2 T_0 - I)x = (-b - e + f)\gamma u - f\gamma w - fu\delta.$$

Hence $f = 0$ and $e = -b$, with a, b, c, d arbitrary, which is exactly the middle lattice in (3.3). These are integral equations with a unit coefficient, so their solution lattice is saturated.

In the ordered basis $(\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta)$,

$$(\textstyle\bigwedge^3 T_0 - I)(a, b, c, d)^t = (d, -c, 0, 0)^t.$$

The fixed lattice is therefore generated by the first two basis vectors and is saturated. Degrees zero and four are fixed because $\det T_0 = 1$. □

5.2 Global coinvariants

Let $G = \langle T_1, T_2\rangle$ and, for a G-lattice L, write $L_G = L/((T_1 - I)L + (T_2 - I)L)$.

Theorem 5.3 (coinvariant certificates)**.** ***[VERIFIED]*** *The common invariant lattices are*

$$V^G = \mathbb{Z}\gamma, \qquad (\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \qquad (\bigwedge^3 V)^G = \mathbb{Z}\gamma uw, \qquad (\bigwedge^4 V)^G = \mathbb{Z}\,\mathrm{vol},$$

where $q = uw + 6\gamma\delta$ *and* $\mathrm{vol} = \gamma uw\delta$*. There are torsion-free identifications*

$$V_G = \mathbb{Z}[\delta], \qquad (\textstyle\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad (\bigwedge^3 V)_G = \mathbb{Z}[uw\delta], \tag{3.5}$$

and $[uw] = 6[\gamma\delta]$ *in degree two.*

Proof. Solving $(T_1 - I)v = 0$ and $(T_2 - I)v = 0$ in (3.1) gives

$$V^{T_1} = \langle \gamma, 2u + w + 3\delta\rangle, \qquad V^{T_2} = \langle \gamma, u + w + 2\delta\rangle.$$

If a vector belongs to both lattices, comparison of its u- and w-coordinates gives simultaneously $2r = t$ and $r = t$ for the two non-γ coefficients; hence $r = t = 0$. Thus $V^G = \mathbb{Z}\gamma$.

The higher common invariants follow from the same integral nullspace calculation. In the ordered bases

$$(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta), \qquad (\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta),$$

the simultaneous equations $(\bigwedge^k T_1 - I)x = (\bigwedge^k T_2 - I)x = 0$ are respectively equivalent to

$$x_1 = x_2 = x_5 = x_6 = 0, \qquad x_3 = 6x_4,$$

and

$$x_2 = x_3 = x_4 = 0.$$

Thus the integral solution lattices are generated by $uw + 6\gamma\delta = q$ and γuw. Both generators are primitive. Finally $\det T_1 = \det T_2 = 1$, so the common degree-four invariant lattice is $\mathbb{Z}\,\mathrm{vol}$.

In V_G, columns of $T_1 - I$ and $T_2 - I$ give, among others,

$$u + w = 0, \qquad u - 3\gamma = 0, \qquad 2\gamma + u + w = 0, \qquad 2u + w = 0.$$

The first two relations give $w = -u$ and $u = 3\gamma$; the last two then give $2\gamma = 0$ and $3\gamma = 0$. Hence $\gamma = u = w = 0$. Every column has zero δ-coordinate, so no relation involves δ. This proves $V_G = \mathbb{Z}[\delta]$, including absence of torsion.

For degree two use the ordered basis

$$(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta).$$

In this basis the twelve columns of the relation map are

$$B_2 = \left[\begin{array}{cccccc|cccccc} -2 & 1 & 1 & -6 & 2 & -8 & -1 & -1 & 1 & -6 & 6 & -3 \\ -1 & -1 & 1 & -6 & 2 & -6 & 1 & -1 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & -6 & 0 & 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -2 & 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 \end{array}\right], \tag{3.6a}$$

where the two blocks are $\bigwedge^2 T_1 - I$ and $\bigwedge^2 T_2 - I$. Its column lattice is

$$\mathrm{im}[\,\textstyle\bigwedge^2 T_1 - I \mid \bigwedge^2 T_2 - I\,] = \langle \gamma u, \gamma w, u\delta, w\delta, uw - 6\gamma\delta \rangle. \tag{3.6}$$

For completeness, the third columns of the two blocks yield γu and $\gamma u + \gamma w$; their difference yields γw. After these are removed, the fifth and sixth columns give $2u\delta + w\delta$ and $u\delta + w\delta$, hence both $u\delta$ and $w\delta$; the remaining nonzero relation is $uw - 6\gamma\delta$. Conversely, substituting the twelve displayed columns of the two exterior powers shows that they all lie in the right side. After ordering the ambient basis as $(\gamma u, \gamma w, u\delta, w\delta, uw, \gamma\delta)$, the five generators have matrix $I_4 \oplus (1, -6)^t$. Its maximal minors have gcd 1, so (3.6) is saturated. The quotient is therefore freely generated by $[\gamma\delta]$, with $[uw] = 6[\gamma\delta]$.

For degree three, let $(a, b, c, d) = (\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta)$. The full relation matrix is

$$B_3 = \left[\begin{array}{cccc|cccc} 0 & 0 & 1 & 2 & 0 & -1 & 0 & -3 \\ 0 & -2 & 1 & -6 & 0 & -1 & -1 & -6 \\ 0 & -1 & -1 & -6 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]. \tag{3.6b}$$

If $c_{i,r}$ denotes column r of the T_i block, then $c_{1,2} - c_{2,3} = -b$, $c_{2,3} = -b - c$, and $c_{1,3} = a + b - c$. Hence a, b, c belong to the column lattice. Conversely every displayed column lies in $\langle a, b, c \rangle$, and the fourth row is zero. The relation lattice is therefore exactly the primitive summand $\langle a, b, c \rangle$, so the quotient is $\mathbb{Z}[uw\delta]$ with no hidden torsion. □

5.3 Finite quotient fibres

Put

$$\eta_1 = 2u + w + 3\delta, \qquad \eta_2 = u + w + 2\delta, \qquad \mathbf{q} = uw + 6\gamma\delta,$$

$m_1 = 3$, $m_2 = 4$, and $v_1 = (1, 2, -4, 0)^t$, $v_2 = -(1, 3, -3, 0)^t$ in Λ.

Proposition 5.4 (finite-fibre pullback indices)**.** ***[VERIFIED]*** *For the free affine actions and finite fibres geometrically realized in theorem 8.7, one has*

$$S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle$$

with $H^*(A_0^{(j)};\mathbb{Z}) = \bigwedge^* V$*, the degree-*$m_j$ *covering* $\pi_j : A_0^{(j)} \to S_j$ *has injective pullback with indices, in degrees* 0 *through* 4*,*

$$(1,3,1,1,3), \qquad (1,4,2,2,4). \tag{3.7}$$

Moreover

$$\pi_1^* H^1(S_1) = \langle 3\gamma, \eta_1 \rangle, \qquad \pi_2^* H^1(S_2) = \langle 4\gamma, \eta_2 \rangle,$$
$$\pi_1^* H^2(S_1) = (\textstyle\bigwedge^2 V)^{T_1}, \qquad \pi_2^* H^2(S_2) = \{a\gamma\eta_2 + b\mathbf{q} : a \equiv b \pmod 2\}. \tag{3.8}$$

The identification of these abstract lattices with the actual finite-fibre stalks is theorem 8.8; hence no geometric hypothesis remains open in this proposition.

Proof. The fundamental group is the extension

$$\Gamma_j = \langle \Lambda, g \mid g\lambda g^{-1} = A_j\lambda,\ g^{m_j} = v_j \rangle.$$

After abelianization,

$$H_1(S_j) = (\Lambda_{A_j} \oplus \mathbb{Z}g)/\mathbb{Z}([v_j], -m_j). \tag{3.9}$$

The matrices $A_j - I$ have Smith form $\mathrm{diag}(1,1,0,0)$, so Λ_{A_j} is free of rank two. The relation in (3.9) is primitive: if a prime divides m_j, its image $[v_j]$ modulo that prime is nonzero because $\gamma((A_j - I)\Lambda) = 0$ and $\gamma(v_1) = 1$, $\gamma(v_2) = -1$. Thus $H_1(S_j)$ is free of rank two. The quotient is a closed oriented four-manifold (the linear parts have determinant one) and $e(S_j) = e(A_0^{(j)})/m_j = 0$. Poincaré duality and the universal coefficient theorem therefore give

$$H_k(S_j;\mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z})$$

with no torsion. Indeed, Poincaré duality gives $H_3(S_j) \simeq H^1(S_j) \simeq \mathbb{Z}^2$ and $H^3(S_j) \simeq H_1(S_j) \simeq \mathbb{Z}^2$. The universal coefficient sequence for H^3 then forces $\mathrm{Ext}(H_2(S_j),\mathbb{Z}) = 0$, so $H_2(S_j)$ is torsion-free; Euler characteristic zero gives its rank as two.

Transfer gives $\pi_{j*}\pi_j^* = m_j$; hence π_j^* is injective and its image lies in the T_j-invariants. The Lyndon–Hochschild–Serre spectral sequence for $1 \to \Lambda \to \Gamma_j \to \mathbb{Z}/m_j \to 1$ [27, Chapter VII, §6] has

$$E_2^{p,q} = H^p(\mathbb{Z}/m_j; \textstyle\bigwedge^q V).$$

It computes $H^*(S_j;\mathbb{Z})$ because the universal cover of this affine torus quotient is $\mathbb{R}^4$, hence contractible. Its transgression on an invariant $\xi \in V$ evaluates the extension class: $d_2\xi = \xi(v_j) \bmod m_j$. Since $V^{T_j} = \langle \gamma, \eta_j \rangle$ and $\eta_j(v_j) = 0$, the degree-one images and indices in (3.8) follow.

In degree two, the integral invariant lattices are

$$(\textstyle\bigwedge^2 V)^{T_j} = \langle \gamma\eta_j, \mathbf{q} \rangle$$

with no omitted finite-index overlattice. Indeed, writing an invariant as (a,b,c,d,e,f) in the ordered basis $(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta)$, the T_1 equations are $e = f = 0$, $a = 2b$, $c = 3b + 6d$, while the T_2 equations are $e = f = 0$, $a = b$, $c = 2b + 6d$. Using $\gamma uw\delta$ as orientation,

$$(\gamma\eta_j)^2 = 0, \quad (\gamma\eta_1)\mathbf{q} = 3, \quad (\gamma\eta_2)\mathbf{q} = 2, \quad \mathbf{q}^2 = 12. \tag{3.10}$$

Let $L_j = \pi_j^* H^2(S_j)$. The intersection lattice of S_j is unimodular, and pullback multiplies its form by m_j. Hence $|\det L_j| = m_j^2$. The two invariant Gram determinants from (3.10) are 9 and 4. If k_j is the index of L_j in the invariant lattice, $k_j^2 |\det(\bigwedge^2 V)^{T_j}| = m_j^2$, so $k_1 = 1$ and $k_2 = 2$.

For $j = 2$, multiplicativity of the spectral sequence gives $d_2(\gamma\eta_2) = -[\eta_2] \neq 0$ in $\langle \gamma, \eta_2\rangle/\langle 4\gamma, 2\eta_2\rangle$. Thus $\gamma\eta_2 \notin L_2$. The remaining index-two possibilities are $L_2 = \langle 2\gamma\eta_2, a\gamma\eta_2 + \mathbf{q}\rangle$, $a \in \{0, 1\}$. The covering is étale, so $\pi_2^* K_{S_2}$ is trivial and $K_{S_2}^{\otimes 4}$ is trivial. Because $H^2(S_2; \mathbb{Z})$ is torsion-free, $c_1(S_2) = 0$ and $w_2(S_2) = 0$; the intersection form is even. Dividing the pullback pairings by four gives

$$\begin{pmatrix} 0 & 1 \\ 1 & a+3 \end{pmatrix},$$

so $a + 3$ is even and $a = 1$. This proves (3.8).

For clarity, the displayed differential contains no suppressed coefficient calculation. For the cyclic group generated by T_2,

$$H^2(\mathbb{Z}/4; V) = V^{T_2}/(I + T_2 + T_2^2 + T_2^3)V = \langle \gamma, \eta_2\rangle/\langle 4\gamma, 2\eta_2\rangle.$$

The extension class gives $d_2\gamma = \gamma(v_2) = -1$ and $d_2\eta_2 = \eta_2(v_2) = 0$; the derivation rule therefore gives $d_2(\gamma\eta_2) = -[\eta_2]$, which is nonzero because $\eta_2 \notin \langle 4\gamma, 2\eta_2\rangle$.

In degree four the index is the covering degree. For degree three, direct exterior calculation gives the primitive invariant bases

$$(\textstyle\bigwedge^3 V)^{T_1} = \langle \gamma uw,\ uw\delta - 4\gamma u\delta - 2\gamma w\delta\rangle,$$
$$(\textstyle\bigwedge^3 V)^{T_2} = \langle \gamma uw,\ uw\delta - 3\gamma u\delta - 3\gamma w\delta\rangle.$$

Against (γ, η_j) their pairing matrices are

$$\begin{pmatrix} 0 & 1 \\ -3 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix},$$

so the absolute determinants are 3 and 2. If the degree-three index is r_j, the degree-one indices already proved are 3 and 4. Pullback multiplies the perfect $H^1(S_j) \times H^3(S_j)$ pairing by m_j, whose rank-two determinant is m_j^2. Hence $3 \cdot 3r_1 = 3^2$ and $2 \cdot 4r_2 = 4^2$; hence $r_1 = 1$, $r_2 = 2$. Together with degrees zero through two this proves (3.7). □

5.4 The terminal presentation and degree-two map

Proposition 5.5 (terminal group, including the nonabelian step)**.** ***[VERIFIED]*** *Consider*

$$G(\ell_0, \ell_1, \ell_2) = \langle c, x, y \mid c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2}\rangle$$

and put $p = 12\ell_0 - 4\ell_1 - 3\ell_2$*. If* $p \neq 0$*, then* $G \cong \mathbb{Z}/|p|\mathbb{Z}$*; if* $p = 0$*, then* $G \cong \mathbb{Z}$*. For* $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ *one has* $p = -1$ *and* $G = 1$*.*

Proof. The relation $xy = c^{\ell_0}$ gives $y = x^{-1}c^{\ell_0}$. Since c is central, x and y commute, so the apparently nonabelian presentation is already abelian. In the ordered additive generators (x, c) the remaining relations form

$$R = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det R = -p.$$

The greatest common divisor of all entries of R is 1 because it divides both 3 and 4. Thus its Smith form is $\mathrm{diag}(1, |p|)$ when $p \neq 0$ and $\mathrm{diag}(1, 0)$ when $p = 0$. The stated groups follow. □

Proposition 5.6 (terminal degree-two lattice)**.** ***[VERIFIED]*** *Let* $A : \mathbb{Z}^6 \to \mathbb{Z}^4$ *have columns*

$$c_1 = (2, -4, -2, 3)^t, \quad c_2 = (1, -2, -2, 3)^t,$$
$$c_3 = (3, 0, -4, 0)^t, \quad c_4 = (0, 1, 0, -1)^t, \qquad c_5 = c_6 = 0.$$

Then $\operatorname{im} A$ *is saturated of rank three,*

$$\operatorname{im} A = \ker(4, 2, 3, 2), \qquad \operatorname{coker} A \cong \mathbb{Z}, \tag{3.11}$$

and $(4, 2, 3, 2)$ *is the primitive left-kernel generator.*

Proof. The row $\Phi = (4, 2, 3, 2)$ annihilates the four nonzero columns:

$$\Phi c_1 = 8 - 8 - 6 + 6 = 0, \quad \Phi c_2 = 4 - 4 - 6 + 6 = 0, \quad \Phi c_3 = 12 - 12 = 0, \quad \Phi c_4 = 2 - 2 = 0.$$

It is primitive since the gcd of its entries is 1, so $\ker \Phi \subset \mathbb{Z}^4$ is a saturated rank-three lattice. The 3×3 minor on rows $(1, 2, 3)$ and columns $(1, 2, 4)$ is 2, whereas the minor on rows $(1, 2, 4)$ and the same columns is -3. Consequently the ideal generated by all maximal minors contains $(2, 3) = (1)$. This proves that A has rank three and saturated image. We have $\operatorname{im} A \subseteq \ker \Phi$ between rank-three lattices. The finite quotient $\ker \Phi / \operatorname{im} A$ injects into the torsion-free group $\mathbb{Z}^4 / \operatorname{im} A$, so it is zero. Thus equality holds, and the primitive surjection $\Phi : \mathbb{Z}^4 \to \mathbb{Z}$ induces (3.11). □

Proposition 5.7 (all-degree terminal homology package)**.** ***[VERIFIED]*** *Let $C_\bullet$ be the degree-by-degree terminal package printed for the glued six-manifold, and assume only the following algebraic outputs of its displayed maps:*

(i) $H_0(C_\bullet) = H_6(C_\bullet) = \mathbb{Z}$ and the degree-one presentation is $G(\ell_0, \ell_1, \ell_2)$ of theorem 5.5;

(ii) in degree two there is an exact sequence $0 \to \mathbb{Z} \to H_2^\circ \to \mathbb{Z} \to 0$, the retained boundary lattice kills the second quotient, and its remaining generator maps to p times a primitive generator of the first copy of $\mathbb{Z}$;

(iii) the package obeys integral Poincaré duality and the universal coefficient theorem in formal dimension six, and its Euler characteristic is 2.

Then, for $p \neq 0$,

$$(H_0(C_\bullet), \dots, H_6(C_\bullet)) \cong (\mathbb{Z}, \mathbb{Z}/|p|, \mathbb{Z}/|p|, \mathbb{Z}/|p|, \mathbb{Z}/|p|, 0, \mathbb{Z}). \tag{3.12}$$

In particular, the printed terminal package has the integral homology of S^6 when $p = -1$.

Proof. By theorem 5.5, the degree-one group is $\mathbb{Z}/|p|$. In degree two, exactness and the stated retained-boundary map first quotient H_2° down to the primitive first summand and then quotient that summand by p; hence $H_2(C_\bullet) = \mathbb{Z}/|p|$. Both groups are finite, so their Betti numbers vanish. Universal coefficients and duality give

$$H_5 \cong H^1 = 0, \qquad H_4 \cong H^2 \cong \operatorname{Ext}(H_1, \mathbb{Z}) = \mathbb{Z}/|p|.$$

The Euler-characteristic identity is therefore $2 = 1 - b_3 + 1$, so $b_3 = 0$. Finally

$$H_3 \cong H^3 \cong \operatorname{Ext}(H_2, \mathbb{Z}) = \mathbb{Z}/|p|,$$

which proves (3.12). □

Geometric boundary. The calculations in this section deliberately isolate the printed lattices, presentations, and matrices. Their geometric realization is proved separately: the finite-band and cusp maps are identified in theorem 44.10, the based fundamental-group calculation is theorem 44.9, and the comparison with the intrinsic Leray transgressions is theorem 44.13.

6 An exact Erdős–Straus/S6 terminal-matrix crosswalk

This section records an arithmetic relation suggested by comparing the candidate's terminal presentation with the divisor-shell coordinates used in the maintained Erdős–Straus corpus [69]. All maps below are proved over the integers. The conclusion is an exact change of relators; it is not a geometric map from the reciprocal cubic surface to the candidate complex threefold.

6.1 The first-half shell and its divisor coordinates

Proposition 6.1 (first-half denominator). ***[VERIFIED]*** *Let $p \equiv 1 \pmod 4$ be an odd prime. If*

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \qquad x, y, z \in \mathbb{Z}_{>0}, \qquad x \le y \le z,$$

then

$$\frac{p}{4} < x \le \frac{p}{2}.$$

Proof. The lower bound follows from $1/x < 4/p$. Suppose $x > p/2$. Clearing denominators and reducing modulo p gives

$$4xyz = p(xy + xz + yz), \qquad p \mid xyz.$$

If $p \mid x$, then $x, y, z \ge p$, so the sum of the three reciprocals is at most $3/p$. If $p \mid y$, then $y, z \ge p$, and

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} < \frac{2}{p} + \frac{2}{p} = \frac{4}{p}.$$

Both alternatives are impossible. Hence $z = kp$ for an integer $k \ge 1$, and substitution gives

$$(4k - 1)xy = kp(x + y).$$

Neither x nor y is divisible by p, so $p \mid 4k - 1$. Write $4k - 1 = \ell p$. Since $p \equiv 1 \pmod 4$, one has $\ell \equiv 3 \pmod 4$, hence $\ell \ge 3$. Cancelling p gives $\ell xy = k(x + y)$. Since $x \le y$,

$$x \le \frac{2k}{\ell} = \frac{\ell p + 1}{2\ell} = \frac{p}{2} + \frac{1}{2\ell} < \frac{p+1}{2}.$$

But the integer $x > p/2$ is at least $(p + 1)/2$, a contradiction. □

Fix henceforth a prime $p \equiv 1 \pmod{12}$, and put

$$L = \frac{p-1}{4}, \qquad R_j = 4j + 3, \qquad a_j = \frac{p + R_j}{4}, \qquad 0 \le j < L.$$

The preceding proposition shows that every solution's smallest denominator lies in the exact candidate set $\{a_j : 0 \le j < L\}$. Notice that

$$4a_j = p + R_j, \qquad \gcd(R_j, p) = \gcd(R_j, a_j) = \gcd(p, a_j) = 1.$$

Proposition 6.2 (complete two-channel divisor selector). ***[VERIFIED]*** *Define*

$$E_j = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + 1\},$$
$$M_j = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + p\}.$$

Then the Erdős–Straus equation for p has a positive integral solution if and only if at least one E_j or M_j is nonempty. For $u \in E_j$ set $\varepsilon = 0$, and for $u \in M_j$ set $\varepsilon = 1$. With

$$T = \frac{4u + p^{\varepsilon}}{R_j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4},$$

the corresponding solution is

$$(x, y, z) = \left(a_j, \frac{p^{\varepsilon} a_j K}{u}, pK\right),$$

up to interchange of y and z.

Proof. Fix $x = a_j$, and abbreviate $R = R_j$, $a = a_j$, and $N = pa$. The remaining equation is

$$Ryz = N(y + z).$$

It is equivalent to

$$(Ry - N)(Rz - N) = N^2.$$

Both factors are positive. Since $\gcd(R, N) = 1$, positive solutions are in bijection with positive divisors $d \mid N^2$ satisfying

$$d \equiv -N \pmod{R};$$

the complementary divisor $e = N^2/d$ then also satisfies $e \equiv -N \pmod{R}$, and

$$y = \frac{N + d}{R}, \qquad z = \frac{N + e}{R}.$$

Because $\gcd(p, a) = 1$, the p-adic exponent of d is 0, 1, or 2. Complementation $d \leftrightarrow e$ exchanges exponents 0 and 2, so choose the exponent-two representative in that pair. If $d = p^2u$, then $u \mid a^2$, and

$$R \mid d + N \iff R \mid pu + a \iff R \mid 4u + 1,$$

where $4a \equiv p \pmod{R}$ and p is invertible modulo R. This is the E_j channel. If $d = pu$, then

$$R \mid d + N \iff R \mid u + a \iff R \mid 4u + p,$$

which is the M_j channel. This proves completeness of the two selector sets.

For either channel, $R_jT = 4u + p^\varepsilon$. Modulo 4, this says $-T \equiv 1$, so $T \equiv 3 \pmod{4}$; the displayed K is therefore a positive integer. Direct calculation using $4a = p + R_j$ gives

$$R_jK = a + p^{1-\varepsilon}u, \qquad Ta = u + p^\varepsilon K.$$

Since $u \mid a^2$, the second identity implies $u \mid p^\varepsilon aK$. Finally,

$$\frac{u}{p^\varepsilon aK} + \frac{1}{pK} = \frac{p^{1-\varepsilon}u + a}{paK} = \frac{R_j}{pa} = \frac{4}{p} - \frac{1}{a},$$

which proves the asserted formula and its converse. □

6.2 The unimodular relation-matrix bijection

Put

$$\lambda(\ell_0, \ell_1, \ell_2) = 12\ell_0 - 4\ell_1 - 3\ell_2.$$

By theorem 5.5, the candidate's ordered terminal relation matrix in the ordered generator basis (x, c), for $\ell = (\ell_0, \ell_1, \ell_2)$, is

$$\mathsf{R}(\ell) = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det \mathsf{R}(\ell) = -\lambda(\ell).$$

For $\varepsilon \in \{0, 1\}$, define the marked selector space

$$\mathcal{W}_\varepsilon(p) = \left\{ (j, u) \in \mathbb{Z} \times \mathbb{Z}_{>0} : \begin{array}{l} 0 \le j < L,\ u \in \mathbb{Z}_{>0},\ u \mid a_j^2, \\ R_j \mid 4u + p^\varepsilon \end{array} \right\}$$

and the normalized divisor-supported terminal slice

$$\mathcal{H}_\varepsilon(p) = \left\{ (j, \ell_0, \ell_1) \in \mathbb{Z}^3 : \begin{array}{l} 0 \le j < L,\ \lambda(\ell_0, \ell_1, 1) = p^\varepsilon, \\ T := 4\ell_0 - 1 > 0, \quad u := \ell_1 + jT > 0, \quad u \mid a_j^2 \end{array} \right\}.$$

Theorem 6.3 (exact marked selector/terminal-slice equivalence)**.** ***[VERIFIED]*** *The map*

$$\Theta_{\varepsilon,p} : \mathcal{W}_\varepsilon(p) \longrightarrow \mathcal{H}_\varepsilon(p), \qquad (j,u) \longmapsto \left(j, \frac{T+1}{4}, u - jT\right), \quad T = \frac{4u+p^\varepsilon}{R_j},$$

is a bijection. Under it, the two relation matrices are related by the explicit unimodular row operation

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1-4\ell_0 \end{pmatrix} = \mathsf{R}(\ell_0, \ell_1, 1),$$

and both determinants equal $-p^\varepsilon$*.*

Proof. For $(j,u) \in \mathcal{W}_\varepsilon(p)$, T is a positive integer and the congruence computation in theorem 6.2 gives $T \equiv 3 \pmod 4$, so $\ell_0 = (T+1)/4$ is integral. Set $\ell_1 = u - jT$. Since $R_j = 4j+3$,

$$\lambda(\ell_0, \ell_1, 1) = 3(T+1) - 4(u - jT) - 3 = (4j+3)T - 4u = p^\varepsilon.$$

The positivity and divisor conditions in the definition of $\mathcal{H}_\varepsilon(p)$ follow from $u = \ell_1 + jT$. Matrix multiplication gives the displayed identity, and

$$\det \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix} = -R_jT + 4u = -p^\varepsilon.$$

Conversely, from $(j, \ell_0, \ell_1) \in \mathcal{H}_\varepsilon(p)$ recover

$$T = 4\ell_0 - 1, \qquad u = \ell_1 + jT.$$

Expanding $\lambda(\ell_0, \ell_1, 1) = p^\varepsilon$ gives $R_jT = 4u + p^\varepsilon$. Thus $R_j \mid 4u + p^\varepsilon$, and the remaining positivity and divisor hypotheses are already part of $\mathcal{H}_\varepsilon(p)$. These constructions are literal inverses. □

The normalization $\ell_2 = 1$ automatically satisfies the candidate's odd ℓ_2 admissibility condition. It also forces $3 \nmid \ell_1$: if $3 \mid \ell_1$, then $\lambda(\ell_0, \ell_1, 1)$ is divisible by 3, whereas $p^\varepsilon \equiv 1 \pmod 3$. Write $\mathrm{ES}(p)$ for the assertion that $4/p$ is a sum of three positive integral unit fractions. Consequently

$$\mathrm{ES}(p) \Longleftrightarrow \mathcal{H}_0(p) \sqcup \mathcal{H}_1(p) \neq \varnothing.$$

The E-channel is the slice with terminal parameter $p_{\mathrm{S6}} = 1$; the M-channel is the slice with $p_{\mathrm{S6}} = p$.

Here the channel label is part of the datum. The raw selector subsets need not be disjoint: for $p = 13$, $j = 0$, and $u = 2$, one has $u \in E_0 \cap M_0$. Accordingly the occurrence statement above means the tagged disjoint union $\{0\} \times \mathcal{H}_0(p) \sqcup \{1\} \times \mathcal{H}_1(p)$. Moreover the map from tagged selector states to *unordered* unit-fraction solutions is not injective in the M-channel: complementation $u \leftrightarrow a_j^2/u$ interchanges y and z. For example, $p = 13$, $j = 0$, and $u = 2, 8$ give the same unordered solution $\{4, 26, 52\}$. The theorem is a bijection with the marked terminal slice, not a bijection with unordered solutions.

The tags are essential. For every eligible prime q, define the fixed-prime forgetful map

$$\mathfrak{f}_q : \mathcal{H}_0(q) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{f}_q(j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

For $p = 13$, the two E-channel selector states satisfy

$$\Theta_{0,13}(0,2) = (0,1,2), \qquad \Theta_{0,13}(1,5) = (1,1,2),$$

so their marked images are distinct, while both are sent by $\mathfrak{f}_{13}$ to $(1,2,1)$. Thus forgetting j destroys injectivity. To compare different prime fibres, define

$$\mathfrak{F} : \bigsqcup_{\substack{q \text{ prime} \\ q \equiv 1 \pmod{12}}} (\{q\} \times \mathcal{H}_0(q)) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{F}(q; j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

At $q = 73$, one has $\Theta_{0,73}(1,5) = (1,1,2)$, so this tagged image is also sent by $\mathfrak{F}$ to $(1,2,1)$. Hence forgetting the external prime loses still more information.

6.3 What the crosswalk proves, and what it does not

The theorem identifies the complete fixed-p, normalized, positively oriented, j-marked divisor selector with an arithmetic slice of the candidate's terminal relation matrices. It explains a genuine structural continuity between the earlier reciprocal-equation work and the terminal algebra used here: the common object is a 2×2 integral presentation, and the passage between the two displayed matrices is an element of $\mathrm{SL}_2(\mathbb{Z})$.

The orientation qualification is substantive. The geometrically validated candidate specialization used elsewhere in this workbench has terminal parameter $p_{\mathrm{S6}} = -1$, whereas the E-channel here has $p_{\mathrm{S6}} = +1$. No result in this section identifies those two oriented terminal data. Thus the exact arithmetic equivalence does not, by itself, place an Erdős–Straus selector state on the particular complex threefold reconstructed in the candidate-specific $p = -1$ branch.

It does not prove that the selector is nonempty for every prime, so it does not prove the Erdős–Straus conjecture. It also does not construct a map from the reciprocal cubic surface to X, identify selector divisors with cycles or attaching maps in the candidate geometry, or transfer the candidate's complex structure to the arithmetic surface. Those would be additional geometric assertions, not consequences of unimodular equivalence of relation matrices.

7 The exact marked Erdős–Straus–S^6 bridge

Result and scope

For each prime $p \equiv 1 \pmod{12}$, the complete first-half Erdős–Straus divisor selector is in explicit bijection with a normalized, positively oriented, divisor-supported, and shell-marked slice of the terminal parameter lattice occurring in the candidate complex structure on S^6. The coordinate change is integral in both directions. More strongly, the selector relation matrix and the S6 terminal relation matrix have exactly the same row lattice: they differ by a displayed element of $\mathrm{SL}_2(\mathbb{Z})$. Composing the bijection with the shell reconstruction gives positive denominators solving $4/p = 1/x + 1/y + 1/z$, and every positive solution is detected by at least one of the two tagged slices.

The nonlinear condition $u \mid a_j^2$, together with the external labels (p, j, ε), is indispensable. If it is removed, each normalized S6 fibre is an infinite affine line and carries no Erdős–Straus existence information. A particularly natural unmarked S6 point produces precisely the already-solved congruence class $p \equiv 13 \pmod{24}$ and produces no selector state when $p \equiv 1 \pmod{24}$. Thus the bridge is an exact arithmetic equivalence of marked presentations, not a proof of universal selector nonemptiness and not a geometric morphism from the reciprocal cubic to the complex threefold.

7.1 All objects and conventions

Fix throughout a prime

$$p \equiv 1 \pmod{12}, \qquad h = \frac{p-1}{12} \in \mathbb{Z}_{>0},$$

and write

$$L = \frac{p-1}{4} = 3h, \qquad J_p = \{j \in \mathbb{Z} : 0 \le j < L\}.$$

For $j \in J_p$, retain the four integers

$$R_j = 4j + 3, \qquad a_j = 3h + j + 1 = \frac{p + R_j}{4}, \qquad N_j = pa_j, \qquad 4a_j = p + R_j. \tag{1.1}$$

No label in (p, j) will be suppressed. The inequalities

$$3 \le R_j \le p - 2, \qquad \frac{p+3}{4} \le a_j \le \frac{p-1}{2} < p$$

give

$$\gcd(p, a_j) = \gcd(p, R_j) = \gcd(a_j, R_j) = 1. \tag{1.2}$$

Indeed p is prime and $0 < a_j < p$; then $R_j = 4a_j - p$ gives the other two equalities.

Let

$$\mathscr{S}(p) = \left\{(x, y, z) \in \mathbb{Z}^3_{>0} : 4xyz = p(xy + xz + yz)\right\}.$$

The equation is symmetric, but whenever completeness is invoked we choose the ordered representative $x \le y \le z$. Write ES(p) for the assertion $\mathscr{S}(p) \neq \varnothing$.

For the S6 terminal parameters put

$$\lambda(\ell_0, \ell_1, \ell_2) = 12\ell_0 - 4\ell_1 - 3\ell_2 \tag{1.3}$$

and define the abstract terminal group

$$G(\ell_0, \ell_1, \ell_2) = \left\langle c, x, y \;\middle|\; c \text{ central},\ xy = c^{\ell_0},\ x^3 = c^{\ell_1},\ y^4 = c^{\ell_2} \right\rangle. \tag{1.4}$$

After eliminating y, and using the ordered generator basis (x, c), its ordered row-relation matrix is

$$\mathsf{R}(\ell_0, \ell_1, \ell_2) = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}, \qquad \det \mathsf{R} = -\lambda(\ell_0, \ell_1, \ell_2). \tag{1.5}$$

The letter p in this note always denotes the Erdős–Straus prime. The signed terminal value is denoted

$$q_\varepsilon = p^\varepsilon \in \{1, p\}, \qquad \varepsilon \in \{0, 1\}.$$

Thus no identification of the external prime p with the terminal value is made in the $\varepsilon = 0$ channel.

For any integral 2×2 matrix A, define its row-relation lattice and row-presented group by

$$\mathrm{Rel}(A) = \{v^{\mathsf{T}} A : v \in \mathbb{Z}^2\} \subseteq \mathbb{Z}^2, \qquad \mathcal{G}(A) = \mathbb{Z}^2 / \mathrm{Rel}(A) = \mathrm{coker}(A^{\mathsf{T}}). \tag{1.6}$$

This convention makes every matrix direction in the bridge explicit.

7.2 The complete first-half shell selector

Lemma 7.1 (first-half denominator)**.** *Let $p \equiv 1 \pmod 4$ be an odd prime. If $(x, y, z) \in \mathscr{S}(p)$ and $x \le y \le z$, then*

$$\frac{p}{4} < x \le \frac{p}{2}.$$

Proof. Positivity of the two reciprocals other than $1/x$ gives $1/x < 4/p$, hence $x > p/4$. Suppose $x > p/2$. Clearing denominators gives

$$4xyz = p(xy + xz + yz),$$

so $p \mid xyz$. If $p \mid x$, then $x, y, z \ge p$ and the reciprocal sum is at most $3/p$. If $p \mid y$, then $y, z \ge p$, while $1/x < 2/p$, so the sum is strictly less than $4/p$. Both cases are impossible. Therefore $z = kp$ for some $k \in \mathbb{Z}_{>0}$, while neither x nor y is divisible by p. Substitution yields

$$(4k - 1)xy = kp(x + y).$$

Reduction modulo p gives $p \mid 4k - 1$. Write $4k - 1 = rp$. Since $p \equiv 1 \pmod 4$, one has $r \equiv 3 \pmod 4$, hence $r \ge 3$. After cancelling p,

$$rxy = k(x + y).$$

As $x \le y$, division by xy gives

$$x \le \frac{2k}{r} = \frac{rp + 1}{2r} = \frac{p}{2} + \frac{1}{2r} < \frac{p + 1}{2}.$$

The integer $x > p/2$ must be at least $(p + 1)/2$, a contradiction. □

For fixed $j \in J_p$, define the ordered residual shell

$$\mathscr{P}_{p,j} = \{(y,z) \in \mathbb{Z}_{>0}^2 : R_j yz = N_j(y+z)\} \tag{2.1}$$

and the oriented divisor shell

$$\mathscr{D}_{p,j} = \{d \in \mathbb{Z}_{>0} : d \mid N_j^2,\ d \equiv -N_j \pmod{R_j}\}. \tag{2.2}$$

Proposition 7.2 (oriented shell bijection)**.** *The maps*

$$\begin{aligned} \delta_{p,j} &: \mathscr{P}_{p,j} \longrightarrow \mathscr{D}_{p,j}, & (y,z) &\longmapsto R_j y - N_j, \\ \rho_{p,j} &: \mathscr{D}_{p,j} \longrightarrow \mathscr{P}_{p,j}, & d &\longmapsto \left(\frac{N_j+d}{R_j}, \frac{N_j+N_j^2/d}{R_j}\right) \end{aligned}$$

are mutually inverse. The involution

$$\iota_{N_j}(d) = \frac{N_j^2}{d}$$

corresponds exactly to $(y,z) \mapsto (z,y)$.

Proof. Abbreviate $R = R_j$ and $N = N_j$. Equation (2.1) is equivalent to

$$(Ry - N)(Rz - N) = N^2. \tag{2.3}$$

Both factors are positive: for example, division of $Ryz = N(y+z)$ by z gives $Ry = N(1+y/z) > N$, and similarly for $Rz - N$. Therefore $d = Ry - N$ is a positive divisor of N^2 and $d \equiv -N \pmod{R}$.

Conversely, let $d \in \mathscr{D}_{p,j}$ and put $e = N^2/d$. By (1.2), d and N are units modulo R; hence

$$e \equiv N^2(-N)^{-1} \equiv -N \pmod{R}.$$

Thus both components of $\rho_{p,j}(d)$ are positive integers. Their two shifted factors are d and e, so (2.3), and therefore (2.1), holds. The formulas visibly compose to the identity. Replacing d by $e = N^2/d$ exchanges the two reconstructed components. □

For $\varepsilon \in \{0,1\}$, define the fixed-shell selector and the fully marked selector by

$$\mathcal{W}_\varepsilon(p,j) = \{u \in \mathbb{Z}_{>0} : u \mid a_j^2,\ R_j \mid 4u + p^\varepsilon\}, \tag{2.4}$$

$$\mathcal{W}_\varepsilon(p) = \{(j,u) : j \in J_p,\ u \in \mathcal{W}_\varepsilon(p,j)\}. \tag{2.5}$$

The tag $\varepsilon = 0$ is the exterior, or E, channel; $\varepsilon = 1$ is the middle, or M, channel. Whenever a disjoint union of the two channels is written below, it means the tagged coproduct

$$\{0\} \times \mathcal{W}_0(p) \ \sqcup\ \{1\} \times \mathcal{W}_1(p).$$

This convention is necessary because the two raw subsets can intersect.

Proposition 7.3 (exact two-channel decomposition)**.** *The maps*

$$\begin{aligned} \iota_E &: \mathcal{W}_0(p,j) \longrightarrow \mathscr{D}_{p,j}, & u &\longmapsto p^2 u, \\ \iota_M &: \mathcal{W}_1(p,j) \longrightarrow \mathscr{D}_{p,j}, & u &\longmapsto pu \end{aligned}$$

are bijections onto the subsets on which the p-adic valuations are 2 *and* 1*, respectively. The valuation-zero subset is* $\iota_{N_j}(\iota_E(\mathcal{W}_0(p,j)))$*. Consequently*

$$\mathscr{D}_{p,j} = \iota_E(\mathcal{W}_0(p,j)) \ \sqcup\ \iota_M(\mathcal{W}_1(p,j)) \ \sqcup\ \iota_{N_j}(\iota_E(\mathcal{W}_0(p,j))), \tag{2.6}$$

and

$$|\mathscr{D}_{p,j}| = 2|\mathcal{W}_0(p,j)| + |\mathcal{W}_1(p,j)|. \tag{2.7}$$

Proof. Because $p \nmid a_j$, every divisor $d \mid p^2 a_j^2$ has $v_p(d) \in \{0, 1, 2\}$.

If $d = p^2 u$, then $u \mid a_j^2$. The shell congruence is

$$R_j \mid d + N_j = p(pu + a_j).$$

After cancelling the unit p modulo R_j and multiplying by 4, the identity $4a_j = p + R_j$ makes this equivalent to $R_j \mid 4u + 1$. This proves the E-channel assertion in both directions. If $d = pu$, the same calculation gives

$$R_j \mid d + N_j \iff R_j \mid u + a_j \iff R_j \mid 4u + p,$$

which proves the M-channel assertion. Complementation preserves the shell congruence by Proposition 7.2, exchanges valuations 0 and 2, and preserves valuation 1. This proves the disjoint decomposition and its cardinality formula. □

For $u \in \mathcal{W}_\varepsilon(p, j)$, define the two exact quotients

$$T = \frac{4u + p^\varepsilon}{R_j}, \qquad K = \frac{p^{1-\varepsilon} T + 1}{4}. \tag{2.8}$$

Since $R_j \equiv 3 \pmod 4$ and $p^\varepsilon \equiv 1 \pmod 4$, the first equation gives $T \equiv 3 \pmod 4$. Hence $K \in \mathbb{Z}_{>0}$. Direct calculation, without dropping the tag, gives

$$R_j K = a_j + p^{1-\varepsilon} u, \qquad T a_j = u + p^\varepsilon K. \tag{2.9}$$

Indeed, multiplication by 4 changes these identities into

$$\begin{aligned} 4R_j K &= p^{1-\varepsilon}(4u + p^\varepsilon) + R_j = 4p^{1-\varepsilon} u + p + R_j, \\ 4T a_j &= T(p + R_j) = pT + 4u + p^\varepsilon = 4u + 4p^\varepsilon K. \end{aligned}$$

Proposition 7.4 (denominator reconstruction)**.** *The map*

$$\Psi_{p,j,\varepsilon} : \mathcal{W}_\varepsilon(p, j) \longrightarrow \mathscr{S}(p), \qquad u \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u} \right) \tag{2.10}$$

is well-defined. Its two oriented shell factors are

$$R_j(pK) - N_j = p^{2-\varepsilon} u, \qquad R_j \left(\frac{p^\varepsilon a_j K}{u} \right) - N_j = p^\varepsilon \frac{a_j^2}{u}. \tag{2.11}$$

Moreover,

$$\mathrm{ES}(p) \quad \iff \quad \big(\{0\} \times \mathcal{W}_0(p)\big) \sqcup \big(\{1\} \times \mathcal{W}_1(p)\big) \neq \varnothing. \tag{2.12}$$

Proof. The last coordinate in (2.10) is integral. To see this without assuming it, reduce the second identity in (2.9) modulo u, multiply by a_j, and use $u \mid a_j^2$: this gives $u \mid p^\varepsilon a_j K$. The two identities (2.11) now follow from the first identity in (2.9). Their product is N_j^2, so Proposition 7.2 shows that (2.10) lies in $\mathscr{S}(p)$.

Conversely, suppose $\mathscr{S}(p) \neq \varnothing$, and sort a solution. Lemma 7.1 shows that its smallest denominator is uniquely of the form $x = a_j$ with $j \in J_p$: explicitly

$$R_j = 4x - p \equiv 3 \pmod 4, \qquad j = \frac{4x - p - 3}{4}.$$

Proposition 7.2 supplies a divisor $d \in \mathscr{D}_{p,j}$, and Proposition 7.3 places either d or its complement in the image of ι_E, or places d in the image of ι_M. Hence one of the two marked selector sets is nonempty. The forward implication follows from (2.10). □

7.3 The exact selector–terminal-lattice equivalence

Define the normalized, positively oriented, divisor-supported terminal slice

$$\mathcal{H}_\varepsilon(p) = \left\{ (j,\ell_0,\ell_1) \in \mathbb{Z}^3 : \begin{array}{l} j \in J_p, \quad \lambda(\ell_0,\ell_1,1) = p^\varepsilon, \\ T(\ell_0) := 4\ell_0 - 1 > 0, \\ u(j,\ell_0,\ell_1) := \ell_1 + jT(\ell_0) > 0, \\ u(j,\ell_0,\ell_1) \mid a_j^2 \end{array} \right\}. \tag{3.1}$$

Let $\mathcal{H}_\varepsilon(p,j)$ be the fibre with the displayed marker j.

For a selector state define its relation matrix

$$\mathsf{A}_\varepsilon(p;j,u) = \begin{pmatrix} R_j & -u \\ 4 & -T \end{pmatrix}, \qquad R_j T = 4u + p^\varepsilon. \tag{3.2}$$

For the marker j, retain both inverse unimodular shears

$$\mathsf{U}_j = \begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{L}_j = \mathsf{U}_j^{-1} = \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}. \tag{3.3}$$

Theorem 7.5 (strong marked bridge theorem)**.** *For every $\varepsilon \in \{0,1\}$, the map*

$$\Theta_{\varepsilon,p} : \mathcal{W}_\varepsilon(p) \longrightarrow \mathcal{H}_\varepsilon(p), \qquad (j,u) \longmapsto \left(j, \frac{T+1}{4}, u - jT \right) \tag{3.4}$$

is a bijection. Its inverse is

$$\Theta_{\varepsilon,p}^{-1}(j,\ell_0,\ell_1) = (j, \ell_1 + j(4\ell_0 - 1)). \tag{3.5}$$

For corresponding points, with $\ell = (\ell_0,\ell_1,1)$, one has the exact matrix identities

$$\boxed{\mathsf{U}_j \mathsf{A}_\varepsilon(p;j,u) = \mathsf{R}(\ell_0,\ell_1,1), \qquad \mathsf{A}_\varepsilon(p;j,u) = \mathsf{L}_j \mathsf{R}(\ell_0,\ell_1,1).} \tag{3.6}$$

In particular,

$$\mathrm{Rel}(\mathsf{A}_\varepsilon(p;j,u)) = \mathrm{Rel}(\mathsf{R}(\ell_0,\ell_1,1)), \qquad \det \mathsf{A}_\varepsilon = \det \mathsf{R} = -p^\varepsilon. \tag{3.7}$$

Thus the bridge intertwines the entire integral row presentation, not only its determinant.

Proof. Take $(j,u) \in \mathcal{W}_\varepsilon(p)$. Section 2 proves that T is a positive integer satisfying $T \equiv 3 \pmod 4$, so $\ell_0 = (T+1)/4 \in \mathbb{Z}$. Put $\ell_1 = u - jT$. Since $R_j = 4j+3$,

$$\begin{aligned} \lambda(\ell_0,\ell_1,1) &= 3(T+1) - 4(u - jT) - 3 \\ &= (4j+3)T - 4u = p^\varepsilon. \end{aligned}$$

The positivity and divisor-support conditions in (3.1) hold because $u = \ell_1 + jT$. Thus (3.4) is well-defined.

Conversely, take $(j,\ell_0,\ell_1) \in \mathcal{H}_\varepsilon(p)$, and set

$$T = 4\ell_0 - 1, \qquad u = \ell_1 + jT.$$

Expanding $\lambda(\ell_0,\ell_1,1) = p^\varepsilon$ gives

$$(4j+3)T - 4u = p^\varepsilon,$$

so $R_j \mid 4u + p^\varepsilon$. All other selector conditions occur explicitly in (3.1). The two constructions recover T, u, ℓ_0, ℓ_1 literally, which proves bijectivity and (3.5).

Finally, direct multiplication gives

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4j+3 & -u \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -u + jT \\ 4 & -T \end{pmatrix} = \begin{pmatrix} 3 & -\ell_1 \\ 4 & 1 - 4\ell_0 \end{pmatrix}.$$

Both shears have determinant 1, so their integral row operations are invertible and the row lattices are equal. The signed determinant is $-R_jT + 4u = -p^\varepsilon$, proving (3.7). □

Corollary 7.6 (presented groups and the Erdős–Straus equation)**.** *For every corresponding selector/S6 point,*

$$\mathcal{G}(\mathsf{A}_\varepsilon(p;j,u)) = \mathcal{G}(\mathsf{R}(\ell_0,\ell_1,1)) \cong G(\ell_0,\ell_1,1) \cong \mathbb{Z}/p^\varepsilon\mathbb{Z}. \tag{3.8}$$

Here $\mathbb{Z}/1\mathbb{Z}$ is the trivial group. Moreover the direct map

$$\widehat{\Psi}_{\varepsilon,p} : \mathcal{H}_\varepsilon(p) \longrightarrow \mathscr{S}(p) \tag{3.9}$$

is obtained by (3.5), (2.8), and

$$(j,\ell_0,\ell_1) \longmapsto \left(a_j,\ pK,\ \frac{p^\varepsilon a_j K}{u}\right), \quad \begin{cases} T = 4\ell_0 - 1, \\ u = \ell_1 + jT, \\ K = (p^{1-\varepsilon}T+1)/4. \end{cases} \tag{3.10}$$

It satisfies

$$\boxed{\mathrm{ES}(p) \Longleftrightarrow (\{0\}\times\mathcal{H}_0(p)) \sqcup (\{1\}\times\mathcal{H}_1(p)) \neq \varnothing.} \tag{3.11}$$

Proof. The relation $xy = c^{\ell_0}$ in (1.4) gives $y = x^{-1}c^{\ell_0}$; since c is central, x and y commute. The remaining two relations are the rows of (1.5). The greatest common divisor of the four entries of R is 1, because it divides both 3 and 4. Its Smith form is therefore $\mathrm{diag}(1, p^\varepsilon)$, proving (3.8). Formula (3.10) is the composition of Theorem 7.5 with Proposition 7.4, and (3.11) follows from (2.12). □

Remark 7.7 (the selector tag and the solution map). The channel tag is not redundant. For $p = 13$, $j = 0$, and $u = 2$, one has $u \in \mathcal{W}_0(p,j) \cap \mathcal{W}_1(p,j)$, but the two tags give different terminal values and different marked terminal points. Nor is the map to unordered denominator triples injective: in the M-channel at $p = 13, j = 0$, the complementary divisors $u = 2$ and $u = 8 = a_0^2/2$ both give the unordered solution $\{4, 26, 52\}$. The bijection in Theorem 7.5 is a bijection of marked selector states with marked terminal states, not a bijection with unordered solutions.

Corollary 7.8 (fixed-fibre affine coordinates and the intrinsic gate)**.** *Put*

$$C_\varepsilon = \frac{p^\varepsilon + 3}{4}, \qquad D_{\varepsilon,j} = j + C_\varepsilon = \frac{R_j + p^\varepsilon}{4}.$$

Every integral point with $\ell_2 = 1$ and $\lambda(\ell_0,\ell_1,1) = p^\varepsilon$ is uniquely

$$(\ell_0,\ell_1,\ell_2) = (n, 3n - C_\varepsilon, 1), \qquad n \in \mathbb{Z}, \tag{3.12}$$

and, for each fixed marker $j \in J_p$, define

$$T = 4n - 1, \qquad u = R_j n - D_{\varepsilon,j}, \qquad K = \frac{p^{1-\varepsilon}T + 1}{4}. \tag{3.13}$$

Then $T \equiv 3 \pmod 4$, so $K \in \mathbb{Z}$, and the congruence equation $R_j T = 4u + p^\varepsilon$ holds identically. On the locus $T > 0$ and $u > 0$, one has

$$\gcd(R_j, u) = 1, \qquad u \mid a_j^2 \Longleftrightarrow u \mid K^2. \tag{3.14}$$

Thus the nonlinear divisor gate can be written entirely in the S6 coordinates, but it does not disappear.

Proof. Equation (3.12) is the direct solution of $12\ell_0 - 4\ell_1 - 3 = p^\varepsilon$. Substitution gives the first two formulas in (3.13), and $T \equiv 3 \pmod 4$ proves that the displayed integer K is well defined. The definitions give

$$4u = 4R_j n - (R_j + p^\varepsilon) = R_j T - p^\varepsilon,$$

so the congruence equation holds for every integral point in (3.12). If a positive integer divides both R_j and u, then the congruence equation makes it divide p^{ε}; (1.2) therefore gives $\gcd(R_j, u) = 1$. Without invoking the selector hypothesis, direct substitution gives

$$4(R_jK - p^{1-\varepsilon}u - a_j) = p^{1-\varepsilon}(R_jT - 4u) - p = 0.$$

Thus $a_j \equiv R_jK \pmod{u}$. Since R_j is a unit modulo u, $a_j^2 \equiv R_j^2K^2 \pmod{u}$, which proves (3.14) in both directions. □

The normalization automatically lies in the admissible residue class used by the S6 terminal presentation:

$$\ell_2 = 1 \text{ is odd}, \qquad \ell_1 \equiv 2 \pmod 3. \tag{3.15}$$

Indeed reduction of $\lambda(\ell_0, \ell_1, 1) = p^{\varepsilon} \equiv 1 \pmod 3$ gives $-\ell_1 \equiv 1 \pmod 3$.

Remark 7.9 (orientation and the constructed S6 candidate). The geometrically reconstructed candidate elsewhere in the audited corpus uses the oriented terminal parameter $\lambda(0, 1, -1) = -1$. The two arithmetic channels in this note use the positive terminal values $+1$ and $+p$. No audited map identifies the -1 terminal datum with either positive datum. Consequently the row shear proved here does not, by itself, place any selector state on the particular complex threefold constructed in the -1 branch.

7.4 The exact remaining arithmetic incidence

The bridge can be expressed without an untyped phrase such as "some divisor of a square." The next factor coordinates are canonical and retain their orientation.

Lemma 7.10 (canonical ordered factor coordinates). *For $a, u \in \mathbb{Z}_{>0}$, the condition $u \mid a^2$ is equivalent to the existence of a unique ordered triple $(r, s, t) \in \mathbb{Z}_{>0}^3$ such that*

$$u = r^2 s, \qquad a = rst, \qquad \gcd(r, t) = 1. \tag{4.1}$$

Explicitly, with $g = \gcd(a, u)$,

$$r = \frac{u}{g}, \qquad s = \frac{g^2}{u}, \qquad t = \frac{a}{g}. \tag{4.2}$$

Proof. Write $v_q(a) = \alpha$ and $v_q(u) = \beta \le 2\alpha$ prime by prime. Then

$$v_q(g) = \min(\alpha, \beta), \qquad v_q(g^2/u) = 2\min(\alpha, \beta) - \beta \ge 0.$$

Thus (4.2) consists of positive integers. If $\beta \le \alpha$, then $q \nmid r$; if $\beta \ge \alpha$, then $q \nmid t$. Hence $\gcd(r, t) = 1$, and direct multiplication gives (4.1).

Conversely, (4.1) gives $a^2/u = st^2 \in \mathbb{Z}_{>0}$. It also gives $\gcd(a, u) = rs$, because $\gcd(r, t) = 1$. Formula (4.2) then recovers (r, s, t), proving uniqueness. □

Proposition 7.11 (factor form of both selector channels). *For the unique factorization $u = r^2 s$, $a_j = rst$, $\gcd(r, t) = 1$, one has*

$$u \in \mathcal{W}_0(p, j) \iff R_j \mid pr + t, \tag{4.3}$$

$$u \in \mathcal{W}_1(p, j) \iff R_j \mid r + t. \tag{4.4}$$

Proof. All of r, s, t are units modulo R_j, because their product is a_j and $\gcd(a_j, R_j) = 1$. For the E-channel,

$$t(4u + 1) = 4r^2st + t = (p + R_j)r + t = (pr + t) + R_jr.$$

Cancelling t modulo R_j proves (4.3). For the M-channel,

$$4u + p \equiv 4(u + a_j) = 4rs(r + t) \pmod{R_j},$$

and cancellation of the unit $4rs$ proves (4.4). □

Theorem 7.12 (necessary-and-sufficient counterexample criterion)**.** *A prime $p \equiv 1 \pmod{12}$ is a counterexample to the Erdős–Straus equation if and only if, for every $j \in J_p$ and every ordered $(r,s,t) \in \mathbb{Z}^3_{>0}$ satisfying*

$$a_j = rst, \qquad \gcd(r,t) = 1,$$

one has simultaneously

$$R_j \nmid pr + t, \qquad R_j \nmid r + t. \tag{4.5}$$

Equivalently, with the tag retained, it is necessary and sufficient that

$$R_j \nmid p^{1-\varepsilon} r + t \tag{4.6}$$

for every $j \in J_p$, every $\varepsilon \in \{0,1\}$, and every such factor triple.

Proof. Lemma 7.10 is a bijection between all allowed divisors in the selector (2.4) and all displayed ordered factor triples. Proposition 7.11 translates the two selector congruences without loss. The assertion now follows directly from (2.12), or equivalently (3.11), by negating the finite existential statement. □

This theorem isolates the unresolved problem exactly: the S6 row shear has not removed the bounded factor-incidence condition (4.5).

7.5 Why the labels and support cannot be forgotten

Lemma 7.13 (full terminal fibres are affine lattices)**.** *The kernel of $\lambda : \mathbb{Z}^3 \to \mathbb{Z}$ is*

$$\ker \lambda = \mathbb{Z}(1,3,0) \oplus \mathbb{Z}(1,0,4). \tag{5.1}$$

If $q \equiv 1 \pmod{12}$ and $m_q = (q+11)/12$, then

$$\lambda^{-1}(q) = \{(m_q + r + s,\ 2 + 3r,\ 1 + 4s) : r,s \in \mathbb{Z}\}. \tag{5.2}$$

In the normalized slice $\ell_2 = 1$, this is an infinite affine line.

Proof. If $12a - 4b - 3c = 0$, reduction modulo 4 gives $c = 4s$. The remaining equation gives $b = 3(a-s)$; putting $r = a - s$ yields $(a,b,c) = r(1,3,0) + s(1,0,4)$, uniquely. Since $\lambda(m_q, 2, 1) = 12m_q - 11 = q$, translation by (5.1) proves (5.2). □

The lemma proves that the mere existence of a terminal parameter with signed value 1 or p is automatic. It cannot imply selector nonemptiness. The following examples show separately why the marker and the divisor-support condition are both part of the theorem's domain.

Example 7.14 (forgetting j and p is noninjective)**.** *For each eligible prime q, define the fixed-prime forgetful map*

$$\mathfrak{f}_q : \mathcal{H}_0(q) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{f}_q(j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

At $p = 13$, the two E-states

$$(j,u,T) = (0,2,3), \qquad (1,5,3)$$

satisfy

$$\Theta_{0,13}(0,2) = (0,1,2), \qquad \Theta_{0,13}(1,5) = (1,1,2),$$

and both images are sent by $\mathfrak{f}_{13}$ to $(1,2,1)$. To compare different prime fibres, form the tagged coproduct and the global forgetful map

$$\mathfrak{F} : \bigsqcup_{\substack{q \text{ prime} \\ q \equiv 1 \pmod{12}}} (\{q\} \times \mathcal{H}_0(q)) \longrightarrow \mathbb{Z}^3, \qquad \mathfrak{F}(q; j, \ell_0, \ell_1) = (\ell_0, \ell_1, 1).$$

The state $(p,j,u) = (73,1,5)$ has $\Theta_{0,73}(1,5) = (1,1,2)$, so its tagged image also maps under $\mathfrak{F}$ to $(1,2,1)$. Therefore neither j nor the external prime p is recoverable after applying the explicitly displayed forgetful maps.

Example 7.15 (ambient positivity does not imply divisor support)**.** *For $p = 13$, $j = 2$, one has $R_j = 11$ and $a_j = 6$. The positive normalized E-line has*

$$(\ell_0, \ell_1) = (n, 3n - 1), \qquad u = 11n - 3,$$

and the positive normalized M-line has

$$(\ell_0, \ell_1) = (n, 3n - 4), \qquad u = 11n - 6.$$

Both lines are infinite. The only positive $u \le a_j^2 = 36$ are

$$E: \ 8, 19, 30, \qquad M: \ 5, 16, 27,$$

and none divides 36. Hence $\mathcal{H}_0(13, 2) = \mathcal{H}_1(13, 2) = \varnothing$, even though both ambient positive S6 lines are nonempty. The prime 13 is solved at other markers, so this is a fixed-shell counterexample to forgetting support, not an Erdős–Straus counterexample.

7.6 The natural unmarked S^6 point and its complete divisor gate

The most direct attempted use of the S6 presentation starts from one point in the signed p-fibre and shears it through every shell. Its exact content can be classified completely.

Write

$$p = 12m - 11, \qquad m = h + 1 \ge 2, \qquad J_p = \{0, \dots, 3m - 4\}. \tag{6.1}$$

Retain the unmarked S6 point

$$\ell^{\mathrm{aff}} = (m, 2, 1), \qquad \lambda(\ell^{\mathrm{aff}}) = p, \tag{6.2}$$

and for $j \in J_p$ put

$$T_0 = 4m - 1, \quad u_j = 2 + jT_0, \quad a_j = 3m + j - 2, \quad R_j = 4j + 3. \tag{6.3}$$

Then

$$R_j T_0 = 4u_j + p, \qquad R_j m = a_j + u_j, \qquad T_0 a_j - u_j = pm, \tag{6.4}$$

and

$$\mathsf{U}_j \begin{pmatrix} R_j & -u_j \\ 4 & -T_0 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 4 & 1 - 4m \end{pmatrix} = \mathsf{R}(m, 2, 1). \tag{6.5}$$

Theorem 7.16 (complete affine-line gate)**.** *For prime $p = 12m - 11$, $m \ge 2$, and $0 \le j < 3m - 3$,*

$$u_j \mid a_j^2 \iff (j = 0 \text{ and } 2 \mid m) \text{ or } (j = 2 \text{ and } 8 \mid m). \tag{6.6}$$

Consequently the single unmarked point (6.2), with all markers restored, produces an Erdős–Straus witness if and only if $p \equiv 13 \pmod{24}$. In that case $j = 0, u = 2, K = m$ gives

$$\frac{4}{p} = \frac{1}{3m - 2} + \frac{1}{pm} + \frac{1}{pm(3m - 2)/2}. \tag{6.7}$$

For $p \equiv 1 \pmod{24}$, this affine S6 point supplies no supported selector state at any $j \in J_p$.

Proof. The identities (6.4) and (6.5) follow by direct expansion. Assume first that $u_j \mid a_j^2$, and put $g = \gcd(u_j, a_j)$. Write $u_j = gu'$ and $a_j = ga'$ with $\gcd(u', a') = 1$. The assumed divisibility gives $u' \mid g(a')^2$, hence $u' \mid g$, and therefore

$$u_j \mid g^2. \tag{6.8}$$

The last identity in (6.4) gives $g \mid pm$. Since $0 < a_j < p$, the primality of p and $g \mid a_j$ imply $p \nmid g$; therefore $g \mid m$. Reducing $u_j = 2 + j(4m-1)$ modulo m now gives

$$g \mid j - 2. \tag{6.9}$$

Suppose $j \neq 0, 2$. Then $j \geq 1$, and (6.8)–(6.9) imply $u_j \leq g^2 \leq (j-2)^2$. But

$$u_j - (j-2)^2 = j(4m + 3 - j) - 2 > 0 \tag{6.10}$$

For $m \geq 2$ and $1 \leq j < 3m - 3$, indeed $4m + 3 - j \geq m + 7$, so the right-hand side of (6.10) is at least $m + 5 > 0$. This is a contradiction. Thus only $j = 0$ or $j = 2$ can occur.

At $j = 0$, $u_0 = 2$ and $a_0 = 3m - 2$, so $u_0 \mid a_0^2$ exactly when m is even. At $j = 2$, $u_2 = 8m$ and $a_2 = 3m$, so

$$u_2 \mid a_2^2 \iff 8m \mid 9m^2 \iff 8 \mid m.$$

These calculations also prove both converse implications in (6.6).

Now $p = 12m - 11$ is congruent to 13 (mod 24) exactly when m is even. The $j = 0$ state lies in the M-channel, and (2.8) gives $T = T_0$ and $K = (T_0 + 1)/4 = m$. Formula (2.10) is precisely (6.7). If m is odd, (6.6) excludes every marker on this particular S6 point. □

The last sentence is deliberately local to (6.2). For example, $p = 73$ has $m = 7$ odd, so the affine point contributes nothing; nevertheless $(j, u, T; \ell_0, \ell_1) = (1, 1, 11; 3, -10)$ is an M-state. Thus failure of the affine point is not failure of the full selector.

7.7 Exact descendant maps transported through the bridge

The bridge also transports a verified selector witness along multiplicative descendant rays. This is a map from an already nonempty domain, not a nonemptiness theorem.

Fix a supported state at (p, j), retain $R = R_j$, and let $b \in \mathbb{Z}_{>0}$. Define

$$a_b = ba_j, \qquad p_b = 4a_b - R = bp + (b-1)R. \tag{7.1}$$

Whenever p_b is again a prime congruent to 1 (mod 12), the same marker j belongs to J_{p_b}, since $R < p \leq p_b$, and the following maps have the displayed domains and codomains:

$$\Phi^E_{b;p} : \mathcal{H}_0(p, j) \longrightarrow \mathcal{H}_0(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, \ell_0, \ell_1), \tag{7.2}$$

$$\Phi^M_{b;p} : \mathcal{H}_1(p, j) \longrightarrow \mathcal{H}_1(p_b, j), \qquad (j, \ell_0, \ell_1) \longmapsto (j, b\ell_0, b\ell_1 - j(b-1)). \tag{7.3}$$

Write

$$\theta_{\varepsilon,p,j} : \mathcal{W}_\varepsilon(p, j) \longrightarrow \mathcal{H}_\varepsilon(p, j), \qquad \theta_{\varepsilon,p,j}(u) = \Theta_{\varepsilon,p}(j, u)$$

for the fixed-marker restriction of the marked bridge.

Proposition 7.17 (intertwining of descendant actions)**.** *The selector maps corresponding to (7.2) and (7.3) are the typed maps*

$$\psi^E_{b;p} : \mathcal{W}_0(p, j) \longrightarrow \mathcal{W}_0(p_b, j), \qquad u \longmapsto u,$$

$$\psi^M_{b;p} : \mathcal{W}_1(p, j) \longrightarrow \mathcal{W}_1(p_b, j), \qquad u \longmapsto bu.$$

They intertwine the fixed-marker bridges exactly:

$$\Phi^\bullet_{b;p} \circ \theta_{\varepsilon,p,j} = \theta_{\varepsilon,p_b,j} \circ \psi^\bullet_{b;p}, \qquad (\bullet, \varepsilon) \in \{(E, 0), (M, 1)\}. \tag{7.4}$$

Equivalently,

$$\theta^{-1}_{0,p_b,j} \Phi^E_{b;p} \theta_{0,p,j}(u) = u, \qquad \theta^{-1}_{1,p_b,j} \Phi^M_{b;p} \theta_{1,p,j}(u) = bu.$$

On the accompanying arithmetic metadata these maps replace (a_j, p, u) *by* (a_b, p_b, u) *and* (a_b, p_b, bu)*, respectively. For the latter,*

$$T_b = bT + b - 1, \qquad u_b = bu, \qquad K_b = bK, \tag{7.5}$$

and

$$\lambda(b\ell_0, b\ell_1 - j(b-1), 1) = p_b. \tag{7.6}$$

For either tag the maps are injective. For $c \in \mathbb{Z}_{>0}$*, put*

$$a_{bc} = bc\, a_j, \qquad p_{bc} = 4a_{bc} - R = bc\, p + (bc-1)R.$$

Whenever p_b *and* p_{bc} *are primes congruent to* $1 \pmod{12}$*, the displayed maps are composable and satisfy*

$$\Phi^{\bullet}_{c;p_b} \circ \Phi^{\bullet}_{b;p} = \Phi^{\bullet}_{bc;p} \tag{7.7}$$

Proof. In the E-channel, $u \mid a_j^2$ implies $u \mid a_b^2$, and the congruence $R \mid 4u+1$ and quotient T are unchanged. Hence ℓ_0, ℓ_1 are unchanged by (3.4).

In the M-channel, $bu \mid b^2 a_j^2 = a_b^2$, and

$$4bu + p_b = b(4u + p) + (b-1)R.$$

Thus the descendant is a selector state and its quotient is $T_b = bT + b - 1$. Since $K = (T+1)/4$ in this channel, $K_b = (T_b + 1)/4 = bK$. Applying (3.4) gives

$$\ell_{0,b} = \frac{T_b + 1}{4} = b\ell_0, \qquad \ell_{1,b} = bu - jT_b = b\ell_1 - j(b-1),$$

which proves (7.3)–(7.6), including the intertwining identity (7.4). Each map is visibly injective for fixed b. Substitution into (7.1)–(7.3), with the source prime retained, proves the typed composition law (7.7). □

Reverse descent need not preserve divisor support: the factor b can introduce divisors absent from a proposed parent. Therefore (7.2)–(7.7) propagate witnesses but cannot create the first witness on an empty ray.

7.8 Sharp verdict

The strongest theorem supported by the audited data is Theorem 7.5 together with the exhaustive equivalence (3.11): the complete tagged Erdős–Straus selector for a fixed prime $p \equiv 1 \pmod{12}$ is exactly the marked divisor-supported S6 terminal slice, and the two integral relation matrices have the same row lattice. The exact remaining obstruction is (4.5).

Three stronger readings are false or unsupported.

1. The abstract group $G(\ell)$, its determinant, or an unmarked terminal triple does not determine selector support. Every E-state has the same trivial presented group, every M-state has group $\mathbb{Z}/p\mathbb{Z}$, and Examples 7.14 and 7.15 exhibit the information lost by forgetting the marker and divisor gate.

2. The row shear is not a geometric map from the reciprocal cubic $4xyz = p(xy + xz + yz)$ to the candidate complex threefold X. No map of varieties or analytic spaces, no cycle map, no attaching map, and no cohomological correspondence is defined by left multiplication of a relation matrix.

3. The existence of the full S6 affine fibre cannot prove universal Erdős–Straus positivity. Those fibres are always nonempty by (5.2), whereas selector nonemptiness is exactly the additional finite nonlinear incidence in (4.5). The natural affine point is completely exhausted by Theorem 7.16 and contributes nothing to the unresolved $p \equiv 1 \pmod{24}$ half.

Accordingly, the bridge is a genuine theorem and is stronger than a shared determinant or a heuristic analogy. It is also exact enough to prove its own boundary: without a new argument forcing one of the bounded factor congruences in (4.5), it neither proves nor disproves the Erdős–Straus conjecture.

8 The analytic threefold constructed in the candidate

This section audits the construction itself. It does not infer existence from the later topology. The period family, the three local fillings, and the gluing are proved in that order. The notation is the candidate's: $B \simeq \mathbb{P}^1$, $B^\circ = B \setminus \{p_0, p_1, p_2\}$, and the local monodromy orders at p_1, p_2 are $m_1 = 3, m_2 = 4$.

8.1 The period family

Let $\Delta \simeq (\mathbb{Z}/3) * (\mathbb{Z}/4)$ be the orientation-preserving $(3, 4, \infty)$-triangle group acting on a simply connected Riemann surface $\mathfrak{h}_z$, with elliptic generators g_1, g_2 and $g_0 = (g_1 g_2)^{-1}$. Let t be the affine coordinate on B with values $0, 1, \infty$ at p_1, p_2, p_0, respectively. Write

$$\pi : \mathfrak{h}_z \longrightarrow B \setminus \{p_0\}$$

for the orbifold quotient map; its stabilizers at chosen lifts $z_j \in \pi^{-1}(p_j)$ are $\langle g_j \rangle$. Let $\rho_V : \Delta \to \mathrm{GL}(V)$ be the printed representation, $\rho_V(g_j) = T_j$, and put

$$M_g := \rho_V(g)^{\mathsf{t}}, \qquad A_g := M_g^{-1}.$$

Thus $M_{gh} = M_h M_g$, $A_{g_j} = A_j$, and $A_{g_0} = M_0$ in the notation of theorem 5.1.

Lemma 8.1 (simultaneous elliptic-orientation normalization)**.** ***[VERIFIED]*** *The generators may, and henceforth do, satisfy*

$$g_1'(z_1) = e^{-2\pi i/3}, \qquad g_2'(z_2) = e^{-2\pi i/4} = -i, \tag{1}$$

while $g_0 = (g_1 g_2)^{-1}$ remains parabolic. More precisely, the Cayley coordinates

$$s_j(z) = \frac{z - z_j}{z - \bar{z}_j} \tag{2}$$

obey

$$s_1 \circ g_1 = e^{-2\pi i/3} s_1, \qquad s_2 \circ g_2 = -i s_2. \tag{3}$$

The two signs in equation (1) *are a simultaneous orientation choice; they are not obtained by shrinking a coordinate disc.*

Proof. Realize Δ as the orientation-preserving subgroup of the reflection group of a hyperbolic triangle with counterclockwise vertices A, B, C, angles $\pi/3, \pi/4, 0$, and opposite sides a, b, c, respectively. Let $\sigma_a, \sigma_b, \sigma_c$ denote reflection in those sides and put

$$x = \sigma_b \sigma_c, \qquad y = \sigma_c \sigma_a, \qquad z = \sigma_a \sigma_b.$$

The product of reflections in two oriented geodesics meeting at angle θ is the rotation through 2θ. Hence x is the counterclockwise rotation about A through $2\pi/3$, y is the counterclockwise rotation about B through $\pi/2$, and z is parabolic at the ideal vertex C; this is the standard triangle-group construction [26, Chapters 7 and 10]. Moreover

$$xyz = \sigma_b \sigma_c \sigma_c \sigma_a \sigma_a \sigma_b = 1.$$

Set

$$z_1 = A, \qquad z_2 = B, \qquad g_1 = x^{-1}, \qquad g_2 = y^{-1}.$$

Both finite rotations are thereby reversed, so their complex derivatives at their fixed points are exactly those in equation (1). Since $g_2 g_1 = y^{-1} x^{-1} = z$, one also has

$$g_1 g_2 = g_1 z g_1^{-1};$$

thus g_1g_2, and therefore $g_0 = (g_1g_2)^{-1}$, is parabolic.

This simultaneous reversal is forced once the orientation of g_1 and the cusp conjugacy class are fixed. Indeed, in $\langle x\rangle * \langle y\rangle \simeq C_3 * C_4$ every parabolic element is conjugate to a nonzero power of the primitive cusp word $z = y^{-1}x^{-1} = y^3x^2$. The two mixed choices would instead give

$$(x^{-1}y)^{-1} = y^{-1}x = y^3x, \qquad (xy^{-1})^{-1} = yx^{-1} = yx^2.$$

Both words are cyclically reduced of length two. A nontrivial power z^n has cyclically reduced length $2|n|$; hence a mixed word could be parabolic only for $|n| = 1$. But neither y^3x nor yx^2 is a cyclic permutation of y^3x^2 or its inverse xy. Thus neither mixed choice is parabolic. The derivative selected at z_1 consequently fixes the derivative at z_2 as well.

Finally, for an elliptic Möbius transformation g fixing $z_0 \in \mathfrak{h}_z$, the Cayley map $s = (z - z_0)/(z - \bar{z}_0)$ conjugates g to a disc automorphism fixing zero. Such an automorphism is $s \mapsto g'(z_0)s$. Applying this to g_1, g_2 proves equation (3). □

Proposition 8.2 (existence and non-degeneracy of the period matrix)**.** ***[VERIFIED]*** *There are holomorphic functions*

$$\tau : \mathfrak{h}_z \to \mathfrak{h}, \qquad \mu, \beta : \mathfrak{h}_z \to \mathbb{C}$$

satisfying

$$\begin{array}{lll} \tau \circ g_0 = \tau - 1, & \mu \circ g_0 = \mu, & \beta \circ g_0 = \beta + 1, \\ \tau \circ g_2 = -1/\tau, & \mu \circ g_2 = 1 + \mu/\tau, & \beta \circ g_2 = \beta - 3 - 6\mu^2/\tau, \end{array} \tag{2.1}$$

and the corresponding order-three transformation laws obtained from $\tau \circ g_1 = (\tau - 1)/\tau$. They may be chosen so that

$$\operatorname{Im}\tau > 0, \qquad D := \operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} < 0. \tag{2.2}$$

Consequently

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix} \tag{2.3}$$

maps the marked rank-four lattice Λ isomorphically onto a lattice in $\mathbb{C}^2$ at every point. The quotients $\mathbb{C}^2/\Pi(z)\Lambda$ form a holomorphic family of compact two-dimensional complex tori over $\mathfrak{h}_z$, equivariant for Δ.

Proof. Normalize the modular function by $j(q) = q^{-1}(1 + O(q))$ and put $J_B = 1728(t \circ \pi)$; the modular facts used here are collected in [25, Chapter VII]. Put

$$\mathfrak{h}_z^\dagger = \mathfrak{h}_z \setminus J_B^{-1}\{0, 1728\}, \qquad \mathfrak{h}_\tau^\dagger = \mathfrak{h} \setminus j^{-1}\{0, 1728\}.$$

Away from the deleted sets, the classical modular map $j : \mathfrak{h}_\tau^\dagger \to \mathbb{C} \setminus \{0, 1728\}$ is an unramified covering with deck group $\mathrm{PSL}_2(\mathbb{Z})$. The deleted set in $\mathfrak{h}_z$ is locally finite. Since $\mathfrak{h}_z$ is simply connected, $\pi_1(\mathfrak{h}_z^\dagger)$ is normally generated by small positive meridians about its points. At a translate of z_1, J_B has local degree three over 0; the corresponding meridian therefore has deck image equal to the third power of the order-three modular generator. At a translate of z_2, J_B has local degree four over 1728; its meridian has deck image equal to the fourth power of the order-two modular generator. Both images are the identity. The covering-space lifting criterion gives a global lift τ with $j(\tau) = J_B$.

Near a deleted point, compose the lift with a biholomorphism $\chi : \mathfrak{h} \to \mathbb{D}$. The bounded map $\chi \circ \tau$ extends by the removable-singularity theorem. Its value at the puncture has modulus strictly less than one: equality would give an interior maximum and force $\chi \circ \tau$ to be constant, although its punctured values lie in $\mathbb{D}$. Thus the extension still lands in $\mathfrak{h}$.

Because J_B is Δ-invariant, $\tau \circ g$ and τ are lifts of the same map through the Galois covering j on the punctured surface. Hence there is a unique deck transformation $\varphi(g) \in \mathrm{PSL}_2(\mathbb{Z})$ such that

$$\tau \circ g = \varphi(g) \circ \tau. \tag{4}$$

The identity extends across the deleted points. Uniqueness and $(\tau\circ g)\circ h=\tau\circ(gh)$ give $\varphi(gh)=\varphi(g)\varphi(h)$, so φ is a homomorphism.

Normalize $\tau(z_1)=\rho=e^{\pi i/3}$. The local cubic expansion of j at ρ, together with the local degree three of J_B, gives a simple zero of $\tau-\rho$ at z_1. At z_2, choose $\gamma\in\mathrm{PSL}_2(\mathbb{Z})$ with $\tau(z_2)=\gamma(i)$. The local quadratic expansion of $j\circ\gamma$ at i, together with the local degree four of J_B, gives a double zero of $\gamma^{-1}\tau-i$ at z_2. At z_1, differentiating equation (4) and using the simple zero gives

$$\varphi(g_1)'(\rho)=g_1'(z_1)=e^{-2\pi i/3}.$$

The stabilizer of ρ is generated by $S_1(\tau)=(\tau-1)/\tau$, whose derivative at ρ is $e^{-2\pi i/3}$; hence $\varphi(g_1)=S_1$, rather than S_1^{-1}. The order-four generator cannot have trivial modular image: otherwise the invariant local power series $\gamma^{-1}\tau-i$ would have order divisible by four, contrary to its order two. Its modular image therefore has order two. If $P=\varphi(g_0)$, the exact relation $g_1g_2g_0=1$ gives

$$\varphi(g_2)=S_1^{-1}P^{-1}. \tag{5}$$

It remains to determine the cusp image. Choose a punctured cusp disc $D_0^*\subset B$ and a component

$$U_0\subset\pi^{-1}(D_0^*)$$

stabilized by $\langle g_0\rangle$. Choose R and shrink D_0^* so that $\tau(U_0)$ lies in a single component

$$K\subset j^{-1}(\{|w|>R\}).$$

Put $P=\varphi(g_0)$. Since $g_0U_0=U_0$,

$$P\tau(U_0)=\tau(g_0U_0)=\tau(U_0).$$

Hence $P(K)=K$: the two components K and $P(K)$ intersect in the nonempty set $\tau(U_0)$. Let c be the rational boundary cusp of K, choose $\gamma_0\in\mathrm{PSL}_2(\mathbb{Z})$ with $\gamma_0(\infty)=c$, and set

$$\tau'=\gamma_0^{-1}\tau|_{U_0},\qquad P'=\gamma_0^{-1}P\gamma_0.$$

Then

$$\tau'\circ g_0=P'\circ\tau',$$

and P' preserves $K'=\gamma_0^{-1}K$, whose boundary cusp is ∞.

Suppose $P'=1$. Then τ' is g_0-invariant and descends to a holomorphic map $\bar\tau':D_0^*\to\mathfrak{h}$. Put

$$q'=\exp(2\pi i\bar\tau').$$

After increasing R, the q-expansion is valid on K', and $0<|q'|<1$. Thus q' extends holomorphically across the puncture. If $|q'(0)|=1$, the maximum-modulus principle makes q', hence J_B, constant. If $q'(0)\neq0$, the expansion $j(q')=(q')^{-1}(1+O(q'))$ makes J_B bounded, contrary to its pole. Therefore $q'(0)=0$. But a holomorphic zero has positive winding on a small positive circle, while q' has the single-valued holomorphic logarithm $2\pi i\bar\tau'$ on D_0^*, so its winding is zero. This contradiction proves $P'\neq1$. Since P' preserves the selected cusp component K', it is parabolic; hence the original P is a nontrivial parabolic.

We now discard the auxiliary conjugation by γ_0 and return to the original normalization

$$\tau(z_1)=\rho,\qquad\varphi(g_1)=S_1.$$

Choose the trace-+2 lift $\widetilde P\in\mathrm{SL}_2(\mathbb{Z})$ of P. For a primitive fixed vector $(p,r)^{\mathsf t}$, it has the unique form

$$\widetilde P=I+k\begin{pmatrix}-pr & p^2\\ -r^2 & pr\end{pmatrix},\qquad p,r\in\mathbb{Z},\quad\gcd(p,r)=1,\quad k\neq0.$$

With

$$\widetilde{S}_1 = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix},$$

one has

$$\operatorname{tr}(\widetilde{S}_1^{-1}\widetilde{P}^{-1}) = 1 + k(p^2 - pr + r^2).$$

Since $S_1^{-1}P^{-1} = \varphi(g_2)$ has projective order two, this trace is zero. Hence

$$-k(p^2 - pr + r^2) = 1,$$

so $k = -1$ and $p^2 - pr + r^2 = 1$. Modulo simultaneous sign, the three solutions are cycled transitively by S_1. Conjugating the entire deck representation by a suitable power of S_1, which fixes ρ and leaves S_1 unchanged, therefore gives

$$P(w) = w - 1, \qquad \varphi(g_2)(w) = -\frac{1}{w}.$$

Rename the resulting period map τ. Since z_2 is fixed by the order-four generator, the new normalization has $\tau(z_2) = i$, and the preceding local ramification calculation says exactly that $\tau - i$ has order two at z_2. Let t_c vanish at p_0 and define anew

$$q = e^{2\pi i\tau}.$$

Since $\tau \circ g_0 = \tau - 1$, q descends to D_0^*. The simple pole of $J_B = j(\tau)$ gives

$$q = t_c u_1(t_c), \qquad u_1(0) \neq 0.$$

Writing

$$S_1(w) = \frac{w-1}{w}, \qquad S(w) = -\frac{1}{w}, \qquad P(w) = w - 1,$$

the normalized deck certificate is therefore

$$\varphi(g_1) = S_1, \qquad \varphi(g_2) = S, \qquad \varphi(g_0) = P, \qquad S_1SP = 1 \quad \text{in } \mathrm{PSL}_2(\mathbb{Z}). \tag{6}$$

Differentiation at z_1, where $\tau - \rho$ has order one, gives $g_1'(z_1) = S_1'(\rho) = \rho^{-2} = e^{-2\pi i/3}$. At z_2, comparison of the quadratic leading terms gives only

$$\bigl(g_2'(z_2)\bigr)^2 = S'(i) = -1.$$

The choice $g_2'(z_2) = -i$, rather than $+i$, is supplied by the simultaneous parabolic normalization in theorem 8.1; it does not follow from the quadratic ramification calculation alone.

We now solve the two affine cocycle problems, including the orbifold and cusp regularity which the abbreviated argument had suppressed. Define $\widetilde{\jmath} : \Delta \times \mathfrak{h}_z \to \mathbb{C}^*$ on the generators by

$$\widetilde{\jmath}(g_1, z) = -\tau(z), \qquad \widetilde{\jmath}(g_2, z) = \tau(z),$$

and extend it by $\widetilde{\jmath}(gh, z) = \widetilde{\jmath}(g, hz)\widetilde{\jmath}(h, z)$. This is well defined: the products around the two finite cyclic subgroups are

$$(-1)^3\tau\frac{\tau-1}{\tau}\frac{-1}{\tau-1} = 1, \qquad \tau\left(-\frac{1}{\tau}\right)\tau\left(-\frac{1}{\tau}\right) = 1.$$

Moreover $\widetilde{\jmath}(g_0, \cdot) = 1$.

For an open $U \subset B$, set $\widetilde{U} = \pi^{-1}(U \setminus \{p_0\})$. Define

$$\mathcal{L}(U) = \Bigl\{\nu \in \mathcal{O}(\widetilde{U}) : \nu \circ g = \nu/\widetilde{\jmath}(g, \cdot) \quad (g \in \Delta)\Bigr\}$$

and let $\mathcal{M}(U)$ be the holomorphic functions on $\widetilde{U}$ satisfying

$$\mu \circ g_1 = \frac{1-\mu}{\tau}, \qquad \mu \circ g_2 = 1 + \frac{\mu}{\tau}. \tag{2.1a}$$

If $p_0 \in U$, both definitions additionally require that the restriction to the distinguished $\langle g_0 \rangle$-invariant cusp component, viewed as a function of t_c, extend holomorphically across $t_c = 0$. Because $\widetilde{\jmath}(g_0) = 1$ and $\mu \circ g_0 = \mu$, this requirement is meaningful and is stable under restriction. Thus $\mathcal{L}$ is a line sheaf and $\mathcal{M}$, if locally nonempty, is an affine $\mathcal{L}$-torsor.

Local nonemptiness is explicit. Over an ordinary small disc, prescribe $\mu = 0$ on one sheet and extend using (2.1a). At the cusp prescribe $\mu = 0$ on the distinguished component. At the elliptic points use

$$\mu_1 = \frac{2-\tau}{3} \quad \text{on } \Delta_1, \qquad \mu_2 = \frac{1-\tau}{2} \quad \text{on } \Delta_2. \tag{2.1b}$$

Indeed

$$\mu_1 \circ g_1 = \frac{\tau+1}{3\tau} = \frac{1-\mu_1}{\tau}, \qquad \mu_2 \circ g_2 = \frac{\tau+1}{2\tau} = 1 + \frac{\mu_2}{\tau}.$$

The cyclic products above, or direct iteration of (2.1a), show that these prescriptions close after three and four steps.

To identify $\mathcal{L}$, use that E_6 has a simple zero at the order-two elliptic orbit and no other zero in $\mathfrak{h}$. Since $\tau - i$ has order two at every translate of z_2, the function $E_6 \circ \tau$ has only even-order zeros on the simply connected surface $\mathfrak{h}_z$. Fix a holomorphic square root h, $h^2 = E_6 \circ \tau$, and put

$$\Theta = \frac{(E_4 \circ \tau)^2 h}{\Delta_{\mathrm{mod}} \circ \tau}.$$

For $j = 1, 2$, the quotient $h(g_j z)/(\tau(z)^3 h(z))$ has square one away from its removable zero set, hence is a constant sign. At z_1, $h(z_1) \neq 0$ and $\rho^3 = -1$, so the sign for g_1 is -1. At z_2, choose the linearizing coordinate with $s_2 \circ g_2 = -i s_2$. The function h has a simple zero there, and therefore $h(g_2 z)/h(z) \to -i = i^3$; the sign for g_2 is $+1$. The modular transformation formulae

$$E_4(\gamma\tau) = (c\tau + d)^4 E_4(\tau), \quad E_6(\gamma\tau) = (c\tau + d)^6 E_6(\tau), \quad \Delta_{\mathrm{mod}}(\gamma\tau) = (c\tau + d)^{12} \Delta_{\mathrm{mod}}(\tau)$$

now give the exact identities

$$\Theta \circ g_1 = -\Theta/\tau, \qquad \Theta \circ g_2 = \Theta/\tau. \tag{2.1e}$$

These are precisely the automorphy laws defining $\mathcal{L}$. Since E_4 has a simple zero at ρ, whereas E_6 has a simple zero at i, the orders of Θ at z_1, z_2 are respectively two and one. At the distinguished cusp, E_4, E_6 are units and $\Delta_{\mathrm{mod}} = q \cdot (\text{unit})$, with $q = t_c \cdot (\text{unit})$, so $\Theta = t_c^{-1} \cdot (\text{unit})$. Taking the stabilizer orders into account, division by Θ identifies the ordinary and elliptic stalks with $\mathcal{O}_B$, while the cusp stalk is $\mathfrak{m}_{p_0}\Theta$. Hence

$$\mathcal{L} \simeq \mathcal{O}_B(-p_0) \simeq \mathcal{O}_{\mathbb{P}^1}(-1).$$

Consequently $H^1(B, \mathcal{L}) = H^0(B, \mathcal{L}) = 0$: the locally nonempty torsor $\mathcal{M}$ has a unique global section. Evaluating at the fixed points gives

$$\mu(z_1) = \frac{2-\rho}{3}, \qquad \mu(z_2) = \frac{1-i}{2},$$

and the cusp condition says precisely that its invariant branch extends holomorphically.

For the second affine problem put

$$\phi_1 = 2 - \frac{6(1-\mu)^2}{\tau}, \qquad \phi_2 = -3 - \frac{6\mu^2}{\tau}.$$

Substitution around the finite orbits gives

$$\sum_{k=0}^{2} \phi_1 \circ g_1^k = 0, \qquad \sum_{k=0}^{3} \phi_2 \circ g_2^k = 0. \tag{2.1c}$$

For example, the three summands in the first identity are

$$2 - \frac{6(1-\mu)^2}{\tau}, \quad 2 - \frac{6(\tau-1+\mu)^2}{\tau(\tau-1)}, \quad 2 + \frac{6\mu^2}{\tau-1},$$

Putting them over the common denominator $\tau(\tau-1)$, the three constant terms contribute $6\tau(\tau-1)$, while the remaining terms contribute $-6\tau(\tau-1)$; hence the total numerator is zero. For the order-four identity the bracket after extracting $6/\tau$ is 2τ, cancelling -12.

Define $\mathcal{B}(U)$ to consist of holomorphic β on $\widetilde{U}$ with $\beta \circ g_j - \beta = \phi_j$; at the cusp require $b = \beta + \tau$ to extend holomorphically in t_c. Differences of sections are ordinary holomorphic functions on U, so $\mathcal{B}$ is a pseudo-torsor under $\mathcal{O}_B$. It is locally nonempty: prescribe $\beta = 0$ on one ordinary sheet; prescribe $\beta = -\tau$ on the distinguished cusp component; and at p_j use

$$\beta_j = \frac{1}{m_j} \sum_{k=0}^{m_j-1} k\,(\phi_j \circ g_j^k). \tag{2.1d}$$

Equation (2.1c) makes the last expression telescope to $\beta_j \circ g_j - \beta_j = \phi_j$. Therefore $\mathcal{B}$ is an $\mathcal{O}_B$-torsor. Since $H^1(B, \mathcal{O}_B) = 0$, it has global sections, and any two differ by a constant; by construction $b = \beta + \tau$ is invariant and holomorphic at the cusp.

The real determinant is

$$\det_{\mathbb{R}} \Pi = -\det \operatorname{Im} Z = \operatorname{Im} \tau\, D. \tag{2.4}$$

For completeness, the equivariance can be checked before any quotient is taken. Regard the rows of Π as the two covectors

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta \quad \text{in } V \otimes \mathbb{C}. \tag{2.3a}$$

Multiplication by the printed matrices gives

$$\begin{aligned} T_1\sigma_1 &= (6\mu - 6)\gamma + (1-\tau)u - \tau w, \\ T_1\sigma_2 &= (\beta + 2)\gamma + (1-\mu)u + (1-\mu)w + \delta, \\ T_2\sigma_1 &= (6\mu + 6\tau)\gamma - u + \tau w, \\ T_2\sigma_2 &= (\beta + 6\mu - 3)\gamma + u + \mu w + \delta. \end{aligned} \tag{2.3b}$$

Using the four affine transformation laws in (2.1a)–(2.1d), coefficient comparison in (2.3b) yields

$$\begin{aligned} \sigma_1 \circ g_1 &= -\tau^{-1} T_1\sigma_1, \quad & \sigma_2 \circ g_1 &= T_1\sigma_2 + (1-\mu)\tau^{-1} T_1\sigma_1, \\ \sigma_1 \circ g_2 &= \tau^{-1} T_2\sigma_1, & \sigma_2 \circ g_2 &= T_2\sigma_2 - \mu\tau^{-1} T_2\sigma_1. \end{aligned} \tag{2.3c}$$

For instance, dividing the first line of (2.3b) by $-\tau$ gives the coefficients $6(1-\mu)/\tau$, $(\tau-1)/\tau$, and 1, exactly those of $\sigma_1(g_1 z)$; subtracting $(1-\mu)$ times that normalized row from $T_1\sigma_2$ gives $\sigma_2(g_1 z)$. The two g_2 identities follow by the same coefficient check. Thus, with

$$R_{g_1} = \begin{pmatrix} -1/\tau & 0 \\ (1-\mu)/\tau & 1 \end{pmatrix}, \qquad R_{g_2} = \begin{pmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{pmatrix},$$

pairing (2.3c) with $\lambda \in \Lambda$ gives

$$\Pi(gz) = R_g(z)\Pi(z)M_g. \tag{2.3d}$$

Since $M_{gh} = M_h M_g$, uniqueness of the left factor gives $R_{gh}(z) = R_g(hz)R_h(z)$; hence these formulae define an action, not merely generator-wise maps. Taking real determinants in (2.3d), using $\det M_g = 1$ and $\operatorname{Im}\tau(g_j z) = \operatorname{Im}\tau/|\tau|^2$, shows $D(gz) = D(z)$. At the cusp,

$$D = \operatorname{Im} b - \operatorname{Im}\tau - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} \longrightarrow -\infty.$$

The descended function is bounded above on the complement of a cusp neighbourhood. Replacing β by $\beta + c_0$ adds $\operatorname{Im} c_0$ to D, so a sufficiently negative imaginary constant makes (2.2) hold globally. Equation (2.4) then makes Π a real isomorphism everywhere. The equivariance identity supplies the claimed Δ-action and completes the construction of the period family. □

For the cusp filling, record the local periods explicitly. Let t_c be the coordinate vanishing at p_0. The simple pole of j gives

$$q = e^{2\pi i\tau} = t_c u_1(t_c), \qquad u_1(0) \neq 0.$$

After choosing a logarithm of u_1, put

$$h = (2\pi i)^{-1}\log u_1, \qquad s = \tau - h, \qquad b = \beta + \tau.$$

Then $e^{2\pi i s} = t_c$, $s \circ g_0 = s - 1$, and μ, b, h are holomorphic functions of t_c at the origin. Consequently

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z = sB_0 + C(t_c), \qquad C(t_c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b(t_c) - h(t_c) & \mu(t_c) \end{pmatrix}, \tag{2.5a}$$

and C is holomorphic on a full cusp disc. In particular, for $\bar\lambda \in \bar\Lambda \simeq \mathbb{Z}^2$,

$$\exp(2\pi i Z\bar\lambda) = t_c^{B_0\bar\lambda} c_{\bar\lambda}(t_c), \qquad c_{\bar\lambda}(t_c) = \exp(2\pi i C(t_c)\bar\lambda). \tag{2.5b}$$

8.2 The toric filling at the cusp

Let $\mathcal{T}_{A_2}$ be the translation-invariant triangulation of $\mathbb{R}^2$ by the two triangles $\{v, v+e_1, v+e_2\}$ and $\{v+e_1, v+e_2, v+e_1+e_2\}$. In $N' = \mathbb{Z}^2 \oplus \mathbb{Z}$, take the cones over all cells at height one, and denote the associated toric threefold by Y. We use the standard orbit–cone and affine-chart constructions of [22, Chapters 1 and 3]; all nonstandard assertions caused by the infinite fan are proved directly below.

For $\bar\lambda \in \bar\Lambda$, define the integral linear map

$$\varphi_{\bar\lambda}(y, y_3) = (y + y_3 B_0\bar\lambda, y_3).$$

On the height-one slice this is translation by $B_0\bar\lambda \in \mathbb{Z}^2$, so it preserves the triangulation and maps the cone over a cell P to the cone over $P + B_0\bar\lambda$. It therefore induces a toric automorphism $\Phi_{\bar\lambda}$ of Y, satisfying

$$\Phi_{\bar\lambda}(x, t) = (t^{B_0\bar\lambda}x, t)$$

on the dense torus. Because t and $C(t)$ are holomorphic on the full cusp model, multiplication by $(c_{\bar\lambda}(t), 1) \in (\mathbb{C}^*)^3$ defines a biholomorphism on all of Y_ϵ. Set

$$\Psi_{\bar\lambda}(p) = (c_{\bar\lambda}(t(p)), 1) \cdot \Phi_{\bar\lambda}(p).$$

The maps $\Phi_{\bar\lambda}$ fix the first two cocharacter directions and preserve t, so they commute with this torus multiplication. The linearity of $C(t)\bar\lambda$ gives $c_{\bar\lambda}c_{\bar\mu} = c_{\bar\lambda+\bar\mu}$; hence $\Psi_{\bar\lambda}\Psi_{\bar\mu} = \Psi_{\bar\lambda+\bar\mu}$. This proves, before taking any quotient, both the holomorphic extension and the action law used below.

For a triangle $\sigma = \{v_0, v_1, v_2\}$, let θ_i^σ be its barycentric coordinates and let $z_i = \chi^{m_i}$ be the coordinates dual to $(v_i, 1)$. Set

$$\sigma^{(\eta)} = \{y : \theta_i^\sigma(y) > -\eta \text{ for } i = 0, 1, 2\}, \quad \mathcal{U}_\sigma(S) = \{|z_i| < S \text{ for all } i\}.$$

Lemma 8.3 (toric chart control)**.** *For a torus point $p = (x, t)$ with $0 < |t| < 1$, put*

$$y(p) = \frac{(\log|x_1|, \log|x_2|)}{\log|t|}.$$

Then

$$p \in \mathcal{U}_\sigma(S) \quad \Longleftrightarrow \quad y(p) \in \sigma^{(\log S/|\log|t||)}. \tag{2.5c}$$

Every point of $\{|t| < 1\} \subset Y$ belongs to $\overline{\mathcal{U}}_\sigma(1) = \{|z_i| \le 1\}$ for some σ. Consequently the sets $\mathcal{U}_\sigma(2)$ cover Y_ϵ, and every compact subset is contained in the union of finitely many members of this cover.

Proof. Since $(y, 1) = \sum_i \theta_i^\sigma(y)(v_i, 1)$, duality gives

$$\log|z_i(p)| = \log|t(p)|\, \theta_i^\sigma(y(p)).$$

Because $\log|t| < 0$, this is exactly (2.5c).

For the boundary assertion let p lie in the toric orbit $O_{\widehat{P}}$ of the cone over a cell P. In a chart with $\sigma \supseteq P$, the coordinates corresponding to vertices of P vanish. On the quotient lattice $N'/(N' \cap \operatorname{span}\widehat{P})$, the remaining coordinates are the characters dual to the image cone $\bar\sigma$. Hence $p \in \overline{\mathcal{U}}_\sigma(1)$ exactly when $-\operatorname{Log}_{\widehat{P}}(p) \in \bar\sigma$. The cones $\bar\sigma$, for triangles $\sigma \supseteq P$, cover the quotient real vector space: lift a vector n, choose a relative interior point $p_0 \in P \times \{1\}$, and for sufficiently small $u > 0$ place the height-one rescaling of $p_0 + un$ in a triangle of the locally finite triangulation. That triangle contains P, and its quotient cone contains n. This proves the asserted cover. Compactness then supplies a finite subcover. □

Theorem 8.4 (the cusp quotient and its central fibre)**.** ***[VERIFIED]*** *For the period functions and holomorphic cusp matrix $C(t)$ constructed above, and sufficiently small $\epsilon > 0$, the maps*

$$\Psi_{\bar\lambda}(x, t) = (c_{\bar\lambda}^-(t) t^{B_0 \bar\lambda} x, t), \qquad c_{\bar\lambda}^-(t) = \exp(2\pi i C(t)\bar\lambda), \tag{2.5}$$

extend to a free properly discontinuous action of $\bar\Lambda \simeq \mathbb{Z}^2$ on $Y_\epsilon = \{|t| < \epsilon\}$. The quotient

$$f_0 : N_0 = Y_\epsilon/\bar\Lambda \longrightarrow \Delta_\epsilon$$

is a proper holomorphic map from a connected smooth complex threefold. Its nonzero fibres are the tori of theorem 8.2. Its full scheme-theoretic central fibre W is reduced and has local equations

$$z_0, \qquad z_0 z_1, \qquad z_0 z_1 z_2. \tag{2.6}$$

The surface W is irreducible and nonnormal. Its normalization is $\nu : E_0 \simeq \mathrm{dP}_6 \to W$, obtained by identifying opposite sides of the toric hexagon; its double locus consists of three rational curves through exactly two triple points. Thus W is a normal-crossings surface, but it is not a simple-normal-crossings divisor. Moreover

$$\omega_{N_0} \simeq \mathcal{O}_{N_0}. \tag{2.6a}$$

Proof. For a bounded cell P, the cone over $P \times \{1\}$ is strongly convex, and two such cones meet in the cone over the common face of the two cells. The locally finite triangulation therefore defines a countable rational fan with support $(\mathbb{R}^2 \times \mathbb{R}_{>0}) \cup \{0\}$. For the determinant calculation, order the height-one generators of the two kinds as

$$((v, 1), (v + e_1, 1), (v + e_2, 1)), \qquad ((v + e_1, 1), (v + e_2, 1), (v + e_1 + e_2, 1)).$$

Their determinants are respectively $+1$ and -1, hence have absolute value one. Every cone is therefore unimodular. The ordinary pairwise affine-chart separation proof for a toric fan applies without a finiteness hypothesis; countability of the fan gives second countability. Thus Y is a connected Hausdorff smooth complex threefold with dense torus.

If m_0, m_1, m_2 is the basis dual to the height-one generators, then the corresponding affine chart is $\mathbb{C}^3$, with coordinates $z_i = \chi^{m_i}$, and

$$t = z_0 z_1 z_2. \tag{2.7}$$

Since every primitive ray generator has height one, the divisor of t is $\sum_v E_v$, all with multiplicity one. Equation (2.7) proves that the scheme-theoretic fibre $t^{-1}(0)$ is reduced and normal crossings. The star of a height-one vertex has rays $e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2$, the complete smooth hexagonal fan; hence every irreducible component E_v is dP_6.

The local canonical trivialization is also intrinsic to this toric model. On the dense torus put

$$\Omega = \frac{dx_1 \wedge dx_2 \wedge dt}{x_1 x_2}.$$

If (z_0, z_1, z_2) is a unimodular toric chart, then

$$\bigwedge_i \frac{dz_i}{z_i} = \pm \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge \frac{dt}{t}, \qquad t = z_0 z_1 z_2,$$

and therefore $\Omega = \pm dz_0 \wedge dz_1 \wedge dz_2$. It extends holomorphically and nowhere vanishingly across every boundary chart. For a deck transformation $\Psi_{\bar\lambda}$, formula (2.5) gives $\Psi_{\bar\lambda}^*(dx_i/x_i) = dx_i/x_i + \alpha_i(t)dt$ and $\Psi_{\bar\lambda}^* dt = dt$; every additional term in the pullback of Ω contains dt twice. Hence Ω is invariant and descends to a nowhere-vanishing three-form on N_0, proving the intrinsic trivialization (2.6a). No global threefold has been invoked at this stage.

Write $R(t) = -2\pi \operatorname{Im} C(t)$, which is bounded because C is holomorphic at zero. Choose ϵ such that $|\log \epsilon| > 2\|R\|_\infty$. At a torus point fixed by a nonzero $\bar\lambda$, taking logarithmic absolute values in (2.5) gives

$$\log|t| B_0 \bar\lambda + R(t)\bar\lambda = 0,$$

impossible because $|B_0\bar\lambda| = |\bar\lambda|$ and the first term has more than twice the norm of the second. On $t = 0$, a cell is translated by $B_0\bar\lambda \neq 0$; a bounded cell cannot be fixed by a nonzero translation. Thus the action is free.

For proper discontinuity put

$$B_t = B_0 + \frac{R(t)}{\log|t|}.$$

Then $\|B_t - B_0\| \le 1/2$, so $|B_t\bar\lambda| \ge |\bar\lambda|/2$. In logarithmic position coordinates, (2.5) is

$$y(\Psi_{\bar\lambda} p) = y(p) + B_t\bar\lambda. \tag{2.8}$$

Put $\eta_0 = \log 2/|\log\epsilon|$. If $\Psi_{\bar\lambda}\mathcal{U}_\sigma(2)$ meets $\mathcal{U}_{\sigma'}(2)$, the intersection is open and hence contains a point of the dense invariant torus. By theorem 8.3, for that point p and its image p',

$$y(p) \in \sigma^{(\eta_0)}, \qquad y(p') \in \sigma'^{(\eta_0)}.$$

Both thickenings are bounded, so (2.8) gives

$$\frac{1}{2}|\bar\lambda| \le |B_t\bar\lambda| = |y(p') - y(p)| \le \sup_{a \in \sigma'^{(\eta_0)},\, b \in \sigma^{(\eta_0)}} |a - b|.$$

Thus only finitely many $\bar\lambda$ can make this fixed pair of charts meet. A compact set is contained in finitely many of the charts from theorem 8.3; taking the finite union over all chart pairs proves proper discontinuity. Freeness and proper discontinuity give a Hausdorff complex-manifold quotient and a holomorphic covering map.

To prove properness, fix $\epsilon_1 < \epsilon$. Let $\mathcal{F}_0$ be the finite set of triangles meeting the closed Euclidean ball of radius $\sqrt{2}$, and put

$$K_0 = \bigcup_{\sigma \in \mathcal{F}_0} \left(\overline{\mathcal{U}}_\sigma(1) \cap \{|t| \le \epsilon_1\} \right).$$

This is compact. For a torus point p with $0 < |t(p)| \le \epsilon_1$, put $u = B_{t(p)}^{-1} y(p)$ and choose $\bar{\lambda} \in \mathbb{Z}^2$ with $|u + \bar{\lambda}|_\infty \le 1/2$. Since $\|B_t\| \le 3/2$, equation (2.8) gives

$$|y(\Psi_{\bar{\lambda}} p)| = |B_t(u + \bar{\lambda})| \le \frac{3}{2\sqrt{2}} < \sqrt{2}.$$

Choose a triangle containing this point. Formula (2.5c) with $S = 1$ places the translate in the corresponding $\overline{\mathcal{U}}_\sigma(1) \subset K_0$.

If p lies in the central orbit belonging to a cell P, translate a vertex of P to zero; this is possible because $B_0 \bar{\Lambda} = \mathbb{Z}^2$. The refined boundary part of theorem 8.3 then places the translate in $\overline{\mathcal{U}}_\sigma(1)$ for a triangle σ containing zero, hence in K_0. Therefore the image of K_0 in the quotient is exactly $f_0^{-1}(\{|t| \le \epsilon_1\})$, which is compact. Every compact subset of the base lies in such a closed subdisc, so f_0 is proper.

For $t \neq 0$, choose s with $e^{2\pi i s} = t$. The multiplier in (2.5) is $\exp(2\pi i Z(s)\bar{\lambda})$, so exponentiation identifies the fibre with

$$\mathbb{C}^2 / (\mathbb{Z}^2 + Z(s)\mathbb{Z}^2).$$

It is compact because $\operatorname{Im} Z(s)$ is invertible by (2.2).

The quotient map is a local biholomorphism; therefore the local equation (2.7), its multiplicity one, and reducedness descend to W. Every cell has a vertex, and $B_0 \bar{\Lambda} = \mathbb{Z}^2$ acts simply transitively on vertices, so every orbit in the central fibre meets E_0. Thus $E_0 \to W$ is surjective. Two interior points cannot be identified; on the boundary, precisely opposite edges and the corresponding hexagon vertices are identified. The fibres have at most three points. Since E_0 is compact, the map is finite; since it is generically injective from a smooth irreducible surface to the reduced surface W, it is the normalization. The two translation-orbits of triangles give the two triple points, and the three unoriented edge directions give the three double curves. Irreducibility follows from the finite surjection by E_0. □

Proposition 8.5 (the cusp punctured-overlap biholomorphism)**.** ***[VERIFIED]*** *Let $D_0^* = \{0 < |t_c| < \epsilon\}$, and shrink it if necessary so that its inverse image has a distinguished component U_0 stabilized exactly by $\langle g_0 \rangle$. Exponentiation induces a fibre-preserving biholomorphism*

$$J|_{D_0^*} \xrightarrow{\sim} N_0|_{D_0^*} = N_0 \setminus W. \tag{2.8i}$$

On the covering space it is given by

$$\mathcal{E} : U_0 \times \mathbb{C}^2 \longrightarrow (\mathbb{C}^*)^2 \times D_0^*, \qquad \mathcal{E}(z, \zeta_1, \zeta_2) = (e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}, e^{2\pi i s(z)}). \tag{2.8ii}$$

Proof. First identify every deck transformation. From $s \circ g_0 = s - 1$, the equality $t_c \circ \pi = e^{2\pi i s}$, and the fact that t_c is a base coordinate, ds never vanishes on U_0. Thus s is a local biholomorphism to the logarithmic cover of D_0^*. The restriction $U_0 \to D_0^*$ is a connected regular cover with deck group $\langle g_0 \rangle \simeq \mathbb{Z}$. Under the classification of connected covers of the punctured disc, it corresponds to a subgroup of $\pi_1(D_0^*) \simeq \mathbb{Z}$; because its deck group is infinite, that subgroup is zero. Hence U_0 is the universal cover. The logarithmic cover is universal as well, and its deck generator acts by $s \mapsto s - 1$. Therefore s is an equivariant biholomorphism onto that logarithmic cover.

The cusp transformation laws give

$$\Pi(g_0 z) = [Z(s-1) \mid I_2] = [Z(s) - B_0 \mid I_2].$$

In the same block decomposition, the explicit matrix in theorem 5.1 gives

$$M_{g_0} = M_0^{-1} = \begin{pmatrix} I_2 & 0 \\ -B_0 & I_2 \end{pmatrix}.$$

Consequently

$$\Pi(g_0 z) = \Pi(z) M_{g_0}.$$

Comparison with (2.3d) and real invertibility of $\Pi(g_0 z)$ yields $R_{g_0} = I_2$. To make the crossed action explicit, for $\lambda \in \Lambda$ and $g \in \Delta$ put

$$T_\lambda(z,\zeta) = (z, \zeta + \Pi(z)\lambda), \qquad \widetilde{g}(z,\zeta) = (gz, R_g(z)\zeta).$$

Equation (2.3d), together with $A_g = M_g^{-1}$, gives

$$\begin{aligned} \widetilde{g} T_\lambda(z,\zeta) &= \big(gz, R_g(z)\zeta + R_g(z)\Pi(z)\lambda\big) \\ &= \big(gz, R_g(z)\zeta + \Pi(gz) A_g \lambda\big) = T_{A_g\lambda}\widetilde{g}(z,\zeta). \end{aligned}$$

Thus

$$\widetilde{g} T_\lambda = T_{A_g\lambda}\widetilde{g}. \tag{2.8ii-a}$$

At the cusp $A_{g_0} = M_0$, so

$$\widetilde{g}_0(z,\zeta) = (g_0 z, \zeta), \qquad \widetilde{g}_0 T_\lambda = T_{M_0\lambda}\widetilde{g}_0. \tag{2.8ii-b}$$

In particular $\mathcal{E}$ is invariant under $\widetilde{g}_0$.

Let $\Lambda_{\mathrm{tor}} = \mathbb{Z}\widehat{w} \oplus \mathbb{Z}\widehat{\delta}$. By theorem 5.2,

$$\Lambda_{\mathrm{tor}} = \ker(M_0 - I) = \mathrm{im}(M_0 - I). \tag{2.8ii-c}$$

Put

$$\mathcal{G}_0 = \langle T_\lambda, \widetilde{g}_0 : \lambda \in \Lambda \rangle \simeq \Lambda \rtimes_{M_0} \langle \widetilde{g}_0 \rangle.$$

Since M_0 fixes Λ_{tor} pointwise, (2.8ii-b) shows that

$$H := \{ T_\nu \widetilde{g}_0^n : \nu \in \Lambda_{\mathrm{tor}},\ n \in \mathbb{Z} \} \simeq \Lambda_{\mathrm{tor}} \times \langle \widetilde{g}_0 \rangle. \tag{2.8ii-d}$$

Moreover, for $\lambda \in \Lambda$ and $\nu \in \Lambda_{\mathrm{tor}}$,

$$\widetilde{g}_0 T_\nu \widetilde{g}_0^{-1} = T_\nu, \qquad T_\lambda T_\nu T_\lambda^{-1} = T_\nu, \qquad T_\lambda \widetilde{g}_0 T_\lambda^{-1} = T_{(I-M_0)\lambda}\widetilde{g}_0 \in H. \tag{2.8ii-e}$$

These generator conjugations prove that $H \triangleleft \mathcal{G}_0$. Translation by Λ_{tor} adds an integral vector to ζ, so it lies in the deck group of $\mathcal{E}$. Conversely, if two points have the same image under $\mathcal{E}$, their s-coordinates differ by an integer and their ζ-coordinates by an element of $\mathbb{Z}^2$. Equivariance and injectivity of the logarithmic coordinate then show that they differ by a unique element of H. Thus $\mathcal{E}$ identifies $(U_0 \times \mathbb{C}^2)/H$ with $(\mathbb{C}^*)^2 \times D_0^*$.

It remains to identify the residual action. Write $\bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}}$, and let λ be any lift of $\bar{\lambda}$. In the period basis, translation by λ adds $Z(s)\bar{\lambda}$ to ζ, up to an integral vector killed by exponentiation. Therefore (2.5b) gives

$$\mathcal{E}(z, \zeta + \Pi(z)\lambda) = \big(c_{\bar{\lambda}}(t_c) t_c^{B_0\bar{\lambda}} x, t_c\big) = \Psi_{\bar{\lambda}}(\mathcal{E}(z,\zeta)). \tag{2.8iii}$$

This is independent of the chosen lift. Moreover M_0 is the identity on $\bar{\Lambda}$, by (2.8ii-c). The crossed multiplication law is

$$(T_\lambda \widetilde{g}_0^n)(T_\mu \widetilde{g}_0^m) = T_{\lambda + M_0^n \mu} \widetilde{g}_0^{n+m}.$$

It follows that

$$\vartheta : \mathcal{G}_0 \longrightarrow \bar{\Lambda}, \qquad \vartheta(T_\lambda \widetilde{g}_0^n) = \bar{\lambda},$$

is a surjective homomorphism with kernel H. Hence $\mathcal{G}_0/H \simeq \bar{\Lambda}$, and (2.8iii) identifies this residual action with the displayed $\Psi_{\bar{\lambda}}$-action. Dividing by it gives

$$(U_0 \times \mathbb{C}^2)/\mathcal{G}_0 \simeq ((\mathbb{C}^*)^2 \times D_0^*)/\bar{\Lambda}.$$

The left side is $J|_{D_0^*}$, while the right side is $N_0|_{D_0^*}$ because $t^{-1}(\mathbb{C}^*)$ is the dense torus of Y. This proves (2.8i), including holomorphicity of the inverse. □

8.3 The finite logarithmic fillings and the global gluing

Set

$$\varepsilon = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad \varepsilon' = \widehat{\gamma} + 3\widehat{u} - 3\widehat{w}. \tag{2.8a}$$

Direct multiplication by the matrices in theorem 5.1 gives $A_1\varepsilon = \varepsilon$ and $A_2\varepsilon' = \varepsilon'$.

A *linearizing pair* (Δ_j, s_j) consists of a simply connected g_j-invariant neighbourhood of z_j, disjoint from all its translates outside $\langle g_j\rangle$, and an injective coordinate satisfying

$$s_j(z_j) = 0, \qquad s_j \circ g_j = \zeta_j s_j, \qquad \zeta_j = e^{-2\pi i/m_j}. \tag{2.8b}$$

Shrinking gives such a pair. Put $D_j = \pi(\Delta_j)$, $\Delta_j^* = \Delta_j \setminus \{z_j\}$, and choose the base coordinate t_j by $t_j \circ \pi = s_j^{m_j}$. Let

$$\mathcal{T} = (\mathfrak{h}_z \times \mathbb{C}^2)/\Lambda, \qquad \mathfrak{h}_z^\circ = \mathfrak{h}_z \setminus (\Delta z_1 \cup \Delta z_2), \qquad J = (\mathcal{T}|_{\mathfrak{h}_z^\circ})/\Delta$$

$$J_j' = \mathcal{T}|_{\Delta_j}, \qquad A_0^{(j)} = \mathbb{C}^2/\Pi(z_j)\Lambda. \tag{2.8c}$$

Lemma 8.6 (the middle period quotient is proper)**.** *The induced Δ-action on $\mathcal{T}|_{\mathfrak{h}_z^\circ}$ is free and properly discontinuous. Consequently J is a Hausdorff complex manifold and its natural map*

$$J \longrightarrow B_{\mathrm{mid}} := \mathfrak{h}_z^\circ/\Delta$$

is a proper holomorphic submersion with compact torus fibres.

Proof. The deleted translates contain every elliptic fixed point, so the Δ-action on $\mathfrak{h}_z^\circ$ is free; it is properly discontinuous because it is the restriction of the Fuchsian action. The action on $\mathcal{T}$ projects equivariantly to this base action. Hence an element fixing a point of $\mathcal{T}|_{\mathfrak{h}_z^\circ}$ fixes its base point and is the identity. If K_1, K_2 are compact subsets of the total space and $gK_1 \cap K_2 \neq \varnothing$, then $g\ \mathrm{pr}(K_1) \cap \mathrm{pr}(K_2) \neq \varnothing$. Base proper discontinuity leaves only finitely many such g, proving proper discontinuity upstairs and hence the Hausdorff manifold assertion.

For properness, take a simply connected open set $U \Subset B_{\mathrm{mid}}$ evenly covered by $\mathfrak{h}_z^\circ$, and one lift $\widetilde{U}$. The quotient over U is identified with $\mathcal{T}|_{\widetilde{U}}$. Before quotienting, projection is a holomorphic submersion, and the quotient maps are local biholomorphisms; hence the displayed map is a holomorphic submersion. The flat real-analytic trivialization

$$\widetilde{U} \times (\Lambda \otimes \mathbb{R})/\Lambda \longrightarrow \mathcal{T}|_{\widetilde{U}}, \qquad (z, x + \Lambda) \longmapsto [z, \Pi(z)x]$$

shows this properness explicitly: if $C \subset U$ is compact, its chosen lift $\widetilde{C} \subset \widetilde{U}$ is compact and the inverse image of C is homeomorphic to $\widetilde{C} \times (\Lambda \otimes \mathbb{R})/\Lambda$, hence compact. Thus the map is proper over U. Properness is local on the target, so the displayed map is proper. □

For $b \in \Lambda \otimes \mathbb{R}$, define the varying holomorphic lift

$$\widetilde{g}_j^b(z, \zeta) = (g_j z, R_{g_j}(z)\zeta + \Pi(g_j z)b). \tag{2.8d}$$

Equation (2.3d) says

$$R_{g_j}(z)\Pi(z) = \Pi(g_j z)A_j, \tag{2.8e}$$

so (2.8d) normalizes the lattice action and descends to J_j'. In the flat real-analytic coordinates

$$\Xi(z, x + \Lambda) = [z, \Pi(z)x]$$

one has, with $\Sigma_k = I + A_j + \cdots + A_j^{k-1}$,

$$\Xi^{-1}(\widetilde{g}_j^b)^k \Xi(z, x) = (g_j^k z, A_j^k x + \Sigma_k b). \tag{2.8f}$$

We call $\widetilde{g}_j^0$ the linear lift and, for $v \in \Lambda^{A_j}$, call $\widetilde{g}_j^{\log} := \widetilde{g}_j^{v/m_j}$ the logarithmic lift.

Proposition 8.7 (finite fillings). ***[VERIFIED]*** *Let A_j be the two finite monodromies and choose*

$$v_1 = \varepsilon, \qquad v_2 = -\varepsilon', \qquad \gamma(v_1) = 1, \quad \gamma(v_2) = -1.$$

On the local torus family over a linearizing disc Δ_j, the affine map

$$(z, x) \longmapsto (g_j z, A_j x + v_j/m_j) \tag{2.9}$$

generates a free action of order m_j. The quotient $f_j : N_j \to D_j$ is a smooth complex threefold with

$$f_j^*(p_j) = m_j S_j, \qquad S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle. \tag{2.10}$$

Over the punctured disc it is biholomorphic to the original period family.

Proof. First prove freeness on the actual local family. For a finite-order integral map A, the affine self-map $x \mapsto Ax + b$ of $(\Lambda \otimes \mathbb{R})/\Lambda$ has a fixed point exactly when

$$\varphi(b) \in \mathbb{Z} \quad \text{for every } \varphi \in \mathrm{Hom}(\Lambda, \mathbb{Z}) \text{ with } \varphi \circ A = \varphi. \tag{2.9a}$$

Necessity is immediate. Conversely, the averaging projector $p = m^{-1} \sum_{r=0}^{m-1} A^r$ has image $\ker(A - I)_{\mathbb{R}}$ and kernel $\mathrm{im}(A - I)_{\mathbb{R}}$; the dual lattice of $p(\Lambda)$ is exactly the invariant integral covector lattice. Thus (2.9a) puts pb in $p(\Lambda)$, and $b \in (A - I)(\Lambda \otimes \mathbb{R}) + \Lambda$, which supplies a fixed point.

The exact matrix calculation gives

$$V^{T_1^k} = \langle \gamma, \psi_1 \rangle, \quad \psi_1 = 2u + w + 3\delta \quad (k = 1, 2),$$

and

$$V^{T_2^k} = \langle \gamma, \psi_2 \rangle, \quad \psi_2 = u + w + 2\delta \quad (k = 1, 2, 3). \tag{2.9b}$$

For $v_1 = \varepsilon$ and $v_2 = -\varepsilon'$,

$$\gamma(v_1) = 1, \quad \psi_1(v_1) = 0, \qquad \gamma(v_2) = -1, \quad \psi_2(v_2) = 0. \tag{2.9c}$$

Because $A_j v_j = v_j$, formula (2.8f) shows that the k-th power of the logarithmic lift has translation kv_j/m_j. For $j = 1$, (2.9a) fails for $k = 1, 2$ already on γ, since $k/3 \notin \mathbb{Z}$. For $j = 2$, it fails for $k = 1, 2, 3$, since $-k/4 \notin \mathbb{Z}$. Away from z_j, the base map g_j^k has no fixed point in Δ_j. Hence every nonidentity power is fixed-point free on J_j'. At $k = m_j$, (2.8f) is translation by $v_j \in \Lambda$, hence the identity; the order is exactly m_j because the map covers g_j.

It follows that

$$N_j = J_j'/\langle \widetilde{g}_j^{\log} \rangle$$

is a smooth complex threefold and the quotient map is finite étale of degree m_j. The function $s_j^{m_j} = t_j \circ \pi$ is invariant and descends to $f_j : N_j \to D_j$. The projection $J_j' \to \Delta_j$ is proper: over a compact subset, the image of the compact set obtained by letting x range over a fixed unit parallelotope in $\Lambda \otimes \mathbb{R}$ is the whole inverse image. Together with finiteness of $\Delta_j \to D_j$ and of the quotient, this proves properness of f_j. Pulling the divisor of t_j back through the etale quotient gives the divisor of $s_j^{m_j}$, namely m_j times the reduced central torus. This proves (2.10).

It remains to identify the punctured filling with the original family and to check that no logarithm branch has entered the answer. On a simply connected $U \subset \Delta_j^*$, choose a branch ℓ of $\log s_j$ and put

$$\sigma_j^{\ell}(z) = \frac{\ell(z)}{2\pi i} \Pi(z) v_j. \tag{2.9d}$$

Changing ℓ by $2\pi i n$ changes (2.9d) by the lattice vector $n\Pi(z)v_j$. The classes therefore glue to a branch-independent holomorphic section σ_j of $J_j'|_{\Delta_j^*}$; translation by that section defines a biholomorphism h_j.

For the conjugacy, use ℓ on U and on g_jU use the branch

$$\ell'(w) = \ell(g_j^{-1}w) - \frac{2\pi i}{m_j}.$$

This is a logarithm of $s_j(w)$ because $s_j \circ g_j = e^{-2\pi i/m_j} s_j$. For $z \in U$, equation (2.8e) and $A_j v_j = v_j$ give

$$\widetilde{g}_j^{\log} h_j^{-1}[z,\zeta] = \Big[g_j z, R_{g_j}(z)\zeta - \Big(\frac{\ell(z)}{2\pi i} - \frac{1}{m_j}\Big)\Pi(g_j z)v_j\Big].$$

But

$$\frac{\ell'(g_j z)}{2\pi i} = \frac{\ell(z)}{2\pi i} - \frac{1}{m_j},$$

so applying h_j cancels the displayed translation and proves the global identity

$$h_j \widetilde{g}_j^{\log} h_j^{-1} = \widetilde{g}_j^0. \tag{2.9e}$$

Finally, by the stabilizer and translate conditions in (2.8b),

$$J|_{D_j^*} = (J_j'|_{\Delta_j^*})/\langle \widetilde{g}_j^0 \rangle.$$

Thus (2.9e) descends to the asserted punctured-disc biholomorphism. □

Lemma 8.8 (radial retraction and the finite-fibre stalk map)**.** ***[VERIFIED]*** *Under theorem 8.7, the finite local model N_j strongly deformation retracts onto its reduced central fibre S_j. After identifying a nearby fibre with the marked real torus $A_s = (\Lambda \otimes \mathbb{R})/\Lambda$ by the flat coordinates Ξ, the specialization map*

$$H^r(S_j;\mathbb{Z}) \longrightarrow H^r(A_s;\mathbb{Z})$$

is the pullback π_j^* of the quotient covering

$$\pi_j : A_0^{(j)} \longrightarrow S_j = A_0^{(j)}/\langle x \mapsto A_j x + v_j/m_j \rangle.$$

Proof. Use the linearizing coordinate to define the radial contraction $r_u : \Delta_j \to \Delta_j$ by

$$s_j(r_u(z)) = (1-u)s_j(z), \qquad 0 \le u \le 1.$$

Because $s_j \circ g_j = \zeta_j s_j$, this contraction commutes with g_j. In the global real-analytic marking

$$\Xi(z, x+\Lambda) = [z, \Pi(z)x],$$

put

$$H_u\big(\Xi(z,x+\Lambda)\big) = \Xi(r_u(z), x+\Lambda).$$

Formula (2.8f) shows directly that H_u commutes with the logarithmic lift: it keeps x fixed while both composites send it to $A_j x + v_j/m_j$. Hence H_u descends to a strong deformation retraction of N_j onto S_j.

Fix $t_j \neq 0$ and one root s with $s^{m_j} = t_j$. Every orbit of the logarithmic lift meets the fibre over s exactly once, so that fibre maps isomorphically onto $f_j^{-1}(t_j)$. Following the descended radial retraction sends the marked point $x \in A_s$ to the orbit of the same x in the central torus, namely $\pi_j(x)$. The induced map from central-fibre cohomology to nearby-fibre cohomology is therefore exactly π_j^*, with no unrecorded conjugacy or index. □

Theorem 8.9 (global analytic construction)**.** ***[VERIFIED]*** *Gluing the period family over B° to N_0, N_1, N_2 by the preceding punctured-disc biholomorphisms produces a connected compact Hausdorff complex threefold X and a proper surjective holomorphic map*

$$f : X \longrightarrow B \simeq \mathbb{P}^1$$

with connected fibres. The fibre over p_0 is exactly the reduced nonnormal surface W of theorem 8.4; those over p_1, p_2 have scheme multiplicities three and four. The open embedding $N_0 \hookrightarrow X$ identifies canonical bundles, so

$$\omega_X|_{N_0} \simeq \omega_{N_0} \simeq \mathcal{O}_{N_0}. \tag{7}$$

Proof. Take the disjoint union of the four manifolds and identify the two copies of each punctured-disc restriction by the displayed biholomorphism. The special discs are pairwise disjoint, so every nonsingleton equivalence class has two points and meets each summand at most once. The saturation of an open set is the union of that set with its images under finitely many overlap biholomorphisms, hence is open. Each summand therefore embeds openly in the quotient, and the only transition maps are biholomorphisms. This gives a complex threefold.

Points over distinct base values have disjoint disc preimages. Points over one base value lie in a common one of the four Hausdorff charts, so the quotient is Hausdorff. The local maps to the base agree on overlaps. Each is proper, for the period-family piece by theorem 8.6 and for the three fillings by their local proper constructions; properness is local on a locally compact Hausdorff base, so the global map is proper. Since the base is compact, $X = f^{-1}(B)$ is compact. All four pieces are connected and every local filling meets the central period-family piece, so X is connected. The fibre statements follow from theorems 8.4 and 8.7; their fibres are connected, and surjectivity follows. Because each summand embeds openly in the quotient, the restriction of the intrinsic canonical bundle ω_X to the open summand N_0 is ω_{N_0}. Equation (2.6a) proves equation (7). □

What has and has not been discharged. The formulas above reconstruct the period functions, both finite fillings, the toric cusp quotient, and all three punctured-overlap biholomorphisms. Thus theorems 8.4 and 8.9 discharge existence, smoothness, properness, the full central-fibre equation, and the normalization assertions formerly labelled (H1)–(H3) in theorem 46.24. This promotion is local to the analytic construction: no collapse, Leray transgression, or Mayer–Vietoris computation is used here. Integral cohomology and sphere recognition remain separate below.

9 Flat spectral geometry on the positive radial cusp

The period family used in this section is the family already constructed and proved nondegenerate in theorem 8.2. In particular, this section introduces no additional period hypothesis. We retain the proved ordered lattices

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \Lambda = V^* = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$$

and the proved period matrix

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}.$$

Thus the columns of Π, in the displayed ordered basis of Λ, are

$$(6\mu, \beta)^{\mathsf{t}}, \quad (\tau, \mu)^{\mathsf{t}}, \quad (1, 0)^{\mathsf{t}}, \quad (0, 1)^{\mathsf{t}}.$$

The canonical cusp calculation has already produced holomorphic functions $\mu(t_c), b(t_c), h(t_c)$ on a full disc about $t_c = 0$, where

$$b = \beta + \tau, \qquad s = \tau - h, \qquad e^{2\pi i s} = t_c,$$

and has proved the exact identity

$$Z = s\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6\mu & h \\ b-h & \mu \end{pmatrix}. \tag{8}$$

Choose T_0 so that $e^{-2\pi T}$ lies in the proved cusp disc whenever $T \geq T_0$. On the positive radial ray

$$t_c = e^{-2\pi T}, \qquad T \geq T_0,$$

we choose the lift $s = iT$. The other lifts differ by the already accounted for integral deck transformation. Put

$$\begin{aligned} \mu_T &:= \mu(e^{-2\pi T}) = \alpha_T + i m_T, \\ h_T &:= h(e^{-2\pi T}) = \eta_T + i H_T, \\ b_T &:= b(e^{-2\pi T}) = \xi_T + i B_T, \\ c_T &:= \xi_T - \eta_T, \qquad L_T := T + H_T, \qquad q_T := L_T - B_T, \\ D_T &:= L_T q_T + 6 m_T^2. \end{aligned} \tag{9}$$

All letters on the right-hand side of equation (9) are real. We reserve

$$m := \operatorname{Im}\mu(0), \qquad H := \operatorname{Im} h(0), \qquad B := \operatorname{Im} b(0) \tag{10}$$

for their limiting values. Holomorphy at the origin gives, with the same statement for the real parts,

$$m_T = m + O(e^{-2\pi T}), \qquad H_T = H + O(e^{-2\pi T}), \qquad B_T = B + O(e^{-2\pi T}). \tag{11}$$

Identify $\mathbb{C}^2$ with $\mathbb{R}^4$ by the ordered coordinates

$$(z_1, z_2) = (x_1 + i y_1, x_2 + i y_2) \longmapsto (x_1, y_1, x_2, y_2). \tag{12}$$

The metric in this section is, by definition, the quotient of $dx_1^2 + dy_1^2 + dx_2^2 + dy_2^2$. This makes the metric choice explicit; the complex period construction itself neither requires nor singles out this Euclidean metric.

Proposition 9.1 (Exact radial realification and dual lattice)**.** *In the domain basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$ and the real coordinate order equation* (12)*, the positive radial period matrix is*

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T & 1 & 0 \\ 6m_T & L_T & 0 & 0 \\ c_T & \alpha_T & 0 & 1 \\ -q_T & m_T & 0 & 0 \end{pmatrix}. \tag{13}$$

Its oriented determinant, positive covolume, and relation with the determinant function in theorem 8.2 are

$$\det P_T = -D_T, \qquad \operatorname{covol}(P_T\mathbb{Z}^4) = D_T > 0, \qquad D_{\mathrm{can},T} = -\frac{D_T}{L_T} < 0. \tag{14}$$

Here $D_{\mathrm{can},T} = \operatorname{Im}\beta_T - 6(\operatorname{Im}\mu_T)^2/\operatorname{Im}\tau_T$ is the canonical determinant function; in particular, D_T is not a new nondegeneracy assumption.

The complete inverse transpose is

$$P_T^{-\mathsf{T}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \dfrac{m_T}{D_T} & \dfrac{q_T}{D_T} & \dfrac{-6m_T\alpha_T - q_T\eta_T}{D_T} & \dfrac{-m_T c_T - q_T\alpha_T}{D_T} \\ 0 & 0 & 0 & 1 \\ -\dfrac{L_T}{D_T} & \dfrac{6m_T}{D_T} & \dfrac{6(L_T\alpha_T - m_T\eta_T)}{D_T} & \dfrac{L_T c_T - 6m_T\alpha_T}{D_T} \end{pmatrix}. \tag{15}$$

Equivalently, for $k = (r, s, n_1, n_2)^{\mathsf{t}} \in \mathbb{Z}^4$, define

$$R_T(k) := r - 6\alpha_T n_1 - c_T n_2, \qquad S_T(k) := s - \eta_T n_1 - \alpha_T n_2. \tag{16}$$

Then the Euclidean dual lattice is exactly

$$\begin{aligned}(P_T\mathbb{Z}^4)^* = \Bigg\{ \Bigg(n_1, \frac{m_T R_T(k) + q_T S_T(k)}{D_T}, n_2, \frac{-L_T R_T(k) + 6m_T S_T(k)}{D_T} \Bigg) : \\ k = (r, s, n_1, n_2) \in \mathbb{Z}^4 \Bigg\}.\end{aligned} \tag{17}$$

Consequently, if

$$F_T := \mathbb{R}^4 / P_T\mathbb{Z}^4$$

and Δ_T is the nonnegative self-adjoint scalar Laplacian on $L^2(F_T, \mathrm{d}\,\mathrm{vol}_{F_T})$, obtained from the closed Dirichlet form

$$\mathfrak{q}_T[f] = \int_{F_T} |\mathrm{d}f|^2 \,\mathrm{d}\,\mathrm{vol}_{F_T}, \qquad \mathrm{Dom}(\mathfrak{q}_T) = H^1(F_T),$$

with operator domain $\mathrm{Dom}(\Delta_T) = H^2(F_T)$, then, with multiplicity,

$$\mathrm{Spec}(\Delta_T) = \{\lambda_T(k) : k \in \mathbb{Z}^4\}, \tag{18}$$

where

$$\begin{aligned}\lambda_T(k) = 4\pi^2 \Bigg(n_1^2 + n_2^2 \\ + \frac{(m_T R_T(k) + q_T S_T(k))^2 + (-L_T R_T(k) + 6m_T S_T(k))^2}{D_T^2} \Bigg).\end{aligned} \tag{19}$$

The eigenfunction belonging to k is the character

$$\chi_{T,k}([x]) = \exp\bigl(2\pi i \langle P_T^{-\mathsf{T}} k, x\rangle\bigr). \tag{20}$$

Proof. Substitution of $s = iT$ into equation (8) gives

$$Z_T = \begin{pmatrix} 6\mu_T & h_T + iT \\ b_T - h_T - iT & \mu_T \end{pmatrix}.$$

Realifying its four columns in the interlaced coordinate order (x_1, y_1, x_2, y_2) gives equation (13); in particular, the two divergent entries have signs $+L_T$ and $-q_T = B_T - L_T$.

Expanding the determinant along the last two columns gives

$$\det P_T = -L_T q_T - 6m_T^2 = -D_T.$$

Moreover,

$$\mathrm{Im}\,\tau_T = L_T, \qquad \mathrm{Im}\,\beta_T = B_T - L_T = -q_T,$$

so the canonical identity $\det_{\mathbb{R}} \Pi_T = \mathrm{Im}\,\tau_T D_{\mathrm{can},T}$ becomes

$$L_T D_{\mathrm{can},T} = L_T \left(-q_T - \frac{6m_T^2}{L_T} \right) = -D_T.$$

The already-proved inequalities $L_T > 0$ and $D_{\mathrm{can},T} < 0$ imply $D_T > 0$.

To compute the inverse transpose without suppressing the real parts, solve $P_T^{\mathsf{T}} x = k$. Its third and fourth equations give

$$x_1 = n_1, \qquad x_3 = n_2.$$

The remaining equations are

$$\begin{pmatrix} 6m_T & -q_T \\ L_T & m_T \end{pmatrix} \begin{pmatrix} x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} R_T(k) \\ S_T(k) \end{pmatrix}. \tag{21}$$

The determinant of the matrix in equation (21) is D_T, and its inverse is

$$\frac{1}{D_T} \begin{pmatrix} m_T & q_T \\ -L_T & 6m_T \end{pmatrix}.$$

This proves both equation (15) and equation (17).

A character $x \mapsto e^{2\pi i\langle \ell, x\rangle}$ descends to F_T precisely when $\langle \ell, P_T k\rangle \in \mathbb{Z}$ for every $k \in \mathbb{Z}^4$, which is precisely $\ell \in P_T^{-\mathsf{T}}\mathbb{Z}^4$. These characters are the complete Fourier basis of the compact abelian group F_T, and applying the positive Euclidean Laplacian gives $4\pi^2\|\ell\|^2$. Inserting equation (17) proves equations (18) to (20). □

Theorem 9.2 (Shortest cusp modes and their first corrections)**.** *Let*

$$\ell_{\gamma,T} := P_T^{-\mathsf{T}}(1,0,0,0)^{\mathsf{t}}, \qquad \ell_{u,T} := P_T^{-\mathsf{T}}(0,1,0,0)^{\mathsf{t}}. \tag{22}$$

For all sufficiently large T, every shortest nonzero vector of $(P_T\mathbb{Z}^4)^*$ belongs to

$$\{\pm\ell_{\gamma,T},\ \pm\ell_{u,T}\}.$$

Their exact squared lengths are

$$\|\ell_{\gamma,T}\|^2 = \frac{L_T^2 + m_T^2}{D_T^2}, \qquad \|\ell_{u,T}\|^2 = \frac{q_T^2 + 36m_T^2}{D_T^2}. \tag{23}$$

Their exact comparison is

$$D_T^2(\|\ell_{\gamma,T}\|^2 - \|\ell_{u,T}\|^2) = B_T(2L_T - B_T) - 35m_T^2. \tag{24}$$

Thus the shortest Fourier indices are $\pm(0,1,0,0)$ if the right-hand side of equation (24) is positive, $\pm(1,0,0,0)$ if it is negative, and both pairs if it is zero.

In the standard basis e_1, e_2, e_3, e_4 of the real coordinate space, the vector expansions are

$$T\ell_{\gamma,T} = -e_4 + \frac{me_2 + (H-B)e_4}{T} + O(T^{-2}), \tag{25}$$

$$T\ell_{u,T} = e_2 + \frac{-He_2 + 6me_4}{T} + O(T^{-2}). \tag{26}$$

Accordingly,

$$4\pi^2\|\ell_{\gamma,T}\|^2 = \frac{4\pi^2}{T^2}\left(1 + \frac{2(B-H)}{T} + O(T^{-2})\right), \tag{27}$$

$$4\pi^2\|\ell_{u,T}\|^2 = \frac{4\pi^2}{T^2}\left(1 - \frac{2H}{T} + O(T^{-2})\right). \tag{28}$$

More generally, every fixed plane index has the first correction

$$\frac{T^2\lambda_T(r,s,0,0)}{4\pi^2} = r^2 + s^2 + \frac{2(B-H)r^2 - 10mrs - 2Hs^2}{T} + O_{r,s}(T^{-2}). \tag{29}$$

In particular, the first positive eigenvalue, including the coefficient after its leading term, is

$$\boxed{\lambda_1(F_T) = \frac{4\pi^2}{T^2}\left(1 + \frac{-2H + 2\min\{B,0\}}{T} + O(T^{-2})\right).} \tag{30}$$

If $B > 0$, the bottom indices are eventually $\pm(0,1,0,0)$; if $B < 0$, they are eventually $\pm(1,0,0,0)$. If $B = 0$, both pairs have the same displayed first correction, and the exact decision is made by equation (24). In particular, if $B = 0$ but $m \neq 0$, the bottom indices are eventually $\pm(1,0,0,0)$. The first positive eigenvalue has complex multiplicity two when exactly one pair minimizes and multiplicity four when both pairs minimize.

Proof. For $k = (r, s, 0, 0)$, define

$$K_T := \frac{T}{D_T} \begin{pmatrix} m_T & q_T \\ -L_T & 6m_T \end{pmatrix}.$$

Equations equations (9) and (11) give

$$K_T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + O_{\mathrm{op}}(T^{-1}). \tag{31}$$

Hence there is a constant C, independent of $(r, s) \in \mathbb{R}^2$, such that

$$(1 - C/T)\sqrt{r^2 + s^2} \leq T\|P_T^{-\mathsf{T}}(r, s, 0, 0)\| \leq (1 + C/T)\sqrt{r^2 + s^2} \tag{32}$$

for all sufficiently large T. Thus among the vectors with $(n_1, n_2) = (0, 0)$, every vector with $r^2 + s^2 \geq 2$ is eventually longer than all four vectors for which $r^2 + s^2 = 1$.

If instead $(n_1, n_2) \neq (0, 0)$, equation (17) gives

$$\|P_T^{-\mathsf{T}} k\|^2 \geq n_1^2 + n_2^2 \geq 1,$$

whereas each of the four unit vectors just identified has length $T^{-1}(1 + O(T^{-1}))$. This proves the asserted exhaustion of the shortest vectors.

Taking the first two columns of equation (15) proves equation (23). Since $q_T = L_T - B_T$, their numerator difference is

$$L_T^2 + m_T^2 - q_T^2 - 36m_T^2 = B_T(2L_T - B_T) - 35m_T^2,$$

which proves equation (24) and the exact mode classification.

Finally,

$$D_T = T^2 + (2H - B)T + H(H - B) + 6m^2 + O(Te^{-2\pi T}). \tag{33}$$

Expanding the first two columns of equation (15) with equation (33) gives equations (25) and (26). Taking their squared norms gives equations (27) and (28). The same multiplication for a fixed (r, s) gives equation (29); explicitly, the scaled two-dimensional matrix is

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \frac{1}{T} \begin{pmatrix} m & -H \\ H - B & 6m \end{pmatrix} + O(T^{-2}),$$

and twice the Euclidean inner product of its leading and first-order terms is $2(B - H)r^2 - 10mrs - 2Hs^2$. The smaller relative correction is

$$\min\{2(B - H), -2H\} = -2H + 2\min\{B, 0\},$$

which proves equation (30). □

Corollary 9.3 (Volume and unit-volume scaling)**.** *The Euclidean volume is exactly*

$$V_T := \mathrm{Vol}(F_T) = D_T = (T + H_T)(T + H_T - B_T) + 6m_T^2, \tag{34}$$

and it has the expansion equation (33). *Define the unit-volume metric*

$$\widehat{g}_T := D_T^{-1/2} g_T. \tag{35}$$

Let $\widehat{\Delta}_T$ *denote the nonnegative self-adjoint scalar Laplacian on* $L^2(F_T, \mathrm{d}\,\mathrm{vol}_{\widehat{g}_T})$ *obtained from the closed Dirichlet form*

$$\widehat{\mathfrak{q}}_T[f] = \int_{F_T} |\mathrm{d}f|^2_{\widehat{g}_T}\, \mathrm{d}\,\mathrm{vol}_{\widehat{g}_T}, \qquad \mathrm{Dom}(\widehat{\mathfrak{q}}_T) = H^1(F_T),$$

with operator domain $\mathrm{Dom}(\widehat{\Delta}_T) = H^2(F_T)$. *Then the first positive eigenvalue of* $\widehat{\Delta}_T$ *is*

$$\boxed{\lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}\left(1 + \frac{-H - B/2 + 2\min\{B, 0\}}{T} + O(T^{-2})\right).} \tag{36}$$

Thus unit-volume normalization does not prevent the first eigenvalue from tending to zero. In the normalized metric the periods e_1, e_3 *have length* $D_T^{-1/4} = T^{-1/2}(1 + O(T^{-1}))$, *while the two periods with indices* $\widehat{\gamma}, \widehat{u}$ *have length* $T^{1/2}(1 + O(T^{-1}))$.

Proof. The volume of a Euclidean lattice quotient is its covolume, so equation (14) proves equation (34). In real dimension four, replacing g by ag multiplies volume by a^2 and every Laplace eigenvalue by a^{-1}. Therefore equation (35) has volume one and

$$\lambda_1(F_T, \widehat{g}_T) = D_T^{1/2} \lambda_1(F_T, g_T).$$

Equation equation (33) gives

$$D_T^{1/2} = T\left(1 + \frac{H - B/2}{T} + O(T^{-2})\right).$$

Multiplication by equation (30) proves equation (36). The stated period-length estimates follow directly from equation (13): e_1, e_3 have original length one, whereas the $\widehat{\gamma}$- and $\widehat{u}$-columns have original length $T(1 + O(T^{-1}))$; lengths are multiplied by $D_T^{-1/4}$. □

Proposition 9.4 (Exact counting formula and its two cusp regimes)**.** *For* $E \geq 0$, *count eigenvalues with their Fourier multiplicities by*

$$N_T(E) := \#\{k \in \mathbb{Z}^4 : \lambda_T(k) \leq E\}. \tag{37}$$

Put

$$R_E^2 := \frac{E}{4\pi^2}, \qquad (x)_+ := \max\{x, 0\}.$$

For every fixed E,

$$N_T(E) = \pi D_T \sum_{n \in \mathbb{Z}^2} (R_E^2 - \|n\|^2)_+ + O_E(T + 1). \tag{38}$$

The sum is finite. Equivalently,

$$\lim_{T \to \infty} \frac{N_T(E)}{D_T} = \pi \sum_{n \in \mathbb{Z}^2} (R_E^2 - \|n\|^2)_+. \tag{39}$$

In the first band, $0 < E < 4\pi^2$, *this reduces to*

$$N_T(E) = \frac{E}{4\pi} D_T + O_E(T + 1). \tag{40}$$

For the unit-volume metric $\widehat{g}_T$, *put*

$$\widehat{N}_T(E) := \#\{k \in \mathbb{Z}^4 : \lambda_T(k; \widehat{g}_T) \leq E\}. \tag{41}$$

For every fixed $E > 0$,

$$\widehat{N}_T(E) = \frac{E}{4\pi} D_T^{1/2} + O_E(D_T^{1/4} + 1), \qquad \lim_{T\to\infty} \frac{\widehat{N}_T(E)}{D_T^{1/2}} = \frac{E}{4\pi}. \tag{42}$$

There is also a finite counting law at the collapsing spectral scale. Define

$$N_T^{\mathrm{coll}}(E) := \#\left\{ k \in \mathbb{Z}^4 : \lambda_T(k) \le \frac{4\pi^2 E}{T^2} \right\}. \tag{43}$$

If E is not a sum of two integer squares, then, for all sufficiently large T,

$$N_T^{\mathrm{coll}}(E) = \#\{(r,s) \in \mathbb{Z}^2 : r^2 + s^2 \le E\}. \tag{44}$$

At a threshold $E \in \{r^2 + s^2 : r, s \in \mathbb{Z}\}$, the sharp statement available from the zeroth-order limit is

$$\begin{aligned} \#\{(r,s) : r^2 + s^2 < E\} &\le \liminf_{T\to\infty} N_T^{\mathrm{coll}}(E) \\ &\le \limsup_{T\to\infty} N_T^{\mathrm{coll}}(E) \le \#\{(r,s) : r^2 + s^2 \le E\}. \end{aligned} \tag{45}$$

For a threshold vector (r,s), the coefficient in equation (29) decides its inclusion whenever that coefficient is nonzero. If it vanishes, the exact eigenvalue formula and, if necessary, higher Taylor coefficients decide the inclusion.

Proof. Set

$$\mathcal{A}_T := \begin{pmatrix} 6m_T & -q_T \\ L_T & m_T \end{pmatrix}, \qquad \theta_T(n_1,n_2) := \begin{pmatrix} 6\alpha_T n_1 + c_T n_2 \\ \eta_T n_1 + \alpha_T n_2 \end{pmatrix}. \tag{46}$$

Then $\det \mathcal{A}_T = D_T$, and the exact eigenvalue formula can be rewritten as

$$\frac{\lambda_T(r,s,n_1,n_2)}{4\pi^2} = \|(n_1,n_2)\|^2 + \left\| \mathcal{A}_T^{-1} \left(\begin{pmatrix} r \\ s \end{pmatrix} - \theta_T(n_1,n_2) \right) \right\|^2. \tag{47}$$

Fix $n = (n_1,n_2)$. If $\|n\|^2 > R_E^2$, this index contributes nothing. Otherwise put $\rho_n = (R_E^2 - \|n\|^2)^{1/2}$. The allowed pairs (r,s) are the integer points in the translated ellipse

$$\theta_T(n) + \mathcal{A}_T \overline{B}_{\mathbb{R}^2}(0,\rho_n). \tag{48}$$

Its area is $\pi D_T \rho_n^2$. Its boundary length is $O_E(T+1)$, because $\|\mathcal{A}_T\|_{\mathrm{op}} = O(T+1)$. If $\rho_n = 0$, the set in equation (48) is one point and contributes either zero or one integer point, which is absorbed in the $O_E(1)$ remainder below. We apply the planar area–perimeter estimate only when $\rho_n > 0$.

For completeness, the elementary planar lattice estimate used here is

$$\#(\mathbb{Z}^2 \cap K) = \mathrm{area}(K) + O(\mathrm{length}(\partial K) + 1)$$

for every translated compact convex set K. Indeed, attach to each integer point its centered unit square and let U be the union of the squares centered at points of $K \cap \mathbb{Z}^2$. A point of $U \triangle K$ lies within distance $\sqrt{2}$ of ∂K: this is immediate for $U \setminus K$, while a point of $K \setminus U$ whose distance from the boundary exceeded $\sqrt{2}$ would have its nearest integer point inside K and would belong to the corresponding square. If $P(K)$ is the perimeter, the outer parallel-set formula for a convex polygon gives $\mathrm{area}(K + B_\rho) - \mathrm{area}(K) = \rho P(K) + \pi\rho^2$: its added pieces are the edge rectangles and vertex sectors, whose angles total 2π. Polygonal approximation gives the same bound for every compact convex K. Applying it to the outer and inner boundary strips with $\rho = \sqrt{2}$ gives $\mathrm{area}(U \triangle K) = O(P(K) + 1)$. Since the unit squares have disjoint interiors and total area $\#(K \cap \mathbb{Z}^2)$, the lattice estimate follows.

Only finitely many n's occur for fixed E. Applying the estimate to equation (48) and summing proves equation (38). Since $D_T \sim T^2$, division by D_T proves equation (39); if $0 < E < 4\pi^2$, only $n = 0$ occurs and equation (40) follows.

For $\widehat{g}_T$, eigenvalues are multiplied by $D_T^{1/2}$, so $\widehat{N}_T(E) = N_T(E/D_T^{1/2})$. For large T, only $n = 0$ can occur in equation (47). Its ellipse has radius parameter $\rho = (E/(4\pi^2 D_T^{1/2}))^{1/2}$, hence area $ED_T^{1/2}/(4\pi)$ and perimeter $O_E(TD_T^{-1/4} + 1) = O_E(D_T^{1/4} + 1)$. The same lattice estimate proves equation (42).

For the collapsing scale, any $(n_1, n_2) \neq (0, 0)$ contributes at least $4\pi^2$, so it contributes nothing for all sufficiently large T. For $n_1 = n_2 = 0$, equation (31) says uniformly that

$$\frac{T^2\lambda_T(r, s, 0, 0)}{4\pi^2} = \|K_T(r, s)^{\mathsf{t}}\|^2 \longrightarrow r^2 + s^2.$$

The lower bound in equation (32) also shows that only a fixed finite set of (r, s)'s can enter a fixed rescaled window. Termwise comparison on that finite set proves equation (44) away from thresholds and the two-sided bound equation (45) at thresholds. □

Theorem 9.5 (Pointed limits and expanding local covers)**.** *Let o_T be the origin of F_T, and set*

$$\mathbb{T}^2_{\mathrm{re}} := (\mathbb{R}e_1 \oplus \mathbb{R}e_3)/(\mathbb{Z}e_1 \oplus \mathbb{Z}e_3). \tag{49}$$

With the product Euclidean metric,

$$(F_T, o_T) \longrightarrow (\mathbb{T}^2_{\mathrm{re}} \times (\mathbb{R}e_2 \oplus \mathbb{R}e_4), (0, 0)) \cong (\mathbb{T}^2 \times \mathbb{R}^2, (0, 0)) \tag{50}$$

in pointed smooth, and hence pointed Gromov–Hausdorff, topology. More precisely, for every $R < \infty$, the radius-R balls in the two sides of equation (50) *are isometric for all sufficiently large T.*

For $M \in \mathbb{Z}_{\geq 1}$, define the marked local sublattice

$$\Lambda_{T,M} := \mathbb{Z}v_{\gamma,T} \oplus \mathbb{Z}v_{u,T} \oplus \mathbb{Z}(Me_1) \oplus \mathbb{Z}(Me_3), \tag{51}$$

where

$$\begin{aligned} v_{\gamma,T} &= (6\alpha_T, 6m_T, c_T, -q_T), \\ v_{u,T} &= (\eta_T, L_T, \alpha_T, m_T). \end{aligned} \tag{52}$$

Then

$$\widetilde{F}_{T,M} := \mathbb{R}^4/\Lambda_{T,M}, \qquad c_{T,M} : \widetilde{F}_{T,M} \longrightarrow F_T \tag{53}$$

is a regular holomorphic local-isometric cover of degree M^2, with deck group $(\mathbb{Z}/M\mathbb{Z})^2$. For every integer-valued function $M(T) \to \infty$,

$$(\widetilde{F}_{T,M(T)}, \widetilde{o}_{T,M(T)}) \longrightarrow (\mathbb{R}^4, 0) \tag{54}$$

in pointed smooth and pointed Gromov–Hausdorff topology. Again, every fixed radius ball is eventually exactly isometric to its Euclidean counterpart. If the cover is instead equipped with the pullback of the unit-volume metric $\widehat{g}_T$, then it converges in pointed smooth topology, and hence in pointed Gromov–Hausdorff topology, to $\mathbb{R}^4$ if and only if

$$\frac{M(T)}{D_T^{1/4}} \longrightarrow \infty, \quad \textit{equivalently} \quad \frac{M(T)}{\sqrt{T}} \longrightarrow \infty. \tag{55}$$

By contrast, if every covering torus is itself rescaled to volume one, no choice of $M(T)$ gives a pointed smooth limit $\mathbb{R}^4$.

Proof. The period lattice is

$$P_T\mathbb{Z}^4 = \mathbb{Z}v_{\gamma,T} \oplus \mathbb{Z}v_{u,T} \oplus \mathbb{Z}e_1 \oplus \mathbb{Z}e_3. \tag{56}$$

First quotient $\mathbb{R}^4$ by $\mathbb{Z}e_1 \oplus \mathbb{Z}e_3$. This gives the proposed limit

$$X_\infty := \mathbb{T}^2_{\mathrm{re}} \times (\mathbb{R}e_2 \oplus \mathbb{R}e_4).$$

The remaining quotient translations are generated by the images of $v_{\gamma,T}, v_{u,T}$. The projection of their integer combination $av_{\gamma,T} + bv_{u,T}$ to the noncompact (e_2, e_4)-plane is

$$Y_T \begin{pmatrix} a \\ b \end{pmatrix}, \qquad Y_T := \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix}. \tag{57}$$

More precisely,

$$Y_T = T \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + \begin{pmatrix} 6m_T & H_T \\ B_T - H_T & m_T \end{pmatrix}. \tag{58}$$

Holomorphy makes the second matrix uniformly bounded for $T \geq T_0$. Choose C above its operator norm. The first matrix is T times an orthogonal matrix, so the reverse triangle inequality gives

$$\left\| Y_T \begin{pmatrix} a \\ b \end{pmatrix} \right\| \geq (T - C)\sqrt{a^2 + b^2}. \tag{59}$$

The displacement in X_∞ of a translation is at least the norm of its noncompact-plane projection. Hence every nonidentity element of the remaining deck group has displacement at least $T - C$. Together with the fixed lattice vectors e_1, e_3, this also proves

$$\mathrm{sys}(P_T\mathbb{Z}^4) = 1, \qquad \mathrm{inj}(F_T, g_T) = \frac{1}{2} \tag{60}$$

for all sufficiently large T.

Fix R. Once $T - C > 4R$, the quotient map $X_\infty \to F_T$ is injective on the radius-R ball about the origin. It is a local isometry, and every minimizing geodesic of length at most R in the quotient lifts from the origin, so the closed radius-R ball maps onto the closed radius-R ball in F_T. Moreover, a nontrivial translate cannot shorten the distance between two points of the ball: their original distance is at most $2R$, while every translated distance is at least $(T - C) - 2R > 2R$. The balls are therefore isometric. This proves equation (50), including its pointed smooth assertion.

Because the four vectors in equation (56) are a lattice basis, the sublattice equation (51) has index M^2. Thus the identity of $\mathbb{R}^4$ induces the regular holomorphic degree-M^2 local-isometric cover equation (53). The quotient of the two lattices is $(\mathbb{Z}/M\mathbb{Z})^2$, which is its deck group. Consider a nonzero vector of $\Lambda_{T,M}$:

$$av_{\gamma,T} + bv_{u,T} + M(pe_1 + qe_3), \qquad a, b, p, q \in \mathbb{Z}. \tag{61}$$

If $(a, b) \neq (0, 0)$, its (e_2, e_4)-projection and equation (59) show that its norm is at least $T - C$. If $(a, b) = (0, 0)$, its norm is at least M. Therefore

$$\mathrm{sys}(\Lambda_{T,M}) \geq \min\{T - C, M\}. \tag{62}$$

Conversely, Me_1 has length M, while either long generator has length at most $T + C'$ for a fixed C'. Hence

$$\min\{T - C, M\} \leq \mathrm{sys}(\Lambda_{T,M}) \leq \min\{T + C', M\}, \qquad \mathrm{inj}(\widetilde{F}_{T,M}) = \tfrac{1}{2}\,\mathrm{sys}(\Lambda_{T,M}). \tag{63}$$

For $M = M(T) \to \infty$, this lower bound tends to infinity. Once it is greater than $4R$, the universal covering map $\mathbb{R}^4 \to \widetilde{F}_{T,M(T)}$ identifies the radius-R balls isometrically. This proves equation (54).

For the pullback of $\widehat{g}_T$, all lengths in equation (62) are multiplied by $D_T^{-1/4}$. The long lower bound becomes $(T-C)D_T^{-1/4} \sim \sqrt{T}$, which diverges automatically. The explicit lattice vector Me_1 has normalized length $MD_T^{-1/4}$. Hence the normalized systole tends to infinity exactly under equation (55). Divergence gives Euclidean balls by the preceding covering argument. We now prove necessity from the particular flat lattices rather than from any general injectivity-radius principle. Put

$$\ell_T := M(T)D_T^{-1/4}, \qquad A_T := (T-C)D_T^{-1/4}.$$

Then $A_T \sim \sqrt{T} \to \infty$. Quotienting the normalized Euclidean cover first by $\ell_T\mathbb{Z}e_1 \oplus \ell_T\mathbb{Z}e_3$ gives

$$X_{\ell_T} := (\mathbb{R}/\ell_T\mathbb{Z})^2 \times \mathbb{R}^2.$$

Every remaining nonidentity deck transformation has displacement at least A_T. Consequently, for each fixed R, once $A_T > 4R$, the closed radius-R ball in the normalized covering torus is exactly the closed radius-R ball in X_{ℓ_T}. If $\ell_T \not\to \infty$, pass to a subsequence with $\ell_T \to \ell < \infty$. When $\ell > 0$, these pointed product spaces converge to $(\mathbb{R}/\ell\mathbb{Z})^2 \times \mathbb{R}^2$; when $\ell = 0$, the two circle factors have diameter tending to zero and the pointed Gromov–Hausdorff limit is $\mathbb{R}^2$. Neither limit is $\mathbb{R}^4$. Thus $\ell_T \to \infty$, which is precisely equation (55), is necessary for both the pointed smooth and the pointed Gromov–Hausdorff conclusion.

Finally, the volume of the covering torus with its unscaled pulled-back metric is M^2D_T. If one instead puts

$$g_{T,M}^{\mathrm{cov},1} := (M^2D_T)^{-1/2} g_T \quad \text{on } \widetilde{F}_{T,M}, \tag{64}$$

then its volume is one and its length scale is multiplied by $(M^2D_T)^{-1/4}$. The explicit periods Me_1 and $v_{\gamma,T}$ consequently have respective lengths

$$\sqrt{\frac{M}{T}}\,(1+o(1)), \qquad \sqrt{\frac{T}{M}}\,(1+o(1)). \tag{65}$$

Their leading factors have product one, so at least one stays bounded. The injectivity radius therefore cannot tend to infinity, which excludes a pointed smooth $\mathbb{R}^4$ limit for the individually unit-volume covers. □

Proposition 9.6 (Exact compact quotient of the native cusp fibre)**.** *Let*

$$\rho : \mathbb{R}^4 \longrightarrow \mathbb{R}^2, \qquad \rho(x_1,x_2,x_3,x_4) = (x_2,x_4), \tag{66}$$

and let Y_T be the matrix in equation (57)*. Then*

$$\det Y_T = D_T > 0. \tag{67}$$

Consequently

$$Q_T := \mathbb{R}^2/Y_T\mathbb{Z}^2 \tag{68}$$

is a compact real two-torus, and ρ induces an exact sequence of compact real Lie groups

$$0 \longrightarrow \mathbb{T}^2_{\mathrm{re}} \longrightarrow F_T \xrightarrow{\pi_T} Q_T \longrightarrow 0. \tag{69}$$

Moreover,

$$\mathrm{sys}(Q_T) \geq T - C, \tag{70}$$

where C is the constant in equation (59)*. Thus the native $\mathbb{T}^2 \times \mathbb{R}^2$ limit separates exactly into a fixed unit two-torus kernel and an expanding two-torus quotient.*

Proof. Direct calculation gives

$$\det Y_T = (6m_T)(m_T) - L_T(-q_T) = L_T q_T + 6m_T^2 = D_T.$$

The period decomposition equation (56) gives

$$\rho(P_T\mathbb{Z}^4) = Y_T\mathbb{Z}^2,$$

because $\rho(e_1) = \rho(e_3) = 0$ and the projected long generators are the two columns of Y_T. Hence

$$\pi_T([x]_{P_T\mathbb{Z}^4}) := [\rho(x)]_{Y_T\mathbb{Z}^2}$$

is well defined and surjective.

It remains to determine its kernel without suppressing the integral lattices. If $\pi_T([x]) = 0$, there are $a, b \in \mathbb{Z}$ such that

$$\rho(x) = Y_T(a,b)^{\mathsf{t}} = \rho(av_{\gamma,T} + bv_{u,T}).$$

Thus

$$x - av_{\gamma,T} - bv_{u,T} \in \mathbb{R}e_1 \oplus \mathbb{R}e_3,$$

so $[x]$ lies in the displayed copy of $\mathbb{T}^2_{\mathrm{re}}$. Conversely that copy is plainly killed by π_T. Finally, if a vector of $P_T\mathbb{Z}^4$ lies in $\mathbb{R}e_1 \oplus \mathbb{R}e_3$, its long coefficients a, b satisfy $Y_T(a,b)^{\mathsf{t}} = 0$. Since $\det Y_T = D_T > 0$, one has $a = b = 0$; the intersection is therefore exactly $\mathbb{Z}e_1 \oplus \mathbb{Z}e_3$. This proves injectivity at the left and exactness. The systolic bound is equation (59). □

Theorem 9.7 (The maximal regular-locus $\mathbb{R}^4$ quotient and its compactification obstruction)**.** ***[VERIFIED]****Let*

$$\Lambda = \mathbb{Z}\langle\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}\rangle, \qquad K_\Lambda = \mathbb{Z}\langle\widehat{w}, \widehat{\delta}\rangle,$$

and define

$$\varepsilon_\gamma(a\widehat{\gamma} + b\widehat{u} + c\widehat{w} + d\widehat{\delta}) = a, \qquad L_\Lambda = \ker\varepsilon_\gamma = \mathbb{Z}\langle\widehat{u}, \widehat{w}, \widehat{\delta}\rangle. \tag{G.1}$$

Then the following statements hold.

(i) Regular group quotient. *Over B°, the period family $J = X^\circ = f^{-1}(B^\circ)$ is a holomorphic family of complex Lie groups, not merely an affine torus bundle. The formula*

$$\chi_\gamma([z, \Pi(z)x]) = ([z], \varepsilon_\gamma(x) \bmod \mathbb{Z}) \tag{71}$$

defines a global smooth real-analytic fibrewise group epimorphism

$$\chi_\gamma : J \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z}). \tag{72}$$

Write

$$\theta_\gamma := \mathrm{pr}_{\mathbb{R}/\mathbb{Z}} \circ \chi_\gamma : J \longrightarrow \mathbb{R}/\mathbb{Z} \tag{73}$$

for its circle coordinate. Thus $\chi_\gamma = (f, \theta_\gamma)$; the two symbols have different codomains and will not be interchanged below. Its kernel is the connected real three-torus family

$$\mathcal{H} := \frac{\mathfrak{h}^\circ_z \times ((L_\Lambda \otimes \mathbb{R})/L_\Lambda)}{\Delta}, \tag{74}$$

where $\Delta = \pi_1(B^\circ)$ acts diagonally through the deck action on $\mathfrak{h}^\circ_z$ and the restricted monodromy on L_Λ. Thus every fibre of $\mathcal{H}$ is $(L_\Lambda \otimes \mathbb{R})/L_\Lambda$; the family need not be a product. The quotient is real and is not a holomorphic quotient of complex tori.

(ii) Exact radial quotient metric. *On the positive radial cusp, the restriction of equation (71) to $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$ is*

$$\chi_{\gamma,T}([y]) = \left\langle P_T^{-\mathsf{T}} e_\gamma, y \right\rangle \bmod \mathbb{Z}, \qquad e_\gamma = (1,0,0,0)^{\mathsf{T}}. \tag{75}$$

For the Riemannian quotient of the Euclidean metric, the target circle has circumference

$$\ell_T = \frac{1}{\|P_T^{-\mathsf{T}} e_\gamma\|} = \frac{D_T}{\sqrt{L_T^2 + m_T^2}} = T + H - B + O(T^{-1}). \tag{76}$$

In particular, $\ell_T \to \infty$.

(iii) Four copies and minimality. *The source*

$$J^{[4]} = J \times_{B^\circ} J \times_{B^\circ} J \times_{B^\circ} J \tag{77}$$

is the actual fourfold fibre product over all of B°. *The product map*

$$\chi_\gamma^{[4]} = (\chi_\gamma, \chi_\gamma, \chi_\gamma, \chi_\gamma) : J^{[4]} \longrightarrow B^\circ \times (\mathbb{R}/\mathbb{Z})^4 \tag{78}$$

has kernel $\mathcal{H}^{[4]} = \mathcal{H} \times_{B^\circ} \cdots \times_{B^\circ} \mathcal{H}$, *whose fibres are* $\mathbb{T}^{12}$. *Along the synchronized positive radial cusp, its quotient fibre, with product quotient metric, is*

$$G_T = (\mathbb{R}/\ell_T\mathbb{Z})^4, \qquad \operatorname{sys}(G_T) = \ell_T. \tag{79}$$

Consequently

$$(G_T, 0) \longrightarrow (\mathbb{R}^4, 0) \tag{80}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. If $\ell_T > 4R$, *the radius-*R *ball about zero is exactly Euclidean.*

Equivalently, the external Cartesian product gives a quotient

$$J^4 \longrightarrow (B^\circ)^4 \times (\mathbb{R}/\mathbb{Z})^4,$$

and $J^{[4]}$ *is its pullback along the diagonal* $B^\circ \to (B^\circ)^4$. *At four independent cusp parameters its target lattice is* $\operatorname{diag}(\ell_{T_1}, \dots, \ell_{T_4})\mathbb{Z}^4$, *so the same pointed limit holds whenever* $\min_i \ell_{T_i} \to \infty$.

Among connected torus quotients obtained from r *copies by a primitive full-monodromy-invariant kernel and whose quotient systole tends to infinity on the radial cusp, the quotient rank is at most* r. *Thus four copies are minimal for a four-dimensional all-expanding quotient, and equation* (78) *attains the bound.*

The cusp-local two-copy map

$$\pi_T \times \pi_T : F_T \times F_T \longrightarrow Q_T \times Q_T \tag{81}$$

from theorem 9.6 also has pointed limit $\mathbb{R}^4$. *It is not a quotient over the full* B°, *however, and no finite base cover makes it one.*

(iv) Finite fillings. *At* p_j, *use the notation of theorem 8.7 and put*

$$(m_1, v_1, \kappa_1) = (3, \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, 1),$$

$$(m_2, v_2, \kappa_2) = (4, -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w}, -1), \qquad \kappa_j = \varepsilon_\gamma(v_j). \tag{G.2}$$

In the un-conjugated flat coordinates, the finite filling generator is

$$\Phi_j(s, x) = \left(e^{-2\pi i/m_j} s, A_j x + \frac{v_j}{m_j} \right). \tag{82}$$

The raw character is not invariant:

$$\varepsilon_\gamma\left(A_j x + \frac{v_j}{m_j} \right) = \varepsilon_\gamma(x) + \frac{\kappa_j}{m_j} \pmod{\mathbb{Z}}. \tag{83}$$

It instead descends equivariantly to the Seifert circle family

$$C_j = \frac{\Delta_j \times (\mathbb{R}/\mathbb{Z})}{(s,\theta) \sim (e^{-2\pi i/m_j} s, \theta + \kappa_j/m_j)}. \tag{84}$$

The total space C_j is a smooth solid torus, the map $C_j \to D_j$ has a multiple central circle of order m_j, and the induced map $N_j \to C_j$ is a smooth proper $\mathbb{T}^3$-bundle. The target translation in equation (84) *is exactly compatible with the logarithmic gluing map h_j.*

For four synchronized copies, the smooth simultaneous-root extension is

$$\widetilde{N}_j^{[4]} = \frac{(J_j')^{\times_{\Delta_j} 4}}{\langle (\widetilde{g}_j^{\log})^{\times 4} \rangle}, \tag{85}$$

with target

$$C_j^{[4]} = \frac{\Delta_j \times (\mathbb{R}/\mathbb{Z})^4}{(s,\theta_1,\dots,\theta_4) \sim (e^{-2\pi i/m_j} s, \theta_1 + \kappa_j/m_j, \dots, \theta_4 + \kappa_j/m_j)}. \tag{86}$$

The map $\widetilde{N}_j^{[4]} \to C_j^{[4]}$ is a smooth proper $\mathbb{T}^{12}$-bundle. There is a canonical finite map

$$\nu_j : \widetilde{N}_j^{[4]} \longrightarrow N_j \times_{D_j} N_j \times_{D_j} N_j \times_{D_j} N_j \tag{87}$$

which is an isomorphism over D_j^*, is the normalization of the reduced ordinary filled fibre product, and has degree m_j^3 on the reduced central fibre. These two degrees are 27 *and* 64*. Gluing these two local normalizations to $J^{[4]}$ by $(h_j)^{\times 4}$, and gluing their targets to the product $B^\circ \times (\mathbb{R}/\mathbb{Z})^4$ by $(h_j^{S^1})^{\times 4}$, gives the corresponding normalization and Seifert target over $B \setminus \{p_0\}$.*

(v) Cusp nonextension. *Neither the literal map χ_γ nor $\chi_\gamma^{[4]}$ extends continuously through the toric cusp fibre W. Thus the four-copy quotient is global over B°, and its finite-filling normalization is global over $B \setminus \{p_0\}$, but the displayed map $\chi_\gamma^{[4]}$ has no continuous extension to the compact fourfold fibre product or to X^4 with the stated regular-fibre restriction, and it does not factor through $X^{\wedge 4} \cong S^{24}$. The cohomology class of χ_γ does extend over N_0; this weaker homotopy statement does not extend the displayed quotient map or its metric.*

(vi) Products, smashes, and sphere constructions. *If the recognition $X \cong_{\mathrm{diff}} S^6$ is used, then*

$$X^2 \cong_{\mathrm{diff}} S^6 \times S^6 \not\simeq S^{12}, \qquad H_6(X^2;\mathbb{Z}) \cong \mathbb{Z}^2, \tag{88}$$

and

$$\begin{aligned} X^4 &\cong_{\mathrm{diff}} (S^6)^4 \not\simeq S^{24}, \\ H_6(X^4;\mathbb{Z}) \cong \mathbb{Z}^4, &\qquad H_{12}(X^4;\mathbb{Z}) \cong \mathbb{Z}^6, \quad H_{18}(X^4;\mathbb{Z}) \cong \mathbb{Z}^4. \end{aligned} \tag{89}$$

The synchronized regular fourfold $J^{[4]}$ has real dimension 18*, not* 24*. Choosing a base point gives based homeomorphisms*

$$X \wedge X \cong S^{12}, \qquad X^{\wedge 4} \cong S^{24}, \tag{90}$$

but these smash spheres do not carry either literal fixed-target *quotient above. The later cone constructions theorems 9.12 to 9.14 change the codomain by retaining the radial parameter and collapsing the obstructing boundary cross-section; they therefore do not contradict this nonextension theorem. Moreover, neither S^{12} nor S^{24}, with any differentiable structure, admits an almost-complex structure [62, §15, Proposition 15.1 and the following remark, p. 434]; hence neither can be promoted to a compact complex manifold. This is an obstruction to a proposed complex S^{12}- or S^{24}-globalization, not to the real quotient or its pointed metric limit. Precisely, $\pi_T \times \pi_T$ is a map on F_T^2, and it fails to factor through $F_T \wedge F_T$; $\chi_{\gamma,T}^{[4]}$ is a map on F_T^4, and it fails to factor through $F_T^{\wedge 4}$. Neither map is a map on X^2 or X^4, and no extension with the stated fibre restriction exists.*

There is one exact but different sphere construction. If $C_T = \mathbb{R}/\ell_T\mathbb{Z}$ is based at zero, postcomposition with the target fat-wedge collapse gives

$$F_T^4 \xrightarrow{\chi^{[4]}_{\gamma,T}} C_T^4 \longrightarrow C_T^{\wedge 4} \cong S^4. \tag{91}$$

This map does factor through $F_T^{\wedge 4}$. With the quotient length metric and base point $[\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2]$, its pointed limit is $\mathbb{R}^4$, with every radius-R ball exactly Euclidean once $\ell_T > 4R$. The metric is singular at the collapsed fat wedge, and the map still does not extend to X^4 or $X^{\wedge 4}$.

Proof. Invariant lattices and the regular group law.

The matrices in theorem 5.1 give

$$(M_0 - I)(a, b, c, d)^{\mathsf{T}} = (0, 0, b, -a)^{\mathsf{T}},$$

so

$$K_\Lambda = \ker(M_0 - I) = \operatorname{im}(M_0 - I). \tag{92}$$

They also give

$$\begin{aligned} &A_1\widehat{u} = -\widehat{w} + \widehat{\delta}, \quad A_1\widehat{w} = \widehat{u} - \widehat{w}, \quad A_1\widehat{\delta} = \widehat{\delta}, \\ &A_2\widehat{u} = \widehat{w}, \qquad\quad A_2\widehat{w} = -\widehat{u} + \widehat{\delta}, \quad A_2\widehat{\delta} = \widehat{\delta}. \end{aligned} \tag{93}$$

Thus L_Λ is invariant. It is the smallest invariant lattice containing K_Λ, because $\widehat{u} = A_1\widehat{w} + \widehat{w}$. Both lattices are primitive.

For a row covector (a, b, c, d), invariance under A_1 gives $b = 2c$ and $d = 3c$, whereas invariance under A_2 gives $b = c$ and $d = 2c$. Hence $b = c = d = 0$, and

$$\operatorname{Hom}(\Lambda, \mathbb{Z})^\Delta = \mathbb{Z}\varepsilon_\gamma. \tag{94}$$

The family over the cover is $(\mathfrak{h}_z^\circ \times \mathbb{C}^2)/\Lambda$, and the action used to form J is

$$[z, \zeta] \longmapsto [gz, R_g(z)\zeta].$$

It is complex linear in every fibre and fixes the zero section. Fibrewise addition, inversion, and zero are therefore equivariant and descend to J. In the flat real-analytic marking

$$\Xi(z, x + \Lambda) = [z, \Pi(z)x],$$

equation (2.3d) makes the action

$$(z, x + \Lambda) \longmapsto (gz, A_g x + \Lambda).$$

Equation equation (94) proves that equation (71) is well defined. Primitivity of ε_γ proves surjectivity and connectedness of the kernel. The induced action on $\Lambda/L_\Lambda \simeq \mathbb{Z}$ is trivial, so the target is the product circle bundle. Since the kernel has odd real dimension, this is not a holomorphic quotient.

The quotient metric and the four-copy limit.

The first column of equation (15) is

$$P_T^{-\mathsf{T}} e_\gamma = \left(0, \frac{m_T}{D_T}, 0, -\frac{L_T}{D_T}\right)^{\mathsf{T}}. \tag{95}$$

The quotient metric on $\mathbb{R}/\mathbb{Z}$ is therefore $\ell_T^2 d\theta^2$. Equivalently,

$$P_T(L_\Lambda \otimes \mathbb{R}) = \operatorname{span}_{\mathbb{R}}\{e_1, e_3, v_{u,T}\}$$

has unit normal

$$\nu_T = \frac{(0, m_T, 0, -L_T)}{\sqrt{L_T^2 + m_T^2}},$$

and

$$\operatorname{dist}(v_{\gamma,T}, P_T(L_\Lambda \otimes \mathbb{R})) = |\langle v_{\gamma,T}, \nu_T \rangle| = \frac{L_T q_T + 6m_T^2}{\sqrt{L_T^2 + m_T^2}} = \ell_T.$$

The expansions in equations (9) and (11) give

$$D_T = T^2 + (2H - B)T + O(1), \qquad \sqrt{L_T^2 + m_T^2} = T + H + O(T^{-1}),$$

which proves equation (76).

After rescaling the circle coordinate by ℓ_T, the product target is $\mathbb{R}^4/\ell_T\mathbb{Z}^4$. If $x, y \in B_R(0)$ and $0 \neq \lambda \in \ell_T\mathbb{Z}^4$, then

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| > \ell_T - 2R > 2R \geq \|x - y\|$$

when $\ell_T > 4R$. Hence the covering projection is an exact isometry on $B_R(0)$, proving all three convergences in equation (80). The quotient map is global over B°; only the metric limit is evaluated along the radial cusp. The separately chosen Euclidean fibre metric is not claimed to be a canonical global metric away from that cusp.

Minimality and the finite-base-cover obstruction.

Let $N \subset \Lambda^{\oplus r}$ be a primitive invariant kernel whose connected quotient has radial systole tending to infinity. Every element of $K_\Lambda^{\oplus r}$ has uniformly bounded physical length, since

$$P_T\widehat{w} = e_1, \qquad P_T\widehat{\delta} = e_3.$$

If such an element survived in the quotient lattice, its quotient norm would be no greater than its source norm and would give a nonzero uniformly bounded period. Therefore $K_\Lambda^{\oplus r} \subset N$. Invariance and equation (93) then force $L_\Lambda^{\oplus r} \subset N$, and

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \leq 4r - 3r = r.$$

This proves the lower bound $r \geq 4$, while $N = L_\Lambda^{\oplus 4}$ proves sharpness.

For the finite-cover assertion, put

$$H_* = A_2A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ -5 & 0 & -1 & 1 \end{pmatrix} = I + N_*. \tag{96}$$

Direct multiplication gives $N_*^2 = 0$, and hence

$$H_*^n\widehat{w} = \widehat{w} + n(\widehat{u} - \widehat{\delta}), \qquad H_*^n\widehat{\delta} = \widehat{\delta}. \tag{97}$$

Thus $H_*^n K_\Lambda \neq K_\Lambda$ for every $n \neq 0$. Every finite-index subgroup of the base fundamental group contains a positive power of the loop with monodromy H_*. Such a subgroup cannot preserve K_Λ, proving that no finite base cover globalizes the native two-torus quotient.

The affine finite fillings and the actual overlap maps.

Equation equation (94) gives $\varepsilon_\gamma A_j = \varepsilon_\gamma$, so equation (83) follows. Define

$$\overline{\Phi}_j(s, \theta) = \left(e^{-2\pi i/m_j} s, \theta + \frac{\kappa_j}{m_j} \right).$$

Then $(s,x) \mapsto (s, \varepsilon_\gamma(x))$ is $(\Phi_j, \overline{\Phi}_j)$-equivariant; denote the descended map by $\overline{\chi}_j : N_j \to C_j$. A nonidentity power which fixes the base must lie over $s = 0$, where it translates the circle by $k\kappa_j/m_j$. Since $\gcd(\kappa_j, m_j) = 1$, this is nonintegral for $0 < k < m_j$. The target action is free; the source action is free by theorem 8.7.

For an explicit solid-torus chart, put $\lambda_j = e^{2\pi i/m_j}$, $u = e^{2\pi i\theta}$, and choose $r_j \in \mathbb{Z}$ with $r_j\kappa_j \equiv 1 \pmod{m_j}$. Then

$$w = su^{r_j}, \qquad U = u^{m_j}$$

are invariant and identify C_j real-analytically with a solid torus. In these coordinates

$$t_j = s^{m_j} = w^{m_j}U^{-r_j},$$

which proves the central multiplicity. Equivariance and the primitive kernel give the proper $\mathbb{T}^3$-bundle $N_j \to C_j$.

It remains to check the actual logarithmic gluing. Let $\Pi_{\mathbb{R}}(z) : \Lambda \otimes \mathbb{R} \to \mathbb{C}^2$ be the underlying real isomorphism. On a logarithm branch ℓ, define $b_j^\ell(z)$ uniquely by

$$\Pi_{\mathbb{R}}(z)b_j^\ell(z) = \frac{\ell(z)}{2\pi i}\Pi(z)v_j, \qquad \beta_j(z) = \varepsilon_\gamma(b_j^\ell(z)) \bmod \mathbb{Z}. \tag{98}$$

Changing ℓ by $2\pi i n$ adds nv_j to b_j^ℓ, hence adds $n\kappa_j \in \mathbb{Z}$ to its ε_γ-value. Thus β_j is branch independent. In flat coordinates the source conjugacy (2.9e) is translation by b_j^ℓ. Applying ε_γ to that conjugacy gives

$$\beta_j(g_jz) = \beta_j(z) - \frac{\kappa_j}{m_j} \pmod{\mathbb{Z}}. \tag{99}$$

Therefore

$$h_j^{S^1}(z,\theta) = (z, \theta + \beta_j(z))$$

conjugates $\overline{\Phi}_j$ to the linear target action $(z,\theta) \mapsto (g_jz, \theta)$, and the overlap identity is

$$\chi_\gamma \circ h_j = h_j^{S^1} \circ \overline{\chi}_j. \tag{100}$$

The diagonal action on $(J_j')^{\times_{\Delta_j} 4}$ is free, and the same coprimality argument proves that $C_j^{[4]}$ is smooth. Every fibre of the resulting map is a $\mathbb{T}^{12}$-torsor; over D_j^* its structure torus is the fourfold fibre product of $\mathcal{H}$. No group zero is asserted at the affine filling. In local etale coordinates along the central fibre, the ordinary filled fibre product has equations

$$s_i^{m_j} = s_1^{m_j}, \qquad i = 2,3,4. \tag{101}$$

It has m_j^3 smooth branches along the reduced central fibre. The simultaneous-root model separates them, is smooth and hence normal, and is an isomorphism away from the central fibre. The map ν_j is proper: both spaces are proper over D_j, and the inverse image of a compact subset of the target is closed in the inverse image of its compact image in D_j. It is quasi-finite, since its fibres are singletons away from the central fibre and have at most m_j^3 points over the central fibre. Hence ν_j is finite. The universal property of normalization now proves that it is the normalization of the reduced ordinary fibre product. On the central fibre the relative root choices form $(\mathbb{Z}/m_j)^4/\operatorname{diag}(\mathbb{Z}/m_j)$, of order m_j^3, which proves the asserted degrees.

On D_j^*, the source identification is $(h_j)^{\times 4}$, and equation (100) identifies the target overlap with $(h_j^{S^1})^{\times 4}$. These are precisely the fourfolds of the one-copy overlap maps and therefore satisfy the cocycle identities. They glue the two local normalizations and Seifert targets to $J^{[4]}$ and to $B^\circ \times (\mathbb{R}/\mathbb{Z})^4$, respectively. Equivalently, the source gluing follows from uniqueness of normalization.

The fixed product-target trivialization does not extend over X, even if base winding is allowed. Indeed, in the presentation of theorem 39.6, let a homomorphism to $\mathbb{Z}$ restrict to $n\varepsilon_\gamma$ on Λ and take values a, b on $\widehat{\rho}_1, \widehat{\rho}_2$. The three attachment relations give

$$3a = n, \qquad 4b = -n, \qquad a + b = 0.$$

The first two give $a = n/3$, $b = -n/4$, while the third gives $n/12 = 0$. Hence $n = 0$. This is the exact global obstruction behind the fact that the two Seifert targets do not combine into a fixed-circle map on X.

The explicit toric cusp obstruction.

On the positive radial ray, write

$$Z_T = iTB_0 + C(t_c), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad t_c = e^{-2\pi T}.$$

For $a = (a_1, a_2)^{\mathsf{T}} \in \mathbb{R}^2$, take the point of the regular fibre represented by $Z_T a$, with zero short-period coordinates, and exponentiate it to the dense torus of the cusp model:

$$x_T(a) = \exp(2\pi i Z_T a) \in (\mathbb{C}^*)^2.$$

Its logarithmic position is

$$y_T(a) = \frac{(\log|x_{T,1}(a)|, \log|x_{T,2}(a)|)}{\log|t_c|} = B_0 a + \frac{\operatorname{Im} C(t_c) a}{T}. \tag{102}$$

The circle character has the constant value

$$\chi_{\gamma,T}(x_T(a)) = a_1 \bmod \mathbb{Z}. \tag{103}$$

Use the affine toric chart of $\sigma = \operatorname{conv}(0, e_1, e_2)$. Its coordinates are

$$z_0 = \frac{t_c}{x_1 x_2}, \qquad z_1 = x_1, \qquad z_2 = x_2. \tag{104}$$

If $B_0 a = (y_1, y_2)$ is in the interior of σ, then $y_1 > 0$, $y_2 > 0$, and $1 - y_1 - y_2 > 0$. For all sufficiently large T, the three absolute values in equation (104) are

$$|t_c|^{1-y_{T,1}-y_{T,2}}, \qquad |t_c|^{y_{T,1}}, \qquad |t_c|^{y_{T,2}},$$

and all tend to zero. Every such path converges to the same toric fixed point and hence to the same triple point of W.

Take

$$a = \left(-\frac{1}{4}, \frac{1}{4}\right), \qquad a' = \left(-\frac{1}{2}, \frac{1}{4}\right).$$

Then

$$B_0 a = \left(\frac{1}{4}, \frac{1}{4}\right), \qquad B_0 a' = \left(\frac{1}{4}, \frac{1}{2}\right)$$

are both in the interior of σ. The paths converge to the same point of W, whereas

$$\chi_{\gamma,T}(x_T(a)) = \frac{3}{4}, \qquad \chi_{\gamma,T}(x_T(a')) = \frac{1}{2}.$$

No continuous extension of the literal map χ_γ exists. Choosing identical marked radial paths in the other three factors and using these two paths in one factor proves the same assertion for $\chi_\gamma^{[4]}$.

The radial family of translations by $\widehat{u}/2$ in $\mathcal{H}$ fails to extend as well. Take two paths whose limiting logarithmic positions are

$$y = \left(\frac{1}{8}, \frac{1}{8}\right), \qquad y' = \left(\frac{3}{8}, \frac{3}{8}\right).$$

Both approach the fixed point of $\mathrm{conv}(0, e_1, e_2)$. Translation by $\widehat{u}/2 \in \mathcal{H}_T$ adds the exact displacement

$$\frac{1}{2} B_0 e_2 + \frac{1}{2T} \operatorname{Im} C(t_c) e_2 \longrightarrow \left(\frac{1}{2}, 0\right)$$

to logarithmic position. The translated limits are

$$\left(\frac{5}{8}, \frac{1}{8}\right) \quad \text{and} \quad \left(\frac{7}{8}, \frac{3}{8}\right).$$

The first lies in the interior of $\mathrm{conv}(0, e_1, e_2)$, while the second lies in the interior of $\mathrm{conv}(e_1, e_2, e_1 + e_2)$. These two triangle types are the two translation orbits in the A_2 triangulation and give the two distinct triple points of W, as proved in theorem 8.4. Thus one regular-fibre translation sends two paths with the same limit to paths with different limits; it has no continuous extension to N_0.

This pointwise obstruction must not be confused with the cohomology class. By theorem 39.6,

$$\pi_1(N_0) = \Lambda / K_\Lambda,$$

and ε_γ kills K_Λ. It therefore defines an integral H^1-class on N_0, hence a map to the Eilenberg–Mac Lane space S^1 whose restriction is homotopic to χ_γ. The explicit paths prove that no such representative agrees with χ_γ on the whole punctured cusp. The phase torus $(K_\Lambda \otimes \mathbb{R})/K_\Lambda$ extends as the compact real torus action of the toric model, generally with boundary stabilizers; the full radial $\widehat{u}/2$-translation and the full quotient do not.

Products and smash quotients.

The integral Kunneth theorem gives

$$H_k(X^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{n}{j}}, & k = 6j, \quad 0 \le j \le n, \\ 0, & \text{otherwise}, \end{cases} \qquad \chi(X^n) = 2^n,$$

which proves equations (88) and (89). The synchronized regular n-fold fibre product has dimension

$$6n - 2(n-1) = 4n + 2,$$

which gives 18 for four copies.

A product map factors through a smash quotient if and only if it is constant on the corresponding fat wedge. On $F_T \times \{0\}$, the local two-copy quotient is

$$x \longmapsto (\pi_T(x), 0),$$

which is nonconstant. Thus the literal conclusion is that $\pi_T \times \pi_T$ does not factor through $F_T \wedge F_T$; it would be a type error to call it a map on $X \wedge X$. The restriction of $\chi_{\gamma,T}^{[4]}$ to $F_T \times \{0\}^3$ is similarly

$$x \longmapsto (\chi_{\gamma,T}(x), 0, 0, 0),$$

so it does not factor through $F_T^{\wedge 4}$.

There is no global extension which repairs the domain mismatch: $H^1(X^n; \mathbb{Z}) = 0$, whereas both local quotient maps pull nonzero target H^1-classes back to nonzero classes on the cusp fibre. Hence no map on X^2 or X^4 has the required restriction, and precomposition with the smash collapses in equation (90) cannot produce the local quotient maps.

Identify the metric quotient circle with $C_T = \mathbb{R}/\ell_T\mathbb{Z}$ by $\theta \mapsto \ell_T\theta$. For equation (91), target-smash collapse is constant whenever one source coordinate is the identity, so the modified map does factor through $F_T^{\wedge 4}$. Its chosen base point has distance $\ell_T/2$ from the collapsed target fat wedge. If two points lie within distance R of it, a route through the collapsed point has length at least $\ell_T - 2R$, while their direct Euclidean distance is at most $2R$. When $\ell_T > 4R$, neither the collapse nor circle wrap-around shortens distances in that ball.

Let $u_i \in H^1(F_T^4;\mathbb{Z})$ be the pullback of the generator from the i-th quotient circle. The orientation class of $S^4 = C_T^{\wedge 4}$ pulls back under equation (91) to

$$\pm u_1 \smile u_2 \smile u_3 \smile u_4 \neq 0.$$

But $H^4(X^4;\mathbb{Z}) = 0$. Thus even this modified S^4-target map does not extend to X^4, and a fortiori does not arise from a map on $X^{\wedge 4} \cong S^{24}$.

For completeness, Borel–Serre prove that S^{2n} admits no almost-complex structure for $n \geq 4$. Their argument applies to every differentiable structure on the sphere: on S^{2n}, all lower Chern classes vanish, while their reduced-power calculation forces the top Chern class to vanish modulo a suitable odd prime $p < n$; on the other hand that class is the Euler class, namely $2h$ for a generator $h \in H^{2n}(S^{2n};\mathbb{Z})$, a contradiction. Taking $n = 6$ and $n = 12$ gives the asserted obstructions for S^{12} and S^{24}. This topological obstruction does not itself prohibit the real four-circle target G_T, the target-smash S^4, or their metric limits.

Finally, $X * X \cong S^{13}$ and $X^{*4} \cong S^{27}$, so joins do not produce the proposed S^{12} or S^{24} dimensions. Suspensions can produce spheres of those dimensions only by discarding the torus-fibration maps. □

9.1 The Kummer cover extends across the toric cusp

This section determines exactly what is true in the assertion that the degree-N^2 cover of the regular fibres extends by a toric power map. The answer is positive for the completed *local* cusp family, but the source action has to be changed, and two different degrees must not be conflated.

Retain the cusp data

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s},$$

where C is holomorphic on a disc Δ_t about zero. Put

$$c_{\bar\lambda}(t) := \exp\bigl(2\pi i C(t)\bar\lambda\bigr) \in (\mathbb{C}^*)^2, \qquad \bar\lambda \in \bar\Lambda \simeq \mathbb{Z}^2.$$

Let Σ be the fan in $N' = \mathbb{Z}^2 \oplus \mathbb{Z}$ consisting of the cones over the cells of the translation-invariant A_2 triangulation at height one, and write $Y = Y_\Sigma$. On the dense torus, with coordinates (x_1, x_2, t), the target action is

$$\Psi_{\bar\lambda}(x,t) = \bigl(c_{\bar\lambda}(t)t^{B_0\bar\lambda}x, t\bigr). \tag{105}$$

Its quotient is the completed cusp family $f_0 \colon N_0 = Y/\bar\Lambda \to \Delta_t$.

Fix an integer $N \geq 1$. Let v be a coordinate on a second disc and make the base change

$$\rho_N \colon \Delta_v \longrightarrow \Delta_t, \qquad \rho_N(v) = v^N, \qquad t = v^N. \tag{106}$$

On a copy $Y^{(N)}$ of the same toric variety, with dense-torus coordinates (ξ_1, ξ_2, v), define

$$C_N(v) := \frac{1}{N}C(v^N), \qquad c_{\bar\lambda}^{(N)}(v) := \exp\bigl(2\pi i C_N(v)\bar\lambda\bigr) \tag{107}$$

and

$$\Psi_{\bar\lambda}^{(N)}(\xi, v) := \bigl(c_{\bar\lambda}^{(N)}(v)v^{B_0\bar\lambda}\xi, v\bigr). \tag{108}$$

Theorem 9.8 (Exact Kummer extension through the cusp)**.** ***[VERIFIED]****After shrinking the v-disc, the following statements hold.*

1. *The maps in* (108) *extend to a free, properly discontinuous action on $Y^{(N)}$. The quotient*

$$f_N\colon N_0^{\langle N\rangle} := Y^{(N)}/\bar{\Lambda} \longrightarrow \Delta_v$$

is a proper holomorphic map from a smooth threefold. Its central fibre W_N is reduced, irreducible, and nonnormal, and its normalization is a toric del Pezzo surface of degree six.

2. *Multiplication by N on the cocharacter lattice defines a finite flat toric morphism*

$$P_N\colon Y^{(N)} \longrightarrow Y, \qquad P_N(\xi_1,\xi_2,v) = (\xi_1^N,\xi_2^N,v^N), \tag{109}$$

of degree N^3. It is equivariant for (108) *and* (105)*, hence descends to a finite flat map*

$$\kappa_N\colon N_0^{\langle N\rangle} \longrightarrow N_0, \qquad f_0\circ\kappa_N = \rho_N\circ f_N, \tag{110}$$

of degree N^3.

3. *The map κ_N is etale over $v\neq 0$, and*

$$\kappa_N^* W = NW_N, \qquad \mathrm{Ram}(\kappa_N) = (N-1)W_N. \tag{111}$$

In particular, the branch support of the map to the original cusp family is the whole central fibre.

4. *Let*

$$N_0[N] := N_0\times_{\Delta_t}\Delta_v, \qquad t = v^N. \tag{112}$$

Then $N_0[N]$ is normal and κ_N factors uniquely as

$$N_0^{\langle N\rangle} \xrightarrow{Q_N} N_0[N] \longrightarrow N_0. \tag{113}$$

The map Q_N has generic degree N^2. For $N>1$ it is quasi-etale but not flat: it is etale over the smooth locus of $N_0[N]$, and its non-etale and nonflat loci are exactly the inverse images of the double and triple strata of the base-changed central fibre. Thus Q_N is not a normalization map.

5. *On the reduced central fibres, the induced map $W_N\to W$ has generic degree N^2. It lifts to a commutative diagram*

$$\begin{array}{ccc} E_N\simeq \mathrm{dP}_6 & \xrightarrow{[N]} & E\simeq \mathrm{dP}_6 \\ {\scriptstyle\nu_N}\big\downarrow & & \big\downarrow{\scriptstyle\nu} \\ W_N & \longrightarrow & W, \end{array} \tag{114}$$

where $[N]$ is toric multiplication by N, of degree N^2, ramified with index N along each of the six toric boundary curves. The diagram respects the paired conductor curves and the two triple-point classes.

6. *Over $v\neq 0$, Q_N is exactly the cover defined by*

$$\mathbb{Z}\langle\widehat{\gamma},\widehat{u},N\widehat{w},N\widehat{\delta}\rangle \subset \mathbb{Z}\langle\widehat{\gamma},\widehat{u},\widehat{w},\widehat{\delta}\rangle.$$

Its deck group is $(\mathbb{Z}/N\mathbb{Z})^2$. The pulled-back cusp monodromy is M_0^N, and in the displayed cover basis it is M_0:

$$S_N^{-1}M_0^N S_N = M_0, \qquad S_N = \mathrm{diag}(1,1,N,N). \tag{115}$$

Proof. We prove the assertions in the order in which the constructions occur.

The source action. For $\bar\lambda \in \bar\Lambda$, the integral linear map

$$\varphi_{\bar\lambda}(y, y_3) = (y + y_3 B_0 \bar\lambda, y_3)$$

translates the height-one slice by the integral vector $B_0\bar\lambda$. It preserves the A_2 triangulation and hence induces a toric automorphism $\Phi_{\bar\lambda}$ of $Y^{(N)}$. On the dense torus this automorphism is

$$\Phi_{\bar\lambda}(\xi, v) = (v^{B_0\bar\lambda}\xi, v).$$

Multiplication by the nowhere-zero holomorphic function $c_{\bar\lambda}^{(N)}(v)$ extends across every toric chart. Since $C_N(v)\bar\lambda$ is additive in $\bar\lambda$, the resulting maps satisfy $\Psi_{\bar\lambda}^{(N)}\Psi_{\bar\mu}^{(N)} = \Psi_{\bar\lambda+\bar\mu}^{(N)}$.

For completeness, freeness and proper discontinuity can be checked directly. Put

$$R_N(v) := -2\pi \operatorname{Im} C_N(v).$$

It is bounded at $v = 0$. At a dense-torus point fixed by $\bar\lambda$, logarithmic absolute values give

$$\log|v| B_0\bar\lambda + R_N(v)\bar\lambda = 0.$$

Choose the disc so small that $|\log|v|| > 2\|R_N\|_\infty$. Because B_0 is orthogonal, the last equality is impossible for $\bar\lambda \neq 0$. On the central fibre, $\Psi_{\bar\lambda}^{(N)}$ translates every bounded cell by the nonzero vector $B_0\bar\lambda$, so it fixes no toric orbit.

For a dense-torus point set

$$y(\xi, v) = \frac{(\log|\xi_1|, \log|\xi_2|)}{\log|v|}, \qquad B_{N,v} = B_0 + \frac{R_N(v)}{\log|v|}.$$

Then

$$y(\Psi_{\bar\lambda}^{(N)}(\xi, v)) = y(\xi, v) + B_{N,v}\bar\lambda, \qquad |B_{N,v}\bar\lambda| \geq \tfrac{1}{2}|\bar\lambda|.$$

The logarithmic-position regions associated with any two fixed toric charts are bounded. Hence only finitely many $\bar\lambda$ can translate one of them to meet the other. Local finiteness of the fan reduces the same assertion for compact sets to finitely many chart pairs, proving proper discontinuity. Finally, reducing $B_{N,v}^{-1}y(\xi, v)$ modulo $\mathbb{Z}^2$ places every orbit meeting a closed subdisc in finitely many closed unit toric charts. Their union is compact. This proves properness of f_N. The central-fibre and normalization assertions follow exactly from the height-one fan: locally v is the product of the coordinates belonging to the rays through a cell, every component occurs with multiplicity one, the lattice action is transitive on vertices, and the star of a vertex is the complete smooth hexagonal fan.

The toric map and equivariance. Multiplication by N on N' sends every cone $\sigma \in \Sigma$ onto itself as a real cone. It therefore defines the toric morphism P_N. For a maximal cone, let z_0, z_1, z_2 be the target characters dual to its three primitive height-one ray generators, and let w_0, w_1, w_2 be the corresponding source characters. The map on convergent local rings is

$$\mathbb{C}\{z_0, z_1, z_2\} \longrightarrow \mathbb{C}\{w_0, w_1, w_2\}, \qquad z_i \longmapsto w_i^N. \tag{116}$$

The target-chart preimage is the corresponding source chart, and the right-hand ring is free of rank N^3 over the left-hand ring, with basis

$$\{w_0^{a_0} w_1^{a_1} w_2^{a_2} : 0 \leq a_i < N\}.$$

Thus P_N is finite flat of degree N^3 on every maximal chart, hence globally.

The choice (107) is forced by equivariance. Indeed, writing powers coordinatewise,

$$P_N\Psi_{\bar\lambda}^{(N)}(\xi,v) = \big((c_{\bar\lambda}^{(N)}(v))^N v^{NB_0\bar\lambda}\xi^N, v^N\big) \tag{117}$$

$$= \big(c_{\bar\lambda}(v^N)(v^N)^{B_0\bar\lambda}\xi^N, v^N\big) \tag{118}$$

$$= \Psi_{\bar\lambda}P_N(\xi,v). \tag{119}$$

Consequently P_N descends to (110). To see finiteness and flatness after the infinite-lattice quotient without any implicit descent theorem, take an evenly covered open set in N_0 and one of its lifts $V \subset Y$. Equivariance gives a unique representative of every source orbit above that open set whose image lies in V; two such representatives cannot differ by a nontrivial lattice translate. The descended map is therefore locally identified with $P_N^{-1}(V) \to V$. It is finite flat of degree N^3 by (116).

In the same chart the base characters are

$$t = z_0z_1z_2, \qquad v = w_0w_1w_2, \qquad P_N^*t = v^N.$$

The Jacobian determinant of (116) is

$$N^3(w_0w_1w_2)^{N-1} = N^3v^{N-1}.$$

This proves (111); the local quotient maps are biholomorphisms, so the calculation is unchanged downstairs.

The relative degree-N^2 factor. The equality $(f_0\circ\kappa_N)([\xi,v]) = v^N = (\rho_N\circ f_N)([\xi,v])$ gives the factorization (113). At a triple point the exact local-ring map is

$$\begin{aligned} &A_N := \frac{\mathbb{C}\{z_0,z_1,z_2,v\}}{(z_0z_1z_2 - v^N)}, \qquad B := \mathbb{C}\{w_0,w_1,w_2\},\\ &A_N \longrightarrow B, \qquad z_i \longmapsto w_i^N, \qquad v \longmapsto w_0w_1w_2. \end{aligned} \tag{120}$$

Let

$$G_N = \{(\varepsilon_0,\varepsilon_1,\varepsilon_2) \in \mu_N^3 : \varepsilon_0\varepsilon_1\varepsilon_2 = 1\} \simeq \mu_N^2$$

act diagonally on B. A monomial $w_0^{a_0}w_1^{a_1}w_2^{a_2}$ is G_N-invariant if and only if

$$a_0 \equiv a_1 \equiv a_2 \pmod N.$$

Writing their common residue as r, with $0 \le r < N$, expresses that monomial uniquely in the form

$$(w_0w_1w_2)^r(w_0^N)^{b_0}(w_1^N)^{b_1}(w_2^N)^{b_2}.$$

The same coefficientwise argument applies to convergent power series. Hence

$$A_N = B^{G_N}. \tag{121}$$

It follows that A_N is normal, that the map has generic degree $|G_N| = N^2$, and that it is a finite quotient rather than a normalization. The corresponding face-chart calculations prove normality along the double and smooth strata as well, so $N_0[N]$ is normal everywhere.

The stabilizer in G_N is trivial when none or exactly one of the w_i vanishes, is isomorphic to μ_N when exactly two vanish, and is all of G_N when all three vanish. Thus Q_N is etale in codimension one and has no ramification divisor, but it is not etale above the double and triple strata. It is also nonflat at every such point when $N > 1$. At a triple point the closed fibre of (120) has length

$$\dim_{\mathbb{C}} \frac{\mathbb{C}\{w_0,w_1,w_2\}}{(w_0^N,w_1^N,w_2^N,w_0w_1w_2)} = N^3 - (N-1)^3 \tag{122}$$

$$= 3N^2 - 3N + 1, \tag{123}$$

which exceeds the generic rank N^2. At a point of a double stratum, localizing at the nonzero third coordinate and choosing its holomorphic Nth root reduces the ramified factor to

$$\frac{\mathbb{C}\{z_0, z_1, v\}}{(z_0 z_1 - v^N)} \longrightarrow \mathbb{C}\{w_0, w_1\}, \qquad z_i \longmapsto w_i^N, \quad v \longmapsto w_0 w_1.$$

Its local generic rank is N, whereas its closed fibre has length

$$\dim_{\mathbb{C}} \frac{\mathbb{C}\{w_0, w_1\}}{(w_0^N, w_1^N, w_0 w_1)} = 2N - 1 > N. \tag{124}$$

The remaining transverse nonzero torus coordinate is etale locally. Equations (123) and (124), together with etaleness elsewhere, identify the nonflat locus exactly.

The regular fibres and monodromy. Choose a logarithm r with $v = e^{2\pi i r}$. The target period matrix at $t = v^N$ is

$$Z(Nr) = NrB_0 + C(v^N),$$

whereas the source residual action is determined by

$$Z_N(r) = rB_0 + \frac{1}{N} C(v^N) = \frac{1}{N} Z(Nr).$$

The source fibre is therefore

$$\frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z_N(r)\mathbb{Z}^2},$$

and multiplication by N on $\mathbb{C}^2$ maps it to the target fibre. Equivalently, in the unscaled additive coordinate it is the identity map

$$\frac{\mathbb{C}^2}{N\mathbb{Z}^2 + Z(Nr)\mathbb{Z}^2} \longrightarrow \frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z(Nr)\mathbb{Z}^2}. \tag{125}$$

The imaginary part of $Z(Nr)$ is invertible on a regular fibre, so its period lattice meets $\mathbb{Z}^2$ trivially in the required real sense; the index in (125) is N^2. Its deck group is $(\mathbb{Z}/N\mathbb{Z})^2$. In the ordered lattice basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$, this is exactly the sublattice obtained by S_N. Since $M_0 = I + R$ with $R^2 = 0$ and $R(a, b, c, d) = (0, 0, b, -a)$, one has

$$M_0^N = I + NR, \qquad S_N^{-1} M_0^N S_N = M_0,$$

proving (115).

The central normalization and conductor. Multiplication by N fixes every ray of Σ. It therefore maps a chosen source component of the toric central fibre to the corresponding target component. The star of that ray is the hexagonal fan, so the restriction is the toric power map $[N]\colon \mathrm{dP}_6 \to \mathrm{dP}_6$. On its dense two-torus it has degree N^2; in each toric boundary chart it is an Nth-power map in the normal coordinate, hence has ramification index N along all six boundary curves. Equation (119) says that the source and target lattice identifications commute with this map. Consequently opposite boundary curves, their conductor identifications, and the two translation-orbits of triangle vertices descend compatibly. This proves (114) and completes the proof. □

Remark 9.9 (Why the unmodified action does not work). If one reuses the target action with v in place of t at the source, then the two sides of equivariance have multiplicative factors

$$c_{\bar{\lambda}}(v)^N v^{NB_0\bar{\lambda}} \quad \text{and} \quad c_{\bar{\lambda}}(v^N) v^{NB_0\bar{\lambda}},$$

respectively. In general $c_{\bar{\lambda}}(v)^N \neq c_{\bar{\lambda}}(v^N)$. Thus the bare formula $(\xi_1, \xi_2, v) \mapsto (\xi_1^N, \xi_2^N, v^N)$ extends on the ambient toric variety but does not descend through the manuscript's quotient unless the source factor (107) is inserted. The explicit logarithm $2\pi i C(v^N)\bar{\lambda}/N$ makes this root holomorphic; no unjustified choice of an Nth root is involved.

Remark 9.10 (Exact scope). The theorem constructs a finite morphism of completed families over the local cusp discs. It is a degree-N^3 map to the original family and a degree-N^2 map only after the target is base changed. It is neither a degree-N^2 self-map over the unchanged disc nor the normalization of the base change.

Nothing here supplies a cover over the entire base B. In fact the displayed sublattice is not preserved by the original full monodromy when $N > 1$: for example, the printed matrix A_1 sends $\widehat{u}$ to a vector whose $\widehat{w}$- and $\widehat{\delta}$-coefficients are -1 and 1. A further global base cover and compatible extensions through both finite fillings would have to be constructed separately. The local Kummer morphism also supplies no map to a product sphere, smash sphere, S^{12}, or S^{24}.

9.2 Cone-compactified sphere carriers for the native cusp quotient

This subsection answers a topological question which is strictly different from the finite-cover construction. No period is enlarged by passing to a cover. Instead, the native expanding quotient is retained on the cusp, the parts on which that quotient is unavailable are collapsed, and the resulting based map is passed through a smash product. The outcome is an exact global map with source S^{12}, and there is an analogous S^{24} map from the full-monodromy-invariant circle quotient. The targets are conically stratified spaces. The Euclidean limits below use the intrinsic metrics inherited from the flat quotient tori on the regular slices; they are not ambient metric limits in one fixed metric on the compact targets.

Fix a closed coordinate disc

$$D_0 = \{t \in \mathbb{C} : |t| \le \epsilon\} \subset B, \qquad p_0 = \{t = 0\}, \qquad N_0 = f^{-1}(D_0), \qquad W = f^{-1}(0), \tag{126}$$

small enough that it contains no singular value other than p_0. Put

$$K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle, \qquad \overline{\Lambda} = \Lambda / K_\Lambda, \qquad \overline{T}^2 = \frac{\overline{\Lambda} \otimes \mathbb{R}}{\overline{\Lambda}}. \tag{127}$$

The ordered basis of $\overline{\Lambda}$ is the image of $(\widehat{\gamma}, \widehat{u})$.

Lemma 9.11 (The native quotient over the whole punctured cusp)**.** *There is a well-defined real-analytic fibrewise group epimorphism*

$$\overline{\pi} : N_0 \setminus W \longrightarrow D_0^* \times \overline{T}^2 \tag{128}$$

whose kernel is the connected two-torus family defined by K_Λ. The target local system is trivial. On the positive radial fibre F_T, $t = e^{-2\pi T}$, the isomorphism

$$\iota_T : \overline{T}^2 \longrightarrow Q_T, \qquad [(a, b)] \longmapsto [Y_T (a, b)^{\mathsf{T}}] \tag{129}$$

satisfies

$$\pi_T = \iota_T \circ \overline{\pi}|_{F_T}. \tag{130}$$

If $\overline{T}^2$ is given the pullback by ι_T of the Euclidean quotient metric on $Q_T = \mathbb{R}^2 / Y_T \mathbb{Z}^2$, then ι_T is an isometry and

$$\mathrm{sys}(\overline{T}^2, g_T) = \mathrm{sys}(Q_T) \ge T - C. \tag{131}$$

Proof. On the universal cover of D_0^*, use the real-analytic flat marking

$$(z, x + \Lambda) \longmapsto [z, \Pi(z)x].$$

The deck generator acts by

$$(z, x + \Lambda) \longmapsto (g_0 z, M_0 x + \Lambda).$$

The displayed matrix M_0 gives

$$(M_0 - I)(a, b, c, d)^{\mathsf{T}} = (0, 0, b, -a)^{\mathsf{T}}, \qquad \mathrm{im}(M_0 - I) = K_\Lambda. \tag{132}$$

Consequently M_0 preserves K_Λ and induces the identity on $\overline{\Lambda}$. The formula

$$[z, \Pi(z)x] \longmapsto ([z], [x \bmod K_\Lambda])$$

is invariant under the deck action and proves equation (128). The lattice K_Λ is primitive, so the kernel is connected. On the radial fibre, projection to the (e_2, e_4)-coordinates sends $a\widehat{\gamma} + b\widehat{u}$ to $Y_T(a, b)^{\mathsf{T}}$ and kills K_Λ. This proves equations (129) and (130). The metric and systolic assertion is the quotient calculation in theorem 9.6. □

Define

$$\mathcal{A} = \frac{[0, \epsilon] \times S^1 \times \overline{T}^2}{\{0\} \times S^1 \times \overline{T}^2 \sim v_0, \quad \{\epsilon\} \times S^1 \times \overline{T}^2 \sim v_1}. \tag{133}$$

Thus

$$\mathcal{A} \cong \mathrm{Susp}(S^1 \times \overline{T}^2) \cong \mathrm{Susp}(T^3), \tag{134}$$

where Susp denotes the unreduced suspension. The extra circle is the angular coordinate of the punctured base. If one deliberately forgets that angle, the evident map $\mathcal{A} \to \mathrm{Susp}(\overline{T}^2)$ gives a smaller target with the same restriction to every fixed radial fibre. We retain $\mathcal{A}$ because it records the entire punctured-cusp quotient.

Theorem 9.12 (Exact localized S^{12} route)**.** *Choose $x_0 \in W$, and base $\mathcal{A}$ at v_0. There is a continuous surjective based quotient map*

$$q_2 : (X, x_0) \longrightarrow (\mathcal{A}, v_0) \tag{135}$$

with the following exact description. For $x \in N_0 \setminus W$, write

$$t(f(x)) = re^{i\phi}, \qquad 0 < r < \epsilon,$$

and let $\overline{\pi}_f(x) \in \overline{T}^2$ be the fibre component of $\overline{\pi}(x)$. Then

$$q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)], \qquad q_2(W) = v_0, \qquad q_2\big(X \setminus \operatorname{int} N_0\big) = v_1. \tag{136}$$

On $f^{-1}(|t| = \epsilon)$, the value is v_1.

The smash square is a global based map

$$q_2^{\wedge 2} : X \wedge X \longrightarrow \mathcal{A} \wedge \mathcal{A}. \tag{137}$$

Using $X \cong_{\mathrm{diff}} S^6$, a based choice of the diffeomorphism identifies its source with

$$X \wedge X \cong S^6 \wedge S^6 \cong S^{12}. \tag{138}$$

For $0 < e^{-2\pi T} < \epsilon$, the canonical product-to-smash quotient maps restrict to embeddings on F_T^2 and on the corresponding interior target-product cross-section. Under those embeddings and ι_T^2, the induced smash map restricts exactly to

$$\pi_T \times \pi_T : F_T \times F_T \longrightarrow Q_T \times Q_T. \tag{139}$$

With the Euclidean ℓ^2-product flat quotient metric,

$$\mathrm{sys}(Q_T \times Q_T) \geq T - C, \qquad (Q_T \times Q_T, (0, 0)) \longrightarrow (\mathbb{R}^4, 0) \tag{140}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology. Once $T - C > 4R$, the radius-R ball is exactly Euclidean.

The integral homology of the targets is

$$\widetilde{H}_k(\mathcal{A};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^3, & k = 2, 3, \\ \mathbb{Z}, & k = 4, \\ 0, & \textit{otherwise}, \end{cases} \tag{141}$$

and

$$\widetilde{H}_k(\mathcal{A} \wedge \mathcal{A};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^9, & k = 4, \\ \mathbb{Z}^{18}, & k = 5, \\ \mathbb{Z}^{15}, & k = 6, \\ \mathbb{Z}^6, & k = 7, \\ \mathbb{Z}, & k = 8, \\ 0, & \textit{otherwise}. \end{cases} \tag{142}$$

In particular, the target is not S^4, S^{12}, or a compact four-manifold.

Proof. The equivalence relation in equation (133) is closed: it is the union of the diagonal with the squares of the two closed end cross-sections. A compact Hausdorff space modulo a closed equivalence relation is compact Hausdorff. Rescaling r to the unit interval gives exactly the unreduced suspension in equation (134).

Well-definedness of the interior formula follows from theorem 9.11. A full turn in ϕ changes the marking by M_0, but M_0 is the identity modulo K_Λ.

We check both collapsed strata. If U is an open neighborhood of v_0, its inverse image in the cylinder is open and contains the compact set $\{0\} \times S^1 \times \overline{T}^2$. Hence, for some $\delta > 0$, it contains $[0, \delta) \times S^1 \times \overline{T}^2$. Therefore $q_2^{-1}(U)$ contains $f^{-1}(|t| < \delta)$, a neighborhood of W. The same compactness argument at the other end shows that every neighborhood of v_1 contains the image of $(\epsilon - \delta, \epsilon] \times S^1 \times \overline{T}^2$. Its preimage together with $X \setminus \operatorname{int} N_0$ is a neighborhood of the outer boundary. Continuity on the interior is equation (128); the closed pasting lemma gives continuity on X.

Every interior point has a preimage because $\overline{\pi}$ is fibrewise surjective. The nonempty sets W and $X \setminus \operatorname{int} N_0$ map onto v_0 and v_1, respectively. Thus q_2 is surjective. A continuous surjection from compact X to Hausdorff $\mathcal{A}$ is a quotient map.

The map is based, so functoriality gives equation (137). The standard CW quotient gives $S^m \wedge S^n \cong S^{m+n}$ [9, Chapter 0, Smash Product], proving equation (138).

The base point lies in W, so F_T^2 is disjoint from the source fat wedge. Its image lies in an interior cross-section of $\mathcal{A}^2$, disjoint from the target fat wedge. The source and target fat wedges are closed in the compact Hausdorff products X^2 and $\mathcal{A}^2$, so both smash quotients are Hausdorff. Their restrictions to the relevant compact subspaces are continuous and one-to-one, and are therefore embeddings. Equations equations (130) and (136) give equation (139).

The period lattice of the product target is $\operatorname{diag}(Y_T, Y_T)\mathbb{Z}^4$. Its systole is the systole of $Y_T\mathbb{Z}^2$, proving the first assertion in equation (140). If $x, y \in B_R(0) \subset \mathbb{R}^4$ and $0 \neq \lambda$ is a product period, then

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| \geq (T - C) - 2R > 2R \geq \|x - y\|$$

when $T - C > 4R$. The quotient projection is an exact isometry on the Euclidean radius-R ball. Conversely, any point of quotient distance less than R has a shortest lattice representative of norm less than R, so this embedded ball is the entire quotient radius-R ball. This proves all three pointed convergences.

Reduced suspension shifts reduced homology by one. Since

$$H_1(T^3) = \mathbb{Z}^3, \qquad H_2(T^3) = \mathbb{Z}^3, \qquad H_3(T^3) = \mathbb{Z},$$

equation (141) follows. All groups are free, so the Kunneth theorem for smash products has no Tor terms. The ranks in degrees $4, 5, 6, 7, 8$ are

$$3 \cdot 3, \quad 2(3 \cdot 3), \quad 2(3 \cdot 1) + 3 \cdot 3, \quad 2(3 \cdot 1), \quad 1,$$

which proves equation (142). □

The localized one-circle construction. The full-monodromy-invariant character also gives a smaller cusp-supported target. Define

$$\mathcal{B} = \frac{[0, \epsilon] \times S^1 \times S^1}{\{0\} \times S^1 \times S^1 \sim b_0, \quad \{\epsilon\} \times S^1 \times S^1 \sim b_1} \cong \mathrm{Susp}(T^2), \tag{143}$$

where the two circle coordinates in an interior cross-section are the base angle and the value of θ_γ.

Proposition 9.13 (Localized S^{24} route with target $\mathrm{Susp}(T^2)^{\wedge 4}$)**.** *There is a continuous surjective based quotient map*

$$q_{1,\mathrm{cusp}} : (X, x_0) \longrightarrow (\mathcal{B}, b_0) \tag{144}$$

given by

$$q_{1,\mathrm{cusp}}(x) = [r, e^{i\phi}, \theta_\gamma(x)] \quad (0 < |t| = r < \epsilon), \qquad q_{1,\mathrm{cusp}}(W) = b_0, \qquad q_{1,\mathrm{cusp}}\big(X \setminus \operatorname{int} N_0\big) = b_1. \tag{145}$$

Its fourth smash power has source $X^{\wedge 4} \cong S^{24}$, and on four synchronized radial fibres it is exactly

$$\chi_{\gamma,T}^{\times 4} : F_T^4 \longrightarrow C_T^4. \tag{146}$$

Consequently its intrinsic target slices satisfy the Euclidean limit equation (158)*. The target homology is*

$$\widetilde{H}_k(\mathcal{B}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^2, & k = 2, \\ \mathbb{Z}, & k = 3, \\ 0, & \textit{otherwise}, \end{cases} \tag{147}$$

and

$$\widetilde{H}_k(\mathcal{B}^{\wedge 4}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{16}, & k = 8, \\ \mathbb{Z}^{32}, & k = 9, \\ \mathbb{Z}^{24}, & k = 10, \\ \mathbb{Z}^{8}, & k = 11, \\ \mathbb{Z}, & k = 12, \\ 0, & \textit{otherwise}. \end{cases} \tag{148}$$

Proof. The monodromy invariance $\varepsilon_\gamma M_0 = \varepsilon_\gamma$ makes the interior formula single-valued after a turn in the base angle. The two endpoint-continuity arguments are identical to those in the proof of theorem 9.12, with compact cross-section T^2 in place of T^3. Fibrewise surjectivity of θ_γ, together with the nonempty inner fibre and exterior, proves surjectivity. Compact-to-Hausdorff makes the map a quotient map.

Since $x_0 \in W$, the fourth smash power exists. The radial regular fibres and their circle targets avoid the two closed fat wedges. The restrictions of the Hausdorff smash quotients to these compact subspaces are continuous and one-to-one, hence embeddings, and the fibre map is exactly equation (146). The intrinsic metric statement is therefore the four-circle calculation equation (158). Finally, suspension shifts the free homology of T^2, giving equation (147). In the fourth smash power, choosing j of the four factors in degree 3 and the other $4 - j$ factors in degree 2 gives degree $8 + j$ and rank $\binom{4}{j} 2^{4-j}$. This is equation (148). □

The full-base circle target before coning. Choose pairwise disjoint closed coordinate discs D_0, D_1, D_2 about the three singular values and put

$$P = B \setminus \bigcup_{j=0}^{2} \operatorname{int} D_j. \tag{149}$$

For $j = 1, 2$, let C_j be the solid-torus Seifert target in equation (84), and let $h_j^{S^1}$ be the target overlap map in equation (100). Glue

$$\mathcal{M} = (P \times S^1) \underset{h_1^{S^1}}{\cup} C_1 \underset{h_2^{S^1}}{\cup} C_2. \tag{150}$$

The gluing is along boundary tori over $\partial D_1, \partial D_2$. The remaining boundary is

$$\partial\mathcal{M} = \partial D_0 \times S^1 \cong T^2. \tag{151}$$

Define

$$\mathcal{C} = \mathcal{M} \cup_{\partial\mathcal{M}} \operatorname{Cone}(\partial\mathcal{M}), \tag{152}$$

and write c_0 for the cone vertex. This changes the codomain at the cusp: the whole boundary two-torus, including the target circle, is collapsed to c_0. It therefore does not contradict the proved nonextension of the literal map χ_γ to the fixed product target $B \times S^1$.

Theorem 9.14 (Exact cone-compactified S^{24} route)**.** *There is a continuous surjective based quotient map*

$$q_1 : (X, x_0) \longrightarrow (\mathcal{C}, c_0), \qquad x_0 \in W, \tag{153}$$

defined as follows:

1. *over P, it is $(f, \theta_\gamma) : f^{-1}(P) \to P \times S^1$, equivalently the restriction of χ_γ to $f^{-1}(P)$;*
2. *over D_j, $j = 1, 2$, it is the finite-filling map $N_j \to C_j$ from equation* (84)*;*
3. *on $N_0 \setminus W$, writing $t = re^{i\phi}$, it is*

$$x \longmapsto [r, (e^{i\phi}, \theta_\gamma(x))] \in \operatorname{Cone}(\partial D_0 \times S^1), \tag{154}$$

   and every point of W maps to c_0.

The fourth smash power is a global based map

$$q_1^{\wedge 4} : X^{\wedge 4} \longrightarrow \mathcal{C}^{\wedge 4}. \tag{155}$$

Using $X \cong_{\mathrm{diff}} S^6$, its source is

$$X^{\wedge 4} \cong (S^6)^{\wedge 4} \cong S^{24}. \tag{156}$$

On four synchronized radial fibres, the canonical product-to-smash quotient maps restrict to embeddings on the source fibre product and the interior target-product cross-section. Under those embeddings, equation (155) *restricts exactly to*

$$\chi_{\gamma,T}^{\times 4} : F_T^4 \longrightarrow C_T^4, \qquad C_T = \mathbb{R}/\ell_T\mathbb{Z}. \tag{157}$$

With its Euclidean ℓ^2-product flat quotient metric,

$$\operatorname{sys}(C_T^4) = \ell_T = T + H - B + O(T^{-1}), \qquad (C_T^4, 0) \longrightarrow (\mathbb{R}^4, 0) \tag{158}$$

in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

The target homology is

$$\widetilde{H}_k(\mathcal{C};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 2, 3, \\ 0, & \textit{otherwise}, \end{cases} \tag{159}$$

and

$$\widetilde{H}_k(\mathcal{C}^{\wedge 4};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 8, 12, \\ \mathbb{Z}^4, & k = 9, 11, \\ \mathbb{Z}^6, & k = 10, \\ 0, & \textit{otherwise}. \end{cases} \tag{160}$$

Proof. The compact spaces $P \times S^1, C_1, C_2$ are glued along closed boundary tori by homeomorphisms, producing a compact orientable three-manifold $\mathcal{M}$ with boundary $\partial D_0 \times S^1$. Attaching the cone along this closed boundary gives a compact Hausdorff finite CW space $\mathcal{C}$.

The overlap identity

$$\chi_\gamma \circ h_j = h_j^{S^1} \circ \overline{\chi}_j$$

proves that the first two pieces of q_1 agree. On ∂D_0, the cone base is glued to the remaining boundary of $P \times S^1$, so the first and third pieces agree. If U is a neighborhood of c_0, its inverse image in the cone contains $[0, \delta) \times T^2$ for some $\delta > 0$, by compactness of the cone base. Therefore $q_1^{-1}(U)$ contains $f^{-1}(|t| < \delta)$, a neighborhood of W. This proves continuity at the new stratum; continuity elsewhere follows from the input maps and the closed pasting lemma.

Every regular circle quotient and both finite Seifert targets are surjective, and W maps onto the cone vertex. Thus q_1 is a continuous surjection from compact X to Hausdorff $\mathcal{C}$, hence a quotient map. It is based, so its fourth smash power exists. Iterating $S^m \wedge S^n \cong S^{m+n}$ gives equation (156).

The base point lies in W, while each radial regular fibre is disjoint from W. Hence F_T^4 and its product-circle image avoid the closed source and target fat wedges. The canonical quotient maps from the source product and the indicated interior target-product cross-section to their smash products restrict to embeddings. Under those embeddings, the induced smash map restricts to $\chi_{\gamma,T}^{\times 4}$, which itself retains its displayed nontrivial torus kernel. The quotient-metric calculation in equation (76) gives ℓ_T. The product lattice is $\ell_T\mathbb{Z}^4$, so the radius-R argument, including the shortest-representative proof that the embedded Euclidean ball is the whole quotient ball, in the proof of theorem 9.12 proves equation (158).

For homology, let ρ_0, ρ_1, ρ_2 be the boundary loops of the pair of pants, with $\rho_0 + \rho_1 + \rho_2 = 0$, and let η be the regular target-circle class. With the boundary-loop and circle orientations used in the finite-filling presentation, the meridian relations are

$$3\rho_1 - \eta = 0, \qquad 4\rho_2 + \eta = 0. \tag{161}$$

Reversing the circle orientation reverses both displayed η-signs and changes none of the following groups. Thus

$$H_1(\mathcal{M};\mathbb{Z}) = \frac{\mathbb{Z}\langle \eta, \rho_1, \rho_2 \rangle}{\langle 3\rho_1 - \eta,\ 4\rho_2 + \eta \rangle} \cong \mathbb{Z}\langle g \rangle, \tag{162}$$

where

$$\rho_0 = -g, \qquad \rho_1 = 4g, \qquad \rho_2 = -3g, \qquad \eta = 12g. \tag{163}$$

In particular, $H_1(\partial\mathcal{M}) = \mathbb{Z}\langle \rho_0, \eta \rangle \to H_1(\mathcal{M})$ is surjective. The orientable three-manifold $\mathcal{M}$ has one torus boundary, $H_3(\mathcal{M}) = 0$, and $\chi(\mathcal{M}) = \frac{1}{2}\chi(\partial\mathcal{M}) = 0$; hence $H_2(\mathcal{M};\mathbb{Q}) = 0$. To obtain the integral statement without suppressing torsion, Poincare–Lefschetz duality and the universal coefficient theorem give

$$H_2(\mathcal{M};\mathbb{Z}) \cong H^1(\mathcal{M}, \partial\mathcal{M};\mathbb{Z}) \cong \mathrm{Hom}(H_1(\mathcal{M}, \partial\mathcal{M};\mathbb{Z}), \mathbb{Z}),$$

because $H_0(\mathcal{M}, \partial\mathcal{M};\mathbb{Z}) = 0$. Thus $H_2(\mathcal{M};\mathbb{Z})$ is free; its rank is zero, so it vanishes.

The cone attachment is the homotopy cofiber of $\partial\mathcal{M} \hookrightarrow \mathcal{M}$. Since the boundary inclusion is a cofibration,

$$\widetilde{H}_k(\mathcal{C};\mathbb{Z}) \cong H_k(\mathcal{M},\partial\mathcal{M};\mathbb{Z}).$$

Poincare–Lefschetz duality and $H^0(\mathcal{M}) = H^1(\mathcal{M}) = \mathbb{Z}$, $H^2(\mathcal{M}) = H^3(\mathcal{M}) = 0$, give equation (159). The homology is free, so the smash Kunneth theorem says that four factors of degree 2 or 3 produce ranks $\binom{4}{j}$ in degree $8+j$, proving equation (160). □

Definition 9.15 (The three radial target-slice embeddings)**.** Put $r_T = e^{-2\pi T}$ and let

$$\vartheta_T : \mathbb{R}/\mathbb{Z} \longrightarrow C_T = \mathbb{R}/\ell_T\mathbb{Z}, \qquad [a] \longmapsto [\ell_T a].$$

For $y_1, y_2 \in Q_T$, define

$$j_T(y_1,y_2) = \left[[r_T, 1, \iota_T^{-1}(y_1)], [r_T, 1, \iota_T^{-1}(y_2)]\right]_\wedge \in \mathcal{A}^{\wedge 2}.$$

For $c_1,\dots,c_4 \in C_T$, define

$$k_T^{\mathcal{B}}(c_1,\dots,c_4) = \left[[r_T, 1, \vartheta_T^{-1}(c_1)], \dots, [r_T, 1, \vartheta_T^{-1}(c_4)]\right]_\wedge \in \mathcal{B}^{\wedge 4},$$

and, using the cusp-cone piece of $\mathcal{C}$,

$$k_T^{\mathcal{C}}(c_1,\dots,c_4) = \bigwedge_{j=1}^{4} [r_T, (1, \vartheta_T^{-1}(c_j))] \in \mathcal{C}^{\wedge 4}.$$

These are embeddings because every displayed factor lies away from the smash basepoint, so equality in the smash quotient is equality in every factor; compact-to-Hausdorff injectivity then gives a topological embedding.

Proposition 9.16 (Intrinsic expansion versus ambient collapse)**.** *Let D_{12}, $D_{24}^{\mathcal{B}}$, and $D_{24}^{\mathcal{C}}$ be arbitrary fixed metrics compatible with the topologies of $\mathcal{A}^{\wedge 2}$, $\mathcal{B}^{\wedge 4}$, and $\mathcal{C}^{\wedge 4}$, respectively. In their corresponding ambient metrics, the radial target slices satisfy*

$$\operatorname{diam}_{\mathrm{amb}}\big(j_T(Q_T^2)\big) \longrightarrow 0, \qquad \operatorname{diam}_{\mathrm{amb}}\big(k_T^{\mathcal{B}}(C_T^4)\big) \longrightarrow 0, \qquad \operatorname{diam}_{\mathrm{amb}}\big(k_T^{\mathcal{C}}(C_T^4)\big) \longrightarrow 0. \tag{164}$$

By contrast, in the intrinsic quotient metrics,

$$\operatorname{diam}(Q_T^2) \geq \tfrac{1}{2}(T-C) \longrightarrow \infty, \qquad \operatorname{diam}(C_T^4) \geq \tfrac{1}{2}\ell_T \longrightarrow \infty. \tag{165}$$

Thus equations (140) *and* (158) *assert intrinsic, flat pointed limits of varying slices, not Sormani–Wenger intrinsic-flat convergence and not an ambient pointed $\mathbb{R}^4$ limit inside one fixed compact metric target.*

Proof. Every neighborhood of the relevant cone vertex contains every sufficiently small radial cross-section, uniformly in all compact angular and torus coordinates. In a compatible metric, every such cross-section eventually lies in an arbitrarily small ball about the vertex. Products followed by the smash quotient preserve this uniform convergence, proving equation (164).

For a flat torus with systole s, choose a shortest primitive period λ. The class of $\lambda/2$ has distance $s/2$ from zero: for every period μ, the nonzero period $\lambda - 2\mu$ has length at least s. Hence the diameter is at least $s/2$. Equations equations (131) and (158) prove equation (165). □

Proposition 9.17 (Exact almost-complex obstruction on S^{12})**.** *Let M be a smooth manifold homeomorphic to S^{12}. There is no bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\operatorname{Id}_{TM}$. This conclusion has two independent proofs which use only the homotopy type of M: the original Borel–Serre reduced-power calculation at the single prime 5, and Bott integrality. For the standard embedded sphere there is also an independent stable-tangent Pontryagin-class contradiction.*

These statements concern an unstabilized complex structure on the actual rank-twelve tangent bundle. A stable complex structure on $TS^{12} \oplus \varepsilon_{\mathbb{R}}^{r}$, a complex structure confined to an open subset or to fibres of a map, or an endomorphism which ceases to be a bundle endomorphism along a singular set is not an almost-complex structure on TS^{12} and therefore lies outside the hypothesis.

Proof. Assume that J exists and put $E = (TM, J)$, a complex vector bundle of rank six. Orient M by the interleaved complex frame

$$(v_1, Jv_1, \dots, v_6, Jv_6),$$

write $[M]_J$ for the resulting fundamental class, and choose $u_J \in H^{12}(M;\mathbb{Z})$ by $\langle u_J, [M]_J \rangle = 1$. Since M has the integral cohomology of S^{12},

$$c_1(E) = c_2(E) = c_3(E) = c_4(E) = c_5(E) = 0. \tag{166}$$

The top-Chern/Euler identity uses exactly the orientation just fixed, so Poincare–Hopf gives

$$c_6(E) = e(E_{\mathbb{R}}) = e(TM) = 2u_J. \tag{167}$$

Thus reversing a previously chosen orientation can change the displayed generator but cannot change the coefficient two.

We first specialize the original calculation of Borel and Serre [62, §12, Proposition 12.4, Lemma 12.5, and Proposition 12.6, p. 430; §15, Proposition 15.1, p. 434]. They write C_{2j} for the universal class now denoted c_j. For $j = 6$, the admissible prime is exactly

$$p = 5 < 6, \qquad 5 \nmid 6,$$

and their identity in $H^*(BU(6);\mathbb{F}_5)$ reads

$$\overline{c}_6 = 6^{-1}\mathcal{P}_5^1(\overline{c}_2) + Q(\overline{c}_1, \dots, \overline{c}_5). \tag{168}$$

Here $6^{-1} = 1$ in $\mathbb{F}_5$. To retain the exact coefficient rather than suppressing the lower-class polynomial, pass to the splitting space, write $x_1, \dots, x_6$ for the degree-two Chern roots, put $s_r = \sum_i x_i^r$, and use $\mathcal{P}_5^1(x_i) = x_i^5$ and the Cartan formula. The resulting universal identity is

$$\begin{aligned}\mathcal{P}_5^1(\overline{c}_2) &= \sum_{i<j}(x_i^5 x_j + x_i x_j^5) = s_5 s_1 - s_6 \\ &= 6\overline{c}_6 + \overline{c}_1^4\overline{c}_2 - \overline{c}_1^3\overline{c}_3 - 4\overline{c}_1^2\overline{c}_2^2 + \overline{c}_1^2\overline{c}_4 + 7\overline{c}_1\overline{c}_2\overline{c}_3 - \overline{c}_1\overline{c}_5 \\ &\quad + 2\overline{c}_2^3 - 6\overline{c}_2\overline{c}_4 - 3\overline{c}_3^2.\end{aligned} \tag{169}$$

Consequently, for the lower-Chern ideal $I = (\overline{c}_1, \dots, \overline{c}_5)$,

$$\mathcal{P}_5^1(\overline{c}_2) \equiv 6\overline{c}_6 = \overline{c}_6 \pmod{I}, \tag{170}$$

which verifies the exact top-class coefficient in equation (168). Pulling back along the classifying map of E, using equation (166) and naturality of $\mathcal{P}_5^1$, gives

$$\overline{c_6(E)} = \mathcal{P}_5^1(\overline{c_2(E)}) = \mathcal{P}_5^1(0) = 0.$$

But equation (167) gives $\overline{c_6(E)} = 2\overline{u}_J \neq 0$ in $H^{12}(M;\mathbb{F}_5)$, the first contradiction.

For the second contradiction, choose an orientation-preserving homotopy equivalence $f : S^{12} \to M$. Bott integrality says that the sixth Chern class of every complex bundle on S^{12} is divisible by $(6-1)! = 120$ [60, Chapter 20, §9, Theorem 9.6 and Corollary 9.8, p. 308]; see also [61, §6]. The coefficient can be checked without normalization ambiguity: Newton's sixth identity and equation (166) give

$$\mathrm{ch}_6(f^*E) = \frac{s_6(f^*E)}{6!} = -\frac{c_6(f^*E)}{5!}.$$

The integral Bott generator therefore forces $c_6(f^*E) = 120k f^*u_J$ for some $k \in \mathbb{Z}$, whereas equation (167) gives $c_6(f^*E) = 2f^*u_J$. Thus $120k = 2$, which has no integral solution.

Finally take $M = S^{12}$ with its standard smooth embedding in $\mathbb{R}^{13}$. There is an explicit stable trivialization

$$TS^{12} \oplus \varepsilon^1_{\mathbb{R}} \xrightarrow{\cong} \varepsilon^{13}_{\mathbb{R}}, \qquad (x; v, a) \longmapsto (x; v + ax), \tag{171}$$

so $p_3(TS^{12}) = 0$. On the other hand,

$$E_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \cong E \oplus \overline{E}, \qquad p_3(E_{\mathbb{R}}) = -c_6(E \oplus \overline{E}).$$

Equation equation (166) and $c_6(\overline{E}) = c_6(E)$ give the exact relation

$$p_3(TS^{12}) = -2c_6(E) = -4u_J, \tag{172}$$

contradicting stable triviality in the torsion-free group $H^{12}(S^{12}; \mathbb{Z})$. This is the Pontryagin–Euler proof recorded independently in [61, Lemma 4.1 and Theorem 4.2].

Every contradiction above requires a globally defined J on the actual bundle TM. Stabilizing that bundle, deleting points, restricting to regular fibres, or allowing J to fail at a singular locus removes that hypothesis rather than producing an exception to it. □

Theorem 9.18 (Sharp obstructions to stronger interpretations)**.** *The sphere routes above have the following exact limitations.*

1. *Let $r \geq 1$, and let $N \subset \Lambda^{\oplus r}$ be a primitive kernel defining a connected fibrewise torus quotient over the full regular base, or over any finite cover of that base. Equip the quotient with the norm induced by the unrescaled marked Euclidean metric on the original fibres. If its quotient systole tends to infinity on the radial cusp, then*

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \leq r. \tag{173}$$

   Thus two copies cannot give a full-base four-torus quotient. The S^{12} map in theorem 9.12 evades the obstruction only by localizing to the cusp and collapsing the exterior.

2. *No nontrivial finite base cover, including one extending with branching over the original singular values, makes K_Λ a sub-local system over the full regular base. The local cyclic change $t = s^M$ used for the auxiliary degree-M^2 cover cannot extend as a connected cover of $B \cong \mathbb{P}^1$ branched only at p_0.*

3. *There is no map from X^2, $X \wedge X$, X^4, or $X^{\wedge 4}$ to a fixed T^4 whose restriction retains all four target character classes of the displayed product quotient. There is no map to a fixed S^4 whose restriction is a degree-one collapse of Q^2_T, or of the four-circle target, for even one regular fibre. The conical targets evade these obstructions because the fibre character and orientation classes do not extend as global target classes.*

4. *Cartesian products do not produce the named spheres:*

$$X^2 \simeq S^6 \times S^6 \not\simeq S^{12}, \qquad X^4 \simeq (S^6)^4 \not\simeq S^{24}. \tag{174}$$

   The two- and four-fold synchronized fibre products have real dimensions 10 *and* 18*, respectively, and generally have singular completed fibres. Joins give*

$$X * X \cong S^{13}, \qquad X^{*4} \cong S^{27}. \tag{175}$$

5. *Since $X \cong S^6$ is simply connected, every connected unbranched finite cover of X, X^r, $X \wedge X \cong S^{12}$, or $X^{\wedge 4} \cong S^{24}$ is trivial. The local degree-M^2 unspooling cover is not the restriction of a nontrivial connected unbranched cover of any of these compact sources. Branched or normalized compactifications are different objects and are not constructed here.*

6. *By theorem 9.17, no smooth manifold homeomorphic to S^{12} admits an almost-complex structure. Nor does S^{24}, with any differentiable structure, by the Borel–Serre theorem [62, §15, Proposition 15.1 and the following remark, p. 434]. Thus the smash sources cannot be promoted to compact complex spheres.*

Proof. Every element of $K_\Lambda^{\oplus r}$ has uniformly bounded physical length on the radial cusp. If one survived in the quotient lattice, it would be a nonzero period of bounded quotient norm. Hence

$$K_\Lambda^{\oplus r} \subset N. \tag{176}$$

For the full monodromy group, $A_1\widehat{w} = \widehat{u} - \widehat{w}$ and invariance force every copy of $\widehat{u}$ into N. Therefore

$$L_\Lambda^{\oplus r} = \mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta}\rangle^{\oplus r} \subset N,$$

which proves equation (173).

The same conclusion holds after a finite base cover. Let H be its finite-index subgroup and put $H_* = A_2A_1 = I + N_*$, $N_*^2 = 0$. Every finite-index subgroup contains H_*^n for some $n > 0$, while

$$H_*^n\widehat{w} = \widehat{w} + n(\widehat{u} - \widehat{\delta}). \tag{177}$$

Equations equations (176) and (177) give $n\widehat{u} \in N$. Primitivity gives $\widehat{u} \in N$, proving the same rank bound. The orbit also gives $H_*^n K_\Lambda \neq K_\Lambda$ for every $n > 0$, so no finite cover of the regular base preserves the native kernel. A compact cover branched over the singular values restricts to such a cover. Finally, a connected finite cover of $\mathbb{P}^1$ branched only at p_0 would restrict to a connected unbranched finite cover of $\mathbb{P}^1 \setminus \{p_0\} \cong \mathbb{C}$, hence has degree one.

For a fixed T^4 target, $H^1(X^r;\mathbb{Z}) = 0$, as does the first cohomology of the smash spheres. A global map therefore pulls every target character class back to zero. On a regular fibre the quotient homomorphisms inject the target character lattice into the character lattice of the source torus, giving four independent nonzero pullbacks, a contradiction.

For the fixed S^4 obstruction, let $\omega_T \in H^2(Q_T;\mathbb{Z})$ be the orientation class. Since $\pi_T^* : H^1(Q_T) \to H^1(F_T)$ injects two independent classes and torus cohomology is their exterior algebra,

$$(\pi_T \times \pi_T)^*(\omega_T \times \omega_T) \neq 0 \quad \text{in } H^4(F_T^2;\mathbb{Z}). \tag{178}$$

A degree-one collapse $Q_T^2 \to S^4$ pulls the orientation generator back to $\pm(\omega_T \times \omega_T)$. But

$$H^4(X^2;\mathbb{Z}) = H^4(S^{12};\mathbb{Z}) = 0,$$

so no global extension has this restriction. The four-circle case is the same: its orientation class pulls back to the exterior product of four independent circle characters, while $H^4(X^4) = H^4(S^{24}) = 0$.

The Kunneth theorem gives $H_6(X^2) = \mathbb{Z}^2$, whereas $H_6(S^{12}) = 0$, and gives nonzero groups of ranks $4, 6, 4$ in degrees $6, 12, 18$ for X^4, whereas S^{24} has none. The real dimension of an r-fold synchronized fibre product is $6r - 2(r-1) = 4r + 2$, giving 10 and 18. The join formula $S^m * S^n \cong S^{m+n+1}$ proves equation (175). The remaining assertions follow from simple connectivity, theorem 9.17, and the cited Borel–Serre theorem for S^{24}. □

Canonicity and exact scope. The lattice $K_\Lambda = \ker(M_0 - I) = \mathrm{im}(M_0 - I)$ and the punctured-cusp quotient are intrinsic to the marked cusp monodromy. The compact map q_2 additionally depends on the chosen cusp coordinate, closed cusp disc, and the decision to collapse its outer boundary and the exterior. The identifications $X^{\wedge 2} \cong S^{12}$ and $X^{\wedge 4} \cong S^{24}$ depend on a based choice of $X \cong S^6$. None of these choices is claimed to be holomorphic or canonical in the algebraic category.

The map $q_2^{\wedge 2}$ is the direct two-copy construction from the native $T^2 \times \mathbb{R}^2$ cusp: it is globally defined on the smash sphere, and its radial regular-fibre quotients are $Q_T \times Q_T$, which converge pointed local-isometrically, smoothly, and Gromov–Hausdorff to $\mathbb{R}^4$. It is not a torus quotient over the full regular base.

The map $q_{1,\mathrm{cusp}}^{\wedge 4}$ is the parallel cusp-supported construction with target $\mathrm{Susp}(T^2)^{\wedge 4}$. The stronger map $q_1^{\wedge 4}$ uses the quotient over every regular and finite-filled fibre before the cusp cone is attached; four copies are minimal for that full-base all-expanding group quotient. Neither construction gives a compact complex S^{12} or S^{24}, an ambient Euclidean limit in a fixed compact metric target, or a claim about the interacting Yang–Mills spectrum.

The argument does not rule out a separately constructed branched or toroidal compactification of the auxiliary finite covers after additional branching away from p_0. Such a construction would have to specify its global base cover, normalize and resolve every pulled-back singular fibre, track the lifted metric, and determine the topology of the resulting total space. No such smooth compact sphere construction follows from the present data.

9.2.1 Optimal sphere exponents and the exact Euclidean boundary

Within the class of native connected fibrewise torus quotients equipped with the induced, unscaled quotient of the marked Euclidean metric, the two sphere dimensions in the preceding constructions are forced by two different integral quotient problems. Cusp-locally, each copy has an expanding quotient of rank two, so two copies are necessary and sufficient within that class. Over the full regular base, monodromy leaves at most one expanding quotient direction per copy, so four copies are necessary and sufficient within the same class. We now prove this scoped optimality, identify the compact conical carriers, and construct a noncompact comparison manifold in which the limiting $\mathbb{R}^4$ is an actual boundary fibre. There is only a continuous comparison map from this manifold to the compact conical carrier: it set-theoretically collapses the new boundary, is a homeomorphism away from that boundary, and is nonproper and nonquotient.

Throughout this subsubsection,

$$B_R^Z(z) := \{w \in Z : d_Z(z,w) \le R\}$$

denotes the closed radius-R ball; when the ambient space and center are clear, superscripts and centers are suppressed.

Theorem 9.19 (Optimality within the native connected torus-quotient class)**.** *Consider only native connected fibrewise torus quotients of copies of the original marked fibres, and equip every such quotient with the induced, unscaled Euclidean quotient metric.*

1. *For a native connected torus quotient of r copies over the punctured cusp, if the quotient systole tends to infinity on the positive radial ray, then its lattice rank is at most $2r$. A rank-four quotient therefore requires $r \ge 2$. Equality is attained for $r = 2$ by*

$$(\Lambda/K_\Lambda)^{\oplus 2}, \qquad K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle,$$

whose physical period lattice is

$$L_T^{(12)} = \mathrm{diag}(Y_T, Y_T)\mathbb{Z}^4.$$

The corresponding smash source is $X^{\wedge 2} \cong S^{12}$.

2. *For a native connected torus quotient of r copies over the full regular base, or over a finite cover of that base, radial divergence of the quotient systole forces its lattice rank to be at most r. A rank-four quotient therefore requires $r \ge 4$. Equality is attained by the four copies of the full-monodromy-invariant character χ_γ; on the radial fibre its period lattice is*

$$L_T^{(24)} = \ell_T \mathbb{Z}^4, \qquad \ell_T = T + H - B + O(T^{-1}),$$

and the corresponding smash source is $X^{\wedge 4} \cong S^{24}$.

Thus, among precisely these native connected torus quotients with their induced unscaled marked Euclidean metrics, S^{12} is the smallest smash sphere produced by the cusp-local rank-two quotient, and S^{24} is the smallest smash sphere produced by a quotient which survives the full regular-base monodromy. No optimality is asserted here among arbitrary continuous quotient maps or after a T-dependent rescaling of the metric.

Proof. For r copies, every element of

$$K_\Lambda^{\oplus r} = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle^{\oplus r}$$

has uniformly bounded physical length on the radial cusp. Let $N \subset \Lambda^{\oplus r}$ be a primitive sublattice defining a connected torus quotient. If some $v \in K_\Lambda^{\oplus r}$ had nonzero class in $\Lambda^{\oplus r}/N$, that nonzero quotient period would have quotient norm at most the original norm of v, uniformly in T. The quotient systole could not diverge. Hence

$$K_\Lambda^{\oplus r} \subset N,$$

and

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \le 4r - 2r = 2r.$$

For $r = 2$, the primitive kernel $K_\Lambda^{\oplus 2}$ attains this bound. The cusp monodromy is the identity modulo K_Λ, so the quotient descends over the punctured cusp. Its radial lattice is exactly $\operatorname{diag}(Y_T, Y_T)\mathbb{Z}^4$, and

$$\operatorname{sys}(L_T^{(12)}) = \operatorname{sys}(Y_T\mathbb{Z}^2) \ge T - C \longrightarrow \infty.$$

For a quotient over the full regular base, monodromy invariance gives a stronger bound. Since $\widehat{w} \in N$ and

$$A_1\widehat{w} = \widehat{u} - \widehat{w},$$

one has $\widehat{u} \in N$. Together with $\widehat{w}, \widehat{\delta} \in N$, this places the rank-three lattice

$$\mathbb{Z}\langle \widehat{u}, \widehat{w}, \widehat{\delta}\rangle^{\oplus r}$$

inside N, and hence

$$\operatorname{rank}(\Lambda^{\oplus r}/N) \le r.$$

The same conclusion holds after a finite base cover. Indeed, if Γ_{mon} is the corresponding finite-index monodromy subgroup and $G_* = A_2A_1 = I + N_*$, then $N_*^2 = 0$, and Γ_{mon} contains G_*^m for some $m > 0$. The exact orbit formula

$$G_*^m\widehat{w} = \widehat{w} + m(\widehat{u} - \widehat{\delta})$$

implies $m\widehat{u} \in N$; primitivity of N implies $\widehat{u} \in N$.

The invariant character χ_γ has precisely the complementary rank-one quotient in each copy. The full-base construction, including the two finite Seifert fillings and the cone at the unipotent cusp, proves that four copies attain rank four. Finally, after choosing the based diffeomorphism $X \cong_{\mathrm{diff}} S^6$ already reconstructed in this workbench, $X^{\wedge r} \cong (S^6)^{\wedge r} \cong S^{6r}$, which gives the two claimed sphere dimensions. □

Theorem 9.20 (Exact sphere-to-sphere carriers with moving four-spheres)**.** *There are continuous surjective based quotient maps*

$$\widehat{q}_3 : (X, x_0) \longrightarrow (S^3, *), \qquad \widehat{q}_\gamma : (X, x_0) \longrightarrow (S^2, *),$$

whose smash powers are continuous surjective quotient maps

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6, \qquad \widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8. \tag{179}$$

Use the marked presentation $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ and put

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})), \qquad c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2.$$

Here c is the quotient collapsing the coordinate one-skeleton. For every sufficiently large T, put

$$\begin{aligned} r_T &= e^{-2\pi T}, \\ K_T^{(1)} &:= \iota_T(K^{(1)}) \subset Q_T, \\ S_T^2 &:= \overline{T}^2/K^{(1)} \cong Q_T/K_T^{(1)}, \\ c_T &:= c \circ \iota_T^{-1} : Q_T \longrightarrow S_T^2, \end{aligned}$$

where the identification of the two displayed quotients is induced by ι_T. Within this theorem and its proof, c_T denotes the displayed collapse map; the earlier scalar $c_T = \xi_T - \eta_T$ does not enter. At reduced-suspension cross-section height r_T, the first target contains a based topologically embedded four-sphere, equipped with its intrinsic quotient length metric,

$$\mathbb{S}_{12,T}^4 = S_T^2 \wedge S_T^2 \hookrightarrow S^6,$$

and the second target contains, at the same cross-section height, a based topologically embedded four-sphere, equipped with its intrinsic quotient length metric,

$$\mathbb{S}_{24,T}^4 = C_T^{\wedge 4} \hookrightarrow S^8.$$

On synchronized radial fibres, the exact restrictions factor as

$$F_T^2 \xrightarrow{\pi_T \times \pi_T} Q_T^2 \xrightarrow{c_T \times c_T} (S_T^2)^2 \xrightarrow{\rho_{12,T}} S_T^2 \wedge S_T^2 = \mathbb{S}_{12,T}^4, \tag{180}$$

$$F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \xrightarrow{\rho_{24,T}} C_T^{\wedge 4} = \mathbb{S}_{24,T}^4, \tag{181}$$

Here c_T collapses $K_T^{(1)}$, and $\rho_{12,T}$ and $\rho_{24,T}$ are the smash quotient maps. Thus the full collapsed locus in Q_T^2 is exactly

$$W_{12,T} = (K_T^{(1)} \times Q_T) \cup (Q_T \times K_T^{(1)}), \tag{182}$$

and the full collapsed locus in C_T^4 is exactly

$$W_{24,T} = \{(x_1, x_2, x_3, x_4) \in C_T^4 : x_j = 0 \pmod{\ell_T} \text{ for some } j\}. \tag{183}$$

The maps $\rho_{12,T} \circ (c_T \times c_T) : Q_T^2 \to S_T^2 \wedge S_T^2$ and $\rho_{24,T} : C_T^4 \to C_T^{\wedge 4}$ send exactly $W_{12,T}$ and $W_{24,T}$, respectively, to the smash basepoint and are injective on their complements. The full composites from F_T^2 and F_T^4 still identify every pair of points identified by $\pi_T \times \pi_T$ and $\chi_{\gamma,T}^{\times 4}$, respectively; in particular, their first arrows retain their nontrivial torus kernels. Their exact smash-basepoint preimages in the full radial sources are

$$(\pi_T^{-1}(K_T^{(1)}) \times F_T) \cup (F_T \times \pi_T^{-1}(K_T^{(1)}))$$

and

$$\{(x_1, x_2, x_3, x_4) \in F_T^4 : \chi_{\gamma,T}(x_j) = 0 \text{ in } C_T \text{ for some } j\},$$

respectively. Outside these basepoint preimages, two source points have the same final image if and only if their images under the corresponding first arrow in equations (180) *and* (181) *are equal.*

Choose the Euclidean Riemannian ℓ^2-product length metrics on Q_T^2 and C_T^4 before taking the smash quotients, and use the successive quotient length metrics specified in the proof. With the specified midpoint basepoints,

$$(\mathbb{S}_{12,T}^4, \zeta_{12,T}) \longrightarrow (\mathbb{R}^4, 0), \qquad (\mathbb{S}_{24,T}^4, \zeta_{24,T}) \longrightarrow (\mathbb{R}^4, 0) \tag{184}$$

in pointed local-isometric and pointed Gromov–Hausdorff topology. Moreover, the convergence is smooth on the eventually Euclidean basepoint neighborhoods. For every $R > 0$, once $T - C > 4R$, respectively once $\ell_T > 4R$, the corresponding quotient map restricts to an isometry from the radius-R ball in Q_T^2, respectively C_T^4, onto *the entire radius-R ball in the indicated smash sphere. The domain ball, and hence the quotient ball, is exactly a Euclidean four-ball.*

Proof. Write $\Sigma_r(Z, z_0)$ for the reduced suspension

$$\frac{[0,\epsilon]\times Z}{(\{0,\epsilon\}\times Z)\cup([0,\epsilon]\times\{z_0\})},$$

where the displayed union is collapsed to one basepoint.

Use the marked presentation $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, let

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z})\times\{0\})\cup(\{0\}\times(\mathbb{R}/\mathbb{Z})),$$

and let

$$c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2$$

be the degree-one quotient obtained by collapsing the one-skeleton. The formula

$$\widehat{p}_2 : \mathcal{A} \longrightarrow \Sigma_r(\overline{T}^2/K^{(1)}, c(K^{(1)})) \cong S^3, \qquad [r, e^{i\phi}, u] \longmapsto [r, c(u)]$$

is well defined: the two suspension ends of $\mathcal{A}$ and the entire cylinder over $K^{(1)}$ map to the reduced-suspension basepoint. It is a continuous surjection. Set $\widehat{q}_3 = \widehat{p}_2 \circ q_2$.

For every sufficiently large T, define the transported one-skeleton and its collapse by

$$\begin{aligned} K_T^{(1)} &:= \iota_T(K^{(1)}) \subset Q_T, \\ S_T^2 &:= \overline{T}^2/K^{(1)} \cong Q_T/K_T^{(1)}, \\ c_T &:= c\circ\iota_T^{-1} : Q_T \longrightarrow S_T^2. \end{aligned} \tag{185}$$

The displayed target identification is the one induced by ι_T, so $c_T^{-1}(*) = K_T^{(1)}$, where $* = c(K^{(1)}) = c_T(K_T^{(1)})$.

Give Q_T its Euclidean quotient length metric and S_T^2 the quotient length metric under c_T. Put

$$y_* = [(1/2, 1/2)] \in \overline{T}^2, \qquad y_T := \iota_T(y_*) \in Q_T, \qquad z_T := c_T(y_T) \in S_T^2.$$

Equivalently, the metric on $\overline{T}^2$ used below is $g_T := \iota_T^* g_{Q_T}$. For every $v \in \mathbb{R}^2$, the exact decomposition

$$Y_T = T\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + E_T, \qquad \|E_T\|_{\mathrm{op}} \le C,$$

gives

$$\|Y_T v\| \ge (T - C)\|v\|.$$

The standard flat distance from y_* to $K^{(1)}$ is $1/2$; therefore

$$d_T := \mathrm{dist}_{Q_T}(y_T, K_T^{(1)}) = \mathrm{dist}_{g_T}(y_*, K^{(1)}) \ge \frac{T - C}{2}. \tag{186}$$

For a length-space quotient which collapses a closed subset K to one point, the distance between two noncollapsed classes represented by x, y is

$$\min\{d(x,y), d(x,K) + d(y,K)\}.$$

The two displayed terms are upper bounds, obtained respectively from a direct path and from paths from x and y to K. Conversely, a rectifiable quotient path either avoids the collapsed point, in which case its compact image lifts to $X \setminus K$ and has length at least $d(x,y)$, or meets that point, in which case its two

pieces have total length at least $d(x,K)+d(y,K)$. Taking infima proves the formula. If $T-C>4R$, then $d_T>2R$. The displayed formula and the systolic bound show that

$$c_T : B_R^{Q_T}(y_T) \longrightarrow B_R^{S_T^2}(z_T)$$

is an isometry onto the entire target ball, and both balls are exactly Euclidean two-balls. Indeed, the second term in the minimum cannot realize a distance at most $2R$, and any point of the target ball has a unique noncollapsed representative; the same formula puts that representative in the source ball.

At reduced-suspension coordinate $r_T = e^{-2\pi T}$, S_T^2 is a based embedded subspace of S^3: its point $*$ is the global reduced-suspension basepoint. Hence

$$\mathbb{S}^4_{12,T} = S_T^2 \wedge S_T^2$$

embeds in $S^3 \wedge S^3 = S^6$. Equip Q_T^2 first with the Euclidean Riemannian ℓ^2-product length metric, equip $(S_T^2)^2$ with the ℓ^2-product of the two quotient length metrics, and let

$$\rho_{12,T} : (S_T^2)^2 \longrightarrow S_T^2 \wedge S_T^2$$

carry the quotient length metric obtained by collapsing the fat wedge

$$\mathcal{W}_{12,T} = (S_T^2 \times \{*\}) \cup (\{*\} \times S_T^2).$$

Since $c_T^{-1}(*) = K_T^{(1)}$, the inverse image of $\mathcal{W}_{12,T}$ under $c_T \times c_T$ is exactly $W_{12,T}$ in equation (182). On $Q_T^2 \setminus W_{12,T}$, the map $c_T \times c_T$ is injective and its image avoids $\mathcal{W}_{12,T}$; the map $\rho_{12,T}$ is injective on the complement of $\mathcal{W}_{12,T}$. Hence their composite identifies all of $W_{12,T}$ to the smash basepoint and makes no other identifications.

The point $\zeta_{12,T} = [z_T, z_T]$ has distance d_T from this fat wedge. The one-factor onto isometry above gives an onto isometry from the radius-R ball about (y_T, y_T) in Q_T^2 to the corresponding ball about (z_T, z_T) in $(S_T^2)^2$. Applying the same collapsed-set distance formula to $\rho_{12,T}$, and using $d_T > 2R$, shows that

$$\rho_{12,T} \circ (c_T \times c_T) : B_R^{Q_T^2}((y_T, y_T)) \xrightarrow{\cong_{\mathrm{isom}}} B_R^{\mathbb{S}^4_{12,T}}(\zeta_{12,T}) \tag{187}$$

is onto as well as isometrically embedded. The systolic bound $\mathrm{sys}(Q_T^2) \geq T - C > 4R$ makes its domain exactly a Euclidean four-ball. On a radial fibre,

$$(\widehat{p}_2 \circ q_2)|_{F_T} = [r_T, c(\iota_T^{-1}(\pi_T(\,\cdot\,)))] = [r_T, c_T(\pi_T(\,\cdot\,))],$$

and taking two factors followed by $\rho_{12,T}$ proves the entire factorization in equation (180).

For the four-copy construction, choose $1 \in S^1$ as character basepoint and define

$$\widehat{p}_1 : \mathcal{B} \longrightarrow \Sigma_r(S^1, 1) \cong S^2, \qquad [r, e^{i\phi}, \theta] \longmapsto [r, \theta].$$

The formula forgets the base angle and collapses the two suspension ends and the entire cylinder with $\theta = 1$ to the reduced-suspension basepoint. It is a continuous surjection. Put $\widehat{q}_\gamma = \widehat{p}_1 \circ q_{1,\mathrm{cusp}}$.

At the interior suspension coordinate $r_T = e^{-2\pi T}$, identify the character circle $C_T = \mathbb{R}/\ell_T\mathbb{Z}$ with $\mathbb{R}/\mathbb{Z} \subset S^2$ by $[x] \mapsto [x/\ell_T]$. It is a based embedded circle, so its fourth smash is an embedded

$$\mathbb{S}^4_{24,T} = C_T^{\wedge 4} \subset (S^2)^{\wedge 4} = S^8.$$

Equip C_T^4 first with the Euclidean Riemannian ℓ^2-product length metric, and let

$$\rho_{24,T} : C_T^4 \longrightarrow C_T^{\wedge 4}$$

carry the quotient length metric obtained by collapsing exactly the fat wedge $W_{24,T}$ of equation (183). Away from that fat wedge the smash quotient is injective, so no further points are identified. The midpoint

$$\zeta_{24,T} = [(\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2)]$$

has distance $\ell_T/2$ from that fat wedge. If $\ell_T > 4R$, neither a nonzero period nor a path through the collapsed fat wedge can shorten a distance between points in the radius-R ball of C_T^4. Conversely, the collapsed-set distance formula shows that every point of the radius-R quotient ball has its unique noncollapsed representative in that product ball. Consequently

$$\rho_{24,T} : B_R^{C_T^4}((\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2)) \xrightarrow{\cong_{\mathrm{isom}}} B_R^{\mathbb{S}^4_{24,T}}(\zeta_{24,T}) \tag{188}$$

is onto as well as isometrically embedded, and its domain is exactly a Euclidean four-ball because $\mathrm{sys}(C_T^4) = \ell_T > 4R$. On the radial source the formula discards only the base angle and then performs this smash quotient, giving equation (181).

Both one-copy maps are continuous surjections from compact X to Hausdorff spheres, hence quotient maps, and they are based. Their smash powers are surjective; compact-to-Hausdorff again makes them quotient maps. Using the same chosen based diffeomorphism $X \cong_{\mathrm{diff}} S^6$, the homeomorphisms

$$X^{\wedge 2} \cong S^{12}, \quad (S^3)^{\wedge 2} \cong S^6, \qquad X^{\wedge 4} \cong S^{24}, \quad (S^2)^{\wedge 4} \cong S^8$$

prove equation (179). The exact Euclidean-ball statements prove all three convergences in equation (184). □

Corollary 9.21 (The two-copy rank-two sphere carrier is null-homotopic)**.** *After the based identifications in equation* (179)*, the continuous surjective quotient*

$$\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6$$

is based null-homotopic. In fact, $g \wedge g : S^{12} \to S^6$ is based null-homotopic for every based map $g : S^6 \to S^3$.

Proof. Apply the full external-smash calculation in theorem 9.30 to $g = \widehat{q}_3$. That theorem proves the stronger assertion that $x \wedge y : S^{12} \to S^6$ is null-homotopic for every pair of based maps $x, y : S^6 \to S^3$. □

Corollary 9.22 (Every repeated smash of the circle-quotient carrier is null)**.** *Put $f = \widehat{q}_\gamma : S^6 \to S^2$ after the chosen based identification $X \cong S^6$. For every $r \geq 2$,*

$$[f^{\wedge r}] = 0 \quad \textit{in} \quad \pi_{6r}(S^{2r}). \tag{189}$$

In particular,

$$[\widehat{q}_\gamma^{\wedge 2}] = 0 \in \pi_{12}(S^4), \qquad [\widehat{q}_\gamma^{\wedge 4}] = 0 \in \pi_{24}(S^8). \tag{190}$$

Thus the displayed surjective quotient $\widehat{q}_\gamma^{\wedge 4} : S^{24} \twoheadrightarrow S^8$ is globally null-homotopic. This does not alter either its point-set surjectivity or the exact intrinsic $\mathbb{R}^4$ limit on the moving four-spheres $\mathbb{S}^4_{24,T}$: those are properties of the specified representative and its radial restrictions, whereas null-homotopy classifies that representative only up to based homotopy.

Proof. Associativity of the smash product gives the exact factorization

$$S^6 \wedge S^6 \xrightarrow{f \wedge \mathrm{id}_{S^6}} S^2 \wedge S^6 \xrightarrow{\mathrm{id}_{S^2} \wedge f} S^2 \wedge S^2. \tag{191}$$

Under the standard based sphere identifications, the first arrow is the sixfold suspension of f, up to the harmless twist homeomorphisms dictated by the left-versus-right suspension convention:

$$f \wedge \mathrm{id}_{S^6} = \Sigma^6 f : S^{12} \longrightarrow S^8. \tag{192}$$

Freudenthal's suspension theorem is already in its isomorphism range here, because $12 \leq 2 \cdot 8 - 2$, and remains in that range after every further suspension. It therefore identifies $\pi_{12}(S^8)$ with the fourth stable stem. The low-stem calculation gives

$$\pi_{12}(S^8) \cong \pi_4^{\mathrm{S}} = 0 \quad \text{[9, Corollary 4.24, p. 360]; [11, Table 1, p. 113].} \tag{193}$$

Hence $\Sigma^6 f$ is based null-homotopic, so the composite in equation (191), namely $f \wedge f$, is based null-homotopic. For $r \geq 2$, the canonical associativity homeomorphisms identify

$$f^{\wedge r} = (f \wedge f) \wedge f^{\wedge(r-2)}, \tag{194}$$

with the convention $f^{\wedge 0} = \mathrm{id}_{S^0}$. Smashing a based null-homotopy with any based map is a based null-homotopy. This proves equations (189) and (190). □

Corollary 9.23 (Literal $\mathbb{R}^4$ open cells in the sphere targets)**.** *Let $*$ denote the smash basepoint and set*

$$U_{12,T} := \mathbb{S}^4_{12,T} \setminus \{*\}, \qquad U_{24,T} := \mathbb{S}^4_{24,T} \setminus \{*\}.$$

The embedded moving four-spheres restrict to topological embeddings

$$U_{12,T} \cong \mathbb{R}^4 \hookrightarrow S^6, \qquad U_{24,T} \cong \mathbb{R}^4 \hookrightarrow S^8.$$

More exactly, the quotient-stage maps restrict to homeomorphisms

$$(Q_T \setminus K_T^{(1)})^2 \xrightarrow{\cong} U_{12,T}, \tag{195}$$

$$(C_T \setminus \{0\})^4 \xrightarrow{\cong} U_{24,T}. \tag{196}$$

These copies of $\mathbb{R}^4$ are nonclosed topological open cells: they are open in their moving four-spheres, not open in the ambient S^6 or S^8, and their closures in those ambient spheres are $\mathbb{S}^4_{12,T}$ and $\mathbb{S}^4_{24,T}$, respectively. They are not globally isometric to Euclidean $\mathbb{R}^4$ for either the assigned compact quotient metrics or the restriction of any fixed compatible metric on the compact sphere targets. The exact midpoint balls in equations (187) *and* (188) *give the precise metric statement: as $T \to \infty$, every fixed Euclidean radius-R ball occurs isometrically and onto the corresponding pointed ball.*

Proof. For a quotient which collapses a closed subset K to a point, restriction to the complement gives a homeomorphism

$$X \setminus K \xrightarrow{\cong} (X/K) \setminus \{*\}.$$

Indeed, the complement of the quotient point is open, its inverse image is the saturated open set $X \setminus K$, and restriction of a quotient map over an open subset of its target is again a quotient map. The restricted map is bijective, and a bijective quotient map is a homeomorphism. Consequently

$$Q_T \setminus K_T^{(1)} \cong S_T^2 \setminus \{*\}.$$

The marked coordinate one-skeleton $K^{(1)}$ is the 1-skeleton of the standard one-vertex CW structure on $\overline{T}^2$, with one 0-cell, two 1-cells, and one 2-cell. Its complement is therefore the open 2-cell $(0,1)^2 \cong \mathbb{R}^2$. Transport by ι_T gives

$$Q_T \setminus K_T^{(1)} \cong \mathbb{R}^2.$$

Likewise $C_T \setminus \{0\}$ is an open interval and hence is homeomorphic to $\mathbb{R}$.

The complement of the smash basepoint is the product of the complements of the factor basepoints. Therefore

$$(S_T^2 \wedge S_T^2) \setminus \{*\} \cong (S_T^2 \setminus \{*\})^2 \cong \mathbb{R}^4,$$

and

$$C_T^{\wedge 4} \setminus \{*\} \cong (C_T \setminus \{0\})^4 \cong \mathbb{R}^4.$$

The full moving four-spheres are topologically embedded in their compact sphere targets, so these restrictions are topological embeddings. In each case the omitted smash basepoint lies in the closure of the open cell, which proves the closure assertions. Finally, every compatible metric on a compact target is bounded, as is each compact quotient metric assigned to a moving four-sphere, whereas Euclidean $\mathbb{R}^4$ is unbounded. Thus no global isometry is possible; the exact metric content is precisely the onto-ball calculation already proved. □

Remark 9.24 (The moving four-spheres are null-homotopic in the targets). For a fixed interior reduced-suspension coordinate, the inclusion $S_T^2 \hookrightarrow S^3$ contracts through based maps by sliding that coordinate to a suspension endpoint; throughout the homotopy the collapsed meridian remains at the global basepoint. Hence $S_T^2 \wedge S_T^2 \hookrightarrow S^6$ is null-homotopic. The identical sliding argument contracts $C_T \hookrightarrow S^2$, and therefore also contracts $C_T^{\wedge 4} \hookrightarrow S^8$. Thus the moving metric four-spheres do not contradict the fixed-target orientation obstruction: their fundamental classes die in the ambient targets.

Remark 9.25 (Homotopical status of the two sphere-to-sphere maps). The maps $S^{12} \to S^6$ and $S^{24} \to S^8$ induce zero on all positive-degree integral homology and cohomology groups and have zero ordinary Hurewicz image. They have no classical degree because source and target dimensions differ. The full four-copy map $\widehat{q}_\gamma^{\wedge 4} : S^{24} \to S^8$ is based null-homotopic by theorem 9.22. The distinct two-copy map $\widehat{q}_3^{\wedge 2} : S^{12} \to S^6$ is also based null-homotopic, by theorem 9.21. In both cases the vanishing is a theorem about the full specified sphere-valued map, not merely its Hurewicz image or the inclusion of a moving four-sphere. The one-copy classes $[\widehat{q}_3] \in \pi_6(S^3)$ and $[\widehat{q}_\gamma] \in \pi_6(S^2)$ are not computed here.

Proposition 9.26 (The compact cone-on-T^4 carriers)**.** *Fix T_0 sufficiently large and put $r_0 = e^{-2\pi T_0} < \epsilon$. Define*

$$\mathcal{D}_{12} = \frac{[0, r_0] \times (\overline{T}^2)^2}{\{0\} \times (\overline{T}^2)^2 \sim d_{12}}, \qquad \mathcal{D}_{24} = \frac{[0, r_0] \times (S^1)^4}{\{0\} \times (S^1)^4 \sim d_{24}}.$$

Both spaces are homeomorphic to $\mathrm{Cone}(T^4)$*. There are topological embeddings*

$$\delta_{12} : \mathcal{D}_{12} \hookrightarrow \mathcal{A} \wedge \mathcal{A}, \qquad [r, u_1, u_2] \longmapsto [\,[r, 1, u_1], [r, 1, u_2],\,], \tag{197}$$

$$\delta_{24} : \mathcal{D}_{24} \hookrightarrow \mathcal{C}^{\wedge 4}, \qquad [r, \theta_1, \dots, \theta_4] \longmapsto \bigwedge_{j=1}^{4} [r, (1, \theta_j)]. \tag{198}$$

Here $1 \in S^1$ fixes the positive radial base angle. For every $T \geq T_0$, put $r_T = e^{-2\pi T}$. The restrictions of the sphere quotient maps to the synchronized radial fibres are exactly the cross-sections at height r_T:

$$F_T^2 \xrightarrow{\pi_T \times \pi_T} Q_T^2 \cong \{r_T\} \times T^4 \subset \mathcal{D}_{12}, \tag{199}$$

$$F_T^4 \xrightarrow{\chi_{\gamma,T}^{\times 4}} C_T^4 \cong \{r_T\} \times T^4 \subset \mathcal{D}_{24}. \tag{200}$$

Every one of these embedded four-torus slices is null-homotopic in its carrier, and hence in the full smash target.

At the cone vertex d_ν, $\nu \in \{12, 24\}$, the local integral homology is

$$H_k(\mathcal{D}_\nu, \mathcal{D}_\nu \setminus \{d_\nu\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^4, & k = 2, \\ \mathbb{Z}^6, & k = 3, \\ \mathbb{Z}^4, & k = 4, \\ \mathbb{Z}, & k = 5, \\ 0, & \textit{otherwise.} \end{cases} \tag{201}$$

In particular, d_ν is not an integral homology-manifold point, with or without boundary, of any dimension. Consequently it is not a topological manifold point, with or without boundary, of any dimension.

Proof. The two displayed domains are cones on $(\overline{T}^2)^2 \cong T^4$ and $(S^1)^4 \cong T^4$. At $r = 0$, every factor on the right of equations (197) and (198) is the relevant smash basepoint, so the formulae descend from the cylinders to the cones. For $0 < r \le r_0$, none of the factors is the smash basepoint. Equality in the smash quotient is then equality in every factor, and the coordinate descriptions of the suspension and cusp cone give equality of r and of all torus coordinates. Thus each δ_ν is injective. Its domain is compact and its target is Hausdorff, so it is an embedding.

The already proved one-factor restriction formulae, with $r_T = e^{-2\pi T}$, are

$$q_2(x) = [r_T, 1, \iota_T^{-1}(\pi_T(x))] \quad (x \in F_T)$$

and, in the cusp-cone piece of $\mathcal{C}$,

$$q_1(x) = [r_T, (1, \vartheta_T^{-1}(\chi_{\gamma,T}(x)))] \quad (x \in F_T).$$

Applying these identities in two and four factors gives equations (199) and (200). Sliding the cone coordinate from r_T to zero contracts each slice to the cone vertex. Finally, excision and the long exact sequence of the pair give, for $L = T^4$,

$$H_k(\mathrm{Cone}(L), \mathrm{Cone}(L) \setminus \{v\}; \mathbb{Z}) \cong \widetilde{H}_{k-1}(L; \mathbb{Z}).$$

The ranks $4, 6, 4, 1$ of H_1, H_2, H_3, H_4 of T^4 prove equation (201). At an interior point of an integral homology n-manifold the local integral homology is $\mathbb{Z}$ in degree n and zero in every other degree, whereas at a boundary point of an integral homology manifold with boundary all these local groups vanish. The simultaneous nonzero groups in degrees $2, 3, 4, 5$, including ranks greater than one, agree with neither pattern for any n. Every topological manifold is locally an integral homology manifold, so the final assertion follows as well. □

Theorem 9.27 (Point-set quotient maps and exact homotopy types)**.** *The smash maps are continuous surjective quotient maps*

$$q_2^{\wedge 2} : S^{12} \twoheadrightarrow \mathcal{A}^{\wedge 2}, \qquad q_1^{\wedge 4} : S^{24} \twoheadrightarrow \mathcal{C}^{\wedge 4}.$$

Their targets have homotopy types

$$\mathcal{A}^{\wedge 2} \simeq \bigvee^{9} S^4 \vee \bigvee^{18} S^5 \vee \bigvee^{15} S^6 \vee \bigvee^{6} S^7 \vee S^8, \tag{202}$$

$$\mathcal{C}^{\wedge 4} \simeq S^8 \vee \bigvee^{4} S^9 \vee \bigvee^{6} S^{10} \vee \bigvee^{4} S^{11} \vee S^{12}. \tag{203}$$

Both maps induce zero on every positive-degree integral homology and cohomology group and have zero ordinary Hurewicz image. Neither map has a continuous homotopy section. These conclusions do not determine whether the unstable homotopy class of either map is zero.

Proof. Let $f : (K, k_0) \to (L, \ell_0)$ be a surjective based map between compact Hausdorff spaces. Every non-basepoint class of $L^{\wedge r}$ has a representative $(y_1, \dots, y_r)$ with each $y_i \neq \ell_0$. Choosing $x_i \in f^{-1}(y_i)$ proves surjectivity of $f^{\wedge r}$; the smash basepoint is also in its image. The source smash is compact. The target fat wedge is closed, hence the target smash is Hausdorff. A continuous surjection from a compact space to a Hausdorff space is a quotient map. This proves the first assertion for q_2 and q_1.

The suspension splitting

$$\Sigma(X_1 \times \cdots \times X_n) \simeq \bigvee_{\varnothing \neq I \subseteq \{1,\dots,n\}} \Sigma\left(\bigwedge_{i \in I} X_i\right)$$

gives

$$\mathcal{A} = \Sigma T^3 \simeq \bigvee^3 S^2 \vee \bigvee^3 S^3 \vee S^4.$$

Finite smash products distribute over finite wedges. The ordered degree sums give multiplicities $9, 18, 15, 6, 1$ in dimensions $4, 5, 6, 7, 8$, proving equation (202).

For $\mathcal{C}$, write $x = \rho_1$, $y = \rho_2$, and $h = \eta$. The pair-of-pants product and the two exact Seifert fillings give

$$\pi_1(\mathcal{M}) = \langle x, y, h \mid [x, h] = [y, h] = 1, x^3 h^{-1} = 1,\ y^4 h = 1 \rangle.$$

The remaining boundary torus is generated by h and $\rho_0 = (xy)^{-1}$. Coning this torus kills their normal closure, and therefore

$$\pi_1(\mathcal{C}) \cong \langle x, y \mid x^3 = 1,\ y^4 = 1,\ xy = 1 \rangle = 1,$$

because $y = x^{-1}$ and $\gcd(3, 4) = 1$. The previously computed relative homology gives

$$\widetilde{H}_2(\mathcal{C}; \mathbb{Z}) = \mathbb{Z}, \qquad \widetilde{H}_3(\mathcal{C}; \mathbb{Z}) = \mathbb{Z},$$

and no other reduced homology. Since $\mathcal{C}$ is a simply connected finite three-complex, Hurewicz supplies a sphere representing the H_2-generator, and the Whitehead exact sequence makes $\pi_3(\mathcal{C}) \to H_3(\mathcal{C})$ surjective. A wedge of representing maps $S^2 \vee S^3 \to \mathcal{C}$ is a homology equivalence between simply connected CW complexes, hence a homotopy equivalence. Thus

$$\mathcal{C} \simeq S^2 \vee S^3.$$

Taking the fourth smash power yields equation (203) with the binomial multiplicities $1, 4, 6, 4, 1$.

The first target has no reduced homology above degree eight and the second has none above degree twelve. The source spheres have no positive homology below degrees twelve and twenty-four. Hence all positive-degree homology and cohomology homomorphisms induced by the two maps vanish. Their Hurewicz images lie in $H_{12}(\mathcal{A}^{\wedge 2}) = 0$ and $H_{24}(\mathcal{C}^{\wedge 4}) = 0$, respectively. A section would split a nonzero lower homology group of the target into the zero lower homology of the source sphere, which is impossible. Higher homotopy groups of the displayed wedges need not vanish, so none of these arguments decides the unstable null-homotopy question. □

Theorem 9.28 (Exact decompactification spaces with boundary $\mathbb{R}^4$)**.** *For $\nu \in \{12, 24\}$, set*

$$A_T^{(12)} = \operatorname{diag}(Y_T, Y_T), \qquad A_T^{(24)} = \ell_T I_4, \qquad s_0 = T_0^{-1}.$$

On $[0, s_0] \times \mathbb{R}^4$, define an equivalence relation $\sim_\nu$ fibrewise over s by

$$(0, x) \sim_\nu (0, y) \iff x = y,$$
$$(s, x) \sim_\nu (s, y) \iff x - y \in A_{1/s}^{(\nu)} \mathbb{Z}^4 \quad (0 < s \le s_0).$$

Let

$$\mathfrak{E}_\nu = ([0, s_0] \times \mathbb{R}^4)/\!\sim_\nu, \qquad p_\nu : \mathfrak{E}_\nu \longrightarrow [0, s_0].$$

Then $\mathfrak{E}_\nu$ is a Hausdorff, second-countable topological five-manifold with boundary, and

$$p_{12}^{-1}(s) \cong Q^2_{1/s} \qquad (s > 0), \qquad p_{12}^{-1}(0) \cong \mathbb{R}^4, \tag{204}$$
$$p_{24}^{-1}(s) \cong C^4_{1/s} \qquad (s > 0), \qquad p_{24}^{-1}(0) \cong \mathbb{R}^4. \tag{205}$$

The other boundary component is the compact four-torus over $s = s_0$.

Give every positive fibre its Euclidean quotient metric and the zero fibre its Euclidean metric. For every $R > 0$, there is $s_R > 0$ such that, for $0 \le s < s_R$, writing $q_{\nu,s}(x) = [s, x]$, the quotient map restricts to an exact isometry

$$q_{\nu,s} : B_R^{\mathbb{R}^4}(0) \xrightarrow{\cong_{\mathrm{isom}}} B_R^{p_\nu^{-1}(s)}([s, 0])$$

onto the entire quotient radius-R ball, not merely onto an embedded subset of that ball. Consequently the positive fibres converge to the zero fibre in pointed local-isometric, pointed smooth, and pointed Gromov–Hausdorff topology.

There are continuous surjective comparison maps which set-theoretically collapse the zero boundary fibres,

$$\mathbb{R}^4 = p_{12}^{-1}(0) \hookrightarrow \mathfrak{E}_{12} \xrightarrow{\beta_{12}} \mathcal{D}_{12} \xhookrightarrow{\delta_{12}} \mathcal{A}^{\wedge 2},$$

and

$$\mathbb{R}^4 = p_{24}^{-1}(0) \hookrightarrow \mathfrak{E}_{24} \xrightarrow{\beta_{24}} \mathcal{D}_{24} \xhookrightarrow{\delta_{24}} \mathcal{C}^{\wedge 4}.$$

For $s > 0$, β_ν is the specified marked homeomorphism from $p_\nu^{-1}(s)$ to the radial torus slice at $r = e^{-2\pi/s}$. Its restriction

$$\beta_\nu : \mathfrak{E}_\nu \setminus p_\nu^{-1}(0) \xrightarrow{\cong} \mathcal{D}_\nu \setminus \{d_\nu\}$$

is a homeomorphism, while

$$\beta_\nu^{-1}(d_\nu) = p_\nu^{-1}(0) \cong \mathbb{R}^4.$$

Neither β_ν is proper or a quotient map. It factors through a continuous bijection

$$\overline{\beta}_\nu : \mathfrak{E}_\nu / p_\nu^{-1}(0) \longrightarrow \mathcal{D}_\nu,$$

but this bijection is not a homeomorphism; equivalently, the topology of $\mathcal{D}_\nu$ is not the quotient topology induced by β_ν. Thus the limiting $\mathbb{R}^4$ is an actual boundary fibre of $\mathfrak{E}_\nu$, whereas the continuous comparison map sends that entire fibre to the cone vertex.

Proof. We give a common proof. Write

$$L_s = A_{1/s}^{(\nu)} \mathbb{Z}^4 \quad (s > 0), \qquad L_0 = \{0\}.$$

The matrices vary continuously for $s > 0$, and

$$\operatorname{sys}(L_s) \longrightarrow \infty \qquad (s \downarrow 0)$$

by $\operatorname{sys}(Y_T \mathbb{Z}^2) \geq T - C$ in the S^{12} case and $\operatorname{sys}(\ell_T \mathbb{Z}^4) = \ell_T$ in the S^{24} case.

The equivalence relation is closed. Indeed, suppose $(s_j, x_j) \sim_\nu (s_j, y_j)$ and $(s_j, x_j, y_j) \to (s, x, y)$. If $s > 0$, write $y_j - x_j = A_{1/s_j}^{(\nu)} n_j$. On a compact neighborhood of s, the inverse matrices are uniformly bounded, so boundedness of $y_j - x_j$ forces n_j to lie in a finite set. Passing to a subsequence makes n_j constant and proves $x \sim_\nu y$. If $s = 0$, any nonzero n_j would give

$$\|y_j - x_j\| \geq \operatorname{sys}(L_{s_j}) \longrightarrow \infty,$$

contrary to convergence. Hence $x = y$.

The quotient projection $q_\nu : [0, s_0] \times \mathbb{R}^4 \to \mathfrak{E}_\nu$ is open. If U is open, its saturation is the union of U with the images, for $n \in \mathbb{Z}^4$, of

$$U \cap ((0, s_0] \times \mathbb{R}^4) \quad \text{under} \quad (s, x) \longmapsto (s, x + A_{1/s}^{(\nu)} n),$$

and every one of these images is open. A closed equivalence relation with an open quotient map has Hausdorff quotient: two inequivalent points admit a product neighborhood disjoint from the relation, and the saturations of the two factor neighborhoods are disjoint open sets. Since the source is second countable and the quotient map is open, images of a countable base form a countable base of the quotient.

Fix $x_0 \in \mathbb{R}^4$. Choose $a > 0$ and $s_1 > 0$ such that $\operatorname{sys}(L_s) > 2a$ for $0 < s < s_1$. The quotient projection is one-to-one on $[0, s_1) \times B_a^\circ(x_0)$, where $B_a^\circ(x_0) = \{x : \|x - x_0\| < a\}$, and its restriction is an open continuous bijection onto an open subset of $\mathfrak{E}_\nu$; it is therefore a homeomorphism. These are half-space

charts at $s = 0$. The usual evenly covered small balls give product charts for $s > 0$, and the same argument at $s = s_0$ gives the second boundary component. This proves the manifold assertion and the fibre identifications.

For the metric assertion, choose s_R so that $\mathrm{sys}(L_s) > 4R$ for $s < s_R$. If $x, y \in B_R(0)$ and $0 \neq \lambda \in L_s$, then

$$\|x - y + \lambda\| \geq \|\lambda\| - \|x - y\| > 4R - 2R \geq \|x - y\|.$$

Thus no translated representative shortens the Euclidean distance on that ball, so $q_{\nu,s}$ is an exact isometric embedding there. It is also onto the full quotient ball. Indeed, if $\xi = [s, x]$ satisfies $d([s, 0], \xi) \leq R$, discreteness of L_s makes the infimum in

$$d([s, 0], [s, x]) = \min_{\lambda \in L_s} \|x + \lambda\|$$

attained. For a minimizing λ_0, the representative $x + \lambda_0$ lies in $B_R^{\mathbb{R}^4}(0)$ and maps to ξ. At $s = 0$, $q_{\nu,0}$ is the identity. This proves the asserted onto isometry, and since it holds for every fixed R, it proves all three pointed convergences.

It remains to define these comparison maps, which set-theoretically collapse the zero fibres. For $s > 0$, put $T = 1/s$ and $r = e^{-2\pi T}$. In the S^{12} case, write $x = (x_1, x_2) \in \mathbb{R}^2 \oplus \mathbb{R}^2$ and set

$$\beta_{12}([s, x]) = \big[r, \iota_T^{-1}([x_1]_{Y_T\mathbb{Z}^2}), \iota_T^{-1}([x_2]_{Y_T\mathbb{Z}^2})\big].$$

In the S^{24} case, write $x = (x_1, \ldots, x_4)$ and set

$$\beta_{24}([s, x]) = [r, [x_1/\ell_T], \ldots, [x_4/\ell_T]].$$

These formulae are well defined on the positive fibres and are continuous there. Define $\beta_\nu([0, x]) = d_\nu$ for every $x \in \mathbb{R}^4$. Every neighborhood of d_ν in the cone contains all cross-sections with sufficiently small r. Since $e^{-2\pi/s} \to 0$ as $s \downarrow 0$, continuity at the zero fibre is uniform in x. Surjectivity follows because each positive fibre maps onto its complete cross-section and the zero fibre maps onto the vertex. The restriction away from $p_\nu^{-1}(0)$ is a homeomorphism because $s \mapsto e^{-2\pi/s}$ is a homeomorphism from $(0, s_0]$ to $(0, r_0]$, and the displayed marked-torus identifications are fibrewise homeomorphisms varying continuously with s. The vertex fibre is exactly $p_\nu^{-1}(0)$, so β_ν is not proper.

It is also not a quotient map. Choose $a \in \mathbb{R}^4 \setminus \mathbb{Z}^4$, for example $a = (1/2, 1/2, 1/2, 1/2)$, choose $s_j \downarrow 0$, and put

$$z_j = [s_j, A^{(\nu)}_{1/s_j} a] \in \mathfrak{E}_\nu.$$

For $\nu = 12$,

$$\inf_{n \in \mathbb{Z}^4} \|A^{(12)}_{1/s_j}(a + n)\| \geq (1/s_j - C)\,\mathrm{dist}(a, \mathbb{Z}^4) \longrightarrow \infty,$$

and for $\nu = 24$ the same infimum equals

$$\ell_{1/s_j}\,\mathrm{dist}(a, \mathbb{Z}^4) \longrightarrow \infty.$$

More explicitly, for every fixed $x_0 \in \mathbb{R}^4$,

$$\inf_{n \in \mathbb{Z}^4} \|A^{(\nu)}_{1/s_j}(a + n) - x_0\| \geq \inf_{n \in \mathbb{Z}^4} \|A^{(\nu)}_{1/s_j}(a + n)\| - \|x_0\| \longrightarrow \infty.$$

Hence $\{z_j : j \geq 1\}$ is closed in $\mathfrak{E}_\nu$: it has no accumulation in a positive fibre and the last estimate shows that it eventually misses every half-space chart about every point of the zero fibre. But

$$\beta_\nu(z_j) = [e^{-2\pi/s_j}, a] \longrightarrow d_\nu,$$

so $\{\beta_\nu(z_j) : j \geq 1\}$ is not closed in $\mathcal{D}_\nu$. Since β_ν is injective off the zero fibre and the latter maps to d_ν, the inverse image of this nonclosed set is the closed set $\{z_j : j \geq 1\}$. Thus β_ν is not a quotient map. This

proves the two diagrams and their exact topological scope. Since β_ν is constant exactly on $p_\nu^{-1}(0)$ and otherwise injective, it induces the continuous bijection $\overline{\beta}_\nu$ displayed in the theorem. Were that bijection a homeomorphism, β_ν would be the composite of the quotient projection with a homeomorphism and hence a quotient map, contrary to the counterexample. Therefore $\overline{\beta}_\nu$ is not a homeomorphism. □

Corollary 9.29 (Sharp global meaning of the Euclidean limit)**.** *The native routes constructed above exhibit the underlying space $\mathbb{R}^4$ in the following three precise senses:*

1. *as each of the nonclosed topologically embedded open cells $U_{12,T} \subset S^6$ and $U_{24,T} \subset S^8$;*
2. *as the pointed intrinsic Euclidean limit of the four-torus slices in the global quotient carriers of S^{12} and S^{24}; and*
3. *as the Euclidean decompactification boundary fibre of $\mathfrak{E}_{12}$ and $\mathfrak{E}_{24}$.*

It does not produce $\mathbb{R}^4$ as the whole compact sphere, as the whole compact conical target, or as a continuous quotient target of either compact space. Nor does the unrescaled restriction of any fixed compatible ambient target metric have $\mathbb{R}^4$ as its slice limit, and the open cells in item 1 are not globally Euclidean-isometric in such a metric.

Proof. The three positive assertions are theorems 9.23, 9.26 and 9.28. For the negative assertions, a continuous image of a compact sphere or compact conical target is compact, so it cannot be all of $\mathbb{R}^4$. The local homology calculation equation (201) shows that the carrier vertex is not an integral homology-manifold point, and hence not a manifold point, of any dimension, with or without boundary. Finally, in every fixed compatible metric on a compact cone target, all sufficiently deep radial slices lie uniformly in an arbitrarily small neighborhood of the vertex, so their ambient diameters tend to zero. Their intrinsic diameters tend to infinity because their intrinsic systoles do. Hence without a T-dependent rescaling no fixed ambient metric realizes the intrinsic Euclidean limit on those slices. The fixed cusp-adapted blow-ups are treated separately in theorem 9.35. The comparison diagrams record the unrescaled distinction exactly: they collapse the new $\mathbb{R}^4$ boundary rather than identify it with the compact target. □

Scope. The constructions above use only the original, unscaled fibre metrics and the quotient maps derived from the native S^6 cusp. They do not use the auxiliary finite-cover unspooling. The conical carriers depend on the chosen cusp coordinate, positive radial ray, and based identification $X \cong S^6$. The spaces $\mathfrak{E}_\nu$ additionally remember the marked Euclidean quotient metrics. None of these data defines a compact complex structure on S^{12} or S^{24}, a four-dimensional physical spacetime, or an interacting Yang–Mills Hamiltonian.

9.3 The full $S^{12} \to S^6$ carrier is null-homotopic

This subsection determines the based homotopy class of the full two-copy sphere carrier. It is important that the calculation concerns the full quotient map, not merely the inclusion of one of its moving four-spheres.

Recall the exact maps from theorems 9.12 and 9.20. The cusp quotient is

$$q_2 : (X, x_0) \longrightarrow \mathcal{A} = \mathrm{Susp}(S^1 \times \overline{T}^2),$$

and, for the coordinate one-skeleton

$$K^{(1)} = ((\mathbb{R}/\mathbb{Z}) \times \{0\}) \cup (\{0\} \times (\mathbb{R}/\mathbb{Z})) \subset \overline{T}^2,$$

the degree-one quotient $c : \overline{T}^2 \to \overline{T}^2/K^{(1)} \cong S^2$ defines

$$\widehat{p}_2([r, e^{i\phi}, u]) = [r, c(u)] \in \Sigma_r S^2 \cong S^3.$$

Thus

$$g := \widehat{q}_3 = \widehat{p}_2 \circ q_2 : (X, x_0) \longrightarrow (S^3, *)$$

is the specified continuous surjective based quotient. After the fixed based identification $X \cong_{\text{diff}} S^6$, its smash square is the specified continuous surjective based quotient

$$g^{\wedge 2} : S^{12} = S^6 \wedge S^6 \longrightarrow S^3 \wedge S^3 = S^6.$$

Theorem 9.30 (The full external smash pairing is zero)**.** *For every pair of based maps $x, y : S^6 \to S^3$,*

$$[x \wedge y] = 0 \qquad in \qquad \pi_{12}(S^6).$$

In particular, $\widehat{q}_3^{\wedge 2} : S^{12} \twoheadrightarrow S^6$ is simultaneously surjective as a point-set quotient and based null-homotopic.

Proof. Write E for reduced suspension, using the ordered smash coordinates on spheres. The external smash product has the exact factorization

$$x \wedge y = (x \wedge \mathrm{id}_{S^3}) \circ (\mathrm{id}_{S^6} \wedge y) : S^6 \wedge S^6 \longrightarrow S^6 \wedge S^3 \longrightarrow S^3 \wedge S^3. \tag{206}$$

Under the displayed ordered sphere identifications, the two factors, in order of application, represent $E^6 y \in \pi_{12}(S^9)$ and $E^3 x \in \pi_9(S^6)$. Equivalently, Toda's Barratt–Hilton reduced-join formula gives, with $p = q = k = h = 3$,

$$[x \wedge y] = (E^3[x]) \circ (E^6[y]). \tag{207}$$

Here Toda's sign is $(-1)^{(p+k)h} = (-1)^{18} = +1$. Reversing the two displayed factors introduces the separate twist sign $(-1)^{ph} = -1$; no such reversal is used here.

The three exact groups needed below are

$$\pi_6(S^3) \cong \mathbb{Z}/12, \qquad \pi_{12}(S^9) \cong \mathbb{Z}/24, \qquad \pi_{12}(S^6) \cong \mathbb{Z}/2. \tag{208}$$

The first and third are the integral low unstable groups displayed in Hatcher's table extracted from Toda; the middle group is in the stable three-stem [9, opening table, p. 339]; [11, Table 1, p. 113].

Put $\beta = E^6[y] \in \pi_{12}(S^9)$. Since $12[y] = 0$, functoriality of suspension gives $12\beta = 0$. In the cyclic group $\mathbb{Z}/24$, the elements annihilated by 12 are exactly the even elements. Hence there is $\gamma \in \pi_{12}(S^9)$ such that $\beta = 2\gamma$. Postcomposition with $E^3[x] : S^9 \to S^6$ induces a group homomorphism

$$(E^3[x])_* : \pi_{12}(S^9) \longrightarrow \pi_{12}(S^6).$$

The target has exponent two, so equation (207) gives

$$[x \wedge y] = (E^3[x])_*(\beta) = 2(E^3[x])_*(\gamma) = 0.$$

This proves the vanishing of the entire external smash pairing and hence the theorem. □

Corollary 9.31 (All higher smash powers)**.** *For every based $g : S^6 \to S^3$ and every $r \geq 2$,*

$$g^{\wedge r} : S^{6r} \longrightarrow S^{3r}$$

is based null-homotopic.

Proof. With the fixed associativity homeomorphisms for smash products,

$$g^{\wedge r} = g^{\wedge 2} \wedge g^{\wedge (r-2)}.$$

Smashing a based null-homotopy with a based map gives a based null-homotopy. □

Remark 9.32 (The remaining one-copy invariant). The proof does not need to determine the one-copy class $[g] \in \pi_6(S^3) \cong \mathbb{Z}/12$. Toda's suspension calculation identifies $\pi_6(S^3)$ with the even subgroup of the stable three-stem; Hatcher gives an integral Serre-spectral-sequence derivation of the same injectivity and identifies twice the stable Hopf–ν generator with a generator of the suspended $\mathbb{Z}/12$-summand [10, Theorem 5.41(b), p. 573, and pp. 577–578]. Thus, after choosing a stable generator ν_9, there is a unique $a \in \mathbb{Z}/12$ such that

$$[E^6 g] = 2a\nu_9 \in \pi_{12}(S^9) \cong \mathbb{Z}/24. \tag{209}$$

Because $a \mapsto 2a$ embeds $\mathbb{Z}/12$ as the even subgroup of $\mathbb{Z}/24$, $E^6 g$ is null-homotopic if and only if g is null-homotopic. Thus the one-copy class and its sixfold suspension are the same unresolved integer modulo 12, not two separate proof obligations.

The exact geometric invariant which computes a is also identifiable. Choose $0 < r_0 < \epsilon$ and $u_0 \in \overline{T}^2 \setminus K^{(1)}$, and let $y_0 = [r_0, c(u_0)]$ lie in the open three-cell of $\Sigma_r S^2$. From the literal formula for q_2,

$$g^{-1}(y_0) = \{x \in f^{-1}(|t| = r_0) : \overline{\pi}_f(x) = u_0\}. \tag{210}$$

The fibre of $\overline{\pi}$ is the two-torus defined by $K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle$, and the cusp monodromy fixes both $\widehat{w}$ and $\widehat{\delta}$. Therefore (210) is its identity mapping torus, $S^1 \times T^2 \cong T^3$. After choosing a smooth chart in the open target cell, the differential of g supplies its normal three-framing. Pontryagin–Thom identifies the stabilized class of g with the framed bordism class of this specifically framed T^3. Since stabilization on $\pi_6(S^3)$ is injective, computing that framing class computes a exactly. The unframed diffeomorphism type T^3, point-set surjectivity, and the zero ordinary homology map do not compute a.

Remark 9.33 (Scope). The theorem changes no point-set or metric statement. A surjective quotient map may be null-homotopic, and the embedded moving four-spheres and their intrinsic pointed $\mathbb{R}^4$-limits remain exactly as constructed. The theorem concerns the sphere-valued postcomposition $\widehat{q}_3^{\wedge 2}$; it does not imply null-homotopy of the earlier conical-target map $q_2^{\wedge 2} : S^{12} \to \mathcal{A}^{\wedge 2}$.

9.4 Fixed-target metric realization and its sharp dimensional scope

This note concerns the codomains of the three based maps

$$q_2^{\wedge 2} : X^{\wedge 2} \cong S^{12} \longrightarrow Z_{12} := \mathcal{A}^{\wedge 2},$$

$$q_{1,\mathrm{cusp}}^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow Z_{24}^{\mathcal{B}} := \mathcal{B}^{\wedge 4}, \qquad q_1^{\wedge 4} : X^{\wedge 4} \cong S^{24} \longrightarrow Z_{24}^{\mathcal{C}} := \mathcal{C}^{\wedge 4}.$$

In particular, S^{12} and S^{24} are the sources, not the spaces in which the metric limits occur. We construct one fixed compatible length metric on each codomain which realizes the native four-dimensional flat limits on the target slices. The realization is sharp: after the same rescaling, the full ambient limits have dimensions 8 and 12, and the collapsed smash base point goes to infinity.

Put

$$r_T = e^{-2\pi T}, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Y_T = TJ + E_T,$$

where E_T is the bounded matrix in the radial cusp expansion. Choose and fix T_0 and $C > 0$ so that $\|E_T\|_{\mathrm{op}} \le C$ for every $T \ge T_0$, in the fixed Euclidean coordinates used below. This notation avoids confusing the error matrix with the quotient circle C_T used below. On the fixed torus

$$\overline{T}^2 = (\overline{\Lambda} \otimes \mathbb{R})/\overline{\Lambda} \cong \mathbb{R}^2/\mathbb{Z}^2$$

let

$$h_T := \iota_T^* g_{Q_T} = dx^{\mathsf{T}} Y_T^{\mathsf{T}} Y_T dx.$$

Thus

$$T^{-2} h_T \longrightarrow h_\infty := dx^{\mathsf{T}} dx \tag{211}$$

in every C^k norm in the fixed flat coordinates. Indeed, the cusp coefficients are holomorphic in $t = e^{-2\pi T}$, so their T-derivatives are exponentially small, while $T^{-1}Y_T = J + T^{-1}E_T$.

Use normalized angular coordinates

$$u, \theta \in \mathbb{R}/\mathbb{Z};$$

u is the argument of the base coordinate and θ is the value of the invariant character. Increase this same T_0 once more, if necessary, so that all radial cusp formulas used below hold for $T \geq T_0$ and $e^{-2\pi T_0} < \epsilon$, and put

$$r_0 = e^{-2\pi T_0}.$$

For $0 < r \leq r_0$, write

$$T(r) := -\frac{1}{2\pi} \log r.$$

Target-slice notation. We use the embeddings $j_T, k_T^{\mathcal{B}}, k_T^{\mathcal{C}}$ defined in theorem 9.15; the normalized angle $u = 0$ used below is the same positive radial base angle denoted by $1 \in S^1$ there.

Theorem 9.34 (Topology-forced collapse for every fixed compatible metric)**.** *Let $D_{12}, D_{24}^{\mathcal{B}}, D_{24}^{\mathcal{C}}$ be arbitrary fixed metrics compatible with the respective compact target topologies, and let $*$ denote the corresponding smash base point. Then*

$$d_H^{D_{12}}\big(j_T(Q_T^2), \{*\}\big) \longrightarrow 0, \tag{212}$$

$$d_H^{D_{24}^{\mathcal{B}}}\big(k_T^{\mathcal{B}}(C_T^4), \{*\}\big) \longrightarrow 0, \tag{213}$$

$$d_H^{D_{24}^{\mathcal{C}}}\big(k_T^{\mathcal{C}}(C_T^4), \{*\}\big) \longrightarrow 0. \tag{214}$$

Consequently, with the metrics restricted from any one fixed ambient target, all three target slices converge in Gromov–Hausdorff topology to a point, not to $\mathbb{R}^4$.

Proof. We prove the first assertion; the other two are identical. Let

$$G : [0, r_0]^2 \times (T^3)^2 \longrightarrow Z_{12}$$

be the product of the two inner-cone quotient maps followed by the smash quotient. It is continuous and $G(0, 0, z_1, z_2) = *$ for every $(z_1, z_2) \in (T^3)^2$. Since the domain is compact, uniform continuity with respect to D_{12} gives

$$\sup_{(z_1,z_2)\in(T^3)^2} D_{12}\big(G(r, r, z_1, z_2), *\big) \longrightarrow 0.$$

The set $j_T(Q_T^2)$ is contained in such a cross-section, which proves the first Hausdorff convergence. Replacing T^3 by T^2 and taking four factors proves the other two. Hausdorff convergence to a singleton implies Gromov–Hausdorff convergence to that singleton. □

The preceding theorem is unrescaled. It does not prevent a blow-up of one fixed metric. The following construction gives the strongest natural positive statement.

Theorem 9.35 (One fixed cusp-adapted metric realizes the vertical $\mathbb{R}^4$ limits)**.** *There are fixed compatible length metrics*

$$D_{12} \text{ on } Z_{12}, \qquad D_{24}^{\mathcal{B}} \text{ on } Z_{24}^{\mathcal{B}}, \qquad D_{24}^{\mathcal{C}} \text{ on } Z_{24}^{\mathcal{C}}$$

with the following properties.

On the inner cone of $\mathcal{A}$, the metric is induced by

$$g_{\mathcal{A}} = dr^2 + r^2 du^2 + \frac{r^2}{T(r)^2} h_{T(r)}. \tag{215}$$

On the cusp cone of either $\mathcal{B}$ or $\mathcal{C}$, it is induced by

$$g_{\mathcal{D}} = dr^2 + r^2(du^2 + d\theta^2), \qquad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}. \tag{216}$$

The metrics on the smash powers are the metric quotients of the product length metrics after the closed fat wedge is collapsed. Write $g_{Z_{12}}$ and $g_{Z_{24}^{\mathcal{D}}}$ for the resulting smooth Riemannian tensors on the open regular strata.

Let

$$p_T^{12} := j_T(0,0), \qquad p_T^{24,\mathcal{D}} := k_T^{\mathcal{D}}(0,0,0,0).$$

Put

$$\Sigma_T^{12} := j_T(Q_T^2), \qquad \Sigma_T^{24,\mathcal{D}} := k_T^{\mathcal{D}}(C_T^4).$$

Let $d_{T,12}^{\mathrm{slice}}$ and $d_{T,24,\mathcal{D}}^{\mathrm{slice}}$ be the intrinsic Riemannian path metrics of the first fundamental forms on these embedded slices. These must be distinguished from the restricted ambient distance functions

$$d_{T,12}^{\mathrm{amb}} := D_{12}|_{\Sigma_T^{12}\times\Sigma_T^{12}}, \qquad d_{T,24,\mathcal{D}}^{\mathrm{amb}} := D_{24}^{\mathcal{D}}|_{\Sigma_T^{24,\mathcal{D}}\times\Sigma_T^{24,\mathcal{D}}}.$$

In general the latter can be smaller, because an ambient path is allowed to leave the slice.

Then the induced first fundamental forms on the target slices satisfy the exact identities

$$\left(\frac{T}{r_T}\right)^2 j_T^* g_{Z_{12}} = g_{Q_T} \oplus g_{Q_T}, \tag{217}$$

and

$$\left(\frac{\ell_T}{r_T}\right)^2 (k_T^{\mathcal{D}})^* g_{Z_{24}^{\mathcal{D}}} = g_{C_T}^{\oplus 4}. \tag{218}$$

Consequently,

$$\left(\Sigma_T^{12}, \frac{T}{r_T} d_{T,12}^{\mathrm{slice}}, p_T^{12}\right) \longrightarrow (\mathbb{R}^4, 0), \tag{219}$$

$$\left(\Sigma_T^{24,\mathcal{D}}, \frac{\ell_T}{r_T} d_{T,24,\mathcal{D}}^{\mathrm{slice}}, p_T^{24,\mathcal{D}}\right) \longrightarrow (\mathbb{R}^4, 0) \tag{220}$$

in pointed local-isometric and pointed Gromov–Hausdorff topology, and smoothly on the exhausting Euclidean slice balls.

The full ambient pointed blow-ups are instead

$$\left(Z_{12}, \frac{T}{r_T} D_{12}, p_T^{12}\right) \longrightarrow (\mathbb{R}^8, 0), \tag{221}$$

$$\left(Z_{24}^{\mathcal{D}}, \frac{\ell_T}{r_T} D_{24}^{\mathcal{D}}, p_T^{24,\mathcal{D}}\right) \longrightarrow (\mathbb{R}^{12}, 0). \tag{222}$$

as pointed metric spaces in pointed Gromov–Hausdorff topology. More precisely, for every fixed radius the corresponding rescaled target ball is eventually contained in an open regular-stratum coordinate neighborhood, and the pulled-back Riemannian tensors converge there in every C^k norm. Thus smooth convergence is asserted on an exhaustion by regular-stratum neighborhoods, not globally on the singular compact targets. Under these limits, the slice $\mathbb{R}^4$ is respectively the coordinate subspace obtained by fixing the two radial and two base-angle coordinates, or by fixing the four radial and four base-angle coordinates.

Proof. Equation (211) shows that the metrics on the compact cross-sections of (215) remain uniformly bounded and converge to a positive definite flat metric. Therefore the cross-section diameter is $O(r)$. Moreover the radial coordinate is 1-Lipschitz and a radial segment from level r to the vertex has length r. The length completion consequently adds exactly one point at $r = 0$, and its topology is the prescribed

cone topology. The same assertions are immediate for (216). Interpolation on a closed annulus away from the vertex extends these metrics to fixed compatible length metrics on the compact spaces $\mathcal{A}, \mathcal{B}, \mathcal{C}$.

If a compact metric space (E, d) and a closed subset $W \subset E$ are given, the metric quotient which collapses W to one point is

$$d_{E/W}([x],[y]) = \min\{d(x,y), d(x,W) + d(y,W)\}.$$

Applying this to the fat wedges in the product length spaces gives the claimed fixed compatible metrics on the smash targets.

On the r_T-slice of $\mathcal{A}$, with $u = 0$, the first fundamental form is

$$\frac{r_T^2}{T^2} h_T.$$

The definition $h_T = \iota_T^* g_{Q_T}$ proves (217). On the r_T-slice of $\mathcal{B}$ or $\mathcal{C}$, with $u = 0$, the form is $r_T^2 d\theta^2$. Under $\vartheta_T : [\theta] \mapsto [\ell_T \theta]$, the native quotient-circle metric is $\ell_T^2 d\theta^2$. This proves (218). The native systole estimates

$$\mathrm{sys}(Q_T^2) \geq T - C, \qquad \mathrm{sys}(C_T^4) = \ell_T$$

then prove (220): every prescribed radius ball is eventually exactly a Euclidean four-ball.

It remains to compute the ambient limits. In one $\mathcal{A}$-factor, put

$$\lambda_T = \frac{T}{r_T}, \qquad s = \lambda_T(r - r_T), \qquad \eta = Tu, \qquad z = Y_T x$$

on the universal cover of the torus directions. On every fixed bounded set in (s, η, z), one has

$$\frac{r}{r_T} = 1 + \frac{s}{T}, \qquad T(r) = T + O(T^{-1}), \qquad Y_{T(r)} Y_T^{-1} = I + O(T^{-2}),$$

with the same estimates after any fixed number of derivatives. Thus $\lambda_T^2 g_{\mathcal{A}}$ converges in these coordinates to

$$ds^2 + d\eta^2 + dz_1^2 + dz_2^2.$$

The inward radial boundary is at distance T, the u-period is T, and the z-period lattice is $Y_T \mathbb{Z}^2$, whose systole is at least $T - C$. The outward edge of the fixed cusp chart is also at scaled distance

$$\frac{T}{r_T}(r_0 - r_T) = T\left(\frac{r_0}{r_T} - 1\right) \longrightarrow \infty.$$

Thus the inward vertex, all periods, and the outward edge recede to infinity. A single factor converges on exhausting regular-stratum coordinate neighborhoods to $\mathbb{R}^4$, and two factors converge to $\mathbb{R}^8$.

For one $\mathcal{B}$- or $\mathcal{C}$-factor, put

$$\lambda_T = \frac{\ell_T}{r_T}, \qquad s = \lambda_T(r - r_T), \qquad \eta = \ell_T u, \qquad \zeta = \ell_T \theta.$$

Then

$$\lambda_T^2 g_{\mathcal{D}} = ds^2 + \left(\frac{r}{r_T}\right)^2 (d\eta^2 + d\zeta^2),$$

which converges smoothly on bounded coordinate sets to the Euclidean metric on $\mathbb{R}^3$. The two circle periods and the inward radial boundary are all at scaled distance $\ell_T \to \infty$, while the outward edge is at scaled distance

$$\frac{\ell_T}{r_T}(r_0 - r_T) = \ell_T\left(\frac{r_0}{r_T} - 1\right) \longrightarrow \infty.$$

Four factors therefore converge on exhausting regular-stratum coordinate neighborhoods to $\mathbb{R}^{12}$.

Finally, a level-r_T factor point has distance exactly r_T from its cone vertex. Indeed, the radial segment has that length. A competing curve contained in the inner chart has length at least r_T, while one which leaves has inner-chart portions of total length at least $(r_0 - r_T) + r_0 > r_T$. The Riemannian-ℓ^2-product distance from a synchronized point to the fat wedge is the minimum of the factor distances, and hence is also r_T.

For the chosen scale put

$$b_T = \lambda_T r_T, \qquad a_T = \lambda_T(r_0 - r_T),$$

and let σ_T be the least scaled period in a factor. Thus $\sigma_T \geq T - C$ in the $\mathcal{A}$ calculation and $\sigma_T = \ell_T$ in the $\mathcal{B},\mathcal{C}$ calculation. The preceding computations give

$$a_T \longrightarrow \infty, \qquad b_T \longrightarrow \infty, \qquad \sigma_T \longrightarrow \infty.$$

Fix R. A curve of rescaled length at most R from p_T meets neither the fat wedge nor $r = r_0$ for all large T, so the radius-R target ball lies in the open regular stratum. If x, y lie in that ball, a competitor through the fat wedge has length at least

$$(b_T - R) + (b_T - R) > 2R,$$

and a competitor which exits the inner chart and returns has length at least

$$2(a_T - R) > 2R.$$

Every remaining competitor of length at most $2R$ stays in $r/r_T = 1 + o_R(1)$. After lifting it to the angular and torus covers, a nonzero deck translation has length at least $(1 - o_R(1))(\sigma_T - 2R) > 2R$ for all large T. Since the path through p_T has length at most $2R$, none of these three kinds of global shortcut computes the distance between x and y. Thus the metric on every fixed target ball is computed entirely in one lifted regular-stratum chart. The C^∞ convergence of the pulled-back tensors on these exhausting charts implies the pointed Gromov–Hausdorff limits in (222). □

Theorem 9.36 (Complete intrinsic scalar-rescaling trichotomy)**.** *For the fixed metrics of theorem 9.35, let $\lambda_T > 0$ be arbitrary and put*

$$\beta_T := \lambda_T r_T.$$

Choose one of the following triples, retaining the indicated ambient metric:

$$(\Sigma_T, d_T^{\mathrm{slice}}, d_T^{\mathrm{amb}}, p_T; D) = (\Sigma_T^{12}, d_{T,12}^{\mathrm{slice}}, d_{T,12}^{\mathrm{amb}}, p_T^{12}; D_{12}),$$
$$(\Sigma_T, d_T^{\mathrm{slice}}, d_T^{\mathrm{amb}}, p_T; D) = (\Sigma_T^{24,\mathcal{D}}, d_{T,24,\mathcal{D}}^{\mathrm{slice}}, d_{T,24,\mathcal{D}}^{\mathrm{amb}}, p_T^{24,\mathcal{D}}; D_{24}^{\mathcal{D}}), \qquad \mathcal{D} \in \{\mathcal{B},\mathcal{C}\}.$$

After passing to a subsequence so that β_T has a limit in $[0,\infty]$, the intrinsic pointed slice limit is

$$(\Sigma_T, \lambda_T d_T^{\mathrm{slice}}, p_T) \longrightarrow \begin{cases} \{\mathrm{pt}\}, & \beta_T \longrightarrow 0, \\ ((\mathbb{R}/L\mathbb{Z})^4, 0), & \beta_T \longrightarrow L \in (0,\infty), \\ (\mathbb{R}^4, 0), & \beta_T \longrightarrow \infty. \end{cases} \tag{223}$$

The convergence in (223) *is pointed Gromov–Hausdorff in all three cases. In the finite positive case the flat metrics converge smoothly under the fixed marked-torus identifications; in the unrolling case the convergence is pointed local-isometric and smooth on the exhausting Euclidean balls. In the middle line the lattice is orthogonally equivalent to the square lattice $L\mathbb{Z}^4$. At the same time,*

$$\lambda_T D(p_T, *) = \beta_T \tag{224}$$

for every one of the displayed choices.

Only in the unrolling regime $\beta_T \to \infty$ do we compare the intrinsic metric with the restricted ambient distance. In that regime, for every $R \in (0,\infty)$,

$$\sup_{x,y \in \overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T,R)} \left| \lambda_T d_T^{\mathrm{amb}}(x,y) - \lambda_T d_T^{\mathrm{slice}}(x,y) \right| \longrightarrow 0. \tag{225}$$

Moreover, there are $\varepsilon_{T,R} \to 0$ such that

$$\overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T,R) \subset \overline{B}_{\lambda_T d_T^{\mathrm{amb}}}(p_T,R) \subset \overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T,(1+\varepsilon_{T,R})R). \tag{226}$$

Thus the restricted ambient metric has the same pointed $\mathbb{R}^4$ limit on these fixed slice balls. No agreement is asserted in the regimes $\beta_T \to 0$ or $\beta_T \to L \in (0,\infty)$, where radial or fat-wedge shortcuts need not disappear. Every rescaling which unrolls the intrinsic slice sends the cone/fat-wedge base point to infinity, so no such limit is a tangent cone based at that point.

In the same regime $\beta_T \to \infty$, the corresponding full ambient bubbles are

$$(Z_{12}, \lambda_T D_{12}, p_T^{12}) \longrightarrow (\mathbb{R}^8, 0), \tag{227}$$

$$(Z_{24}^{\mathcal{D}}, \lambda_T D_{24}^{\mathcal{D}}, p_T^{24,\mathcal{D}}) \longrightarrow (\mathbb{R}^{12}, 0), \qquad \mathcal{D} \in \{\mathcal{B}, \mathcal{C}\}. \tag{228}$$

These are pointed Gromov–Hausdorff limits of the singular metric targets. The convergence of tensors is C^∞ only under coordinate identifications on exhausting open regular-stratum neighborhoods; no global smooth convergence of the singular targets is claimed.

Proof. For the Q_T^2 route, the rescaled period matrix in each two-dimensional factor is

$$\frac{\lambda_T r_T}{T} Y_T = \beta_T \left(J + \frac{E_T}{T} \right).$$

If $\beta_T \to L \in (0,\infty)$, this lattice converges to $LJ\mathbb{Z}^2$; two factors give the square four-torus in (223). If $\beta_T \to 0$, the lengths of a lattice basis tend to zero, so the diameter tends to zero. If $\beta_T \to \infty$, the reverse triangle inequality gives systole

$$\frac{\beta_T}{T}(T - C) = \beta_T \left(1 - \frac{C}{T} \right) \longrightarrow \infty,$$

so every fixed-radius ball is eventually Euclidean.

For a C_T^4 slice, the four periods after rescaling have the common length

$$\lambda_T r_T = \beta_T,$$

which gives the same three intrinsic cases directly.

Consider first one factor. A curve from a level-r_T point to the cone vertex which remains in the inner chart has length at least r_T, because the dr^2-term makes r 1-Lipschitz there. If it leaves the chart, it has a first crossing of $r = r_0$ and, before ending at the vertex, a last re-entry through $r = r_0$; its two inner-chart portions have total length at least $(r_0 - r_T) + r_0 > r_T$. The radial segment has length r_T, so the factor distance to its based vertex is exactly r_T. The Riemannian-ℓ^2-product distance from every synchronized slice point to the fat wedge is the minimum of these factor distances and hence is also r_T. The metric-quotient formula in the preceding proof therefore gives (224).

It remains to prove the restricted-ambient comparison, so assume $\beta_T \to \infty$ and fix R. Two points in $\overline{B}_{\lambda_T d_T^{\mathrm{slice}}}(p_T,R)$ have intrinsic distance at most $2R/\lambda_T$. The slice path is an ambient competitor, so

$$d_T^{\mathrm{amb}}(x,y) \le d_T^{\mathrm{slice}}(x,y).$$

The quotient shortcut through the fat wedge has rescaled length $2\lambda_T r_T = 2\beta_T$, and hence is longer than $2R$ for all large T. A path which leaves the fixed cusp chart must reach $r = r_0$. Its outward cost is controlled by

$$\lambda_T(r_0 - r_T) = \beta_T \left(\frac{r_0}{r_T} - 1 \right) \longrightarrow \infty, \tag{229}$$

so no path of rescaled length at most $2R$ can use the interpolating annulus or the global part of a factor.

Consider now an ambient competitor of length at most $2R/\lambda_T + o(\lambda_T^{-1})$ which stays in the cusp chart. In every factor its radial coordinate satisfies

$$|r - r_T| \leq \frac{2R + o(1)}{\lambda_T}, \qquad \frac{r}{r_T} = 1 + O_R(\beta_T^{-1}).$$

On this tube, the horizontal tensors in the directions tangent to the slice are uniformly $(1+\varepsilon_{T,R})$-bilipschitz to their values at $r = r_T$, where $\varepsilon_{T,R} \to 0$. For $\mathcal{B}$ and $\mathcal{C}$ this follows directly from $(r/r_T)^2 = 1 + O_R(\beta_T^{-1})$. For $\mathcal{A}$, the same conclusion follows, without changing the fixed coordinates or the matrix Y_T, from

$$\begin{aligned} T(r) - T &= O_R(\beta_T^{-1}), \\ \frac{T(r)}{T} &= 1 + O_R((\beta_T T)^{-1}), \\ Y_{T(r)} Y_T^{-1} &= I + O_R((\beta_T T)^{-1}). \end{aligned}$$

Projecting the competitor to the torus or character directions, and discarding its radial and base-angle components, produces a path between the same slice endpoints. The bilipschitz estimate therefore gives

$$d_T^{\mathrm{slice}}(x, y) \leq (1 + \varepsilon_{T,R}) d_T^{\mathrm{amb}}(x, y).$$

Together with the opposite inequality and $\lambda_T d_T^{\mathrm{slice}}(x, y) \leq 2R$, this proves (225). The first inclusion in (226) follows from $d_T^{\mathrm{amb}} \leq d_T^{\mathrm{slice}}$. For a point x in the middle ball, apply the same fat-wedge and outward exclusions to a near-minimizing ambient path from p_T to x, whose rescaled length is at most $R + o(1)$, and project that path to the slice directions. This gives

$$\lambda_T d_T^{\mathrm{slice}}(p_T, x) \leq (1 + \varepsilon_{T,R}) R$$

after enlarging $\varepsilon_{T,R}$ by a quantity tending to zero, and proves the second inclusion.

For completeness, the full ambient bubbles at an arbitrary unrolling scale are obtained without changing any coordinate or orientation. In one $\mathcal{A}$-factor use

$$s = \lambda_T(r - r_T), \qquad \eta = \beta_T u, \qquad z = \frac{\beta_T}{T} Y_T x.$$

Writing $A_T(r) = Y_{T(r)} Y_T^{-1}$, one obtains the exact expression

$$\lambda_T^2 g_{\mathcal{A}} = ds^2 + \left(\frac{r}{r_T} \right)^2 d\eta^2 + \left(\frac{r}{r_T} \right)^2 \left(\frac{T}{T(r)} \right)^2 dz^{\mathsf{T}} A_T(r)^{\mathsf{T}} A_T(r) dz.$$

On every fixed bounded coordinate set,

$$\frac{r}{r_T} = 1 + O_R(\beta_T^{-1}), \qquad T(r) - T = O_R(\beta_T^{-1}), \qquad A_T(r) = I + O_R((\beta_T T)^{-1}),$$

with the same estimates after every fixed number of derivatives. Hence the tensor converges smoothly to the Euclidean metric on $\mathbb{R}^4$. The η-period is β_T, and the z-period lattice is

$$\frac{\beta_T}{T} Y_T \mathbb{Z}^2, \qquad \mathrm{sys}\left(\frac{\beta_T}{T} Y_T \mathbb{Z}^2 \right) \geq \beta_T \left(1 - \frac{C}{T} \right) \longrightarrow \infty.$$

In one $\mathcal{B}$- or $\mathcal{C}$-factor use

$$s = \lambda_T(r - r_T), \qquad \eta = \beta_T u, \qquad \zeta = \beta_T \theta.$$

Then

$$\lambda_T^2 g_{\mathcal{D}} = ds^2 + \left(\frac{r}{r_T}\right)^2 (d\eta^2 + d\zeta^2),$$

and both angular periods are $\beta_T \to \infty$.

In all three cases the inward boundary and the fat wedge are at scaled distance β_T, while a lift starting at radius r_T incurs the one-way radial cost in (229) before its first outward crossing of $r = r_0$. Fix R. For large T, all these quantities and all nonzero period lengths exceed $4R$. If x, y lie in the ambient rescaled radius-R ball, their distances from the fat wedge are at least $\beta_T - R$; hence a quotient path through the smash base point has length at least $2(\beta_T - R) > 2R$. Every remaining competitor lifts to the complement of the fat wedge in the product. If such a lift exits the inner product chart, then some factor has a first crossing of $r = r_0$; the radial variation before that crossing is at least $\lambda_T(r_0 - r_T) - R$, and returning to the lift of y incurs the same lower bound. Its length is therefore greater than $2R$. Finally, a path of length at most $2R$ remains in the tube $r/r_T = 1 + O_R(\beta_T^{-1})$; there every nonzero deck translation has length tending to infinity, so such a path cannot wrap around an angular or torus period. The displayed coordinate neighborhoods therefore exhaust the pointed balls without quotient or global shortcuts. Their smooth tensor convergence proves the pointed Gromov–Hausdorff limits in (228), with smooth convergence confined to this regular-stratum exhaustion. □

Corollary 9.37 (Unit-volume normalization does not give $\mathbb{R}^4$)**.** *The four-volume of Q_T^2 is $(\det Y_T)^2 \sim T^4$. Its unit-volume length factor is therefore $(\det Y_T)^{-1/2} \sim T^{-1}$, and its unit-volume limit is a square flat four-torus, not $\mathbb{R}^4$. Likewise, the four-volume of C_T^4 is ℓ_T^4, and its unit-volume normalization is exactly the square unit four-torus.*

Proposition 9.38 (What vertex tangents and compact asymptotic cones can mean)**.** *The compact target topology alone determines no metric tangent cone. For an exact conic length metric built from fixed piecewise-Riemannian metrics on the links, however, the vertex links have topological and Hausdorff dimensions 7 for Z_{12} and 11 for both Z_{24} targets. The resulting vertex cones have topological and Hausdorff dimensions 8 and 12, and are not $\mathbb{R}^4$.*

More precisely, writing $L_3 = T^3$ and $L_2 = T^2$, one has the based topological models

$$Z_{12} \cong \mathrm{Susp}(L_3 * L_3), \qquad Z_{24}^{\mathcal{B}} \cong \mathrm{Susp}(L_2 * L_2 * L_2 * L_2).$$

At the suspension base point their conic links are respectively $L_3 * L_3$ and L_2^{*4}. Also

$$(\mathcal{C}, c_0) \cong (\mathcal{M}/\partial\mathcal{M}, [\partial\mathcal{M}]), \qquad (Z_{24}^{\mathcal{C}}, *) \cong (\mathcal{M}^4/\partial(\mathcal{M}^4), [\partial(\mathcal{M}^4)]),$$

so an exact conic metric at the collapsed boundary has link $\partial(\mathcal{M}^4)$. At the collapsed point of $Z_{24}^{\mathcal{C}}$, the complete integral local-homology calculation is

$$H_i(Z_{24}^{\mathcal{C}}, Z_{24}^{\mathcal{C}} \setminus \{*\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{i-1}}, & 2 \le i \le 5, \\ 0, & i = 6, 7, \\ \mathbb{Z}^{\binom{4}{i-8}}, & 8 \le i \le 12, \\ 0, & \text{otherwise.} \end{cases} \tag{230}$$

In particular, its local homology is not concentrated in top degree. For the first two targets the top local homology occurs in degree 8 and 12, respectively; for the third target (230) includes a top $\mathbb{Z}$ in degree 12. None has the local homology of $\mathbb{R}^4$.

Finally, the standard asymptotic cone obtained by blowing down any fixed compact metric target is a point. Indeed, if $R_n \to \infty$, then

$$\operatorname{diam}(Z, R_n^{-1} D) = R_n^{-1} \operatorname{diam}(Z, D) \longrightarrow 0.$$

Thus the limits in theorem 9.35 are moving-base-point blow-ups, often called bubbles. They are neither vertex-based tangent cones nor asymptotic cones of the compact targets.

Proof. For compact spaces V_i with closed subspaces A_i, there is a canonical homeomorphism

$$(V_1/A_1) \wedge \cdots \wedge (V_m/A_m) \cong \frac{V_1 \times \cdots \times V_m}{\bigcup_i V_1 \times \cdots \times A_i \times \cdots \times V_m}.$$

Take $V_i = \operatorname{Cone}(L_i)$ and $A_i = L_i$, its outer boundary. The product of m cones is homeomorphic to the cone on the iterated join $L_1 * \cdots * L_m$, and its displayed union of faces is the outer link. Collapsing that link gives its suspension. This proves the first two models and the link dimensions

$$\dim(T^3 * T^3) = 3 + 3 + 1 = 7, \qquad \dim((T^2)^{*4}) = 4 \cdot 2 + 3 = 11.$$

We next prove the based homeomorphism for $\mathcal{C}$, rather than identifying the two presentations without a map. Put $L = \partial\mathcal{M}$, write the attached cone as

$$\operatorname{Cone}(L) = L \times [0,1]/(L \times \{0\}),$$

with base $L \times \{1\}$, and choose a collar $\kappa : L \times [0,1] \hookrightarrow \mathcal{M}$ satisfying $\kappa(x,0) = x$. There is a homeomorphism

$$\Phi : \mathcal{C} \setminus \{c_0\} \longrightarrow \operatorname{int} \mathcal{M}$$

given on the punctured cone, the inner collar, and the remaining core by

$$\Phi([x,t]) = \kappa(x, t/e) \quad (0 < t \le 1), \qquad \Phi(\kappa(x,s)) = \kappa(x, e^{s-1}) \quad (0 \le s \le 1),$$

$$\Phi(q) = q \quad \text{for} \quad q \in \mathcal{M} \setminus \kappa(L \times [0,1)).$$

The first two formulas agree at the glued points $[x,1] = \kappa(x,0)$, and the second and third agree at $\kappa(x,1)$. Their collar-coordinate images are respectively $L \times (0, e^{-1}]$ and $L \times [e^{-1}, 1]$; hence the displayed piecewise map and its piecewise inverse are continuous.

The compact Hausdorff spaces $\mathcal{C}$ and $\mathcal{M}/L$ are both one-point compactifications of $\operatorname{int}\mathcal{M}$: for $\mathcal{C}$ use Φ, and for $\mathcal{M}/L$ use the quotient map on $\mathcal{M} \setminus L = \operatorname{int}\mathcal{M}$. Uniqueness of the one-point compactification therefore extends Φ to a based homeomorphism

$$(\mathcal{C}, c_0) \xrightarrow{\cong} (\mathcal{M}/\partial\mathcal{M}, [\partial\mathcal{M}]). \tag{231}$$

Only after establishing (231) do we apply the quotient identity at the start of the proof. It gives

$$\mathcal{C}^{\wedge 4} \cong \frac{\mathcal{M}^4}{\bigcup_{a=1}^{4} \mathcal{M}^{a-1} \times \partial\mathcal{M} \times \mathcal{M}^{4-a}} = \mathcal{M}^4/\partial(\mathcal{M}^4),$$

and carries the smash base point to the collapsed boundary.

Put $N = \mathcal{M}^4$. Rounding its product corners preserves the homeomorphism type of the pair $(N, \partial N)$ and makes ∂N an eleven-dimensional closed manifold. A boundary collar descends in $N/\partial N$ to a cone neighborhood of the collapsed point, so the local link is exactly ∂N and

$$H_i(N/\partial N, (N/\partial N) \setminus \{*\}; \mathbb{Z}) \cong \widetilde{H}_{i-1}(\partial N; \mathbb{Z}). \tag{232}$$

The integral homology calculation in the proof of theorem 9.14 for the specific compact orientable three-manifold $\mathcal{M}$ gives

$$H_j(\mathcal{M};\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & j = 0, 1, \\ 0, & \text{otherwise.} \end{cases}$$

All groups are free, so the integral Kunneth theorem gives

$$H_j(N;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{j}}, & 0 \le j \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

Poincare–Lefschetz duality, with the product orientation retained, gives

$$H_j(N, \partial N;\mathbb{Z}) \cong H^{12-j}(N;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{j-8}}, & 8 \le j \le 12, \\ 0, & \text{otherwise.} \end{cases}$$

The long exact sequence of $(N, \partial N)$ now has disjoint absolute and relative degree ranges. It gives

$$\widetilde{H}_j(\partial N;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}^{\binom{4}{j}}, & 1 \le j \le 4, \\ 0, & j = 5, 6, \\ \mathbb{Z}^{\binom{4}{j-7}}, & 7 \le j \le 11, \\ 0, & \text{otherwise.} \end{cases}$$

Substitution in (232) proves (230).

An exact metric cone is invariant under every radial blow-up at its vertex, and its Hausdorff dimension is one plus that of its fixed piecewise-Riemannian link. For the first target,

$$H_8\big(\mathrm{Cone}(T^3 * T^3), T^3 * T^3;\mathbb{Z}\big) \cong \mathbb{Z};$$

the analogous top local group for the $\mathcal{B}$ target is $\mathbb{Z}$ in degree 12, while the full $\mathcal{C}$ calculation is (230). Euclidean $\mathbb{R}^4$ has local homology $\mathbb{Z}$ only in degree 4, so none of these conic vertex models is $\mathbb{R}^4$. The diameter calculation proves the final assertion. □

Consequences for the three sphere routes. The exact positive statement is that one fixed metric on each compact codomain contains the native $\mathbb{R}^4$ decompactification as a vertical four-plane in a moving-point Euclidean bubble. The whole bubbles are $\mathbb{R}^8$ for the S^{12} route and $\mathbb{R}^{12}$ for the S^{24} routes. The factor $\beta_T = \lambda_T r_T$ proves that this dimension discrepancy cannot be removed by a different scalar rescaling of the same fixed cusp-adapted metric: the slice unrolls only when the smash base point has already gone to infinite scaled distance.

No metric on a source sphere is supplied by these quotient constructions. If an independent smooth Riemannian metric is placed on S^{12} or S^{24}, its ordinary tangent spaces are $\mathbb{R}^{12}$ or $\mathbb{R}^{24}$. Pulling a codomain metric back through $q_2^{\wedge 2}$ or $q_1^{\wedge 4}$, on the regular strata where the maps are differentiable, gives a degenerate quadratic form along the positive-dimensional quotient kernels, not a Riemannian source metric.

The moving four-sphere carriers

$$\mathbb{S}^4_{12,T} = S^2_T \wedge S^2_T \subset S^6, \qquad \mathbb{S}^4_{24,T} = C_T^{\wedge 4} \subset S^8$$

of theorem 9.20 carry separately assigned metrics. Namely, S^2_T first receives the quotient length metric obtained from $(\overline{T}^2, h_T) \cong (Q_T, g_{Q_T})$ by collapsing $K^{(1)} \subset \overline{T}^2$, equivalently by collapsing $\iota_T(K^{(1)}) \subset Q_T$. Then $\mathbb{S}^4_{12,T}$ receives the quotient length metric of the Riemannian ℓ^2-product $(S^2_T)^2$ after its fat wedge is collapsed. Likewise, $\mathbb{S}^4_{24,T}$ receives the quotient length metric of the Riemannian ℓ^2-product C^4_T after its

fat wedge is collapsed. These metrics are assigned directly to the moving carriers; they are unrescaled intrinsic quotient metrics, not restrictions of D_{12} or $D_{24}^{\mathcal{B}}$, whose domains are the conical targets, and no fixed ambient metric on S^6 or S^8 is used to define them. Their Euclidean balls are based at the midpoint classes

$$\begin{aligned} y_* &= [(1/2, 1/2)] \in \overline{T}^2, \qquad z_T = c(y_*), \\ \zeta_{12,T} &= [z_T, z_T], \\ \zeta_{24,T} &= [(\ell_T/2, \ell_T/2, \ell_T/2, \ell_T/2)]. \end{aligned}$$

not at the images of p_T^{12} or $p_T^{24,\mathcal{B}}$, which belong to the subsequently collapsed loci. In particular, the trichotomy in theorem 9.36 is the trichotomy of the uncollapsed intrinsic flat-torus slices; it is not a finite-β_T description of the singular quotient-metric moving four-spheres. Indeed, the restriction of any one compatible metric on the compact sphere S^6 or S^8 has diameter bounded by the diameter of that sphere. By contrast, in the separately assigned moving metrics the midpoint distances to the collapsed loci are at least $(T - C)/2$ and exactly $\ell_T/2$, respectively, and hence tend to infinity. Thus the assigned unrescaled metrics cannot themselves be restrictions of one fixed ambient metric. This does not preclude a separately proved realization after a scalar blow-up of another fixed quotient metric.

Finally, the rank obstruction for a full-base torus quotient must retain its actual metric hypothesis: *quotient systole tends to infinity* means the quotient norm induced by the unrescaled Euclidean fibre metric. Without that qualification the assertion is false, since an arbitrary T-dependent scalar multiplication can make the systole of any fixed nonzero quotient tend to infinity.

9.5 The exceptional first-coordinate elliptic quotient

The native real quotient in equation (69) exists throughout the radial cusp. A holomorphic quotient obtained from the first complex coordinate is a different and much more restrictive construction. We determine its locus without replacing it by an ambient coefficient-space path.

For $z \in \mathfrak{h}_z$, write

$$\begin{aligned} v_{\gamma,z} &= (6\mu(z), \beta(z))^{\mathsf{t}}, \quad v_{u,z} = (\tau(z), \mu(z))^{\mathsf{t}}, \\ v_{w,z} &= (1, 0)^{\mathsf{t}}, \qquad\qquad v_{\delta,z} = (0, 1)^{\mathsf{t}}, \end{aligned} \tag{233}$$

and put

$$\mathcal{A}_z := \mathbb{C}^2/\mathbb{Z}\langle v_{\gamma,z}, v_{u,z}, v_{w,z}, v_{\delta,z}\rangle, \qquad E_{\tau(z)} := \mathbb{C}/(\mathbb{Z}\tau(z) + \mathbb{Z}). \tag{234}$$

These are the marked fibres of theorem 8.2 and the elliptic curves selected by the same period τ.

Proposition 9.39 (Exact first-coordinate descent and connected kernel)**.** *The linear map* $\mathrm{pr}_1 : \mathbb{C}^2 \to \mathbb{C}$ *induces a surjective holomorphic homomorphism*

$$\overline{\mathrm{pr}}_{1,z} : \mathcal{A}_z \longrightarrow E_{\tau(z)} \tag{235}$$

if and only if there is a necessarily unique pair $(a, b) \in \mathbb{Z}^2$ *such that*

$$6\mu(z) = a\tau(z) + b. \tag{236}$$

On this locus define

$$\tau_f(z) := a\mu(z) - \beta(z). \tag{237}$$

Then

$$\operatorname{Im} \tau_f(z) = -D_{\mathrm{can}}(z) > 0, \qquad D_{\mathrm{can}}(z) := \operatorname{Im} \beta(z) - \frac{6(\operatorname{Im} \mu(z))^2}{\operatorname{Im} \tau(z)}, \tag{238}$$

and the kernel in equation (235) *is connected and isomorphic to*

$$E_{\tau_f(z)} = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau_f(z)). \tag{239}$$

Proof. The four first coordinates in equation (233) are $6\mu(z), \tau(z), 1, 0$. Thus the linear projection descends precisely when $6\mu(z) \in \mathbb{Z}\tau(z) + \mathbb{Z}$, which is equation (236). The pair is unique because $\operatorname{Im}\tau(z) > 0$. The periods $v_{u,z}$ and $v_{w,z}$ map onto the two generators of the target lattice, so the induced map is surjective.

Replace the marked lattice basis by the unimodular basis whose first vector is

$$v_{\gamma,z} - a v_{u,z} - b v_{w,z} = (0, \beta(z) - a\mu(z))^{\mathsf{t}} = (0, -\tau_f(z))^{\mathsf{t}}. \tag{240}$$

Together with $v_{\delta,z} = (0,1)^{\mathsf{t}}$, these are exactly the vertical periods. Indeed, after subtracting integral multiples of $v_{u,z}$ and $v_{w,z}$, every point in the kernel has first coordinate zero; and an integral combination of $v_{u,z}, v_{w,z}$ has first coordinate zero only when both coefficients vanish. The kernel is therefore the connected one-dimensional complex torus in equation (239).

Taking imaginary parts in equation (236) gives $6\operatorname{Im}\mu = a\operatorname{Im}\tau$. Hence

$$\operatorname{Im}\tau_f = a\operatorname{Im}\mu - \operatorname{Im}\beta = \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} - \operatorname{Im}\beta = -D_{\mathrm{can}} > 0$$

by theorem 8.2. □

For $(a,b) \in \mathbb{Z}^2$, define the marked analytic set

$$\mathcal{D}_{a,b} := \{z \in \mathfrak{h}_z : 6\mu(z) - a\tau(z) - b = 0\}, \qquad \mathcal{D} := \bigcup_{(a,b)\in\mathbb{Z}^2} \mathcal{D}_{a,b}. \tag{241}$$

Theorem 9.40 (Monodromy, cusp exclusion, and finiteness of the descent locus)**.** *The transformation laws of theorem 8.2 act on the integer labels in equation* (241) *by*

$$g_0 : (a,b) \longmapsto (a, a+b), \qquad g_2 : (a,b) \longmapsto (-b, 6+a). \tag{242}$$

Consequently $\mathcal{D}$ is invariant under the triangle group and has a well-defined image in the completed orbifold base $B \simeq \mathbb{P}^1$. That image is finite and is disjoint from some punctured neighborhood of the distinguished cusp p_0.

Proof. Suppose $z \in \mathcal{D}_{a,b}$. From $\tau \circ g_0 = \tau - 1$ and $\mu \circ g_0 = \mu$,

$$6\mu(g_0 z) = a(\tau(g_0 z) + 1) + b = a\tau(g_0 z) + (a+b),$$

so $g_0 z \in \mathcal{D}_{a,a+b}$. Similarly, $\tau \circ g_2 = -1/\tau$ and $\mu \circ g_2 = 1 + \mu/\tau$ give

$$6\mu(g_2 z) = 6 + a + \frac{b}{\tau(z)} = -b\tau(g_2 z) + (6+a),$$

so $g_2 z \in \mathcal{D}_{-b,6+a}$. Both label maps are bijections; and g_0, g_2 generate the triangle group because $g_0 = (g_1 g_2)^{-1}$. This proves invariance.

We next exclude the cusp with the integer labels allowed to vary. If no punctured cusp neighborhood were disjoint from $\mathcal{D}$, there would be points $z_j \in \mathcal{D}_{a_j,b_j}$ in the distinguished cusp component with $t_c(z_j) \to 0$. Put $T_j = \operatorname{Im}\tau(z_j)$ and $m_j = \operatorname{Im}\mu(z_j)$. The cusp construction in theorem 8.2 and the formulas preceding equation (13) give $T_j \to \infty$ while m_j stays bounded. Taking imaginary parts yields

$$a_j = \frac{6m_j}{T_j} \longrightarrow 0. \tag{243}$$

Because a_j is integral, $a_j = 0$ for all sufficiently large j. Then $6\mu(z_j) = b_j$. The cusp extension of μ is bounded, so the integers b_j are bounded; after taking a subsequence, they equal one integer b. The holomorphic function $6\mu - b$ extends across $t_c = 0$ and has zeros accumulating there. It therefore vanishes on the cusp disc and, by the identity theorem, on the connected surface $\mathfrak{h}_z$.

More generally, no identity $6\mu = a\tau + b$ with fixed integers a, b can hold on $\mathfrak{h}_z$. Applying g_0 to such an identity and subtracting it from the original one gives $a = 0$. Then $\mu = b/6$ is constant, whereas the g_2-law would give

$$\frac{b}{6} = 1 + \frac{b}{6\tau}. \tag{244}$$

This is impossible for the nonconstant map τ. The preceding conclusion $6\mu - b \equiv 0$ is therefore a contradiction, proving cusp exclusion.

Choose a closed fundamental polygon for the triangle group and remove the excluded cusp neighborhood. Its closure K is compact in $\mathfrak{h}_z$, including the two elliptic fixed points. The functions τ, μ are holomorphic there and $\inf_K \operatorname{Im}\tau > 0$. At a point of $\mathcal{D}_{a,b} \cap K$,

$$a = \frac{6\operatorname{Im}\mu}{\operatorname{Im}\tau}, \qquad b = 6\mu - a\tau. \tag{245}$$

Thus only finitely many integer pairs (a, b) occur on K. For each of them, $6\mu - a\tau - b$ is a nonzero holomorphic function on a neighborhood of the compact set K, so its zero set meets K in finitely many points. The $(3, 4, \infty)$ triangle group has one cusp orbit. Hence cusp exclusion holds at every lift of the chosen cusp neighborhood. The set $\mathcal{D}$ is triangle-group invariant by the already proved label permutation, and it is disjoint from all those lifted cusp neighborhoods. Consequently every triangle-group orbit contained in $\mathcal{D}$ meets K. Hence the image of $\mathcal{D}$ in the completed orbifold base is finite. □

Remark 9.41 (Exact scope of the elliptic classification). The result classifies only the quotient obtained from the specified first coordinate and the specified target lattice $\mathbb{Z}\tau + \mathbb{Z}$. Its cusp exclusion does not prove that $\mathcal{A}_z$ has no other elliptic quotient or elliptic subtorus, that its Néron–Severi group vanishes, or that it carries no smooth integral magnetic line bundle. Those are different typed questions. In particular, the first-coordinate elliptic construction cannot supply the gap-closing cusp sequence, whereas the native scalar and classical gauge soft modes proved above do not use this exceptional locus.

Theorem 9.42 (Base-changed cover monodromy and exact momentum sectors)**.** *In the ordered integral basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$, put*

$$\mathsf{S}_M := \operatorname{diag}(1, 1, M, M), \qquad \Lambda^{(M)} := \mathsf{S}_M \mathbb{Z}^4. \tag{246}$$

Then $P_T \Lambda^{(M)} = \Lambda_{T,M}$, and the cover in equation (53) *is induced by the lattice inclusion*

$$P_T \mathsf{S}_M \mathbb{Z}^4 \subset P_T \mathbb{Z}^4.$$

Writing $M_0 = I + R$, one has

$$R^2 = 0, \qquad \ker R = \operatorname{im} R = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \tag{247}$$

and

$$M_0^M = I + MR, \qquad \mathsf{S}_M^{-1} M_0^M \mathsf{S}_M = M_0. \tag{248}$$

Thus, after the cyclic base change $t_c = v^M$, the degree-M^2 fibre covers form a regular relative cover over $v \neq 0$. This particular statement proves only punctured descent. The distinct completed local extension, with the required modified source action and the separate degree-M^3 and degree-M^2 maps, is proved in theorem 9.8.

Equip $\widetilde{F}_{T,M}$ with the pullback Euclidean metric. Its Riemannian volume is $M^2 D_T$, so normalized Haar measure is exactly

$$\mathrm{d}\widetilde{\nu}_{T,M} := (M^2 D_T)^{-1} \mathrm{d}\operatorname{vol}_{\widetilde{F}_{T,M}}.$$

Let $\widetilde{\Delta}_{T,M}$ be the nonnegative self-adjoint scalar Laplacian on $L^2(\widetilde{F}_{T,M}, \mathrm{d}\widetilde{\nu}_{T,M})$ obtained from the closed Dirichlet form

$$\widetilde{\mathfrak{q}}_{T,M}[f] := \int_{\widetilde{F}_{T,M}} |\mathrm{d}f|^2 \, \mathrm{d}\widetilde{\nu}_{T,M}, \qquad \operatorname{Dom}(\widetilde{\mathfrak{q}}_{T,M}) = H^1(\widetilde{F}_{T,M}),$$

with operator domain $\mathrm{Dom}(\widetilde{\Delta}_{T,M}) = H^2(\widetilde{F}_{T,M})$. *Its complete orthonormal character basis is*

$$\widetilde{\chi}_{T,M,k}([x]) = \exp\Big(2\pi i \left\langle P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} k, x \right\rangle\Big), \qquad k \in \mathbb{Z}^4, \tag{249}$$

and

$$\mathrm{Spec}(\widetilde{\Delta}_{T,M}) = \left\{ 4\pi^2 \left| P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} k \right|^2 : k \in \mathbb{Z}^4 \right\} \tag{250}$$

with multiplicity. For this covering map one has

$$c_{T,M}^* \chi_{T,\ell} = \widetilde{\chi}_{T,M,\mathsf{S}_M \ell}. \tag{251}$$

Consequently the pullback, or equivalently deck-invariant, sector consists exactly of indices

$$k \in \mathsf{S}_M \mathbb{Z}^4 \quad \Longleftrightarrow \quad k_3 \equiv k_4 \equiv 0 \pmod{M}. \tag{252}$$

There is a constant $C_{\mathrm{mom}} > 0$*, independent of sufficiently large* T *and of* $M \geq 1$*, such that the Euclidean covering radius of the cover momentum lattice satisfies*

$$\sup_{p \in \mathbb{R}^4} \mathrm{dist}\Big(p, P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} \mathbb{Z}^4\Big) \leq C_{\mathrm{mom}} \left(\frac{1}{T} + \frac{1}{M} \right). \tag{253}$$

In particular, if $T_j, M_j \to \infty$*, the cover momentum lattices become locally dense in all of* $\mathbb{R}^4$*, and every* $E \geq 0$ *is a limit of scalar eigenvalues in equation* (250)*.*

Proof. The equality $P_T \Lambda^{(M)} = \Lambda_{T,M}$ follows by multiplying the last two period columns e_1, e_3 by M. The matrix in theorem 5.1 gives

$$R(a,b,c,d)^{\mathsf{t}} = (0,0,b,-a)^{\mathsf{t}}.$$

This proves equation (247) and $R^2 = 0$. The binomial theorem then gives $M_0^M = I + MR$. For every $(a,b,c,d)^{\mathsf{t}}$,

$$\mathsf{S}_M^{-1} M_0^M \mathsf{S}_M (a,b,c,d)^{\mathsf{t}} = (a, b, c+b, d-a)^{\mathsf{t}} = M_0 (a,b,c,d)^{\mathsf{t}},$$

which proves equation (248). In particular $M_0^M \Lambda^{(M)} = \Lambda^{(M)}$, which is precisely the descent condition after $t_c = v^M$.

The dual of $P_T \mathsf{S}_M \mathbb{Z}^4$ is

$$(P_T \mathsf{S}_M)^{-\mathsf{T}} \mathbb{Z}^4 = P_T^{-\mathsf{T}} \mathsf{S}_M^{-1} \mathbb{Z}^4.$$

The standard Fourier theorem for a flat torus proves equations (249) and (250). Substitution of $k = \mathsf{S}_M \ell$ proves equation (251). A character on the cover descends through $c_{T,M}$ exactly when it is invariant under the deck group, and the displayed equality shows that this occurs exactly for $k \in \mathsf{S}_M \mathbb{Z}^4$, proving equation (252).

It remains to prove the quantitative density claim. Let $b_{j,T,M}$ be the four columns of $P_T^{-\mathsf{T}} \mathsf{S}_M^{-1}$. The exact matrix equation (15), the boundedness of $\alpha_T, \eta_T, c_T, m_T$, and

$$L_T, q_T = O(T), \qquad D_T \asymp T^2$$

give, after increasing a uniform constant C_0,

$$|b_{1,T,M}| + |b_{2,T,M}| \leq \frac{C_0}{T}, \qquad |b_{3,T,M}| + |b_{4,T,M}| \leq \frac{C_0}{M}.$$

Given $p \in \mathbb{R}^4$, express it uniquely as $p = \sum_j t_j b_{j,T,M}$, and choose $n_j \in \mathbb{Z}$ with $|t_j - n_j| \leq \frac{1}{2}$. Then

$$\left| p - \sum_j n_j b_{j,T,M} \right| \leq \frac{1}{2} \sum_j |b_{j,T,M}| \leq \frac{C_0}{2} \left(\frac{1}{T} + \frac{1}{M} \right).$$

This proves equation (253). For $E > 0$, apply it to any p with $4\pi^2 |p|^2 = E$; continuity gives eigenvalues tending to E. For $E = 0$, use $k = (1,0,0,0)$, whose momentum norm is $O(T^{-1})$. □

Scope and residual data. The conclusions above use the explicitly declared Euclidean quotient metric. The canonical period theorem proves the lattice family but does not itself specify a Riemannian metric, so no assertion about a different metric follows without an explicit comparison.

The constants H and B determine the first relative T^{-1}-correction. When $B = 0$, they do not by themselves determine which of the two unit pairs is strictly shortest at the next order. The canonical functions do determine it through the exact expression $B_T(2L_T - B_T) - 35m_T^2$ in equation (24); if both $B = 0$ and $m = 0$, its sign can depend on higher Taylor coefficients of b and μ. This is an indeterminacy of the two requested zeroth-order constants, not a missing proof obligation. The proved construction permits an additive constant in β and chooses its imaginary part only by the qualitative requirement that the canonical determinant be negative. It records no numerical additive normalization. Accordingly, the formulas retain the resulting value $B = \operatorname{Im} b(0)$ and do not silently impose a sign on it.

Finally, the degree-M^2 covers in equation (53) are local in the marked radial family, exactly as stated. The sublattice $\mathbb{Z}\langle\widehat{\gamma}, \widehat{u}, M\widehat{w}, M\widehat{\delta}\rangle$ is not invariant under the full cusp monodromy when $M > 1$: the proved monodromy sends $\widehat{\gamma}$ to $\widehat{\gamma} - \widehat{\delta}$ and $\widehat{u}$ to $\widehat{u} + \widehat{w}$. Therefore these particular degree-M^2 covers do not, without additional construction, descend around the entire punctured cusp. Its M-th power does preserve the sublattice, because it adds $-M\widehat{\delta}$ to $\widehat{\gamma}$ and $M\widehat{w}$ to $\widehat{u}$. Equivalently, $M_0^M = I + M(M_0 - I)$. Thus, after the degree-M cyclic base change $t_c = v^M$, the cover forms a regular relative degree-M^2 cover over $v \neq 0$. The pointed-limit proof itself uses no central-fibre extension. Nevertheless, theorem 9.8 proves that a corrected source action extends this cover through $v = 0$ as a finite morphism of completed *local* cusp families. It remains true that no cover over the entire original base, and no cover of the compact threefold X, is used or constructed here.

10 Classical gauge fluctuation operators on the positive radial cusp

Retain the positive radial flat tori $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$, their Euclidean quotient metrics, and the real quantities $D_T, L_T, q_T, m_T, B_T, H, B$ from theorems 9.1 and 9.2. Thus

$$D_T = L_T q_T + 6m_T^2 > 0, \qquad L_T = \operatorname{Im}\tau_T, \qquad q_T = L_T - B_T = -\operatorname{Im}\beta_T,$$

and D_T is the positive covolume. Let

$$\mathbf{e}_\gamma = (1,0,0,0)^{\mathsf{t}}, \qquad \mathbf{e}_u = (0,1,0,0)^{\mathsf{t}};$$

then the already-proved dual covectors are

$$\ell_{\gamma,T} = P_T^{-\mathsf{T}}\mathbf{e}_\gamma, \qquad \ell_{u,T} = P_T^{-\mathsf{T}}\mathbf{e}_u,$$

with exact squared lengths

$$a_{\gamma,T} := \|\ell_{\gamma,T}\|^2 = \frac{L_T^2 + m_T^2}{D_T^2}, \qquad a_{u,T} := \|\ell_{u,T}\|^2 = \frac{q_T^2 + 36m_T^2}{D_T^2}. \tag{254}$$

For all sufficiently large T, the shortest nonzero dual vectors are precisely the minimizing members of $\{\pm\ell_{\gamma,T}, \pm\ell_{u,T}\}$. Put

$$G_T = P_T^{\mathsf{T}}P_T. \tag{255}$$

The matrix G_T is the tangent Gram matrix in the lattice coordinates $y \in \mathbb{R}^4/\mathbb{Z}^4$, where $x = P_T y$.

10.1 The continuum Yang–Mills Hessian

Let G be a compact Lie group with real Lie algebra $\mathfrak{g}$. Fix a positive-definite $\mathrm{Ad}(G)$-invariant inner product on $\mathfrak{g}$, put

$$d_G = \dim_{\mathbb{R}}\mathfrak{g}, \qquad \mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C},$$

and fix a coupling $g_{\mathrm{YM}} > 0$. On the trivial principal bundle

$$\mathcal{P}_T = F_T \times G \longrightarrow F_T$$

let A_0 be the product connection. The Yang–Mills functional is normalized by

$$\mathrm{YM}_T(A) = \frac{1}{2g_{\mathrm{YM}}^2} \int_{F_T} |F_A|^2 \, \mathrm{d}\,\mathrm{vol}_{F_T} . \tag{256}$$

Proposition 10.1 (Hessian at the product connection)**.** *For smooth $a, b \in \Omega^1(F_T; \mathfrak{g})$,*

$$\mathrm{Hess}_{A_0} \mathrm{YM}_T(a, b) = \frac{1}{g_{\mathrm{YM}}^2} \langle \mathrm{d}a, \mathrm{d}b \rangle_{L^2}. \tag{257}$$

Relative to the unscaled L^2 inner product on one-forms, the associated closed Hessian form is

$$\mathfrak{l}_T[a] = g_{\mathrm{YM}}^{-2} \|\mathrm{d}_1 a\|_{L^2}^2, \qquad \mathrm{Dom}(\mathfrak{l}_T) = \mathrm{Dom}(\mathrm{d}_1), \tag{258}$$

on $L^2\Omega^1(F_T; \mathfrak{g})$, where d_1 is the maximal closed exterior derivative

$$\mathrm{Dom}(\mathrm{d}_1) = \{a \in L^2\Omega^1(F_T; \mathfrak{g}) : \mathrm{d}a \in L^2\Omega^2(F_T; \mathfrak{g}) \text{ distributionally}\}.$$

The associated nonnegative self-adjoint operator is

$$\mathcal{L}_T = g_{\mathrm{YM}}^{-2} \mathrm{d}_1^* \mathrm{d}_1, \qquad \mathrm{Dom}(\mathcal{L}_T) = \{a \in \mathrm{Dom}(\mathrm{d}_1) : \mathrm{d}_1 a \in \mathrm{Dom}(\mathrm{d}_1^*)\}. \tag{259}$$

Its differential expression on smooth one-forms is $g_{\mathrm{YM}}^{-2} \mathrm{d}^* \mathrm{d}$.

Proof. With the convention $F_{A_0+a} = \mathrm{d}a + \frac{1}{2}[a \wedge a]$, one has

$$F_{A_0+sa+tb} = s\,\mathrm{d}a + t\,\mathrm{d}b + O((|s| + |t|)^2).$$

The coefficient of st in the squared norm is $2\langle \mathrm{d}a, \mathrm{d}b \rangle_{L^2}$. Taking the mixed derivative in equation (256) proves equation (257). The maximal distributional exterior derivative is closed and densely defined. Hence its graph-norm square is a closed nonnegative form, and the first representation theorem identifies its operator with $g_{\mathrm{YM}}^{-2} \mathrm{d}_1^* \mathrm{d}_1$ on precisely the domain displayed in equation (259). □

10.2 The finite cubical cochain complex

Fix $N \geq 2$ and put

$$\Lambda_N = (\mathbb{Z}/N\mathbb{Z})^4.$$

The vertex $n \in \Lambda_N$ is realized in F_T as the class of $P_T(n/N)$. Let $E = (\mathbb{R}^4)^*$, with its Euclidean inner product, and let $\mathbf{e}_1, \ldots, \mathbf{e}_4$ be the standard coordinate columns in the lattice-coordinate space, also used as the coefficient columns of its dual coordinate basis. Define the physical coframe

$$\vartheta_T^j = P_T^{-\mathsf{T}} \mathbf{e}_j \in E, \qquad 1 \leq j \leq 4. \tag{260}$$

Thus $\vartheta_T^j = \mathrm{d}y^j$ in physical x-coordinates, and the Gram matrix of this coframe is G_T^{-1}.

For $0 \leq p \leq 4$, define

$$C_N^p(T; \mathfrak{g}_{\mathbb{C}}) = \mathrm{Map}\left(\Lambda_N, \bigwedge^p E_{\mathbb{C}} \otimes \mathfrak{g}_{\mathbb{C}}\right). \tag{261}$$

These spaces are finite-dimensional. Every finite differential, adjoint, Hodge operator, projection, and reduced Hessian below therefore acts on its entire displayed finite-dimensional source space; no additional

operator domain is implicit. The coefficient of $\vartheta_T^{i_1}\wedge\cdots\wedge\vartheta_T^{i_p}$ at n is the coefficient assigned to the positively oriented coordinate p-cell based at n. Let

$$(\mathsf{T}_j a)(n) = a(n+\mathbf{e}_j)$$

and define the forward and backward differences by

$$D^{+}_{N,j} = N(\mathsf{T}_j - \mathrm{id}), \qquad (D^{-}_{N,j}a)(n) = N\big(a(n) - a(n-\mathbf{e}_j)\big). \tag{262}$$

The finite exterior derivative is

$$\mathrm{d}_N a = \sum_{j=1}^{4} \varepsilon(\vartheta_T^j) D^{+}_{N,j} a. \tag{263}$$

Lemma 10.2 (Cochain property and induced exterior metric)**.** *The maps in*

$$0 \longrightarrow C_N^0 \xrightarrow{\mathrm{d}_N} C_N^1 \xrightarrow{\mathrm{d}_N} C_N^2 \xrightarrow{\mathrm{d}_N} C_N^3 \xrightarrow{\mathrm{d}_N} C_N^4 \longrightarrow 0 \tag{264}$$

form a cochain complex. Equip C_N^p with

$$\langle a, b\rangle_{N,p,T} = \frac{D_T}{N^4} \sum_{n\in\Lambda_N} \langle a(n), b(n)\rangle_{\wedge^p E\otimes\mathfrak{g}_{\mathbb{C}}}. \tag{265}$$

For increasing multi-indices $I=(i_1,\dots,i_p)$ and $J=(j_1,\dots,j_p)$, the exterior metric is exactly

$$\Big\langle \vartheta_T^{i_1}\wedge\cdots\wedge\vartheta_T^{i_p}, \vartheta_T^{j_1}\wedge\cdots\wedge\vartheta_T^{j_p}\Big\rangle = \det\Big((G_T^{-1})_{i_a j_b}\Big)_{a,b=1}^{p}. \tag{266}$$

Proof. The differences $D^{+}_{N,j}$ commute, whereas the exterior multiplications $\varepsilon(\vartheta_T^j)$ anticommute. Therefore

$$\mathrm{d}_N^2 = \sum_{j,\ell} \varepsilon(\vartheta_T^j)\varepsilon(\vartheta_T^\ell) D^{+}_{N,j} D^{+}_{N,\ell} = 0,$$

which proves that equation (264) is a complex. The factor D_T/N^4 in equation (265) is the exact Riemann-sum weight: D_T is the covolume of $P_T\mathbb{Z}^4$. The Euclidean inner product of $\vartheta_T^i = P_T^{-\mathsf{T}}\mathbf{e}_i$ and $\vartheta_T^j = P_T^{-\mathsf{T}}\mathbf{e}_j$ is $(G_T^{-1})_{ij}$. The induced inner product on an exterior power is the determinant of the matrix of pairings, proving equation (266). □

Remark 10.3 (Integrated cellular cochains give the same operator). The convention above stores form coefficients. If $c_I(n)$ instead stores the integral over a p-cell, put $a_I(n) = N^p c_I(n)$. The unscaled cellular coboundary $\mathsf{T}_j - \mathrm{id}$, together with the inner product

$$\langle c, c'\rangle^{\mathrm{int}}_{N,p,T} = D_T N^{2p-4} \sum_{n\in\Lambda_N} \langle c(n), c'(n)\rangle_{\wedge^p E\otimes\mathfrak{g}_{\mathbb{C}}},$$

is carried isometrically to equations (263) and (265). Thus the two conventions give identical Hessian symbols and eigenvalues.

Proposition 10.4 (Adjoint and Fourier wedge–contraction symbols)**.** *The adjoint of $\mathrm{d}_N : C_N^p \to C_N^{p+1}$ is*

$$\mathrm{d}_N^{*} = -\sum_{j=1}^{4} \iota_{(\vartheta_T^j)^\sharp} D^{-}_{N,j}. \tag{267}$$

For $k \in \Lambda_N$, define

$$\chi_{N,k}(n) = \exp(2\pi i k \cdot n/N),$$
$$\mathfrak{d}_N(k)_j = N\left(e^{2\pi i k_j/N} - 1\right), \tag{268}$$
$$\zeta_{N,T}(k) = \sum_{j=1}^{4} \mathfrak{d}_N(k)_j \vartheta_T^j = P_T^{-\mathsf{T}} \mathfrak{d}_N(k) \in E_{\mathbb{C}}. \tag{269}$$

For every constant exterior amplitude α,

$$\mathrm{d}_N(\chi_{N,k}\alpha) = \chi_{N,k}\varepsilon(\zeta_{N,T}(k))\alpha, \tag{270}$$
$$\mathrm{d}_N^*(\chi_{N,k}\alpha) = \chi_{N,k}\iota_{\overline{\zeta_{N,T}(k)}^\sharp}\alpha, \tag{271}$$
$$(\mathrm{d}_N\mathrm{d}_N^* + \mathrm{d}_N^*\mathrm{d}_N)(\chi_{N,k}\alpha) = |\zeta_{N,T}(k)|^2\chi_{N,k}\alpha. \tag{272}$$

Consequently the harmonic p-cochains are exactly the constant cochains and have complex dimension $\binom{4}{p}d_G$.

Proof. Translation is unitary for equation (265); hence

$$(D_{N,j}^+)^* = N(\mathsf{T}_j^{-1} - \mathrm{id}) = -D_{N,j}^-.$$

For the induced exterior metric, the adjoint of exterior multiplication by ϑ_T^j is contraction by $(\vartheta_T^j)^\sharp$. This proves equation (267). On $\chi_{N,k}$, the forward difference has eigenvalue $\mathfrak{d}_N(k)_j$, whereas $-D_{N,j}^-$ has eigenvalue $\overline{\mathfrak{d}_N(k)_j}$. Summing proves equations (270) and (271). The Hermitian exterior-algebra identity

$$\iota_{\overline{\zeta}^\sharp}\varepsilon(\zeta) + \varepsilon(\zeta)\iota_{\overline{\zeta}^\sharp} = |\zeta|^2\,\mathrm{id}$$

proves equation (272). Finally, $\mathfrak{d}_N(k) = 0$ exactly when $k = 0$ in Λ_N, and P_T is invertible. Thus the Hodge symbol vanishes exactly on the zero Fourier mode, where the entire constant exterior amplitude is harmonic. □

Define the finite reduced Coulomb slice by

$$\mathscr{C}_{N,T} = \ker(\mathrm{d}_N^* : C_N^1 \to C_N^0) \cap (\ker \Delta_{N,T}^{(1)})^\perp, \qquad \Delta_{N,T}^{(1)} = \mathrm{d}_N\mathrm{d}_N^* + \mathrm{d}_N^*\mathrm{d}_N. \tag{273}$$

Equivalently, one deletes the zero Fourier mode and, for each $k \neq 0$, restricts the amplitude to

$$\mathcal{T}_{N,T,k} = \ker \iota_{\overline{\zeta_{N,T}(k)}^\sharp} \subset E_{\mathbb{C}}. \tag{274}$$

The exact orthogonal projection is

$$\Pi_{N,T,k}\alpha = \alpha - \zeta_{N,T}(k)\frac{\iota_{\overline{\zeta_{N,T}(k)}^\sharp}\alpha}{|\zeta_{N,T}(k)|^2}. \tag{275}$$

Thus the projection uses the physical covector $P_T^{-\mathsf{T}}\mathfrak{d}_N(k)$, including every skew-metric cross term.

The finite quadratic gauge form and its associated operator are

$$Q_{N,T}(a) = \frac{1}{2g_{\mathrm{YM}}^2}\|\mathrm{d}_N a\|_{N,2,T}^2, \qquad \mathcal{L}_{N,T} = g_{\mathrm{YM}}^{-2}\mathrm{d}_N^*\mathrm{d}_N. \tag{276}$$

This is the exact linearized, Lie-algebra-valued cubical discretization of equation (257); it is not a nonlinear compact-link Wilson action.

Corollary 10.5 (Exact finite Coulomb spectrum)**.** *On the $k \neq 0$ Fourier fibre of $\mathscr{C}_{N,T}$, $\mathcal{L}_{N,T}$ has the single eigenvalue*

$$\lambda^{\mathrm{YM}}_{N,T}(k) = \frac{1}{g^2_{\mathrm{YM}}} |P_T^{-\mathsf{T}} \mathfrak{d}_N(k)|^2 = \frac{1}{g^2_{\mathrm{YM}}} \mathfrak{d}_N(k)^* G_T^{-1} \mathfrak{d}_N(k) \tag{277}$$

with complex multiplicity $3d_G$.

Proof. On $\mathcal{T}_{N,T,k}$, the contraction term in

$$\iota_{\overline{\zeta}^\sharp} \varepsilon(\zeta)\alpha = |\zeta|^2 \alpha - \varepsilon(\zeta) \iota_{\overline{\zeta}^\sharp} \alpha$$

vanishes. This proves the eigenvalue. Since $\zeta_{N,T}(k) \neq 0$, the contraction has rank one on $E_{\mathbb{C}}$, and its kernel has dimension three. Tensoring with $\mathfrak{g}_{\mathbb{C}}$ gives the stated multiplicity. □

10.3 Exact coordinate-mode eigenvalues and all mesh regimes

Set

$$\rho_N = 4N^2 \sin^2\left(\frac{\pi}{N}\right). \tag{278}$$

Because $\mathfrak{d}_N(\mathbf{e}_\gamma)$ and $\mathfrak{d}_N(\mathbf{e}_u)$ are the common scalar $N(e^{2\pi i/N} - 1)$ times the corresponding coordinate vector, equations (254) and (277) give the exact eigenvalues

$$\lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_\gamma) = \frac{\rho_N}{g^2_{\mathrm{YM}}} \frac{L_T^2 + m_T^2}{D_T^2}, \tag{279}$$

$$\lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_u) = \frac{\rho_N}{g^2_{\mathrm{YM}}} \frac{q_T^2 + 36 m_T^2}{D_T^2}. \tag{280}$$

Each individual complex Fourier fibre contributes multiplicity $3d_G$. For real cochains, the pair $\{\mathbf{e}_j, -\mathbf{e}_j\}$ contributes $6d_G$ dimensions when $N > 2$. When $N = 2$, these two labels coincide, so its real contribution is $3d_G$.

Theorem 10.6 (Finite gap and simultaneous cusp–mesh limits)**.** *There is T_1, independent of $N \geq 2$, such that for $T \geq T_1$ the first eigenvalue of $\mathcal{L}_{N,T}$ on $\mathscr{C}_{N,T}$ is*

$$\gamma_{N,T} = \min\left\{ \lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_\gamma), \lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_u) \right\}. \tag{281}$$

Uniformly for $N \geq 2$,

$$\gamma_{N,T} = \frac{\rho_N}{g^2_{\mathrm{YM}} T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2})\right). \tag{282}$$

Let

$$I_{N,T} = \{j \in \{\gamma, u\} : \lambda^{\mathrm{YM}}_{N,T}(\mathbf{e}_j) = \gamma_{N,T}\}, \qquad \eta_N = \begin{cases} 1, & N = 2, \\ 2, & N > 2. \end{cases}$$

The real multiplicity is exactly

$$3 d_G \eta_N \, \# I_{N,T}. \tag{283}$$

The limiting conclusions are:

(i) For fixed $N \geq 2$,

$$T^2 \gamma_{N,T} \longrightarrow \frac{\rho_N}{g^2_{\mathrm{YM}}}, \qquad \gamma_{N,T} \longrightarrow 0. \tag{284}$$

(ii) For every integer-valued $N(T) \geq 2$, with no growth assumption, there are constants $0 < c < C < \infty$, independent of $N(T)$, such that

$$\frac{c}{T^2} \leq \gamma_{N(T),T} \leq \frac{C}{T^2} \tag{285}$$

for all large T. Hence $\gamma_{N(T),T} \to 0$, although the scaled gaps need not converge when $N(T)$ oscillates.

(iii) If $N(T) \to \infty$, then

$$\gamma_{N(T),T} = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2} \left(1 + O(T^{-1}) + O(N(T)^{-2})\right). \tag{286}$$

(iv) The largest physical elementary edge has length

$$\max_{1 \leq j \leq 4} \frac{\|P_T \mathbf{e}_j\|}{N(T)} = \frac{T}{N(T)}(1 + O(T^{-1})). \tag{287}$$

Thus the varying continuum torus is resolved if and only if $N(T)/T \to \infty$. This refinement condition is not needed for the algebraic soft-mode conclusion in (ii).

Proof. Let

$$\sigma_N = 2N \sin(\pi/N), \qquad \rho_N = \sigma_N^2.$$

The least nonzero modulus of a coordinate of $\mathfrak{d}_N(k)$ is σ_N, attained precisely at frequency ± 1, with the two signs equal for $N = 2$. If $\mathfrak{d}_N(k)$ is supported in its first two components but $k \notin \{\pm \mathbf{e}_\gamma, \pm \mathbf{e}_u\}$, then

$$|\mathfrak{d}_N(k)_1|^2 + |\mathfrak{d}_N(k)_2|^2 \geq 2\sigma_N^2. \tag{288}$$

Indeed, either both coordinates are nonzero, or the single nonzero frequency is not ± 1. In the latter case

$$\frac{\sin(2\pi/N)}{\sin(\pi/N)} = 2\cos(\pi/N) \geq \sqrt{2}$$

for $N \geq 4$, and the cases $N = 2, 3$ are immediate.

For $z = (z_1, z_2, 0, 0)^{\mathsf{t}} \in \mathbb{C}^4$, the exact inverse transpose in theorem 9.1 gives

$$P_T^{-\mathsf{T}} z = \left(0, \frac{m_T z_1 + q_T z_2}{D_T}, 0, \frac{-L_T z_1 + 6 m_T z_2}{D_T}\right). \tag{289}$$

The scaled 2×2 matrix

$$\frac{T}{D_T} \begin{pmatrix} m_T & q_T \\ -L_T & 6m_T \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + O_{\mathrm{op}}(T^{-1})$$

has all singular values $1 + O(T^{-1})$, also on complex vectors. Thus the factor $\sqrt{2}$ in equation (288) separates every other soft-plane frequency from the coordinate frequencies uniformly in N for large T.

For a general $z \in \mathbb{C}^4$, the full inverse transpose in theorem 9.1 has first and third components z_3, z_4. Hence a finite frequency with nonzero third or fourth coordinate satisfies

$$\|P_T^{-\mathsf{T}} \mathfrak{d}_N(k)\|^2 \geq |\mathfrak{d}_N(k)_3|^2 + |\mathfrak{d}_N(k)_4|^2 \geq \sigma_N^2,$$

whereas either coordinate candidate is $O(\sigma_N^2/T^2)$. This proves the exact minimizing list and equation (281); the reality discussion before the theorem proves equation (283).

Multiplying the two expansions in theorem 9.2 by $\rho_N / g_{\mathrm{YM}}^2$ proves equation (282). The elementary sine bounds $2x/\pi \leq \sin x \leq x$ for $0 < x \leq \pi/2$ give

$$16 \leq \rho_N \leq 4\pi^2. \tag{290}$$

Together with equation (282), these bounds prove (i) and (ii). The Taylor expansion $\rho_N = 4\pi^2(1 + O(N^{-2}))$ proves (iii).

The first two columns of P_T, displayed in theorem 9.1, have Euclidean lengths $T(1 + O(T^{-1}))$, while its last two columns have length one. Division by $N(T)$ proves equation (287) and (iv). □

Remark 10.7 (Every fixed reduced model remains positively gapped). For each fixed T, the reduced continuum operator has a positive first eigenvalue. For each fixed pair (T, N), the reduced finite operator also has a positive first eigenvalue. What fails is a positive lower bound uniform as $T \to \infty$. Before reduction, gauge-longitudinal directions and constant harmonic one-forms are separate Hessian-zero directions.

10.4 Continuum Coulomb reduction and exact spectrum

The infinitesimal gauge directions $\mathrm{d}\Omega^0(F_T;\mathfrak{g})$ and the harmonic one-forms are Hessian-zero directions. Coulomb gauge removes the former but not the latter. The reduced Coulomb space is therefore

$$\mathscr{C}_T = \left\{ a \in L^2\Omega^1(F_T;\mathfrak{g}) : \mathrm{d}^* a = 0 \text{ distributionally},\ a \perp \mathcal{H}^1(F_T;\mathfrak{g}) \right\}. \tag{291}$$

The subspace $\mathscr{C}_T$ reduces $\mathcal{L}_T$. On it the closed Hessian form has domain $H^1\Omega^1(F_T;\mathfrak{g}) \cap \mathscr{C}_T$, and the restricted self-adjoint operator has domain $H^2\Omega^1(F_T;\mathfrak{g}) \cap \mathscr{C}_T$. Indeed, on the distributionally coclosed slice the graph norm of d_1, together with the L^2-norm, is the Hodge graph norm

$$\|a\|_{L^2}^2 + \|\mathrm{d}a\|_{L^2}^2 + \|\mathrm{d}^* a\|_{L^2}^2.$$

The Gaffney estimate on the compact flat torus identifies its completion with H^1, and elliptic regularity for the Hodge Laplacian identifies the operator domain with H^2. Orthogonality to the finite-dimensional harmonic space is preserved by the Fourier multiplier.

For $k \in \mathbb{Z}^4$, write

$$\xi_{T,k} = P_T^{-\mathsf{T}} k, \qquad \chi_{T,k}([x]) = \exp\bigl(2\pi i \langle \xi_{T,k}, x \rangle\bigr).$$

For a covector ξ, let $\varepsilon(\xi)$ denote exterior multiplication by ξ, and use the Euclidean metric to define $\xi^\sharp$ and contraction $\iota_{\xi^\sharp}$.

Proposition 10.8 (Continuum Fourier symbol, slice, and multiplicity)**.** *For $\alpha \in (\mathbb{R}^4)^*_{\mathbb{C}} \otimes \mathfrak{g}_{\mathbb{C}}$,*

$$\mathrm{d}(\chi_{T,k}\alpha) = 2\pi i\, \chi_{T,k}\varepsilon(\xi_{T,k})\alpha, \tag{292}$$

$$\mathrm{d}^*(\chi_{T,k}\alpha) = -2\pi i\, \chi_{T,k}\iota_{\xi_{T,k}^\sharp}\alpha, \tag{293}$$

$$\mathrm{d}^*\mathrm{d}(\chi_{T,k}\alpha) = 4\pi^2 \chi_{T,k}\left(|\xi_{T,k}|^2 \alpha - \varepsilon(\xi_{T,k})\iota_{\xi_{T,k}^\sharp}\alpha \right). \tag{294}$$

For $k \neq 0$, the Coulomb polarization space is

$$\mathcal{T}_{T,k} = \ker \iota_{\xi_{T,k}^\sharp} \subset (\mathbb{R}^4)^*_{\mathbb{C}}. \tag{295}$$

It has complex dimension three. On $\mathcal{T}_{T,k} \otimes \mathfrak{g}_{\mathbb{C}}$, the Hessian eigenvalue is

$$\lambda_T^{\mathrm{YM}}(k) = \frac{4\pi^2}{g_{\mathrm{YM}}^2} |P_T^{-\mathsf{T}} k|^2 \tag{296}$$

with complex multiplicity $3d_G$ on that signed Fourier fibre. If

$$K_T(\lambda) = \{ k \in \mathbb{Z}^4 \setminus \{0\} : \lambda_T^{\mathrm{YM}}(k) = \lambda \},$$

the total complex multiplicity, and also the dimension of the real eigenspace, is

$$3d_G \, \# K_T(\lambda). \tag{297}$$

Equivalently, a conjugate pair $\{k, -k\}$ contributes $6d_G$ real dimensions.

Proof. Differentiating the character proves equation (292). On the Euclidean cover,

$$\mathrm{d}^* = -\sum_{j=1}^{4} \iota_{\partial_{x_j}} \partial_{x_j},$$

which proves equation (293). The exterior algebra identity

$$\iota_{\xi^\sharp}\varepsilon(\xi) + \varepsilon(\xi)\iota_{\xi^\sharp} = |\xi|^2 \,\mathrm{id}$$

proves equation (294). For $k \neq 0$, the invertibility of P_T gives $\xi_{T,k} \neq 0$. Contraction by $\xi_{T,k}^\sharp$ has rank one on one-forms, so its kernel has dimension three. On that kernel the second term of equation (294) vanishes, proving the eigenvalue and its per-fibre multiplicity. The zero Fourier mode consists of the constant harmonic one-forms and has already been removed in equation (291). Fourier completeness gives all eigenvectors. Complex conjugation exchanges k with $-k$; the reality condition $\alpha_{-k} = \overline{\alpha_k}$ turns each conjugate pair of complex $3d_G$-dimensional spaces into a real $6d_G$-dimensional space. This proves equation (297). □

Remark 10.9 (The Coulomb projection uses the physical covector metric). In lattice-coordinate one-form coefficients a_y, the mode k obeys

$$k^{\mathsf{T}} G_T^{-1} a_y = 0, \qquad \Pi_{T,k} a_y = a_y - k \frac{k^{\mathsf{T}} G_T^{-1} a_y}{k^{\mathsf{T}} G_T^{-1} k}. \tag{298}$$

After conversion to physical x-coefficients, this is Euclidean projection perpendicular to $P_T^{-\mathsf{T}} k$. It is generally incorrect to impose Euclidean coefficient orthogonality to k, or to delete one coordinate merely because k is a coordinate lattice vector.

Theorem 10.10 (Exact reduced continuum Hessian gap)**.** *For all sufficiently large T, the first eigenvalue of $\mathcal{L}_T$ on $\mathscr{C}_T$ is*

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2} \min\left\{ \frac{L_T^2 + m_T^2}{D_T^2}, \frac{q_T^2 + 36 m_T^2}{D_T^2} \right\}. \tag{299}$$

It has the radial expansion

$$\gamma_T = \frac{4\pi^2}{g_{\mathrm{YM}}^2 T^2} \left(1 + \frac{-2H + 2\min\{B, 0\}}{T} + O(T^{-2}) \right). \tag{300}$$

Let

$$I_T = \{ j \in \{\gamma, u\} : a_{j,T} = \min\{a_{\gamma,T}, a_{u,T}\} \}.$$

The real multiplicity of γ_T is exactly

$$6 d_G \, \# I_T. \tag{301}$$

Proof. By theorem 10.8, minimizing the reduced Hessian eigenvalue is exactly the shortest-nonzero-dual-vector problem. The shortest-mode theorem theorem 9.2 says that, for all large T, the only possible minimizers are the two signed pairs with exact squared lengths given in equation (254). This proves equation (299). The expansions already proved in theorem 9.2 give relative corrections $2(B - H)$ and $-2H$; their minimum is $-2H + 2\min\{B, 0\}$, proving equation (300). Each minimizing signed pair contributes $6d_G$ real dimensions by theorem 10.8, which proves the multiplicity formula. □

10.5 The physical boundary and the exact no-gap bridge

The operators proved above are a classical Hessian and a finite quadratic cochain approximation on the internal flat manifolds F_T. They do not construct a quantum gauge theory on standard spacetime $\mathbb{R}^4$, and they do not determine the mass spectrum of such a theory. The precise boundaries are as follows.

(1) The unscaled cusp retains two compact geometric directions. The period parameter T has not been identified with a spacetime box size, a lattice spacing, a correlation length, or a renormalization scale.

(2) The operator $g_{\mathrm{YM}}^{-2}\mathrm{d}^*\mathrm{d}$ is the quadratic classical fluctuation operator at the product connection. It contains no nonperturbative interacting measure, dimensional transmutation, or confinement argument.

(3) The spaces $C_N^p(T;\mathfrak{g}_{\mathbb{C}})$ are linear Lie-algebra cochains. They are not compact-group link variables, and equation (276) is not a Wilson plaquette action.

(4) No Euclidean time direction or reflection, reflection-positive measure, transfer matrix, Osterwalder–Schrader reconstruction, Gauss-law physical quotient, local observable algebra, thermodynamic limit, or continuum limit in fixed physical units has been constructed from these operators.

(5) An eigenvalue of a four-dimensional elliptic Hessian has units of squared inverse length. Without a time reconstruction and a scale conversion, it is not an eigenvalue of a physical Hamiltonian. In particular, a dimensionless transfer-matrix gap can behave as $a_t m \to 0$ while the physical mass $m > 0$ remains fixed.

The exact extra statement that would transfer the auxiliary soft sequence to one already constructed physical Hamiltonian is the following. It must use the full zero-energy sector rather than only a chosen vacuum vector.

Proposition 10.11 (Typed state-and-energy no-gap bridge)**.** *Let $\mathcal{H}_{\mathrm{phys}}$ be a Hilbert space and let $H \geq 0$ be self-adjoint on it. Its closed energy form is*

$$\mathfrak{h}[\psi] = \|H^{1/2}\psi\|^2, \qquad \mathcal{D}(\mathfrak{h}) = \mathcal{D}(H^{1/2}). \tag{302}$$

Let

$$P_0 = \mathbf{1}_{\{0\}}(H), \qquad \mathcal{V} = \ker H = P_0\mathcal{H}_{\mathrm{phys}}, \qquad \mathcal{H}_{\mathrm{exc}} = (\mathrm{id} - P_0)\mathcal{H}_{\mathrm{phys}}, \tag{303}$$

and fix a normalized vacuum vector $\Omega \in \mathcal{V}$. Then

$$\mathcal{H}_{\mathrm{phys}} = \mathcal{V} \oplus \mathcal{H}_{\mathrm{exc}} \tag{304}$$

reduces H. Denote its excitation restriction by

$$H_{\mathrm{exc}} = H|_{\mathcal{D}(H)\cap\mathcal{H}_{\mathrm{exc}}}.$$

For every n, let $\mathcal{K}_n$ be an auxiliary Hilbert space with a closed nonnegative quadratic form $\mathfrak{q}_n$. Let $u_n \in \mathcal{D}(\mathfrak{q}_n)$ satisfy

$$\|u_n\|_{\mathcal{K}_n} = 1, \qquad \mathfrak{q}_n[u_n] \longrightarrow 0. \tag{305}$$

Suppose there are linear maps

$$J_n : \mathcal{D}(\mathfrak{q}_n) \longrightarrow \mathcal{D}(\mathfrak{h}), \qquad \psi_n = (\mathrm{id} - P_0)J_n u_n, \tag{306}$$

and numbers $c > 0$, $C_n \geq 0$, and $\varepsilon_n \geq 0$ such that

$$\|\psi_n\| \geq c, \tag{307}$$

$$\mathfrak{h}[\psi_n] \leq C_n\mathfrak{q}_n[u_n] + \varepsilon_n, \qquad C_n\mathfrak{q}_n[u_n] + \varepsilon_n \longrightarrow 0. \tag{308}$$

Then

$$\inf \operatorname{Spec}(H_{\mathrm{exc}}) = 0 \tag{309}$$

and, for every $\epsilon > 0$,

$$\mathbf{1}_{(0,\epsilon)}(H) \neq 0. \tag{310}$$

Moreover,

$$0 \in \operatorname{Spec}(H_{\mathrm{exc}}), \qquad \ker H_{\mathrm{exc}} = \{0\}. \tag{311}$$

Thus zero is a continuous spectral threshold, essential under the usual Weyl convention, and H *has no positive gap above its full zero-energy sector.*

If the vacuum is unique, so that $\mathcal{V} = \mathbb{C}\Omega$*, one may replace the projection in equation* (306) *by the requirement* $J_n u_n \perp \Omega$*. Without uniqueness, orthogonality only to* Ω *is insufficient.*

Proof. The operator P_0 is a spectral projection of H, so it commutes with the resolvent and with the spectral calculus of H. Hence equation (304) is a reducing decomposition and

$$\mathcal{D}(\mathfrak{h}|_{\mathcal{H}_{\mathrm{exc}}}) = \mathcal{D}(H^{1/2}) \cap \mathcal{H}_{\mathrm{exc}}.$$

The normalized vectors

$$v_n = \frac{\psi_n}{\|\psi_n\|}$$

belong to this excitation form domain. By equations (307) and (308),

$$\mathfrak{h}[v_n] \leq c^{-2}\big(C_n \mathfrak{q}_n[u_n] + \varepsilon_n\big) \longrightarrow 0.$$

The form version of the spectral variational identity is

$$\inf \operatorname{Spec}(H_{\mathrm{exc}}) = \inf_{0 \neq v \in \mathcal{D}(H^{1/2}) \cap \mathcal{H}_{\mathrm{exc}}} \frac{\mathfrak{h}[v]}{\|v\|^2}.$$

Nonnegativity and the sequence v_n prove equation (309).

If $\mathbf{1}_{(0,\epsilon)}(H) = 0$ for some $\epsilon > 0$, the spectral measure of every excitation vector would be supported in $[\epsilon, \infty)$, and therefore

$$\mathfrak{h}[v] \geq \epsilon \|v\|^2$$

throughout the excitation form domain. This contradicts $\mathfrak{h}[v_n] \to 0$, proving equation (310). Since the spectrum of a self-adjoint operator is closed, equation (309) puts zero in $\operatorname{Spec}(H_{\mathrm{exc}})$. Its kernel is trivial by the definition of $\mathcal{H}_{\mathrm{exc}}$. A self-adjoint operator has no residual spectrum, and equation (310) makes zero an accumulation point. This proves equation (311) and the essential-threshold assertion.

If $\mathcal{V} = \mathbb{C}\Omega$, orthogonality to Ω is exactly membership in $\mathcal{H}_{\mathrm{exc}}$. If the kernel is larger, a zero-energy vector can be orthogonal to the selected Ω, so that orthogonality alone does not produce an excitation vector. This proves the final assertion. □

Remark 10.12 (The present geometry supplies only the auxiliary side). Choose any $T_n \to \infty$, any $N_n \geq 2$, and a unit real transverse eigenvector $u_n \in \mathscr{C}_{N_n,T_n}$ at a minimizing coordinate mode. With

$$\mathcal{K}_n = \mathscr{C}_{N_n,T_n}, \qquad \mathfrak{q}_n[u] = g_{\mathrm{YM}}^{-2} \|\mathrm{d}_{N_n} u\|^2,$$

theorem 10.6 gives $\mathfrak{q}_n[u_n] = \gamma_{N_n,T_n} \to 0$. Thus the auxiliary soft-state data in equation (305) are present.

What is absent is one already constructed, gauge-invariant, positive-metric physical Hilbert space for four-dimensional quantum Yang–Mills theory, its Hamiltonian H, and maps J_n satisfying equations (307) and (308) with constants controlled in physical units after gauge reduction, reconstruction, renormalization, and all limiting operations. Consequently theorem 10.11 cannot be applied to infer a quantum mass statement from the cusp geometry. Even if such maps were supplied, the proposition would establish only spectrum arbitrarily close to zero for that specified H; it would not by itself identify a one-particle state, establish scattering, or construct the required quantum gauge theory.

11 Cubical holonomy and finite gauge systems

This section records three constructions on the regular cusp fibres while keeping two independent integers separate throughout: M_{cov} is the index used to enlarge two periods of a fibre, whereas N_{mesh} is the number of cubical subdivisions in each coordinate direction. In particular, no mesh refinement is identified with a change of covering space.

Let

$$P_T = \begin{pmatrix} 6\,\mathrm{Re}\,\mu_T & \mathrm{Re}\,\tau_T & 1 & 0 \\ 6\,\mathrm{Im}\,\mu_T & \mathrm{Im}\,\tau_T & 0 & 0 \\ \mathrm{Re}\,\beta_T & \mathrm{Re}\,\mu_T & 0 & 1 \\ \mathrm{Im}\,\beta_T & \mathrm{Im}\,\mu_T & 0 & 0 \end{pmatrix}, \qquad F_T = \mathbb{R}^4/P_T\mathbb{Z}^4. \tag{312}$$

Write

$$L_T = \mathrm{Im}\,\tau_T, \qquad m_T = \mathrm{Im}\,\mu_T, \qquad q_T = -\,\mathrm{Im}\,\beta_T, \qquad D_T = L_T q_T + 6m_T^2 > 0. \tag{313}$$

Expanding the determinant along the last two columns gives

$$\det P_T = L_T\,\mathrm{Im}\,\beta_T - 6m_T^2 = -D_T.$$

Along the positive radial cusp the source data give

$$L_T = T + O(1), \qquad q_T = T + O(1), \qquad m_T = O(1), \tag{314}$$

and all three real parts occurring in (312) are bounded. Equivalently, after ordering physical coordinates as $(x_1, x_3; x_2, x_4)$, the first two columns have block form

$$\begin{pmatrix} A_T \\ B_T \end{pmatrix}, \qquad B_T = \begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix} = T\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} + C_T, \qquad \sup_T \|C_T\|_{\mathrm{op}} < \infty. \tag{315}$$

For $M_{\mathrm{cov}} \geq 1$, put

$$S_{M_{\mathrm{cov}}} = \mathrm{diag}(1, 1, M_{\mathrm{cov}}, M_{\mathrm{cov}}), \qquad P_{T,M_{\mathrm{cov}}} = P_T S_{M_{\mathrm{cov}}}, \tag{316}$$

and define the degree-M_{cov}^2 cover

$$\widetilde{F}_{T,M_{\mathrm{cov}}} = \mathbb{R}^4/P_{T,M_{\mathrm{cov}}}\mathbb{Z}^4 \longrightarrow F_T. \tag{317}$$

Indeed, $S_{M_{\mathrm{cov}}}\mathbb{Z}^4 \subset \mathbb{Z}^4$ and

$$\mathbb{Z}^4/S_{M_{\mathrm{cov}}}\mathbb{Z}^4 \cong (\mathbb{Z}/M_{\mathrm{cov}}\mathbb{Z})^2,$$

so the identity on $\mathbb{R}^4$ induces the asserted cover and its deck group has order M_{cov}^2. Its Euclidean quotient metric, written on the coordinate torus $\mathbb{R}^4/\mathbb{Z}^4$, has Gram matrix and covolume

$$G_{T,M_{\mathrm{cov}}} = P_{T,M_{\mathrm{cov}}}^{\mathsf{T}} P_{T,M_{\mathrm{cov}}}, \qquad \mathcal{D}_{T,M_{\mathrm{cov}}} = \sqrt{\det G_{T,M_{\mathrm{cov}}}} = M_{\mathrm{cov}}^2 D_T. \tag{318}$$

11.1 The transported cubical subdivision and continuum holonomy

Fix an integer $N_{\mathrm{mesh}} \geq 2$, and let $K_{N_{\mathrm{mesh}}}$ be the standard cubical subdivision of $\mathbb{R}^4/\mathbb{Z}^4$ obtained by dividing every coordinate circle into N_{mesh} equal intervals. Its vertex set and positively oriented coordinate edges are

$$V_{N_{\mathrm{mesh}}} = (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \qquad E_{N_{\mathrm{mesh}}} = V_{N_{\mathrm{mesh}}} \times \{1, 2, 3, 4\}. \tag{319}$$

Proposition 11.1 (Exact cubical realization and the torsion vertices)**.** *The linear map $P_{T,M_{\mathrm{cov}}}$ induces a diffeomorphism*

$$\Phi_{T,M_{\mathrm{cov}}} : \mathbb{R}^4/\mathbb{Z}^4 \longrightarrow \widetilde{F}_{T,M_{\mathrm{cov}}}, \qquad [y] \longmapsto [P_{T,M_{\mathrm{cov}}} y]. \tag{320}$$

Transporting $K_{N_{\mathrm{mesh}}}$ by this diffeomorphism gives an honest cubical cell decomposition of the whole torus. On vertices its restriction is the bijection

$$\begin{aligned} \iota_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \quad & (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4 \xrightarrow{\cong} \widetilde{F}_{T,M_{\mathrm{cov}}}[N_{\mathrm{mesh}}], \\ & [n] \longmapsto \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}} n\right]. \end{aligned} \tag{321}$$

Here the codomain is the finite N_{mesh}-torsion subgroup, not the entire torus. The coordinate edge (n, j) is the path

$$\gamma_{n,j}(t) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}}(n + t\mathbf{e}_j)\right], \qquad 0 \le t \le 1. \tag{322}$$

More generally, for an ordered subset $I = \{i_1, \dots, i_p\} \subset \{1, 2, 3, 4\}$, the corresponding closed p-cell is parametrized by

$$c_{n,I}(t_1, \dots, t_p) = \left[\frac{1}{N_{\mathrm{mesh}}} P_{T,M_{\mathrm{cov}}} \left(n + \sum_{a=1}^{p} t_a \mathbf{e}_{i_a}\right)\right], \qquad (t_1, \dots, t_p) \in [0, 1]^p. \tag{323}$$

Proof. If $y - y' \in \mathbb{Z}^4$, then $P_{T,M_{\mathrm{cov}}}(y - y')$ belongs to the quotient lattice, so (320) is well defined. The matrix $P_{T,M_{\mathrm{cov}}}$ is invertible, and its inverse induces the inverse map on the two quotients. Hence it is a diffeomorphism and carries every open cell of the standard subdivision diffeomorphically onto an open cell of the target.

Every point displayed in (321) is killed by N_{mesh}. If it is zero, then

$$\frac{P_{T,M_{\mathrm{cov}}} n}{N_{\mathrm{mesh}}} \in P_{T,M_{\mathrm{cov}}} \mathbb{Z}^4.$$

Invertibility gives $n \in N_{\mathrm{mesh}}\mathbb{Z}^4$, proving injectivity. Conversely, if $[x]$ is killed by N_{mesh}, choose a lift $x \in \mathbb{R}^4$. There is $n \in \mathbb{Z}^4$ such that $N_{\mathrm{mesh}} x = P_{T,M_{\mathrm{cov}}} n$, and therefore $[x] = [P_{T,M_{\mathrm{cov}}} n/N_{\mathrm{mesh}}]$. This proves surjectivity. Formula (322) is simply the transport of the corresponding coordinate edge; at a wrap-around endpoint the extra vector $P_{T,M_{\mathrm{cov}}}\mathbf{e}_j$ is a period. □

Let G be a compact matrix Lie group and use the trivial principal bundle over $\widetilde{F}_{T,M_{\mathrm{cov}}}$. A connection is represented by $A \in \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g})$. For an oriented path γ, define $U_A(\gamma)$ by

$$\frac{dU}{dt} = -A_{\gamma(t)}(\dot{\gamma}(t))U(t), \qquad U(0) = 1, \qquad U_A(\gamma) = U(1). \tag{324}$$

Proposition 11.2 (Typed continuum-to-link holonomy map)**.** *The formula*

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)_{n,j} = U_A(\gamma_{n,j}) \tag{325}$$

defines a map

$$\mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} : \Omega^1(\widetilde{F}_{T,M_{\mathrm{cov}}}; \mathfrak{g}) \longrightarrow G^{E_{N_{\mathrm{mesh}}}}. \tag{326}$$

For the convention

$$A^g = g^{-1}Ag + g^{-1}dg, \qquad g \in C^\infty(\widetilde{F}_{T,M_{\mathrm{cov}}}, G), \tag{327}$$

it is equivariant for evaluation of g at the vertices:

$$\mathrm{Hol}(A^g)_{n,j} = g(\gamma_{n,j}(1))^{-1} \mathrm{Hol}(A)_{n,j} g(\gamma_{n,j}(0)). \tag{328}$$

Proof. The smooth matrix ODE (324) has a unique solution on every edge, which proves that (326) is defined. Let U solve that ODE for A, set $g_t = g(\gamma_{n,j}(t))$, and put $V(t) = g_t^{-1}U(t)g_0$. Direct differentiation gives

$$V'(t) = -(g_t^{-1}A(\dot\gamma_{n,j})g_t + g_t^{-1}g_t')V(t), \qquad V(0) = 1.$$

This is exactly the parallel-transport equation for A^g. Uniqueness and evaluation at $t = 1$ prove (328). Notice that the endpoint inverse in this formula is forced by the joint conventions (324) and (327). □

11.2 An explicit compact-link SU(2) flux

The ordered plaquette convention used below is most naturally written for the inverse-transport link variable

$$\mathcal{U}(A)_{n,j} = \mathrm{Hol}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(A)^{-1}_{n,j}. \tag{329}$$

Equation (328) then becomes

$$\mathcal{U}(A^g)_{n,j} = g(\gamma_{n,j}(0))^{-1}\mathcal{U}(A)_{n,j}g(\gamma_{n,j}(1)). \tag{330}$$

Thus the product in (336) transforms by conjugation at its base vertex. The symbols $U_j(n)$ in the remainder of this subsection use this inverse-transport convention.

Use

$$\mathfrak{su}(2) = \{X \in M_2(\mathbb{C}) : X^* = -X,\ \mathrm{tr}\, X = 0\}, \qquad \langle X, Y\rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\,\mathrm{tr}(XY), \tag{331}$$

on real elements, and fix

$$H_{\mathrm{link}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \qquad \|H_{\mathrm{link}}\|^2 = 1, \qquad \exp(2\pi H_{\mathrm{link}}) = 1. \tag{332}$$

Write $N = N_{\mathrm{mesh}}$ temporarily inside the next displayed link formulas, and represent n_1, n_2 by integers in $\{0, \dots, N-1\}$.

Theorem 11.3 (A periodic compact-link flux and its exact action)**.** *On the oriented edges of K_N, define*

$$U_1(n) = \exp\left(-\frac{2\pi n_2}{N^2}H_{\mathrm{link}}\right), \tag{333}$$

$$U_2(n) = \begin{cases} 1, & 0 \le n_2 \le N-2, \\ \exp(2\pi n_1 H_{\mathrm{link}}/N), & n_2 = N-1, \end{cases} \tag{334}$$

$$U_3(n) = U_4(n) = 1. \tag{335}$$

For

$$U_{ij}(n) = U_i(n)U_j(n+\mathbf{e}_i)U_i(n+\mathbf{e}_j)^{-1}U_j(n)^{-1}, \tag{336}$$

one has

$$\begin{aligned} &U_{12}(n) = \exp(2\pi H_{\mathrm{link}}/N^2) && \text{for every } n, \\ &U_{ij}(n) = 1 && \text{for } 1 \le i < j \le 4, \quad (i,j) \ne (1,2). \end{aligned} \tag{337}$$

Let Log *be the principal logarithm on the displayed plaquette. It is unambiguous because $0 < 2\pi/N^2 \le \pi/2$. Define the logarithmic curvature two-form and its metric-weighted action by*

$$\mathcal{F}_N^{\log}(n) = N^2\,\mathrm{Log}(U_{12}(n))\,dy^1 \wedge dy^2, \tag{338}$$

$$S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2N^4}\sum_{n\in V_N}\|\mathcal{F}_N^{\log}(n)\|^2_{G^{-1}_{T,M_{\mathrm{cov}}}\otimes\mathfrak{su}(2)}. \tag{339}$$

Then

$$\mathcal{F}_N^{\log}(n) = 2\pi H_{\mathrm{link}}\, dy^1 \wedge dy^2, \qquad S^{\log}_{T,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}. \tag{340}$$

In particular, the native-fibre value $M_{\mathrm{cov}} = 1$ is $2\pi^2/(g_{\mathrm{YM}}^2 D_T)$, independently of the mesh.

There is no canonical scalar Wilson action for a general nonorthogonal metric obtained merely by assigning independent character weights to all coordinate plaquettes. For this configuration, however, the exact continuum-matched $(1,2)$-sector Wilson functional is unambiguous if it is defined explicitly by the following formula. Its ordered plaquette holonomy and real trace are the Wilson lattice-action construction [64]; the metric coefficient and normalization here are the present definition, and their consequences are proved below rather than imported from that source:

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}\kappa^{12}_{T,M_{\mathrm{cov}}}}{2g_{\mathrm{YM}}^2} \sum_{n\in V_N} \left(2 - \operatorname{Re}\operatorname{tr} U_{12}(n)\right), \tag{341}$$

where

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = (G^{-1}_{T,M_{\mathrm{cov}}})_{11}(G^{-1}_{T,M_{\mathrm{cov}}})_{22} - (G^{-1}_{T,M_{\mathrm{cov}}})^2_{12} = D_T^{-2}. \tag{342}$$

Its exact value and continuum-mesh limit are

$$S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{M_{\mathrm{cov}}^2 N^4}{2g_{\mathrm{YM}}^2 D_T}\left(2 - 2\cos\frac{2\pi}{N^2}\right), \tag{343}$$

$$= \frac{M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}\left(2\pi^2 - \frac{2\pi^4}{3N^4} + \frac{4\pi^6}{45N^8} + O(N^{-12})\right), \tag{344}$$

$$\lim_{N\to\infty} S^{W,12}_{T,M_{\mathrm{cov}},N}(U) = \frac{2\pi^2 M_{\mathrm{cov}}^2}{g_{\mathrm{YM}}^2 D_T}. \tag{345}$$

Proof. All links lie in the same diagonal circle subgroup, so they commute. If $0 \le n_2 \le N-2$, the two U_2-terms in (336) are trivial and

$$U_{12}(n) = \exp(-2\pi n_2 H_{\mathrm{link}}/N^2)\exp(2\pi(n_2+1)H_{\mathrm{link}}/N^2) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

If $n_2 = N-1$, then $U_1(n+\mathbf{e}_2) = 1$. Moreover,

$$U_2(n+\mathbf{e}_1)U_2(n)^{-1} = \exp(2\pi H_{\mathrm{link}}/N).$$

For $n_1 = N-1$, the exponent computed with representatives differs from $2\pi H_{\mathrm{link}}/N$ by $-2\pi H_{\mathrm{link}}$, whose exponential is one; hence the formula also holds across that wrap boundary. It follows that

$$U_{12}(n) = \exp\left(-\frac{2\pi(N-1)}{N^2}H_{\mathrm{link}}\right)\exp\left(\frac{2\pi}{N}H_{\mathrm{link}}\right) = \exp(2\pi H_{\mathrm{link}}/N^2).$$

The functions U_1, U_2 are independent of n_3, n_4, and $U_3 = U_4 = 1$, proving every remaining assertion in (337). Reversing the orientation gives $U_{21}(n) = U_{12}(n)^{-1}$; it is not an additional trivial plaquette.

The lower-right 2×2 principal block of the cover Gram matrix is $M_{\mathrm{cov}}^2 I_2$. Jacobi's complementary-minor identity therefore gives

$$\kappa^{12}_{T,M_{\mathrm{cov}}} = \frac{\det(M_{\mathrm{cov}}^2 I_2)}{\det G_{T,M_{\mathrm{cov}}}} = \frac{M_{\mathrm{cov}}^4}{M_{\mathrm{cov}}^4 D_T^2} = D_T^{-2},$$

which proves (342). By (332) and the principal-log choice, $N^2 \operatorname{Log} U_{12} = 2\pi H_{\mathrm{link}}$. Hence every one of the N^4 summands in (339) equals $4\pi^2 D_T^{-2}$. Substitution of $\mathcal{D}_{T,M_{\mathrm{cov}}} = M_{\mathrm{cov}}^2 D_T$ proves (340).

Finally, H_{link} has eigenvalues $\pm i$, so

$$\operatorname{Re}\operatorname{tr}\exp(2\pi H_{\mathrm{link}}/N^2) = 2\cos(2\pi/N^2).$$

There are N^4 equal plaquettes. Substitution proves (343). Applying

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + \frac{x^6}{720} + O(x^8)$$

with $x = 2\pi/N^2$ proves (344) and (345). □

11.3 Finite Gaussian quantization with independent cover and mesh

Let

$$\vartheta^j_{T,M_{\mathrm{cov}}} = P^{-\mathsf{T}}_{T,M_{\mathrm{cov}}}\mathbf{e}_j \in (\mathbb{R}^4)^*, \qquad 1 \le j \le 4. \tag{346}$$

For $0 \le p \le 4$, define the real finite cochain spaces

$$C^p_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}}) = \mathrm{Map}\left(V_{N_{\mathrm{mesh}}}, \bigwedge^p (\mathbb{R}^4)^* \otimes \mathfrak{su}(2)\right). \tag{347}$$

The exterior powers carry the metric induced by the physical coframe (346); explicitly, for increasing multi-indices $I = (i_1, \dots, i_p)$ and $J = (j_1, \dots, j_p)$,

$$\left\langle \vartheta^{i_1} \wedge \cdots \wedge \vartheta^{i_p}, \vartheta^{j_1} \wedge \cdots \wedge \vartheta^{j_p} \right\rangle = \det\left((G^{-1}_{T,M_{\mathrm{cov}}})_{i_a j_b}\right)^p_{a,b=1}. \tag{348}$$

Together with (331), this defines the pointwise tensor-product metric. The cochain inner product is

$$\langle a, b\rangle_p = \frac{\mathcal{D}_{T,M_{\mathrm{cov}}}}{N^4_{\mathrm{mesh}}} \sum_{n \in V_{N_{\mathrm{mesh}}}} \langle a(n), b(n)\rangle. \tag{349}$$

Let $\mathsf{T}_j a(n) = a(n + \mathbf{e}_j)$, set

$$D^+_{N_{\mathrm{mesh}},j} = N_{\mathrm{mesh}}(\mathsf{T}_j - 1), \qquad D^-_{N_{\mathrm{mesh}},j} a(n) = N_{\mathrm{mesh}}(a(n) - a(n - \mathbf{e}_j)), \tag{350}$$

and define

$$d_{N_{\mathrm{mesh}}} a = \sum_{j=1}^4 \varepsilon(\vartheta^j_{T,M_{\mathrm{cov}}}) D^+_{N_{\mathrm{mesh}},j} a. \tag{351}$$

The commuting differences and anticommuting exterior multiplications give $d^2_{N_{\mathrm{mesh}}} = 0$. Its adjoint is

$$d^*_{N_{\mathrm{mesh}}} = -\sum_{j=1}^4 \iota_{(\vartheta^j_{T,M_{\mathrm{cov}}})^\sharp} D^-_{N_{\mathrm{mesh}},j}. \tag{352}$$

Define the real reduced Coulomb space

$$\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = \ker(d^*_{N_{\mathrm{mesh}}} : C^1 \to C^0) \cap \left(\ker \Delta^{(1)}_{N_{\mathrm{mesh}}}\right)^\perp, \tag{353}$$

$$\Delta^{(1)}_{N_{\mathrm{mesh}}} = d_{N_{\mathrm{mesh}}} d^*_{N_{\mathrm{mesh}}} + d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{354}$$

On this finite-dimensional space put

$$\mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = g^{-2}_{\mathrm{YM}} d^*_{N_{\mathrm{mesh}}} d_{N_{\mathrm{mesh}}}. \tag{355}$$

Proposition 11.4 (Exact finite Hessian spectrum)**.** *For*

$$\mathfrak{d}_{N_{\mathrm{mesh}}}(k)_j = N_{\mathrm{mesh}}\left(e^{2\pi i k_j/N_{\mathrm{mesh}}} - 1\right), \qquad k \in (\mathbb{Z}/N_{\mathrm{mesh}}\mathbb{Z})^4, \tag{356}$$

the positive spectrum of $\mathcal{L}$ is obtained from

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) = \frac{1}{g^2_{\mathrm{YM}}} \left\| P_T^{-\mathsf{T}} S^{-1}_{M_{\mathrm{cov}}} \mathfrak{d}_{N_{\mathrm{mesh}}}(k)\right\|^2, \qquad k \neq 0. \tag{357}$$

Each complex Fourier fibre has transverse multiplicity $3 \dim \mathfrak{su}(2) = 9$. In particular,

$$\lambda_{\min}(\mathcal{L}) = \min_{k \neq 0} \lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(k) > 0. \tag{358}$$

For $\mathbf{e}_1 = (1,0,0,0)^{\mathsf{T}}$, the exact mode value is

$$\lambda_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}(\mathbf{e}_1) = \frac{4N^2_{\mathrm{mesh}} \sin^2(\pi/N_{\mathrm{mesh}})}{g^2_{\mathrm{YM}}} \frac{L^2_T + m^2_T}{D^2_T}; \tag{359}$$

it is independent of M_{cov}.

Proof. Translation is unitary for (349), so $(D_j^+)^* = -D_j^-$. Exterior multiplication by ϑ^j has adjoint contraction by $(\vartheta^j)^\sharp$, proving (352). On the Fourier character $\chi_k(n) = e^{2\pi i k\cdot n/N_{\mathrm{mesh}}}$, the derivative symbol is exterior multiplication by

$$\zeta(k) = P_{T,M_{\mathrm{cov}}}^{-\mathsf{T}}\mathfrak{d}_{N_{\mathrm{mesh}}}(k) = P_T^{-\mathsf{T}}S_{M_{\mathrm{cov}}}^{-1}\mathfrak{d}_{N_{\mathrm{mesh}}}(k).$$

The Hermitian exterior-algebra identity

$$\iota_{\overline{\zeta}^\sharp}\varepsilon(\zeta) + \varepsilon(\zeta)\iota_{\overline{\zeta}^\sharp} = \|\zeta\|^2\,\mathrm{id}$$

shows that on the Coulomb polarization $\ker \iota_{\overline{\zeta}^\sharp} \subset (\mathbb{C}^4)^*$, the operator d^*d is multiplication by $\|\zeta\|^2$. For $k \neq 0$, $\mathfrak{d}_{N_{\mathrm{mesh}}}(k) \neq 0$, so invertibility of $P_{T,M_{\mathrm{cov}}}$ gives $\zeta(k) \neq 0$. The contraction has rank one, and its kernel has dimension three. The zero Fourier fibre consists exactly of constant harmonic cochains and was removed in (353). Fourier completeness now proves (357)–(358).

For $k = \mathbf{e}_1$, the only nonzero component of $\mathfrak{d}_{N_{\mathrm{mesh}}}(k)$ has modulus

$$2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}}),$$

and $S_{M_{\mathrm{cov}}}^{-1}\mathbf{e}_1 = \mathbf{e}_1$. Solving $P_T^{\mathsf{T}}\xi = \mathbf{e}_1$, or directly inverting the lower 2×2 block in (312), gives

$$\|P_T^{-\mathsf{T}}\mathbf{e}_1\|^2 = \frac{L_T^2 + m_T^2}{D_T^2}.$$

Substitution proves (359). □

Let da denote Lebesgue measure on $\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}$ induced by (349) and define

$$\mathcal{H}^{\mathrm{Gauss}}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = L^2(\mathscr{C}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}, da). \tag{360}$$

Theorem 11.5 (Finite Gaussian oscillator and a resolved cusp sequence)**.** *The Friedrichs realization on* (360) *of*

$$\widehat{H}^{\mathrm{Gauss}}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}} = -\frac{1}{2}\Delta_{\mathscr{C}} + \frac{1}{2}\langle a, \mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}}a\rangle \tag{361}$$

has a unique Gaussian ground state and spectral gap

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) = \sqrt{\lambda_{\min}(\mathcal{L}_{T,M_{\mathrm{cov}},N_{\mathrm{mesh}}})}. \tag{362}$$

Consequently,

$$\mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) \le \frac{2N_{\mathrm{mesh}}\sin(\pi/N_{\mathrm{mesh}})}{g_{\mathrm{YM}}}\frac{\sqrt{L_T^2 + m_T^2}}{D_T}. \tag{363}$$

There is an explicit diagonal sequence for which the cover geometry tends locally to $\mathbb{R}^4$, the cubical mesh resolves that geometry, and the Gaussian gap tends to zero. Explicitly, for integers $j \to \infty$, take

$$T_j = j^2, \qquad M_{\mathrm{cov},j} = j, \qquad N_{\mathrm{mesh},j} = j^3. \tag{364}$$

Then

$$\begin{aligned} \mathrm{sys}(\widetilde{F}_{T_j,M_{\mathrm{cov},j}}) &\longrightarrow \infty,\\ \max_r \frac{\|P_{T_j,M_{\mathrm{cov},j}}\mathbf{e}_r\|}{N_{\mathrm{mesh},j}} &\longrightarrow 0,\\ \mathrm{gap}(\widehat{H}^{\mathrm{Gauss}}) &\longrightarrow 0,\\ S^{\log}_{T_j,M_{\mathrm{cov},j},N_{\mathrm{mesh},j}}(U) &\longrightarrow 0. \end{aligned} \tag{365}$$

Thus the pointed cover tori converge locally isometrically to $\mathbb{R}^4$ while their finite linearized Gaussian systems are simultaneously refined.

Proof. The real finite-dimensional operator $\mathcal{L}$ is positive definite on the reduced Coulomb space. Choose an orthonormal eigenbasis with eigenvalues $0 < \lambda_1 \le \cdots \le \lambda_r$. In the resulting coordinates, (361) is the tensor sum

$$\sum_{s=1}^{r} \left(-\frac{1}{2}\frac{\partial^2}{\partial x_s^2} + \frac{1}{2}\lambda_s x_s^2 \right).$$

The normalized Hermite basis has eigenvalues $\sum_s (n_s + \frac{1}{2})\sqrt{\lambda_s}$, with $n_s \in \mathbb{Z}_{\ge 0}$. The all-zero multi-index is the unique ground state, and the smallest possible positive excitation is $\sqrt{\lambda_1}$. This proves (362); applying the minimum to the explicit $\mathbf{e}_1$-mode in (359) proves (363).

For completeness, reorder physical coordinates as in (315). Every vector of the cover lattice has the form

$$\begin{pmatrix} A_T n + M_{\mathrm{cov}} m \\ B_T n \end{pmatrix}, \qquad n, m \in \mathbb{Z}^2.$$

Let $C = \sup_T \|C_T\|_{\mathrm{op}}$. If $n \neq 0$, then $\|B_T n\| \ge (T - C)\|n\|$; if $n = 0$ and the lattice vector is nonzero, its norm is at least M_{cov}. Hence

$$\mathrm{sys}(\widetilde{F}_{T,M_{\mathrm{cov}}}) \ge \min\{T - C, M_{\mathrm{cov}}\}. \tag{366}$$

This tends to infinity for the first two choices in (364). For any fixed $R > 0$, once the systole exceeds $4R$, the quotient map from $\mathbb{R}^4$ is injective and distance-preserving on the radius-R ball: for a nonzero period λ and x, y in that ball,

$$\|x - y + \lambda\| \ge \|\lambda\| - \|x - y\| > 4R - 2R \ge \|x - y\|.$$

This proves the pointed local-isometric $\mathbb{R}^4$ limit.

The first two columns of P_T have norm $O(T)$, while the last two columns of $P_{T,M_{\mathrm{cov}}}$ have norm exactly M_{cov}. Therefore the largest physical edge in the N_{mesh}-subdivision has length $O(\max\{T, M_{\mathrm{cov}}\}/N_{\mathrm{mesh}})$, which is $O(j^{-1})$ for (364). Finally, (314) gives $\sqrt{L_T^2 + m_T^2}/D_T = T^{-1}(1 + O(T^{-1}))$, while $2N \sin(\pi/N) \le 2\pi$. The right-hand side of (363) is therefore $O(j^{-2})$. Moreover, (340) is $O(j^{-2})$, since $M_{\mathrm{cov},j}^2/D_{T_j} = j^2/(j^4 + O(j^2))$. This proves every assertion in (365). □

11.4 The exponential map and the exact physical boundary

Let $C^1 = C^1_{N_{\mathrm{mesh}}}(T, M_{\mathrm{cov}})$, regarded only as a finite-dimensional Lie-algebra cochain space, and define componentwise

$$\mathrm{Exp}_{N_{\mathrm{mesh}}} : C^1 \longrightarrow \mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, \qquad a \longmapsto (\exp(a_e/N_{\mathrm{mesh}}))_e\,. \tag{367}$$

Here, if $a(n) = \sum_{j=1}^{4} \vartheta^j_{T,M_{\mathrm{cov}}} \otimes a_j(n)$, then $a_e = a_j(n)$ for the oriented edge $e = (n, j)$.

Proposition 11.6 (Local chart, not a global embedding)**.** *The map* (367) *is smooth and a local diffeomorphism at zero, but it is not globally injective. Under left trivialization of every tangent space at the identity,*

$$D\,\mathrm{Exp}_{N_{\mathrm{mesh}}}(0)\,a = N_{\mathrm{mesh}}^{-1} a. \tag{368}$$

If S_W is any smooth compact-link action for which the identity configuration is critical, then

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0[a, b] = N_{\mathrm{mesh}}^{-2} D^2(S_W)_{\mathbf{1}}[a, b] \tag{369}$$

under the same tangent identification. Thus, for a prescribed cochain quadratic form Q,

$$D^2(S_W \circ \mathrm{Exp}_{N_{\mathrm{mesh}}})_0 = Q$$

holds exactly when

$$D^2(S_W)_{\mathbf{1}} = N_{\mathrm{mesh}}^2 Q$$

under the coefficient identification. Both this scaling and the tensor structure must be verified for the chosen compact-link action; neither follows from the exponential map alone.

Proof. For the Lie-group exponential, $D\exp_0$ is the identity. Applying the chain rule to $a_e \mapsto a_e/N_{\mathrm{mesh}}$ gives (368), which is invertible, so the inverse function theorem gives a local diffeomorphism. It is not globally injective: a cochain which is zero on every edge except one, where it equals $2\pi N_{\mathrm{mesh}} H_{\mathrm{link}}$, has the same image as the zero cochain by (332). Finally, the second-derivative chain rule contains a term involving $DS_W(\mathbf{1})$, which vanishes by criticality; the two remaining first derivatives of Exp each supply the factor N_{mesh}^{-1}. This proves (369). □

The Gaussian Hilbert space in (360) is not the compact-link Yang–Mills Hilbert space. At this finite lattice, the latter kinematic and gauge-invariant spaces would be

$$\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}} = L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU), \qquad \mathcal{H}^{\mathrm{inv}}_{N_{\mathrm{mesh}}} = \left(\mathcal{H}^{\mathrm{link}}_{N_{\mathrm{mesh}}}\right)^{\mathrm{SU}(2)^{V_{N_{\mathrm{mesh}}}}}, \tag{370}$$

where dU is product normalized Haar measure and the superscript denotes the nonlinear vertex-gauge action. Here "gauge-invariant" refers only to the link-configuration space of this four-dimensional cell complex. Without a chosen time direction or transfer matrix, it is not asserted to be a canonical Hamiltonian physical Hilbert space. Lüscher's construction of a selfadjoint, strictly positive transfer matrix [65] and Menotti–Pelissetto's reflection-positivity theorem for the Wilson action [66] concern the standard Wilson lattice setting after the requisite reflection/time structure has been fixed; they do not supply those omitted choices for the nonorthogonal sector functional (341). The Osterwalder–Schrader reconstruction boundary is likewise the one stated in the primary Euclidean axioms papers [67, 68], not a consequence of a finite Gaussian oscillator calculation. The differences are structural:

1. $\mathcal{H}^{\mathrm{Gauss}}$ is an L^2-space on a noncompact linear Coulomb slice with Lebesgue measure; $\mathcal{H}^{\mathrm{inv}}$ is a gauge-invariant subspace of an L^2-space on compact link variables with Haar measure.

2. The local exponential map is neither a global coordinate chart nor a measure-preserving bijection. No unitary map between the two displayed Hilbert spaces has been constructed.

3. A link configuration U determines the Dirac distribution δ_U, but

$$\delta_U \in \mathcal{D}'(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}), \qquad \delta_U \notin L^2(\mathrm{SU}(2)^{E_{N_{\mathrm{mesh}}}}, dU). \tag{371}$$

   Indeed, product Haar measure is nonatomic, so a point mass is singular with respect to it and has no square-integrable density.

4. No nonlinear compact-link Hamiltonian, Gauss-law spectral problem, time direction, reflection-positive transfer matrix, renormalized-scale limit, or Osterwalder–Schrader reconstruction is supplied here.

Accordingly, (362) is an exact theorem about the finite quadratic oscillator defined by the Coulomb-restricted operator $g_{\mathrm{YM}}^{-2} d^* d$. It is not an eigenvalue statement about a nonperturbative physical Yang–Mills Hamiltonian, and the small classical action in (340) is not by itself a physical-state Rayleigh quotient.

12 Integral magnetic sectors and a reducible $SU(2)$ critical point on the radial cusp

Retain the positive radial flat tori $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$, their Euclidean quotient metrics, and the real quantities $L_T, q_T, m_T, D_T, H, B, m$ from equations (9) and (33) and theorem 9.1. In particular,

$$L_T = \operatorname{Im}\tau_T, \qquad q_T = -\operatorname{Im}\beta_T, \qquad D_T = L_T q_T + 6m_T^2 = \mathrm{Vol}(F_T) > 0.$$

The real coordinate order is the already fixed order

$$x = (x_1, y_1, x_2, y_2)^{\mathsf{t}}.$$

12.1 The integral coframe

Let $e_1, \dots, e_4$ be the standard basis of $\mathbb{R}^4$, put $u = P_T^{-1}x$, and define

$$\theta_T^j := \mathrm{d}u_j = (P_T^{-\mathsf{T}} e_j)^{\mathsf{t}} (\mathrm{d}x_1, \mathrm{d}y_1, \mathrm{d}x_2, \mathrm{d}y_2)^{\mathsf{t}}. \tag{372}$$

Proposition 12.1 (Integral coframe and the distinguished unit-flux plane)**.** *The forms θ_T^j are the integral coframe dual to the ordered marked lattice basis:*

$$\int_{[0,1] \ni s \mapsto [x + sP_T e_k]} \theta_T^j = \delta_{jk}.$$

The first two members are exactly

$$\theta_T^1 = \frac{m_T}{D_T}\,\mathrm{d}y_1 - \frac{L_T}{D_T}\,\mathrm{d}y_2, \qquad \theta_T^2 = \frac{q_T}{D_T}\,\mathrm{d}y_1 + \frac{6m_T}{D_T}\,\mathrm{d}y_2, \tag{373}$$

and hence

$$\theta_T^1 \wedge \theta_T^2 = \frac{1}{D_T}\,\mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad |\theta_T^1 \wedge \theta_T^2|^2 = \frac{1}{D_T^2}. \tag{374}$$

Proof. For $n \in \mathbb{Z}^4$,

$$u(x + P_T n) = u(x) + n,$$

which proves the integral-period assertion. The first two columns of the already-proved matrix $P_T^{-\mathsf{T}}$ in equation (15) give equation (373). Their wedge coefficient is

$$\frac{6m_T^2 + L_T q_T}{D_T^2} = \frac{1}{D_T}.$$

The coordinate forms $\mathrm{d}y_1, \mathrm{d}y_2$ are orthonormal for the fixed quotient metric, proving equation (374). □

12.2 The unit-flux line and the topologically trivial rank-two sum

The following construction is made in the marked coordinates u of (372); it therefore applies to the actual skew lattice $P_T\mathbb{Z}^4$, not to a rectangular replacement.

Proposition 12.2 (Explicit factor of automorphy and connection)**.** *For $n \in \mathbb{Z}^4$, set*

$$j_n(u) = \exp(-2\pi i n_1 u_2). \tag{375}$$

The quotient

$$\mathscr{L}_T = (\mathbb{R}^4 \times \mathbb{C})/\mathbb{Z}^4, \qquad n \cdot (u, z) = (u + n, j_n(u) z),$$

is a Hermitian line bundle over F_T. The connection on its pullback to $\mathbb{R}^4$,

$$\widetilde{\nabla}^{\mathscr{L}_T} = \mathrm{d} + 2\pi i u_1\,\mathrm{d}u_2, \tag{376}$$

descends, and its curvature is

$$F_{\mathscr{L}_T} = 2\pi i\,\mathrm{d}u_1 \wedge \mathrm{d}u_2 = 2\pi i\,\theta_T^1 \wedge \theta_T^2. \tag{377}$$

With the unitary Chern–Weil convention $c_1 = [\,iF/(2\pi)\,]$, one has $c_1(\mathscr{L}_T) = -[\theta_T^1 \wedge \theta_T^2]$.

Proof. For $m, n \in \mathbb{Z}^4$,

$$j_m(u+n)j_n(u) = j_{m+n}(u)\exp(-2\pi i m_1 n_2) = j_{m+n}(u).$$

Thus (375) is a factor of automorphy. Its values are unitary, so the standard fibre metric descends. If $\mathcal{A} = 2\pi i u_1 \,\mathrm{d}u_2$, then

$$\mathcal{A}(u+n) = \mathcal{A}(u) + 2\pi i n_1 \,\mathrm{d}u_2, \qquad j_n^{-1}\mathrm{d}j_n = -2\pi i n_1 \,\mathrm{d}u_2.$$

Hence $\mathcal{A}(u+n)+j_n^{-1}\mathrm{d}j_n = \mathcal{A}(u)$, which is precisely the descent identity for $\mathrm{d}+\mathcal{A}$ under the stated quotient convention. Taking $\mathrm{d}\mathcal{A}$ gives (377). The Chern–Weil statement follows without a sign suppression:

$$\frac{i}{2\pi}F_{\mathscr{L}_T} = -\theta_T^1 \wedge \theta_T^2.$$

□

Let $\mathscr{L}_T^{-1} = \mathscr{L}_T^*$ carry the dual metric and dual connection, and set

$$\mathscr{E}_T = \mathscr{L}_T \oplus \mathscr{L}_T^{-1}. \tag{378}$$

Proposition 12.3 (Determinant, c_2, and explicit $SU(2)$-triviality)**.** *The factors of automorphy of $\mathscr{E}_T$ are*

$$G_n(u) = \begin{pmatrix} j_n(u) & 0 \\ 0 & j_n(u)^{-1} \end{pmatrix} \in SU(2).$$

Evaluation gives a canonical parallel determinant trivialization

$$\det \mathscr{E}_T = \mathscr{L}_T \otimes \mathscr{L}_T^{-1} \xrightarrow{\mathrm{ev}} F_T \times \mathbb{C}.$$

Moreover

$$c_1(\mathscr{E}_T) = 0, \qquad c_2(\mathscr{E}_T) = 0 \quad \text{in } H^4(F_T;\mathbb{Z}),$$

and the underlying principal $SU(2)$-bundle is smoothly trivial. The last assertion follows from the explicit nullhomotopy below and does not rely only on the vanishing of a de Rham Chern form.

Proof. The determinant assertion follows pointwise from $j_n j_n^{-1} = 1$. The induced determinant connection has connection form

$$\mathrm{tr}\begin{pmatrix} 2\pi i u_1 \,\mathrm{d}u_2 & 0 \\ 0 & -2\pi i u_1 \,\mathrm{d}u_2 \end{pmatrix} = 0,$$

so the evaluation trivialization is parallel.

Let $c = c_1(\mathscr{L}_T)$. The integral Whitney identity gives

$$c(\mathscr{E}_T) = (1+c)(1-c) = 1 - c^2.$$

The factor j_n depends only on (u_1, u_2), so $\mathscr{L}_T$ is the pullback of a line bundle L_0 on $\mathbb{R}^2/\mathbb{Z}^2$ under the marked projection $[u_1, u_2, u_3, u_4] \mapsto [u_1, u_2]$. Hence

$$c^2 = \pi_{12}^*(c_1(L_0)^2) = 0$$

integrally because $H^4(\mathbb{T}^2;\mathbb{Z}) = 0$. This proves the two Chern-class identities on the four-torus.

It remains to trivialize the $SU(2)$-bundle itself. On the fundamental square $0 \le u_1, u_2 \le 1$ for the first two marked coordinates, the only nontrivial seam is

$$(0, t, v) \sim (1, t, \gamma(t)v), \qquad \gamma(t) = \begin{pmatrix} e^{-2\pi i t} & 0 \\ 0 & e^{2\pi i t} \end{pmatrix}.$$

For $0 \le r \le 1$, define

$$a(r,t) = (1-r)e^{-2\pi i t} + r, \qquad s(r) = \sin(\pi r), \qquad N(r,t) = \big(|a(r,t)|^2 + s(r)^2\big)^{1/2},$$

and

$$U(r,t) = \frac{1}{N(r,t)} \begin{pmatrix} a(r,t) & -s(r) \\ s(r) & a(r,t) \end{pmatrix}. \tag{379}$$

If $0 < r < 1$, then $s(r) > 0$; at $r = 0, 1$, one has $|a(r,t)| = 1$. Thus N never vanishes. The two columns in (379) are orthonormal and its determinant is one, so $U(r,t) \in SU(2)$. It is smooth and periodic in t, and

$$U(0,t) = \gamma(t), \qquad U(1,t) = I_2.$$

This is an explicit nullhomotopy of the clutching loop. To see directly that it produces a smooth global frame, choose a smooth $\chi : [0,1] \to [0,1]$ which is zero near 0 and one near 1, and put

$$K(\sigma,t) = U(1-\chi(\sigma),t).$$

Then $K = I_2$ near $\sigma = 0$, $K = \gamma$ near $\sigma = 1$, and $K(\sigma,0) = K(\sigma,1)$. The required trivialization, written in the direction from the clutched bundle to the product bundle, is

$$[\sigma,t,v] \longmapsto \big([\sigma,t], K(\sigma,t)^{-1}v\big). \tag{380}$$

Indeed, the quotient convention gives $[0,t,v] = [1,t,\gamma(t)v]$, and the two representatives have the same image because

$$K(1,t)^{-1}\gamma(t)v = \gamma(t)^{-1}\gamma(t)v = v = K(0,t)^{-1}v.$$

At the other seam, $K(\sigma,0)^{-1} = K(\sigma,1)^{-1}$. The constant collars used in the definition of χ make the local matrix in equation (380) constant after applying the seam transition, so the map and its inverse are smooth across the seam. It is fibrewise unitary and determinant one. Extending it independently of u_3, u_4 and pulling it through the marking gives the required global $SU(2)$-frame of $\mathscr{E}_T$. □

12.3 A non-flat reducible Yang–Mills critical connection

The symbol $H = \operatorname{Im} h(0)$ is already fixed by equation (10). For the requested Lie-algebra element, fix instead the unambiguous notation

$$\langle X, Y\rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\operatorname{Tr}(XY), \qquad H_{\mathrm{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}. \tag{381}$$

Theorem 12.4 (Reducible critical point and exact action)**.** *The direct-sum connection on $\mathscr{E}_T$ has, in the lifted splitting,*

$$\nabla^T = \mathrm{d} + 2\pi u_1\,\mathrm{d}u_2\,H_{\mathrm{mag}}, \qquad F_{\nabla^T} = 2\pi H_{\mathrm{mag}}\,\theta_T^1 \wedge \theta_T^2 = \frac{2\pi}{D_T} H_{\mathrm{mag}}\,\mathrm{d}y_1 \wedge \mathrm{d}y_2. \tag{382}$$

After any of the global $SU(2)$-trivializations from theorem 12.3, this is a connection on the trivial principal $SU(2)$-bundle. It is non-flat and satisfies

$$\mathrm{d}^*_{\nabla^T} F_{\nabla^T} = 0.$$

For the normalization

$$\mathrm{YM}_T(\nabla) = \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_T} |F_\nabla|^2\,\mathrm{d}\,\mathrm{vol}_{F_T}, \tag{383}$$

its action is exactly

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T}. \tag{384}$$

Its first inverse-power correction, in the canonical scalar notation $H = \operatorname{Im} h(0)$, $B = \operatorname{Im} b(0)$, is

$$\mathrm{YM}_T(\nabla^T) = \frac{2\pi^2}{g_{\mathrm{YM}}^2 T^2}\left(1 + \frac{B - 2H}{T} + O(T^{-2})\right). \tag{385}$$

Proof. The connection matrix is

$$\begin{pmatrix} 2\pi i u_1 \,\mathrm{d}u_2 & 0 \\ 0 & -2\pi i u_1 \,\mathrm{d}u_2 \end{pmatrix} = 2\pi u_1 \,\mathrm{d}u_2 \, H_{\mathrm{mag}},$$

so equation (382) follows from theorem 12.2 and equation (374). The diagonal endomorphism H_{mag} is a globally defined parallel section of $\operatorname{ad} \mathscr{E}_T$, because every transition G_n commutes with it. The forms θ_T^j are parallel constant forms for the descended flat metric. Thus $\nabla^T F_{\nabla^T} = 0$, which implies $\mathrm{d}^*_{\nabla^T} F_{\nabla^T} = 0$. The curvature is nonzero because $D_T > 0$.

The chosen Lie-algebra normalization gives

$$|H_{\mathrm{mag}}|^2 = -\frac{1}{2}\operatorname{Tr}(H_{\mathrm{mag}}^2) = 1.$$

Equations equations (34) and (374) therefore give

$$\int_{F_T} |F_{\nabla^T}|^2 \,\mathrm{d}\operatorname{vol}_{F_T} = 4\pi^2 D_T \frac{1}{D_T^2} = \frac{4\pi^2}{D_T},$$

which proves equation (384). Finally, equation (33) and the elementary reciprocal expansion

$$\frac{1}{T^2 + aT + O(1)} = \frac{1}{T^2}\left(1 - \frac{a}{T} + O(T^{-2})\right)$$

with $a = 2H - B$ prove equation (385). □

12.4 Constant-curvature magnetic Laplacians for arbitrary integral nondegenerate flux

We next give the Hilbert space, closed form, self-adjoint realization, and oscillator spectrum in a form valid for every nondegenerate integral alternating form on the marked four-torus.

Let

$$\mathsf{E} = \sum_{1 \le j < k \le 4} E_{jk} \,\mathrm{d}u_j \wedge \mathrm{d}u_k \in \bigwedge^2 (\mathbb{Z}^4)^*, \qquad E_{jk} \in \mathbb{Z},$$

and let the same symbol denote its skew matrix. For $n \in \mathbb{Z}^4$, put

$$J_n^{\mathsf{E}}(u) = \exp\left(-2\pi i \sum_{j<k} E_{jk} n_j u_k\right). \tag{386}$$

The integer in

$$\sum_{j<k} E_{jk} m_j n_k$$

shows directly that $J_m^{\mathsf{E}}(u+n) J_n^{\mathsf{E}}(u) = J_{m+n}^{\mathsf{E}}(u)$. Thus (386) defines a Hermitian line bundle $\mathscr{L}_{\mathsf{E},T}$. On its pullback, set

$$\nabla^{\mathsf{E}} = \mathrm{d} + 2\pi i \sum_{j<k} E_{jk} u_j \,\mathrm{d}u_k. \tag{387}$$

Exactly as in theorem 12.2, the connection descends and

$$F_{\nabla^{\mathsf{E}}} = 2\pi i\,\mathsf{E}. \tag{388}$$

In u-coordinates the flat metric matrix is

$$G_T = P_T^{\mathsf{T}} P_T, \qquad \sqrt{\det G_T} = D_T. \tag{389}$$

Define the Hilbert space concretely by

$$\begin{aligned} \mathcal{H}_{\mathsf{E},T} &= \{\, f \in L^2_{\mathrm{loc}}(\mathbb{R}^4) : f(u+n) = J_n^{\mathsf{E}}(u) f(u) \text{ for every } n \in \mathbb{Z}^4 \}, \\ \|f\|^2 &= \int_{[0,1]^4} |f(u)|^2 \sqrt{\det G_T}\, \mathrm{d}^4 u. \end{aligned} \tag{390}$$

Let $\mathcal{C}_{\mathsf{E},T}$ be the smooth functions with the same automorphy. On this dense subspace define

$$q_{\mathsf{E},T}[f] = \int_{[0,1]^4} (G_T^{-1})^{jk} \overline{\nabla_j^{\mathsf{E}} f}\, \nabla_k^{\mathsf{E}} f \sqrt{\det G_T}\, \mathrm{d}^4 u. \tag{391}$$

Theorem 12.5 (Friedrichs realization and oscillator spectrum)**.** *The form* (391) *is nonnegative and closable. Its closed form domain is*

$$\operatorname{Dom} \overline{q}_{\mathsf{E},T} = H^1(F_T; \mathscr{L}_{\mathsf{E},T}).$$

Its Friedrichs operator is the magnetic Laplacian

$$\Delta_{\mathsf{E},T} = (\nabla^{\mathsf{E}})^* \nabla^{\mathsf{E}}, \qquad \operatorname{Dom}(\Delta_{\mathsf{E},T}) = H^2(F_T; \mathscr{L}_{\mathsf{E},T}),$$

and it has compact resolvent.

Assume now that E *is nondegenerate. Let* $b_1, b_2 > 0$ *be defined by the eigenvalues*

$$\operatorname{Spec}\Big(G_T^{-1/2} (2\pi \mathsf{E}) G_T^{-1/2} \Big) = \{ i b_1, -i b_1, i b_2, -i b_2 \}. \tag{392}$$

Then, as a set (with finite multiplicities),

$$\operatorname{Spec}(\Delta_{\mathsf{E},T}) = \{ (2n_1+1) b_1 + (2n_2+1) b_2 : n_1, n_2 \in \mathbb{Z}_{\geq 0} \}. \tag{393}$$

In particular, the ground-to-first-distinct-level spacing is

$$\operatorname{gap}(\Delta_{\mathsf{E},T}) = 2 \min(b_1, b_2). \tag{394}$$

No multiplicity formula is needed for, or asserted in, these identities.

Proof. In local unitary trivializations, the integrand in (391) is the squared norm of a first-order unitary covariant derivative. A finite trivializing cover and a partition of unity show that the form norm $\|f\|^2 + q_{\mathsf{E},T}[f]$ is equivalent to the usual local H^1 norm. Its completion is therefore $H^1(F_T; \mathscr{L}_{\mathsf{E},T})$, proving closability and identifying the closed domain. The representation theorem for closed nonnegative forms gives the Friedrichs operator. Its local principal symbol is $G_T^{-1}(\xi, \xi)$, so it is uniformly elliptic. Standard interior elliptic regularity on the compact manifold without boundary identifies the operator domain with H^2. The compact inclusion $H^1 \hookrightarrow L^2$ implies compact resolvent.

The real skew-normal-form theorem supplies a constant orthonormal frame $v_1, \dots, v_4$ on the Euclidean cover in which the real two-form $2\pi\mathsf{E}$ is

$$b_1 v^1 \wedge v^2 + b_2 v^3 \wedge v^4.$$

Changing the sign of v^2 or v^4, if necessary, makes both frequencies positive. Let ∇_a denote covariant differentiation along v_a. Each ∇_a is formally skew-adjoint on the common invariant core $\mathcal{C}_{\mathsf{E},T}$; the closures used below are the ones determined by the closed magnetic form. On that smooth core one has

$$[\nabla_1, \nabla_2] = ib_1, \qquad [\nabla_3, \nabla_4] = ib_2,$$

and all other commutators vanish. Put

$$C_1 = \nabla_1 - i\nabla_2, \qquad C_2 = \nabla_3 - i\nabla_4, \qquad \Delta_j = -(\nabla_{2j-1}^2 + \nabla_{2j}^2).$$

Direct multiplication, retaining the commutator signs, gives

$$C_j^* C_j = \Delta_j - b_j, \qquad C_j C_j^* = \Delta_j + b_j, \qquad [\Delta_{\mathsf{E},T}, C_j] = -2b_j C_j. \tag{395}$$

Because the resolvent is compact, the lowest eigenvalue has a smooth eigenfunction $\psi_0 \neq 0$. If $C_j\psi_0 \neq 0$, the last identity in (395) would make it an eigenfunction with a strictly smaller eigenvalue. Hence $C_1\psi_0 = C_2\psi_0 = 0$, and the first identity gives

$$\Delta_{\mathsf{E},T}\psi_0 = (b_1 + b_2)\psi_0.$$

Conversely, repeated application of C_1^*, C_2^* produces nonzero eigenfunctions with eigenvalues $(2n_1+1)b_1 + (2n_2+1)b_2$. Nonvanishing follows from $C_jC_j^* = C_j^*C_j + 2b_j$.

It remains to exclude other eigenvalues. For an eigenfunction ψ with eigenvalue $\lambda > b_1 + b_2$, the operator identity

$$(\lambda - b_1 - b_2)\psi = C_1^*C_1\psi + C_2^*C_2\psi$$

expresses ψ as a sum of raising operators applied to the lower eigenfunctions $C_j\psi$. Induction over the finite set of eigenvalues below λ reduces every eigenfunction to raised ground states. This proves (393). Raising once in a least-frequency plane proves (394). □

Corollary 12.6 (Frequency product and cusp spacing bound)**.** *Let*

$$d = |\operatorname{Pf}(\mathsf{E})| = |E_{12}E_{34} - E_{13}E_{24} + E_{14}E_{23}| > 0.$$

Then

$$b_1 b_2 = \frac{4\pi^2 d}{D_T}, \tag{396}$$

and

$$\begin{aligned} \operatorname{gap}(\Delta_{\mathsf{E},T}) &\le 4\pi\sqrt{\frac{d}{D_T}} \\ &= \frac{4\pi\sqrt{d}}{T}\left(1 + \frac{B - 2H}{2T} + O(T^{-2})\right). \end{aligned} \tag{397}$$

Proof. The determinant of the skew matrix in (392) is both $b_1^2 b_2^2$ and

$$\frac{\det(2\pi\mathsf{E})}{\det G_T} = \frac{(2\pi)^4 \operatorname{Pf}(\mathsf{E})^2}{D_T^2}.$$

Taking the positive square root proves (396). Since $\min(b_1, b_2) \le \sqrt{b_1 b_2}$, Theorem 12.5 gives (397). Expanding the square root of equation (33) gives the displayed first correction. □

12.5 A background-gauge negative mode

We finish with the classical background-gauge Jacobi operator at the reducible critical connection. Every sign below uses exactly the inner product and the element H_{mag} fixed in equation (381).

Put

$$\mathscr{H}_T^{\mathrm{ad},1} := L^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_T).$$

Put also

$$\mathscr{H}_T^{\mathrm{ad},0} := L^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_T), \qquad D_T^{\max} := \mathrm{d}_{\nabla^T}^{*,\max}.$$

For the resolvent argument these real Hilbert spaces are complexified; all operators commute with the real structure, so their restrictions to real sections are unchanged. Here the maximal distributional adjoint is the closed, densely defined operator whose domain and value are specified without an implicit regularity assumption by

$$\begin{aligned}\operatorname{Dom}(D_T^{\max}) := \{a \in \mathscr{H}_T^{\mathrm{ad},1} :\ &\text{there is a } u \in \mathscr{H}_T^{\mathrm{ad},0} \text{ such that}\\ &\langle a, \mathrm{d}_{\nabla^T}\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \text{ for every } \phi \in C^\infty(F_T; \operatorname{ad}\mathscr{E}_T)\},\\ D_T^{\max}a := u.&\end{aligned} \tag{398}$$

The test space is dense, so u is unique. The definition also proves closedness directly: if $a_n \to a$ and $D_T^{\max}a_n \to u$ in L^2, passage to the limit in the displayed testing identity gives $a \in \operatorname{Dom}(D_T^{\max})$ and $D_T^{\max}a = u$. Consequently the background slice

$$\mathscr{S}_T := \ker D_T^{\max} \subset \mathscr{H}_T^{\mathrm{ad},1}. \tag{399}$$

is a closed subspace.

The scalar covariant Laplacian used below is fixed as follows. On $C^\infty(F_T; \operatorname{ad}\mathscr{E}_T)$ start with the nonnegative form

$$q_T^{(0)}[\phi, \psi] := \langle \mathrm{d}_{\nabla^T}\phi, \mathrm{d}_{\nabla^T}\psi\rangle_{L^2}.$$

Its closure has domain $H^1\Omega^0(F_T; \operatorname{ad}\mathscr{E}_T)$, and its associated Friedrichs operator is

$$\Delta_{\nabla^T}^{(0)} := \mathrm{d}_{\nabla^T}^*\mathrm{d}_{\nabla^T}, \qquad \operatorname{Dom}(\Delta_{\nabla^T}^{(0)}) = H^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_T). \tag{400}$$

The H^2 domain follows from elliptic regularity on the compact flat torus; in particular, this notation always denotes the nonnegative self-adjoint realization, not merely its differential expression.

For an adjoint-valued one-form a, define the smooth bundle endomorphism

$$(V_T a)_\nu := -2\sum_{\mu=1}^{4}[F_{\nu\mu}, a_\mu].$$

It is symmetric: after interchanging μ, ν, the identities $F_{\mu\nu} = -F_{\nu\mu}$ and skew-adjointness of $\operatorname{ad}(F_{\nu\mu})$ give $\langle V_T a, b\rangle = \langle a, V_T b\rangle$. On the full one-form Hilbert space define

$$\mathfrak{j}_T^{\mathrm{gf}}[a, b] := \langle \nabla^T a, \nabla^T b\rangle_{L^2} + \langle V_T a, b\rangle_{L^2}, \qquad \operatorname{Dom}(\mathfrak{j}_T^{\mathrm{gf}}) = H^1\Omega^1(F_T; \operatorname{ad}\mathscr{E}_T). \tag{401}$$

Because V_T is a bounded symmetric endomorphism on the compact torus, this form is closed and bounded below. Its associated self-adjoint operator $\mathcal{J}_T^{\mathrm{gf}}$ has

$$\operatorname{Dom}(\mathcal{J}_T^{\mathrm{gf}}) = H^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_T) \tag{402}$$

and differential expression

$$(\mathcal{J}_T^{\mathrm{gf}} a)_\nu = -\sum_{\mu=1}^{4}\nabla_{e_\mu}\nabla_{e_\mu}a_\nu - 2\sum_{\mu=1}^{4}[F_{\nu\mu}, a_\mu].$$

Proposition 12.7 (Intertwining and the reducing Coulomb slice)**.** *At the Yang–Mills connection ∇^T, the following formal identities hold on smooth sections:*

$$\mathcal{J}_T^{\mathrm{gf}}\mathrm{d}_{\nabla^T} = \mathrm{d}_{\nabla^T}\Delta^{(0)}_{\nabla^T}, \qquad \mathrm{d}^*_{\nabla^T}\mathcal{J}_T^{\mathrm{gf}} = \Delta^{(0)}_{\nabla^T}\mathrm{d}^*_{\nabla^T}. \tag{403}$$

For every $z \in \mathbb{C}\setminus\mathbb{R}$, the correctly typed closed-operator resolvent identity is

$$D_T^{\max}(\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f = (\Delta^{(0)}_{\nabla^T} - z)^{-1} D_T^{\max} f, \qquad f \in \mathrm{Dom}(D_T^{\max}). \tag{404}$$

In particular, $\mathscr{S}_T$ reduces $\mathcal{J}_T^{\mathrm{gf}}$. The part of the full operator in this subspace and its closed form are therefore

$$\mathcal{J}_T := \mathcal{J}_T^{\mathrm{gf}}\big|_{\mathscr{S}_T}, \quad \mathrm{Dom}(\mathcal{J}_T) = H^2\Omega^1(F_T;\mathrm{ad}\,\mathscr{E}_T)\cap\mathscr{S}_T, \quad \mathrm{Dom}(\mathfrak{j}_T) = H^1\Omega^1(F_T;\mathrm{ad}\,\mathscr{E}_T)\cap\mathscr{S}_T \tag{405}$$

with $\mathfrak{j}_T = \mathfrak{j}_T^{\mathrm{gf}}|_{\mathrm{Dom}(\mathfrak{j}_T)}$. The operator $\mathcal{J}_T$ is self-adjoint on $\mathscr{S}_T$, and $\mathfrak{j}_T$ is its closed, lower-bounded form.

Proof. Write $\nabla_\mu = \nabla^T_{e_\mu}$ in the constant orthonormal frame, so $[\nabla_\mu,\nabla_\nu]\phi = [F_{\mu\nu},\phi]$. For a smooth adjoint-valued scalar ϕ, commuting the derivatives without suppressing either curvature contribution gives

$$\begin{aligned}
&\big(\mathcal{J}_T^{\mathrm{gf}}\mathrm{d}_{\nabla^T}\phi - \mathrm{d}_{\nabla^T}\Delta^{(0)}_{\nabla^T}\phi\big)_\nu \\
&\quad = -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu},\nabla_\mu\phi] \\
&\quad = -[\nabla_\mu F_{\mu\nu},\phi] = [(\mathrm{d}^*_{\nabla^T}F_{\nabla^T})_\nu,\phi].
\end{aligned}$$

The last member vanishes by theorem 12.4. This proves the first identity in (403). Since both Laplacians and $\mathcal{J}_T^{\mathrm{gf}}$ are formally symmetric, pairing it against arbitrary smooth zero- and one-forms proves the second identity.

Fix $z \notin \mathbb{R}$, take $f \in \mathrm{Dom}(D_T^{\max})$, and put $u = (\mathcal{J}_T^{\mathrm{gf}} - z)^{-1}f$. The full operator domain gives $u \in H^2$, hence $D_T^{\max}u \in H^1$. Moreover $(\mathcal{J}_T^{\mathrm{gf}} - z)u = f$ and the distributional extension of the second formal identity imply

$$(\Delta^{(0)}_{\nabla^T} - z)D_T^{\max}u = D_T^{\max}f \quad \text{in } \mathcal{D}'(F_T;\mathrm{ad}\,\mathscr{E}_T). \tag{406}$$

The right side is in L^2. Elliptic regularity for the scalar Laplacian upgrades $D_T^{\max}u$ from H^1 to H^2, so it belongs to $\mathrm{Dom}(\Delta^{(0)}_{\nabla^T})$. Applying the bounded scalar resolvent to (406) proves (404); no extension of $D_T^{\max}$ to all of L^2 is asserted or needed.

If $f \in \mathscr{S}_T$, the right side of (404) is zero, hence both resolvents at z and at $\overline{z}$ preserve $\mathscr{S}_T$. For $g \in \mathscr{S}_T^{\perp}$ and $s \in \mathscr{S}_T$, self-adjointness of the full operator gives

$$\langle(\mathcal{J}_T^{\mathrm{gf}} - z)^{-1}g, s\rangle = \langle g, (\mathcal{J}_T^{\mathrm{gf}} - \overline{z})^{-1}s\rangle = 0.$$

Thus the resolvent preserves both $\mathscr{S}_T$ and its orthogonal complement, which is precisely the reducing property. The standard part of a self-adjoint operator on a reducing subspace is self-adjoint, and its domain is the intersection in (405). The reducing orthogonal projection commutes with the spectral calculus of the lower-bounded full operator, including its form square root; this proves the asserted form domain and completes the proof. □

Theorem 12.8 (Exact charged Landau negative mode)**.** *For the self-adjoint Yang–Mills Jacobi operator $\mathcal{J}_T$ on the closed background slice $\mathscr{S}_T$ defined above, the differential expression in a constant orthonormal frame $(e_\mu)_{\mu=1}^4$ is*

$$(\mathcal{J}_T a)_\nu = -\sum_{\mu=1}^{4}\nabla_{e_\mu}\nabla_{e_\mu}a_\nu - 2\sum_{\mu=1}^{4}[F_{\nu\mu},a_\mu]. \tag{407}$$

There exists a nonzero smooth real $\mathfrak{su}(2)$-valued one-form a such that

$$\|a\|_{L^2} = 1, \qquad \mathrm{d}^*_{\nabla^T} a = 0, \qquad \mathcal{J}_T a = -\frac{4\pi}{D_T} a. \tag{408}$$

Equivalently, for the action normalization equation (383)*,*

$$\mathrm{YM}''_T\big|_{\nabla^T}(a,a) = -\frac{4\pi}{g^2_{\mathrm{YM}} D_T}. \tag{409}$$

Thus the reducible non-flat critical connection has a strictly negative classical second-variation direction for every finite T.

Proof. First isolate the exact two-dimensional magnetic problem. Let

$$W = \{x_1 = x_2 = 0\} \subset \mathbb{R}^4,$$

so that (y_1, y_2) are oriented orthonormal coordinates on W. By equation (13), orthogonal projection to W sends the first two marked lattice generators to

$$v_1 = (6m_T, -q_T), \qquad v_2 = (L_T, m_T),$$

and sends the last two to zero. Since $\det(v_1, v_2) = D_T > 0$, projection descends to

$$\varpi_T : F_T \longrightarrow \Sigma_T := W/(\mathbb{Z}v_1 + \mathbb{Z}v_2).$$

The factors and connection of $\mathscr{L}^2_T$ depend only on (u_1, u_2). They therefore define a Hermitian line bundle with unitary connection $(\mathscr{M}_T, \nabla^{\mathscr{M}_T})$ on Σ_T such that

$$(\mathscr{L}^2_T, \nabla^{\mathscr{L}^2_T}) = \varpi_T^*(\mathscr{M}_T, \nabla^{\mathscr{M}_T}). \tag{410}$$

Let $\Delta_+ = (\nabla^{\mathscr{M}_T})^* \nabla^{\mathscr{M}_T}$ be the nonnegative self-adjoint Bochner Laplacian on $L^2(\Sigma_T; \mathscr{M}_T)$, defined by its closed H^1 form and with operator domain $H^2(\Sigma_T; \mathscr{M}_T)$. More precisely, if $c_T > 0$ is the constant Riemannian volume of a fibre of ϖ_T, then

$$U_T : L^2(\Sigma_T; \mathscr{M}_T) \longrightarrow L^2(F_T; \mathscr{L}^2_T), \qquad U_T s = c_T^{-1/2} \varpi_T^* s,$$

is an isometry. Its image

$$\mathscr{B}_T := U_T L^2(\Sigma_T; \mathscr{M}_T)$$

is the closed fibre-basic subspace. The pulled-back realization is, by definition,

$$\Delta_{+,\mathrm{bas}} := U_T \Delta_+ U_T^{-1} \quad \text{on } \mathscr{B}_T, \qquad \mathrm{Dom}(\Delta_{+,\mathrm{bas}}) = U_T H^2(\Sigma_T; \mathscr{M}_T),$$

and its closed form domain is $U_T H^1(\Sigma_T; \mathscr{M}_T)$. Only this operator on $\mathscr{B}_T$ is self-adjoint. In the formulas below we abbreviate $\Delta_{+,\mathrm{bas}}$ again to Δ_+; no self-adjoint operator on the full $L^2(F_T; \mathscr{L}^2_T)$ is thereby defined or asserted. On this root line the curvature is

$$F_{\mathscr{L}^2_T} = 4\pi i\, \theta^1_T \wedge \theta^2_T = i\kappa_T \,\mathrm{d}y_1 \wedge \mathrm{d}y_2, \qquad \kappa_T = \frac{4\pi}{D_T} > 0. \tag{411}$$

We now construct an exact lowest Landau section on this actual projected lattice. In the marked coordinates put

$$\mathcal{D}_1 := \partial_{u_1}, \qquad \mathcal{D}_2 := \partial_{u_2} + 4\pi i u_1,$$

and define

$$\zeta_T := \frac{q_T - 6im_T}{m_T + iL_T}, \qquad v_T := -\operatorname{Im} \zeta_T = \frac{D_T}{m_T^2 + L_T^2} > 0. \tag{412}$$

Indeed,

$$\operatorname{Im}((q_T - 6im_T)(m_T - iL_T)) = -(L_T q_T + 6m_T^2) = -D_T,$$

which also checks the sign $v_T > 0$. The series

$$\Theta_T(u_1, u_2) = \sum_{\ell \in \mathbb{Z}} \exp\Big(-2\pi i \zeta_T (\ell + u_1)^2 + 4\pi i \ell u_2\Big) \tag{413}$$

converges normally with all derivatives, because the absolute value of its ℓ-th summand is $\exp(-2\pi v_T(\ell + u_1)^2)$. Reindexing the series and using $\ell \in \mathbb{Z}$ give

$$\Theta_T(u_1 + 1, u_2) = e^{-4\pi i u_2}\Theta_T(u_1, u_2), \qquad \Theta_T(u_1, u_2 + 1) = \Theta_T(u_1, u_2). \tag{414}$$

Thus Θ_T is a smooth section of $\mathscr{L}_T^2$. Termwise differentiation gives

$$(\mathcal{D}_1 + \zeta_T \mathcal{D}_2)\Theta_T = 0. \tag{415}$$

The exact period block on the (y_1, y_2)-plane is

$$\begin{pmatrix} 6m_T & L_T \\ -q_T & m_T \end{pmatrix}.$$

Its inverse transpose gives the physical covariant derivatives

$$\nabla^+_{y_1} = \frac{m_T \mathcal{D}_1 + q_T \mathcal{D}_2}{D_T}, \qquad \nabla^+_{y_2} = \frac{-L_T \mathcal{D}_1 + 6m_T \mathcal{D}_2}{D_T}. \tag{416}$$

Consequently

$$\nabla^+_{y_1} - i\nabla^+_{y_2} = \frac{m_T + iL_T}{D_T}(\mathcal{D}_1 + \zeta_T \mathcal{D}_2). \tag{417}$$

Orthogonality of the u_2-Fourier modes also gives the nonzero norm

$$\int_0^1 \int_0^1 |\Theta_T|^2 \, \mathrm{d}u_1 \, \mathrm{d}u_2 = \sum_{\ell \in \mathbb{Z}} \int_0^1 e^{-4\pi v_T (\ell + u_1)^2} \, \mathrm{d}u_1 = \frac{1}{2\sqrt{v_T}}.$$

Set

$$\phi_T := \left(\frac{2\sqrt{v_T}}{D_T}\right)^{1/2} \Theta_T$$

and pull it back by ϖ_T. Since the marked-coordinate volume element is $D_T \, \mathrm{d}^4 u$, it has unit $L^2(F_T)$-norm. Its covariant derivatives in the x_1, x_2 directions vanish, and

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})\phi_T = 0, \qquad \Delta_+ \phi_T = \kappa_T \phi_T, \qquad \Delta_+ \phi_T = -\big((\nabla^+_{y_1})^2 + (\nabla^+_{y_2})^2\big)\phi_T. \tag{418}$$

Indeed, equation (416) and $[\mathcal{D}_1, \mathcal{D}_2] = 4\pi i$ give

$$[\nabla^+_{y_1}, \nabla^+_{y_2}] = \frac{m_T(6m_T) - q_T(-L_T)}{D_T^2} \, 4\pi i = i\kappa_T.$$

Direct multiplication then gives

$$(\nabla^+_{y_1} - i\nabla^+_{y_2})^*(\nabla^+_{y_1} - i\nabla^+_{y_2}) = \Delta_+ - \kappa_T,$$

and the second equation in (418) follows from the first.

In $\mathfrak{sl}_2(\mathbb{C})$, put

$$X_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad X_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$[H_{\mathrm{mag}}, X_+] = 2iX_+, \qquad [H_{\mathrm{mag}}, X_-] = -2iX_-.$$

The X_+-root line in $\mathrm{ad}(\mathscr{L}_T \oplus \mathscr{L}_T^{-1}) \otimes \mathbb{C}$ is $\mathscr{L}_T^2$, so

$$a^+ = \phi_T X_+(\mathrm{d}y_1 - i\,\mathrm{d}y_2)$$

is a globally defined complexified adjoint-valued one-form. The first equation in (418) gives

$$\mathrm{d}^*_{\nabla^T} a^+ = -(\nabla^+_{y_1} - i\nabla^+_{y_2})\phi_T\, X_+ = 0.$$

Put $\mathcal{B}_T := 2\pi/D_T$, so that $\kappa_T = 2\mathcal{B}_T$ and $F_{y_1y_2} = \mathcal{B}_T H_{\mathrm{mag}}$. The two nonzero components of a^+ are $a^+_{y_1} = \phi_T X_+$ and $a^+_{y_2} = -i\phi_T X_+$. Substitution into (407), with no suppressed charge factor, gives

$$\begin{aligned}(\mathcal{J}_T a^+)_{y_1} &= \kappa_T\phi_T X_+ - 2[\mathcal{B}_T H_{\mathrm{mag}}, -i\phi_T X_+] = -\kappa_T\phi_T X_+,\\ (\mathcal{J}_T a^+)_{y_2} &= -i\kappa_T\phi_T X_+ - 2[-\mathcal{B}_T H_{\mathrm{mag}}, \phi_T X_+] = i\kappa_T\phi_T X_+.\end{aligned}$$

Thus $\mathcal{J}_T a^+ = -\kappa_T a^+$.

For completeness, formula (407) follows by expanding $F_{\nabla^T + ta} = F_{\nabla^T} + t\,\mathrm{d}_{\nabla^T} a + \frac{1}{2}t^2[a \wedge a]$, integrating the quadratic form by parts on the background slice, and using the flat-base Weitzenbock identity. The Hodge Laplacian contributes $-[F_{\nu\mu}, a_\mu]$, and the curvature term in the second variation contributes the identical term, giving the coefficient -2.

Finally take the anti-Hermitian real form

$$a := \frac{1}{\sqrt{2}}(a^+ - (a^+)^\dagger) = \frac{1}{\sqrt{2}}\left(\phi_T X_+(\mathrm{d}y_1 - i\,\mathrm{d}y_2) - \overline{\phi_T} X_-(\mathrm{d}y_1 + i\,\mathrm{d}y_2)\right). \tag{419}$$

It is nonzero, traceless, and anti-Hermitian. For $X(z) := zX_+ - \overline{z}X_-$, the fixed inner product gives $-\frac{1}{2}\operatorname{Tr}(X(z)^2) = |z|^2$. The two orthonormal one-form components in equation (419) therefore have total pointwise norm $|a|^2 = |\phi_T|^2$, so $\|a\|_{L^2} = 1$. Complex conjugation and Hermitian adjunction preserve both the gauge equation and the real Jacobi operator, so (408) follows from the calculation for a^+. The Hessian identity follows from $\mathrm{YM}''_T(a,a) = g_{\mathrm{YM}}^{-2}\langle a, \mathcal{J}_T a\rangle_{L^2}$ on the background slice. □

13 The native full-flux magnetic sector

This section retains the canonical radial parameter, period matrix, Euclidean quotient metric, marked lattice, and every sign already fixed in section 9. It treats the monodromy-invariant integral class that uses both elementary alternating blocks. It is therefore distinct from the unit-flux plane treated in section 12. No metric rescaling, finite covering, rectangular replacement of the lattice, or orientation suppression occurs below.

For $T > T_0$, retain

$$\begin{aligned} L_T = T + H_T = \operatorname{Im}\tau_T, \qquad q_T = L_T - B_T = -\operatorname{Im}\beta_T,\\ m_T = \operatorname{Im}\mu_T, \qquad D_T = L_T q_T + 6m_T^2 > 0,\end{aligned} \tag{420}$$

and write

$$\eta_T^{\mathrm{re}} := \operatorname{Re} h_T.$$

This scalar is denoted by η_T in section 9; the superscript is retained throughout this section so that it cannot be confused with the native integral two-form introduced below. The complete period matrix is

$$P_T = \begin{pmatrix} 6\alpha_T & \eta_T^{\mathrm{re}} & 1 & 0\\ 6m_T & L_T & 0 & 0\\ c_T & \alpha_T & 0 & 1\\ -q_T & m_T & 0 & 0\end{pmatrix}, \qquad \det P_T = -D_T. \tag{421}$$

Thus $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$ has the unrescaled Euclidean quotient metric and volume D_T. In the marking coordinates $y \in \mathbb{R}^4$, the map $y \mapsto x = P_T y$ pulls the Euclidean metric back to $G_T = P_T^{\mathsf{T}} P_T$.

The limiting cusp constants remain

$$m = \operatorname{Im}\mu(0), \qquad H = \operatorname{Im} h(0), \qquad B = \operatorname{Im} b(0), \tag{422}$$

so holomorphy at $e^{-2\pi T} = 0$ gives

$$\begin{aligned} m_T &= m + O(e^{-2\pi T}), \quad H_T = H + O(e^{-2\pi T}), \\ B_T &= B + O(e^{-2\pi T}), \quad D_T = T^2 + (2H - B)T \\ &\qquad + (H^2 - BH + 6m^2) + O(Te^{-2\pi T}). \end{aligned} \tag{423}$$

13.1 The invariant class, its cocycle, and both orientations

In the ordered marking $\mathbb{Z}\langle\gamma, u, w, \delta\rangle$, put

$$\eta = u \wedge w + 6\gamma \wedge \delta. \tag{424}$$

The two finite monodromy generators act by

$$\begin{aligned} &T_1\gamma = \gamma, && T_1 u = -u - w, && T_1 w = -6\gamma + u, && T_1\delta = 2\gamma + u + w + \delta, \\ &T_2\gamma = \gamma, && T_2 u = 6\gamma + w, && T_2 w = -u, && T_2\delta = -3\gamma + u + \delta. \end{aligned}$$

Directly, without passing to a normalized representative,

$$\begin{aligned} (\textstyle\bigwedge^2 T_1)\eta &= (-u - w) \wedge (-6\gamma + u) + 6\gamma \wedge (2\gamma + u + w + \delta) \\ &= u \wedge w + 6\gamma \wedge \delta, \\ (\textstyle\bigwedge^2 T_2)\eta &= (6\gamma + w) \wedge (-u) + 6\gamma \wedge (-3\gamma + u + \delta) \\ &= u \wedge w + 6\gamma \wedge \delta. \end{aligned}$$

Hence η is invariant under both generators.

On $Y = \mathbb{R}^4/\mathbb{Z}^4$, with ordered coordinates y^1, y^2, y^3, y^4, its constant representative and alternating matrix are

$$\eta_T = \mathrm{d}y^2 \wedge \mathrm{d}y^3 + 6\,\mathrm{d}y^1 \wedge \mathrm{d}y^4, \qquad E_\eta = \begin{pmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -6 & 0 & 0 & 0 \end{pmatrix}. \tag{425}$$

In the paired basis $(e_2, e_3; e_1, e_4)$, the alternating elementary divisors are $(1, 6)$; consequently

$$\operatorname{SNF}(E_\eta) = \operatorname{diag}(1, 1, 6, 6), \qquad \operatorname{Pf}(E_\eta) = 6, \qquad \det E_\eta = 36. \tag{426}$$

The actual Hermitian line bundle is the quotient of $\mathbb{R}^4 \times \mathbb{C}$ by

$$\begin{aligned} &(y, z) \longmapsto (y + e_1, e^{12\pi i y^4} z), && (y, z) \longmapsto (y + e_2, e^{2\pi i y^3} z), \\ &(y, z) \longmapsto (y + e_3, z), && (y, z) \longmapsto (y + e_4, z). \end{aligned} \tag{427}$$

The only possibly nontrivial generator commutators acquire the factors $e^{12\pi i} = 1$ and $e^{2\pi i} = 1$, so these maps define a $\mathbb{Z}^4$-action. Denote the quotient by $\mathscr{L}_{\eta,T} \to F_T$, using the marking $[y] \mapsto [P_T y]$. The connection

$$\nabla_{\eta,T} = \mathrm{d} - 2\pi i(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3) \tag{428}$$

descends through the displayed cocycle. For instance translation by e_1 changes its connection form by $-12\pi i\,\mathrm{d}y^4$, which equals the negative logarithmic derivative of $e^{12\pi i y^4}$; the e_2 check is identical. Therefore

$$F_{\nabla_{\eta,T}} = -2\pi i \eta_T, \qquad c_1(\mathscr{L}_{\eta,T}) = \frac{i}{2\pi} F_{\nabla_{\eta,T}} = \eta_T. \tag{429}$$

There are two opposite orientations which must not be conflated. Write

$$\mathrm{or}_{\mathrm{mark}} = \mathrm{d}y^1 \wedge \mathrm{d}y^2 \wedge \mathrm{d}y^3 \wedge \mathrm{d}y^4 \tag{430}$$

and let or_x be the native complex orientation of $(z_1, z_2) = (x_1 + iy_1, x_2 + iy_2)$. Then

$$P_T^* \,\mathrm{or}_x = -D_T \,\mathrm{or}_{\mathrm{mark}}, \tag{431}$$

whereas

$$\frac{1}{2}\eta_T^2 = 6\,\mathrm{or}_{\mathrm{mark}}\,. \tag{432}$$

Consequently

$$\frac{1}{2}\langle \eta_T^2, [F_T]_{\mathrm{mark}} \rangle = 6, \qquad \frac{1}{2}\langle \eta_T^2, [F_T]_{x\text{-complex}} \rangle = -6. \tag{433}$$

13.2 The exact physical frequencies

In the physical Euclidean coordinates the two-form matrix is

$$\Omega_T = P_T^{-\mathsf{T}} E_\eta P_T^{-1} = \frac{1}{D_T} \begin{pmatrix} 0 & -q_T & 0 & -6m_T \\ q_T & 0 & 6m_T & 0 \\ 0 & -6m_T & 0 & 6L_T \\ 6m_T & 0 & -6L_T & 0 \end{pmatrix}. \tag{434}$$

After the exact permutation $(x^1, x^2, x^3, x^4) \mapsto (x^1, x^3; x^2, x^4)$, its off-diagonal block is

$$K_T = \begin{pmatrix} -q_T & -6m_T \\ -6m_T & 6L_T \end{pmatrix}, \qquad \det K_T = -6D_T. \tag{435}$$

Define the magnetic scalars

$$S_T := 6L_T - q_T, \qquad \Xi_T := \sqrt{(6L_T + q_T)^2 + 144 m_T^2}. \tag{436}$$

Here S_T has no argument and is distinct from the affine residual $S_T(k)$ used in the dual-lattice calculation of theorem 9.1. The signed eigenvalues of K_T are

$$\kappa_{T,\pm} = \frac{S_T \pm \Xi_T}{2}. \tag{437}$$

Direct expansion, with no asymptotic replacement, gives

$$\Xi_T^2 - S_T^2 = 24(L_T q_T + 6m_T^2) = 24 D_T > 0. \tag{438}$$

Thus $\kappa_{T,+} > 0 > \kappa_{T,-}$. The two positive charge-one magnetic frequencies are

$$\boxed{\mathcal{B}_{T,+} = \frac{\pi(S_T + \Xi_T)}{D_T}, \qquad \mathcal{B}_{T,-} = \frac{\pi(\Xi_T - S_T)}{D_T}.} \tag{439}$$

They obey the exact identities

$$\mathcal{B}_{T,+}\mathcal{B}_{T,-} = \frac{24\pi^2}{D_T}, \tag{440}$$

$$\mathcal{B}_{T,+} + \mathcal{B}_{T,-} = \frac{2\pi\Xi_T}{D_T}, \qquad \mathcal{B}_{T,+} - \mathcal{B}_{T,-} = \frac{2\pi S_T}{D_T}, \tag{441}$$

$$\mathcal{B}_{T,+} = \frac{24\pi}{\Xi_T - S_T}, \qquad \mathcal{B}_{T,-} = \frac{24\pi}{\Xi_T + S_T}. \tag{442}$$

The last line follows by multiplying by the conjugate factor and using equation (438).

Keeping the three independent limiting coefficients H, B, m, one obtains

$$\mathcal{B}_{T,-} = \frac{2\pi}{T} - \frac{2\pi H}{T^2} + \frac{2\pi}{T^3}\left(H^2 - \frac{6}{7}m^2\right) + O(T^{-4}), \tag{443}$$

$$\mathcal{B}_{T,+} = \frac{12\pi}{T} + \frac{12\pi(B-H)}{T^2} + \frac{12\pi}{T^3}\left((B-H)^2 - \frac{36}{7}m^2\right) + O(T^{-4}), \tag{444}$$

$$\rho_T := \frac{\mathcal{B}_{T,+}}{\mathcal{B}_{T,-}} = 6 + \frac{6B}{T} + \frac{6B^2 - 6BH - 180m^2/7}{T^2} + O(T^{-3}). \tag{445}$$

These follow by substituting equation (423) into equation (442) and expanding the reciprocals; no sign of H, B, m is assumed. In particular, on a sufficiently deep cusp tail,

$$S_T > 0, \qquad \mathcal{B}_{T,+} > \mathcal{B}_{T,-} > 0, \qquad 5 < \rho_T < 7. \tag{446}$$

13.3 Joint Landau levels and their exact multiplicities

Let $\Delta_{\eta,T} = \nabla^*_{\eta,T}\nabla_{\eta,T}$ be the Friedrichs magnetic Laplacian on $L^2(F_T, \mathscr{L}_{\eta,T})$. Its form domain is $H^1(F_T; \mathscr{L}_{\eta,T})$, its operator domain is $H^2(F_T; \mathscr{L}_{\eta,T})$, and compactness of the torus gives compact resolvent. We now prove the multiplicity rather than infer it from the Pfaffian alone.

Define A^{pol}_T on the marked tangent space by

$$G_T(A^{\mathrm{pol}}_T v, w) = 2\pi E_\eta(v, w), \qquad Q^{\mathrm{pol}}_T = \left(-(A^{\mathrm{pol}}_T)^2\right)^{1/2}, \qquad J_{\eta,T} = A^{\mathrm{pol}}_T (Q^{\mathrm{pol}}_T)^{-1}. \tag{447}$$

The endomorphism A^{pol}_T is G_T-skew-adjoint and invertible, so Q^{pol}_T is positive, commutes with A^{pol}_T, and

$$J^2_{\eta,T} = -1, \qquad E_\eta(J_{\eta,T}v, J_{\eta,T}w) = E_\eta(v, w), \qquad 2\pi E_\eta(v, J_{\eta,T}v) = G_T(Q^{\mathrm{pol}}_T v, v) > 0. \tag{448}$$

For every signed charge $q \in \mathbb{Z} \setminus \{0\}$, use the convention

$$\mathscr{L}^{\otimes q}_{\eta,T} = \begin{cases} (\mathscr{L}_{\eta,T})^{\otimes q}, & q > 0, \\ (\mathscr{L}^\vee_{\eta,T})^{\otimes(-q)}, & q < 0, \end{cases} \qquad A^{\mathrm{pol}}_{q,T} = qA^{\mathrm{pol}}_T, \qquad Q^{\mathrm{pol}}_{q,T} = \left(-(A^{\mathrm{pol}}_{q,T})^2\right)^{1/2} = |q| Q^{\mathrm{pol}}_T, \tag{449}$$

and define

$$J_{q,T} = A^{\mathrm{pol}}_{q,T}(Q^{\mathrm{pol}}_{q,T})^{-1} = \operatorname{sgn}(q) J_{\eta,T}. \tag{450}$$

All these coefficients are constant, hence the Nijenhuis tensor of $J_{q,T}$ vanishes and the marking transports it to a complex structure on F_T. The induced connection $\nabla^{(q)}_{\eta,T}$ has

$$F_{\nabla^{(q)}_{\eta,T}} = -2\pi i q \eta_T, \qquad 2\pi q E_\eta(v, J_{q,T}v) = |q| G_T(Q^{\mathrm{pol}}_T v, v) > 0. \tag{451}$$

Thus its curvature is of type $(1,1)$, and $(\nabla^{(q)}_{\eta,T})^{0,1}$ makes $\mathscr{L}^{\otimes q}_{\eta,T}$ a positive Appell–Humbert line bundle on the auxiliary complex torus $(F_T, J_{q,T})$.

The auxiliary orientation is fixed, rather than repaired by an absolute value. For $q > 0$, compatibility identifies the $J_{q,T}$-complex orientation with the E_η-symplectic orientation; for $q < 0$, replacing J by $-J$ preserves complex orientation in complex dimension two. Hence

$$[F_T]_{J_{q,T}} = [F_T]_{\mathrm{mark}} = -[F_T]_{x\text{-complex}}. \tag{452}$$

This auxiliary positive structure is not the native complex structure inherited from the S^6 construction. Indeed, with $E_\eta = \operatorname{Im} H_{\eta,T}$, the native Appell–Humbert calculation is

$$G_{\eta,T} = \begin{pmatrix} 2L_T & 2m_T \\ 2m_T & -q_T/3 \end{pmatrix}, \qquad \det G_{\eta,T} = -\frac{2}{3} D_T < 0, \qquad H_{\eta,T} = 2G_{\eta,T}^{-1}. \tag{453}$$

Consequently $H_{\eta,T}$, and also $qH_{\eta,T}$ for every nonzero q, has signature $(1,1)$. The native line bundle is holomorphic because $q\eta_T$ is of type $(1,1)$, but it is indefinite and is not the positive bundle just constructed.

The tangent and canonical bundles of the auxiliary complex torus are trivial. Kodaira vanishing and Hirzebruch–Riemann–Roch therefore give

$$\begin{aligned} &\dim_{\mathbb{C}} H^0((F_T, J_{q,T}), \mathscr{L}_{\eta,T}^{\otimes q}) = \int_{[F_T]_{J_{q,T}}} \frac{(q\eta_T)^2}{2} = \int_{[F_T]_{\mathrm{mark}}} \frac{(q\eta_T)^2}{2} = 6q^2, \\ &\dim_{\mathbb{C}} H^0((F_T, J_{\eta,T}), \mathscr{L}_{\eta,T}) = 6, \\ &\dim_{\mathbb{C}} H^0((F_T, J_{2,T}), \mathscr{L}_{\eta,T}^{\otimes 2}) = 24. \end{aligned} \tag{454}$$

No absolute value occurs in this auxiliary Hirzebruch–Riemann–Roch integral. Equivalently, the positive polarization has elementary divisors $(|q|, 6|q|)$, and the Appell–Humbert theta theorem gives their product $6q^2$; see Beauville, Theorems 2.6 (pp. 107–108) and 3.5 (p. 112). On the native complex torus the same signed Hirzebruch–Riemann–Roch calculation instead gives

$$\chi((F_T, J_{\mathrm{native},T}), \mathscr{L}_{\eta,T}^{\otimes q}) = \int_{[F_T]_{x\text{-complex}}} \frac{(q\eta_T)^2}{2} = -6q^2, \tag{455}$$

and positivity vanishing is unavailable there. Thus the positive auxiliary section count and the native signed Euler characteristic concern distinct Dolbeault problems and distinct complex orientations.

Choose a G_T-orthonormal frame $e_+, f_+ = J_{\eta,T}e_+, e_-, f_- = J_{\eta,T}e_-$ which diagonalizes Q_T^{pol}, labelled so its positive eigenvalues are $\mathcal{B}_{T,+}, \mathcal{B}_{T,-}$. Put

$$a_\sigma = \frac{\nabla_{e_\sigma} + i\nabla_{f_\sigma}}{\sqrt{2\mathcal{B}_{T,\sigma}}}, \qquad a_\sigma^* = \frac{-\nabla_{e_\sigma} + i\nabla_{f_\sigma}}{\sqrt{2\mathcal{B}_{T,\sigma}}}, \qquad \sigma \in \{+,-\}. \tag{456}$$

The curvature commutators give

$$[a_\sigma, a_\tau^*] = \delta_{\sigma\tau}, \qquad [a_\sigma, a_\tau] = [a_\sigma^*, a_\tau^*] = 0, \tag{457}$$

and, for $N_\sigma = a_\sigma^* a_\sigma$,

$$\Delta_{\eta,T} = \mathcal{B}_{T,+}(2N_+ + 1) + \mathcal{B}_{T,-}(2N_- + 1). \tag{458}$$

The simultaneous vacuum $\ker a_+ \cap \ker a_- = H^0((F_T, J_{\eta,T}), \mathscr{L}_{\eta,T})$ has complex dimension six by equation (454).

Theorem 13.1 (Complete native joint Landau decomposition)**.** *For every $n_+, n_- \in \mathbb{Z}_{\geq 0}$, the joint eigenspace*

$$\mathcal{H}_{T;n_+,n_-} = \ker(N_+ - n_+) \cap \ker(N_- - n_-) \tag{459}$$

has complex dimension six and energy

$$E_{T;n_+,n_-} = (2n_+ + 1)\mathcal{B}_{T,+} + (2n_- + 1)\mathcal{B}_{T,-}. \tag{460}$$

Moreover

$$L^2(F_T, \mathscr{L}_{\eta,T}) = \widehat{\bigoplus}_{n_+,n_-\geq 0} \mathcal{H}_{T;n_+,n_-}, \tag{461}$$

and the map from the vacuum to the joint level is the unitary

$$\frac{(a_+^*)^{n_+}(a_-^*)^{n_-}}{\sqrt{n_+!n_-!}}. \tag{462}$$

For a numerical eigenvalue, the multiplicity is six times the number of joint pairs in equation (460) *producing that number.*

Proof. The elliptic operator has compact resolvent, and the two commuting number operators preserve its finite-dimensional eigenspaces; they can therefore be diagonalized simultaneously. If $N_\sigma\psi = \lambda_\sigma\psi$, then $\lambda_\sigma \geq 0$. Repeated application of a_σ lowers the Laplacian energy by $2\mathcal{B}_{T,\sigma} > 0$, so it terminates. If $a_\sigma^r\psi \neq 0$ and $a_\sigma^{r+1}\psi = 0$, then

$$0 = \|a_\sigma^{r+1}\psi\|^2 = (\lambda_\sigma - r)\|a_\sigma^r\psi\|^2,$$

and hence $\lambda_\sigma = r \in \mathbb{Z}_{\geq 0}$. Lowering both indices reaches the simultaneous vacuum.

The commutator identities give, for a vacuum vector v,

$$\|(a_\sigma^*)^n v\|^2 = n!\|v\|^2, \qquad a_\sigma^n(a_\sigma^*)^n v = n!v.$$

On the n-th level they also give $(a_\sigma^*)^n a_\sigma^n = n!$. Operators in the two planes commute, so equation (462) is unitary with inverse $a_+^{n_+}a_-^{n_-}/\sqrt{n_+!n_-!}$. This proves the dimension, energy, and completeness assertions. Orthogonal joint eigenspaces with coincident numerical energies must be added, which proves the final sentence. □

The compact-resolvent lowering argument proves that no additional joint eigenspaces occur. Independently, this is exactly the case $m = 2$, $k = 1$ of Gordon–Kirwin–Schueth–Webb, Proposition 4.5, article pp. 9–10 (volume pp. 175–176), doi:10.1090/pspum/084/1355. Their signed tensor-power statement gives every oscillator tuple multiplicity $|k|^m V_{E_\eta} = |k|^2 \operatorname{Pf}(E_\eta) = 6k^2$, so it also supplies the external completeness check for every $k \neq 0$.

The exact ground energy and first distinct spacing are therefore

$$E_0(T) = \mathcal{B}_{T,+} + \mathcal{B}_{T,-} = \frac{2\pi\Xi_T}{D_T} = \frac{14\pi}{T} + \frac{\pi(12B - 14H)}{T^2} + O(T^{-3}), \tag{463}$$

$$g_0(T) = 2\min\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\} = \frac{2\pi(\Xi_T - |S_T|)}{D_T} = \frac{48\pi}{\Xi_T + |S_T|}. \tag{464}$$

On the deep tail equation (446), this is $g_0(T) = 2\mathcal{B}_{T,-}$, with expansion

$$g_0(T) = \frac{4\pi}{T} - \frac{4\pi H}{T^2} + \frac{4\pi}{T^3}\left(H^2 - \frac{6}{7}m^2\right) + O(T^{-4}). \tag{465}$$

Corollary 13.2 (Heat trace and exact fixed-energy count)**.** *For $t > 0$,*

$$\operatorname{Tr}(e^{-t\Delta_{\eta,T}}) = \frac{3}{2\sinh(t\mathcal{B}_{T,+})\sinh(t\mathcal{B}_{T,-})} = \frac{D_T}{16\pi^2 t^2}(1 + O_t(T^{-2})). \tag{466}$$

For $E > 0$, let $N_{\eta,T}(E)$ count complex eigenvalues not exceeding E, with multiplicity. Then

$$N_{\eta,T}(E) = \begin{cases} 0, & E < E_0(T), \\ 6 \displaystyle\sum_{n_+=0}^{\lfloor (E-E_0(T))/(2\mathcal{B}_{T,+})\rfloor} \left(1 + \left\lfloor \frac{E - E_0(T) - 2\mathcal{B}_{T,+}n_+}{2\mathcal{B}_{T,-}} \right\rfloor\right), & E \geq E_0(T). \end{cases} \tag{467}$$

For fixed $E > 0$, as $T \to \infty$,

$$N_{\eta,T}(E) = \frac{D_T}{32\pi^2}(E - E_0(T))^2 + O_E(T) = \frac{E^2 D_T}{32\pi^2} + O_E(T). \tag{468}$$

Proof. The heat trace is the product of the two geometric series in equation (460), multiplied by six. Expanding both hyperbolic sines and using equation (440) proves its asymptotic. The counting formula enumerates the lattice points

$$(n_+, n_-) \in \mathbb{R}^2_{\geq 0}, \qquad 2\mathcal{B}_{T,+}n_+ + 2\mathcal{B}_{T,-}n_- \leq E - E_0(T).$$

Its triangular area is $(E - E_0(T))^2/(8\mathcal{B}_{T,+}\mathcal{B}_{T,-})$. The difference between the lattice count and the area is bounded by a constant times the perimeter plus one, hence is $O_E(T)$. Multiplying by six and using the exact frequency product gives the first asymptotic; the second follows from $E_0(T) = O(T^{-1})$ and $D_T = O(T^2)$. □

13.4 The split $SU(2)$ critical connection

Put

$$\mathscr{E}_{\eta,T} = \mathscr{L}_{\eta,T} \oplus \mathscr{L}_{\eta,T}^{-1}, \qquad A_{\eta,T} = \nabla_{\eta,T} \oplus \nabla_{\eta,T}^{-1}. \tag{469}$$

The determinant is canonically trivial and

$$c(\mathscr{E}_{\eta,T}) = (1 + \eta_T)(1 - \eta_T) = 1 - \eta_T^2, \qquad c_2(\mathscr{E}_{\eta,T}) = -\eta_T^2. \tag{470}$$

Thus the orientation-sensitive Chern numbers are

$$\langle c_2(\mathscr{E}_{\eta,T}), [F_T]_{\text{mark}} \rangle = -12, \qquad \langle c_2(\mathscr{E}_{\eta,T}), [F_T]_{x\text{-complex}} \rangle = +12. \tag{471}$$

The cohomology class $c_2 = -\eta_T^2$ is nonzero, so this split $SU(2)$-bundle is not smoothly trivial. This is another exact distinction from the unit-flux split bundle of theorem 12.3.

With

$$H_{\text{mag}} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \langle X, Y \rangle_{\mathfrak{su}(2)} = -\frac{1}{2}\operatorname{Tr}(XY), \tag{472}$$

one has $|H_{\text{mag}}| = 1$ and

$$F_{A_{\eta,T}} = -2\pi\eta_T H_{\text{mag}}. \tag{473}$$

This curvature is parallel; hence $d^*_{A_{\eta,T}} F_{A_{\eta,T}} = 0$, and $A_{\eta,T}$ is Yang–Mills. For

$$\mathrm{YM}_T(A) = \frac{1}{2g_{\mathrm{YM}}^2} \int_{F_T} |F_A|^2 \, \mathrm{d}\,\mathrm{vol}_{g_T}, \tag{474}$$

diagonalizing equation (435) gives the exact action

$$\mathrm{YM}_T(A_{\eta,T}) = \frac{D_T}{2g_{\mathrm{YM}}^2}(\mathcal{B}_{T,+}^2 + \mathcal{B}_{T,-}^2) \tag{475}$$

$$= \frac{2\pi^2}{g_{\mathrm{YM}}^2 D_T}(q_T^2 + 36L_T^2 + 72m_T^2) \tag{476}$$

$$= \frac{24\pi^2}{g_{\mathrm{YM}}^2} + \frac{2\pi^2 S_T^2}{g_{\mathrm{YM}}^2 D_T}. \tag{477}$$

Indeed $q_T^2 + 36L_T^2 + 72m_T^2 = 12D_T + (6L_T - q_T)^2$. Consequently

$$\mathrm{YM}_T(A_{\eta,T}) = \frac{74\pi^2}{g_{\mathrm{YM}}^2} + \frac{70\pi^2 B}{g_{\mathrm{YM}}^2 T} + O(T^{-2}). \tag{478}$$

13.5 The closed Jacobi operator and exact Coulomb reduction

We next define the operator before computing its spectrum. Let

$$\mathscr{H}_T^1 = L^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_{\eta,T}), \qquad \mathscr{H}_T^0 = L^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_{\eta,T}) \tag{479}$$

be real Hilbert spaces. Their complexifications carry the root decomposition used below, and every operator constructed here commutes with the anti-linear real structure.

Define the maximal distributional codifferential by

$$\begin{aligned} \operatorname{Dom} D_T^{\max} = \{a \in \mathscr{H}_T^1 : &\text{ there is } u \in \mathscr{H}_T^0 \text{ such that} \\ &\langle a, d_{A_{\eta,T}}\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \text{ for every smooth } \phi\}, \end{aligned} \tag{480}$$

$$D_T^{\max} a = u, \qquad \mathscr{S}_T := \ker D_T^{\max}. \tag{481}$$

Uniqueness follows from density of the test space. Passing to the limit in the defining identity proves that $D_T^{\max}$ is closed, hence the Coulomb slice $\mathscr{S}_T$ is a closed subspace.

On adjoint-valued scalars, let $\Delta^{(0)}_{A_{\eta,T}}$ be the nonnegative self-adjoint operator of the closed form

$$\mathfrak{q}_T^{(0)}[\phi,\psi] = \langle d_{A_{\eta,T}}\phi, d_{A_{\eta,T}}\psi\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{q}_T^{(0)} = H^1\Omega^0. \tag{482}$$

Elliptic regularity gives

$$\Delta^{(0)}_{A_{\eta,T}} = d^*_{A_{\eta,T}} d_{A_{\eta,T}}, \qquad \operatorname{Dom}\Delta^{(0)}_{A_{\eta,T}} = H^2\Omega^0. \tag{483}$$

In a constant orthonormal frame, define the symmetric zeroth-order endomorphism

$$(V_T a)_\nu = -2\sum_{\mu=1}^{4} [F_{A_{\eta,T},\nu\mu}, a_\mu]. \tag{484}$$

Symmetry follows from invariance of the compact Lie-algebra inner product, antisymmetry of $F_{\nu\mu}$, and an exchange of ν,μ. Consider the gauge-fixed form

$$\mathfrak{j}_T^{\mathrm{gf}}[a,b] = \langle \nabla_{A_{\eta,T}} a, \nabla_{A_{\eta,T}} b\rangle_{L^2} + \langle V_T a, b\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{j}_T^{\mathrm{gf}} = H^1\Omega^1. \tag{485}$$

Proposition 13.3 (Closed gauge-fixed realization)**.** *The form equation* (485) *is densely defined, closed, symmetric, and bounded below. Its self-adjoint operator $\mathcal{J}_T^{\mathrm{gf}}$ has compact resolvent, domain $H^2\Omega^1$, and expression*

$$(\mathcal{J}_T^{\mathrm{gf}} a)_\nu = -\sum_{\mu=1}^{4} \nabla_{A_{\eta,T},e_\mu}\nabla_{A_{\eta,T},e_\mu} a_\nu - 2\sum_{\mu=1}^{4}[F_{A_{\eta,T},\nu\mu}, a_\mu]. \tag{486}$$

It is exactly the Hessian operator with the background-gauge term, before the overall factor g_{YM}^{-2}.

Proof. The first summand in equation (485), together with the L^2-norm, is an equivalent H^1-norm. The smooth endomorphism V_T is bounded and symmetric. The bounded perturbation theorem for closed forms gives closedness and a lower bound, and the representation theorem gives a self-adjoint operator. Its principal symbol is $|\xi|^2$ times the identity. Elliptic regularity identifies its maximal L^2-domain with $H^2\Omega^1$; direct substitution supplies the reverse inclusion. The compact embedding $H^1 \hookrightarrow L^2$ gives compact resolvent.

For the final assertion, define $[a\wedge b]_{\mu\nu} = [a_\mu, b_\nu] - [a_\nu, b_\mu]$. The exact two-parameter curvature expansion is

$$F_{A_{\eta,T}+sa+tb} = F_{A_{\eta,T}} + s d_A a + t d_A b + \frac{s^2}{2}[a\wedge a] + st[a\wedge b] + \frac{t^2}{2}[b\wedge b].$$

It yields

$$\mathrm{YM}_T''\big|_{A_{\eta,T}}(a,b) = \frac{1}{g_{\mathrm{YM}}^2}\left(\langle d_A a, d_A b\rangle + \langle F_A, [a\wedge b]\rangle\right). \tag{487}$$

On the flat torus, commuting the covariant derivatives with the retained curvature convention gives

$$\langle d_A a, d_A b\rangle + \langle d_A^* a, d_A^* b\rangle = \langle \nabla_A a, \nabla_A b\rangle - \sum_{\nu,\mu}\langle a_\nu, [F_{\nu\mu}, b_\mu]\rangle,$$
$$\langle F_A, [a\wedge b]\rangle = -\sum_{\nu,\mu}\langle a_\nu, [F_{\nu\mu}, b_\mu]\rangle.$$

The factor $1/2$ in the two-form inner product is included in the second identity. Adding $g_{\mathrm{YM}}^{-2}\langle d_A^* a, d_A^* b\rangle$ to equation (487) gives exactly $g_{\mathrm{YM}}^{-2}\mathfrak{j}_T^{\mathrm{gf}}[a,b]$, including the coefficient -2 in equation (484). □

Proposition 13.4 (Resolvent intertwining and the reducing Coulomb slice)**.** *On smooth sections,*

$$\mathcal{J}_T^{\mathrm{gf}} d_{A_{\eta,T}} = d_{A_{\eta,T}}\Delta^{(0)}_{A_{\eta,T}}, \qquad d^*_{A_{\eta,T}}\mathcal{J}_T^{\mathrm{gf}} = \Delta^{(0)}_{A_{\eta,T}} d^*_{A_{\eta,T}}. \tag{488}$$

For $z\in\mathbb{C}\setminus\mathbb{R}$ and $f\in\operatorname{Dom} D_T^{\max}$,

$$D_T^{\max}(\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f = (\Delta^{(0)}_{A_{\eta,T}} - z)^{-1} D_T^{\max} f. \tag{489}$$

Thus $\mathscr{S}_T$ reduces $\mathcal{J}_T^{\mathrm{gf}}$, and its part

$$\mathcal{J}_T = \mathcal{J}_T^{\mathrm{gf}}\big|_{\mathscr{S}_T}, \qquad \operatorname{Dom}\mathcal{J}_T = H^2\Omega^1\cap\mathscr{S}_T \tag{490}$$

is self-adjoint. Its closed form is the restriction of $\mathfrak{j}_T^{\mathrm{gf}}$ to $H^1\Omega^1\cap\mathscr{S}_T$, and the Yang–Mills Hessian on this slice is

$$\mathrm{YM}_T''\big|_{A_{\eta,T}} = g_{\mathrm{YM}}^{-2}\mathfrak{j}_T. \tag{491}$$

Proof. For a smooth adjoint-valued scalar ϕ, write $\nabla_\mu = \nabla_{A_{\eta,T},e_\mu}$. Retaining both curvature terms when commuting three derivatives gives

$$\begin{aligned}(\mathcal{J}_T^{\mathrm{gf}} d_A\phi - d_A\Delta_A^{(0)}\phi)_\nu &\\ &= -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu},\nabla_\mu\phi]\\ &= -[\nabla_\mu F_{\mu\nu},\phi] = [(d_A^* F_A)_\nu,\phi] = 0.\end{aligned}$$

This proves the first identity in equation (488); formal adjunction gives the second.

Let $u = (\mathcal{J}_T^{\mathrm{gf}} - z)^{-1} f$. The operator domain gives $u\in H^2$, hence $D_T^{\max}u\in H^1$. Distributionally applying the second intertwining identity to $(\mathcal{J}_T^{\mathrm{gf}} - z)u = f$ yields

$$(\Delta_A^{(0)} - z) D_T^{\max} u = D_T^{\max} f.$$

The right side is in L^2; scalar elliptic regularity upgrades $D_T^{\max}u$ to H^2, and the scalar resolvent proves equation (489). Both resolvents of the self-adjoint gauge-fixed operator preserve $\mathscr{S}_T$; adjunction shows that they also preserve its orthogonal complement. Therefore the slice is reducing and its operator part is self-adjoint with the displayed domain. The reducing projection commutes with the spectral calculus, so the form domain is exactly $H^1\Omega^1\cap\mathscr{S}_T$. The Hessian identity is the restriction of the last assertion of theorem 13.3. □

13.6 Charge two, all polarizations, and the Morse index

Choose root vectors with $[H_{\mathrm{mag}}, E_\pm] = \pm 2iE_\pm$. Then

$$(\operatorname{ad}\mathscr{E}_{\eta,T})_{\mathbb{C}} = \underline{\mathbb{C}}H_{\mathrm{mag}} \oplus \mathscr{L}_{\eta,T}^{\otimes 2}E_+ \oplus \mathscr{L}_{\eta,T}^{\otimes(-2)}E_-. \tag{492}$$

The E_+ root connection has curvature $-4\pi i\eta_T$. Replacing E_η by $2E_\eta$ gives $J_{2,T} = J_{\eta,T}$, doubles both frequencies, and changes the vacuum dimension from six to twenty-four by equation (454). The E_- root has $q = -2$, uses the conjugate auxiliary structure $J_{-2,T} = -J_{\eta,T}$, and has the same positive frequencies and the same dimension $6(-2)^2 = 24$. Repeating the complete ladder proof of theorem 13.1 therefore gives, for either root, the joint charge-two scalar levels

$$E^{(2)}_{T;p,q} = 2(2p+1)\mathcal{B}_{T,+} + 2(2q+1)\mathcal{B}_{T,-}, \qquad p, q \in \mathbb{Z}_{\geq 0}, \tag{493}$$

each of complex dimension

$$|\operatorname{Pf}(2E_\eta)| = 24. \tag{494}$$

Coincident joint pairs must again be summed.

Choose $O_T \in SO(2)$ with $O_T^{\mathsf{T}} K_T O_T = \operatorname{diag}(\kappa_{T,+}, \kappa_{T,-})$, and define the constant orthonormal coframe

$$\begin{pmatrix} \mathrm{d}X_+ \\ \mathrm{d}X_- \end{pmatrix} = O_T^{\mathsf{T}} \begin{pmatrix} \mathrm{d}x^1 \\ \mathrm{d}x^3 \end{pmatrix}, \qquad \begin{pmatrix} \mathrm{d}Y_+ \\ \mathrm{d}Y_- \end{pmatrix} = O_T^{\mathsf{T}} \begin{pmatrix} \mathrm{d}x^2 \\ \mathrm{d}x^4 \end{pmatrix}. \tag{495}$$

Then

$$\eta_T = \frac{\mathcal{B}_{T,+}}{2\pi} \mathrm{d}X_+ \wedge \mathrm{d}Y_+ - \frac{\mathcal{B}_{T,-}}{2\pi} \mathrm{d}X_- \wedge \mathrm{d}Y_-. \tag{496}$$

The odd grouping permutation in equation (495) reverses orientation; it is used only to expose the two blocks and never to infer a Chern sign.

On the E_+ root line, the four constant circular one-forms diagonalize the spin endomorphism. Direct substitution into equation (484), with the charge-two root normalization retained, gives

polarization	V_T-shift	$\mathcal{J}_T^{\mathrm{gf}}$-eigenvalue on (p,q)
$\mathrm{d}X_+ + i\mathrm{d}Y_+$	$-4\mathcal{B}_{T,+}$	$2[(2p-1)\mathcal{B}_{T,+} + (2q+1)\mathcal{B}_{T,-}]$
$\mathrm{d}X_+ - i\mathrm{d}Y_+$	$+4\mathcal{B}_{T,+}$	$2[(2p+3)\mathcal{B}_{T,+} + (2q+1)\mathcal{B}_{T,-}]$
$\mathrm{d}X_- - i\mathrm{d}Y_-$	$-4\mathcal{B}_{T,-}$	$2[(2p+1)\mathcal{B}_{T,+} + (2q-1)\mathcal{B}_{T,-}]$
$\mathrm{d}X_- + i\mathrm{d}Y_-$	$+4\mathcal{B}_{T,-}$	$2[(2p+1)\mathcal{B}_{T,+} + (2q+3)\mathcal{B}_{T,-}]$

The E_- table is its anti-linear conjugate. The Cartan sector is the ordinary nonnegative coefficientwise Laplacian on one-forms.

Theorem 13.5 (Exact negative spectrum, walls, and real Morse index)**.** *Put*

$$\mathcal{B}_{T,\max} = \max\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \quad \mathcal{B}_{T,\min} = \min\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \quad \rho_T^* = \frac{\mathcal{B}_{T,\max}}{\mathcal{B}_{T,\min}} = \frac{\Xi_T + |S_T|}{\Xi_T - |S_T|}. \tag{497}$$

For every regular T, the negative spectrum of the self-adjoint Coulomb operator is exactly

$$\left\{ \nu_{T,n} = 2[(2n+1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}] = \frac{4\pi}{D_T}[n\Xi_T - (n+1)|S_T|] : 2n+1 < \rho_T^* \right\}. \tag{498}$$

Every indicated band has real multiplicity 48*, and*

$$\operatorname{ind}_{\mathbb{R}}(\mathcal{J}_T) = 48 \left\lceil \frac{\rho_T^* - 1}{2} \right\rceil. \tag{499}$$

At the n-th wall the strict inequality becomes equality. Equivalently,

$$\rho_T^* = 2n+1 \iff n\Xi_T = (n+1)|S_T| \iff 24n^2 D_T = (2n+1)S_T^2. \tag{500}$$

The corresponding branch is then a zero branch and is not counted in the Morse index. If one removes the absolute value by choosing the $\mathcal{B}_{T,+} > \mathcal{B}_{T,-}$ side, the polynomial equation in equation (500) *must be accompanied by $S_T > 0$; otherwise squaring introduces the opposite side as an extraneous solution.*

On the sufficiently deep tail of equation (446)*, the negative branches are exactly* $n = 0, 1, 2$*,*

$$\nu_{T,n} = \frac{4\pi}{D_T}[n\Xi_T - (n+1)S_T], \qquad n = 0, 1, 2, \tag{501}$$

each of real multiplicity 48*. Therefore*

$$\operatorname{ind}_{\mathbb{R}}(\mathcal{J}_T) = 144, \tag{502}$$

and

$$\begin{aligned} \nu_{T,0} &= -\frac{20\pi}{T} + O(T^{-2}), & \nu_{T,1} &= -\frac{12\pi}{T} + O(T^{-2}), \\ \nu_{T,2} &= -\frac{4\pi}{T} + O(T^{-2}), & \nu_{T,3} &= +\frac{4\pi}{T} + O(T^{-2}). \end{aligned} \tag{503}$$

The Hessian operator for equation (474) *has eigenvalues* $\nu_{T,n}/g_{\mathrm{YM}}^2$ *and the same index.*

Proof. Assume first $\mathcal{B}_{T,+} > \mathcal{B}_{T,-}$. In the first row of the polarization table, $p \geq 1$ makes the eigenvalue positive. At $p = 0$ it is

$$2[-\mathcal{B}_{T,+} + (2q+1)\mathcal{B}_{T,-}],$$

which is negative exactly when $2q + 1 < \rho_T^*$. The second and fourth rows are positive, and the third has minimum $2(\mathcal{B}_{T,+} - \mathcal{B}_{T,-}) > 0$. When $\mathcal{B}_{T,-} > \mathcal{B}_{T,+}$, the identical argument with the two planes exchanged shows that only the $q = 0$ part of the third row can be negative and gives the same formula with the ordered frequencies. If the frequencies agree, both candidate lowest branches are zero and none is negative. This proves the exhaustive list for the gauge-fixed operator.

Every negative gauge-fixed eigenform is automatically Coulomb. Elliptic regularity makes it smooth; if $\mathcal{J}_T^{\mathrm{gf}} a = \lambda a$ with $\lambda < 0$, then equation (488) gives

$$\Delta^{(0)}_{A_{\eta,T}}(d^*_{A_{\eta,T}} a) = \lambda d^*_{A_{\eta,T}} a.$$

Nonnegativity of the scalar operator forces $d_A^* a = 0$. Hence the full and Coulomb negative spectral spaces coincide.

One joint charge-two scalar space contributes complex dimension 24 in the E_+ sector. Its conjugate contributes complex dimension 24 in the complexification of the real adjoint problem. A complexified real eigenspace has complex dimension equal to the original real dimension, so the real multiplicity is $24 + 24 = 48$, not 24 or 96.

The count of integers $n \geq 0$ satisfying $2n + 1 < \rho_T^*$ is $\lceil (\rho_T^* - 1)/2 \rceil$, including the correct exclusion at an odd integer wall. Substitution of the ordered frequency formulas gives equation (498). The wall equation is $n\Xi_T = (n+1)|S_T|$; squaring and using $\Xi_T^2 - S_T^2 = 24 D_T$ gives the final polynomial form, and nonnegativity of both sides gives the converse. Finally, equation (446) leaves precisely $n = 0, 1, 2$, and the leading frequency terms give $\nu_{T,n} = (8n - 20)\pi/T + O(T^{-2})$. □

The lowest branch and the action excess are tied by the exact identities

$$\nu_{T,0} = -2|\mathcal{B}_{T,+} - \mathcal{B}_{T,-}| = -\frac{4\pi|S_T|}{D_T}, \qquad \mathrm{YM}_T(A_{\eta,T}) - \frac{24\pi^2}{g_{\mathrm{YM}}^2} = \frac{D_T}{8g_{\mathrm{YM}}^2}\nu_{T,0}^2. \tag{504}$$

13.7 The fixed and varying radial direct integrals

Choose $T_1 > T_0$ so that the complete deep-tail inequalities in equation (446) hold on

$$I := (T_1, \infty).$$

Let $Y = \mathbb{R}^4/\mathbb{Z}^4$. Denote by $\mathscr{L}_\eta \to Y$ the fixed marked Hermitian line bundle defined by the factor of automorphy in equation (427). The marking diffeomorphism and its transported bundle isomorphism are

$$\Phi_T : Y \longrightarrow F_T, \qquad [y] \longmapsto [P_T y], \qquad \widehat{\Phi}_T : \mathscr{L}_\eta \longrightarrow \mathscr{L}_{\eta,T}.$$

They are fixed by requiring

$$\widehat{\Phi}_T^* \nabla_{\eta,T} = \mathrm{d} - 2\pi i(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3),$$

so no gauge, metric, lattice, or orientation normalization is hidden in the identification. Put

$$\mathscr{K}_T := L^2(F_T, \mathscr{L}_{\eta,T}; \mathrm{d}\,\mathrm{vol}_{g_T}).$$

The pullback of the physical volume is

$$\Phi_T^*(\mathrm{d}\,\mathrm{vol}_{g_T}) = \sqrt{\det(P_T^{\mathsf{T}} P_T)}\,\mathrm{d}^4 y = D_T\,\mathrm{d}^4 y. \tag{505}$$

Put

$$\mathscr{H}_* := L^2(Y, \mathscr{L}_\eta; \mathrm{d}^4 y), \qquad \mathscr{D}_* := H^2(Y; \mathscr{L}_\eta),$$

where the Sobolev space is defined using any one fixed smooth reference metric and connection on the fixed compact bundle. Define

$$U_T : \mathscr{K}_T \longrightarrow \mathscr{H}_*, \qquad U_T s := D_T^{1/2} \widehat{\Phi}_T^* s. \tag{506}$$

Then

$$\|U_T s\|^2_{\mathscr{H}_*} = D_T \int_Y |\widehat{\Phi}_T^* s|^2\,\mathrm{d}^4 y = \|s\|^2_{\mathscr{K}_T},$$

and

$$U_T^{-1} v = D_T^{-1/2} (\widehat{\Phi}_T^{-1})^* v.$$

Thus U_T is an onto unitary, not merely an isometric injection.

Transport the fibre operator to the fixed space:

$$A_T := U_T \Delta_{\eta,T} U_T^{-1}, \qquad \mathrm{Dom}(A_T) = \mathscr{D}_*. \tag{507}$$

Its closed quadratic form is

$$\widetilde{\mathfrak{a}}_T[v, w] = \int_Y (G_T^{-1})^{jk} \langle \nabla_{\eta,j} v, \nabla_{\eta,k} w \rangle\,\mathrm{d}^4 y, \qquad G_T = P_T^{\mathsf{T}} P_T. \tag{508}$$

Here $\nabla_{\eta,j}$ is the $\mathrm{d}y^j$-component of the fixed marked connection displayed above. The factor $D_T^{1/2}$ in U_T is constant in y, and it cancels the factor D_T in (505); hence no unrecorded factor appears in (508).

Lemma 13.6 (Common domain and measurable resolvents)**.** *For every compact interval $K \Subset I$, the family*

$$T \longmapsto A_T \in \mathcal{B}(\mathscr{D}_*, \mathscr{H}_*)$$

is norm-continuous, in fact real analytic. For every $z \in \mathbb{C} \setminus \mathbb{R}$, the resolvents $(A_T - z)^{-1}$ are locally norm-continuous as operators on $\mathscr{H}_*$, and hence form a measurable operator field.

Proof. On K, the matrices G_T^{-1} are uniformly positive definite, and their entries are real analytic in T. In fixed local bundle trivializations, (508) therefore gives a uniformly elliptic second-order expression whose coefficients depend real analytically on T. Elliptic regularity identifies every operator domain with the same set $\mathscr{D}_*$, and the standard local coordinate estimate for a differential operator of order two gives norm-continuity in $\mathcal{B}(\mathscr{D}_*, \mathscr{H}_*)$.

Uniform ellipticity on K gives

$$\|v\|_{\mathscr{D}_*} \le C_K(\|A_T v\|_{\mathscr{H}_*} + \|v\|_{\mathscr{H}_*}), \qquad T \in K.$$

For $v = (A_T - z)^{-1} f$, self-adjointness gives $\|v\| \le |\operatorname{Im} z|^{-1}\|f\|$, while $A_T v = f + zv$. Consequently

$$\sup_{T\in K} \|(A_T - z)^{-1}\|_{\mathscr{H}_*\to\mathscr{D}_*} < \infty. \tag{509}$$

The resolvent identity, with its exact sign,

$$(A_T - z)^{-1} - (A_R - z)^{-1} = -(A_T - z)^{-1}(A_T - A_R)(A_R - z)^{-1},$$

together with (509), proves local norm-resolvent continuity. A countable compact exhaustion of I gives measurability on all of I. □

We now define the varying measurable Hilbert field rather than presuppose one. A section $T \mapsto s_T \in \mathscr{K}_T$ is declared measurable precisely when

$$T \longmapsto U_T s_T \in \mathscr{H}_*$$

is strongly measurable. This is the measurable field pulled back from the constant separable field $I \times \mathscr{H}_*$. Its direct integral is

$$\mathscr{H}_{\mathrm{var}} := \int_I^{\oplus} \mathscr{K}_T \,\mathrm{d}T.$$

The fixed radial Hilbert space is

$$\mathscr{H}_{\mathrm{rad}} := L^2(I, \mathrm{d}T; \mathscr{H}_*).$$

By construction,

$$\mathscr{U} := \int_I^{\oplus} U_T \,\mathrm{d}T : \mathscr{H}_{\mathrm{var}} \longrightarrow \mathscr{H}_{\mathrm{rad}}, \qquad (\mathscr{U} s)(T) = U_T s_T, \tag{510}$$

is an isometry. It is onto: if $\Psi \in \mathscr{H}_{\mathrm{rad}}$, then $s_T = U_T^{-1}\Psi(T)$ is measurable by the very definition of the varying field and has the same squared norm integral.

On the fixed direct integral, define the maximal multiplication operator

$$(\widetilde{\mathscr{M}_\eta}\Psi)(T) := A_T \Psi(T) \tag{511}$$

with the exact domain

$$\begin{aligned} \operatorname{Dom}(\widetilde{\mathscr{M}_\eta}) := \{\Psi \in \mathscr{H}_{\mathrm{rad}} : {} & \Psi(T) \in \mathscr{D}_* \text{ for almost every } T, \\ & T \mapsto A_T\Psi(T) \text{ is strongly measurable}, \\ & \int_I \|A_T\Psi(T)\|_{\mathscr{H}_*}^2 \,\mathrm{d}T < \infty\}. \end{aligned} \tag{512}$$

On the varying direct integral, define

$$\begin{aligned} \operatorname{Dom}(\mathscr{M}_\eta) := \{s \in \mathscr{H}_{\mathrm{var}} : {} & s_T \in \operatorname{Dom}(\Delta_{\eta,T}) \text{ for almost every } T, \\ & T \mapsto U_T\Delta_{\eta,T} s_T \text{ is strongly measurable}, \\ & \int_I \|\Delta_{\eta,T} s_T\|_{\mathscr{K}_T}^2 \,\mathrm{d}T < \infty\}, \end{aligned} \tag{513}$$

and

$$(\mathscr{M}_\eta s)_T := \Delta_{\eta,T} s_T \quad \text{for almost every } T. \tag{514}$$

Every expression in (511) and (514) is now typed on its actual fibre. Equations (507), (510), (512), and (513) give both the domain equality and the exact intertwining identity

$$\mathscr{U}\operatorname{Dom}(\mathscr{M}_\eta) = \operatorname{Dom}(\widetilde{\mathscr{M}_\eta}), \qquad \mathscr{U}\mathscr{M}_\eta\mathscr{U}^{-1} = \widetilde{\mathscr{M}_\eta}. \tag{515}$$

13.7.1 Density and self-adjointness on the maximal domain

Proposition 13.7 (Maximal-domain realization)**.** *The operators $\widetilde{\mathscr{M}_\eta}$ and $\mathscr{M}_\eta$ are densely defined, nonnegative, and self-adjoint on the domains* (512) *and* (513)*, respectively.*

Proof. For $\chi \in C_c(I)$ and $v \in \mathscr{D}_*$, local continuity of $T \mapsto A_T v$ makes it bounded on $\operatorname{supp}\chi$. Thus $\chi v \in \operatorname{Dom}(\widetilde{\mathscr{M}_\eta})$. Since $\mathscr{D}_*$ is dense in $\mathscr{H}_*$, the algebraic tensor product

$$C_c(I) \odot \mathscr{D}_*$$

is dense in $L^2(I;\mathscr{H}_*)$. Hence the fixed operator is densely defined. Fibrewise self-adjointness and Cauchy–Schwarz give, for Ψ, Θ in its domain,

$$\langle \widetilde{\mathscr{M}_\eta}\Psi, \Theta\rangle = \int_I \langle A_T\Psi(T), \Theta(T)\rangle \,\mathrm{d}T = \langle \Psi, \widetilde{\mathscr{M}_\eta}\Theta\rangle,$$

so it is symmetric.

Fix $z \in \mathbb{C}\setminus\mathbb{R}$. For $F \in \mathscr{H}_{\mathrm{rad}}$, define pointwise

$$(\mathcal{R}_z F)(T) := (A_T - z)^{-1} F(T). \tag{516}$$

The locally norm-continuous resolvent field, approximation of F by simple functions, and

$$\|(A_T - z)^{-1}\|_{\mathscr{H}_*\to\mathscr{H}_*} \le |\operatorname{Im} z|^{-1}$$

show that $\mathcal{R}_z F$ is strongly measurable and

$$\|\mathcal{R}_z F\|_{\mathscr{H}_{\mathrm{rad}}} \le |\operatorname{Im} z|^{-1}\|F\|_{\mathscr{H}_{\mathrm{rad}}}. \tag{517}$$

Pointwise, $(A_T - z)^{-1}F(T) \in \mathscr{D}_*$. The exact identity

$$A_T(A_T - z)^{-1} = I + z(A_T - z)^{-1} \tag{518}$$

gives the global uniform bound

$$\|A_T(A_T - z)^{-1}\| \le 1 + \frac{|z|}{|\operatorname{Im} z|}.$$

The right-hand side of (518), applied to $F(T)$, is strongly measurable and square-integrable. Therefore $\mathcal{R}_z F$ belongs to the full maximal domain (512); this is the domain step that pointwise inversion alone does not prove.

Fibrewise identities now yield

$$(\widetilde{\mathscr{M}_\eta} - z)\mathcal{R}_z = I \quad \text{on } \mathscr{H}_{\mathrm{rad}}, \qquad \mathcal{R}_z(\widetilde{\mathscr{M}_\eta} - z) = I \quad \text{on } \operatorname{Dom}(\widetilde{\mathscr{M}_\eta}).$$

In particular, the ranges of $\widetilde{\mathscr{M}_\eta} - i$ and $\widetilde{\mathscr{M}_\eta} + i$ are both all of $\mathscr{H}_{\mathrm{rad}}$. The range criterion for a densely defined symmetric operator proves self-adjointness. Moreover,

$$\langle \widetilde{\mathscr{M}_\eta}\Psi, \Psi\rangle = \int_I \langle A_T\Psi(T), \Psi(T)\rangle \,\mathrm{d}T \ge 0,$$

so it is nonnegative. The unitary equivalence (515) transfers density, self-adjointness, and nonnegativity to the varying-fibre operator. □

13.7.2 Measurable eigenvectors, including at band crossings

Lemma 13.8 (Explicit measurable eigenvector selection)**.** *For every fixed* $\mathbf{n} = (n_+, n_-) \in \mathbb{Z}^2_{\geq 0}$*, there is a strongly measurable map*

$$I \longrightarrow \mathscr{D}_*, \qquad T \longmapsto \phi_{\mathbf{n}}(T),$$

such that

$$\|\phi_{\mathbf{n}}(T)\|_{\mathscr{H}_*} = 1, \qquad A_T\phi_{\mathbf{n}}(T) = E_{\mathbf{n}}(T)\phi_{\mathbf{n}}(T) \tag{519}$$

for every $T \in I$*. The construction remains valid when* $E_{\mathbf{n}}(T)$ *crosses one or more other bands.*

Proof. Set

$$C_{\mathbf{n},T} := A_T - E_{\mathbf{n}}(T), \qquad \mathrm{Dom}(C_{\mathbf{n},T}) = \mathscr{D}_*.$$

The band $E_{\mathbf{n}}$ is continuous. The proof of Lemma 13.6, applied also to the scalar shift, shows that $(C_{\mathbf{n},T} - z)^{-1}$ is a measurable resolvent field for every nonreal z.

Let

$$\mathsf{P}_{\mathbf{n}}(T) := \mathbf{1}_{\{0\}}(C_{\mathbf{n},T}) = \mathbf{1}_{\{E_{\mathbf{n}}(T)\}}(A_T).$$

This singleton projection is not assumed continuous at a crossing. Instead, the spectral theorem gives the pointwise strong limit

$$\mathsf{P}_{\mathbf{n}}(T) = \operatorname*{s-lim}_{k\to\infty}(I + kC^2_{\mathbf{n},T})^{-1}, \tag{520}$$

and the exact resolvent formula

$$(I + kC^2_{\mathbf{n},T})^{-1} = \frac{1}{2i\sqrt{k}}\left[(C_{\mathbf{n},T} - i/\sqrt{k})^{-1} - (C_{\mathbf{n},T} + i/\sqrt{k})^{-1}\right] \tag{521}$$

shows that every approximating field is strongly measurable. Therefore $T \mapsto \mathsf{P}_{\mathbf{n}}(T)v$ is strongly measurable for every $v \in \mathscr{H}_*$. Proposition 13.1 says that $\mathsf{P}_{\mathbf{n}}(T) \neq 0$ for every T, including crossings.

Choose a countable dense sequence $(v_k)_{k\geq 1}$ in the separable space $\mathscr{H}_*$, and define the measurable partition

$$X_k = \{T : \|\mathsf{P}_{\mathbf{n}}(T)v_k\| > 0\} \cap \bigcap_{\ell<k}\{T : \|\mathsf{P}_{\mathbf{n}}(T)v_\ell\| = 0\}.$$

The sets X_k are pairwise disjoint and cover I: a nonzero bounded projection cannot annihilate a dense set. On X_k, set

$$\phi_{\mathbf{n}}(T) = \frac{\mathsf{P}_{\mathbf{n}}(T)v_k}{\|\mathsf{P}_{\mathbf{n}}(T)v_k\|}.$$

Countable measurable patching gives a strongly measurable unit vector. The range of the singleton spectral projection lies in $\mathrm{Dom}(A_T)$ and satisfies $(A_T - E_{\mathbf{n}}(T))\phi_{\mathbf{n}}(T) = 0$, which proves (519). □

13.7.3 Full spectrum and spectral type

Theorem 13.9 (Purely continuous full half-line spectrum)**.** *For both the fixed-fibre and varying-fibre maximal radial multiplication operators,*

$$\begin{aligned} \sigma(\widetilde{\mathscr{M}_\eta}) &= \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}) = \sigma_{\mathrm{c}}(\widetilde{\mathscr{M}_\eta}) = [0,\infty), \\ \sigma_{\mathrm{p}}(\widetilde{\mathscr{M}_\eta}) &= \varnothing, \qquad \ker \widetilde{\mathscr{M}_\eta} = \{0\}, \end{aligned} \tag{522}$$

and

$$\begin{aligned} \sigma(\mathscr{M}_\eta) &= \sigma_{\mathrm{ess}}(\mathscr{M}_\eta) = \sigma_{\mathrm{c}}(\mathscr{M}_\eta) = [0,\infty), \\ \sigma_{\mathrm{p}}(\mathscr{M}_\eta) &= \varnothing, \qquad \ker \mathscr{M}_\eta = \{0\}. \end{aligned} \tag{523}$$

Here σ_{ess} *is the Weyl essential spectrum.*

Proof. It is enough to prove (522), because (515) is a unitary equivalence. Proposition 13.7 and nonnegativity give

$$\sigma(\widetilde{\mathscr{M}_\eta}) \subset [0,\infty).$$

The zero-energy singular sequence. Let $a_j = T_1 + 2j + 1$. Using the measurable normalized ground eigenvector from Lemma 13.8, define

$$\Psi_j(T) := \mathbf{1}_{[a_j, a_j+1]}(T)\phi_{(0,0)}(T).$$

These vectors have norm one and have pairwise disjoint supports, hence are orthonormal and converge weakly to zero. They belong to the maximal domain because

$$A_T\Psi_j(T) = \mathbf{1}_{[a_j, a_j+1]}(T)E_0(T)\phi_{(0,0)}(T)$$

is strongly measurable and square-integrable. From (463),

$$\|\widetilde{\mathscr{M}_\eta}\Psi_j\|^2 = \int_{a_j}^{a_j+1} E_0(T)^2\,\mathrm{d}T = \frac{196\pi^2}{a_j^2} + O(a_j^{-3}) \longrightarrow 0.$$

Thus (Ψ_j) is a singular Weyl sequence at zero, and

$$0 \in \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}).$$

Positive energy: escaping crossings before disjoint supports. Fix $\lambda > 0$, and fix once and for all $T_\bullet = T_1 + 1 \in I$. For $n \in \mathbb{Z}_{\geq 0}$, retain the specific canonical band

$$e_n(T) := E_{(0,n)}(T) = \mathcal{B}_{T,+} + (2n+1)\mathcal{B}_{T,-}. \tag{524}$$

Since $\mathcal{B}_{T_\bullet,-} > 0$,

$$e_n(T_\bullet) \longrightarrow \infty \quad (n \to \infty).$$

For each fixed n, equations (443) and (444) give

$$e_n(T) \longrightarrow 0 \quad (T \to \infty).$$

Hence, for every sufficiently large n, the intermediate value theorem gives at least one crossing parameter $r_n > T_\bullet$ satisfying

$$e_n(r_n) = \lambda. \tag{525}$$

We prove explicitly that every such choice of crossing parameters escapes to infinity. Fix $M > T_\bullet$. Positivity and continuity give

$$c_M := \min_{T \in [T_\bullet, M]} \mathcal{B}_{T,-} > 0.$$

For every $T \in [T_\bullet, M]$,

$$e_n(T) \geq (2n+1)c_M.$$

Thus, once $(2n+1)c_M > \lambda$, equation (525) has no solution in $[T_\bullet, M]$, and the chosen r_n must satisfy $r_n > M$. Since M was arbitrary,

$$r_n \longrightarrow \infty. \tag{526}$$

Only now, before choosing any support, pass to a subsequence $n_j \to \infty$ such that, with $r_j := r_{n_j}$,

$$r_{j+1} > r_j + 2. \tag{527}$$

Continuity of e_{n_j} at r_j permits the subsequent choice of $\varepsilon_j \in (0, \frac{1}{4})$ such that

$$J_j := (r_j - \varepsilon_j, r_j + \varepsilon_j) \Subset I, \qquad \sup_{T \in J_j} |e_{n_j}(T) - \lambda| < \frac{1}{j}.$$

The order of construction matters: the separation (527) makes the intervals J_j pairwise disjoint.

Take the measurable normalized eigenvector $\phi_{(0,n_j)}(T)$ from Lemma 13.8; the singleton projection used there remains valid if the selected band crosses another one inside J_j. Define

$$\Theta_j(T) := |J_j|^{-1/2}\mathbf{1}_{J_j}(T)\phi_{(0,n_j)}(T). \tag{528}$$

The sections Θ_j have norm one and are orthogonal by disjointness of their supports. They lie in the maximal domain because

$$A_T\Theta_j(T) = |J_j|^{-1/2}\mathbf{1}_{J_j}(T)e_{n_j}(T)\phi_{(0,n_j)}(T)$$

is strongly measurable and square-integrable. Moreover,

$$\|(\widetilde{\mathscr{M}_\eta} - \lambda)\Theta_j\|^2 = \frac{1}{|J_j|}\int_{J_j} |e_{n_j}(T) - \lambda|^2 \,\mathrm{d}T < \frac{1}{j^2}.$$

Every orthonormal sequence is weakly null, so (Θ_j) is a singular Weyl sequence at λ. Thus every $\lambda > 0$ belongs to the Weyl essential spectrum. Together with the zero sequence and nonnegativity, this proves

$$\sigma(\widetilde{\mathscr{M}_\eta}) = \sigma_{\mathrm{ess}}(\widetilde{\mathscr{M}_\eta}) = [0,\infty). \tag{529}$$

Real analyticity, absence of flat bands, and no eigenvalues. Equation (438) writes

$$\Xi_T = \sqrt{S_T^2 + 24D_T}$$

with a strictly positive real-analytic radicand. The positive square-root function is real analytic on $(0,\infty)$. Hence Ξ_T, $\mathcal{B}_{T,+}$, $\mathcal{B}_{T,-}$, and every $E_{\mathbf{n}}(T)$ are real analytic on the connected interval I. For each fixed $\mathbf{n}$,

$$E_{\mathbf{n}}(T) > 0 \quad (T < \infty), \qquad \lim_{T\to\infty} E_{\mathbf{n}}(T) = 0.$$

It follows that no band is constant: a constant band would have constant value equal to its limit zero, contradicting strict positivity at every finite T. This proves the absence of flat bands, rather than assuming it.

Fix $\lambda \in \mathbb{R}$. For every $\mathbf{n}$, the real-analytic function $E_{\mathbf{n}} - \lambda$ is not identically zero, so its zero set in I is discrete and has Lebesgue measure zero. Since $\mathbb{Z}^2_{\geq 0}$ is countable,

$$Z_\lambda := \{T \in I : \lambda \in \sigma_{\mathrm{p}}(A_T)\} = \bigcup_{\mathbf{n}\in\mathbb{Z}^2_{\geq 0}} \{T : E_{\mathbf{n}}(T) = \lambda\} \tag{530}$$

is a null set.

If $\widetilde{\mathscr{M}_\eta}\Psi = \lambda\Psi$, then $(A_T - \lambda)\Psi(T) = 0$ for almost every T. At every point where $\Psi(T) \neq 0$, Proposition 13.1 forces $T \in Z_\lambda$. Thus the essential support of Ψ is contained in the null set (530), and $\Psi = 0$. A self-adjoint operator has only real eigenvalues, so

$$\sigma_{\mathrm{p}}(\widetilde{\mathscr{M}_\eta}) = \varnothing, \qquad \ker \widetilde{\mathscr{M}_\eta} = \{0\}.$$

Self-adjoint operators have empty residual spectrum. Combining this with (529) proves that the entire spectrum is continuous and gives (522). Unitary equivalence gives (523). □

13.7.4 Exact interpretive boundary

Remark 13.10 (No radial derivative and no interacting Yang–Mills claim). The variable T in (511) and (514) is a direct-integral parameter. Neither operator contains a T-derivative. In particular, there is no term such as $-\partial_T^2$, no radial Sobolev condition, no cusp boundary condition, and no interaction between distinct T-fibres. This is why characteristic functions of radial intervals legitimately occur in the maximal-domain Weyl sequences above.

The theorem is therefore a complete spectral statement for the specified auxiliary decomposable family of native full-flux magnetic Laplacians. It does not construct, identify, or determine the spectrum or mass gap of an interacting quantum Yang–Mills Hamiltonian, and it makes no Osterwalder–Schrader reconstruction claim.

14 The native full-flux split-Cartan $SU(r)$ extension

This section fixes an integer $r \geq 2$ and extends the full-flux construction of section 13 from its already proved split $SU(2)$ case to arbitrary split-Cartan $SU(r)$ backgrounds. Every object remains on the same regular flat four-torus $F_T = \mathbb{R}^4/P_T\mathbb{Z}^4$, with the unrescaled Euclidean quotient metric. In particular, the integral two-form η_T, the line bundle $\mathscr{L}_{\eta,T}$, the two orientations, and the exact quantities

$$\begin{aligned} L_T, \quad q_T, \quad m_T, \quad D_T = L_T q_T + 6m_T^2 > 0, \\ S_T = 6L_T - q_T, \qquad \Xi_T = \sqrt{(6L_T + q_T)^2 + 144m_T^2}. \end{aligned}$$

are precisely those of equations (420) and (436). The charge-one frequencies are

$$\mathcal{B}_{T,+} = \frac{\pi(S_T + \Xi_T)}{D_T}, \qquad \mathcal{B}_{T,-} = \frac{\pi(\Xi_T - S_T)}{D_T}, \tag{531}$$

and we shall use the globally ordered notation

$$\begin{aligned} \mathcal{B}_{T,\max} = \max\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \quad \mathcal{B}_{T,\min} = \min\{\mathcal{B}_{T,+}, \mathcal{B}_{T,-}\}, \\ \rho_T^* = \frac{\mathcal{B}_{T,\max}}{\mathcal{B}_{T,\min}}. \end{aligned} \tag{532}$$

Thus every assertion below remains valid on either side of $S_T = 0$.

14.1 The split bundle, determinant, and Chern number

Fix a nonzero trace-zero integral weight vector

$$\mathbf{m} = (m_1, \ldots, m_r) \in \mathbb{Z}^r, \qquad \sum_{j=1}^r m_j = 0, \qquad s_2(\mathbf{m}) := \sum_{j=1}^r m_j^2. \tag{533}$$

Define

$$\mathscr{E}_{\mathbf{m},T} := \bigoplus_{j=1}^r \mathscr{L}_{\eta,T}^{\otimes m_j}, \qquad H_{\mathbf{m}} := i\,\mathrm{diag}(m_1, \ldots, m_r) \in \mathfrak{su}(r). \tag{534}$$

Proposition 14.1 (Parallel determinant trivialization and exact c_2)**.** *The ordered splitting in equation* (534) *has a canonical parallel determinant trivialization. With $c = [\eta_T]$,*

$$c_1(\mathscr{E}_{\mathbf{m},T}) = 0, \qquad c_2(\mathscr{E}_{\mathbf{m},T}) = -\frac{s_2(\mathbf{m})}{2}c^2. \tag{535}$$

The marked and physical-complex Chern numbers are respectively

$$\langle c_2, [F_T]_{\mathrm{mark}} \rangle = -6s_2(\mathbf{m}), \qquad k_{\mathbf{m}} := \langle c_2, [F_T]_{x\text{-complex}} \rangle = 6s_2(\mathbf{m}) > 0. \tag{536}$$

Moreover $s_2(\mathbf{m})$ is even. Hence this direct-sum family realizes only $k_{\mathbf{m}} \in 12\mathbb{Z}_{>0}$, not all topological $SU(r)$ sectors.

Proof. Let $j_n^\eta(y)$ denote the factor of automorphy in equation (427). The transition matrix of the ordered sum is

$$\mathrm{diag}\big((j_n^\eta)^{m_1}, \ldots, (j_n^\eta)^{m_r}\big).$$

Its determinant is $(j_n^\eta)^{\sum_j m_j} = 1$ identically. The tensor product of the ordered unit frames therefore defines a global determinant section, and the determinant connection has connection form proportional to $\sum_j m_j = 0$; the section is parallel.

The splitting principle gives

$$c(\mathscr{E}_{\mathbf{m},T}) = \prod_{j=1}^{r}(1+m_j c), \qquad c_2 = \sum_{i<j} m_i m_j c^2.$$

Since

$$2\sum_{i<j} m_i m_j = \left(\sum_j m_j\right)^2 - \sum_j m_j^2 = -s_2(\mathbf{m}),$$

this proves equation (535). The two evaluations follow from equation (433); retaining its orientation reversal is essential for the sign. Finally

$$s_2(\mathbf{m}) = \sum_j m_j^2 \equiv \sum_j m_j \equiv 0 \pmod 2,$$

because $a^2 \equiv a \pmod 2$. This proves the divisibility statement. □

The diagonal connection induced by the summands is

$$\nabla^{\mathbf{m},T} = \mathrm{d} - 2\pi(6y^1\,\mathrm{d}y^4 + y^2\,\mathrm{d}y^3)H_{\mathbf{m}}, \qquad F_{\mathbf{m},T} = -2\pi H_{\mathbf{m}}\eta_T. \tag{537}$$

We retain the invariant inner product and functional normalization

$$\begin{aligned} \langle X, Y\rangle_{\mathfrak{su}(r)} = -\frac{1}{2}\operatorname{Tr}(XY), \qquad \|H_{\mathbf{m}}\|^2 = \frac{1}{2}s_2(\mathbf{m}), \\ \mathrm{YM}_T(A) = \frac{1}{2g_{\mathrm{YM}}^2}\int_{F_T} |F_A|^2\,\mathrm{d}\,\mathrm{vol}_T\,. \end{aligned} \tag{538}$$

Theorem 14.2 (Yang–Mills criticality and exact action)**.** *The connection $\nabla^{\mathbf{m},T}$ is non-flat, reducible, and Yang–Mills. Its action is*

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T}) = \frac{\pi^2 s_2(\mathbf{m})}{g_{\mathrm{YM}}^2 D_T}(q_T^2 + 36L_T^2 + 72m_T^2) \tag{539}$$

$$= \frac{2\pi^2 k_{\mathbf{m}}}{g_{\mathrm{YM}}^2} + \frac{\pi^2 s_2(\mathbf{m}) S_T^2}{g_{\mathrm{YM}}^2 D_T}. \tag{540}$$

On the canonical cusp tail,

$$\mathrm{YM}_T(\nabla^{\mathbf{m},T}) = \frac{\pi^2 s_2(\mathbf{m})}{g_{\mathrm{YM}}^2}\left(37 + \frac{35B}{T} + O(T^{-2})\right). \tag{541}$$

Proof. Both $H_{\mathbf{m}}$ and η_T are parallel for the diagonal connection and flat quotient metric, and their brackets vanish. Hence $d^*_{\nabla^{\mathbf{m},T}} F_{\mathbf{m},T} = 0$. Non-flatness follows from $\mathbf{m} \neq 0$ and $\eta_T \neq 0$.

In the physical orthonormal frame used in equation (496),

$$|\eta_T|^2 = \frac{\mathcal{B}_{T,+}^2 + \mathcal{B}_{T,-}^2}{4\pi^2}.$$

Using equation (538), $\mathrm{vol}(F_T) = D_T$, and the exact frequency square sum from the native module yields equation (539). The identity

$$q_T^2 + 36L_T^2 + 72m_T^2 = 12(L_T q_T + 6m_T^2) + (6L_T - q_T)^2 = 12D_T + S_T^2$$

and $k_{\mathbf{m}} = 6s_2(\mathbf{m})$ give equation (540). Substituting the retained expansions of L_T, q_T, m_T, D_T, without suppressing H, B, m, gives

$$\frac{q_T^2 + 36L_T^2 + 72m_T^2}{D_T} = 37 + \frac{35B}{T} + O(T^{-2}),$$

which proves the last formula. □

14.2 Every nonzero integral root charge

The native module proves the charge-one ladder decomposition and the charge-two specialization used by its $SU(2)$ theorem. The following lemma records, and proves, the exact integer-charge extension needed here.

Lemma 14.3 (Charge-d Landau decomposition)**.** *Let $d \in \mathbb{Z} \setminus \{0\}$, put $q = |d|$, and equip $\mathscr{L}_{\eta,T}^{\otimes d}$ with its tensor-power connection. Its scalar covariant Laplacian has joint levels*

$$q\big((2a+1)\mathcal{B}_{T,+} + (2b+1)\mathcal{B}_{T,-}\big), \qquad a, b \in \mathbb{Z}_{\geq 0}, \tag{542}$$

and every joint space has complex dimension $6q^2$. If different pairs (a, b) give the same numerical value, their orthogonal joint spaces are to be summed.

Proof. The curvature is $-2\pi i d\eta_T$. In the polar construction of theorem 13.1, replace the auxiliary complex structure by $J_{d,T} = \operatorname{sgn}(d) J_{\eta,T}$. Then $d\eta_T$ is positive of type $(1,1)$, so the vacuum is the holomorphic-section space of the positive bundle $\mathscr{L}_{\eta,T}^{\otimes d}$ on that auxiliary complex torus. Its tangent and canonical bundles are trivial. Kodaira vanishing and Hirzebruch–Riemann–Roch give

$$\dim_{\mathbb{C}} H^0 = \frac{1}{2}\int_{[F_T]_{d\eta\text{-symp}}} (d\eta_T)^2 = \frac{1}{2} q^2 \int_{[F_T]_{\mathrm{mark}}} \eta_T^2 = 6q^2.$$

Here changing d to $-d$ does not change the four-dimensional symplectic orientation. Scaling the curvature by d scales both positive oscillator frequencies by q, while the two canonical commutation relations and the lowering argument in theorem 13.1 are unchanged after this rescaling. Thus every joint level is unitarily obtained from the vacuum and has the asserted energy and dimension. Orthogonal joint spaces remain a direct sum when their numerical energies coincide. □

14.3 Closed gauge-fixed and Coulomb operators

Write $A = \nabla^{\mathbf{m},T}$ in this subsection. Let

$$\mathscr{H}^1_{\mathbf{m},T} = L^2\Omega^1(F_T; \operatorname{ad}\mathscr{E}_{\mathbf{m},T}), \qquad \mathscr{H}^0_{\mathbf{m},T} = L^2\Omega^0(F_T; \operatorname{ad}\mathscr{E}_{\mathbf{m},T}) \tag{543}$$

as real Hilbert spaces. Define the maximal distributional codifferential by

$$\begin{aligned} \operatorname{Dom} D^{\max}_{\mathbf{m},T} = \{a \in \mathscr{H}^1_{\mathbf{m},T} : \exists u \in \mathscr{H}^0_{\mathbf{m},T} \text{ such that} \\ \langle a, d_A\phi\rangle_{L^2} = \langle u, \phi\rangle_{L^2} \quad \text{for every} \\ \phi \in C^\infty(F_T; \operatorname{ad}\mathscr{E}_{\mathbf{m},T})\}, \end{aligned} \tag{544}$$

$$D^{\max}_{\mathbf{m},T} a = u, \qquad \mathscr{S}_{\mathbf{m},T} := \ker D^{\max}_{\mathbf{m},T}. \tag{545}$$

Density of smooth test scalars makes u unique. If $a_j \to a$ and $D^{\max}_{\mathbf{m},T} a_j \to u$ in L^2, passage to the limit in the defining identity gives $D^{\max}_{\mathbf{m},T} a = u$. Thus $D^{\max}_{\mathbf{m},T}$ is closed and $\mathscr{S}_{\mathbf{m},T}$ is a closed subspace.

Let $\Delta_A^{(0)}$ be the nonnegative self-adjoint scalar operator of the closed form

$$\mathfrak{q}^{(0)}_{\mathbf{m},T}[\phi, \psi] = \langle d_A\phi, d_A\psi\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{q}^{(0)}_{\mathbf{m},T} = H^1\Omega^0. \tag{546}$$

It has domain $H^2\Omega^0$ and equals $d_A^* d_A$. In a constant orthonormal frame set

$$(V_{\mathbf{m},T} a)_\nu = -2\sum_{\mu=1}^{4} [F_{\mathbf{m},T,\nu\mu}, a_\mu]. \tag{547}$$

Invariance of the compact Lie-algebra inner product and $F_{\nu\mu} = -F_{\mu\nu}$ show directly that $V_{\mathbf{m},T}$ is a bounded symmetric endomorphism. Define

$$\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T}[a, b] = \langle \nabla_A a, \nabla_A b\rangle_{L^2} + \langle V_{\mathbf{m},T} a, b\rangle_{L^2}, \qquad \operatorname{Dom}\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T} = H^1\Omega^1. \tag{548}$$

Proposition 14.4 (Closed realization and exact Coulomb reduction)**.** *The form in equation* (548) *is densely defined, closed, symmetric, and bounded below. Its self-adjoint operator has compact resolvent, domain $H^2\Omega^1$, and expression*

$$(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}a)_\nu = -\sum_{\mu=1}^{4}\nabla_{A,e_\mu}\nabla_{A,e_\mu}a_\nu - 2\sum_{\mu=1}^{4}[F_{\mathbf{m},T,\nu\mu},a_\mu]. \tag{549}$$

For every $z\in\mathbb{C}\setminus\mathbb{R}$ and $f\in\operatorname{Dom}D^{\max}_{\mathbf{m},T}$,

$$D^{\max}_{\mathbf{m},T}(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}-z)^{-1}f = (\Delta^{(0)}_A - z)^{-1}D^{\max}_{\mathbf{m},T}f. \tag{550}$$

Consequently $\mathscr{S}_{\mathbf{m},T}$ reduces $\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}$, and

$$\mathcal{J}_{\mathbf{m},T} := \mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}\big|_{\mathscr{S}_{\mathbf{m},T}}, \qquad \operatorname{Dom}\mathcal{J}_{\mathbf{m},T} = H^2\Omega^1\cap\mathscr{S}_{\mathbf{m},T} \tag{551}$$

is self-adjoint, with compact resolvent and closed form $\mathfrak{j}_{\mathbf{m},T}$ equal to the restriction of $\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T}$ to $H^1\Omega^1\cap\mathscr{S}_{\mathbf{m},T}$. The physical Yang–Mills Hessian on this slice is

$$\mathrm{YM}''_T\big|_A = g_{\mathrm{YM}}^{-2}\mathfrak{j}_{\mathbf{m},T}. \tag{552}$$

Thus an eigenvalue ν of $\mathcal{J}_{\mathbf{m},T}$ is the physical Hessian eigenvalue ν/g^2_{YM}, while its sign and Morse-index contribution are unchanged.

Proof. The first term of equation (548), after addition of the L^2-norm, is an equivalent H^1-norm. The second term is a bounded symmetric form. The bounded perturbation theorem for closed forms and the representation theorem therefore give a lower-bounded self-adjoint operator. Its principal symbol is $|\xi|^2$ times the identity, so elliptic regularity gives the H^2 domain. Rellich compactness on the compact torus gives compact resolvent.

The two-parameter curvature expansion and the flat Weitzenböck identity, with the inner product and functional normalization in equation (538), give

$$\mathrm{YM}''_T\big|_A(a,b) + g_{\mathrm{YM}}^{-2}\langle d^*_A a, d^*_A b\rangle = g_{\mathrm{YM}}^{-2}\mathfrak{j}^{\mathrm{gf}}_{\mathbf{m},T}[a,b].$$

This proves the Hessian normalization after restriction to Coulomb gauge.

It remains to prove that the restriction is an operator rather than a formal constraint. For a smooth adjoint scalar ϕ, write $\nabla_\mu = \nabla_{A,e_\mu}$. Commuting all three derivatives and using the Yang–Mills equation gives

$$\begin{aligned}&(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}d_A\phi - d_A\Delta^{(0)}_A\phi)_\nu\\ &\quad= -\nabla_\mu\nabla_\mu\nabla_\nu\phi + \nabla_\nu\nabla_\mu\nabla_\mu\phi - 2[F_{\nu\mu},\nabla_\mu\phi]\\ &\quad= -[\nabla_\mu F_{\mu\nu},\phi] = [(d^*_A F_A)_\nu,\phi] = 0.\end{aligned}$$

Formal adjunction yields, on smooth one-forms,

$$\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}d_A = d_A\Delta^{(0)}_A, \qquad d^*_A\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T} = \Delta^{(0)}_A d^*_A. \tag{553}$$

For $u = (\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}-z)^{-1}f$, the operator domain gives $u\in H^2$, hence $D^{\max}_{\mathbf{m},T}u\in H^1$. Applying the second identity distributionally to $(\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}-z)u = f$ gives

$$(\Delta^{(0)}_A - z)D^{\max}_{\mathbf{m},T}u = D^{\max}_{\mathbf{m},T}f.$$

Scalar elliptic regularity and the scalar resolvent give equation (550). The two conjugate resolvents preserve $\mathscr{S}_{\mathbf{m},T}$; adjunction then makes its orthogonal complement invariant as well. The slice is reducing, so the self-adjoint part has the operator and form domains displayed above. □

14.4 Rootwise spectrum, coincidences, index, and nullity

For $i \neq j$, the matrix unit E_{ij} obeys

$$[H_{\mathbf{m}}, E_{ij}] = i(m_i - m_j)E_{ij},$$

so its complex root line is

$$\mathrm{Hom}(\mathscr{L}_{\eta,T}^{m_j}, \mathscr{L}_{\eta,T}^{m_i}) = \mathscr{L}_{\eta,T}^{m_i - m_j}, \qquad q_{ij} := |m_i - m_j|. \tag{554}$$

The factor q_{ij} is the complete charge normalization; for the existing $SU(2)$ weights $(1, -1)$, it gives $q_{12} = 2$.

Let the distinct numerical values among $m_1, \ldots, m_r$ have multiplicities $r_1, \ldots, r_\ell$. The neutral centralizer is

$$\mathfrak{z}_{\mathbf{m}} = \{X \in \mathfrak{su}(r) : [H_{\mathbf{m}}, X] = 0\} \cong \mathfrak{s}\left(\bigoplus_{a=1}^{\ell} \mathfrak{u}(r_a)\right), \qquad \dim_{\mathbb{R}} \mathfrak{z}_{\mathbf{m}} = \sum_{a=1}^{\ell} r_a^2 - 1. \tag{555}$$

Theorem 14.5 (Complete negative spectrum and numerical multiplicities)**.** *Put*

$$N_-(T) := \#\{n \in \mathbb{Z}_{\geq 0} : 2n + 1 < \rho_T^*\} = \left\lceil \frac{\rho_T^* - 1}{2} \right\rceil. \tag{556}$$

For each positive root $i < j$ with $q_{ij} > 0$, the complete negative spectrum of the normalized Coulomb operator is

$$\nu_{ij,n}(T) = q_{ij}\big((2n + 1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big), \qquad 0 \leq n < N_-(T). \tag{557}$$

The contribution of that positive-root pair to each displayed band has real dimension $12q_{ij}^2$. For an actual numerical eigenvalue $\lambda < 0$, its full real multiplicity is therefore

$$\mathrm{mult}_{\mathbb{R}}(\lambda) = \sum_{\substack{i<j,\ q_{ij}>0,\ 0\leq n<N_-(T) \\ \nu_{ij,n}(T)=\lambda}} 12q_{ij}^2. \tag{558}$$

In particular, coincident root charges or accidental coincidences between different q_{ij}, n add rather than being counted as one band. The real Morse index is

$$\boxed{\mathrm{ind}_{\mathbb{R}}\, \mathcal{J}_{\mathbf{m},T} = 12N_-(T) \sum_{i<j} (m_i - m_j)^2 = 12rN_-(T)s_2(\mathbf{m}).} \tag{559}$$

Every eigenvalue in equation (557) *is divided by g_{YM}^2 for the physical Hessian.*

Proof. On the E_{ij} line, the scalar oscillator is the charge-q_{ij} operator of theorem 14.3. In a skew-normal orthonormal coframe, the four circular one-form polarizations diagonalize equation (547); the two polarizations in a plane of charge-q_{ij} frequency $q_{ij}\mathcal{B}$ shift the scalar energy by $\pm 2q_{ij}\mathcal{B}$. The unfavorable polarizations are positive. For the favorable polarization in the high-frequency plane, high-plane oscillator number zero leaves

$$q_{ij}\big((2n + 1)\mathcal{B}_{T,\min} - \mathcal{B}_{T,\max}\big).$$

Increasing its high-plane oscillator number makes it positive. The favorable low-frequency polarization has bottom $q_{ij}(\mathcal{B}_{T,\max} - \mathcal{B}_{T,\min}) \geq 0$, and increasing either number cannot lower it. This proves that the list is exhaustive.

Every joint scalar space on one complex root line has dimension $6q_{ij}^2$. The opposite root is its anti-linear conjugate. The pair is the complexification of a real root subspace, so its real dimension is $12q_{ij}^2$. Root subspaces and joint oscillator spaces are orthogonal; hence equality of numerical eigenvalues produces precisely the direct sum in equation (558).

The strict inequality in equation (556) counts the integers correctly even when ρ_T^* is an odd integer. Summing dimensions gives the first index formula. Finally,

$$\sum_{i<j}(m_i-m_j)^2 = r\sum_i m_i^2 - \left(\sum_i m_i\right)^2 = rs_2(\mathbf{m}),$$

which gives the second formula.

It remains to justify Coulomb gauge. If $\mathcal{J}^{\mathrm{gf}}_{\mathbf{m},T}a = \lambda a$ with $\lambda < 0$, elliptic regularity and equation (553) give

$$\Delta_A^{(0)} d_A^* a = \lambda d_A^* a.$$

The scalar operator is nonnegative, so $d_A^* a = 0$. Thus the negative gauge-fixed and Coulomb eigenspaces coincide, completing the proof. □

Theorem 14.6 (Exact neutral and wall nullities)**.** *Away from an odd wall, the neutral Coulomb kernel has real dimension*

$$\mathrm{null}_{\mathrm{neutral}}\,\mathcal{J}_{\mathbf{m},T} = 4\left(\sum_{a=1}^{\ell} r_a^2 - 1\right). \tag{560}$$

At a wall $\rho_T^* = 2n+1 > 1$, the nonzero-root part adds

$$12\sum_{i<j} q_{ij}^2 = 12rs_2(\mathbf{m}) \tag{561}$$

real zero modes. At the equality wall $S_T = 0$, equivalently $\rho_T^* = 1$, both favorable plane polarizations vanish and the nonzero root contribution doubles to

$$24\sum_{i<j} q_{ij}^2 = 24rs_2(\mathbf{m}). \tag{562}$$

The total nullity at a wall is the sum of the neutral term and the stated root term.

Proof. On $\mathfrak{z}_{\mathbf{m}}$, both the induced connection commutator and the spin endomorphism vanish. The operator is therefore the ordinary coefficientwise Laplacian on one-forms. Its kernel on a flat four-torus is the space of constant one-forms, all of which are Coulomb. Tensoring the four-dimensional real harmonic-one-form space with $\mathfrak{z}_{\mathbf{m}}$ proves equation (560).

If $\rho_T^* = 2n+1 > 1$, the high-plane favorable polarization at high oscillator number zero and low oscillator number n is the unique zero branch for each nonzero root. Its positive-root-pair real dimension is $12q_{ij}^2$. If $S_T = 0$, the two frequencies agree and the lowest favorable polarization in each of the two planes is zero, so this dimension doubles. These charged zero modes are Coulomb: the intertwining identity shows that their divergences lie in the kernel of the charged scalar Laplacian, but that scalar Laplacian has strictly positive ground energy by equation (542). Summing over positive roots and using $\sum_{i<j} q_{ij}^2 = rs_2(\mathbf{m})$ proves the result. □

For $n \geq 1$, the n-th odd wall is equivalently

$$\rho_T^* = 2n+1 \iff n\Xi_T = (n+1)|S_T| \iff 24n^2 D_T = (2n+1)S_T^2. \tag{563}$$

The final polynomial equation must be accompanied by the unsigned ordering in the middle expression if one needs to distinguish the two sides $S_T > 0$ and $S_T < 0$. The $n = 0$ wall is exactly $S_T = 0$.

On the sufficiently deep cusp tail, $S_T > 0$, $5 < \rho_T^* < 7$, and $N_-(T) = 3$. Hence

$$\mathrm{ind}_{\mathbb{R}}\,\mathcal{J}_{\mathbf{m},T} = 36rs_2(\mathbf{m}) = 6rk_{\mathbf{m}}, \tag{564}$$

and the three negative branches have the retained asymptotics

$$\nu_{ij,n}(T) = \frac{q_{ij}\pi}{T}(4n-10) + O(T^{-2}), \qquad n = 0,1,2. \tag{565}$$

For example, the genuinely $SU(3)$ data are

weights	nonzero q_{ij}	$k_{\mathbf{m}}$	deep index	neutral nullity
$(1,0,-1)$	$1,1,2$	12	216	8
$(1,1,-2)$	$3,3$	36	648	16
$(2,1,-3)$	$1,4,5$	84	1512	8

The middle row exhibits both required edge phenomena: one neutral root from the repeated weight and coincident numerical bands from the two charge-three positive roots. The specialization $r = 2$, $\mathbf{m} = (1,-1)$ is exactly the already proved native $SU(2)$ theorem and is not reproduced here.

14.5 Stability boundary, possible wall bifurcation, and exact scope

Proposition 14.7 (Where the split background is, and is not, unstable)**.** *If $S_T \neq 0$, then $\rho_T^* > 1$, $N_-(T) \geq 1$, and every nonzero positive root contributes a negative Coulomb direction. Since a nonzero trace-zero weight vector has at least one nonzero root, $\nabla^{\mathbf{m},T}$ is then not a local minimum modulo gauge. This applies in particular throughout the sufficiently deep cusp tail.*

If $S_T = 0$, the negative root spectrum is empty. In that case equation (540) *attains the topological value $2\pi^2 k_{\mathbf{m}}/g_{\mathrm{YM}}^2$, while equation* (562) *supplies root zero modes. No assertion of strict stability follows because the Hessian is degenerate.*

Proof. The first paragraph follows immediately from equation (532) and theorem 14.5. For the second, $S_T = 0$ makes the two frequencies equal and hence $\rho_T^* = 1$; the strict negative-band inequality has no solution. The action and nullity statements are exactly equation (540) and theorem 14.6. □

A negative eigenspace is a space of descent directions, not a Lyapunov–Schmidt bifurcation kernel. A local bifurcation analysis must instead be posed at one of the zero-mode parameters in theorem 14.6. It must first quotient or slice the ever-present centralizer-valued harmonic kernel, then identify the remaining kernel and cokernel of the gauge-fixed nonlinear Yang–Mills map, and finally compute the parameter transversality and nonlinear obstruction coefficients. The present linear calculation supplies the exact candidate kernel at each wall; it does not assert existence of a nearby irreducible branch.

The theorem proved here is strictly lower-rung and fibrewise. It constructs explicit reducible split-Cartan $SU(r)$ Yang–Mills critical points on the already existing regular flat four-torus fibres and determines their topology, action, normalized Jacobi spectrum, physical Hessian scaling, Morse index, and nullity. It constructs no new base, spacetime, cone, S^{12}, S^{24}, transfer matrix, or interacting quantum Hamiltonian. The radial multiplication operator already treated in the canonical native module is not repeated here.

There is also an exact obstruction to interpreting the nonzero-flux bundle as the restriction of a bundle on a proposed higher carrier. If $i : F_T \to Y$ is any map and $H^4(Y;\mathbb{Z}) = 0$, then every complex bundle $\mathscr{V} \to Y$ has $c_2(\mathscr{V}) = 0$, so naturality gives

$$c_2(i^*\mathscr{V}) = i^* c_2(\mathscr{V}) = 0.$$

This contradicts equations (535) and (536) for $\mathscr{E}_{\mathbf{m},T}$. Therefore this nonzero-c_2 bundle cannot extend over any carrier with vanishing fourth integral cohomology, including a space diffeomorphic to S^6, S^{12}, or S^{24}. Changing the gauge group changes the root system on the fibre; it does not by itself produce the proposed higher rung or any conclusion about a physical mass gap.

15 The higher-rung SU(3) Gram cone

This section isolates the part of the proposed higher-rung construction that is independent of the candidate complex threefold and proves it from the underlying linear action. The compact object used here is the representation sphere

$$\mathrm{Mat}_{3\times 2}(\mathbb{C})^+,$$

the one-point compactification of a twelve-real-dimensional vector space. It must not be identified with a cusp-induced S^{12} construction unless an additional based identification has first been chosen. The proofs below also distinguish three structures that otherwise look deceptively similar: the Lorentzian determinant on $\mathrm{Herm}_2(\mathbb{C})$, the positive-semidefinite Gram cone inside that vector space, and a separately chosen Euclidean metric on the positive-definite cone.

15.1 Conventions and the Gram morphism

Put

$$\mathscr{V} = \mathrm{Mat}_{3\times 2}(\mathbb{C}), \qquad \Gamma : \mathscr{V} \longrightarrow \mathrm{Herm}_2(\mathbb{C}), \qquad \Gamma(A) = A^{\dagger}A. \tag{566}$$

Write $A = (a\ b)$, with $a, b \in \mathbb{C}^3$, and let $\mathrm{SU}(3)$ act on the left:

$$U \cdot A = UA.$$

For $r = 0, 1, 2$, let

$$\mathscr{V}_{(r)} = \{A \in \mathscr{V} : \mathrm{rank}_{\mathbb{C}}\, A = r\}.$$

The Hermitian products are linear in the second argument. We orient $\mathscr{V}$ by the complex orientation, in the real coordinate order

$$\begin{aligned}(&\operatorname{Re} A_{11}, \operatorname{Im} A_{11}, \operatorname{Re} A_{12}, \operatorname{Im} A_{12}, \operatorname{Re} A_{21}, \operatorname{Im} A_{21},\\ &\operatorname{Re} A_{22}, \operatorname{Im} A_{22}, \operatorname{Re} A_{31}, \operatorname{Im} A_{31}, \operatorname{Re} A_{32}, \operatorname{Im} A_{32}).\end{aligned} \tag{567}$$

On $\mathrm{Herm}_2(\mathbb{C})$, use the oriented real-linear coordinates

$$\Phi(t, x, y, z) = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix}, \qquad \mathrm{or}_{\mathrm{Herm}} = dt \wedge dx \wedge dy \wedge dz. \tag{568}$$

Thus, if $\Gamma(A)_{12} = \langle a, b\rangle$, then

$$\begin{aligned} t &= \frac{1}{2}(\|a\|^2 + \|b\|^2), & z &= \frac{1}{2}(\|a\|^2 - \|b\|^2),\\ x &= \operatorname{Re}\langle a, b\rangle, & y &= -\operatorname{Im}\langle a, b\rangle. \end{aligned} \tag{569}$$

15.2 Exact orbit classification

Theorem 15.1 (The Gram quotient)**.** *The morphism Γ is $\mathrm{SU}(3)$-invariant, has image exactly $\mathrm{Herm}_2(\mathbb{C})_{\geq 0}$, and separates $\mathrm{SU}(3)$-orbits. It induces a proper homeomorphism*

$$\overline{\Gamma} : \mathscr{V}/\mathrm{SU}(3) \xrightarrow{\ \cong\ } \mathrm{Herm}_2(\mathbb{C})_{\geq 0}. \tag{570}$$

Proof. For $U \in \mathrm{SU}(3)$,

$$\Gamma(UA) = A^{\dagger}U^{\dagger}UA = A^{\dagger}A,$$

so Γ is invariant. Every matrix $A^{\dagger}A$ is positive semidefinite. Conversely, for $H \in \mathrm{Herm}_2(\mathbb{C})_{\geq 0}$, let $H^{1/2}$ be its unique positive-semidefinite square root and define

$$s(H) = \begin{pmatrix} H^{1/2} \\ 0 \quad 0 \end{pmatrix} \in \mathrm{Mat}_{3\times 2}(\mathbb{C}). \tag{571}$$

Then $s(H)^{\dagger}s(H) = H$, proving surjectivity and giving a continuous section.

It remains to prove exact orbit separation. Suppose $A^{\dagger}A = B^{\dagger}B$. Define

$$T : \operatorname{im} A \longrightarrow \operatorname{im} B, \qquad T(Au) = Bu.$$

If $Au = 0$, then

$$0 = \|Au\|^2 = u^\dagger A^\dagger A u = u^\dagger B^\dagger B u = \|Bu\|^2,$$

so $Bu = 0$, and T is well defined. For $u, v \in \mathbb{C}^2$,

$$\langle T(Au), T(Av)\rangle = u^\dagger B^\dagger B v = u^\dagger A^\dagger A v = \langle Au, Av\rangle.$$

Thus T is a unitary isomorphism of the image subspaces. Extend it to some $U_0 \in \mathrm{U}(3)$ satisfying $U_0 A = B$. Since $\operatorname{rank} B \le 2$, the orthogonal complement of $\operatorname{im} B$ has positive complex dimension. Choose a unitary map V which fixes $\operatorname{im} B$ pointwise and whose determinant is $(\det U_0)^{-1}$. Then

$$U = V U_0 \in \mathrm{SU}(3), \qquad UA = VB = B.$$

Hence equal Gram matrices are exactly equal $\mathrm{SU}(3)$-orbits.

The action is proper because $\mathrm{SU}(3)$ is compact; hence the orbit space is Hausdorff. The continuous map $H \mapsto [s(H)]$ is inverse to the continuous induced map $\overline{\Gamma}$, so (570) is a homeomorphism. Finally,

$$\|A\|_F^2 = \operatorname{tr}(A^\dagger A) = \operatorname{tr} \Gamma(A). \tag{572}$$

If K is compact in the positive-semidefinite cone, then $\Gamma^{-1}(K)$ is closed and bounded by (572), hence compact. Thus Γ is proper. For compact K, $\overline{\Gamma}^{-1}(K)$ is the continuous image in the orbit space of the compact set $\Gamma^{-1}(K)$, so $\overline{\Gamma}$ is proper as well. □

Proposition 15.2 (Stabilizers and orbit-type strata)**.** *For $A \in \mathscr{V}_{(r)}$,*

$$\operatorname{Stab}_{\mathrm{SU}(3)}(A) \cong \mathrm{SU}(3 - r). \tag{573}$$

The exact dimensions are

r	$\dim_{\mathbb{R}} \mathscr{V}_{(r)}$	$\operatorname{Stab}(A)$	$\dim_{\mathbb{R}}(\mathrm{SU}(3) \cdot A)$	$\dim_{\mathbb{R}}(\mathscr{V}_{(r)}/\mathrm{SU}(3))$
2	12	$\{1\}$	8	4
1	8	$\mathrm{SU}(2)$	5	3
0	0	$\mathrm{SU}(3)$	0	0.

(574)

Consequently the quotient has topological dimension four, but the action is free only on the rank-two stratum.

Proof. The equality $UA = A$ is equivalent to the assertion that U fixes $\operatorname{im} A$ pointwise. Restriction to its orthogonal complement identifies the stabilizer with $\mathrm{SU}(3 - r)$, proving (573). The rank-two locus is open in the twelve-real-dimensional vector space. The complex rank-one locus in the space of 3×2 matrices has complex dimension

$$1(3 + 2 - 1) = 4,$$

hence real dimension eight away from the origin. Subtracting $8 - \dim_{\mathbb{R}} \operatorname{Stab}(A)$ gives the displayed orbit and quotient dimensions. They agree with the target strata: positive-definite Hermitian matrices form a four-dimensional open cone, while every nonzero rank-one positive-semidefinite matrix has a unique expression $\lambda v v^\dagger$ with $\lambda > 0$ and $v \in S^3$, modulo $v \sim e^{i\theta} v$. Hence that target stratum is

$$(0, \infty) \times \mathbb{CP}^1 \cong (0, \infty) \times S^2,$$

of real dimension three. The rank-zero stratum is the origin. □

15.3 The free locus, its explicit trivialization, and orientation

Theorem 15.3 (Determinant-one completion)**.** *There is an* $\mathrm{SU}(3)$*-equivariant diffeomorphism*

$$\mathscr{V}_{(2)} \xrightarrow{\cong} \mathrm{SU}(3) \times \mathrm{Herm}_2(\mathbb{C})_{>0} \tag{575}$$

under which Γ *is projection to the second factor and the group acts by left multiplication on the first. Thus the principal bundle over the free quotient has a global section and is trivial.*

Proof. For $A \in \mathscr{V}_{(2)}$, set

$$H = A^\dagger A > 0, \qquad Q = AH^{-1/2} = (q_1\ q_2).$$

Then $Q^\dagger Q = I_2$, so (q_1, q_2) is an ordered orthonormal two-frame in $\mathbb{C}^3$. With $\varepsilon_{123} = 1$, define

$$(q_3)_i = \sum_{j,k=1}^{3} \varepsilon_{ijk} \overline{(q_1)_j}\, \overline{(q_2)_k}. \tag{576}$$

Antisymmetry gives $q_1^\dagger q_3 = q_2^\dagger q_3 = 0$. The Gram–Plücker identity, proved independently in theorem 15.4, gives $\|q_3\| = 1$. Finally, contraction of the two alternating tensors gives

$$\begin{aligned} \det[q_1\ q_2\ q_3] &= \sum_{i,j,k} \varepsilon_{ijk} (q_1)_i (q_2)_j (q_3)_k \\ &= \|q_1\|^2 \|q_2\|^2 - |q_1^\dagger q_2|^2 = 1. \end{aligned}$$

Thus $U_A = [q_1\ q_2\ q_3] \in \mathrm{SU}(3)$, and

$$A = U_A s(H),$$

where s is (571). The positive square root, inverse square root, and the polynomial-conjugate formula (576) are smooth on the positive-definite locus. Consequently

$$A \longmapsto (U_A, A^\dagger A), \qquad (U, H) \longmapsto U s(H)$$

are smooth inverse maps. Their equivariance is immediate. □

The order of factors in (575) does not by itself fix an orientation. We now fix and calculate it. Give $\mathfrak{su}(3)$ the ordered real basis

$$\begin{gathered} X_{12}, Y_{12}, X_{13}, Y_{13}, X_{23}, Y_{23}, D_1, D_2, \\ X_{jk} = E_{jk} - E_{kj}, \qquad Y_{jk} = i(E_{jk} + E_{kj}), \\ D_1 = i\,\mathrm{diag}(1, -1, 0), \qquad D_2 = i\,\mathrm{diag}(0, 1, -1). \end{gathered} \tag{577}$$

At

$$A_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

use the oriented Hermitian basis

$$K_t = I_2, \quad K_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad K_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{578}$$

and horizontal lifts $B_\alpha = \frac{1}{2} A_0 K_\alpha$. Directly, $d\Gamma_{A_0}(B_\alpha) = K_\alpha$. Relative to (567), the twelve columns

$$(X_{12}A_0, Y_{12}A_0, X_{13}A_0, Y_{13}A_0, X_{23}A_0, Y_{23}A_0, D_1A_0, D_2A_0, B_t, B_x, B_y, B_z)$$

form the exact real coordinate matrix

$$M_{\mathrm{or}} := \begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}. \tag{579}$$

Laplace expansion gives

$$\det M_{\mathrm{or}} = \frac{1}{2} > 0. \tag{580}$$

Since both factors in (575) are connected, the sign is constant on the free locus. Therefore the quotient orientation determined by

$$\mathrm{or}_{\mathscr{V}} = \mathrm{or}_{\mathrm{SU}(3)} \wedge \mathrm{or}_{\mathscr{V}_{(2)}/\mathrm{SU}(3)}$$

agrees under Γ with $dt \wedge dx \wedge dy \wedge dz$.

15.4 Lorentz determinant and Plücker norm

For $H = \Phi(t, x, y, z)$, direct expansion gives

$$\operatorname{tr} H = 2t, \qquad \det H = t^2 - x^2 - y^2 - z^2, \tag{581}$$

and the eigenvalues are $t \pm \sqrt{x^2 + y^2 + z^2}$. Hence

$$\begin{aligned}
H > 0 &\iff t > \sqrt{x^2 + y^2 + z^2}, \\
H \geq 0 &\iff t \geq \sqrt{x^2 + y^2 + z^2}, \\
\operatorname{rank} H = 1 &\iff t = \sqrt{x^2 + y^2 + z^2} > 0.
\end{aligned} \tag{582}$$

Thus det has signature $(1, 3)$ when positive directions are listed first, while $-\det$ has signature $(3, 1)$. The quotient image is not all of this Lorentz vector space: it is precisely the closed future causal cone, and its free locus is the open future timelike cone.

Proposition 15.4 (Gram–Plücker identity)**.** *For $A = (a\ b)$,*

$$\det \Gamma(A) = \|a\|^2 \|b\|^2 - |\langle a, b\rangle|^2 = \sum_{1 \leq i < j \leq 3} |a_i b_j - a_j b_i|^2 = \|a \wedge b\|^2. \tag{583}$$

Proof. Expanding gives

$$\begin{aligned}
\|a\|^2 \|b\|^2 - |\langle a, b\rangle|^2 &= \sum_{i,j} |a_i|^2 |b_j|^2 - \sum_{i,j} \overline{a_i} b_i a_j \overline{b_j} \\
&= \sum_{i<j} \left(|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - \overline{a_i} b_i a_j \overline{b_j} - \overline{a_j} b_j a_i \overline{b_i} \right) \\
&= \sum_{i<j} |a_i b_j - a_j b_i|^2.
\end{aligned}$$

The last expression is the squared Hermitian norm of $a \wedge b$ in the ordered basis $(e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3)$. □

The determinant is only one invariant and does not classify orbits. For example,

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} \sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \\ 0 & 0 \end{pmatrix}$$

satisfy $\det \Gamma(A_1) = \det \Gamma(A_2) = 1$, but their Gram matrices are I_2 and $\mathrm{diag}(2, 1/2)$, so theorem 15.1 shows that they lie in distinct orbits.

15.5 The four-coefficient form is additional input

Let $r = (r_1, r_2, r_3, r_4)^{\mathsf{T}} \in \mathbb{R}^4$, and define the orthogonal Hadamard coordinates

$$\begin{pmatrix} h \\ x \\ y \\ z \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{pmatrix}. \tag{584}$$

The proposed coefficient-space form

$$Q_4(r) = \frac{1}{2} \left(\sum_{i=1}^{4} r_i \right)^2 - \sum_{i=1}^{4} r_i^2 \tag{585}$$

satisfies the exact identity

$$Q_4(r) = h^2 - x^2 - y^2 - z^2. \tag{586}$$

Indeed, orthogonality gives $\sum r_i^2 = h^2 + x^2 + y^2 + z^2$ and $\sum r_i = 2h$.

The same Hermitian matrix can be written in the alternative convention

$$\Psi(h, x, y, z) = \begin{pmatrix} h + y & x + iz \\ x - iz & h - y \end{pmatrix}. \tag{587}$$

It agrees exactly with Φ under

$$(t, x, y_\Phi, z_\Phi) = (h, x, -z, y).$$

The transformation $(y, z) \mapsto (-z, y)$ has determinant $+1$, so the two conventions give the same matrix, Lorentz form, and orientation.

Nothing in the SU(3) orbit calculation selects the coefficient $1/2$. More generally,

$$Q_{4,\lambda}(r) = \lambda \left(\sum_i r_i \right)^2 - \sum_i r_i^2 = (4\lambda - 1)h^2 - x^2 - y^2 - z^2. \tag{588}$$

Every $\lambda > 1/4$ therefore has Lorentz signature $(1, 3)$, after a rescaling of h. The value $\lambda = 1/2$ is the unique member for which the fixed orthogonal coordinates in (584) have equal unit coefficients. This is a normalization choice in coefficient space, not a theorem forced by the Gram quotient.

Nor does a nonnegative coefficient tuple automatically determine a point of the positive Gram cone. The tuple $(1, 0, 0, 0)$ has $Q_4 = -1/2$. A coefficient tuple enters the Gram quotient through (587) only when that Hermitian matrix is positive semidefinite, equivalently when $h \geq 0$ and $Q_4 \geq 0$. The hypersurface $z = 0$ is a real-symmetric coefficient slice, but that fact alone does not supply a coordinatewise nonnegative embedding of a three-component mass domain.

15.6 The compact representation quotient

Theorem 15.5 (The S^{12} representation quotient)**.** *Let*

$$S^{12}_{\mathrm{col}} = \mathscr{V}^+ = \mathscr{V} \cup \{\infty\}$$

be the one-point compactification, and extend the $\mathrm{SU}(3)$ *action by fixing* ∞*. Then*

$$S^{12}_{\mathrm{col}}/\mathrm{SU}(3) \cong \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^+ \cong B^4. \tag{589}$$

The orbit-type strata are

source stratum	*quotient stratum*	*stabilizer*
$\operatorname{rank} A = 2$	$\operatorname{int} B^4$	$\{1\}$
$\operatorname{rank} A = 1$	$\partial B^4 \setminus \{p_0, p_\infty\}$	$\mathrm{SU}(2)$
$A = 0$	$\{p_0\}$	$\mathrm{SU}(3)$
$A = \infty$	$\{p_\infty\}$	$\mathrm{SU}(3)$.

(590)

The ball identification is topological on the full stratified quotient. On its smooth interior, it may be chosen to carry the quotient orientation to the orientation fixed in (568)*.*

Proof. As a real vector space, $\mathscr{V} \cong \mathbb{R}^{12}$, so its one-point compactification is S^{12}. On $\mathscr{V} \cong \mathbb{C}^3 \oplus \mathbb{C}^3$, the action of $U \in \mathrm{SU}(3)$ has complex determinant $(\det U)^2 = 1$, hence real determinant one; it preserves the standard orientation before and after one-point compactification.

The properness proved from (572) implies that Γ extends continuously to

$$\Gamma^+ : \mathscr{V}^+ \longrightarrow \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^+, \qquad \Gamma^+(\infty) = \infty.$$

It separates the finite orbits by theorem 15.1, and the added point is a distinct fixed orbit. Hence it induces a continuous bijection

$$\mathscr{V}^+/\mathrm{SU}(3) \longrightarrow \left(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}\right)^+.$$

The source is compact Hausdorff because a compact group acts on the compact Hausdorff space $\mathscr{V}^+$. The target is Hausdorff because the positive-semidefinite cone is locally compact Hausdorff. The continuous bijection is therefore a homeomorphism.

By (582), the map

$$(t, \mathbf{x}) \longmapsto (t - \|\mathbf{x}\|, \mathbf{x})$$

is a homeomorphism from the closed future cone to the closed half-space $\mathbb{R}_{\geq 0} \times \mathbb{R}^3$. For $u = (s, \mathbf{x})$ in that half-space, inverse stereographic projection is

$$u \longmapsto \frac{1}{1 + \|u\|^2}(2s, 2\mathbf{x}, \|u\|^2 - 1) \in S^4 \subset \mathbb{R}^5. \tag{591}$$

Its image is the closed hemisphere with first coordinate nonnegative, minus the north pole. The point at infinity supplies that pole. A closed hemisphere is a closed four-ball, proving (589).

Positive-definite matrices correspond to the open half-space and hence the ball interior. Nonzero rank-one matrices correspond to $\{0\} \times (\mathbb{R}^3 \setminus \{0\})$. The zero matrix and the point at infinity give two distinct endpoints in the compactification of that three-dimensional boundary. Combining this with theorem 15.2 proves (590). The orientation statement on the connected interior is the calculation following (580). □

The notation S^{12}_{col} records the construction. There is a canonical pointed homeomorphism of representation spheres

$$(\mathbb{C}^3)^+ \wedge (\mathbb{C}^3)^+ \cong (\mathbb{C}^3 \oplus \mathbb{C}^3)^+ = S^{12}_{\mathrm{col}}.$$

If a different based six-sphere X is used, a based homeomorphism $X \cong (\mathbb{C}^3)^+$ transports the action to $X \wedge X$. The transported action depends on that choice. Without an independently proved equivariance statement, it is not induced by a torus fibration, cusp monodromy, period matrix, or fibrewise flux on X.

15.7 Exact gauge-theoretic boundary

Theorem 15.6 (No native nonzero S^4 sector from the Gram quotient)**.** *The construction above supplies no native principal* SU(3) *bundle of nonzero second Chern number on* S^4*. More precisely:*

1. *the orbit projection over the rank-two locus is the product bundle in* (575)*, so its second Chern class is zero;*
2. *on the rank-one boundary the orbit projection is not a principal* SU(3)*-bundle, because its stabilizer is* SU(2)*;*
3. *the full compact quotient is* B^4*, not* S^4*; and*
4. *replacing* B^4 *by the one-point compactification of its interior is a different construction whose clutching class is additional data.*

Proof. The first two assertions are theorems 15.2 and 15.3; the third is theorem 15.5. For the fourth, the matrix logarithm is a diffeomorphism

$$\log : \mathrm{Herm}_2(\mathbb{C})_{>0} \xrightarrow{\cong} \mathrm{Herm}_2(\mathbb{C}) \cong \mathbb{R}^4.$$

Its one-point compactification is therefore S^4, but this operation collapses the entire finite rank-one boundary of the full Gram quotient into the new point at infinity. Principal SU(3)-bundles on this new S^4 are obtained by clutching two trivial bundles over hemispheres with a map

$$\kappa : S^3 \longrightarrow \mathrm{SU}(3).$$

The fibration $\mathrm{SU}(2) \to \mathrm{SU}(3) \to S^5$, together with $\pi_4(S^5) = \pi_3(S^5) = 0$, shows that

$$\pi_3(\mathrm{SU}(3)) \cong \pi_3(\mathrm{SU}(2)) = \pi_3(S^3) \cong \mathbb{Z}.$$

Choose the generator so that its associated bundle has second Chern number $+1$. A clutching class $k \in \mathbb{Z}$ then has second Chern number k. Extending the explicit product framing in theorem 15.3 by the same framing gives the constant clutching map and $k = 0$. Neither $A^\dagger A$ nor $\|a \wedge b\|^2$ selects a nonconstant κ. Thus $k \neq 0$ is valid additional gauge data, but it is not produced by the Gram construction. □

There are two further cohomological checks. If a candidate total space is diffeomorphic to S^6, then

$$H^2(S^6; \mathbb{Z}) = H^4(S^6; \mathbb{Z}) = 0.$$

Consequently no nonzero fibrewise degree-two class can be the restriction of a degree-two class on that total space, and no c_2 on the total space can restrict to a nonzero degree-four class on a fibre. This does not forbid independently chosen higher characteristic data; it proves exactly that the degree-two and degree-four classes in question do not extend.

Likewise, for every continuous map $f : S^{12} \to S^4$ and generator $u_4 \in H^4(S^4; \mathbb{Z})$,

$$f^* u_4 = 0$$

because $H^4(S^{12}; \mathbb{Z}) = 0$. Therefore the pullback along any such map of an SU(3) bundle with nonzero $c_2 = k u_4$ has $c_2 = 0$. This statement does not assert that every intrinsic SU(3) bundle on S^{12} is trivial; it says that an S^4 second-Chern charge is not inherited by pullback to the representation sphere.

15.8 The log-Euclidean metric and Yang–Mills theory are extra choices

The logarithm identifies the open positive cone with a real vector space. One may therefore declare

$$X = \log H, \qquad g_{\log} = \frac{1}{2}\operatorname{tr}(dX\, dX), \tag{592}$$

which is a complete flat Euclidean metric in the X-coordinates. This metric is not induced by the Frobenius metric on $\mathscr{V}$. To see the difference without discarding any scale factor, set

$$H(s) = sI_2, \qquad A(s) = \sqrt{s}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad s > 0.$$

For $\xi \in \mathfrak{su}(3)$,

$$\operatorname{Re}\operatorname{tr}(\dot{A}^{\dagger}\xi A) = \frac{\dot{s}}{2}\operatorname{Re}(\xi_{11} + \xi_{22}) = 0,$$

so $\dot{A}$ is horizontal for the Frobenius quotient. Hence

$$\|\dot{H}\|^2_{\mathrm{quot}} = \|\dot{A}\|^2_F = \frac{\dot{s}^2}{2s}. \tag{593}$$

By contrast, $X(s) = (\log s)I_2$, and

$$\|\dot{H}\|^2_{\log} = \frac{1}{2}\operatorname{tr}(\dot{X}^2) = \frac{\dot{s}^2}{s^2}. \tag{594}$$

The ratio of (593) and (594) depends on s, so no global constant rescaling identifies the metrics. The Lorentz form (581) is a third object and is indefinite.

After choosing (592), one may define classical Yang–Mills theory on the trivial product bundle supplied by theorem 15.3. The additional choice of the zero connection has $F_0 = 0$, so it solves the nonlinear classical equation $d_0^* F_0 = 0$; its Coulomb Hessian is the transverse free Laplacian. These are the standard zero-charge vacuum statements on a chosen Euclidean $\mathbb{R}^4$. The Gram morphism itself supplies no nonzero clutching class, no nonflat connection, no interacting Hamiltonian, no quantum Hilbert-space identification, and no mass-gap estimate. Thus the exact logical implication is

four-dimensional Gram/Lorentz quotient: proved;
$S^{12}_{\mathrm{col}}/\mathrm{SU}(3) \cong B^4$: proved;
nonzero native SU(3) sector on S^4 : not produced;
interacting or quantum Yang–Mills inheritance: not produced.

16 The three-column SU(3) Gram cone and its rank-loss S^8

This section proves the topology forced by the left actions of U(3) and SU(3) on three complex columns. The calculation produces a canonical topological S^8, but it does not equip that sphere with a complex structure, a fibration, a gauge bundle, or a physical interpretation.

16.1 Conventions

Put

$$\mathscr{V}_3 = \mathrm{Mat}_{3\times 3}(\mathbb{C}), \qquad \|A\|^2_F = \operatorname{tr}(A^{\dagger}A). \tag{595}$$

This is a real vector space of dimension 18. Both U(3) and SU(3) act on the left:

$$U \cdot A = UA. \tag{596}$$

Let

$$\mathcal{P}_3 = \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0\}, \qquad \mathcal{K}_3 = \{H \in \mathcal{P}_3 : \operatorname{tr} H = 1\}. \tag{597}$$

Write $S_F^{17} \subset \mathscr{V}_3$ for the Frobenius unit sphere and $\mathscr{V}_3^+ = \mathscr{V}_3 \cup \{\infty\} \cong S^{18}$ for the one-point compactification, with ∞ fixed by each group.

16.2 The U(3) orbit space

Theorem 16.1 (Exact Gram classification)**.** *The map*

$$\Gamma_U : \mathscr{V}_3 \longrightarrow \mathcal{P}_3, \qquad \Gamma_U(A) = A^\dagger A \tag{598}$$

is U(3)*-invariant, is surjective, and separates* U(3)*-orbits. It induces a homeomorphism*

$$\overline{\Gamma}_U : \mathscr{V}_3/\mathrm{U}(3) \xrightarrow{\sim} \mathcal{P}_3. \tag{599}$$

Proof. Invariance is the identity

$$(UA)^\dagger(UA) = A^\dagger U^\dagger U A = A^\dagger A.$$

For $H \succeq 0$, its positive-semidefinite square root satisfies

$$(H^{1/2})^\dagger H^{1/2} = H,$$

so the map is surjective.

Suppose $A^\dagger A = B^\dagger B$. Define

$$T : \operatorname{im} A \longrightarrow \operatorname{im} B, \qquad T(Ax) = Bx.$$

If $Ax = 0$, then

$$\|Bx\|^2 = x^\dagger B^\dagger B x = x^\dagger A^\dagger A x = \|Ax\|^2 = 0,$$

so T is well-defined. For $x, y \in \mathbb{C}^3$,

$$\langle T(Ax), T(Ay)\rangle = x^\dagger B^\dagger B y = x^\dagger A^\dagger A y = \langle Ax, Ay\rangle.$$

Thus T is a complex-linear isometry. Extend it to a unitary $U \in \mathrm{U}(3)$. Then $UA = B$, so the Gram matrix separates orbits.

Finally,

$$\|A\|_F^2 = \operatorname{tr}(A^\dagger A).$$

Hence Γ_U is proper. The induced map $\overline{\Gamma}_U$ is a proper continuous bijection between locally compact Hausdorff spaces and is therefore a homeomorphism. □

Lemma 16.2 (Explicit ball chart on the trace-one Gram body)**.** *Let*

$$E = \{X \in \mathrm{Herm}_3(\mathbb{C}) : \operatorname{tr} X = 0\} \tag{600}$$

with Frobenius inner product. Then $\dim_{\mathbb{R}} E = 8$*, and there is an explicit homeomorphism*

$$\chi : \mathcal{K}_3 \xrightarrow{\sim} B(E) \cong B^8 \tag{601}$$

which sends the singular locus in $\mathcal{K}_3$ *onto* $S(E)$*.*

Proof. Put $c = I_3/3$. If $u \in S(E)$, then $u \neq 0$ and $\operatorname{tr} u = 0$, so $\lambda_{\min}(u) < 0$. Define

$$R(u) = -\frac{1}{3\lambda_{\min}(u)} > 0. \tag{602}$$

The eigenvalues of $c + su$ are $1/3 + s\lambda_j(u)$. Therefore, for $s \geq 0$,

$$c + su \succeq 0 \quad \Longleftrightarrow \quad 0 \leq s \leq R(u), \tag{603}$$

and $c + R(u)u$ is singular.

Every $H \in \mathcal{K}_3 \setminus \{c\}$ has the unique expression

$$H = c + su, \qquad s = \|H - c\|_F, \qquad u = \frac{H - c}{\|H - c\|_F}.$$

Define

$$\chi(c) = 0, \qquad \chi(c + su) = \frac{s}{R(u)}u, \tag{604}$$

and

$$\chi^{-1}(0) = c, \qquad \chi^{-1}(ru) = c + rR(u)u \quad (0 < r \leq 1). \tag{605}$$

The minimum-eigenvalue function is continuous. Since it is strictly negative on the compact sphere $S(E)$, R is continuous, finite, positive, and bounded above and below. Equations (604) and (605) are therefore continuous also at the origin and are inverse homeomorphisms. Equation (603) shows that

$$\chi\big(\{H \in \mathcal{K}_3 : \operatorname{rank} H \leq 2\}\big) = S(E).$$

□

Corollary 16.3 (Unit-sphere U(3) quotient)**.** *There is a homeomorphism of pairs*

$$\Big(S_F^{17}/\mathrm{U}(3), \{[A] : \operatorname{rank} A \leq 2\}\Big) \cong (B^8, S^7). \tag{606}$$

Proof. On S_F^{17},

$$\operatorname{tr}(A^\dagger A) = \|A\|_F^2 = 1.$$

Apply Theorem 16.1 and Lemma 16.2. □

Theorem 16.4 (Full U(3) cone and compactification)**.** *There are homeomorphisms*

$$\mathscr{V}_3/\mathrm{U}(3) \cong \mathcal{P}_3 \cong \mathbb{R}^8 \times \mathbb{R}_{\geq 0} \tag{607}$$

and

$$\mathscr{V}_3^+/\mathrm{U}(3) \cong \mathcal{P}_3^+ \cong B^9. \tag{608}$$

Under the second homeomorphism,

$$\partial B^9 = \{[A] : \operatorname{rank} A \leq 2\} \cup \{[\infty]\} \cong S^8. \tag{609}$$

Proof. For $H \neq 0$, write

$$H = r\rho, \qquad r = \operatorname{tr} H > 0, \qquad \rho = \frac{H}{\operatorname{tr} H} \in \mathcal{K}_3.$$

Thus $\mathcal{P}_3$ is the infinite cone on $\mathcal{K}_3 \cong B^8$.

We give the compactification map explicitly. Put

$$t = \frac{r - 1}{r + 1} \in (-1, 1)$$

and define

$$\mathcal{D}^9 = \{(y,t) \in E \oplus \mathbb{R} : \|y\|_F + |t| \le 1\}. \tag{610}$$

Define

$$\begin{aligned} \Xi_U(r\rho) &= \big((1-|t|)\chi(\rho), t\big), \\ \Xi_U(0) &= (0,-1), \\ \Xi_U(\infty) &= (0,1). \end{aligned} \tag{611}$$

For $|t| < 1$, the inverse is

$$x = \frac{y}{1-|t|}, \qquad \rho = \chi^{-1}(x), \qquad r = \frac{1+t}{1-t}, \qquad H = r\rho. \tag{612}$$

As $r \downarrow 0$ or $r \to \infty$, the factor $1-|t|$ tends uniformly to zero. A subset of $\mathcal{P}_3$ is unbounded if and only if its trace is unbounded. Thus Ξ_U is a homeomorphism $\mathcal{P}_3^+ \to \mathcal{D}^9$.

For $p = (y,t) \neq 0$, put

$$\nu(p) = \sqrt{\|y\|_F^2 + t^2}, \qquad \gamma(p) = \|y\|_F + |t|$$

and define

$$\Theta(p) = \frac{\gamma(p)}{\nu(p)} p, \qquad \Theta(0) = 0. \tag{613}$$

Then $\|\Theta(p)\|_2 = \gamma(p) \le 1$. For nonzero $q \in B(E \oplus \mathbb{R})$, the inverse is

$$\Theta^{-1}(q) = \frac{\|q\|_2}{\|q_E\|_F + |q_{\mathbb{R}}|}\, q. \tag{614}$$

Hence $\Theta : \mathcal{D}^9 \to B^9$ is a homeomorphism and takes $\partial\mathcal{D}^9$ onto S^8.

For finite $H = r\rho$, Equation (611) lies on $\partial\mathcal{D}^9$ exactly when $\|\chi(\rho)\|_F = 1$, equivalently when H is singular. The points 0 and ∞ are the two boundary tips. This proves (609).

The canonical map

$$\mathscr{V}_3^+/\mathrm{U}(3) \longrightarrow (\mathscr{V}_3/\mathrm{U}(3))^+$$

is a continuous bijection: the finite orbits agree and the added orbit is $\{\infty\}$. Its source is compact and its target is Hausdorff, hence it is a homeomorphism. This proves (608).

Finally, $\Theta \circ \Xi_U$ sends ∞ to $N = (0,\dots,0,1) \in \partial B^9$. The Cayley map

$$(x', x_9) \longmapsto \left(\frac{2x'}{\|x'\|^2 + (1-x_9)^2}, \frac{1-\|x\|^2}{\|x'\|^2 + (1-x_9)^2} \right) \tag{615}$$

is a homeomorphism

$$B^9 \setminus \{N\} \longrightarrow \mathbb{R}^8 \times \mathbb{R}_{\ge 0}.$$

Its inverse is

$$(v,s) \longmapsto \left(\frac{2v}{\|v\|^2 + (s+1)^2}, \frac{\|v\|^2 + s^2 - 1}{\|v\|^2 + (s+1)^2} \right). \tag{616}$$

This proves (607). □

16.3 The SU(3) orbit space

Theorem 16.5 (Gram matrix plus alternating determinant)**.** *Define*

$$\Gamma_{SU} : \mathscr{V}_3 \longrightarrow \mathrm{Herm}_3(\mathbb{C}) \times \mathbb{C}, \qquad \Gamma_{SU}(A) = (A^\dagger A, \det A). \tag{617}$$

Its image is exactly

$$\mathcal{Y}_3 = \{(H, z) \in \mathcal{P}_3 \times \mathbb{C} : |z|^2 = \det H\}. \tag{618}$$

It separates SU(3)*-orbits and induces a homeomorphism*

$$\mathscr{V}_3/\mathrm{SU}(3) \xrightarrow{\sim} \mathcal{Y}_3. \tag{619}$$

Proof. The map is SU(3)-invariant, and

$$\det(A^\dagger A) = \overline{\det A} \det A = |\det A|^2.$$

Thus its image is contained in $\mathcal{Y}_3$.

Conversely, take $(H, z) \in \mathcal{Y}_3$. If H is singular, then $z = 0$, and $A = H^{1/2}$ realizes the pair. If $H \succ 0$, put

$$\omega = \frac{z}{\sqrt{\det H}} \in S^1, \qquad Q_\omega = \mathrm{diag}(\omega, 1, 1) \in \mathrm{U}(3), \qquad A = Q_\omega H^{1/2}.$$

Then $A^\dagger A = H$ and $\det A = z$. This proves surjectivity.

Suppose

$$A^\dagger A = B^\dagger B, \qquad \det A = \det B.$$

Theorem 16.1 gives $U_0 \in \mathrm{U}(3)$ with $U_0 A = B$. If A is invertible, then

$$\det U_0 = \frac{\det B}{\det A} = 1,$$

so $U_0 \in \mathrm{SU}(3)$.

If $\operatorname{rank} A \le 2$, choose a unitary V which is the identity on $\operatorname{im} B$ and has determinant $(\det U_0)^{-1}$ on $(\operatorname{im} B)^\perp$. This is possible because the orthogonal complement has positive complex dimension. Then

$$VB = B, \qquad \det(VU_0) = 1, \qquad (VU_0)A = B.$$

Thus the invariant separates SU(3)-orbits.

Since $\operatorname{tr} H = \|A\|_F^2$, the invariant map is proper. The induced continuous bijection onto the closed Hausdorff set $\mathcal{Y}_3$ is therefore a homeomorphism. □

Theorem 16.6 (The unit-sphere SU(3) quotient)**.** *Let*

$$\mathcal{Y}_{3,1} = \{(H, z) \in \mathcal{Y}_3 : \operatorname{tr} H = 1\}. \tag{620}$$

There is an explicit homeomorphism

$$S_F^{17}/\mathrm{SU}(3) \cong \mathcal{Y}_{3,1} \xrightarrow{\sim} S(E \oplus \mathbb{C}) \cong S^9. \tag{621}$$

Proof. The first homeomorphism follows from Theorem 16.5. To construct the second, let

$$x = \chi(H) \in B(E), \qquad r = \|x\|_F.$$

If $r < 1$, then $H \succ 0$ and $z \neq 0$. Define

$$\Omega(H, z) = \left(x, \sqrt{1 - r^2}\,\frac{z}{|z|}\right) \quad (r < 1), \qquad \Omega(H, 0) = (x, 0) \quad (r = 1). \tag{622}$$

Its norm is exactly one:

$$\|x\|_F^2 + \left|\sqrt{1-r^2}\,\frac{z}{|z|}\right|^2 = r^2 + (1-r^2) = 1.$$

The map is continuous on the full-rank locus. As H approaches the singular locus, $r \to 1$, so its complex coordinate tends to zero regardless of phase. Hence it is continuous everywhere.

Given $(x, w) \in S(E \oplus \mathbb{C})$, set $H = \chi^{-1}(x)$. If $\|x\|_F < 1$, then $w \neq 0$, and the unique inverse value is

$$z = \sqrt{\det H}\,\frac{w}{|w|}. \tag{623}$$

If $\|x\|_F = 1$, then $w = 0$, H is singular, and $z = 0$. Thus Ω is a continuous bijection from a compact space to a Hausdorff space and hence is a homeomorphism. □

Theorem 16.7 (The weighted cone and its compactification)**.** *There is a homeomorphism*

$$\Phi_{SU} : \mathcal{Y}_3 \xrightarrow{\sim} E \oplus \mathbb{C} \cong \mathbb{R}^{10} \tag{624}$$

and therefore

$$\mathscr{V}_3^+/\mathrm{SU}(3) \cong S^{10}. \tag{625}$$

Proof. For $(H, z) \neq (0, 0)$, set

$$s = \sqrt{\operatorname{tr} H}, \qquad h = \frac{H}{s^2}, \qquad \zeta = \frac{z}{s^3}. \tag{626}$$

The exponents are forced by scalar multiplication in the original source: for every real $\lambda > 0$,

$$(\lambda A)^\dagger(\lambda A) = \lambda^2 A^\dagger A, \qquad \det(\lambda A) = \lambda^3 \det A, \qquad s(\lambda A) = \lambda s(A). \tag{627}$$

Moreover,

$$\operatorname{tr} h = 1, \qquad |\zeta|^2 = \frac{|z|^2}{s^6} = \frac{\det H}{s^6} = \det h.$$

Thus $(h, \zeta) \in \mathcal{Y}_{3,1}$. Define

$$\Phi_{SU}(0, 0) = 0, \qquad \Phi_{SU}(H, z) = s\,\Omega(h, \zeta) \quad (s > 0). \tag{628}$$

Since $\Omega(h, \zeta)$ has norm one,

$$\|\Phi_{SU}(H, z)\| = s.$$

For nonzero $v \in E \oplus \mathbb{C}$, the inverse is obtained from

$$s = \|v\|, \qquad (h, \zeta) = \Omega^{-1}\left(\frac{v}{\|v\|}\right), \qquad H = s^2 h, \qquad z = s^3\zeta.$$

The norm identity proves continuity at the origin in both directions, so Φ_{SU} is a homeomorphism.

As in the proof of Theorem 16.4, compactness of SU(3) gives

$$\mathscr{V}_3^+/\mathrm{SU}(3) \cong (\mathscr{V}_3/\mathrm{SU}(3))^+.$$

The one-point compactification of $\mathbb{R}^{10}$ is S^{10}. □

16.4 Rank strata, the standard S^8, and determinant winding

Proposition 16.8 (Stabilizers and dimensions)**.** *Let*

$$\mathscr{V}_{3,(r)} = \{A \in \mathscr{V}_3 : \operatorname{rank}_{\mathbb{C}} A = r\}.$$

For $A \in \mathscr{V}_{3,(r)}$,

$$\operatorname{Stab}_{\mathrm{U}(3)}(A) \cong \mathrm{U}(3-r), \qquad \operatorname{Stab}_{\mathrm{SU}(3)}(A) \cong \mathrm{SU}(3-r), \tag{629}$$

with the rank-three and SU(1) *stabilizers understood to be trivial. The quotient-stratum dimensions are*

r	$\dim_{\mathbb{R}} \mathscr{V}_{3,(r)}$	$\dim U(3)\cdot A$	$\dim \mathscr{V}_{3,(r)}/U(3)$	$\dim SU(3)\cdot A$	$\dim \mathscr{V}_{3,(r)}/SU(3)$
3	*18*	*9*	*9*	*8*	*10*
2	*16*	*8*	*8*	*8*	*8*
1	*10*	*5*	*5*	*5*	*5*
0	*0*	*0*	*0*	*0*	*0*

On the unit sphere, the nonzero quotient-stratum dimensions are

r	3	2	1
$U(3)$	8	7	4
$SU(3)$	9	7	4.

The trace-one rank-one locus is $\mathbb{C}P^2$.

Proof. A left stabilizer fixes $\operatorname{im} A$ pointwise. Its unrestricted unitary action on $(\operatorname{im} A)^{\perp}$ gives $\mathrm{U}(3-r)$. Imposing determinant one gives $\mathrm{SU}(3-r)$. The real dimension of the complex rank-r matrix stratum is

$$2r(6-r).$$

Subtracting the orbit dimensions gives the table; intersecting a nonzero rank stratum with the unit sphere subtracts one more dimension.

A trace-one positive-semidefinite matrix has rank one exactly when it has the form $vv^{\dagger}$ with $\|v\| = 1$. The phase of v does not change $vv^{\dagger}$, so this locus is

$$S^5/S^1 \cong \mathbb{C}P^2.$$

□

Theorem 16.9 (The exact rank-loss pair and complement)**.** *Let*

$$\mathcal{Y}_{3,\mathrm{sing}} = \{(H,0) \in \mathcal{Y}_3 : \operatorname{rank} H \le 2\}. \tag{630}$$

The homeomorphism Φ_{SU} *gives homeomorphisms of pairs*

$$(\mathcal{Y}_3, \mathcal{Y}_{3,\mathrm{sing}}) \cong (E \oplus \mathbb{C}, E \oplus \{0\}) \tag{631}$$

and

$$(\mathcal{Y}_3^+, \mathcal{Y}_{3,\mathrm{sing}}^+) \cong (S^{10}, S^8_{\mathrm{lin}}). \tag{632}$$

Consequently the compactified rank-loss locus is a standard, topologically locally flat codimension-two $S^8 \subset S^{10}$*, and*

$$S^{10} \setminus S^8 \cong E \times (\mathbb{C} \setminus \{0\}) \cong \mathbb{R}^8 \times (\mathbb{C} \setminus \{0\}) \simeq S^1. \tag{633}$$

The map

$$\vartheta(H,z) = \frac{z}{|z|} = \frac{\det A}{|\det A|} \tag{634}$$

is the angular deformation-retraction coordinate and represents a generator of

$$H^1(S^{10} \setminus S^8; \mathbb{Z}) \cong \pi_1(S^{10} \setminus S^8) \cong \mathbb{Z}$$

after choosing the standard positive orientation on S^1.

Proof. In Equation (622), H is singular exactly when $\|\chi(H)\|_F = 1$, in which case the complex coordinate is zero. Conversely, a zero complex coordinate on the unit S^9 has its E-coordinate on $S(E)$, and its inverse Gram matrix is singular. Radial extension through Equation (628) therefore gives exactly (631).

The homeomorphism Φ_{SU} is proper and extends to one-point compactifications, proving (632). The second member is the one-point compactification of the linear pair

$$(E \oplus \mathbb{C}, E \oplus \{0\}).$$

It is the standard topological pair $(S^{10}, S^8_{\text{lin}})$; therefore the S^8 is topologically locally flat at every point.

This local-flatness statement is not a smooth-stratification assertion. The unit rank-deficient S^7 contains the rank-one orbit stratum $\mathbb{C}P^2$, with rank-two complement $S^7 \setminus \mathbb{C}P^2$. The homeomorphism χ uses the minimum eigenvalue and need not be smooth across eigenvalue-collision strata.

Removing the linear subspace from the affine pair gives (633). Under Φ_{SU}, the phase of its complex coordinate is $z/|z|$, so ϑ is the usual angular projection

$$E \times (\mathbb{C} \setminus \{0\}) \longrightarrow S^1.$$

This is a deformation retraction. The explicit quotient loop

$$A_\theta = \frac{1}{\sqrt{3}} \operatorname{diag}(e^{i\theta}, 1, 1), \qquad 0 \le \theta \le 2\pi, \tag{635}$$

satisfies

$$A_\theta^\dagger A_\theta = \frac{1}{3} I_3, \qquad \det A_\theta = \frac{e^{i\theta}}{3\sqrt{3}}.$$

Thus $\vartheta([A_\theta]) = e^{i\theta}$ winds once and is a generator. □

Corollary 16.10 (Residual determinant-phase circle)**.** *The residual group*

$$\mathrm{U}(3)/\mathrm{SU}(3) \cong S^1$$

acts on $E \oplus \mathbb{C}$ *by*

$$e^{i\theta} \cdot (x, w) = (x, e^{i\theta} w). \tag{636}$$

Its fixed set after compactification is the standard $S^8 \subset S^{10}$*, and*

$$S^{10}/S^1 \cong B^9, \qquad \operatorname{Fix}(S^1) = S^8 = \partial B^9. \tag{637}$$

Proof. If $U \in \mathrm{U}(3)$, left multiplication fixes $H = A^\dagger A$ and sends $z = \det A$ to $(\det U)z$. Every phase occurs as a determinant. Equation (622) therefore converts the residual action into (636). On the unit sphere,

$$S(E \oplus \mathbb{C})/S^1 \cong \{(x, |w|) : \|x\|_F^2 + |w|^2 = 1,\ |w| \ge 0\} \cong B(E).$$

On the affine cone,

$$(E \oplus \mathbb{C})/S^1 \cong E \times \mathbb{R}_{\ge 0}.$$

One-point compactification gives B^9, and the fixed set is $(E \oplus \{0\})^+ \cong S^8$, which maps to the boundary. □

16.5 The exact rank-two/rank-three distinction

Theorem 16.11 (Quadratic Lorentz cone versus cubic rank-three cone)**.** *For* $\mathscr{V}_2 = \mathrm{Mat}_{3\times 2}(\mathbb{C})$*, the left* SU(3)*- and* U(3)*-orbit relations coincide, and*

$$\mathscr{V}_2/\mathrm{SU}(3) \cong \mathrm{Herm}_2(\mathbb{C})^{\succeq 0}. \tag{638}$$

In the coordinates

$$H_2(t, x_1, x_2, x_3) = \begin{pmatrix} t + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & t - x_3 \end{pmatrix}, \tag{639}$$

$$\operatorname{tr} H_2 = 2t, \qquad \det H_2 = t^2 - x_1^2 - x_2^2 - x_3^2, \tag{640}$$

and

$$H_2 \succeq 0 \quad \Longleftrightarrow \quad t \geq 0, \qquad t^2 \geq x_1^2 + x_2^2 + x_3^2. \tag{641}$$

Thus the two-column quotient is exactly the closed future $3+1$*-dimensional Lorentz cone; its one-point compactification is* B^4*, with rank-one-plus-infinity boundary* S^3*.*

By contrast, the three-column U(3) *quotient is the nine-real-dimensional, Jordan-rank-three cone* $\mathrm{Herm}_3(\mathbb{C})^{\succeq 0}$*. Its determinant is cubic, its trace-one section is* B^8*, and its compactified rank-deficient-plus-infinity boundary is* S^8*. It is not a Lorentz cone.*

Proof. Every 3×2 matrix has rank at most two, so the determinant-adjustment step in the proof of Theorem 16.5 converts every left U(3) equivalence into a left SU(3) equivalence. The Gram proof of Theorem 16.1, with two columns, gives (638).

Direct expansion of (639) gives (640). A 2×2 Hermitian matrix is positive semidefinite exactly when its trace and determinant are nonnegative, which gives (641). Its trace-one section is

$$t = \frac{1}{2}, \qquad x_1^2 + x_2^2 + x_3^2 \leq \frac{1}{4},$$

a closed 3-ball. The compactification argument of Theorem 16.4, with E of dimension three, gives B^4 with boundary S^3.

For three columns, Theorem 16.4 gives the B^9/S^8 statements. The determinant of a 3×3 Hermitian matrix is homogeneous of degree three, so this rank-three cone is not the rank-two quadratic Lorentz cone. □

16.6 The alternating invariant and the precise nonclaim

The extra invariant retained by the square-matrix SU(3) quotient is

$$z = \det A = \varepsilon_{ijk} A_{i1} A_{j2} A_{k3}. \tag{642}$$

It is exactly the complex alternating volume of the three columns:

$$\det(UA) = \det(U) \det(A).$$

It is invariant for $U \in \mathrm{SU}(3)$, its phase is rotated by the residual $\mathrm{U}(3)/\mathrm{SU}(3)$, and it vanishes exactly when the three columns are linearly dependent.

This invariant theory supplies exact carrier and orbit data for a physical translation, but does not by itself complete that translation. The next construction must specify a quark field, colour-charge and baryon observables, a principal SU(3) bundle with connection and characteristic classes, and an interacting Yang–Mills sector, together with maps relating each object to Equation (642). Likewise, Theorem 16.4 supplies a distinguished S^8, while an almost-complex or integrable-complex structure, a three-fibre structure, and

analogues of the candidate S^6 special-fibre attaching maps remain exact construction problems rather than consequences already proved by the compactification theorem.

The exact conclusion is the topological carrier theorem

$$\boxed{\mathscr{V}_3^+/\mathrm{U}(3) \cong B^9, \quad \partial B^9 \cong S^8, \quad \mathscr{V}_3^+/\mathrm{SU}(3) \cong S^{10}, \quad S^{10} \setminus S^8 \simeq S^1.} \tag{643}$$

Any gauge-theoretic or geometric use of this carrier requires additional explicit descent data.

17 A four-branch toroidal lift and the obstruction to a complex S^8 total space

17.1 The exact meaning of the three-way cusp

The cusp in the audited threefold is controlled by the $\mathbb{Z}^2$-periodic A_2-triangulation of $\mathbb{R}^2$. Its quotient has one vertex, three edges, and two triangles. The central fibre is globally irreducible; its normalization is the degree-six del Pezzo surface, whose six toric boundary curves are paired into three double curves. At either of the two deepest points the smooth total threefold maps to the disc by

$$(z_0, z_1, z_2) \longmapsto t = z_0 z_1 z_2. \tag{644}$$

Thus the three relevant statements are distinct: there are three local branches at a deepest point, three global double curves, and only one global irreducible component. The generalized dual intersection complex is

$$\mathcal{T}_{A_2}/\mathbb{Z}^2 \cong \mathbb{R}^2/\mathbb{Z}^2 = T^2. \tag{645}$$

These baseline data are transcribed from the audited canonical-source snapshot of 620,782 bytes and 6,315 lines, with SHA–256 digest

`0696B6B7FAB83804DFEC947D45607601E5D7F57D64C6148487969FCCB6C67D2D.`

line 320 records the periodic A_2 triangulation and its $(1, 3, 2)$ quotient cell counts; lines 372 and 377 record the normalization, three double curves, and two triple points; lines 1431–1461 construct the fan and its unimodular toric charts; lines 1647–1678 give the quotient local models and orbit strata; and line 3966 gives the deepest central-fibre equation $\{z_0 z_1 z_2 = 0\}$. The line references and digest fix the exact source edition without introducing a machine-local dependency.

The higher-dimensional construction below raises the rank of this periodic triangulation from two to three. It does not replace three line bundles by four line bundles.

17.2 The rank-three periodic fan

Let

$$L = \mathbb{Z}^3, \qquad L_{\mathbb{R}} = \mathbb{R}^3, \qquad \mathbf{1} = e_1 + e_2 + e_3.$$

For $m \in L$ and $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$, put

$$\sigma_{m,\pi} = \mathrm{conv}\left\{m,\ m + e_{\pi_1},\ m + e_{\pi_1} + e_{\pi_2},\ m + \mathbf{1}\right\}. \tag{646}$$

Inside the cube $m + [0, 1]^3$, this is the region in which the coordinates, in the order π_1, π_2, π_3, are weakly decreasing. The six possible orders partition the cube, and the induced partitions agree on common faces. Their integral translates therefore give a locally finite L-periodic triangulation $\mathcal{T}_3$ of $\mathbb{R}^3$.

Put

$$N' = L \oplus \mathbb{Z}e_4$$

and, for every cell P of $\mathcal{T}_3$, define

$$\widehat{P} = \mathrm{cone}\{(v,1) : v \in P\} \subset N'_{\mathbb{R}}. \tag{647}$$

Because all cells are bounded, $\widehat{P} \cap \widehat{Q} = \widehat{P \cap Q}$. Hence the cones $\widehat{P}$, together with the zero cone, form a fan Σ_3.

Lemma 17.1 (Unimodularity)**.** *Every maximal cone of Σ_3 is unimodular.*

Proof. For the four lifted vertices of $\sigma_{m,\pi}$, subtract the first column $(m,1)$ from the other three columns. Up to the orientation sign, the determinant is

$$\det\left(e_{\pi_1},\ e_{\pi_1} + e_{\pi_2},\ \mathbf{1}\right) = \pm 1.$$

The four primitive ray generators therefore form a basis of N'. Every face is unimodular as well. □

Let $Y_3 = X_{\Sigma_3}$ be the associated, generally non-quasi-compact, toric fourfold. It is smooth by theorem 17.1. Let

$$M' = \mathrm{Hom}(N', \mathbb{Z}), \qquad h = (0,0,0,1) \in M',$$

and define the toric character

$$t = \chi^h : Y_3 \longrightarrow \mathbb{C}. \tag{648}$$

The character h has value one on every primitive ray generator $(v,1)$. Consequently

$$\mathrm{div}(t) = \sum_{v \in L} E_v \tag{649}$$

with multiplicity one on every toric divisor E_v.

Proposition 17.2 (Four-branch local equation)**.** *If a tetrahedron σ has vertices v_0, v_1, v_2, v_3, its toric chart is $\mathbb{C}^4$, and in its dual-character coordinates (z_0, z_1, z_2, z_3) one has*

$$t = z_0 z_1 z_2 z_3. \tag{650}$$

In particular, four distinct components of the divisor upstairs meet transversely at the torus fixed point of $\widehat{\sigma}$.

Proof. Let m_0, m_1, m_2, m_3 be the basis of M' dual to $(v_0,1), \ldots, (v_3,1)$, and set $z_i = \chi^{m_i}$. Unimodularity identifies the affine chart with $\mathbb{C}^4$. Since h takes value one on every ray generator,

$$h = m_0 + m_1 + m_2 + m_3.$$

Taking characters gives (650). □

17.3 The analytic quotient and the abelian-threefold fibres

For $\lambda \in L$, define the integral shear

$$\varphi_\lambda : N' \longrightarrow N', \qquad \varphi_\lambda(y,r) = (y + r\lambda, r). \tag{651}$$

It sends $\widehat{P}$ to $\widehat{P + \lambda}$, so it induces a toric automorphism Φ_λ of Y_3. These automorphisms preserve t and satisfy $\Phi_\lambda \Phi_\mu = \Phi_{\lambda+\mu}$. On the dense torus and over $t \neq 0$, their formula is

$$\Phi_\lambda(x_1, x_2, x_3, t) = (t^{\lambda_1} x_1, t^{\lambda_2} x_2, t^{\lambda_3} x_3, t). \tag{652}$$

The lattice map (651), rather than the expression with possibly negative powers in (652), proves that the action extends over the toric boundary.

Fix $0 < \epsilon < 1$ and set

$$Y_{3,\epsilon} = \{p \in Y_3 : |t(p)| < \epsilon\}.$$

Define

$$N_3 = Y_{3,\epsilon}/L, \qquad \pi_3 : N_3 \longrightarrow \Delta_\epsilon. \tag{653}$$

Theorem 17.3 (The rank-three toroidal cusp)**.** *The L-action on $Y_{3,\epsilon}$ is free and properly discontinuous. The quotient N_3 is a smooth complex fourfold and π_3 is proper. For $q \neq 0$, its fibre is the abelian threefold*

$$N_{3,q} \cong \mathbb{C}^3/(\mathbb{Z}^3 + \tau\mathbb{Z}^3) \cong E_\tau^3, \qquad q = e^{2\pi i\tau}, \quad \tau \in \mathbb{H}.$$

Proof. A boundary point belongs to a unique torus orbit $O_{\widehat{P}}$. If Φ_λ fixes it, then $O_{\widehat{P+\lambda}} = O_{\widehat{P}}$, whence $P + \lambda = P$. Since P is bounded, $\lambda = 0$. On a fibre $t = q \neq 0$, (652) is free because $|q| < 1$ and the equalities $q^{\lambda_i} = 1$ for all i force $\lambda = 0$.

For proper discontinuity and properness, use on the dense fibre the logarithmic coordinate

$$\operatorname{trop}_q(x) = \left(\frac{\log|x_1|}{\log|q|}, \frac{\log|x_2|}{\log|q|}, \frac{\log|x_3|}{\log|q|}\right). \tag{654}$$

Equation (652) translates this coordinate by λ. Every orbit can therefore be moved into $[0,1]^3$. If

$$r = \sum_{i=0}^{3} \alpha_i v_i$$

belongs to $\sigma = \operatorname{conv}(v_0, v_1, v_2, v_3)$, then $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$, and the dual-character coordinates of theorem 17.2 obey

$$\frac{\log|z_i|}{\log|q|} = \alpha_i, \qquad |z_i| = |q|^{\alpha_i} \leq 1.$$

For $\epsilon' < \epsilon$, let $F_{\epsilon'}$ be the closure of the points with $0 < |q| \leq \epsilon'$ and $\operatorname{trop}_q(x) \in [0,1]^3$. For each of the six tetrahedra in that cube, the preceding identities put these points in the closed unit polydisc $\{|z_0|, \dots, |z_3| \leq 1\}$ of its $\mathbb{C}^4$-chart. The finite union of those polydiscs is compact and closed in the Hausdorff toric variety; $F_{\epsilon'}$, being closed in that union, is compact. Its translates cover the dense locus, and the same holds on $t = 0$ because every cell has an L-translate meeting the fundamental cube.

It remains to verify proper discontinuity at the boundary, where trop_q itself is undefined. Let $K \subset Y_{3,\epsilon}$ be compact. Local compactness and the toric affine cover give finitely many maximal-cone charts U_{σ_j} and compact subsets $K_j \subset U_{\sigma_j}$ whose interiors cover K. On the dense part of one such compact set, write

$$r = \sum_{i=0}^{3} \alpha_i v_i, \qquad \alpha_i = \frac{\log|z_i|}{\log|t|}, \qquad \sum_i \alpha_i = 1.$$

The coordinate moduli are uniformly bounded above. When $|t|$ is small, this gives a uniform lower bound for every α_i, and the identity $\sum_i \alpha_i = 1$ gives a uniform upper bound. When $|t|$ is bounded away from zero, the product $t = z_0 z_1 z_2 z_3$ and the upper bounds give uniform positive lower bounds for the $|z_i|$, hence again uniform bounds for every α_i. Since only finitely many vertex sets occur, all dense points of K consequently have tropical coordinates in one bounded set $C_K \subset \mathbb{R}^3$.

If $p, \Phi_\lambda(p) \in K$ are dense, then $\lambda \in (C_K - C_K) \cap L$, a finite set. If p lies on the boundary, its orbit cell and that of $\Phi_\lambda(p)$ lie among the finitely many faces of the σ_j, and they satisfy $P + \lambda = Q$; again only finitely many λ occur. Thus a compact set meets only finitely many of its translates, proving proper discontinuity. Finally, the image of $F_{\epsilon'}$ is the entire inverse image of $\{|q| \leq \epsilon'\}$, which proves that π_3 is proper. Freeness on the smooth $Y_{3,\epsilon}$ makes N_3 smooth.

For $q \neq 0$, all non-dense toric orbits lie in $t = 0$; hence $Y_{3,q} = (\mathbb{C}^*)^3$, and

$$N_{3,q} = (\mathbb{C}^*)^3/q^{\mathbb{Z}^3}.$$

The map

$$[w_1, w_2, w_3] \longmapsto [e^{2\pi i w_1}, e^{2\pi i w_2}, e^{2\pi i w_3}]$$

induces the asserted isomorphism from $\mathbb{C}^3/(\mathbb{Z}^3 + \tau\mathbb{Z}^3)$. □

The preceding construction is the untwisted $B_0 = I$, $C(t) = 0$ instance of the Tate–Mumford construction. The general analytic construction is due to Mumford, *An analytic construction of degenerating abelian varieties over complete rings*, Compositio Math. **24** (1972), 239–272. No appeal to that generality is needed for the displayed quotient.

17.4 Monodromy and the cusp inclusion on fundamental groups

Use the integral basis

$$(a_1, a_2, a_3, b_1, b_2, b_3)$$

of $H_1(N_{3,q}; \mathbb{Z})$, where a_i is the unit period and b_i is the τ-period in the i-th factor. A positive loop about $q = 0$ sends τ to $\tau + 1$.

Proposition 17.4 (Cusp monodromy and filling map)**.** *The monodromy is*

$$M_0 = \begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix}, \qquad R = M_0 - I = \begin{pmatrix} 0 & I_3 \\ 0 & 0 \end{pmatrix}. \tag{655}$$

It satisfies

$$R^2 = 0, \qquad \operatorname{rank} R = 3, \qquad \ker R = \operatorname{im} R = \mathbb{Z}\langle a_1, a_2, a_3 \rangle.$$

Moreover

$$\pi_1(N_3) \cong L \cong \mathbb{Z}^3,$$

and the inclusion of a smooth fibre induces

$$a_i \longmapsto 0, \qquad b_i \longmapsto e_i \in L. \tag{656}$$

Proof. The period change $b_i \mapsto b_i + a_i$, $a_i \mapsto a_i$, gives (655); the assertions about R follow by direct block multiplication.

The primitive rays

$$(0, 1), \quad (e_1, 1), \quad (e_2, 1), \quad (e_3, 1)$$

generate N'. We record the toric fundamental-group calculation needed here. If Σ_0 is a finite subfan of this smooth fan and N_{Σ_0} is the sublattice generated by its primitive ray vectors, then

$$\pi_1(X_{\Sigma_0}) \cong N'/N_{\Sigma_0}.$$

Indeed, the dense torus has fundamental group N'. Adjoining a neighbourhood of the generic point of the divisor belonging to a primitive ray v fills precisely the circle v, so van Kampen quotients by $\langle v \rangle$. The remaining orbit strata have complex codimension at least two; general position for loops and disc homotopies in the smooth real manifold shows that adding them changes no fundamental group. Applying van Kampen over all rays gives the displayed quotient.

Every loop in Y_3 has compact image and is therefore contained in the union of finitely many toric affine charts. Enlarge their finite subfan by the four rays displayed above. Those rays generate N', so the finite subfan formula makes this enlarged toric open set simply connected. The loop bounds there, and hence $\pi_1(Y_3) = 0$. For $0 < u \leq 1$, the real one-parameter torus element $\vartheta_u = (1, 1, 1, u)$ satisfies

$$t(\vartheta_u p) = u\, t(p).$$

Any loop in $Y_{3,\epsilon}$ bounds a disc in Y_3. Choosing u small moves that entire disc into $Y_{3,\epsilon}$, while $\vartheta_{u'}$, $u \leq u' \leq 1$, gives a homotopy back to the original loop. Hence $\pi_1(Y_{3,\epsilon}) = 0$. The quotient map in (653) is therefore the universal cover with deck group L, so $\pi_1(N_3) = L$.

The loop a_i is the angular loop in the i-th $\mathbb{C}^*$-coordinate and is killed by the corresponding toric divisor. The path representing b_i closes precisely by the deck transformation Φ_{e_i}. This proves (656). □

17.5 The central fibre, its normalization, and all strata

Let

$$W_3 = \pi_3^{-1}(0) = \left(\bigcup_{v \in L} E_v \right) / L. \tag{657}$$

The toric orbit–cone correspondence identifies its open strata with L-orbits of cells of $\mathcal{T}_3$. Its generalized dual intersection complex is therefore

$$\mathcal{T}_3 / L \cong \mathbb{R}^3 / L = T^3. \tag{658}$$

Theorem 17.5 (Forced stratum counts)**.** *The quotient cell structure and the open torus stratification of W_3 are*

cell	*number*	*open stratum*	*local branches*
vertex	1	$(\mathbb{C}^*)^3$	1
edge	7	$(\mathbb{C}^*)^2$	2
triangle	12	$\mathbb{C}^*$	3
tetrahedron	6	pt	4.

In particular,

$$\chi(W_3) = 6. \tag{659}$$

Proof. There is one vertex orbit. A fundamental parallelepiped of L has normalized three-volume $3! = 6$, whereas every unimodular tetrahedron has normalized volume one. Hence there are six tetrahedron orbits. Every triangular face is incident to two tetrahedra, so

$$f_2 = \frac{4 f_3}{2} = 12.$$

Since the quotient is T^3,

$$0 = \chi(T^3) = f_0 - f_1 + f_2 - f_3 = 1 - f_1 + 12 - 6,$$

and therefore $f_1 = 7$. Explicit edge representatives are

$$[0, e_1],\ [0, e_2],\ [0, e_3],\ [0, e_1 + e_2],\ [0, e_1 + e_3],\ [0, e_2 + e_3],\ [0, \mathbf{1}].$$

A k-cell gives a torus orbit of complex dimension $3 - k$ and an intersection of $k + 1$ local branches. Finally every positive-dimensional $(\mathbb{C}^*)^j$ has Euler characteristic zero. Additivity over the finite stratification leaves the six point strata and proves (659). □

Theorem 17.6 (Normalization)**.** *W_3 is globally irreducible. The quotient map induces its normalization*

$$\nu : E_0 \longrightarrow W_3.$$

The variety E_0 is the smooth three-dimensional permutohedral toric variety. It has fourteen toric boundary surfaces, thirty-six invariant curves, and twenty-four fixed points. Under ν, these map respectively two-to-one, three-to-one, and four-to-one onto seven double surfaces, twelve triple curves, and six quadruple points.

Proof. The lattice L acts simply transitively on the vertices and therefore transitively on the components E_v. The quotient W_3 is irreducible. Every cell has a vertex and can be translated to one containing zero, so the map $E_0 \to W_3$ is surjective. Only finitely many cells contain zero, hence the map is finite, and it is injective on the dense torus. The star fan at zero is complete and unimodular, so E_0 is smooth, compact, and normal. The finite birational map is consequently the normalization.

The rays of the star fan are

$$\pm\sum_{i\in I} e_i, \qquad \varnothing \neq I \subseteq \{1,2,3\}.$$

These are the fourteen rays of the A_3 braid fan. Its maximal cones are indexed by the twenty-four permutations of four symbols; hence the fan is the permutohedral fan. There are 24 maximal cones. Each has three two-dimensional faces and each such face is shared by two maximal cones, so there are $24 \cdot 3/2 = 36$ two-dimensional cones. Together with the fourteen rays this gives the asserted orbit counts.

An edge has two vertices, a triangle has three, and a tetrahedron has four. The normalization preimage of the corresponding generic quotient stratum therefore has respectively two, three, or four points. Division of the normalization counts by these degrees gives

$$14/2 = 7, \qquad 36/3 = 12, \qquad 24/4 = 6.$$

□

Remark 17.7 (Local branches are not global components). At each quadruple point the four local branches are four germs of the same global irreducible central fibre. The conductor in the normalization is the union of fourteen toric boundary surfaces, paired into seven quotient surfaces. No canonical four-term decomposition of a vector bundle or line bundle follows from this local branch number.

More generally, let $\bar{L} \cong \mathbb{Z}^3$ and let $B_0 : \bar{L} \hookrightarrow L$ have finite index d. Replacing λ by $B_0\lambda$ in (651) gives d vertex orbits and hence d global central components. The stratum counts are

$$(f_0, f_1, f_2, f_3) = d(1, 7, 12, 6). \tag{660}$$

Corollary 17.8 (Euler characteristic as the determinant of the shear)**.** *Choose oriented integral bases of $\bar{L}$ and L, and let $[B_0] \in \mathrm{Mat}_{3\times 3}(\mathbb{Z})$ be the matrix of the injective shear. The associated central fibre satisfies*

$$\chi(W_{3,B_0}) = 3!\,[L : B_0\bar{L}] = 3!\,|\det[B_0]|. \tag{661}$$

The right side is unchanged by independent unimodular changes of the two marked bases.

Proof. The index of the image of a full-rank integral matrix is $[L : B_0\bar{L}] = |\det[B_0]|$. By (660), the quotient has $6d$ tetrahedron orbits. Their zero-dimensional torus strata are the only strata with nonzero Euler characteristic, so $\chi(W_{3,B_0}) = 6d = 3!d$. Replacing the source and target bases multiplies the determinant by units in $\mathbb{Z}$, and hence does not change its absolute value. □

If a polarization additionally identifies $[B_0]$ with a symmetric positive-definite matrix, then its determinant is the cubic Jordan norm on that real symmetric subcone of $\mathrm{Herm}_3(\mathbb{C})_{\geq 0}$. Without those additional hypotheses $[B_0]$ is only an integral lattice map, not a positive Hermitian matrix. In either case the scalar in (661) does not determine the fan, the period correction, the singular filling, or the total space.

Thus four global components occur only after the independent choice $d = 4$; four local branches already occur for $d = 1$. A nontrivial Fable-style twist would additionally require holomorphic translations $c_\lambda(t)$ satisfying the exact group cocycle, extension, freeness, and properness identities.

17.6 Two independent obstructions to a complex S^8 globalization

Theorem 17.9 (Euler obstruction to the literal three-special-fibre pattern)**.** *Let $g : M^8 \to S^2$ be a proper triangulable map which is a locally trivial T^6-bundle away from finitely many critical values. If one singular fibre is W_3 and every other singular fibre is a finite free quotient of T^6, then*

$$\chi(M) = 6.$$

In particular, M is not homeomorphic to S^8.

Proof. Choose a triangulation for g in which every critical value is a vertex. Additivity of compactly supported Euler characteristic over the open regular stratum and the critical singleton strata, proved cell by cell in this finite triangulation, gives

$$\chi(M) = \chi_c(S^2 \setminus \Sigma)\chi(T^6) + \sum_{p\in\Sigma} \chi(g^{-1}(p)).$$

Here multiplicativity on the regular stratum follows by subdividing its locally trivial bundle over base cells. Since M is compact, $\chi_c(M) = \chi(M)$, and since $\chi(T^6) = 0$, the first term vanishes. By (659), the cusp contributes 6. A finite free quotient by a group G has Euler characteristic $\chi(T^6)/|G| = 0$. Therefore $\chi(M) = 6$, whereas $\chi(S^8) = 2$. □

Consequently, retaining the cusp W_3 in a real S^8 candidate would require additional genuinely singular fibres whose Euler characteristics sum to -4. Smooth multiple torus fibres cannot supply them.

Theorem 17.10 (No smooth complex S^8 total space)**.** *A smooth manifold homotopy equivalent to S^8 admits no almost-complex structure. Hence a complex structure on the regular locus of a putative T^6-fibration on S^8 cannot extend through all its singular fillings.*

Proof. Suppose $E = (TM, J)$ is a complex rank-four bundle whose underlying real bundle is TM. Orient M by J, and let $u \in H^8(M;\mathbb{Z})$ evaluate to one. Since

$$H^2(M;\mathbb{Z}) = H^4(M;\mathbb{Z}) = H^6(M;\mathbb{Z}) = 0,$$

one has

$$c_1(E) = c_2(E) = c_3(E) = 0, \qquad c_4(E) = e(TM) = \chi(M)u = 2u.$$

For the underlying real bundle,

$$p_2(TM) = c_2(E)^2 - 2c_1(E)c_3(E) + 2c_4(E) = 4u. \tag{662}$$

On the other hand $H^4(M) = 0$ gives $p_1(TM) = 0$, and the middle cohomology vanishes, so $\operatorname{sign}(M) = 0$. Hirzebruch's eight-dimensional signature formula gives

$$0 = \left\langle \frac{7p_2(TM) - p_1(TM)^2}{45}, [M] \right\rangle,$$

which forces $p_2(TM) = 0$. This contradicts (662). For the standard sphere the same vanishing follows immediately from $TS^8 \oplus \varepsilon^1_{\mathbb{R}} \cong \varepsilon^9_{\mathbb{R}}$. □

If every local filling were a smooth complex toroidal fourfold and every overlap map were biholomorphic, these local complex tangent bundles would glue to an almost-complex structure on all of S^8, contradicting theorem 17.10. At least one filling, transition, or total-space singularity must therefore leave the smooth complex category.

17.7 An explicit integral monodromy candidate and its failure

Put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad Q = S^{-1}T^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}.$$

Then

$$S^4 = I, \qquad Q^6 = I, \qquad SQ = T^{-1}. \tag{663}$$

On $\Lambda = \mathbb{Z}^3 \otimes \mathbb{Z}^2 \cong \mathbb{Z}^6$, define

$$M_1 = I_3 \otimes S, \qquad M_2 = I_3 \otimes Q, \qquad M_0 = I_3 \otimes T. \tag{664}$$

They are integral and symplectic, and

$$M_1 M_2 M_0 = I.$$

After reordering the tensor-product basis as $(a_1, a_2, a_3, b_1, b_2, b_3)$, M_0 is exactly the cusp matrix (655). Thus (664) is an explicit local system over a thrice-punctured sphere with finite orders $4, 6$ and a rank-three unipotent cusp.

Proposition 17.11 (Fixed-point obstruction)**.** *No translation makes either finite linear action M_1 or M_2 free on T^6.*

Proof. Neither S nor Q has eigenvalue one. More explicitly,

$$\det(I - S) = 2, \qquad \det(I - Q) = 1.$$

Consequently $I - M_j : T^6 \to T^6$ is a surjective isogeny for $j = 1, 2$. For any translation parameter $c \in T^6$, the fixed-point equation for

$$x \longmapsto M_j x + c$$

is

$$(I - M_j)x = c,$$

and it always has a solution. Therefore every affine lift has a fixed point. □

This is an integral symplectic homology local system compatible with a diagonal rank-three period ansatz. The three matrices alone do not define a polarized variation of Hodge structure or a holomorphic family E_τ^3; those require an equivariant period map in addition. Any isotrivial affine quotient filling built from either displayed finite generator is singular by the fixed-point calculation. A successful alternative would need finite monodromies with invariant torus directions that admit fixed-point-free affine lifts, while retaining the product relation and the rank-three cusp. It would still have to overcome both obstructions in section 17.6.

17.8 Exact replication requirements for any real singular $T^6 \to S^8 \to S^2$ candidate

The preceding results exclude the direct smooth holomorphic construction, not an arbitrary real stratified map. Let

$$\Sigma = \{p_0, \dots, p_s\} \subset S^2, \qquad B^\circ = S^2 \setminus \Sigma.$$

A candidate must first specify an affine integral monodromy

$$\widetilde{\rho} : \pi_1(B^\circ) \longrightarrow \mathrm{GL}(6, \mathbb{Z}) \ltimes T^6 \tag{665}$$

with based peripheral matrices satisfying

$$M_0 M_1 \cdots M_s = I.$$

The corresponding affine peripheral transformations $\widetilde{M}_i = (M_i, c_i)$ must satisfy the stronger identity

$$\widetilde{M}_0 \widetilde{M}_1 \cdots \widetilde{M}_s = (I, 0), \qquad c_0 + M_0 c_1 + \cdots + M_0 \cdots M_{s-1} c_s = 0 \quad \text{in } T^6. \tag{666}$$

Choose pairwise disjoint small closed discs D_i around the p_i, and put

$$B = S^2 \setminus \bigcup_i \operatorname{int} D_i, \qquad J = g^{-1}(B).$$

It must then give each compact filling $N_i = g^{-1}(D_i)$, its boundary mapping torus

$$A_i = J \cap N_i,$$

and the actual based inclusion maps

$$A_i \longrightarrow J, \qquad A_i \longrightarrow N_i.$$

Here one fixes a base point in J, a base point in each A_i, and a whisker in J joining them; changing a whisker only conjugates the corresponding homomorphism and does not change the pushout below.

Proposition 17.12 (Fundamental-group replication map)**.** *The fundamental group of the glued total space is the iterated pushout*

$$\pi_1(M) \cong \pi_1(J) *_{\pi_1(A_0)} \pi_1(N_0) *_{\pi_1(A_1)} \cdots *_{\pi_1(A_s)} \pi_1(N_s), \tag{667}$$

where both homomorphisms out of each $\pi_1(A_i)$ are the literal based inclusion homomorphisms. If, in addition, every $\pi_1(A_i) \to \pi_1(N_i)$ is surjective, then

$$\pi_1(M) \cong \pi_1(J) \Big/ \Big\langle\!\Big\langle (j_{J,i})_* \ker\{\pi_1(A_i) \to \pi_1(N_i)\} : 0 \le i \le s \Big\rangle\!\Big\rangle. \tag{668}$$

Consequently a proposed S^8 construction must either compute the full pushout in (667)*, or prove the stated surjectivities and prove that the displayed normal generators kill $\pi_1(J)$.*

Proof. Apply van Kampen successively to

$$M = J \cup N_0 \cup \cdots \cup N_s$$

with connected collar intersections A_i. Each attachment quotients by the relations identifying the two images of every element of $\pi_1(A_i)$, which is precisely the iterated amalgamated product (667). If the map to $\pi_1(N_i)$ is surjective, then every generator coming from $\pi_1(N_i)$ can be represented by an element of $\pi_1(A_i)$. Eliminating those generators leaves exactly the normal closure of the image in $\pi_1(J)$ of the kernel of the map to $\pi_1(N_i)$. Iterating proves (668). □

For homology, define the literal Mayer–Vietoris map

$$\alpha_q = \bigoplus_i ((j_{J,i})_*, -(j_{N_i})_*) : \bigoplus_i H_q(A_i;\mathbb{Z}) \longrightarrow H_q(J;\mathbb{Z}) \oplus \bigoplus_i H_q(N_i;\mathbb{Z}). \tag{669}$$

Theorem 17.13 (Integral homology-sphere criterion)**.** *Assume that J and the N_i are compact, connected, collar-compatible eight-manifolds with nonempty boundary, so their top homology vanishes. The glued space has the integral homology of S^8 if and only if*

q	*condition on* α_q
0	α_0 *is injective and* $\operatorname{coker}\alpha_0 \cong \mathbb{Z}$,
$1 \le q \le 6$	α_q *is an isomorphism,*
7	α_7 *is surjective and* $\ker\alpha_7 \cong \mathbb{Z}$.

Proof. Insert (669) into the Mayer–Vietoris long exact sequence. For every $1 \le q \le 7$, exactness gives the short exact sequence

$$0 \longrightarrow \operatorname{coker}\alpha_q \longrightarrow H_q(M;\mathbb{Z}) \longrightarrow \ker\alpha_{q-1} \longrightarrow 0.$$

Thus simultaneous vanishing in degrees $1,\dots,7$ is equivalent to α_0 being injective, α_q being an isomorphism for $1 \le q \le 6$, and α_7 being surjective. In degree zero, $H_0(M) \cong \operatorname{coker}\alpha_0$, which gives the remaining degree-zero condition. Since the pieces have zero H_8, exactness at

$$0 \longrightarrow H_8(M) \longrightarrow \bigoplus_i H_7(A_i) \xrightarrow{\alpha_7} H_7(J) \oplus \bigoplus_i H_7(N_i) \longrightarrow H_7(M) \longrightarrow 0$$

identifies $H_8(M)$ with $\ker\alpha_7$. Hence $H_8(M) \cong \mathbb{Z}$ is exactly the last kernel condition. □

If A_i is the mapping torus of a torus automorphism with integral monodromy M_i, its integral homology is fixed by the Wang exact sequence:

$$0 \longrightarrow \operatorname{coker}\left(I - \bigwedge^{q} M_i\right) \longrightarrow H_q(A_i;\mathbb{Z}) \longrightarrow \ker\left(I - \bigwedge^{q-1} M_i\right) \longrightarrow 0. \tag{670}$$

The right-hand group is free, so this sequence splits as a sequence of abelian groups, though not canonically. Equations (665), (667), (669), and (670), together with the actual tangent clutching maps, are the minimum replication data. For an S^8 total space those clutching maps must give

$$p_1 = p_2 = 0, \qquad e = 2u,$$

while preventing the regular-locus complex structure from extending, as required by theorem 17.10.

17.9 The Hermitian determinant boundary is not the four-branch fibre

Let

$$\mathcal{C}_3 = \operatorname{Herm}_3(\mathbb{C})_{\geq 0}.$$

The real dimension of the positive-semidefinite rank-r stratum is

$$2 \cdot 3r - r^2.$$

Indeed, write $H = BB^*$ with B a full-rank complex $3 \times r$ matrix; the entire redundancy is the right action of $U(r)$. Thus the rank $3, 2, 1, 0$ strata have real dimensions

$$9, \quad 8, \quad 5, \quad 0. \tag{671}$$

The rank-two stratum is smooth because the derivative of the determinant is the trace pairing with $\operatorname{adj}(H)$, which is nonzero exactly in rank two. The singular locus is the rank-at-most-one locus.

Proposition 17.14 (The real Hermitian boundary compactifies to S^8)**.** *The determinant-zero boundary*

$$\partial\mathcal{C}_3 = \{H \in \operatorname{Herm}_3(\mathbb{C}) : H \geq 0,\ \det H = 0\}$$

is homeomorphic to $\mathbb{R}^8$. Its one-point compactification is therefore S^8, with the rank strata in (671) *giving a non-smooth stratification.*

Proof. Intersect $\mathcal{C}_3$ with the affine hyperplane $\operatorname{tr} H = 1$:

$$K = \{H \geq 0 : \operatorname{tr} H = 1\}.$$

This is a compact convex body with nonempty interior in the eight-dimensional affine space $\{H = H^* : \operatorname{tr} H = 1\}$. Radial projection from the interior point $I_3/3$ gives a homeomorphism from K to the closed eight-ball. Explicitly, after choosing a Euclidean norm on the trace-zero translation space, every ray

$$I_3/3 + sU, \qquad \|U\| = 1, \quad s \geq 0,$$

meets ∂K once, at a radius $r(U) > 0$; compactness and convexity make r continuous, and $(\rho, U) \mapsto I_3/3 + \rho r(U)U$, $0 \leq \rho \leq 1$, is the required radial homeomorphism after collapsing $\{0\} \times S^7$ to the centre. Thus $\partial K \cong S^7$.

Every nonzero $H \in \partial\mathcal{C}_3$ has the unique expression

$$H = rP, \qquad r = \operatorname{tr} H > 0, \qquad P \in \partial K.$$

Adding $H = 0$ identifies

$$\partial\mathcal{C}_3 \cong ([0,\infty) \times S^7)/(\{0\} \times S^7) \cong \mathbb{R}^8.$$

The Alexandroff one-point compactification of $\mathbb{R}^8$ is S^8. □

Theorem 17.15 (Non-SNC determinant boundary)**.** *At a rank-one point, the semialgebraic boundary of $\mathcal{C}_3$ has a quadratic-cone normal slice rather than a union of smooth crossing faces. The ambient real determinant hypersurface, and its complexification, are not normal-crossings hypersurfaces there. Neither that rank-one germ nor the germ at the zero matrix is the four-branch germ in* (650)*.*

Proof. Unitary conjugation and positive rescaling reduce any rank-one point to

$$H_0 = \operatorname{diag}(1, 0, 0).$$

Near H_0, write

$$H = \begin{pmatrix} a & b^* \\ b & C \end{pmatrix}, \qquad a \in \mathbb{R}, \quad b \in \mathbb{C}^2, \quad C \in \operatorname{Herm}_2(\mathbb{C}).$$

On $a > 0$, the real-analytic coordinate change

$$(a, b, C) \longmapsto (a, b, S), \qquad S = C - a^{-1} b b^*$$

is a local diffeomorphism. The Schur-complement identities are exact:

$$\det H = a \det S, \qquad H \geq 0 \quad \Longleftrightarrow \quad S \geq 0. \tag{672}$$

Writing

$$S = \begin{pmatrix} u & x + iy \\ x - iy & v \end{pmatrix}$$

gives the exact semialgebraic normal boundary

$$uv - x^2 - y^2 = 0, \qquad u \geq 0, \quad v \geq 0. \tag{673}$$

The defining quadratic form has rank four. A product of two real linear forms has quadratic rank at most two, so the ambient real-algebraic hypersurface is not a union of smooth real hyperplane branches; imposing the displayed positive-semidefinite inequalities cannot turn its quadratic cone into transverse smooth faces.

After complexification, the normal slice is

$$s_{11} s_{22} - s_{12} s_{21} = 0.$$

All four first derivatives vanish only at the origin. A product of two linear forms would be singular along their codimension-two intersection, so the quadratic is irreducible and is not a coordinate crossing. At the zero matrix the 3×3 determinant is homogeneous of degree three and therefore has multiplicity three. The germ $z_0 z_1 z_2 z_3 = 0$ has multiplicity four. Neither germ is the other. □

There is also a dimension obstruction. The toroidal central fibre W_3 has complex dimension three and real dimension six, while the dense rank-two boundary of $\mathcal{C}_3$ has real dimension eight. The one-point compactification in theorem 17.14 is a stratified S^8, but it is not the central divisor of the complex fourfold N_3.

The two occurrences of the number three are independent: $\det_{\mathrm{Herm}_3}$ has three eigenvalue factors, whereas a semistable degeneration of complex three-tori has $3 + 1 = 4$ local branches at a deepest point because one includes the base parameter. Moving to Herm_4 makes the determinant quartic but changes the real ambient dimension to 16 and the natural compactified boundary dimension to 15; it does not produce an S^8.

17.10 Exact conclusion and permissible carriers

The local higher-rung toroidal object is exact:

$$\boxed{\text{rank-three periodic fan} \quad \Longrightarrow \quad t = z_0 z_1 z_2 z_3, \quad \mathcal{D}(W_3) \cong T^3, \quad (f_0, f_1, f_2, f_3) = (1, 7, 12, 6).}$$

It lives on the smooth noncompact complex fourfold $N_3 \to \Delta$, not on a complex S^8. A compact complex globalization must have different topology. A real S^8 may retain the complex-torus structure only on its regular locus and requires explicitly non-complex or singular fillings satisfying every map in section 17.8. The Hermitian Gram cone provides a separate real stratified S^8 boundary, not an SNC toroidal fibre. Stable complex data are also permissible because

$$TS^8 \oplus \varepsilon^2_{\mathbb{R}} \cong \varepsilon^{10}_{\mathbb{R}},$$

but this stable structure cannot destabilize to an almost-complex structure on TS^8.

Finally, the six tetrahedron orbits are naturally indexed by the six permutations in S_3; neither this integer six nor the rank-six lattice $H_1(T^6; \mathbb{Z})$ is by itself a cyclic group $\mathbb{Z}/6\mathbb{Z}$. Any cyclic-order-six or gauge-centre interpretation requires an additional exact homomorphism.

18 An exact toroidal elliptic bridge to the candidate cusp

This section asks for the smallest typed morphism between the rank-three toroidal construction and the reconstructed Fabel–Alpöge threefold. It also tests whether that morphism is supplied by the nine-dimensional Hermitian Gram cone. The conclusions are distinct:

1. a quotient of integral lattices extends to a morphism of toric cusp families and commutes with monodromy and the cusp filling maps;
2. the quotient has an explicit unimodular splitting as the candidate cusp plus one elliptic cusp;
3. the order-three and order-four affine peripheral actions admit exact rank-six lattice lifts;
4. none of these maps is induced by the determinant-rank stratification or the rank-one $\mathbb{CP}^2$ normal cones of the Gram construction.

The last assertion is an obstruction to a structure-preserving identification. It is not the false assertion that no continuous map between the underlying spheres exists.

18.1 The candidate lattice and filling data

Retain the candidate map

$$f : X \longrightarrow B \simeq \mathbb{P}^1, \qquad B^\circ = B \setminus \{p_0, p_1, p_2\}.$$

Its smooth fibre has marked first homology

$$\Lambda_F = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle.$$

Put

$$A_F = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad B_F = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u} \rangle.$$

Thus $\Lambda_F = A_F \oplus B_F$, with the decomposed order $(\widehat{w}, \widehat{\delta}; \widehat{\gamma}, \widehat{u})$. Define

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : B_F \longrightarrow A_F. \tag{TEB.1}$$

The cusp monodromy is

$$M_F(a,b) = (a + B_0 b, b), \tag{TEB.2}$$

or, in the manuscript order $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$,

$$[M_F] = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{TEB.3}$$

In particular,

$$(M_F - I)^2 = 0, \qquad \ker(M_F - I) = \operatorname{im}(M_F - I) = A_F. \tag{TEB.4}$$

The toric cusp neighbourhood retracts onto its central fibre W. In the marking above, the induced filling homomorphism is

$$c_F : H_1(T^4;\mathbb{Z}) = A_F \oplus B_F \longrightarrow H_1(W;\mathbb{Z}) = B_F, \qquad c_F(a,b) = b. \tag{TEB.5}$$

Thus the two toric angular cycles are killed and the two deck cycles survive. The exact homology of the central fibre is

$$H_*(W;\mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}). \tag{TEB.6}$$

At the other two special points, the finite monodromies are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \tag{TEB.7}$$

in the manuscript order. They satisfy

$$A_1^3 = I, \qquad A_2^4 = I, \qquad A_1 A_2 M_F = I. \tag{TEB.8}$$

The invariant primitive vectors

$$v_1 = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad v_2 = -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w} \tag{TEB.9}$$

define free affine actions

$$x \longmapsto A_j x + v_j/m_j, \qquad (m_1, m_2) = (3,4). \tag{TEB.10}$$

18.2 The rank-three cusp

Let

$$L_3 = \mathbb{Z}\langle e_1, e_2, e_3 \rangle.$$

For $m \in L_3$ and $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$, put

$$\sigma_{m,\pi} = \operatorname{conv}\{m, m + e_{\pi_1}, m + e_{\pi_1} + e_{\pi_2}, m + e_1 + e_2 + e_3\}. \tag{TEB.11}$$

These tetrahedra give the periodic Kuhn triangulation $\mathcal{T}_3$ of $L_{3,\mathbb{R}}$. Coning all cells at height one gives a smooth toric fourfold Y_3, because every maximal cone is unimodular. The height character gives

$$t : Y_3 \longrightarrow \mathbb{C},$$

and a maximal affine chart has the exact equation

$$t = z_0 z_1 z_2 z_3. \tag{TEB.12}$$

For the untwisted quotient, a noncentral fibre is

$$\mathbb{C}^3/(s\mathbb{Z}^3+\mathbb{Z}^3), \qquad t=e^{2\pi i s},$$

and its monodromy, in angular–deck order, is

$$M_3=\begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix}, \qquad M_3(a,b)=(a+b,b). \tag{TEB.13}$$

The toric filling map is

$$c_3 : H_1(T^6;\mathbb{Z}) = L_3 \oplus \overline{L}_3 \longrightarrow \overline{L}_3, \qquad c_3(a,b)=b. \tag{TEB.14}$$

Indeed, each angular generator bounds a toric disc, while a deck generator closes only by the corresponding lattice transformation and survives in the filling group.

18.3 The exact A_2 quotient of the Kuhn fan

In A_F define

$$n_1=\widehat{w}, \qquad n_2=\widehat{\delta}-\widehat{w}, \qquad n_3=-\widehat{\delta}. \tag{TEB.15}$$

They are primitive and satisfy $n_1+n_2+n_3=0$. Define

$$P : L_3 \longrightarrow A_F, \qquad P(e_i)=n_i. \tag{TEB.16}$$

In the bases (e_1,e_2,e_3) and $(\widehat{w},\widehat{\delta})$,

$$P=\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \ker P=\mathbb{Z}(1,1,1). \tag{TEB.17}$$

Define the deck-lattice quotient

$$Q=B_0^{-1}P=\begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix} : \overline{L}_3 \longrightarrow B_F. \tag{TEB.18}$$

Then

$$B_0Q=P, \qquad \ker Q=\mathbb{Z}(1,1,1). \tag{TEB.19}$$

Proposition 18.1 (The Kuhn-to-A_2 fan morphism)**.** *The map P sends every tetrahedron of $\mathcal{T}_3$ onto a triangle of the candidate's A_2 triangulation. Consequently*

$$\widehat{P} : L_3 \oplus \mathbb{Z} \longrightarrow A_F \oplus \mathbb{Z}, \qquad \widehat{P}(x,r)=(Px,r), \tag{TEB.20}$$

is a map of height-one fans and induces a toric morphism

$$F : Y_3 \longrightarrow Y_2, \qquad t \circ F = t. \tag{TEB.21}$$

Proof. The four vertices of $\sigma_{m,\pi}$ map to

$$Pm, \quad Pm+n_{\pi_1}, \quad Pm+n_{\pi_1}+n_{\pi_2}=Pm-n_{\pi_3}, \quad Pm. \tag{TEB.22}$$

Write $(n_1,n_2,n_3)=(e_1,e_2-e_1,-e_2)$ in the target basis. For the six permutations, the nonrepeated images are

$$\begin{array}{c|c} (1,2,3) & \{v,\, v+e_1,\, v+e_2\} \\ (2,1,3) & \{v,\, v-e_1+e_2,\, v+e_2\} \\ (1,3,2) & \{v,\, v+e_1,\, v+e_1-e_2\} \\ (3,1,2) & \{v,\, v-e_2,\, v+e_1-e_2\} \\ (2,3,1) & \{v,\, v-e_1+e_2,\, v-e_1\} \\ (3,2,1) & \{v,\, v-e_2,\, v-e_1\}. \end{array} \tag{TEB.23}$$

These are precisely the six triangles incident to $v = Pm$ in the periodic triangulation by

$$\mathrm{conv}\{u, u+e_1, u+e_2\}, \quad \mathrm{conv}\{u+e_1, u+e_2, u+e_1+e_2\}.$$

Thus every source cone maps into a target cone. The toric fan criterion gives F; the last coordinate in (TEB.20) is unchanged, so the height character is unchanged. □

The kernel $\mathbb{Z}(1,1,1)$ is invariant under permutations of the three coordinates. Hence the quotient lattice is natural up to the Weyl action on the A_2 target. The ordered basis and orientation used in (TEB.1) choose one representative of that Weyl orbit.

18.4 Monodromy and filling compatibility

Theorem 18.2 (Exact local lattice bridge)**.** *The homomorphism*

$$\Phi = P \oplus Q : L_3 \oplus \overline{L}_3 \longrightarrow A_F \oplus B_F, \qquad \Phi(a,b) = (Pa, Qb), \tag{TEB.24}$$

is surjective and obeys

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3. \tag{TEB.25}$$

Thus the diagram

$$\begin{array}{ccc} H_1(T^6;\mathbb{Z}) & \xrightarrow{\Phi} & H_1(T^4;\mathbb{Z}) \\ {\scriptstyle c_3}\downarrow & & \downarrow{\scriptstyle c_F} \\ \overline{L}_3 & \xrightarrow{Q} & B_F \end{array} \tag{TEB.26}$$

commutes.

Proof. Both P and Q are surjective. Using (TEB.13),

$$\Phi M_3(a,b) = (P(a+b), Qb).$$

Using (TEB.2) and $B_0 Q = P$,

$$M_F \Phi(a,b) = (Pa + B_0 Qb, Qb) = (Pa + Pb, Qb).$$

This proves monodromy equivariance. The filling identity is the literal calculation

$$c_F \Phi(a,b) = Qb = Q c_3(a,b).$$

□

18.5 The actual period correction and the cusp-family morphism

The candidate period matrix at the cusp has the exact form

$$Z_2(t) = sB_0 + C(t), \qquad t = e^{2\pi i s}, \tag{TEB.27}$$

where $C(t)$ is holomorphic and bounded at $t = 0$. Fix the integral right inverse

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad PS = I_2, \tag{TEB.28}$$

and define

$$C_3(t) = SC(t)Q, \qquad Z_3(t) = sI_3 + C_3(t). \tag{TEB.29}$$

Theorem 18.3 (Holomorphic toroidal elliptic bridge)**.** *After shrinking the cusp disc,* $\operatorname{Im} Z_3(t)$ *is invertible and*

$$PZ_3(t) = Z_2(t)Q. \tag{TEB.30}$$

Consequently $P : \mathbb{C}^3 \to \mathbb{C}^2$ *descends on every punctured fibre to a surjective holomorphic homomorphism*

$$\frac{\mathbb{C}^3}{Z_3(t)\mathbb{Z}^3 + \mathbb{Z}^3} \longrightarrow \frac{\mathbb{C}^2}{Z_2(t)\mathbb{Z}^2 + \mathbb{Z}^2}, \tag{TEB.31}$$

with connected elliptic kernel and integral homology map Φ*.*

Define the source and target toric actions by

$$\Psi^{(3)}_\lambda(x,t) = \big(\exp(2\pi i C_3(t)\lambda)t^\lambda x, t\big), \qquad \lambda \in \mathbb{Z}^3, \tag{TEB.32}$$

$$\Psi^{(2)}_\mu(y,t) = \big(\exp(2\pi i C(t)\mu)t^{B_0\mu} y, t\big), \qquad \mu \in \mathbb{Z}^2. \tag{TEB.33}$$

Then

$$F\Psi^{(3)}_\lambda = \Psi^{(2)}_{Q\lambda} F. \tag{TEB.34}$$

The actions are free and properly discontinuous on sufficiently small toric cusp neighbourhoods, and (TEB.21) descends to a holomorphic map

$$\overline{F} : N_3^C \longrightarrow N_0 \tag{TEB.35}$$

over the cusp disc. Its smooth-fibre and filling maps are (TEB.24) and (TEB.26).

Proof. Equations (TEB.19), (TEB.28), and (TEB.29) give

$$PZ_3 = sP + PSCQ = sB_0Q + CQ = Z_2Q.$$

Because $\operatorname{Im} s \to +\infty$ at the cusp while $C_3(t)$ is bounded, the determinant of $\operatorname{Im} Z_3(t)$ has leading term $(\operatorname{Im} s)^3$; it is nonzero on a sufficiently small punctured disc. Hence (TEB.31) is a well-defined homomorphism of complex tori.

Both kernel lattices in (TEB.17) and (TEB.19) are generated by $k = (1,1,1)$, and $C_3(t)k = 0$. The kernel is therefore

$$\mathbb{C}k/(\mathbb{Z}k + s\mathbb{Z}k),$$

an elliptic curve. Saturation of the primitive kernel proves connectedness.

On the dense torus, (TEB.30) gives

$$\begin{aligned} F\Psi^{(3)}_\lambda(x,t) &= \big(\exp(2\pi i P C_3(t)\lambda)t^{P\lambda}F(x), t\big) \\ &= \big(\exp(2\pi i C(t)Q\lambda)t^{B_0Q\lambda}F(x), t\big), \end{aligned}$$

which is (TEB.34). Proposition 18.1 extends the first map over every toric stratum.

For completeness, freeness at a boundary point follows from orbit labels. A nonzero λ translates the bounded cell labelling its toric orbit to that cell plus λ; equality would force $\lambda = 0$. On a smooth fibre a fixed point would give $Z_3(t)\lambda \in \mathbb{Z}^3$, and taking imaginary parts gives $\operatorname{Im} Z_3(t)\lambda = 0$, again forcing $\lambda = 0$. Here is the required uniform estimate for proper discontinuity and properness. Put $y = \operatorname{Im} s = -\log|t|/(2\pi)$, and, on a punctured fibre, use

$$\operatorname{trop}_t(x) = \frac{1}{\log|t|}\big(\log|x_1|, \log|x_2|, \log|x_3|\big).$$

The exact transformation law is

$$\operatorname{trop}_t(\Psi^{(3)}_\lambda x) = \operatorname{trop}_t(x) + A(t)\lambda, \qquad A(t) = I_3 + \frac{\operatorname{Im} C_3(t)}{y}. \tag{TEB.35a}$$

Let

$$K = \sup_{|t|\le\epsilon} \|\operatorname{Im} C_3(t)\|_{\mathrm{op}} < \infty.$$

After shrinking ϵ, $K/y \le 1/2$, and hence

$$\frac{1}{2}\|\lambda\| \le \|A(t)\lambda\| \le \frac{3}{2}\|\lambda\| \tag{TEB.35b}$$

uniformly. If a compact set meets its λ-translate in the dense locus, its tropical coordinates lie in one bounded set C, so $\|A(t)\lambda\| \le \operatorname{diam}(C)$ and (TEB.35b) leaves only finitely many integral λ. On the toric boundary the cell label is translated by λ, because the nonzero factor $\exp(2\pi i C_3(t)\lambda)$ does not change an orbit label; finitely many cells again allow only finitely many λ.

For properness, given a tropical point r, choose $\lambda \in \mathbb{Z}^3$ by rounding the three coordinates of $A(t)^{-1}r$. Then $r - A(t)\lambda$ lies in $A(t)[0,1]^3$, a uniformly bounded set by (TEB.35b). Over a smaller closed disc, the closures of these representatives lie in the finite union of the closed toric polydiscs belonging to the six Kuhn tetrahedra, exactly as in the untwisted rank-three quotient. That union is compact. Thus the action is properly discontinuous, the inverse image of every smaller closed disc has compact quotient, and the quotient and descended map in (TEB.35) exist. The homology assertions follow from Theorem 18.2. □

18.6 A simultaneous unimodular splitting

Proposition 18.4 (Candidate cusp plus one elliptic cusp)**.** *Let*

$$r = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix},$$

$$A = \begin{pmatrix} P \\ r \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} Q \\ r \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{TEB.36}$$

and

$$D = B_0 \oplus [1] = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{TEB.37}$$

Then

$$\det A = \det B = 1, \qquad A = DB. \tag{TEB.38}$$

For $J = \operatorname{diag}(A, B) \in SL(6,\mathbb{Z})$,

$$JM_3J^{-1} = \begin{pmatrix} I_3 & D \\ 0 & I_3 \end{pmatrix}. \tag{TEB.39}$$

In the order $(A_F, k_A; B_F, k_B)$, this is the literal matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

In the coordinates

$$(A_F, k_A; B_F, k_B) = (Pa, ra; Qb, rb),$$

and after reordering them as

$$(A_F, B_F; k_A, k_B),$$

the right side of (TEB.39) is

$$M_F \oplus C_0, \qquad C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{TEB.40}$$

Explicitly, in the reordered basis it is

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} M_F & 0 \\ 0 & C_0 \end{pmatrix},$$

where M_F on the right is in the decomposed order $(A_F; B_F)$. Projection to the candidate block is exactly $\Phi = P \oplus Q$.

Proof. The matrix A is upper triangular with unit diagonal. Expanding B along its last row gives

$$\det B = \det \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} = 1.$$

Direct multiplication gives $DB = A$. Hence

$$JM_3J^{-1} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} I & I \\ 0 & I \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ 0 & B^{-1} \end{pmatrix} = \begin{pmatrix} I & AB^{-1} \\ 0 & I \end{pmatrix} = \begin{pmatrix} I & D \\ 0 & I \end{pmatrix}.$$

The first two rows of A and B are P and Q; their last row records the primitive diagonal kernel coordinate. Since $D = B_0 \oplus 1$, the asserted permutation separates the candidate and kernel blocks exactly. □

This proposition proves integrally, rather than only rationally, that the rank-three cusp module is the candidate cusp module plus one standard rank-one cusp. It also identifies both kernel cycles with the primitive diagonal vector $(1, 1, 1)$.

18.7 The complete finite-order lattice lift

Define

$$C_1 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{TEB.41}$$

Direct multiplication gives

$$C_1^3 = I, \qquad C_2^4 = I, \qquad C_1C_2 = C_0^{-1}, \qquad (C_0 - I)^2 = 0, \quad \operatorname{rank}(C_0 - I) = 1. \tag{TEB.42}$$

Proposition 18.5 (Rank-six lift of the $(3, 4, \infty)$ peripheral actions)**.** *On $\widetilde{\Lambda} = \Lambda_F \oplus \mathbb{Z}^2$, put*

$$\widetilde{A}_1 = A_1 \oplus C_1, \qquad \widetilde{A}_2 = A_2 \oplus C_2, \qquad \widetilde{M}_0 = M_F \oplus C_0. \tag{TEB.43}$$

Then

$$\widetilde{A}_1^3 = I, \qquad \widetilde{A}_2^4 = I, \qquad \widetilde{A}_1\widetilde{A}_2\widetilde{M}_0 = I, \tag{TEB.44}$$

and $\widetilde{M}_0 - I$ has square zero, rank three, and equal kernel and image of rank three. Projection

$$\mathrm{pr} : \widetilde{\Lambda} \longrightarrow \Lambda_F$$

intertwines all three monodromies.

The affine maps

$$\widetilde{g}_j(x, y) = (A_j x + v_j/m_j, C_j y) \quad \text{on } T^4 \times T^2 \tag{TEB.45}$$

have orders $m_1 = 3, m_2 = 4$ and are free. Their quotient maps project to the candidate finite-fibre quotient models.

Proof. Equations (TEB.8) and (TEB.42) prove (TEB.44) blockwise. The nilpotent part of $\widetilde{M}_0$ is the direct sum of the candidate rank-two nilpotent and the rank-one nilpotent $C_0 - I$, which proves all cusp assertions. Block projection gives equivariance.

The vectors $(v_j, 0)$ are invariant under $\widetilde{A}_j$, so the m_j-th iterate of (TEB.45) is translation by the integral vector $(v_j, 0)$, hence the identity on the torus. If a nonidentity iterate had a fixed point, its projection to T^4 would be a fixed point of the corresponding nonidentity iterate of (TEB.10). The latter actions are free. Thus (TEB.45) is free and has exact order m_j. Projection descends because it is equivariant. □

Proposition 18.5 proves that the orders $3, 4$, affine freeness, and a rank-three cusp are mutually compatible at the integral topological level. It does not yet identify these block lifts with one holomorphic period family on the full triangle-group cover. That is part of the remaining problem below.

18.8 The six toric chambers and the diagonal Gram boundary

The numerical agreement between the six deepest toric strata and the six A_2 Weyl chambers has the following exact, but purely fan-theoretic, form. Let

$$\mathfrak{a} = \{\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3) : \lambda_1 + \lambda_2 + \lambda_3 = 0\} \subset \mathrm{Herm}_3^0(\mathbb{C})$$

and put

$$\nu_i = E_{ii} - \frac{1}{3} I_3, \qquad \nu_1 + \nu_2 + \nu_3 = 0. \tag{TEB.45a}$$

Because $n_1 + n_2 + n_3 = 0$, there is a unique real-linear isomorphism

$$\jmath : A_{F,\mathbb{R}} \xrightarrow{\sim} \mathfrak{a}, \qquad \jmath(n_i) = \nu_i. \tag{TEB.45b}$$

For $H \in \mathrm{Herm}_3^0(\mathbb{C})$, let

$$\Phi_{\mathrm{G}}(H) = H - \lambda_{\min}(H) I_3.$$

This is the $SU(3)$-equivariant homeomorphism onto the singular positive-semidefinite Gram cone used in the preceding higher-rung analysis.

Proposition 18.6 (The Kuhn strata are the marked A_2 chambers)**.** *For every $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$,*

$$\jmath(P(\sigma_{0,\pi})) = \mathrm{conv}\{0, \nu_{\pi_1}, -\nu_{\pi_3}\}, \tag{TEB.45c}$$

and the cone on this triangle is the closed Weyl chamber

$$\boxed{\mathbb{R}_{\geq 0}\nu_{\pi_1} + \mathbb{R}_{\geq 0}(-\nu_{\pi_3}) = \{\lambda_{\pi_1} \geq \lambda_{\pi_2} \geq \lambda_{\pi_3}\}.} \tag{TEB.45d}$$

Consequently the six tetrahedron orbits, hence the six quadruple-point strata of the untwisted toroidal central fibre, are identified S_3-equivariantly with the six marked diagonal Weyl chambers.

On this chamber the Gram-boundary homeomorphism is the explicit piecewise-linear map

$$\Phi_{\mathrm{G}}(H) = H - \lambda_{\pi_3}(H) I_3. \tag{TEB.45e}$$

In particular,

$$\Phi_{\mathrm{G}}(\nu_{\pi_1}) = E_{\pi_1\pi_1}, \qquad \Phi_{\mathrm{G}}(-\nu_{\pi_3}) = I_3 - E_{\pi_3\pi_3}. \tag{TEB.45f}$$

Proof. Equation (TEB.22) with $m = 0$ gives the three distinct image vertices $0, n_{\pi_1}, -n_{\pi_3}$, proving (TEB.45c). For $a, b \geq 0$, the diagonal entries of $a\nu_{\pi_1} - b\nu_{\pi_3}$ satisfy

$$\lambda_{\pi_1} - \lambda_{\pi_2} = a, \qquad \lambda_{\pi_2} - \lambda_{\pi_3} = b.$$

Conversely these two nonnegative differences recover a, b, since the trace is zero. This proves (TEB.45d). Both the permutation action on the Kuhn labels and the Weyl action on diagonal entries are the same action of S_3, so the correspondence is equivariant. On the displayed chamber the least eigenvalue is λ_{π_3}, which proves (TEB.45e). Substitution of the two extreme rays gives (TEB.45f). □

This is stronger than the bare equality $6 = |S_3|$, but it is not an identification of singular spaces. After quotienting the target fan by A_F, the six triangles incident to a marked vertex form two global triangle orbits. Moreover $\Phi_{\mathrm{G}} \circ \jmath$ is a map on the real cocharacter space, not a holomorphic map of toroidal total spaces. It neither supplies the period correction nor changes the analytic factorization comparison below.

18.9 The exact Jordan determinant and Lorentz sections

The determinant–index identity in Corollary 17.8 has a precise extension inside the nine-dimensional Hermitian cone. It is an algebraic extension of the scalar invariant; it is not a map from the toroidal degeneration to the Hermitian cone.

Write

$$\mathcal{J}_3 = \mathrm{Herm}_3(\mathbb{C}), \qquad \mathcal{J}_3^{\succeq 0} = \{H \in \mathcal{J}_3 : H \succeq 0\}, \qquad \Delta(H) = \det H.$$

On the ambient vector spaces, the inclusion

$$\iota : \mathrm{Sym}_3(\mathbb{R}) \hookrightarrow \mathcal{J}_3, \qquad \iota(S) = S + i0 \tag{TEB.45g}$$

is an injective unital homomorphism of real Jordan algebras for the product $A \circ B = (AB + BA)/2$. Its restriction sends the positive-definite cone into the positive-definite Hermitian cone. The ambient formulation matters: a positive cone is not itself a vector space or a Jordan algebra. The map also uses the retained marking of the integral lattice; it is not a coordinate-free map from an unmarked congruence class of polarizations to a fixed matrix space.

For $a = (a_1, a_2, a_3)^{\mathsf{T}} \in \mathbb{R}^3$, fix the oriented cross-product matrix

$$[a]_\times = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}. \tag{TEB.45h}$$

Every $H \in \mathcal{J}_3$ has one and only one decomposition

$$H = S + i[a]_\times, \qquad S \in \mathrm{Sym}_3(\mathbb{R}), \quad a \in \mathbb{R}^3. \tag{TEB.45i}$$

Theorem 18.7 (Hermitian determinant splitting)**.** *For every real symmetric S and every $a \in \mathbb{R}^3$,*

$$\boxed{\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.} \tag{TEB.45j}$$

If S is positive definite, then

$$S + i[a]_\times \succeq 0 \quad \Longleftrightarrow \quad a^{\mathsf{T}} S a \le \det S, \tag{TEB.45k}$$

and positive semidefiniteness and the weak scalar inequality may both be replaced by strict positivity and a strict scalar inequality.

Proof. Direct expansion in the fixed coordinates (TEB.45h) gives

$$\det(S + i[a]_\times) = \det S - a^{\mathsf{T}} S a.$$

For the order statement let $M = S^{-1/2}$, the positive-definite symmetric inverse square root. The identity

$$M[a]_\times M^{\mathsf{T}} = \det(M)[M^{-\mathsf{T}} a]_\times \tag{TEB.45l}$$

follows by applying both sides to two vectors and evaluating against a third with the fixed oriented volume form. Hence

$$S^{-1/2}(S + i[a]_\times)S^{-1/2} = I_3 + i[b]_\times, \qquad b = \det(S)^{-1/2} S^{1/2} a,$$

and

$$\|b\|^2 = \frac{a^{\mathsf{T}} S a}{\det S}.$$

The three eigenvalues of $I_3 + i[b]_\times$ are $1 - \|b\|, 1, 1 + \|b\|$. Congruence preserves inertia, so (TEB.45k) and its strict version follow. □

For a fixed marked positive-definite symmetric monodromy form B, define the injective real-linear map

$$\ell_B : \mathbb{R} \oplus \mathbb{R}^3 \longrightarrow \mathcal{J}_3, \qquad \ell_B(\lambda, a) = \lambda B + i[a]_\times. \tag{TEB.45m}$$

Theorem 18.7, applied first to $\lambda > 0$ and then directly at $\lambda = 0$, gives

$$\det \ell_B(\lambda, a) = \lambda(\lambda^2 \det B - a^{\mathsf{T}} B a), \tag{TEB.45n}$$

and the exact inverse image

$$\boxed{\ell_B^{-1}(\mathcal{J}_3^{\geq 0}) = \{(\lambda, a) : \lambda \geq 0,\ a^{\mathsf{T}} B a \leq \lambda^2 \det B\}.} \tag{TEB.45o}$$

Indeed, if $\lambda < 0$, then $\operatorname{tr} \ell_B(\lambda, a) = \lambda \operatorname{tr} B < 0$, which is impossible for a positive-semidefinite matrix. Thus every marked positive monodromy ray determines a literal future $(3+1)$-Lorentz cone with quadratic form

$$q_B(\lambda, a) = \lambda^2 \det B - a^{\mathsf{T}} B a. \tag{TEB.45p}$$

At $B = I_3$ this is $\lambda^2 - a_1^2 - a_2^2 - a_3^2$. The equality

$$\chi(W_{3,I_3}) = 6 = 6 q_{I_3}(1, 0) \tag{TEB.45q}$$

is therefore exact, while remaining an equality of scalar invariants rather than a morphism of degenerations.

There is also an exact order-cone lift of the rank-two Lorentz cone from Theorem 16.11. In its coordinates

$$H_2(t, x_1, x_2, x_3) = \begin{pmatrix} t + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & t - x_3 \end{pmatrix},$$

put

$$\Theta(H_2(t, x_1, x_2, x_3)) = \begin{pmatrix} t & -i x_3 & -i x_2 \\ i x_3 & t & -i x_1 \\ i x_2 & i x_1 & t \end{pmatrix}. \tag{TEB.45r}$$

Proposition 18.8 (Order-cone lift and its complete rank behaviour)**.** *The map Θ is a real-linear bijection from* $\mathrm{Herm}_2(\mathbb{C})$ *onto* $\mathbb{R} I_3 \oplus i\mathfrak{so}(3)$*, and*

$$H_2 \succeq 0 \quad \Longleftrightarrow \quad \Theta(H_2) \succeq 0. \tag{TEB.45s}$$

Moreover

$$\boxed{\det_3 \Theta(H_2) = \frac{\operatorname{tr} H_2}{2} \det_2 H_2.} \tag{TEB.45t}$$

On the positive-semidefinite cone, Θ sends ranks $2, 1, 0$ respectively to ranks $3, 2, 0$.

Proof. Put $r = (x_1^2 + x_2^2 + x_3^2)^{1/2}$. The eigenvalues of H_2 are $t - r, t + r$, while those of $\Theta(H_2)$ are $t - r, t, t + r$. Both matrices are positive semidefinite exactly when $t \geq r$. Multiplication of the displayed eigenvalues gives

$$\det_3 \Theta(H_2) = t(t^2 - r^2) = \frac{\operatorname{tr} H_2}{2} \det_2 H_2.$$

The same eigenvalues give all three rank assertions. The displayed matrix also makes real-linearity, injectivity, and the stated image immediate. □

If P is a rank-one projection in $\mathrm{Herm}_2(\mathbb{C})$, then its Lorentz coordinates satisfy $t = r = 1/2$, so $\Theta(P)$ has eigenvalues $0, 1/2, 1$ and is not an idempotent. Thus Θ is not a Jordan-algebra homomorphism. Globally, the pullback of the cubic determinant hypersurface is

$$\Theta^{-1}\{\det_3 = 0\} = \{\operatorname{tr} H_2 = 0\} \cup \{\det_2 H_2 = 0\}; \tag{TEB.45u}$$

only on the positive cone does the first set add no points beyond the origin. Neither ℓ_B nor Θ is a holomorphic toric map, a map of integral period lattices, or a singular-fibre filling map. For the actual one-parameter toric degeneration the monodromy form is fixed at $B = I_3$ while the local parameter in $z_0z_1z_2z_3 = t$ tends to zero. Consequently the induced Jordan datum is the constant point I_3, not a path to $\det B = 0$. Degeneration at fixed monodromy and rank loss of the monodromy form are different operations.

Finally, the cyclic group $C_3 \subset S_3$ does have two orbits on the six Kuhn labels. Under P, the even labels $(123), (231), (312)$ give translates of the lower target triangle $\mathrm{conv}\{u, u+e_1, u+e_2\}$, while the odd labels $(132), (213), (321)$ give translates of the upper target triangle $\mathrm{conv}\{u+e_1, u+e_2, u+e_1+e_2\}$. This action is by coordinate permutations and the exact bridge lowers lattice rank through P; it is not translation by an index-three superlattice. No such translation-lattice realization is constructed by the packet supplying (TEB.45j)–(TEB.45t). More decisively, for any rank-three lattice Λ', a fundamental domain has normalized volume $3! = 6$, so every Λ'-unimodular triangulation of that domain has six maximal tetrahedra. The canonical six-to-two construction is instead the rank-lowering fan map $P : L_3 \to A_F$ in (TEB.16)–(TEB.23). It must not be replaced by an unproved semistable translation quotient.

Audit record 18.9 (Completed Jordan–toric packet, 3 September 2026)**.** The source archive has SHA-256 `8BA0BD7FD7BADEAAF2C44E2F2B4EFD42C25DA9119DA3180017FC488CA364E96F`. Its supplied symbolic identities were replayed independently, and separate Euler/stratification, morphism/type, fan/Weyl, and theorem-map audits were performed. Equations (TEB.45j)–(TEB.45t) are the surviving exact content. The positive-cone Jordan typing, the normalization stratum attribution, and the index-three compression were corrected rather than copied. The audit reports and exact replay script are retained under `agent_logs/` and `agent_scratch/jordan_toric_symbolic_replay_20260903/` in the local research record.

18.10 The Gram determinant does not supply this bridge

Let

$$V = \mathrm{Mat}_3(\mathbb{C}), \qquad \mathcal{G} = V^+/SU(3), \qquad \mathcal{D} = \{\det = 0\}^+/SU(3).$$

Here $V^+ = V \cup \{\infty\}$ is the one-point compactification and $\{\det = 0\}^+ = \{\det = 0\} \cup \{\infty\}$. The exact Gram classification gives

$$(\mathcal{G}, \mathcal{D}) \cong (S^{10}, S^8), \qquad \mathcal{G} \setminus \mathcal{D} \cong \mathbb{R}^9 \times S^1. \tag{TEB.46}$$

The real fixed form gives

$$\Big(\mathrm{Mat}_3(\mathbb{R})^+/SO(3), \{\det = 0\}^+/SO(3)\Big) \cong (S^6, S^5). \tag{TEB.47}$$

Theorem 18.10 (No determinant-stratified identification)**.** *There is no homeomorphism of pairs from the real Gram pair (TEB.47) to*

$$\Big(X, f^{-1}\{p_0, p_1, p_2\}\Big). \tag{TEB.48}$$

There is no homotopy equivalence from the complex Gram complement in (TEB.46) to the candidate regular part $f^{-1}(B^\circ)$. The rank-one trace-one Gram stratum $\mathbb{CP}^2$ is not homotopy equivalent to any of the three candidate special fibres. Finally, no equivalence of function germs identifies its normal determinant with any of the three candidate local filling maps.

Proof. The determinant-zero set in (TEB.47) is a connected codimension-one S^5, and its complement has two components. The set in (TEB.48) is the disjoint union of three connected real four-dimensional fibres, hence has codimension two and three components. Its complement is connected because it is a torus bundle over the connected pair of pants B°. This proves the first assertion.

The complex Gram complement has fundamental group $\mathbb{Z}$. The homotopy sequence of the candidate regular torus bundle gives

$$1 \longrightarrow \mathbb{Z}^4 \longrightarrow \pi_1\big(f^{-1}(B^\circ)\big) \longrightarrow \pi_1(B^\circ) = F_2 \longrightarrow 1. \tag{TEB.49}$$

Its fundamental group surjects onto the nonabelian group F_2, so it is not $\mathbb{Z}$.

The rank-one trace-one stratum has

$$H_*(\mathbb{CP}^2;\mathbb{Z}) = (\mathbb{Z},0,\mathbb{Z},0,\mathbb{Z}), \qquad \chi(\mathbb{CP}^2) = 3.$$

The cusp fibre has (TEB.6), so $b_1 = 2, b_2 = 4, \chi = 2$. Each finite reduced fibre is a free finite quotient of T^4, hence has $\chi = 0$, and its audited first homology has free rank two. These invariants rule out all three homotopy equivalences.

At a rank-one Gram matrix, Schur-complement coordinates put the normal determinant in the exact form

$$uv - x^2 - y^2, \tag{TEB.50}$$

a nondegenerate quadratic form of signature $(1,3)$. Its complexification is the irreducible quadratic determinant

$$s_{11}s_{22} - s_{12}s_{21}. \tag{TEB.51}$$

At a deepest candidate cusp point, the map is instead

$$(z_0,z_1,z_2) \longmapsto z_0z_1z_2, \tag{TEB.52}$$

with three smooth complex branches. The source dimensions, multiplicities, and number of irreducible branches in (TEB.50)–(TEB.52) disagree. The two finite transverse maps have multiplicities three and four, while the Gram determinant has multiplicity two along the rank-one stratum and multiplicity three at the zero matrix. It has no order-four stratum. Orders of vanishing and analytic factorization are invariant under equivalence of function germs, which proves the final assertion. □

There are two further typed obstructions. First, the determinant phase map

$$\arg\det : \mathcal{G} \setminus \mathcal{D} \longrightarrow S^1$$

has trivial page monodromy after the $SU(3)$ quotient. Before quotienting, its page is diffeomorphic to $\mathbb{R}_{>0} \times SL(3,\mathbb{C})$, hence is simply connected, and its geometric monodromy is the scalar central element of order three. It supplies no $\mathbb{Z}^4$- or $\mathbb{Z}^6$-valued first-homology local system and no order-four peripheral loop.

Second, the trace-one rank-one locus is the homogeneous space

$$\mathbb{CP}^2 = SU(3)/S(U(1)\times U(2)).$$

Any $SU(3)$-invariant map from it to the candidate base, on which no $SU(3)$-action is part of the construction, is constant. Choosing a line $\ell \in \mathbb{CP}^2$ does define the exact positive compression

$$\mathrm{Herm}_3(\mathbb{C})_{\geq 0} \longrightarrow \mathrm{Herm}(\ell^\perp)_{\geq 0}, \qquad H \longmapsto P_{\ell^\perp} H|_{\ell^\perp}, \tag{TEB.53}$$

but its target is a four-real-dimensional Lorentz cone, not $\mathbb{P}^1$, and it contains no integral torus lattice. An integral quotient such as P requires the additional choice of maximal-torus and weight-lattice data. Irreducibility of the fundamental $SU(3)$ representation forbids an $SU(3)$-equivariant rank-two linear quotient of $\mathbb{C}^3$.

In particular, the homomorphism (TEB.14) belongs to the toroidal degeneration, not to the Gram cone. A generic conjugacy orbit in the trace-normalized Hermitian cone is $SU(3)/T^2$, which is simply connected, and the determinant page is simply connected as well. Neither has first homology $\mathbb{Z}^6$. Calling any map from it a T^6 filling map before constructing a torus bundle would be a type error.

18.11 The remaining global problem

The local bridge and the finite lattice lift isolate the remaining question. One must construct a proper complex, or explicitly stratified-real, map

$$\widetilde{f} : \widetilde{X}^8 \longrightarrow \mathbb{P}^1$$

and a map over the same base

$$\Pi : \widetilde{X} \longrightarrow X \tag{TEB.54}$$

with all of the following literal properties:

1. on every regular fibre, Π is the elliptic extension (TEB.31) and its marked integral map is Φ;
2. over the cusp disc, Π is the toric morphism (TEB.35), including the central strata and the filling square (TEB.26);
3. over each finite-order disc, the affine monodromy and translation are those in (TEB.43)–(TEB.45), and its punctured overlap agrees with the regular period family;
4. the actual based van Kampen maps and integral Mayer–Vietoris maps are computed from these inclusions, rather than inferred from the displayed monodromy matrices.

Equations (TEB.24)–(TEB.45) prove that there is no integral monodromy or affine-freeness obstruction to these requirements. They do not supply the global period and overlap construction in item 3.

There is also an unavoidable topological conclusion. The rank-three toroidal central fibre has Euler characteristic six. The two finite fibres in (TEB.45) are free finite quotients of T^6, so each has Euler characteristic zero. If these are the only three special fibres, Euler integration gives

$$\chi(\widetilde{X}) = 6. \tag{TEB.55}$$

Therefore this direct complex globalization cannot be S^8, whose Euler characteristic is two. Independently, S^8 admits no almost-complex structure. A real S^8 construction would have to add non-complex singular filling data with total Euler contribution -4, then pass the full fundamental-group, integral-homology, and tangent-clutching tests.

The nine-dimensional Gram cone remains a valid independent representation-theoretic carrier. Relating it to (TEB.54) requires a new, explicit morphism from determinant-rank data to period, lattice, and filling data. Neither the determinant, the Cartan cubic, nor the $\mathbb{CP}^2$-family of normal Lorentz cones supplies such a morphism.

19 The rank-six cusp lattice inside the nine-dimensional cone program

The rank-three toroidal construction contains a free rank-six lattice before any finite quotient is taken. This section identifies that lattice, its integral symmetric-group action, its monodromy, and the exact part of it seen by the Hermitian Gram boundary. It also separates it from the different rank-six group which classifies bundle defects on the regular locus of the reconstructed S^6.

19.1 The two rank-six groups have different types

Retain the notation of section 18.2. The lattice native to the rank-three toroidal cusp is

$$\Lambda_{\mathrm{tor}} := H_1(T^6;\mathbb{Z}) = L_3 \oplus \overline{L}_3 = \mathbb{Z}\langle e_1, e_2, e_3\rangle \oplus \mathbb{Z}\langle \bar{e}_1, \bar{e}_2, \bar{e}_3\rangle. \tag{674}$$

Its first summand consists of angular cycles and its second summand consists of deck cycles. The cusp monodromy and filling map are

$$M_3(a,b) = (a+b,b), \qquad c_3(a,b) = b. \tag{675}$$

The bundle-defect lattice on the original threefold has a different source and a different cohomological degree. With $D = W \sqcup S_1 \sqcup S_2$ and $U = X \setminus D$, it is

$$L_{\mathrm{def}} := H^4(U;\mathbb{Z}) \cong H_1(W;\mathbb{Z}) \oplus H_1(S_1;\mathbb{Z}) \oplus H_1(S_2;\mathbb{Z}) \cong (\mathbb{Z}^2)^3. \tag{676}$$

The map $\Phi = P \oplus Q$ below is defined on Λ_{tor}, whereas the fibre-degree character of theorem 45.1 is defined on L_{def}. Equality of their ranks supplies no map between them. In particular, neither group is the cyclic group $\mathbb{Z}/6\mathbb{Z}$.

There is also no component-permuting symmetric-group action on L_{def} supplied by the marked candidate fibration. The three deleted fibres have, respectively, unipotent, order-three, and order-four local monodromy. An automorphism of the marked fibration which permutes the punctures must preserve these conjugacy-order types and hence fixes all three punctures. One may impose an abstract permutation of the three displayed summands in (676) after choosing three identifications with $\mathbb{Z}^2$, but that is a re-marking, not an automorphism furnished by the geometry. This argument does not classify automorphisms fixing the base pointwise or internal actions on the individual summands; in particular it does not assert that the abstract group admits no S_3-action.

19.2 The integral S_3-module of the cusp

Let R_σ permute the three standard basis vectors by $R_\sigma e_i = e_{\sigma(i)}$, and let S_3 act diagonally on $\Lambda_{\mathrm{tor}} = L_3 \oplus \overline{L}_3$. Put

$$k = e_1 + e_2 + e_3, \qquad K = \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,\bar{k}), \quad \bar{k} = \bar{e}_1 + \bar{e}_2 + \bar{e}_3. \tag{677}$$

The group S_3 fixes K pointwise, commutes with M_3, and makes c_3 equivariant for the permutation action on $\overline{L}_3$: $c_3(R_\sigma a, R_\sigma b) = R_\sigma c_3(a,b)$.

Recall

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \qquad \Phi = P \oplus Q. \tag{678}$$

For $\sigma \in S_3$, define integral actions on A_F and B_F by

$$\rho_A(\sigma)(Px) = P(R_\sigma x), \qquad \rho_B(\sigma)(Qx) = Q(R_\sigma x). \tag{679}$$

These definitions are well posed because $\ker P = \ker Q = \mathbb{Z}k$.

Theorem 19.1 (The exact rank-six S_3-module sequence)**.** *The map Φ fits into an exact sequence of integral S_3-modules*

$$0 \longrightarrow K \longrightarrow \Lambda_{\mathrm{tor}} \xrightarrow{\Phi} A_F \oplus B_F \longrightarrow 0. \tag{680}$$

It is simultaneously equivariant for the cusp monodromies:

$$\Phi M_3 = M_F \Phi, \qquad M_3|_K = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{in the basis } ((k,0),(0,\bar{k})). \tag{681}$$

As a sequence of free abelian groups with monodromy it splits integrally, as shown by theorem 18.4. As a sequence of integral S_3-modules it does not split. After tensoring with $\mathbb{Q}$,

$$\Lambda_{\mathrm{tor},\mathbb{Q}} \cong \mathbf{21} \oplus 2V_{\mathrm{std}}, \qquad (A_F \oplus B_F)_{\mathbb{Q}} \cong 2V_{\mathrm{std}}, \qquad K_{\mathbb{Q}} \cong \mathbf{21}. \tag{682}$$

Proof. The kernels of P and Q are both the primitive diagonal lattice $\mathbb{Z}k$, and both maps are surjective. This proves exactness. Equations (679) make P, Q, and hence Φ, equivariant. Since $R_\sigma(a+b) = R_\sigma a + R_\sigma b$, the source action commutes with M_3. Moreover

$$M_3(xk, y\bar{k}) = ((x+y)k, y\bar{k}),$$

which gives (681). The target equivariance with M_F is equivalently the already verified identity $\Phi M_3 = M_F \Phi$.

It remains to prove the integral non-splitting. Any S_3-equivariant homomorphism $r : \mathbb{Z}^3 \to \mathbb{Z}$, where the target is trivial, has the same value $a \in \mathbb{Z}$ on all three standard basis vectors. Thus $r(k) = 3a$, so no such map retracts the inclusion $\mathbb{Z}k \hookrightarrow \mathbb{Z}^3$. For two copies, every entry of the matrix obtained by restricting an equivariant map $(\mathbb{Z}^3)^2 \to \mathbb{Z}^2$ to $(\mathbb{Z}k)^2$ is divisible by three. That restriction cannot be the identity matrix. Hence (680) does not split over $\mathbb{Z}[S_3]$.

Over $\mathbb{Q}$, the invariant diagonal line has the equivariant complement $\{x_1 + x_2 + x_3 = 0\}$, the standard representation. Applying this to both copies gives (682). □

For reference, if s_{12} and s_{23} are the adjacent transpositions, the induced target matrices are

$$\rho_A(s_{12}) = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}, \quad \rho_A(s_{23}) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \tag{683}$$

and

$$\rho_B(s_{12}) = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}, \quad \rho_B(s_{23}) = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}. \tag{684}$$

Direct multiplication gives

$$\rho_A(\sigma)P = PR_\sigma, \qquad \rho_B(\sigma)Q = QR_\sigma, \qquad \rho_A(\sigma)B_0 = B_0\rho_B(\sigma). \tag{685}$$

The quotient $\mathbb{Z}^3/\mathbb{Z}k$ is the A_2 weight lattice, not the A_2 root lattice. Indeed, the augmentation-zero lattice is generated by $e_1 - e_2$ and $e_2 - e_3$, whose images under P are the columns of

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{686}$$

Its determinant is three. Thus the root lattice is the index-three sublattice of A_F generated by the differences of the three weights n_1, n_2, n_3. The integral non-splitting in theorem 19.1 is the same index-three phenomenon viewed before passing to the quotient.

19.3 The six homology generators are not the six chambers

There are three related but distinct appearances of the integer six.

Theorem 19.2 (The three sixes)**.** *For the rank-three toroidal cusp and every injective integral shear $\mathcal{B} : \overline{L}_3 \hookrightarrow L_3$,*

$$\operatorname{rk} \Lambda_{\mathrm{tor}} = 2 \operatorname{rk} L_3 = 6, \tag{687}$$

$$\#\{\textit{Kuhn maximal cells modulo } L_3\} = |S_3| = 3! = 6, \tag{688}$$

$$\chi(W_{3,\mathcal{B}}) = |S_3|[L_3 : \mathcal{B}\overline{L}_3] = 3! \,|\det \mathcal{B}|. \tag{689}$$

The equalities (688) *and* (689) *have the same toric source: the maximal cells are indexed by orderings of three angular coordinates. Equation* (687) *has a different source: first homology contains one angular and one deck copy of the rank-three lattice. For a general shear $\mathcal{B}$, this cell count does not assert that the diagonal S_3-action descends to the sheared quotient; that requires the additional commutation equations $R_\sigma \mathcal{B} = \mathcal{B} R_\sigma$.*

In particular, the six standard basis vectors of Λ_{tor} are not S_3-equivariantly identifiable with the six Kuhn cells. The basis vectors form two orbits of size three; the maximal cells form one free orbit of size six.

Proof. The rank formula follows from (674). A Kuhn cell in one lattice cube is labelled by the order in which the three coordinate increments are taken, hence by a unique element of S_3. Left multiplication is free and transitive on these labels, proving (688). The normalization and orbit calculation in theorem 17.8 gives (689).

Under the diagonal permutation action, the angular basis $\{e_1, e_2, e_3\}$ and the deck basis $\{\bar{e}_1, \bar{e}_2, \bar{e}_3\}$ are two distinct transitive three-element sets. The cell labels form the regular six-element S_3-set. No equivariant bijection exists. Equivalently, over $\mathbb{Q}$ the character of the basis permutation module on the conjugacy classes $(1), (12), (123)$ is

$$(6, 2, 0),$$

whereas the regular representation has character $(6, 0, 0)$. The representations are not isomorphic even over $\mathbb{Q}$. □

This distinction also locates the chamber construction: the Kuhn fan uses the angular summand L_3, while the deck summand is needed for periods, monodromy, and filling. The map P produces the chamber fan; the different map Q makes the filling square commute. Replacing both by a single untyped "six-direction map" loses precisely the information that makes the cusp morphism valid.

19.4 The exact map into the Gram boundary

Let

$$\mathfrak{a} = \{\mathrm{diag}(\lambda_1, \lambda_2, \lambda_3) : \lambda_1 + \lambda_2 + \lambda_3 = 0\} \subset \mathrm{Herm}_3^0(\mathbb{C}),$$

let $\jmath : A_{F,\mathbb{R}} \to \mathfrak{a}$ be the marked isomorphism $\jmath(n_i) = E_{ii} - I_3/3$, and let

$$\Phi_{\mathrm{G}}(H) = H - \lambda_{\min}(H) I_3.$$

Write $\mathrm{pr}_{\mathrm{ang}}(a, b) = a$. The maps which actually exist are

$$\Psi = \jmath \circ P \circ \mathrm{pr}_{\mathrm{ang}} : \Lambda_{\mathrm{tor},\mathbb{R}} \longrightarrow \mathfrak{a}, \tag{690}$$

and

$$\Gamma = \Phi_{\mathrm{G}} \circ \Psi : \Lambda_{\mathrm{tor},\mathbb{R}} \longrightarrow \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0,\ \det H = 0\}. \tag{691}$$

Proposition 19.3 (Exact scope of the Gram-boundary map)**.** *The map Ψ is linear and S_3-equivariant, with*

$$\ker \Psi = \mathbb{R}(k, 0) \oplus (0, \overline{L}_{3,\mathbb{R}}). \tag{692}$$

The map Γ is continuous, positively homogeneous, S_3-equivariant, and piecewise linear. On the six projected Kuhn cones it is exactly the six diagonal Weyl-chamber map of theorem 18.6.

The map Γ is not additive, is not proper, ignores the complete deck summand, and does not descend to a map from a smooth torus fibre. Thus it is an exact continuous piecewise-linear chamber map on a real cocharacter space into the diagonal part of the nine-dimensional Gram carrier's eight-dimensional boundary, not a toric fan morphism and not a morphism from the toroidal total space to the Hermitian Gram cone.

Proof. Linearity and equivariance of Ψ follow from (685) and the Weyl permutation of diagonal entries. Since $\jmath$ is injective and $\ker P = \mathbb{R}k$, its kernel is exactly (692). The least-eigenvalue function is continuous, positively homogeneous, and linear on each closed Weyl chamber, which proves the corresponding assertions for Γ. The chamber formula is theorem 18.6.

The function $H \mapsto H - \lambda_{\min}(H) I_3$ is not additive across different Weyl chambers. The kernel in (692) has positive dimension, so Γ is not proper. It ignores the deck summand by definition. Finally, translation by the angular lattice does not leave it invariant: for example $\Gamma(e_1, 0) \neq \Gamma(0, 0)$. Hence it cannot descend through the quotient of the universal cover by Λ_{tor}. □

There is one further integral check on what the elliptic quotient does not preserve. On Λ_{tor}, define the standard alternating form

$$E_3((a,b),(a',b')) = a^{\mathsf{T}}b' - b^{\mathsf{T}}a'. \tag{693}$$

It is invariant under S_3 and M_3, but on the kernel it satisfies

$$E_3((k,0),(0,\bar{k})) = 3. \tag{694}$$

Every alternating form pulled back along Φ vanishes whenever one argument lies in K. Therefore E_3 is not the pullback of an alternating form on $A_F \oplus B_F$. In particular, the holomorphic elliptic quotient in theorem 18.3 is not a morphism preserving this chosen principal lattice form. This is an integral statement about (693); it does not assert that either nonalgebraic torus family carries a polarization.

Audit record 19.4 (Rank-six conclusion)**.** The native higher-rung object is the monodromy lattice $\Lambda_{\mathrm{tor}} = H_1(T^6;\mathbb{Z})$, not a cyclic group. Its exact relation to the nine-dimensional Hermitian construction factors through the angular quotient to the A_2 weight lattice and then through the marked Cartan/Weyl fan. The deck half is indispensable for the period and filling squares but invisible to the Gram-boundary chamber map. The separate six-coordinate group $H^4(U;\mathbb{Z})$ remains the correct classifier for bundle defects on the original S^6 regular locus; no identification of these two rank-six groups is used.

20 The chamber torsor inside the nine-dimensional matrix carrier

The preceding section separates the two three-element basis orbits of Λ_{tor} from the regular six-element set of Kuhn chambers. There is nevertheless a natural place where the regular six-set occurs. It is not the lattice Λ_{tor} itself, but the off-diagonal part of the rank-nine bilinear carrier

$$\mathcal{M}_{\mathbb{Z}} := \mathrm{Hom}(\overline{L}_3, L_3) \cong \mathrm{Mat}_{3\times 3}(\mathbb{Z}).$$

This is the integral real form underlying the nine-real-dimensional Hermitian carrier. The distinction between the full carrier and its positive-semidefinite cone will be essential.

Retain the marked bases e_1, e_2, e_3 of L_3 and $\bar{e}_1, \bar{e}_2, \bar{e}_3$ of $\overline{L}_3$. Let E_{ij} denote the homomorphism which sends $\bar{e}_j$ to e_i and annihilates the other two basis vectors. The diagonal S_3-action on the two rank-three lattices induces

$$\sigma \cdot X = R_\sigma X R_\sigma^{-1}, \qquad \sigma \cdot E_{ij} = E_{\sigma(i),\sigma(j)}. \tag{695}$$

Put

$$\Delta = \bigoplus_{i=1}^{3} \mathbb{Z}E_{ii}, \qquad \mathcal{O} = \bigoplus_{i\neq j} \mathbb{Z}E_{ij}. \tag{696}$$

Theorem 20.1 (The regular chamber torsor in rank nine)**.** *There is an integral S_3-module decomposition*

$$\mathcal{M}_{\mathbb{Z}} = \Delta \oplus \mathcal{O}, \qquad \Delta \cong \mathbb{Z}[S_3/S_2], \qquad \mathcal{O} \cong \mathbb{Z}[S_3]. \tag{697}$$

If a chamber is indexed in one-line notation by $\pi = (\pi_1\pi_2\pi_3) \in S_3$, then

$$\vartheta(c_\pi) = E_{\pi_1,\pi_3} \tag{698}$$

is an S_3-equivariant bijection from the six Kuhn chambers at a marked vertex to the six-element basis of $\mathcal{O}$. Equivalently, these are the six ordered mixed pairs

$$\big((e_i,0),(0,\bar{e}_j)\big), \qquad i \neq j. \tag{699}$$

They are exactly the mixed basis pairs which are orthogonal for the integral alternating form E_3, since

$$E_3\big((e_i,0),(0,\bar{e}_j)\big) = \delta_{ij}. \tag{700}$$

The three excluded diagonal pairs $i = j$ comprise the basis of Δ and are the symplectically dual mixed pairs.

Proof. Equation (696) partitions the nine matrix units and is preserved by (695), proving the direct-sum assertion. The action is transitive on the three diagonal units, and the stabilizer of E_{11} is the subgroup permuting 2 and 3; hence $\Delta \cong \mathbb{Z}[S_3/S_2]$.

The action on ordered pairs (i,j) with $i \neq j$ is free: a permutation fixing two distinct letters fixes the remaining letter and is therefore the identity. It is transitive because any ordered pair of distinct letters can be carried to any other. Thus $\mathcal{O}$ is the regular permutation module. For each ordered pair (i,j), there is a unique remaining letter k, and hence a unique one-line permutation (i,k,j). This proves that (698) is a bijection. Moreover,

$$\vartheta(c_{\sigma\pi}) = E_{\sigma(\pi_1),\sigma(\pi_3)} = \sigma \cdot E_{\pi_1,\pi_3},$$

so it is equivariant. Finally, (700) is immediate from the defining formula $E_3((a,b),(a',b')) = a^{\mathsf{T}}b' - b^{\mathsf{T}}a'$. □

20.1 The signed quotient and the exact chamber incidence

The repeated indices on the angular and deck bases determine a marking $\iota : \overline{L}_3 \to L_3$, $\iota(\bar{e}_i) = e_i$. Define

$$s : A_F \oplus B_F \longrightarrow A_F, \qquad s(x,y) = x - B_0 y, \qquad D = s \circ (P \oplus Q). \tag{701}$$

Since $B_0 Q = P$, this means, in the retained bases,

$$D(a,b) = Pa - B_0 Q b = P(a - \iota b), \qquad [D] = [P \mid -P]. \tag{702}$$

The formula is typed: Q is applied to the deck lattice and B_0 then carries its value from B_F to A_F.

Theorem 20.2 (Unimodular pair–Kuhn–Weyl incidence)**.** *The map D is surjective, with saturated kernel*

$$\ker D = \mathbb{Z}\langle (e_1,\bar{e}_1), (e_2,\bar{e}_2), (e_3,\bar{e}_3), (k,0)\rangle, \qquad k = e_1 + e_2 + e_3. \tag{703}$$

For $i \neq j$, put

$$U_{ij} = \mathbb{Z}(e_i,0) \oplus \mathbb{Z}(0,\bar{e}_j).$$

Then U_{ij} is primitive and isotropic for E_3, and

$$D|_{U_{ij}} : U_{ij} \xrightarrow{\sim} A_F, \qquad (e_i,0) \longmapsto n_i, \quad (0,\bar{e}_j) \longmapsto -n_j \tag{704}$$

is an integral isomorphism.

Let k_{ij} be the remaining index and $\pi_{ij} = (i,k_{ij},j)$. Under the Cartan marking $\jmath(n_i) = \nu_i = E_{ii} - I_3/3$, the positive quadrant in $U_{ij,\mathbb{R}}$ maps isomorphically to the Weyl chamber

$$\mathcal{C}_{ij} = \mathbb{R}_{\geq 0}\nu_i + \mathbb{R}_{\geq 0}(-\nu_j) = \{\mathrm{diag}(\lambda) : \lambda_i \geq \lambda_{k_{ij}} \geq \lambda_j,\ \textstyle\sum_r \lambda_r = 0\}. \tag{705}$$

The assignments

$$((e_i,0),(0,\bar{e}_j)) \longmapsto \pi_{ij} \longmapsto [\sigma_{0,\pi_{ij}}] \longmapsto \mathcal{C}_{ij} \tag{706}$$

are equivariant bijections of S_3-torsors, and

$$P(\sigma_{0,\pi_{ij}}) = \mathrm{conv}\{0, n_i, -n_j\}. \tag{707}$$

The vector

$$D(e_i,\bar{e}_j) = n_i - n_j \tag{708}$$

is the interior A_2-root bisector of this chamber. With the source and target orientations fixed by the displayed bases,

$$\det[e_i \mid e_i + e_{k_{ij}} \mid k] = \mathrm{sgn}(\pi_{ij}) = \det[n_i \mid -n_j]. \tag{709}$$

Proof. Since P is surjective, so is D. Equation (702) gives $D(a,b)=0$ precisely when $a-\iota b=tk$ for a unique $t\in\mathbb{Z}$. Hence

$$(a,b)=\sum_{r=1}^{3} b_r(e_r,\bar{e}_r)+t(k,0),$$

and uniqueness gives (703). Surjectivity implies that the kernel is saturated.

The plane U_{ij} is spanned by two members of a basis of Λ_{tor}, hence is primitive. Equation (700) makes it isotropic exactly when $i\neq j$. In its displayed basis the restriction of D has columns $[n_i \mid -n_j]$. Directly from

$$n_1=\begin{pmatrix}1\\0\end{pmatrix},\qquad n_2=\begin{pmatrix}-1\\1\end{pmatrix},\qquad n_3=\begin{pmatrix}0\\-1\end{pmatrix},$$

one obtains $\det[n_i \mid -n_j]=\pm 1$ for $i\neq j$, proving (704).

For $x,y\geq 0$, the diagonal matrix $H=x\nu_i-y\nu_j$ satisfies

$$\lambda_i-\lambda_{k_{ij}}=x,\qquad \lambda_{k_{ij}}-\lambda_j=y.$$

Conversely these two differences, together with trace zero, recover H, which proves (705) and the cone isomorphism. The four vertices of $\sigma_{0,(i,k_{ij},j)}$ have P-images

$$0,\quad n_i,\quad n_i+n_{k_{ij}}=-n_j,\quad n_1+n_2+n_3=0,$$

proving (707). Ordered distinct pairs, one-line permutations, Kuhn-cell translation classes, and Weyl chambers are all free transitive S_3-sets, and the displayed assignments commute with the coordinate action. At $x=y=1$, $\jmath(n_i-n_j)=E_{ii}-E_{jj}$, whose ordered diagonal entries in positions (i,k_{ij},j) are $(1,0,-1)$; this is the interior bisector.

Finally, relative to the ordered basis $(e_i,e_{k_{ij}},e_j)$, the columns $(e_i,e_i+e_{k_{ij}},k)$ form an upper unitriangular matrix. Their determinant in the fixed source basis is therefore $\operatorname{sgn}(\pi_{ij})$. For $(i,k_{ij},j)=(1,2,3)$, $[n_i \mid -n_j]=I_2$; equivariance and $\det\rho_A(\sigma)=\operatorname{sgn}(\sigma)$ prove the second equality in (709). □

Proposition 20.3 (Exact marking and monodromy boundary)**.** *There is no endomorphism $T:A_F\to A_F$ satisfying*

$$DM_3=TD. \tag{710}$$

Moreover, for every $c\in\mathbb{Z}$, the shear

$$S_c(a,b)=(a+c\iota b,b) \tag{711}$$

is an integral automorphism which fixes the angular lattice and its Kuhn fan, preserves c_3 and E_3, and commutes with M_3 and the diagonal S_3-action, but

$$DS_c(a,b)=D(a,b)+cP(\iota b). \tag{712}$$

Thus the pair–chamber correspondence is canonical relative to the retained angular/deck bases, their common indexing, P,Q,B_0, and the sign in (701)*; it is not an invariant of the unmarked data (M_3,E_3,c_3) together with the angular fan.*

Proof. The vector $(e_i,\bar{e}_i)$ lies in $\ker D$, whereas

$$M_3(e_i,\bar{e}_i)=(2e_i,\bar{e}_i),\qquad D(2e_i,\bar{e}_i)=n_i\neq 0.$$

Thus $\ker D$ is not M_3-invariant, which is necessary for (710).

The inverse of S_c is S_{-c}. Direct substitution gives

$$S_cM_3(a,b)=(a+(c+1)\iota b,b)=M_3S_c(a,b),$$

and proves preservation of c_3 and commutation with the diagonal coordinate action. Symmetry of the dot product gives

$$E_3(S_c(a,b),S_c(a',b')) = (a+c\iota b)^{\mathsf{T}}b' - b^{\mathsf{T}}(a'+c\iota b') = E_3((a,b),(a',b')).$$

Equation (712) follows from (702). For $c \neq 0$, the deck complement is moved, so the retained marked pair set and D are changed. □

The parity and opposition operations also retain exact information. The even and odd chamber triples are

$$\vartheta\{c_{123},c_{231},c_{312}\} = \{E_{13},E_{21},E_{32}\}, \tag{713}$$

$$\vartheta\{c_{132},c_{213},c_{321}\} = \{E_{12},E_{23},E_{31}\}. \tag{714}$$

Thus they are the two orientations of the directed three-cycle. Reversal of one-line notation, $(i,k,j) \mapsto (j,k,i)$, is carried by ϑ to matrix transposition $E_{ij} \mapsto E_{ji}$. This is the exact chamber opposition involution; it exchanges the two parity triples.

20.2 The real Hermitian representation and its integral defect

Let $\mathrm{Herm}_3(\mathbb{C})$ carry the S_3-action by conjugation with the real permutation matrices. Define

$$\Xi : \mathrm{Mat}_{3\times 3}(\mathbb{R}) \longrightarrow \mathrm{Herm}_3(\mathbb{C}), \qquad \Xi(X) = \frac{X+X^{\mathsf{T}}}{2} + \frac{i(X-X^{\mathsf{T}})}{2}. \tag{715}$$

Proposition 20.4 (Exact representation bridge and the positivity boundary)**.** *The map Ξ is an S_3-equivariant real-linear isomorphism. Its inverse sends a Hermitian matrix $H = S + iA$, with S real symmetric and A real skew-symmetric, to $S + A$. Consequently*

$$\mathrm{Herm}_3(\mathbb{C}) \cong_{\mathbb{R}[S_3]} 2\cdot\mathbf{1} \oplus 3\cdot V_{\mathrm{std}} \oplus \mathbf{sgn}. \tag{716}$$

For each unordered pair $\{i,j\}$, the symmetric and skew generators $E_{ij}+E_{ji}$ and $E_{ij}-E_{ji}$ span an index-two sublattice of $\mathbb{Z}E_{ij} \oplus \mathbb{Z}E_{ji}$. Hence the symmetric–skew off-diagonal splitting has total integral index

$$2^3 = 8. \tag{717}$$

In particular, Ξ is a real representation isomorphism, not an identification of the two preferred integral lattices.

Moreover, for $i \neq j$, $\Xi(E_{ij})$ has eigenvalues $2^{-1/2}, -2^{-1/2}, 0$. Thus the regular chamber torsor is located in the nine-dimensional Hermitian representation, but its matrix-unit image is not contained in the positive-semidefinite Gram cone or its boundary.

Proof. Every real matrix has the unique decomposition $X = S + A$, where $S = (X+X^{\mathsf{T}})/2$ is symmetric and $A = (X-X^{\mathsf{T}})/2$ is skew-symmetric. Equation (715) sends this pair to $S + iA$, which is Hermitian, and the displayed inverse proves bijectivity. Permutation conjugation commutes with transpose and with these two projections, proving equivariance.

The real permutation representation on three letters is $\mathbf{1} \oplus V_{\mathrm{std}}$. Therefore

$$\mathrm{Mat}_{3\times 3}(\mathbb{R}) \cong (\mathbf{1} \oplus V_{\mathrm{std}})^{\otimes 2} \cong 2\cdot\mathbf{1} \oplus 3\cdot V_{\mathrm{std}} \oplus \mathbf{sgn},$$

where $V_{\mathrm{std}} \otimes V_{\mathrm{std}} \cong \mathbf{1} \oplus V_{\mathrm{std}} \oplus \mathbf{sgn}$. For one unordered pair, the change-of-basis matrix from (E_{ij}, E_{ji}) to $(E_{ij}+E_{ji}, E_{ij}-E_{ji})$ has determinant -2. Multiplication over the three unordered pairs proves (717).

Finally, on the i,j coordinate plane, $\Xi(E_{ij})$ has block

$$\begin{pmatrix} 0 & (1+i)/2 \\ (1-i)/2 & 0 \end{pmatrix},$$

whose eigenvalues are $\pm|1+i|/2 = \pm 2^{-1/2}$; the remaining coordinate has eigenvalue zero. Hence it is indefinite. □

20.3 The positive six-set is a different equivariant realization

Although the matrix units themselves are indefinite, the spectral-floor map of the preceding section gives a second realization of the same chamber torsor in the rank-two positive-semidefinite boundary. If $\pi = (i, k, j)$, then

$$G_\pi = 2E_{ii} + E_{kk} \tag{718}$$

has diagonal entries equal to a permutation of $(2, 1, 0)$. The assignment

$$E_{ij} \longmapsto 2E_{ii} + E_{kk}, \qquad \{i, j, k\} = \{1, 2, 3\}, \tag{719}$$

is therefore an equivariant bijection of the two marked six-element sets. It extends uniquely by linearity to a homomorphism $\mathcal{O} \to \Delta$, but that extension is neither the map Ξ nor the nonlinear spectral-floor map on all of $\mathrm{Herm}_3^0(\mathbb{C})$. Thus (719) records exact chamber incidence; it does not manufacture a positive monodromy-compatible morphism from the toroidal family.

Audit record 20.5 (What the native $\mathbb{Z}^6$ result becomes)**.** The six Kuhn chambers are not the six generators of $H_1(T^6; \mathbb{Z})$. They are canonically, relative to the retained marking, the six symplectically orthogonal ordered angular–deck pairs and hence the six off-diagonal basis elements in the associated rank-nine bilinear carrier. This places the regular S_3-torsor exactly inside the nine-dimensional program while preserving the different roles of angular cycles, deck cycles, monodromy, and the positive-semidefinite Gram boundary. The construction is a representation and incidence theorem; positivity and global toroidal descent remain separate questions and are not inferred from the equality of dimensions or orbit cardinalities.

21 The Lorentz observation morphism and the full nine-dimensional fibre

The rank-three Hermitian carrier and the four-real-dimensional Lorentz cone are not identical objects. The required next step is therefore to compute their exact relation. This section constructs a split positive observation morphism, determines every one of its fibres, identifies the five retained directions beyond the observed four-vector, and proves the complete cubic coupling among the resulting Lorentz-vector, spinor, and scalar coordinates. No direction is removed from the source calculation.

21.1 The marked angular epimorphism and its exact splitting

Retain the angular lattice matrix and its primitive kernel vector:

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad Pu = 0. \tag{720}$$

The standard Hermitian products in these displayed marked bases give

$$C := PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det C = 3. \tag{721}$$

Define the corresponding orthogonal right inverse and the projection onto the displayed kernel line by

$$R := P^* C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}, \qquad \Pi := \frac{uu^*}{3}, \qquad Q_u := I_3 - \Pi. \tag{722}$$

The choice of R uses the displayed marking and its standard Hermitian products. All formulas below retain that choice explicitly.

Lemma 21.1 (Exact splitting identities)**.** *The matrices in* (720)–(722) *satisfy*

$$PR = I_2, \qquad u^*R = 0, \qquad R^*R = C^{-1}, \qquad RP = Q_u, \qquad \Pi R = 0, \qquad P\Pi = 0. \tag{723}$$

Moreover, with

$$X = (R\ u) = \begin{pmatrix} 2/3 & 1/3 & 1 \\ -1/3 & 1/3 & 1 \\ -1/3 & -2/3 & 1 \end{pmatrix}, \tag{724}$$

one has

$$\det X = 1, \qquad X^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}. \tag{725}$$

Proof. Direct multiplication, with every denominator retained, gives

$$PR = \frac{1}{3}\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix} = I_2$$

and

$$RP = \frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = I_3 - \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = Q_u.$$

The identities involving u, R^*R, and Π follow from the same displayed matrices and from $R = P^*(PP^*)^{-1}$. Finally,

$$\det(3X) = \det\begin{pmatrix} 2 & 1 & 3 \\ -1 & 1 & 3 \\ -1 & -2 & 3 \end{pmatrix} = 27,$$

so $\det X = 27/3^3 = 1$. Both products

$$\begin{pmatrix} P \\ u^*/3 \end{pmatrix}(R\ u) \quad \text{and} \quad (R\ u)\begin{pmatrix} P \\ u^*/3 \end{pmatrix}$$

are the identity by (723); this proves (725). □

21.2 A positive observation morphism with all fibres retained

Put

$$\mathcal{J}_3 = \mathrm{Herm}_3(\mathbb{C}), \qquad \mathcal{J}_2 = \mathrm{Herm}_2(\mathbb{C}), \qquad \mathcal{P}_r = \{H \in \mathcal{J}_r : H \succeq 0\}.$$

Define real-linear maps

$$\mathcal{O}_P : \mathcal{J}_3 \longrightarrow \mathcal{J}_2, \qquad \mathcal{O}_P(H) = PHP^*, \tag{726}$$

and

$$\jmath_P : \mathcal{J}_2 \longrightarrow \mathcal{J}_3, \qquad \jmath_P(G) = RGR^*. \tag{727}$$

Theorem 21.2 (Split observation and the exact five-dimensional kernel)**.** *The observation morphism is surjective and positive, and its displayed section is a positive linear embedding:*

$$\mathcal{O}_P\jmath_P = \mathrm{id}_{\mathcal{J}_2}, \qquad \jmath_P\mathcal{O}_P(H) = Q_uHQ_u. \tag{728}$$

Consequently

$$\mathcal{J}_3 = \operatorname{im}\jmath_P \oplus \ker\mathcal{O}_P, \qquad \dim_{\mathbb{R}}\operatorname{im}\jmath_P = 4, \qquad \dim_{\mathbb{R}}\ker\mathcal{O}_P = 5. \tag{729}$$

The kernel is explicitly

$$\ker \mathcal{O}_P = \{a\,uu^* + uv^* + vu^* : a \in \mathbb{R},\ v \in u^\perp\}. \tag{730}$$

It is exactly the tangent space at $H_0 = uu^*$ to the nonzero rank-one positive-semidefinite stratum

$$\mathcal{R}_1 = \{ww^* : w \in \mathbb{C}^3 \setminus \{0\}\} \subset \mathcal{P}_3. \tag{731}$$

On the positive cone, the complete zero fibre is the retained ray

$$\mathcal{O}_P^{-1}(0) \cap \mathcal{P}_3 = \{\lambda uu^* : \lambda \geq 0\}. \tag{732}$$

Proof. Equations (723) give

$$\mathcal{O}_P(\jmath_P(G)) = PRGR^*P^* = G$$

and

$$\jmath_P(\mathcal{O}_P(H)) = RPHP^*R^* = Q_uHQ_u.$$

Both maps preserve positive semidefiniteness because they are congruences. The first identity proves surjectivity, injectivity of $\jmath_P$, and the direct sum in (729). Since $\dim_{\mathbb{R}} \mathcal{J}_3 = 9$ and $\dim_{\mathbb{R}} \mathcal{J}_2 = 4$, the dimensions follow.

The condition $PHP^* = 0$ is equivalent to $Q_uHQ_u = 0$. Relative to $\mathbb{C}^3 = u^\perp \oplus \mathbb{C}u$, such a Hermitian operator has a zero $u^\perp$-to-$u^\perp$ block and therefore has the unique form (730). The differential at u of $F(w) = ww^*$ is

$$dF_u(\dot{w}) = u\dot{w}^* + \dot{w}u^*.$$

Writing $\dot{w} = (a/2)u + v + ibu$, with $a, b \in \mathbb{R}$ and $v \in u^\perp$, shows that its image is exactly (730); the imaginary radial direction ibu is the one-real-dimensional kernel of dF_u. Thus the image has dimension five and is the stated tangent space.

Finally let $H \succeq 0$ and $PHP^* = 0$. For every $\xi \in \mathbb{C}^2$,

$$0 = \xi^*PHP^*\xi = \|H^{1/2}P^*\xi\|^2.$$

Hence $H^{1/2}$ annihilates $\operatorname{im} P^* = u^\perp$. The range of H is contained in $\mathbb{C}u$, and Hermitian positivity then gives $H = \lambda uu^*$ for one $\lambda \geq 0$. The converse follows from $Pu = 0$. □

The observation morphism is not a dimension count. It factors through the already proved two-column Gram carrier. For $H \in \mathcal{P}_3$, put

$$\mathcal{Z}_P(H) = H^{1/2}P^* \in \mathrm{Mat}_{3\times 2}(\mathbb{C}). \tag{733}$$

Then, with no omitted scalar,

$$\mathcal{Z}_P(H)^*\mathcal{Z}_P(H) = PHP^* = \mathcal{O}_P(H). \tag{734}$$

21.3 Complete block coordinates and the cubic coupling

Theorem 21.3 (Lorentz-vector–spinor–scalar coordinates)**.** *Every $H \in \mathcal{J}_3$ has a unique expression*

$$H = X\begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix}X^* = RGR^* + R\psi u^* + u\psi^*R^* + \lambda uu^*, \tag{735}$$

where

$$G = PHP^*, \qquad \psi = \frac{1}{3}PHu, \qquad \lambda = \frac{1}{9}u^*Hu, \tag{736}$$

with $G \in \mathcal{J}_2$, $\psi \in \mathbb{C}^2$, and $\lambda \in \mathbb{R}$. Thus the displayed marking gives the exact real-vector-space decomposition

$$\mathcal{J}_3 \xrightarrow{\sim} \mathcal{J}_2 \oplus \mathbb{C}^2 \oplus \mathbb{R}, \qquad 9 = 4 + 4 + 1. \tag{737}$$

For $G = \left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$, *retain*

$$\operatorname{adj}(G) = \begin{pmatrix} d & -b \\ -\bar{b} & a \end{pmatrix}. \tag{738}$$

Then the complete determinant is

$$\boxed{\det H = \lambda \det G - \psi^* \operatorname{adj}(G)\psi.} \tag{739}$$

For $G \succeq 0$, *let* G^+ *be its Moore–Penrose inverse on* $\operatorname{im} G$, *extended by zero on* $\ker G$. *The complete positive fibre is*

$$\mathcal{O}_P^{-1}(G) \cap \mathcal{P}_3 = \left\{ X \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} X^* : \psi \in \operatorname{im} G, \; \lambda \geq \psi^* G^+ \psi \right\}. \tag{740}$$

In particular,

$$H \succ 0 \quad \Longleftrightarrow \quad G \succ 0 \;\; \textit{and} \;\; \lambda > \psi^* G^{-1} \psi. \tag{741}$$

If G *is singular and* $H \succeq 0$, *then* H *is singular. If* $G \succ 0$, *the boundary of its fibre is the graph* $\lambda = \psi^* G^{-1} \psi$.

Proof. Equation (725) gives

$$X^{-1} H X^{-*} = \begin{pmatrix} PHP^* & PHu/3 \\ u^*HP^*/3 & u^*Hu/9 \end{pmatrix},$$

which proves uniqueness, (735), and (736). Because $\det X = 1$, direct expansion of the 3×3 block determinant gives

$$\det H = \lambda(ad - |b|^2) - \begin{pmatrix} \bar{\psi}_1 & \bar{\psi}_2 \end{pmatrix} \begin{pmatrix} d & -b \\ -\bar{b} & a \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix},$$

which is (739).

Invertible congruence by X preserves positivity. Suppose the block matrix is positive semidefinite. Its upper-left principal block gives $G \succeq 0$. If $v \in \ker G$, positivity of the restriction to $\mathbb{C}(v, 0) \oplus \mathbb{C}(0, 1)$ forces $v^*\psi = 0$; hence $\psi \perp \ker G$, or equivalently $\psi \in \operatorname{im} G$. Write $\psi = G^{1/2} w$ with $w \in \operatorname{im} G$. Completing the square gives

$$\begin{pmatrix} x \\ s \end{pmatrix}^* \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} \begin{pmatrix} x \\ s \end{pmatrix} = \|G^{1/2}x + sw\|^2 + |s|^2(\lambda - \|w\|^2).$$

Here $\|w\|^2 = \psi^* G^+ \psi$. This proves necessity and sufficiency of (740). The strict case is the ordinary Schur-complement criterion (741).

If G is singular, then $\psi \in \operatorname{im} G$ and $\operatorname{adj}(G)$ has image in $\ker G$. Thus $\psi^* \operatorname{adj}(G)\psi = 0 = \det G$, so (739) gives $\det H = 0$. If $G \succ 0$, the same formula becomes

$$\det H = \det G \, (\lambda - \psi^* G^{-1} \psi),$$

which proves the last assertion. □

21.4 The embedded Lorentz group and the meaning of $9 = 4 + 4 + 1$

For $g \in GL_2(\mathbb{C})$, define

$$T_P(g) = RgP + \Pi = X \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix} X^{-1}. \tag{742}$$

Theorem 21.4 (Exact Lorentz-vector, Weyl-spinor, and scalar action)**.** *The map $T_P : GL_2(\mathbb{C}) \to GL_3(\mathbb{C})$ is an injective group homomorphism satisfying*

$$T_P(g)u = u, \qquad PT_P(g) = gP, \qquad T_P(g)R = Rg, \qquad \det T_P(g) = \det g. \tag{743}$$

Under congruence by $T_P(g)$, the block coordinates transform exactly as

$$(G, \psi, \lambda) \longmapsto (gGg^*, g\psi, \lambda). \tag{744}$$

Consequently

$$\mathcal{O}_P(T_P(g)HT_P(g)^*) = g\mathcal{O}_P(H)g^*, \qquad T_P(g)\jmath_P(G)T_P(g)^* = \jmath_P(gGg^*). \tag{745}$$

For $g \in SL_2(\mathbb{C})$, congruence on $\mathcal{J}_2$ is the standard double-cover action on Minkowski space. In the retained coordinates

$$G(t, x_1, x_2, x_3) = \begin{pmatrix} t + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & t - x_3 \end{pmatrix}, \tag{746}$$

one has

$$\det G = t^2 - x_1^2 - x_2^2 - x_3^2, \qquad G \succeq 0 \Longleftrightarrow t \geq 0, \quad t^2 \geq x_1^2 + x_2^2 + x_3^2. \tag{747}$$

The induced Lorentz-vector homomorphism

$$\rho_{\mathrm{vec}} : SL_2(\mathbb{C}) \longrightarrow GL(\mathcal{J}_2), \qquad \rho_{\mathrm{vec}}(g)(G) = gGg^*, \tag{748}$$

has kernel $\{I_2, -I_2\}$ and image $SO^+(1,3)$. The embedding T_P, and hence the action on the full $4+4+1$ carrier, is faithful. Indeed, on the full nine-dimensional carrier $-I_2$ fixes G and λ but sends $\psi \mapsto -\psi$. Thus the full carrier retains the spin double cover, and (737) *is exactly the real representation sum*

$$\boxed{\textit{Lorentz four-vector} \ \oplus \ \textit{defining complex spin representation } \mathbb{C}^2 \ \oplus \ \textit{real scalar.}} \tag{749}$$

With the convention that g acts on undotted spinor indices, the middle summand is called a left Weyl spinor; reversing that convention reverses the chirality name but not the displayed action $\psi \mapsto g\psi$.

Proof. The second expression in (742) and (725) prove the homomorphism, determinant, and injectivity assertions. Equations (723) give the other identities directly. Moreover,

$$T_P(g)X = X(g \oplus 1),$$

so congruence of (735) gives (744). The two equivariance identities follow.

For $g \in SL_2(\mathbb{C})$,

$$\det(gGg^*) = |\det g|^2 \det G = \det G,$$

and positive semidefiniteness is preserved. Hence congruence defines a homomorphism to the identity-component (proper orthochronous) Lorentz group of (747). If $gGg^* = G$ for every Hermitian G, then $gg^* = I_2$ by taking $G = I_2$; the remaining equations say that g commutes with every Hermitian matrix. Thus g is scalar, and $\det g = 1$ gives $g = \pm I_2$.

The differential has zero kernel: if $AG + GA^* = 0$ for all Hermitian G, then $A + A^* = 0$ at $G = I_2$, and the remaining equation is $[A, G] = 0$ for every Hermitian G. Hence A is scalar; $\operatorname{tr} A = 0$ gives $A = 0$. Both $SL_2(\mathbb{C})$ and $SO^+(1,3)$ have real dimension six, so the image is an open subgroup of the connected group $SO^+(1,3)$, hence the whole group. Finally, (744) gives the action of the central element and identifies the three standard real representations. □

The cubic in (739) is invariant under the full embedded $SL_2(\mathbb{C})$ action. Indeed,

$$\operatorname{adj}(gGg^*) = g^{-*} \operatorname{adj}(G) g^{-1} \qquad (g \in SL_2(\mathbb{C})), \tag{750}$$

so $(g\psi)^* \operatorname{adj}(gGg^*)(g\psi) = \psi^* \operatorname{adj}(G)\psi$. For the larger $GL_2(\mathbb{C})$ action it is instead a relative invariant:

$$\det(T_P(g)HT_P(g)^*) = |\det g|^2 \det H. \tag{751}$$

Thus no GL_2-invariance is being inferred from the SL_2 formula.

21.5 The Lorentz cone inside the rank-loss S^8 and its normal opening

Let

$$e_2(K) = \frac{1}{2}\big((\operatorname{tr} K)^2 - \operatorname{tr}(K^2)\big) \qquad (K \in \mathcal{J}_3). \tag{752}$$

Theorem 21.5 (Proper Lorentz-cone section and determinant opening)**.** *For every $G \in \mathcal{J}_2$,*

$$\operatorname{rank} \jmath_P(G) = \operatorname{rank} G, \qquad \det \jmath_P(G) = 0, \qquad e_2(\jmath_P(G)) = \frac{1}{3}\det G. \tag{753}$$

Hence $\jmath_P$ maps the entire future Lorentz cone properly and injectively into

$$\partial\mathcal{P}_3 = \{H \succeq 0 : \det H = 0\} = \mathcal{P}_{3,\le 2}. \tag{754}$$

It extends to an embedding of one-point compactifications

$$\jmath_P^+ : \mathcal{P}_2^+ \cong B^4 \hookrightarrow (\partial\mathcal{P}_3)^+ \cong S^8. \tag{755}$$

The timelike, nonzero null, and zero strata map respectively to rank two, rank one, and rank zero.

For every real ε, the affine slice through $H_0 = uu^*$ satisfies the exact identity

$$\boxed{\det\big(uu^* + \varepsilon \jmath_P(G)\big) = \varepsilon^2 \det G.} \tag{756}$$

For $\varepsilon > 0$ and $G \succeq 0$, timelike G maps to the interior of $\mathcal{P}_3$, nonzero null G maps to its rank-two boundary, and $G = 0$ maps to H_0. The restricted determinant Hessian is

$$D^2(\det)_{H_0}\big(\jmath_P(G), \jmath_P(K)\big) = \operatorname{tr} G \operatorname{tr} K - \operatorname{tr}(GK), \tag{757}$$

so

$$\frac{1}{2}D^2(\det)_{H_0}\big(\jmath_P(G), \jmath_P(G)\big) = \det G = t^2 - x_1^2 - x_2^2 - x_3^2. \tag{758}$$

The surjection $\mathcal{O}_P : \partial\mathcal{P}_3 \to \mathcal{P}_2$ is not proper and therefore does not extend by sending infinity to infinity. Thus (755) *is an actual compactified embedding, whereas the reverse compactified arrow is not supplied by $\mathcal{O}_P$.*

Proof. Because R has complex rank two, congruence by R preserves the rank of a 2×2 Hermitian matrix. Its image has rank at most two, so its 3×3 determinant vanishes. When G is invertible, the product of the two nonzero eigenvalues of RGR^* equals

$$\det(GR^*R) = \det G \det(C^{-1}) = \frac{1}{3}\det G.$$

That product is $e_2(RGR^*)$. Both sides are polynomial in the four real entries of G, so the identity extends to singular G, proving (753). The map is a linear embedding between finite-dimensional spaces, so its restriction to the closed cone is proper. The compactification statement now follows from the previously proved identifications $\mathcal{P}_2^+ \cong B^4$ and $(\partial\mathcal{P}_3)^+ \cong S^8$.

Using (724),

$$uu^* + \varepsilon \jmath_P(G) = X \begin{pmatrix} \varepsilon G & 0 \\ 0 & 1 \end{pmatrix} X^*.$$

Since $\det X = 1$, taking determinants proves (756). The rank and positivity statements follow from this same block matrix. Polarizing $\det(sG + tK)$ gives

$$\det(sG + tK) = s^2 \det G + st\big(\operatorname{tr} G \operatorname{tr} K - \operatorname{tr}(GK)\big) + t^2 \det K,$$

which proves (757) and (758).

For nonproperness choose $G_0 = \operatorname{diag}(1,0)$ and

$$H_n = \jmath_P(G_0) + nuu^*, \qquad n \in \mathbb{N}.$$

The two summands have orthogonal ranges, so $H_n \succeq 0$ has rank two and lies in $\partial\mathcal{P}_3$. Its norm tends to infinity, while $\mathcal{O}_P(H_n) = G_0$ for every n. A continuous extension taking the source point at infinity to the target point at infinity would therefore send this convergent compactified sequence both to G_0 and to infinity, which is impossible. □

21.6 The ten-dimensional determinant-phase carrier in the same coordinates

The square-matrix $SU(3)$-orbit theorem already identifies its affine quotient with

$$\mathcal{Y}_3 = \{(H,\omega) : H \in \mathcal{P}_3,\ |\omega|^2 = \det H\}. \tag{759}$$

Substitution of (735) gives an exact reformulation.

Corollary 21.6 (Lorentz-spinor-scalar-phase form of the $SU(3)$ quotient)**.** *There is a homeomorphism*

$$\mathcal{Y}_3 \cong \left\{(G,\psi,\lambda,\omega) : \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} \succeq 0, \quad |\omega|^2 = \lambda \det G - \psi^* \operatorname{adj}(G)\psi \right\}. \tag{760}$$

The $SL_2(\mathbb{C})$-action is

$$(G,\psi,\lambda,\omega) \longmapsto (gGg^*, g\psi, \lambda, \omega). \tag{761}$$

The complex determinant coordinate is invariant, and its zero locus is exactly the rank-deficient cone:

$$\omega = 0 \quad \Longleftrightarrow \quad \det H = 0. \tag{762}$$

After the proved one-point compactifications, this is the distinguished S^8 inside the ten-real-dimensional S^{10} carrier. The ray $\{(G,\psi,\omega) = (0,0,0), \lambda \geq 0\}$ is retained inside that zero locus.

Proof. The block-coordinate map is an invertible real-linear congruence, so it preserves positivity and is a homeomorphism. Equation (739) gives the displayed scalar relation. Equations (744) and (750) prove invariance. The equivalence (762) follows directly from $|\omega|^2 = \det H$. The compactification and the identification of its zero-determinant locus are the previously proved $SU(3)$-orbit and rank-loss compactification theorems. □

This corollary makes the relation among the dimensions explicit. The nine-real-dimensional Hermitian carrier has the $4+4+1$ decomposition in (737). Adding the complex determinant coordinate and retaining its one-real-modulus equation gives $9 + 2 - 1 = 10$ real dimensions. The determinant-zero locus removes the phase and imposes the cubic boundary equation, leaving the eight-real-dimensional rank-loss cone whose one-point compactification is S^8.

21.7 The holomorphic toric seed and its Lorentz-orbit completion

The local four-branch chart has the already typed monomial map

$$q(z_0,z_1,z_2,z_3) = (z_0z_3, z_1, z_2).$$

Define, without identifying angular and filling lattices,

$$\kappa(w_0,w_1,w_2) = \operatorname{diag}(w_0, w_1w_2), \qquad K = \kappa q. \tag{763}$$

Then

$$K(z) = \operatorname{diag}(z_0z_3, z_1z_2), \qquad \boxed{\det K(z) = z_0z_1z_2z_3.} \tag{764}$$

This is a holomorphic toric morphism into the diagonal complex matrices. On the retained positive real locus put

$$a = z_0 z_3, \qquad b = z_1 z_2, \qquad p_0 = \frac{a+b}{2}, \qquad p_3 = \frac{a-b}{2}, \qquad p_1 = p_2 = 0.$$

Then $K(z) = G(p) \succeq 0$ and

$$p_0^2 - p_3^2 = ab = z_0 z_1 z_2 z_3.$$

Thus the toric products are exactly the two light-cone coordinates $p_0 + p_3$ and $p_0 - p_3$.

For a retained $\mu > 0$, define the affine section

$$\Psi_\mu(G) = \mu u u^* + \jmath_P(G). \tag{765}$$

Equation (756) gives

$$\det \Psi_\mu(G) = \mu \det G, \qquad \Psi_\mu(G) \succeq 0 \Longleftrightarrow G \succeq 0, \qquad \operatorname{rank} \Psi_\mu(G) = 1 + \operatorname{rank} G. \tag{766}$$

Every $G \succeq 0$ has a spectral decomposition $G = v \operatorname{diag}(a, b) v^*$ with $a, b \geq 0$ and $v \in SU(2)$: a scalar phase changes the determinant of a chosen unitary diagonalizer without changing its congruence action. Hence

$$\Theta_\mu : SL_2(\mathbb{C}) \times \mathbb{R}^4_{\geq 0} \longrightarrow \Psi_\mu(\mathcal{P}_2), \qquad \Theta_\mu(g, z) = \Psi_\mu(g K(z) g^*) \tag{767}$$

is surjective. It obeys

$$\det \Theta_\mu(g, z) = \mu z_0 z_1 z_2 z_3. \tag{768}$$

On $(\mathbb{C}^*)^4$, the complete auxiliary fibre of K is

$$(\rho, \sigma) \cdot (z_0, z_1, z_2, z_3) = (\rho z_0, \sigma z_1, \sigma^{-1} z_2, \rho^{-1} z_3), \qquad (\rho, \sigma) \in (\mathbb{C}^*)^2. \tag{769}$$

Thus the local toric chart supplies a diagonal $1 + 1$ seed, its $SL_2(\mathbb{C})$-orbit supplies all future causal directions, and the two-complex-dimensional fibre in (769) is retained as the exact redundancy of this parametrization.

21.8 An exact colour-invariant Lorentz momentum and constrained dynamics

The preceding results supply a pointwise carrier with commuting colour and spin actions and a complete Lorentz-covariant orbit coordinate. Let

$$\mathscr{V} = \operatorname{Mat}_{3\times 2}(\mathbb{C}), \qquad U \cdot Z = UZ \quad (U \in SU(3)), \qquad g \triangleright Z = Z g^* \quad (g \in SL_2(\mathbb{C})). \tag{770}$$

These two left actions commute. Retain positive conversion constants $\alpha > 0$ and $c > 0$, with c carrying speed dimension, and define

$$p_\alpha(Z) = \alpha Z^* Z \in \mathcal{P}_2, \qquad m^2_{\alpha,c}(Z) = \frac{\det p_\alpha(Z)}{c^2}. \tag{771}$$

If p_α is assigned momentum dimension, then $[\alpha] = [p]/[Z]^2$ and $[m^2_{\alpha,c}] = [p]^2/[c]^2 = [\text{mass}]^2$; no value of α or c is selected or suppressed. Equivalently, $\det p_\alpha = m^2_{\alpha,c} c^2$, as required when the time component of the Hermitian momentum matrix is E/c.

Theorem 21.7 (Complete colour-orbit momentum and its mass invariant)**.** *The map p_α is $SU(3)$-invariant, is covariant for the spin action, and separates $SU(3)$-orbits:*

$$p_\alpha(UZ) = p_\alpha(Z), \qquad p_\alpha(g \triangleright Z) = g p_\alpha(Z) g^*. \tag{772}$$

It induces a homeomorphism

$$\mathscr{V}/SU(3) \xrightarrow{\sim} \mathcal{P}_2, \qquad [Z] \longmapsto p_\alpha(Z). \tag{773}$$

Writing the rows of Z as three two-component covectors, one has

$$\boxed{m^2_{\alpha,c}(Z) = \frac{\alpha^2}{c^2} \sum_{1\le a<b\le 3} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2.} \tag{774}$$

It is invariant under both $SU(3)$ and $SL_2(\mathbb{C})$, and

$$m^2_{\alpha,c}(Z) > 0 \Longleftrightarrow \operatorname{rank}_{\mathbb{C}} Z = 2, \qquad m^2_{\alpha,c}(Z) = 0 \Longleftrightarrow \operatorname{rank}_{\mathbb{C}} Z \le 1. \tag{775}$$

For every rank-two Z_0,

$$m^2_{\alpha,c}(rZ_0) = r^4 m^2_{\alpha,c}(Z_0) \qquad (r > 0), \tag{776}$$

so the unrestricted kinematic carrier contains positive masses arbitrarily close to zero.

Proof. The covariance equations follow directly from (770). The orbit-separation and homeomorphism statement is the exact two-column $SU(3)$ Gram quotient, with multiplication by the retained positive scalar α. The 2×2 Gram determinant is the Cauchy–Binet identity

$$\det(Z^* Z) = \sum_{a<b} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2,$$

which proves (774). Congruence by an $SL_2(\mathbb{C})$ matrix preserves the determinant, while left $SU(3)$ preserves Z^*Z. The sum vanishes exactly when all 2×2 minors vanish, which is equivalent to rank at most one. Finally $p_\alpha(rZ) = r^2 p_\alpha(Z)$, and taking the 2×2 determinant gives (776). □

For a retained mass $m_0 \ge 0$, the constraint

$$\mathscr{C}_{m_0,\alpha,c} = \{Z \in \mathscr{V} : \det p_\alpha(Z) = m_0^2 c^2\} \tag{777}$$

is invariant under both actions, and for $m_0 > 0$ its $SU(3)$-orbit space is exactly the future mass hyperboloid

$$\{p \in \mathcal{P}_2 : \det p = m_0^2 c^2\}. \tag{778}$$

Retain the Lorentz bilinear pairing

$$\langle A, B\rangle_L = \frac{1}{2}\big(\operatorname{tr} A \operatorname{tr} B - \operatorname{tr}(AB)\big), \qquad \langle A, A\rangle_L = \det A. \tag{779}$$

For paths $x : I \to \mathcal{J}_2$, $Z : I \to \mathscr{V}$, and a real multiplier $e : I \to \mathbb{R}$, define the first-order constrained action

$$\mathcal{S}_{m_0,\alpha,c}[x, Z, e] = \int_I \Big(\langle p_\alpha(Z), \dot{x}\rangle_L - \frac{e}{2}\big(\det p_\alpha(Z) - m_0^2 c^2\big) \Big)\, d\tau. \tag{780}$$

It is invariant under arbitrary τ-dependent left $SU(3)$ rotations of Z and under the constant $SL_2(\mathbb{C})$ action

$$x \longmapsto g x g^*, \qquad Z \longmapsto Z g^*. \tag{781}$$

If x has length dimension and τ retains its chosen parameter dimension, then $[e] = [x]/([\tau][p])$, so both summands in the integrand have dimension $[p][x]/[\tau]$ and the action has dimension $[p][x]$. Variation of e gives (777); compactly supported variation of x gives

$$\frac{d}{d\tau} p_\alpha(Z) = 0. \tag{782}$$

For a variation $\delta Z : I \to \mathscr{V}$, retain the exact differential

$$\delta p_\alpha = \alpha(\delta Z^* Z + Z^* \delta Z). \tag{783}$$

The Z-Euler–Lagrange equation is the weak identity

$$\big\langle \alpha\big(\delta Z^* Z + Z^* \delta Z\big), \dot{x} - e p_\alpha(Z) \big\rangle_L = 0 \quad \text{for every } \delta Z \in \mathscr{V}. \tag{784}$$

On the rank-two stratum this is equivalent to

$$\dot{x} = e p_\alpha(Z). \tag{785}$$

Indeed, for arbitrary $A = A^* \in \mathcal{J}_2$, the variation

$$\delta Z_A = \frac{1}{2\alpha} Z (Z^* Z)^{-1} A \tag{786}$$

satisfies $\delta p_\alpha = A$. Hence the differential of the Gram map is surjective there, and nondegeneracy of $\langle\ ,\ \rangle_L$ proves (785). On the rank-at-most-one stratum the differential need not be surjective, so the complete equation is (784); it is not replaced by the rank-two formula. The invariance of (779) follows by polarizing $\det(gAg^*) = \det A$. The multiplier and position equations follow by direct variation and integration by parts, while (784) follows from $D(\det)_p[A] = 2\langle p, A\rangle_L$.

Equation (780) is a complete constrained kinematic dynamics with an independently retained mass parameter. It does not dynamically select $m_0 > 0$: choosing $m_0 = 0$ is equally compatible, and (776) proves that the unrestrained carrier has no positive kinematic threshold. The next physical construction is thereby specified: introduce a field-theoretic kinetic operator and interaction on the full $(G, \psi, \lambda, \omega)$ carrier, derive rather than insert the selected spectral sector, and compare its gauge-invariant spectrum with the measured regime. The exact morphisms and representations proved above are the data that such a construction must preserve.

22 Exact presymplectic reduction of the $SU(3)$-Gram worldline model

The constrained Hamiltonian framework used below goes back to Dirac's analysis of singular Lagrangians and constrained Hamiltonian systems [70]. The separation of constraint surface, characteristic distribution, and reduced phase space is the finite-dimensional presymplectic constraint problem treated systematically by Gotay–Nester–Hinds [71]; the regular quotient step is the symplectic reduction mechanism of Marsden–Weinstein [72]. These references supply the general framework only. Every constraint rank, orbit, stabilizer, kernel, one-form, two-form, quotient map, sign, equation of motion, and singular counterexample used for this particular (SU(3))-Gram model is computed explicitly below.

22.1 Objects, constants, and equations

Let

$$\mathcal{J}_2 = \{A \in \mathrm{Mat}_{2\times 2}(\mathbb{C}) : A = A^*\}, \qquad \langle A, B\rangle_L = \frac{1}{2}\big(\mathrm{tr}\, A\, \mathrm{tr}\, B - \mathrm{tr}(AB)\big).$$

Then $\langle A, A\rangle_L = \det A$, and the pairing is nondegenerate. Write

$$\mathcal{P}_2 = \{p \in \mathcal{J}_2 : p \succeq 0\}, \quad \mathcal{P}_2^{++} = \{p \in \mathcal{J}_2 : p \succ 0\}, \quad \mathcal{N}_+^{\times} = \{p \in \mathcal{P}_2 : \mathrm{rank}_{\mathbb{C}}\, p = 1\}.$$

Retain

$$\mathscr{V} = \mathrm{Mat}_{3\times 2}(\mathbb{C}), \quad \alpha > 0, \quad c > 0, \quad m_0 \geq 0,$$

and define

$$P_\alpha(Z) = \alpha Z^* Z, \tag{787}$$

$$DP_{\alpha,Z}(\delta Z) = \alpha(\delta Z^* Z + Z^* \delta Z), \tag{788}$$

$$\Phi_{m_0,\alpha,c}(Z) = \frac{1}{2}\left(\det P_\alpha(Z) - m_0^2 c^2\right). \tag{789}$$

The history configuration space on a retained parameter interval I is

$$\mathfrak{C}_I = C^\infty(I, \mathcal{J}_2) \times C^\infty(I, \mathscr{V}) \times C^\infty(I, \mathbb{R}),$$

with coordinates (x, Z, e). Because the Lagrangian is first order in x, contains no $\dot{Z}$, and contains no $\dot{e}$, the corresponding instantaneous presymplectic phase carrier is

$$\mathcal{E} = \mathcal{J}_{2x} \times \mathscr{V}_Z \times \mathbb{R}_e.$$

Define

$$\theta = \langle P_\alpha(Z), \mathrm{d}x \rangle_L, \qquad \Omega = -\mathrm{d}\theta, \qquad \mathcal{H} = e\Phi_{m_0,\alpha,c}. \tag{790}$$

For tangent vectors $v = (v_x, v_Z, v_e)$ and $w = (w_x, w_Z, w_e)$,

$$\Omega(v, w) = \langle v_x, DP_{\alpha,Z}(w_Z) \rangle_L - \langle w_x, DP_{\alpha,Z}(v_Z) \rangle_L. \tag{791}$$

Proposition 22.1 (Exact presymplectic Euler–Lagrange equation)**.** *A path (x, Z, e) is stationary under compactly supported variations if and only if*

$$\det P_\alpha(Z) = m_0^2 c^2, \tag{792}$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau} P_\alpha(Z) = 0, \tag{793}$$

$$\langle DP_{\alpha,Z}(\delta Z), \dot{x} - eP_\alpha(Z) \rangle_L = 0 \quad \text{for every } \delta Z \in \mathscr{V}. \tag{794}$$

Equivalently, $\iota_{\dot\gamma}\Omega = \mathrm{d}\mathcal{H}$.

Proof. Since

$$D(\det)_p(A) = 2\langle p, A \rangle_L,$$

one has

$$\mathrm{d}\mathcal{H}(w) = \Phi_{m_0,\alpha,c} w_e + e\langle P_\alpha(Z), DP_{\alpha,Z}(w_Z) \rangle_L.$$

Insert (791) into $\iota_{\dot\gamma}\Omega = \mathrm{d}\mathcal{H}$ and test successively w_x, w_e, w_Z. This gives (793), (792), and (794). □

22.2 Colour quotient and orbit types

The left action is $U \cdot (x, Z, e) = (x, UZ, e)$.

Lemma 22.2 (Local colour is a zero-moment null symmetry)**.** *For $\xi \in \mathfrak{su}(3)$, let $\xi^\# = (0, \xi Z, 0)$. Then*

$$DP_{\alpha,Z}(\xi Z) = 0, \qquad \iota_{\xi^\#}\theta = 0, \qquad \iota_{\xi^\#}\Omega = 0.$$

Thus the zero map to $\mathfrak{su}(3)^*$ is a presymplectic moment map. The action is invariant under every τ-dependent $U : I \to SU(3)$, with no connection term.

Proof. Because $\xi^* = -\xi$,

$$DP_{\alpha,Z}(\xi Z) = \alpha(Z^*\xi^* Z + Z^*\xi Z) = 0.$$

The form identities follow from (790) and (791). Pointwise, $(UZ)^*(UZ) = Z^*Z$, even when U depends on τ. □

Theorem 22.3 (Gram quotient, fibres, and stabilizers)**.** *The map P_α induces a homeomorphism*

$$\mathscr{V}/SU(3) \xrightarrow{\sim} \mathcal{P}_2, \qquad [Z] \longmapsto P_\alpha(Z).$$

Its orbit types are

$\operatorname{rank}_{\mathbb{C}} Z$	*fibre*	$\operatorname{Stab}_{SU(3)}(Z)$
2	$SU(3)$	$\{I_3\}$
1	$SU(3)/SU(2) \cong S^5$	$SU(2)$
0	*a point*	$SU(3)$

The rank-two quotient is smooth and free. Rank one is a smooth nonprincipal orbit-type stratum with nontrivial stabilizer; the zero vertex is the singular apex of the total stratified quotient.

Proof. If $Z_1^* Z_1 = Z_2^* Z_2$, the map $Z_1 v \mapsto Z_2 v$ is an isometry between their images and extends to a unitary map of $\mathbb{C}^3$. Its action on the orthogonal complement can be adjusted to make its determinant one. In rank two the complement is one-dimensional and the adjustment is unique, hence the stabilizer is trivial. In rank one the stabilizer is $SU(2)$ on the two-dimensional complement. At zero it is all of $SU(3)$. Surjectivity has the retained section

$$Z_p = \alpha^{-1/2} \begin{pmatrix} p^{1/2} & \\ 0 & 0 \end{pmatrix}.$$

Compactness of $SU(3)$ and continuity of the positive square root give the quotient homeomorphism. On $\mathcal{P}_2^{++}$ the section is smooth. On $\mathcal{N}_+^{\times}$, the rank-one identity

$$p^{1/2} = \frac{p}{\sqrt{\operatorname{tr} p}}$$

shows directly that the same section is smooth; together with the constant orbit type $SU(3)/SU(2)$, this gives the stated smooth nonprincipal stratum. □

Lemma 22.4 (Exact ranks of the Gram differential)**.** (i) *If* $\operatorname{rank}_{\mathbb{C}} Z = 2$*, then*

$$\operatorname{im} DP_{\alpha,Z} = \mathcal{J}_2, \qquad \ker DP_{\alpha,Z} = T_Z(SU(3) \cdot Z),$$

of real dimensions 4 *and* 8*.*

(ii) *If* $\operatorname{rank}_{\mathbb{C}} Z = 1$ *and* $p = P_\alpha(Z)$*, then*

$$\operatorname{im} DP_{\alpha,Z} = p^{\perp_L} = T_p \mathcal{N}_+^{\times}$$

has real dimension 3*. The full ambient kernel has dimension* 9*, but*

$$\ker(DP_{\alpha,Z}|_{T_Z \mathscr{V}_1}) = T_Z(SU(3) \cdot Z)$$

has dimension 5*, where* $\mathscr{V}_1 = \{Z : \operatorname{rank}_{\mathbb{C}} Z = 1\}$*. The remaining four real kernel directions are*

$$\operatorname{Hom}_{\mathbb{C}}(\ker Z, (\operatorname{im} Z)^{\perp});$$

they raise rank and are not colour gauge.

(iii) *At* $Z = 0$*,* $DP_{\alpha,0} = 0$*.*

Proof. For rank two and arbitrary $A = A^*$, retain

$$\delta Z_A = \frac{1}{2\alpha} Z (Z^* Z)^{-1} A.$$

Substitution into (788) gives $DP_{\alpha,Z}(\delta Z_A) = A$. Hence the kernel has dimension $12 - 4 = 8$, equal to the free colour-orbit tangent.

For rank one write, without fixing either norm,

$$Z = ab^*, \qquad a \in \mathbb{C}^3 \setminus \{0\}, \quad b \in \mathbb{C}^2 \setminus \{0\}, \quad r = a^* a > 0.$$

Then $p = \alpha r b b^*$. If $w = \delta Z^* a$, every $w \in \mathbb{C}^2$ occurs by taking $\delta Z = a w^*/r$, and

$$\operatorname{im} DP_{\alpha,Z} = \{\alpha(wb^* + bw^*) : w \in \mathbb{C}^2\}.$$

The kernel of $w \mapsto wb^* + bw^*$ is $\{itb : t \in \mathbb{R}\}$, so the image has dimension three. It is contained in $p^{\perp_L}$, and equality follows by dimension.

Use the exact orthogonal splittings

$$\mathbb{C}^2 = (\ker Z)^\perp \oplus \ker Z, \qquad \mathbb{C}^3 = \operatorname{im} Z \oplus (\operatorname{im} Z)^\perp.$$

Retaining the nonzero singular value $\sigma > 0$, adapted unit vectors give

$$Z = \begin{pmatrix} \sigma & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \delta Z = \begin{pmatrix} a_0 & b_0 \\ c_1 & d_1 \\ c_2 & d_2 \end{pmatrix},$$

and

$$\alpha^{-1} DP_{\alpha,Z}(\delta Z) = \begin{pmatrix} 2\sigma \operatorname{Re} a_0 & \sigma b_0 \\ \sigma \overline{b_0} & 0 \end{pmatrix}.$$

The ambient kernel has $\operatorname{Re} a_0 = b_0 = 0$ and dimension nine. The linearized rank-one equations also impose $d_1 = d_2 = 0$. The five remaining directions are the $SU(3)/SU(2)$-orbit tangent; the d_1, d_2 block is the displayed complex Hom-space. The zero case follows from (788). □

22.3 Forms after colour reduction

Define

$$\overline{\mathcal{E}} = \mathcal{J}_{2x} \times \mathcal{P}_{2p} \times \mathbb{R}_e \cong \mathcal{E}/SU(3), \qquad q(x, Z, e) = (x, P_\alpha(Z), e).$$

Proposition 22.5 (Reduced one-form and two-form)**.** *The forms are basic for the colour action and satisfy*

$$\theta = q^* \overline{\theta}, \qquad \Omega = q^* \overline{\Omega},$$

where

$$\overline{\theta} = \langle p, \mathrm{d}x \rangle_L, \tag{795}$$

$$\overline{\Omega} = -\mathrm{d}\overline{\theta}, \tag{796}$$

$$\overline{\Omega}((v_x, v_p), (w_x, w_p)) = \langle v_x, w_p \rangle_L - \langle w_x, v_p \rangle_L. \tag{797}$$

After omitting the multiplier coordinate, this is symplectic on $\mathcal{J}_2 \times \mathcal{P}_2^{++}$. On $\mathcal{J}_2 \times \mathcal{N}_+^\times$, its kernel is exactly

$$\ker \overline{\Omega}_{(x,p)} = \operatorname{span}_{\mathbb{R}}\{(p, 0)\}.$$

On $\mathcal{J}_2 \times \{0\}$, the stratum-restricted two-form is zero.

Proof. Basicness follows from Lemma 22.2; the formulas follow by differentiation. The rank-two assertion uses that $\mathcal{P}_2^{++}$ is open in $\mathcal{J}_2$. At nonzero rank-one p, one has

$$T_p \mathcal{N}_+^\times = p^{\perp_L}.$$

If (v_x, v_p) annihilates (797), testing arbitrary w_x gives $v_p = 0$, and testing $w_p \in p^{\perp_L}$ gives $v_x \in (p^{\perp_L})^{\perp_L} = \operatorname{span}_{\mathbb{R}}\{p\}$. At $p = 0$, the zero stratum has no momentum tangent direction. □

22.4 Positive mass

For $m_0 > 0$, define the retained future shell

$$\mathcal{H}^+_{m_0,c} = \{p \in \mathcal{P}_2^{++} : \det p = m_0^2 c^2\}.$$

The parameter α remains present in the exact quotient map $p = \alpha Z^* Z$; the displayed subset of momentum space itself depends on m_0, c.

Theorem 22.6 (Exact massive reduction)**.** *Let $m_0 > 0$. Then:*

(a) *The constraint surface*

$$\Sigma_{m_0,\alpha,c} = \mathcal{J}_2 \times \{Z : \det P_\alpha(Z) = m_0^2 c^2\} \times \mathbb{R}_e$$

is smooth and the colour action on it is free.

(b) *Its colour quotient is*

$$\overline{\Sigma}_{m_0,c} = \mathcal{J}_2 \times \mathcal{H}^+_{m_0,c} \times \mathbb{R}_e.$$

Before colour reduction the kernel of the restricted two-form is the direct sum of the eight-dimensional colour-orbit tangent, $\mathrm{span}_{\mathbb{R}}\{(p,0)\}$*, and* $\mathrm{span}_{\mathbb{R}}\{\partial_e\}$*. After colour reduction and omission of e, only* $\mathrm{span}_{\mathbb{R}}\{(p,0)\}$ *remains.*

(c) *The equations are*

$$\dot{p} = 0, \qquad \dot{x} = e\,p, \qquad \det p = m_0^2 c^2,$$

while $\dot{e}$ and motion in the colour fibre are arbitrary.

(d) *The physical phase space*

$$\mathcal{R}_{m_0,c} = \frac{\mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}}{(x,p) \sim (x + sp, p),\ s \in \mathbb{R}}$$

is canonically symplectomorphic to $T^*\mathcal{H}^+_{m_0,c}$.

(e) *The colour-reduced positive-mass model is exactly the standard future-directed spinless first-order relativistic particle of mass m_0 and speed constant c.*

Proof. The constraint implies $\mathrm{rank}_{\mathbb{C}}\, Z = 2$. By Lemma 22.4, P_α is a submersion there, and

$$D(\det)_p = 2\langle p, \cdot\rangle_L \neq 0.$$

This proves smoothness. Theorem 22.3 gives freeness and the colour quotient. Since

$$T_p\mathcal{H}^+_{m_0,c} = p^{\perp_L},$$

the proof of Proposition 22.5 gives the characteristic line. Pullback through the Gram submersion adds exactly the colour-orbit kernel, and the multiplier adds ∂_e.

Surjectivity of $DP_{\alpha,Z}$ turns (794) into $\dot{x} = ep$; the other equations are (792) and (793). A path with constant p lies in one free Gram fibre and differs locally from a constant lift by a τ-dependent colour transformation.

For the physical identification, if $A \in T_p\mathcal{H}^+_{m_0,c}$, define

$$\xi_{[x],p}(A) = -\langle x, A\rangle_L.$$

This is unchanged by $x \mapsto x + sp$. Conversely, the annihilator of $p^{\perp_L}$ is $\mathrm{span}_{\mathbb{R}}\{p\}$, so every cotangent covector has one and only one class $[x]$. Hence

$$([x], p) \longmapsto (p, \xi_{[x],p})$$

is a fibrewise linear bijection to $T^*\mathcal{H}^+_{m_0,c}$. The characteristic $\mathbb{R}$-action is free because every shell momentum is nonzero. It is proper: choose any Euclidean norm on $\mathcal{J}_2$. For compact sets $K_1, K_2 \subset \mathcal{J}_2 \times \mathcal{H}^+_{m_0,c}$, the momenta occurring in them have a positive minimum Euclidean norm δ; if $(x,p) \in K_1$ and $(x+sp,p) \in K_2$, boundedness of the two position projections gives $|s|\delta \le C(K_1,K_2)$. Thus the orbit space is smooth. The smooth vector-bundle map

$$(x,p) \longmapsto (p,\ A \mapsto -\langle x, A\rangle_L)$$

is fibrewise onto with kernel $\operatorname{span}_{\mathbb{R}}\{p\}$, and therefore descends with a smooth inverse. The displayed bijection is consequently a diffeomorphism.

The basic one-form on the shell is

$$\theta_{\text{basic}} = -\langle x, \mathrm{d}p\rangle_L. \tag{798}$$

It is the pullback of the tautological cotangent one-form, and

$$\overline{\theta} = \mathrm{d}\langle p, x\rangle_L + \theta_{\text{basic}}. \tag{799}$$

Therefore $\overline{\Omega} = -\mathrm{d}\theta_{\text{basic}}$, the canonical cotangent symplectic form under the convention $\omega_{\text{can}} = -\mathrm{d}\lambda_{\text{can}}$. The original $\overline{\theta}$ is not horizontal on the massive characteristic because $\overline{\theta}(p,0) = m_0^2c^2$; the exact boundary term in (799) produces the reduced one-form.

Finally the reduced action is

$$\int_I \left(\langle p, \dot{x}\rangle_L - \frac{e}{2}(\det p - m_0^2c^2) \right) \mathrm{d}\tau.$$

Because $\mathcal{P}_2^{++}$ is open in $\mathcal{J}_2$, p has every infinitesimal Hermitian variation. This is the standard covariant first-order spinless particle action on the future-timelike sector, with no residual internal symplectic factor. □

Proposition 22.7 (Exact Hamiltonian gauge quasi-symmetry)**.** *For a smooth real $\varepsilon(\tau)$, set*

$$\delta_\varepsilon x = \varepsilon P_\alpha(Z), \qquad \delta_\varepsilon Z = 0, \qquad \delta_\varepsilon e = \dot{\varepsilon}.$$

Then

$$\delta_\varepsilon L = \frac{\mathrm{d}}{\mathrm{d}\tau}\left[\frac{\varepsilon}{2}(\det P_\alpha(Z) + m_0^2c^2) \right]. \tag{800}$$

For compactly supported ε this is an exact gauge quasi-symmetry, whose phase-space part generates the shell characteristic where that characteristic is regular.

Proof. Put $p = P_\alpha(Z)$ and $q = \det p = \langle p, p\rangle_L$. Since $\dot{q} = 2\langle p, \dot{p}\rangle_L$,

$$\begin{aligned} \delta_\varepsilon L &= \langle p, \dot{\varepsilon} p + \varepsilon \dot{p}\rangle_L - \frac{\dot{\varepsilon}}{2}(q - m_0^2c^2) \\ &= \frac{\dot{\varepsilon}}{2}(q + m_0^2c^2) + \frac{\varepsilon}{2}\dot{q}, \end{aligned}$$

which is (800). □

22.5 Massless rank one

Set

$$\mathcal{C}_{0,\alpha,c} = \{Z : \det P_\alpha(Z) = 0\} = \{Z : \operatorname{rank}_{\mathbb{C}} Z \le 1\}.$$

The constant $c > 0$ is retained in the model, although it cancels from the set-theoretic $m_0 = 0$ equation.

Lemma 22.8 (The pulled-back massless constraint is irregular)**.** *Writing the rows of Z as $Z_{a\bullet}$,*

$$\det P_\alpha(Z) = \alpha^2 \sum_{1\le a<b\le 3} |Z_{a1}Z_{b2} - Z_{a2}Z_{b1}|^2. \tag{801}$$

Consequently,

$$\mathrm{d}(\det P_\alpha)_Z = 0 \quad \textit{for every } Z \in \mathcal{C}_{0,\alpha,c}.$$

The single real multiplier equation describes the rank-at-most-one set, but its differential supplies none of the four real normal equations on the smooth rank-one stratum.

Proof. Equation (801) is Cauchy–Binet with the factor α^2 retained. On its zero set every complex minor vanishes, so differentiation of the sum of squared absolute values gives zero. Equivalently, at rank one,

$$\mathrm{d}(\det P_\alpha)_Z(\delta Z) = 2\langle p, DP_{\alpha,Z}(\delta Z)\rangle_L = 0$$

by Lemma 22.4; at $Z = 0$ use $DP_{\alpha,0} = 0$. □

Theorem 22.9 (Nonzero massless equations and physical quotient)**.** *Suppose $m_0 = 0$ and $\operatorname{rank}_{\mathbb{C}} Z = 1$, and put $p = P_\alpha(Z) \in \mathcal{N}_+^\times$. Then:*

(a) *The colour fibre is $SU(3)/SU(2)$, with stabilizer $SU(2)$.*

(b) *The complete equations are*

$$\dot p = 0, \qquad \dot x = \lambda(\tau)p \quad \textit{for some real } \lambda(\tau), \qquad e(\tau) \textit{ arbitrary.} \tag{802}$$

No equation identifies λ with the written multiplier e.

(c) *On the intrinsic smooth rank-one shell, the kernel before colour reduction is the direct sum of*

$$T_Z(SU(3)\cdot Z), \qquad \operatorname{span}_{\mathbb{R}}\{(p,0)\}, \qquad \operatorname{span}_{\mathbb{R}}\{\partial_e\}.$$

After colour reduction and omission of e, only the characteristic $\operatorname{span}_{\mathbb{R}}\{(p,0)\}$ remains.

(d) *The physical phase space*

$$\mathcal{R}_0^\times = \frac{\mathcal{J}_2 \times \mathcal{N}_+^\times}{(x,p) \sim (x + sp, p)}$$

is canonically symplectomorphic to $T^*\mathcal{N}_+^\times$, with reduced one-form $-\langle x, \mathrm{d}p\rangle_L$ and its canonical two-form. No quotient by positive rescaling of p occurs.

(e) *Thus the unparameterized spacetime curves and physical symplectic quotient agree with the standard future-directed nonzero spinless massless particle. The original triples (x, Z, e), their off-shell neighbourhood, and their multiplier equations are not exactly the standard massless system.*

Proof. The fibre statement is Theorem 22.3. By Lemma 22.4, equation (794) says

$$\dot x - ep \in (p^{\perp_L})^{\perp_L} = \operatorname{span}_{\mathbb{R}}\{p\}.$$

Since ep lies in this line, this is equivalent to $\dot x = \lambda p$, with no relation between λ and e. Equation (793) gives $\dot p = 0$. A path in a fixed rank-one Gram fibre differs locally from a constant representative by a τ-dependent $SU(3)$ transformation; the $SU(2)$ stabilizer makes this gauge description reducible.

On the intrinsic rank-one manifold, Lemma 22.4 identifies the vertical kernel with the five colour directions. Proposition 22.5 adds the one spacetime characteristic, and e adds ∂_e.

Since $T_p\mathcal{N}_+^\times = p^{\perp_L}$, the massive cotangent construction applies without division by $\det p$:

$$([x], p) \longmapsto (p,\ A \longmapsto -\langle x, A\rangle_L)\,.$$

Every $p \in \mathcal{N}_+^\times$ is nonzero. The freeness, properness, and smooth vector-bundle argument in the massive proof applies to compact subsets of $\mathcal{J}_2 \times \mathcal{N}_+^\times$ verbatim, and proves that this map is a diffeomorphism to $T^*\mathcal{N}_+^\times$. Moreover, (799) identifies the canonical form. Reparametrizing the worldline changes x along p; it does not identify distinct positive magnitudes of p. Lemma 22.8 and the absence of $\lambda = e$ establish the final inequivalence statement. □

Counterexample 22.10 (Rank-one multiplier mismatch)**.** *Retain arbitrary $s > 0$, and set*

$$Z_s = \begin{pmatrix} s & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad p_s = \alpha \begin{pmatrix} s^2 & 0 \\ 0 & 0 \end{pmatrix}.$$

The path

$$Z(\tau) = Z_s, \qquad x(\tau) = \tau p_s, \qquad e(\tau) = 0$$

satisfies all equations in Proposition 22.1 for $m_0 = 0$. The standard action with momentum independently varied in $\mathcal{J}_2$ and the same $e = 0$ would require $\dot{x} = ep_s = 0$.

Under Proposition 22.7, a solution $\dot{x} = \lambda p$, $\dot{p} = 0$, transforms by

$$\lambda \longmapsto \lambda + \dot{\varepsilon}, \qquad e \longmapsto e + \dot{\varepsilon}.$$

Thus $\lambda - e$ is invariant under the stated Hamiltonian gauge; the discrepancy remains when the original multiplier is retained.

Counterexample 22.11 (Four non-colour first-order null directions)**.** *Retain arbitrary $s > 0$ and $t \in \mathbb{C} \setminus \{0\}$, and put*

$$Z_s = \begin{pmatrix} s & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \Delta_t = \begin{pmatrix} 0 & 0 \\ 0 & t \\ 0 & 0 \end{pmatrix}.$$

Then

$$DP_{\alpha,Z_s}(\Delta_t) = 0, \qquad \det P_\alpha(Z_s + \varepsilon\Delta_t) = \alpha^2 s^2 |t|^2 \varepsilon^2 \quad (\varepsilon \in \mathbb{R}). \tag{803}$$

The vector Δ_t is not tangent to a colour orbit, since every ξZ_s has zero second column. It is not tangent to the rank-one variety, because the perturbed matrix has rank two for $\varepsilon \neq 0$. Allowing both complex coordinates in $(\operatorname{im} Z_s)^\perp$ gives the four real transverse directions of Lemma 22.4. They are second-order shell-opening directions, not gauge.

22.6 The zero-momentum vertex

Theorem 22.12 (Additional singular branch)**.** *Let $m_0 = 0$ and $Z = 0$. Then $p = 0$, the stabilizer is all of $SU(3)$, and both θ and Ω vanish on $\mathcal{J}_2 \times \{0\} \times \mathbb{R}_e$. Every pair of paths*

$$x : I \to \mathcal{J}_2, \qquad e : I \to \mathbb{R},$$

with $Z(\tau) \equiv 0$, satisfies the Euler–Lagrange equations.

The phase part of the Hamiltonian gauge vanishes at $p = 0$, while the presymplectic kernel contains every x-direction. Quotienting by the actual colour and Hamiltonian-gauge orbits therefore leaves the whole $\mathcal{J}_2$ branch, whereas quotienting by all characteristic leaves would collapse it to one point. This is an exact failure of regular Dirac reduction at the vertex.

Proof. At $Z = 0$, one has $p = 0$ and $DP_{\alpha,0} = 0$. Every equation in Proposition 22.1 is identically satisfied, independent of $\dot{x}, e, \dot{e}$. The form and gauge statements follow from (790) and $\delta_\varepsilon x = \varepsilon p = 0$. □

Counterexample 22.13 (Stationary zero momentum with nonconstant position)**.** *The retained paths*

$$x(\tau) = \begin{pmatrix} \tau^2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Z(\tau) = 0, \qquad e(\tau) = \sin\tau$$

are stationary for the present $m_0 = 0$ action. The standard independent-momentum action gives $\dot{x} = ep = 0$ at $p = 0$, and therefore excludes this path. Removing $p = 0$, as is conventional for physical massless phase space, removes this vertex counterexample but does not regularize the off-shell rank-one constraint.

22.7 Kernel dimensions and exact verdict

Proposition 22.14 (Every kernel jump)**.** *Omit the always-null multiplier direction for the table. On the ambient* 16*-real-dimensional $\mathcal{J}_{2x} \times \mathcal{V}_Z$,*

$\operatorname{rank}_{\mathbb{C}} Z$	$\operatorname{rank}_{\mathbb{R}} DP_{\alpha,Z}$	$\operatorname{rank}_{\mathbb{R}} \Omega$	$\dim_{\mathbb{R}} \ker \Omega$
2	4	8	8
1	3	6	10
0	0	0	16

On the 12*-real-dimensional intrinsic rank-one stratum $\mathcal{J}_{2x} \times \mathcal{V}_1$, the pulled-back form has rank* 6 *and nullity* 6*: five colour directions and one spacetime characteristic.*

Proof. Using the Lorentz pairing to identify $\mathcal{J}_2 \cong \mathcal{J}_2^*$, the matrix of Ω is

$$\begin{pmatrix} 0 & G_L DP_{\alpha,Z} \\ -(DP_{\alpha,Z})^{\mathsf{T}} G_L & 0 \end{pmatrix}.$$

Its rank is twice the rank of $DP_{\alpha,Z}$. Apply Lemma 22.4 in the ambient space and on $T_Z\mathcal{V}_1$. □

Theorem 22.15 (Final comparison)**.** *The strongest correct result is:*

(1) *for $m_0 > 0$, colour reduction gives exactly the standard future-timelike first-order spinless particle, including its equations, constraint characteristic, and physical cotangent symplectic space;*

(2) *for $m_0 = 0$ and $p \neq 0$, colour and characteristic reduction gives exactly the standard future-null physical phase space and unparameterized null trajectories;*

(3) *the full $m_0 = 0$ pulled-back multiplier action is not the standard massless variational system: its constraint has zero differential, its rank-one equation is $\dot{x} = \lambda p$ rather than $\dot{x} = ep$, its ambient kernel contains four non-colour transverse directions, and its vertex admits arbitrary $x(\tau)$.*

The exact relating morphism is $P_\alpha(Z) = \alpha Z^* Z$; the exact failure locus is the singular differential of $\det \circ P_\alpha$ *on the rank-at-most-one variety.*

Proof. Combine Theorem 22.6, Theorem 22.9, Lemma 22.8, and Counterexamples 22.10–22.13. □

22.8 Deterministic verification companion

The companion program `verify_worldline_reduction.py` checks the determinant polarization, the rank-two Gram right inverse, all Gram and presymplectic ranks, all colour-orbit ranks, the intrinsic rank-one tangent calculation, the zero gradient and quadratic transverse opening of the massless constraint, the cotangent-form sign, and (800), using exact symbolic arithmetic.

22.9 Comparison with the extended exact Lorentz packet

This subsection records a bounded line-specific audit of source lines 875–1654. The audited source file is

`exact_lorentz_observation_bridge.tex`.

It is stored beneath `supporting_materials/workbench/evidence` in the directory

`worldline_presymplectic_reduction_20260904`.

Its file SHA-256 is

`2BE16E1DC71B2E5DC04238F0FBE440138258D0825A4841AB2F29662587681599`.

That packet adds a third column $\phi \in \mathbb{C}^3$, writes $B = [Z\ \phi]$, and uses

$$G = Z^* Z, \quad z = Z^* \phi, \quad \lambda = \phi^* \phi, \quad \omega = \det B.$$

Its mass convention is $m^2_{\mathrm{packet}} = \det p$, with physical mass $M_{\mathrm{phys}} = \sqrt{\det p}/c$. The convention in the audited S^6 module is $m^2_{\alpha,c} = \det p / c^2$. The exact dictionary is therefore

$$m^2_{\mathrm{packet}} = c^2 m^2_{\alpha,c}, \qquad M_{\mathrm{phys}} = m_{\alpha,c};$$

there is no mathematical disagreement.

22.9.1 Claim ledger

Source lines	Verdict	Exact scope
875–950	validated	$p = \alpha Z^* Z$ is a proper surjection, its fibres are the $SU(3)$-orbits, and $\det p = \alpha^2 \sum_{a<b} \lvert\Delta_{ab}(Z)\rvert^2$ has range $[0, \infty)$. The zero fibre is rank at most one.
952–1027	validated as a multiplication-operator model	With the stated pushforward measure, the observer-energy and mass-squared multiplication operators have essential ranges $[0, \infty)$, no point spectrum, and no positive gap after a vacuum is adjoined. This is a kinematic/free spectral statement, not a statement about an interacting field Hamiltonian.
1029–1476	validated	For $\lambda_0 > 0$, $\omega_0 \neq 0$, the constrained quotient, associated-ball description, momentum fibres, and sharp lower bound $\det p \geq \alpha^2 \lvert\omega_0\rvert^2 / \lambda_0$ are algebraic and differential-topological statements. Since $\omega_0 \neq 0$ forces B invertible and Z rank two, the massless rank-one defect proved above does not invalidate these claims.
1482–1531	valid invariance and constraints, incomplete dynamics	The displayed action is invariant and its multiplier and x variations give the stated constraints and $\dot{p} = 0$. Its omitted (Z, ϕ) equations materially change the dynamical interpretation; Theorem 22.16 below computes them intrinsically.
1533–1603	validated with scope qualification	The multiplication spectra and the direct-sum/Fock vacuum gap equal the displayed threshold. The threshold is inserted by fixing $\lambda = \lambda_0$ and $\omega = \omega_0 \neq 0$; it is not a dynamically generated interacting mass gap and is not a Yang–Mills mass-gap theorem.
1605–1633	validated and scope-sharpening	The two explicit degenerating sequences prove that controlling only one of λ and ω gives no positive threshold. Thus the positive lower endpoint is exactly constraint-selected.
1635–1653	validated	The diagonal boost excludes a positive-definite quadratic form on the spinor carrier invariant under all of $SL_2(\mathbb{C})$.

The independent replay and line-by-line claim ledger for this packet are retained under the project-relative path `supporting_materials/workbench/evidence/exact_lorentz_observation_bridge_audit_20260904`. The packet checker passes 33/33 assertions and the separately written checker passes 30/30. The audit-report SHA–256 is `c8d01af28836340798c2fa7e27f8e097d67e0b8d7c13f75271af473958d391df`; the promotion manifest, which verifies all retained audit artifacts, has SHA–256 `52f53746558fa219260b224abdb71c86d2cd6c13192774d0d2fb1a1ed71bcf35`.

22.9.2 The full dynamics of the packet's constrained action

Retain the packet's constants

$$\lambda_0 > 0, \qquad \omega_0 \in \mathbb{C}^*, \qquad \alpha > 0$$

and its smooth colour quotient

$$\mathcal{Q}_{\lambda_0,\omega_0} \cong \mathrm{Herm}_2^0(\mathbb{C}) \times \{w \in \mathbb{C}^2 : \|w\|^2 < \lambda_0\}.$$

Put

$$\mu(w) = \frac{|\omega_0|}{\sqrt{\lambda_0 - \|w\|^2}}, \qquad G(A,w) = \mu(w)e^A, \qquad p(A,w) = \alpha G(A,w). \tag{804}$$

Eliminating the three regular multiplier constraints from source lines 1484–1495 gives the intrinsic action

$$S_{\mathrm{packet}}[x,A,w] = \int_I \langle p(A,w), \dot{x}\rangle_L \,\mathrm{d}\tau \tag{805}$$

on $\mathcal{J}_{2x} \times \mathcal{Q}_{\lambda_0,\omega_0}$. The use of the intrinsic constrained action loses no Euler–Lagrange equation: the constraint map is a submersion on $\omega_0 \neq 0$, so the Lagrange-multiplier theorem identifies ambient stationarity with stationarity against every tangent vector of the constrained manifold. This is a statement about classical stationary paths; it does not assert equality of quantum measures or path integrals after formally integrating out the multipliers.

Theorem 22.16 (Complete reduced dynamics of the constrained packet)**.** *For the momentum projection in* (804)*:*

(i) *If $w \neq 0$, then $Dp_{(A,w)}$ has real rank* 4*, and its kernel has dimension* 3*, tangent to the S^3 momentum fibre. The Euler–Lagrange equations of* (805) *are*

$$\dot{p} = 0, \qquad \dot{x} = 0,$$

with motion along the fixed S^3 fibre presymplectically null.

(ii) *If $w = 0$, then*

$$\det p = \frac{\alpha^2|\omega_0|^2}{\lambda_0},$$

$Dp_{(A,0)}$ has rank 3*, its image is $p^{\perp_L}$, and its kernel consists of all four real w-directions. The pointwise equations are*

$$\dot{p} = 0, \qquad \dot{x} = \lambda(\tau)p.$$

Every actual solution remaining on the threshold has constant (A,w), because the momentum fibre over a threshold momentum is a single point.

(iii) *Consequently, the source action is not a family of ordinary massive particle actions over all masses above the lower endpoint. Only its threshold stratum has the ordinary worldline characteristic; strict-interior configurations have $\dot{x} = 0$. The observer multiplication operator of source lines 1533–1603 is a separate positive-energy representation and is not the Hamiltonian generating τ-evolution in* (805)*.*

Proof. For fixed w, varying A varies p through the three-dimensional fixed-determinant future hyperboloid, so the image of the A-differential is $p^{\perp_L}$. Direct differentiation without suppressing any constant gives

$$D\mu_w(\delta w) = \frac{|\omega_0|\,\mathrm{Re}\langle w, \delta w\rangle}{(\lambda_0 - \|w\|^2)^{3/2}}. \tag{806}$$

When $w \neq 0$, a radial δw makes (806) nonzero and changes $\det p = \alpha^2\mu(w)^2$. This supplies a fourth direction transverse to $p^{\perp_L}$. The remaining w-directions satisfy $\mathrm{Re}\langle w, \delta w\rangle = 0$, form a three-dimensional space, and are exactly tangent to the S^3 momentum fibre.

At $w = 0$, equation (806) vanishes in every direction, so the four w-directions lie in $\ker Dp$ and the image is only $p^{\perp_L}$. These directions open the mass above its minimum at second order:

$$\det p(A, \varepsilon v) = \frac{\alpha^2|\omega_0|^2}{\lambda_0 - \varepsilon^2\|v\|^2}.$$

For an action $\int \langle p(q), \dot{x} \rangle_L \mathrm{d}\tau$, variation of x gives $\dot{p} = 0$, while variation of q gives

$$\langle Dp_q(\delta q), \dot{x} \rangle_L = 0 \quad \text{for all } \delta q.$$

If $w \neq 0$, the image is all of $\mathcal{J}_2$, hence $\dot{x} = 0$. If $w = 0$, the image is $p^{\perp_L}$, hence $\dot{x} \in \mathrm{span}_{\mathbb{R}}\{p\}$. The packet's own momentum-fibre theorem states that the fibre over a threshold momentum is a singleton, which completes the solution statement. □

22.9.3 Relation to the rank-one irregularity

The packet's fixed-$\omega_0 \neq 0$ constrained manifold never meets $\mathrm{rank}_{\mathbb{C}} Z \leq 1$, so the zero differential of $\det \circ P_\alpha$ does not invalidate its topology or positive lower bound. The exact relation is instead:

1. on the unrestricted carrier, the rank-one and zero strata give the massless irregularities proved in Theorems 22.9 and 22.12;
2. on the packet's constrained carrier, Z remains rank two, but the momentum projection develops a different four-real-dimensional critical kernel at its positive-mass threshold $w = 0$;
3. the algebraic inequality is a sharp, externally constraint-selected threshold; a dynamically generated gap would require a specified interacting Hamiltonian or field operator whose spectrum produces the scale rather than receiving λ_0, ω_0 as fixed input.

23 The Hermitian determinant normal slice as a Lorentzian host

This independently extractable section restates one already verified finite-dimensional identity and constructs exact physical models from it. When it is placed after the nine-dimensional observation construction, the notation agrees by $C_{A_2} = C$, $R_P = R$, $H_0 = uu^*$, $Q_\perp = Q_u$, and the same map $\jmath_P(G) = R_P G R_P^*$. The positive conclusion is not merely that the identity resembles Minkowski space: the affine normal slice has a canonically polarized Lorentz metric, the displayed $SL_2(\mathbb{C})$ congruence subgroup induces the spin double cover of the proper orthochronous Lorentz group, and after one explicit choice of length scale it supports a completely specified Klein–Gordon theory and finite Gaussian regulators. The same host also supports classical Yang–Mills theory for any separately chosen compact gauge group. In addition, the standard metrics entering the Moore–Penrose section select one exact compact $SU(2)$-type representation internal to the normal data; promoting its pointwise action to a local gauge symmetry gives the gauge–scalar model below and is an additional construction. The proofs identify exactly which structures come from the determinant and which are additional physical choices.

23.1 Fixed matrices and the exact normal metric

Put

$$u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad C_{A_2} = PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \tag{807}$$

and use the standard Hermitian products in these displayed bases. The standard-metric Moore–Penrose right inverse is

$$R_P = P^*(PP^*)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}. \tag{808}$$

The denominators in (808) are essential: this is a metric-selected complex-linear section, not a homomorphism splitting the displayed integral lattices. Set

$$H_0 = uu^*, \qquad \mathcal{N}_P = \{X \in \mathrm{Herm}_3(\mathbb{C}) : Xu = 0\}, \qquad \jmath_P(G) = R_P G R_P^*. \tag{809}$$

Proposition 23.1 (Exact identification of the normal slice)**.** *The following identities hold without a change of normalization:*

$$PR_P = I_2, \qquad u^* R_P = 0, \qquad R_P^* R_P = C_{A_2}^{-1}, \qquad R_P P = I_3 - \frac{1}{3} u u^*. \tag{810}$$

The real-linear map $\jmath_P : \mathrm{Herm}_2(\mathbb{C}) \to \mathcal{N}_P$ *is an isomorphism. Its inverse is*

$$\jmath_P^{-1}(X) = PXP^*. \tag{811}$$

For all $G \in \mathrm{Herm}_2(\mathbb{C})$ *and* $\varepsilon \in \mathbb{R}$*,*

$$\det\bigl(H_0 + \varepsilon \jmath_P(G)\bigr) = \varepsilon^2 \det G. \tag{812}$$

Proof. The first three equalities in (810) follow by direct multiplication. The last member is the orthogonal projection onto the row space of P, which is $u^\perp$; direct multiplication gives the displayed matrix as well. Hence $P\jmath_P(G)P^* = G$. Conversely, if $X = X^*$ and $Xu = 0$, then also $u^*X = 0$, and therefore

$$R_P(PXP^*)R_P^* = (R_P P)X(R_P P) = X.$$

This proves the isomorphism and its inverse.

For the determinant, put $v_0 = u/\sqrt{3}$, choose an orthonormal complement $U_\perp$, and let $U = (v_0\ U_\perp)$ and $L = U_\perp^* R_P$. Then

$$U^* H_0 U = \mathrm{diag}(3, 0, 0), \qquad U^* \jmath_P(G) U = 0 \oplus LGL^*,$$

while

$$|\det L|^2 = \det(L^*L) = \det(R_P^* R_P) = \det(C_{A_2}^{-1}) = \frac{1}{3}.$$

Taking the determinant in this orthonormal frame gives

$$\det(H_0 + \varepsilon \jmath_P(G)) = 3\varepsilon^2 |\det L|^2 \det G = \varepsilon^2 \det G.$$

□

Put $X_P = (R_P\ u)$. The same block computation also records the exact position of this module inside the full nine-real-dimensional carrier:

$$H_0 + \varepsilon \jmath_P(G) = X_P \begin{pmatrix} \varepsilon G & 0 \\ 0 & 1 \end{pmatrix} X_P^*.$$

Thus, in the full (G, ψ, λ) block coordinates, this section uses the strict sublocus $(\psi, \lambda) = (0, 1)$. Its field models live on the observed four-vector section; they do not yet define dynamics for the retained spinor ψ and scalar λ directions.

For $G, K \in \mathrm{Herm}_2(\mathbb{C})$, define

$$\eta_2(G, K) = \frac{1}{2}\bigl(\det(G + K) - \det G - \det K\bigr). \tag{813}$$

Transport this form to $\mathcal{N}_P$ by

$$\eta_P(\jmath_P(G), \jmath_P(K)) = \eta_2(G, K). \tag{814}$$

Theorem 23.2 (The determinant Hessian is exactly Minkowski)**.** *For tangent vectors $X = \jmath_P(G)$ and $Y = \jmath_P(K)$,*

$$\eta_P(X, Y) = \frac{1}{2} D^2(\det)_{H_0}[X, Y]. \tag{815}$$

Writing

$$G(t, x, y, z) = \begin{pmatrix} t + y & x + iz \\ x - iz & t - y \end{pmatrix}, \tag{816}$$

Relative to the convention $\left(\begin{smallmatrix} t+x_3 & x_1-ix_2 \\ x_1+ix_2 & t-x_3 \end{smallmatrix}\right)$ used in the observation construction, the exact coordinate dictionary is $(x_1, x_2, x_3) = (x, -z, y)$. Thus the two displayed Pauli conventions differ only by this stated spatial relabelling and sign, not by a change of Lorentz form. one has

$$\eta_2(G(t, x, y, z), G(t', x', y', z')) = tt' - xx' - yy' - zz'. \tag{817}$$

Thus $(\mathcal{N}_P, \eta_P)$ has signature $(1, 3)$. Its distinguished future cone is

$$\mathcal{C}_P^+ = \jmath_P(\mathrm{Herm}_2(\mathbb{C})_{\geq 0}) = \{\jmath_P(G(t, \mathbf{x})) : t \geq |\mathbf{x}|\}. \tag{818}$$

Its nonzero null boundary consists exactly of the images of rank-one positive semidefinite matrices.

Proof. Proposition 23.1, applied to $sG + rK$, gives

$$\det(H_0 + s\jmath_P(G) + r\jmath_P(K)) = \det(sG + rK).$$

The coefficient of sr on the right is $2\eta_2(G, K)$. Taking one derivative in each variable proves (815). Direct expansion of (816) gives

$$\det G(t, x, y, z) = t^2 - x^2 - y^2 - z^2,$$

and polarization gives (817). A two-by-two Hermitian matrix is positive semidefinite precisely when its trace is nonnegative and its determinant is nonnegative, which here is $t \geq 0$ and $t^2 \geq |\mathbf{x}|^2$. On the nonzero boundary its determinant vanishes and its rank is one. □

The corresponding tetrad inside $\mathrm{Herm}_3(\mathbb{C})$ is completely explicit. If $E_0 = \jmath_P(G(1, 0, 0, 0))$, $E_1 = \jmath_P(G(0, 1, 0, 0))$, $E_2 = \jmath_P(G(0, 0, 1, 0))$, and $E_3 = \jmath_P(G(0, 0, 0, 1))$, then

$$E_0 = \frac{1}{9}\begin{pmatrix} 5 & -1 & -4 \\ -1 & 2 & -1 \\ -4 & -1 & 5 \end{pmatrix}, \qquad E_1 = \frac{1}{9}\begin{pmatrix} 4 & 1 & -5 \\ 1 & -2 & 1 \\ -5 & 1 & 4 \end{pmatrix}, \tag{819}$$

$$E_2 = \frac{1}{3}\begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \qquad E_3 = \frac{1}{3}\begin{pmatrix} 0 & i & -i \\ -i & 0 & i \\ i & -i & 0 \end{pmatrix}. \tag{820}$$

Every row and column of each E_μ sums to zero, and

$$\eta_P(E_\mu, E_\nu) = \mathrm{diag}(1, -1, -1, -1)_{\mu\nu}. \tag{821}$$

These identities follow by multiplying the matrices in (808) and then applying (817); no orthonormalization of the marked lattice is being suppressed.

The metric is a genuine normal Hessian. It is not a positive energy form: it takes both signs. Energy appears only after choosing a time function and passing to a Hamiltonian formulation below.

Because $\det(H_0 + \jmath_P(G)) = \det G$ on the entire affine slice, not just to second order, its restriction there is quadratic. Hence the polarized Hessian η_P is constant at every point of the affine slice; affine translations really are isometries of this metric.

23.2 Exact Lorentz and affine symmetry

For $A \in SL_2(\mathbb{C})$, define

$$\widehat{A}_P = \frac{1}{3}uu^* + R_P A P \in SL_3(\mathbb{C}). \tag{822}$$

Theorem 23.3 (The spin action inside the normal slice)**.** *The matrix $\widehat{A}_P$ is invertible, with*

$$\widehat{A}_P^{-1} = \frac{1}{3}uu^* + R_P A^{-1} P, \qquad \det \widehat{A}_P = \det A = 1. \tag{823}$$

It fixes u and satisfies

$$\widehat{A}_P H_0 \widehat{A}_P^* = H_0, \qquad \widehat{A}_P \jmath_P(G) \widehat{A}_P^* = \jmath_P(AGA^*). \tag{824}$$

Consequently $SL_2(\mathbb{C})$ acts on $\mathcal{N}_P$ by future-cone preserving η_P-isometries. The kernel is $\{I_2, -I_2\}$, and the induced map

$$PSL_2(\mathbb{C}) \xrightarrow{\sim} SO^+(1,3) \tag{825}$$

is an isomorphism of real Lie groups.

The kernel $\{\pm I_2\}$ is a statement about the normal-vector action. On the full (G, ψ, λ) carrier the same embedded $SL_2(\mathbb{C})$-action is faithful, because $-I_2$ fixes G and λ but sends $\psi \mapsto -\psi$.

Proof. Let $Q_0 = uu^*/3$ and $Q_\perp = R_P P = I_3 - Q_0$. The relations $Q_0 R_P = 0$, $P Q_0 = 0$, and $P R_P = I_2$ show that the two matrices in (823) multiply to $Q_0 + Q_\perp = I_3$. In the ordered decomposition $\mathbb{C}u \oplus R_P \mathbb{C}^2$, the map is $1 \oplus A$, proving its determinant. The same relations prove (824). Since $\det(AGA^*) = |\det A|^2 \det G = \det G$, the action preserves η_P; congruence preserves positive semidefiniteness, hence the chosen future cone.

If $AGA^* = G$ for every Hermitian G, taking $G = I_2$ makes A unitary, after which A commutes with every Hermitian matrix. Hence A is scalar, and determinant one gives $A = \pm I_2$. The differential of the induced map has zero kernel: if $XG + GX^* = 0$ for every Hermitian G, then $G = I_2$ makes X skew-Hermitian and the remaining equations make it scalar; trace zero then gives $X = 0$. Domain and target Lie algebras both have real dimension six, so the image contains a neighborhood of the identity. It is therefore an open subgroup of the connected group $SO^+(1,3)$, hence the whole group. □

Translations are also intrinsic to the affine slice. If $B \in \mathrm{Herm}_2(\mathbb{C})$, define

$$(B, A) \cdot (H_0 + \jmath_P(G)) = H_0 + \jmath_P(B + AGA^*). \tag{826}$$

With multiplication $(B, A)(C, D) = (B + ACA^*, AD)$, this is the spin double cover of the proper orthochronous Poincaré action. Its kernel is $\{(0, I_2), (0, -I_2)\}$. Under the dimensional chart (848) introduced below, a physical translation by $b \in \mathbb{R}^{1,3}$ is the element $B = G(b/\ell)$.

Proposition 23.4 (Ambient $SU(3)$ is not an internal color action)**.** *The subgroup of $SU(3)$ preserving H_0 is*

$$\mathrm{Stab}_{SU(3)}(H_0) = \{g \in SU(3) : gu = e^{i\theta}u\} \cong S(U(1) \times U(2)). \tag{827}$$

It preserves $\mathcal{N}_P$. A general element of $SU(3)$ moves H_0 and hence moves the chosen affine normal slice to a different one. Furthermore, $\widehat{A}_P$ in (822) *is unitary exactly when*

$$A^* C_{A_2}^{-1} A = C_{A_2}^{-1}; \tag{828}$$

this compact subgroup is conjugate to $SU(2)$, whereas generic Lorentz boosts are not unitary. Thus the Lorentz spin action supplied by the normal slice is not the ambient $SU(3)$ orbit action and does not itself define an internal $SU(3)$ gauge symmetry.

Proof. The equality $gH_0g^* = H_0$ is equivalent to equality of the rank-one lines and norms, hence to $gu = e^{i\theta}u$. The orthogonal complement is then invariant, giving the displayed block stabilizer and preservation of Hermitian operators supported on $u^\perp$. On $R_P\mathbb{C}^2 = u^\perp$, the standard norm pulls back to the form $R_P^*R_P = C_{A_2}^{-1}$. Therefore $\widehat{A}_P$ is unitary precisely when (828) holds. Positive-definite congruence identifies this group with $SU(2)$. □

The proposition concerns the congruence action of the ambient $SU(3)$ on $\mathrm{Herm}_3(\mathbb{C})$. It is compatible with, but distinct from, the left $SU(3)$-action on $\mathrm{Mat}_{3\times2}(\mathbb{C})$ whose invariant Gram map is $Z \mapsto Z^*Z$: the source spaces and actions are different, and no identification of them is being made here.

23.3 The compact gauge–scalar carrier selected by the standard metrics

The Lorentz congruence group is noncompact, so it cannot preserve a positive field-space norm. The standard metrics already used in (808) do, however, select the positive matrix and compact subgroup

$$W_P := R_P^*R_P = C_{A_2}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad K_P := \{a \in SL_2(\mathbb{C}) : a^*W_Pa = W_P\}. \tag{829}$$

Because $W_P = C_{A_2}^{-1}$, inversion of the defining equality gives

$$a^*W_Pa = W_P \quad \Longleftrightarrow \quad aC_{A_2}a^* = C_{A_2}.$$

Accordingly, K_P is exactly the Lorentz little group of the future timelike vector C_{A_2}, equivalently the observer stabilizer selected by the standard metrics. Its promotion to a local internal gauge symmetry below is mathematically valid but is an additional physical interpretation.

Proposition 23.5 (Exact compact representation on the normal slice)**.** *Conjugation $a \mapsto W_P^{1/2}aW_P^{-1/2}$ is an isomorphism $K_P \cong SU(2)$. The real bilinear form*

$$E_P(G,K) := \mathrm{tr}\big(\jmath_P(G)\jmath_P(K)\big) = \mathrm{tr}(GW_PKW_P) \tag{830}$$

is a positive-definite K_P-invariant inner product on $\mathrm{Herm}_2(\mathbb{C})$*. For $a \in K_P$, the lift*

$$\widehat{a}_P = \frac{1}{3}uu^* + R_PaP \tag{831}$$

lies in $SU(3)$, fixes H_0, and obeys

$$\widehat{a}_P\jmath_P(G)\widehat{a}_P^* = \jmath_P(aGa^*). \tag{832}$$

After whitening, $\widetilde{G} = W_P^{1/2}GW_P^{1/2}$, the action is ordinary unitary congruence. Therefore, as a real K_P-module,

$$\mathrm{Herm}_2(\mathbb{C}) = \mathbb{R}W_P^{-1} \oplus \{G \in \mathrm{Herm}_2(\mathbb{C}) : \mathrm{tr}(W_PG) = 0\} \cong \mathbf{1} \oplus \mathbf{3}. \tag{833}$$

The centre $\{\pm I_2\}$ acts trivially. Thus the metric-selected normal field is a singlet plus an adjoint triplet; it is neither a fundamental weak doublet nor an $SU(3)$ color triplet.

Proof. The defining identity in (829) is equivalent to $(W_P^{1/2}aW_P^{-1/2})^*(W_P^{1/2}aW_P^{-1/2}) = I_2$; determinants are one, so this identifies K_P with $SU(2)$. Cyclicity of the trace and $R_P^*R_P = W_P$ give the second equality in (830). Moreover $E_P(G,G) = \mathrm{tr}(\jmath_P(G)^2)$, which is positive for nonzero Hermitian $\jmath_P(G)$. Equation (824) and unitarity of $\widehat{a}_P$ prove invariance. Proposition 23.4 gives $\widehat{a}_P \in U(3)$, and the block action on $u^\perp \oplus \mathbb{C}u$ gives $\det\widehat{a}_P = \det a = 1$. The whitened formula is direct multiplication. Unitary congruence fixes the scalar line and acts through the adjoint rotation representation on traceless Hermitian matrices; the central elements act by $G \mapsto (\pm I_2)G(\pm I_2)^* = G$. □

Theorem 23.6 (A fully typed positive gauge–scalar realization)**.** *Let (M,g) be an oriented Riemannian d-manifold and use the trivial principal K_P-bundle. Take*

$$\mathcal{A}\in\Omega^1(M;\mathfrak{k}_P),\qquad G\in C^\infty(M;\mathrm{Herm}_2(\mathbb{C})) \tag{834}$$

as the connection and normal scalar. For $a\in C^\infty(M,K_P)$, define

$$G^a=aGa^*,\qquad \mathcal{A}^a=a\mathcal{A}a^{-1}-(da)a^{-1}, \tag{835}$$

$$D_{\mathcal{A}}G=dG+\mathcal{A}G+G\mathcal{A}^*,\qquad F_{\mathcal{A}}=d\mathcal{A}+\mathcal{A}\wedge\mathcal{A}. \tag{836}$$

Choose on $\mathfrak{k}_P$ the positive $\mathrm{Ad}(K_P)$-invariant form transported from $-\operatorname{Re}\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$. For $g_{\mathrm{YM}},Z_G>0$ and $m_G,\lambda_4\ge 0$, the Euclidean action

$$\begin{aligned}S^{\mathrm{cpt}}_{g_{\mathrm{YM}},Z_G,m_G,\lambda_4}[\mathcal{A},G]=\int_M\Big(&\frac{1}{2g_{\mathrm{YM}}^2}|F_{\mathcal{A}}|^2+\frac{Z_G}{2}|D_{\mathcal{A}}G|^2_{E_P}+\frac{m_G^2}{2}|G|^2_{E_P}\\&+\frac{\lambda_4}{4}(\det G)^2\Big)\,d\operatorname{vol}_g\end{aligned} \tag{837}$$

is nonnegative and gauge invariant. The associated normal-matrix field

$$H_\varepsilon=H_0+\varepsilon\jmath_P(G) \tag{838}$$

is equivariant under the lifted action $H^a_\varepsilon=\widehat{a}_PH_\varepsilon\widehat{a}_P^*$. If $\widehat{\mathcal{A}}=R_P\mathcal{A}P$, then

$$D_{\widehat{\mathcal{A}}}\jmath_P(G)=\jmath_P(D_{\mathcal{A}}G),\qquad D_{\widehat{\mathcal{A}}}H_0=0. \tag{839}$$

On $M=\mathbb{R}^d$, the scalar Hessian at $(\mathcal{A},G)=(0,0)$ is the nonnegative self-adjoint operator

$$\begin{aligned}\mathcal{K}_{Z_G,m_G}&=(-Z_G\Delta+m_G^2)\otimes I_{\mathrm{Herm}_2},\\ \mathrm{Dom}(\mathcal{K}_{Z_G,m_G})&=H^2(\mathbb{R}^d;\mathrm{Herm}_2(\mathbb{C})),\\ \mathrm{Spec}(\mathcal{K}_{Z_G,m_G})&=[m_G^2,\infty).\end{aligned} \tag{840}$$

Thus the fixed normal data plus the standard metrics select the compact representation $K_P\curvearrowright\mathrm{Herm}_2(\mathbb{C})$ and its positive fibre metric. After the additional choice to gauge that pointwise action, the formulas above give an exact compact gauge–scalar model, but they still do not select $m_G,Z_G,g_{\mathrm{YM}},\lambda_4$, a quantum state, or an interacting gap.

Proof. Direct substitution gives $D_{\mathcal{A}^a}G^a=a(D_{\mathcal{A}}G)a^*$ and $F_{\mathcal{A}^a}=aF_{\mathcal{A}}a^{-1}$. Invariance of the two positive inner products and $\det(aGa^*)=|\det a|^2\det G=\det G$ prove gauge invariance; every term in (837) is nonnegative. Equation (832) gives equivariance of (838). Using $PR_P=I_2$, $Pu=0$, and $u^*R_P=0$, direct multiplication proves (839). Fourier transformation gives the self-adjoint operator, its full domain, and its spectrum in (840). Finally, $(\det G)^2$ is homogeneous of degree four, so it has zero Hessian at $G=0$. □

Corollary 23.7 (The selected normal field cannot screen a fundamental centre charge)**.** *Every tensor power and every polynomial composite of the $\mathbf{1}\oplus\mathbf{3}$ normal representation has trivial action by $-I_2\in K_P$. The fundamental two-dimensional representation has action $-I_2$. Therefore the field G in Theorem 23.6 cannot by itself screen a fundamental centre charge; such a model requires an additional centre-nontrivial matter field and its independently specified kinetic, mass, and interaction terms. This non-screening statement is confined to the G-only normal section. The full (G,ψ,λ) carrier already contains the centre-nontrivial spinor ψ; what remains missing is a specified kinetic and interaction theory for that retained field.*

Proof. The centre acts trivially on G, hence trivially on every tensor power and subquotient generated from it. It acts as minus the identity in the fundamental representation, so that representation cannot occur in this tensor category. □

23.4 The candidate cusp matrix is an exact spacelike vector

The preceding construction applies to every Hermitian two-by-two matrix. For the particular inverse-dual matrix retained in the cusp calculation, let

$$G_{\eta,T} = \begin{pmatrix} 2L_T & 2m_T \\ 2m_T & -q_T/3 \end{pmatrix}, \qquad D_T = L_T q_T + 6m_T^2 > 0. \tag{841}$$

Proposition 23.8 (Exact candidate-specific Lorentz coordinates)**.** *In the coordinates* (816)*,*

$$(t_T, x_T, y_T, z_T) = \left(L_T - \frac{q_T}{6},\ 2m_T,\ L_T + \frac{q_T}{6},\ 0 \right), \tag{842}$$

and

$$\eta_2(G_{\eta,T}, G_{\eta,T}) = \det G_{\eta,T} = -\frac{2}{3} D_T < 0. \tag{843}$$

If $L_T/T \to 1$, $q_T/T \to 1$, and $m_T/T \to 0$, then

$$\frac{1}{T}(t_T, x_T, y_T, z_T) \longrightarrow \left(\frac{5}{6}, 0, \frac{7}{6}, 0 \right). \tag{844}$$

Thus this particular candidate vector is spacelike for every T. It is not a positive-mass four-momentum. This statement concerns the indefinite inverse-dual matrix $G_{\eta,T}$; it does not concern the different positive Gram momentum $p_\alpha(Z) = \alpha Z^* Z$ used in the retained-c worldline construction. The latter is positive semidefinite by definition, and its image is all of $\mathrm{Herm}_2(\mathbb{C})_{\geq 0}$*: for every $G \succeq 0$, the block matrix $Z = \alpha^{-1/2} \left(\begin{smallmatrix} G^{1/2} \\ 0_{1\times 2} \end{smallmatrix} \right) \in \mathrm{Mat}_{3\times 2}(\mathbb{C})$ satisfies $p_\alpha(Z) = G$. Hence*

$$G_{\eta,T} \notin \operatorname{im} p_\alpha \qquad (T > 0,\ \alpha > 0).$$

Positive rescaling preserves the determinant sign and $GL_2(\mathbb{C})$ congruence preserves inertia, so neither operation repairs this mismatch. The Gram momentum lies on $\det p_\alpha = m_0^2 c^2$ *only after that mass-shell constraint and the parameters α, c, m_0 are supplied. Interpreting $G_{\eta,T}$ as a field value, a normal deformation, or a spacelike wave-vector requires an additional map; the determinant calculation alone does not choose among those interpretations.*

Proof. Solving $t + y = 2L_T$, $t - y = -q_T/3$, and $x + iz = 2m_T$ gives (842). Substitution gives

$$t_T^2 - x_T^2 - y_T^2 - z_T^2 = (2L_T)(-q_T/3) - 4m_T^2 = -\frac{2}{3} D_T.$$

The limit is componentwise. □

Theorem 23.9 (Exact determinant-level orbit geometry)**.** *Fix $\varepsilon > 0$ and write*

$$\mathbb{M}_\varepsilon := H_0 + \varepsilon \jmath_P(\mathrm{Herm}_2(\mathbb{C})).$$

For every $r > 0$, the future timelike determinant shell

$$\mathscr{P}^+_{r,\varepsilon} := \{ H_0 + \varepsilon \jmath_P(G) : G \succ 0,\ \det G = r^2 \} \cong SL_2(\mathbb{C})/SU(2) \tag{845}$$

is one orbit of the lifted congruence action. It equals

$$\{ H \in \mathbb{M}_\varepsilon : H \succ 0,\ \det H = \varepsilon^2 r^2 \}.$$

The nonzero future-null locus

$$\mathscr{N}^+_\varepsilon := \{ H_0 + \varepsilon \jmath_P(G) : G \succeq 0,\ \operatorname{rank} G = 1 \} \tag{846}$$

is also one $SL_2(\mathbb{C})$-orbit; each of its matrices is positive semidefinite of ambient rank two. Finally, with $r_T = \sqrt{2D_T/3}$, the candidate-specific spacelike shell is

$$\mathscr{S}_{T,\varepsilon} := \{H_0 + \varepsilon \jmath_P(G) : \det G = -r_T^2\} \cong SL_2(\mathbb{C})/SU(1,1), \tag{847}$$

and every matrix in it has inertia $(2,1)$. Consequently the normal slice contains every timelike mass shell and the future null cone after a unit dictionary is supplied, but it selects neither r nor a physical mass.

Proof. If $G \succ 0$ and $\det G = r^2$, then $A = (G/r)^{1/2}$ lies in $SL_2(\mathbb{C})$ and $A(rI_2)A^* = G$. The stabilizer of rI_2 is $SU(2)$, proving (845); positivity and the ambient determinant follow from the block decomposition in the proof of Proposition 23.1. Every nonzero positive rank-one G is vv^*. Choose $w \in \mathbb{C}^2$ with $\det(v\ w) = 1$; then $A = (v\ w) \in SL_2(\mathbb{C})$ and $A\,\mathrm{diag}(1,0)A^* = vv^*$. This proves null transitivity, and the orthogonal block decomposition gives ambient rank $1 + 1 = 2$.

Let $J = \mathrm{diag}(1,-1)$. A Hermitian G with $\det G = -r_T^2$ has inertia $(1,1)$, so Sylvester congruence gives $G = B(r_T J)B^*$ for some $B \in GL_2(\mathbb{C})$. Determinants give $|\det B| = 1$. If $\det B = e^{i\theta}$, then $D = \mathrm{diag}(e^{-i\theta}, 1)$ stabilizes J, and $A = BD$ has determinant one. Thus the action is transitive, and the stabilizer is

$$\{A \in SL_2(\mathbb{C}) : AJA^* = J\} = SU(1,1).$$

Equation (843) puts the retained candidate on this shell. The block decomposition gives one positive eigenvalue from H_0 and one of each sign from G, proving the inertia claim. □

23.5 Units and the minimal continuum field model

All entries in (807)–(809) are dimensionless. A physical spacetime requires a scale. For $\ell > 0$, set

$$\kappa_\ell(x^0, x^1, x^2, x^3) = H_0 + \jmath_P\left(G\left(\frac{x^0}{\ell}, \frac{x^1}{\ell}, \frac{x^2}{\ell}, \frac{x^3}{\ell}\right)\right). \tag{848}$$

Equip the affine slice with the constant metric $g_{P,\ell} = \ell^2 \eta_P$. Then

$$\kappa_\ell^* g_{P,\ell} = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2, \qquad \det \kappa_\ell(x) = \ell^{-2}\eta_{\mu\nu}x^\mu x^\nu. \tag{849}$$

For the scaled normal displacement $H_0 + \varepsilon \jmath_P(G(x/\ell))$, the determinant is multiplied by ε^2. For $\varepsilon \neq 0$, the corresponding isometric metric rescaling is $(\ell/|\varepsilon|)^2\eta_P$. Hence ε fixes no absolute energy or length by itself.

The exact unit dictionary is:

symbol	role	status from the determinant data
$P, u, R_P, \jmath_P$	marked linear algebra	fixed and dimensionless
ε	normal-displacement amplitude	fixed only when supplied
ℓ	conversion from matrix coordinates to length	independent choice
$x^0 = ct$	time coordinate	requires the independent constant c
m_{KG}	inverse Compton length in the scalar model	independent choice
$m_{\mathrm{phys}} = \hbar m_{\mathrm{KG}}/c$	physical scalar mass	requires $\hbar, c$
$L = Na$	infrared box size and ultraviolet mesh	regulator choices
K, g_{YM}	compact gauge group and coupling	K arbitrary; $K_P \cong SU(2)$ is the metric-selected little group and its gauging is optional

The retained worldline mass m_0, the Klein–Gordon inverse length m_{KG}, and the dimensionless determinant-shell radius r are independent inputs. Only after declaring that the worldline and field theory describe the same physical species may one impose the unit conversion

$$m_{\mathrm{KG}} = \frac{m_0 c}{\hbar}, \qquad \Delta E_{\mathrm{free}} = \hbar c\, m_{\mathrm{KG}} = m_0 c^2.$$

No determinant identity derives this identification.

Use natural units $c = \hbar = 1$ in the remaining formulas. The minimal field placed on the Lorentzian host is a real scalar $\phi \in C_c^\infty(\mathbb{R}^{1,3}; \mathbb{R})$ with action

$$S_{m_{\mathrm{KG}}}[\phi] = \frac{1}{2} \int_{\mathbb{R}^{1,3}} \left(\partial_\mu \phi\, \partial^\mu \phi - m_{\mathrm{KG}}^2 \phi^2 \right) d^4x, \qquad m_{\mathrm{KG}} \geq 0. \tag{850}$$

Its Euler–Lagrange operator is $\Box + m_{\mathrm{KG}}^2$. The Poincaré action (826) acts on fields by pullback. The action, the stress-energy tensor

$$T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \eta_{\mu\nu} (\partial_\alpha \phi \partial^\alpha \phi - m_{\mathrm{KG}}^2 \phi^2), \tag{851}$$

and the mass shell $p_\mu p^\mu = m_{\mathrm{KG}}^2$ are invariant. Smeared fields $\phi(f) = \int \phi f\, d^4x$ are the basic linear observables.

On $x^0 = 0$, smooth compactly supported initial data (ϕ_0, π_0) carry the symplectic form

$$\sigma((\phi, \pi), (\psi, \rho)) = \int_{\mathbb{R}^3} (\phi\rho - \pi\psi)\, d^3x \tag{852}$$

and energy

$$E_{m_{\mathrm{KG}}}(\phi, \pi) = \frac{1}{2} \int_{\mathbb{R}^3} (|\pi|^2 + |\nabla \phi|^2 + m_{\mathrm{KG}}^2 |\phi|^2)\, d^3x. \tag{853}$$

For $m_{\mathrm{KG}} > 0$, the energy completion is $H^1(\mathbb{R}^3) \oplus L^2(\mathbb{R}^3)$; for $m_{\mathrm{KG}} = 0$, it is $\dot{H}^1(\mathbb{R}^3) \oplus L^2(\mathbb{R}^3)$. The symplectic pairing (852) is exact on the dense smooth, compactly supported data. In the massless homogeneous completion it need not extend continuously to every pair without an additional dual-space choice.

Theorem 23.10 (Exact free-field spectral alternative)**.** *Let*

$$K_{m_{\mathrm{KG}}} = -\Delta + m_{\mathrm{KG}}^2, \quad \mathrm{Dom}(K_{m_{\mathrm{KG}}}) = H^2(\mathbb{R}^3), \quad \omega_{m_{\mathrm{KG}}} = K_{m_{\mathrm{KG}}}^{1/2} \tag{854}$$

on $L^2(\mathbb{R}^3)$. On the bosonic Fock space $\mathscr{F}_s(L^2(\mathbb{R}^3))$, define the vacuum-subtracted Hamiltonian

$$\mathbb{H}_{m_{\mathrm{KG}}} = d\Gamma(\omega_{m_{\mathrm{KG}}}). \tag{855}$$

Explicitly, on the orthogonal decomposition $\mathscr{F}_s = \bigoplus_{n=0}^\infty L^2(\mathbb{R}^3)^{\otimes_s n}$, its domain consists of the vectors $\Psi = (\Psi_n)$ for which each Ψ_n belongs to the domain of $\sum_{j=1}^n I \otimes \cdots \otimes \omega_{m_{\mathrm{KG}}} \otimes \cdots \otimes I$ and the squared norms of those images are summable. For every $m_{\mathrm{KG}} \geq 0$, the Fock vacuum is the unique zero-energy vector. Its excitation gap is exactly

$$\inf \mathrm{Spec}(\mathbb{H}_{m_{\mathrm{KG}}}|_{\Omega^\perp}) = m_{\mathrm{KG}}. \tag{856}$$

Thus the same determinant normal geometry admits a positive gap of any chosen value $m_{\mathrm{KG}} > 0$ and a gapless model at $m_{\mathrm{KG}} = 0$.

Proof. Fourier transform identifies $K_{m_{\mathrm{KG}}}$ with multiplication by $|p|^2 + m_{\mathrm{KG}}^2$ and $\omega_{m_{\mathrm{KG}}}$ with multiplication by $\sqrt{|p|^2 + m_{\mathrm{KG}}^2}$. The latter has zero kernel even when $m_{\mathrm{KG}} = 0$, since the singleton $p = 0$ has Lebesgue measure zero. Hence the only Fock vector annihilated by $d\Gamma(\omega_{m_{\mathrm{KG}}})$ is the vacuum. On the n-particle sector, the Hamiltonian is the sum of n copies of $\omega_{m_{\mathrm{KG}}}$, so the nonvacuum spectral bottom is at least m_{KG}. The one-particle sector is reducing and has spectral bottom m_{KG}, proving equality. For $m_{\mathrm{KG}} = 0$, normalized Fourier-space functions supported in balls $|p| < 1/n$ have energy expectation at most $1/n$, which makes the absence of a positive gap explicit. □

There is no contradiction between the absence of a positive kinematic threshold on the unrestricted Gram carrier and the free Fock gap m_{KG}: the former allows the rescaling $Z \mapsto rZ$, whereas the latter first fixes an external mass coefficient and quantizes the chosen free Hamiltonian.

23.6 A complete finite regulator and its two limits

Let $L > 0$, $N \geq 2$, $a = L/N$, and $V_N = (\mathbb{Z}/N\mathbb{Z})^3$. Give $Q_{L,N} = \mathbb{R}^{V_N}$ the inner product

$$\langle q, r\rangle_a = a^3 \sum_{v \in V_N} q(v) r(v). \tag{857}$$

The positive periodic lattice Laplacian is

$$(-\Delta_a q)(v) = a^{-2} \sum_{j=1}^{3} (2q(v) - q(v + e_j) - q(v - e_j)). \tag{858}$$

Let $d\lambda_a$ be Lebesgue measure associated with the Euclidean space $(Q_{L,N}, \langle\cdot,\cdot\rangle_a)$, and let $\Delta_{Q_{L,N}}$ denote its coordinate Laplacian, so that $-\Delta_{Q_{L,N}}$ is nonnegative. For $m_{\mathrm{KG}} > 0$, define on $L^2(Q_{L,N}, d\lambda_a)$

$$\mathbb{H}_{m_{\mathrm{KG}},L,N} = \frac{1}{2}\Big(-\Delta_{Q_{L,N}} + \langle q, (-\Delta_a + m_{\mathrm{KG}}^2) q\rangle_a\Big) - E_{0,m_{\mathrm{KG}},L,N}, \tag{859}$$

where $E_{0,m_{\mathrm{KG}},L,N}$ is its finite oscillator ground energy. More precisely, before subtracting $E_{0,m_{\mathrm{KG}},L,N}$, this is the self-adjoint operator associated with the closed form

$$\mathfrak{h}_{m_{\mathrm{KG}},L,N}[\Psi] = \frac{1}{2} \int_{Q_{L,N}} \Big(|\nabla \Psi(q)|_a^2 + \langle q, (-\Delta_a + m_{\mathrm{KG}}^2) q\rangle_a |\Psi(q)|^2\Big)\, d\lambda_a(q), \tag{860}$$

whose domain is

$$H^1(Q_{L,N}) \cap L^2\Big(Q_{L,N}, \langle q, (-\Delta_a + m_{\mathrm{KG}}^2) q\rangle_a \, d\lambda_a(q)\Big). \tag{861}$$

Theorem 23.11 (Exact regulated spectrum and continuum limits)**.** *For $n = (n_1, n_2, n_3) \in V_N$, the squared normal-mode frequencies are*

$$\omega_{m_{\mathrm{KG}},L,N}(n)^2 = m_{\mathrm{KG}}^2 + \frac{4}{a^2} \sum_{j=1}^{3} \sin^2\Big(\frac{\pi n_j}{N}\Big). \tag{862}$$

The operator $\mathbb{H}_{m_{\mathrm{KG}},L,N}$ is self-adjoint on the tensor-product oscillator domain, has a unique vacuum, and has exact gap m_{KG}. The subtracted constant is

$$E_{0,m_{\mathrm{KG}},L,N} = \frac{1}{2} \sum_{n \in V_N} \omega_{m_{\mathrm{KG}},L,N}(n). \tag{863}$$

For $m_{\mathrm{KG}} = 0$, the constant mode is a free particle and the unreduced operator has no normalized ground state. On the mean-zero configuration space

$$Q_{L,N}^0 = \{q \in Q_{L,N} : \textstyle\sum_v q(v) = 0\}, \tag{864}$$

the vacuum-subtracted oscillator has a unique ground state and gap

$$\Delta_{L,N}^0 = \frac{2}{a} \sin\Big(\frac{\pi}{N}\Big). \tag{865}$$

At fixed L, for every fixed $r \in \mathbb{Z}^3$,

$$\omega_{m_{\mathrm{KG}},L,N}(r)^2 \longrightarrow m_{\mathrm{KG}}^2 + \frac{4\pi^2 |r|^2}{L^2} \qquad (N \to \infty), \tag{866}$$

and

$$\Delta^0_{L,N} \longrightarrow \frac{2\pi}{L}. \tag{867}$$

After the thermodynamic limit $L \to \infty$, the massive gap remains m_{KG}, whereas the massless positive-mode gap tends to zero.

Proof. The characters $v \mapsto L^{-3/2} e^{2\pi i n\cdot v/N}$ diagonalize $-\Delta_a$. Substitution gives (862). Orthogonal diagonalization of the real configuration space turns (859) into a tensor sum of one-dimensional harmonic oscillators with frequencies $\omega_{m_{\mathrm{KG}},L,N}(n)$. For $m_{\mathrm{KG}} > 0$, their minimum is attained at $n = 0$ and equals m_{KG}, proving the ground-state and gap claims.

When $m_{\mathrm{KG}} = 0$, the constant coordinate has zero restoring force, hence gives the free-particle operator and no L^2 ground state. Removing that coordinate leaves the smallest character with one component congruent to ± 1 and the others zero, giving (865). With $a = L/N$, the elementary limit $N \sin(\pi r_j/N) \to \pi r_j$ proves (866) and (867). The last claims follow by sending L to infinity. □

This regulator supplies a literal configuration space, state Hilbert space, quadratic Hamiltonian, observables generated by the coordinates and momenta, and exact fixed-mode ultraviolet and lowest-frequency infrared limits. The displayed limits do not assert strong- or norm-resolvent convergence, convergence of Gaussian measures, or convergence of Fock representations. The mass parameter is not hidden in P, R_P, ε, or the determinant: it is the independent coefficient in (850).

23.7 An exact classical gauge-theory completion

Let K be any compact matrix Lie group with Lie algebra $\mathfrak{k}$ and an $\mathrm{Ad}\, K$-invariant positive inner product. On the trivial bundle over the Lorentzian host, take a smooth compactly supported gauge potential $A \in \Omega^1_c(\mathbb{R}^{1,3}; \mathfrak{k})$, with curvature

$$F_A = dA + A \wedge A. \tag{868}$$

The gauge group of smooth maps $\mathbb{R}^{1,3} \to K$ that equal the identity off a compact set acts by

$$A^g = gAg^{-1} - (dg)g^{-1}, \qquad F_{A^g} = gF_Ag^{-1}. \tag{869}$$

For an independently chosen coupling $g_{\mathrm{YM}} > 0$, the classical action is

$$S_{K,g_{\mathrm{YM}}}[A] = -\frac{1}{4g^2_{\mathrm{YM}}} \int \langle F_{\mu\nu}, F^{\mu\nu} \rangle \, d^4x. \tag{870}$$

It is invariant under (869) and the Poincaré action. For a closed curve γ and a finite-dimensional representation ρ of K,

$$W_{\gamma,\rho}(A) = \mathrm{tr}\, \rho(\mathrm{Hol}_A(\gamma)) \tag{871}$$

is gauge invariant.

On the spatial regulator $\Sigma_L = (\mathbb{R}/L\mathbb{Z})^3$, the smooth Hamiltonian phase space before reduction consists of $(A, E) \in \Omega^1(\Sigma_L; \mathfrak{k})^2$, with gauge action

$$(A, E)^g = (gAg^{-1} - (dg)g^{-1},\, gEg^{-1}). \tag{872}$$

Its Gauss constraint and energy are

$$d^*_A E = 0, \qquad \mathcal{E}_{K,g}(A, E) = \frac{1}{2g^2_{\mathrm{YM}}} \int_{\Sigma_L} (|E|^2 + |F_A|^2) \, d^3x, \tag{873}$$

and the reduced classical state space is the constraint locus modulo $C^\infty(\Sigma_L, K)$. At this smooth level it is a set-theoretic orbit space, generally stratified by stabilizer type; making it a smooth symplectic space requires Sobolev completions and hypotheses on the action. Equations (869) and (873) show

directly that the constraint locus, energy, and Wilson observables descend to this quotient: in particular, $d^*_{A^g} E^g = g(d^*_A E)g^{-1}$, because the Hodge star acts only on the base and the adjoint action is orthogonal. In temporal gauge the canonical momentum is $\Pi_E = g_{\mathrm{YM}}^{-2} E$. Equivalently, the symplectic form on tangent vectors is

$$\Omega_{(A,E)}\big((a,e),(a',e')\big) = \frac{1}{g_{\mathrm{YM}}^2}\int_{\Sigma_L} \big(\langle e, a'\rangle - \langle e', a\rangle\big)\, d\,\mathrm{vol}\,. \tag{874}$$

Proposition 23.12 (The exact nonzero-mode Gaussian gauge gap)**.** *At $(A,E) = (0,0)$, impose the Coulomb conditions $d^*A = d^*E = 0$ and remove harmonic one-forms. Relative to the unweighted L^2 configuration metric, the energy Hessian is $g_{\mathrm{YM}}^{-2}\Delta_1$, where the generalized squared-frequency operator is*

$$\Delta_1 = d^*d, \quad \mathrm{Dom}(\Delta_1) = H^2\Omega^1(\Sigma_L;\mathfrak{k}) \cap \ker d^* \cap (\ker \Delta_1)^{\perp}. \tag{875}$$

on the closed Hilbert space

$$\mathscr{K}_L^{\mathrm{tr}} = \big(L^2\Omega^1(\Sigma_L;\mathfrak{k}) \cap \ker d^*\big) \cap (\ker \Delta_1)^{\perp}. \tag{876}$$

Its spectrum is

$$\left\{ \frac{4\pi^2|n|^2}{L^2} : n \in \mathbb{Z}^3 \setminus \{0\} \right\}, \tag{877}$$

with transverse-polarization and $\dim \mathfrak{k}$ multiplicities. On the unreduced transverse Gaussian Fock space after this explicit harmonic-mode truncation, the gap is $2\pi/L$, which tends to zero as $L \to \infty$. Constant gauge transformations remain after Coulomb gauge, so this statement does not assert that its one-particle level survives restriction to a gauge-invariant Fock subspace.

Proof. The curvature linearization at zero is $A \mapsto dA$. On the Coulomb slice, the configuration part of the energy has quadratic form $g_{\mathrm{YM}}^{-2}\|dA\|^2/2$, whose Hessian relative to unweighted L^2 is $g_{\mathrm{YM}}^{-2} d^*d$ with the displayed elliptic domain. Dividing by the kinetic metric $g_{\mathrm{YM}}^{-2} I$ gives the squared-frequency operator $\Delta_1 = d^*d$. Fourier characters on the three-torus give (877); for each nonzero wave-vector, the condition $d^*A = 0$ leaves its two orthogonal polarizations. Gaussian quantization replaces each Hessian eigenvalue λ by the oscillator frequency $\sqrt{\lambda}$: in canonical variables a single mode has Hamiltonian $\frac{1}{2}(g_{\mathrm{YM}}^2 P_Q^2 + \lambda g_{\mathrm{YM}}^{-2} Q^2)$, whose frequency is $\sqrt{\lambda}$. The smallest nonzero wave-vector has length $2\pi/L$, proving the result. □

Thus the determinant slice is an exact four-dimensional Lorentzian host for classical gauge theory, not merely an analogy. It does not select K: the same construction works for $U(1)$, $SU(2)$, $SU(3)$, or any other compact matrix group. In particular, Proposition 23.4 prevents the ambient $SU(3)$ orbit label from being silently reused as a local color gauge group. Nor does the free Hessian establish the nonlinear four-dimensional Yang–Mills mass gap; that problem concerns the physical gauge-invariant quantum Hamiltonian after a continuum construction [63].

A cubical mesh gives the exact holonomy regulator of Wilson [64]. To type the state dependence, let V be the vertices, choose one orientation E^+ of the edges, and put

$$\mathcal{C}_E = K^{E^+}, \qquad \mathcal{G}_V = K^V, \qquad (h\cdot U)_e = h_{s(e)} U_e h_{t(e)}^{-1}. \tag{878}$$

Reverse orientation means $U_{\bar{e}} = U_e^{-1}$. For a closed edge path $\gamma = e_1 \cdots e_r$ and a nontrivial irreducible representation ρ, use the normalized Wilson observable

$$w_{\gamma,\rho}(U) = \frac{1}{\dim \rho}\,\mathrm{tr}\,\rho(U_{e_1}\cdots U_{e_r}). \tag{879}$$

Proposition 23.13 (Link kinematics selects no confinement state)**.** *There are two $\mathcal{G}_V$-invariant probability measures on the same configuration space $\mathcal{C}_E$ such that*

$$\mathbb{E}_{\mathrm{pg}}[w_{\gamma,\rho}] = 1 \quad \textit{for every closed } \gamma, \qquad \mathbb{E}_{\mathrm{Haar}}[w_{\gamma,\rho}] = 0 \tag{880}$$

whenever γ contains an unoriented edge exactly once in the second formula. Their convex mixtures give every expectation between zero and one. Thus the link configuration and gauge action do not determine an area law or string tension.

Proof. For the first measure, give K^V product Haar measure and push it forward under $h \mapsto U(h)$, where $U(h)_e = h_{s(e)} h_{t(e)}^{-1}$. Left invariance of vertex Haar measure makes the push-forward gauge invariant, and every closed holonomy telescopes to the identity, proving the first value. For the second, give the independent variables U_e, $e \in E^+$, product Haar measure. Bi-invariance makes this measure gauge invariant. Integrating first over an edge that occurs exactly once inserts $\int_K \rho(k)\, dk = 0$, because a nontrivial irreducible representation has no invariant vector. This proves the second value. Convexity proves the remaining assertion. □

The link system is a valid ultraviolet regulator only after a time direction, Euclidean rotation, coupling normalization, action, and probability measure have been chosen. Proposition 23.13 shows why the normal-slice geometry and link space alone do not supply confinement.

23.8 Exact obstructions and the sufficient spectral bridge

Proposition 23.14 (No Lorentz-invariant positive quadratic energy)**.** *There is no positive-definite quadratic form on $\mathcal{N}_P$ invariant under the full $PSL_2(\mathbb{C})$ action of Theorem 23.3. Consequently the determinant Hessian cannot itself be a nonnegative Hamiltonian.*

Proof. Suppose a positive-definite norm were invariant. In the coordinates (816), Lorentz boosts send $(1,0,0,0)$ to $(\cosh s, \sinh s, 0, 0)$. Invariance would keep the positive-definite norm of this orbit constant. Every positive-definite norm on a finite-dimensional space is equivalent to the Euclidean norm, whereas the Euclidean norms of these vectors diverge with $|s|$. This is a contradiction. □

The next two models make the role of entanglement exact. They use bipartite tensor-product Hilbert kinematics but have opposite gap behavior.

Proposition 23.15 (Entanglement determines neither positivity nor magnitude of a gap)**.** *Let $\Omega = (e_0 \otimes e_0 + e_1 \otimes e_1)/\sqrt{2}$ in $\mathbb{C}^2 \otimes \mathbb{C}^2$. Then*

$$H_{\mathrm{gap}} = I - |\Omega\rangle\langle\Omega| \tag{881}$$

is bounded, nonnegative, and self-adjoint, has the entangled vector Ω as its unique vacuum, and has gap one.

On $\ell^2(\mathbb{N}_0) \otimes \ell^2(\mathbb{N}_0)$, put

$$\Omega_0 = \frac{e_0 \otimes e_0 + e_1 \otimes e_1}{\sqrt{2}}, \qquad \psi_n = \frac{e_{2n} \otimes e_{2n} + e_{2n+1} \otimes e_{2n+1}}{\sqrt{2}} \quad (n \geq 1). \tag{882}$$

Define a bounded self-adjoint operator by

$$H_{\mathrm{ent}}\Omega_0 = 0, \qquad H_{\mathrm{ent}}\psi_n = n^{-1}\psi_n, \qquad H_{\mathrm{ent}} = I \text{ on } \overline{\mathrm{span}}\{\Omega_0, \psi_n : n \geq 1\}^{\perp}. \tag{883}$$

Put

$$\mathcal{K}_{\mathrm{ent}} = \overline{\mathrm{span}}\{\psi_n : n \geq 1\}.$$

Every nonzero vector in $\mathcal{K}_{\mathrm{ent}}$ is entangled, $\mathcal{K}_{\mathrm{ent}}$ reduces H_{ent}, and

$$\inf \mathrm{Spec}(H_{\mathrm{ent}}|_{\mathcal{K}_{\mathrm{ent}}}) = 0.$$

Thus even a closed, completely entangled reducing sector need not have a positive energy floor.

Proof. Equation (881) is an orthogonal projection with kernel $\mathbb{C}\Omega$. The vectors in (882) are mutually orthonormal and each has Schmidt rank two. Equation (883) is a bounded real diagonal operator in the indicated orthogonal decomposition. Its kernel is $\mathbb{C}\Omega_0$, while $1/n \to 0$. Under the standard coefficient-operator identification of a bipartite tensor, a nonzero vector $\sum_{n\geq 1} c_n \psi_n$ has, for every n with $c_n \neq 0$, two equal nonzero diagonal entries on the coordinates $2n, 2n+1$. Its Schmidt rank is therefore at least two. The displayed diagonal action also proves reduction of $\mathcal{K}_{\mathrm{ent}}$ and spectral bottom zero. □

Finally, let A_{phys} be a lower-semibounded self-adjoint physical Hamiltonian on $\mathcal{H}_{\mathrm{phys}}$, put $E_{\mathrm{vac}} = \inf \mathrm{Spec}(A_{\mathrm{phys}})$, and assume that E_{vac} is an eigenvalue. Let $P_0 = \mathbf{1}_{\{E_{\mathrm{vac}}\}}(A_{\mathrm{phys}})$, let $\mathfrak{a}_{\mathrm{phys}}$ be its closed quadratic form, and put $\mathcal{H}_{\mathrm{exc}} = (I - P_0)\mathcal{H}_{\mathrm{phys}}$. This is a reducing subspace. Let $\mathcal{H}_{\mathrm{aux}}$ be another Hilbert space and let $B \geq 0$ be self-adjoint on it, with closed form $\mathfrak{b}$ satisfying

$$\mathfrak{b}[v] \geq \gamma \|v\|_{\mathrm{aux}}^2 \quad (v \in \mathrm{Dom}\, \mathfrak{b}) \tag{884}$$

for some $\gamma > 0$.

Theorem 23.16 (Full-domain coercive transfer)**.** *Suppose $\mathcal{H}_{\mathrm{exc}} \neq \{0\}$ and a linear map*

$$R : \mathrm{Dom}(\mathfrak{a}_{\mathrm{phys}}) \cap \mathcal{H}_{\mathrm{exc}} \longrightarrow \mathrm{Dom}(\mathfrak{b}) \tag{885}$$

is defined on the entire physical excitation form domain and, for fixed $c_R, C_R > 0$, satisfies

$$\|Ru\|_{\mathrm{aux}} \geq c_R \|u\|_{\mathrm{phys}}, \qquad \mathfrak{b}[Ru] \leq C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}}\|u\|^2). \tag{886}$$

Then

$$\inf \mathrm{Spec}(A_{\mathrm{phys}}|_{\mathcal{H}_{\mathrm{exc}}}) - E_{\mathrm{vac}} \geq \frac{\gamma c_R^2}{C_R}. \tag{887}$$

An intertwiner from only one gapped subspace cannot replace (885)*.*

Proof. For every vector in the domain of (885),

$$C_R(\mathfrak{a}_{\mathrm{phys}}[u] - E_{\mathrm{vac}}\|u\|^2) \geq \mathfrak{b}[Ru] \geq \gamma \|Ru\|_{\mathrm{aux}}^2 \geq \gamma c_R^2 \|u\|_{\mathrm{phys}}^2.$$

The variational characterization of the spectral bottom gives (887). For the final assertion, take $B = I$ on $\mathbb{C}$ and

$$A = 0 \oplus I \oplus M_x \quad \text{on} \quad \mathbb{C}\Omega \oplus \mathbb{C} \oplus L^2((0,1), dx),$$

where $(M_x f)(x) = x f(x)$. The isometry $Jz = 0 \oplus z \oplus 0$ obeys $AJ = JB$, yet A has unique vacuum Ω and excitation spectral bottom zero because of the multiplication summand. □

23.9 Exact conclusion

The determinant identity therefore supplies all of the following, exactly: a canonical dimensionless affine Minkowski space, its Lorentz metric and future cone, the spin and Poincaré actions, a natural time orientation, and a dimensionless normal coordinate system. After the independent choice of one length/time dictionary this carrier becomes a spacetime model supporting a standard continuum Klein–Gordon model, an exact finite periodic regulator, and any chosen classical compact-group gauge theory. The standard metrics also select the compact $K_P \cong SU(2)$ singlet–adjoint representation; Theorem 23.6 gives one exact gauging of that representation. Those are substantive physical constructions with typed configuration spaces, states, actions, observables, spectra, and limits.

The determinant normal-slice data alone do not select a preferred inertial observer, an internal gauge interpretation, a coupling, a mass, a probability measure, a quantum reconstruction, or a nonlinear spectral gap. A separate two-column Gram construction supplies a left $SU(3)$-action and the positive momentum $p_\alpha(Z) = \alpha Z^* Z$, but it retains the independent parameters α, c, m_0 and does not dynamically select $m_0 > 0$. These limitations are exact rather than semantic: the inverse-dual candidate cusp vector (842) is spacelike; no positive quadratic energy is Lorentz invariant; the same host has the gapped and gapless scalar realizations of Theorem 23.10; the massless gauge Gaussian gap closes as $L \to \infty$; and entangled states occur in both gapped and gapless Hamiltonians. A claim about a physical mass gap must therefore be proved from the physical spectrum or from a global comparison controlling its entire excitation domain; Theorem 23.16 is one sufficient such criterion.

24 Exact constraints on an S^8 higher-rung analogue

The phrase "an S^8 analogue" can denote three mathematically different proposals, which must not be conflated. A literal smooth complex-fourfold analogue is impossible because a homotopy 8-sphere is not almost complex. A merely real map to S^2 with generic fibre T^6 is not excluded if it has genuine singular fibres, but its local fillings must satisfy the exact Euler, fundamental-group, Wang, and Mayer–Vietoris tests below. Finally, explicit real stratified quotient models with boundary or total space S^8 do exist; they do not thereby acquire a complex-fourfold structure.

There is also a different, well-typed alternative: retain the four-real- dimensional torus fibres of the reconstructed S^6 candidate and place $SU(3)$ bundle data on its regular locus. Subsection 24.8 computes the exact extension obstruction. No gauge-theoretic or physical conclusion is asserted without a specified bundle, connection, trace convention, boundary condition, and analytic operator statement.

24.1 The literal complex-fourfold proposal

24.1.1 Characteristic-class obstruction

Let M be a smooth manifold homotopy equivalent to S^8, and suppose that $J\colon TM \to TM$ satisfies $J^2 = -1$. Then

$$E = (TM, J)$$

is a complex vector bundle of rank four whose underlying oriented real bundle is TM. Since

$$H^2(M;\mathbb{Z}) = H^4(M;\mathbb{Z}) = H^6(M;\mathbb{Z}) = 0,$$

one has

$$c_1(E) = c_2(E) = c_3(E) = 0.$$

Orient M by J, and choose $u \in H^8(M;\mathbb{Z})$ with $\langle u, [M]\rangle = 1$. The top-Chern–Euler identity and the Poincaré–Hopf theorem give

$$c_4(E) = e(TM) = \chi(M)u = 2u. \tag{888}$$

For a complex bundle, the second Pontryagin class of the underlying real bundle is

$$p_2(E_{\mathbb{R}}) = c_2(E)^2 - 2c_1(E)c_3(E) + 2c_4(E).$$

Consequently,

$$p_2(TM) = 4u. \tag{889}$$

On the other hand, $H^4(M;\mathbb{Z}) = 0$ gives $p_1(TM) = 0$, and the middle cohomology vanishes, so $\operatorname{sign}(M) = 0$. Hirzebruch's eight-dimensional signature formula is

$$\operatorname{sign}(M) = \left\langle \frac{7p_2(TM) - p_1(TM)^2}{45}, [M] \right\rangle.$$

It forces $\langle p_2(TM), [M]\rangle = 0$, contradicting (889). For the standard embedded sphere, the signature step may instead be replaced by the explicit stable trivialisation

$$TS^8 \oplus \varepsilon^1_{\mathbb{R}} \cong \varepsilon^9_{\mathbb{R}},$$

which gives $p_2(TS^8) = 0$ directly. Hence

$$\boxed{S^8 \text{ admits no almost-complex structure.}} \tag{890}$$

Integrability is stronger than being almost complex, so a complex-fourfold structure is excluded a fortiori.

24.1.2 Independent arithmetic checks

Borel and Serre prove that S^{2n} admits no almost-complex structure for $n \geq 4$, including arbitrary differentiable structures on the sphere. In the S^8 specialization, their mod-3 reduced-power identity is

$$\mathcal{P}^1(c_2) = c_1^2 c_2 - 2c_2^2 - c_1 c_3 + 4c_4.$$

All lower Chern classes vanish on S^8, so it would follow that $c_4 = 0 \pmod 3$, whereas (888) gives $c_4 = \pm 2u \neq 0 \pmod 3$.

Bott integrality gives a second independent check. For a complex bundle over S^{2n}, the top Chern number is divisible by $(n-1)!$. At $n = 4$,

$$6 \mid \langle c_4(E), [S^8] \rangle,$$

which is incompatible with the Euler number ± 2.

24.1.3 Exact scope of the obstruction

The obstruction rules out a smooth complex fourfold diffeomorphic to S^8; a proper holomorphic family of complex three-tori over $\mathbb{P}^1$ whose smooth total space is diffeomorphic to S^8; and an almost-complex structure on all of TS^8, whether or not integrable. It also rules out a symplectic or Kähler structure: if ω were symplectic, then $H^2(S^8; \mathbb{R}) = 0$ would give $\omega = d\alpha$, while

$$\int_{S^8} \omega^4 = \int_{S^8} d(\alpha \wedge \omega^3) = 0,$$

contradicting symplectic volume.

It does not rule out a complex structure on an open subset, a stable complex structure, a singular complex analytic space whose tangent sheaf is not a vector bundle, or a real stratified quotient whose underlying boundary is S^8. The standard sphere is stably complex because

$$TS^8 \oplus \varepsilon_{\mathbb{R}}^2 \cong \varepsilon_{\mathbb{R}}^{10},$$

but this is not a complex structure on the actual rank-eight tangent bundle. Moreover, the usual link of an isolated complex singularity of complex dimension d has real dimension $2d - 1$. Thus an S^8 obtained as a quotient boundary is not, merely for that reason, an ordinary complex-analytic link.

24.2 Requirements for a real Fabel-style S^8 fibration

The reconstructed S^6 candidate has a map

$$f \colon X^6 \longrightarrow B \cong \mathbb{P}^1$$

whose regular fibre is a complex two-torus, hence a real T^4, and whose three special fibres comprise one non-normal toric degeneration and two multiple bielliptic fibres. The direct dimension analogue on a real eight-manifold would be a map

$$g \colon M^8 \longrightarrow S^2$$

with regular fibre T^6. If M is required to be a smooth complex fourfold and the regular fibres are complex three-tori, the obstruction in Subsection 24.1 ends the proposal. If g is only a real smooth or stratified map, the proposal remains topologically meaningful.

The adjective "singular" is essential. There is no fibre bundle

$$T^6 \longrightarrow S^8 \longrightarrow S^2.$$

Indeed, its long exact homotopy sequence would contain

$$\pi_2(S^8) = 0 \longrightarrow \pi_2(S^2) = \mathbb{Z} \xrightarrow{\partial} \pi_1(T^6) = \mathbb{Z}^6 \longrightarrow \pi_1(S^8) = 0.$$

Exactness would require $\partial\colon \mathbb{Z} \to \mathbb{Z}^6$ to be surjective, which is impossible. The Euler identity for a fibre bundle gives the same conclusion:

$$\chi(S^8) = 2 \neq \chi(S^2)\chi(T^6) = 2 \cdot 0 = 0.$$

24.3 Euler constraint for a finite multipole model

Let $g\colon M^8 \to S^2$ be a proper triangulable map with finite critical-value set $\Sigma = \{p_1, \dots, p_s\}$, and suppose that every fibre over $S^2 \setminus \Sigma$ is T^6. Euler integration for a constructible map gives

$$\begin{aligned} \chi(M) &= \chi(S^2 \setminus \Sigma)\chi(T^6) + \sum_{i=1}^{s} \chi(g^{-1}(p_i)) \\ &= \sum_{i=1}^{s} \chi(F_i). \end{aligned} \tag{891}$$

Equivalently, one may cut out small discs, use multiplicativity on the regular bundle, and note that each boundary mapping torus has Euler characteristic zero. For $M \cong S^8$ one therefore needs

$$\boxed{\sum_{i=1}^{s} \chi(F_i) = 2.} \tag{892}$$

Multiple fibres that are finite free quotients of tori have Euler characteristic zero and cannot account for S^8. At least one special fibre must contain a genuine collapse or fixed stratum with nonzero Euler characteristic. The numerical pattern consisting of two Euler-zero multiple fibres and one fibre of Euler characteristic two is compatible with (892), but that numerical compatibility is not a construction.

24.4 Fundamental group and monodromy

Put $B^\circ = S^2 \setminus \Sigma$, $E = g^{-1}(B^\circ)$, and $\Lambda = H_1(T^6; \mathbb{Z}) \cong \mathbb{Z}^6$. The regular bundle has monodromy

$$\rho\colon \pi_1(B^\circ) \longrightarrow \mathrm{GL}(\Lambda).$$

If the fibres and total space are oriented compatibly, its image lies in $\mathrm{SL}(6, \mathbb{Z})$. For oriented peripheral loops $\gamma_1, \dots, \gamma_s$ satisfying $\gamma_1 \cdots \gamma_s = 1$, the matrices $M_i = \rho(\gamma_i)$ satisfy

$$M_1 M_2 \cdots M_s = I. \tag{893}$$

When $s \geq 1$, B° has $\pi_2(B^\circ) = 0$. The homotopy sequence of the regular bundle therefore gives the exact extension

$$1 \longrightarrow \Lambda \longrightarrow \pi_1(E) \longrightarrow \pi_1(B^\circ) \longrightarrow 1. \tag{894}$$

After choosing a section and a compatible basepoint system, (894) becomes an affine semidirect presentation; without those choices it remains the displayed exact extension.

Let $N_i = g^{-1}(D_i)$ be the filling over a small closed disc D_i about p_i, and let $A_i = E \cap N_i$, a T^6 mapping torus. Van Kampen gives

$$\pi_1(M) \cong \pi_1(E) \Big/ \Big\langle\!\Big\langle \ker\bigl(\pi_1(A_i) \to \pi_1(N_i)\bigr),\ 1 \leq i \leq s \Big\rangle\!\Big\rangle. \tag{895}$$

Thus an S^8 construction must derive the actual local inclusion maps and prove that their kernels normally generate all of (894). Monodromy matrices or Smith forms alone do not prove that (895) is trivial.

24.5 Integral homology: the exact gluing tests

Replace B° by a compact sphere with s open discs removed and replace E, N_i by compact collar-compatible pieces. Define the actual Mayer–Vietoris maps

$$\alpha_q = \bigoplus_{i=1}^{s}((j_{E,i})_*, -(j_{N,i})_*)\colon \bigoplus_i H_q(A_i;\mathbb{Z}) \longrightarrow H_q(E;\mathbb{Z}) \oplus \bigoplus_i H_q(N_i;\mathbb{Z}). \tag{896}$$

Because the pieces have nonempty boundary, their top homology vanishes. The Mayer–Vietoris sequence and $H_q(S^8;\mathbb{Z}) = 0$ for $1 \le q \le 7$ imply the following necessary and sufficient conditions:

q	condition on α_q
0	α_0 is injective and $\operatorname{coker}\alpha_0 \cong \mathbb{Z}$,
$1 \le q \le 6$	α_q is an isomorphism,
7	α_7 is surjective and $\ker\alpha_7 \cong \mathbb{Z}$.

(897)

The generator of $\ker\alpha_7$ is the boundary of the fundamental class of the closed eight-manifold obtained by gluing the pieces. Equations (896)–(897), using the actual inclusion maps, are the integral replication test for a proposed S^8 multipole construction.

Each boundary group is constrained by monodromy. If A_i is the mapping torus of $\phi_i\colon T^6 \to T^6$ and $M_i = (\phi_i)_*|_{H_1(T^6;\mathbb{Z})}$, the Wang sequence gives

$$0 \longrightarrow \operatorname{coker}\bigl(I - \textstyle\bigwedge^q M_i\bigr) \longrightarrow H_q(A_i;\mathbb{Z}) \longrightarrow \ker\bigl(I - \textstyle\bigwedge^{q-1} M_i\bigr) \longrightarrow 0. \tag{898}$$

The right group is free, so (898) splits as an abelian group, although not canonically. Since the regular base retracts to a wedge of $s-1$ circles, the homological Serre spectral sequence gives the filtration

$$0 \longrightarrow H_0\bigl(B^\circ;\textstyle\bigwedge^q \Lambda\bigr) \xrightarrow{\iota_q} H_q(E;\mathbb{Z}) \xrightarrow{\pi_q} H_1\Bigl(B^\circ;\textstyle\bigwedge^{q-1}\Lambda\Bigr) \longrightarrow 0. \tag{899}$$

To type both edge terms explicitly, put $V_q = \bigwedge^q \Lambda$, choose oriented free generators $\gamma_1, \dots, \gamma_{s-1}$ of $\pi_1(B^\circ)$, and write $M_j = \rho(\gamma_j)$. In the Wang sign convention the cellular differential is

$$d_q\colon \bigoplus_{j=1}^{s-1} V_q \longrightarrow V_q, \qquad (v_j)_j \longmapsto \sum_{j=1}^{s-1} (I - \textstyle\bigwedge^q M_j)\, v_j, \tag{900}$$

so the two homological edge groups are

$$H_0(B^\circ;V_q) = \operatorname{coker} d_q = (V_q)_\rho = \frac{V_q}{\sum_{j=1}^{s-1}(I - \bigwedge^q M_j)V_q}, \qquad H_1(B^\circ;V_{q-1}) = \ker d_{q-1}. \tag{901}$$

Thus the coinvariants are the Serre filtration subobject and the base-one-cycle group is the quotient. Since $\ker d_{q-1}$ is a subgroup of a free abelian group, it is free, and (899) splits noncanonically as an abstract abelian group.

Dually, set $V_q^\vee = \bigwedge^q \Lambda^\vee$; its monodromy along γ_j is $\bigwedge^q(M_j^{-\mathsf{T}})$. The cohomological filtration is

$$0 \longrightarrow H^1(B^\circ;V_{q-1}^\vee) \longrightarrow H^q(E;\mathbb{Z}) \longrightarrow H^0(B^\circ;V_q^\vee) \longrightarrow 0, \qquad \begin{aligned} H^0(B^\circ;V_q^\vee) &= (V_q^\vee)^\rho \\ &= \bigcap_{j=1}^{s-1} \ker\Bigl(I - \textstyle\bigwedge^q(M_j^{-\mathsf{T}})\Bigr). \end{aligned} \tag{902}$$

Accordingly, each boundary inclusion in α_q must send its Wang coinvariant filtration term through ι_q, and its base-one-cycle term must be tested after π_q; neither exact sequence supplies a canonical splitting. The corrected sequence identifies $H_q(E;\mathbb{Z})$ noncanonically with the direct sum of its two edge groups, so a rank or torsion count made from that direct sum survives. The natural edge maps do not: every map-level Mayer–Vietoris test must use (899) in the displayed order. Equations (896)–(902) retain every lattice, monodromy, kernel, coinvariant, invariant, and inclusion that a correct proof must compute.

24.6 The torus-action bound and a maximal real orbit model

For a smooth torus action on a compact manifold,

$$\chi(M) = \chi(M^T).$$

Since $\chi(S^8) = 2$, every torus action on S^8 has a fixed point. Equip S^8 with a T-invariant Riemannian metric. At a fixed point the isotropy representation is a sum of four real two-dimensional weight planes and possibly trivial summands. If the global action is effective, this tangent representation is faithful: an isometry fixing a point with identity derivative is the identity on the connected manifold. Four weights span a character lattice of rank at most four. Hence

$$\boxed{\operatorname{rank} T \le 4.} \tag{903}$$

In particular, a generic T^6 fibration of S^8 cannot be the orbit map of a global effective torus action. This does not rule out a monodromic T^6-bundle over the regular part of S^2, because such a bundle need not supply a global T^6 action.

The bound is sharp. Write

$$S^8 = \left\{ (z_1, z_2, z_3, z_4, y) \in \mathbb{C}^4 \oplus \mathbb{R} : \sum_{i=1}^{4} |z_i|^2 + y^2 = 1 \right\},$$

and let T^4 rotate the four complex coordinates. The orbit map is

$$q(z_1, z_2, z_3, z_4, y) = (|z_1|, |z_2|, |z_3|, |z_4|, y),$$

with image

$$Q = \left\{ (r_1, r_2, r_3, r_4, y) : r_i \ge 0, \sum r_i^2 + y^2 = 1 \right\}.$$

Two points have the same image exactly when they lie in the same orbit, so compact–Hausdorff descent gives $S^8/T^4 \cong Q$. Away from the two poles, write $r_i = \sqrt{1 - y^2}\, u_i$. Then

$$(u_1, u_2, u_3, u_4) \in \{ u_i \ge 0,\ \sum u_i^2 = 1 \} \cong \Delta^3,$$

and the values $y = \pm 1$ collapse the two ends. Therefore

$$\boxed{S^8/T^4 \cong \operatorname{Susp}(\Delta^3) \cong B^4.} \tag{904}$$

The locus on which all $r_i > 0$ has free T^4 orbit. Exactly j zero complex coordinates give isotropy T^j, and the two poles are fixed by the whole torus. This is a rigorous two-pole singular torus compactification of S^8, but it has four-dimensional quotient and generic T^4 orbit. It is not a three-special-fibre map to S^2.

24.7 A three-column Gram quotient whose quotient boundary is S^8

Let

$$V = \operatorname{Mat}_{3\times 3}(\mathbb{C}), \qquad \dim_{\mathbb{R}} V = 18,$$

and let $U(3)$ act on the left. Define

$$\Gamma \colon V \longrightarrow \operatorname{Herm}_3(\mathbb{C})_{\ge 0}, \qquad \Gamma(A) = A^*A. \tag{905}$$

The map is surjective because every $Q \ge 0$ is the Gram matrix of $Q^{1/2}$. It separates $U(3)$-orbits. Indeed, if $A^*A = B^*B$, the rule $Ax \mapsto Bx$ is a well-defined isometry from $\operatorname{im} A$ to $\operatorname{im} B$: equality of the Gram

matrices implies both equality of kernels and preservation of the Hermitian inner product. Extend this isometry to a unitary U of $\mathbb{C}^3$; then $B = UA$. Hence

$$V/U(3) \cong \mathrm{Herm}_3(\mathbb{C})_{\geq 0}. \tag{906}$$

For the Frobenius unit ball $\mathbb{B}(V) = \{A : \mathrm{tr}(A^*A) \leq 1\}$, (905) gives

$$\mathbb{B}(V)/U(3) \cong K := \{Q \in \mathrm{Herm}_3(\mathbb{C}) : Q \geq 0, \mathrm{tr}\, Q \leq 1\}. \tag{907}$$

The real vector space of Hermitian 3×3 matrices has dimension nine. The set K is compact, convex, and has nonempty interior $\{Q > 0, \mathrm{tr}\, Q < 1\}$. Radial projection from an interior point gives a homeomorphism of pairs $(K, \partial K) \cong (B^9, S^8)$, and therefore

$$\boxed{\partial(\mathbb{B}^{18}/U(3)) \cong S^8.} \tag{908}$$

Two qualifications are essential. First, the quotient of the source sphere is not the boundary in (908):

$$S^{17}/U(3) \cong \{Q \geq 0 : \mathrm{tr}\, Q = 1\}.$$

The space on the right is an eight-dimensional compact convex body and hence is homeomorphic to B^8. The rest of ∂K consists of rank-deficient Gram matrices with $\mathrm{tr}\, Q < 1$. Thus taking the boundary and taking the orbit quotient do not commute in this construction.

Second, strict $SU(3)$ is different. For invertible matrices, A^*A does not record the phase of $\det A$, while left multiplication by $SU(3)$ preserves $\det A$. The strict quotient is described by

$$(Q, d) = (A^*A, \det A), \qquad Q \geq 0, \qquad |d|^2 = \det Q. \tag{909}$$

It has generic real dimension ten, not nine. The construction in (908) therefore uses

$$U(3) \cong (SU(3) \times U(1))/\mu_3,$$

not colour $SU(3)$ alone. It is a precise S^8 real quotient boundary, but no morphism has been produced from its Gram or rank strata to the period matrix, torus bundle, logarithmic twists, or three special fibres of the Fabel–Alpöge construction.

24.8 Retaining the $T^4/\mathbb{R}^4$ fibre geometry and placing $SU(3)$ on the regular locus

This subsection classifies the required topological bundle data and its obstruction to crossing the singular fibres. It makes no physical identification.

24.8.1 The regular locus and its fourth cohomology

Let $X \cong_{\mathrm{diff}} S^6$ be the reconstructed candidate and put

$$D = W \sqcup S_1 \sqcup S_2, \qquad U = X \setminus D = f^{-1}(B^\circ), \qquad B^\circ = \mathbb{P}^1 \setminus \{p_0, p_1, p_2\}.$$

Here W is the non-normal central fibre and S_1, S_2 are the underlying smooth bielliptic surfaces of the two multiple fibres. The reconstructed integral calculations give

$$H_1(W; \mathbb{Z}) \cong \mathbb{Z}^2, \qquad H_1(S_1; \mathbb{Z}) \cong H_1(S_2; \mathbb{Z}) \cong \mathbb{Z}^2.$$

The compact analytic set D is triangulable, so Alexander duality in $X \cong S^6$ applies:

$$\widetilde{H}^i(U; \mathbb{Z}) \cong \widetilde{H}_{5-i}(D; \mathbb{Z}).$$

At $i = 4$, it gives

$$\boxed{H^4(U;\mathbb{Z}) \cong H_1(D;\mathbb{Z}) \cong \mathbb{Z}^6.} \tag{910}$$

There is an independent fibrewise description. The pair of pants B° retracts to a wedge of two circles, and the T^4 bundle retracts to its restriction over that graph. Hence U has the homotopy type of a CW complex of dimension at most five. If $\mathcal{V}^q = R^q f_* \mathbb{Z}$ is the monodromy local system, the Serre spectral sequence has only columns zero and one and yields

$$0 \longrightarrow H^1(B^\circ;\mathcal{V}^3) \longrightarrow H^4(U;\mathbb{Z}) \xrightarrow{\mathrm{res}_F} H^0(B^\circ;\mathcal{V}^4) \longrightarrow 0. \tag{911}$$

Holomorphic fibre monodromy preserves the real orientation, so $H^0(B^\circ;\mathcal{V}^4) \cong \mathbb{Z}$. Thus one coordinate of (910) is the constant evaluation on a smooth T^4 fibre; the remaining rank-five kernel in (911) consists of mixed base–fibre classes.

24.8.2 Exact classification of $SU(3)$ bundles on U

Let $P \to U$ be a principal $SU(3)$ bundle and $E_P = P \times_{SU(3)} \mathbb{C}^3$ its determinant-trivial rank-three complex bundle. Through degree five,

$$\pi_i(BSU(3)) = 0 \quad (i = 1, 2, 3, 5), \qquad \pi_4(BSU(3)) \cong \mathbb{Z},$$

with universal generator c_2. Because U has CW dimension at most five, obstruction theory gives a bijection

$$\boxed{\{\text{principal } SU(3) \text{ bundles on } U\}/\cong \xrightarrow{\ c_2\ } H^4(U;\mathbb{Z}) \cong \mathbb{Z}^6.} \tag{912}$$

The fibre integer

$$k_F = \langle c_2(E_P)|_{F_b}, [F_b] \rangle, \qquad b \in B^\circ, \tag{913}$$

is independent of b. A nonzero k_F is only one possible obstruction: a class with $k_F = 0$ may still have a nonzero mixed component in the kernel of (911).

24.8.3 Extension across the three special fibres

The cohomology sequence of the pair (X, U) contains

$$H^4(X;\mathbb{Z}) \longrightarrow H^4(U;\mathbb{Z}) \xrightarrow{\delta} H^5(X,U;\mathbb{Z}) \longrightarrow H^5(X;\mathbb{Z}).$$

Both outer groups vanish because $X \cong S^6$. Therefore

$$\boxed{\delta\colon H^4(U;\mathbb{Z}) \xrightarrow{\ \cong\ } H^5(X,U;\mathbb{Z}).} \tag{914}$$

Every $SU(3)$ bundle on all of X has $c_2 = 0$, so its restriction to U is trivial by (912). Conversely, the trivial bundle on U extends as the trivial bundle on X. Thus, for unframed topological bundles,

$$\boxed{P \to U \text{ extends across } D \iff c_2(P) = 0 \iff \delta c_2(P) = 0.} \tag{915}$$

Under Alexander duality, (914) identifies the obstruction with a six-integer class supported on the one-cycles of $W \sqcup S_1 \sqcup S_2$. Because the three components are disjoint and $H_1(D)$ is their direct sum, nonzero components do not vanish merely by declaring that charges at distinct special fibres cancel. They cancel as an extension obstruction only when the complete class in $\mathbb{Z}^6$ is zero.

This is a statement about the bundle. Even when $c_2 = 0$, a specified connection on the trivial bundle may fail to extend smoothly because of its holonomy or growth near D; this is an analytic boundary-value problem not decided by (915).

24.8.4 The distinct six-dimensional clutching charge

Principal $SU(3)$ bundles on the complete sphere are classified by

$$[S^6, BSU(3)] \cong \pi_5(SU(3)) \cong \mathbb{Z}.$$

If k is the clutching integer, Bott integrality fixes the top Chern class as

$$c_3(E_P) = \pm 2k\, u \in H^6(S^6; \mathbb{Z}). \tag{916}$$

Every such global bundle restricts to the trivial bundle on U, because its restricted c_2 vanishes and (912) is complete. Thus a global six-dimensional $SU(3)$ clutching charge can be concentrated in the gluing across the deleted fibres, but it is a $c_3/\pi_5(SU(3))$ class, not a four-dimensional c_2 instanton class on the regular fibres.

Alexander duality also gives $H^5(U; \mathbb{Z}) \cong \widetilde{H}_0(D; \mathbb{Z}) \cong \mathbb{Z}^2$. The pair sequence therefore yields

$$0 \longrightarrow \mathbb{Z}^2 \longrightarrow H^6(X, U; \mathbb{Z}) \longrightarrow \mathbb{Z} \longrightarrow 0,$$

and hence

$$H^6(X, U; \mathbb{Z}) \cong \mathbb{Z}^3. \tag{917}$$

After choosing the complex orientations of the three fibre components, the three relative top classes map to the global fundamental class by the augmentation

$$(n_0, n_1, n_2) \longmapsto n_0 + n_1 + n_2. \tag{918}$$

The rank-two kernel records redistribution among the three supports with the same global total. To promote (917)–(918) from exact cohomological data to a relative $SU(3)$ bundle theorem with prescribed framings, one must give the three clutching maps and compute their homotopy classes; arbitrary relative cohomology integers are not silently declared realised.

24.8.5 Data required before a gauge-theory conclusion

A mathematical colour or gauge proposal on U must specify:

1. a principal $SU(3)$ bundle $P \to U$, equivalently its complete class $c_2(P) \in \mathbb{Z}^6$;
2. a connection A and a fixed trace convention for its curvature Chern–Weil form;
3. tubular boundary conditions or framings at each of W, S_1, S_2;
4. whether the invariant under study is the fibre number (913), the extension defect (914), or the six-dimensional clutching number (916); and
5. an explicit analytic statement about a Yang–Mills operator, action, or spectrum.

The topology alone proves no mass gap, confinement statement, particle identification, or Yang–Mills dynamics. It supplies a carrier and an exact obstruction ledger on which such a calculation can be posed.

24.9 Consequences for the higher-rung search

Replacing S^6 by S^8 does not permit a literal smooth complex-fourfold version of the Fabel–Alpöge construction. A real T^6-over-S^2 singular fibration could still exist, but it cannot arise as a global effective torus action and must pass the map-level tests (892), (895), and (897). The standard T^4 action gives an exact and economical real S^8 multipole model, but its base is B^4, not S^2, and its singular locus is a face stratification rather than three isolated fibres.

The 3×3 $U(3)$ Gram construction gives an exact quotient boundary S^8 and is the strongest precise realisation presently obtained for the "one rung higher" intuition. Strict $SU(3)$ retains the determinant phase and does not give the same quotient. Keeping the original T^4 fibres and placing $SU(3)$ on $X \setminus D$ is instead mathematically well typed: its unframed bundle classes form $\mathbb{Z}^6$, every nonzero c_2 is an exact nonextendable divisor defect, and global $SU(3)$ bundles on S^6 carry the even-valued c_3 charge in (916) while restricting trivially to the regular locus.

None of these statements identifies a lepton, quark, or four-dimensional Yang–Mills theory.

Sources. The almost-complex obstruction is cross-checked against A. Borel and J.-P. Serre, *Groupes de Lie et puissances réduites de Steenrod*, Amer. J. Math. **75** (1953), 409–448, Proposition 15.1, `doi:10.2307/2372495`, and P. Konstantis and M. Parton, *Almost complex structures on spheres*, Differential Geom. Appl. **57** (2018), 10–22, `doi:10.1016/j.difgeo.2017.10.010`. The fixed-point identity used in (903) is from S. Kobayashi, *Fixed points of isometries*, Nagoya Math. J. **13** (1958), 63–68, `doi:10.1017/S0027763000023497`. For singular torus fibrations and the coordinate sphere action, see respectively Y. Matsumoto, *Torus fibrations over the 2-sphere with the simplest singular fibers*, J. Math. Soc. Japan **37** (1985), 605–636, `doi:10.2969/jmsj/03740605`, and H. Ishida, Y. Fukukawa, and M. Masuda, *Topological toric manifolds*, Moscow Math. J. **13** (2013), 57–98, `doi:10.17323/1609-4514-2013-13-1-57-98`. The theorem-level calculations above are given in full and do not rely on these citations as substitutes for omitted morphisms.

24.10 The Koide cone and the canonical period discriminant

For $r = (r_1, r_2, r_3) \in \mathbb{R}^3$, set

$$Q_K(r) := \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2) \tag{919}$$

and introduce the orthogonal coordinates

$$\begin{pmatrix} h \\ x \\ y \end{pmatrix} = \mathcal{F} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}, \qquad \mathcal{F} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix}. \tag{920}$$

We use exactly the canonical period convention

$$Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}, \qquad \Pi = [Z \mid I_2]. \tag{921}$$

Write

$$t = \operatorname{Im}\tau, \qquad m = \operatorname{Im}\mu, \qquad \varrho = \operatorname{Im}\beta.$$

Relative to the period-lattice basis

$$(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$$

and the interleaved real codomain basis

$$(\operatorname{Re} z_1, \operatorname{Im} z_1, \operatorname{Re} z_2, \operatorname{Im} z_2),$$

the realification of Π is

$$P(\tau, \mu, \beta) = \begin{pmatrix} 6\operatorname{Re}\mu & \operatorname{Re}\tau & 1 & 0 \\ 6m & t & 0 & 0 \\ \operatorname{Re}\beta & \operatorname{Re}\mu & 0 & 1 \\ \varrho & m & 0 & 0 \end{pmatrix}. \tag{922}$$

For comparison with the Lorentz cone, define the auxiliary symmetric coefficient matrix

$$S(t,m,\varrho) := \begin{pmatrix} t & \sqrt{6}\,m \\ \sqrt{6}\,m & \varrho \end{pmatrix}. \tag{923}$$

Thus S represents the same coefficient discriminant, but it is not the canonical 4×4 real period matrix P.

Theorem 24.1 (Ambient coefficient-space discriminant isomorphism)**.** *The matrix $\mathcal{F}$ is orthogonal, and*

$$Q_K(r) = h^2 - x^2 - y^2. \tag{924}$$

The real-linear map

$$\mathcal{K}(h,x,y) = (\operatorname{Im}\tau, \operatorname{Im}\mu, \operatorname{Im}\beta) = \left(h+y, \frac{x}{\sqrt{6}}, h-y\right) \tag{925}$$

is an isomorphism from $\mathbb{R}^3$ onto the ambient real coefficient space, with inverse

$$\mathcal{K}^{-1}(t,m,\varrho) = \left(\frac{t+\varrho}{2}, \sqrt{6}\,m, \frac{t-\varrho}{2}\right). \tag{926}$$

For every choice of the real parts of τ,μ,β,

$$\det P(\tau,\mu,\beta) = \det S(t,m,\varrho) = t\varrho - 6m^2. \tag{927}$$

Consequently, if $(t,m,\varrho) = \mathcal{K}(h,x,y)$, then

$$\det P(\tau,\mu,\beta) = \det S(\mathcal{K}(h,x,y)) = Q_K(r). \tag{928}$$

Moreover, for $r \in \mathbb{R}^3_{\geq 0}\setminus\{0\}$, the equation $Q_K(r)=0$ is equivalent to the assertion that

$$S(\mathcal{K}(\mathcal{F}(r))) = \begin{pmatrix} h+y & x \\ x & h-y \end{pmatrix} \tag{929}$$

is positive semidefinite of rank one.

Proof. The rows of $\mathcal{F}$ have norm one and pairwise scalar product zero, so $\mathcal{F}\mathcal{F}^{\mathsf{T}} = I_3$. Hence

$$r_1^2 + r_2^2 + r_3^2 = h^2 + x^2 + y^2, \qquad r_1 + r_2 + r_3 = \sqrt{3}\,h.$$

Substitution in equation (919) proves equation (924).

Adding and subtracting the first and third entries of equation (925), and multiplying the second by $\sqrt{6}$, proves equation (926). Expanding equation (922) along its last two columns gives

$$\det P = -\det\begin{pmatrix} 6m & t \\ \varrho & m \end{pmatrix} = t\varrho - 6m^2.$$

Equivalently, in the exact canonical convention, $\det P = -\det \operatorname{Im} Z$. Taking the determinant in equation (923) proves the other equality in equation (927), while

$$(h+y)(h-y) - 6\left(\frac{x}{\sqrt{6}}\right)^2 = h^2 - x^2 - y^2$$

proves equation (928).

Finally, let $r \in \mathbb{R}^3_{\geq 0}\setminus\{0\}$. Then $h = (r_1+r_2+r_3)/\sqrt{3} > 0$. If $Q_K(r)=0$, the symmetric matrix in equation (929) has determinant zero and trace $2h>0$; its eigenvalues are therefore 0 and $2h$, so it is positive semidefinite of rank one. Conversely, rank one forces its determinant, and hence $Q_K(r)$, to vanish. □

Corollary 24.2 (The Koide cone is not a family of regular tori)**.** *The identification in theorem 24.1 is only a real-linear isomorphism of ambient coefficient spaces together with their quadratic discriminants. It neither supplies the real parts of the periods nor imposes the holomorphic and equivariance conditions of the canonical period family. At every point of the Koide cone, every matrix of the canonical form equation* (922) *with coefficient triple equation* (925) *satisfies* $\det P = 0$*. Its columns therefore do not span* $\mathbb{R}^4$*, do not generate a full lattice, and do not define a regular compact complex two-torus. Thus the ambient discriminant identity is not a map from Koide triples to regular fibres of the canonical* S^6 *family.*

Corollary 24.3 (Separation from the actual canonical cusp)**.** *In the notation*

$$L_T = T + H_T, \qquad q_T = L_T - B_T, \qquad D_T = L_T q_T + 6m_T^2$$

of the positive radial cusp, the already-proved determinant formula equation (14) *gives*

$$\det P_T = -D_T < 0. \tag{930}$$

Moreover, with $H = \operatorname{Im} h(0)$ *and* $B = \operatorname{Im} b(0)$*, equation* (33) *gives*

$$D_T = T^2 + (2H - B)T + O(1), \qquad \det P_T \sim -T^2. \tag{931}$$

Thus every actual radial cusp fibre is regular and has nonzero oriented period determinant, whereas every point in the Koide image has determinant zero.

Proof. The first assertion is exactly equation (14); the stated asymptotic follows from equation (33). Since $D_T > 0$, the matrix P_T is invertible and its columns form the full period lattice of the corresponding regular fibre. □

25 Exact arithmetic of the Koide defect-one shell

This section isolates an arithmetic result that does not use the existence of the proposed complex structure on S^6, any period-map assertion, or any Yang–Mills interpretation. All coordinates, chambers, local factors, and normalizations used in the final constant are fixed explicitly below.

For $r = (r_1, r_2, r_3)$, define the Koide quadratic form

$$Q_K(r_1, r_2, r_3) := \frac{2}{3}(r_1 + r_2 + r_3)^2 - (r_1^2 + r_2^2 + r_3^2). \tag{932}$$

For integral coordinates put

$$\Delta(x, y, z) := 4(xy + xz + yz) - (x^2 + y^2 + z^2). \tag{933}$$

Expanding (932) gives the exact identity

$$\Delta(x, y, z) = 3Q_K(x, y, z). \tag{934}$$

Thus the equation studied here, $\Delta = -1$, is the nonzero level $Q_K = -1/3$; it is not an affine translate of the null cone $Q_K = 0$.

25.1 Vieta reduction and exact ternary coordinates

Regarded as a monic quadratic equation in z, the defect equation is

$$z^2 - 4(x + y)z + x^2 + y^2 - 4xy - 1 = 0. \tag{935}$$

Consequently the other root is

$$z' = 4(x+y) - z. \tag{936}$$

The analogous substitutions in x and y are integral involutions preserving Δ.

Let e_1, e_2, e_3 be the standard basis and let B_K be the integral bilinear form with Gram matrix

$$A_K = \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}, \qquad B_K(u,v) = u^{\mathsf{T}} A_K v. \tag{937}$$

Thus $\Delta(u) = B_K(u,u)$. The eigenvalue of A_K on $\mathbb{R}(1,1,1)$ is 3, and its eigenvalue on the orthogonal complement is -3, so this integral presentation has signature $(1,2)$.

Proposition 25.1 (Vieta moves are Lorentzian reflections)**.** *For $i = 1,2,3$, define*

$$s_i(u) = u - \frac{2B_K(u,e_i)}{B_K(e_i,e_i)} e_i. \tag{938}$$

Then $s_i \in \mathrm{O}(B_K, \mathbb{Z})$, $s_i^2 = 1$, and $\det s_i = -1$. Explicitly,

$$s_1(x,y,z) = (-x + 4y + 4z, y, z), \tag{939}$$
$$s_2(x,y,z) = (x, -y + 4x + 4z, z), \tag{940}$$
$$s_3(x,y,z) = (x, y, -z + 4x + 4y). \tag{941}$$

Consequently these are exactly the three Vieta root substitutions for $\Delta = -1$, now identified as reflections of the unchanged Lorentzian lattice $(\mathbb{Z}^3, B_K)$.

Proof. Since $B_K(e_i,e_i) = -1$, formula (938) is integral and direct evaluation gives (939)– (941). The standard reflection calculation gives

$$B_K(s_i(u), s_i(v)) = B_K(u,v), \qquad s_i^2 = 1.$$

The map fixes the hyperplane $e_i^{\perp}$ pointwise and sends e_i to $-e_i$, hence has determinant -1. Finally, when the equation is viewed as a monic quadratic in one coordinate, the displayed replacement is the sum of its two roots minus the original root, proving the last assertion. □

Proposition 25.2 (Strict Vieta reduction)**.** *Let $0 < x \le y \le z$ be integers satisfying $\Delta(x,y,z) = -1$. If $z < 4(x+y)$, then the other root in* (936) *satisfies $0 < z' < y$. If $z \ge 4(x+y)$, then $z' \le 0$.*

Proof. In the first case $z' > 0$ follows immediately from (936). Let $P(Z)$ be the polynomial on the left of (935). Since $x \le y$,

$$P(y) = x^2 - 8xy - 2y^2 - 1 \le xy - 8xy - 2y^2 - 1 < 0.$$

But $P(y) = (y-z)(y-z')$, and $y - z \le 0$; hence $y - z' > 0$, so $z' < y$. The second assertion again follows directly from (936). □

Define the reduced-root set and its height count by

$$\mathcal{R} := \{(x,y,z) \in \mathbb{Z}^3_{>0} : x \le y \le z,\ \Delta(x,y,z) = -1,\ z \ge 4(x+y)\}, \tag{942}$$
$$R(X) := \#\{(x,y,z) \in \mathcal{R} : z \le X\}. \tag{943}$$

Proposition 25.2 shows that the wall $z \ge 4(x+y)$ is exactly the terminal side of the decreasing Vieta move after the coordinates have been ordered.

Proposition 25.3 (The positive Vieta graph is a rooted forest)**.** *Let $\mathcal{S}^+$ be the set of unordered triples of positive integers satisfying $\Delta = -1$. Join two vertices in a simple graph when one is obtained from the other by a Vieta reflection s_i and the reflected coordinate is positive. Every connected component of this graph is a tree with exactly one root, and the roots are precisely the unordered triples whose ordered representatives lie in $\mathcal{R}$. Equivalently, every positive integral solution has a unique reduced representative, and it is obtained by repeatedly ordering the coordinates and reflecting the largest coordinate whenever it lies below the terminal wall.*

Proof. First, no Vieta reflection fixes a point of $\Delta = -1$. Indeed, a fixed reflection in a coordinate w, with the other two coordinates denoted by u, v, would give $w = 2(u + v)$, and substitution in the unchanged form gives

$$\Delta(u, v, 2(u + v)) = 3(u^2 + 4uv + v^2) > 0.$$

Thus the sum of the three coordinates changes strictly across every edge.

Now order a vertex as $0 < x \le y \le z$. In fact $y < z$, because $y = z$ would give

$$\Delta(x, y, y) = -x^2 + 8xy + 2y^2 > 0.$$

Reflecting either of the two smaller coordinates produces a new coordinate strictly larger than z:

$$4(y + z) - x > z, \qquad 4(x + z) - y > z.$$

If $z < 4(x + y)$, Proposition 25.2 says that reflection in the largest coordinate produces $0 < z' < y$ and strictly decreases the coordinate sum. If $z \ge 4(x + y)$, that reflection is nonpositive and hence is not an edge of $\mathcal{S}^+$. Therefore every nonterminal vertex has exactly one neighbour of smaller coordinate sum, whereas a terminal vertex has none.

Iteration of the unique decreasing move terminates because the coordinate sum is a positive integer. The endpoint is represented by an element of $\mathcal{R}$. The graph has no cycle: on a putative simple cycle choose a vertex of largest coordinate sum; its two neighbours on the cycle would be two distinct smaller neighbours, contrary to uniqueness. Thus each component is a tree. A tree in which following the unique decreasing edge always terminates has exactly one terminal vertex, proving both uniqueness and the characterization of the roots. □

For an ordered triple define

$$n = z - 4(x + y), \qquad q = y - x, \qquad a = x + y - 4n, \tag{944}$$

and let

$$\mathcal{T} := \left\{ (a, q, n) \in \mathbb{Z}^3 : \begin{array}{c} a > 0,\ q \ge 0,\ n \ge 0,\ a + 4n > q, \\ a^2 - 3q^2 - 18n^2 = -2 \end{array} \right\}. \tag{945}$$

Theorem 25.4 (Integral ternary-coordinate bijection)**.** *The map in* (944) *restricts to a bijection $\Phi : \mathcal{R} \to \mathcal{T}$. Its inverse is*

$$(a, q, n) \longmapsto \left(\frac{a + 4n - q}{2}, \frac{a + 4n + q}{2}, 4a + 17n \right). \tag{946}$$

For arbitrary integral (x, y, z) and the associated (a, q, n),

$$2\Delta(x, y, z) = a^2 - 3q^2 - 18n^2, \qquad z = 4a + 17n. \tag{947}$$

In particular,

$$R(X) = \#\{(a, q, n) \in \mathcal{T} : 4a + 17n \le X\}. \tag{948}$$

Proof. Put $p = x + y$. Since $q = y - x$, one has $2x = p - q$ and $2y = p + q$, and direct substitution into (933) gives

$$2\Delta = p^2 - 3q^2 + 8pz - 2z^2.$$

The definition $n = z - 4p$ gives $z = 4p + n$, whence

$$2\Delta = p^2 - 3q^2 - 8pn - 2n^2.$$

Finally $a = p - 4n$, equivalently $p = a + 4n$, gives both identities in (947).

Suppose first that $(x, y, z) \in \mathcal{R}$. Then $q, n \geq 0$, while $a + 4n = p > q$ because $p - q = 2x > 0$. The defect equation and (947) give $a^2 - 3q^2 - 18n^2 = -2$. If $n = 0$, then $a = p > 0$. If $n \geq 1$, then

$$a^2 = 3q^2 + 18n^2 - 2 \geq 18n^2 - 2 \geq 16n^2.$$

Were $a \leq 0$, this inequality would force $a \leq -4n$, contrary to $a + 4n > q \geq 0$. Hence $a > 0$, and the forward map lands in $\mathcal{T}$.

Conversely, reducing the ternary equation modulo 2 gives $a \equiv q \pmod 2$, so all three coordinates in (946) are integral. Put $p = a + 4n$. The chamber inequality gives

$$x = \frac{p - q}{2} > 0, \qquad y = \frac{p + q}{2} \geq x.$$

Moreover

$$z = 4a + 17n = 4p + n \geq 4p > p > y.$$

The identities (947) now give $\Delta = -1$ and $z - 4(x + y) = n \geq 0$. Thus the inverse map lands in $\mathcal{R}$. The displayed formulae show directly that the two compositions are the identity. The last assertion follows from the height identity in (947). □

Proposition 25.5 (Complete Pell classification of the chamber faces)**.** *The integral points of $\mathcal{T}$ on its two coordinate faces are the following two disjoint families.*

On the terminal Vieta wall $n = 0$, define integers a_k, q_k by

$$a_k + q_k\sqrt{3} = (1 + \sqrt{3})(2 + \sqrt{3})^k \qquad (k \geq 0). \tag{949}$$

Then the points of $\mathcal{T}$ with $n = 0$ are exactly $(a_k, q_k, 0)$ for $k \geq 1$, and their reduced roots are

$$\left(\frac{a_k - q_k}{2}, \frac{a_k + q_k}{2}, 4a_k\right). \tag{950}$$

The first is $(1, 4, 20)$.

On the symmetry face $q = 0$, define integers A_k, N_k by

$$A_k + N_k\sqrt{18} = (4 + \sqrt{18})(17 + 4\sqrt{18})^k \qquad (k \geq 0). \tag{951}$$

Then the points of $\mathcal{T}$ with $q = 0$ are exactly $(A_k, 0, N_k)$ for $k \geq 0$, and their reduced roots are

$$\left(\frac{A_k + 4N_k}{2}, \frac{A_k + 4N_k}{2}, 4A_k + 17N_k\right). \tag{952}$$

The first is $(4, 4, 33)$.

In the projectivized real quadric these families tend, respectively, to the two chamber-boundary rays

$$(a, q, n) = \mathbb{R}_{>0}(\sqrt{3}, 1, 0), \qquad (a, q, n) = \mathbb{R}_{>0}(3\sqrt{2}, 0, 1). \tag{953}$$

Each family contributes only $O(\log X)$ points to $R(X)$.

Proof. On $n=0$, the equation of $\mathcal{T}$ is $a^2-3q^2=-2$. Here $q=0$ is impossible. If $q=1$, positivity gives $a=1$, which fails the strict chamber inequality $a>q$. For any positive solution with $q\geq 2$, multiplication by the norm-one unit $2-\sqrt{3}$ gives

$$a'+q'\sqrt{3}=(a+q\sqrt{3})(2-\sqrt{3}),\qquad a'=2a-3q,\quad q'=2q-a. \tag{954}$$

The unchanged equation implies

$$a<2q,\qquad a>\frac{3}{2}q,\qquad a>q\quad (q\geq 2),$$

because its squares differ from $4q^2$, $9q^2/4$, and q^2 by, respectively, q^2+2, $3q^2/4-2>0$, and $2q^2-2>0$. Thus $a',q'>0$ and $q'<q$. Repeated descent reaches $(1,1)$, and reversing the descent proves (949) is the complete positive solution set. The recurrence

$$a_{k+1}=2a_k+3q_k,\qquad q_{k+1}=a_k+2q_k$$

shows inductively that a_k,q_k are both odd. The chamber excludes only $k=0$, and Theorem 25.4 now gives (950).

On $q=0$, one has $a^2-18n^2=-2$. Necessarily $n>0$. Direct substitution for $1\leq n\leq 5$ gives the sole solution $(a,n)=(4,1)$. If $n\geq 6$, multiplication by the norm-one unit $17-4\sqrt{18}$ gives

$$a'+n'\sqrt{18}=(a+n\sqrt{18})(17-4\sqrt{18}),\qquad a'=17a-72n,\quad n'=17n-4a. \tag{955}$$

Both new coordinates are positive. Indeed,

$$\frac{a}{n}=\sqrt{18-\frac{2}{n^2}}>\frac{72}{17}\quad (n\geq 6),\qquad \frac{a}{n}<\sqrt{18}<\frac{17}{4}.$$

Moreover $a>4n$ and hence $0<n'<n$. Descent therefore reaches $(4,1)$, and reversal proves (951) is the complete positive solution set. Its recurrence,

$$A_{k+1}=17A_k+72N_k,\qquad N_{k+1}=4A_k+17N_k,$$

keeps A_k even and N_k odd, so (952) follows from the inverse coordinate formula.

Conjugating the two identities defining the sequences shows

$$\frac{a_k}{q_k}\longrightarrow\sqrt{3},\qquad \frac{A_k}{N_k}\longrightarrow\sqrt{18}=3\sqrt{2},$$

which proves (953). Finally, the dominant factors $2+\sqrt{3}>1$ and $17+4\sqrt{18}>1$ give exponential growth in k, whereas the heights in (950) and (952) are positive linear forms in the displayed coordinates. Hence either face contains only $O(\log X)$ points of height at most X. □

25.2 The affine quadric and its local densities

Put

$$F(a,q,n):=a^2-3q^2-18n^2,\qquad \mathscr{X}:=\{F=-2\}\subset\mathbb{A}^3. \tag{956}$$

The matrix of F is $A=\operatorname{diag}(1,-3,-18)$, so $\det A=54$, and $-(-2)\det A=108$ is not a square. The integral point $x_0=(1,1,0)$ lies on $\mathscr{X}$.

For every prime p, define

$$N_{p,k}:=\#\{(a,q,n) \bmod p^k : F(a,q,n)\equiv -2 \pmod{p^k}\},\qquad \sigma_p:=\lim_{k\to\infty}p^{-2k}N_{p,k}. \tag{957}$$

Theorem 25.6 (Exact local densities)**.** *The limits in* (957) *exist and are*

$$\sigma_2 = \frac{3}{2}, \qquad \sigma_3 = 2, \qquad \sigma_p = \frac{1 - p^{-2}}{1 - \left(\frac{3}{p}\right) p^{-1}} \quad (p \geq 5). \tag{958}$$

Writing χ_{12} for the primitive real character modulo 12, the singular series, with the primes taken in increasing order (equivalently, with Borovoi's convergence-factor prescription), is therefore

$$\mathfrak{S}(\mathscr{X}) := \prod_p \sigma_p = \frac{9}{2} \frac{L(1, \chi_{12})}{\zeta(2)} = \frac{9}{2} \frac{\log(2 + \sqrt{3})}{\sqrt{3}\, \zeta(2)}. \tag{959}$$

Proof. We compute the two bad primes first.

For $p = 3$, once $q, n \bmod 3^k$ have been chosen, the equation for a is

$$a^2 \equiv -2 + 3q^2 + 18n^2 \equiv 1 \pmod 3.$$

The two roots modulo 3 are nonsingular, and each has a unique Hensel lift modulo every 3^k. Thus

$$N_{3,k} = 2 \cdot 3^{2k}, \tag{960}$$

which gives $\sigma_3 = 2$.

For $p = 2$, reduction modulo 4 leaves exactly two parity branches:

$$(a, q, n) \equiv (0, 0, 1) \pmod 2, \qquad (a, q, n) \equiv (1, 1, 0) \pmod 2. \tag{961}$$

Fix $k \geq 4$ and put $j = k - 1$. On the first branch write $a = 2A$, $q = 2Q$, and put $r = 3n$. Then

$$F(a, q, n) + 2 = -2\big(r^2 - 2A^2 + 6Q^2 - 1\big). \tag{962}$$

Hence we must solve

$$r^2 \equiv 1 + 2A^2 - 6Q^2 \pmod{2^j}.$$

An odd unit is a square modulo 2^j, $j \geq 3$, precisely when it is $1 \pmod 8$, in which case it has four roots. Here that condition is equivalent to A, Q both being even. There are 2^{2j-2} such pairs $(A, Q) \bmod 2^j$, four choices of $r \bmod 2^j$, and, because 3 is a unit, two odd choices of $n \bmod 2^k$ above each such r. This branch therefore contributes

$$2^{2j-2} \cdot 4 \cdot 2 = 2^{2k-1} \tag{963}$$

solutions modulo 2^k.

On the second branch write $n = 2m$ and set

$$v = \frac{a - q}{2}, \qquad r = \frac{-a + 3q}{2}. \tag{964}$$

Then r, v have opposite parity, the inverse formula is $a = r + 3v$, $q = r + v$, and

$$F(a, q, n) + 2 = -2\big(r^2 - 3v^2 + 36m^2 - 1\big). \tag{965}$$

First suppose $v = 2V$ and r is odd. For fixed $V \bmod 2^{j-1}$ and $m \bmod 2^j$, the equation for r is

$$r^2 \equiv 1 + 12V^2 - 36m^2 \pmod{2^j}.$$

Its right side is $1 \pmod 8$ exactly when V, m have the same parity. There are $2 \cdot 2^{j-2} 2^{j-1} = 2^{2j-2}$ such pairs, and four roots r, so this subbranch gives 2^{2j} triples $(r, v, m) \bmod 2^j$.

Now suppose $r = 2R$ and v is odd. The congruence in (965), solved for v^2, has an odd square on the left; its right side is $1 \pmod 8$ exactly when R, m have opposite parity. Again there are 2^{2j-2} admissible pairs and four square roots, hence 2^{2j} transformed triples.

It remains to account for the divided-coordinate map. Each class $(r, v) \bmod 2^j$ of opposite parity has exactly two inverse classes $(a, q) \bmod 2^k$: shifting either r or v by 2^j changes both $a = r + 3v$ and $q = r + v$ by 2^j, while shifting both changes neither modulo 2^k. The class $m \bmod 2^j$ gives one even class $n = 2m \bmod 2^k$. Thus the second branch contributes

$$2(2^{2j} + 2^{2j}) = 2^{2k}. \tag{966}$$

Combining (963) and (966) gives

$$N_{2,k} = 3 \cdot 2^{2k-1} \qquad (k \geq 4), \tag{967}$$

and hence $\sigma_2 = 3/2$.

For $p \geq 5$, one has $p \nmid 108$. Borovoi's good-prime formula [51, Section 1.3] gives

$$\sigma_p = \frac{1 - p^{-2}}{1 - \chi(p)p^{-1}}, \qquad \chi(p) = \left(\frac{-(-2)\det A}{p}\right) = \left(\frac{108}{p}\right) = \left(\frac{3}{p}\right),$$

which proves the remaining formula in (958). Since $\chi(p) = \chi_{12}(p)$ for $p \geq 5$,

$$\prod_{p \geq 5} \sigma_p = L(1, \chi_{12}) \prod_{p \geq 5} (1 - p^{-2}) = \frac{3}{2} \frac{L(1, \chi_{12})}{\zeta(2)}.$$

Multiplication by $\sigma_2 \sigma_3 = 3$ gives the first equality in (959). Finally, the residue classes on which χ_{12} is nonzero give

$$\begin{aligned} L(1, \chi_{12}) &= \sum_{m \geq 0} \left(\frac{1}{12m+1} - \frac{1}{12m+5} - \frac{1}{12m+7} + \frac{1}{12m+11} \right) \\ &= \int_0^1 \frac{1 - x^2}{x^4 - x^2 + 1} \, \mathrm{d}x = \frac{\log(2 + \sqrt{3})}{\sqrt{3}}, \end{aligned}$$

where the middle equality follows by summing the geometric series and the last by an elementary partial-fraction integration. This completes the local computation. □

25.3 The archimedean chamber factor

The Poincaré-residue gauge form on the whole smooth affine quadric is the unique regular 2-form ω_∞ whose local representatives satisfy

$$\omega_\infty \wedge \mathrm{d}F = \mathrm{d}a \wedge \mathrm{d}q \wedge \mathrm{d}n, \tag{968}$$

Explicitly, on the three standard charts it is

$$\omega_\infty = \frac{\mathrm{d}q \wedge \mathrm{d}n}{2a} \quad (a \neq 0), \qquad \omega_\infty = \frac{\mathrm{d}a \wedge \mathrm{d}n}{6q} \quad (q \neq 0), \qquad \omega_\infty = -\frac{\mathrm{d}a \wedge \mathrm{d}q}{36n} \quad (n \neq 0). \tag{969}$$

These expressions agree on overlaps after restriction to the tangent bundle of $F = -2$. Since the gradient $(2a, -6q, -36n)$ cannot vanish on that level, the three charts cover $\mathscr{X}$, so ω_∞ is globally defined and nowhere vanishing. Let $m_\infty := |\omega_\infty|$ be its associated positive measure. On the sheet $a > 0$, used for the chamber count, the first expression in (969) applies. Let

$$\mathcal{C} := \{(a, q, n) \in \mathbb{R}^3 : a > 0,\ q \geq 0,\ n \geq 0,\ a + 4n > q\}. \tag{970}$$

Theorem 25.7 (Exact archimedean factor)**.** *As $X \to \infty$,*

$$m_\infty\{u \in \mathscr{X}(\mathbb{R}) \cap \mathcal{C} : 4a + 17n \le X\} \sim I_\infty X, \qquad I_\infty = \frac{1}{2\sqrt{3}} \log\!\left(\frac{3}{2\sqrt{2}}\right). \tag{971}$$

Proof. First, the wall $a+4n > q$ removes only a bounded part of the positive orthant. Indeed, if $q \ge a+4n$, then $a \le q - 4n$; using $a^2 = 3q^2 + 18n^2 - 2$ gives

$$3q^2 + 18n^2 - 2 \le (q-4n)^2, \qquad \text{hence} \qquad q^2 + 4qn + n^2 \le 1.$$

Thus the removed real region has bounded m_∞-measure. Its only integral point on $\mathscr{X}$ with $a > 0, q, n \ge 0$ is $(a, q, n) = (1, 1, 0)$.

Introduce polar coordinates in the negative plane by

$$\sqrt{3}\, q = r\cos\theta, \qquad 3\sqrt{2}\, n = r\sin\theta, \qquad 0 \le \theta \le \frac{\pi}{2}. \tag{972}$$

On $F = -2$, one has $a = \sqrt{r^2 - 2}$ and $r\,\mathrm{d}r = a\,\mathrm{d}a$. The Jacobian in (972) therefore gives

$$\mathrm{d}m_\infty = \frac{1}{6\sqrt{6}}\,\mathrm{d}a\,\mathrm{d}\theta. \tag{973}$$

At fixed θ, the height boundary is

$$4a + \frac{17}{3\sqrt{2}}\sqrt{a^2+2}\sin\theta = X. \tag{974}$$

Let $A_X(\theta)$ be its nonnegative solution when one exists, and put it equal to zero otherwise. Then $0 \le A_X(\theta)/X \le 1/4$, and for every fixed θ,

$$\frac{A_X(\theta)}{X} \longrightarrow \frac{1}{4 + \frac{17}{3\sqrt{2}}\sin\theta}.$$

Dominated convergence and (973) give

$$I_\infty = \frac{1}{6\sqrt{6}} \int_0^{\pi/2} \frac{\mathrm{d}\theta}{4 + \frac{17}{3\sqrt{2}}\sin\theta}. \tag{975}$$

For completeness, set $t = \tan(\theta/2)$. If $\alpha = 2\sqrt{2}/3$ and $\beta = 3/(2\sqrt{2})$, then $\alpha\beta = 1$, and

$$\int_0^{\pi/2} \frac{\mathrm{d}\theta}{4 + \frac{17}{3\sqrt{2}}\sin\theta} = \frac{1}{2}\int_0^1 \frac{\mathrm{d}t}{(t+\alpha)(t+\beta)} = 3\sqrt{2}\log\!\left(\frac{3}{2\sqrt{2}}\right).$$

Substitution in (975) proves (971). □

25.4 A genuine norm and the homogeneous counting theorem

Lemma 25.8 (A genuine norm realizing the height)**.** *Define*

$$\|(a,q,n)\|_* := \max\{|4a + 17n|, |q|, |a|\}. \tag{976}$$

Then $\|\cdot\|_*$ is a norm on $\mathbb{R}^3$, and on the chamber $\mathcal{C}$ it satisfies

$$\|(a,q,n)\|_* = 4a + 17n. \tag{977}$$

For every integer $X \ge 0$,

$$R(X) = \#\{u \in \mathscr{X}(\mathbb{Z}) \cap \mathcal{C} : \|u\|_* < X + 1\}. \tag{978}$$

Proof. The linear map

$$(a, q, n) \longmapsto (4a + 17n, q, a)$$

has determinant -17. Thus (976) is the pullback of the supremum norm by an invertible linear map and is a genuine norm. If $(a, q, n) \in \mathcal{C}$, then

$$0 < a < 4a + 17n, \qquad 0 \le q < a + 4n < 4a + 17n,$$

which proves (977). Since $4a + 17n$ is integral on $\mathscr{X}(\mathbb{Z})$, equation (978) follows from (948). □

Lemma 25.9 (Verification of the Gorodnik–Oh–Shah hypotheses)**.** *Let* $\mathbf{G} = \mathrm{Spin}(F)$*, acting through its standard three-dimensional orthogonal representation on* $W = \mathbb{R}^3$*, and let* $x_0 = (1, 1, 0)$*. Then*

$$\mathscr{X}(\mathbb{R}) = \mathbf{G}(\mathbb{R})^\circ x_0.$$

This orbit is a closed affine symmetric variety; $\mathbf{G}$ *is connected and absolutely almost simple over* $\mathbb{Q}$*, is isotropic over* $\mathbb{R}$*, and acts irreducibly on* W *over* $\mathbb{R}$*; and the stabilizer* $\mathbf{H} = \mathrm{Stab}_{\mathbf{G}}(x_0)$ *has no nontrivial* $\mathbb{Q}$*-characters. The cone* $\mathcal{C}$ *in* (970) *is admissible and generic for [52, Theorem 1.6].*

Proof. The form F has signature $(1, 2)$. Hence $\mathbf{G}$ is a connected absolutely almost simple group of type A_1, it is $\mathbb{R}$-isotropic, and its standard three-dimensional representation is $\mathbb{R}$-irreducible. The nonzero level $\mathscr{X}$ is Zariski closed. With

$$Q = \sqrt{3}\, q, \qquad N = 3\sqrt{2}\, n,$$

its real equation becomes

$$Q^2 + N^2 - a^2 = 2.$$

The parametrization

$$(a, \theta) \longmapsto (a, \sqrt{a^2 + 2}\cos\theta, \sqrt{a^2 + 2}\sin\theta), \qquad (a, \theta) \in \mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z}),$$

shows that $\mathscr{X}(\mathbb{R})$ is connected. Witt extension first produces an element $t \in \mathrm{O}(F)(\mathbb{R})$ taking any prescribed point of the level set to any other. If $\det t = -1$, choose an anisotropic vector in the orthogonal complement of the target point and compose t on the left with its reflection. That reflection fixes the target point and has determinant -1; hence the composite lies in $\mathrm{SO}(F)(\mathbb{R})$. Thus $\mathrm{SO}(F)(\mathbb{R})$ is transitive on the level set. The identity-component orbits have the full dimension of the level surface and are open; because their complements are unions of open orbits, each is also closed. Connectedness therefore leaves only one such orbit, proving $\mathscr{X}(\mathbb{R}) = \mathbf{G}(\mathbb{R})^\circ x_0$. The stabilizer of a nonisotropic vector is the fixed subgroup of the involution induced by reflection in that vector, so the orbit is affine symmetric.

The orthogonal complement of x_0 is

$$x_0^\perp = \{(3s, s, t) : s, t \in \mathbb{Q}\}, \qquad F(3s, s, t) = 6(s^2 - 3t^2). \tag{979}$$

Thus $\mathbf{H}$ is the norm-one torus of $K = \mathbb{Q}(\sqrt{3})$. Its character lattice is $\mathbb{Z}$, with the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$ acting by -1; consequently it has no nonzero Galois-invariant character and no nontrivial $\mathbb{Q}$-character. The integral point x_0 supplies the required nonempty integral locus.

It remains to check the sector rather than merely the affine orbit. For the given representation and any Euclidean sphere in W, the Satake boundary is

$$\mathscr{X}^\infty = \{F = 0\} \cap S(W) = \mathcal{O}_+ \sqcup \mathcal{O}_-, \qquad \mathcal{O}_\pm := \{F = 0,\ \pm a > 0\} \cap S(W), \tag{980}$$

and $\mathbf{G}(\mathbb{R})^\circ$ is transitive on each displayed circle. For a nonzero null vector with $a, q, n \ge 0$,

$$a = \sqrt{3q^2 + 18n^2} > q - 4n.$$

Hence both the closure and the relative interior of $\mathcal{C}$ meet exactly $\mathcal{O}_+$, not $\mathcal{O}_-$. The intersection of $\partial\mathcal{C}$ with the Satake boundary consists of the two radial directions

$$(a,q,n) = (\sqrt{3},1,0), \qquad (a,q,n) = (3\sqrt{2},0,1). \tag{981}$$

These are exactly the two projective limits of the integral Pell families in Proposition 25.5. This two-point intersection has measure zero for the smooth measure class on the boundary orbit. This is precisely the admissibility and genericity condition required by the cited sector theorem. □

25.5 The 2-adic orbit sign and Borovoi's coefficient

Write $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$. For the nonsquare ternary quadric under consideration, Borovoi associates to every local point an orbit sign $\nu_v \in \{\pm 1\}$, constant on $\mathrm{Spin}(F)(\mathbb{Q}_v)$-orbits and equal to $+1$ at almost every finite place. Thus $\nu_f((x_p)_p) = \prod_p \nu_p(x_p)$ is defined on $\mathscr{X}(\widehat{\mathbb{Z}})$. Because $\mathscr{X}(\mathbb{R})$ is connected, his normalization reads

$$c_{\mathscr{X}} = \frac{2m_f\{x \in \mathscr{X}(\widehat{\mathbb{Z}}) : \nu_f(x) = +1\}}{m_f(\mathscr{X}(\widehat{\mathbb{Z}}))}, \tag{982}$$

where m_f is the finite Tamagawa measure attached to the gauge form (968) and Borovoi's convergence factors.

Proposition 25.10 (Finite adelic sign balance)**.** *For the affine quadric $\mathscr{X} = \{F = -2\}$, the coefficient $c_{\mathscr{X}}$ in Borovoi's counting formula [51, Theorem 1.4.1 and Proposition 1.5.1] is exactly*

$$c_{\mathscr{X}} = 1. \tag{983}$$

Proof. Continue to write $A = \mathrm{diag}(1,-3,-18)$. Put $e = (1,0,0)$, $v = (1,0,1)$, and for every anisotropic vector w let

$$r_w(u) = u - \frac{2u^{\mathsf{T}}Aw}{w^{\mathsf{T}}Aw}\, w$$

be its orthogonal reflection. The element $g = r_v r_e$ is

$$g = \begin{pmatrix} -19/17 & 0 & -36/17 \\ 0 & 1 & 0 \\ -2/17 & 0 & -19/17 \end{pmatrix}. \tag{984}$$

Direct multiplication gives

$$g^{\mathsf{T}}Ag = A, \qquad \det g = 1. \tag{985}$$

Because $17 \in \mathbb{Z}_2^\times$, both g and g^{-1} have entries in $\mathbb{Z}_2$, so $g \in \mathrm{SO}(F,\mathbb{Z}_2)$. Its spinor norm is

$$\vartheta(g) = F(v)F(e) = -17 \quad \text{in } \mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}. \tag{986}$$

We identify the stabilizer's spinor-norm image. By (979), over $\mathbb{Q}_2$ the orthogonal complement of x_0 is

$$F|_{x_0^\perp} = 6(s^2 - 3t^2) = 6N_{K_2/\mathbb{Q}_2}(s + t\sqrt{3}), \qquad K_2 = \mathbb{Q}_2(\sqrt{3}). \tag{987}$$

The odd unit 3 is not a square in $\mathbb{Q}_2$, so K_2 is a quadratic field and this binary plane is anisotropic. Put $u_0 = (3,1,0)$, so $F(u_0) = 6$. If $h \in \mathrm{SO}(F|_{x_0^\perp})(\mathbb{Q}_2)$, then hr_{u_0} is a two-dimensional orthogonal transformation of determinant -1, hence is a reflection r_w in some nonisotropic vector $w = (3s,s,t)$. Therefore $h = r_w r_{u_0}$ and

$$\vartheta(h) = F(w)F(u_0) = 36N_{K_2/\mathbb{Q}_2}(s + t\sqrt{3}) \mod \mathbb{Q}_2^{\times 2}.$$

Conversely these products realize every norm square class as w ranges through the nonzero vectors of the anisotropic plane. Hence

$$\vartheta\Big(\mathrm{Stab}_{\mathrm{SO}(F)}(x_0)(\mathbb{Q}_2)\Big) = N_{K_2/\mathbb{Q}_2}(K_2^\times) \mod \mathbb{Q}_2^{\times 2}. \tag{988}$$

For a quadratic extension, its norm group is the kernel of the Hilbert symbol with the defining square class [59, Corollary 13.4.6]. The odd-unit formula [59, Equation 12.4.14] gives

$$(-17,3)_2 = (-1)^{((-17-1)/2)((3-1)/2)} = -1. \tag{989}$$

Thus $-17 \notin N_{K_2/\mathbb{Q}_2}(K_2^\times)$.

By Witt extension, every $x \in \mathscr{X}(\mathbb{Q}_2)$ is tx_0 for some $t \in \mathrm{O}(F)(\mathbb{Q}_2)$. If $\det t = -1$, choose an anisotropic vector in $x^\perp$ and compose on the left with its reflection; this fixes x and changes the determinant. Hence every such x has the form $x = sx_0$ with $s \in \mathrm{SO}(F)(\mathbb{Q}_2)$. The image of $\mathrm{Spin}(F)(\mathbb{Q}_2)$ in $\mathrm{SO}(F)(\mathbb{Q}_2)$ is the kernel of the spinor norm, as used in [51, Lemma 3.1]. Consequently the $\mathrm{Spin}(F)(\mathbb{Q}_2)$-orbit of $x = sx_0$ is determined by

$$\vartheta(s) \in \mathbb{Q}_2^\times/(\mathbb{Q}_2^{\times 2} N_{K_2/\mathbb{Q}_2}(K_2^\times)), \tag{990}$$

a quotient of order two. Equations (986) and (989) show that multiplication by g exchanges these two orbits. In Borovoi's convention, the orbit of x_0 has sign $\nu_2 = +1$ and the other orbit has sign $\nu_2 = -1$. Hence

$$\nu_2(gx) = -\nu_2(x) \qquad (x \in \mathscr{X}(\mathbb{Q}_2)). \tag{991}$$

Apply g at the prime 2 and the identity at every other finite prime. This gives a bijection of $\mathscr{X}(\widehat{\mathbb{Z}})$. It preserves the finite Tamagawa measure: at 2 this follows from invariance of the local measure, equivalently from $g^*F = F$ and $\det g = 1$, and at all other primes the map is the identity. Equation (991) shows that it exchanges the finite sign loci $\nu_f = +1$ and $\nu_f = -1$. They therefore have equal measure. Since $\mathscr{X}(\mathbb{R})$ is connected, Borovoi's formula becomes

$$c_{\mathscr{X}} = \frac{2m_f(\mathscr{X}(\widehat{\mathbb{Z}})_+)}{m_f(\mathscr{X}(\widehat{\mathbb{Z}}))} = 1.$$

The argument uses a measure-preserving, sign-reversing bijection; it does not require g to be an involution (indeed $g^2 \neq I$). □

25.6 Calibration and the counting constant

Theorem 25.11 (Calibrated reduced-root asymptotic)**.** *The number of reduced positive defect-one roots satisfies*

$$R(X) \sim C_0 X \qquad (X \to \infty), \tag{992}$$

where

$$\begin{aligned} C_0 &= \mathfrak{S}(\mathscr{X}) I_\infty \\ &= \frac{9}{2\pi^2} \log(2+\sqrt{3}) \log\!\left(\frac{3}{2\sqrt{2}}\right) \\ &= 0.035362047799731853\ldots. \end{aligned} \tag{993}$$

Proof. Let μ_{GOS} denote the invariant measure on $\mathscr{X}(\mathbb{R})$ with the normalization appearing in the Gorodnik–Oh–Shah sector theorem [52, Theorem 1.6]. The gauge measure m_∞ in (968) is also a nonzero $\mathbf{G}(\mathbb{R})^\circ$-invariant Radon measure. Because $\mathscr{X}(\mathbb{R})$ is a single homogeneous orbit, uniqueness of invariant measure gives

$$\mu_{\mathrm{GOS}} = \kappa m_\infty \tag{994}$$

for some $\kappa > 0$.

The scalar is fixed by applying both counting theorems to the same Euclidean balls and the full cone W. Gorodnik–Oh–Shah give

$$\#\{x \in \mathscr{X}(\mathbb{Z}) : |x|_2 < T\} \sim \kappa\, m_\infty\{x \in \mathscr{X}(\mathbb{R}) : |x|_2 < T\}.$$

Borovoi's Theorem 1.4.1 and Proposition 1.5.1, together with Proposition 25.10, give for the same balls

$$\#\{x \in \mathscr{X}(\mathbb{Z}) : |x|_2 < T\} \sim \mathfrak{S}(\mathscr{X})\, m_\infty\{x \in \mathscr{X}(\mathbb{R}) : |x|_2 < T\}.$$

The real volume here is asymptotic to a positive constant times T. Comparison therefore gives the invariant-measure calibration

$$\kappa = \mathfrak{S}(\mathscr{X}). \tag{995}$$

Lemma 25.8 and Lemma 25.9 now allow the sector theorem to be applied with the actual norm and chamber of the reduced-root problem:

$$\begin{aligned} R(X) &\sim \mu_{\mathrm{GOS}}\{u \in \mathscr{X}(\mathbb{R}) \cap \mathcal{C} : \|u\|_* \le X\} \\ &= \mathfrak{S}(\mathscr{X})\, m_\infty\{u \in \mathscr{X}(\mathbb{R}) \cap \mathcal{C} : 4a + 17n \le X\} \\ &\sim \mathfrak{S}(\mathscr{X}) I_\infty X. \end{aligned}$$

The switch between strict and weak inequalities is harmless by a $(1 \pm \varepsilon)X$ squeeze; on integral points it also follows directly from (978). Since $R(X) = R(\lfloor X \rfloor)$, the integer-height asymptotic extends to all real $X \to \infty$. Finally, substituting (959) and (971), and using $\zeta(2) = \pi^2/6$, gives exactly (993). □

25.7 Exact computation and the remaining open problem

The first five reduced roots are

$$(1, 4, 20), \quad (4, 4, 33), \quad (4, 8, 49), \quad (4, 15, 76), \quad (8, 24, 129). \tag{996}$$

The first root is the first admissible term of the $n = 0$ Pell family and the second is the first term of the $q = 0$ Pell family in Proposition 25.5. Exact integer enumeration in the ternary coordinates gives the following values. The residual column uses the full constant in (993), not a fitted constant.

X	$R(X)$	$R(X) - C_0 X$
10,000	353	−0.62048
20,000	715	7.75904
50,000	1,755	−13.10239
75,000	2,654	1.84642
100,000	3,535	−1.20478
125,000	4,460	39.74403
150,000	5,294	−10.30717
175,000	6,195	6.64164
200,000	7,109	36.59044

The enumeration loops over $n \ge 0$ and $a > 0$ with $4a + 17n \le X$, tests whether

$$q^2 = \frac{a^2 - 18n^2 + 2}{3}$$

is the square of a nonnegative integer satisfying $q < a + 4n$, and then reconstructs (x, y, z) with (946). Every retained triple is rechecked in the original polynomial (933) and against all ordering and wall inequalities. Separate exhaustive congruence enumerations agree with the formulae above at $p = 2, 3, 5, 7, 11, 13$ through the computed powers. These finite checks verify the implementation; the asymptotic theorem rests on the proofs above, not on numerical fitting.

The reproducibility script is

`supporting_materials/workbench/reproduction/t1_koide_exact/verify_koide_narain_root_density.py`.

Running it with `-max-z 200000` emits the coordinate, local-density, 2-adic isometry, and bounded-enumeration certificates recorded in

`supporting_materials/workbench/reproduction/t1_koide_exact/koide_verification_200000.json`.

The script uses vectorized square detection only to propose candidates; it rechecks every retained candidate with integer arithmetic in both coordinate systems and in the original polynomial Δ.

Validation, repair, or disproof test 25.12 (Square-root discrepancy)**.** **[OPEN]**Determine whether, for every $\varepsilon > 0$,

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}). \tag{997}$$

The homogeneous-dynamics argument above and the exact effective-well-roundedness verification in theorem 32.2 below prove, for some fixed $\eta > 0$,

$$R(X) = C_0 X + O(X^{1-\eta}),$$

but do not prove (997). No unproved spectral gap, GRH assumption, or implicit "conditional theorem" is being used here: the sharper error term is an open problem in this workbench; see the primary-source comparison in section 37.1.

26 Exact ideal and unit-orbit parametrization of the unchanged sharp Koide count

Throughout this section the variables, equation, chamber, and height are exactly

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0, \quad q \geq 0, \quad n \geq 0, \quad a + 4n > q, \qquad 4a + 17n \leq X. \tag{998}$$

Every auxiliary symbol introduced below is connected to these same integer points by an explicitly proved map and inverse. No count for a normalized form, a rescaled lattice, or a substituted cutoff is used in place of (998). Thus

$$R(X) = \#\left\{(a, q, n) \in \mathbb{Z}^3 : \text{all conditions in (998) hold}\right\}. \tag{999}$$

The already calibrated leading constant is

$$C_0 = \frac{9}{2\pi^2} \log(2 + \sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right). \tag{1000}$$

The exact parametrization proved below has four pieces of data for each fixed $n \geq 1$: a positive integer C, a positive integer D, a square root E of 3 modulo D, and a nonnegative Pell exponent k. There is no quotient left implicit: equations (1018), (1019), and (1021) construct (a, q, n) by finite integer operations, while (1013) and the descent (1017) recover (C, D, E, k) from (a, q, n). The face $n = 0$ is recorded separately and exactly.

This parametrization gives the exact sharp-cutoff formula (1035). It also separates the discrepancy into the radial shifted-divisor sum and the unit-phase discrepancy in (1042). The square-root estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}) \tag{1001}$$

is not proved or refuted here. What is proved unconditionally, by applying the effective symmetric-variety theorem to the exact chamber, is that there is a constant $\eta > 0$ for which

$$R(X) = C_0 X + O(X^{1-\eta}). \tag{1002}$$

The cited general theorem does not supply $\eta \geq 1/2$, and its proof does not justify replacing (1002) by (1001).

27 Arithmetic of $\mathbb{Z}[\sqrt{3}]$

Put

$$K = \mathbb{Q}(\sqrt{3}), \qquad \mathcal{O} = \mathbb{Z}[\sqrt{3}], \qquad \epsilon = 2 + \sqrt{3}. \tag{1003}$$

For $\alpha = u + v\sqrt{3} \in K$, write

$$\alpha' = u - v\sqrt{3}, \qquad \mathrm{N}(\alpha) = \alpha\alpha' = u^2 - 3v^2. \tag{1004}$$

Lemma 27.1 (Ring, class number, and totally positive units)**.** *The ring of integers of K is $\mathcal{O}$. Every nonzero integral ideal of $\mathcal{O}$ is principal. Every unit has norm $+1$, and the totally positive unit group is*

$$\mathcal{O}^{\times,+} = \{\epsilon^j : j \in \mathbb{Z}\}. \tag{1005}$$

Proof. Since $3 \equiv 3 \pmod 4$, the standard integral-basis calculation for a quadratic field gives $\mathcal{O}_K = \mathbb{Z}[\sqrt{3}]$ and discriminant 12. Minkowski's bound says that every ideal class in a real quadratic field of discriminant 12 has an integral representative of norm at most $\frac{2!}{2^2}\sqrt{12} = \sqrt{3} < 2$. Its norm must therefore be 1, so it is $\mathcal{O}$ and the ordinary class number is one.

The norm of a unit is ± 1. The equation $u^2 - 3v^2 = -1$ is impossible modulo 3, because it would give $u^2 \equiv 2 \pmod 3$. Hence every unit has norm $+1$. Dirichlet's unit theorem makes the positive units an infinite cyclic group. To identify its least generator, suppose $\gamma = u + v\sqrt{3} > 1$ and $\gamma' > 0$. Replacing γ by its inverse if necessary makes $v \geq 0$. If $v = 0$, then $\gamma = 1$. If $v \geq 1$, then $u^2 - 3v^2 = 1$ forces $u \geq 2$, and consequently $\gamma \geq 2 + \sqrt{3} = \epsilon$. Since ϵ itself has norm one, it is the least positive unit greater than one and generates the group. □

For $m \geq 1$ define

$$\chi_{12}(d) = \left(\frac{12}{d}\right), \qquad \rho(m) = \sum_{d \mid m} \chi_{12}(d). \tag{1006}$$

Here the symbol is Kronecker's symbol; in particular

$$\chi_{12}(d) = \begin{cases} 0, & 2 \mid d \text{ or } 3 \mid d, \\ 1, & d \equiv 1, 11 \pmod{12}, \\ -1, & d \equiv 5, 7 \pmod{12}. \end{cases} \tag{1007}$$

Lemma 27.2 (Ideal number)**.** *The number of integral ideals of $\mathcal{O}$ having absolute norm m is $\rho(m)$.*

Proof. The Dedekind zeta function factors as $\zeta_K(s) = \zeta(s)L(s, \chi_{12})$. Comparing Dirichlet coefficients gives the assertion. Equivalently, the equality follows prime by prime. At 2 and 3 there is one ramified prime ideal, so there is one ideal of norm p^e for every $e \geq 0$. If $\chi_{12}(p) = 1$, then p splits and there are $e+1$ ideals of norm p^e. If $\chi_{12}(p) = -1$, then p is inert, so there is one such ideal when e is even and none when e is odd. These are respectively the local values of $\sum_{j=0}^{e} \chi_{12}(p)^j$. Consequently, if $m = 2^{e_2}3^{e_3}\prod_{p \geq 5} p^{e_p}$, then

$$\rho(m) = \begin{cases} \displaystyle\prod_{\substack{p \mid m,\ p \geq 5 \\ p \equiv 1,11 \ (12)}} (e_p + 1), & e_p \text{ is even for every } p \equiv 5, 7 \pmod{12}, \\ 0, & \text{otherwise.} \end{cases} \tag{1008}$$

□

28 Hermite-normal-form parameters for every ideal

For $m \geq 1$ let

$$\mathcal{P}(m) = \left\{ (C, D, E) : \begin{array}{ll} C, D \in \mathbb{Z}_{>0}, & C^2 D = m, \\ 0 \leq E < D, & E^2 \equiv 3 \pmod{D} \end{array} \right\}. \tag{1009}$$

Proposition 28.1 (Exact HNF parametrization of ideals)**.** *The map*

$$(C, D, E) \longmapsto \mathfrak{a}(C, D, E) := C\big(D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z}\big) \tag{1010}$$

is a bijection from $\mathcal{P}(m)$ to the integral ideals of $\mathcal{O}$ of norm m. Consequently

$$\#\mathcal{P}(m) = \sum_{C^2 D = m} \#\{E \bmod D : E^2 \equiv 3 \pmod{D}\} = \rho(m). \tag{1011}$$

Proof. Relative to the basis $1, \sqrt{3}$ of $\mathcal{O}$, the lattice in (1010) has column basis

$$\begin{pmatrix} CD & CE \\ 0 & C \end{pmatrix}.$$

Its index is $C^2 D = m$. It is stable under multiplication by $\sqrt{3}$ because

$$\sqrt{3}(CD) = D\,C(E + \sqrt{3}) - E(CD),$$
$$\sqrt{3}\,C(E + \sqrt{3}) = E\,C(E + \sqrt{3}) + \frac{3 - E^2}{D}(CD),$$

and the last coefficient is integral. It is therefore an integral ideal of norm m.

Conversely, write the unique column Hermite normal form of an integral ideal $\mathfrak{a}$ as

$$\mathfrak{a} = A\mathbb{Z} + (B + C\sqrt{3})\mathbb{Z}, \qquad A, C > 0, \quad 0 \leq B < A.$$

Stability of A under multiplication by $\sqrt{3}$ implies $C \mid A$ and $C \mid B$. Write $A = CD$ and $B = CE$. Stability of $B + C\sqrt{3}$ then implies $D \mid 3 - E^2$. The HNF inequality gives $0 \leq E < D$, and the index is $AC = C^2 D$. HNF uniqueness proves injectivity and surjectivity.

For completeness, the last equality in (1011) can be checked without zeta functions. Let $r_3(D) = \#\{E \bmod D : E^2 \equiv 3 \pmod{D}\}$. For $p \geq 5$, Hensel's lemma gives $r_3(p^j) = 2$ for every $j \geq 1$ if $\chi_{12}(p) = 1$, and $r_3(p^j) = 0$ if $\chi_{12}(p) = -1$. Also $r_3(2) = r_3(3) = 1$ and $r_3(2^j) = r_3(3^j) = 0$ for $j \geq 2$. Thus, for every prime power p^e,

$$\sum_{2c + j = e} r_3(p^j) = \sum_{j=0}^{e} \chi_{12}(p)^j.$$

Multiplicativity proves (1011). □

The HNF parameters can be recovered directly from a generator. Suppose $\alpha = a + q\sqrt{3}$ has positive norm m, with $a > 0$ and $q \geq 0$. Put

$$C = \gcd(a, q), \qquad A = a/C, \qquad B = q/C, \qquad D = m/C^2. \tag{1012}$$

If $D = 1$, put $E = 0$. If $D > 1$, then

$$\gcd(B, D) = 1, \qquad E \equiv AB^{-1} \pmod{D}, \qquad 0 \leq E < D. \tag{1013}$$

Indeed, $\gcd(B, D) = \gcd(B, A^2 - 3B^2) = \gcd(B, A^2) = 1$, and $E^2 \equiv 3 \pmod{D}$. The principal ideal (α) is exactly $\mathfrak{a}(C, D, E)$: the element $A + B\sqrt{3}$ lies in $D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z}$, so the principal ideal is contained in the HNF ideal, while both ideals have norm D after division by C.

29 Complete integral parametrization of the original equation

For $n \geq 1$ retain the exact integer

$$m_n = 18n^2 - 2. \tag{1014}$$

It satisfies $m_n > 0$ and $m_n \equiv 1 \pmod 3$.

Lemma 29.1 (Every ideal of norm m_n has positive-norm generators)**.** *If $\mathfrak{a}$ is an integral ideal of $\mathcal{O}$ of norm m_n, then every generator has norm $+m_n$, rather than $-m_n$.*

Proof. Lemma 27.1 makes $\mathfrak{a}$ principal. A generator $u + v\sqrt{3}$ has norm $\pm m_n$. Negative norm would give

$$u^2 - 3v^2 = -m_n,$$

and reduction modulo 3 would give $u^2 \equiv -m_n \equiv 2 \pmod 3$, which is impossible. Thus the norm is $+m_n$. □

For each $\mathbf{p} = (C, D, E) \in \mathcal{P}(m_n)$, choose a generator of $\mathfrak{a}(C, D, E)$. By Lemma 29.1 its norm is m_n. Change its sign so that both real embeddings are positive, and multiply by a unique power of ϵ so as to obtain the unique generator

$$\alpha_{n,\mathbf{p}} = a_{n,\mathbf{p},0} + q_{n,\mathbf{p},0}\sqrt{3}, \qquad \sqrt{m_n} \leq \alpha_{n,\mathbf{p}} < \epsilon\sqrt{m_n}. \tag{1015}$$

Existence and uniqueness follow from (1005), because the half-open intervals $[\epsilon^j\sqrt{m_n}, \epsilon^{j+1}\sqrt{m_n})$ partition $\mathbb{R}_{>0}$. Since $\alpha'_{n,\mathbf{p}} = m_n/\alpha_{n,\mathbf{p}}$, the lower inequality in (1015) implies $q_{n,\mathbf{p},0} \geq 0$.

Lemma 29.2 (Purely integral reduced seed and descent)**.** *For an integral pair (a, q) satisfying*

$$a^2 - 3q^2 = m_n, \qquad a > 0, \qquad q \geq 0,$$

the half-open inequalities

$$\sqrt{m_n} \leq a + q\sqrt{3} < \epsilon\sqrt{m_n}$$

are equivalent to the purely integral inequalities

$$a > 2q \geq 0. \tag{1016}$$

If $a \leq 2q$, multiplication by $\epsilon^{-1} = 2 - \sqrt{3}$ is the integral inverse step

$$(a, q) \longmapsto (a^-, q^-) = (2a - 3q, 2q - a). \tag{1017}$$

It preserves the norm, has $a^- > 0$ and $0 \leq q^- < q$, and can therefore be iterated until the unique pair satisfying (1016) *is reached.*

Proof. Put $u = (a + q\sqrt{3})/\sqrt{m_n}$. Norm positivity and the sign conditions give $u \geq 1$. Moreover

$$\frac{a}{\sqrt{m_n}} = \frac{u + u^{-1}}{2}, \qquad \frac{q\sqrt{3}}{\sqrt{m_n}} = \frac{u - u^{-1}}{2}.$$

For $u \geq 1$, the ratio

$$\frac{a}{q} = \sqrt{3}\,\frac{u^2 + 1}{u^2 - 1}$$

(with value $+\infty$ when $q = 0$) is strictly decreasing. At $u = \epsilon$ it equals 2, since $\epsilon + \epsilon^{-1} = 4$. Thus $u < \epsilon$ is equivalent to $a > 2q$; the strict inequality exactly matches the half-open upper endpoint.

If $a \leq 2q$, norm positivity gives $a > \sqrt{3}q$, and hence

$$a^- = 2a - 3q > (2\sqrt{3} - 3)q > 0, \qquad q^- = 2q - a \geq 0.$$

Also $a > \sqrt{3}q > q$ gives $q^- = 2q - a < q$. Thus the positive integer q strictly decreases at every inverse step, so the process terminates. A terminal pair has $a > 2q$. Conversely, for such a pair the second coordinate after one inverse step is negative, so no further inverse step stays in the nonnegative-q region. This proves uniqueness as well as termination. □

The preceding seed can be reconstructed from the HNF data by a bounded integer search; thus the parametrization does not require an unspecified choice of a generator. More precisely, for $\mathbf{p} = (C, D, E) \in \mathcal{P}(m_n)$ there is a unique pair (A, B) with

$$\begin{gathered} A, B \in \mathbb{Z}, \qquad A > 0, \qquad 0 \le B, \qquad B^2 < D, \\ A^2 - 3B^2 = D, \qquad A \equiv EB \pmod{D}. \end{gathered} \tag{1018}$$

It gives the generator in (1015) by the exact integer formula

$$(a_{n,\mathbf{p},0}, q_{n,\mathbf{p},0}) = (CA, CB). \tag{1019}$$

Indeed, existence follows by writing the reduced generator as $C(A + B\sqrt{3})$. The seed inequality $A > 2B$ gives $D = A^2 - 3B^2 > B^2$. Conversely, the conditions in (1018) imply $A > 2B$ and put $A + B\sqrt{3}$ in $D\mathbb{Z} + (E + \sqrt{3})\mathbb{Z}$. Its principal ideal is contained in that HNF ideal, and both have norm D, so they are equal. Lemma 29.2 and HNF uniqueness give uniqueness. In particular one needs to test only the integers $0 \le B \le \lfloor \sqrt{D-1} \rfloor$ and the positive square $A = \sqrt{D + 3B^2}$. The phase introduced below is therefore also explicit in the same integer data:

$$\tau_{n,(C,D,E)} = \frac{\log((A + B\sqrt{3})/\sqrt{D})}{\log \epsilon}. \tag{1020}$$

Here the factor C cancels because $m_n = C^2 D$; no ideal generator remains to be chosen.

For $k \ge 0$ define integers $a_{n,\mathbf{p},k}$ and $q_{n,\mathbf{p},k}$ by

$$a_{n,\mathbf{p},k} + q_{n,\mathbf{p},k}\sqrt{3} = \alpha_{n,\mathbf{p}} \epsilon^k. \tag{1021}$$

Equivalently,

$$\begin{pmatrix} a_{n,\mathbf{p},k+1} \\ q_{n,\mathbf{p},k+1} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a_{n,\mathbf{p},k} \\ q_{n,\mathbf{p},k} \end{pmatrix}. \tag{1022}$$

Theorem 29.3 (Integral parametrization, with inverse and multiplicity)**.** *The map*

$$(n, \mathbf{p}, k) \longmapsto (a_{n,\mathbf{p},k}, q_{n,\mathbf{p},k}, n), \qquad n \ge 1, \quad \mathbf{p} \in \mathcal{P}(m_n), \quad k \ge 0, \tag{1023}$$

is a bijection onto all integral solutions of

$$a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0,\ q \ge 0,\ n \ge 1. \tag{1024}$$

Every such solution automatically satisfies $a + 4n > q$.

The inverse is as follows. From (a, q, n) form $\alpha = a + q\sqrt{3}$ and recover (C, D, E) by (1012)–(1013)*. There is a unique $k \ge 0$ such that*

$$\alpha = \alpha_{n,(C,D,E)} \epsilon^k. \tag{1025}$$

Thus every point occurs with multiplicity exactly one.

Proof. Equation (1021) preserves the norm, so its image satisfies

$$a^2_{n,\mathbf{p},k} - 3q^2_{n,\mathbf{p},k} = m_n,$$

which is exactly the equation in (1024). Both embeddings of the generator are positive, hence $a > 0$; and $\alpha_{n,\mathbf{p}} \ge \sqrt{m_n}$ and $k \ge 0$ imply $q \ge 0$.

Conversely, a solution in (1024) has $\mathrm{N}(a + q\sqrt{3}) = m_n > 0$. Since $a > 0$ and $q \ge 0$, the inequality $a^2 - 3q^2 = m_n > 0$ gives $a > \sqrt{3}q$; hence both embeddings are positive. Its principal ideal has unique HNF data (C, D, E). The quotient of $a + q\sqrt{3}$ by the canonical generator of that ideal is a totally positive unit,

hence is ϵ^k for a unique $k \in \mathbb{Z}$. Lemma 29.2 supplies the completely integral recovery of k: iterate (1017) precisely while $a \le 2q$; the number of steps is k, and the terminal pair is $(a_{n,\mathbf{p},0}, q_{n,\mathbf{p},0})$. In particular $k \ge 0$. This proves surjectivity and the stated inverse. HNF uniqueness, seed uniqueness, and unit-exponent uniqueness prove multiplicity one.

It remains to prove the chamber wall. If a point with $n \ge 1$ had $q \ge a + 4n$, then $a \le q - 4n$, and the unchanged equation would give

$$3q^2 + 18n^2 - 2 = a^2 \le (q - 4n)^2.$$

After moving terms and dividing by 2, this is

$$q^2 + 4qn + n^2 \le 1.$$

For nonnegative integers with $n \ge 1$, equality could occur only at $(q, n) = (0, 1)$, but that pair does not satisfy $q \ge a + 4n$ with $a > 0$. Thus the assumed wall inequality is impossible. □

The congruence needed by the original inverse map to the (x, y, z) coordinates is retained automatically: reducing $a^2 - 3q^2 - 18n^2 = -2$ modulo 2 gives

$$a \equiv q \pmod 2. \tag{1026}$$

No parity class has been lost or added in Theorem 29.3.

Proposition 29.4 (The face $n = 0$)**.** *All points satisfying the original equation and chamber with $n = 0$ are*

$$a_k + q_k\sqrt{3} = (1 + \sqrt{3})\epsilon^k, \qquad k \ge 1. \tag{1027}$$

They have height $4a_k$ and occur with multiplicity one.

Proof. When $n = 0$, the norm is -2. Its parity implies $a \equiv q \pmod 2$, and therefore

$$\frac{a + q\sqrt{3}}{1 + \sqrt{3}} = \frac{3q - a}{2} + \frac{a - q}{2}\sqrt{3} \in \mathcal{O}.$$

The quotient has norm one, so it is $\pm\epsilon^k$ by Lemma 27.1. The conditions $a > 0$ and $q \ge 0$ select the positive sign. At $k = 0$ one gets $(a, q) = (1, 1)$, which is excluded by the strict chamber wall $a > q$. At every $k \ge 1$, the recurrence (1022) gives $a_{k+1} - q_{k+1} = a_k + q_k > 0$. For $k \le -1$, direct use of $a = (t - 2/t)/2$ with $t = (1 + \sqrt{3})\epsilon^k \le \sqrt{3} - 1 < \sqrt{2}$ gives $a < 0$. Hence precisely the exponents $k \ge 1$ occur. □

30 Exact divisor and sharp-height formulas

The number of unit-orbits at a fixed positive n has an especially simple divisor formula.

Proposition 30.1 (Orbit divisor sum and shifted factorization)**.** *For $n \ge 1$, the number of orbits under $(a, q) \mapsto (2a + 3q, a + 2q)$ among the solutions with $a > 0$ and $q \ge 0$ is*

$$\rho(m_n) = \sum_{d \mid 18n^2 - 2} \chi_{12}(d) = \rho(3n - 1)\rho(3n + 1). \tag{1028}$$

Proof. The first equality is Theorem 29.3 and Proposition 28.1. For the second, write

$$m_n = 2(3n - 1)(3n + 1).$$

The two displayed linear factors have greatest common divisor dividing 2. The function $\rho = 1 * \chi_{12}$ is multiplicative and $\rho(2^j) = 1$ for all $j \ge 0$. Removing all powers of 2 therefore makes the two factors coprime without changing any of the three ρ values, proving the product identity. □

For $\mathbf{p} \in \mathcal{P}(m_n)$ define its exact unit phase by

$$\tau_{n,\mathbf{p}} = \frac{\log(\alpha_{n,\mathbf{p}}/\sqrt{m_n})}{\log \epsilon}, \qquad 0 \le \tau_{n,\mathbf{p}} < 1. \tag{1029}$$

For $n \ge 1$ put

$$Y_n(X) = \frac{X - 17n}{4}. \tag{1030}$$

Whenever $Y_n(X) \ge \sqrt{m_n}$, put

$$L_n(X) = \frac{1}{\log \epsilon} \log\left(\frac{Y_n(X) + \sqrt{Y_n(X)^2 - m_n}}{\sqrt{m_n}} \right). \tag{1031}$$

Let

$$\mathcal{N}(X) = \{n \in \mathbb{Z}_{\ge 1} : Y_n(X) \ge \sqrt{m_n}\}. \tag{1032}$$

This is the exact finite range

$$X \ge 17n + 4\sqrt{18n^2 - 2}; \tag{1033}$$

no asymptotic replacement of this boundary is made.

For the $n = 0$ face define

$$R_0(X) = \max\left\{ 0, \left\lfloor \frac{\log\left(\dfrac{X/4 + \sqrt{(X/4)^2 + 2}}{1 + \sqrt{3}} \right)}{\log \epsilon} \right\rfloor \right\}. \tag{1034}$$

Theorem 30.2 (Exact sharp-height floor formula)**.** *For every real $X \ge 0$,*

$$R(X) = R_0(X) + \sum_{n \in \mathcal{N}(X)} \sum_{\mathbf{p} \in \mathcal{P}(m_n)} \left(\lfloor L_n(X) - \tau_{n,\mathbf{p}} \rfloor + 1 \right). \tag{1035}$$

Every equality case in $4a + 17n \le X$ is included by the floor in (1035)*.*

Proof. For a fixed positive n and parameter $\mathbf{p}$, put $t_k = \alpha_{n,\mathbf{p}} \epsilon^k$. Since $t_k \ge \sqrt{m_n}$,

$$a_{n,\mathbf{p},k} = \frac{1}{2}\left(t_k + \frac{m_n}{t_k} \right) \tag{1036}$$

is nondecreasing, and is strictly increasing after the possible initial case $q = 0$. The unchanged height inequality is $a_{n,\mathbf{p},k} \le Y_n(X)$. It has no solution unless $Y_n(X) \ge \sqrt{m_n}$. In that range, solving the quadratic inequality in t_k gives

$$t_k \le Y_n(X) + \sqrt{Y_n(X)^2 - m_n}. \tag{1037}$$

After division by $\sqrt{m_n}$ and logarithms, this is precisely

$$k + \tau_{n,\mathbf{p}} \le L_n(X). \tag{1038}$$

Because $L_n(X) \ge 0$ and $0 \le \tau_{n,\mathbf{p}} < 1$, the number of integers $k \ge 0$ satisfying (1038) is exactly $\lfloor L_n(X) - \tau_{n,\mathbf{p}} \rfloor + 1$; this expression is zero, not negative, when $L_n(X) < \tau_{n,\mathbf{p}}$. Equality is retained. Theorem 29.3 now proves the positive-n sum.

For $n = 0$, put $t = (1 + \sqrt{3})\epsilon^k$. Since the norm is -2, $a_k = (t - 2/t)/2$. The inequality $4a_k \le X$ is equivalent to

$$t \le X/4 + \sqrt{(X/4)^2 + 2}.$$

Counting the integers $k \ge 1$ gives (1034) and completes the proof. □

Write

$$J_n(X) = \lfloor L_n(X) \rfloor, \qquad \lambda_n(X) = L_n(X) - J_n(X) \in [0,1), \tag{1039}$$

and define the exact phase distribution

$$A_n(t) = \#\{\mathbf{p} \in \mathcal{P}(m_n) : \tau_{n,\mathbf{p}} \le t\}, \qquad 0 \le t < 1, \tag{1040}$$

with discrepancy

$$\Delta_n(t) = A_n(t) - t\rho(m_n). \tag{1041}$$

Since $\lfloor J + \lambda - \tau \rfloor + 1 = J + \mathbf{1}_{\{\tau \le \lambda\}}$, Theorem 30.2 gives the exact identity

$$R(X) - C_0 X = R_0(X) + \left[\sum_{n \in \mathcal{N}(X)} \rho(m_n) L_n(X) - C_0 X \right] + \sum_{n \in \mathcal{N}(X)} \Delta_n(\lambda_n(X)). \tag{1042}$$

This identity includes the two unbounded coordinate faces and every sharp height boundary point. Formula (1034) also gives $R_0(X) = O(\log(2+X))$, so the $n = 0$ face is below the requested $X^{1/2+\varepsilon}$ scale; the two positive-n brackets are the genuine analytic terms.

31 The phase term as exact Hecke-character data

The last sum in (1042) is not determined by the divisor count $\rho(m_n)$. It records where the generator of each ideal lies inside a fundamental interval for the positive units.

For $\ell \in \mathbb{Z}$ and a nonzero integral ideal $\mathfrak{a} = (\gamma)$, define

$$\xi_\ell(\mathfrak{a}) = \exp\!\left(\frac{\pi i \ell}{\log \epsilon} \log \left| \frac{\gamma}{\gamma'} \right| \right). \tag{1043}$$

This is independent of the generator: replacing γ by $\pm\epsilon^j \gamma$ changes the logarithm by $2j \log \epsilon$, and the exponential by $e^{2\pi i \ell j} = 1$. It is multiplicative, hence is the unramified ideal part of a unitary Hecke character of K. For $m = m_n$ and the canonical generator,

$$\xi_\ell(\mathfrak{a}(C,D,E)) = e^{2\pi i \ell \tau_{n,(C,D,E)}}. \tag{1044}$$

Define its norm-m coefficient by

$$S_\ell(m) = \sum_{\mathrm{N}\,\mathfrak{a} = m} \xi_\ell(\mathfrak{a}). \tag{1045}$$

Then $S_0(m) = \rho(m)$, whereas the nonzero $S_\ell(m_n)$ are the Fourier coefficients of the phase multiset.

The coefficient $S_\ell(m)$ also has a completely explicit Euler-factor description. It is multiplicative in m. If $p \ge 5$ is split, choose one prime ideal $\mathfrak{p}$ above p and put $\omega_{\ell,p} = \xi_\ell(\mathfrak{p})$. Since $\mathfrak{p}\overline{\mathfrak{p}} = (p)$ and $\xi_\ell((p)) = 1$, one has

$$S_\ell(p^e) = \sum_{j=0}^{e} \omega_{\ell,p}^{2j-e} \qquad (\chi_{12}(p) = 1). \tag{1046}$$

If $p \ge 5$ is inert, then

$$S_\ell(p^e) = \begin{cases} 1, & 2 \mid e, \\ 0, & 2 \nmid e. \end{cases} \qquad (\chi_{12}(p) = -1). \tag{1047}$$

Finally the unique primes above 2 and 3 are generated by $1 + \sqrt{3}$ and $\sqrt{3}$, respectively. Since

$$\left| \frac{1+\sqrt{3}}{1-\sqrt{3}} \right| = \epsilon, \qquad \left| \frac{\sqrt{3}}{-\sqrt{3}} \right| = 1,$$

their factors are

$$S_\ell(2^e) = (-1)^{\ell e}, \qquad S_\ell(3^e) = 1. \tag{1048}$$

These formulas enumerate every ideal factor with multiplicity one. At $\ell = 0$ they reduce prime by prime to the divisor sum $\rho(m)$; at $\ell \neq 0$ they show exactly what angular information the unweighted divisor sum omits.

They also retain the shifted factorization of m_n at every Fourier mode. Writing $3n-1 = 2^r u$ and $3n+1 = 2^s v$ with u, v odd, one has $\gcd(u, v) = 1$. Multiplicativity and (1048) therefore give the exact identity

$$S_\ell(18n^2 - 2) = (-1)^\ell S_\ell(3n-1) S_\ell(3n+1). \tag{1049}$$

Indeed, the exponent of 2 on the left is $1 + r + s$, while the product on the right has total exponent $r + s$; all odd prime factors are disjoint. Thus the angular obstruction is itself a shifted correlation, now of nontrivial Hecke coefficients rather than merely of their dimensions.

Here is an endpoint-exact Fourier statement. Suppose $0 < \lambda < 1$, and let

$$b_n(t) = \#\{\mathbf{p} \in \mathcal{P}(m_n) : \tau_{n,\mathbf{p}} = t\}. \tag{1050}$$

Fejér summation of the Fourier series of $\mathbf{1}_{[0,\lambda]} - \lambda$ gives

$$\begin{aligned}\Delta_n(\lambda) = {} & \frac{1}{2}\big(b_n(0) + b_n(\lambda)\big) \\ & + \lim_{M\to\infty} \sum_{0<|\ell|\le M} \left(1 - \frac{|\ell|}{M+1}\right) \frac{1 - e^{-2\pi i \ell \lambda}}{2\pi i \ell} S_\ell(m_n).\end{aligned} \tag{1051}$$

The two half-weights correct the midpoint convention at the circular endpoint 0 and at λ. When $\lambda = 0$, the exact value is simply $\Delta_n(0) = b_n(0)$.

Combining this with (1049) makes the aggregate obstruction a literal shifted correlation. Define the exact endpoint contribution

$$\mathcal{E}(X) = \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)=0}} b_n(0) + \frac{1}{2} \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)>0}} \big(b_n(0) + b_n(\lambda_n(X))\big). \tag{1052}$$

Since $\mathcal{N}(X)$ is finite, summation over n commutes with the Fejér limit and gives

$$\begin{aligned}\sum_{n\in\mathcal{N}(X)} \Delta_n(\lambda_n(X)) = {} & \mathcal{E}(X) \\ & + \lim_{M\to\infty} \sum_{0<|\ell|\le M} \left(1 - \frac{|\ell|}{M+1}\right) \frac{(-1)^\ell}{2\pi i \ell} \\ & \times \sum_{\substack{n\in\mathcal{N}(X)\\ \lambda_n(X)>0}} \big(1 - e^{-2\pi i \ell \lambda_n(X)}\big) S_\ell(3n-1) S_\ell(3n+1).\end{aligned} \tag{1053}$$

Every coefficient and endpoint in this identity comes from the original weak height inequality; no smoothing has been inserted.

Equations (1042) and (1051) identify the analytic problem without discarding the original height. Its radial part is the sharply weighted shifted correlation

$$\sum_{n\in\mathcal{N}(X)} \rho(3n-1)\rho(3n+1) L_n(X), \tag{1054}$$

and its angular part requires cancellation in the polynomially sampled Hecke coefficients

$$S_\ell(18n^2 - 2), \tag{1055}$$

uniformly over the Fourier modes needed to approximate the discontinuous threshold $\tau \le \lambda_n(X)$. The endpoint (1033) gives $L_n(X) = 0$ and has a square-root singularity in its derivative; it is not a fixed smooth weight. A bound for the unweighted orbit total $\rho(m_n)$ therefore does not bound the actual sharp-height count.

32 An unconditional effective exponent for the exact sector

This section records the power saving that is presently justified without assuming the square-root estimate.

Let

$$F(a,q,n) = a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} = \{F = -2\}, \qquad \mathbf{G} = \mathrm{Spin}(F). \tag{1056}$$

Let $x_0 = (1,1,0)$. Its orthogonal complement is

$$x_0^{\perp} = \{(3s,s,t) : s,t \in \mathbb{Q}\}, \qquad F(3s,s,t) = 6(s^2 - 3t^2), \tag{1057}$$

so the identity component of its stabilizer is the norm-one torus of $\mathbb{Q}(\sqrt{3})$. Its rational character group is zero because Galois conjugation acts by -1 on its geometric character lattice.

The form F is anisotropic over $\mathbb{Q}$. Indeed, suppose a primitive integral triple satisfies $F(a,q,n) = 0$. Modulo 3 one first gets $3 \mid a$. Writing $a = 3a_1$ and dividing the equation by 3 gives $3a_1^2 - q^2 - 6n^2 = 0$, hence $3 \mid q$. Write $q = 3q_1$ and divide by 3 again to get

$$a_1^2 - 3q_1^2 - 2n^2 = 0. \tag{1058}$$

Modulo 3, the sum $a_1^2 + n^2$ is zero, which forces $3 \mid a_1$ and $3 \mid n$. Thus 3 divides a, q, n, contradicting primitivity. In particular the associated arithmetic Fuchsian quotient is cocompact.

For $X > 0$ let B_X be the exact subset of $\mathscr{X}(\mathbb{R})$ cut out by

$$a > 0, \quad q \ge 0, \quad n \ge 0, \quad a + 4n > q, \quad 4a + 17n \le X. \tag{1059}$$

With the gauge measure

$$dm_\infty = \frac{dq\,dn}{2a} \qquad (a > 0), \tag{1060}$$

put

$$I_\infty = \frac{1}{2\sqrt{3}} \log\!\left(\frac{3}{2\sqrt{2}}\right). \tag{1061}$$

Lemma 32.1 (Exact-sector volume with a power-saving remainder)**.** *As $X \to \infty$,*

$$m_\infty(B_X) = I_\infty X + O(1). \tag{1062}$$

Moreover, the family B_X is effectively well-rounded in the sense of Benoist–Oh, Definition 13.1 [53, p. 1931].

Proof. Introduce exact polar coordinates on the negative plane by

$$\sqrt{3}q = r\cos\theta, \qquad 3\sqrt{2}n = r\sin\theta, \qquad 0 \le \theta \le \pi/2. \tag{1063}$$

On $F = -2$ one has $r = \sqrt{a^2 + 2}$ and

$$dm_\infty = \frac{1}{6\sqrt{6}}\, da\, d\theta. \tag{1064}$$

Write $b(\theta) = 17\sin\theta/(3\sqrt{2})$. In the positive orthant the outer boundary is the unique nonnegative solution $A_X(\theta)$ of

$$4A_X(\theta) + b(\theta)\sqrt{A_X(\theta)^2 + 2} = X. \tag{1065}$$

For all sufficiently large X, uniformly in θ,

$$A_X(\theta) = \frac{X}{4 + b(\theta)} + O(X^{-1}). \tag{1066}$$

To see this without a formal expansion, subtract $(4+b)A_X$ from (1065) and use

$$\sqrt{A_X^2 + 2} - A_X = \frac{2}{\sqrt{A_X^2 + 2} + A_X} = O(X^{-1}),$$

where (1065) uniformly gives $A_X \asymp X$. Integration of (1066) in (1064) gives

$$\frac{X}{6\sqrt{6}} \int_0^{\pi/2} \frac{d\theta}{4 + \frac{17}{3\sqrt{2}} \sin\theta} + O(X^{-1}) = I_\infty X + O(X^{-1}).$$

For the asserted exact value, set $c = 17/(3\sqrt{2})$, $d = \sqrt{c^2 - 16} = 1/(3\sqrt{2})$, and $t = \tan(\theta/2)$. Then

$$\begin{aligned} \int_0^{\pi/2} \frac{d\theta}{4 + c\sin\theta} &= \int_0^1 \frac{dt}{2t^2 + ct + 2} \\ &= \frac{1}{d}\left[\log\left(\frac{4t + c - d}{4t + c + d}\right)\right]_{t=0}^{t=1} \\ &= 3\sqrt{2}\log\left(\frac{3}{2\sqrt{2}}\right). \end{aligned}$$

Multiplication by $1/(6\sqrt{6})$ is exactly the constant in (1061). If the strict chamber wall fails, then the calculation in the proof of Theorem 29.3, now over $\mathbb{R}$, gives $q^2 + 4qn + n^2 \le 1$. The removed real subset is therefore compact. For all large X it contributes a fixed finite measure, proving (1062).

It remains to check effective well-roundedness for the actual sector. Let U_δ be the image in the fixed vector representation of the group ball used by Benoist–Oh [53, Definition 13.1]. For every linear functional ℓ on $\mathbb{R}^3$ there is a constant, independent of x and δ, such that

$$|\ell(xg) - \ell(x)| \le C_\ell \delta \|x\| \qquad (g \in U_\delta). \tag{1067}$$

Apply this to the five literal boundary functionals

$$a, \qquad q, \qquad n, \qquad a + 4n - q, \qquad 4a + 17n.$$

On the unbounded part of the positive sheet, $\|x\| \asymp a + 1$, and the last functional is also comparable to $\|x\|$. Thus membership in the outer height inequality can change only in a shell of radial width $O(\delta X)$. Membership in $q \ge 0$ or $n \ge 0$ can change only in angular strips of width $O(\delta)$ about $\theta = \pi/2$ or $\theta = 0$. The boundary $a = 0$ is compact. The boundary $a + 4n - q = 0$ is compact by the calculation preceding (1062); moreover its restriction to $\mathscr{X}$ is regular, because proportionality of $(1, -1, 4)$ to $(2a, -6q, -36n)$ is incompatible with $a, q, n \ge 0$. Consequently its group-action boundary tube has measure $O(\delta)$. By (1064), all these regions together have measure $O(\delta X)$. The same analysis applies to the inner intersection that defines $B_{X,\delta}^-$. Hence

$$m_\infty(B_{X,\delta}^+ \setminus B_{X,\delta}^-) \ll \delta\, m_\infty(B_X), \tag{1068}$$

which is precisely Definition 13.1(2), with boundary exponent 1, for the group-action thickenings rather than merely Euclidean tubes.

For Definition 13.1(3) of [53], take the archimedean height to be the Euclidean norm in the displayed integral basis. It is bounded below on $\mathscr{X}$. Uniformly for $0 < \delta < 1$, the thickening $B_{X,\delta}^+$ lies in a fixed

finite enlargement of the polar sector; there $H_\infty(x) \asymp 1+a$ and the measure in a radial interval is $O(da)$. Hence for every $s>0$, uniformly in δ,

$$\int_{B_X^+} \|u\|^{-s}\, dm_\infty(u) \ll \begin{cases} X^{1-s}, & 0<s<1,\\ \log X, & s=1,\\ 1, & s>1. \end{cases}$$

Each line is $O(m_\infty(B_X)^{1-\delta_s})$, for example with $\delta_s = \min(s/2, 1/2) > 0$. This is the remaining condition. Since $S=\{\infty\}$, Definition 13.1(1), concerning a finite-place compact open subgroup, is void. The choices of strict or weak inequality on the finitely many piecewise smooth boundary arcs do not change these neighborhood estimates. □

Theorem 32.2 (Unconditional effective conclusion)**.** *There exists $\eta>0$, depending only on the fixed form, lattice, chamber, and height in* (998)*, such that*

$$R(X) = C_0 X + O(X^{1-\eta}). \tag{1069}$$

Proof. Use the standard vector representation $\rho : \mathrm{Spin}(F) \to \mathrm{SO}(F) \subset \mathrm{GL}_3$, whose kernel is finite, and the displayed integral basis of the exact lattice $\mathbb{Z}^3$. Let

$$\mathbf{G}_\mathbb{Z} = \{g \in \mathbf{G}(\mathbb{Q}) : \rho(g)\mathbb{Z}^3 = \mathbb{Z}^3\}.$$

This is the integral group attached to the chosen representation and lattice in the Benoist–Oh setup [53, pp. 1892–1893]; take their congruence subgroup Γ_S to be the level-one group $\mathbf{G}_\mathbb{Z}$ itself. The group $\mathbf{G}$ is connected, simply connected, and almost $\mathbb{Q}$-simple. It is noncompact over $\mathbb{R}$ because F has signature $(1,2)$. Thus $S=\{\infty\}$ contains every noncompact archimedean place and G_S is noncompact. The orbit $x_0\mathbf{G}$ is the affine symmetric variety $\mathscr{X}$, and its stabilizer has no nontrivial rational character by (1057). These verify every standing hypothesis on pp. 1892–1893 of [53]. Their Theorem 12.5 supplies the required effective equidistribution of the symmetric-stabilizer translates.

The Borel–Harish-Chandra finiteness theorem quoted in [53, p. 1892] p. 1892 says that $\mathscr{X}(\mathbb{Z})$ is a finite union of $\mathbf{G}_\mathbb{Z}$-orbits. A rational representative of any such orbit again has a symmetric stabilizer with no rational character: its orthogonal complement is anisotropic over $\mathbb{Q}$, since an isotropic vector in it would be an isotropic vector for F. Apply Benoist–Oh, Theorem 13.2 [53, pp. 1931–1932], to each orbit and to the exact effectively well-rounded family B_X of Lemma 32.1. For each orbit it gives its invariant-volume main term with error $O(m_\infty(B_X)^{1-\eta_j})$ for some $\eta_j>0$. There are finitely many orbits, so their sum has the same form with $\eta = \min_j \eta_j > 0$.

All invariant measures on the same real homogeneous variety are proportional. Hence the sum of the finitely many orbit main terms is $K\, m_\infty(B_X)$ for one fixed $K>0$. Lemma 32.1 and the already established asymptotic $R(X) \sim C_0 X$ force $K I_\infty = C_0$. Therefore

$$K\, m_\infty(B_X) = K(I_\infty X + O(1)) = C_0 X + O(1).$$

Since $m_\infty(B_X) \asymp X$, summing the effective errors proves (1069); the $O(1)$ term is absorbed after decreasing η if necessary. The theorem is applied first to the sequence of positive integral X. For real X, the original height $4a+17n$ is an integer, so $R(X) = R(\lfloor X \rfloor)$; replacing X by $\lfloor X \rfloor$ changes $C_0 X$ by $O(1)$ and proves the same estimate. □

Remark 32.3 (What exponent has actually been obtained). Theorem 32.2 is an unconditional exponent strictly smaller than 1: the error is $O(X^{1-\eta})$ for some $\eta>0$. Neither the statement nor the proof of [53, Theorem 13.2] identifies an exponent $1-\eta \le 1/2$ for this sharp sector. Tracing their Sobolev and mixing constants would make a positive η effective in principle, but that calculation has not been carried out in this section and cannot be replaced by the assertion $\eta = 1/2$.

33 Exact obstruction to the square-root claim

There are two complementary descriptions of the remaining obstruction; the first is an exact identity, while the second explains the relevant spectral mechanisms without asserting an unproved intertwining formula.

First, the elementary ideal description gives the literal identity (1042). Thus the square-root claim is equivalent to a bound of that size for the displayed sum of the radial bracket, the aggregate phase discrepancy, and $R_0(X)$. Controlling the radial and phase terms separately is a sufficient route, though the exact identity also permits cancellation between them. Formula (1051) shows that a separate estimate of the latter entails uniform cancellation in the nonzero archimedean Hecke modes (1055). Merely using $\#\mathcal{P}(m_n) = \rho(m_n)$, or replacing each phase distribution by Lebesgue measure, deletes a term that is exactly present in $R(X)$.

Second, F is rationally anisotropic, so $\mathbf{G}$ gives a cocompact arithmetic Fuchsian quotient. A negative-norm vector represents an oriented geodesic, and its stabilizer is the split real torus appearing in (1057). In the standard radial torus-orbit counting kernel, the spectral expansion has two possible sources larger than the square-root scale:

1. an exceptional eigenvalue $\lambda = s(1-s) < 1/4$ can contribute $c_s X^s$ with $s > 1/2$, unless the exact torus period and exact sector transform make $c_s = 0$;

2. after exceptional terms are absent or removed, estimating the tempered spectral sum for a discontinuous cutoff at the conjectural $X^{1/2+\varepsilon}$ scale requires cancellation beyond the general effective-mixing argument.

For the present nonradial chamber, one must first construct a remainder-preserving sector spectral expansion before these candidate terms can even be asserted with their exact coefficients. An actual refutation by a secondary term would then require proving both $s > 1/2$ and $c_s \neq 0$ for this fixed arithmetic quotient and exact chamber. No such construction or nonvanishing has been established here. Conversely, absence of exceptional terms alone would not estimate the full tempered sharp-cutoff sum. This is why the exact parametrization neither proves nor refutes (1001), while it does isolate the data a future proof must bound.

For comparison only, not as a theorem about the present nonradial sector, the classical hyperbolic lattice-point problem for a conjugacy class has an unconditional $O(X^{2/3})$ remainder due to Good [54, Theorem 4], with the modern proof of Chatzakos–Petridis [55, Theorem 1.1 and Conjecture 5.7], and a conjectural $O_\varepsilon(X^{1/2+\varepsilon})$ scale. That published $O(X^{2/3})$ theorem counts a different radial kernel. An exact intertwining of its kernel with the simultaneous chamber and height in (998) has not been proved, so $O(X^{2/3})$ is not claimed here for $R(X)$.

The independent finite replay used to check the HNF multiplicities, seed reduction, inverse descent, and divisor factorization for $1 \leq n \leq 200$ is retained at `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/parametrization/verify_parametrization.py`; its SHA–256 digest is `061c5e2225828169fd1ecb8bcc5e7cf8b8d19bfdb918650668c3e90f2f9730d`. That replay is a finite consistency check. The proofs above, rather than the finite range, establish the parametrization for every admissible integer.

34 Boundary ledger

For clarity, every boundary used above has the following exact treatment.

- $a > 0$ is strict. There is no omitted integral solution with $a = 0$, because the original equation would give a multiple of 3 equal to 2.

- $q = 0$ is included. It corresponds to phase $\tau = 0$, is included by $\tau \leq \lambda$, and is retained in the endpoint term $b_n(0)/2$ in (1051).

- $n = 0$ is included and is counted exactly by $R_0(X)$; the sole solution $(a, q, n) = (1, 1, 0)$ on $a+4n = q$ is excluded.
- $a + 4n = q$ is strict. Its real intersection is bounded and its sole integral point in the nonnegative chamber is $(1, 1, 0)$.
- $4a + 17n = X$ is included. Equality is included in (1037), (1038), and every floor in (1035).
- The canonical unit interval is half-open, $[\sqrt{m_n}, \epsilon\sqrt{m_n})$. Consequently no generator is counted twice at a unit boundary.
- Conjugate ideals are distinct unless the ideal is fixed by conjugation. They yield distinct canonical phases and distinct $q \geq 0$ unit-orbits; no division by 2 is made. When $q = 0$, the ideal is self-conjugate and still occurs once by HNF uniqueness.

35 Exact unit-orbit census through height 10^8

The census retains exactly the same equation, chamber, strict wall, and height as the preceding theorems:

$$(a, q, n) \in \mathbb{Z}^3, \qquad a^2 - 3q^2 - 18n^2 = -2, \qquad a > 0, \quad q, n \geq 0, \quad a + 4n > q, \qquad 4a + 17n \leq X. \tag{1070}$$

It does not infer an asymptotic from finite data. Its exact fixed-n equation and factorization are

$$a^2 - 3q^2 = m_n, \qquad m_n = 18n^2 - 2 = 2(3n - 1)(3n + 1). \tag{1071}$$

For each $n \geq 1$, every ideal of norm m_n is enumerated by its Hermite-normal-form triple

$$(C, D, E) \in \mathbb{Z}^2_{>0} \times \mathbb{Z}, \qquad C^2 D = m_n, \qquad 0 \leq E < D, \qquad E^2 \equiv 3 \pmod{D}. \tag{1072}$$

The generator is reduced by exact multiplication by the fundamental unit $2 + \sqrt{3}$ to the unique seed $a_0 > 2q_0 \geq 0$. Its forward unit orbit is

$$\begin{pmatrix} a_{k+1} \\ q_{k+1} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a_k \\ q_k \end{pmatrix}, \qquad k \geq 0. \tag{1073}$$

Each emitted point is reconstructed by

$$x = \frac{a + 4n - q}{2}, \qquad y = \frac{a + 4n + q}{2}, \qquad z = 4a + 17n, \tag{1074}$$

and is rechecked in both coordinate systems and in the original polynomial $\Delta(x, y, z) = -1$. Thus the accelerated enumeration does not replace the original objects by a floating-point or rescaled surrogate.

35.1 Complete result and checkpoint table

The complete integer-domain audit for $1 \leq X \leq 100{,}000{,}000$ gives

$$\boxed{R(100{,}000{,}000) = 3{,}536{,}002.} \tag{1075}$$

The endpoint in n is 2,943,725; the enumeration contains 2,327,513 canonical unit orbits. The following checkpoints are exact integer counts. The residuals use the proved constant C_0 in (993).

X	$R(X)$	$R(X)-C_0X$	$(R(X)-C_0X)/\sqrt{X}$
200,000	7,109	36.590440053629	0.081818711287
250,000	8,854	13.488050067037	0.026976100134
500,000	17,695	13.976100134073	0.019765190359
1,000,000	35,381	18.952200268147	0.018952200268
2,000,000	70,637	−87.095599463706	−0.061585888992
5,000,000	176,943	132.761001340734	0.059372524752
10,000,000	353,598	−22.477997318532	−0.007108166877
20,000,000	707,458	217.044005362937	0.048532515010
50,000,000	1,768,230	127.610013407342	0.018046781166
100,000,000	3,536,002	−202.779973185317	−0.020277997319

35.2 Certified extrema over every integer

Write $E(X)=R(X)-C_0X$. Between consecutive occupied heights, $R(X)$ is constant and both $E(X)$ and $E(X)/\sqrt{X}$ are strictly decreasing, since

$$\frac{d}{dX}\left(\frac{R}{\sqrt{X}}-C_0\sqrt{X}\right)=-\frac{R}{2X^{3/2}}-\frac{C_0}{2\sqrt{X}}<0. \tag{1076}$$

Checking both integer endpoints of every constant-count interval therefore certifies extrema over every integer in the audited range, rather than at selected checkpoints. On $1\le X\le 10^8$, the maximum signed residual is

$$\begin{aligned} E(77{,}078{,}404)&=721.79342495712975696566815535386910933 79\ldots,\\ R(77{,}078{,}404)&=2{,}726{,}372, \end{aligned} \tag{1077}$$

and the minimum signed, hence largest absolute, residual is

$$\begin{aligned} E(32{,}461{,}294)&=-812.83006914880688970919660067974916559 33\ldots,\\ R(32{,}461{,}294)&=1{,}147{,}085. \end{aligned} \tag{1078}$$

The maximum of $|E(X)|/\sqrt{X}$ over the full audited range is attained at $X=160$, where

$$\frac{|E(160)|}{\sqrt{160}}=0.18515707693408214138401018551531723 3816\ldots.$$

On $10{,}000\le X\le 10^8$, the maximum is

$$\begin{aligned} \frac{|E(867{,}240)|}{\sqrt{867{,}240}}&=0.17140019280881455972894965803234555 2502\ldots,\\ R(867{,}240)&=30{,}827, \end{aligned} \tag{1079}$$

and the root entering at that height is

$$(a,q,n)=(199997,115061,3956),\qquad (x,y,z)=(50380,165441,867240). \tag{1080}$$

The bound $|E(X)|\le 0.171\sqrt{X}$ first fails on this tail at $X=867{,}216$, where $R(X)=30{,}826$ and

$$E(X)=159.46635530774122203304493532343198171 99\ldots.$$

No counterexample to the finite statement with constant 0.172 occurs through 10^8. These are bounded computations, not a claim that either constant persists.

The audit finds 12,916 sign changes. The 3,536,002 roots occupy 2,472,015 distinct heights; the maximum shell multiplicity is 35, at $z=92{,}923{,}120$, and the largest gap between consecutive occupied heights is 564, from 71,586,276 to 71,586,840.

35.3 Independent replay and exact algorithm

An independent pass decompressed and rechecked all 3,536,002 six-coordinate rows, recomputed every checkpoint, verified the coordinate maps, equations, ordering, chamber, height, nonduplication, and gzip CRC, and returned `PASS`. It also compared the literal (n, a) scan through $X = 20{,}000$, every generalized-Pell seed for $1 \le n \le 1000$, and the full six-coordinate point set through $X = 200{,}000$.

The reproducibility sources and receipts are retained project-relatively under `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/census`. The two 10^8-range outputs and the full coordinate receipt are in its `final_100000000` subdirectory. The principal digests are

artifact	SHA-256
`koide_fast_enum.py`	`dddb73b828b63be08e2ff0b5346edb32 31662005f83d5b2448c06fef6ba31017`
`koide_discrepancy_100000000.json`	`da92f0866074e530b8174a48915caa0f 65cd4cd8957e209306adc200739d1961`
`koide_roots_through_100000000.csv.gz`	`dfc313492aedb396dcfa10520bfefc699 ba601357bc9933d921dfa3884310bd4`
`coordinate_audit_receipt.json`	`3010363108e7b460b7331a92f3263bbd 77992aedfc3064973750e16d707da80f`

The exact algorithm first finds the largest admissible n by integer binary search, then factors the unchanged integers $3n - 1$ and $3n + 1$ with a smallest-prime-factor sieve. It enumerates every HNF triple in (1072), constructs all roots of 3 modulo D by prime-power lifting and the Chinese remainder theorem, reduces each generator to the unique seed, and iterates (1073) until the literal height fails. For each n, the independently checked identity

$$\#\mathcal{P}(m_n) = \rho(m_n) = \rho(3n-1)\rho(3n+1)$$

certifies the orbit multiplicity. The work is

$$X^{1+o(1)} + O(R(X))$$

integer operations before the optional height sort. This complexity and the finite census use no conjectural discrepancy estimate.

36 Exact quaternion bridge: order, shell, and hyperbolic conjugacy classes

The proposed quaternionic realization is correct. The displayed lattice is an order in an indefinite rational quaternion division algebra of discriminant 6. The parity-restricted ternary shell is exactly the reduced-trace-4, reduced-norm-1 locus in that order. Every resulting element is primitive hyperbolic, with eigenvalues $2 \pm \sqrt{3}$ and translation length $2\log(2 + \sqrt{3})$.

The centralizer statement is correct when written intrinsically as

$$\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}].$$

The entire integral shell is a finite union of conjugacy classes under $\Gamma = \mathcal{O}^1$. This gives an exact reduction of the Koide count to a finite sum of sharp, non-radial sector counts inside primitive hyperbolic conjugacy classes of a cocompact arithmetic Fuchsian group.

It does **not** prove the open estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}).$$

Rather, it converts the count exactly into a sharp sector problem on a finite union of hyperbolic conjugacy classes. This places it beside, but does not prove it equivalent to, the standard hyperbolic conjugacy-class lattice-point problem. Even for the radial Huber cutoff, the general pointwise theorem gives an $O(X^{2/3})$ remainder after all exceptional spectral terms are included, while the $O(X^{1/2+\varepsilon})$ remainder is conjectural. The Koide cutoff is a different sharp polyhedral-sector height, so the standard radial theorem cannot be imported without a new transfer or trace-formula argument.

36.1 Quaternion algebra and reduced invariants

Let

$$B = (18, -6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18,\ J^2 + 6,\ IJ + JI), \qquad K = IJ.$$

Then

$$K^2 = -I^2J^2 = 108.$$

For

$$x = x_0 + x_1 I + x_2 J + x_3 K,$$

the standard involution, reduced trace, and reduced norm are

$$\bar{x} = x_0 - x_1 I - x_2 J - x_3 K, \qquad \mathrm{trd}(x) = x + \bar{x} = 2x_0,$$

and

$$\mathrm{nrd}(x) = x\bar{x} = x_0^2 - 18x_1^2 + 6x_2^2 - 108x_3^2. \tag{1081}$$

All signs in (1081) are forced by $\mathrm{nrd}(I) = -18$, $\mathrm{nrd}(J) = 6$, and $\mathrm{nrd}(K) = -108$.

36.2 The proposed lattice is an order

Put

$$E = \frac{I + J}{2}, \qquad F_0 = \frac{I - J}{2}, \qquad G = \frac{K}{2},$$

and let

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}E \oplus \mathbb{Z}F_0 \oplus \mathbb{Z}G. \tag{1082}$$

The displayed elements are a rational basis of B, because

$$I = E + F_0, \qquad J = E - F_0, \qquad K = 2G.$$

Direct multiplication gives the complete nontrivial table

	E	F_0	G
E	3	$6 - G$	$6E - 3F_0$
F_0	$6 + G$	3	$3E - 6F_0$
G	$-6E + 3F_0$	$-3E + 6F_0$	$27.$

(1083)

Every structure constant in (1083) is integral. Thus (1082) is a unital subring and a full rank-four $\mathbb{Z}$-lattice in B, hence an order. In particular, the unital ring sometimes denoted $\mathbb{Z}[1, E, F_0, G]$ is exactly the direct sum in (1082).

An arbitrary order element is

$$x = r + sE + tF_0 + uG = r + \frac{s+t}{2}I + \frac{s-t}{2}J + \frac{u}{2}K \qquad (r, s, t, u \in \mathbb{Z}), \tag{1084}$$

and (1081) gives

$$\mathrm{trd}(x) = 2r, \qquad \mathrm{nrd}(x) = r^2 - 3s^2 - 12st - 3t^2 - 27u^2. \tag{1085}$$

The reduced-trace pairing on the basis $(1, E, F_0, G)$ is

$$\big(\mathrm{trd}(b_i b_j)\big)_{i,j} = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 6 & 12 & 0 \\ 0 & 12 & 6 & 0 \\ 0 & 0 & 0 & 54 \end{pmatrix},$$

with determinant

$$-11664 = -108^2. \tag{1086}$$

Thus the reduced discriminant of this order is 108, not 6. The algebra has discriminant 6, proved below, so $\mathcal{O}$ is nonmaximal and has index 18 in a maximal order containing it. This index is not an Eichler level: the order is nonmaximal at the ramified primes 2 and 3, so it is not an Eichler order in the standard sense. Any optimal-embedding or trace-formula result that assumes a maximal order must account for this exact nonmaximal order instead of replacing $\mathcal{O}$.

36.3 Trace-zero lattice and exact shell bijection

Equation (1085) gives

$$\mathcal{O}^0 := \{x \in \mathcal{O} : \mathrm{trd}(x) = 0\} = \mathbb{Z}E \oplus \mathbb{Z}F_0 \oplus \mathbb{Z}G.$$

Writing

$$q = s + t, \qquad a = s - t, \qquad n = u$$

gives the exact description

$$\boxed{\mathcal{O}^0 = \left\{ \frac{qI + aJ + nK}{2} : a, q, n \in \mathbb{Z},\ a \equiv q \pmod 2 \right\}.} \tag{1087}$$

The inverse formulas are

$$s = \frac{q+a}{2}, \qquad t = \frac{q-a}{2}, \qquad u = n,$$

which are integral exactly when $a \equiv q \pmod 2$.
Define

$$F(a, q, n) = a^2 - 3q^2 - 18n^2, \qquad v(a, q, n) = \frac{qI + aJ + nK}{2},$$

and

$$\gamma(a, q, n) = 2 + v(a, q, n). \tag{1088}$$

Distinct pure quaternion basis elements anticommute, so

$$v(a,q,n)^2 = \frac{18q^2 - 6a^2 + 108n^2}{4} = -\frac{3}{2}F(a,q,n), \tag{1089}$$

and

$$\mathrm{nrd}(v) = \frac{3}{2}F(a,q,n).$$

Since v has reduced trace zero,

$$\mathrm{trd}(\gamma) = 4, \qquad \mathrm{nrd}(\gamma) = 4 + \mathrm{nrd}(v) = 4 + \frac{3}{2}F(a,q,n). \tag{1090}$$

Therefore (1088) defines a bijection

$$\boxed{\left\{(a,q,n) \in \mathbb{Z}^3 : F(a,q,n) = -2,\ a \equiv q \pmod 2\right\} \xrightarrow{\sim} \{\gamma \in \mathcal{O} : \mathrm{trd}(\gamma) = 4,\ \mathrm{nrd}(\gamma) = 1\}.} \tag{1091}$$

The inverse sends

$$\gamma = 2 + sE + tF_0 + uG$$

to

$$(a,q,n) = (s - t, s + t, u).$$

There is no hidden congruence loss. On the integral equation $F = -2$, reduction modulo 2 gives

$$a^2 - q^2 \equiv 0 \pmod 2,$$

so $a \equiv q \pmod 2$ is automatic. The parity condition in (1091) records the order lattice exactly but removes no integral shell point.

Finally,

$$(\gamma - 2)^2 = 3, \qquad \gamma^2 - 4\gamma + 1 = 0, \qquad \gamma^{-1} = 4 - \gamma = 2 - v \in \mathcal{O}. \tag{1092}$$

Thus every element on the right of (1091) lies in $\Gamma = \mathcal{O}^1$.

36.4 Division over $\mathbb{Q}$ and splitting over $\mathbb{R}$

Square factors may be removed from a Hilbert-symbol presentation, so

$$(18, -6)_{\mathbb{Q}} \simeq (2, -6)_{\mathbb{Q}}.$$

At $p = 3$, write

$$18 = 3^2 \cdot 2, \qquad -6 = 3 \cdot (-2).$$

The odd-prime Hilbert-symbol formula gives

$$(18, -6)_3 = (-1)^{2\cdot 1\cdot (3-1)/2} \left(\frac{2}{3}\right)\left(\frac{-2}{3}\right)^2 = -1. \tag{1093}$$

Thus $B \otimes \mathbb{Q}_3$ is division, and B is a division algebra over $\mathbb{Q}$.

At $p = 2$, write

$$18 = 2 \cdot 9, \qquad -6 = 2 \cdot (-3).$$

The 2-adic formula

$$(2^\alpha u, 2^\beta w)_2 = (-1)^{((u-1)/2)((w-1)/2)+\alpha(w^2-1)/8+\beta(u^2-1)/8}$$

gives

$$(18, -6)_2 = -1. \tag{1094}$$

At every odd prime other than 3, both valuations are zero and the Hilbert symbol is $+1$. At the real place, $18 > 0$, so the real Hilbert symbol is $+1$. Consequently

$$\mathrm{Ram}(B) = \{2, 3\}, \qquad \mathrm{disc}(B) = 6, \tag{1095}$$

and

$$B \otimes_{\mathbb{Q}} \mathbb{R} \simeq M_2(\mathbb{R}).$$

An explicit real splitting is

$$\rho(I) = \begin{pmatrix} 3\sqrt{2} & 0 \\ 0 & -3\sqrt{2} \end{pmatrix}, \qquad \rho(J) = \begin{pmatrix} 0 & 6 \\ -1 & 0 \end{pmatrix}. \tag{1096}$$

These matrices square to 18 and -6 and anticommute. Therefore

$$\rho(K) = \begin{pmatrix} 0 & 18\sqrt{2} \\ 3\sqrt{2} & 0 \end{pmatrix}.$$

For the element associated with (a, q, n),

$$\rho(\gamma) = \begin{pmatrix} 2 + \frac{3\sqrt{2}}{2} q & 3a + 9\sqrt{2}\, n \\ -\frac{1}{2} a + \frac{3\sqrt{2}}{2} n & 2 - \frac{3\sqrt{2}}{2} q \end{pmatrix}. \tag{1097}$$

Its determinant is the reduced norm in (1090).

Let $\mathcal{M}$ be a maximal overorder containing $\mathcal{O}$. Since $[\mathcal{M} : \mathcal{O}] = 18$, the finite abelian group $\mathcal{M}/\mathcal{O}$ has order 18. Lagrange's theorem therefore shows that multiplication by 18 annihilates every element of this additive quotient; equivalently, the ideal $(18) = 18\mathbb{Z}$ annihilates $\mathcal{M}/\mathcal{O}$, and $18\mathcal{M} \subset \mathcal{O}$. Hence the principal congruence subgroup

$$\mathcal{M}^1(18\mathcal{M}) := \{u \in \mathcal{M}^1 : u - 1 \in 18\mathcal{M}\}$$

satisfies the exact inclusions

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1.$$

Thus $\mathcal{O}^1$ has finite index in the norm-one group of a maximal order and contains a principal congruence subgroup. This is the required finite-index and congruence bridge; it does not replace the local type of $\mathcal{O}$ by the local type of $\mathcal{M}$.

The image

$$\bar{\Gamma} = \rho(\mathcal{O}^1)/\{\pm 1\} \subset PSL_2(\mathbb{R})$$

is a discrete congruence arithmetic Fuchsian group. Since the defining rational quaternion algebra is division, $\bar{\Gamma}\backslash\mathbb{H}$ is compact; see [59, Lemma 38.1.2, Proposition 38.1.3, and Definition 38.7.2].

36.5 Centralizer, optimal quadratic order, and primitivity

Fix a shell point and set

$$v = \gamma - 2.$$

Equation (1092) gives $v^2 = 3$, hence

$$\mathbb{Q}(\gamma) = \mathbb{Q}(v) \cong \mathbb{Q}(\sqrt{3}). \tag{1098}$$

Since γ is noncentral, the centralizer theorem for quaternion algebras gives

$$C_B(\gamma) = \mathbb{Q}(\gamma). \tag{1099}$$

Moreover,

$$\mathbb{Z}[v] \subseteq \mathcal{O} \cap \mathbb{Q}(v).$$

Every element of $\mathcal{O}$ is integral over $\mathbb{Z}$, so every element of the intersection is an algebraic integer in $\mathbb{Q}(\sqrt{3})$. Because

$$\mathcal{O}_{\mathbb{Q}(\sqrt{3})} = \mathbb{Z}[\sqrt{3}]$$

and $\mathbb{Z}[v]$ is already isomorphic to this maximal quadratic order, both inclusions force

$$\boxed{\mathcal{O} \cap \mathbb{Q}(\gamma) = \mathbb{Z}[v] = \mathbb{Z}[\gamma - 2] \cong \mathbb{Z}[\sqrt{3}].} \tag{1100}$$

Thus each shell point gives an optimal embedding

$$\iota_\gamma : \mathbb{Z}[\sqrt{3}] \hookrightarrow \mathcal{O}, \qquad \sqrt{3} \longmapsto \gamma - 2. \tag{1101}$$

Conversely, every embedding (1101) gives a trace-4, norm-1 element $2 + \iota(\sqrt{3})$, hence one integral ternary point. The shell is exactly the set of these optimal embeddings before quotienting by conjugation.

The norm-one units of $\mathbb{Z}[\sqrt{3}]$ are

$$\{\pm(2 + \sqrt{3})^k : k \in \mathbb{Z}\}. \tag{1102}$$

For completeness, $2 + \sqrt{3}$ is the smallest positive norm-one unit greater than 1. If $x + y\sqrt{3} > 1$ has norm 1 and $y > 0$, then $y \geq 1$, $x^2 = 1 + 3y^2$, and $x + y\sqrt{3} \geq 2 + \sqrt{3}$. Multiplication by a suitable power of $2 - \sqrt{3}$ reduces any positive norm-one unit to $[1, 2 + \sqrt{3})$, proving (1102).

Equations (1099)–(1102) imply

$$C_\Gamma(\gamma) = \{\pm\gamma^k : k \in \mathbb{Z}\}. \tag{1103}$$

Hence γ is primitive in Γ and its image is primitive in $\bar{\Gamma}$. Indeed, a root of γ, or a root of its class modulo the center, would commute with γ and therefore be a unit from (1102); the exponent equation $rk = 1$ has no solution with $k \geq 2$.

The reduced characteristic polynomial is

$$T^2 - 4T + 1,$$

so the real eigenvalues are

$$2 + \sqrt{3}, \qquad 2 - \sqrt{3}. \tag{1104}$$

For a hyperbolic element of $PSL_2(\mathbb{R})$,

$$2\cosh(\ell/2) = |\operatorname{tr}(\rho(\gamma))| = 4.$$

Thus every shell element has primitive translation length

$$\boxed{\ell_0 = 2\operatorname{arcosh}(2) = 2\log(2+\sqrt{3}).} \tag{1105}$$

36.6 Integral adjoint action and finitely many classes

Let

$$\mathscr{S} = \{\gamma \in \Gamma : \operatorname{trd}(\gamma) = 4\}.$$

By (1091), $\mathscr{S}$ is exactly the entire integral ternary shell. By (1105), every element of $\mathscr{S}$ is hyperbolic with translation length ℓ_0.

Put

$$L = \{(a,q,n) \in \mathbb{Z}^3 : a \equiv q \pmod 2\}.$$

Conjugation preserves $\mathcal{O}$, $\mathcal{O}^0$, and reduced norm. Because

$$\operatorname{nrd}(v(a,q,n)) = \frac{3}{2}F(a,q,n),$$

it gives an exact orthogonal representation on the actual parity lattice

$$\operatorname{Ad} : \Gamma/\{\pm 1\} \hookrightarrow SO^+(L,F). \tag{1106}$$

The kernel before dividing by the center is $\Gamma \cap \mathbb{Q}^\times = \{\pm 1\}$. No action on the larger lattice $\mathbb{Z}^3$, and no finite-index equality with its full integral orthogonal group, is asserted.

There are only finitely many Γ-conjugacy classes in $\mathscr{S}$. Choose a compact fundamental set $D \subset \mathbb{H}$ for $\bar{\Gamma}$, and fix $z_0 \in \mathbb{H}$. For each conjugacy class, conjugate a representative so that its invariant axis meets D. If $z \in D$ lies on that axis, then

$$d(z, \gamma z) = \ell_0.$$

Writing

$$R_D = \max_{w \in D} d(z_0, w),$$

the triangle inequality gives

$$d(z_0, \gamma z_0) \le 2R_D + \ell_0.$$

Proper discontinuity leaves only finitely many elements of $\bar{\Gamma}$ moving z_0 by at most this amount. Hence only finitely many $\bar{\Gamma}$-classes occur. This also proves finiteness of the corresponding Γ-classes: if two images are conjugate in $\bar{\Gamma}$, lifting a conjugator gives

$$\delta\gamma_1\delta^{-1} = \pm\gamma_2.$$

Both sides before the possible sign have reduced trace 4; the minus sign would have reduced trace -4, so it is impossible.

Choose representatives $\gamma_1, \dots, \gamma_h$. Then

$$\boxed{\mathscr{S} = \coprod_{j=1}^{h} \{\delta\gamma_j\delta^{-1} : [\delta] \in \Gamma/C_\Gamma(\gamma_j)\}, \qquad C_\Gamma(\gamma_j) = \{\pm\gamma_j^k : k \in \mathbb{Z}\}.} \tag{1107}$$

This proves finiteness but does not compute h or a representative list. Those finite arithmetic data are needed by a class-by-class spectral formula whose coefficients contain geodesic periods.

36.7 Exact reformulation of the Koide chamber count

For $\gamma \in \mathscr{S}$, let

$$c(\gamma) = (a(\gamma), q(\gamma), n(\gamma))$$

be the inverse coordinate map in section 36.3. Let

$$\mathcal{C} = \{(a, q, n) \in \mathbb{R}^3 : a > 0,\ q \geq 0,\ n \geq 0,\ a + 4n > q\},$$

and retain the canonical height

$$H(a, q, n) = 4a + 17n.$$

The integral bijections in theorems 25.4 and 29.3 give

$$\boxed{R(X) = \sum_{j=1}^{h} \#\left\{ [\delta] \in \Gamma/C_\Gamma(\gamma_j) : \begin{array}{l} c(\delta\gamma_j\delta^{-1}) \in \mathcal{C}, \\ H(c(\delta\gamma_j\delta^{-1})) \leq X \end{array} \right\}.} \tag{1108}$$

There is no hidden finite-to-one factor in (1108): cosets by the centralizer parametrize the elements of each class exactly once, and (1091) parametrizes shell elements by triples exactly once. Put $C_j = C_\Gamma(\gamma_j)$. Here Γ/C_j is the set of left cosets δC_j, not the set of right cosets $C_j\delta$, and the exact orbit bijection is

$$\Phi_j : \Gamma/C_j \longrightarrow \{\delta\gamma_j\delta^{-1} : \delta \in \Gamma\}, \qquad \delta C_j \longmapsto \delta\gamma_j\delta^{-1}.$$

Indeed, right multiplication of δ by an element of C_j does not change its conjugate, so Φ_j is well defined and surjective; and

$$\Phi_j(\delta_1 C_j) = \Phi_j(\delta_2 C_j) \iff \delta_2^{-1}\delta_1 \in C_j \iff \delta_1 C_j = \delta_2 C_j,$$

so it is injective. If $\pi : \Gamma \to \bar{\Gamma}$ is the quotient map, then

$$\Gamma/C_j \cong \bar{\Gamma}/C_{\bar{\Gamma}}(\bar{\gamma}_j)$$

via $\delta C_j \mapsto \pi(\delta)C_{\bar{\Gamma}}(\bar{\gamma}_j)$. To prove the required equality of stabilizers, observe first that $\pi(C_j) \subseteq C_{\bar{\Gamma}}(\bar{\gamma}_j)$. Conversely, if $\pi(\delta)$ centralizes $\bar{\gamma}_j$, then $\delta\gamma_j\delta^{-1} = \pm\gamma_j$. Reduced-trace invariance and $\mathrm{trd}(\gamma_j) = 4$ exclude the minus sign, so $\delta \in C_j$. Thus $\pi(C_j) = C_{\bar{\Gamma}}(\bar{\gamma}_j)$; together with $\ker\pi = \{\pm 1\} \subset C_j$, this proves the displayed bijection and shows that passage to PSL_2 introduces no factor 2. Positive trace selects the lift sign, but the chamber is not invariant under $\gamma \mapsto \gamma^{-1}$; inverse classes and their orientations must still be retained separately.

Under the splitting (1096),

$$\|\rho(\gamma)\|_{\mathrm{F}}^2 = 8 + 9q^2 + \frac{37}{4}a^2 + \frac{105\sqrt{2}}{2}an + \frac{333}{2}n^2. \tag{1109}$$

For $i \in \mathbb{H}$,

$$2\cosh d(i,\rho(\gamma)i) = \|\rho(\gamma)\|_{\mathrm{F}}^2.$$

Since $\sinh(\ell_0/2) = \sqrt{3}$, the standard Huber parameter is

$$X_i(\gamma)^2 = \frac{\|\rho(\gamma)\|_{\mathrm{F}}^2 - 2}{12}. \tag{1110}$$

The shell equation simplifies (1109) exactly:

$$\|\rho(\gamma)\|_{\mathrm{F}}^2 = 14 + \frac{(7a + 15\sqrt{2}\,n)^2}{4},$$

and hence

$$\boxed{X_i(\gamma)^2 = 1 + \frac{(7a + 15\sqrt{2}\,n)^2}{48}.} \tag{1111}$$

Thus, in the positive chamber, the radial cutoff has the exact irrational linear height $7a + 15\sqrt{2}\,n$. The Koide height $H = 4a + 17n$ is a different linear height. The two sizes are comparable:

$$a \le \frac{H}{4}, \qquad n \le \frac{H}{17}, \qquad 0 \le q < a + 4n \le \frac{33}{68}H,$$

and conversely

$$H \le \sqrt{305}\sqrt{a^2 + q^2 + n^2}.$$

Norm equivalence in the trace-zero space then compares H coarsely with X_i. Such comparability identifies the growth scale but does not transfer a main coefficient or a remainder between the sharply bounded regions.

36.8 Consequence for the open square-root discrepancy

The quaternion construction proves:

1. the ternary shell is exactly a finite union of primitive hyperbolic conjugacy classes in one explicit cocompact arithmetic Fuchsian group;
2. $\log(2 + \sqrt{3}) = \ell_0/2$, so its occurrence in the main constant is tied to the primitive translation length;
3. the conjectural exponent $1/2$ is the same exponent proposed for the standard hyperbolic lattice-point problem in conjugacy classes, giving a precise analytic analogy but not an equivalence of the two cutoff problems.

The third item classifies the remaining analytic problem; it does not prove the error estimate. Chatzakos–Petridis [55, Theorem 1.1] give

$$E(\mathcal{H}, X; z) = O(X^{2/3})$$

for the standard radial count after the main expression, including every small-eigenvalue term, is subtracted. Their Conjecture 5.7 proposes

$$E(\mathcal{H}, X; z) = O_\varepsilon(X^{1/2+\varepsilon}).$$

The cutoff in (1108) uses both a sharp chamber and $4a + 17n$, not the radial height $7a + 15\sqrt{2}\,n$ in (1111). Their symmetric difference can have main-order size, so even the conjectured radial bound would not by itself prove the Koide bound. Therefore the status of

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon})$$

remains **OPEN**. Closing it requires a spectral or automorphic argument for the exact finite sum (1108), with its exact sector and height.

36.9 Missing mathematics before an error theorem can be claimed

The bridge does not supply the following results.

1. **Class data.** Determine h, representatives γ_j, and orientation/inversion conventions. Finiteness does not compute the periods in a spectral expansion.

2. **Exact nonmaximal order.** The order has reduced discriminant 108, is nonmaximal at the ramified primes, and is not an Eichler order in the standard sense. A result for maximal or specified Eichler orders must be checked at this order or transferred through finite-index groups with all coset multiplicities retained. Index and discriminant alone do not determine the precise local order type required by a trace formula.

3. **Central quotient.** Track the passage from $\Gamma \subset SL_2(\mathbb{R})$ to $\bar{\Gamma} \subset PSL_2(\mathbb{R})$, including ± 1, inverse classes, and possible orbifold torsion.

4. **Exact cutoff.** Prove a trace formula or effective orbital theorem for all four inequalities defining $\mathcal{C}$ and for $4a + 17n \le X$. The radial Huber cutoff is different.

5. **Sharp-boundary control.** Remove smoothing with an explicit estimate at the required scale. The two Pell faces contribute only $O(\log X)$, as already proved canonically, but this alone does not control every smoothing strip adjacent to all chamber walls.

6. **Exceptional spectrum.** Once an appropriate sector spectral expansion is constructed, one must determine whether it contains X^{s_j} terms for $\lambda_j = s_j(1 - s_j) < 1/4$, weighted by closed-geodesic periods. To subtract only $C_0 X$ with a $O(X^{1/2+\varepsilon})$ remainder, every relevant exceptional term must be shown to vanish or cancel, or it must be displayed and subtracted. No such vanishing is proved here.

7. **Pointwise square-root estimate.** After secondary terms and the exact sector are handled, the needed pointwise $1/2 + \varepsilon$ bound remains at the conjectural strength of the classical problem. Mean-square results do not give it for every X.

8. **Main coefficient.** A spectral derivation must reproduce

$$C_0 = \frac{9}{2\pi^2} \log(2 + \sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right)$$

   using the same lattice, chamber, strict walls, height, and Haar measure. The existing local-density proof fixes this constant but does not calibrate a new trace formula automatically.

9. **Transfer between the two exact heights.** Formula (1111) already gives the exact standard radial parameter. What is missing is a remainder-level transfer between its height $7a + 15\sqrt{2}\, n$ and the canonical height $4a + 17n$, together with the chamber walls. Coarse norm comparability does not preserve a discrepancy estimate.

These are concrete mathematical tasks. None invalidates the quaternion bridge; together they explain why it does not close the open claim.

36.10 Reproducibility and references

The canonical quaternion replay is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/verify_quaternion_bridge.py` (SHA-256 `6675ba2fc4fa8c2b10e14558e412178b6b5b579c9c96afca78e1db9bcdf5c6b4`). Its recorded successful output is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/verification.log` (SHA-256 `66853c6191a000c7c827aacd1e0e33dcf34dbe3dd61d746f66fa4d805f34c91e`), and its local manifest is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/quaternion/SHA256SUMS.txt` (SHA-256 `f9d044db0d4b003e7aef7649b3394e6b299c09a6b2eb7adca095f8f9100ba623`). The checkpoint-wide manifest is `supporting_materials/workbench/evidence/koide_exact_checkpoint_20260904/SHA256SUMS` (SHA-256 `8d108eb928e5d07093badde169f443a4a38b6fcf3eb59d5192bdd33ecef8d7a4`). The replay checks the basis multiplication table; the trace, norm, and coordinate formulas for $(r, s, t, u) \in \{-4, -3, \ldots, 4\}^4$; the shell identities for $(a, q, n) \in \{-24, -23, \ldots, 24\}^2 \times \{-8, -7, \ldots, 8\}$, containing 48 shell points; the inverse identity on those points; the radial-height identity on that box; and the Hilbert-symbol formulas at 2 and 3. These finite ranges are consistency checks, not proofs outside the stated boxes. The exact canonical sources used for the counting statements imported into this bridge are:

- `supporting_materials/workbench/research/koide_counting_repair.tex`, SHA-256 `965b0a3dca8bfe4f22b9e9208368fa04697070fd25bd06bd67f2481d17cb9d9d`;
- `supporting_materials/workbench/research/koide_exact_parametrization.tex`, SHA-256 `a457709dcce0fce0774eb9326b84b00d5d8144c2087f480da704974e499bcdd3`;
- `supporting_materials/workbench/research/koide_exact_census.tex`, SHA-256 `18ed5968b7e67d9a0b23d8675b40c37ba46170ad29efdbb9ce10ed632c99600a`.

The bounded coordinate-bijection replay, whose scope does not include the quaternion multiplication table, Hilbert symbols, centralizer, conjugacy finiteness, or spectral conclusions, is `supporting_materials/workbench/reproduction/t1_koide_exact/verify_koide_narain_root_density.py` (SHA-256 `e4bf2982b26b07add6057b5d2d5c771a52ef52c629b5ad5f0aab24e440e41986`). Its stored height-200000 output is `supporting_materials/workbench/reproduction/t1_koide_exact/koide_verification_200000.json` (SHA-256 `b907f0ee5bdd3ad80622b669722d218b79a11a5baab0d4571e141936100706b7`); the exact invocation and scope are stated in `supporting_materials/workbench/reproduction/t1_koide_exact/README.md` (SHA-256 `5e9cdcb9e67954c37d8fde69e13549172aa03245674b358e4a1d25f8d9d3bff0`). The quaternion identities and deductions in this section are proved in the displayed algebraic calculations themselves, not inferred from that bounded replay.

References:

- John Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, especially Section 10.1, Lemma 38.1.2, and Proposition 38.1.3; DOI.
- Dimitrios Chatzakos and Yiannis N. Petridis, *The hyperbolic lattice point problem in conjugacy classes*, arXiv:1504.01307v2, Theorem 1.1 and Conjecture 5.7; arXiv.

- The exact Koide chamber, height, main term, and Pell-face estimates are in Sections 25–35 of this workbench.

37 Primary-source theorem audit of effective counting for the fixed Koide ternary quadric sector

Date: 4 September 2026
Exact counting function:

$$R(X) = \#\left\{ (a,q,n) \in \mathbb{Z}^3 : \begin{array}{l} a^2 - 3q^2 - 18n^2 = -2, \\ a > 0,\ q \geq 0,\ n \geq 0,\ a + 4n > q, \\ 4a + 17n \leq X \end{array} \right\}.$$

The leading constant proved in theorem 25.11 is

$$C_0 = \frac{9}{2\pi^2} \log(2+\sqrt{3}) \log\left(\frac{3}{2\sqrt{2}}\right).$$

The canonical project-relative mathematical sources used by this audit are:

- `supporting_materials/workbench/research/koide_counting_repair.tex`, which proves the coordinate bijection, local densities, invariant-volume factor, Gorodnik–Oh–Shah hypotheses, and calibrated constant, SHA-256 `965b0a3dca8bfe4f22b9e9208368fa04697070fd25bd06bd67f2481d17cb9d9d`;
- `supporting_materials/workbench/research/koide_exact_parametrization.tex`, which proves the exact parametrization and the Benoist–Oh power-saving application, SHA-256 `a457709dcce0fce0774eb9326b84b00d5d8144c2087f480da704974e499bcdd3`;
- `supporting_materials/workbench/research/koide_quaternion_bridge.tex`, which proves the exact order, shell, centralizer, conjugacy-class, and radial-height statements used below, SHA-256 `93759f7ba99870c84c9144a373a53d2e9a8bc3ae909518e47991373ace0b47bb`;
- `supporting_materials/workbench/research/koide_exact_census.tex`, which records the separately bounded finite census, SHA-256 `18ed5968b7e67d9a0b23d8675b40c37ba46170ad29efdbb9ce10ed632c99600a`.

The primary publications themselves are identified by exact versions, dates where version-sensitive, locators, and DOI or arXiv links in section 37.11. They are not claimed to be stored in a separate canonical evidence manifest.

37.1 Determination

Among the source versions enumerated in section 37.11, no theorem proves, for this exact fixed level, full integral lattice, sharp chamber, and sharp polyhedral height, the unconditional estimate

$$R(X) - C_0 X = O_\varepsilon(X^{1/2+\varepsilon}).$$

The strongest exact-domain conclusion obtained from the general theorems enumerated there is

$$\boxed{R(X) = C_0 X + O(X^{1-\eta}) \quad \text{for some unspecified } \eta > 0.}$$

It follows by applying Benoist–Oh, Definition 13.1 and Theorem 13.2, orbit by orbit after directly verifying effective well-roundedness for the present family. Benoist–Oh do not give a numerical lower bound for η, and their proof does not imply $\eta \geq 1/2$.

The superficially stronger exponents in nearby problems do not answer the question:

- Good's sharp hyperbolic conjugacy-class theorem, as formulated and reproved by Chatzakos–Petridis, gives $O(X^{2/3})$ only for a different radial cutoff and only after every small-eigenvalue term X^{s_j} has been placed in the main expression. The same paper states $O_\varepsilon(X^{1/2+\varepsilon})$ as Conjecture 5.7.

- Liu–Sarnak obtain a smooth radial error $O_\varepsilon(X^{1/2+\theta+\varepsilon})$, hence $O_\varepsilon(X^{39/64+\varepsilon})$ from $\theta = 7/64$, and square-root strength only under the generalized Selberg/Ramanujan-at-infinity conjecture for the relevant Jacquet–Langlands transfers. Their paper's global affine-sieve setup assumes $t\det(f)$ square-free, but the analytic Proposition 3.2 itself is formulated for a variable cocompact lattice with a spectral gap. After the exact quaternionic identification it applies class by class to this lattice. What it still does not supply is a sharp cutoff, the coordinate chamber, or the nonradial height $4a + 17n$.
- Gorodnik–Nevo's published sharp polar-sector theorem loses powers through set thickening. For $\dim G = 3$, it gives $O(X^{19/20+\epsilon})$ even with tempered spectrum; the Kim–Sarnak complementary-series threshold, which does apply to the exact congruence arithmetic lattice by Jacquet–Langlands, gives the still weaker $O(X^{39/40+\epsilon})$. This remains a comparison theorem because Gorodnik–Nevo's fixed polar-sector family has not been identified with the exact polyhedron, including all integral orbits.
- Square-root-scale statements in Chatzakos–Petridis, Chatzakos, Voskou, and Lekkas–Petridis are mean-square, averaged, Omega, or conjectural statements. They are not pointwise sharp-sector upper bounds.
- Recent split-stabilizer theorems assume $-m\det(F)$ is a square and have a $T\log T$-type main term. Here $-(-2)\det(F) = 108$ is not a square, so they concern the opposite arithmetic regime.

The two independent missing inputs are therefore (i) a sharp-boundary spectral or arithmetic estimate for the actual height and chamber, and (ii) control or elimination of any complementary-spectrum secondary terms for the actual sector. The Pell faces $q = 0$ and $n = 0$, which contribute only $O(\log X)$, do not solve the moving outer-boundary problem.

37.2 Exact arithmetic and geometric classification

37.2.1 The rational group and stabilizer

Let

$$F(a, q, n) = a^2 - 3q^2 - 18n^2, \qquad \mathscr{X} = \{F = -2\}.$$

The matrix of F is $\mathrm{diag}(1, -3, -18)$, so $\det(F) = 54$ and

$$-(-2)\det(F) = 108 \notin \mathbb{Q}^{\times 2}.$$

The homogeneous form F is anisotropic over $\mathbb{Q}$. Indeed, suppose a primitive integral triple satisfies

$$a^2 - 3q^2 - 18n^2 = 0.$$

Modulo 3, one gets $3 \mid a$; write $a = 3a_1$. Dividing by 3 and reducing modulo 3 gives $3 \mid q$; write $q = 3q_1$. The resulting equation is

$$a_1^2 = 3q_1^2 + 2n^2.$$

Modulo 3, this says $a_1^2 + n^2 \equiv 0 \pmod 3$, whence $3 \mid a_1, n$. Thus $3 \mid a, q, n$, contradicting primitivity. Put $G = \mathrm{Spin}(F)$ and $x_0 = (1, 1, 0)$. The orthogonal complement of x_0 is

$$x_0^\perp = \{(3s, s, t) : s, t \in \mathbb{Q}\}, \qquad F(3s, s, t) = 6(s^2 - 3t^2).$$

Consequently the connected stabilizer $H = \mathrm{Stab}_G(x_0)^\circ$ is the norm-one torus

$$\mathrm{Res}^1_{\mathbb{Q}(\sqrt{3})/\mathbb{Q}}\,\mathbb{G}_m.$$

It is split over $\mathbb{R}$, anisotropic over $\mathbb{Q}$, and has no nontrivial $\mathbb{Q}$-character. Under the quaternionic identification in section 36.5, x_0 corresponds to $v_0 = (I + J)/2$, with $v_0^2 = 3$, and

$$\mathcal{O} \cap \mathbb{Q}(v_0) = \mathbb{Z}[v_0], \qquad H(\mathbb{Q}) \cap \Gamma = \{\pm(2 + v_0)^k : k \in \mathbb{Z}\}.$$

Thus $(H(\mathbb{Q}) \cap \Gamma)/\{\pm 1\}$ is infinite cyclic. Under $H(\mathbb{R})^\circ \cong \mathbb{R}_{>0}$, the logarithm sends its positive part to $\log(2 + \sqrt{3})\mathbb{Z} \subset \mathbb{R}$; the quotient is compact, and in particular $(H \cap \Gamma)\backslash H(\mathbb{R})$ has finite volume. Since F is $\mathbb{Q}$-anisotropic, the arithmetic Fuchsian quotient attached to G is cocompact as well. There is no ambient hyperbolic cusp.

This is the finite-volume stabilizer regime associated with a closed geodesic. It is not the rationally split regime with divergent stabilizer in which one obtains $T \log T$ growth.

37.2.2 The exact height really is a norm-sector height

Define

$$\|(a, q, n)\|_* = max\{|4a + 17n|, |q|, |a|\}.$$

This is a genuine norm because it is the pullback of the sup norm under the invertible linear map

$$(a, q, n) \longmapsto (4a + 17n, q, a),$$

whose determinant is -17. On the target chamber,

$$0 < a < 4a + 17n, \qquad 0 \le q < a + 4n < 4a + 17n,$$

and therefore $\|(a, q, n)\|_* = 4a + 17n$. Thus the outer cutoff is not merely comparable to a norm: it is exactly the restriction of a fixed ambient norm.

The additional wall $a + 4n > q$ removes only a bounded real set. If $q \ge a + 4n$, then

$$a^2 + 2 = 3q^2 + 18n^2 \ge 3(a + 4n)^2 + 18n^2,$$

so $2a^2 + 24an + 66n^2 \le 2$. Its only integral point in the positive orthant on $\mathscr{X}$ is $(1, 1, 0)$. The unbounded faces $q = 0$ and $n = 0$ are Pell families and contribute $O(\log X)$ integral points, but their real boundary strips still have to be included in a sharp-domain argument.

37.2.3 Invariant volume with a strong enough secondary estimate

Use

$$\sqrt{3}q = r\cos\theta, \qquad 3\sqrt{2}n = r\sin\theta, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

On $F = -2$, $r = \sqrt{a^2 + 2}$, and the gauge measure is

$$dm_\infty = \frac{1}{6\sqrt{6}}\, da\, d\theta.$$

Writing $c = 17/(3\sqrt{2})$, the outer boundary is

$$4a + c\sqrt{a^2 + 2}\sin\theta = X.$$

If $A_X(\theta)$ is its positive solution, then uniformly in θ,

$$A_X(\theta) = \frac{X}{4 + c\sin\theta} + O(X^{-1}).$$

Indeed $A_X(\theta) \asymp X$ uniformly and $\sqrt{A^2+2} = A + O(A^{-1})$. Once X contains the bounded wall excision, it follows that

$$m_\infty(B_X) = I_\infty X + O(1), \qquad I_\infty = \frac{1}{2\sqrt{3}} \log\!\left(\frac{3}{2\sqrt{2}}\right).$$

This $O(1)$ refinement matters: a theorem whose main term is the exact invariant volume can be converted to the displayed $C_0 X$ main term without spending a power of X.

37.2.4 The full integral locus is not assumed to be one orbit

The exact lattice must not be silently replaced by a convenient single orbit or by a commensurable maximal-order lattice. Borel–Harish-Chandra finiteness, in the form quoted by Benoist–Oh on journal p. 1892, says that $G(\mathbb{Z})$ has finitely many orbits on $\mathscr{X}(\mathbb{Z})$. Thus

$$\mathscr{X}(\mathbb{Z}) = \bigsqcup_{j=1}^{h} x_j \Gamma$$

for the integral spin group Γ attached to the chosen lattice and finitely many integral representatives x_j. An effective theorem for one orbit must be applied to every $x_j\Gamma$, and the finitely many main-measure scalars must be summed. The full-lattice scalar is calibrated from Borovoi's local/Brauer–Manin density formula in theorems 25.10 and 25.11; it is not obtained by declaring that all integral points form the orbit of x_0.

37.2.5 Exact quaternionic lattice and its congruence status

For the spectral comparisons below, the precise arithmetic lattice can be retained. Put

$$B = (18,-6)_{\mathbb{Q}} = \mathbb{Q}\langle I, J\rangle/(I^2 - 18,\ J^2 + 6,\ IJ + JI), \qquad K = IJ,$$

and

$$\mathcal{O} = \mathbb{Z}1 \oplus \mathbb{Z}\frac{I+J}{2} \oplus \mathbb{Z}\frac{I-J}{2} \oplus \mathbb{Z}\frac{K}{2}.$$

The complete integral multiplication table is (1083), and its reduced-trace Gram determinant is -108^2 by (1086); hence $\operatorname{discrd}(\mathcal{O}) = 108$. The Hilbert-symbol calculations (1093) and (1094), together with the unramified-place and real-place calculation summarized in (1095), show that the ramification set is exactly $\{2, 3\}$. Thus B is the indefinite quaternion division algebra of discriminant 6. The map

$$(a, q, n) \longmapsto \gamma(a, q, n) = 2 + \frac{qI + aJ + nK}{2}$$

is the bijection (1091) from $F(a, q, n) = -2$ to the reduced-trace-4, reduced-norm-one locus in $\mathcal{O}$. Indeed, the parity condition $a \equiv q \pmod 2$, which is exactly what membership in $\mathcal{O}$ requires, follows automatically from $F = -2$ modulo 2, and

$$\operatorname{nrd}(\gamma) = 4 + \frac{3}{2}F(a, q, n) = 1.$$

Thus the acting Fuchsian lattice is exactly $\Gamma = \mathcal{O}^1$, not a substituted maximal-order lattice. If $\mathcal{M} \supset \mathcal{O}$ is maximal, the reduced-discriminant index formula gives

$$[\mathcal{M} : \mathcal{O}] = \frac{108}{6} = 18, \qquad 18\mathcal{M} \subset \mathcal{O},$$

and consequently

$$\mathcal{M}^1(18\mathcal{M}) \subset \mathcal{O}^1 \subset \mathcal{M}^1.$$

Here $\mathcal{M}^1(18\mathcal{M}) = \{u \in \mathcal{M}^1 : u - 1 \in 18\mathcal{M}\}$. Therefore $\mathcal{O}^1$ is a congruence arithmetic Fuchsian group in the exact sense used by Chu–Li. The number 108 is the order's reduced discriminant, not an Eichler level: the order is nonmaximal at the ramified primes $2, 3$. This non-Eichler feature changes the finite compact-open subgroup and hence which automorphic representations have fixed vectors and the dimensions of those fixed-vector spaces; it does not change the Jacquet–Langlands/Kim–Sarnak archimedean bound.

37.3 Result matrix

Source and exact locator	What is actually proved	Status for the target
Duke–Rudnick–Sarnak (1993), Theorem 1.2, p. 145; Example 1.5, pp. 146–147	One arithmetic orbit on an affine symmetric variety: count $\sim$ invariant volume; ternary anisotropic stabilizer gives linear rather than $T \log T$ growth	Correct qualitative regime, no rate and no exact sharp-sector theorem
Eskin–McMullen (1993), Theorem 1.4, author PDF p. 3	Every well-rounded sequence has count asymptotic to volume	Qualitative only; no exponent
Gorodnik–Oh–Shah (2009), definitions before Theorem 1.6 and Theorem 1.6, p. 5; Theorem 1.18, pp. 8–9	Any norm and an admissible generic cone: full integral count $\sim$ volume	Direct qualitative theorem for the exact norm and cone, whose hypotheses are proved in theorems 25.8 and 25.9; no error term
Gorodnik–Oh–Shah (2010), Theorem 1.13, author PDF pp. 4–5	Well-rounded polar sectors and their asymptotic	Qualitative only
Benoist–Oh (2012), Theorem 1.3, p. 1894	Standard product/Euclidean height ball: relative $O(T^{-\delta})$ for some $\delta > 0$	Theorem 1.3 alone is not the exact polyhedron
Benoist–Oh (2012), Definition 13.1, p. 1931, and Theorem 13.2, pp. 1931–1932	Any independently verified effectively well-rounded family: $\mathrm{vol}(B)(1 + O(\mathrm{vol}(B)^{-\delta_0}))$	Applies to the exact family after the proof in section 37.4.2; yields only unspecified $O(X^{1-\eta})$
Gorodnik–Nevo (2012), Definition 3.1, p. 13; Theorem 8.1 and Corollary 8.3, pp. 34–35	Sharp fixed polar sectors: $O(\mathrm{vol}^{1-\zeta+\epsilon})$, $\zeta = (2n_e(p))^{-1}(1 + 3\dim G)^{-1}$	Explicit but much weaker; target polyhedron and finite orbit union require an additional reduction
Liu–Sarnak (2010), equations (2.3)–(2.8), pp. 4–5; Proposition 3.2, pp. 10–11; equation (3.65), p. 16	Smooth radial count for a cocompact lattice containing the fixed cyclic stabilizer: $O_\epsilon(T^{1/2+\theta+\epsilon})$	Exact lattice and finite class sum are available; unconditional 39/64, conditional 1/2, but fixed-width smoothing and the radial height do not match the sharp target
Good (1983), Theorem 4, p. 116, as explicitly restated and reproved in Chatzakos–Petridis (2016), Theorem 1.1, p. 3	Sharp radial hyperbolic conjugacy-class count: $O(X^{2/3})$ after all X^{s_j} small-spectrum terms	Closest sharp spectral model, not the exact polyhedron; square root is CP Conjecture 5.7

Source and exact locator	What is actually proved	Status for the target
Bourgain–Kontorovich–Sarnak (2010), Theorem 1.1, p. 2	Fixed angular Fourier modes in a sharp group norm ball: $O_{n,k}(T^{4/3})$ after T^{2s_j} terms	With affine height $X \asymp T^2$, again $X^{2/3}$; constants are not uniform enough to recover a sharp interval for free
Browning–Gorodnik (2017), Theorem 2.1, p. 11; Corollaries 2.5–2.6, pp. 24–25	Full Euclidean balls, finite adelic/orbit sums, and congruence conditions: unspecified $O(\mathrm{vol}^{1-\varrho})$	Correct nonsquare homogeneous regime, wrong height geometry and no explicit exponent
Cao–Huang (2022), Theorem 1.5, p. 4; Theorem 3.4, pp. 21–22	Arithmetic purity for anisotropic nonsquare ternary quadrics; effective input for Euclidean balls has an unspecified power	Qualitative full-lattice theorem, not a square-root sector result
Cao–Huang–Zhang (arXiv:2509.23320), Setting 2.1 and Theorem 2.1	Product Euclidean/max height balls with congruence-uniform $O(\mathbf{N}(\mathfrak{l})^{\kappa}\,\mathrm{vol}^{1-\delta})$	Current extension, but δ is unspecified and the growing real factor is the standard height ball, not this cone
Oh–Shah (2014), Theorem 1.2, p. 3; Theorem 6.1, pp. 13–14	Divergent-geodesic/split-stabilizer $T\log T$ count; smooth error $O(T)$, sharp relative logarithmic error	Excluded by the finite-volume anisotropic stabilizer
Xu–Zhang (2023), main theorem; Huang (arXiv:2412.03350), Theorem 1.1; Zhang (arXiv:2603.21080), Theorem 1.1	Split-stabilizer results assuming $-m\det(F)$ is a square, with smooth or radial $T\log T$-type expansions	108 is nonsquare; wrong main-term regime
Huang (arXiv:2412.05903), Theorems 1.1–1.2	Smooth compact real weight while the level varies as $m_0p_0^{2h}$; error of logarithmic relative strength	Varying level and smooth weight, not the fixed-level sharp count
Li–Liu (arXiv:2501.15033v2), Theorems 1.1–1.2, pp. 2–3; Lemma 2.3, pp. 5–6	Almost-prime results using a smooth radial orbit count with $O_\epsilon(d^{1+\epsilon}T^{1/2+\theta+\epsilon})$	Current use of the Liu–Sarnak input; globally assumes $t\det(f)$ square-free and supplies no sharp-sector count
Voskou (arXiv:2407.03134v3), Theorem 3 and Conjecture 1, p. 3	Four sharp orientation sectors in a closed-geodesic double-coset problem: $O(X^{2/3})$; square root conjectured	Current sharp comparator, not the target
Lekkas–Petridis (arXiv:2509.12902, version dated 24 Aug. 2026), Theorem 1.2, Theorem 1.5, Conjecture 1.6	Sharp closed-geodesic double-coset $O(X^{2/3})$, square-root mean square, pointwise square-root conjecture	Version-specific confirmation that mean square does not furnish the requested pointwise bound

37.4 Benoist–Oh: exact applicability and exact limitation

Among the general effective theorems enumerated in section 37.3, this is the one whose stated hypotheses can be verified for the actual family without changing its height or chamber.

37.4.1 What the theorem assumes

Benoist–Oh work with a connected semisimple simply connected almost K-simple group G, a finite-kernel rational representation, a symmetric subgroup H, and the hypothesis that H° has no nontrivial K-character; see journal pp. 1892–1893. For a congruence subgroup Γ_S, Theorem 12.5, journal p. 1929, supplies the effective equidistribution of the translates of $(\Gamma_S \cap H_S)\backslash H_S$ used in Theorem 13.2. Example (A), journal p. 1896, explicitly treats ternary quadratic forms provided the homogeneous form does not represent zero over $\mathbb{Q}$.

All of these hypotheses hold here:

- $G = \mathrm{Spin}(F)$ is simply connected and absolutely almost simple of type A_1, and $G(\mathbb{R})$ is noncompact.
- The standard three-dimensional orthogonal representation has finite kernel.
- The stabilizer of a nonisotropic vector is symmetric.
- The mod-3 descent in section 37.2.1 proves that F does not represent zero over $\mathbb{Q}$.
- H° is the $\mathbb{Q}$-anisotropic norm-one torus of $\mathbb{Q}(\sqrt{3})$, so it has no nontrivial $\mathbb{Q}$-character and its arithmetic quotient is compact.
- With $S = \{\infty\}$, the full integral spin group is itself a congruence subgroup in Benoist–Oh's sense.

There is no omitted wavefront or properness condition: symmetry of H is exactly what invokes their effective wavefront argument in Lemma 12.3 and Theorem 12.5. Cocompactness makes escape in the automorphic quotient simpler, but does not remove Definition 13.1(3), which still has to be checked on the growing affine set.

Here "congruence" is an exact property of the chosen integral group, not an inference from mere commensurability. Under the quaternionic realization of G, the finite-integrality conditions define a compact-open subgroup of $G(\mathbb{A}_f)$; equivalently, after placing the corresponding order $\mathcal{O}$ in a maximal order $\mathcal{O}_{\max}$, some principal subgroup $\{u \in \mathcal{O}^1_{\max} : u \equiv 1 \pmod{N\mathcal{O}_{\max}}\}$ lies in the exact integral spin group. This distinction matters for spectral-gap transfer: an arbitrary finite-index noncongruence subgroup need not inherit the Kim–Sarnak gap.

37.4.2 Definition 13.1 for the actual B_X

Let $B_X \subset \mathscr{X}(\mathbb{R})$ be the exact chamber cut out by the five linear conditions and $4a + 17n \le X$. In Benoist–Oh's notation, $B^+_{X,\epsilon} = B_X B_\epsilon$ and $B^-_{X,\epsilon} = \bigcap_{u \in B_\epsilon} B_X u$.

Condition (1) is void because there is no finite-place factor.

For condition (2), if u lies in an ϵ-ball about the identity, every fixed linear functional L defining the chamber or height satisfies

$$|L(xu) - L(x)| \ll \epsilon \|x\|_2.$$

On the unbounded portion of this chamber, $\|x\|_2 \asymp 1 + a \asymp 1 + (4a + 17n)$. Hence a membership change under u is confined to:

1. an angular strip of width $O(\epsilon)$ at $q = 0$ or $n = 0$;
2. an $O(\epsilon X)$-thick radial shell at $4a + 17n = X$; or
3. an $O(\epsilon)$-neighborhood of one of the compact boundary pieces $a = 0$ or $a + 4n = q$.

Since $dm_\infty = (6\sqrt{6})^{-1} da\, d\theta$, their total measure is $O(\epsilon X)$. Thus

$$m_\infty(B^+_{X,\epsilon} \setminus B^-_{X,\epsilon}) \ll \epsilon\, m_\infty(B_X),$$

so Definition 13.1(2) holds with $\kappa = 1$.

For condition (3), Benoist–Oh's Euclidean height is comparable to $1 + a$ throughout a fixed group-neighborhood enlargement of the chamber. For every $k > 0$, uniformly in $0 < \epsilon < 1$,

$$\int_{B^+_{X,\epsilon}} H_\infty(x)^{-k}\, dm_\infty(x) \ll_k \int_0^{O(X)} (1+a)^{-k}\, da.$$

The right side is $O(X^{1-k})$ for $0 < k < 1$, $O(\log X)$ for $k = 1$, and $O(1)$ for $k > 1$. As $m_\infty(B_X) \asymp X$, this is $O(m_\infty(B_X)^{1-\delta(k)})$ for some $\delta(k) > 0$ in every case. Definition 13.1(3) is therefore satisfied.

Proposition 14.2 of Benoist–Oh verifies only their standard product height balls. It should not be cited as though it automatically covers this polyhedron. The preceding direct verification is the necessary bridge to Theorem 13.2.

37.4.3 Finite-orbit passage and main-term normalization

Benoist–Oh state on journal p. 1892 that $G(\mathcal{O}_S)$ has finitely many orbits on $\mathscr{X}(\mathcal{O}_S)$. Apply Theorem 13.2 to each exact integral orbit. If $\eta_j > 0$ is the exponent for the j-th orbit, take $\eta = \min_j \eta_j > 0$. The finite sum has the form

$$R(X) = \kappa_{\mathrm{orb}}\, m_\infty(B_X) + O(X^{1-\eta}).$$

Here κ_{orb} is the sum of the orbit-volume normalization constants. The comparison with Borovoi's Theorem 1.4.1 and Proposition 1.5.1 is carried out in theorem 25.11; specifically, (959), (971), and (993) give

$$\kappa_{\mathrm{orb}} = \mathfrak{S}(\mathscr{X}) = \frac{9\sqrt{3}}{\pi^2} \log(2+\sqrt{3}), \qquad \kappa_{\mathrm{orb}} I_\infty = C_0.$$

Together with $m_\infty(B_X) = I_\infty X + O(1)$, this yields

$$R(X) = C_0 X + O(X^{1-\eta}).$$

This counts the actual integral lattice. No assertion that $\mathscr{X}(\mathbb{Z})$ is one orbit, and no replacement by a maximal order, is involved.

37.4.4 Why the exponent remains unspecified

The proof of Benoist–Oh Theorem 13.2 balances an effective-equidistribution/Sobolev error and a boundary-thickening error:

$$c_1 \epsilon^{-p} v_X^{1-\delta} + c_2 \epsilon^{\kappa} v_X, \qquad v_X = \mathrm{vol}(B_X).$$

The resulting saving is of the form

$$\eta = \frac{\delta\kappa}{p+\kappa} > 0.$$

Neither the Sobolev exponent p nor the effective equidistribution exponent δ is made numerically strong enough in the theorem to show $\eta \geq 1/2$. Thus Benoist–Oh supplies a genuine exact-family power saving, but not the requested square-root bound.

37.5 The sharp hyperbolic frontier

37.5.1 Good and Chatzakos–Petridis

For a cocompact or cofinite Fuchsian group and a fixed hyperbolic conjugacy class $\mathcal{H}$, Chatzakos–Petridis define, in equations (1.4)–(1.6),

$$M(\mathcal{H}, X; z) = \sum_{1/2<s_j\le 1} A(s_j)\widehat{u}_j u_j(z) X^{s_j},$$

where $\lambda_j = s_j(1-s_j) < 1/4$ and $\widehat{u}_j$ is the period of u_j along the invariant closed geodesic. Their Theorem 1.1, author manuscript p. 3, restates Good's Theorem 4, p. 116, and proves

$$N(\mathcal{H}, X; z) - M(\mathcal{H}, X; z) = O(X^{2/3}).$$

The 2/3 arises from the sharp majorant/minorant error

$$O(XY^{-1/2} + Y + X^{1/2}\log X)$$

after choosing $Y = X^{2/3}$. Theorems 1.2–1.3 are mean-square results, not pointwise improvements. Conjecture 5.7, author manuscript p. 22, explicitly asks for

$$O_\varepsilon(X^{1/2+\varepsilon}).$$

There are two independent failures of direct transfer to $R(X)$:

1. the Huber cutoff is radial, whereas $4a + 17n \le X$ plus the chamber is a nonradial sharp polyhedron; norm comparability does not preserve the leading constant or error because the symmetric difference can have volume of order X;
2. even in the radial problem, the theorem subtracts every nonzero small-eigenvalue contribution, not only the linear term.

For the exact Koide sector no spectral expansion establishing actual X^{s_j} terms has yet been produced. They are therefore a necessary obstruction to check, not terms that this audit claims are already known to occur. A proof of $R(X) - C_0 X = O(X^{1/2+\varepsilon})$ would have to show absence or vanishing of all relevant complementary-spectrum coefficients, or explicitly subtract them and then prove they vanish in the target formulation.

37.5.2 Current sharp refinements remain at 2/3

Voskou, Theorem 3 of arXiv:2407.03134v3, proves $O(X^{2/3})$ for four prescribed orientation/side sectors in a closed-geodesic-to-closed-geodesic problem, with all small-spectrum period terms included in the main expression. Conjecture 1 on the same page proposes $O_\varepsilon(X^{1/2+\varepsilon})$, while Theorem 4 is a mean-square result.

Lekkas–Petridis, in the 24 August 2026 version of arXiv:2509.12902, Theorem 1.2, likewise prove

$$N(X, l) = \sum_{1/2<s_j\le 1} \frac{2}{\pi}\gamma_1(s_j)|\widehat{u}_j|^2 X^{s_j} + O(X^{2/3})$$

for a closed-geodesic double-coset count. Their Theorem 1.5 gives

$$\frac{1}{X}\int_X^{2X} |E(x, l)|^2\, dx \ll X\log^2 X,$$

and Conjecture 1.6 proposes the pointwise $X^{1/2+\epsilon}$ bound. Their introduction states that Good's 2/3 has not been improved for any of the nine point/geodesic/cusp double-coset configurations. This is strong current evidence about the sharp frontier, but neither paper treats the exact Koide height.

37.5.3 Fixed angular harmonics do not give a sharp interval for free

Bourgain–Kontorovich–Sarnak, Theorem 1.1, prove for fixed left and right K-Fourier modes n, k

$$\sum_{\|\gamma\|<T} e^{2in\theta_1(\gamma)}e^{2ik\theta_2(\gamma)} = \mathbf{1}_{n=0}\mathbf{1}_{k=0}c_0T^2 + \sum_j c_j(n,k)T^{2s_j} + O_{n,k}(T^{4/3}).$$

On the adjoint orbit, affine height is quadratic in the group norm, so $T^{4/3}$ again corresponds to $X^{2/3}$. Recovering a sharp angular interval requires summing many Fourier modes, while the theorem's implied constants depend on n, k. It is not a uniform desmoothing theorem for this chamber.

37.6 Smooth spectral counting

Liu–Sarnak's paper is the closest source with a square-root-plus-gap exponent. Its global arithmetic setup and the hypotheses of its analytic proposition must be separated.

The introduction, author PDF p. 2, fixes an indefinite anisotropic integral ternary form f, a nonzero level t, and assumes $t\det(f)$ is square-free. Thus the paper's affine-sieve theorem, as globally stated, does not cover $t\det(f) = -108$. Equations (2.3)–(2.4), author PDF p. 4, fix one arithmetic orbit $\mathcal{O}$ and a smooth K-invariant radial weight F_T satisfying

$$F_T = 1 \quad (|x| \le T/c_0), \qquad F_T = 0 \quad (|x| \ge c_0T),$$

with $c_0 > 1$ fixed. In Fermi coordinates, Proposition 3.2 uses a smooth even function equal to one for $|\rho| \le R-1$, zero for $|\rho| \ge R+1$, and with uniformly bounded derivatives. Unlike the global sieve statement, Proposition 3.2 is formulated for any variable subgroup $\Gamma < \mathrm{SL}_2(\mathbb{R})$ satisfying its uniform local condition (3.1), the spectral gap (2.7), and containing the fixed primitive cyclic group Γ_ξ. It proves

$$H_R(z) = \frac{X(R)}{\mathrm{Area}(\Gamma\backslash\mathbb{H})} + O_\varepsilon(e^{(1/2+\theta+\varepsilon)R}), \qquad X(R) \asymp e^R.$$

Equation (3.60) has $T = \sinh R$, and equation (3.65) transfers this to the one-sheeted affine orbit. Thus the smooth affine error is

$$O_\varepsilon(T^{1/2+\theta+\varepsilon}).$$

Equation (2.7) assumes $\lambda_1 \ge 1/4 - \theta^2$. The paper invokes Jacquet–Langlands and Kim–Sarnak to take $\theta = 7/64$, giving

$$O_\varepsilon(T^{39/64+\varepsilon})$$

unconditionally in its congruence-arithmetic setting; $\theta = 0$, and hence square root, is conditional on the generalized Selberg/Ramanujan-at-infinity conjecture for the relevant Jacquet–Langlands transfers.

The numerical input itself is Kim–Sarnak, Appendix 2, Proposition 2, printed p. 176. It applies to an arbitrary cuspidal automorphic representation of $\mathrm{GL}_2/\mathbb{Q}$: for a spherical real component, $|\Re\mu_{j,\infty}| \le 7/64$, independently of finite conductor. Jacquet–Langlands preserves the real component because the quaternion algebra is split at infinity. Chu–Li, proof of Corollary 1.2, equation (5.2), printed p. 5126, records the resulting

$$\lambda_1 \ge \frac{975}{4096}$$

for every congruence arithmetic Fuchsian group. Thus the nonmaximal or non-Eichler structure of the exact finite order does not invalidate $s \le 39/64$. It changes the finite compact-open subgroup and hence which automorphic representations have fixed vectors and the dimensions of those fixed-vector spaces; it does not change this archimedean bound. This validates the spectral-gap input for the exact lattice.

It does not provide sharp-boundary control or force the periods of possible complementary-spectrum eigenfunctions to vanish.

For the exact order lattice $\Gamma = \mathcal{O}^1$, each primitive trace-four conjugacy class supplies the cyclic subgroup in Proposition 3.2, and there are only finitely many such classes. Hence the proposition and equation (3.65) do give an unconditional 39/64 estimate after summing the corresponding **smooth standard-radial** counts. The square-free hypothesis is not an obstruction to this isolated analytic application.

This still does not desmooth to the target. A transition of fixed width in R is a fixed multiplicative annulus in T, whose invariant volume is of order T, not $T^{1/2}$. Shrinking the transition makes the derivative/Sobolev constants in Proposition 3.2 grow, and the paper gives no uniform estimate that balances this loss at square-root scale. It also has no sharp coordinate chamber, and its radial height is not $4a + 17n$. These geometric failures remain even after the exact lattice and finite class sum have been supplied.

The same distinction applies to any informal statement that a tempered spectral component has square-root size. Duke–Rudnick–Sarnak's discussion on p. 155 estimates individual tempered frequencies; it does not control the complete spectral sum evaluated at a point with a shrinking approximation to a sharp characteristic function.

37.7 General effective sector and full-lattice theorems

37.7.1 Gorodnik–Nevo's explicit generic exponent

Gorodnik–Nevo Definition 3.1 defines

$$n_e(2) = 1, \qquad n_e(p) = \text{the least even integer at least } p/2 \quad (p > 2).$$

Their Section 8 considers

$$S_t(\Phi, \Psi) = \{g : \log \|gv_0\| \le t\} \cap \Phi A^+ \Psi.$$

Theorem 8.1, author PDF p. 34, assumes L^{p+} matrix coefficients and Lipschitz-well-rounded fixed angular sets; Corollary 8.3, p. 35, passes to an orbit when $(\Gamma \cap H)\backslash H$ is compact. The exponent is

$$\zeta = \frac{1}{2n_e(p)(1 + 3\dim G)}.$$

For $\dim G = 3$,

$$\zeta = \frac{1}{20n_e(p)}.$$

Even tempered spectrum has $p = 2$, $n_e(p) = 1$, and hence absolute error $O(X^{19/20+\epsilon})$. For the exact congruence arithmetic lattice, Jacquet–Langlands and Kim–Sarnak give $s \le 39/64$. The corresponding matrix coefficients are L^{p+} for $p = 64/25$, so $n_e(p) = 2$ and the theorem gives only $O(X^{39/40+\epsilon})$.

This numerical calculation is an upper benchmark for the published generic polar-sector theorem, not a claim that Corollary 8.3 literally counts $R(X)$. A fixed polar sector contains every radial parameter in fixed polar directions. An arbitrary affine linear cone and its finite-radius polyhedral height need an exact polar decomposition or a quantitatively controlled symmetric-difference argument; all integral orbits must then be included. None of those identifications appears in the theorem statement.

Gorodnik–Nevo themselves note after Corollary 5.3, author PDF p. 23, that their sharp-set counting mechanism remains worse than square-root volume error even for tempered spectrum, whereas square-root quality is available there only for a smooth weighted problem.

37.7.2 Browning–Gorodnik and subsequent full-lattice work

Browning–Gorodnik Theorem 2.1 gives, for a Euclidean ball B_H in one congruence orbit,

$$|\Gamma_\ell y \cap B_H| = \frac{m_{Z_{y,\ell}}(Z_{y,\ell})}{m_{X_\ell}(X_\ell)} m_Y(B_H) + O\Big(\ell^{\dim H + \dim G} m_Y(B_H)^{1-\varrho}\Big)$$

for some unspecified $\varrho > 0$. Corollaries 2.5–2.6 sum adelic and integral orbits and retain the Brauer/Kottwitz density, at the cost of $\ell^{\dim H + 2\dim G}$. This is a useful exact-lattice precedent, but its archimedean set is the full norm ball, not the present sector, and no square-root exponent is asserted.

Cao–Huang Theorem 1.5 proves arithmetic purity of the Hardy–Littlewood property for an anisotropic ternary affine quadric with $-m\det(q)$ nonsquare. Its Theorem 3.4 invokes Nevo–Sarnak, Gorodnik–Nevo, and Browning–Gorodnik to furnish an unspecified positive exponent, denoted β_X there with X meaning the affine variety; write it here as $\beta_{\mathscr{X}} > 0$ to distinguish it from the height variable. It is not a sharp-sector remainder theorem.

The current Cao–Huang–Zhang preprint, arXiv:2509.23320, Setting 2.1 and Theorem 2.1, gives a congruence-uniform power saving

$$O\Big(\mathbf{N}(\mathfrak{l})^{\kappa} \operatorname{vol}(\mathbb{B}_S(N))^{1-\delta}\Big)$$

for standard product Euclidean/max height balls and a compact piecewise-smooth factor. The exponent δ is unspecified, and the unbounded real factor in the target is neither compact nor their standard Euclidean ball. This preprint extends the general effective framework but does not improve the conclusion of the direct Benoist–Oh argument in sections 37.4.2 and 37.4.3.

37.8 Results in different arithmetic regimes

37.8.1 Square discriminant and divergent stabilizer

Oh–Shah Theorem 1.2 assumes

$$\operatorname{vol}((H \cap \Gamma)\backslash H) = \infty$$

and obtains a smoothed $cT\log T + O(T)$ count. Its sharp version has only a logarithmic relative error. Theorem 6.1 allows an open cone and arbitrary norm, but retains the logarithmic main-factor characteristic of its divergent-stabilizer regime; that regime is excluded here.

Xu–Zhang, *Quarterly Journal of Mathematics* 74 (2023), 659–685, assumes $-m\det(f)$ is a square over the number field. Huang, arXiv:2412.03350, Theorem 1.1, assumes the same split condition and a fixed smooth compact weight; its principal error is $B(\log B)^{1983/1984}$ in a $B\log B$ count. Zhang, arXiv:2603.21080, Section 1.1 and Theorem 1.1, again assumes $-m\det(q) \in (k^\times)^2$ and proves a smooth K_0-invariant expansion with an error $O(T^{l_1+l_2-\delta})$, $0 < \delta < 1/2$; Remark 1.2 obtains an authentic count for $\Gamma = \mathrm{SL}_2(\mathcal{O}_k)$. These are important current results, but all lie in the rationally split torus regime excluded by $108 \notin \mathbb{Q}^{\times 2}$, and none uses the target polyhedral height.

37.8.2 Varying level, isotropic forms, and special rotational quadrics

Huang, arXiv:2412.05903, studies $F(x) = m_0 N$ with $N = p_0^{2h} \to \infty$, a smooth compactly supported real weight $w(x/\sqrt{N})$, and prescribed local conditions. Theorems 1.1–1.2 have a $\sqrt{N}/(\log N)^A$ error. This is a Linnik-type varying-level theorem, not a fixed affine shell.

Li–Liu, arXiv:2501.15033v2 (18 April 2025), Theorems 1.1–1.2, prove new almost-prime conclusions for anisotropic ternary equations. Their global assumption remains $t\det(f)$ square-free. The quantitative counting input in Lemma 2.3, pp. 5–6, is

$$O_\varepsilon(d^{1+\varepsilon}T^{1/2+\theta+\varepsilon})$$

for the same kind of smooth radial orbit weight inherited from Liu–Sarnak. It is a current application of the 7/64 gap, not a new unsmoothed sector theorem.

Lowry-Duda's thesis, arXiv:1704.02376, Theorem 5.5.4, treats the special rotational form $x_1^2+x_2^2 = x_3^2+h$ in a full Euclidean ball. Written in ordinary radius T, its three-dimensional sharp error is $O(T^{21/22+\epsilon})$. The homogeneous form is $\mathbb{Q}$-isotropic, and neither the lattice nor the height sector is the target one.

Results on values of irrational ternary forms, shrinking sets while the discriminant varies, or positive-definite ternary representation numbers answer different problems. Lindelöf or generalized Lindelöf hypotheses occurring there do not yield the present fixed-level pointwise sector bound.

37.9 Spectral-gap, GRH, smoothing, and averaging taxonomy

Classification	Source result	Why it does not prove the target
Unconditional, exact sharp family	Benoist–Oh Theorem 13.2 plus the direct Definition 13.1 verification	Gives only $O(X^{1-\eta})$ with $\eta > 0$ unspecified
Unconditional, exact norm/cone, full lattice	Gorodnik–Oh–Shah Theorem 1.6, with the exact hypotheses proved in theorems 25.8 and 25.9	Qualitative $C_0X + o(X)$ only
Unconditional, sharp radial closed-geodesic count	Good / Chatzakos–Petridis	$O(X^{2/3})$, different height, and main expression includes small-spectrum terms
Unconditional, smooth radial count	Liu–Sarnak Proposition 3.2 with $\theta = 7/64$, summed over the exact finite conjugacy-class decomposition	39/64 for the exact lattice, but fixed multiplicative smoothing and a different radial height
Selberg/Ramanujan conditional	Liu–Sarnak with $\theta = 0$	Square-root only for the smooth radial problem; still not the exact count
GRH/Lindelöf conditional	None among the source versions in section 37.11 proves this fixed sharp-sector estimate	The occurrences listed there concern varying levels/forms, shrinking-set equidistribution, or lower/Omega results
Smoothed	Liu–Sarnak; Oh–Shah; Huang I; Zhang 2026	Smoothing or split arithmetic changes the theorem; no square-root desmoothing supplied
Averaged/mean square	Chatzakos–Petridis Theorems 1.2–1.3; Voskou Theorem 4; Lekkas–Petridis Theorem 1.5	Does not imply a uniform pointwise estimate at every X
Omega/lower-bound	Chatzakos (2019), including a cocompact geodesic-average $\Omega(X^{1/2}\log\log\log X)$	Evidence for scale, not an upper bound
Isotropic or split	Oh–Shah; Xu–Zhang; Huang I; Zhang 2026; Lowry-Duda	Different stabilizer, quotient, main order, form, or level
Varying level	Huang II	Counts $F = m_0p_0^{2h}$, not $F = -2$

In particular, "assuming GRH" is not a substitute for the missing theorem. The conditional hypothesis that occurs naturally in the closest smooth spectral result is the Selberg/Ramanujan eigenvalue conjecture, and even it only reaches square root before the sharp-boundary unsmoothing problem is addressed.

37.10 Citation-ready conclusion

Subject to the bounded-search qualification in section 37.12, the following paragraph records the conclusion supported by the enumerated source versions:

> Among the source versions enumerated in section 37.11, none implies the unconditional sharp estimate $R(X) - C_0X = O_\varepsilon(X^{1/2+\varepsilon})$ for the fixed ternary quadric $a^2 - 3q^2 - 18n^2 = -2$ in the chamber $a > 0$, $q, n \geq 0$, $a + 4n > q$, with height $4a + 17n \leq X$. Gorodnik–Oh–Shah, Theorem 1.6, applies to the exact admissible generic cone and an arbitrary norm—with those exact hypotheses proved in theorems 25.8 and 25.9—but gives only a qualitative asymptotic. Benoist–Oh, Definition 13.1 and Theorem 13.2, applies after the direct effective-well-roundedness verification in section 37.4.2 and, after summing the finitely many integral spin-group orbits, yields $R(X) = C_0X + O(X^{1-\eta})$ for some unspecified $\eta > 0$, not $\eta = 1/2$. In the closest sharp closed-geodesic problem, Good's theorem, as restated and reproved by Chatzakos–Petridis, Theorem 1.1, gives $O(X^{2/3})$ only after all small-eigenvalue terms X^{s_j} are included; Chatzakos–Petridis, Conjecture 5.7, states the $O_\varepsilon(X^{1/2+\varepsilon})$ improvement as a conjecture. Liu–Sarnak, Proposition 3.2, applies class by class to the exact congruence quaternionic lattice and proves the smooth radial bound $O_\varepsilon(X^{1/2+\theta+\varepsilon})$, unconditionally with $\theta = 7/64$ and conditionally with $\theta = 0$, but its fixed-width smoothing and radial height do not match the exact sharp count. Thus complementary-spectrum terms and sharp nonradial boundary control remain separate unresolved inputs.

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21. Marios Voskou, “Refined Counting of Geodesic Segments in the Hyperbolic Plane.”

arXiv:2407.03134v3; Theorems 3–4 and Conjecture 1, pp. 3–4.

22. Dimitrios Lekkas and Yiannis N. Petridis, “A relative trace formula and counting geodesic arcs in the hyperbolic plane,” arXiv:2509.12902, version dated 24 August 2026. Theorem 1.2, Theorem 1.5, and Conjecture 1.6.

23. Henry H. Kim, “Functoriality for the exterior square of GL_4 and the symmetric fourth of GL_2,” with Appendix 2 by Henry H. Kim and Peter Sarnak, “Refined estimates towards the Ramanujan and Selberg conjectures,” *Journal of the American Mathematical Society* **16** (2003), 139–183. DOI, official AMS PDF. Appendix 2, Proposition 2, printed p. 176: $|\Re\mu_{j,\infty}| \le 7/64$ for a spherical real component of any cuspidal $\mathrm{GL}_2/\mathbb{Q}$ representation, and the displayed consequence $\lambda_1 \ge 975/4096$.

24. Michelle Chu and Han Li, “Small generators of cocompact arithmetic Fuchsian groups,” *Proceedings of the American Mathematical Society* **144** (2016), 5121–5127. DOI, arXiv. Section 3.1, p. 5122, defines congruence arithmetic Fuchsian groups; proof of Corollary 1.2, equation (5.2), p. 5126, records $\lambda_1 \ge 975/4096$ for every such group via Jacquet–Langlands and Kim–Sarnak.

25. Hervé Jacquet and Robert P. Langlands, *Automorphic Forms on* GL(2), Lecture Notes in Mathematics 114, Springer, 1970. DOI, IAS-hosted text. Chapter III, especially §16, supplies the global quaternionic transfer; at split places, including the real place here, the local component is preserved.

26. John Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, Springer, 2021. DOI, open author version. Proposition 15.5.2 (maximal superorders); §23.1.3 (Eichler orders at ramified primes); equations (27.6.6)–(27.6.7) (compact-open finite adelic order/unit group); Theorem 28.2.10 (strong approximation for indefinite B^1); Definition 38.7.2 (principal congruence groups).

27. Jiamin Li and Jianya Liu, “Anisotropic quadratic equations in three variables.”

arXiv:2501.15033v2; Theorems 1.1–1.2, pp. 2–3; Lemma 2.3, pp. 5–6. The global form-level hypothesis is $t\det(f)$ square-free; the quantitative input is a smooth radial, congruence-orbit estimate with

$$O_\varepsilon\Big(d^{1+\varepsilon}T^{1/2+\theta+\varepsilon}\Big).$$

28. David Lowry-Duda, *On Some Variants of the Gauss Circle Problem*, arXiv:1704.02376. Theorem 5.5.4, thesis pp. 97–98.

37.12 Audit limitations

The claim-by-claim source versions and locators used in this bounded audit are enumerated in section 37.11. The canonical tree does not preserve byte-identical snapshots or retrieval hashes for those external publications, so this file does not assert a reproducible local source corpus. Good’s 1983 book was the sole exception to direct full-text access: its theorem locator and statement were checked through the explicit restatement and independent proof in Chatzakos–Petridis (2016) and the version-dated Lekkas–Petridis treatment.

The statement that no theorem was located is a result about the source versions enumerated here, not a proof that no unpublished or differently indexed result exists. It does establish that none of the enumerated works may be cited as a proof of the exact unconditional square-root discrepancy.

38 The exact gapless auxiliary operator and the physical boundary

The preceding flat-torus calculation produces eigenvalues tending to zero along the radial cusp. This section identifies a single self-adjoint operator for which those numbers genuinely form a gapless spectrum,

proves that complex and Hodge-theoretic data alone cannot determine such a spectrum, and states the exact additional morphism needed to transfer an auxiliary Rayleigh sequence to a physical Hamiltonian. In particular, the word *gapless* below is attached to a specified operator and Hilbert space, not to the complex threefold in isolation.

38.1 A direct-integral scalar Laplacian with zero essential threshold

Use the angular coordinates $y \in \mathbb{R}^4/\mathbb{Z}^4$ of theorem 9.1. The map $y \mapsto P_T y$ identifies this fixed torus with F_T, and normalized Haar measure on F_T pulls back to dy. Write

$$e_k(y) := e^{2\pi i k\cdot y}, \qquad \lambda_k(T) := 4\pi^2 |P_T^{-\mathsf{T}} k|^2, \qquad k \in \mathbb{Z}^4.$$

By theorem 9.1, these are exactly the scalar-Laplacian eigenfunctions and eigenvalues on (F_T, g_T), normalized in L^2 for Haar probability measure.

Fix T_0 large enough that all radial formulas of theorems 9.1 and 9.2 hold, and put

$$\mathscr{H}_{\mathrm{aux}} := L^2\Big([T_0,\infty), dT; L^2_0(\mathbb{R}^4/\mathbb{Z}^4, dy)\Big). \tag{1112}$$

Fourier expansion gives the concrete unitary identification

$$\mathscr{H}_{\mathrm{aux}} \cong \bigoplus_{k\in\mathbb{Z}^4\setminus\{0\}} L^2([T_0,\infty), dT). \tag{1113}$$

Define the multiplication operator

$$(\mathscr{L}_{\mathrm{aux}} u)_k(T) := \lambda_k(T) u_k(T) \tag{1114}$$

on the exact domain

$$D(\mathscr{L}_{\mathrm{aux}}) := \left\{ u \in \mathscr{H}_{\mathrm{aux}} : \sum_{k\neq 0} \int_{T_0}^{\infty} \lambda_k(T)^2 |u_k(T)|^2 \, dT < \infty \right\}. \tag{1115}$$

Thus $\mathscr{L}_{\mathrm{aux}}$ is, by definition and not by analogy, the direct integral of the mean-zero scalar Laplacians Δ_{F_T}.

Theorem 38.1 (A proved gapless auxiliary operator)**.** *The operator $\mathscr{L}_{\mathrm{aux}}$ in equation* (1114) *is nonnegative and self-adjoint. It has no zero eigenvector, but*

$$0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}}). \tag{1116}$$

In particular, this specified auxiliary operator has no positive spectral gap above zero.

Proof. For each $k \neq 0$, $\lambda_k(T)$ is a positive continuous real function. Multiplication by it on

$$\{v \in L^2([T_0,\infty)) : \lambda_k v \in L^2([T_0,\infty))\}$$

is self-adjoint and nonnegative: its adjoint domain is obtained by testing against compactly supported bounded functions, which forces $\lambda_k v \in L^2$, and the adjoint acts by the same real multiplier. The orthogonal sum of these multiplication operators has precisely the domain in equation (1115); hence it is self-adjoint and nonnegative. Since every $\lambda_k(T)$ is strictly positive, $\mathscr{L}_{\mathrm{aux}} u = 0$ implies $u_k = 0$ almost everywhere for every $k \neq 0$, so its kernel is zero.

Let $k_0 = (1,0,0,0)$. The exact formula equation (23) gives

$$\lambda_{k_0}(T) = 4\pi^2 \frac{L_T^2 + m_T^2}{D_T^2} = \frac{4\pi^2}{T^2}\big(1 + O(T^{-1})\big). \tag{1117}$$

Choose integers $n > T_0$, and define $u^{(n)}$ by

$$u_{k_0}^{(n)}(T) = \mathbf{1}_{[n,n+1]}(T), \qquad u_k^{(n)}(T) = 0 \quad (k \neq k_0). \tag{1118}$$

The vectors $u^{(n)}$ lie in $D(\mathscr{L}_{\mathrm{aux}})$, are orthonormal, and converge weakly to zero because their T-supports escape to infinity. Moreover,

$$\|\mathscr{L}_{\mathrm{aux}} u^{(n)}\|^2 = \int_n^{n+1} \lambda_{k_0}(T)^2 \, dT \leq \sup_{T \in [n,n+1]} \lambda_{k_0}(T)^2 \longrightarrow 0$$

by equation (1117). This orthonormal singular sequence proves $0 \in \sigma_{\mathrm{ess}}(\mathscr{L}_{\mathrm{aux}})$. Explicitly, if zero were outside the essential spectrum, then, because the kernel is zero, some interval $[0, \varepsilon)$ would have finite-dimensional spectral projection. Its complement would satisfy $\|\mathscr{L}_{\mathrm{aux}} v\| \geq \varepsilon \|v\|$; weak convergence would make the components of $u^{(n)}$ in the finite-dimensional spectral subspace tend strongly to zero, contradicting $\|\mathscr{L}_{\mathrm{aux}} u^{(n)}\| \to 0$ and $\|u^{(n)}\| = 1$. □

Remark 38.2 (The unit-volume normalization is a different operator family). The theorem uses the unscaled Euclidean quotient metrics g_T. By theorem 9.3, the first positive scalar eigenvalue for $\widehat{g}_T = D_T^{-1/2} g_T$ still tends to zero, but at the different rate

$$\lambda_1(F_T, \widehat{g}_T) = \frac{4\pi^2}{T}(1 + O(T^{-1})),$$

rather than the unscaled T^{-2} rate. Rescaling a metric changes its Laplacian, so the two operator families and their asymptotic scales must not be identified.

38.2 Complex data do not select a spectral gap

Proposition 38.3 (Fixed complex torus, fixed volume, arbitrarily small gap)**.** *Let*

$$A = (\mathbb{C}/(\mathbb{Z} + i\mathbb{Z}))^2$$

with fixed complex coordinates $z_j = x_j + iy_j$. For $R \geq 1$, the metric

$$g_R = R^2(dx_1^2 + dy_1^2) + R^{-2}(dx_2^2 + dy_2^2) \tag{1119}$$

is a flat Kähler metric of volume one on the same complex torus, while

$$\lambda_1(\Delta_{g_R}) = \frac{4\pi^2}{R^2} \longrightarrow 0. \tag{1120}$$

Consequently the complex structure, Hodge decomposition, integral cohomology, and coherent direct-image data of a trivial family do not, even together with total volume, determine a positive lower bound for the scalar Laplacian gap.

Proof. The associated two-form

$$\omega_R = R^2 dx_1 \wedge dy_1 + R^{-2} dx_2 \wedge dy_2$$

is closed, positive, and J-invariant, so g_R is Kähler for the fixed complex structure. In the ordered real coordinates (x_1, y_1, x_2, y_2), its matrix is $\mathrm{diag}(R^2, R^2, R^{-2}, R^{-2})$, whose determinant is one; therefore the volume is one. The Fourier character

$$e_{a,b,c,d} = \exp(2\pi i(ax_1 + by_1 + cx_2 + dy_2))$$

has eigenvalue

$$4\pi^2\left(R^{-2}(a^2+b^2)+R^2(c^2+d^2)\right).$$

For $R \geq 1$, the minimum over nonzero integer quadruples is $4\pi^2 R^{-2}$, attained at the four vectors

$$(a,b,c,d) \in \{(1,0,0,0),(-1,0,0,0),(0,1,0,0),(0,-1,0,0)\}.$$

This proves equation (1120). None of the listed complex or cohomological objects depends on R, and for a trivial holomorphic family their coherent direct images are likewise unchanged. □

38.3 The exact bridge to a physical Hamiltonian

The mathematical mass gap in the Clay formulation is a property of the Hamiltonian of a constructed four-dimensional quantum Yang–Mills theory, not of an internal scalar Laplacian or of a classical Hessian [63]. Euclidean input reaches such a Hamiltonian only after a positive-metric reconstruction satisfying, in particular, the Osterwalder–Schrader requirements [67, 68]. The following proposition isolates the exact operator-theoretic statement that any proposed transfer must prove.

Proposition 38.4 (Typed Rayleigh-sequence transfer)**.** *Let $H \geq 0$ be self-adjoint on a Hilbert space $\mathcal{H}_{\mathrm{phys}}$. Define*

$$\mathcal{V}_0 := \ker H, \qquad P_0 := \mathbf{1}_{\{0\}}(H), \qquad \mathcal{H}_{\mathrm{exc}} := (I-P_0)\mathcal{H}_{\mathrm{phys}}, \tag{1121}$$

and let

$$\mathfrak{h}[\psi] := \|H^{1/2}\psi\|^2, \qquad \mathcal{Q}(H) := D(H^{1/2}). \tag{1122}$$

For each n, let $\mathcal{K}_n$ be a Hilbert space, $\mathfrak{q}_n$ a nonnegative quadratic form on a domain $D(\mathfrak{q}_n) \subseteq \mathcal{K}_n$, and $0 \neq u_n \in D(\mathfrak{q}_n)$. Suppose there is a linear map

$$J_n : \mathrm{span}\{u_n\} \longrightarrow \mathcal{Q}(H),$$

and put $\psi_n = (I-P_0)J_n u_n$. If there are $c > 0$, $C_n \geq 0$, and $\varepsilon_n \geq 0$ such that

$$\|\psi_n\| \geq c\|u_n\|, \tag{1123}$$

$$\mathfrak{h}[\psi_n] \leq C_n \mathfrak{q}_n[u_n] + \varepsilon_n \|u_n\|^2, \tag{1124}$$

$$C_n \frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2} + \varepsilon_n \longrightarrow 0. \tag{1125}$$

Then the excitation Hamiltonian

$$H_{\mathrm{exc}} := H\big|_{D(H)\cap\mathcal{H}_{\mathrm{exc}}}$$

is self-adjoint and satisfies

$$\inf \sigma(H_{\mathrm{exc}}) = 0, \qquad 0 \in \sigma_{\mathrm{ess}}(H_{\mathrm{exc}}). \tag{1126}$$

Proof. The spectral projection P_0 commutes with H and $H^{1/2}$, so

$$\mathcal{H}_{\mathrm{phys}} = \ker H \oplus \mathcal{H}_{\mathrm{exc}}$$

is a reducing decomposition. Hence H_{exc} is self-adjoint, nonnegative, has form domain $\mathcal{Q}(H)\cap\mathcal{H}_{\mathrm{exc}}$, and has zero kernel. Define $w_n = \psi_n/\|\psi_n\|$. Then $\|w_n\| = 1$, $w_n \in \mathcal{Q}(H)\cap\mathcal{H}_{\mathrm{exc}}$, and

$$\begin{aligned}\mathfrak{h}[w_n] &= \frac{\mathfrak{h}[\psi_n]}{\|\psi_n\|^2} \\ &\leq c^{-2}\left(C_n\frac{\mathfrak{q}_n[u_n]}{\|u_n\|^2}+\varepsilon_n\right) \longrightarrow 0.\end{aligned}$$

The variational characterization of the bottom of a nonnegative self-adjoint operator gives

$$\inf \sigma(H_{\mathrm{exc}}) = \inf_{0 \neq \psi \in \mathcal{Q}(H) \cap \mathcal{H}_{\mathrm{exc}}} \frac{\mathfrak{h}[\psi]}{\|\psi\|^2} = 0.$$

Since the spectrum is closed, zero belongs to it; since $\ker H_{\mathrm{exc}} = 0$, zero is not an eigenvalue. It is therefore a continuous spectral point, and hence belongs to the essential spectrum under the standard definition that removes isolated eigenvalues of finite multiplicity. □

Remark 38.5 (Why one vacuum vector is not enough). If the vacuum is unique, choose a normalized vector Ω spanning $\ker H$. Then $P_0 = |\Omega\rangle\langle\Omega|$. Without uniqueness, merely requiring $J_n u_n \perp \Omega$ does not project away every zero-energy state. For example,

$$H = \mathrm{diag}(0, 0, 1), \qquad \Omega = (1, 0, 0), \qquad \psi = (0, 1, 0)$$

has $\psi \perp \Omega$ and zero Rayleigh quotient, although the genuine excitation space $(\ker H)^{\perp}$ is the third coordinate and has gap one. Thus equation (1121) must use the full kernel unless vacuum uniqueness is independently proved.

Audit record 38.6 (What the S^6 calculation proves)**.** The radial period matrix, scalar spectra, classical gauge Hessian, finite cochain operator, magnetic Laplacian, and reducible $SU(2)$ Jacobi operator are explicit auxiliary geometric or classical operators. They prove the operator statements made in theorems 10.6, 10.10, 12.5, 12.8 and 38.1. They do not construct a common gauge-invariant positive-metric physical Hilbert space $\mathcal{H}_{\mathrm{phys}}$, a quantum Hamiltonian H, or maps J_n satisfying equations (1123) to (1125) after continuum, infinite-volume, and renormalization limits. Nor do the period, Hodge, Leray, or direct-image calculations determine those data, as theorem 38.3 already disproves at the level of geometric Laplacians. Therefore the auxiliary gaplessness is **[VERIFIED]**, while a conclusion about the four-dimensional quantum Yang–Mills Hamiltonian is **[OPEN]**.

39 Integral topology without the cusp-collapse argument

This section follows the independent nearby-cycle/Leray route. It does not use the candidate's cellular collapse, its Mayer–Vietoris homology matrix, or the asserted equality between those two constructions. All coefficient groups are integral.

Put

$$B = \mathbb{P}^1, \quad \Sigma = \{p_0, p_1, p_2\}, \quad B^\circ = B \setminus \Sigma, \quad j : B^\circ \hookrightarrow B,$$

and let F be a smooth fibre. Retain

$$V = H^1(F) = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \quad q = uw + 6\gamma\delta, \quad \mathrm{vol} = \gamma u w \delta.$$

39.1 Integral specialization at the cusp

Theorem 39.1 (nearby-cycle specialization)**.** ***[VERIFIED]*** *For the proper holomorphic cusp filling $f_0 : N_0 \to \Delta$ of theorem 8.4, with its proved local normal-crossings equations and normalization combinatorics, the specialization map of the central fibre W,*

$$\mathrm{sp}^r : H^r(W; \mathbb{Z}) \longrightarrow H^r(F; \mathbb{Z})$$

is injective and

$$\mathrm{im}\, \mathrm{sp}^r = H^r(F; \mathbb{Z})^{T_0} \tag{5.1}$$

for $0 \le r \le 4$. In particular, the image is saturated and there is no unrecorded finite-index defect at p_0.

Proof. Let $\widetilde{\Delta}^* \to \Delta^*$ be the universal cover, put $\widetilde{N}_0^* = N_0 \times_\Delta \widetilde{\Delta}^*$, and let

$$k : \widetilde{N}_0^* \longrightarrow N_0, \qquad i : W \hookrightarrow N_0$$

be the natural maps. The analytic nearby-cycle complex used here is

$$\psi = \psi_{f_0}\mathbb{Z} := i^* R k_* k^* \mathbb{Z}_{N_0} \in D^b_c(W). \tag{5.0}$$

Dimca defines analytic nearby cycles in [32, Definition 4.2.1]; the Milnor-fibre description of their stalks is [32, Proposition 4.2.2]. The proper-direct-image theorem for nearby cycles is [32, Proposition 4.2.11]; applied to f_0 and the identity map of the disc, it gives

$$\mathbb{H}^n(W;\psi) \simeq H^n(F;\mathbb{Z})$$

and identifies the morphism induced by the unit $\mathbb{Z}_W \to \psi$ with specialization. Proper base change itself is [32, Theorem 2.3.26]. Dimca's coefficient hypothesis is a commutative noetherian ring of finite global dimension, which $\mathbb{Z}$ satisfies. Thus the general integral analytic formalism is available here. The candidate-specific existence and local description of f_0 are proved in theorems 8.4 and 8.5. Kashiwara–Schapira [33, §8.6] corroborates the analytic formalism, while SGA 7, Exposé XIII [31], is its algebraic antecedent.

At a point where $f = z_0 \cdots z_k$, $k = 0, 1, 2$, the Milnor fibre deformation retracts onto $(S^1)^k$: in polar coordinates the radial variables form a contractible open simplex and the phases satisfy one linear relation. Hence

$$\mathcal{H}^0\psi = \mathbb{Z}_W, \qquad \mathcal{H}^1\psi =: \mathscr{V}, \qquad \mathcal{H}^2\psi = \mathbb{Z}_P \oplus \mathbb{Z}_Q, \qquad \mathcal{H}^b\psi = 0 \ (b \geq 3), \tag{5.2}$$

where $\mathscr{V}$ is supported on the three double curves $\bar{C}_1, \bar{C}_2, \bar{C}_3 \simeq \mathbb{P}^1$.

Orient each vanishing circle by the displayed order of its two branches in the normalization. At P, the branches of $(\bar{C}_1, \bar{C}_2, \bar{C}_3)$ are respectively

$$(b_0, b_1), \qquad (b_0, b_2), \qquad (b_1, b_2).$$

The three restrictions are therefore evaluations on

$$e_0 - e_1, \qquad e_0 - e_2, \qquad e_1 - e_2,$$

whose alternating sum is zero. At Q, the ordered branch pairs are

$$(c_2, c_1), \qquad (c_0, c_1), \qquad (c_0, c_2),$$

so the evaluations are

$$e_2 - e_1, \qquad e_0 - e_1, \qquad e_0 - e_2,$$

and again the first minus the second plus the third is zero. The global normalization labels C_k and C_{k+3} distinguish the two branches of each double curve and therefore fix these signs simultaneously. Thus

$$0 \longrightarrow \mathscr{V} \longrightarrow \bigoplus_{k=1}^{3} i_{k*}\mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0,$$

$$\delta_1(n_1, n_2, n_3) = (s, s), \qquad s = n_1 - n_2 + n_3. \tag{5.3}$$

The image $\mathbb{Z}(1,1)$ is primitive. Taking cohomology of (5.3) gives free groups with ranks

$$\operatorname{rk} H^a(W;\mathscr{V}) = (2, 1, 3, 0) \quad (a = 0, 1, 2, 3). \tag{5.4}$$

The normalization resolution for the constant sheaf is

$$0 \longrightarrow \mathbb{Z}_W \longrightarrow \nu_*\mathbb{Z}_{\mathrm{dP}_6} \longrightarrow \bigoplus_{k=1}^{3} i_{k*}\mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \longrightarrow 0. \tag{5.5}$$

This is exact stalkwise. At a smooth point it is the identity $\mathbb{Z} \to \mathbb{Z}$. At a generic double point it is the augmented complex

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^2 \xrightarrow{(x_0,x_1)\mapsto x_0-x_1} \mathbb{Z} \longrightarrow 0.$$

At a triple point, in either of the branch orders just fixed, it is

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}^3 \xrightarrow{d_0} \mathbb{Z}^3 \xrightarrow{d_1} \mathbb{Z} \longrightarrow 0,$$

where at P

$$d_0(x_0, x_1, x_2) = (x_0 - x_1, x_0 - x_2, x_1 - x_2), \qquad d_1(n_1, n_2, n_3) = n_1 - n_2 + n_3,$$

while at Q

$$d_0(x_0, x_1, x_2) = (x_2 - x_1, x_0 - x_1, x_0 - x_2), \qquad d_1(n_1, n_2, n_3) = n_1 - n_2 + n_3.$$

These augmented simplicial complexes are exact, which proves (5.5), including its endpoint signs.

On H^2, the only nontrivial map is

$$d : H^2(\mathrm{dP}_6; \mathbb{Z}) \longrightarrow \mathbb{Z}^3, \qquad d(L) = (L \cdot C_k - L \cdot C_{k+3})_{k=1}^3.$$

Write dP_6 as the blow-up of $\mathbb{P}^2$ at three noncollinear points, with basis H, E_1, E_2, E_3, and order its hexagon as

$$E_1, H - E_1 - E_2, E_2, H - E_2 - E_3, E_3, H - E_1 - E_3.$$

For $L = aH - \sum b_i E_i$, direct intersection gives

$$d(L) = (s, -s, s), \qquad s = b_1 + b_2 + b_3 - a. \tag{5.6}$$

Thus $\operatorname{im} d = \mathbb{Z}(1, -1, 1)$ is primitive, $\ker d \simeq \mathbb{Z}^3$, and $\operatorname{coker} d \simeq \mathbb{Z}^2$. The resolution (5.5) has width two and the cohomology of its smooth strata is concentrated in even degrees. The only possible higher differential would be $d_2 : E_2^{0,q} \to E_2^{2,q-1}$, whose target is an odd row and is zero; differentials with $r \geq 3$ leave the width-two range. Therefore

$$H^a(W; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}) \quad (a = 0, 1, 2, 3, 4), \tag{5.7}$$

with no torsion.

The hypercohomology spectral sequence

$$E_2^{a,b} = H^a(W; \mathcal{H}^b\psi) \Longrightarrow H^{a+b}(F; \mathbb{Z}) \tag{5.8}$$

has the following ranks, all on free groups:

$b = 2$	2	0	0	0	0
$b = 1$	2	1	3	0	0
$b = 0$	1	2	4	2	1
	$a = 0$	$a = 1$	$a = 2$	$a = 3$	$a = 4.$

Let the possible differential ranks be

$$\alpha = \operatorname{rk} d_2^{0,1}, \quad \beta = \operatorname{rk} d_2^{1,1}, \quad \gamma' = \operatorname{rk} d_2^{2,1}, \quad \delta' = \operatorname{rk} d_2^{0,2}, \quad \epsilon = \operatorname{rk} d_3^{0,2}.$$

Comparison with the torus Betti numbers $(1, 4, 6, 4, 1)$ in total degrees one, four, and two gives

$$\alpha = 0, \qquad \gamma' = 0, \qquad \beta + \delta' + \epsilon = 1. \tag{5.9}$$

We determine the remaining ambiguity geometrically, keeping the supported specialization map explicit. Choose a compact T_f-orbit $K = T_f \cdot w_0 \simeq T^2$ in the smooth stratum $W^\circ \simeq (\mathbb{C}^*)^2$. On a relatively compact neighbourhood of K, average a Riemannian metric over T_f and horizontally lift $\partial_{\mathrm{Re}\, t}$ and $\partial_{\mathrm{Im}\, t}$. If ξ_1^h, ξ_2^h are those lifts, put

$$\xi_t = (\mathrm{Re}\, t)\xi_1^h + (\mathrm{Im}\, t)\xi_2^h.$$

Choose a relatively compact T_f-invariant open neighbourhood K' with $K \Subset K'$. For some $r_0 > 0$, the time-one flow

$$\Theta : K' \times \Delta_{r_0} \longrightarrow N_0, \qquad \Theta(w,t) = \Phi_1^{\xi_t}(w), \tag{5.9a}$$

is a T_f-equivariant open embedding over Δ_{r_0}, with $\Theta(w,0) = w$. Indeed $df_0(\xi_t) = t$, so $f_0(\Theta(w,t)) = t$; its differential is invertible along $K' \times \{0\}$, and injectivity follows first by applying f_0 and then from uniqueness for the time-one flow. Shrinking K' and r_0 gives the claimed embedding.

For $0 < r < r_0$, set

$$A_r = f_0^{-1}(\Delta_r), \qquad Z_r = \Theta(K \times \Delta_r), \qquad K_t = \Theta(K \times \{t\}).$$

The subset Z_r is closed in A_r. Indeed, if a sequence in Z_r converges in A_r, compactness of K gives a convergent subsequence of its K-coordinates; its base coordinates converge to the image under f_0, which still lies in Δ_r, and the embedding Θ then identifies the limit with a point of Z_r.

For $0 < |t| < r$, functoriality of relative cohomology gives the two commutative squares

$$\begin{array}{ccc} H^3(A_r, A_r \setminus Z_r) & \longrightarrow & H^3(A_r) \\ {\scriptstyle u_0\ (\simeq)}\big\downarrow & & \big\downarrow{\scriptstyle \rho_0} \\ H^3(W, W \setminus K) & \longrightarrow & H^3(W) \end{array} \qquad \begin{array}{ccc} H^3(A_r, A_r \setminus Z_r) & \longrightarrow & H^3(A_r) \\ {\scriptstyle u_t\ (\simeq)}\big\downarrow & & \big\downarrow{\scriptstyle \rho_t} \\ H^3(F_t, F_t \setminus K_t) & \longrightarrow & H^3(F_t). \end{array} \tag{5.9b}$$

The two left vertical arrows are isomorphisms. By excision they are the restrictions to the two slices of

$$H^3\big(K' \times \Delta_r, (K' \setminus K) \times \Delta_r\big),$$

and homotopy invariance makes both restrictions isomorphisms. Taking the direct limit over shrinking r, the maps ρ_0 induce the proper-base-change isomorphism

$$\varinjlim_{r \downarrow 0} H^3(A_r; \mathbb{Z}) \xrightarrow{\sim} H^3(W; \mathbb{Z}).$$

Choose the points t on one radial ray and identify the F_t with the fixed marked fibre F by parallel transport. Restriction to F_t, followed by this identification, is compatible under shrinking r. The nearby-cycle specialization is precisely the composite

$$\mathrm{sp}^3 : H^3(W; \mathbb{Z}) \xleftarrow[\sim]{\rho_0} \varinjlim_{r \downarrow 0} H^3(A_r; \mathbb{Z}) \xrightarrow{\rho_{\mathrm{near}}} H^3(F; \mathbb{Z});$$

this is the neighbourhood realization of the unit $\mathbb{Z}_W \to \psi_{f_0}\mathbb{Z}$ in the comparison cited above. Taking the direct limit in the two functorial squares therefore proves, rather than assumes, that

$$H^3(W, W \setminus K) \longrightarrow H^3(W) \xrightarrow{\mathrm{sp}^3} H^3(F_t) \tag{5.9c}$$

is the transported supported class followed by the Gysin map for $K_t \hookrightarrow F_t$.

Under the marking of the torus fibre,

$$\Lambda_{\mathrm{tor}} = \mathbb{Z}\widehat{w} \oplus \mathbb{Z}\widehat{\delta}$$

is a saturated direct summand of Λ, and K_t is a translate of $(\Lambda_{\mathrm{tor}} \otimes \mathbb{R})/\Lambda_{\mathrm{tor}}$. Choose a complementary summand Λ', and write $F_t \simeq T_f \times T_B$ as oriented real tori, where $T_B = (\Lambda' \otimes \mathbb{R})/\Lambda'$, with the product

orientation chosen to agree with that of F_t. The real rank-two normal bundle of the translate K_t is trivial and oriented. The Thom isomorphism gives

$$H^3(F_t, F_t \setminus K_t) \simeq H^1(T_f),$$

and its map to absolute cohomology is

$$\iota_!(a) = a \otimes [T_B]^\vee \in H^1(T_f) \otimes H^2(T_B) \subset H^3(F_t). \tag{5.9d}$$

This is an injective primitive rank-two Künneth direct summand. On the central smooth stratum, $W^\circ \simeq (\mathbb{C}^*)^2 \simeq T_f \times \mathbb{R}^2$ as an oriented real manifold; hence the real rank-two normal bundle of K is also trivial and oriented. The same Thom calculation gives $H^3(W, W \setminus K) \simeq H^1(K) \simeq \mathbb{Z}^2$. Hence (5.9c) has primitive rank-two image. Since $\operatorname{rk} H^3(W) = 2$, specialization in degree three is injective and has precisely that primitive image. In (5.9), this forces $\beta = \epsilon = 0$, and consequently $\delta' = 1$.

We make the integral saturation argument explicit. The unit $\mathbb{Z}_W \to \psi$ is a morphism from the complex concentrated in cohomological degree zero to the Postnikov filtration of ψ. It therefore induces a morphism of hypercohomology spectral sequences from the single row

$$H^a(W; \mathbb{Z}) = E_2^{a,0}(\mathbb{Z}_W)$$

to (5.8). On the $b = 0$ row the induced E_2-map is the identity, because $\mathcal{H}^0\psi = \mathbb{Z}_W$. Consequently, in total degree r, specialization factors through the last filtration step

$$H^r(W; \mathbb{Z}) \xrightarrow{\sim} E_\infty^{r,0} = F^r H^r(F; \mathbb{Z}) \hookrightarrow H^r(F; \mathbb{Z}) \tag{5.9e}$$

whenever no differential $d_k : E_k^{r-k,k-1} \to E_k^{r,0}$ enters on any page. For $r = 1$ this is automatic; for $r = 2$ the only possible entering differential is $d_2^{0,1}$, and $\alpha = 0$. Thus (5.9e) is an isomorphism onto the specialization image in degrees one and two. The quotient by F^r has the finite filtration whose graded groups are the remaining $E_\infty^{a,r-a}$, $a < r$. In degree one these are kernels of maps of free abelian groups. In degree two they are $E_\infty^{1,1} = \mathbb{Z}$ and $E_\infty^{0,2} = \ker d_2^{0,2}$; here the possible subsequent map is $d_3^{0,2}$, and it is zero because its already determined rank ϵ is zero. Both groups are therefore free. Successive extensions of free abelian groups split, so $H^r(F; \mathbb{Z})/F^r$ is free for $r = 1, 2$. Hence the specialization images in those degrees are primitive.

In degree three, (5.9c)–(5.9d) give a primitive rank-two lattice contained in $\operatorname{im} \mathrm{sp}^3$. Since $H^3(W; \mathbb{Z})$ itself has rank two, the specialization image has the same rational span; a primitive lattice has no proper finite-index overlattice in that span inside $H^3(F; \mathbb{Z})$. Therefore the two images coincide, proving both injectivity and primitivity in degree three. In degree four, $E_\infty^{4,0} = H^4(F; \mathbb{Z})$, so specialization is an isomorphism. Thus specialization is injective with primitive image in every degree. Monodromy invariance gives

$$\operatorname{im} \mathrm{sp}^r \subseteq H^r(F; \mathbb{Z})^{T_0}.$$

By theorem 5.2, the invariant ranks are $(1, 2, 4, 2, 1)$, exactly those in (5.7), and each invariant lattice is the kernel of an integral endomorphism, hence primitive. Two primitive sublattices of the same rational subspace coincide: both equal the intersection of that subspace with the ambient lattice. This proves (5.1). □

39.2 The global E_2-page

The Leray spectral sequence used in this subsection is [32, Corollary 2.3.4]. Let $R^r = R^r f_* \mathbb{Z}$ and $\mathcal{L}^r = j^* R^r$. The finite-fibre lattices have already been proved as abstract lattice statements in theorem 5.4. Their identification with the stalks of the proposed family requires the explicit radial-stalk compatibility below. This compatibility is treated separately in theorem 8.8 and is not silently absorbed here.

Proposition 39.2 (all Leray terms used below)**.** ***[VERIFIED]*** *The specialization theorem theorem 39.1, the finite fillings of theorem 8.7, and theorem 8.8 give, for $j = 1, 2$ and $0 \le r \le 4$, the finite-filling compatibility*

$$N_j \text{ strongly deformation retracts onto } S_j, \qquad H^r(S_j;\mathbb{Z}) \longrightarrow H^r(F;\mathbb{Z})^{T_j} \text{ is } \pi_j^* \tag{FF}$$

under the marked punctured-fibre identification. Consequently,

$$H^0(B,R^1) = \mathbb{Z}(12\gamma), \quad H^0(B,R^2) = \mathbb{Z}(2q), \quad H^0(B,R^3) = \mathbb{Z}(2\gamma uw), \tag{5.10}$$

$$H^1(B,R^1) = H^1(B,R^2) = 0, \tag{5.11}$$

and, with ω a generator of $H^2(B;\mathbb{Z})$,

$$H^2(B,R^1) = \mathbb{Z}\omega[\delta], \qquad H^2(B,R^2) = \mathbb{Z}\omega[\gamma\delta], \qquad H^2(B,R^4) = \mathbb{Z}\omega[\mathrm{vol}]. \tag{5.12}$$

These identifications are compatible with exterior products in V.

Proof. Under the stated hypotheses, proper base change identifies the stalk of R^r at p_0 with $H^r(W;\mathbb{Z})$, while $(j_*\mathcal{L}^r)_{p_0} = H^r(F;\mathbb{Z})^{T_0}$. The stalk map is specialization and is an isomorphism by theorem 39.1; hence its cokernel at p_0 is zero. At p_j, condition (FF) identifies the stalk with $H^r(S_j;\mathbb{Z})$, the target with $H^r(F;\mathbb{Z})^{T_j}$, and the stalk map with π_j^*. The finite-fibre calculation theorem 5.4 makes each of these maps injective. Away from the three special points the map is the identity of the local system. Stalkwise kernels and cokernels therefore give the exact sequence

$$0 \longrightarrow R^r \longrightarrow j_*\mathcal{L}^r \longrightarrow Q^r \longrightarrow 0, \tag{5.13}$$

with

$$Q^1 = (\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}, \quad Q^2 = (\mathbb{Z}/2)_{p_2}, \quad Q^3 = (\mathbb{Z}/2)_{p_2}, \quad Q^4 = (\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}.$$

The degree-one quotients have orders three and four and are generated by the class of γ. In degree two the p_1-quotient is zero, while

$$(\textstyle\bigwedge^2 V)^{T_2}/\{a\gamma\eta_2 + bq : a \equiv b \pmod 2\} \simeq \mathbb{Z}/2$$

is generated by the class of q. In degree three the p_1-quotient is zero and the p_2-quotient has order two, generated by the class of γuw. This last identification uses the actual twist $v_2 = -\varepsilon'$, for which the source's admissibility quantity $\eta_2(v_2)$ is zero and hence divisible by four. In degree four, pullback multiplies the orientation class by the covering degree m_j, giving the displayed Q^4. Hence the skyscraper sheaves have exactly the asserted stalks; no claim for an arbitrary twist is being imported. By theorem 5.3,

$$V^G = \mathbb{Z}\gamma, \quad (\textstyle\bigwedge^2 V)^G = \mathbb{Z}q, \quad (\textstyle\bigwedge^3 V)^G = \mathbb{Z}\gamma uw, \tag{5.14}$$

and

$$V_G = \mathbb{Z}[\delta], \qquad (\textstyle\bigwedge^2 V)_G = \mathbb{Z}[\gamma\delta], \qquad [q] = 12[\gamma\delta]. \tag{5.15}$$

Since $\det T_1 = \det T_2 = 1$, the action on $\bigwedge^4 V = \mathbb{Z}\,\mathrm{vol}$ is trivial and

$$(\textstyle\bigwedge^4 V)_G = \mathbb{Z}[\mathrm{vol}]. \tag{5.15a}$$

The maps from the invariant groups in (5.14) onto the corresponding skyscraper fibres are surjective, with kernels $12\mathbb{Z}\gamma$, $2\mathbb{Z}q$, and $2\mathbb{Z}\gamma uw$. This proves (5.10).

Because each Q^r is a skyscraper, it has no positive cohomology; the long exact sequence of (5.13) therefore identifies $H^2(B,R^r)$ with $H^2(B,j_*\mathcal{L}^r)$. For any free local system $\mathcal{L}$ on B°, with stalk M, the quotient $j_*\mathcal{L}/j_!\mathcal{L}$ is a skyscraper. Hence

$$H^2(B,j_*\mathcal{L}) \simeq H^2_c(B^\circ,\mathcal{L}) \simeq H_0(B^\circ,\mathcal{L}) = M_G, \tag{5.16}$$

where the middle isomorphism is Poincaré duality with local coefficients [9, §3.H]; see also G. E. Bredon, *Sheaf Theory*, 2nd ed., Chapter VI. Cap product with the locally finite fundamental class makes (5.16) natural for coefficient pairings, proving (5.12) and the product assertion.

It remains to prove (5.11), for which we spell out the parabolic-cohomology quotient. The low-degree Leray sequence for $j : B^\circ \hookrightarrow B$ contains

$$0 \longrightarrow H^1(B, j_*\mathcal{L}) \longrightarrow H^1(B^\circ, \mathcal{L}) \longrightarrow H^0(B, R^1 j_*\mathcal{L}).$$

If $I_i = \langle t_i \rangle$ is the local inertia group at p_i, then

$$(R^1 j_*\mathcal{L})_{p_i} = H^1(I_i, M) = M/(T_i - I)M.$$

Consequently the last arrow sends the class of a cocycle c to the three classes of $c(t_i)$, and $H^1(B, j_*\mathcal{L})$ is exactly the kernel defined by the three parabolic conditions below. Use left cocycles, so $c(gh) = c(g) + g c(h)$. A cocycle on $\pi_1(B^\circ) = F(t_1, t_2)$ is determined by $x = c(t_1)$ and $y = c(t_2)$. Extension across the three punctures is the parabolic condition

$$x \in (T_1 - I)M, \qquad y \in (T_2 - I)M, \qquad c(t_0) \in (T_0 - I)M, \qquad t_0 = (t_1 t_2)^{-1}.$$

Write $y = (T_2 - I)m$ and subtract the coboundary of m; this normalizes $y = 0$. A further coboundary preserves the normalization precisely when its defining element lies in M^{T_2}, and then changes x by an element of $(T_1 - I)M^{T_2}$. With $y = 0$, $c(t_0) = -T_0 x$. Since T_0 preserves $(T_0 - I)M$, the last parabolic condition is equivalent to $x \in (T_0 - I)M$. Therefore

$$H^1(B, j_*\mathcal{L}) \simeq \frac{(T_1 - I)M \cap (T_0 - I)M}{(T_1 - I)M^{T_2}}. \tag{5.17}$$

For $M = V$,

$$(T_1 - I)V = \langle 2\gamma + u + w, 2\gamma - u \rangle, \qquad (T_0 - I)V = \langle \gamma, u \rangle,$$

and their intersection is $\mathbb{Z}(2\gamma - u) = (T_1 - I)\mathbb{Z}\eta_2$. For $M = \bigwedge^2 V$,

$$\begin{aligned}(\textstyle\bigwedge^2 T_1 - I)\bigwedge^2 V = \langle &\gamma u, \gamma w, 2u\delta + w\delta, \\ &uw + u\delta - w\delta - 6\gamma\delta \rangle,\end{aligned}$$

whereas

$$(\textstyle\bigwedge^2 T_0 - I)\bigwedge^2 V = \langle \gamma u, \gamma w + u\delta \rangle.$$

Their intersection is $\mathbb{Z}\gamma u$, equal to $(\bigwedge^2 T_1 - I)\mathbb{Z}(\gamma\eta_2)$. Thus both groups in (5.17) vanish. The maps $H^0(B, j_*\mathcal{L}^r) \to H^0(B, Q^r)$ were proved surjective above for $r = 1, 2$, and skyscraper sheaves have no positive cohomology. The long exact sequence of (5.13) therefore gives (5.11) for R^1, R^2. □

39.3 The three transgressions

Lemma 39.3 (the primitive first transgression)**.** ***[VERIFIED]*** *For the constructed map and the based meridians of theorem 39.6, one may orient ω so that*

$$d_2^{0,1}(12\gamma) = \omega. \tag{5.18}$$

Proof. Choose disjoint discs D_i about the special points and a pair-of-pants neighbourhood V_0 of their complement. The class $\theta = 12\gamma$ is a section of $R^1 f_*\mathbb{Z}$ over each D_i: at p_0 this is theorem 39.1, and at p_j it is the degree-one finite-fibre lattice. It is also a section over V_0, because it is globally monodromy invariant. For every member $U \in \{D_0, D_1, D_2, V_0\}$, the local Leray five-term sequence contains

$$H^1(f^{-1}(U); \mathbb{Z}) \longrightarrow H^0(U, R^1 f_*\mathbb{Z}) \longrightarrow H^2(U, R^0 f_*\mathbb{Z}) = H^2(U; \mathbb{Z}) = 0.$$

Thus the section θ has a cohomological lift $\Theta_U \in H^1(f^{-1}(U);\mathbb{Z})$ on every member of the cover. On an overlap $D_i \cap V_0$, the two lifts have the same image in $H^0(R^1f_*\mathbb{Z})$; local five-term exactness therefore identifies their difference with a class in $H^1(D_i \cap V_0, R^0f_*\mathbb{Z}) = H^1(D_i \cap V_0;\mathbb{Z})$. These three differences form the Čech connecting cocycle used below.

Use clockwise meridians. The affine deck relation at p_j is

$$\widehat{\rho}_j^{\,m_j} = t_{v_j}. \tag{5.19}$$

Put $\Theta_j = \Theta_{D_j}$. Since Θ_j is integral and restricts to θ on the fibre, (5.19) gives $m_j\Theta_j(\widehat{\rho}_j) = \Theta_j(t_{v_j}) = \theta(v_j)$. Therefore

$$\Theta_1(\widehat{\rho}_1) = 12\gamma(v_1)/3 = 4, \qquad \Theta_2(\widehat{\rho}_2) = 12\gamma(v_2)/4 = -3. \tag{5.20}$$

By theorem 39.6, the toric meridian $\widetilde{a}_0$ is the boundary of the holomorphic disc $\{(1,1,t) : |t| \le \epsilon\}$. It is therefore null-homotopic and the local lift evaluates to zero.

Write $\alpha_i = \Theta_{V_0}(\widehat{\rho}_i)$. By the exact based cusp relation (5.23a), the three annular discrepancies have meridian values

$$4-\alpha_1, \qquad -3-\alpha_2, \qquad \alpha_1+\alpha_2, \tag{5.21}$$

whose sum is one. Mayer–Vietoris on the base identifies

$$H^1\Big(\coprod_i (D_i \cap V_0)\Big) \,/\, \operatorname{im} H^1(V_0) \simeq H^2(B;\mathbb{Z}).$$

Under meridian evaluation the numerator is $\mathbb{Z}^3$, the image is the sum-zero lattice, and the connecting map is the signed sum of the three coordinates.

We now identify this connecting class with the Leray differential rather than taking that identification as an extra hypothesis. Put $K = Rf_*\mathbb{Z}_X$. The canonical truncation triangle [36, Tag 08J5] is

$$R^0f_*\mathbb{Z}_X \longrightarrow \tau_{\le 1}K \longrightarrow R^1f_*\mathbb{Z}_X[-1] \xrightarrow{\kappa} R^0f_*\mathbb{Z}_X[1]. \tag{5.21a}$$

After applying $R\Gamma(B,-)$, the connecting homomorphism

$$H^0(B, R^1f_*\mathbb{Z}) \xrightarrow{H^0(\kappa)} H^2(B, R^0f_*\mathbb{Z}) \tag{5.21b}$$

is precisely $d_2^{0,1}$ in the hypercohomology, hence Leray, spectral sequence; this is the canonical-truncation construction of that spectral sequence [36, Tag 0FLJ]. On the chosen disc–pair-of-pants cover, functorial Mayer–Vietoris for the triangle [36, Tag 0F16] computes $H^0(\kappa)$ by choosing the local degree-one lifts Θ_U and taking their differences on the annular overlaps. Those differences are exactly the cocycle (5.21). Thus the signed sum just computed is $d_2^{0,1}(\theta)$. Equation (5.21) therefore represents a primitive generator; orient ω to obtain (5.18). □

Lemma 39.4 (the higher transgressions)**.** ***[VERIFIED]*** *In the integral Leray spectral sequence of the constructed map,*

$$d_2^{0,2}(2q) = \omega[\delta], \qquad d_2^{0,3}(2\gamma uw) = \omega[\gamma\delta]. \tag{5.22}$$

Proof. Use the multiplicative Leray spectral sequence obtained from a filtered multiplicative cochain resolution. Its E_2-product is the natural cup product

$$H^p(B,R^q) \otimes H^{p'}(B,R^{q'}) \longrightarrow H^{p+p'}(B,R^{q+q'}),$$

and every d_r is a graded derivation. This is the standard filtered-DGA construction; compare the filtered-complex spectral sequence [36, Tags 012K and 0FLJ] with the derived cup product [36, §20.31]. The

generator identifications and their products are natural by the last sentence of theorem 39.2. Put $\theta = 12\gamma$, $a = 2q$, and $b = 2\gamma uw$. By theorem 39.2, write

$$d_2 a = r\omega[\delta], \qquad d_2 b = s\omega[\gamma\delta].$$

In the E_2-ring,

$$\theta a = 12b, \qquad ab = 0, \qquad [q] = 12[\gamma\delta].$$

The derivation rule and $|\theta| = 1$ give

$$12 d_2 b = d_2(\theta a) = d_2\theta \cdot a - \theta \cdot d_2 a,$$

hence $s = 2 - r$. Since $|a| = 2$, $0 = d_2(ab) = d_2 a \cdot b + a \cdot d_2 b$. The exterior products are

$$\delta \wedge 2\gamma uw = -2\,\mathrm{vol}, \qquad 2q \wedge \gamma\delta = 2\,\mathrm{vol}\,.$$

Therefore $-2r + 2s = 0$, so $r = s$. Together with $s = 2 - r$, this gives $r = s = 1$, proving (5.22). □

Theorem 39.5 (integral cohomology in low degrees)**.** ***[VERIFIED]*** *For the constructed holomorphic map,*

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \qquad H_2(X;\mathbb{Z}) = 0. \tag{5.23}$$

Proof. The Leray terms in total degree two are

$$E_2^{2,0} = \mathbb{Z}\omega, \qquad E_2^{1,1} = 0, \qquad E_2^{0,2} = \mathbb{Z}(2q).$$

By theorems 39.3 and 39.4, the incoming map to the first term and the outgoing map from the last term are isomorphisms. Thus all three E_∞-pieces vanish and $H^2(X) = 0$. The first map has zero kernel, giving $H^1(X) = 0$. In total degree three, the quotient of $E_2^{2,1}$ by the image of $E_2^{0,2}$ is zero, $E_2^{1,2} = 0$, and the outgoing map from $E_2^{0,3}$ is injective by (5.22). Hence $H^3(X) = 0$. No later differential can meet these terms because the base has cohomological dimension two.

The universal coefficient sequences show both $\mathrm{Hom}(H_2(X), \mathbb{Z}) = 0$ and $\mathrm{Ext}(H_2(X), \mathbb{Z}) = 0$. Since $H_2(X)$ is finitely generated, it has neither a free nor a torsion summand, so it is zero. □

39.4 Based van Kampen and sphere recognition

Proposition 39.6 (the geometric fundamental-group presentation)**.** ***[VERIFIED]*** *For the punctured-overlap gluings of theorems 8.5 and 8.7, the based van Kampen presentation is*

$$\pi_1(X) \simeq \langle c, x, y \mid c \text{ central},\ xy = 1,\ x^3 = c,\ y^4 = c^{-1} \rangle \simeq 1. \tag{5.24}$$

Proof. Choose b_* in a pair-of-pants core of B°, put x_* on the holomorphic zero section above it, and choose three nonintersecting base whiskers from b_* to clockwise boundary circles. Lift all whiskers by the zero section. With the ordering compatible with $g_1 g_2 g_0 = 1$, the resulting based loops obey

$$\widehat{\rho}_1 \widehat{\rho}_2 \widetilde{a}_0 = 1. \tag{5.23a}$$

The period-family action is linear and hence preserves the zero section. It splits the homotopy sequence over the pair of pants, giving

$$\pi_1(f^{-1}(B^\circ)) = \Lambda \rtimes F(\widehat{\rho}_1, \widehat{\rho}_2),$$

with conjugation actions A_1, A_2. The clockwise meridian at p_j lifts to the affine generator in (2.9), including its sign and based translation. Indeed, on a logarithm branch put $\alpha = \log(s_j)/(2\pi i)$. The overlap map h_j translates flat coordinates by $+\alpha v_j$, so the inverse image of the period-family zero section

is $x(z) = -\alpha(z)v_j$. Clockwise continuation takes z to $g_j z$ and $\alpha(g_j z) = \alpha(z) - 1/m_j$. Because $A_j v_j = v_j$, the affine generator sends the initial point to

$$A_j(-\alpha v_j) + v_j/m_j = -(\alpha - 1/m_j)v_j,$$

exactly the continued endpoint. Thus the based zero-section meridian is the affine generator, not its inverse or a translated conjugate. Formula (2.8f) gives

$$\widehat{\rho}_j^{\,m_j} = t_{v_j}. \tag{5.23b}$$

The radial model of theorem 8.8 shows that attaching N_j imposes exactly this relation. Since the global and local whiskers are the same zero-section path under the overlap, conjugating back to x_* adds no fibre drift.

For completeness, the cusp covering has no hidden fundamental group. A loop in the infinite toric variety has compact image and therefore lies in the open $X_{\Sigma_{\mathrm{tor}}}$ associated with a finite subfan Σ_{tor}. Enlarge that subfan by the rays through $(0,1), (e_1,1), (e_2,1)$. Write

$$N_{\Sigma_{\mathrm{tor}}} := \langle |\Sigma_{\mathrm{tor}}| \cap N' \rangle_{\mathbb{Z}}.$$

The three displayed primitive generators span N', so the classical formula $\pi_1(X_{\Sigma_{\mathrm{tor}}}) = N'/N_{\Sigma_{\mathrm{tor}}}$ [23, Theorem 12.1.10] makes the resulting finite toric open simply connected; the original loop therefore has a null-homotopy in that open. Thus Y is simply connected. The one-parameter subgroup in the third torus coordinate acts by $t \mapsto rt$, $r > 0$. Given the compact image of a null-homotopy, choose r so small that its translate lies in $|t| < \epsilon$; the path $r_s = (1-s) + sr$ gives a homotopy from the original loop to its translate inside Y_ϵ. Hence Y_ϵ is also simply connected. The verified free quotient $N_0 = Y_\epsilon/\bar{\Lambda}$ therefore has universal cover Y_ϵ, so $\pi_1(N_0) = \bar{\Lambda}$. Equation (2.8iii) identifies a fibre-lattice loop $\lambda \in \Lambda$ with the deck transformation $\bar{\lambda}$; hence the inclusion of a nearby fibre is exactly

$$\Lambda \longrightarrow \bar{\Lambda}, \qquad \lambda \longmapsto \bar{\lambda}, \tag{5.23c}$$

with kernel $\Lambda_{\mathrm{tor}} = \langle \widehat{w}, \widehat{\delta} \rangle$. On the zero section the same overlap sends $(z,0)$ to $(1,1,t_c)$. In the toric chart of the triangle $\{0, e_1, e_2\}$, whose coordinates may be written $(t_c/(x_1x_2), x_1, x_2)$, this is the holomorphic disc $t_c \mapsto (t_c, 1, 1)$. Thus the cusp meridian $\widetilde{a}_0$ is null-homotopic, with no fibre translation. Combining this fact with (5.23a), the cusp attachment imposes precisely $\widehat{\rho}_1\widehat{\rho}_2 = 1$.

Van Kampen now gives

$$\left\langle \Lambda, \widehat{\rho}_1, \widehat{\rho}_2 \;\middle|\; \widehat{\rho}_j \lambda \widehat{\rho}_j^{-1} = A_j\lambda, \widehat{\rho}_1^3 = t_{v_1}, \widehat{\rho}_2^4 = t_{v_2}, \widehat{\rho}_1\widehat{\rho}_2 = 1, \Lambda_{\mathrm{tor}} = 1 \right\rangle.$$

The normal closure of Λ_{tor} under A_1, A_2 is $\ker\gamma = \langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$; for example $A_1\widehat{w} = \widehat{u} - \widehat{w}$. The image of Λ is therefore generated by $c = [\widehat{\gamma}]$, which is central because γ is invariant. Since $\gamma(v_1) = 1$ and $\gamma(v_2) = -1$, the presentation becomes (5.24). It is the case $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ of theorem 5.5, where $p = -1$; hence the group is trivial. □

Corollary 39.7 (topology and the third Chern number)**.** ***[VERIFIED]*** *The underlying smooth manifold of the constructed threefold is diffeomorphic to the standard six-sphere, and*

$$\langle c_3(T_X), [X] \rangle = 2. \tag{5.25}$$

Proof. Combine theorems 39.5 and 39.6. The universal coefficient theorem and Poincaré duality give the integral homology of S^6, exactly as in theorem 44.15. The complete recognition argument of theorem 44.16 then gives $X \cong_{\mathrm{diff}} S^6$. Its Euler characteristic is two. On a complex threefold the top Chern class of the complex tangent bundle is the Euler class of the underlying oriented real tangent bundle, proving (5.25). □

Independent role of the collapse route. The proof above is independent of the cellular route. The candidate's cusp collapse, its integral homology map, and its equality with nearby-cycle specialization are now independently verified in theorems 44.5 to 44.7. The actual finite-band and cusp Mayer–Vietoris maps, their primitive blocks, and their commuting comparison with the differentials proved in this section are now theorems 44.10, 44.11 and 44.13. This supplies a second complete integral topology certificate; it remains logically independent of the sphere-recognition conclusion already obtained in this section.

40 The direct algebraic-reduction proof

The calculation in this section is independent of the integral topology of X. It uses the period family of theorem 8.2 and the proper connected fibration of theorem 8.9; it does not use $H^2(X;\mathbb{Z})$, the sphere-recognition argument, or any theorem of Campana–Demailly–Peternell.

Lemma 40.1 (algebraic reduction of a two-dimensional torus)**.** *Let T be a complex torus of dimension two. Suppose*

$$\mathrm{NS}(T) = \mathbb{Z}\eta, \qquad \eta^2 \neq 0,$$

and the Appell–Humbert Hermitian form H_η is nondegenerate and indefinite. Then $a(T) = 0$.

Proof. The algebraic reduction of a compact complex torus is a quotient $q : T \to A = T/K$ by a connected complex subtorus, with A an abelian variety and $\dim A = a(T)$ [18, Chapter 2, §6]; see also [21, §8, p. 2437].

Assume $a(T) > 0$. If $a(T) = 2$, then T is an abelian surface. An ample line bundle has first Chern class $n\eta$, $n \neq 0$, and its Appell–Humbert form nH_η is positive definite. This contradicts the indefiniteness of H_η.

If $a(T) = 1$, then A is an elliptic curve. For a point $p \in A$, put

$$\varphi = q^* c_1(\mathcal{O}_A(p)) \in \mathrm{NS}(T).$$

The class is nonzero: for any translation-invariant Kahler form ω on T, the projection formula gives $\int_T \varphi \wedge \omega = \int_K \omega > 0$. But $\varphi^2 = 0$, since A is a curve. Writing $\varphi = n\eta$, $n \neq 0$, yields $0 = n^2\eta^2 \neq 0$, a contradiction. Both possible positive algebraic dimensions are excluded. □

Let

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle, \qquad \eta = u \wedge w + 6\gamma \wedge \delta.$$

For an integral two-class $E \in \bigwedge^2 V$, write its coordinates in the ordered basis $(\gamma u, \gamma w, \gamma\delta, uw, u\delta, w\delta)$ as $E_{\gamma u}, \dots, E_{w\delta}$, and set

$$\begin{aligned} P_E = {} & E_{\gamma u} - E_{\gamma w}\tau - E_{\gamma\delta}\mu + 6E_{uw}\mu + E_{u\delta}\beta \\ & + E_{w\delta}(6\mu^2 - \tau\beta). \end{aligned} \tag{3.1}$$

Proposition 40.2 (very general Neron–Severi group)**.** ***[VERIFIED]*** *For the descended period family of theorem 8.2, outside a countable subset of B° the smooth fibre F_b satisfies*

$$\mathrm{NS}(F_b) = \mathbb{Z}\eta, \qquad a(F_b) = 0. \tag{3.2}$$

Proof. The two holomorphic one-forms defining the Hodge subspace are

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta.$$

The determinant inequality (2.2) gives $V_{\mathbb{C}} = F^1 \oplus \overline{F^1}$, where $F^1 = \langle \sigma_1, \sigma_2 \rangle$. For a real two-class E, the Hodge decomposition therefore gives

$$E \in H^{1,1}(F_b) \quad \Longleftrightarrow \quad E \wedge \sigma_1 \wedge \sigma_2 = 0.$$

Expanding the wedge product gives

$$E \wedge \sigma_1 \wedge \sigma_2 = P_E \, \gamma \wedge u \wedge w \wedge \delta. \tag{3.3}$$

We next determine exactly which P_E vanish identically. Suppose

$$a + b\tau + c\mu + d\beta + e(6\mu^2 - \tau\beta) \equiv 0. \tag{3.4}$$

Apply the cusp transformation $(\tau, \mu, \beta) \mapsto (\tau - 1, \mu, \beta + 1)$ and subtract. This gives

$$-b + d + e(\beta - \tau + 1) = 0.$$

Because $\beta - \tau$ is translated by two, $e = 0$, and then $d = b$. Apply the order-four transformation in (2.1), subtract (3.4), and multiply by $-\tau$. One obtains

$$6b\mu^2 + c(\tau - 1)\mu + b(\tau^2 + 3\tau + 1) - c\tau = 0.$$

Subtract its cusp translate:

$$c\mu + 2b(\tau + 1) - c = 0. \tag{3.5}$$

Translate (3.5) once around the cusp. Since μ is invariant and $\tau \mapsto \tau - 1$, subtracting the translated identity from (3.5) gives $2b = 0$. Thus $b = 0$, and (3.5) becomes $c(\mu - 1) = 0$. If $c \neq 0$, then $\mu \equiv 1$, whereas the order-four law would give $1 = \mu \circ g_2 = 1 + 1/\tau$, an impossibility. Hence $c = 0$, then $b = d = 0$, and finally $a = 0$. Thus the five functions

$$1, \quad \tau, \quad \mu, \quad \beta, \quad 6\mu^2 - \tau\beta$$

are linearly independent over $\mathbb{C}$.

Reading the coefficients in (3.1) now gives

$$P_E \equiv 0 \quad \Longleftrightarrow \quad E \in \mathbb{Z}\eta. \tag{3.6}$$

For each other integral class, the zero set of P_E on the connected Riemann surface $\mathfrak{h}_z$ is discrete, hence countable. There are only countably many integral classes. The class $\eta = q = uw + 6\gamma\delta$ is monodromy invariant by theorem 5.3, so it defines a class on every marked fibre after descent. The image in B° of the countable union of the remaining discrete zero sets is still countable. A base point outside that image has no additional integral $(1,1)$-class on any lift, proving the Neron–Severi statement on the quotient family.

Finally,

$$\eta^2 = 12 \, \gamma \wedge u \wedge w \wedge \delta. \tag{3.7}$$

For a nondegenerate alternating form E, write $E^\vee$ for its inverse alternating form on the dual real vector space. If $G_\eta = (i\eta^\vee(\sigma_a, \overline{\sigma_b}))_{a,b}$, direct inversion of the alternating-form matrix and substitution give

$$G_\eta = \begin{pmatrix} 2\operatorname{Im}\tau & 2\operatorname{Im}\mu \\ 2\operatorname{Im}\mu & \frac{1}{3}\operatorname{Im}\beta \end{pmatrix},$$

$$\det G_\eta = \frac{2}{3}\operatorname{Im}\tau \left(\operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} \right) < 0. \tag{3.8}$$

Here is the Appell–Humbert bridge needed to read the inertia of this matrix. With the convention $E = \operatorname{Im} H_E$, one has

$$E = \frac{1}{2i} \sum_{a,b} (H_E)_{ab} \, \sigma_a \wedge \overline{\sigma_b}.$$

Thus, in the ordered basis $(\sigma_1, \sigma_2, \overline{\sigma_1}, \overline{\sigma_2})$, the upper-right block of the alternating-form matrix of E is $A = -(i/2)H_E$. Inverting the block matrix gives

$$E^\vee(\sigma_a, \overline{\sigma_b}) = [-(A^t)^{-1}]_{ab} = -2i(\overline{H_E}^{-1})_{ab}, \qquad G_E = 2\overline{H_E}^{-1}. \tag{3.8a}$$

The displayed G_η is real symmetric, so $H_\eta = 2G_\eta^{-1}$. Inversion and multiplication by a positive scalar preserve inertia. Hence H_η is nondegenerate and indefinite, while (3.7) is nonzero. Apply theorem 40.1. $\square$

Lemma 40.3 (meromorphic descent)**.** *Let $f : X \to B$ be a proper surjective holomorphic map from a connected complex threefold to a connected compact Riemann surface, with connected fibres, and a submersion over $B^\circ = B \setminus \Sigma$, where Σ is finite and $B^\circ \neq \varnothing$. If $g \in \mathcal{M}(X)$ is constant on the fibres over an uncountable subset of B°, then there is a unique $\varphi \in \mathcal{M}(B)$ such that $g = f^*\varphi$.*

Proof. Let P be the polar divisor of g, and put $\Omega = f^{-1}(B^\circ) \setminus |P|$. A proper analytic subset of a connected complex manifold has real codimension at least two and does not disconnect it. First note that $f^{-1}(B^\circ)$ is connected. Indeed, each of its connected components is open, and its image is open because f is a submersion and closed because the base change $f^{-1}(B^\circ) \to B^\circ$ is proper. Since B° is connected, every component would meet every fibre; connectedness of the fibres therefore permits only one component. Removing $|P|$ now leaves Ω connected. Only finitely many fibres are components of $|P|$. For every other b, the complement $F_b \cap \Omega$ of a proper analytic subset in the connected manifold F_b is connected.

On Ω, define the analytic set

$$Z = \{dg \wedge f^*(dt) = 0\},$$

where t is any local coordinate on B. If $g|_{F_b}$ is constant, then $F_b \cap \Omega \subset Z$. If $Z \neq \Omega$, then, because Ω is second countable, Z has at most countably many irreducible components. Each irreducible two-dimensional fibre contained in Z is one of those components, and distinct fibres give distinct components. This contradicts the uncountable hypothesis. Therefore $Z = \Omega$.

It follows that dg annihilates $\ker df$. Connectedness of $F_b \cap \Omega$, together with local holomorphic sections of the submersion, gives $g = \varphi \circ f$ on Ω, for a holomorphic φ away from finitely many points of B.

Fix such a point t_0. Choose a general holomorphic curve germ $\kappa : (\Delta, 0) \to (X, x)$, not contained in $|P|$, with

$$f \circ \kappa(s) = t_0 + s^k u(s), \qquad u(0) \neq 0.$$

If $g \circ \kappa$ has pole order at most M, then $|g \circ \kappa(s)| \leq C|s|^{-M}$. After shrinking the disc, boundedness above and below of $|u|$ gives constants $c, C_1 > 0$ such that

$$c|t - t_0|^{1/k} \leq |s| \leq C_1|t - t_0|^{1/k}$$

on each local inverse branch. Therefore

$$|\varphi(t)| \leq C_2|t - t_0|^{-M/k}$$

away from the image of a discrete exceptional set, and then everywhere on the punctured disc by holomorphy. For any integer $N > M/k$, the function $(t - t_0)^N\varphi(t)$ is bounded and extends holomorphically. Thus φ is meromorphic at t_0. Repeating at the finitely many exceptional points produces $\varphi \in \mathcal{M}(B)$. Equality on the dense open set Ω gives equality of meromorphic functions on X, and surjectivity of f gives uniqueness. □

Theorem 40.4 (algebraic dimension and algebraic reduction)**.** ***[VERIFIED]*** *For the global holomorphic family of theorem 8.9,*

$$\mathcal{M}(X) = f^*\mathcal{M}(\mathbb{P}^1) = f^*\mathbb{C}(t), \qquad a(X) = 1, \tag{3.9}$$

and f is the algebraic reduction.

Proof. By theorem 40.2, all but countably many smooth fibres have algebraic dimension zero. Let $g \in \mathcal{M}(X)$. Only finitely many fibres are components of its polar divisor, so on uncountably many of the preceding fibres the restriction of g is a meromorphic function on a torus of algebraic dimension zero and is therefore constant. By theorem 40.3, $g = f^*\varphi$ for a unique $\varphi \in \mathcal{M}(\mathbb{P}^1)$. The reverse inclusion is immediate, and (3.9) follows. Since f is proper, surjective, has connected fibres, and induces the equality of meromorphic function fields, it is the algebraic reduction. □

Disposition. The earlier label "algebraic reduction open" was too broad. The direct proof above validates the assertion independently of Section 7 and of the CDP comparison. Its provenance is the candidate's Lemmas 9.2–9.6 and Proposition 9.7, reconstructed here with an explicit Appell–Humbert matrix bridge, a proof that the smooth locus remains connected, and the omitted root estimate in the final meromorphic-extension step. This does not by itself prove the claimed topology of X; that is a separate calculation.

41 Structure-sheaf direct images and uniform non-torsion

This section reconstructs the analytic inputs used by the candidate's Proposition 10.8. The proof differs at one important point from the candidate's route: extension across the cusp is obtained from the intrinsic nearby-cycle isomorphism of theorem 39.1, not from the cellular collapse. The two cusp maps are now independently proved equal in theorem 44.7; the argument below nevertheless retains its logically independent intrinsic formulation.

Put

$$e = d\zeta_1 \wedge d\zeta_2$$

on the marked torus family. Retain the local coordinates and notation of section 8:

$$s_j \circ g_j = \zeta_j s_j, \qquad (m_1, m_2) = (3, 4), \qquad t_j = s_j^{m_j}, \qquad f^{-1}(p_j) = m_j S_j.$$

41.1 The Hodge line and the finite normal characters

Lemma 41.1 (determinant characters and the Hodge line)**.** ***[VERIFIED]*** *Let*

$$\mathfrak{K} = f_* \omega_{X/B}.$$

Then $\mathfrak{K}$ is invertible and

$$\mathfrak{K} \simeq \mathcal{O}_B(1). \tag{9.1}$$

More precisely,

$$\det R_{g_1} = -\frac{1}{\tau}, \qquad \det R_{g_2} = \frac{1}{\tau}, \qquad \det R_{g_0} = 1, \tag{9.2}$$

and at the fixed points

$$\det R_{g_j}(z_j) = \zeta_j^{a_j}, \qquad (a_1, a_2) = (2, 1). \tag{9.3}$$

The form e generates $\mathfrak{K}$ at the cusp, while a unit times $s_j^{-a_j} e$ generates it at p_j.

Proof. Equation (2.3d) in theorem 8.2 gives

$$R_g(z)\Pi(z)\lambda = \Pi(gz) A_g \lambda.$$

Using

$$A_1 \widehat{w} = \widehat{u} - \widehat{w}, \quad A_1 \widehat{\delta} = \widehat{\delta}, \qquad A_2 \widehat{w} = -\widehat{u} + \widehat{\delta}, \quad A_2 \widehat{\delta} = \widehat{\delta},$$

and writing $\tau^\sharp = \tau(g_j z)$, $\mu^\sharp = \mu(g_j z)$, direct multiplication gives

$$R_{g_1} = \begin{pmatrix} \tau^\sharp - 1 & 0 \\ \mu^\sharp & 1 \end{pmatrix}, \qquad R_{g_2} = \begin{pmatrix} -\tau^\sharp & 0 \\ 1 - \mu^\sharp & 1 \end{pmatrix}.$$

The transformation laws $\tau \circ g_1 = (\tau - 1)/\tau$ and $\tau \circ g_2 = -1/\tau$ prove (9.2) at g_1, g_2; the cusp formulas give $R_{g_0} = I$. Since $\widehat{\delta}$ is fixed, one eigenvalue is 1. At $\tau(z_1) = \rho = e^{\pi i/3}$, the other is $\rho - 1 = e^{2\pi i/3} = \zeta_1^2$; at $\tau(z_2) = i$, it is $-i = \zeta_2$. This proves (9.3).

We determine the extension over a finite filling without suppressing the ramification. On J_j', the nowhere-zero form $ds_j \wedge e$ transforms by

$$(\widetilde{g}_j^{\log})^*(ds_j \wedge e) = \chi_j(s_j)\, ds_j \wedge e, \qquad \chi_j = \zeta_j \det R_{g_j}. \tag{9.4}$$

Iteration m_j times shows $\prod_{r=0}^{m_j-1} \chi_j(\zeta_j^r s) = 1$. Write $\chi_j(s) = \chi_j(0)e^{\psi(s)}$, with $\psi(0) = 0$. The product identity says that the power series of ψ has no terms whose degree is divisible by m_j. Hence

$$\varphi(s) - \varphi(\zeta_j s) = \psi(s)$$

has a holomorphic solution, obtained coefficientwise by division by $1 - \zeta_j^n$. Therefore

$$s^{k_j} e^{\varphi(s)} ds \wedge e, \qquad k_j = m_j - 1 - a_j, \tag{9.5}$$

is invariant: $\chi_j(0) = \zeta_j^{1+a_j}$, so invariance is equivalent to $k_j + 1 + a_j \equiv 0 \pmod{m_j}$. We now distinguish the descended canonical section from the generator of its direct image. Let

$$q_j : J_j' \longrightarrow N_j, \qquad \widetilde{f}_j : J_j' \longrightarrow \Delta_j, \qquad \varpi_j : \Delta_j \longrightarrow D_j$$

denote the finite quotient, the torus-family projection, and the cyclic base quotient, respectively, so that

$$f_j \circ q_j = \varpi_j \circ \widetilde{f}_j, \qquad t_j = s_j^{m_j}.$$

Put $\mathcal{A}_j = \widetilde{f}_j^{-1}(0) = \{s_j = 0\}$ and $G_j = \langle \widetilde{g}_j^{\log} \rangle$. The action is free by theorem 8.7; hence q_j is finite étale and $q_j^{-1}(S_j) = \mathcal{A}_j$.

The invariant form in (9.5), now denoted

$$\sigma_j = s_j^{k_j} e^{\varphi(s_j)} ds_j \wedge e, \qquad k_j = m_j - 1 - a_j, \tag{9.5a}$$

descends uniquely to $\overline{\sigma}_j \in H^0(N_j, K_{N_j})$ with $q_j^* \overline{\sigma}_j = \sigma_j$. Since $e^{\varphi(s_j)}$ is a unit and $ds_j \wedge e$ is nowhere vanishing,

$$\operatorname{div}_{J_j'}(\sigma_j) = k_j \mathcal{A}_j.$$

Because q_j is étale,

$$q_j^* \operatorname{div}_{N_j}(\overline{\sigma}_j) = \operatorname{div}_{J_j'}(q_j^* \overline{\sigma}_j) = k_j \mathcal{A}_j = q_j^*(k_j S_j).$$

Faithfulness of finite surjective pullback on Cartier divisors therefore gives

$$\operatorname{div}_{N_j}(\overline{\sigma}_j) = k_j S_j. \tag{9.5b}$$

Thus (9.5a) is a canonical section with that divisor, not a nowhere-vanishing frame when $k_j > 0$; explicitly, $k_1 = 0$ and $k_2 = 2$.

For the direct-image generator, let

$$\vartheta_j = (ds_j \wedge e) \otimes (dt_j)^{-1} \tag{9.5c}$$

be the nowhere-vanishing frame of

$$q_j^* \omega_{N_j/D_j} = K_{J_j'} \otimes (f_j \circ q_j)^* K_{D_j}^{-1}.$$

Here $(dt_j)^{-1}$ is the pulled-back frame of the inverse canonical line of D_j, not division by the differential of $s_j^{m_j}$. Put $\vartheta_j' = e^{\varphi(s_j)} \vartheta_j$. The defining equation for φ, together with $\chi_j(s) = \chi_j(0)e^{\psi(s)}$ and $\chi_j(0) = \zeta_j^{1+a_j}$, gives

$$(\widetilde{g}_j^{\log})^* \vartheta_j' = \zeta_j^{1+a_j} \vartheta_j'. \tag{9.5d}$$

Every holomorphic coefficient of a relative top form is pulled back from Δ_j, because the fibres of $\widetilde{f}_j$ are connected compact complex tori. Hence

$$\widetilde{f}_{j*}q_j^*\omega_{N_j/D_j} = \mathcal{O}_{\Delta_j}\vartheta'_j.$$

For $F(s_j) \in \mathcal{O}_{\Delta_j}$, the section $F(s_j)\vartheta'_j$ is G_j-invariant exactly when

$$F(\zeta_j s_j)\zeta_j^{1+a_j} = F(s_j). \tag{9.5e}$$

Writing $F(s_j) = \sum_{n\geq 0} c_n s_j^n$, this says that $c_n = 0$ unless

$$n + 1 + a_j \equiv 0 \pmod{m_j}, \qquad \text{equivalently} \qquad n \equiv k_j \pmod{m_j}.$$

Consequently,

$$\Big((\varpi_j)_*\widetilde{f}_{j*}q_j^*\omega_{N_j/D_j}\Big)^{G_j} = \mathcal{O}_{D_j}\cdot s_j^{k_j}\vartheta'_j,$$

and finite descent gives the precise invariant-module identity

$$f_{j*}\omega_{N_j/D_j} = \mathcal{O}_{D_j}\cdot s_j^{k_j}\vartheta'_j. \tag{9.5f}$$

Finally, over Δ_j^*, the differential identity

$$dt_j = m_j s_j^{m_j-1} ds_j$$

identifies this generator with

$$s_j^{k_j}\vartheta'_j = \frac{e^{\varphi(s_j)}}{m_j} s_j^{k_j-(m_j-1)} e = \frac{e^{\varphi(s_j)}}{m_j} s_j^{-a_j} e. \tag{9.5g}$$

Thus a unit times $s_j^{-a_j}e$ generates $f_{j*}\omega_{N_j/D_j} = (f_*\omega_{X/B})|_{D_j}$, whereas the canonical section (9.5a) itself has divisor $k_j S_j$.

At the cusp, in every smooth toric chart the unimodular monomial change of coordinates turns

$$\Omega_0 = \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge dt_c$$

into a unit times the ordinary coordinate volume form. Thus it generates the canonical line. It is invariant under translations and under the unimodular shears defining the quotient. Under the punctured-overlap map, $\Omega_0 = (2\pi i)^2 dt_c \wedge e$, so e generates $\mathfrak{K}$ at p_0.

It remains to compute the degree. The modular-form calculation already used in the proof of theorem 8.2 gives

$$\Theta = \left(\frac{E_4^2 E_6^{1/2}}{\Delta_{\mathrm{mod}}}\right)\circ\tau.$$

The square root exists because the zeros of $E_6 \circ \tau$ have even order. The weights $8, 3, 12$ give the same automorphy factor as e; the sign at z_2 is fixed by its linearizing coordinate. The orders are

$$\mathrm{ord}_{z_1}\Theta = 2 = a_1, \qquad \mathrm{ord}_{z_2}\Theta = 1 = a_2, \qquad \Theta = t_c^{-1}\cdot(\text{unit}) \tag{9.6}$$

at the cusp. Consequently $\mathsf{e} = \Theta^{-1}e$ is single-valued. At p_j it is a unit times $s_j^{-a_j}e$; at p_0 it is a unit times $t_c e$. Hence it is a global section of $\mathfrak{K}$ with the single simple zero p_0. This proves $\mathfrak{K} \simeq \mathcal{O}_B(p_0) \simeq \mathcal{O}_B(1)$. Finally, $\mathfrak{K}$ is torsion free of generic rank one on the smooth curve B, hence is invertible. □

Lemma 41.2 (exact order of the finite-fibre normal line)**.** ***[VERIFIED]*** *For $j = 1, 2$,*

$$\mathcal{O}_X(S_j)|_{S_j} = N_{S_j/X}$$

has exact order m_j in $\mathrm{Pic}(S_j)$.

Proof. On the finite cover of theorem 8.7, the central torus $A_0^{(j)} = \{s_j = 0\}$ maps to S_j as a connected principal $G_j \simeq \mathbb{Z}/m_j$-cover. Its normal line is trivial, and a generator of G_j acts on the fibre by $s_j \mapsto \zeta_j s_j$. Thus $N_{S_j/X}$ is associated to this principal cover by the character ζ_j, and its m_j-th power is trivial.

Suppose its k-th power were trivial for some $0 < k < m_j$. Pulling a nowhere-zero trivializing section back to $A_0^{(j)}$ would give a nowhere-zero holomorphic function h satisfying

$$h(gx) = \zeta_j^{-k} h(x).$$

Every holomorphic function on the compact connected torus $A_0^{(j)}$ is constant. The displayed equation would force $\zeta_j^k = 1$, contrary to $0 < k < m_j$. Hence the order is exactly m_j. □

41.2 The cusp structure sheaf

Lemma 41.3 (structure-sheaf cohomology of the cusp fibre)**.** ***[VERIFIED]*** *The reduced normal-crossings fibre W satisfies*

$$h^q(W, \mathcal{O}_W) = (1, 2, 1) \qquad (q = 0, 1, 2). \tag{9.7}$$

The natural map

$$H^1(W; \mathbb{C}) \longrightarrow H^1(W, \mathcal{O}_W) \tag{9.8}$$

is an isomorphism. In degree two the corresponding map is surjective and is injective on the kernel of pullback to the normalization $\nu : \mathrm{dP}_6 \to W$. Finally,

$$\bigwedge^2 H^1(W, \mathcal{O}_W) \longrightarrow H^2(W, \mathcal{O}_W) \tag{9.9}$$

is an isomorphism.

Proof. Let $D_1, D_2, D_3 \simeq \mathbb{P}^1$ be the double curves and P, Q the two triple points, with the branch order fixed in the proof of theorem 39.1. The normalization complex is

$$0 \longrightarrow \mathcal{O}_W \longrightarrow \nu_*\mathcal{O}_{\mathrm{dP}_6} \xrightarrow{\delta_0} \bigoplus_{k=1}^{3} \mathcal{O}_{D_k} \xrightarrow{\delta_1} \mathbb{C}_P \oplus \mathbb{C}_Q \longrightarrow 0, \tag{9.10}$$

where δ_0 is the difference of the two branch restrictions and

$$\delta_1(g_1, g_2, g_3) = (g_1(P) - g_2(P) + g_3(P), g_1(Q) - g_2(Q) + g_3(Q)).$$

For completeness, exactness is stalkwise. At a generic double point it is the fibre-product sequence

$$0 \to \mathbb{C}\{x, y, z\}/(xy) \to \mathbb{C}\{y, z\} \oplus \mathbb{C}\{x, z\} \to \mathbb{C}\{z\} \to 0.$$

At a triple point it is the augmented branch complex for $\mathbb{C}\{x, y, z\}/(xyz)$: three branch rings, three pairwise intersection rings, and the common residue field, with alternating differences. The augmented two-simplex complex is exact; successively lifting compatible power series proves exactness of the analytic rings. Away from the singular locus the first arrow is an isomorphism.

The same formulas give an exact complex with $\mathbb{C}$ in place of $\mathcal{O}$. Since dP_6 is rational and each D_k is a rational curve, the positive structure-sheaf cohomology of every term on the right of (9.10) vanishes. Hypercohomology therefore reduces to

$$\mathbb{C} \xrightarrow{0} \mathbb{C}^3 \xrightarrow{d_1} \mathbb{C}^2, \qquad d_1(a_1, a_2, a_3) = (s, s), \quad s = a_1 - a_2 + a_3. \tag{9.11}$$

Its cohomology has dimensions $1, 2, 1$, proving (9.7).

For the constant-sheaf complex, the row of degree zero is again (9.11), the row of degree one vanishes, and the remaining degree-two row begins with $H^2(\mathrm{dP}_6; \mathbb{C})$. The comparison induced by $\mathbb{C}_W \hookrightarrow \mathcal{O}_W$ is the identity on (9.11) and zero on the positive rows of the target. It follows that (9.8) is an isomorphism, while in degree two it maps the dual-complex summand $\ker \nu^*$ isomorphically onto $H^2(W, \mathcal{O}_W)$.

Choose the basis $\gamma_W, u_W \in H^1(W; \mathbb{Z})$ whose intrinsic specializations are γ, u, respectively; it exists and is unique by theorem 39.1. Their cup product is nonzero, because its specialization is $\gamma \wedge u \neq 0 \in H^2(F; \mathbb{Z})$. Its pullback to dP_6 is zero since $H^1(\mathrm{dP}_6; \mathbb{C}) = 0$. The degree-two comparison therefore sends it to a nonzero element of $H^2(W, \mathcal{O}_W)$. The images of γ_W, u_W form a basis by (9.8), and both sides of (9.9) are one-dimensional. Hence (9.9) is an isomorphism. □

41.3 Direct images, base change, and the Picard line

Theorem 41.4 (structure-sheaf direct images)**.** ***[VERIFIED]*** *For the constructed map $f : X \to B = \mathbb{P}^1$,*

$$f_*\mathcal{O}_X = \mathcal{O}_B, \qquad R^1 f_*\mathcal{O}_X \simeq \mathcal{O}_B \oplus \mathcal{O}_B(-1), \qquad R^2 f_*\mathcal{O}_X \simeq \mathcal{O}_B(-1). \tag{9.12}$$

All three direct images are locally free and every base-change morphism

$$(R^q f_*\mathcal{O}_X) \otimes k(b) \longrightarrow H^q(X_b, \mathcal{O}_{X_b}), \qquad 0 \leq q \leq 2, \tag{9.13}$$

is an isomorphism, including at the two nonreduced fibres. Moreover the invariant class γ defines the trivial summand in $R^1 f_*\mathcal{O}_X$ and is nowhere zero on B.

Proof. The proper map has connected fibres and B is normal, so Stein factorization gives $f_*\mathcal{O}_X = \mathcal{O}_B$.

The map is flat. A local parameter on B pulls back to a nonzero element of the regular local domain $\mathcal{O}_{X,x}$; if it vanished identically near one point, analytic continuation on the connected manifold would make f constant. Thus $\mathcal{O}_{X,x}$ is torsion-free, hence flat, over the discrete valuation ring $\mathcal{O}_{B,f(x)}$.

Over B°, the fibres are two-dimensional complex tori and their structure-sheaf cohomology dimensions are $1, 2, 1$. At the cusp the same dimensions hold by theorem 41.3. Grauert's theorem therefore gives local freeness of ranks $1, 2, 1$ and base change on a neighbourhood of $B^\circ \cup \{p_0\}$; the precise analytic statement is [6, Chapter III, §4, Theorem 4.12], originally due to Grauert [7].

At p_j, use the finite cover $J_j' \to N_j$. If $\widetilde{f} : J_j' \to \Delta_j$ is the smooth torus family and $\pi(s) = s^{m_j}$, exactness of finite-group invariants in characteristic zero gives

$$R^q f_*\mathcal{O}_{N_j} = \left(\pi_* R^q \widetilde{f}_*\mathcal{O}_{J_j'}\right)^{G_j}. \tag{9.14}$$

After averaging, a free eigenframe v may be chosen for each character ν. The invariant sections are exactly $s^r v\, \mathcal{O}_{D_j}$, where $0 \leq r < m_j$ is determined by $\nu\zeta_j^r = 1$. Hence (9.14) is free of ranks $1, 2, 1$. Higher direct images vanish by the fibre-dimension bound, so these are all the nonzero direct images. Since all direct images are locally free, the analytic cohomology-and-base-change theorem, applied by descending induction from $q = 2$, proves (9.13) at the finite fibres as well. In the notation of [6, Chapter III, §3, Theorem 3.4, Corollaries 3.5, 3.7, and 3.10]. Corollary 3.10 says that, for each fixed q, freeness of $R^p f_*\mathcal{O}_X$ for every $p \geq q$ is equivalent to cohomological flatness in all dimensions $p \geq q$. Applying this successively with $q = 2, 1, 0$ gives the displayed base-change maps in every relevant degree. This is a local statement on the base and so applies equally to the nonreduced scheme-theoretic fibres.

Write $G = R^1 f_*\mathcal{O}_X$. We construct its distinguished section without using a cellular collapse. On B°, the monodromy-invariant integral class γ maps to a holomorphic section $\gamma^{0,1}$ of G. It is nowhere zero there: if its image in $H^{0,1}(F)$ vanished, then $\gamma \in H^{1,0}(F) \cap H^1(F; \mathbb{R}) = 0$. On a finite cover it is a constant invariant class and remains nonzero at $s_j = 0$, so it descends across p_j.

At the cusp, let $j : \Delta^* \hookrightarrow \Delta$. The natural specialization morphism

$$R^1 f_*\mathbb{C} \longrightarrow j_* R^1 f_*\mathbb{C}|_{\Delta^*}$$

is an isomorphism: away from 0 this is tautological, while at 0 it is exactly the isomorphism from $H^1(W;\mathbb{C})$ onto the T_0-invariant subspace proved in theorem 39.1. Hence the invariant classes γ, u extend as sections of $R^1f_*\mathbb{C}$ on the whole disc. The natural morphism induced by $\mathbb{C} \hookrightarrow \mathcal{O}$,

$$R^1f_*\mathbb{C} \otimes \mathcal{O}_\Delta \longrightarrow G|_\Delta,$$

sends them at the central point to the two classes in $H^1(W, \mathcal{O}_W)$. These form a basis by theorem 41.3. Thus $\gamma^{0,1}$ extends across the cusp and is nowhere zero everywhere. In particular $\mathcal{O}_B\gamma^{0,1} \subset G$ is a line subbundle.

Relative duality applies to the proper flat Gorenstein map $f\colon X$ is smooth, the base is a smooth curve, and $\omega_{X/B}$ is invertible. Using theorem 41.1, it gives

$$R^2f_*\mathcal{O}_X \simeq \mathfrak{K}^\vee \simeq \mathcal{O}_B(-1). \tag{9.15}$$

The relative theorem is the one of Ramis–Ruget–Verdier [5, pp. 261–283]; the absolute singular dualizing-complex comparison is independently recorded in [4, Theorems 1–2 and Proposition 2].

The relative cup product

$$\bigwedge^2 G \longrightarrow R^2f_*\mathcal{O}_X \tag{9.16}$$

is an isomorphism. It is so on every smooth torus fibre. At the cusp it is so by (9.13) and theorem 41.3. At p_j, choose on the cover an eigenframe v_1, v_2, with $v_1 = \gamma^{0,1}$ invariant. Its invariant exponent is $r_1 = 0$. The action on $v_1 \wedge v_2$ is dual to the Hodge character in (9.3), so its invariant generator is $s_j^{a_j}(v_1 \wedge v_2)$. The invariant generator for the second line is consequently $s_j^{a_j}v_2$, because $0 < a_j < m_j$. Wedge product sends $v_1 \wedge (s_j^{a_j}v_2)$ to exactly the invariant generator of the right side of (9.16), with no wrap modulo m_j.

Consequently $\det G \simeq R^2f_*\mathcal{O}_X \simeq \mathcal{O}_B(-1)$. The quotient of G by its trivial subbundle is $\mathcal{O}_B(-1)$, and the extension splits because

$$\operatorname{Ext}^1(\mathcal{O}_B(-1), \mathcal{O}_B) = H^1(B, \mathcal{O}_B(1)) = 0.$$

This proves (9.12) and identifies the trivial summand. □

Corollary 41.5 (the Picard parameter and fibrewise restriction)**.** ***[VERIFIED]*** *One has*

$$H^1(X, \mathcal{O}_X) = \mathbb{C}, \qquad \operatorname{Pic}(X) = \operatorname{Pic}^0(X) \simeq (\mathbb{C}, +). \tag{9.17}$$

For every scheme-theoretic fibre X_b, restriction induces an injective complex-linear map

$$\iota_b : H^1(X, \mathcal{O}_X) \longrightarrow H^1(X_b, \mathcal{O}_{X_b}). \tag{9.18}$$

On a smooth fibre its image is the complex line generated by the $(0,1)$-image of γ.

Proof. The Leray spectral sequence and $H^1(B, \mathcal{O}_B) = 0$ identify

$$H^1(X, \mathcal{O}_X) = H^0(B, R^1f_*\mathcal{O}_X) = \mathbb{C}$$

by theorem 41.4. Since the verified integral topology gives $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, the analytic exponential sequence

$$0 \to \mathbb{Z} \to \mathcal{O}_X \xrightarrow{u \mapsto e^{2\pi i u}} \mathcal{O}_X^* \to 1$$

identifies $H^1(X, \mathcal{O}_X)$ with $\operatorname{Pic}(X)$ and shows that every line bundle is topologically trivial. Under Leray, restriction is evaluation of a global section of $R^1f_*\mathcal{O}_X$, followed by base change. The only global section is a scalar multiple of the nowhere-zero section $\gamma^{0,1}$, so evaluation is injective at every point and has the stated image on a smooth fibre. □

41.4 One line bundle non-torsion on every fibre

Theorem 41.6 (uniform non-torsion)**.** ***[VERIFIED]*** *There is a line bundle $L \in \operatorname{Pic}(X)$ such that $L|_{X_b}$ is non-torsion for every scheme-theoretic fibre $X_b = f^*(b)$, and $L|_{S_j}$ is non-torsion for $j = 1, 2$. More precisely, identify $\operatorname{Pic}(X)$ with $\mathbb{C}$, $z \mapsto L_z$. Every z outside the union of $\mathbb{R}$ with a countable set has these properties.*

Proof. The pullback $f^*\mathcal{O}_B(1)$ is nontrivial: by the projection formula and $f_*\mathcal{O}_X = \mathcal{O}_B$,

$$h^0(X, f^*\mathcal{O}_B(1)) = h^0(B, \mathcal{O}_B(1)) = 2,$$

whereas $h^0(X, \mathcal{O}_X) = 1$. Rescale the additive coordinate in theorem 41.5 so that

$$L_{12} = f^*\mathcal{O}_B(1). \tag{9.19}$$

Fix a smooth fibre F_q. Put

$$\Lambda_q = \iota_q^{-1}\Big(\operatorname{im}[H^1(F_q; \mathbb{Z}) \to H^1(F_q, \mathcal{O}_{F_q})]\Big) \subset \mathbb{C}. \tag{9.20}$$

The integral image is a lattice in the underlying real vector space of $H^1(F_q, \mathcal{O}_{F_q})$, so Λ_q is discrete. It contains 12, because (9.19) is trivial on every fibre. The exponential sequence on F_q shows

$$L_z|_{F_q} \text{ is torsion} \quad \Longleftrightarrow \quad z \in \mathbb{Q}\Lambda_q. \tag{9.21}$$

If $\operatorname{rk}\Lambda_q = 1$, its nonzero element 12 forces $\Lambda_q \subset \mathbb{R}$; all its bad parameters are real.

Suppose $\operatorname{rk}\Lambda_q = 2$. By theorem 41.5, the image of ι_q is the line generated by the image of γ. Thus there is an integral class

$$\lambda = a\gamma + bu + cw + d\delta \notin \mathbb{Q}\gamma \tag{9.22}$$

whose $(0,1)$-image lies on that line. On the covering family, the four marked lattice columns are

$$\pi_{\widehat{\gamma}} = (6\mu, \beta), \quad \pi_{\widehat{u}} = (\tau, \mu), \quad \pi_{\widehat{w}} = (1, 0), \quad \pi_{\widehat{\delta}} = (0, 1). \tag{9.23}$$

Every real cohomology class has a unique expression

$$2\operatorname{Re}(\bar{\xi}\, d\zeta_1 + \bar{v}\, d\zeta_2).$$

Evaluating it on (9.23) gives

$$\begin{aligned} 2\operatorname{Re}\bar{\xi} &= c, & 2\operatorname{Re}\bar{v} &= d, \\ 2\operatorname{Re}(\bar{\xi}\tau + \bar{v}\mu) &= b, & 2\operatorname{Re}(6\bar{\xi}\mu + \bar{v}\beta) &= a. \end{aligned} \tag{9.24}$$

For γ, the first two right sides vanish. The third equation then shows that its $(0,1)$-line is represented, up to a nonzero scalar, by $(\operatorname{Im}\mu, -\operatorname{Im}\tau)$; the fourth equation only fixes the scalar. Write $\bar{\xi} = c/2 + ir$, $\bar{v} = d/2 + is$. The condition that the class in (9.22) have $(0,1)$-image on the γ-line is

$$\bar{\xi}\operatorname{Im}\tau + \bar{v}\operatorname{Im}\mu = 0.$$

Its real part, together with the third equation of (9.24), is equivalent to the single holomorphic equation

$$c\tau + d\mu - b = 0. \tag{9.25}$$

Neither a nor β occurs, as adding a multiple of γ must not change the condition.

The functions $1, \tau, \mu$ are linearly independent. Applying the cusp transformation $(\tau, \mu) \mapsto (\tau - 1, \mu)$ to a constant linear relation first kills the coefficient of τ. If the remaining relation made μ constant,

the order-four law $\mu \circ g_2 = 1 + \mu/\tau$, with nonconstant τ, would force an impossible identity. Hence all coefficients vanish.

For every integral $\lambda \notin \mathbb{Q}\gamma$, equation (9.25) is therefore a nonzero holomorphic function on the covering Riemann surface. Its zero set is discrete and countable. There are countably many λ, so their union, and its image in B°, are countable. This image contains every q for which $\operatorname{rk} \Lambda_q = 2$. Each such fibre contributes the countable bad set $\mathbb{Q}\Lambda_q$; all rank-one fibres together contribute only parameters in $\mathbb{R}$. Thus the smooth fibres exclude only $\mathbb{R}$ together with a countable set.

There remain the three scheme-theoretic special fibres. By theorems 41.4 and 41.5, restriction

$$\mathbb{C} \longrightarrow H^1(X_b, \mathcal{O}_{X_b})$$

is complex-linear and injective for each of them. The exponential sequence and countability of $H^1(X_b; \mathbb{Z})$ show that the parameters for which $L_z|_{X_b}$ is torsion form a countable set.

Finally consider the reductions S_j. The restriction $\varphi_j : \mathbb{C} \to H^1(S_j, \mathcal{O}_{S_j})$ is complex-linear. It is nonzero. Write $\mathcal{O}_X(S_j) = L_{z_j}$. Since

$$\mathcal{O}_X(m_j S_j) = f^*\mathcal{O}_B(1) = L_{12},$$

the additive parameter gives $z_j = 12/m_j$, while theorem 41.2 says that

$$L_{12/m_j}|_{S_j} = \mathcal{O}_X(S_j)|_{S_j}$$

is nontrivial. If φ_j were zero, the exponential sequence and the vanishing of the restricted first Chern class would make every $L_z|_{S_j}$ trivial, a contradiction. Thus φ_j is injective, and the same exponential-sequence argument makes its torsion parameters countable. Removing these two additional countable sets proves the theorem. □

Remark 41.7 (provenance of the repair). The period equation $c\tau + d\mu - b = 0$ and the countability strategy are the candidate's. The exact normal-character proof above makes the order of $\mathcal{O}_X(S_j)|_{S_j}$ explicit, while the cusp extension in theorem 41.4 replaces the candidate's collapse-dependent argument by the independently verified intrinsic nearby-cycle map. Identifying the restriction line with the global $\gamma^{0,1}$-summand also removes the candidate's separate continuity argument for a moving integral lift.

41.5 A local analytic obstruction to a cusp-only Picard argument

The relative Picard space of the cusp family cannot by itself force the normalised central restriction to vanish. The obstruction is visible directly in the toric model of theorem 8.4. We give the complete factor-of-automorphy calculation, including the descent cocycle and the sign of the generic Chern class.

Retain the cusp notation of theorems 8.2 and 8.4:

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad C(t) = \begin{pmatrix} 6\mu(t) & h(t) \\ b(t) - h(t) & \mu(t) \end{pmatrix}, \qquad t = e^{2\pi i s}, \tag{5.17t}$$

$$c_\lambda(t) = \exp(2\pi i C(t)\lambda), \qquad \exp(2\pi i Z(s)\lambda) = t^{B_0\lambda} c_\lambda(t) \quad (\lambda \in \bar{\Lambda} \simeq \mathbb{Z}^2), \tag{5.17u}$$

and

$$\Phi_\lambda(x,t) = (t^{B_0\lambda}x, t), \qquad \Psi_\lambda(p) = (c_\lambda(t(p)), 1) \cdot \Phi_\lambda(p), \qquad \Psi_\lambda\Psi_\mu = \Psi_{\lambda+\mu}. \tag{5.17v}$$

Write E_v, $v \in \mathbb{Z}^2$, for the toric prime divisors of Y_ϵ, so that $E_0 \simeq \mathrm{dP}_6$, and let

$$\pi : Y_\epsilon \longrightarrow N_0 = Y_\epsilon/\bar{\Lambda}, \qquad \nu = \pi|_{E_0} : E_0 \longrightarrow W$$

be the quotient and the normalization of the central fibre. Put

$$N' = \mathbb{Z}^2 \oplus \mathbb{Z}, \qquad M' = \operatorname{Hom}(N', \mathbb{Z}).$$

For $m=(\ell,k)\in M'=\mathbb{Z}^2\oplus\mathbb{Z}$, fix the character convention

$$\chi^m(x,t)=x^\ell t^k,\qquad \operatorname{div}(\chi^m)=\sum_{v\in\mathbb{Z}^2}\langle m,(v,1)\rangle E_v. \tag{5.17w}$$

Proposition 41.8 (an explicit nonzero local cusp restriction)**.** ***[VERIFIED]*** *Let*

$$H=\begin{pmatrix}1&0\\0&-6\end{pmatrix},\qquad P=B_0^{\mathsf t}H=\begin{pmatrix}0&6\\1&0\end{pmatrix},\qquad Q(v)=v^{\mathsf t}Hv=v_1^2-6v_2^2. \tag{5.17x}$$

Then

$$\mathcal{D}_Q=\sum_{v\in\mathbb{Z}^2}Q(v)E_v \tag{5.17y}$$

is a locally finite Cartier divisor on Y_ϵ. The line bundle $\mathcal{O}_{Y_\epsilon}(\mathcal{D}_Q)$ has an explicit $(\bar\Lambda,\Psi)$-equivariant structure and descends to a holomorphic line bundle $\mathcal{L}_Q$ on N_0. If

$$(a_1,a_2,a_3,a_4,a_5,a_6)=(e_1,e_2,e_2-e_1,-e_1,-e_2,e_1-e_2),\qquad C_i=E_0\cap E_{a_i},$$

then

$$\nu^*(\mathcal{L}_Q|_W)\simeq\mathcal{O}_{E_0}\Big(\sum_{i=1}^6 Q(a_i)C_i\Big),\qquad \big(\deg_{C_i}\nu^*(\mathcal{L}_Q|_W)\big)_{i=1}^6=(-12,2,0,-12,2,0). \tag{5.17z}$$

In particular $\nu^*(\mathcal{L}_Q|_W)$ is nontrivial.

On every nonzero fibre

$$F_t=\mathbb{C}^2/(Z(s)\mathbb{Z}^2+\mathbb{Z}^2),$$

the first Chern class of $\mathcal{L}_Q|_{F_t}$, in the ordered lattice basis $(\widehat\gamma,\widehat u,\widehat w,\widehat\delta)$ indexing the columns of $[Z\mid I_2]$, is exactly

$$c_1(\mathcal{L}_Q|_{F_t})=-2q,\qquad q=u\wedge w+6\gamma\wedge\delta. \tag{5.17aa}$$

Proof. Local finiteness and Cartier status. The fan of Y consists of the cones over the cells of the locally finite A_2-triangulation at height one. If $U_\sigma\simeq\mathbb{C}^3$ is the affine chart belonging to a triangle $\sigma=\{v_0,v_1,v_2\}$, then U_σ meets precisely E_{v_0},E_{v_1},E_{v_2}. Indeed, by the orbit–cone correspondence, $U_\sigma\cap E_v\neq\varnothing$ exactly when $\mathbb{R}_{\ge0}(v,1)$ is a face of the cone over σ. Thus $\{E_v\}_{v\in\mathbb{Z}^2}$ is locally finite and $\mathcal{D}_Q$ is a well-defined locally finite Weil divisor.

Every triangle of the fan is unimodular, as proved in theorem 8.4. If z_0,z_1,z_2 are the characters dual to $(v_0,1),(v_1,1),(v_2,1)$, then

$$\mathcal{D}_Q|_{U_\sigma}=\operatorname{div}\Big(z_0^{Q(v_0)}z_1^{Q(v_1)}z_2^{Q(v_2)}\Big). \tag{5.17ab}$$

Negative exponents merely make the displayed monomial meromorphic, as permitted for a local equation of a Cartier divisor. Hence $\mathcal{D}_Q$ is Cartier on every chart and therefore on Y_ϵ.

The signed divisor pullback identity. For $\lambda\in\bar\Lambda=\mathbb{Z}^2$, set

$$b_\lambda=B_0\lambda,\qquad \ell_\lambda=2Hb_\lambda,\qquad k_\lambda=Q(b_\lambda),\qquad m_\lambda=(\ell_\lambda,k_\lambda)\in M'. \tag{5.17ac}$$

The fan automorphism sends E_v to E_{v+b_λ}. Consequently $\Phi_\lambda^*E_w=E_{w-b_\lambda}$, and

$$\begin{aligned}\Phi_\lambda^*\mathcal{D}_Q-\mathcal{D}_Q&=\sum_{v\in\mathbb{Z}^2}\big(Q(v+b_\lambda)-Q(v)\big)E_v\\&=\sum_{v\in\mathbb{Z}^2}\big(2v^{\mathsf t}Hb_\lambda+b_\lambda^{\mathsf t}Hb_\lambda\big)E_v=\operatorname{div}(\chi^{m_\lambda}).\end{aligned}$$

The t-dependent horizontal torus translation $p \mapsto (c_\lambda(t(p)), 1) \cdot p$ preserves every torus-invariant prime divisor E_v. Hence

$$\boxed{\Psi_\lambda^* \mathcal{D}_Q = \mathcal{D}_Q + \operatorname{div}(\chi^{m_\lambda})}. \tag{5.17ad}$$

The complete Ψ-cocycle. The matrices in (5.17t) and (5.17x) give

$$PC(t) = \begin{pmatrix} 6(b(t) - h(t)) & 6\mu(t) \\ 6\mu(t) & h(t) \end{pmatrix}, \tag{5.17ae}$$

which is symmetric. Thus

$$A(t) := 2PC(t) \tag{5.17af}$$

is a symmetric holomorphic matrix on the full cusp disc. Define the holomorphic units h_λ and meromorphic functions f_λ by

$$h_\lambda(t) = \exp\bigl(\pi i \lambda^{\mathsf{t}} A(t) \lambda\bigr), \qquad f_\lambda = h_\lambda(t) \chi^{m_\lambda}. \tag{5.17ag}$$

Equation (5.17ad) gives

$$\operatorname{div}(f_\lambda) = \Psi_\lambda^* \mathcal{D}_Q - \mathcal{D}_Q. \tag{5.17ah}$$

The cocharacter map underlying Φ_μ is

$$\varphi_\mu(y, y_3) = (y + y_3 b_\mu, y_3),$$

so its pullback on characters is

$$\varphi_\mu^*(\ell, k) = (\ell, k + \ell^{\mathsf{t}} b_\mu). \tag{5.17ai}$$

The quadratic identity for Q gives

$$m_{\lambda+\mu} = m_\mu + \varphi_\mu^* m_\lambda. \tag{5.17aj}$$

Indeed, the first coordinates add, while the last coordinate on the right is

$$Q(b_\mu) + Q(b_\lambda) + 2 b_\lambda^{\mathsf{t}} H b_\mu = Q(b_\lambda + b_\mu).$$

Using (5.17u), (5.17v), and (5.17ai), one obtains

$$\begin{aligned} \Psi_\mu^* \chi^{m_\lambda} &= \chi^{m_\lambda}(c_\mu(t), 1) \chi^{\varphi_\mu^* m_\lambda} \\ &= \exp\bigl(2\pi i \ell_\lambda^{\mathsf{t}} C(t) \mu\bigr)\, \chi^{\varphi_\mu^* m_\lambda} \\ &= \exp\bigl(2\pi i \lambda^{\mathsf{t}} A(t) \mu\bigr)\, \chi^{\varphi_\mu^* m_\lambda}. \end{aligned} \tag{5.17ak}$$

The last equality is exact because

$$\ell_\lambda^{\mathsf{t}} C \mu = (2 H B_0 \lambda)^{\mathsf{t}} C \mu = 2 \lambda^{\mathsf{t}} B_0^{\mathsf{t}} H C \mu = 2 \lambda^{\mathsf{t}} P C \mu = \lambda^{\mathsf{t}} A \mu.$$

Since $A = A^{\mathsf{t}}$, (5.17ag), (5.17aj), and (5.17ak) imply

$$\begin{aligned} f_\mu\, \Psi_\mu^* f_\lambda &= h_\mu h_\lambda \exp(2\pi i \lambda^{\mathsf{t}} A \mu) \chi^{m_\mu + \varphi_\mu^* m_\lambda} \\ &= \exp\bigl(\pi i (\lambda + \mu)^{\mathsf{t}} A (\lambda + \mu)\bigr)\, \chi^{m_{\lambda+\mu}} = f_{\lambda+\mu}. \end{aligned} \tag{5.17al}$$

Use the divisor-sheaf convention

$$\mathcal{O}(D)(U) = \{ g \in \mathcal{M}_Y(U) : \operatorname{div}(g) + D|_U \geq 0 \}.$$

If $D' = D + \operatorname{div}(f)$, multiplication by f maps $\mathcal{O}(D')$ isomorphically to $\mathcal{O}(D)$. Thus (5.17ad) and (5.17ag) define

$$\theta_\lambda : \Psi_\lambda^* \mathcal{O}(\mathcal{D}_Q) \xrightarrow{\times f_\lambda} \mathcal{O}(\mathcal{D}_Q). \tag{5.17am}$$

Because $(\Psi_\lambda\Psi_\mu)^* = \Psi_\mu^*\Psi_\lambda^*$, equation (5.17al) is exactly

$$\theta_{\lambda+\mu} = \theta_\mu \circ \Psi_\mu^*\theta_\lambda. \tag{5.17an}$$

This proves a complete equivariance datum, not merely invariance of the divisor class.

Analytic descent. By theorem 8.4, the action on Y_ϵ is free and properly discontinuous. Every point therefore has an open neighbourhood U whose distinct translates are disjoint and on which $\pi|_U$ is a biholomorphism onto $\pi(U)$. Transport $\mathcal{O}(\mathcal{D}_Q)|_U$ to $\pi(U)$, and on overlaps arising from translates glue by (5.17am). Equation (5.17an) is precisely the triple-overlap condition. The resulting holomorphic line bundle $\mathcal{L}_Q$ on N_0 satisfies

$$\pi^*\mathcal{L}_Q \simeq \mathcal{O}_{Y_\epsilon}(\mathcal{D}_Q). \tag{5.17ao}$$

The normalized central restriction. Let $j_0 : E_0 \hookrightarrow Y_\epsilon$. Since $\nu = \pi \circ j_0$, (5.17ao) gives

$$\nu^*(\mathcal{L}_Q|_W) = j_0^*\pi^*\mathcal{L}_Q \simeq \mathcal{O}_{E_0}(\mathcal{D}_Q|_{E_0}). \tag{5.17ap}$$

The only E_v, $v \neq 0$, meeting E_0 are those indexed by the six a_i in the statement, and the intersections are the six toric (-1)-curves C_i. Since $Q(0) = 0$, there is no $E_0|_{E_0}$ term, so

$$\mathcal{D}_Q|_{E_0} = \sum_{i=1}^{6} Q(a_i)C_i, \qquad (Q(a_1), \dots, Q(a_6)) = (1, -6, -5, 1, -6, -5). \tag{5.17aq}$$

The hexagonal intersection table is

$$C_i^2 = -1, \qquad C_i \cdot C_{i\pm1} = 1, \qquad C_i \cdot C_j = 0 \quad (j \notin \{i, i \pm 1\}), \tag{5.17ar}$$

with indices modulo six. Therefore

$$\deg_{C_i} \mathcal{O}_{E_0}\left(\sum_j Q(a_j)C_j\right) = -Q(a_i) + Q(a_{i-1}) + Q(a_{i+1}),$$

and direct substitution gives

i	1	2	3	4	5	6
$-Q(a_i) + Q(a_{i-1}) + Q(a_{i+1})$	-12	2	0	-12	2	0.

(5.17as)

The degree on $C_1 = C_{e_1}$ is -12, proving nontriviality.

The generic-fibre factor and its exact Chern class. Fix $t \neq 0$, choose s with $t = e^{2\pi i s}$, and write $x = e^{2\pi i\zeta}$, $\zeta \in \mathbb{C}^2$. Equation (5.17u) identifies the deck transformation indexed by $\lambda \in \mathbb{Z}^2$ with

$$\zeta \longmapsto \zeta + Z(s)\lambda$$

modulo the horizontal translations $\zeta \mapsto \zeta + n$, $n \in \mathbb{Z}^2$. The matrix identities

$$HB_0 = P^{\mathsf{t}}, \qquad B_0^{\mathsf{t}}HB_0 = PB_0 \tag{5.17at}$$

give

$$\chi^{m_\lambda}(x,t) = x^{2P^{\mathsf{t}}\lambda}t^{\lambda^{\mathsf{t}}PB_0\lambda}, \qquad h_\lambda(t) = \exp(2\pi i\lambda^{\mathsf{t}}PC(t)\lambda).$$

Since $Z = sB_0 + C$, it follows that

$$f_\lambda(\zeta) = \exp\!\left(2\pi i\lambda^{\mathsf{t}}PZ\lambda + 4\pi i\lambda^{\mathsf{t}}P\zeta\right). \tag{5.17au}$$

The matrix $PZ = sPB_0 + PC$ is symmetric:

$$PB_0 = \operatorname{diag}(-6, 1),$$

and PC is symmetric by (5.17ae).

Under the descent convention (5.17am), a pulled-back local section written as $\varphi(\zeta)$ in the canonical frame satisfies

$$\varphi(\zeta + Z\lambda) = f_\lambda(\zeta)^{-1}\varphi(\zeta).$$

Thus the factor of automorphy for the lattice translation $\zeta \mapsto \zeta + Z\lambda + n$ is

$$a_{(\lambda,n)}(\zeta) = \exp(2\pi i g_{(\lambda,n)}(\zeta)), \qquad g_{(\lambda,n)}(\zeta) = -\lambda^{\mathsf{t}} PZ\lambda - 2\lambda^{\mathsf{t}} P\zeta. \tag{5.17av}$$

This is independent of n as a factor of automorphy: replacing ζ by $\zeta + n$ changes g by the integer $-2\lambda^{\mathsf{t}} Pn$.

For $l = (\lambda, n)$, $l' = (\mu, m)$, the exponential connecting cocycle is

$$\begin{aligned} c(l, l') &= g_l(\zeta + Z\mu + m) + g_{l'}(\zeta) - g_{l+l'}(\zeta) \\ &= -2\lambda^{\mathsf{t}} Pm \in \mathbb{Z}, \end{aligned} \tag{5.17aw}$$

where the symmetry of PZ cancels the two mixed Z-terms. The corresponding alternating integral form is

$$E(l, l') = c(l, l') - c(l', l) = -2\lambda^{\mathsf{t}} Pm + 2\mu^{\mathsf{t}} Pn. \tag{5.17ax}$$

This is $c_1(\mathcal{L}_Q|_{F_t})$. In the ordered column basis of $[Z \mid I_2]$, the cross-block of the class $q = u \wedge w + 6\gamma \wedge \delta$ from $(\widehat{\gamma}, \widehat{u})$ to $(\widehat{w}, \widehat{\delta})$ is

$$\begin{pmatrix} q(\widehat{\gamma}, \widehat{w}) & q(\widehat{\gamma}, \widehat{\delta}) \\ q(\widehat{u}, \widehat{w}) & q(\widehat{u}, \widehat{\delta}) \end{pmatrix} = \begin{pmatrix} 0 & 6 \\ 1 & 0 \end{pmatrix} = P. \tag{5.17ay}$$

Hence (5.17ax) is exactly $-2q$, as asserted. □

Remark 41.9 (the exact obstruction to the purely analytic route). The proposition proves that the local homomorphism

$$\mathrm{Pic}(N_0) \longrightarrow \mathrm{Pic}(E_0), \qquad \mathcal{L} \longmapsto \nu^*(\mathcal{L}|_W),$$

is not zero. Thus the toric cusp geometry, conductor descent on W, or local relative-Picard representability cannot alone prove the desired global vanishing

$$\nu^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(E_0).$$

There is no contradiction with theorems 46.14 and 46.18: the local bundle above has nonzero generic class $-2q$, whereas the two global proofs use the order-three and order-four fillings and their common boundary relation to exclude every nonzero multiple of q.

More generally, Bingener proves that if a holomorphic map is proper, flat, and cohomologically flat in dimension zero, then its relative Picard functor is represented by a *not necessarily separated* complex space [37, Introduction, p. 318; Satz 6.2, pp. 336–337]. The same introduction explains that the exponential-sequence quotient works directly under topological local triviality and cohomological flatness in dimension one, but that passage to the quotient fails in general. The cusp map of theorem 8.4 is not topologically locally trivial across 0. Bingener's theorem therefore supplies representability but neither separatedness nor the component-specialization assertion needed here.

Consequently a purely analytic completion must add global compatibility with both finite fillings and must control the boundary support function; neither relative-Picard representability nor conductor descent supplies those statements. The later calculation in theorems 46.19 to 46.21 now supplies them: it first forces the generic Appell–Humbert form to vanish using the two finite fillings, and then proves that its mixed block is the polarization of the three normalized central degrees. Thus this proposition remains the sharp cusp-local obstruction, while the global purely analytic route is complete.

42 Hodge numbers, cotangent direct images, and the Frölicher page

This section first derives the one-parameter Hodge table and the complete Frölicher spectral sequence from the candidate-specific structure-sheaf calculation and the verified sphere topology. It then performs the missing logarithmic direct-image calculation through all three special fibres. The cyclic invariant lattices, logarithmic Kodaira–Spencer divisor, residue cancellation, and one remaining global scalar give $h^{1,2} = 1$ and $h^{1,1} = 2$; thus no Hodge number remains undetermined.

42.1 Vanishing of holomorphic forms

We first place the holomorphic-form calculation inside the workbench rather than using it as an unexpanded assertion from the reconstructed source.

Proposition 42.1 (candidate-specific holomorphic-form vanishing)**.** ***[VERIFIED]*** *For the constructed threefold,*

$$H^0(X, \Omega_X^p) = 0 \qquad (p = 1, 2, 3).$$

Proof. Over the marked cover of the smooth part, write the fibre coordinates as $\zeta = (\zeta_1, \zeta_2)$ and

$$\Pi = [Z \mid I_2], \qquad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}.$$

Translation by $\lambda \in \Lambda$ sends $d\zeta$ to $d\zeta + \dot{\Pi}\lambda\, dz$. Compactness of each torus fibre therefore puts every translation-invariant holomorphic form into one of the following normal forms:

$$\begin{aligned} &p = 1: \quad a(z)\, dz + c(z) \cdot d\zeta, && c(z)\dot{Z}(z) = 0, \\ &p = 2: \quad dz \wedge (b(z) \cdot d\zeta), \\ &p = 3: \quad c(z)\, dz \wedge d\zeta_1 \wedge d\zeta_2. \end{aligned}$$

Indeed, fibrewise periodic coefficients are constant. A possible affine part in a coefficient vanishes after restriction to the real span of the two constant columns of Π; in degree two this also kills the putative $d\zeta_1 \wedge d\zeta_2$ term because $\dot{\tau}$ is nonzero on the smooth base.

The finite fillings are obtained from smooth torus families by free cyclic quotients, so the displayed coefficients extend holomorphically across the two elliptic lifts. At the cusp use the verified toric charts $t = w_0 w_1 w_2$. Substitution of

$$d\zeta_i = (2\pi i)^{-1} dx_i/x_i, \qquad ds = (2\pi i)^{-1} dt/t$$

and restriction successively to the coordinate lines and planes shows that the coefficients of $d\zeta_i$, ds, $ds \wedge d\zeta_i$, and $ds \wedge d\zeta_1 \wedge d\zeta_2$ vanish to order at least one in t. Indeed, on every unimodular monomial chart the triple $(d\log x_1, d\log x_2, d\log t)$ is a constant invertible linear combination of the $d\log w_j$. A coefficient $F(t)$ multiplying a nonzero logarithmic residue is holomorphic only if its constant term vanishes: restricting to $w_j = 0$ gives that residue as $F(0)$, so $F(t) = tG(t)$. Invertibility of the monomial change of basis applies this argument to each displayed coefficient. Thus every coefficient used below descends with the asserted regularity at all three points of the base.

The linear part of a deck transformation has the form

$$R_g = \begin{pmatrix} r_g & 0 \\ * & 1 \end{pmatrix}, \qquad \jmath_g := r_g^{-1} = c_g\tau + d_g, \qquad g^*(d\tau) = \jmath_g^{-2} d\tau.$$

For a one-form, the second component c_2 is a holomorphic invariant function on the base and vanishes at the cusp, hence is zero. The second column of $c\dot{Z} = 0$ then gives $c_1\dot{\tau} = 0$, so $c_1 = 0$. What remains is a holomorphic one-form on $\mathbb{P}^1$, and hence vanishes.

For a two-form, $b_2\,dz$ descends to a holomorphic one-form on $\mathbb{P}^1$ and is zero. The remaining form $\psi = b_1\,dz$ satisfies $g^*\psi = \jmath_g\psi$. Consequently

$$\psi^{\otimes 2} \otimes d\tau$$

is invariant. At the order-three and order-four points it descends as a cubic differential with a pole of order at most two, while the toric cusp estimate gives a pole of order at most one. It is therefore a section of

$$(\Omega^1_{\mathbb{P}^1})^{\otimes 3}(2p_1 + 2p_2 + p_0) \simeq \mathcal{O}_{\mathbb{P}^1}(-1),$$

so it is zero. Since $\dot\tau$ is nonzero on a dense open set, $\psi = 0$. The coefficient of a three-form obeys the same weight-one law and the same argument applies. Hence all three spaces vanish. □

42.2 The exact one-parameter Hodge diamond

Theorem 42.2 (Hodge diamond determined up to one integer)**.** ***[VERIFIED]*** *Put $m = h^{1,2}(X)$. Then $m \in \mathbb{Z}_{\geq 0}$ and the only nonzero Hodge numbers are*

$$h^{0,0} = h^{0,1} = h^{3,2} = h^{3,3} = 1, \qquad h^{1,1} = h^{2,2} = m + 1, \qquad h^{1,2} = h^{2,1} = m.$$

Equivalently,

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	m	$m+1$	1
1	1	$m+1$	m	0
0	1	0	0	0

where the row index is q and the column index is p.

Proof. The independently reconstructed structure-sheaf direct images are

$$f_*\mathcal{O}_X = \mathcal{O}, \qquad R^1 f_*\mathcal{O}_X = \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^2 f_*\mathcal{O}_X = \mathcal{O}(-1)$$

by theorem 41.4. Coherent Leray on $\mathbb{P}^1$ therefore gives

$$(h^{0,0}, h^{0,1}, h^{0,2}, h^{0,3}) = (1, 1, 0, 0).$$

Together with theorem 42.1 and Serre duality [43, §13, Theorem 4, p. 22], this fixes every entry except the four middle entries, with $h^{1,2} = h^{2,1} = m$ and $h^{1,1} = h^{2,2}$.

The verified integral sphere topology gives $c_1(X) = c_2(X) = 0$ and $\int_X c_3(X) = 2$. In complex dimension three,

$$\operatorname{td}(X) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1c_2,$$
$$\operatorname{ch}(\Omega^1_X) = 3 - c_1 + \frac{1}{2}(c_1^2 - 2c_2) - \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3).$$

Hirzebruch–Riemann–Roch for arbitrary compact complex manifolds [42, Theorem 3, p. 427] hence gives

$$\chi(\Omega^1_X) = -\frac{1}{2}\int_X c_3 = -1.$$

The already known zero entries reduce this identity to $-h^{1,1} + h^{1,2} = -1$, proving $h^{1,1} = m + 1$. □

This specializes the general constraints of Gray and Ugarte; a convenient published synthesis is Angella [47], while the second-page symmetry and inequalities originate in [50]. The argument above uses the stronger candidate-specific direct images and form vanishings, rather than assuming a general Hodge pattern for a hypothetical complex S^6.

Theorem 42.3 (the complete Frölicher page)**.** ***[VERIFIED]*** *The Frölicher spectral sequence of X does not degenerate at E_1, but it does degenerate at E_2. Apart from the endpoint copies of $\mathbb{C}$, the two nonzero $d_1 = \partial$ rows are exact complexes*

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathbb{C}^{m+1} \longrightarrow \mathbb{C}^{m} \longrightarrow 0,$$

$$0 \longrightarrow \mathbb{C}^{m} \longrightarrow \mathbb{C}^{m+1} \longrightarrow \mathbb{C} \longrightarrow 0.$$

Their differential ranks are respectively $(1, m)$ and $(m, 1)$. Consequently

$$\dim_{\mathbb{C}} E_1 = 6 + 4m, \qquad \dim_{\mathbb{C}} E_2 = 2, \qquad E_2 = E_\infty.$$

Proof. Frölicher's original construction identifies E_1 with Dolbeault cohomology and E_∞ with the associated graded de Rham cohomology [41, p. 643, Lemmas 1–2 and Theorem 3]. The table in theorem 42.2 leaves only the displayed $q = 1$ and $q = 2$ row complexes between the endpoint classes. Since $H^k_{\mathrm{dR}}(X; \mathbb{C}) = 0$ for $1 \le k \le 5$, every middle $E_\infty^{p,q}$ is zero. There is only one a priori possible higher differential between middle positions, $d_2 : E_2^{0,2} \to E_2^{2,1}$, but already $E_1^{0,2} = H^2(X, \mathcal{O}_X) = 0$. Every other d_r with $r \ge 2$ has zero source or target by the same Hodge table. Thus $E_2 = E_\infty$, so the two rows must be exact. The ranks follow from their dimensions. □

The theorem explains why neither topology nor the Frölicher sequence alone can select a value of m: the two exact row complexes exist numerically for every $m \ge 0$. The singular-fibre calculation below supplies the missing analytic information.

42.3 Generic maximal variation and its elliptic zero

The most tempting shortcut to the remaining Hodge number is to treat the Kodaira–Spencer matrix as invertible everywhere. The following calculation shows precisely what is true and what is false.

Proposition 42.4 (determinant of the printed period derivative)**.** ***[VERIFIED]*** *With dots denoting differentiation in the coordinate on $\mathfrak{h}_z$, set*

$$K = \det(\dot{Z}) = 6\dot{\mu}^{\,2} - \dot{\tau}\dot{\beta}.$$

Then K is not identically zero, so the Kodaira–Spencer map has rank two on a dense open set. However $K(z_1) = 0$ at the chosen order-three elliptic point. In particular, the assertion $\dot{Z} \neq 0$ does not imply that $\dot{Z}$ is invertible, and direct images across the finite fibres require their actual cyclic invariant lattices.

Proof. At the cusp, theorem 8.2 gives

$$t = e^{2\pi i s}, \qquad Z = sB_0 + C(t), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

with C holomorphic at $t = 0$. Hence

$$\frac{dZ}{ds} = B_0 + 2\pi i\, tC'(t), \qquad \det\!\left(\frac{dZ}{ds}\right) = 1 + O(t).$$

The cusp coordinate s is locally biholomorphic on the marked cover, so K is nonzero near the punctured cusp and therefore is not identically zero.

Now put $\rho = e^{\pi i/3} = \tau(z_1)$. The difference between μ and the local affine solution

$$\mu_1 = \frac{2 - \tau}{3}$$

vanishes to order at least two at z_1: the homogeneous line has local generator of order two there. Since $\tau-\rho$ has a simple zero,

$$\dot\mu(z_1)=-\frac{1}{3}\dot\tau(z_1).$$

Write

$$\phi_1=2-\frac{6(1-\mu)^2}{\tau},\qquad \beta\circ g_1=\beta+\phi_1.$$

Using $\rho^2-\rho+1=0$ and the preceding derivative gives

$$\dot\phi_1(z_1)=-\frac{2}{3}(1+\rho)\dot\tau(z_1).$$

The derivative of the linearized order-three action at its fixed point is $g_1'(z_1)=-\rho$. Differentiating the transformation law for β therefore yields

$$-(1+\rho)\dot\beta(z_1)=\dot\phi_1(z_1),\qquad \dot\beta(z_1)=\frac{2}{3}\dot\tau(z_1).$$

Substitution gives

$$K(z_1)=6\left(-\frac{1}{3}\dot\tau(z_1)\right)^2-\dot\tau(z_1)\left(\frac{2}{3}\dot\tau(z_1)\right)=0.$$

□

The zero divisor of K cannot be read from the single equality $K(z_1)=0$. It is determined below only after the invariant lattices at the two multiple fibres have been inserted. This distinction is essential: direct image does not commute with taking the dual lattice at a ramified coarse fibre.

42.4 Logarithmic cotangent direct images

Put

$$B=\mathbb{P}^1,\qquad \Sigma=p_0+p_1+p_2,\qquad D=f^{-1}(\Sigma)_{\mathrm{red}}=W+S_1+S_2,$$

and set

$$\mathcal{L}=\Omega^1_B(\log\Sigma)\simeq\mathcal{O}_B(1),\qquad \mathcal{A}=\Omega^1_X(\log D),\qquad \mathcal{E}=\mathcal{A}/f^*\mathcal{L}. \tag{11.1}$$

At the cusp the local equations $t=z_1\cdots z_r$, $1\le r\le 3$, show directly that $\mathcal{A}$ and $\mathcal{E}$ are locally free. At a finite fibre the covering family has divisor $s=0$, the cyclic action is free on the total space, and $dt/t=m\,ds/s$; hence its locally free logarithmic sequence descends through the finite ètale quotient. Thus everywhere one has the logarithmic conormal sequence

$$0\longrightarrow f^*\mathcal{L}\longrightarrow\mathcal{A}\longrightarrow\mathcal{E}\longrightarrow 0. \tag{11.2}$$

The local $d\log$ bases and their relative form, including $t=\prod z_i^{e_i}$ and analytic base change, are the logarithmic conormal formalism of [40, Chapter II, §3.2–3.3, pp. 72–75]; the display above also follows directly from the stated local equations.

Proposition 42.5 (the three logarithmic relative direct images)**.** ***[VERIFIED]*** *One has*

$$\begin{aligned} f_*\mathcal{E}&\simeq\mathcal{O}\oplus\mathcal{O}(-1),\\ R^1f_*\mathcal{E}&\simeq\mathcal{O}^{\oplus 2}\oplus\mathcal{O}(-1)^{\oplus 2},\\ R^2f_*\mathcal{E}&\simeq\mathcal{O}\oplus\mathcal{O}(-1).\end{aligned}$$

All three sheaves are locally free and commute with base change.

Proof. At the cusp, the toric coordinates give

$$\mathcal{E}|_{N_0} = \mathcal{O}_{N_0}[d\log x_1] \oplus \mathcal{O}_{N_0}[d\log x_2]. \tag{11.3}$$

Indeed a quotient shear changes each $d\log x_i$ only by a holomorphic multiple of $d\log t$, which is zero in $\mathcal{E}$. The already proved structure-sheaf calculation therefore gives free local direct images of ranks $2, 4, 2$ at p_0.

We now keep the ramification at p_j. Write

$$A = \mathbb{C}\{t\}, \qquad C = \mathbb{C}\{s\}, \qquad t = s^m, \qquad (m, a) = (3, 2) \text{ or } (4, 1).$$

Cartan's analytic quotient theorem identifies local holomorphic functions on the quotient with invariants [38, §1, Theorem 1, pp. 1–2]. The action on the total space is free, so the quotient map is finite étale; the corresponding locally free sheaves and their equivariant structures descend by finite étale descent. This sheaf statement is not being attributed to Cartan's theorem on invariant functions. Finite pushforward is exact, and Reynolds averaging $v \mapsto |G_j|^{-1} \sum_{g \in G_j} gv$ makes the invariant functor exact in characteristic zero. Hence

$$R^q f_* \mathcal{E}|_{D_j} = \left(\pi_* R^q \widetilde{f}_* \widetilde{\mathcal{E}}\right)^{G_j}. \tag{11.3a}$$

Choose equivariant frames e_0, e_1 of the relative cotangent and b_0, b_1 of $R^1 \widetilde{f}_* \mathcal{O}$ whose semilinear characters are respectively

$$(1, \zeta^a), \qquad (1, \zeta^{-a});$$

b_0 is the nowhere-zero class $\gamma^{0,1}$. These may be chosen as holomorphic constant-character frames, rather than merely as eigenvectors on the central fibre. Indeed, if w lifts a ν-eigenvector at $s = 0$, then

$$P_\nu(w) = \frac{1}{m} \sum_{k=0}^{m-1} \nu^{-k} (g^*)^k w$$

satisfies $g^* P_\nu(w) = \nu P_\nu(w)$ and has the same nonzero central value; after shrinking the disc, the projected eigenvectors form a frame. On the marked torus family the projection formula gives

$$R^q \widetilde{f}_* \widetilde{\mathcal{E}} \simeq (\mathcal{O}e_0 \oplus \mathcal{O}e_1) \otimes \bigwedge^q (\mathcal{O}b_0 \oplus \mathcal{O}b_1), \qquad 0 \le q \le 2.$$

Solving $\nu\zeta^r = 1$ for the invariant exponent gives the following *actual* A-bases:

$$\begin{aligned} f_*\mathcal{E} :&\quad e_0,\ s^{m-a}e_1, \\ R^1 f_*\mathcal{E} :&\quad e_0 b_0,\ s^a e_0 b_1,\ s^{m-a} e_1 b_0,\ e_1 b_1, \\ R^2 f_*\mathcal{E} :&\quad s^a e_0 (b_0 \wedge b_1),\ e_1 (b_0 \wedge b_1). \end{aligned} \tag{11.4}$$

For comparison, the variable tangent lattice is generated by $s^a v_1$. Thus evaluation of the descended variable cotangent on the descended variable tangent is $s^m = t$, not a unit. This proves explicitly why

$$f_*\mathcal{E} \longrightarrow (f_* T_{X/B})^\vee$$

is multiplication by $\mathrm{diag}(1, t)$ locally, although $\mathcal{E}^\vee \simeq T_{X/B}$ before direct image. This last identity is an identity of locally free relative sheaves, not an assertion that formation of the kernel commutes with taking a tangent-space fibre. For example, at a triple chart

$$R = \mathbb{C}\{x, y, z\}, \qquad t = xyz,$$

one has

$$T_{X/B} = \ker\left[R^3 \xrightarrow{(yz, xz, xy)} R\right].$$

If $yz\,u + xz\,v + xy\,w = 0$, reduction modulo x and y gives $u = xU$ and $v = yV$, and cancellation in the domain gives $w = -z(U + V)$. Hence

$$T_{X/B} = R(x\partial_x - z\partial_z) \oplus R(y\partial_y - z\partial_z) = \mathcal{E}^\vee.$$

The kernel of the pointwise linear map df_x at the triple point is three-dimensional, whereas the fibre of this sheaf kernel is two-dimensional: tensoring the kernel sequence with the residue field is not left exact because the image ideal (xy, xz, yz) is not flat. Thus a section can be nowhere zero in $T_{X/B} = \mathcal{E}^\vee$ even when its image in the ambient tangent-space fibre vanishes.

The invariant vertical field ξ gives a trivial subline of $V := f_*T_{X/B}$; the quotient has the reciprocal automorphy of the Hodge line $\mathfrak{K} \simeq \mathcal{O}(1)$ from theorem 41.1. Since $H^1(\mathcal{O}(1)) = 0$, the resulting extension splits:

$$V \simeq \mathcal{O}\xi \oplus \mathcal{O}(-1), \qquad V^\vee \simeq \mathcal{O} \oplus \mathcal{O}(1).$$

Evaluation gives a generically invertible map $f_*\mathcal{E} \to V^\vee$. Formula (11.4) shows that its elementary divisors are $(1, t)$ at each of p_1, p_2 and $(1, 1)$ elsewhere. Thus its cokernel has total length two and $\deg(f_*\mathcal{E}) = 1 - 2 = -1$. Evaluation on ξ is stalkwise surjective: the covector dual to the fixed eigenvector is a unit generator at the cusp and in both finite invariant bases. Its kernel is consequently a line bundle of degree -1, and

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow f_*\mathcal{E} \xrightarrow{\mathrm{ev}_\xi} \mathcal{O} \longrightarrow 0.$$

Because $H^1(\mathcal{O}(-1)) = 0$, this extension also splits, proving

$$f_*\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

Also, by theorem 41.4,

$$G := R^1 f_*\mathcal{O}_X = \mathcal{O} \oplus \mathcal{O}(-1), \qquad R^2 f_*\mathcal{O}_X = \det G = \mathcal{O}(-1). \tag{11.5}$$

Here is the global filtration implicit in those bases. Choose the degree-zero summand in the preceding split sequence so that its generator evaluates to one on ξ; locally this is the fixed eigen-covector e_0. Together with the fixed class $b_0 = \gamma^{0,1}$, take the corresponding saturated sublines in $f_*\mathcal{E}$ and G. Formula (11.3) at the cusp and the bases (11.4) at the finite points show that these sublines are locally generated by e_0, b_0 and that their quotients are locally generated by $s^{m-a}e_1, s^a b_1$; hence both inclusions and both quotients are locally free. The evaluation splitting and Hodge-line calculation identify the two filtrations as

$$0 \subset \mathcal{O} \subset f_*\mathcal{E}, \qquad 0 \subset \mathcal{O} \subset G, \qquad \mathrm{gr}(f_*\mathcal{E}) = \mathrm{gr}(G) = \mathcal{O} \oplus \mathcal{O}(-1).$$

Over the smooth locus the cup-product identification is $R^1 f_*\mathcal{E} = (f_*\mathcal{E}) \otimes G$. Formula (11.4) extends its four character lines as saturated rank-one subsheaves over all of B. Three associated-graded lines are generated locally by $e_0 b_0, s^a e_0 b_1, s^{m-a} e_1 b_0$ and have degrees $0, -1, -1$. On the fourth, the separately descended factors would give

$$(s^{m-a}e_1)(s^a b_1) = t(e_1 b_1),$$

whereas $e_1 b_1$ itself is invariant. Thus that graded line is the upper modification of $\mathcal{O}(-2)$ at $p_1 + p_2$, hence is $\mathcal{O}$. Similarly, the two graded lines of $R^2 f_*\mathcal{E}$ are $\mathcal{O}(-1)$ and $\mathcal{O}(-2)(p_1 + p_2) = \mathcal{O}$. Every extension between the resulting degree-zero and degree-minus-one lines splits: its Ext^1 is one of $H^1(\mathcal{O}(-1))$, $H^1(\mathcal{O})$, or $H^1(\mathcal{O}(1))$, all zero. This proves the three asserted splittings. This is also the elementary filtered form of the Birkhoff–Grothendieck classification [45, Theorem 2.1, p. 126]; the divisor modifications and Ext^1 identifications may equivalently be read in [36, Tags 0C6K, 0B3A, and 01XT].

Finally, (11.3a)–(11.4) make every direct image free of ranks $2, 4, 2$ at the finite points; (11.3) does the same at the cusp, and the smooth locus is standard. The multiple-fibre base change can also be read

directly from the invariant modules. The Reynolds idempotent is A-linear, so invariants commute with arbitrary finite A-base change. For the scheme-theoretic special fibre,

$$C \otimes_A A/(t) = C/(s^m), \qquad \big((C/(s^m))v\big)^G = \mathbb{C}\, s^r v$$

on a character line whose invariant exponent satisfies $0 \le r < m$. Every basis vector in (11.4) therefore survives reduction modulo t, and the special-fibre dimensions are exactly $2, 4, 2$. No R^q with $q > 2$ occurs. Descending induction in analytic cohomology and base change then makes every base-change map an isomorphism, including for the nonreduced finite fibres; precisely the same criterion, with citation, is recorded in the proof of theorem 41.4; see in particular [6, Chapter III, §3, Theorem 3.4 and Corollary 3.10]. □

Proposition 42.6 (the logarithmic Kodaira–Spencer divisor)**.** ***[VERIFIED]*** *The connecting maps of (11.2) are*

$$\kappa_0 : f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^1 f_* \mathcal{O}_X,$$
$$\kappa_1 : R^1 f_*\mathcal{E} \longrightarrow \mathcal{L} \otimes R^2 f_* \mathcal{O}_X \simeq \mathcal{O}.$$

Then

$$\operatorname{div}(\det \kappa_0) = p_1 + p_2, \qquad \kappa_1 \text{ is surjective as a sheaf.} \tag{11.6}$$

In particular κ_0 is injective and $\operatorname{coker} \kappa_0 = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$*. Moreover the raw period determinant has*

$$\operatorname{ord}_{z_1} K = 2, \qquad \operatorname{ord}_{z_2} K = 0. \tag{11.7}$$

Proof. On the smooth locus, the connecting construction and its exterior-product Leibniz rule are the Gauss–Manin/Kodaira–Spencer formalism of [46, §2, equations (5)–(10), pp. 202–204]. Katz–Oda work with a smooth algebraic morphism; the citation is not being used as a logarithmic analytic special-fibre theorem. In the present family, Deligne's local monomial calculation supplies the locally split logarithmic sequence (11.2) [40, Chapter II, §3.3]. Taking a local splitting and a Čech representative, the connecting cochain is the difference of the two splittings. Since exterior product of cochains obeys

$$\delta(\alpha \wedge \beta) = \delta\alpha \wedge \beta + (-1)^{\deg \alpha} \alpha \wedge \delta\beta,$$

the two displayed connecting maps extend across every logarithmic chart and satisfy the same Leibniz rule. Thus the singular-fibre formula below is a direct calculation from (11.2), not an extrapolation of the Katz–Oda hypotheses. The determinant is a section of

$$\det(\mathcal{L} \otimes G) \otimes \det(f_*\mathcal{E})^{-1} = \mathcal{L}^{\otimes 2} \otimes \det G \otimes \det(f_*\mathcal{E})^{-1} \simeq \mathcal{O}(2) \otimes \mathcal{O}(-1) \otimes \mathcal{O}(1) \simeq \mathcal{O}(2). \tag{11.8}$$

At the cusp, $Z = sB_0 + C(t)$ and $dZ/ds = B_0 + O(t)$, so it is a unit. At a finite point use $\ell = dt/t = m\, ds/s$. In invariant determinant frames with no normalization imposed against the raw marked frames, direct calculation gives

$$\det \kappa_0 = \varepsilon(t) m^{-2} s^{m+2-2a} K(s), \qquad \varepsilon(0) \neq 0. \tag{11.9}$$

Here ε is the nowhere-zero invariant holomorphic unit recording the two frame normalizations; compatible eigenframes may be normalized so that $\varepsilon = 1$. The dot in K means d/ds in the chosen linearizing coordinate. Replacing s by another such coordinate changes only the unit, so every divisor and order calculation below is coordinate-independent. Equivariance gives

$$\operatorname{ord}_s K \equiv 2a - 2 \pmod{m}. \tag{11.10}$$

At $(m, a) = (3, 2)$, theorem 42.4 supplies $K(z_1) = 0$; hence the right side of (11.9) has coarse order at least one. At $(m, a) = (4, 1)$, the prefactor is $s^4 = t$, so the coarse order is again at least one. The two forced

zeros exhaust the degree in (11.8). They are therefore simple, there are no others, and (11.7) follows from (11.9)–(11.10).

For completeness, we derive the coarse matrices rather than only recording their shape. Let

$$D_{\mathrm{KS}}(s) = (d_{rs}(s))_{0\le r,s\le 1}$$

be the Kodaira–Spencer matrix on the marked cover after the common factor ds has been removed. Since $t = s^m$ and $ds = (s/m)\ell$, passage from the marked frames to the actual invariant frames in (11.4) gives

$$A = \frac{s}{m}d_{00}, \qquad B = \frac{s^{1-a}}{m}d_{01}, \qquad C = \frac{s^{1-a}}{m}d_{10}, \qquad D_0 = \frac{s^{m+1-2a}}{m}d_{11}. \tag{11.10a}$$

The character of each expression in (11.10a) is trivial, so each is a holomorphic function of t. In the bases $(e_0, s^{m-a}e_1)$ and $(b_0, s^a b_1)$ one consequently has the exact matrix

$$[\kappa_0] = \begin{pmatrix} A & tB \\ C & D_0 \end{pmatrix}. \tag{11.10b}$$

Its determinant is (11.9). Applying the exterior-product Leibniz rule to the four ordered generators

$$e_0 b_0, \quad s^a e_0 b_1, \quad s^{m-a} e_1 b_0, \quad e_1 b_1$$

now gives, without a change of basis,

$$[\kappa_1] = (-C,\ A,\ -D_0,\ B) \tag{11.11}$$

into the invariant generator $s^a(b_0 \wedge b_1)\ell$. If (11.11) were not surjective at a finite point, all four entries would lie in (t) and $\det \kappa_0$ would have order at least two, contrary to (11.6). Away from p_1, p_2, the determinant divisor (11.6) makes κ_0 invertible, so the exterior-product row (11.11) contains a unit and is surjective there as well. At the cusp this reduction is not an additional choice. Use the logarithmic base frame $\ell = ds = (2\pi i)^{-1}dt/t$, the relative cotangent frame $(e_0, e_1) = ([d\log x_1], [d\log x_2])$ of (11.3), and the quotient frame (γ, u) of G, which extends across the cusp by theorem 41.4. In these frames the smooth-locus Kodaira–Spencer matrix is the transpose of the derivative of the period matrix: for $h = ae_0 + be_1$,

$$\kappa_0(h) = (6a\dot{\mu} + b\dot{\beta})\gamma + (a\dot{\tau} + b\dot{\mu})u. \tag{11.11a}$$

Here the dot is d/ds. The cusp expansion $Z = sB_0 + C(t)$, $t = e^{2\pi i s}$, gives

$$\frac{dZ}{ds} = B_0 + 2\pi i\, tC'(t), \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

These logarithmic frames extend over $t = 0$, so (11.11a) reduces exactly to B_0^{t}. Write the cusp matrix, separately from the finite-fibre normalization above, as $[\kappa_0]_{p_0} = \left(\begin{smallmatrix} A & E \\ C & D \end{smallmatrix}\right)$. In the same ordered exterior basis used for (11.11), its Leibniz row is $(-C, A, -D, E)$. Substituting $(A, E, C, D) = (0, -1, 1, 0)$ gives the second reduction. Thus, in the named frames,

$$[\kappa_0] \equiv \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \pmod{t}, \qquad [\kappa_1] \equiv (-1, 0, 0, -1) \pmod{t},$$

exactly. Under arbitrary holomorphic changes of source and target frames the matrices change by invertible left and right factors; in particular invertibility of κ_0 and the existence of a unit entry in the row of κ_1 are unchanged. Thus κ_1 is surjective everywhere. □

42.5 Residues and the absolute cotangent direct images

Let $i_D : D \hookrightarrow X$, and let

$$\nu : \widetilde{D} = \mathrm{dP}_6 \sqcup S_1 \sqcup S_2 \longrightarrow D$$

be the normalization. In formulas below, $\nu_*\mathcal{O}_{\widetilde{D}}$ abbreviates $(i_D \circ \nu)_*\mathcal{O}_{\widetilde{D}}$ as a sheaf on X. The ordinary/logarithmic local calculation for $z_1 \cdots z_r = 0$ glues to

$$0 \longrightarrow \Omega^1_X \longrightarrow \mathcal{A} \xrightarrow{\mathrm{res}} \nu_*\mathcal{O}_{\widetilde{D}} \longrightarrow 0. \tag{11.12}$$

The normalization in the target is indispensable at the self-crossing cusp. This is the degree-one residue sequence associated with the normalized strata of a normal-crossings divisor [40, Chapter II, §3.4–3.7, pp. 76–78]; the displayed local calculation fixes its convention in the present self-crossing model.

Proposition 42.7 (absolute cotangent direct images)**.** ***[VERIFIED]*** *Put $\mathcal{K} = \ker \kappa_1$. Then*

$$f_*\Omega^1_X \simeq \mathcal{O}(-2), \qquad R^1 f_*\Omega^1_X \simeq \mathcal{K}, \qquad R^2 f_*\Omega^1_X \simeq \mathcal{O} \oplus \mathcal{O}(-1). \tag{11.13}$$

None of these sheaves has torsion.

Proof. The three normalized components have

$$R^0 f_*\nu_*\mathcal{O}_{\widetilde{D}} = \mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \quad R^1 f_*\nu_*\mathcal{O}_{\widetilde{D}} = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \quad R^2 = 0. \tag{11.14}$$

Indeed dP_6 is rational and S_1, S_2 are bielliptic, with $h^1(\mathcal{O}_{S_j}) = 1$ and $h^2(\mathcal{O}_{S_j}) = 0$.

The long direct-image sequence of (11.2), together with theorem 42.6, gives

$$f_*\mathcal{A} = \mathcal{O}(1), \qquad 0 \longrightarrow T \longrightarrow R^1 f_*\mathcal{A} \longrightarrow \mathcal{K} \longrightarrow 0, \qquad R^2 f_*\mathcal{A} = \mathcal{O} \oplus \mathcal{O}(-1), \tag{11.15}$$

where $T = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$. The residue of the base logarithmic form is weighted evaluation:

$$\mathcal{O}(1) \longrightarrow \mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \qquad q \longmapsto (q(p_0),\, 3q(p_1),\, 4q(p_2)). \tag{11.16}$$

It is surjective as a sheaf and has kernel $\mathcal{O}(1)(-p_0 - p_1 - p_2) = \mathcal{O}(-2)$.

It remains to identify the torsion in (11.15). At p_j, the residue map on $G \otimes \mathbb{C}_{p_j}$ sends

$$xb_0 + ys^a b_1 \longmapsto m_j x[b_0] \in H^1(S_j, \mathcal{O}_{S_j}). \tag{11.17}$$

More precisely, after trivializing $\mathcal{L}$ by $\ell = dt/t$, the logarithmic conormal and residue maps form the exact commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & f^*\mathcal{L} & \longrightarrow & \mathcal{A} & \longrightarrow & \mathcal{E} & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \mathrm{res}_B} & & \downarrow{\scriptstyle \mathrm{res}_{S_j}} & & \downarrow{\scriptstyle 0} & & \\ 0 & \longrightarrow & i_{j*}\mathcal{O}_{S_j} & \xrightarrow{=} & i_{j*}\mathcal{O}_{S_j} & \longrightarrow & 0 & & \end{array} \tag{11.16a}$$

with

$$\mathrm{res}_B(q\ell) = m_j q|_{S_j}. \tag{11.16b}$$

Applying Rf_* to (11.16a) gives the typed composite

$$f_*\mathcal{E}|_{p_j} \xrightarrow{\kappa_0} (\mathcal{L} \otimes G)|_{p_j} \xrightarrow{\mathrm{res}_{1,j}} H^1(S_j, \mathcal{O}_{S_j}), \qquad \mathrm{res}_{1,j}\,\kappa_0 = 0, \tag{11.16c}$$

and (11.16b) gives exactly

$$\mathrm{res}_{1,j}(xb_0 + ys^a b_1) = m_j x[b_0].$$

Thus its kernel is the one-dimensional line $\mathbb{C}\, s^a b_1$. The simple zero of $\det \kappa_0$ proved in theorem 42.6 implies that $\kappa_0 \bmod t$ has rank one; the zero composite in (11.16c) therefore forces

$$\mathrm{im}(\kappa_0 \bmod t) = \ker \mathrm{res}_{1,j} = \mathbb{C}\, s^a b_1. \tag{11.17a_{fin}}$$

Consequently the induced map $\operatorname{coker} \kappa_0 \to H^1(S_j, \mathcal{O}_{S_j})$ is an isomorphism at each finite fibre. Thus each summand $T_{p_j} \to H^1(S_j, \mathcal{O}_{S_j})$ is an isomorphism. In the long sequence of (11.12), (11.16) first removes the entire degree-zero residue, and (11.17) then cancels T. What remains is precisely (11.13). □

Remark 42.8 (the cusp is not the torsion locus). At p_0, if $Q = \nu_*\mathcal{O}_{\mathrm{dP}_6}/\mathcal{O}_W$, then

$$0 \longrightarrow \Omega^1_{N_0/\Delta} \longrightarrow \mathcal{E}|_{N_0} \longrightarrow i_*Q \longrightarrow 0, \qquad h^\bullet(Q) = (2,1,0). \tag{11.17a}$$

This relative sequence is the right column of the following exact normalization–residue diagram; all sheaves in the bottom row are regarded as sheaves on N_0 by i_*:

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & f^*\Omega^1_\Delta & \longrightarrow & \Omega^1_{N_0} & \longrightarrow & \Omega^1_{N_0/\Delta} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & f^*\Omega^1_\Delta(\log 0) & \longrightarrow & \Omega^1_{N_0}(\log W) & \longrightarrow & \mathcal{E}|_{N_0} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \mathrm{res}_B} & & \downarrow{\scriptstyle \mathrm{res}_W} & & \downarrow{\scriptstyle \rho} & & \\
0 & \longrightarrow & i_*\mathcal{O}_W & \longrightarrow & i_*\nu_*\mathcal{O}_{\mathrm{dP}_6} & \longrightarrow & i_*Q & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array} \tag{11.17a$'$}$$

The first two columns are the base and normalized-divisor residue sequences; the bottom row is the definition of Q. Hence the snake lemma identifies the right column with (11.17a), including its displayed map ρ.

The exactness assertion is also visible stalkwise. At a double chart $t = xy$, with one coordinate along the double stratum and logarithmic generator e, the image of ordinary relative differentials in $\mathcal{E}$ contains the stratum differential and has transverse part $(x, y)e$; its cokernel is therefore $R/(x, y)$. At a triple chart $t = xyz$, write $\mathcal{E} = Re_1 \oplus Re_2$ with $d\log x = e_1$, $d\log y = e_2$, and $d\log z = -e_1 - e_2$. The image is generated by

$$xe_1, \qquad ye_2, \qquad z(e_1 + e_2),$$

Writing a normalized branch value as a triple modulo the diagonal, the quotient map is explicitly

$$e_1 \longmapsto [(1,0,0)], \qquad e_2 \longmapsto [(0,1,0)]. \tag{11.17a$''$}$$

Its kernel is exactly $\langle xe_1, ye_2, z(e_1 + e_2)\rangle$, so this is the standard local presentation of the normalization quotient Q. Away from the double locus the map is an isomorphism. This proves (11.17a) stalkwise. Here are the maps, rather than only their ranks. Quotienting the normalization complex (9.10) by $\mathcal{O}_W$ gives

$$0 \longrightarrow Q \longrightarrow \bigoplus_{k=1}^{3} \mathcal{O}_{D_k} \xrightarrow{\delta_1} \mathbb{C}_P \oplus \mathbb{C}_Q \longrightarrow 0.$$

Because $D_k \simeq \mathbb{P}^1$, its cohomology is the cohomology of

$$\mathbb{C}^3 \longrightarrow \mathbb{C}^2, \qquad (a_1, a_2, a_3) \longmapsto (s, s), \qquad s = a_1 - a_2 + a_3.$$

Thus

$$H^0(Q) = \{(a_1, a_2, a_3) : a_1 - a_2 + a_3 = 0\} \simeq \mathbb{C}^2, \qquad H^1(Q) = \mathbb{C}^2/\mathbb{C}(1,1) \simeq \mathbb{C}, \tag{11.17b}$$

and $H^2(Q) = 0$. Since dP_6 is rational, the connecting maps for $0 \to \mathcal{O}_W \to \nu_*\mathcal{O}_{\mathrm{dP}_6} \to Q \to 0$ are isomorphisms

$$\delta_0 : H^0(Q) \xrightarrow{\sim} H^1(W, \mathcal{O}_W), \qquad \delta_1 : H^1(Q) \xrightarrow{\sim} H^2(W, \mathcal{O}_W). \tag{11.17c}$$

At a triple chart $t = z_0z_1z_2$, a relative logarithmic form is represented by $\eta = \sum a_i\, d\log z_i$, with holomorphic coefficient triple (a_0, a_1, a_2) taken modulo $\mathcal{O}(1,1,1)$. Restricting the branch differences to the three double strata gives

$$\rho(\eta) = (a_0 - a_1,\ a_0 - a_2,\ a_1 - a_2). \tag{11.17d}$$

Formula (11.17d) is written in the P-ordering of the globally labelled double curves. The Q-point sign is fixed by a separate calculation, not by an implicit relabelling. In the common e_1, e_2-plane, use the ordered branch vertices

$$(b_0, b_1, b_2) = (0, e_1, e_2), \qquad (c_0, c_1, c_2) = (e_1, e_1 + e_2, e_2)$$

at P and Q, respectively. The coefficient triples of $d\log x_1, d\log x_2$ are

$$(0,1,0),\ (0,0,1) \quad \text{at } P, \qquad (1,1,0),\ (0,1,1) \quad \text{at } Q.$$

The globally labelled branch-difference maps are

$$d_P(a) = (a_0 - a_1, a_0 - a_2, a_1 - a_2), \qquad d_Q(a) = (a_2 - a_1, a_0 - a_1, a_0 - a_2). \tag{11.17d$'$}$$

Direct substitution gives, at both triple points, the same two global sections

$$q_1 = (-1, 0, 1), \qquad q_2 = (0, -1, -1). \tag{11.17d$''$}$$

For constant coefficients this induces an isomorphism $\mathbb{C}^3/\mathbb{C}(1,1,1) \to H^0(Q)$. In the toric chart $(z_0, z_1, z_2) = (t/(x_1x_2), x_1, x_2)$ it sends

$$d\log x_1 \longmapsto (-1, 0, 1), \qquad d\log x_2 \longmapsto (0, -1, -1),$$

which is a basis of the hyperplane in (11.17b); unimodular changes of cusp chart merely change this basis by $\mathrm{GL}_2(\mathbb{Z})$.

Write $\eta_1 = d\log x_1$, $\eta_2 = d\log x_2$, take the q_i of (11.17d$''$), and put $\alpha_i = \delta_0(q_i)$. By (11.17c), α_1, α_2 form a basis of $H^1(W, \mathcal{O}_W)$. Naturality of the connecting maps fixes a sign that must be retained. With the standard Čech differential $(\delta c)_{i_0 \dots i_{r+1}} = \sum_k (-1)^k c_{i_0 \dots \widehat{i_k} \dots i_{r+1}}$, lift q_i to the normalization complex. The total differential of a degree-one cocycle α times that lift is $(-1)^1 \alpha \smile \delta\widetilde{q}_i$. The projection formula therefore gives exactly

$$\delta_1\big(\rho_*(\alpha \otimes \eta_i)\big) = -\alpha \smile \delta_0(q_i) = -\alpha \smile \alpha_i.$$

The cup product in theorem 41.3 is an isomorphism. Define, with this convention,

$$\theta = -\delta_1^{-1}(\alpha_1 \smile \alpha_2).$$

Graded commutativity now gives, with no residual sign ambiguity,

$$\rho_* = (0, 1, -1, 0) : \mathbb{C}\langle \alpha_1\eta_1, \alpha_1\eta_2, \alpha_2\eta_1, \alpha_2\eta_2 \rangle \longrightarrow \mathbb{C}\theta. \tag{11.17e}$$

In particular the map is surjective. Applying direct images to (11.17a), the first map is reduction modulo t followed by the isomorphism (11.17d), and the second is (11.17e). Consequently

$$f_*\Omega^1_{N_0/\Delta} = t\mathcal{O}^2, \quad R^1 f_*\Omega^1_{N_0/\Delta} = \ker(\mathcal{O}^4 \to \mathbb{C}_{p_0}), \quad R^2 f_*\Omega^1_{N_0/\Delta} = \mathcal{O}^2.$$

For the row in (11.17e), the middle kernel is generated by $e_1, e_4, e_2 + e_3, te_2$. Thus $\Omega^1_{X/B}$ is torsion-free in a neighbourhood of the cusp, but non-locally-free along its xy and xyz strata. Globally its torsion is supported precisely on the two multiple fibres, as the following local module calculation proves. On the smooth cover J'_j, let $R = \mathcal{O}_{J'_j, x}$ and choose coordinates (s, y_1, y_2) smooth over the s-disc. Relative to the original base B, where $t = s^{m_j}$, one has

$$\Omega^1_{J'_j/B,x} = \frac{R\,ds \oplus R\,dy_1 \oplus R\,dy_2}{(m_j s^{m_j - 1} ds)} \simeq R\,dy_1 \oplus R\,dy_2 \oplus \big(R/(s^{m_j - 1})\big)ds. \tag{11.17f}$$

Thus the last summand is exactly the torsion submodule. The quotient map $q : J'_j \to N_j$ is finite étale, so the relative cotangent sequence gives

$$q^*\Omega^1_{N_j/B} \simeq \Omega^1_{J'_j/B}.$$

Faithful flatness descends both the presence of the displayed torsion and its absence off $s = 0$. Hence ordinary relative torsion is supported exactly on the two multiple fibres and is absent at the cusp. The reconstructed manuscript's assertion that torsion occurs at all three singular fibres is false.

42.6 The scalar and the exact Hodge diamond

Theorem 42.9 (the remaining scalar is nonzero)**.** ***[VERIFIED]*** *The induced map*

$$H^0(B, R^1 f_*\mathcal{E}) \simeq \mathbb{C}^2 \xrightarrow{H^0(\kappa_1)} H^0(B, \mathcal{O}) \simeq \mathbb{C}$$

has rank one.

Proof. Every matrix R_g in (2.3d) has second column $(0,1)^{\mathsf{t}}$. Consequently the marked vertical field

$$\xi = \frac{\partial}{\partial \zeta_2}$$

is invariant under every deck transformation. The affine translations in theorem 8.7 have identity differential in this direction, and the corresponding toric logarithmic direction extends across the cusp charts of theorem 8.4. Thus ξ is a global section nowhere zero in the locally free relative sheaf $T_{X/B} = \mathcal{E}^\vee$ and is the section used in the evaluation sequence

$$0 \longrightarrow \mathcal{O}(-1) \longrightarrow f_*\mathcal{E} \xrightarrow{\mathrm{ev}_\xi} \mathcal{O} \longrightarrow 0.$$

This sequence splits, and the lift h of $1 \in H^0(\mathcal{O})$ is unique: the difference of two lifts is a global section of $\mathcal{O}(-1)$, while $H^0(B, \mathcal{O}(-1)) = 0$. On the marked cover write that unique lift as

$$h = (a(z), 1)\begin{pmatrix} d\zeta_1 \\ d\zeta_2 \end{pmatrix}. \tag{11.18}$$

The function a is holomorphic on each finite cover because h is a section of the actual invariant lattice, not merely of its generic fibre. For g_2, invariance of (11.18) and $R_{g_2} = \left(\begin{smallmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{smallmatrix}\right)$ mean exactly

$$(a(z), 1) = (a(g_2 z), 1) R_{g_2}(z), \qquad \text{hence} \qquad a(g_2 z) = \tau(z) a(z) + \mu(z). \tag{11.18a}$$

At the fixed point z_2, the normalized values $\tau(z_2) = i$ and $\mu(z_2) = (1-i)/2$ therefore give

$$a(z_2) = \frac{1}{2}. \tag{11.18b}$$

The cup-product map $f_*\mathcal{E} \otimes G \to R^1 f_*\mathcal{E}$ sends $h \otimes \gamma$ to the unit generator $e_0 b_0$ of the degree-zero character line in (11.4). This identifies the precise entry of the row (11.11) that must be evaluated.

Normalize the order-four linearizing coordinate by

$$s = (z_2 - \bar{z}_2) s_2, \qquad s \circ g_2 = -is, \qquad \frac{ds}{dz}(z_2) = 1.$$

Write a dot for d/ds, and put $\ell = dt/t = 4\,ds/s$. In the quotient basis (γ, u) of G, the marked period derivative and $ds = (s/4)\ell$ give the exact logarithmic map

$$\kappa_0(h) = \frac{s}{4}\Big[(6a\dot{\mu} + \dot{\beta})\gamma + (a\dot{\tau} + \dot{\mu})u\Big] \otimes \ell. \tag{11.19}$$

For $(m, a) = (4, 1)$, the actual invariant target generator from (11.4) is $s(b_0 \wedge b_1)\ell$. Normalize b_1 so that

$$\gamma \wedge b_1 = u_2(s)(\gamma \wedge u), \qquad u_2(0) = 1;$$

equivalently, use the invariant frame

$$\omega_2 = s\, u_2(s)(\gamma \wedge u)\ell. \tag{11.19f}$$

The uncorrected expression $s(\gamma \wedge u)\ell$ is only invariant at $s = 0$: indeed $g_2^*(\gamma \wedge u) = \tau(z)(\gamma \wedge u)$, so its pullback factor is $(-i)\tau(z)$. Formula (11.11), in the genuine invariant frame (11.19f), sends the first ordered source generator $e_0 b_0$ to minus the second coefficient of $\kappa_0(e_0)$. Since $u_2(0) = 1$, (11.19) therefore gives the exact signed coarse value

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\big(a(z_2)\dot{\tau}(z_2) + \dot{\mu}(z_2)\big). \tag{11.19a}$$

At the order-four lift z_2, the quotient coordinate has order four while the modular invariant has order two at $\tau = i$; see [25, Chapter VII]. Equivalently, $j(\tau) - 1728$ has a double zero at i and $j(\tau(z)) - 1728$ has order four at z_2. Hence

$$\operatorname{ord}_{z_2}(\tau - i) = 2, \qquad \dot{\tau}(z_2) = 0. \tag{11.20}$$

By (11.7), $K(z_2) \neq 0$; since $K = 6\dot{\mu}^2 - \dot{\tau}\dot{\beta}$, (11.20) gives $K(z_2) = 6\dot{\mu}(z_2)^2 \neq 0$. Substitution in (11.19a) therefore gives

$$\kappa_1(h \otimes \gamma)(p_2) = -\frac{1}{4}\dot{\mu}(z_2) \neq 0. \tag{11.20a}$$

The image is a global section of $\mathcal{L} \otimes R^2 f_* \mathcal{O}_X \simeq \mathcal{O}_B$; because it is nonzero at p_2, it is a nonzero constant. Hence $H^0(\kappa_1)$ has rank one. □

Corollary 42.10 (exact Hodge diamond)**.** ***[VERIFIED]*** *The last parameter in theorem 42.2 is*

$$h^{1,2}(X) = 1.$$

Consequently $h^{1,1} = 2$*, and the Hodge table is*

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	1
1	1	2	1	0
0	1	0	0	0.

The two nontrivial Frölicher d_1 *rows are both exact complexes* $0 \to \mathbb{C} \to \mathbb{C}^2 \to \mathbb{C} \to 0$*, and* $\dim E_1 = 10$*,* $\dim E_2 = 2$*.*

Proof. Put $\mathcal{K} = \ker \kappa_1$. Since

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{O}^{\oplus 2} \oplus \mathcal{O}(-1)^{\oplus 2} \xrightarrow{\kappa_1} \mathcal{O} \longrightarrow 0$$

and the global map has rank one, its restriction to the two trivial summands is a nonzero constant row (the negative summands have no global sections). After a constant change of basis it is the identity on one copy of $\mathcal{O}$. Subtracting the remaining component from that summand makes the map the projection, so its kernel is

$$\mathcal{K} \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}.$$

By theorem 42.7 and coherent Leray,

$$h^{1,2} = h^1(B, \mathcal{K}) + h^0(B, \mathcal{O} \oplus \mathcal{O}(-1)) = 0 + 1 = 1.$$

The remaining assertions follow from theorems 42.2 and 42.3. □

42.7 Bott–Chern and Aeppli dimensions

For a compact complex manifold write

$$H^{p,q}_{BC}(X)=\frac{\ker\partial\cap\ker\bar\partial\subset A^{p,q}(X)}{\operatorname{im}\partial\bar\partial},\qquad H^{p,q}_{A}(X)=\frac{\ker\partial\bar\partial\subset A^{p,q}(X)}{\operatorname{im}\partial+\operatorname{im}\bar\partial}.$$

Theorem 42.11 (exact Bott–Chern and Aeppli diamonds)**.** ***[VERIFIED]*** *The Bott–Chern dimensions of the candidate are*

$q\backslash p$	0	1	2	3
3	0	0	1	1
2	0	1	2	1
1	0	2	1	0
0	1	0	0	0,

(11.21)

and its Aeppli dimensions are

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	0
1	1	2	1	0
0	1	1	0	0.

(11.22)

In total degree $k=0,\dots,6$ this is

$$(h^k_{BC})=(1,0,2,2,2,2,1),\qquad (h^k_{A})=(1,2,2,2,2,0,1).\tag{11.23}$$

Consequently, set

$$\Delta^k=h^k_{BC}+h^k_{A}-2b_k$$

for the degree-k non-$\partial\bar\partial$ defects. The Angella–Tomassini inequality [44, Theorem A] gives $\Delta^k\geq 0$ for every compact complex manifold; here the exact values are

$$(\Delta^k)=(0,2,4,4,4,2,0).\tag{11.24}$$

Proof. McHugh associates to a compact complex manifold the finite-dimensional quotient

$$K^{p,q}=\frac{\ker\partial\cap\ker\bar\partial\subset A^{p,q}}{(\operatorname{im}\partial\cap\ker\bar\partial)+(\operatorname{im}\bar\partial\cap\ker\partial)},\qquad k^{p,q}=\dim_{\mathbb{C}}K^{p,q}.$$

Put

$$Z^{p,q}=\ker\partial\cap\ker\bar\partial\subset A^{p,q},\qquad B^{p,q}_K=(\operatorname{im}\partial\cap\ker\bar\partial)+(\operatorname{im}\bar\partial\cap\ker\partial).$$

The natural map $Z^{p,q}\to H^{p+q}_{\mathrm{dR}}(X;\mathbb{C})$ need not descend to an injection from $K^{p,q}$; the dimension estimate has the opposite quotient mechanism. If a pure form $\alpha\in Z^{p,q}$ is d-exact, the (p,q) component of a primitive gives

$$\alpha=\partial\beta^{p-1,q}+\bar\partial\beta^{p,q-1}.$$

Because $\bar\partial\alpha=0$ and $\partial\alpha=0$, respectively, $\partial\beta^{p-1,q}$ is $\bar\partial$-closed and $\bar\partial\beta^{p,q-1}$ is ∂-closed. Thus $\ker(Z^{p,q}\to H^{p+q}_{\mathrm{dR}})\subset B^{p,q}_K$, so the de Rham image of $Z^{p,q}$ surjects onto $K^{p,q}$. Hence

$$k^{p,q}\leq b_{p+q}.\tag{11.25}$$

Because X has the integral cohomology of S^6, (11.25) gives $k^{p,q}=0$ whenever $0<p+q<6$.

For a compact complex threefold, McHugh's Theorem 6.3, Proposition 6.4, equation (7), and the ensuing table give

$$\begin{aligned} h^{1,1}_{BC} &= 2h^{0,1} - b_1 + \dim_{\mathbb{R}} H^{1,1}(X;\mathbb{R}), \\ h^{1,2}_{BC} &= h^{1,2} + \dim_{\mathbb{C}} E_2^{3,1} - k^{2,0}, \\ h^{2,2}_{BC} &= -h^{1,1}_{BC} + h^{0,1} - h^{0,2} - h^{1,0} + h^{1,1} + h^{1,2} + h^{2,0} \\ &\qquad + 2\dim_{\mathbb{C}} E_2^{3,1} + k^{1,1} - k^{1,2} - 2k^{2,0}, \end{aligned}$$

and his threefold table gives the remaining entries [48, Theorem 6.3, Proposition 6.4, equation (7), §7.1]; see also the independent synthesis in [47, §4]. Here $H^{1,1}(X;\mathbb{R})$ denotes the subspace of $H^2_{\mathrm{dR}}(X;\mathbb{R})$ represented by real d-closed forms of type $(1,1)$, and it is zero because $H^2_{\mathrm{dR}}(X;\mathbb{R}) = 0$, while theorem 42.3 gives $E_2 = E_\infty$; since $b_4 = 0$, this in particular gives $E_2^{3,1} = 0$. Substitution of

$$h^{0,1} = 1, \quad h^{0,2} = h^{1,0} = h^{2,0} = 0, \quad h^{1,1} = 2, \quad h^{1,2} = 1$$

therefore gives

$$h^{1,1}_{BC} = 2, \qquad h^{1,2}_{BC} = h^{2,1}_{BC} = 1, \qquad h^{2,2}_{BC} = 2.$$

The same table gives $h^{2,3}_{BC} = h^{3,2}_{BC} = 1$, the two endpoint dimensions equal to one, and every other entry zero. This proves (11.21). The Hodge-star duality of Schweitzer [49, §2.c, pp. 9–10], $h^{p,q}_A = h^{3-p,3-q}_{BC}$, proves (11.22). Summing along diagonals and using $b_0 = b_6 = 1$ and $b_k = 0$ for $1 \le k \le 5$ proves (11.23)–(11.24). □

The dimension calculation above is completed by the explicit representative and product theorem in the following section.

43 Period-coordinate Dolbeault classes and the Bott–Chern algebra

43.1 Fixed input and conventions

Let $f\colon X \to B = \mathbb{P}^1$ be the compact complex threefold of theorem 8.9. We use $d = \partial + \bar\partial$ and, following [49, §1.a],

$$H^{p,q}_{BC}(X) = \frac{\ker\partial \cap \ker\bar\partial \subset \mathcal{A}^{p,q}(X)}{\operatorname{im}(\partial\bar\partial)}. \tag{1127}$$

The natural map to Dolbeault cohomology is denoted

$$\pi^{p,q}_{\bar\partial}\colon H^{p,q}_{BC}(X) \longrightarrow H^{p,q}_{\bar\partial}(X). \tag{1128}$$

Exterior multiplication makes the direct sum of the Bott–Chern groups a bigraded algebra, and complex conjugation interchanges bidegrees (p,q) and (q,p) [49, §1.a].

By theorem 42.10, the two nonzero rows of the Frölicher differential are the exact complexes

$$0 \longrightarrow H^{0,1}_{\bar\partial}(X) \xrightarrow{d_1} H^{1,1}_{\bar\partial}(X) \xrightarrow{d_1} H^{2,1}_{\bar\partial}(X) \longrightarrow 0, \tag{1129}$$

$$0 \longrightarrow H^{1,2}_{\bar\partial}(X) \xrightarrow{d_1} H^{2,2}_{\bar\partial}(X) \xrightarrow{d_1} H^{3,2}_{\bar\partial}(X) \longrightarrow 0, \tag{1130}$$

and each row has dimensions $1, 2, 1$. The exact Bott–Chern calculation of theorem 42.11, obtained from the compact-threefold formulae of [48], gives

(p,q)	$(0,0)$	$(1,1)$	$(2,1)$	$(1,2)$	$(2,2)$	$(3,2)$	$(2,3)$	$(3,3)$
$h^{p,q}_{BC}$	1	2	1	1	2	1	1	1.

(1131)

All other Bott–Chern groups vanish. By theorem 39.7, $b_k(X) = 0$ for $1 \le k \le 5$.

On a marked smooth fibre retain the integral basis

$$V = \mathbb{Z}\langle \gamma, u, w, \delta \rangle$$

without renaming its element u, and put

$$\eta = u \wedge w + 6\gamma \wedge \delta, \qquad \eta^2 = 12\gamma \wedge u \wedge w \wedge \delta. \tag{1132}$$

The holomorphic Hodge rows supplied by theorems 8.2 and 40.2 are

$$\sigma_1 = 6\mu\gamma + \tau u + w, \qquad \sigma_2 = \beta\gamma + \mu u + \delta. \tag{1133}$$

Thus no global Dolbeault or Bott–Chern generator below is denoted by u or x; those letters remain reserved for the printed fibre basis and for local real linear functions.

43.2 Exact fibre representatives

Write

$$y = \operatorname{Im}\tau, \qquad m = \operatorname{Im}\mu, \qquad r = \operatorname{Im}\beta, \qquad \Delta = r - \frac{6m^2}{y} < 0. \tag{1134}$$

The four marked lattice columns in the fibre coordinate $\zeta = (\zeta_1, \zeta_2)$ are

$$(6\mu, \beta), \quad (\tau, \mu), \quad (1, 0), \quad (0, 1), \tag{1135}$$

in the order $\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}$. On the universal cover of a marked fibre define the real linear functions

$$x_\gamma = \frac{y\,\operatorname{Im}\zeta_2 - m\,\operatorname{Im}\zeta_1}{y\Delta}, \qquad x_u = \frac{r\,\operatorname{Im}\zeta_1 - 6m\,\operatorname{Im}\zeta_2}{y\Delta}. \tag{1136}$$

For the first two columns in equation (1135), the numerators of the changes in x_γ are respectively $yr - 6m^2 = y\Delta$ and $ym - my = 0$, while the numerators of the changes in x_u are $6rm - 6mr = 0$ and $ry - 6m^2 = y\Delta$. The last two columns are real and change neither imaginary part. Hence

$$\begin{aligned} (\Delta_{\widehat{\gamma}} x_\gamma, \Delta_{\widehat{u}} x_\gamma, \Delta_{\widehat{w}} x_\gamma, \Delta_{\widehat{\delta}} x_\gamma) &= (1, 0, 0, 0), \\ (\Delta_{\widehat{\gamma}} x_u, \Delta_{\widehat{u}} x_u, \Delta_{\widehat{w}} x_u, \Delta_{\widehat{\delta}} x_u) &= (0, 1, 0, 0). \end{aligned} \tag{1137}$$

Their differentials, although not the functions themselves, therefore descend to the fibre. Their $(0,1)$ parts are

$$\Gamma := \bar\partial_F x_\gamma = \frac{i}{2\Delta}\left(d\bar\zeta_2 - \frac{m}{y} d\bar\zeta_1\right), \tag{1138}$$

$$U := \bar\partial_F x_u = \frac{i}{2y\Delta}\left(r\, d\bar\zeta_1 - 6m\, d\bar\zeta_2\right). \tag{1139}$$

They represent respectively the $(0,1)$ images of the integral classes γ and u.

The identity

$$\eta = u \wedge \sigma_1 + 6\gamma \wedge \sigma_2 = -\sigma_1 \wedge u - 6\sigma_2 \wedge \gamma \tag{1140}$$

holds in $\bigwedge^2 V_{\mathbb{C}}$. Since η is of type $(1,1)$ on every marked fibre, equation (1140) gives its exact Dolbeault representative

$$Q_F = -d\zeta_1 \wedge U - 6 d\zeta_2 \wedge \Gamma. \tag{1141}$$

Equivalently, with the Appell–Humbert convention $\eta = (2i)^{-1} \sum (H_\eta)_{ab} \sigma_a \wedge \bar\sigma_b$, the matrix calculation in the proof of theorem 40.2 gives

$$H_\eta = \frac{1}{y\Delta}\begin{pmatrix} r & -6m \\ -6m & 6y \end{pmatrix}, \tag{1142}$$

and therefore

$$Q_F = \frac{1}{2iy\Delta}\Big(r\,d\zeta_1 \wedge d\bar{\zeta}_1 - 6m\,d\zeta_1 \wedge d\bar{\zeta}_2 - 6m\,d\zeta_2 \wedge d\bar{\zeta}_1 + 6y\,d\zeta_2 \wedge d\bar{\zeta}_2\Big). \tag{1143}$$

Direct exterior multiplication, with holomorphic differentials ordered before antiholomorphic differentials, gives

$$Q_F \wedge \Gamma = \frac{1}{4y\Delta} d\zeta_1 \wedge d\bar{\zeta}_1 \wedge d\bar{\zeta}_2 \neq 0, \tag{1144}$$

$$Q_F^2 = \frac{3}{y\Delta} d\zeta_1 \wedge d\zeta_2 \wedge d\bar{\zeta}_1 \wedge d\bar{\zeta}_2 \neq 0. \tag{1145}$$

Indeed, $\det H_\eta = (6ry - 36m^2)/(y^2\Delta^2) = 6/(y\Delta)$, and the coefficient in equation (1145) is $\frac{1}{2}\det H_\eta$ in the displayed ordering. The identities Equations (1144) and (1145) will be used only in fibre cohomology.

43.3 Extension of the two Dolbeault inputs

Lemma 43.1 (the degree-one input)**.** ***[VERIFIED]*** *There is a distinguished class $\mathfrak{g} \in H^{0,1}_{\bar\partial}(X)$ whose restriction to every smooth marked fibre is represented by Γ.*

Proof. By theorem 41.4, the monodromy-invariant integral class γ maps to the nowhere-zero holomorphic section $\gamma^{0,1}$ of $R^1 f_*\mathcal{O}_X \simeq \mathcal{O} \oplus \mathcal{O}(-1)$. Coherent Leray gives

$$H^1(X, \mathcal{O}_X) = H^0(B, R^1 f_*\mathcal{O}_X) = \mathbb{C}\gamma^{0,1}.$$

Let $\mathfrak{g}$ be the class mapping to $\gamma^{0,1}$. Equation (1138) is the exact period-coordinate representative of its fibre restriction. This is the distinguished coordinate of theorem 41.5 before any analytic rescaling. □

Lemma 43.2 (extension of the invariant $(1,1)$ class)**.** ***[VERIFIED]*** *The fibre class η extends to a nonzero holomorphic section*

$$\boldsymbol{\eta} \in H^0(B, R^1 f_*\Omega^1_X). \tag{1146}$$

Consequently there is a class $\mathfrak{q} \in H^{1,1}_{\bar\partial}(X)$ whose coherent-Leray edge value is $\boldsymbol{\eta}$. For every smooth fibre inclusion $i_b\colon F_b \to X$,

$$i_b^*\mathfrak{q} = [Q_F]_{\bar\partial} \in H^{1,1}_{\bar\partial}(F_b). \tag{1147}$$

Proof. Over the smooth marked locus, η is flat, monodromy invariant, and of type $(1,1)$ identically. Equation Equation (1140) identifies the corresponding holomorphic section of the relative group as

$$-\sigma_1 \wedge u^{0,1} - 6\sigma_2 \wedge \gamma^{0,1}. \tag{1148}$$

At a finite filling write $t = s^M$, with $(M, a) = (3, 2)$ or $(4, 1)$. The covering family is smooth at $s = 0$, and equation (1140) holds identically there, so equation (1148) remains holomorphic. It is invariant under the cyclic action. In equivariant frames e_0, e_1 for the relative cotangent and b_0, b_1 for $R^1\mathcal{O}$, the actual invariant $\mathbb{C}\{t\}$-basis from theorem 42.5 is

$$e_0b_0, \quad s^a e_0 b_1, \quad s^{M-a} e_1 b_0, \quad e_1 b_1. \tag{1149}$$

Exactness of cyclic invariants in characteristic zero therefore makes equation (1148) a holomorphic descended section at both finite points.

At the cusp, γ and u extend as sections of $R^1 f_*\mathbb{C}$, and their $(0,1)$ images form a basis of $H^1(W, \mathcal{O}_W)$ by theorem 41.4. On the dense torus,

$$\sigma_i = d\zeta_i = (2\pi i)^{-1} d\log x_i.$$

The classes of these logarithmic forms extend in the relative logarithmic sheaf as

$$\mathcal{E}|_{N_0} = \mathcal{O}_{N_0}[d\log x_1] \oplus \mathcal{O}_{N_0}[d\log x_2] \tag{1150}$$

by theorem 42.5. A cusp quotient shear changes $d\log x_i$ only by a holomorphic multiple of $d\log t$, which is zero in the relative logarithmic sheaf. Thus equation (1148) also extends holomorphically through the cusp. This concerns classes in the relative logarithmic sheaf; it does not assert that the $d\log x_i$ are global ordinary one-forms on X.

The Gauss–Manin/Kodaira–Spencer map

$$\kappa_1 \colon R^1 f_* \mathcal{E} \longrightarrow \Omega^1_B(\log \Sigma) \otimes R^2 f_* \mathcal{O}_X$$

annihilates equation (1148) on the smooth locus: a flat class which remains of type $(1,1)$ has zero $(0,2)$ Higgs component. Both the section and κ_1 extend holomorphically, so the identity theorem gives $\kappa_1(\boldsymbol{\eta}) = 0$ on B. The ordinary-cotangent calculation in theorems 42.6, 42.7 and 42.9 identifies

$$R^1 f_* \Omega^1_X = \ker \kappa_1 =: \mathcal{K}, \qquad \mathcal{K} \simeq \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}. \tag{1151}$$

This proves equation (1146). Its value on every smooth fibre is nonzero in Dolbeault cohomology, not merely as a differential form: by equation (1132) its square is the integral top class $12\gamma \wedge u \wedge w \wedge \delta$, and therefore

$$\int_{(F_b, \mathrm{or}_{\mathrm{mark}})} Q_F^2 = 12, \qquad \int_{(F_b, \mathrm{or}_{\mathbb{C}})} Q_F^2 = -12 \qquad (b \in B^\circ), \tag{1152}$$

where $\mathrm{or}_{\mathrm{mark}}$ is the orientation determined by (γ, u, w, δ) and $\mathrm{or}_{\mathbb{C}}$ is the complex orientation. They are opposite because the determinant of the four marked lattice columns in real coordinates is $y\Delta < 0$. Thus $[Q_F]_{\bar\partial} \neq 0$, so the holomorphic section $\boldsymbol{\eta}$ is nonzero. Explicitly, if $Q_F = \bar\partial_F \alpha$ for a $(1,0)$-form α, then $Q_F^2 = \bar\partial_F(\alpha \wedge Q_F)$; integration and Stokes' theorem would contradict equation (1152). Here $\alpha \wedge Q_F$ has bidegree $(2,1)$, so its ∂_F is zero in complex dimension two and the displayed $\bar\partial_F$-boundary is also a d-boundary.

Finally, coherent Leray over the curve gives the exact edge sequence

$$0 \longrightarrow H^1(B, f_*\Omega^1_X) \longrightarrow H^1(X, \Omega^1_X) \longrightarrow H^0(B, R^1 f_* \Omega^1_X) \longrightarrow 0, \tag{1153}$$

where $f_*\Omega^1_X \simeq \mathcal{O}(-2)$ by theorem 42.7. Choose $\mathfrak{q}$ over $\boldsymbol{\eta}$. Base change on a smooth fibre gives equation (1147). Any two choices differ by the one-dimensional left term in equation (1153). That term restricts to zero on every smooth fibre. The nonzero class $d_1\mathfrak{g}$ also restricts to zero, because its restriction is $[\partial_F \Gamma]_{\bar\partial} = 0$; since $H^1(B, \mathcal{O}(-2))$ is one-dimensional, the left term is exactly $\mathrm{im}(d_1 \colon H^{0,1}_{\bar\partial} \to H^{1,1}_{\bar\partial})$. □

Remark 43.3 (what is and is not explicit). Choose global $\bar\partial$-closed forms

$$G \in \mathcal{A}^{0,1}(X), \qquad Q \in \mathcal{A}^{1,1}(X) \tag{1154}$$

representing $\mathfrak{g}$ and $\mathfrak{q}$. Equation (1147) says that $i_b^*[Q]_{\bar\partial}$ is the Dolbeault class of the explicit translation-invariant form displayed in equation (1143); it does not say that an arbitrary representative Q restricts literally to that form. A literal global coefficient formula would require additional C^∞ gauge choices and horizontal correction forms on the disc–pair-of-pants cover, compatible with the two affine cyclic conjugacies and the toric cusp. Those choices are not part of the canonical data. No such literal gauge is needed for the class and product calculations below.

43.4 The factor 12 and the Bott–Chern first Chern class

Let

$$\exp\colon H^1(X,\mathcal{O}_X)\longrightarrow \operatorname{Pic}(X)$$

be induced by $z\mapsto e^{2\pi i z}$. By theorem 39.2,

$$H^0(B,R^1f_*\mathbb{Z})=\mathbb{Z}(12\gamma). \tag{1155}$$

The exponential–Leray comparison gives a line bundle M on B such that

$$\exp(12\mathfrak{g})=f^*M,\qquad c_1(M)=d_2(12\gamma). \tag{1156}$$

For completeness, the first assertion follows because equation (1155) says that the restriction of $12\mathfrak{g}$ to every fibre lies in the image of integral cohomology and hence exponentiates to the trivial bundle. The low-degree Leray sequence for $\mathcal{O}^*$ then makes the resulting bundle a pullback. On a disc–pair-of-pants cover of B, choose local integral lifts of 12γ. Their differences on the annular overlaps form a Cech one-cochain, and its Cech coboundary is simultaneously the two-cocycle defining the Leray transgression and the two-cocycle of logarithms defining $c_1(M)$. This proves the second assertion with the same sign convention; it is the exponential-sequence version of the local-lift calculation in the proof of theorem 39.3.

Orient the generator $\omega\in H^2(B;\mathbb{Z})$ by

$$\omega=c_1(\mathcal{O}_B(1)). \tag{1157}$$

Then theorem 39.3 and equation (1156) give

$$\exp(12\mathfrak{g})=f^*\mathcal{O}_B(1). \tag{1158}$$

If the opposite orientation is used for the primitive generator in theorem 39.3, the corresponding statement is $\exp(-12\mathfrak{g})=f^*\mathcal{O}_B(1)$. Thus the canonical integral data fixes the absolute factor 12, while its sign is the orientation choice fixed in equation (1157).

Put

$$a=\partial G,\qquad a^\dagger=\bar\partial\bar G,\qquad \mathsf{A}=[a]_{BC},\qquad \mathsf{B}=[a^\dagger]_{BC}. \tag{1159}$$

Both forms are d-closed. We now compute their Chern-class normalization without choosing local metric weights. For $z\in\mathbb{C}$, equip the smooth trivial line with a global frame s and the holomorphic structure

$$\bar\partial_z s=-2\pi i z G\otimes s. \tag{1160}$$

It is integrable because $\bar\partial G=0$. If $G=\bar\partial\phi_i$ on a good cover, then $e_i=e^{2\pi i z\phi_i}s$ is $\bar\partial_z$-holomorphic and

$$e_j=e^{2\pi i z(\phi_j-\phi_i)}e_i.$$

The holomorphic cocycle $\phi_j-\phi_i$ represents $\mathfrak{g}$ under the Cech–Dolbeault comparison, so equation (1160) is precisely the line bundle $\exp(z\mathfrak{g})$ for the displayed exponential convention.

Give s unit length. The unit-metric Chern connection is

$$\nabla_z=d-2\pi i\bigl(zG+\bar z\,\bar G\bigr). \tag{1161}$$

Indeed, its $(0,1)$ part is equation (1160), and its connection one-form is purely imaginary, so it is metric compatible; uniqueness identifies it with the Chern connection. Since $\bar\partial G=0$ and $\partial\bar G=0$, its curvature is the global $(1,1)$ form

$$F_{\nabla_z}=-2\pi i\bigl(z\,\partial G+\bar z\,\bar\partial\bar G\bigr)=-2\pi i\bigl(za+\bar z\,a^\dagger\bigr). \tag{1162}$$

With the Bott–Chern convention

$$c_{1,BC}(L,h)=\left[\frac{i}{2\pi}F_{\nabla^h}\right]_{BC} \tag{1163}$$

of [49, §8.a], equation (1162) gives, for every $z \in \mathbb{C}$,

$$c_{1,BC}\big(\exp(z\mathfrak{g})\big) = z\mathsf{A} + \bar{z}\,\mathsf{B}. \tag{1164}$$

Consequently, if $\varepsilon = 1$ for the orientation fixed by equation (1157) and $\varepsilon = -1$ for its opposite, then

$$c_{1,BC}\big(f^*\mathcal{O}_B(1)\big) = 12\varepsilon(\mathsf{A} + \mathsf{B}). \tag{1165}$$

This is the convention-consistent form of the factor-12 identity; no analytic rescaling of the Picard coordinate changes its integral content.

43.5 All Bott–Chern generators

Using equation (1154), define the following d-closed forms and their classes:

class	representative	bidegree
A	$a = \partial G$	$(1,1)$
B	$a^\dagger = \bar\partial \bar G$	$(1,1)$
C	$c = \partial Q$	$(2,1)$
$\overline{\mathsf{C}}$	$c^\dagger = \bar\partial \bar Q$	$(1,2)$
D	$d_0 = \partial(G \wedge Q)$	$(2,2)$
$\overline{\mathsf{D}}$	$d_0^\dagger = \bar\partial(\bar G \wedge \bar Q)$	$(2,2)$
E	$e = \partial(Q^2)$	$(3,2)$
$\overline{\mathsf{E}}$	$e^\dagger = \bar\partial(\bar Q^2)$	$(2,3)$
T	$\nu, \quad \int_X \nu = 1$	$(3,3)$.

(1166)

Here ν is any smooth positive top-degree form divided by its integral. The chosen Q cannot be real: its fibre class is real, but the proof below shows that ∂Q is nonzero globally, whereas a real $\bar\partial$-closed Q would also be ∂-closed.

Proposition 43.4 (nonvanishing and bases)**.** ***[VERIFIED]*** *Together with the unit $\mathbf{1} \in H^{0,0}_{BC}(X)$, the entries in equation* (1166) *give bases of all the nonzero Bott–Chern groups in equation* (1131)*: A, B form the basis in bidegree $(1,1)$ and $\mathsf{D}, \overline{\mathsf{D}}$ form the basis in bidegree $(2,2)$; each other listed bidegree is one-dimensional and is generated by its displayed class.*

Proof. Because X is compact and connected, a smooth function annihilated by both ∂ and $\bar\partial$ is constant. There is no $\partial\bar\partial$ image in bidegree $(0,0)$, so $\mathbf{1} := [1]_{BC}$ generates $H^{0,0}_{BC}(X) = \mathbb{C}\mathbf{1}$.

The Dolbeault image of A is $d_1\mathfrak{g} \neq 0$ by injectivity in equation (1129). Thus ∂G is a nonzero Bott–Chern class. The Dolbeault image of B is zero because $a^\dagger = \bar\partial \bar G$, while $\mathsf{B} = \overline{\mathsf{A}} \neq 0$. Hence A, B are independent, and equation (1131) makes them a basis.

The classes $d_1\mathfrak{g}$ and $\mathfrak{q}$ are independent in $H^{1,1}_{\bar\partial}(X)$: the first restricts to zero on every smooth torus fibre, while the second restricts to the nonzero class by equations (1147) and (1152). Exactness of equation (1129) therefore gives

$$d_1\mathfrak{q} = [\partial Q]_{\bar\partial} \neq 0. \tag{1167}$$

Thus $\mathsf{C} \neq 0$, and conjugation gives $\overline{\mathsf{C}} \neq 0$.

Set

$$v = [G \wedge Q]_{\bar\partial} \in H^{1,2}_{\bar\partial}(X). \tag{1168}$$

Its fibre restriction is the class represented by $\Gamma \wedge Q_F$. This class is nonzero, rather than merely represented by a nonzero invariant form: wedging with the holomorphic one-form $d\zeta_2$ and using equation (1144) gives

$$\int_{F_b} Q_F \wedge \Gamma \wedge d\zeta_2 = \frac{1}{4y\Delta} \int_{F_b} d\zeta_1 \wedge d\zeta_2 \wedge d\bar\zeta_1 \wedge d\bar\zeta_2 = -1 \tag{1169}$$

for the complex orientation: the middle integral equals $-4y\Delta > 0$. Thus the displayed constant top form has nonzero integral. Dolbeault–Serre duality on F_b therefore gives $[\Gamma \wedge Q_F]_{\bar\partial} \neq 0$. Equivalently, if $Q_F \wedge \Gamma = \bar\partial_F \Psi$ with Ψ of bidegree $(1,1)$, then, since $\bar\partial_F d\zeta_2 = 0$,

$$Q_F \wedge \Gamma \wedge d\zeta_2 = \bar\partial_F(\Psi \wedge d\zeta_2),$$

and Stokes' theorem would make the integral in equation (1169) zero. Here $Q_F \wedge \Gamma$ has bidegree $(1,2)$, $d\zeta_2$ has bidegree $(1,0)$, and $\Psi \wedge d\zeta_2$ has bidegree $(2,1)$; its ∂_F is zero in complex dimension two, so the displayed $\bar\partial_F$-boundary is a d-boundary. Thus $v \neq 0$, and it spans $H^{1,2}_{\bar\partial}(X)$. Injectivity in equation (1130) gives

$$d_1 v = [\partial(G \wedge Q)]_{\bar\partial} \neq 0. \tag{1170}$$

Hence $\mathsf{D} \neq 0$, and conjugation gives $\overline{\mathsf{D}} \neq 0$. The latter has zero Dolbeault image because its representative is $\bar\partial$-exact, so it is independent of D; equation (1131) gives the asserted $(2,2)$ basis.

The class $[Q^2]_{\bar\partial}$ has nonzero fibre restriction by equation (1152). Since v spans $H^{1,2}_{\bar\partial}(X)$, the image of d_1 is spanned by $d_1 v$. On a smooth fibre,

$$i_b^*(d_1 v) = d_1[\Gamma \wedge Q_F]_{\bar\partial} = [\partial_F(\Gamma \wedge Q_F)]_{\bar\partial} = 0,$$

because Γ and Q_F have constant fibre coefficients. Thus every class in $\operatorname{im}(d_1 \colon H^{1,2}_{\bar\partial} \to H^{2,2}_{\bar\partial})$ restricts to zero. Therefore $[Q^2]_{\bar\partial}$ is independent of $d_1 v$. Exactness of equation (1130) yields

$$d_1[Q^2]_{\bar\partial} = [\partial(Q^2)]_{\bar\partial} \neq 0. \tag{1171}$$

Hence $\mathsf{E} \neq 0$, and conjugation gives $\overline{\mathsf{E}} \neq 0$. Finally, $\int_X \nu = 1$ proves $\mathsf{T} \neq 0$. The dimensions in equation (1131) complete the proof. □

43.6 Every Bott–Chern product

Theorem 43.5 (square-zero positive ideal)**.** ***[VERIFIED]*** *Let $N \subset H^{*,*}_{BC}(X)$ be the span of the nine positive-degree generators in equation* (1166)*. Then*

$$H^{*,*}_{BC}(X) = \mathbb{C}\mathbf{1} \oplus N, \qquad N^2 = 0. \tag{1172}$$

Equivalently, every product of two positive-degree Bott–Chern classes is zero.

Proof. We first calculate the products landing below top degree. Since $\bar\partial G = \bar\partial Q = 0$ and, after conjugation, $\partial \bar G = \partial \bar Q = 0$, the graded Leibniz rule gives the exact identities

$$\partial\bar\partial(G \wedge \bar G) = -a \wedge a^\dagger, \tag{1173}$$

$$\partial\bar\partial(G \wedge \bar Q) = -a \wedge c^\dagger, \qquad \partial\bar\partial(\bar G \wedge Q) = a^\dagger \wedge c. \tag{1174}$$

Consequently

$$\mathsf{AB} = 0, \qquad \mathsf{A}\overline{\mathsf{C}} = 0, \qquad \mathsf{BC} = 0. \tag{1175}$$

Choose the pullback Fubini–Study metric on $f^*\mathcal{O}_B(1)$. Its Chern form is the pullback of a $(1,1)$ form on the curve B, so its square vanishes pointwise. Squaring equation (1165) and using equation (1175) gives

$$\mathsf{A}^2 + \mathsf{B}^2 = 0. \tag{1176}$$

Equation (1176) alone does not separate the two squares. Apply $\pi^{2,2}_{\bar\partial}$. The class B^2 has zero Dolbeault image because

$$(a^\dagger)^2 = \bar\partial(\bar G \wedge a^\dagger). \tag{1177}$$

It follows from equation (1176) that $[a^2]_{\bar\partial} = 0$. On the other hand,

$$[a^2]_{\bar\partial} = d_1[G \wedge a]_{\bar\partial}, \qquad \partial(G \wedge a) = a^2. \tag{1178}$$

The first arrow in equation (1130) is injective, so equation (1178) implies $[G \wedge a]_{\bar\partial} = 0$. There is therefore a form $\Phi \in \mathcal{A}^{1,1}(X)$ such that

$$G \wedge a = \bar\partial\Phi. \tag{1179}$$

Applying ∂ gives

$$a^2 = \partial\bar\partial\Phi, \tag{1180}$$

and hence $\mathsf{A}^2 = 0$. Conjugation gives $\mathsf{B}^2 = 0$.

It remains among the products below top bidegree to prove $\mathsf{AC} = 0$; its conjugate will give $\mathsf{B}\overline{\mathsf{C}} = 0$. First, the natural map

$$\pi^{3,2}_{\bar\partial} \colon H^{3,2}_{BC}(X) \longrightarrow H^{3,2}_{\bar\partial}(X) \tag{1181}$$

is an isomorphism. Indeed, both spaces have dimension one by equation (1131) and theorem 42.10, and equation (1171) says that E has nonzero image. Pair $\pi^{3,2}_{\bar\partial}(\mathsf{AC})$ with the generator $[G]_{\bar\partial} \in H^{0,1}_{\bar\partial}(X)$. By equation (1179), $\bar\partial c = 0$, and Stokes' theorem,

$$\int_X a \wedge c \wedge G = -\int_X G \wedge a \wedge c = -\int_X \bar\partial\Phi \wedge c = -\int_X \bar\partial(\Phi \wedge c) = 0. \tag{1182}$$

Dolbeault–Serre duality makes

$$H^{3,2}_{\bar\partial}(X) \times H^{0,1}_{\bar\partial}(X) \longrightarrow \mathbb{C}, \qquad ([\alpha],[\rho]) \longmapsto \int_X \alpha \wedge \rho, \tag{1183}$$

a perfect pairing. Since $H^{0,1}_{\bar\partial}(X) = \mathbb{C}[G]_{\bar\partial}$, equation (1182) implies $\pi^{3,2}_{\bar\partial}(\mathsf{AC}) = 0$; injectivity in equation (1181) gives

$$\mathsf{AC} = 0, \qquad \mathsf{B}\overline{\mathsf{C}} = 0. \tag{1184}$$

This step uses Dolbeault–Serre duality together with the independently proved injectivity of equation (1181). The Bott–Chern–Aeppli pairing of [49, §2.c] by itself would supply a perfect pairing but would not annihilate this product.

The only products not yet considered which can have bidegree at most $(3,3)$ are

$$\mathsf{AD}, \quad \mathsf{A}\overline{\mathsf{D}}, \quad \mathsf{BD}, \quad \mathsf{B}\overline{\mathsf{D}}, \quad \mathsf{C}\overline{\mathsf{C}}, \tag{1185}$$

all of bidegree $(3,3)$. Every non-top generator in equation (1166) is de Rham exact:

$$\begin{gathered} a = dG, \qquad a^\dagger = d\bar G, \qquad c = dQ, \qquad c^\dagger = d\bar Q, \\ d_0 = d(G \wedge Q), \qquad d_0^\dagger = d(\bar G \wedge \bar Q), \qquad e = d(Q^2), \qquad e^\dagger = d(\bar Q^2). \end{gathered} \tag{1186}$$

Each factor paired with an exact factor is d-closed, so every product in equation (1185) is d-exact and has integral zero. Integration induces an isomorphism

$$H^{3,3}_{BC}(X) \xrightarrow{\int_X} \mathbb{C}. \tag{1187}$$

It is well defined by Stokes' theorem, the source has dimension one by equation (1131), and it is nonzero on T. Therefore all five products in equation (1185) vanish in Bott–Chern cohomology.

All other products of positive-degree generators have $p > 3$ or $q > 3$ and vanish already as differential forms. Equations (1175), (1180), (1184), (1185) and (1187) exhaust every possible positive-degree product and prove equation (1172). □

Remark 43.6 (literal global gauge boundary). The Bott–Chern algebra equation (1172) and the fibre-coordinate restrictions displayed in equations (1138), (1139), (1141) and (1143) to (1145) are complete. The canonical data do not select a preferred literal global coefficient atlas for the smooth forms G and Q. Such an atlas would require explicit smooth local trivializations and horizontal correction functions through both finite affine fillings and the cusp. After fixing $\mathfrak{g}$ and one coherent-Leray lift $\mathfrak{q}$, changing those smooth gauges changes G and Q by Dolbeault coboundaries but changes none of the cohomology classes, nonvanishing arguments, or products proved above. This is a limitation only on a preferred global coefficient presentation, not on the representatives as cohomology classes or on the Bott–Chern algebra.

Audit record 43.7 (H2: independent replication of the cotangent calculation)**.** **[OPEN]** A further context-isolated AI derivation has independently recovered the cyclic invariant lattices (11.4), the determinant divisor (11.6), the residue cancellation (11.17), the scalar test (11.19)–(11.20), and the splittings

$$R^0 f_* \Omega^1_X = \mathcal{O}(-2), \quad R^1 f_* \Omega^1_X = \mathcal{O} \oplus \mathcal{O}(-1)^{\oplus 2}, \quad R^2 f_* \Omega^1_X = \mathcal{O} \oplus \mathcal{O}(-1).$$

It also independently gives $h^{1,2} = 1$ and $h^{1,1} = 2$. Independent human specialist replication remains outstanding, so the record stays **[OPEN]**; this external-replication boundary is not an unresolved premise of the theorem above.

44 The alternate collapse and Mayer–Vietoris route

The nearby-cycle/Leray proof in section 39 gives one verified route to the topology of X. This section audits the manuscript's independent cellular-collapse/Mayer–Vietoris route. It now closes the entire alternate certificate: the equivariant cusp retraction, its primitive integral cellular map, equality with analytic nearby-cycle specialization, both geometrically realized Mayer–Vietoris stages, and the comparison of their obstruction maps with Leray.

We first correct a conflation in the earlier audit. For any two-set decomposition $Y = A \cup B$, our Mayer–Vietoris convention is

$$\Delta_q = (i_{A*}, -i_{B*}) : H_q(A \cap B; \mathbb{Z}) \longrightarrow H_q(A; \mathbb{Z}) \oplus H_q(B; \mathbb{Z}). \tag{4.1}$$

The manuscript does not identify its printed α_q with a one-shot map for three collars. It uses two stages. First,

$$X^\circ = X_1 \cup X_2, \qquad X_1 \cap X_2 \simeq F, \qquad \alpha_q = (\pi_{1*}, -\pi_{2*}). \tag{4.2}$$

Second, it attaches the cusp through

$$X = N_0' \cup X^\circ, \qquad N_0' \cap X^\circ = M, \qquad K_q = \ker(H_q(M) \to H_q(W)). \tag{4.3}$$

The proof below realizes both stages by actual oriented inclusions and then also records the equivalent one-shot degree-one collar matrix.

44.1 The abstract cusp-retraction lemma

Proposition 44.1 (equivariant topological spine)**.** ***[VERIFIED]*** *Let A be a contractible manifold with boundary and let $A_0 \subset \partial A$ be a contractible boundary component. Let a group Γ act freely and properly discontinuously on (A, A_0). Assume $Q = A/\Gamma$ and $H = A_0/\Gamma$ are manifolds, Q is compact, $H \hookrightarrow Q$ is collared, and the two quotient maps are connected coverings with deck group Γ. If the inclusion identifies the two deck groups by the identity automorphism of Γ, then $H \hookrightarrow Q$ is a strong deformation retract. The retraction lifts uniquely from the identity lift to a Γ-equivariant strong deformation retraction of A onto A_0.*

Proof. Because A and A_0 are contractible universal covers, $\pi_n(Q) = \pi_n(H) = 0$ for $n \geq 2$, while $\pi_1(Q) \cong \Gamma \cong \pi_1(H)$. By hypothesis the inclusion induces the identity on Γ, so it is a weak homotopy equivalence. Manifolds have the homotopy type of CW complexes; Whitehead's theorem [9, Theorem 4.5] therefore makes the inclusion a homotopy equivalence.

The collar makes (Q, H) an NDR pair, hence the inclusion is a cofibration. We record the standard cofibration argument. Choose a homotopy inverse $r_0 : Q \to H$. Since $r_0|_H \simeq \mathrm{id}_H$, the homotopy extension property modifies r_0 through maps $Q \to H$ to a retraction r satisfying $r|_H = \mathrm{id}_H$. The maps id_Q and $i \circ r$ are homotopic and agree on H. Applying the homotopy extension property once more makes that homotopy stationary on H. It is therefore a strong deformation retraction.

Let $P : A \to Q$ be the covering. Lift the deformation homotopy from the initial lift id_A. Since the base homotopy fixes H and the initial lift fixes A_0, uniqueness of path lifting makes the lifted homotopy fix A_0. For $\gamma \in \Gamma$, both $\widetilde{H}_t \circ \gamma$ and $\gamma \circ \widetilde{H}_t$ lift the same map and agree at $t = 0$; uniqueness gives equality for every t. The lift is therefore Γ-equivariant and ends in A_0. □

44.2 The candidate-specific equivariant collapse

We now verify every hypothesis of theorem 44.1 in the candidate. Retain the toric threefold Y, the compact torus

$$T_c = (S^1)^3, \qquad T_f = (S^1)^2 \times \{1\} \subset T_c,$$

and the free properly discontinuous action $\Gamma = \{\Psi_{\bar{\lambda}} : \bar{\lambda} \in \bar{\Lambda}\}$ from theorem 8.4. Put

$$Y_{\leq\eta} = \{|t| \leq \eta\}, \qquad Y_{\geq 0} = \overline{(\mathbb{R}_{>0})^3} \subset Y, \qquad A = (Y_{\leq\eta})_{\geq 0}, \qquad A_0 = (Y_0)_{\geq 0}. \tag{C.1}$$

For a torus point write $y = \log|x| / \log|t|$. The period matrix on a punctured cusp fibre is

$$Z_t = sB_0 + C(t), \qquad s = \frac{\log t}{2\pi i}, \qquad B_t = B_0 + \frac{R(t)}{\log|t|}, \quad R(t) = -2\pi \operatorname{Im} C(t). \tag{C.2}$$

Here $B_0 : \bar{\Lambda} \to \Lambda_{\mathrm{tor}}$ is unimodular.

Lemma 44.2 (straightening the twist)**.** ***[VERIFIED]*** *For all sufficiently small $\eta > 0$ there is a homeomorphism $G : Y_{\leq\eta} \to Y_{\leq\eta}$, fixing Y_0 pointwise and preserving t, which conjugates Γ to the action Γ^0 obtained by replacing $C(t)$ by $C_0 = C(0)$. It commutes with T_f and, on every smooth quotient fibre, induces a real-linear isomorphism which is the identity on the marked lattice Λ.*

Proof. Shrink η until B_t and $B_t^0 = B_0 + R(0)/\log|t|$ are invertible for $0 < |t| \leq \eta$. In logarithmic coordinates $\xi = (2\pi i)^{-1} \log x$ define

$$L_t(\xi) = \xi - (C(t) - C_0)(\operatorname{Im} Z_t)^{-1} \operatorname{Im} \xi. \tag{C.3}$$

Every ξ has the unique expression $\xi = Z_t a + b$, with $a, b \in \mathbb{R}^2$, and then $L_t(\xi) = Z_t^0 a + b$. Thus L_t is invertible, fixes $\mathbb{R}^2$ and its sublattice $\mathbb{Z}^2$, and sends each deck translation $Z_t\bar{\lambda}$ to $Z_t^0\bar{\lambda}$. It therefore descends on $Y_t = \mathbb{C}^2/\mathbb{Z}^2$ and conjugates the two $\bar{\Lambda}$-actions. In toric coordinates it is

$$G(x, t) = \big(\exp(-2\pi i (C(t) - C_0) B_t^{-1} y) x, t\big), \qquad G|_{Y_0} = \mathrm{id}\,. \tag{C.4}$$

It commutes with T_f because y is T_f-invariant.

Only continuity at Y_0 is nonformal. Around any $p \in Y_0$, choose one of the bounded-position subcharts used in the properness proof of theorem 8.4. On that subchart y stays bounded, B_t^{-1} stays bounded, and $C(t) - C_0 \to 0$. Hence the torus factor in (C.4) tends to 1. The same argument applies to the inverse, obtained by interchanging $C(t)$ and C_0. Thus G is a homeomorphism. Because it fixes the real lattice and sends the two period columns $Z_t\bar{\lambda}$ to $Z_t^0\bar{\lambda}$, it induces the identity on the marked $H_1 = \Lambda$. □

Write $c^0_{\bar\lambda} = u_{\bar\lambda} a_{\bar\lambda}$ for its polar decomposition, with $u_{\bar\lambda} \in T_f$ and $a_{\bar\lambda} = \exp(R(0)\bar\lambda) \in (\mathbb{R}_{>0})^2$. Let Γ^+ be the positive action

$$\Psi^+_{\bar\lambda} = (a_{\bar\lambda}, 1)\Phi_{\bar\lambda}. \tag{C.5}$$

Lemma 44.3 (positive quotient and honeycomb)**.** ***[VERIFIED]*** *The pair (A, A_0) in (C.1) consists of contractible topological manifolds, A_0 being a boundary component of A. The action Γ^+ is free and properly discontinuous, and*

$$Q^+ = A/\Gamma^+, \qquad H = A_0/\Gamma^+ \tag{C.6}$$

are respectively a compact 3-manifold with boundary and one of its boundary 2-tori. The inclusion $H \hookrightarrow Q^+$ is a strong deformation retract, and its retraction lifts to a Γ^+-equivariant strong deformation retraction P of A onto A_0.

Proof. Polar multiplication $m : T_c \times Y_{\geq 0} \to Y$ is a closed quotient map: in every smooth unimodular toric chart it is the usual map $(S^1)^3 \times [0,\infty)^3 \to \mathbb{C}^3$, and T_c is compact. If q is in the orbit indexed by a face C of the A_2 triangulation, its stabilizer is the connected compact subtorus on

$$N_{\widehat{C}} = \bigoplus_{v \in V(C)} \mathbb{Z}(v, 1).$$

Unimodularity makes this lattice saturated, and direct intersection with Λ_{tor} gives

$$\mathrm{Stab}_{T_c}(q) \cap T_f = \begin{cases} 1, & C \text{ is a vertex,} \\ S^1_{v'-v}, & C = \{v, v'\} \text{ is an edge,} \\ T_f, & C \text{ is a triangle.} \end{cases} \tag{C.7}$$

There are no hidden finite components. The lattices $N_{\widehat{C}}$ and Λ_{tor} are saturated, and every face contains a vertex $(v, 1)$, so $N_{\widehat{C}} + \Lambda_{\mathrm{tor}} = N'$. Thus the displayed intersections are the connected subtori attached to their primitive intersection lattices.

In a toric chart the positive set A is

$$\{(r_0, r_1, r_2) \in [0,\infty)^3 : r_0 r_1 r_2 \leq \eta\}.$$

Its corner links are discs, so it is a topological 3-manifold with boundary $A_0 \sqcup \{|t| = \eta\}_{>0}$. Its interior is $\mathbb{R}^2_y \times (0, \eta)$. Brown's collaring theorem [28] makes the inclusion of the interior a homotopy equivalence; therefore A is contractible.

The nonnegative part of each component $E_v \simeq \mathrm{dP}_6$ is its moment hexagon. These hexagons meet along exactly the arcs prescribed by the dual honeycomb $\mathcal{H}$ of the A_2 triangulation. Map the two vertex orbits of $\mathcal{H}$ to the corresponding torus fixed points, choose homeomorphisms on representatives of the three edge orbits, and extend the resulting circle homeomorphism over one hexagon. Propagating by $B_0\bar\Lambda$ defines a Γ^+-equivariant continuous bijection

$$\vartheta : \mathbb{R}^2 \xrightarrow{\sim} A_0, \qquad \vartheta(z + B_0\bar\lambda) = \Psi^+_{\bar\lambda}\vartheta(z). \tag{C.8}$$

Local finiteness of the fan makes this bijection proper, hence a homeomorphism. Thus A_0 is contractible.

The action in (C.5) is the already verified cusp action with constant twist $C^+ = (2\pi i)^{-1}R(0)$, since $e^{2\pi i C^+\bar\lambda} = a_{\bar\lambda}$; hence it is free and properly discontinuous. Properness of its quotient over the disc, proved by the same bounded fundamental-domain argument as theorem 8.4, makes Q^+ compact. Its two boundary components are $H \simeq \mathbb{R}^2/B_0\bar\Lambda$ and $\mathbb{R}^2/B^+_\eta\bar\Lambda$, both tori. Since a free properly discontinuous action preserves the boundary charts, Q^+ is a topological 3-manifold with boundary; Brown's collaring theorem makes $H \hookrightarrow Q^+$ collared. The covers $A \to Q^+$ and $A_0 \to H$ have the same deck group, and the equivariant inclusion identifies that group by the identity. The result is now theorem 44.1. □

Lemma 44.4 (spreading the positive retraction by phases)**.** ***[VERIFIED]*** *The lifted retraction P of theorem 44.3 spreads to a Γ^0- and T_c-equivariant strong deformation retraction of $Y_{\le\eta}$ onto Y_0. Conjugating by G gives a Γ- and T_f-equivariant strong deformation retraction, and hence a strong deformation retraction*

$$r_s : N_0' = Y_{\le\eta}/\Gamma \longrightarrow W = Y_0/\Gamma. \tag{C.9}$$

Proof. For $\phi \in T_c$ and $q \in A$ set

$$R_s(\phi q) = \phi P_s(q). \tag{C.10}$$

This is well defined. On the open orbit and the outer boundary the stabilizer is trivial; on A_0 the homotopy fixes q, so (C.7) causes no ambiguity. It is continuous because its pullback along the closed quotient map $m \times \mathrm{id}_{[0,1]}$ is continuous.

The frozen and positive actions satisfy

$$\Psi^0_{\bar\lambda}(\phi q) = u_{\bar\lambda}\varphi_{\bar\lambda}(\phi)\Psi^+_{\bar\lambda}(q), \tag{C.11}$$

where $\varphi_{\bar\lambda}$ is the compact-torus automorphism induced by $(z, z_3) \mapsto (z + z_3 B_0\bar\lambda, z_3)$. Equation (C.11) and Γ^+-equivariance of P prove Γ^0-equivariance of R. The remaining deformation-retraction identities follow directly from those of P. Finally theorem 44.2 conjugates R to the original action and permits descent to (C.9). □

Proposition 44.5 (fixed-fibre cellular collapse)**.** ***[VERIFIED]*** *Fix t_0 with $0 < |t_0| \le \eta$ and put $F = f^{-1}(t_0)$. The retraction in theorem 44.4 can be chosen so that its end map $r : N_0' \to W$ restricts to a cellular collapse $c = r|_F$ with the following description. There is a splitting*

$$F \xrightarrow[\bar G]{\sim} F^0_{t_0} \simeq T_B \times T_f, \tag{C.12}$$

where $\bar G$ is the identity on the marking Λ, and the quotient honeycomb $\Theta \subset T_B$ has two vertices and three edges e_k. Over a vertex, c collapses all of T_f to a triple point of W; over e_k° it collapses the primitive circle $S^1_{n_k}$ and maps onto the double curve $\bar C_k$; over $T_B \setminus \Theta$ it is injective onto $W \setminus D$.

Proof. Let $S \subset A$ be the positive slice $|t| = |t_0|$ and put

$$B^+_\rho = B_0 + \frac{R(0)}{\log\rho}, \qquad \ell_\rho = B_0(B^+_\rho)^{-1}.$$

The map $y : S \to \mathbb{R}^2$ identifies the positive action with translations $y \mapsto y + B^+_{|t_0|}\bar\lambda$, while $\ell_{|t_0|}$ conjugates these to $z \mapsto z + B_0\bar\lambda$. Define on the frozen fibre

$$\widetilde c((\phi, \arg t_0)q) = (\phi, \arg t_0)\,\vartheta(\ell_{|t_0|}y(q)), \qquad \phi \in T_f, q \in S. \tag{C.13}$$

The stabilizer on S is trivial, so the formula is well defined. Equations (C.8) and (C.11) make it Γ^0-equivariant, hence it descends to $c_\vartheta : F^0_{t_0} \to W$.

Writing

$$K_{t_0} = \mathrm{Re}\, C_0 + \frac{\arg t_0}{2\pi}B_0,$$

the frozen action on $T_f \times \mathbb{R}^2_y$ is

$$\bar\lambda(\phi, y) = (\phi e^{2\pi i K_{t_0}\bar\lambda}, y + B^+_{|t_0|}\bar\lambda).$$

The change of variables

$$\Xi(\phi, y) = (\phi e^{-2\pi i K_{t_0}(B^+_{|t_0|})^{-1}y}, y) \tag{C.14}$$

conjugates this to the product action and gives (C.12). Formula (C.7) now proves the asserted cell description, including primitivity of the collapsed circle.

It remains to force the end map of an actual strong deformation retraction to equal (C.13), rather than merely be homotopic to it. In the chart $\vartheta : \mathbb{R}^2 \to A_0$ put

$$\widehat{a} = \vartheta^{-1} P_1.$$

Put $\rho_0 = |t_0|$. If $\rho_0 < \eta$, choose a closed positive band $\{\rho_- \le |t| \le \rho_+\}$, disjoint from A_0, with ρ_0 in its interior. If $\rho_0 = \eta$, instead use the one-sided band $\{\rho_- \le |t| \le \eta\}$. Define $\widehat{b} = \ell_{|t|} y$ only on the chosen band. Wherever they are defined, both maps satisfy $\widehat{a}(\Psi^+_{\bar\lambda} q) = \widehat{a}(q) + B_0 \bar\lambda$ and the same identity for $\widehat{b}$. Choose a continuous tent function $\chi(|t|)$, extended by zero off the band, with $\chi(\rho_0) = 1$ and vanishing near every boundary of the band across which it must be pasted; in the one-sided case no vanishing is required at the ambient boundary $|t| = \eta$. With $w(q) = \chi(|t(q)|)$ define

$$K_u(q) = \vartheta\big((1 - uw(q))\widehat{a}(q) + uw(q)\widehat{b}(q)\big) \tag{C.15}$$

inside the band and $K_u = P_1$ outside. Because the transition action is by translations, affine interpolation preserves Γ^+-equivariance. The weight vanishes near every gluing boundary and near A_0, so (C.15) pastes continuously, fixes A_0, and takes values in A_0.

Concatenate the first half of P with K_u. The result P' still satisfies all four properties used in theorem 44.4, and its end map on S is exactly $\vartheta \ell_{|t_0|} y$. Spreading P' by phases and conjugating by G therefore yields a strong deformation retraction whose restriction to F is

$$r|_F = c_\vartheta \circ \bar G.$$

This proves the proposition, including continuity across all toric strata. □

44.3 The integral collapse map

Let $D = \bar C_1 \cup \bar C_2 \cup \bar C_3$ be the double locus. Choose primitive orientations of its three curve classes and of the honeycomb edges so that the annulus degrees below have the displayed signs. These orientations are compared with the clockwise collar convention in theorems 44.10 and 44.11. In $\Lambda_{\mathrm{tor}} = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle$ put

$$n_1 = \widehat{w}, \qquad n_2 = \widehat{\delta} - \widehat{w}, \qquad n_3 = -\widehat{\delta}, \qquad \nu_k(v) = \det(n_k, v). \tag{C.16}$$

Proposition 44.6 (integral homology of the cellular collapse)**.** ***[VERIFIED]*** *The map c_* of theorem 44.5 is surjective in every degree, and*

$$0 \longrightarrow \mathrm{im}(\textstyle\bigwedge^q M_0 - I) \longrightarrow \bigwedge^q \Lambda \xrightarrow{c_*} H_q(W;\mathbb{Z}) \longrightarrow 0 \tag{C.17}$$

is an exact sequence of free abelian groups. In the splitting $\Lambda = \bar\Lambda \oplus \Lambda_{\mathrm{tor}}$,

$$c_* : \Lambda \to H_1(W) = \bar\Lambda$$

is projection, and in degree two

$$c_*(\widehat{\gamma}\widehat{u}) = [\mathbb{T}_B], \qquad c_*(\widehat{w}\widehat{\delta}) = 0, \tag{C.18}$$

$$c_*(\beta \otimes v) = \sum_{k=1}^{3} \nu_k(B_0\beta)\nu_k(v)[\bar C_k]. \tag{C.19}$$

Equivalently, in the ordered source basis

$$(\widehat{\gamma}\widehat{u}, \widehat{\gamma}\widehat{w}, \widehat{\gamma}\widehat{\delta}, \widehat{u}\widehat{w}, \widehat{u}\widehat{\delta}, \widehat{w}\widehat{\delta})$$

and target basis $([\mathbb{T}_B], [\bar C_1], [\bar C_2], [\bar C_3])$,

$$[c_*]_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{C.20}$$

In degree three,

$$c_*(\widehat{\gamma}\,\widehat{u}\,v) = v \quad (v \in \Lambda_{\mathrm{tor}}), \qquad c_*(\beta\,\widehat{w}\widehat{\delta}) = 0 \quad (\beta \in \bar{\Lambda}), \tag{C.21}$$

where the first formula fixes the displayed orientation of $H_3(W)$.

Proof. First identify the target integrally. The normalization $\nu : \mathrm{dP}_6 \to W$ identifies opposite sides of the anticanonical hexagon. A plumbing neighbourhood of that hexagon descends to a regular neighbourhood of D. If P_0 is its complement in dP_6, then $P_0 \simeq (\mathbb{C}^*)^2 \simeq T^2$, and excision followed by Lefschetz duality gives

$$H_k(W, D) \simeq H_k(P_0, \partial P_0) \simeq H^{4-k}(T^2). \tag{C.22}$$

The class $[\mathbb{T}_B] = c_*[T_B \times \{*\}]$ pairs once with a compact T_f-orbit in $W \setminus D$, so it lifts the generator of $H_2(W, D)$. Likewise the cycles $c_*(T_B \times S^1_{\widehat{w}})$ and $c_*(T_B \times S^1_{\widehat{\delta}})$ pair unimodularly with the two coordinate circles in T_f. Thus the boundary maps of the pair vanish and

$$H_*(W) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}), \tag{C.23}$$

with $H_2(W) = \mathbb{Z}\langle[\bar{C}_1], [\bar{C}_2], [\bar{C}_3], [\mathbb{T}_B]\rangle$ and $H_3(W)$ generated by the two cycles in (C.21).

The cell description gives projection in degree one and (C.18). Give T_B the CW structure with one open hexagonal 2-cell and 1-skeleton Θ. A class $\beta \in H_1(T_B)$ has the unique cellular representative

$$\sum_{k=1}^{3} \nu_k(B_0\beta) e_k.$$

On $e_k \times S^1_v$, the collapse kills $S^1_{n_k}$ and maps the quotient circle to $\bar{C}_k$ with degree $\nu_k(v)$. This proves (C.19), without a rank inference. The three choices $(B_0\beta, v) = (\widehat{w}, \widehat{w}), (\widehat{\delta}, \widehat{\delta}), (\widehat{w}, \widehat{\delta})$ give coefficient vectors

$$(0, 1, 1), \qquad (1, 1, 0), \qquad (0, 1, 0),$$

whose determinant is a unit. Hence degree two is surjective. Formula (C.21) follows directly by crossing the base 2-cell with a primitive fibre circle; it proves degree-three surjectivity and records the orientations. In degree four the top-dimensional open stratum $(T_B \setminus \Theta) \times T_f$ maps homeomorphically to the open 4-stratum of W, so the map is multiplication by a unit after choosing the target orientation. Degree zero is immediate.

Let h be geometric monodromy around the cusp. Fibre transport in N_0' is a homotopy from the inclusion of F to its composite with h. Composing with r yields $c \simeq c \circ h$, and therefore

$$\mathrm{im}(\textstyle\bigwedge^q M_0 - I) \subseteq \ker c_*. \tag{C.24}$$

The inclusion is an equality integrally, not only rationally. In the ordered basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$, direct exterior multiplication gives the primitive image lattices

q	$\mathrm{im}(\bigwedge^q M_0 - I)$
$0, 4$	0
1	$\langle \widehat{w}, \widehat{\delta} \rangle$
2	$\langle \widehat{w}\widehat{\delta}, \widehat{\gamma}\widehat{w} + \widehat{u}\widehat{\delta} \rangle$
3	$\langle \widehat{\gamma}\widehat{w}\widehat{\delta}, \widehat{u}\widehat{w}\widehat{\delta} \rangle$.

(C.25)

Each displayed subgroup is a direct summand. Its rank is $\binom{4}{q} - b_q(W)$ by (C.23), while surjectivity gives the same rank for $\ker c_*$. The quotient $H_q(W)$ is free, so $\ker c_*$ is saturated. Two saturated sublattices of the same rank and rational span coincide, proving (C.17). □

44.4 Comparison with analytic nearby cycles

Theorem 44.7 (collapse equals nearby-cycle specialization)**.** ***[VERIFIED]*** *With the radial marking of the punctured cusp family, for every $0 \le q \le 4$ one has an equality of integral homomorphisms*

$$\mathrm{sp}^q = i_F^* r^* = c^* : H^q(W;\mathbb{Z}) \longrightarrow H^q(F;\mathbb{Z}). \tag{C.26}$$

Consequently the cellular calculation gives

$$\operatorname{im} c^* = (\textstyle\bigwedge^q V)^{T_0}$$

as a saturated lattice. In particular, the degree-three orientation in the cellular computation is the degree-three orientation of intrinsic specialization; there is no independent sign or index.

Proof. Let $N_\rho = f^{-1}(|t| \le \rho) \subset N_0'$ and let $i_{W,\rho} : W \hookrightarrow N_\rho$ be the central-fibre inclusion. The closed tubes are cofinal among neighbourhoods of W: if an open set $\mathcal{O} \subset N_0'$ contains W, properness makes $f(N_0' \setminus \mathcal{O})$ closed and disjoint from 0, so $N_\rho \subset \mathcal{O}$ for all sufficiently small ρ. For each such ρ, choose t_ρ on the radial ray through the fixed point t_0 whose fibre is F, with $0 < |t_\rho| < \rho$, and write $i_{t_\rho} : F_{t_\rho} \hookrightarrow N_\rho$. Identify $H^q(F_{t_\rho};\mathbb{Z})$ with $H^q(F;\mathbb{Z})$ by parallel transport along that ray. The neighborhood realization of the unit $\mathbb{Z}_W \to \psi_f\mathbb{Z}$, together with proper base change, identifies specialization with

$$H^q(W;\mathbb{Z}) \xleftarrow[\sim]{\rho_0} \varinjlim_{\rho \downarrow 0} H^q(N_\rho;\mathbb{Z}) \xrightarrow{\rho_{\mathrm{near}}} H^q(F;\mathbb{Z}), \tag{C.27}$$

where ρ_0 is induced by $i_{W,\rho}^*$ and ρ_{near} is induced by the transported maps $i_{t_\rho}^*$. These restriction maps are compatible under shrinking ρ. This is the same analytic integral construction used in theorem 39.1; see [32, Definition 4.2.1 and Propositions 4.2.2, 4.2.11].

Applying the construction of theorem 44.4 at radius ρ shows that $i_{W,\rho}^*$ is an isomorphism; no compatibility of those radius-dependent deformation homotopies is needed. Let $j_\rho : N_\rho \hookrightarrow N_0'$ be inclusion. The single retraction $r : N_0' \to W$ satisfies

$$i_{W,\rho}^* j_\rho^* r^* = (r \circ j_\rho \circ i_{W,\rho})^* = \mathrm{id}_{H^q(W;\mathbb{Z})}.$$

Hence $j_\rho^* r^*$ is the unique inverse of $i_{W,\rho}^*$. These inverses are compatible under shrinking because they are restrictions of the single map $r^* : H^q(W;\mathbb{Z}) \to H^q(N_0';\mathbb{Z})$. Consequently $\rho_0^{-1}(\alpha)$ in (C.27) is represented, for every sufficiently small ρ, by $j_\rho^* r^* \alpha$.

The restrictions of one class on N_0' to the fibres along the punctured radial ray form a section of the Gauss–Manin local system. Therefore, after the parallel-transport identifications used to define ρ_{near}, $i_{t_\rho}^* j_\rho^* r^* \alpha$ equals $i_F^* r^* \alpha$. It follows that the composite in (C.27) is $i_F^* r^*$. Since $c = r|_F$, functoriality gives $i_F^* r^* = (r \circ i_F)^* = c^*$.

Changing the lift of the radial marking composes the target with monodromy. By (C.24), c^* lands in the monodromy-invariant lattice, so the equality is independent of that harmless choice. Finally theorem 39.1 identifies the common image with the full saturated invariant lattice. □

Validation, repair, or disproof test 44.8 (resolved cusp-collapse falsifiers)**.** **[VERIFIED]** The audit tested the failure modes previously attached to C1–C3: continuity at every toric stratum, finite stabilizer components, equivariance, primitive edge circles, degree-two and degree-three orientations, torsion and saturation, and functorial equality with the nearby-cycle unit. The proofs above exclude each one. This status concerns the cusp collapse only; it does not infer the global Mayer–Vietoris matrices.

Audit record 44.9 (V1: based van Kampen inclusions)**.** **[VERIFIED]** The actual overlap coordinate maps, with the zero section as base locus and clockwise meridians, show that the finite fillings impose $\widehat{\rho}_j^{\,m_j} = t_{v_j}$ with the same sign, the cusp filling kills Λ_{tor} and its toric meridian, and the tautological gluing has zero translation drift. Thus the abstract terminal presentation is the actual $\pi_1(X)$. The deck generators, common clockwise convention, zero-section disc, normal closure, boundary inclusion $\Lambda \to \Lambda/\Lambda_{\mathrm{tor}}$, $\widetilde{a}_0 \mapsto 1$, and absence of extra whisker drift are proved in theorem 39.6.

44.5 The actual finite-band Mayer–Vietoris maps

Fix the ordered orientation

$$H_1(F) = \Lambda = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle$$

dual to (γ, u, w, δ), use the complex orientations of F, S_1, S_2, and orient every base meridian clockwise. Put

$$\eta_1 = 2u + w + 3\delta, \qquad \eta_2 = u + w + 2\delta, \qquad q = uw + 6\gamma\delta. \tag{MV.1}$$

Proposition 44.10 (actual finite-band maps in primitive bases)**.** ***[VERIFIED]*** *In the decomposition $X^\circ = X_1 \cup X_2$, the maps denoted α_r in the manuscript are the actual Mayer–Vietoris maps*

$$\alpha_r = (\pi_{1*}, -\pi_{2*}) : H_r(F;\mathbb{Z}) \longrightarrow H_r(S_1;\mathbb{Z}) \oplus H_r(S_2;\mathbb{Z}). \tag{MV.2}$$

There is no conjugacy, unrecorded index, or independent orientation sign in (MV.2).

Choose primitive bases

$$H_1(S_1) = \mathbb{Z}\langle g_1, h_1 \rangle, \quad [\widehat{\gamma}]_1 = 3g_1, \quad h_1 = \pi_{1*}\widehat{w},$$

$$H_1(S_2) = \mathbb{Z}\langle g_2, h_2 \rangle, \quad [\widehat{\gamma}]_2 = -4g_2, \quad h_2 = \pi_{2*}\widehat{w}.$$

Then, in the source basis $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$ and the target basis (g_1, h_1, g_2, h_2),

$$[\alpha_1] = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \\ 4 & 0 & 0 & 0 \\ 0 & -1 & -1 & -2 \end{pmatrix}. \tag{MV.3}$$

It has Smith form $\operatorname{diag}(1,1,1,0)$ *and*

$$\ker \alpha_1 = \mathbb{Z}\nu_0, \qquad \nu_0 = -\widehat{u} - \widehat{w} + \widehat{\delta}. \tag{MV.4}$$

Define cohomology bases (a_j, b_j) by

$$\pi_1^* a_1 = \gamma\eta_1, \qquad \pi_1^* b_1 = q - 2\gamma\eta_1,$$

$$\pi_2^* a_2 = 2\gamma\eta_2, \qquad \pi_2^* b_2 = q - 3\gamma\eta_2, \tag{MV.5}$$

and let $(a_j^\sharp, b_j^\sharp)$ be the Kronecker-dual homology bases. In the ordered source basis

$$(\widehat{\gamma}\widehat{u}, \widehat{\gamma}\widehat{w}, \widehat{\gamma}\widehat{\delta}, \widehat{u}\widehat{w}, \widehat{u}\widehat{\delta}, \widehat{w}\widehat{\delta})$$

and target basis $(a_1^\sharp, b_1^\sharp, a_2^\sharp, b_2^\sharp)$,

$$[\alpha_2] = \begin{pmatrix} 2 & 1 & 3 & 0 & 0 & 0 \\ -4 & -2 & 0 & 1 & 0 & 0 \\ -2 & -2 & -4 & 0 & 0 & 0 \\ 3 & 3 & 0 & -1 & 0 & 0 \end{pmatrix}. \tag{MV.6}$$

This matrix also has Smith form $\operatorname{diag}(1,1,1,0)$*; its primitive left kernel is*

$$\Phi = (4, 2, 3, 2). \tag{MV.7}$$

The class $\mathcal{A} \in H^2(X^\circ;\mathbb{Z})$ represented by Φ restricts to $2q$, because

$$2q = 4\pi_1^* a_1 + 2\pi_1^* b_1 = 3\pi_2^* a_2 + 2\pi_2^* b_2. \tag{MV.8}$$

If the degree-three bases are Poincaré dual to the degree-one cohomology bases just fixed, the projection formula makes $[\alpha_3]$ the same matrix as (MV.3). Finally

$$[\alpha_0] = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad [\alpha_4] = \begin{pmatrix} 3 \\ -4 \end{pmatrix}. \tag{MV.9}$$

Every image in (MV.3), (MV.6), and (MV.9) is saturated.

Proof. Choose closed discs D_1', D_2' whose intersection is a contractible band disjoint from the three special points. Over the annular shell between D_j' and the smaller filling disc, f is a proper submersion. An Ehresmann connection lifts a base deformation retraction that is stationary near the smaller disc; adjoining the identity there gives a deformation retraction $X_j \to N_j$. The verified local radial retraction of theorem 8.8 keeps the marked flat coordinate x fixed and sends its endpoint to the orbit $\pi_j(x) \in S_j$. Since the band is contractible, choose its marking by the same radial paths. The composite of the actual band inclusion with $X_j \to N_j \to S_j$ is therefore literally π_j. The standard chain-level Mayer–Vietoris injection is $z \mapsto (i_{1*}z, -i_{2*}z)$ [9, §2.2], proving (MV.2) with its sign.

The affine fundamental-group presentations of the finite fibres and the exact finite pullback lattices of theorem 5.4 give the stated primitive degree-one bases and (MV.3). Its 3×3 minors include 3 and 4, so their gcd is one; direct elimination gives (MV.4). The two bases in (MV.5) are unimodular changes of

$$\langle \gamma\eta_1, q \rangle, \qquad \langle 2\gamma\eta_2, \gamma\eta_2 + q \rangle,$$

the exact integral images of π_j^* in degree two. Evaluating them on the six source tori gives (MV.6). Its rank-three minors include 2 and -3, so the image is saturated; multiplication verifies (MV.7) and (MV.8).

For degree three, if $x \in H_3(F)$ and $\theta \in H^1(S_j)$, then

$$\langle \theta, \pi_{j*}x \rangle = \langle \pi_j^*\theta, x \rangle.$$

Thus in the Poincaré-dual bases the degree-three matrix equals the degree-one matrix. In degree four, the holomorphic quotient coverings have positive degrees 3 and 4; the sole minus sign is the Mayer–Vietoris minus sign. This proves (MV.9) and completes the integral certificate. □

44.6 The cusp attachment and its primitive blocks

Proposition 44.11 (actual cusp kernels and global blocks)**.** ***[VERIFIED]*** *For the second decomposition $X = N_0' \cup X^\circ$, the map $H_r(M) \to H_r(W)$ is surjective and*

$$K_1 = \mathbb{Z}[\widetilde{a}_0], \qquad K_2 = \mathbb{Z}\langle \xi_{\widehat{w}}, \xi_{\widehat{\delta}} \rangle, \tag{MV.10}$$

where

$$\xi_\nu = a_*([S^1_\nu] \times [\widetilde{a}_0])$$

uses the oriented fibrewise translation action and the clockwise toric meridian. Both displayed subgroups are primitive.

Let ζ be the generator of $H_1(X^\circ)$ normalized by $\Phi_1(\zeta) = 1$. In the source basis $(\widehat{\gamma}, \widehat{u}, \widetilde{a}_0)$ of $H_1(M)$ and target basis $(\bar{\gamma}, \bar{u}, \zeta)$ of $H_1(W) \oplus H_1(X^\circ)$, the actual map $(c_*, -j_*)$ is

$$D_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -12 & 0 & -p \end{pmatrix}, \qquad \det D_1 = -p. \tag{MV.11}$$

Orient the connecting map by

$$\partial(j_*\xi_\nu) = (I - A_2)\nu = (I - A_1^{-1})\nu \quad (\nu \in \Lambda_{\mathrm{tor}}). \tag{MV.12}$$

Let e be the primitive generator of $C_2 := \operatorname{coker}\alpha_2$ with $\langle \mathcal{A}, e\rangle = 1$, and put $z = j_*\xi_{\widehat{w}}$. Then (e, z) is a basis of $H_2(X^\circ)$ and

$$j_*\xi_{\widehat{\delta}} = -p\,e, \qquad [j_*|_{K_2}]^{(e,z)}_{(\xi_{\widehat{\delta}},\xi_{\widehat{w}})} = \begin{pmatrix} -p & 0 \\ 0 & 1 \end{pmatrix}. \tag{MV.13}$$

After splitting the surjections onto $H_r(W)$, the full degree-one and degree-two Mayer–Vietoris maps are integrally equivalent to

$$I_2 \oplus [p], \qquad I_4 \oplus \begin{pmatrix} -p & 0 \\ 0 & 1 \end{pmatrix}, \tag{MV.14}$$

respectively. They are unimodular for the candidate $p = -1$.

Proof. The composite $H_r(F) \to H_r(M) \to H_r(W)$ is the verified collapse c_*, which is surjective by theorem 44.6; hence the second arrow is surjective. Under the Wang projection, the base class $[\widetilde{a}_0]$ is primitive. It bounds the explicit zero-section disc in the toric filling, proving the first formula in (MV.10).

For K_2, use the toric chart $t = z_1z_2z_3$. The literal solid tori

$$\{|z_1| \le \rho,\ |z_2| = r,\ z_3 = c\} \quad \text{and} \quad \{|z_2| \le \rho,\ |z_1| = r,\ z_3 = c\}$$

have oriented boundaries $\xi_{\widehat{w}}$ and $\xi_{\widehat{\delta}}$. Under

$$H_2(M) \twoheadrightarrow \ker(M_0 - I|\Lambda) = \Lambda_{\mathrm{tor}},$$

these classes map to $\widehat{w}$ and $\widehat{\delta}$. Their span is therefore primitive; the Wang ranks and the verified homology of W show that it is all of K_2.

The actual meridian calculation gives

$$j_*\widehat{\gamma} = 12\zeta, \qquad j_*\widehat{u} = 0, \qquad j_*[\widetilde{a}_0] = p\zeta,$$

while c_* is projection onto $\mathbb{Z}\langle\bar{\gamma}, \bar{u}\rangle$. This is (MV.11). Because $A_1A_2 = M_0^{-1}$ acts trivially on Λ_{tor}, the two expressions in (MV.12) agree. Since $A_2\widehat{w} = -\widehat{u} + \widehat{\delta}$ and $A_2\widehat{\delta} = \widehat{\delta}$,

$$\partial(j_*\xi_{\widehat{w}}) = -\nu_0, \qquad \partial(j_*\xi_{\widehat{\delta}}) = 0. \tag{MV.15}$$

Thus z is a primitive lift, up to the fixed unit sign, of the generator of $\ker\alpha_1$.

It remains to determine the coefficient of e. Normalize contraction by

$$\iota_v(\theta_1 \wedge \cdots \wedge \theta_k) = \sum_r (-1)^{r-1}\theta_r(v)\theta_1 \wedge \cdots \widehat{\theta_r} \cdots \wedge \theta_k.$$

The globally free $S^1_{\widehat{\delta}}$-action and the sweep/slant identity give

$$\langle \mathcal{A}, j_*\xi_{\widehat{\delta}}\rangle = \langle \iota_{\widehat{\delta}}\mathcal{A}, j_*[\widetilde{a}_0]\rangle, \qquad \iota_{\widehat{\delta}}(2q) = -12\gamma. \tag{MV.16}$$

Let $\Theta \in H^1(X^\circ)$ be the primitive class restricting to 12γ. Restriction to F is injective, so (MV.16) implies $\iota_{\widehat{\delta}}\mathcal{A} = -\Theta$. The based meridian relation gives

$$\Theta(j_*[\widetilde{a}_0]) = p.$$

Hence the left side of (MV.16) is $-p$, which proves (MV.13). Splitting the surjective cusp blocks gives (MV.14). □

Remark 44.12 (one-shot H_1 check). The preceding two-stage calculation is the manuscript's argument. There is also a useful one-shot check. Let U be the torus bundle over the pair-of-pants core and $N = N_0' \sqcup N_1' \sqcup N_2'$. In the target basis

$$(c, \rho_1, \rho_2; \bar{\gamma}, \bar{u}; g_1, h_1; g_2, h_2)$$

and domain basis

$$(\gamma_0, u_0, a_0; \gamma_1, w_1, \rho_1; \gamma_2, w_2, \rho_2),$$

the actual map $(i_{U*}, -i_{N*})$ for the candidate is

$$\Delta_1 = \begin{pmatrix} 1 & 0 & \ell_0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}. \tag{MV.17}$$

For example, a_0 maps to $\ell_0 c - \rho_1 - \rho_2$ on the pair-of-pants side and to zero on the cusp side; γ_1 maps to c and $3g_1$, while γ_2 maps to c and $-4g_2$. Thus every column is an actual primitive collar cycle. Exact elimination gives

$$\det \Delta_1 = 1 - 12\ell_0 = -p$$

for $(\ell_1, \ell_2) = (1, -1)$. At $\ell_0 = 0$, this determinant is one.

44.7 Exact comparison with the Leray transgressions

Theorem 44.13 (Mayer–Vietoris/Leray comparison)**.** ***[VERIFIED]*** *With the preceding primitive bases and clockwise meridians, the actual Mayer–Vietoris obstruction maps commute with the intrinsic Leray transgressions in degrees one, two, and three. If $\omega_{\mathrm{MV}} \in H^2(B; \mathbb{Z})$ is the generator fixed below, then*

$$d_2^{0,1}(12\gamma) = -p\,\omega_{\mathrm{MV}}, \tag{L.1}$$

$$d_2^{0,2}(2q) = -p\,\omega_{\mathrm{MV}}[\delta], \qquad d_2^{0,3}(2\gamma uw) = -p\,\omega_{\mathrm{MV}}[\gamma\delta]. \tag{L.2}$$

For the candidate $p = -1$, all three maps are multiplication by $+1$ in these bases.

Proof. Let V be the pair-of-pants neighbourhood in the base and let D_i be the three special discs. Order the annuli as $(D_1 \cap V, D_2 \cap V, D_0 \cap V)$ and orient their meridians clockwise. Define

$$\omega_{\mathrm{MV}} = \partial_B[(1, 0, 0)].$$

The chain-level Mayer–Vietoris sequence on the base gives

$$\partial_B(n_1, n_2, n_0) = (n_1 + n_2 + n_0)\omega_{\mathrm{MV}}, \tag{L.3}$$

because the image of $H^1(V)$ is the sum-zero sublattice.

Choose a lift Θ_V of 12γ over the pair of pants and write $b_i = \Theta_V(\widehat{\rho}_i)$. The finite local lifts have meridian values $4\ell_1$ and $3\ell_2$, while the cusp lift vanishes on $\widetilde{a}_0$. For the general affine cusp regluing used in this comparison, let t_μ denote translation by the class $\mu \in \Lambda/\Lambda_{\mathrm{tor}}$ and set $\ell_0 = \gamma(\mu)$. Thus $\Theta_V(t_\mu) = 12\ell_0$.

The tautological gluing of the constructed candidate has $\mu = 0$ and $\ell_0 = 0$; in that case the following relation has no vertical drift. In general,

$$\widetilde{a}_0 = t_\mu \widehat{\rho}_2^{-1} \widehat{\rho}_1^{-1},$$

the local-minus-pair-of-pants discrepancy is

$$(4\ell_1 - b_1,\ 3\ell_2 - b_2,\ -(12\ell_0 - b_1 - b_2)). \tag{L.4}$$

Its signed sum is $-p$. The canonical truncation triangle identifies this very cocycle, functorially on the induced total-space cover, with Leray d_2; this is the construction proved in theorem 39.3. Equations (L.3)–(L.4) prove (L.1).

This is more than equality of the terminal integer. The transpose of the oriented collar map applied to the tuple of local lifts has meridian values

$$(b_1 - 4\ell_1,\ b_2 - 3\ell_2,\ 12\ell_0 - b_1 - b_2),$$

the negative of (L.4). Consequently the square

$$\begin{array}{ccc} \mathbb{Z}\langle 12\gamma \rangle & \xrightarrow{\ d_2^{0,1}\ } & \mathbb{Z}\langle \omega_{\mathrm{MV}} \rangle \\ \cong\big\downarrow & & \big\downarrow\cong \\ H^1(X^\circ;\mathbb{Z}) & \xrightarrow{(j_*|_{K_1})^\vee} & K_1^\vee \end{array} \tag{L.5}$$

commutes when the left vertical map sends 12γ to Θ and the right sends ω_{MV} to $-\widetilde{a}_0^\vee$. The lower map is multiplication by p, while the upper map is multiplication by $-p$.

For degree two, quotient the primitive lift in (MV.15):

$$\overline{K}_2 = K_2/\mathbb{Z}\xi_{\widehat{w}}, \qquad C_2 = \operatorname{coker}\alpha_2.$$

Equation (MV.13) gives an actual reduced map

$$\bar{j} : \overline{K}_2 \longrightarrow C_2, \qquad [\xi_{\widehat{\delta}}] \longmapsto -p\, e. \tag{L.6}$$

Under the primitive identifications

$$2q \longmapsto \Phi \in C_2^\vee, \qquad \omega_{\mathrm{MV}}[\delta] \longmapsto \xi_{\widehat{\delta}}^{\ \vee} \in \overline{K}_2^\vee,$$

the square

$$\begin{array}{ccc} \mathbb{Z}\langle 2q \rangle & \xrightarrow{\ d_2^{0,2}\ } & \mathbb{Z}\langle \omega_{\mathrm{MV}}[\delta] \rangle \\ \cong\big\downarrow & & \big\downarrow\cong \\ C_2^\vee & \xrightarrow{\ \bar{j}^\vee\ } & \overline{K}_2^\vee \end{array} \tag{L.7}$$

commutes, and both horizontal maps are multiplication by $-p$. This proves the first formula in (L.2).

The final formula has no new sign choice. Write

$$d_2(2q) = r\,\omega_{\mathrm{MV}}[\delta], \qquad d_2(2\gamma uw) = s\,\omega_{\mathrm{MV}}[\gamma\delta].$$

Using

$$(12\gamma)(2q) = 12(2\gamma uw), \qquad (2q)(2\gamma uw) = 0,$$

the derivation rule and the already fixed degree-one sign give

$$s = 2(-p) - r.$$

The identities

$$\delta \wedge 2\gamma uw = -2\,\mathrm{vol}, \qquad 2q \wedge \gamma\delta = 2\,\mathrm{vol}$$

applied to the second product give $r = s$. Hence $r = s = -p$, proving the second formula in (L.2).

Changing every meridian orientation reverses p and ω_{MV} simultaneously. Reversing the $S^1_{\widehat{\delta}}$ orientation reverses both $\xi_{\widehat{\delta}}$ and its dual coefficient generator. Thus all remaining convention changes are common units and cannot change an index. □

Validation, repair, or disproof test 44.14 (resolved global topological falsifiers)**.** **[VERIFIED]** The actual finite-band maps are (MV.2), their degree-one and degree-two images are saturated, the multiple-fibre top class is $(3, -4)$, the cusp kernels are literal primitive collar cycles, the reduced cusp blocks are (MV.11) and (MV.13), and the comparison is the pair of commuting squares (L.5) and (L.7), not a comparison of absolute determinants. The former MV1 and L1 failure modes are therefore excluded for the candidate.

44.8 Complete recognition theorem

Lemma 44.15 (cohomological closure in dimension six)**.** ***[VERIFIED]*** *Let X be a smooth, closed, connected, oriented six-manifold. If*

$$\pi_1(X) = 0, \qquad H^2(X;\mathbb{Z}) = H^3(X;\mathbb{Z}) = 0, \tag{4.4}$$

then $H_k(X;\mathbb{Z})$ is $\mathbb{Z}$ for $k = 0, 6$ and zero for $1 \le k \le 5$.

Proof. The Hurewicz map in degree one identifies $H_1(X;\mathbb{Z}) = \pi_1(X)^{\mathrm{ab}} = 0$. The universal coefficient sequence

$$0 \longrightarrow \mathrm{Ext}(H_1(X),\mathbb{Z}) \longrightarrow H^2(X;\mathbb{Z}) \longrightarrow \mathrm{Hom}(H_2(X),\mathbb{Z}) \longrightarrow 0$$

shows that $\mathrm{Hom}(H_2(X),\mathbb{Z}) = 0$, so the finitely generated group $H_2(X)$ is torsion. The next universal coefficient sequence contains an injection

$$\mathrm{Ext}(H_2(X),\mathbb{Z}) \hookrightarrow H^3(X;\mathbb{Z}) = 0.$$

For a finitely generated torsion group $A = \bigoplus_i \mathbb{Z}/n_i$, one has $\mathrm{Ext}(A,\mathbb{Z}) = \bigoplus_i \mathbb{Z}/n_i$; hence $H_2(X) = 0$. Poincaré duality gives

$$H_3(X) \simeq H^3(X) = 0, \quad H_4(X) \simeq H^2(X) = 0, \quad H_5(X) \simeq H^1(X) = 0.$$

Finally connectedness and orientation give $H_0(X) = H_6(X) = \mathbb{Z}$. □

Theorem 44.16 (sphere recognition from the exact inputs)**.** ***[VERIFIED]*** *Let X be a smooth, closed, connected six-manifold. If*

$$\pi_1(X) = 0, \qquad H_k(X;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \le k \le 5, \end{cases} \tag{4.5}$$

then X is diffeomorphic to the standard S^6.

Proof. A closed smooth manifold has the homotopy type of a finite CW complex. We first prove successively that $\pi_i(X) = 0$ for $1 \le i \le 5$. The case $i = 1$ is assumed. Suppose $2 \le n \le 5$ and $\pi_i(X) = 0$ for $i < n$. Then X is $(n-1)$-connected, so the Hurewicz homomorphism

$$h_n : \pi_n(X) \longrightarrow H_n(X;\mathbb{Z})$$

is an isomorphism by the Hurewicz theorem [9, Theorem 4.32]. The target vanishes by (4.5), hence $\pi_n(X) = 0$. Induction proves the assertion.

Now X is 5-connected. Hurewicz in degree six gives $\pi_6(X) \cong H_6(X;\mathbb{Z}) \cong \mathbb{Z}$. Choose a map $g : S^6 \to X$ whose homotopy class maps to a chosen orientation generator. It has degree one. Hence g_* is an isomorphism in degrees zero and six; in every other degree both integral homology groups vanish. Thus g is an integral homology equivalence. Both spaces are simply connected CW complexes, so the homological Whitehead theorem [9, Corollary 4.33] makes g a homotopy equivalence. Therefore X is a smooth homotopy 6-sphere.

Smale's generalized Poincaré theorem in dimensions greater than four [14, Theorem A] identifies the underlying topological manifold with S^6. The smooth endpoint is stated directly by Smale: every homotopy 6-sphere is diffeomorphic to S^6 [15, Corollary 1.2]. Equivalently, smooth homotopy 6-spheres up to oriented h-cobordism form Θ_6; Kervaire–Milnor's opening table gives $|\Theta_6| = 1$, and their following remark identifies h-cobordism with diffeomorphism in dimension six [16, opening table and following Remark, pp. 504–505]. Hence there is no exotic smooth structure in dimension six, and X is diffeomorphic to the standard sphere. □

Corollary 44.17 (abstract application theorem)**.** ***[VERIFIED]*** *Let X be a smooth, closed, connected, oriented six-manifold. Suppose V1 identifies the actual based van Kampen quotient with $G(\ell_0, \ell_1, \ell_2)$ of theorem 5.5, and suppose an independently constructed MV1/L1 comparison proves*

$$H^2(X;\mathbb{Z}) \cong H^3(X;\mathbb{Z}) \cong \mathbb{Z}/|p|\mathbb{Z}, \qquad p = 12\ell_0 - 4\ell_1 - 3\ell_2. \tag{4.6}$$

If $|p| = 1$, then $X \cong_{\text{diff}} S^6$.

Proof. For $|p| = 1$, theorem 5.5 and the assumed geometric identification give $\pi_1(X) = 0$, while (4.6) gives $H^2(X) = H^3(X) = 0$. Theorem 44.15 converts these inputs into the homology condition (4.5), and theorem 44.16 gives the claimed diffeomorphism. □

Corollary 44.18 (independent Mayer–Vietoris certificate for the candidate)**.** ***[VERIFIED]*** *For the tautological candidate $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$, the two-stage geometric Mayer–Vietoris calculation gives*

$$H_k(X;\mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \le k \le 5. \end{cases}$$

Together with the independently verified based van Kampen calculation, this proves $X \cong_{\text{diff}} S^6$ without using the Leray computation of the same groups.

Proof. Here $p = -1$. Thus (MV.11) and the reduced block (MV.13) are unimodular. The two Mayer–Vietoris sequences give $H_1(X) = H_2(X) = 0$ integrally, with no torsion hidden by a rank calculation. The verified collapse homology (C.23) gives $e(W) = 1 - 2 + 4 - 2 + 1 = 2$. Away from W, the map to the punctured base is stratified by smooth four-tori and the two bielliptic fibres, all of Euler characteristic zero. Additivity therefore gives $e(X) = e(W) = 2$.

Poincaré duality and universal coefficients give $H_5(X) = H_4(X) = 0$. Moreover

$$H_3(X) \cong H^3(X) \cong \operatorname{Hom}(H_3(X), \mathbb{Z}),$$

because $H_2(X) = 0$; hence $H_3(X)$ is free. The equality $e(X) = 2$ forces its rank to be zero. The based van Kampen result theorem 39.6 gives $\pi_1(X) = 1$, and theorem 44.16 completes the proof. □

Thus C1–C3, V1, MV1, and L1 are all verified for the candidate. The nearby-cycle/Leray route and the collapse/Mayer–Vietoris route are now two separate integral certificates for the same sphere-recognition input.

45 Fibre degree and the cyclic-six centre

This section supplies the theorem–proof module behind the fibre-degree and cyclic-centre formulas recorded in the orientation material. There are three different objects in the calculation:

$$H^4(U;\mathbb{Z}), \qquad H^4(U;\mathbb{Z}/6), \qquad Z(SU(6)) = \mu_6.$$

The first is a free rank-six characteristic lattice, the second has six independent residue coordinates, and the third is a single cyclic group. The candidate determines one particular surjection from the first object to the third. It does not identify all three objects, and it does not construct a gauge connection or a spectral operator.

45.1 Spaces, markings, and the finite-band presentation

Retain the reconstructed map

$$f : X \longrightarrow \mathbb{P}^1, \qquad X \cong_{\mathrm{diff}} S^6,$$

and write

$$D_0 = W, \qquad D_1 = S_1, \qquad D_2 = S_2, \qquad D = D_0 \sqcup D_1 \sqcup D_2, \qquad U = X \setminus D.$$

Here W is the reduced non-normal fibre and $3S_1, 4S_2$ are the two multiple fibres. The verified local homology computations give

$$H_1(W;\mathbb{Z}) = \mathbb{Z}\langle a_0, b_0\rangle, \qquad H_1(S_j;\mathbb{Z}) = \mathbb{Z}\langle a_j, b_j\rangle \quad (j = 1, 2).$$

The marking is

$$a_0 = [\widehat{\gamma}], \quad b_0 = [\widehat{u}], \quad a_j = [\widehat{g}_j],$$

where $\widehat{g}_j$ is the clockwise affine deck generator. The second generator b_j is the invariant coinvariant coordinate. With the marking used in the finite-band calculation one may take $b_1 = \pi_{1*}[\widehat{w}]$ and $b_2 = \pi_{2*}[\widehat{w}]$; in the second coinvariant quotient $\pi_{2*}[\widehat{u}] = \pi_{2*}[\widehat{w}]$, so this agrees with the alternative representative used elsewhere. The two specializations of the marked $\widehat{\gamma}$-cycle are

$$\pi_{1*}[\widehat{\gamma}] = 3a_1, \qquad \pi_{2*}[\widehat{\gamma}] = -4a_2. \tag{1188}$$

Fix Alexander duality using the complex orientation of X, outward-first boundary orientation, and the convention that the Poincare dual of a positive nearby fibre is the negative covector of its clockwise meridian. It identifies

$$L := H^4(U;\mathbb{Z}) \xrightarrow{\sim} H_1(D;\mathbb{Z}) = \mathbb{Z}\langle a_0, b_0; a_1, b_1; a_2, b_2\rangle \cong \mathbb{Z}^6. \tag{1189}$$

Indeed, D is compact and locally contractible, so

$$\widetilde{H}^q(S^6 \setminus D;\mathbb{Z}) \cong \widetilde{H}_{5-q}(D;\mathbb{Z}).$$

Take $q = 4$ and use the disjoint-union decomposition of D.

Let X° be the space obtained after the two finite fillings and before the central toric filling. For logarithmic twist integers ℓ_1, ℓ_2, its verified degree-one Mayer–Vietoris calculation is the presentation

$$H_1(X^\circ;\mathbb{Z}) = \frac{\mathbb{Z}\langle c, \bar{g}_1, \bar{g}_2\rangle}{\mathbb{Z}\langle 3\bar{g}_1 - \ell_1 c, 4\bar{g}_2 - \ell_2 c\rangle}. \tag{1190}$$

Thus $c, \bar{g}_1, \bar{g}_2$ are three marked generators, not a basis. The relation matrix is

$$R_1 = \begin{pmatrix} -\ell_1 & 3 & 0 \\ -\ell_2 & 0 & 4 \end{pmatrix}.$$

For admissible twists $3 \nmid \ell_1$ and $2 \nmid \ell_2$, the gcd of its rank-two minors

$$3\ell_2, \qquad -4\ell_1, \qquad 12$$

is one. Therefore the presented group is torsion-free of rank one. The homomorphism

$$\begin{aligned} \Phi_1 : H_1(X^\circ;\mathbb{Z}) &\longrightarrow \mathbb{Z}, \\ \Phi_1(c) = 12, \quad \Phi_1(\bar g_1) &= 4\ell_1, \quad \Phi_1(\bar g_2) = 3\ell_2 \end{aligned} \tag{1191}$$

annihilates both relations. Its three displayed values have gcd one under the same admissibility hypotheses; hence it is an isomorphism. For the candidate

$$(\ell_0,\ell_1,\ell_2) = (0,1,-1), \qquad \Phi_1(c) = 12, \quad \Phi_1(\bar g_1) = 4, \quad \Phi_1(\bar g_2) = -3. \tag{1192}$$

Put

$$\zeta := \bar g_1 + \bar g_2. \tag{1193}$$

Then $\Phi_1(\zeta) = 1$. Since Φ_1 is an isomorphism, ζ is the positive basis element of $H_1(X^\circ;\mathbb{Z})$ selected by Φ_1, and

$$c = 12\zeta, \qquad \bar g_1 = 4\zeta, \qquad \bar g_2 = -3\zeta. \tag{1194}$$

Let

$$\overline{q}_{\mathrm{col}} : H_1(D;\mathbb{Z}) \longrightarrow H_1(X^\circ;\mathbb{Z})$$

be the homomorphism sending a_0, a_1, a_2 to $c, \bar g_1, \bar g_2$, respectively, and sending every b_i to zero. Define $q_{\mathrm{col}} = \overline{q}_{\mathrm{col}} \circ \mathrm{AD}$, where AD is the fixed Alexander isomorphism above. Thus the marked collar homomorphism is well typed:

$$\begin{aligned} q_{\mathrm{col}} &: L \longrightarrow H_1(X^\circ;\mathbb{Z}), \\ q_{\mathrm{col}}(a_0) &= c, & q_{\mathrm{col}}(b_0) &= 0, \\ q_{\mathrm{col}}(a_1) &= \bar g_1, & q_{\mathrm{col}}(b_1) &= 0, \\ q_{\mathrm{col}}(a_2) &= \bar g_2, & q_{\mathrm{col}}(b_2) &= 0. \end{aligned} \tag{1195}$$

The central-band projection sends $a_0 = [\widehat{\gamma}]$ to c and $b_0 = [\widehat{u}]$ to zero. In the two finite-band cokernels, the actual Mayer–Vietoris map is $(\pi_{1*}, -\pi_{2*})$; it sends the affine generators to $\bar g_1, \bar g_2$ and kills the invariant coordinates b_1, b_2. These statements derive the six displayed values from the marked collar calculation. The map is surjective because its images contain the three presentation generators in (1190). It is not an inclusion-induced map $H_1(D) \to H_1(X^\circ)$: no inclusion of the deleted central fibre W into X° exists.

45.2 The exact fibre-degree row

Let $F \cong T^4$ be a complex-oriented smooth fibre and choose $\omega_F \in H^4(F;\mathbb{Z})$ with $\langle \omega_F, [F]\rangle = 1$. If $i_F : F \hookrightarrow U$ is the inclusion, put $\mathrm{res}_F = i_F^* : H^4(U;\mathbb{Z}) \to H^4(F;\mathbb{Z})$. Define

$$\mathrm{res}_F(\alpha) = \deg_F(\alpha)\omega_F, \qquad \deg_F : L \longrightarrow \mathbb{Z}. \tag{1196}$$

Theorem 45.1 (Fibre-degree row)**.** *With the fixed Alexander-duality convention above, one has*

$$\boxed{\deg_F(n_0a_0 + r_0b_0 + n_1a_1 + r_1b_1 + n_2a_2 + r_2b_2) = 12n_0 + 4n_1 - 3n_2.} \tag{1197}$$

Equivalently,

$$[\deg_F]_{(a_0,b_0,a_1,b_1,a_2,b_2)} = (12,0,4,0,-3,0).$$

Moreover,

$$\boxed{\deg_F = \Phi_1 \circ q_{\mathrm{col}}} \tag{1198}$$

for the candidate twists, and

$$\boxed{\ker \deg_F = \mathbb{Z}\langle b_0, b_1, b_2, a_0 - 3a_1, 3a_1 + 4a_2\rangle.} \tag{1199}$$

This kernel is saturated of rank five.

Proof. We retain every sign in the one-shot collar calculation. In the ordered boundary-source basis

$$(\gamma_0, u_0, \widetilde{a}_0; \gamma_1, w_1, \rho_1; \gamma_2, w_2, \rho_2)$$

and ordered target basis

$$(c, \bar{\rho}_1, \bar{\rho}_2; a_0, b_0; a_1, b_1; a_2, b_2),$$

the actual map $(i_{U*}, -i_{N*})$ at $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ is

$$\Delta_1 = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}. \tag{1200}$$

For example, the columns say

$$\gamma_0 \mapsto c - a_0, \quad u_0 \mapsto -b_0, \quad \widetilde{a}_0 \mapsto -\bar{\rho}_1 - \bar{\rho}_2,$$
$$\gamma_1 \mapsto c - 3a_1, \quad w_1 \mapsto -b_1, \quad \rho_1 \mapsto \bar{\rho}_1 - a_1,$$
$$\gamma_2 \mapsto c + 4a_2, \quad w_2 \mapsto -b_2, \quad \rho_2 \mapsto \bar{\rho}_2 - a_2.$$

Thus the entries $-3, +4$ are exactly (1188); all other displayed minus signs are the second-inclusion signs in the Mayer–Vietoris convention. Direct integral elimination gives $\det \Delta_1 = 1$.

Let $\widetilde{a}_0^\vee$ be the integral covector on the ordered boundary-source group that is one on $\widetilde{a}_0$ and zero on the other eight basis elements. Pullback of target covectors is represented by Δ_1^{T}. By the fixed orientation convention, the target covector extending the Poincare dual of a complex-oriented nearby fibre must pull back to $-\widetilde{a}_0^\vee$. Since Δ_1 is unimodular, there is a unique integral extension. Its last six entries are precisely the Alexander coordinates of the restriction functional on $H_1(D)$. Consequently they are obtained from

$$\Delta_1^{\mathsf{T}} y = -\widetilde{a}_0^\vee. \tag{1201}$$

Writing

$$y = (A, B, C; X_0, Y_0; X_1, Y_1; X_2, Y_2),$$

the nine equations are

$$A = X_0, \quad Y_0 = Y_1 = Y_2 = 0, \quad A = 3X_1, \quad A = -4X_2,$$
$$B = X_1, \qquad C = X_2, \qquad -B - C = -1.$$

Their unique solution is

$$y = (12, 4, -3; 12, 0; 4, 0; -3, 0).$$

The last six entries prove (1197). Equations (1191), (1192), and (1195) give the same row on the six basis elements, proving the typed equality (1198).

The row is primitive because $\gcd(12, 4, 3) = 1$; explicitly, $\deg_F(a_1 + a_2) = 1$. Each of the five displayed elements in (1199) lies in the kernel. Conversely, if $12n_0 + 4n_1 - 3n_2 = 0$, reduction modulo four gives $n_2 = 4v$, and then $n_1 = 3v - 3n_0$. Therefore

$$n_0 a_0 + n_1 a_1 + n_2 a_2 = n_0(a_0 - 3a_1) + v(3a_1 + 4a_2).$$

The three b_i-coordinates are free, and the five displayed generators are independent. They therefore form a basis of the kernel. Finally, a kernel of a surjection from a free abelian group to $\mathbb{Z}$ is saturated, proving the last assertion. □

Reversing the global Alexander-duality convention negates all six entries of (1197) simultaneously. It cannot change the sign of only one component. In particular, with the convention fixed here, $q_{\mathrm{col}}(a_2) = +\bar{g}_2$ while the coordinate of $\bar{g}_2$ in the positive generator of $H_1(X^\circ;\mathbb{Z})$ is -3.

For completeness, if $j : M \hookrightarrow X^\circ$ denotes the central collar inclusion, the toric attaching meridian obeys

$$j_*[\widetilde{a}_0] = \ell_0 c - \bar{g}_1 - \bar{g}_2,$$

and hence

$$\Phi_1(j_*[\widetilde{a}_0]) = 12\ell_0 - 4\ell_1 - 3\ell_2 = p.$$

At the candidate values this is $p = -1$. Thus the same correctly typed functional controls the smooth-fibre degree and the terminal meridian, but the two statements use the maps displayed above rather than an untyped identification of their source groups.

45.3 Normal-line holonomy and the two finite reductions

The transverse disc in the order-m_j local quotient is acted on by $e^{-2\pi i/m_j}$, with $m_1 = 3, m_2 = 4$. Therefore the normal-line holonomy characters, in the marked bases, are

$$\begin{aligned} h_1 &: H_1(S_1;\mathbb{Z}) \longrightarrow \mu_3, \quad h_1(a_1) = e^{-2\pi i/3}, \quad h_1(b_1) = 1, \\ h_2 &: H_1(S_2;\mathbb{Z}) \longrightarrow \mu_4, \quad h_2(a_2) = e^{-2\pi i/4}, \quad h_2(b_2) = 1. \end{aligned} \tag{1202}$$

Via (1189), write each $x \in L$ uniquely as $x = (x_0, x_1, x_2)$ with $x_i \in H_1(D_i;\mathbb{Z})$. Regard μ_3 and μ_4 as their standard subgroups of μ_{12}. The inverse on the second factor below records the candidate twist $\ell_2 = -1$, rather than a discretionary change of sign. Define

$$\begin{aligned} \chi_{12} &: L \longrightarrow \mu_{12}, \quad \chi_{12}(x_0, x_1, x_2) = h_1(x_1)h_2(x_2)^{-1}, \\ \chi_6 &: L \longrightarrow \mu_6, \qquad\qquad \chi_6(x) = \chi_{12}(x)^2. \end{aligned} \tag{1203}$$

Proposition 45.2 (The order-twelve and order-six reductions)**.** *Put*

$$\rho_m : \mathbb{Z} \longrightarrow \mu_m, \qquad \rho_m(k) = e^{-2\pi i k/m}.$$

Then

$$\boxed{\chi_{12} = \rho_{12} \circ \deg_F, \qquad \chi_6 = \rho_6 \circ \deg_F\,.} \tag{1204}$$

Both characters are surjective, and

$$L/\ker\chi_{12} \cong \mu_{12}, \qquad L/\ker\chi_6 \cong \mu_6. \tag{1205}$$

Proof. For $x = \sum_{i=0}^{2}(n_i a_i + r_i b_i)$, (1202) gives, with $\zeta_{12} = e^{-2\pi i/12}$,

$$\chi_{12}(x) = \zeta_{12}^{\,4n_1 - 3n_2}.$$

Equation (1197) differs in its exponent only by $12n_0$, which vanishes modulo twelve. This proves the first identity. Squaring it is reduction of the same integer modulo six and proves the second. Since $\deg_F(a_1 + a_2) = 1$, both reductions are surjective. The first isomorphism theorem gives (1205). □

The kernels can be read without suppressing any coordinate:

$$\ker\chi_{12} = \mathbb{Z}a_0 \oplus \mathbb{Z}b_0 \oplus 3\mathbb{Z}a_1 \oplus \mathbb{Z}b_1 \oplus 4\mathbb{Z}a_2 \oplus \mathbb{Z}b_2,$$

$$\ker\chi_6 = \mathbb{Z}a_0 \oplus \mathbb{Z}b_0 \oplus 3\mathbb{Z}a_1 \oplus \mathbb{Z}b_1 \oplus 2\mathbb{Z}a_2 \oplus \mathbb{Z}b_2.$$

The first follows by reducing $4n_1 - 3n_2$ modulo three and four; the second follows by reducing it modulo three and two. In particular, the natural combined character has order twelve before its order-six quotient.

45.4 Coefficient reduction is not the distinguished quotient

Alexander duality also gives

$$H^5(U;\mathbb{Z}) \cong \widetilde{H}_0(D;\mathbb{Z}) \cong \mathbb{Z}^2.$$

The cohomological universal-coefficient sequence with coefficients $\mathbb{Z}/6$ is therefore

$$0 \longrightarrow H^4(U;\mathbb{Z}) \otimes \mathbb{Z}/6 \longrightarrow H^4(U;\mathbb{Z}/6) \longrightarrow \operatorname{Tor}_1^{\mathbb{Z}}(H^5(U;\mathbb{Z}),\mathbb{Z}/6) \longrightarrow 0,$$

whose last term is zero. Thus the marked coefficient group is

$$\boxed{H^4(U;\mathbb{Z}/6) \cong L/6L \cong (\mathbb{Z}/6)^6.} \tag{1206}$$

If $r_6 : L \to L/6L$ is coefficient reduction, then χ_6 factors as

$$L \xrightarrow{r_6} (\mathbb{Z}/6)^6 \xrightarrow{\overline{d}_F} \mathbb{Z}/6 \xrightarrow{k\mapsto e^{-2\pi ik/6}} \mu_6,$$

where, in the six marked residue coordinates,

$$[\overline{d}_F] = (0,0,4,0,3,0) \pmod 6. \tag{1207}$$

Thus

$$\overline{d}_F(\bar{n}_0,\bar{r}_0,\bar{n}_1,\bar{r}_1,\bar{n}_2,\bar{r}_2) = 4\bar{n}_1 + 3\bar{n}_2 \in \mathbb{Z}/6.$$

It is surjective because $\overline{d}_F(0,0,1,0,1,0) = 1$ in $\mathbb{Z}/6$. Consequently

$$L/\ker\chi_6 \cong (L/6L)/\ker\overline{d}_F \cong \mathbb{Z}/6.$$

This is one distinguished cyclic quotient of the six-coordinate group in (1206); it is not that group itself.

45.5 The tensor subgroup and the genuine common-centre quotient

Theorem 45.3 (Tensor subgroup and centres)**.** *Tensor product defines a faithful closed homomorphism*

$$\iota : SU(2) \times SU(3) \hookrightarrow SU(6), \qquad \iota(A,B) = A \otimes B. \tag{1208}$$

Its restriction to the centres is the isomorphism

$$\mu_2 \times \mu_3 \xrightarrow{\sim} \mu_6 = Z(SU(6)), \qquad (\lambda,\mu) \longmapsto \lambda\mu. \tag{1209}$$

The image of ι is a proper subgroup of $SU(6)$, of real dimension $3+8=11$.

Proof. The matrix $A \otimes B$ is unitary and

$$\det(A \otimes B) = (\det A)^3(\det B)^2 = 1,$$

so ι has the stated codomain. Suppose more generally that $A \otimes B = sI_6$. Choose an eigenvector w of B, with eigenvalue β. Then

$$(A \otimes B)(v \otimes w) = \beta Av \otimes w = sv \otimes w$$

for every v, so $A = (s/\beta)I_2$; substitution shows that B is scalar as well. Applying this scalar–tensor lemma with $s = 1$ gives

$$A = \lambda I_2, \qquad B = \mu I_3, \qquad \lambda\mu = 1.$$

The determinant conditions give $\lambda^2 = 1$ and $\mu^3 = 1$. Hence $\lambda = \mu^{-1} \in \mu_2 \cap \mu_3 = \{1\}$, proving injectivity. A continuous injective homomorphism from a compact group into a Hausdorff group has closed image.

On centres the tensor matrix is $(\lambda\mu)I_6$. Multiplication is injective because $\mu_2 \cap \mu_3 = \{1\}$, and both its domain and target have six elements, proving (1209). Explicitly, the inverse sends $s \in \mu_6$ to (s^3, s^{-2}). Finally the dimension of the image is $11 < 35 = \dim SU(6)$, so it is proper. □

The isomorphisms proved above give the explicit chain

$$L/\ker\chi_6 \xrightarrow{\overline{\chi}_6} \mu_6 \xrightarrow{\lambda\mapsto\lambda I_6} Z(SU(6)). \tag{1210}$$

This is an isomorphism of finite groups. It does not provide transition functions for an $SU(6)$-bundle. Also, ι does not factor through the quotient of $SU(2)\times SU(3)$ by its full centre: that centre maps isomorphically, rather than trivially, to $Z(SU(6))$.

Put $\zeta_6 = e^{2\pi i/6}$. In the integer $U(1)$-charge normalization, the common-centre generator used for the Standard-Model global form is

$$(\zeta_6, -I_2, e^{2\pi i/3} I_3).$$

If $\omega = e^{2\pi i/3} = \zeta_6^2$, then

$$\iota(-I_2, \omega I_3) = (-\omega) I_6 = \zeta_6^{-1} I_6;$$

the first coordinate ζ_6 cancels this scalar under the tensor–scalar map below. This agrees with the convention in David Tong, *Line Operators in the Standard Model*, JHEP 07 (2017) 104, arXiv:1705.01853v2. The following matrix theorem proves the quotient directly and does not rely on a physical interpretation of that reference.

Theorem 45.4 (Common-centre quotient with $U(1)$)**.** *Define*

$$\delta : \mu_6 \hookrightarrow U(1)\times SU(2)\times SU(3), \qquad \delta(\xi) = (\xi, \xi^3 I_2, \xi^2 I_3), \tag{1211}$$

and

$$\widehat{\iota} : U(1)\times SU(2)\times SU(3) \longrightarrow U(6), \qquad \widehat{\iota}(z, A, B) = z(A\otimes B). \tag{1212}$$

Then

$$\ker\widehat{\iota} = \delta(\mu_6), \tag{1213}$$

so the induced map is a faithful closed embedding

$$\frac{U(1)\times SU(2)\times SU(3)}{\delta(\mu_6)} \hookrightarrow U(6). \tag{1214}$$

Moreover,

$$\boxed{\det\widehat{\iota}(z, A, B) = z^6,} \qquad \boxed{\operatorname{im}\widehat{\iota}\cap SU(6) = \iota(SU(2)\times SU(3)).} \tag{1215}$$

Proof. For $\xi\in\mu_6$, one has

$$\det(\xi^3 I_2) = \xi^6 = 1, \qquad \det(\xi^2 I_3) = \xi^6 = 1.$$

Thus δ has the stated codomain. It is central, is a homomorphism, and is injective because its first coordinate is ξ.

For $\xi\in\mu_6$,

$$\widehat{\iota}(\delta(\xi)) = \xi(\xi^3 I_2 \otimes \xi^2 I_3) = \xi^6 I_6 = I_6,$$

so $\delta(\mu_6)$ lies in the kernel. Conversely, suppose $z(A\otimes B) = I_6$. The eigenvalue argument in the preceding proof, now applied to $A\otimes B = z^{-1} I_6$, gives

$$A = \lambda I_2, \qquad B = \mu I_3, \qquad \lambda^2 = 1, \quad \mu^3 = 1, \quad z\lambda\mu = 1.$$

The Chinese-remainder isomorphism

$$\mu_6 \longrightarrow \mu_2\times\mu_3, \qquad \xi \longmapsto (\xi^3, \xi^2)$$

provides a unique $\xi\in\mu_6$ with $\lambda = \xi^3$ and $\mu = \xi^2$. Then $z = (\lambda\mu)^{-1} = \xi$, because $\xi^{-5} = \xi$. Hence the kernel is exactly $\delta(\mu_6)$. Compactness makes the induced faithful image closed.

The determinant formula is

$$\det(z(A \otimes B)) = z^6(\det A)^3(\det B)^2 = z^6.$$

Every tensor matrix lies in the intersection by taking $z = 1$. Conversely, if $z(A \otimes B)$ lies in $SU(6)$, then $z^6 = 1$. By (1209), choose $(\lambda, \mu) \in \mu_2 \times \mu_3$ with $\lambda\mu = z$. Then

$$z(A \otimes B) = (\lambda A) \otimes (\mu B),$$

and $\lambda A \in SU(2)$, $\mu B \in SU(3)$. This proves the reverse inclusion in (1215). □

Writing the quotient in (1214) as G_6, determinant therefore gives the exact sequence

$$1 \longrightarrow SU(2) \times SU(3) \xrightarrow{\iota} G_6 \xrightarrow{\det \circ \widehat{\iota}} U(1) \longrightarrow 1.$$

Here the middle arrow means the tensor embedding followed by the quotient identification with $\operatorname{im} \widehat{\iota}$; surjectivity on the right follows because $z \mapsto z^6$ is surjective on $U(1)$, and its kernel is (1215).

The quotient in (1214) is the genuine common-centre quotient: the $U(1)$ factor cancels the nontrivial scalar produced by the two special-unitary centres. The displayed tensor–scalar image in $U(6)$ is not contained in $SU(6)$; by (1215), its intersection with $SU(6)$ is exactly the tensor subgroup. The quotient is not isomorphic to $SU(6)$: their real dimensions are 12 and 35.

45.6 Exact boundary of the result

The construction proves the primitive fibre-degree map, its saturated kernel, its order-twelve and order-six finite reductions, the distinction between the full coefficient group and the selected cyclic quotient, and the two matrix-group embeddings. It does not produce a degree-changing morphism from

$$H^4(U; \mathbb{Z}) \longrightarrow H^2(M; \mu_6)$$

for a four-dimensional spacetime M. In particular, χ_6 is not, merely by its codomain, the obstruction to lifting a common-centre quotient bundle. No physical gauge bundle, connection, Yang–Mills Hamiltonian, confinement statement, or mass-gap conclusion is inferred here.

46 The non-normal fibre and the CDP obstruction

Campana–Demailly–Peternell prove that a compact complex threefold with $H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = 0$, algebraic dimension one, and holomorphic algebraic reduction has $c_3(X) \leq 0$ [2, Theorem 2.2, p. 680]. Their corollary gives $a(X) = 0$ for a compact complex threefold homeomorphic to S^6 [2, Corollary 2.3, p. 680]. The candidate asserts all those hypotheses together with $c_3(X) = 2$ and $a(X) = 1$. This section separates:

(i) validity of one printed inference on p. 692 of [2];

(ii) validity of the published theorem by any other argument;

(iii) validity of the candidate's proposed non-normal-fibre mechanism; and

(iv) truth of the candidate's upstream construction and topology.

46.1 The exact p. 692 inference

Let S be the reduction of a fibre in a smooth threefold $\mathcal{X}$, and let A denote the line-bundle twist after duality. The two relevant sequences are

$$0 \longrightarrow N^{*}_{S/\mathcal{X}} \otimes A \longrightarrow \Omega^{1}_{\mathcal{X}}|_{S} \otimes A \xrightarrow{\rho} \Omega^{1}_{S} \otimes A \longrightarrow 0 \tag{5.1}$$

and

$$0 \longrightarrow \mathcal{T}(\Omega^{1}_{S}) \otimes A \longrightarrow \Omega^{1}_{S} \otimes A \longrightarrow \widetilde{\Omega}^{1}_{S} \otimes A \longrightarrow 0, \tag{5.2}$$

where $\mathcal{T}$ denotes sections supported on a nowhere dense analytic set. The opening reduction of [2, §7, p. 692] treats the vanishing of the global sections of the left end of (5.1) and the right end of (5.2) as sufficient for the vanishing of the middle term of (5.1). That inference leaves untreated a section whose image under ρ is a nonzero torsion section. The omitted sheaf can be identified exactly. Put

$$\mathscr{T}_{S} := \ker(\Omega^{1}_{S} \to j_{*}\Omega^{1}_{S_{\mathrm{reg}}}), \qquad \widetilde{\Omega}^{1}_{S} = \Omega^{1}_{S}/\mathscr{T}_{S},$$

and

$$\mathscr{K}_{S} := \ker(\Omega^{1}_{X}|_{S} \longrightarrow \widetilde{\Omega}^{1}_{S}).$$

Proposition 46.1 (the missing cotangent filtration)**.** *Let S be a reduced effective Cartier divisor in a smooth complex threefold X. There are natural short exact sequences*

$$0 \longrightarrow N^{*}_{S/X} \longrightarrow \mathscr{K}_{S} \longrightarrow \mathscr{T}_{S} \longrightarrow 0,$$

$$0 \longrightarrow \mathscr{K}_{S} \longrightarrow \Omega^{1}_{X}|_{S} \longrightarrow \widetilde{\Omega}^{1}_{S} \longrightarrow 0.$$

Proof. The conormal sequence is right exact. It is also left exact here. Locally write $S = (g = 0)$ with g reduced. If $h\,dg|_{S} = 0$, then h vanishes at the generic smooth point of every irreducible component, because dg is nonzero there. The ring $\mathcal{O}_{S}$ is reduced, so the intersection of its minimal primes is zero; hence $h = 0$. Thus

$$0 \to N^{*}_{S/X} \to \Omega^{1}_{X}|_{S} \to \Omega^{1}_{S} \to 0$$

is exact. By definition, $\mathscr{K}_{S}$ is the inverse image of $\mathscr{T}_{S}$ under the last surjection. Restriction to that inverse image is onto $\mathscr{T}_{S}$ and has kernel $N^{*}_{S/X}$, proving the first sequence. Composing with the quotient by $\mathscr{T}_{S}$ proves the second. □

Proposition 46.2 (the exact global-section criterion)**.** *Let A be a line bundle on S. If*

$$H^{0}(S, \widetilde{\Omega}^{1}_{S} \otimes A) = 0,$$

then

$$H^{0}(S, \Omega^{1}_{X}|_{S} \otimes A) = H^{0}(S, \mathscr{K}_{S} \otimes A).$$

If also $H^{0}(S, N^{*}_{S/X} \otimes A) = 0$, then this group is canonically

$$\ker\Big[H^{0}(S, \mathscr{T}_{S} \otimes A) \xrightarrow{\delta_{A}} H^{1}(S, N^{*}_{S/X} \otimes A)\Big].$$

Hence the two vanishings printed in the p. 692 reduction imply the desired ambient vanishing if and only if δ_{A} is injective.

Proof. Tensoring the second sequence of theorem 46.1 by the locally free sheaf A preserves exactness; its global-section sequence gives the first equality. Tensor the first sequence and take global sections. If the conormal H^{0} vanishes, exactness identifies the middle group with the displayed kernel. No condition on the torsion-free quotient controls that kernel. □

46.2 Local algebra of a reduced normal-crossings fibre

Let $g : \mathcal{X} \to \Delta$ be a proper holomorphic map from a smooth complex threefold to a disc, and put $S = g^{-1}(0)$. Assume S is reduced and has normal crossings: at every point there are coordinates (x, y, z) in which

$$g = ux, \qquad g = uxy, \qquad \text{or} \qquad g = uxyz \tag{5.3}$$

for a unit u. Let $\eta : \widetilde{S} \to S$ be the normalization, $D = S_{\mathrm{sing,red}}$, and $\widetilde{D} = \eta^{-1}(D)_{\mathrm{red}}$.

Lemma 46.3 (the conormal map is injective)**.** ***[VERIFIED]*** *In the setting above, $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, and its generator maps to $dg|_S$. Multiplication by $dg|_S$ is injective, so*

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{h \mapsto h\,dg} \Omega^1_{\mathcal{X}}|_S \longrightarrow \Omega^1_S \longrightarrow 0 \tag{5.4}$$

is exact.

Proof. The fibre ideal is the principal ideal (g) and is reduced, so $(g)/(g^2) \simeq \mathcal{O}_S$. Multiplication by a unit does not affect injectivity, and the term $f\,du$ vanishes after restriction to $f = 0$. It therefore suffices to use $f = x, xy, xyz$.

For $f = xy$, if a lift $\widetilde{h}$ satisfies $\widetilde{h}(y\,dx + x\,dy) = 0$ modulo (xy), then $\widetilde{h}y, \widetilde{h}x \in (xy)$. Hence $\widetilde{h} \in (x) \cap (y) = (xy)$ in the regular local ring, so $h = 0$ on S. For $f = xyz$, the three coefficient conditions say $\widetilde{h}yz, \widetilde{h}xz, \widetilde{h}xy \in (xyz)$; hence $\widetilde{h} \in (x) \cap (y) \cap (z) = (xyz)$. The case $f = x$ is immediate. Thus the conormal map is injective in every local model, proving (5.4). □

Lemma 46.4 (normalization–conductor exact sequence)**.** ***[VERIFIED]*** *The sequence*

$$0 \longrightarrow \mathcal{O}_S \xrightarrow{\alpha} \eta_*\mathcal{O}_{\widetilde{S}} \oplus \mathcal{O}_D \xrightarrow{\beta} \eta_*\mathcal{O}_{\widetilde{D}} \longrightarrow 0, \tag{5.5}$$

$$\alpha(h) = (\eta^* h, h|_D), \qquad \beta(\widetilde{h}, k) = \widetilde{h}|_{\widetilde{D}} - \eta^* k,$$

is exact.

Proof. The assertion is local. Away from D, η is an isomorphism. At a double point $S = \{xy = 0\}$, a member of $\ker \beta$ is

$$((h_1(y, z), h_2(x, z)), k(z)), \qquad h_1(0, z) = h_2(0, z) = k(z).$$

The function

$$h = h_1(y, z) + h_2(x, z) - h_1(0, z)$$

on $\{xy = 0\}$ restricts to h_1, h_2 on the two branches and to k on their intersection. This proves middle exactness in the double model.

At a triple point $S = \{xyz = 0\}$, write the branch functions as $h_1(y, z)$, $h_2(x, z)$, $h_3(x, y)$. Membership in $\ker \beta$ is exactly

$$h_1(0, z) = h_2(0, z), \quad h_1(y, 0) = h_3(0, y), \quad h_2(x, 0) = h_3(x, 0).$$

Then

$$\begin{aligned} h = {} & h_1(y, z) + h_2(x, z) + h_3(x, y) \\ & - h_1(0, z) - h_1(y, 0) - h_2(x, 0) + h_1(0, 0) \end{aligned}$$

restricts to the prescribed function on every branch. Injectivity of α follows because S is reduced and η is an isomorphism over a dense open set. Surjectivity of β follows locally by lifting a function on the branch conductor to the smooth branch and taking $k = 0$. This proves exactness in every local model. □

Theorem 46.5 (conductor-supported ambient differential). ***[VERIFIED]*** *In the setting (5.3), let A be a line bundle on S with $\eta^*A \simeq \mathcal{O}_{\widetilde{S}}$, and choose a nowhere-vanishing section e of η^*A. There is a unique nonzero*

$$s \in H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A) \tag{5.6}$$

such that

$$\eta^* s = \eta^*(dg|_S) \otimes e, \qquad s|_D = 0. \tag{5.7}$$

Its image $\rho(s)$ in $\Omega^1_S \otimes A$ is supported on D and therefore torsion; its image in $\widetilde{\Omega}^1_S \otimes A$ is zero. Moreover:

(a) $\rho(s) = 0$ if and only if $e = \eta^*\sigma$ for a nowhere-vanishing $\sigma \in H^0(S, A)$;

(b) if S is connected and $A \not\simeq \mathcal{O}_S$, then $H^0(S, A) = 0$ and s does not come from the conormal term in (5.1).

Proof. In the double model, $d(xy) = y\,dx + x\,dy$ has coefficients in the reduced ideal (x, y) of D. In the triple model,

$$d(xyz) = yz\,dx + xz\,dy + xy\,dz$$

has coefficients in (xy, xz, yz), the reduced ideal of the union of the three double curves. Unit factors in (5.3) do not change this after restriction to S. Hence $\eta^*(dg|_S)$ restricts to zero on $\widetilde{D}$. Tensor (5.5) by the locally free sheaf $E = \Omega^1_{\mathcal{X}}|_S \otimes A$. The compatible pair

$$(\eta^*(dg|_S) \otimes e, 0)$$

therefore descends uniquely to s. On the smooth locus of S, $dg|_S$ is a nonzero conormal vector and e is nowhere zero, so $s \neq 0$.

On $S \setminus D$, the normalization is an isomorphism and $s = dg|_S \otimes e$. By (5.4) its intrinsic image vanishes there. Thus $\rho(s)$ is supported on D and lies in the torsion subsheaf; its image modulo torsion is zero.

If $\rho(s) = 0$, exactness of (5.4) after tensoring by A gives $s = dg|_S \otimes \sigma$ for a section σ of A. Pulling back and comparing with (5.7) on the dense open set where $dg \neq 0$ gives $\eta^*\sigma = e$ there, hence everywhere by holomorphic continuation. Since e never vanishes and η is surjective, σ never vanishes and trivializes A. The converse is immediate.

Finally suppose S is connected and $\sigma \in H^0(S, A)$. On each compact connected component of $\widetilde{S}$, write $\eta^*\sigma = c_i e$; the holomorphic function c_i is constant. If one c_i is zero and another is nonzero, connectedness of the finite union of irreducible components of S provides two such components meeting at a point. The value of σ at that point would be both zero and nonzero, a contradiction. Hence either every $c_i = 0$, so $\sigma = 0$, or every $c_i \neq 0$, so σ is nowhere zero and A is trivial. When A is nontrivial only the first case occurs. Since $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, the conormal global-section space is $H^0(S, A) = 0$, while $s \neq 0$. □

46.3 A proper counterexample to the printed inference

Theorem 46.6 (nodal-cubic test family). ***[REFUTED]*** *The p. 692 implication described after (5.2) is false as a formal sheaf-theoretic statement, even for a proper family with smooth total space and an irreducible reduced normal-crossings central fibre.*

Proof. Let Δ be a sufficiently small disc with coordinate t and put

$$\mathcal{C} = \{([x : y : z], t) \in \mathbb{P}^2 \times \Delta : y^2 z = x^2(x + z) + tz^3\}. \tag{5.8}$$

This is proper over Δ. Its total space is smooth. On $z \neq 0$, the partial derivative of the defining equation with respect to t is $-z^3 \neq 0$. On $z = 0$, the equation forces $x = 0$; then $y \neq 0$ and the z-derivative is $y^2 \neq 0$. The central fibre C_0 is the irreducible nodal cubic $y^2 z = x^2(x + z)$, with node $[0 : 0 : 1]$.

Set

$$\mathcal{X} = \mathcal{C} \times \mathbb{P}^1, \qquad S = C_0 \times \mathbb{P}^1,$$

and let $g : \mathcal{X} \to \Delta$ be projection. Then $\mathcal{X}$ is a smooth threefold, g is proper, and S is an irreducible reduced normal-crossings surface. Its normalization is $\widetilde{S} \simeq \mathbb{P}^1 \times \mathbb{P}^1$.

Let p, q be the two inverse images of the node in the normalization of C_0. Fix $\lambda \in \mathbb{C}^*$ which is not a root of unity (for example, $\lambda = 2$). Glue the fibres of the trivial line bundle on the normalization at p and q by the scalar λ. This defines a line bundle $A_\lambda \in \mathrm{Pic}^0(C_0)$ whose pullback is trivial. It has infinite order: its nth tensor power has gluing scalar λ^n, which is never one. It is not trivial: a global section pulls back to a constant c satisfying $c = \lambda c$, so

$$H^0(C_0, A_\lambda) = 0. \tag{5.9}$$

Put $A = \mathrm{pr}_1^* A_\lambda$ on S. Then $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$ and $H^0(S, A) = 0$. Because S is the full fibre, $N^*_{S/\mathcal{X}} \simeq \mathcal{O}_S$, so the conormal-twist global sections vanish.

The natural map $\widetilde{\Omega}^1_S \to \eta_* \Omega^1_{\widetilde{S}}$ is injective: its kernel is supported where normalization is not an isomorphism, while the source is torsion-free by definition. Projection formula and $\eta^* A \simeq \mathcal{O}_{\widetilde{S}}$ therefore give

$$H^0(S, \widetilde{\Omega}^1_S \otimes A) \hookrightarrow H^0(\widetilde{S}, \Omega^1_{\widetilde{S}}) = 0, \tag{5.10}$$

the last equality following from $\widetilde{S} = \mathbb{P}^1 \times \mathbb{P}^1$. Nevertheless theorem 46.5 produces

$$0 \neq s \in H^0(S, \Omega^1_{\mathcal{X}}|_S \otimes A).$$

Thus both vanishings invoked in the printed reduction hold while the desired ambient vanishing fails. This test chooses the line bundle A directly on S; it does not assert that A extends to a line bundle on the total space $\mathcal{X}$. It therefore refutes the claimed *formal implication between the three sheaf-cohomology groups* for a non-torsion A, but does not by itself instantiate the narrower global-line-bundle situation on p. 692. The family (5.8) also does not satisfy all hypotheses of CDP Theorem 2.2, so this is not a counterexample to that theorem. □

46.4 Cartier-divisor duality without a normality hypothesis

Theorem 46.7 (complete Cartier-fibre duality)**.** ***[VERIFIED]*** *Let X be a compact smooth complex threefold, $i : W \hookrightarrow X$ an effective Cartier divisor, and E a holomorphic vector bundle. Put $F = E|_W$ and*

$$\omega_W = (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

For $q = 0, 1, 2$ there are natural perfect pairings

$$H^q(W, F)^\vee \simeq H^{2-q}(W, F^\vee \otimes \omega_W). \tag{5.11}$$

Moreover $H^q(W, F) = 0$ for $q > 2$. No normality or reducedness hypothesis on W is used.

Proof. The Cartier divisor gives two short exact sequences

$$0 \to E(-W) \to E \to i_* F \to 0 \tag{5.12}$$

and, by adjunction,

$$0 \to E^\vee \otimes \omega_X \to E^\vee \otimes \omega_X(W) \to i_*(F^\vee \otimes \omega_W) \to 0. \tag{5.13}$$

Smooth Serre duality on the compact threefold gives functorial isomorphisms

$$H^r(X, G)^\vee \simeq H^{3-r}(X, G^\vee \otimes \omega_X) \tag{5.14}$$

for every vector bundle G [8, Theorem 9.37]. Naturality of the trace pairing says that the dual of the map on cohomology induced by $E(-W) \to E$ is the map induced by $E^\vee \otimes \omega_X \to E^\vee \otimes \omega_X(W)$.

Fix $q \in \{0,1,2\}$. Dualize the relevant five terms of the long exact sequence of (5.12) and use (5.14). They become

$$\begin{aligned} H^{2-q}(X, E^\vee \otimes \omega_X) &\longrightarrow H^{2-q}(X, E^\vee \otimes \omega_X(W)) \\ &\longrightarrow H^q(W, F)^\vee \\ &\longrightarrow H^{3-q}(X, E^\vee \otimes \omega_X) \longrightarrow H^{3-q}(X, E^\vee \otimes \omega_X(W)). \end{aligned} \tag{5.15}$$

The long exact sequence of (5.13) is the same five-term sequence with its middle group replaced by $H^{2-q}(W, F^\vee \otimes \omega_W)$. Naturality supplies a commutative diagram whose four outer vertical arrows are identities. The five lemma (or exactness at the three middle terms) gives the unique middle isomorphism (5.11). Finite dimensionality makes the pairing perfect.

For $q = 3$, the map $H^3(X, E(-W)) \to H^3(X, E)$ is dual under (5.14) to the injective map

$$H^0(X, E^\vee \otimes \omega_X) \hookrightarrow H^0(X, E^\vee \otimes \omega_X(W)).$$

It is therefore surjective. The long exact sequence of (5.12) gives $H^3(W, F) = 0$. For $q > 3$ both adjacent ambient cohomology groups vanish by the dimension-three Dolbeault resolution, so (5.12) gives $H^q(W, F) = 0$. This also agrees with analytic duality for the Gorenstein hypersurface W in Ramis–Ruget [4, Theorems 1–2 and Proposition 2]. □

Proposition 46.8 (absolute duality on a compact Cartier fibre)**.** ***[VERIFIED]*** *Let X be a smooth complex threefold, not necessarily compact, and let $i : W \hookrightarrow X$ be a compact effective Cartier divisor that is the full scheme-theoretic fibre of a holomorphic map $f : X \to B$ to a smooth curve. For every vector bundle E on X there is a functorial perfect pairing*

$$H^2(W, E|_W) \times H^0(W, E^\vee|_W \otimes \omega_X|_W) \longrightarrow \mathbb{C}.$$

Equivalently,

$$H^2(W, E|_W)^\vee \simeq H^0(W, E^\vee|_W \otimes \omega_X|_W). \tag{5.15a}$$

Neither normality of W nor compactness of the ambient threefold is required.

Proof. An effective Cartier divisor in a smooth threefold is locally a two-dimensional hypersurface quotient of a regular local ring. Hence W is Cohen–Macaulay and Gorenstein. If $K_W^\bullet$ denotes its analytic dualizing complex, the complete-intersection calculation of Ramis–Ruget gives

$$K_W^\bullet \simeq \omega_W[2]$$

with ω_W invertible [4, Proposition 2 and the complete-intersection calculation, pp. 90–91]. Cartier adjunction gives

$$\omega_W \simeq (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

Because W is the full fibre over $b \in B$, one has $\mathcal{O}_X(W) \simeq f^*\mathcal{O}_B(b)$ locally along W; its restriction to W is the pullback of a one-dimensional vector space. Choosing a nonzero vector identifies $\mathcal{O}_X(W)|_W \simeq \mathcal{O}_W$ and $\omega_W \simeq \omega_X|_W$. A different choice rescales the final trace pairing by a nonzero scalar.

Ramis–Ruget duality pairs $H^p_c(W, \mathcal{G})$ with $\operatorname{Ext}^{-p}_W(\mathcal{G}, K_W^\bullet)$ [4, Theorem 2, pp. 82–83]. Since W is compact, $H^p_c = H^p$. Taking $p = 2$ and $\mathcal{G} = E|_W$, the shift and local freeness give

$$\operatorname{Ext}^{-2}_W(\mathcal{G}, K_W^\bullet) \simeq \operatorname{Hom}_W(\mathcal{G}, \omega_W) = H^0(W, E^\vee|_W \otimes \omega_W).$$

Grauert finiteness makes both coherent cohomology spaces finite-dimensional [7]; the strong analytic duality is therefore the ordinary algebraic perfect pairing (5.15a). Its variance is contravariant in $E|_W$ by precomposition on the Hom term. This proves the assertion. □

46.5 Flatness and top-degree analytic base change

Lemma 46.9 (automatic properness, flatness, and Cartier fibres)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a nonconstant holomorphic map from a connected complex manifold to a Riemann surface. Then f is flat, every holomorphic vector bundle on X is f-flat, and every full scheme-theoretic fibre is an effective Cartier divisor of pure codimension one. If X is compact, then f is proper.*

Proof. Fix $x \in X$, $c = f(x)$, and put $R = \mathcal{O}_{B,c}$, $S = \mathcal{O}_{X,x}$. The ring R is a discrete valuation ring and S is a regular local domain. The homomorphism $R \to S$ is injective. Indeed, if a nonzero germ $h \in R$ mapped to zero, $h \circ f$ would vanish on a neighbourhood of x. The zero set of a nonzero one-variable holomorphic germ is isolated, so f would be constant on that neighbourhood and therefore, by the identity theorem on connected X, constant everywhere.

Thus every nonzero element of R acts on the domain S as a nonzero element and hence as a nonzerodivisor. Therefore S is torsion-free as an R-module. A module over a discrete valuation ring is flat if and only if it is torsion-free [36, Tag 0AUW]; hence S is R-flat. This holds at every point, so f is flat. A vector bundle is locally a finite free $\mathcal{O}_X$-module, and a finite direct sum of R-flat modules is flat; hence it is f-flat. If t is a local parameter at c, then $f^{*}t$ is a nonzero element of the domain S and thus a nonzerodivisor. The full scheme-theoretic fibre is therefore locally cut out by one nonzerodivisor, so it is an effective Cartier divisor of pure codimension one. Finally, when X is compact, the underlying continuous map to the Hausdorff surface B is proper. □

Proposition 46.10 (the exact analytic base-change input)**.** ***[VERIFIED]*** *Let $f : X \to B$ be proper between complex spaces, let $\mathcal{F}$ be coherent and f-flat, and fix $c \in B$. If*

$$H^{3}(X_c, \mathcal{F}|_{X_c}) = 0, \tag{5.16}$$

then the canonical fibre map

$$\beta_c^{2} : (R^{2}f_{*}\mathcal{F})_c \otimes_{\mathcal{O}_{B,c}} \mathbb{C}(c) \longrightarrow H^{2}(X_c, \mathcal{F}|_{X_c}) \tag{5.17}$$

is an isomorphism.

Proof. Apply Bănică–Stănăşilă, Chapter III, §3, Corollary 3.5, with $q = 3$ [6, pp. 120–121]. Its hypotheses are precisely properness, coherence, and flatness. Condition (5.16) first forces the degree-three direct image to vanish near c. The corollary then applies the equivalent base-change conditions of its preceding Theorem 3.4 in degree $q-1 = 2$ and states that the canonical map (5.17) is bijective. This is the top-degree argument used in [3, Lemma 1.5]; it does not require the stronger, false assertion that all direct images are locally free. □

Corollary 46.11 (uniform top-degree base change for surface fibres)**.** ***[VERIFIED]*** *Let $f : X \to B$ be a proper flat holomorphic map to a Riemann surface whose full scheme-theoretic fibres have pure complex dimension two, and let E be a holomorphic vector bundle on X. Then*

$$R^{q}f_{*}E = 0 \quad (q > 2) \tag{5.17-V}$$

and, for every $b \in B$, including a point with nonreduced fibre, the canonical map

$$(R^{2}f_{*}E)_b \otimes_{\mathcal{O}_{B,b}} \mathbb{C}(b) \xrightarrow{\sim} H^{2}(X_b, E|_{X_b}) \tag{5.17-T}$$

is an isomorphism. No constancy of h^{2}, local freeness of $R^{2}f_{*}E$, or reducedness of X_b is required.

Proof. On a sufficiently small neighbourhood U of b, the finite-complex theorem used in analytic cohomology and base change represents $Rf_{*}E|_U$ by a bounded complex $K^{\bullet}$ of finite free $\mathcal{O}_U$-modules [6, Chapter III,

§3, Theorem 3.4]. Flatness of E over B identifies its derived fibre with the ordinary restriction to the full scheme-theoretic fibre. Over the discrete valuation ring $A = \mathcal{O}_{B,b}$, the universal-coefficient sequence is

$$0 \longrightarrow H^q(K^\bullet)_b \otimes_A \mathbb{C}(b) \longrightarrow H^q(K^\bullet \otimes_A \mathbb{C}(b)) \longrightarrow \operatorname{Tor}_1^A(H^{q+1}(K^\bullet)_b, \mathbb{C}(b)) \longrightarrow 0. \tag{5.17-U}$$

If some $R^q f_* E$ with $q > 2$ were nonzero, choose the largest such q and a point of its support. Nakayama's lemma makes the left term of (5.17-U) nonzero there, while maximality kills the Tor term. This would give nonzero coherent cohomology in degree $q > 2$ on a compact two-dimensional complex space, a contradiction. Thus (5.17-V) holds. For $q = 2$, the only obstruction in (5.17-U) is $\operatorname{Tor}_1^A((R^3 f_* E)_b, \mathbb{C}(b))$, which is zero by (5.17-V), proving (5.17-T). This is the explicit obstruction calculation behind theorem 46.10; it was independently supplied, with eight named local certificates, in Reviewer B's candidate-specific follow-up [35]. □

46.6 A topology-independent Pic^0 route and three all-Pic routes

Let $i : W \hookrightarrow X$ denote the cusp-fibre inclusion and $\eta : \widetilde{W} \simeq \mathrm{dP}_6 \to W$ its normalization.

Proposition 46.12 (topologically trivial twists pull back trivially)**.** ***[VERIFIED]*** *Assume that ω_X is trivial on a neighbourhood of W. For every $L \in \mathrm{Pic}^0(X)$, put*

$$A_L = (L^\vee \otimes \omega_X)|_W.$$

Then

$$\eta^* A_L \simeq \mathcal{O}_{\mathrm{dP}_6}. \tag{5.17a}$$

No assertion about $H^2(X;\mathbb{Z})$ is used.

Proof. Every line bundle in $\mathrm{Pic}^0(X)$ has $c_1(L) = 0$. Naturality therefore gives $c_1(\eta^* i^* L) = 0$. The degree-six del Pezzo surface is the blow-up of $\mathbb{P}^2$ in three noncollinear points, so $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$. The exponential sequence consequently makes

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \longrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

injective. Hence $\eta^* i^* L$ is trivial. The assumed neighbourhood trivialization of ω_X also pulls back trivially, which proves (5.17a). □

Proposition 46.13 (the exact all-line-bundle restriction obstruction)**.** ***[VERIFIED]*** *Define*

$$r = \eta^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6).$$

Under the same local canonical trivialization,

$$\eta^* A_L \simeq r(L)^\vee \qquad (L \in \mathrm{Pic}(X)). \tag{5.17b}$$

Thus the trivial-pullback hypothesis used by theorem 46.5 holds simultaneously for every line bundle exactly when $r = 0$. It always holds on $\mathrm{Pic}^0(X)$.

Proof. Pullback commutes with tensor product and duals, while the local trivialization gives $\eta^* i^* \omega_X \simeq \mathcal{O}$. This is (5.17b), and its uniform equivalence with $r = 0$ is immediate. The final assertion is theorem 46.12. □

Proposition 46.14 (Leray-edge vanishing of normalized cusp restrictions)**.** ***[VERIFIED]*** *For the constructed map $f : X \to B$ and every $L \in \mathrm{Pic}(X)$, the degree-two Leray edge section satisfies*

$$e(c_1(L)) = 0 \in H^0(B, R^2 f_* \mathbb{Z}), \qquad \eta^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}. \tag{5.17c}$$

This proof does not invoke the packaged conclusion $H^2(X;\mathbb{Z}) = 0$.

Proof. First record explicitly the compatibility used here. For every $b \in B$, proper base change and functoriality of the filtered-complex spectral sequence give a commutative square

$$\begin{array}{ccc} H^2(X;\mathbb{Z}) & \xrightarrow{\ e\ } & H^0(B, R^2 f_*\mathbb{Z}) \\ {\scriptstyle i_b^*}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{ev}_b} \\ H^2(X_b;\mathbb{Z}) & = & H^2(X_b;\mathbb{Z}). \end{array} \tag{5.17d}$$

Indeed, pull the Leray filtered complex back along $\{b\} \to B$; the spectral sequence over a point has only column zero, so its edge is the identity, while the right vertical map is the proper-base-change identification. This uses the functoriality and convergence statements in [36, Tags 012O, 012U and 012W] and analytic proper base change [32, Theorem 2.3.26]. The connecting homomorphism in the exponential sequence is natural under i_b^*, so (5.17d) sends $c_1(L)$ to $c_1(L|_{X_b})$.

Put $s_L = e(c_1(L))$. The image of an edge morphism consists of permanent cycles, hence

$$s_L \in E_\infty^{0,2} \subseteq \ker(d_2^{0,2} : E_2^{0,2} \to E_2^{2,1}); \tag{5.17e}$$

see [36, Tags 011N, 011O and 012U]. For the constructed candidate, theorems 39.2 and 39.4 give

$$E_2^{0,2} = \mathbb{Z}(2q), \qquad E_2^{2,1} = \mathbb{Z}\,\omega[\delta], \qquad d_2^{0,2}(2q) = \omega[\delta].$$

Thus $d_2^{0,2}$ is an isomorphism and (5.17e) forces $s_L = 0$. Taking $b = p_0$ in (5.17d) gives $c_1(i^*L) = 0$, and hence $c_1(\eta^* i^* L) = 0$. Finally, the injectivity of $c_1 : \mathrm{Pic}(\mathrm{dP}_6) \to H^2(\mathrm{dP}_6;\mathbb{Z})$ proved in theorem 46.12 makes $\eta^* i^* L$ trivial. □

Lemma 46.15 (the pair-of-pants defect class)**.** ***[VERIFIED]*** *Choose the compact pair-of-pants core $P \subset B^\circ$ and the zero-section whiskers used in theorem 39.6, and put $J_P = f^{-1}(P)$. Orient its boundary meridians clockwise. Write ρ_0 for the projection of the zero-section cusp meridian $\widetilde{a}_0$. The constructed candidate uses the tautological cusp gluing, whose affine regluing class is $\mu = 0$ (equivalently $\ell_0 = \gamma(\mu) = 0$); hence the flat-marked cusp lift has no vertical translation. The base loops and their chosen lifts therefore satisfy*

$$\rho_1\rho_2\rho_0 = 1, \qquad \widehat{\rho}_1\widehat{\rho}_2\widetilde{a}_0 = 1. \tag{5.17g}$$

For $L \in \mathrm{Pic}(X)$, write $E = c_1(L|_F) \in \bigwedge^2 V$. Then $E \in (\bigwedge^2 V)^G$, and there is a crossed homomorphism

$$z : \pi_1(P) \longrightarrow V_{\mathbb{R}}, \qquad z(gh) = z(g) + T_g z(h), \tag{5.17h}$$

whose class records $c_1(L|_{J_P}) - [\Omega_E]$. Here Ω_E is the closed, translation- invariant vertical two-form induced by E in the flat marking. If $d_i = z(\rho_i)$ and a bar denotes the image in $V_G \otimes \mathbb{R}$, then

$$\bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0. \tag{5.17i}$$

The three barred classes, and the number $\langle d_0, \widehat{\delta}\rangle$, do not depend on the chosen crossed- homomorphism representative. More precisely, use the base-first convention for the edge injection in (5.17j). For any oriented loop $\rho \subset P$ and any A_ρ-invariant $\nu \in \Lambda$, its flat-marked sweep satisfies

$$\langle z(\rho), \nu\rangle = -\langle c_1(L|_{J_P}) - [\Omega_E], a_*([S^1_\nu] \times [\widetilde{\rho}])\rangle. \tag{5.17j'}$$

Here a is the fibrewise translation action and $\widetilde{\rho}$ is the flat-marked lift of ρ.

Proof. The flat marking identifies the pullback of J_P to the universal cover of P with $\widetilde{P} \times (\Lambda_{\mathbb{R}}/\Lambda)$. Since E is the restriction of a class on the total space, it is fixed by every monodromy T_g. The constant vertical form representing E is therefore invariant under the deck action and descends to the asserted form Ω_E on J_P.

The pair of pants retracts onto a graph. Hence the Serre spectral sequence of the marked real torus bundle has only columns zero and one and gives the short exact edge sequence

$$0 \longrightarrow H^1(P; V_{\mathbb{R}}) \longrightarrow H^2(J_P; \mathbb{R}) \longrightarrow H^0(P; \textstyle\bigwedge^2 V_{\mathbb{R}}) \longrightarrow 0. \tag{5.17j}$$

Both $c_1(L|_{J_P})$ and $[\Omega_E]$ map to E, so their difference is the image of a unique class in $H^1(P; V_{\mathbb{R}})$. The standard group- cohomology model for a graph identifies this group with crossed homomorphisms modulo principal crossed homomorphisms [27, Chapter I, §5]; this gives (5.17h).

For completeness, the slant convention in (5.17j') is part of this identification, not an additional geometric assertion. Give an oriented edge of a graph spine of P a coordinate s increasing with its based loop and represent the corresponding $V_{\mathbb{R}}$-valued cellular one-cochain by a base-first form $ds \wedge \beta_s$. On the sweep of an A_ρ-invariant fibre circle S^1_ν, Fubini's formula gives

$$\int_{S^1_s \times S^1_\nu} ds \wedge \beta_s = \int_{S^1_s} \langle \beta_s, \nu \rangle, ds = \langle z(\rho), \nu \rangle.$$

The sweep in (5.17j') is written in the fibre-first order $[S^1_\nu] \times [\widetilde{\rho}]$, which differs from this base-first cross-product orientation by $(-1)^{1\cdot 1} = -1$; this is the displayed minus sign. Changing the cellular representative by a principal cochain changes both sides by the same boundary term, which vanishes because $A_\rho \nu = \nu$. This proves (5.17j').

Applying z to (5.17g) gives

$$d_1 + T_1 d_2 + T_1 T_2 d_0 = 0.$$

All T_g act trivially on the global coinvariants, proving (5.17i). A change of representative has the form $z(g) \mapsto z(g) + (T_g - I)a$. Its image in $V_G \otimes \mathbb{R}$ is zero. Moreover the displayed matrices in theorem 5.1 fix $\widehat{\delta}$ under every dual action A_g. Therefore

$$\langle (T_g - I)a, \widehat{\delta} \rangle = \langle a, (A_g^{-1} - I)\widehat{\delta} \rangle = 0,$$

which proves the final assertion. □

Lemma 46.16 (the finite-cover norm identity)**.** ***[VERIFIED]*** *Retain the notation of theorems 8.7 and 46.15. For $j = 1, 2$, let*

$$N_{T_j} = I + T_j + \cdots + T_j^{m_j - 1}.$$

With the clockwise conventions of (5.17g), the pullback to the m_j-fold s_j-cover gives

$$N_{T_j} d_j = \iota_{v_j} E \quad \text{in } V_{\mathbb{R}}, \qquad m_j \bar{d}_j = [\iota_{v_j} E] \quad \text{in } V_G \otimes \mathbb{R}. \tag{5.17k}$$

Proof. Restriction of the crossed homomorphism to the subgroup generated by $\rho_j^{m_j}$ has value

$$z(\rho_j^{m_j}) = \sum_{r=0}^{m_j - 1} T_j^r z(\rho_j) = N_{T_j} d_j. \tag{5.17l}$$

This is the cohomological pullback of the annular defect to the s_j-cover; the norm, rather than unqualified multiplication by m_j, is essential before passage to global coinvariants. A principal change $d_j \mapsto d_j + (T_j - I)a$ does not alter this value, since $N_{T_j}(T_j - I)a = (T_j^{m_j} - I)a = 0$.

Let $Q_j : J'_j \to N_j$ be the finite quotient map. The global flat marking Ξ over the full disc Δ_j makes the constant vertical form Ω_E well defined on J'_j. Radial retraction onto the central torus makes

$$H^2(J'_j; \mathbb{R}) \longrightarrow H^2(A_0^{(j)}; \mathbb{R})$$

an isomorphism. On the central fibre the quotient is exactly π_j, and naturality gives

$$(Q_j|_{A_0^{(j)}})^* c_1(L|_{S_j}) = \pi_j^* c_1(L|_{S_j}) = E. \tag{5.17l'}$$

The second equality is the actual radial/stalk identification in theorem 8.8; on a nearby fibre the gluing map h_j is a translation and therefore acts trivially on H^2 of that fibre. Consequently the two classes $c_1(Q_j^*(L|_{N_j}))$ and $[\Omega_E]$ have the same central restriction, and the retraction isomorphism proves

$$c_1(Q_j^*(L|_{N_j})) - [\Omega_E] = 0 \quad \text{in } H^2(J_j'; \mathbb{R}). \tag{5.17l"}$$

On Δ_j^*, the source identity $h_j \widetilde{g}_j^{\log} h_j^{-1} = \widetilde{g}_j^0$ shows that h_j descends from the logarithmic quotient N_j^* to the linear/global quotient J^*. Thus

$$Q_j^*(L|_{N_j})|_{\Delta_j^*} \simeq h_j^*(L|_{J^*}). \tag{5.17l"'}$$

The same h_j is translation in flat coordinates by αv_j, where

$$\alpha = \frac{\log s_j}{2\pi i}.$$

For any constant vertical two-form, direct substitution gives

$$h_j^* \Omega_E - \Omega_E = d\alpha \wedge \iota_{v_j} E. \tag{5.17m}$$

Along a clockwise loop in the s_j-plane one has $\int d\alpha = -1$. Thus the cross component contributed by (5.17m) is $-\iota_{v_j} E$. The remaining cross component is precisely (5.17l), while the total class extends over the disc and has zero cross component. Therefore $N_{T_j} d_j - \iota_{v_j} E = 0$, which is the first identity in (5.17k). In V_G every T_j is the identity, so the norm becomes multiplication by m_j and the second identity follows. □

Lemma 46.17 (the cusp boundary defect and Chern-class specialization)**.** ***[VERIFIED]*** *In the notation of theorem 46.15, one has*

$$\bar{d}_0 = 0 \quad \textit{in } V_G \otimes \mathbb{R}. \tag{5.17n}$$

Moreover, for every line bundle L on the cusp neighbourhood N_0, nearby-cycle specialization is compatible with first Chern classes:

$$\mathrm{sp}^2(c_1(L|_W)) = c_1(L|_F). \tag{5.17o}$$

Proof. Let $\widetilde{a}_0$ be the flat-marked clockwise zero-section meridian in the lifted relation of (5.17g), and let

$$\xi_{\widehat{\delta}} = a_*([S^1_{\widehat{\delta}}] \times [\widetilde{a}_0])$$

be its sweep under fibrewise translation by $\widehat{\delta}$. The boundary ρ_0 of P was chosen to be its projection. This is the zero-section sweep: there is no hidden vertical drift in $\widetilde{a}_0$, by the explicit overlap calculation in theorem 39.6. Hence the vertical form Ω_E vanishes on the swept torus, and the slant identity (5.17j') gives

$$\langle d_0, \widehat{\delta} \rangle = -\langle c_1(L|_{J_P}) - [\Omega_E], \xi_{\widehat{\delta}} \rangle = -\langle c_1(L|_{J_P}), \xi_{\widehat{\delta}} \rangle. \tag{5.17p}$$

Combine the exponential overlap coordinates (x_1, x_2, t) of (2.8ii) with the unimodular cusp chart

$$(u, x_1, x_2) = \left(\frac{t}{x_1 x_2}, x_1, x_2 \right).$$

The literal solid torus

$$\mathbb{S}_{\widehat{\delta}} = \{ |u| = \rho,\ x_1 = 1,\ |x_2| \le 1 \}$$

has boundary parametrization

$$(\theta, \phi) \longmapsto (\rho e^{i(\theta - \phi)}, 1, e^{i\phi}) \tag{5.17p0}$$

and hence $t = \rho e^{i\theta}$; its ϕ-circle is translation by $\widehat{\delta}$. Direct evaluation in the proved cusp marking therefore identifies $\partial \mathbb{S}_{\widehat{\delta}}$ with $\xi_{\widehat{\delta}}$. Reversing θ supplies the fixed clockwise convention and only changes the common

orientation sign. No reduced Mayer–Vietoris block or Leray comparison is used here. Since L extends across N_0, its Chern class evaluates to zero on that boundary. Thus the left side of (5.17p) is zero. By theorem 5.3, $V_G = \mathbb{Z}[\delta]$, and evaluation on the globally invariant dual vector $\widehat{\delta}$ identifies $V_G \otimes \mathbb{R}$ with $\mathbb{R}$. This proves (5.17n).

For (5.17o), let $\ell = c_1(L|_{N_0})$, let $\iota_0 : W \hookrightarrow N_0$, and let $\iota_t : F_t \hookrightarrow N_0$ be a sufficiently near smooth cusp fibre. The neighbourhood realization of the nearby-cycle unit $\mathbb{Z}_W \to \psi_{f_0}\mathbb{Z}$ recalled in the proof of theorem 39.1 gives the commutative square

$$\begin{array}{ccc} H^2(N_0;\mathbb{Z}) & \xrightarrow{\iota_0^*} & H^2(W;\mathbb{Z}) \\ {\scriptstyle \iota_t^*}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{sp}^2} \\ H^2(F_t;\mathbb{Z}) & = & H^2(F_t;\mathbb{Z}). \end{array} \tag{5.17p'}$$

Applying this square to ℓ gives $\mathrm{sp}^2(c_1(L|_W)) = c_1(L|_{F_t})$. Flat parallel transport in the marked smooth family identifies the latter with $c_1(L|_F) = E$. Equivalently, c_1 is the connecting morphism in the exponential sequence, and restriction to the shrinking neighbourhoods and to F_t commutes with that morphism before passage to the direct limit. This proves the claimed naturality without using any global Leray differential. □

Proposition 46.18 (direct boundary-Picard vanishing)**.** ***[VERIFIED]*** *For the constructed threefold and every analytic line bundle $L \in \mathrm{Pic}(X)$,*

$$c_1(L|_F) = 0, \qquad \eta^* i^* L \simeq \mathcal{O}_{\mathrm{dP}_6}. \tag{5.17q}$$

This proof is independent of the global Leray transgressions $d_2^{0,q}$ and of the packaged conclusion $H^2(X;\mathbb{Z}) = 0$; it uses only the actual finite-filling lattices, their logarithmic overlap maps, the based boundary relation, the local cusp solid torus, and the local integral specialization theorem.

Proof. The common invariant calculation in theorem 5.3 first gives

$$E = c_1(L|_F) = kq.$$

Because L extends across the order-four filling, its fibre class belongs to the actual specialization lattice

$$\pi_2^* H^2(S_2;\mathbb{Z}) = \{a\gamma\eta_2 + bq : a \equiv b \pmod 2\}$$

of theorems 5.4 and 8.8. Taking coefficient independence in the displayed invariant basis forces $a = 0$ and $b = k$; hence k is even. Write $E = 2nq$.

Contraction with the two affine translation vectors is an exact integral calculation:

$$\iota_{v_1}(2q) = 4\eta_1, \qquad \iota_{v_2}(2q) = -6\eta_2. \tag{5.17r}$$

Indeed $v_1 = (1,2,-4,0)^t$, $v_2 = (-1,-3,3,0)^t$ in the basis dual to (γ,u,w,δ), while $q = uw + 6\gamma\delta$. In the global coinvariants,

$$[\eta_1] = 3[\delta], \qquad [\eta_2] = 2[\delta].$$

Applying theorem 46.16 with $m_1 = 3$, $m_2 = 4$ therefore gives

$$\bar{d}_1 = 4n[\delta], \qquad \bar{d}_2 = -3n[\delta]. \tag{5.17s}$$

Now combine the balance relation (5.17i) with the cusp vanishing (5.17n):

$$0 = \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = n[\delta].$$

The coinvariant group is torsion-free, so $n = 0$ and hence $E = 0$.

Finally (5.17o) and the injectivity of the local integral specialization in theorem 39.1 imply $c_1(L|_W) = 0$. Pulling back to the normalization gives $c_1(\eta^* i^* L) = 0$. The injectivity of $c_1 : \mathrm{Pic}(\mathrm{dP}_6) \to H^2(\mathrm{dP}_6;\mathbb{Z})$ proved in theorem 46.12 yields the second assertion of (5.17q). □

46.7 An analytic completion of the all-Picard certificate

The local example in theorem 41.8 shows that the cusp neighbourhood alone cannot annihilate normalized central restrictions. We now close the stronger analytic route by retaining the order-three and order-four fillings. The proof has two independent parts. First, logarithms of the actual $\mathcal{O}^*$-factors give the boundary cocycle and its two finite-filling norm identities. Second, a candidate-specific support-function calculation reads every possible central restriction directly from the mixed Appell–Humbert block on a nearby torus. No global Leray differential, nearby-cycle specialization, or Chern-class evaluation on a cusp bounding chain occurs.

Proposition 46.19 (analytic Appell–Humbert boundary cocycle)**.** ***[VERIFIED]*** *Let $P \subset B^\circ$ be the compact pair-of-pants core, orient its boundary meridians clockwise, and retain the zero-section lifts of theorem 39.6. Thus*

$$\rho_1\rho_2\rho_0 = 1, \qquad \widehat{\rho}_1\widehat{\rho}_2\widetilde{a}_0 = 1, \tag{5.17-A1}$$

and the second relation has no terminal vertical translation. For $L \in \mathrm{Pic}(X)$, let

$$E = c_1(L|_F) \in \mathrm{Alt}^2(\Lambda, \mathbb{Z}) = \bigwedge^2 V$$

be its fibrewise Appell–Humbert form. Put $V_{\mathbb{R}} = \mathrm{Hom}(\Lambda, \mathbb{R})$, and use

$$A_g = (T_g^{-1})^{\mathsf{t}}, \qquad \langle T_g\alpha, \lambda\rangle = \langle \alpha, A_g^{-1}\lambda\rangle. \tag{5.17-A2}$$

There is a crossed homomorphism, unique up to a principal crossed homomorphism,

$$d : \pi_1(P) \longrightarrow V_{\mathbb{R}}, \qquad d(gh) = d(g) + T_g d(h), \qquad d(g) \sim d(g) + (T_g - I)a. \tag{5.17-A3}$$

If $d_i = d(\rho_i)$, then

$$d_1 + T_1 d_2 + T_1T_2 d_0 = 0, \qquad \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = 0 \quad \text{in } V_G \otimes \mathbb{R}. \tag{5.17-A4}$$

For the finite filling of order m_j, $j = 1, 2$,

$$N_{T_j} d_j = \iota_{v_j} E, \qquad m_j \bar{d}_j = [\iota_{v_j} E], \qquad N_{T_j} = I + T_j + \cdots + T_j^{m_j - 1}, \tag{5.17-A5}$$

where $(\iota_v E)(\lambda) = E(v, \lambda)$. At the cusp,

$$\langle d_0, \widehat{\delta}\rangle = 0, \qquad \bar{d}_0 = 0 \quad \text{in } V_G \otimes \mathbb{R}. \tag{5.17-A6}$$

These identities are obtained from holomorphic transition factors and the argument principle.

Proof. Choose an open Riemann-surface neighbourhood $P^{\mathrm{an}} \Subset B^\circ$ which deformation retracts onto P, and pull the period family to its universal cover. Before quotienting by the fibre lattice, the resulting space is

$$\widetilde{\mathcal{J}} = \widetilde{P}^{\mathrm{an}} \times \mathbb{C}^2.$$

It is Stein and contractible, so the pullback of L is holomorphically trivial. In a nowhere-zero frame, write the factor belonging to a deck transformation α as

$$a_\alpha \in \mathcal{O}^*(\widetilde{\mathcal{J}}), \qquad a_{\alpha\beta}(x) = a_\alpha(\beta x) a_\beta(x). \tag{5.17-A7}$$

The space is simply connected, hence $a_\alpha = e^{2\pi i h_\alpha}$ for a holomorphic logarithm. Therefore

$$c(\alpha, \beta) = h_\alpha(\beta x) + h_\beta(x) - h_{\alpha\beta}(x) \in \mathbb{Z} \tag{5.17-A8}$$

is independent of x, and associativity makes c an integral group two-cocycle.

Let τ_λ denote fibre translation by $\Pi(z)\lambda$, $\lambda \in \Lambda$, and let $\widehat{g}$ be the zero-section lift of a base deck transformation. Period equivariance gives

$$\widehat{g}\,\tau_\lambda\,\widehat{g}^{-1} = \tau_{A_g\lambda}. \tag{5.17-A9}$$

On fibre translations, the commutator of (5.17-A8) is

$$E(\lambda,\mu) = c(\tau_\lambda,\tau_\mu) - c(\tau_\mu,\tau_\lambda). \tag{5.17-A10}$$

The four oriented sides of the (λ,μ)-parallelogram identify this integer with the exponential connecting class of $L|_F$; this is the Appell–Humbert convention used in (5.17av)–(5.17ax), and agrees with the standard factor-of-automorphy calculation [18, Chapter 1, §2, pp. 5–8, especially Theorem 2.1, p. 7][19, equations (3.3)–(3.10), pp. 3–7]. Conjugating the commutator by $\widehat{g}$ yields

$$E(A_g\lambda, A_g\mu) = E(\lambda,\mu). \tag{5.17-A11}$$

Push the logarithmic central extension defined by c out along $\mathbb{Z} \hookrightarrow \mathbb{R}$, and denote its central element of weight r by $\mathbf{c}(r)$. Choose lifts U_λ of the fibre translations satisfying the real Appell–Humbert normalization

$$U_\lambda U_\mu = \mathbf{c}\Big(\frac{1}{2}E(\lambda,\mu)\Big)\,U_{\lambda+\mu}. \tag{5.17-A12}$$

To see that this normalization exists, start with arbitrary lifts. The difference between their real cocycle and $E/2$ has zero commutator, so it defines an abelian extension of the free abelian group Λ by $\mathbb{R}$. Choose lifts of an integral basis and extend multiplicatively; this splits that abelian extension and gives (5.17-A12). If U'_λ is a second normalized system, then $U'_\lambda U_\lambda^{-1}$ is central, say $\mathbf{c}(\langle a,\lambda\rangle)$. Comparing the two copies of (5.17-A12) proves additivity in λ, so $a \in V_{\mathbb{R}}$. Thus every two normalized systems differ by exactly the character change used below.

For $g \in \pi_1(P)$, choose a lift G_g of $\widehat{g}$ and define $d(g) \in V_{\mathbb{R}}$ by

$$G_g U_{A_g^{-1}\lambda} G_g^{-1} = \mathbf{c}\big(\langle d(g),\lambda\rangle\big)U_\lambda. \tag{5.17-A13}$$

The exponent is additive in λ: conjugate (5.17-A12) and use (5.17-A11). The lifts of gh and G_gG_h differ centrally, so

$$\begin{aligned}\langle d(gh),\lambda\rangle &= \langle d(g),\lambda\rangle + \langle d(h),A_g^{-1}\lambda\rangle\\ &= \langle d(g) + T_g d(h),\lambda\rangle.\end{aligned}$$

This proves the crossed-homomorphism law. Replacing U_λ by $\mathbf{c}(\langle a,\lambda\rangle)U_\lambda$ changes (5.17-A13) by $(T_g - I)a$. Replacing G_g by a central multiple does not change (5.17-A13), so this proves the stated independence and uniqueness. Applying the crossed law to (5.17-A1) proves the first identity in (5.17-A4); passage to coinvariants proves the second.

For the order-m_j filling, distinguish every covering level. Put

$$\widetilde{J}_j = \Delta_j \times \mathbb{C}^2, \qquad r_j : \widetilde{J}_j \longrightarrow J'_j$$

for the lattice quotient, whose deck translations are τ_λ. On the vector cover define

$$Q_{j,\log} := \widetilde{g}_j^{\,v_j/m_j}, \qquad Q_{j,\mathrm{lin}} := \widetilde{g}_j^{\,0}.$$

They satisfy

$$\begin{aligned} Q_{j,\log}\tau_\lambda Q_{j,\log}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\log})^{m_j} = \tau_{v_j},\\ Q_{j,\mathrm{lin}}\tau_\lambda Q_{j,\mathrm{lin}}^{-1} &= \tau_{A_j\lambda}, \quad (Q_{j,\mathrm{lin}})^{m_j} = 1.\end{aligned} \tag{5.17-A14}$$

The positive sign is fixed by $s_j \circ g_j = e^{-2\pi i/m_j}s_j$ and (2.8f): after exactly m_j clockwise iterations, the logarithmic vector-cover lift translates by $+v_j$.

Let $q_{j,\log}, q_{j,\mathrm{lin}}$ be the descended automorphisms of J'_j, and put $J'^*_j = J'_j|_{\Delta^*_j}$. Both descended automorphisms have order dividing m_j, because τ_{v_j} is trivial after the lattice quotient; each covers g_j, whose order is exactly m_j, so each descended order is exactly m_j. The branch-independent torus translation

$$h_j : J'^*_j \longrightarrow J'^*_j$$

constructed in (2.9d) satisfies, on this quotient level,

$$h_j q_{j,\log} h_j^{-1} = q_{j,\mathrm{lin}}.$$

Define

$$p_{j,\log} : J'^*_j \longrightarrow N^*_j := J'^*_j/\langle q_{j,\log}\rangle,$$
$$p_{j,\mathrm{lin}} : J'^*_j \longrightarrow J^*_j := J'^*_j/\langle q_{j,\mathrm{lin}}\rangle.$$

The conjugacy induces a biholomorphism

$$\bar h_j : N^*_j \longrightarrow J^*_j, \qquad p_{j,\mathrm{lin}} \circ h_j = \bar h_j \circ p_{j,\log}.$$

The gluing of X identifies $L|_{N^*_j} \simeq \bar h_j^*(L|_{J^*_j})$; hence the exactly typed line-bundle square gives

$$p^*_{j,\log}(L|_{N^*_j}) \simeq h_j^* p^*_{j,\mathrm{lin}}(L|_{J^*_j}). \tag{5.17-A14a}$$

For the factor extension return to the filled vector cover. Let

$$P_{j,\log} : J'_j \longrightarrow N_j, \qquad \pi_j^{\mathrm{vec}} := P_{j,\log} \circ r_j : \widetilde J_j \longrightarrow N_j.$$

Here $P_{j,\log}$ is the full quotient and $P_{j,\log}|_{J'^*_j} = p_{j,\log}$. The deck group is

$$\Gamma_j^{\log} = \langle \tau_\lambda, Q_{j,\log} : \lambda \in \Lambda\rangle.$$

The pullback $(\pi_j^{\mathrm{vec}})^*(L|_{N_j})$ is trivial on the contractible Stein manifold $\widetilde J_j$; its factors define an integral central extension, which we push out along $\mathbb{Z} \hookrightarrow \mathbb{R}$. Over Δ^*_j, the line-bundle square (5.17-A14a) identifies each fibre with the global factor extension by a translation, so its commutator form is E. That form is integer-valued and locally constant on the connected disc Δ_j; therefore it is E on the filled cover as well. Choose vertical lifts $U^{(j)}_\lambda$ normalized by

$$U^{(j)}_\lambda U^{(j)}_\mu = \mathbf{c}\Big(\frac{1}{2}E(\lambda,\mu)\Big)\, U^{(j)}_{\lambda+\mu}.$$

For $\alpha \in \Gamma_j^{\log}$, define A_α, T_α by

$$\alpha \tau_\lambda \alpha^{-1} = \tau_{A_\alpha \lambda}, \qquad T_\alpha = (A_\alpha^{-1})^{\mathsf t}.$$

For a lift $\widetilde G_\alpha$, define $\widetilde d_j(\alpha) \in V_{\mathbb{R}}$ through

$$\widetilde G_\alpha U^{(j)}_{A_\alpha^{-1}\lambda} \widetilde G_\alpha^{-1} = \mathbf{c}\Big(\langle \widetilde d_j(\alpha), \lambda\rangle\Big)\, U^{(j)}_\lambda. \tag{5.17-A14c}$$

Indeed, before naming $\widetilde d_j(\alpha)$, the quotient of the two sides of (5.17-A14c) is central because both cover the same translation. Conjugating the normalized product and using

$$E(A_\alpha\lambda, A_\alpha\mu) = E(\lambda,\mu)$$

shows that its exponent $r_\alpha(\lambda)$ satisfies

$$r_\alpha(\lambda+\mu) = r_\alpha(\lambda) + r_\alpha(\mu).$$

Thus it is the pairing with a unique element of $V_{\mathbb{R}}$, as asserted. Since $A_{\alpha\beta} = A_\alpha A_\beta$, choose the lift of $\alpha\beta$ to be $\widetilde{G}_\alpha \widetilde{G}_\beta$; any other choice differs centrally. Then

$$\begin{aligned}&\widetilde{G}_\alpha \widetilde{G}_\beta U^{(j)}_{A_\beta^{-1} A_\alpha^{-1} \lambda} \widetilde{G}_\beta^{-1} \widetilde{G}_\alpha^{-1} \\ &\quad = \mathbf{c}\Big(\langle \widetilde{d}_j(\beta), A_\alpha^{-1}\lambda \rangle + \langle \widetilde{d}_j(\alpha), \lambda \rangle \Big) U^{(j)}_\lambda .\end{aligned}$$

The defining dual action $\langle T_\alpha v, \lambda \rangle = \langle v, A_\alpha^{-1}\lambda \rangle$ therefore proves the crossed law

$$\widetilde{d}_j(\alpha\beta) = \widetilde{d}_j(\alpha) + T_\alpha \widetilde{d}_j(\beta). \tag{5.17-A14d}$$

Put

$$\widetilde{J}_j^* := \Delta_j^* \times \mathbb{C}^2, \qquad r_j^* := r_j|_{\widetilde{J}_j^*} : \widetilde{J}_j^* \longrightarrow J_j'^*$$

and define

$$\begin{aligned}\Phi_{j,\log} &:= p_{j,\log} \circ r_j^*, \\ \Phi_{j,\mathrm{glob}} &:= p_{j,\mathrm{lin}} \circ h_j \circ r_j^* = \bar{h}_j \circ \Phi_{j,\log}.\end{aligned}$$

Both maps are invariant under $\Gamma_j^{\log}$. This is immediate for every τ_λ, while

$$\begin{aligned}\Phi_{j,\log} Q_{j,\log} &= p_{j,\log} q_{j,\log} r_j^* = \Phi_{j,\log}, \\ \Phi_{j,\mathrm{glob}} Q_{j,\log} &= p_{j,\mathrm{lin}} h_j q_{j,\log} r_j^* = p_{j,\mathrm{lin}} q_{j,\mathrm{lin}} h_j r_j^* = \Phi_{j,\mathrm{glob}}.\end{aligned}$$

Thus $\widetilde{J}_j^*$ is the same $\Gamma_j^{\log}$-cover of the two biholomorphic quotients. Pulling the gluing isomorphism to this cover gives

$$\Theta_j : \Phi_{j,\log}^*(L|_{N_j^*}) \xrightarrow{\sim} \Phi_{j,\mathrm{glob}}^*(L|_{J_j^*}).$$

The filled factor extension restricts to this cover, and the global factor extension pulls back to it through the fixed boundary whisker. Because Θ_j intertwines descent, it induces an isomorphism of the resulting pushed-out $\Gamma_j^{\log}$-factor extensions which sends $\mathbf{c}(r)$ to $\mathbf{c}(r)$ for every $r \in \mathbb{R}$.

Let $\gamma_j^J : [0,1] \to J_j^*$ be the zero-section image of the unwhiskered clockwise boundary meridian, and let ω_j be the fixed zero-section whisker from the global basepoint to $\gamma_j^J(0)$. Thus the based total-space loop retained in (5.17-A1) is

$$\widehat{\rho}_j = \omega_j * \gamma_j^J * \omega_j^{-1}.$$

Choose $x_j \in J_j'^*$ such that $p_{j,\mathrm{lin}}(h_j(x_j)) = \gamma_j^J(0)$, put $x_{j,\mathrm{lin}} = h_j(x_j)$, and let $\widetilde{\gamma}_{j,\mathrm{lin}}$ be the $p_{j,\mathrm{lin}}$-lift of γ_j^J beginning at $x_{j,\mathrm{lin}}$. For the chosen clockwise generator,

$$\widetilde{\gamma}_{j,\mathrm{lin}}(1) = q_{j,\mathrm{lin}}(x_{j,\mathrm{lin}}).$$

Set

$$\gamma_j^N := \bar{h}_j^{-1} \circ \gamma_j^J, \qquad \widetilde{\gamma}_{j,\log} := h_j^{-1} \circ \widetilde{\gamma}_{j,\mathrm{lin}}.$$

Then the quotient square gives the point-valued identities

$$\begin{aligned}p_{j,\log} \circ \widetilde{\gamma}_{j,\log} &= \gamma_j^N, \\ \widetilde{\gamma}_{j,\log}(0) &= x_j, \\ \widetilde{\gamma}_{j,\log}(1) &= q_{j,\log}(x_j).\end{aligned}$$

Choose $\widetilde{x}_j \in \widetilde{J}_j^*$ with $r_j^*(\widetilde{x}_j) = x_j$, and let $\widetilde{\gamma}_{j,\mathrm{vec}}$ be the r_j^*-lift of $\widetilde{\gamma}_{j,\log}$ beginning at $\widetilde{x}_j$. On adjacent sheets choose compatible logarithm branches ℓ, ℓ' with

$$\ell'(w) = \ell(g_j^{-1} w) - \frac{2\pi i}{m_j}.$$

The calculation in (2.9e) gives the point-valued endpoint formula

$$\widetilde{\gamma}_{j,\mathrm{vec}}(1) = Q_{j,\log}(\widetilde{x}_j),$$

excluding both $Q_{j,\log}^{-1}(\widetilde{x}_j)$ and every $\tau_\lambda Q_{j,\log}(\widetilde{x}_j)$ with $\lambda \neq 0$. Fibre loops remain τ_λ, since h_j is a fibre translation.

The factor of L along the fixed lift of ω_j identifies the target factor over the local boundary segment with the globally based factor. Let $G_{\rho_j}^{(j)}$ be the target lift of the local boundary deck element whose whiskered image is the chosen global lift G_{ρ_j}. Thus it covers $\widehat{\rho}_j$ after whiskering and, by (5.17-A13), satisfies

$$G_{\rho_j}^{(j)} U_{A_j^{-1}\lambda} (G_{\rho_j}^{(j)})^{-1} = \mathbf{c}(\langle d_j, \lambda\rangle) U_\lambda, \qquad d_j = d(\rho_j).$$

Consequently, for some $a_j \in V_{\mathbb{R}}$ and $b_j \in \mathbb{R}$,

$$\Theta_{j,*}(U_\lambda^{(j)}) = \mathbf{c}(\langle a_j, \lambda\rangle) U_\lambda, \qquad \Theta_{j,*}(\widetilde{G}_{Q_{j,\log}}) = \mathbf{c}(b_j) G_{\rho_j}^{(j)}.$$

Here U_λ is the normalized global system in (5.17-A12), and $\mathbf{c}(b_j)$ disappears under conjugation. Therefore

$$\begin{aligned} \langle \widetilde{d}_j(Q_{j,\log}), \lambda\rangle + \langle a_j, \lambda\rangle &= \langle d_j, \lambda\rangle + \langle a_j, A_j^{-1}\lambda\rangle \\ &= \langle d_j + T_j a_j, \lambda\rangle, \end{aligned}$$

so

$$\widetilde{d}_j(Q_{j,\log}) = d_j + (T_j - I) a_j. \tag{5.17-A14e}$$

For a translation choose $\widetilde{G}_{\tau_v} = U_v^{(j)}$. The filled normalization above gives

$$U_v^{(j)} U_\lambda^{(j)} (U_v^{(j)})^{-1} = \mathbf{c}(E(v,\lambda)) U_\lambda^{(j)},$$

and hence

$$\widetilde{d}_j(\tau_v) = \iota_v E, \qquad (\iota_v E)(\lambda) = E(v,\lambda). \tag{5.17-A14f}$$

Here

$$A_{Q_{j,\log}} = A_j, \qquad T_{Q_{j,\log}} = T_j.$$

Since $N_{T_j}(T_j - I) = T_j^{m_j} - I = 0$, equations (5.17-A14), (5.17-A14d), and (5.17-A14e) give

$$\begin{aligned} N_{T_j} d_j &= N_{T_j} \widetilde{d}_j(Q_{j,\log}) \\ &= \widetilde{d}_j((Q_{j,\log})^{m_j}) \\ &= \widetilde{d}_j(\tau_{v_j}) \\ &= \iota_{v_j} E. \end{aligned} \tag{5.17-A15}$$

Changing a line-bundle frame gives an isomorphic factor extension; changing $\widetilde{G}_\alpha$ by a central factor does not change conjugation; and every second normalized vertical system differs by a character, changing $\widetilde{d}_j(\alpha)$ by $(T_\alpha - I)a$. Finally, $\ell \mapsto \ell + 2\pi i n$ changes a vector-cover lift of h_j by

$$H_j \longmapsto H_j' = \tau_{n v_j} H_j.$$

At the factor-extension level this means

$$\Theta_{j,*}' = \mathrm{Ad}(U_{n v_j}) \circ \Theta_{j,*}.$$

Consequently the comparison character changes by $n\iota_{v_j} E$, because

$$U_{n v_j} (\mathbf{c}(\langle a_j, \lambda\rangle) U_\lambda) U_{n v_j}^{-1} = \mathbf{c}(\langle a_j + n\iota_{v_j} E, \lambda\rangle) U_\lambda.$$

Since $A_j v_j = v_j$ and E is A_j-invariant,

$$\langle T_j(\iota_{v_j} E), \lambda\rangle = E(v_j, A_j^{-1}\lambda) = E(A_j v_j, \lambda) = E(v_j, \lambda),$$

so this branch change leaves both (5.17-A14e) and (5.17-A15) unchanged. The translation τ_{nv_j} also commutes with $Q_{j,\mathrm{lin}}$ on the vector cover because $A_j v_j = v_j$; on the lattice quotient it is already the identity. Hence it changes only the central lift choice there. The descended $h_j, \bar{h}_j$, and the quotient square are branch independent. Thus (5.17-A15) is choice-independent and proves (5.17-A5), including its sign.

It remains to obtain (5.17-A6) analytically. Since $A_0\widehat{\delta} = \widehat{\delta}$, equation (5.17-A13) says that

$$[G_0, U_{\widehat{\delta}}] = \mathbf{c}(\langle d_0, \widehat{\delta}\rangle). \tag{5.17-A16}$$

This logarithmic commutator is the clutching winding with the base loop first. Thus, if b and f denote the oriented base and fibre loops, respectively,

$$\mathrm{wind}_{b\wedge f}(L) = \langle d_0, \widehat{\delta}\rangle, \qquad \mathrm{wind}_{\xi_{\widehat{\delta}}}(L) = -\langle d_0, \widehat{\delta}\rangle. \tag{5.17-A17}$$

The second equality uses the fibre-first convention $\xi_{\widehat{\delta}} = [S^1_{\widehat{\delta}}] \times [\rho_0]$, hence the single sign change. Use the actual unimodular cusp chart

$$(u, x_1, x_2) = \left(\frac{t}{x_1 x_2}, x_1, x_2\right), \qquad t = u x_1 x_2. \tag{5.17-A18}$$

For $0 < \rho \ll 1$, the boundary parametrization

$$(\theta, \phi) \longmapsto (\rho e^{i(\theta-\phi)}, 1, e^{i\phi}), \qquad t = \rho e^{i\theta}, \tag{5.17-A19}$$

has base coordinate θ and fibre translation $\widehat{\delta}$ in the ϕ-direction. The latter fixed-base loop does *not* itself bound the x_2-disc. To identify the correct meridian, put $\alpha = \theta - \phi$, let a_u be the positively oriented u-longitude, and let m_{x_2} be the positively oriented x_2-meridian at fixed u. In the basis (a_u, m_{x_2}), the base and fibre loops are

$$b = a_u, \qquad f = m_{x_2} - a_u, \qquad b \wedge f = a_u \wedge m_{x_2}. \tag{5.17-A19a}$$

The change-of-basis determinant is $+1$, so this preserves the base-first orientation in (5.17-A17). Now m_{x_2} does bound the analytic disc

$$z \longmapsto (\rho e^{i\alpha_0}, 1, z), \qquad |z| \le 1. \tag{5.17-A19b}$$

Choose a thin annulus A_u about $|u| = \rho$, a disc D_{x_1} about 1, and a disc D_{x_2} containing $\{|x_2| \le 1\}$, small enough that

$$A_u \times D_{x_1} \times D_{x_2} \subset \{|u x_1 x_2| < \epsilon\}. \tag{5.17-A19c}$$

This neighbourhood is Stein and retracts onto S^1, so its second integral cohomology vanishes. Cartan's theorem B and the exponential sequence therefore give trivial Picard group [39, §2, fundamental theorem, p. 81][8, Proposition 6.28 and equations (6.35)–(6.36), pp. 180–182]. Pull $L|_{N_0}$ back to the cusp cover and choose one holomorphic frame on this entire neighbourhood. Compute the clutching cocycle with this extending frame on both sides of a cut in the u-longitude. Its transition unit on D_{x_2} is then identically $F = 1$, so by (5.17-A19a) its base–fibre commutator winding is

$$\frac{1}{2\pi i}\int_{|x_2|=1} \frac{dF}{F} = 0. \tag{5.17-A20}$$

This computes the same alternating logarithmic cocycle as (5.17-A16). Indeed, changing from the global Appell–Humbert frame to the extending chart frame changes c by a group coboundary. Put

$$g = \widehat{g}_0, \qquad h = \tau_{\widehat{\delta}}$$

in the underlying deck group. Equation (5.17-A9) and $A_0\widehat{\delta} = \widehat{\delta}$ give $gh = hg$. If $c' = c + \delta\beta$ is the cocycle after a frame change, then

$$[c'(g,h) - c'(h,g)] - [c(g,h) - c(h,g)] = \beta(hg) - \beta(gh) = 0.$$

Thus the alternating cocycle is frame-invariant without assuming that its central-extension lifts commute. Equations (5.17-A17), (5.17-A19a), and (5.17-A20) therefore give $\langle d_0, \widehat{\delta}\rangle = 0$. This is a calculation in $\mathcal{O}^*$, not an evaluation of a Chern class on a bounding chain. Finally, the matrices A_1, A_2 fix $\widehat{\delta}$, so the evaluation is unchanged by every principal change in (5.17-A3). Since $V_G = \mathbb{Z}[\delta]$ and $\delta(\widehat{\delta}) = 1$, its vanishing is equivalent to $\bar{d}_0 = 0$. This proves (5.17-A6). □

Lemma 46.20 (the toric support function detects central restriction)**.** ***[VERIFIED]*** *Let $M \in \mathrm{Pic}(N_0)$, let $\pi : Y_\epsilon \to N_0$ be the cusp covering, and let $\nu : E_0 \to W$ be the normalization. If the Appell–Humbert form of $M|_{F_t}$, $t \neq 0$, is E_M, then the three independent degrees of $\nu^*(M|_W)$ determine an integral symmetric form $B : \mathbb{Z}^2 \times \mathbb{Z}^2 \to \mathbb{Z}$ satisfying*

$$E_M\big((\lambda,0),(0,m)\big) = -B(B_0\lambda, m) \qquad (\lambda, m \in \mathbb{Z}^2). \tag{5.17-A21}$$

Consequently

$$E_M = 0 \quad \Longrightarrow \quad \nu^*(M|_W) \simeq \mathcal{O}_{E_0}. \tag{5.17-A22}$$

Proof. Put $\mathcal{M} = \pi^* M$. The identity $\pi \circ \Psi_\lambda = \pi$ gives $\mathcal{M}$ its genuine Ψ-equivariant descent structure. In the notation of theorem 41.8, order the six rays in the star of the origin as

$$(r_1, r_2, r_3, r_4, r_5, r_6) = (e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2), \qquad C_i = E_0 \cap E_{r_i}. \tag{5.17-A23}$$

The quotient identifies opposite conductor branches. Therefore, for $A = \mathcal{M}|_{E_0} = \nu^*(M|_W)$,

$$d_i = \deg_{C_i} A, \qquad (\deg_{C_1} A, \dots, \deg_{C_6} A) = (d_1, d_2, d_3, d_1, d_2, d_3). \tag{5.17-A24}$$

Define the integer-valued quadratic function

$$a(x,y) = (d_2 + d_3)\binom{x}{2} + d_3 xy + (d_1 + d_3)\binom{y}{2}. \tag{5.17-A25}$$

Its polarization is

$$B(v,w) = a(v+w) - a(v) - a(w), \qquad [B] = \begin{pmatrix} d_2 + d_3 & d_3 \\ d_3 & d_1 + d_3 \end{pmatrix}. \tag{5.17-A26}$$

The locally finite divisor

$$D_a = \sum_{v \in \mathbb{Z}^2} a(v) E_v \tag{5.17-A27}$$

is Cartier: on every maximal chart, the three height-one ray generators form an integral basis, so the three assigned coefficients are the values of one integral character. In the standard toric convention the associated support function has ray values $-a(v)$; the divisor coefficients here are the values $a(v)$, consistent with $\mathrm{div}(\chi^m) = \sum_\rho \langle m, u_\rho\rangle D_\rho$ [22, §3.3, pp. 60–63; §3.4, pp. 63–72][23, Propositions 4.1.1–4.1.2, p. 171; Theorems 4.2.8, p. 179 and 4.2.12, p. 182]. This is the same chartwise Cartier argument as in (5.17ab), and it applies to an arbitrary integer-valued a.

Direct substitution gives

$$(a(r_1), \dots, a(r_6)) = (0, 0, d_2, d_2 + d_3, d_1 + d_3, d_1). \tag{5.17-A28}$$

Using $C_i^2 = -1$ and $C_i C_{i\pm1} = 1$, the degree vector of $\mathcal{O}(D_a)|_{E_0}$ is exactly the vector in (5.17-A24). More generally, restrict the principal identity $\operatorname{div}(t) = \sum_w E_w$ to E_v. Since E_v meets precisely E_{v+r_i}, it gives

$$\mathcal{O}(D_a)|_{E_v} \simeq \mathcal{O}_{E_v}\left(\sum_{i=1}^{6}(a(v+r_i) - a(v))C_{v,i}\right). \tag{5.17-A29}$$

The coefficient difference is

$$a(v+r_i) - a(v) = a(r_i) + B(v, r_i). \tag{5.17-A30}$$

The second summand in (5.17-A30) is the divisor of the toric character with exponent $B(v,-)$. Thus every restriction in (5.17-A29) has the same six degrees as $\mathcal{O}(D_a)|_{E_0}$.

Because $B_0\bar{\Lambda} = \mathbb{Z}^2$, the deck action is transitive on the components E_v. The genuine descent structure on $\mathcal{M}$ therefore transports (5.17-A24), with its ordered six boundary directions, to every E_v. It follows that

$$\mathcal{R} := \mathcal{M} \otimes \mathcal{O}(-D_a) \quad \text{satisfies} \quad \mathcal{R}|_{E_v} \simeq \mathcal{O}_{E_v} \quad (v \in \mathbb{Z}^2). \tag{5.17-A31}$$

Indeed, the six toric curves detect $\operatorname{Pic}(\mathrm{dP}_6)$: in the blow-up basis H, E_1, E_2, E_3, vanishing of the degrees on the three exceptional curves kills the three E_i-coefficients, and vanishing on any strict transform $H - E_i - E_j$ kills the remaining H-coefficient. Since $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$, equality of those degrees implies isomorphism, not merely numerical equivalence.

For $b_\lambda = B_0\lambda$, the torus-translation part of Ψ_λ preserves each toric prime divisor, while its fan part translates the index by b_λ. Hence

$$\Psi_\lambda^* D_a - D_a = \sum_v (a(v+b_\lambda) - a(v))E_v = \operatorname{div}(\chi^{m_\lambda}), \tag{5.17-A32}$$

where

$$m_\lambda = (\ell_\lambda, k_\lambda), \qquad \ell_\lambda = B(b_\lambda, -), \qquad k_\lambda = a(b_\lambda). \tag{5.17-A33}$$

Thus multiplication by χ^{m_λ} gives an individual isomorphism $\Psi_\lambda^*\mathcal{O}(D_a) \simeq \mathcal{O}(D_a)$. No cocycle assertion for these individual isomorphisms is needed: we use them only to separate the horizontal winding of the genuine descent factor of $\mathcal{M}$.

For a ray $\mathbb{R}_{\geq 0}(v,1)$, its toric open set in Y_ϵ is

$$U_v \simeq \Delta_\epsilon \times (\mathbb{C}^*)^2, \qquad E_v^\circ = \{0\} \times (\mathbb{C}^*)^2. \tag{5.17-A34}$$

Indeed $(e_1, 0), (e_2, 0), (v, 1)$ is an integral basis and t is the character dual to the last vector; compare [22, §2.1, pp. 28–29] and [23, Example 1.2.21, p. 31]. The set U_v is Stein and retracts onto E_v°. By (5.17-A31), $c_1(\mathcal{R}|_{U_v}) = 0$; the exponential sequence and Cartan's theorem B therefore trivialize $\mathcal{R}|_{U_v}$. Choose a global nowhere-zero frame s_v of $\mathcal{R}|_{E_v}$. A frame on U_v can be multiplied by a unit pulled back from $(\mathbb{C}^*)^2$ so that its restriction to E_v° agrees with $s_v|_{E_v^\circ}$; denote the resulting frame by e_v.

We also need to compare these frames on the generic torus. The direction of the toric affine-open inclusions is essential here. If v, w are adjacent vertices, choose a triangle $\sigma = \{v, w, u\}$ containing them, write z_v, z_w, z_u for the coordinates dual to its three height-one ray generators, and put

$$S_\sigma := U_\sigma \cap Y_\epsilon = \{(z_v, z_w, z_u) \in \mathbb{C}^3 : |z_v z_w z_u| < \epsilon\}. \tag{5.17-A34a}$$

Because $\rho_v < \sigma$, localization of the maximal affine chart gives

$$U_{\rho_v} = D(z_w z_u) \subset U_\sigma, \qquad U_{\rho_w} = D(z_v z_u) \subset U_\sigma, \qquad U_{\rho_v} \cap U_{\rho_w} = D(z_v z_w z_u) = U_0. \tag{1216}$$

Thus U_σ is not contained in $U_{\rho_v} \cap U_{\rho_w}$, and the coordinate discs in S_σ cannot be applied directly to the ratio e_w/e_v.

The space S_σ is nevertheless the required comparison chart. It is the closed graph

$$\{(z_v, z_w, z_u, t) \in \mathbb{C}^3 \times \Delta_\epsilon : t = z_v z_w z_u\}$$

in the Stein space $\mathbb{C}^3 \times \Delta_\epsilon$, and hence is Stein. Radial scaling of the three z-coordinates contracts it to the origin. The exponential sequence and Cartan's theorem B therefore give $\mathrm{Pic}(S_\sigma) = 0$. Choose a frame e_σ of $\mathcal{R}|_{S_\sigma}$.

Let s_v be the global frame of $\mathcal{R}|_{E_v} \simeq \mathcal{O}_{E_v}$ used to normalize e_v. On $U_{\rho_v} \cap Y_\epsilon$ put

$$A_{\sigma,v} = e_v / e_\sigma.$$

At $t = 0$, this is the restriction to $(\mathbb{C}^*)^2$ of the holomorphic unit

$$G_{\sigma,v} = \frac{s_v}{e_\sigma|_{E_v \cap S_\sigma}} \in \mathcal{O}^*(E_v \cap S_\sigma), \qquad E_v \cap S_\sigma = \{z_v = 0\} \simeq \mathbb{C}^2.$$

Consequently its winding on every horizontal cocharacter is zero at $t = 0$. Under $U_{\rho_v} \cap Y_\epsilon \simeq \Delta_\epsilon \times (\mathbb{C}^*)^2$, that winding is an integer locally constant in t, and is therefore zero on every nearby nonzero fibre. The same argument applies to $A_{\sigma,w}$. On their actual common domain $U_0 \cap Y_\epsilon$,

$$e_w / e_v = A_{\sigma,w} / A_{\sigma,v},$$

so this ratio has zero horizontal winding. The one-skeleton of the A_2-triangulation is connected, and multiplication along an edge path now gives

$$\mathrm{wind}_m(e_w / e_v) = 0 \qquad (v, w, m \in \mathbb{Z}^2) \tag{5.17-A34b}$$

on every nonzero fibre.

Let $\theta_\lambda^{\mathcal{M}} : \Psi_\lambda^* \mathcal{M} \to \mathcal{M}$ be the genuine descent isomorphism, let $\mu_\lambda^+ : \Psi_\lambda^* \mathcal{O}(D_a) \to \mathcal{O}(D_a)$ be multiplication by χ^{m_λ} from (5.17-A32), and define the inverse-dual isomorphism

$$\mu_\lambda^- := ((\mu_\lambda^+)^{-1})^\vee : \Psi_\lambda^* \mathcal{O}(-D_a) \longrightarrow \mathcal{O}(-D_a), \qquad s \longmapsto \chi^{-m_\lambda} s.$$

The ordinary dual of μ_λ^+ has the opposite arrow; the inverse-dual above is the map required by the pullback tensor product. Then

$$\theta_\lambda^{\mathcal{R}} := \theta_\lambda^{\mathcal{M}} \otimes \mu_\lambda^- : \Psi_\lambda^* \mathcal{R} \longrightarrow \mathcal{R}.$$

Since $\Psi_\lambda(U_v) = U_{v+b_\lambda}$, define $r_{\lambda,v} \in \mathcal{O}^*(U_v)$ by the typed identity

$$\theta_\lambda^{\mathcal{R}}(\Psi_\lambda^* e_{v+b_\lambda}) = r_{\lambda,v} e_v. \tag{1217}$$

At $t = 0$ it extends to an isomorphism between two trivial line bundles on the compact surface E_v. Its factor in the chosen global frames is therefore a nonzero constant. The winding of $r_{\lambda,v}(t, -)$ around every horizontal one-parameter subgroup is an integer locally constant in t; at $t = 0$ it is zero, hence it is zero for every nearby $t \neq 0$.

To express this isomorphism in one frame on the generic torus, put $q_{\lambda,v} = e_{v+b_\lambda} / e_v$. Equation (5.17-A34b) gives zero horizontal winding for $q_{\lambda,v}$; because $\Psi_\lambda|_{Y_t}$ is multiplication by a fixed torus point, it induces the identity on horizontal H_1, and $\Psi_\lambda^* q_{\lambda,v}$ also has zero winding. Relative to the single frame e_v, the residual multiplier is therefore

$$h_{\lambda,v} = \frac{r_{\lambda,v}}{\Psi_\lambda^* q_{\lambda,v}}, \qquad \mathrm{wind}_m(h_{\lambda,v}) = 0. \tag{5.17-A34c}$$

On U_0 let $\mathbf{1}_a$ be the canonical toric frame of $\mathcal{O}(D_a)$, and put $F_v = e_v \otimes \mathbf{1}_a$. Equations equation (1217) and (5.17-A32) give

$$\theta_\lambda^{\mathcal{M}}(\Psi_\lambda^* F_v) = h_{\lambda,v} \chi^{m_\lambda} F_v.$$

On a nonzero fibre, $\chi^{m_\lambda} = x^{\ell_\lambda} t^{k_\lambda}$. With the section-descent convention (5.17am), the factor of automorphy is therefore

$$\alpha_\lambda = (h_{\lambda,v}\chi^{m_\lambda})^{-1}.$$

The residual factor has zero horizontal winding by (5.17-A34c), and hence

$$\mathrm{wind}_m(\alpha_\lambda) = -\ell_\lambda(m) = -B(B_0\lambda, m). \tag{1218}$$

This winding is exactly the mixed Chern-class entry. Pull F_v back to the universal cover

$$p_t : \mathbb{C}^2 \longrightarrow Y_t, \qquad \zeta \longmapsto (e^{2\pi i\zeta}, t).$$

The period indexed by $(\lambda, n) \in \mathbb{Z}^2 \oplus \mathbb{Z}^2$ acts by $\zeta \mapsto \zeta + Z(s)\lambda + n$. The horizontal factor is one, because F_v is already defined on Y_t. Choose a logarithm $g_\lambda \in \mathcal{O}(\mathbb{C}^2)$ with

$$p_t^*\alpha_\lambda = e^{2\pi i g_\lambda}.$$

For every horizontal period n,

$$g_\lambda(\zeta + n) - g_\lambda(\zeta) = \mathrm{wind}_n(\alpha_\lambda).$$

In the exponential connecting cocycle

$$c(l, l') = g_l(\zeta + l') + g_{l'}(\zeta) - g_{l+l'}(\zeta),$$

take the horizontal logarithm to be zero and the logarithm for the combined period (λ, n) to be g_λ. Then

$$c((\lambda, 0), (0, n)) = \mathrm{wind}_n(\alpha_\lambda), \qquad c((0, n), (\lambda, 0)) = 0.$$

Therefore the alternating exponential connecting integer between the vertical deck translation indexed by λ and the horizontal loop indexed by m is

$$E_M((\lambda, 0), (0, m)) = -\ell_\lambda(m) = -B(B_0\lambda, m),$$

which proves (5.17-A21). This sign also recovers the explicit check in theorem 41.8: for $a(v) = v^{\mathsf{t}}Hv$, one has $B = 2H$, so (5.17-A21) is the cross-block $-2B_0^{\mathsf{t}}H = -2P$, namely $-2q$.

If $E_M = 0$, equation (5.17-A21) and the unimodularity of B_0 force $B = 0$. The matrix in (5.17-A26) then gives $d_3 = 0$, $d_2 = 0$, and $d_1 = 0$. Thus all six degrees of A vanish, and the Picard detection argument following (5.17-A31) gives $A \simeq \mathcal{O}_{E_0}$. This is (5.17-A22). □

Theorem 46.21 (purely analytic vanishing of the global restriction)**.** ***[VERIFIED]*** *For the constructed threefold,*

$$r = \eta^* i^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(\mathrm{dP}_6)$$

is the zero homomorphism. This proof uses only holomorphic $\mathcal{O}^*$-factors, the two finite logarithmic fillings, the cusp argument principle, and the Ψ-equivariant toric support function. It uses neither a global integral Leray transgression nor local integral nearby-cycle specialization.

Proof. Let $L \in \mathrm{Pic}(X)$. By theorem 5.3, its invariant fibre class is

$$E = c_1(L|_F) = kq, \qquad k \in \mathbb{Z}, \qquad q = uw + 6\gamma\delta. \tag{5.17-A35}$$

The exact contractions, now used without an integral specialization lattice, are

$$\iota_{v_1} q = 2\eta_1, \qquad \iota_{v_2} q = -3\eta_2, \qquad [\eta_1] = 3[\delta], \qquad [\eta_2] = 2[\delta]. \tag{5.17-A36}$$

Apply (5.17-A5) with $(m_1, m_2) = (3, 4)$. In the real global coinvariants,

$$\bar{d}_1 = 2k[\delta], \qquad \bar{d}_2 = -\frac{3}{2}k[\delta]. \tag{5.17-A37}$$

Equations (5.17-A4) and (5.17-A6) therefore give

$$0 = \bar{d}_1 + \bar{d}_2 + \bar{d}_0 = \frac{k}{2}[\delta]. \tag{5.17-A38}$$

Since $V_G \otimes \mathbb{R} = \mathbb{R}[\delta]$, one has $k = 0$, so $E = 0$. Apply theorem 46.20 to $M = L|_{N_0}$. It gives

$$\eta^* i^* L = \nu^*(L|_W) \simeq \mathcal{O}_{E_0}.$$

This holds for every L, proving $r = 0$. □

Corollary 46.22 (candidate nonvanishing for Pic^0 without topology)**.** ***[VERIFIED]*** *For the constructed threefold and every $L \in \mathrm{Pic}^0(X)$,*

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0, \qquad H^2(W, (T_X \otimes L)|_W) \neq 0,$$

and

$$R^2 f_*(T_X \otimes L) \neq 0. \tag{5.17f}$$

This proof uses the analytic construction of the cusp fibre but no global homology, Mayer–Vietoris, or Leray calculation.

Proof. The cusp theorem theorem 8.4 supplies the full compact reduced normal-crossings fibre, its dP_6 normalization, and the trivialization of ω_X near W. By theorem 46.12, $\eta^* A_L$ is trivial, so theorem 46.5 gives the first nonvanishing. Apply theorem 46.8 to $E = T_X \otimes L$; its dual space is exactly the preceding nonzero section space, giving the fibrewise H^2 nonvanishing. The fibre has complex dimension two, so its coherent H^3 vanishes. The global construction makes X compact and f nonconstant; theorem 46.9 therefore gives properness and f-flatness of $T_X \otimes L$. The base-change isomorphism of theorem 46.10 now identifies the fibre of the R^2 stalk with the nonzero H^2 group. Hence the stalk, and therefore the direct-image sheaf, is nonzero. □

Remark 46.23 (reconciliation of the three restriction routes). The Pic^0 argument proves (5.17f) without global homology, Mayer–Vietoris, or Leray. The all-Pic statement theorem 46.14 avoids citing the packaged theorem $H^2(X; \mathbb{Z}) = 0$, but it reuses the same verified global integral Leray and specialization machinery from which that theorem is deduced. By contrast, theorem 46.18 supplies a second all-Pic proof without any global Leray differential: it recovers the primitive $4 - 3 = 1$ obstruction directly from the finite logarithmic gluing and the cusp boundary. It still uses the local integral cusp-specialization theorem. The third proof in theorem 46.21 replaces both that specialization and the cusp Chern-class bounding-chain evaluation by logarithmic $\mathcal{O}^*$ factors, the argument principle, and the exact Ψ-equivariant support function. Thus the purely analytic strengthening is now complete; the three proofs have genuinely different dependency sets.

46.8 Abstract conductor theorem for the candidate fibre

Theorem 46.24 (the candidate conductor mechanism, with all hypotheses)**.** ***[VERIFIED]*** *Assume:*

(H1) X is a connected compact smooth complex threefold and $f : X \to B$ is a nonconstant holomorphic map to a smooth compact curve;

(H2) $W = f^{-1}(p_0)$ is the full scheme-theoretic fibre, is reduced, and has the normal-crossings local models (5.3);

(H3) the normalization $\eta : \widetilde{W} \to W$ is the degree-six del Pezzo surface dP_6*;*

(H4) $H^2(X;\mathbb{Z}) = 0$.

Then, for every $L \in \mathrm{Pic}(X)$*,*

$$H^2(W, (T_X \otimes L)|_W) \neq 0 \quad \textit{and} \quad R^2 f_*(T_X \otimes L) \neq 0. \tag{5.18}$$

Proof. Fix L and put

$$\mathcal{F}_L = T_X \otimes L, \qquad A_L = (L^\vee \otimes \omega_X)|_W.$$

Hypothesis (H4) gives $c_1(L^\vee \otimes \omega_X) = 0$. Pullback therefore gives $c_1(\eta^* A_L) = 0$. The surface dP_6 is the blow-up of $\mathbb{P}^2$ at three noncollinear points; the standard blow-up calculation gives $H^1(\mathrm{dP}_6, \mathcal{O}) = 0$ [17, Chapter V]. In the exponential sequence, the kernel of

$$c_1 : \mathrm{Pic}(\mathrm{dP}_6) \longrightarrow H^2(\mathrm{dP}_6; \mathbb{Z})$$

is the image of $H^1(\mathrm{dP}_6, \mathcal{O})$ and is therefore zero. Consequently

$$\eta^* A_L \simeq \mathcal{O}_{\widetilde{W}}. \tag{5.19}$$

Theorem 46.5 now gives

$$H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0. \tag{5.20}$$

Because $W = f^*(p_0)$, one has $\mathcal{O}_X(W) = f^*\mathcal{O}_B(p_0)$, whose restriction to the fibre is trivial. Adjunction therefore gives $\omega_W \simeq \omega_X|_W$. Apply theorem 46.7 to $E = \mathcal{F}_L$:

$$\begin{aligned} H^2(W, \mathcal{F}_L|_W)^\vee &\simeq H^0(W, (\mathcal{F}_L|_W)^\vee \otimes \omega_W) \\ &= H^0(W, \Omega^1_X|_W \otimes A_L). \end{aligned}$$

The right side is nonzero by (5.20), proving the first assertion of (5.18). The same duality theorem gives $H^3(W, \mathcal{F}_L|_W) = 0$.

Since X is compact, f is proper. By theorem 46.9, f is flat and the vector bundle $\mathcal{F}_L$ is f-flat. Thus every hypothesis of theorem 46.10 is satisfied at p_0, and

$$(R^2 f_* \mathcal{F}_L)_{p_0} \otimes \mathbb{C}(p_0) \simeq H^2(W, \mathcal{F}_L|_W) \neq 0.$$

The stalk, and hence the coherent direct-image sheaf, is nonzero. This is the second assertion of (5.18). □

Corollary 46.25 (application to the constructed threefold)**.** ***[VERIFIED]*** *For the global construction of theorem 8.9 and every* $L \in \mathrm{Pic}(X)$*,*

$$H^2(W, (T_X \otimes L)|_W) \neq 0, \qquad R^2 f_*(T_X \otimes L) \neq 0.$$

Proof. The cusp construction gives the compact full Cartier fibre, its dP_6 normalization, and the local trivialization of ω_X. By the local canonical trivialization and theorem 46.18, $\eta^* A_L$ is trivial for every L. The conductor proposition then gives $H^0(W, \Omega^1_X|_W \otimes A_L) \neq 0$; absolute Cartier-fibre duality turns this into the displayed H^2 nonvanishing, and the uniform top-degree base-change isomorphism turns it into a nonzero R^2 stalk. This is exactly the chain proved in theorem 46.22, now with the all-Pic restriction supplied directly by theorem 46.18. The global Leray-edge proof in theorem 46.14 is an independent certificate of the same restriction statement. □

46.9 A second printed issue: abelianization of split extensions

Lemma 46.26 (abelianization of a semidirect product)**.** ***[VERIFIED]*** *If* $1 \to K \to G \to Q \to 1$ *is split, then*

$$G^{\mathrm{ab}} \simeq Q^{\mathrm{ab}} \oplus (K^{\mathrm{ab}})_Q, \tag{5.21}$$

where the subscript denotes coinvariants for the conjugation action of Q*.*

Proof. Choose a splitting and write $G = K \rtimes Q$. Let $N \lhd K$ be generated by $[K, K]$ and by all elements $qkq^{-1}k^{-1}$ with $q \in Q$, $k \in K$. It is normal in G: $[K, K]$ makes it stable under conjugation by K, while conjugation by Q permutes the displayed cross-commutators. By construction $K/N = (K^{\mathrm{ab}})_Q$, and Q acts trivially on this quotient. Hence

$$G/N \simeq (K^{\mathrm{ab}})_Q \times Q.$$

Since $N \subseteq [G, G]$, abelianizing the last display gives (5.21). □

The sentence in [3, Lemma 3.2, pp. 84–85] that passes from a split fundamental-group extension to a direct sum with $H_1(F; \mathbb{Z})$ is therefore valid only when monodromy acts trivially after abelianization. A minimal counterexample retaining a free quotient group is

$$G = \mathbb{Z} \rtimes_{-1} \mathbb{Z}.$$

The extension splits, but $G^{\mathrm{ab}} \simeq \mathbb{Z} \oplus \mathbb{Z}/2$, whereas the printed direct-product formula gives $\mathbb{Z}^2$. For the candidate's printed monodromy, theorem 5.3 gives

$$H^1(F; \mathbb{Q})^{\pi_1(B^\circ)} = \mathbb{Q}\gamma,$$

so the rational coinvariant rank is one, not $b_1(F) = 4$.

If C° is a sphere punctured at s points, the correct rank formula is

$$b_1(X^\circ) = s - 1 + \dim_{\mathbb{Q}} H_1(F; \mathbb{Q})_{\pi_1(C^\circ)}. \tag{5.22}$$

Thus the proof of CDP98 Theorem 3.1 reaches only the coinvariant rank unless an additional theorem proves trivial monodromy. CDP20 Lemma 4.2 repeats the direct-sum formula, and CDP20 Corollary 5.2 imports CDP98 Theorem 3.1. Repairing the local-freeness issue discussed there does not repair this distinct monodromy error. The direct-product step in the torus branch of the CDP20 proof is therefore refuted, independently of the singular-fibre defect above.

46.10 The corrected theorem that survives both defects

The original hypotheses do not force the CDP conclusion: the constructed threefold above is a counterexample. There is nevertheless a sharp useful replacement. It isolates the exact extra global vanishing needed by the intact part of the argument, and the following fibrewise criterion replaces the invalid p. 692 reduction.

Lemma 46.27 (generic endpoint vanishing)**.** ***[VERIFIED]*** *Let* Y *be a connected compact complex manifold, let* D *be a nonzero effective Cartier divisor, and suppose* $\mathcal{O}_Y(D) \in \mathrm{Pic}^0(Y)$*. For every holomorphic vector bundle* B*, a general* $P \in \mathrm{Pic}^0(Y)$ *satisfies*

$$H^0(Y, B \otimes P) = 0, \qquad H^{\dim Y}(Y, B \otimes P) = 0. \tag{5.23}$$

Proof. The inclusions $\mathcal{O}_Y(-(k+1)D) \hookrightarrow \mathcal{O}_Y(-kD)$ give a descending chain

$$V_k = H^0(Y, B \otimes \mathcal{O}_Y(-kD)) \subseteq H^0(Y, B).$$

It stabilizes because the last vector space is finite-dimensional. A section in the stable value belongs to every V_k. In a local equation $z = 0$ for D, every coefficient of that section is divisible by z^k for all k, and hence is zero. The section vanishes on a neighbourhood of D, and therefore on connected Y by analytic continuation. Thus $V_k = 0$ for some k. Since $\mathcal{O}_Y(-kD) \in \mathrm{Pic}^0(Y)$, upper semicontinuity in a local universal family of topologically trivial line bundles shows that

$$\{P : h^0(Y, B \otimes P) > 0\}$$

is a proper analytic subset of $\mathrm{Pic}^0(Y)$. This proves the first vanishing for general P. Serre duality identifies the dual of the second group in (5.23) with

$$H^0(Y, B^\vee \otimes \omega_Y \otimes P^\vee).$$

Apply the first part to $B^\vee \otimes \omega_Y$ and use that inversion is an automorphism of $\mathrm{Pic}^0(Y)$. □

Theorem 46.28 (corrected CDP global-vanishing criterion)**.** ***[VERIFIED]*** *Let X be a compact connected complex threefold such that*

$$H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0, \qquad a(X) = 1,$$

and suppose its algebraic reduction is a holomorphic map $f : X \to C$. If there exists $L_0 \in \mathrm{Pic}(X)$ with

$$H^2(X, T_X \otimes L_0) = 0, \tag{5.24}$$

then, for M in a dense open subset of $\mathrm{Pic}(X)$*,*

$$H^i(X, T_X \otimes M) = 0 \quad (i \neq 1), \tag{5.25}$$

$$\chi(X, T_X \otimes M) \leq 0, \qquad \langle c_3(T_X), [X] \rangle \leq 0. \tag{5.26}$$

Proof. The exponential sequence and the two integral vanishings give

$$\mathrm{Pic}(X) = \mathrm{Pic}^0(X) \simeq H^1(X, \mathcal{O}_X). \tag{5.27}$$

Thus this Picard group is a connected irreducible complex vector group. Any fibre divisor D of f is nonzero and effective, and its line bundle lies in $\mathrm{Pic}^0(X)$ by $H^2(X;\mathbb{Z}) = 0$. Applying theorem 46.27 to $B = T_X$ shows that the degree-zero and degree-three groups in (5.25) vanish away from two proper analytic subsets of $\mathrm{Pic}(X)$.

The cohomology-jump locus

$$\{M \in \mathrm{Pic}(X) : h^2(X, T_X \otimes M) \geq 1\}$$

is analytic by upper semicontinuity. It is proper because (5.24) gives a point outside it. Its complement therefore meets the two endpoint vanishing loci in a dense open set. This proves (5.25), and on that set

$$\chi(X, T_X \otimes M) = -h^1(X, T_X \otimes M) \leq 0. \tag{5.28}$$

Every $M \in \mathrm{Pic}(X)$ has rationally trivial first Chern class. The hypotheses also give $c_1(T_X) = 0$ in rational cohomology and, by Poincaré duality, $H^4(X;\mathbb{Q}) = 0$, so $c_2(T_X) = 0$ rationally. The degree-six term of $\mathrm{ch}(T_X \otimes M)\,\mathrm{td}(X)$ is consequently $\frac{1}{2}c_3(T_X)$. Hirzebruch–Riemann–Roch turns (5.28) into

$$\chi(X, T_X \otimes M) = \frac{1}{2}\,\langle c_3(T_X), [X] \rangle \leq 0,$$

which proves (5.26). □

Corollary 46.29 (intact relative criterion)**.** ***[VERIFIED]*** *In the setting of theorem 46.28, it is enough that one line bundle $L \in \mathrm{Pic}(X)$ satisfy*

$$R^2 f_*(T_X \otimes L) = 0. \tag{5.29}$$

This is the valid content of the sufficient criterion printed as [2, Proposition 2.4, p. 680].

Proof. Let A be a positive-degree line bundle on the compact Riemann surface C, and put $E = T_X \otimes L$. Projection formula gives

$$R^2 f_*(E \otimes f^* A^m) = R^2 f_* E \otimes A^m = 0.$$

For $m \gg 0$, Serre vanishing on C gives

$$H^1(C, R^1 f_* E \otimes A^m) = 0.$$

The Leray spectral sequence has only columns zero and one, so its total-degree-two terms yield

$$H^2(X, E \otimes f^* A^m) = 0.$$

Thus $L_0 = L \otimes f^* A^m$ satisfies (5.24), and theorem 46.28 applies. □

Theorem 46.30 (exact reduced-fibre repair of the p. 692 step)**.** ***[VERIFIED]*** *Retain the setting of theorem 46.28, assume every full scheme-theoretic fibre $W_c = f^{-1}(c)$ is reduced, and fix $L \in \mathrm{Pic}(X)$. Put*

$$A_c = (\omega_X \otimes L^\vee)|_{W_c}, \qquad \mathscr{T}_c = \ker\Big(\Omega^1_{W_c} \longrightarrow j_* \Omega^1_{(W_c)_{\mathrm{reg}}}\Big).$$

For each c, let

$$\delta_c : H^0(W_c, \mathscr{T}_c \otimes A_c) \longrightarrow H^1(W_c, N^*_{W_c/X} \otimes A_c) \tag{5.30}$$

be the connecting map from the missing cotangent filtration. If, for every c,

$$H^0(W_c, \widetilde{\Omega}^1_{W_c} \otimes A_c) = 0, \qquad H^0(W_c, N^*_{W_c/X} \otimes A_c) = 0, \qquad \delta_c \text{ is injective}, \tag{5.31}$$

then (5.29), (5.25), and (5.26) hold.

Proof. Apply theorem 46.2 to the whole reduced Cartier fibre W_c, not separately to its components. Conditions (5.31) give

$$H^0(W_c, \Omega^1_X|_{W_c} \otimes A_c) = \ker \delta_c = 0. \tag{5.32}$$

The map f is flat: a local parameter at c pulls back to a non-zero-divisor in every regular local ring of the connected manifold X, and torsion-freeness over the base DVR is flatness. Moreover, the normal line of the full fibre is the pullback of the one-dimensional vector space $T_c C$, hence is trivial. Adjunction therefore identifies $\omega_{W_c} \simeq \omega_X|_{W_c}$. With $E = T_X \otimes L$, absolute Cartier-fibre duality now gives

$$H^2(W_c, E|_{W_c})^\vee \simeq H^0(W_c, \Omega^1_X|_{W_c} \otimes A_c) = 0.$$

The uniform top-degree base-change isomorphism of theorem 46.11 therefore gives

$$(R^2 f_* E)_c \otimes_{\mathcal{O}_{C,c}} \mathbb{C}(c) = 0$$

for every c. Nakayama's lemma kills every stalk of the coherent sheaf $R^2 f_* E$, proving (5.29). The remaining conclusions follow from theorem 46.29. □

Remark 46.31 (scope and sharpness of the repair). Under the two endpoint vanishings in (5.31), injectivity of δ_c is equivalent to the desired ambient vanishing (5.32); it is not a politeness assumption hiding the original claim. For a reducible reduced fibre the map must be computed on the whole Cartier fibre. A nonreduced fibre requires additional nilpotent-thickening layers and is not covered by theorem 46.30. The global criterion theorem 46.28 and the relative criterion theorem 46.29 do allow nonreduced fibres.

The reconstructed candidate violates both added conditions rather than being silently excluded by terminology. Since $c_3(X) = 2$, the contrapositive of theorem 46.28 proves

$$H^2(X, T_X \otimes L_0) \neq 0 \quad \text{for every } L_0 \in \operatorname{Pic}(X).$$

Independently, theorem 46.25 proves

$$R^2 f_*(T_X \otimes L) \neq 0 \quad \text{for every } L \in \operatorname{Pic}(X),$$

and its cusp conductor supplies a nonzero element of the relevant $\ker \delta_c$. The first conclusion rules out the global hypothesis; the second independently rules out the stronger relative sufficient condition. Thus the corrected theorem is compatible with, and precisely separates, the verified counterexample to CDP20 Theorem 2.2 as stated.

The exact monodromy repair. The split-extension argument can reach only the invariant integral lattice $H^1(F;\mathbb{Z})^{\pi_1(C^\circ)}$. Its final complex-linear inference is valid precisely when

$$\operatorname{Span}_{\mathbb{C}} \operatorname{im}\Big(H^1(F;\mathbb{Z})^{\pi_1(C^\circ)} \longrightarrow H^1(F, \mathcal{O}_F)\Big) = H^1(F, \mathcal{O}_F). \tag{5.33}$$

Trivial monodromy on $H_1(F;\mathbb{Z})$ is a stronger sufficient hypothesis. Neither (5.33) nor trivial monodromy follows from the hypotheses of CDP20 Theorem 2.2. The corrected theorem above bypasses this rank route entirely; it does not relabel the false direct-sum formula as proved.

46.11 Exact present disposition

1. **[REFUTED]** The p. 692 reduction in [2] is not a formal consequence of the two cotangent sequences. Theorem 46.6 is a proper geometric counterexample to that exact sheaf-level inference for a non-torsion bundle on the central fibre; no extension of that bundle to the displayed total space is claimed.

2. **[REFUTED]** The direct-product homology formula in [3, Lemma 3.2] omits monodromy coinvariants. This leaves the torus branch imported by [2] without its printed rank argument.

3. **[VERIFIED]** The normalization–conductor construction, Cartier-fibre duality, flatness over the base curve, and top-degree analytic base change are proved above under their stated hypotheses.

4. **[VERIFIED]** The period-family descent, toric and finite fillings, global gluing, and integral analytic nearby-cycle/Leray bridges are proved in sections 8 and 39. Therefore the candidate's hypotheses (H1)–(H4) hold and $R^2 f_*(T_X \otimes L) \neq 0$ for every L by theorem 46.25.

5. **[VERIFIED]** Those same construction and topology results, together with theorem 40.4, hold for the constructed compact complex threefold. Hence $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$, $a(X) = 1$, f is a holomorphic algebraic reduction, and $c_3(X) = 2$. The resulting X satisfies every stated hypothesis of CDP Theorem 2.2 and contradicts its conclusion; it is therefore a counterexample to that theorem as stated. This theorem-level conclusion uses the complete construction and topology proofs; it is not inferred from the two printed proof defects alone.

6. **[VERIFIED]** The corrected global criterion theorem 46.28 bypasses both refuted dependencies: existence of one L_0 with $H^2(X, T_X \otimes L_0) = 0$ implies generic off-degree-one vanishing and $c_3(X) \leq 0$. The intact relative sufficient condition is $R^2 f_*(T_X \otimes L) = 0$, and theorem 46.30 replaces the invalid reduced-fibre step by the exact injectivity condition on δ_c. The original theorem without these added hypotheses remains refuted by the constructed threefold. Independent specialist replication remains desirable as external adjudication, but it is neither a mathematical hypothesis nor a gate on the present proof program.

Validation, repair, or disproof test 46.32 (decisive theorem-level outcomes)**.** The current workbench records two exact defects in the printed CDP chain, a completed route from the constructed threefold to a theorem-level counterexample, and the strongest corrected conclusion justified by the intact argument. The global vanishing in theorem 46.28 is false for every L_0 by contraposition from $c_3(X) = 2$. Independently, the cusp-conductor calculation forces $R^2 f_*(T_X \otimes L) \neq 0$ for every L, so the stronger relative sufficient condition fails as well. Thus the repair does not contradict the counterexample or recover the original theorem by renaming its conclusion. Any further attempt to prove the original statement must falsify a named step in the candidate reconstruction or derive one of the new vanishing hypotheses from genuinely stronger geometry. Merely invoking the published theorem does neither.

47 Confirmed local repairs kept separate from the theorem

These corrections are mathematically definite but do not settle the central construction.

1. **[VERIFIED]** Proposition 4.7(iv)'s closed-image conclusion is correct, although its printed justification is not. If $s : B \to E$ is the section of $f : E \to B$, then

$$s(B) = \{x \in E : s(f(x)) = x\},$$

the equalizer of id_E and $s \circ f$. Since E is Hausdorff, its diagonal is closed, so this equalizer is closed. Thus the explicit holomorphic section, its cusp disc, and the null-homotopy used in theorems 39.3 and 39.6 require no additional chartwise or local-finiteness hypothesis.

2. **[REFUTED]** Remark 7.28 is false for a fixed triple. For example,

$$(\ell_0, \ell_1, \ell_2) = (100, 1, -1), \qquad p = 12(100) - 4(1) - 3(-1) = 1199 \neq \pm 1.$$

The intended statement must quantify existentially over ℓ_0 for fixed admissible (ℓ_1, ℓ_2).

3. **[VERIFIED]** Appendix Lemma A.3 prints cohomological arrows while labelling them as homology. The sequence must be written

$$H^k(W) \longrightarrow H^k(\widetilde{W}) \oplus H^k(D) \longrightarrow H^k(\widetilde{D}) \longrightarrow H^{k+1}(W),$$

followed by universal coefficients when homology ranks are required.

4. **[VERIFIED]** Appendix Remark A.6 must read $\pi_1(N_0) = \bar{\Lambda} = \pi_1(W)$, not $\pi_1(N_0) = \Lambda$.

5. **[VERIFIED]** The modular assembler has a page-71 portability defect under the tested MiKTEX route; the canonical monolithic source renders correctly. This is a build defect, not a mathematical counterexample.

6. **[REFUTED]** The manuscript's assertion that $\Omega^1_{X/B}$ has torsion at all three singular fibres is false literally. The explicit $t = xy$ and $t = xyz$ modules show that it is torsion-free but non-locally-free at the cusp; torsion occurs only at the two multiple fibres.

7. **[VERIFIED]** In the finite-filling Hodge-line computation, the invariant canonical section

$$\sigma_j = s_j^{m_j-1-a_j} e^{\varphi(s_j)} ds_j \wedge e$$

has divisor $(m_j - 1 - a_j)S_j$ after descent and therefore is not a nowhere-vanishing frame when $j = 2$. The correct direct-image statement is the invariant-module identity

$$f_{j*}\omega_{N_j/D_j} = \mathcal{O}_{D_j} \cdot s_j^{m_j-1-a_j} \vartheta'_j, \qquad \vartheta'_j = e^{\varphi(s_j)}(ds_j \wedge e) \otimes (dt_j)^{-1}.$$

On the punctured cover its generator is a unit times $s_j^{-a_j} e$. This repairs the local wording while leaving $f_*\omega_{X/B} \simeq \mathcal{O}_{\mathbb{P}^1}(1)$, the structure-sheaf direct images, and uniform non-torsion unchanged.

48 Tool and computational resource disclosure

The source manuscript available for this audit was reconstructed from a PDF in ten consecutive source strips—eight strips of eleven pages and two of ten pages—in ChatGPT Professional-mode sessions and then consolidated into a single LaTeX source. Two separate ChatGPT Professional-mode sessions produced the first full-manuscript proof audits. The present consolidation, citation audit, exact-arithmetic reproduction, and validate/repair/disprove workbench were prepared with Codex using ChatGPT 5.6 Sol in Ultra mode. The exact computations reported here use integer and rational arithmetic; their mathematical certificates are reproduced in the body rather than delegated to software output.

These systems are disclosed as tools, not treated as mathematical authorities or authors. Every imported claim is checked against a publicly resolvable human source, and every new assertion remains subject to human verification. This disclosure and the point-of-use citation policy follow the recommendations of the Leiden Declaration on Artificial Intelligence and Mathematics [1]. Responsibility for any version submitted or published, including the adequacy of its proofs and references, belongs to the human author or editor who releases that version.

The explicit candidate and its original theorem claims remain attributed to the reported Claude/Fabel source. The purpose of the present work is narrower but substantive: to make the construction's load-bearing maps and calculations explicit, reproduce them independently where possible, repair identified proof gaps when the geometry permits, and isolate the exact boundary of what still requires outside adjudication. This is offered as a verification aid that may save expert reconstruction time; it is not a claim of priority for the candidate, a transfer of authorship, or a substitute for specialist review.

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